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New NCERT Textbook (Edition 2018-19) Exercises with Solutions
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WITH MIND MAPS

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Includes :

- Previous Years' Questions
- Mind Maps
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WITH MIND MAPS

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OSWAAL LEARNING TOOLS

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PREFACE

In order to maintain a **Uniform Education System** across the country, many **State Boards** are implementing a **Single Curriculum System** via National Council of Educational Research & Training **(NCERT).** In the best interest of Secondary and Senior Secondary students, the Department of Education in Science & Mathematics (DESM) and NCERT have together developed Exemplar Problems in Science & Mathematics. These include practice questions of various typologies and difficulty levels which aid in in depth learning of concepts. They also contain conceptual problems which are a part of the CBSE Board Syllabus as well as the Syllabus of various Competitive Exams like IIT JEE, NEET, AIIMS, etc.

Considering the diversity and the varied difficulty level of questions in these two subjects, we, at **Oswaal Books** have launched **OSWAAL NCERT SOLUTIONS.** They are a compilation of all the Questions of the **Latest Editions of NCERT Textbook & NCERT Exemplar** in a Chapter-wise & Topic-wise format along with their complete solutions. These also include **Previous Years' CBSE Board Examination Questions** fully solved with their respective sources.

We believe that **OSWAAL NCERT SOLUTIONS** will help the students in school and after school in practicing and preparing extensively for both, Final Examinations as well as Competitive Examinations with utmost confidence!

Some Special Features of Oswaal NCERT Solutions are:

- ***Chapter-wise & Topic-wise*** presentation
- ***Chapter Objectives :*** A sneak peek into the chapter
- ***Mind Map :*** A single page snapshot of the entire chapter
- ***Quick Review :*** Concept-based study material
- ***Tips & Tricks :*** Useful guidelines for attempting each question perfectly
- ***Some Commonly Made Errors :*** Most common and unidentified errors made by students discussed
- ***Expert Advice :*** Oswaal Expert Advice on how to score more!
- ***Oswaal QR Codes :*** For Quick Revision on your Mobile Phones & Tablets

We hope that **OSWAAL NCERT SOLUTIONS** will help you at every step as you move closer to your educational goals. We wish you all great success ahead!!

All the Best !

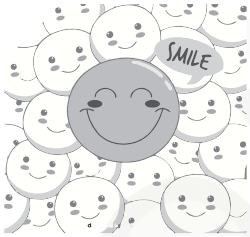
TEAM OSWAAL

Be Happy - It's all that matters !!

Students, remember that most of your stress comes from the way that you respond to different situations in life. This includes preparing yourself for your school/board examinations. We at Oswaal Books bring you our Success Mantra that will train you to adjust your attitude and get rid of all that unnecessary stress.

Begin with a positive attitude

You must feel blessed that your life has a meaning. You should be privileged to have access to unlimited knowledge, good books, good teachers and loving parents.

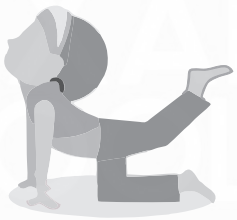


Be happy & believe in yourself

A happy child learns better and performs better. If you believe in yourself you can be more helpful towards your fellow students/classmates and most importantly, you will realize that you have the courage to step over your insecurities and succeed.

Being physically active

Active students are happier, healthier and more satisfied with life. Children must play outdoor games in order to get stronger muscles & bone, increased stamina, better sleep and a sharp mind.



Perseverance & Passion

These are the two secrets of academic success. Gritty students are more likely to achieve their goals before others.

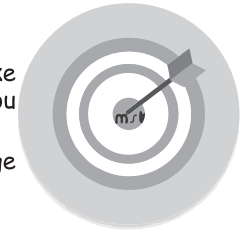
How to be Grittier

(a) Pick your Goal.

(b) Make mistakes - Every mistake made increases your chance to make progress. No matter how many mistakes you make or how slow you progress, you are still ahead of everyone who is not trying

(c) Accept that failures are a part of hardwork. Don't let them discourage you.

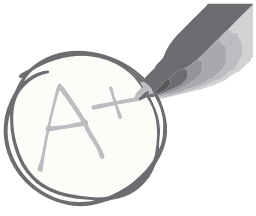
(d) Never give up.



Focus on studying, rather than getting good grades

Why?

Students who study to understand do better than those who aim at getting good scores. Students who study to understand hit the books more often, approach various information sources, and feel less anxious about examinations. If, as a student, you keep in mind the above aspects it would definitely keep the negative emotions at bay and add to your happiness level.



Writing a Perfect Answer Simplified !

OSWAAL EXAM TOOLS

Imagine you are seated in the exam hall, waiting to get your hands on that much anticipated piece of paper- THE EXAMINATION PAPER!! You are loaded with a ton of information in your brain. Your finger tips are numb and your face blank! It is the time to get down to serious business!

As another Exam Tool, Oswaal Books teaches you how to give every Question your best shot and write the PERFECT ANSWER everytime!!

- Student should write the data / information as given in the question carefully

- Student should write the concept / formula which is applicable in the given question
- If the question asked is related to a particular law / phenomena, then the student should write the name of that law / phenomena carefully

- Check the nature of the question –

  If the question is based on a **Theoretical Concept** :

  - ➢ After writing all the details given in the question, the student should be able to identify the concept applicable in the question
  - ➢ The student should then explain all the required points as asked in given question

  If the question is based on a **Numerical** :

  - ➢ Using the required Formula, the student should insert the particular data / values which are given / assumed
  - ➢ Then the numerical can be solved to get the final answer

We would love to hear from you !

We believe that this book will make your Learning Simple. There are a lot of ways in which you can share your thoughts and experiences about this book with millions of current and future students. Here is how—

- ☞ You can log on to **Amazon.in**, search for this book and write a Review.
- ☞ You can search for Oswaal Books on **Google** and write a Review there.
- ☞ You can also share it on your **Facebook Page**
- ☞ You can email us your suggestions / feedback on **contact@oswaalbooks.com**

POSITIVE AFFIRMATIONS

Our mind starts believing what we repeatedly think or say. We, at Oswaal Books resonate with this belief. So, we want all our readers to create their own affirmations ! A positive affirmation is something spoken aloud that you want to believe or want to be true. Repeating positive affirmations daily can help shift your internal dialogue from negative to positive.

So lets get started !

1. I enjoy the subject I am studying.
2. During exams, I recall information quickly and easily.
3. Getting good grades is natural for me.
4. .....
5. .....
6. .....
7. .....
8. .....
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16. .....
17. .....
18. .....
19. .....
20. .....

MIND MAPS : LEARNING MADE SIMPLE

Learning Made Simple
"A Winning Combination"

Result

Presenting words / concepts
as pictures !!

What ?

When ?

Anytime, as frequently as
you like, till it becomes a habit !

Mind Map
An Interactive Magical Tool

How ?

With a blank sheet of paper,
coloured pens &
your creative imagination !!

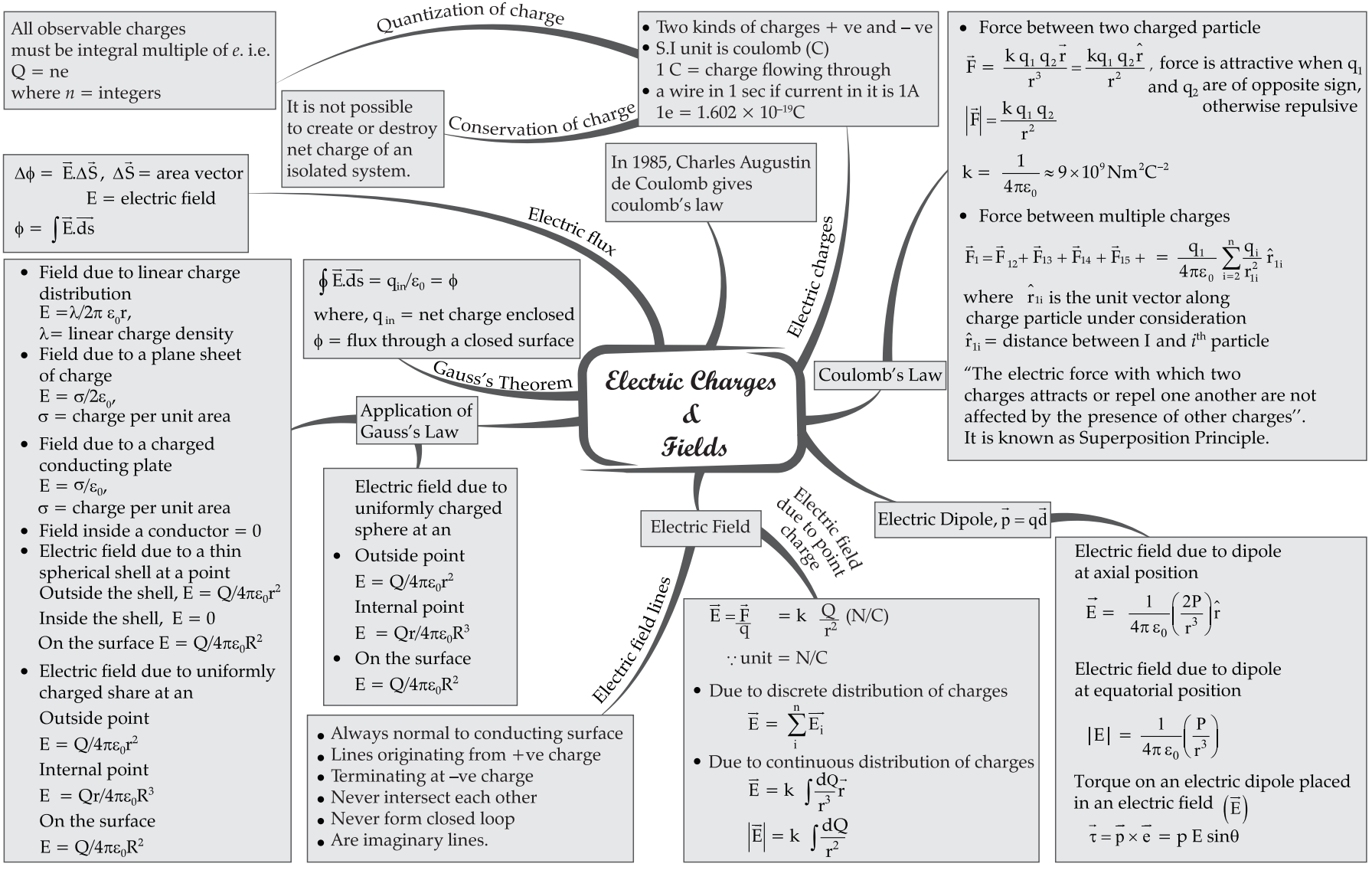
Why ?

Helps
- Study
- Learn
- Memorize
- Revise
- Plan &
- Create

"Mind Maps" are a novel feature of these Question Banks. Mind maps have been given for each chapter in the beginning, to help students understand all the topics & concepts better.

MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 1



MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 2

Electrostatic Potential & Capacitance

Electric Potential

- Work done per unit test charge by an external agent in moving the test charge from reference point to the desired point S.I. unit J/C
  V_0 = Work done/charge

Electric potential due to a point charge

$V_P = Q/4\pi\varepsilon_0 r$

Potential due to a system of charges

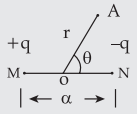
$$V = \frac{1}{4\pi\varepsilon_0}\sum\frac{Q_i}{r_i}$$

Potential due to a dipole

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

Where $p = qd$

$\theta = \angle AON$



At $\theta = 0$, $V = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$ [axial position]

At, $\theta = 90°$, $V = 0$ [equitorial position]

In 1774, Alessandro Volta wrote treatise '' on the forces of attraction of electric fire''

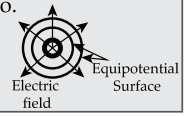
Conductors and Insulators

Conductors	Insulators
Such a material Which when placed in an electric field, the free electrons move in a direction opposite to the field.	Such a material in which electrons are tightly bound, & when exposed in an electric field, Electrons does not move *i.e.* having no free electrons

- Electric field inside a conductor is zero
- Electric field is always perpendicular to the charged surface
- In static state, there will be no additional charge in a conductor

Equipotential Surface

- Potential is same at all the points of the surface
- Component of electric field parallel to an equipotential surface is zero.



Dielectric

- $F = \frac{q_1 q_2}{4\pi K\varepsilon_0 r^2}$, K = dielectric constant of medium
- A dielectric is an electrical insulator that can be polarized by an applied electric field.

Capacitance, $C = \frac{Q}{V}$

Capacitance of a parallel plate capacitor
$C = K\varepsilon_0 A/d$, K = dielectric constant

Capacitance when material slab inserted between them
$C = K\varepsilon_0 A/[Kd - x(K-1)]$
where x = thickness of the slab inserted

Capacitance of a spherical capacitor
$C = 4\pi\varepsilon_0/\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
For isolated sphere
$C = 4\pi\varepsilon_0 R$

Energy stored in a capacitor
$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$

Series grouping of capacitors
$\frac{1}{C} = \frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+---+\frac{1}{C_n}$
For two, $C = \frac{C_1C_2}{C_1+C_2}$

Parallel grouping of capacitors
$C = C_1 + C_2 + C_3 + ---- + C_n$
for two, $C = C_1 + C_2$

Electric Potential Energy

- It is negative of work done by the electric force as the configuration of the system changes.
- $Ur_2 - Ur_1 = -W = \frac{q_1 \cdot q_2}{4\pi\varepsilon_0}\left(\frac{1}{r_2}-\frac{1}{r_1}\right)$
- If the separation between charges is 'r' then $U_{(r)} = \frac{q_1 q_2}{4\pi\varepsilon r}$
- Potential Difference,
  $V_B - V_A = \frac{U_B - U_A}{q}$
  $U_B - U_A$ = change in Potential Energy
  q = test charge

Potential Energy of a dipole

$dU = pE\sin\theta d\theta$
If we choose P.E. of dipole to be zero when
$\theta = 90°$ then
$U_\theta - U_{90°} = \int_{90°}^{\theta} pE\sin\theta d\theta$
$U_\theta = -pE\cos\theta = -\vec{p}\times\vec{E}$
If it is rotated through angle θ against the torque

MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 3

Current Electricity

Current density

$\vec{J} = \frac{\Delta i}{\Delta s \cos\theta} = \sigma E$

Δs = area of cross-section

θ = angle between Area vector & current flow

σ = conductivity

Electric Current

Rate of flow of charge through an area.

$i = \frac{Q}{t}$

$i = \int \vec{J}.\vec{ds}$

In 1827 Georg Simon Ohm gave his complete theory of electricity in his book "Die galvanische Kette, mathematisch bearbeitet

Drift velocity

$V_d = \frac{i}{neA} = \frac{J}{ne}$

Mobility

$\mu = \frac{V_d}{E} = \frac{e\tau}{m}$

τ = average collision time

Ohm's Law

$J = \sigma E$ or $V = IR$

Resistance

$R = \rho \frac{l}{A} \tau$

Resistivity

$\rho = \frac{1}{\sigma} \Omega m$

Conductivity

$\sigma = J/E$

Electric cell

It is a device which maintains a potential difference between its two terminals.

E MF

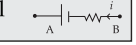
emf = $E = \frac{W}{q} = \frac{F_b}{d}$

It is equal to the potential difference between the terminals when the terminals are not connected externally.

Internal Resistance

$V_A - V_B = E - ir$

r = internal resistance of the cell



Grouping of Cell

Parallel grouping

$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$, $i = \frac{E_{eq}}{R + \frac{r_1 r_2}{r_1 + r_2}}$

Series grouping

$i = \frac{E_1 + E_2}{R + (r_1 + r_2)} = \frac{E_0}{R_1 + R_2}$

$E_0 = E_1 + E_2$

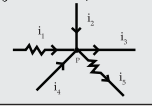
Kirchhoff's Law

Junction Law

$i_1 + i_2 + i_4 = i_3 + i_5$

algebraic sum of all the currents directed towards a point (junction) is zero.

$\Sigma I_p = 0$



Loop Law

The algebraic sum of all the potential difference along a closed loop is zero.

Grouping of Resistance

Series grouping

$R = R_1 + R_2 + R_3 + \text{- - - } + R_n$

For two resistors

$R = R_1 + R_2$

Parallel grouping

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \text{- - -} + \frac{1}{R_n}$

For two resistors

$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

Temperature Dependence of Resistance

$R = R_0 [1 + \alpha (T - T_0)]$

$\rho = \int [1 + \alpha (T - T_0)]$

R = resistance at temperature T

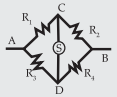
R_0 = resistance at temperature T_0

α = coefficient of resistivity

Wheatstone Bridge

• is balanced

If $\frac{R_1}{R_2} = \frac{R_3}{R_4}$



• $V_c = V_D$

No current through G

Meter Bridge

Unknown resistance

$S = \frac{R(100-l)}{l}$

l = balancing length

R = known resistance

Potentiometer

$P = \frac{V_A - V_B}{l} = \frac{\text{potential difference}}{\text{length of wire}}$

Comparison of emf, $\frac{E_1}{E_2} = \frac{l_1}{l_2}$

Internal Resistance (r) = $R\left(\frac{l_1}{l_2} - 1\right)$

l_1 l_2 are lengths of null point

MIND MAP : LEARNING MADE SIMPLE CHAPTER - 4

Moving Charges & Magnetism

Magnetic Field $(\vec{B})$

- It is a region around a magnet or current carrying conductor or a moving charge in which its magnetic effect can be felt
- SI unit is Tesla(T) = weber/m^2
- 1 Gauss = 10^{-4} Tesla

Magnetic Force on moving charge

$\vec{F} = q\vec{v}\times\vec{B}$

$= qvB\sin\theta$

- For $\theta = 0$, $\vec{F} = 0$ along the magnetic field
- For $\theta = 90°$, i.e. if charge's velocity is perpendicular to field direction, Force is perpendicular to both field & velocity

$F = qvB = \frac{mv^2}{r}$

$r = \frac{mv}{qB}$ = radius of the circle in which charge rotates

Time period (T) = $\frac{2\pi m}{qB}$

ν(frequency) = $\frac{1}{T} = \frac{qB}{2\pi m}$

If $\theta \neq 0, 180°, 90°$

Then, $F = qvB\sin\theta$

And the charge particle will follow helix path whose

$r = \frac{m v_\perp}{q B}$ and pitch = $V_\parallel \times T = V_\parallel \times \frac{2\pi m}{qB}$

Biot-Savart Law

$d\vec{B} = \frac{1}{4\pi\varepsilon_0 c^2}\frac{id\vec{l}\times\vec{r}}{r^3}$

$d\vec{B} = \frac{\mu_0}{4\pi}\frac{id\vec{l}\times\vec{r}}{r^3}$ θ = angle between $d\vec{l}$ and $\vec{r}$

$dB = \frac{\mu_0}{4\pi}\frac{idl\sin\theta}{r^2}$ Direction of field will be perpendicular to plane containing current element and the point of observation.

where $\mu_0 = \frac{1}{\varepsilon_0 c^2}$

$= 4\pi \times 10^{-7}\ TmA^{-1}$

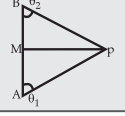
Oersted Law

In April, 1820, Hans Christian Oersted discovered that flow of current in a wire could deflect a nearby magnetic compass needle

Magnetic field due to straight wire current

$B = \frac{\mu_0 i}{4\pi d}(\cos\theta_1 - \cos\theta_2)$

where θ_1 and θ_2 are the angle corresponding to the lower and upper ends respectively



Field due to a long straight wire current

i.e $\theta_1 = 0$

$\theta_2 = \pi$

$B = \frac{\mu_0 i}{2\pi d}$

Field due to a circular current

Field at the centre

$B = \frac{\mu_0 i}{2a}$

Field at an axial point

$B = \frac{\mu_0 i\, a^2}{2(a^2+d^2)^{3/2}}$

Field at a point far away from the centre

i.e. for d>>>a

$B = \frac{\mu_0 i\, a^2}{2\, d^3}$

Ampere's Law

$\oint \vec{B}.d\vec{l} = \mu_0 i$

where i = total current crossing the area bounded by closed curve.

Solenoid

- Magnetic filed at a point inside due to a long solenoid $B = \mu_0 ni$
- And at point on one end $B = \frac{\mu_0 ni}{2}$

where n = no. of turns per unit length along the length of solenoid.

Magnetic field due to Toroid

$B = \frac{\mu_0 Ni}{2\pi r}$, N = total no. of turns

i = current in toroid

Force on a current carrying conductor

$d\vec{F} = i\, d\vec{l}\times\vec{B}$, $F = ilB$

Force between parallel currents

$F = \frac{\mu_0 i_1 i_2}{2\pi d}$

Torque experienced by a loop in uniform magnetic field

$\vec{\tau} = MB\sin\theta\,\hat{n} = \vec{M}\times\vec{B}$

Definition of Ampere

If two parallel wire kept 1 m apart, if $F = 2\times10^{-7}$, then current = 1A in each.

Sensitivity of moving coil galvanometer is nBA/k

Gal.to ammeter

$S = \frac{Ig}{I_T - Ig}G$

Gal.to voltameter

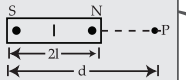
$R = \frac{V}{Ig} - G$

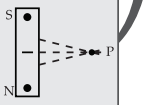
MIND MAP : LEARNING MADE SIMPLE

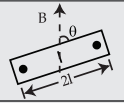
CHAPTER - 5

Magnetism & Matter

- Current loop of area A = πr^2 carrying current *i* may be replaced by a magnetic dipole of dipole moment $\mu = md = iA$
- Circular Loop
- Revolving Electron
- Solenoid
- Dipole moment
  - $|\vec{\mu}| = iA = \frac{ev}{2\pi r} \times \pi r^2$
  - $= evr/2$
  - $\vec{\mu} = \frac{-e}{2m_e}\vec{L}$
  - $\mu = md$
  - m = magnetic charge or pole strength
  - d = separation between poles
  - r = radius of circular orbit
  - e = charge on electron
  - v = speed of electron;
  - L = angular momentum
  - m_e = mass of electron
- $\mu = NIA$,
  N = no. of turns
  I = current
  A = area
- In 1833, Carl Friedrich Gauss and Wilhelm Weber made the first Electromagnetic Telegraph
- Earth's Magnetic field
  - Earth is a natural source of magnetic field having geometric north and geometric south pole
  - Declination
    - Angle b/w magnetic meridian and geographical meridian.
  - Inclination or dip (δ)
    - Angle made by earth's magnetic field with horizontal in the magnetic direction
      $\delta_{equator} = 0, \delta_{pole} = 90°$
  - Horizontal component
    - $B_H = B\cos\theta$
    - B_H (at pole) = 0
    - B_H(at equator) is maximum
- Magnetic field Lines
  - Imaginary lines which continuously represent the direction of magnetic field.
  -

- Megnetic Terms
- Magnetic Substance
  - Para-magnetism
    - Tendency to increase the magnetic field due to magnetization,
      e.g. Al, Mn I, X_m, $\mu_r > 1$
  - Ferromagnetism
    - Tendency of strongly magnetization in the direction of magnetic field.
      e.g. Fe, Co, Ni, μ_r, I, $\chi_m >> 1$
  - Diamagnetism
    - Tendency to magnetise in a direction opposite to direction of magnetic field
      e.g. Bi, Cu, Hg, Ni.
      μ_r, I, χ_m
- Electromagnets
  - Magnet in which the magnetic field is produced by an electric current. Magnetic field disappears when current is turned off.
- Bar Magnet
  - Magnetic field Intensity
    - End–on–position
      $= \frac{\mu_0}{4\pi}\frac{2M}{d^3}$
      M = 2ml
      d >>> l
      

    - Broadside –on Position
      $B = \frac{\mu_0}{4\pi}\frac{M}{(d^2+l^2)^{3/2}}$
      For d >>> l
      $B = \mu_0 M/4\pi d^3$
      

  - Potential Energy of a Field–Magnet System
    - $U_\theta = U_\theta - U_{90°} = -MB\cos\theta = -\vec{M}.\vec{B}$
      we take P.E. at $\theta = 90°$ to be zero
  - Torque
    - $\vec{\tau} = \vec{M} \times \vec{B} = MB\sin\theta$
      M = 2ml
      B = Magnetic field
      

- Magnetic Intensity
  - $H = \frac{B_0}{\mu}$
- Magnetic permeability
  - $\mu = \mu_0(1+\chi_m)$
- Magnetic susceptibility
  - $\chi = \frac{I}{H}$
- Intensity of magnetization
  - $I = \frac{M}{V}$
- Time period oscillating bar magnet
  - $(T) = 2\pi\sqrt{\frac{I}{MB}}$

MIND MAP : LEARNING MADE SIMPLE CHAPTER - 6

Electromagnetic Induction

Faraday's Law of Electromagnetic Induction

Whenever flux of magnetic field through the area bounded by a closed conducting loop changes, an emf is produced in the loop

The emf is given by $E = -d\phi/dt$

where $\phi = \int \vec{B}.\vec{ds}$ is the flux of the magnetic field through the area.

Contribution

In 1831, Michael Faraday discovered electromagnetic induction and James Clerk Maxwell mathematically described it.

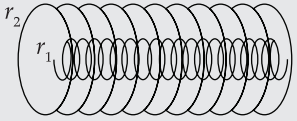
Mutual Inductance

$\phi = Mi$

$\frac{d\phi}{dt} = -M\frac{di}{dt}$

$M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$

$M_{21} = \mu_0 n_1 n_2 \pi r_2^2 l$



Emf induced in an AC generator, $E = NBA\omega \sin \omega t$

Lenz's Law

The direction of the induced current is such that it opposes the changes that has induced it.

Self Induction

If a solenoid of N turns, the flux through each each turn $= \phi = \int \vec{B}.\vec{ds}$

emf induced between the ends of coil $= E = -N\frac{d}{dt}\int \vec{B}.\vec{ds}$

Self Inductance of long Solenoid

$L = \mu_0 n^2 \pi r^2 l$

n = no. of turns per unit length

ϕ = flux = $(\mu_0 ni)\ \pi r^2$

r = radius of each loop of solenoid

- **Growth of current in LR Circuit**

  $i = \frac{E}{R}(1 - e^{tR/L}) = i_0(1 - e^{-t/r})$

- **Decay of current**

  $i = i_0 e^{-t/r}$

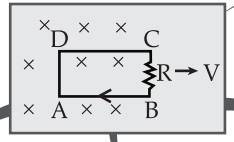
- **Energy stored in an Inductor**

  $U = \frac{1}{2}Li^2$

Eddy Current

It is induced when magnetic flux linked with the conductor changes

Rectangular loop



emf induced $E = vBl$

$i = \frac{vBl}{R}$

Magnetic force on the loop $F = B^2 l^2 v/R$ = force required to move the loop with constant velocity (v)

Thermal power developed in the loop is $P = \frac{v^2 B^2 l^2}{R}$

Induced current

$I = \frac{E}{R} = -\frac{1}{R}\frac{d\phi}{dt}$

Induced EMF

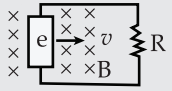
$E = \frac{d\phi}{dt}$

Motional EMF

$E = \left|\frac{d\phi}{dt}\right| = \left|Bl\frac{dx}{dt}\right|$

$= Blv$

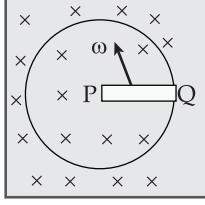
$i = vbl/(R+r)$



r = resistance of rod moving with velocity v in uniform magnetic field $\vec{B}$

EMF induced in a rotating conductor

$E = \frac{1}{2}B\omega l^2$



MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 7

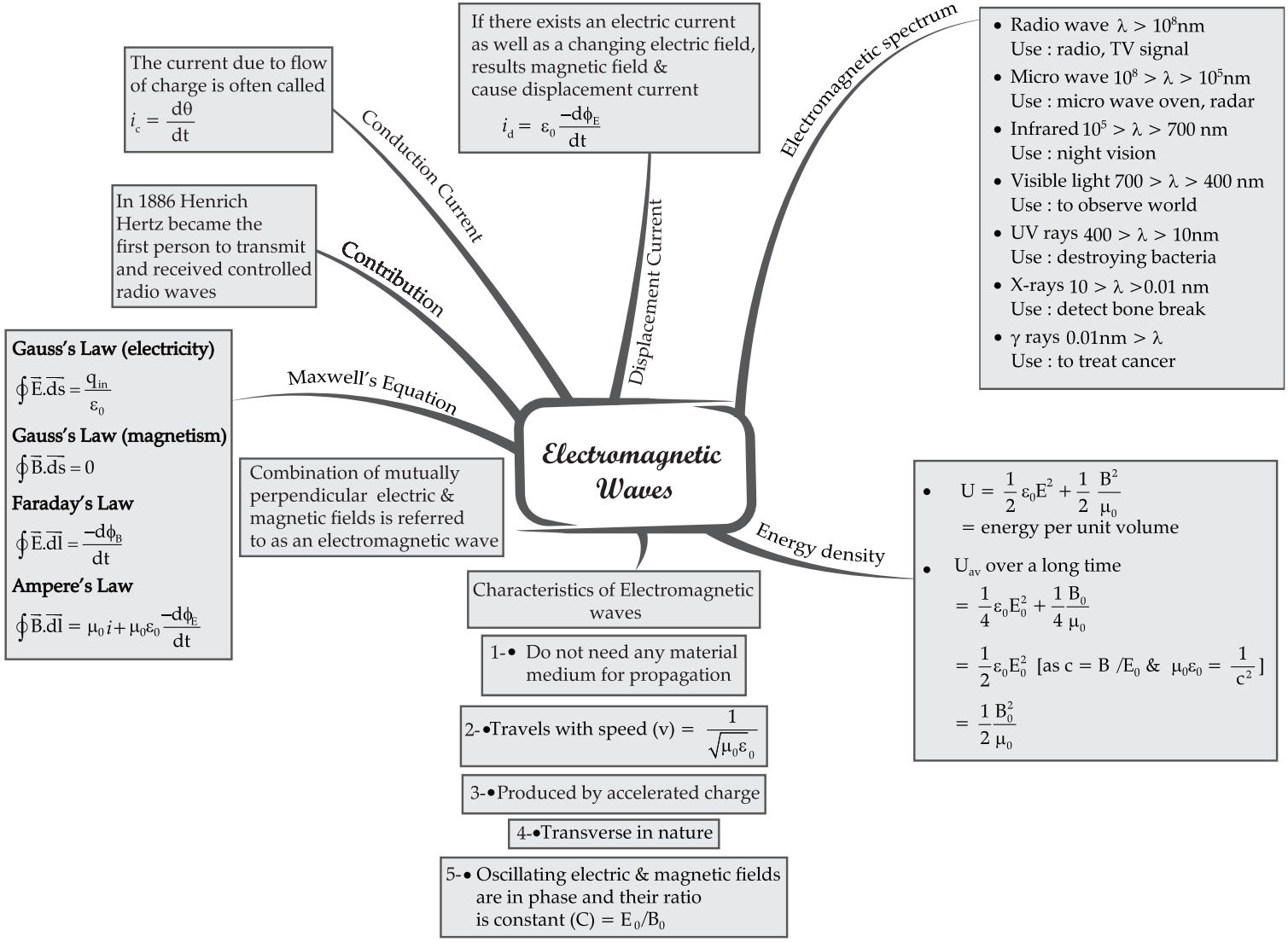
Alternating Current

- $i_{rms} = i_0/\sqrt{2}$, $v_{rms} = v_0/\sqrt{2}$ — RMS value or root mean square value
- $i_{mean} = 2i_0/\pi$, $v_{rms} = 2v_0/\pi$ — Mean value or Average value
- $i_0 = \sqrt{2}\, i_{rms}$, $v_0 = \sqrt{2}\, v_{rms}$ — Peak value or Max^m value
- Direction of current changes alternatively
  $i = i_0 \sin\omega t$
  $v = v_0 \sin\omega t$
  — AC Current
- In 1883 Nikola Tesla constructed the first induction motor at the age of 27 years only. — Contribution
- A device generating alternating current having elements magnet, Armature and graphite Brushes emf induced through coil
  $E = NBA\,\omega \sin\omega t$
  $i = E/R = NBA\,\omega \sin\omega t/R$
  $E_2 = -N_2E_1/N_1$
  $i_2 = -N_1E_1/N_2$
  If $N_2 > N_1$ it is called step–up Transfomer
  If $N_1 > N_2$, it is called step–down Transfomer
  — A C Generator
- It is used to obtain a high AC voltage from a Low–voltage AC source or vice–versa — Transformer
  - η = output power/Input power
  - 99% efficiency can easily achieved
  — Efficiency of a transformer
- The ratio of emf applied and the current produced in an AC circuit $Z = E_0/i_0$ Ohm — Impedance
- The hindrance offered by inductance or capacitance or both to the flow of AC in AC denoted by (X) Ohm — Reactance
- $P = E_{rms}\, i_{rms} \cos\phi = \frac{E_0}{\sqrt{2}} \times \frac{i_0}{\sqrt{2}} \cos\phi = \frac{1}{2} E_0 i_0 \cos\phi$ — Power in AC Circuit

A C Circuits

- $i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$
  Phase difference b/w V& I, $\phi = \pi/2$
  Power factor $\cos(\phi) = 0$
  Voltage leads current by $\pi/2$
  $P = 0$
  — Inductive (L) circuit
- $i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$
  $\phi = \pi/2$ or $-\pi/2$
  $\cos\phi = 0$
  Power(P) $= 0$
  Current leads the voltage by $\pi/2$
  — Capacitive (C) circuit
- $i = i_0 \sin(\omega t)$
  $\phi = 0°$
  $\cos\phi = 1$
  $P = v_0 I_0/2$
  Current & voltage both are in same phase
  — Resistive (R) circuit
- $i = i_0 \sin(\omega t + \phi)$
  $Z = \sqrt{R^2 + X_c^2}$
  $X_c = 1/\omega C$
  $\tan\phi = 1/\omega CR$
  $i_0 = E_0/Z$
  - Power factor $\cos\phi = \frac{R^2}{\sqrt{R^2 + X_c^2}}$
  - current leading.
  — R–C circuit
- $i = i_0 \sin\left(\omega t \pm \frac{\pi}{2}\right)$
  $X = X_L - X_C$
  $\phi = \pi/2$
  - Power factor = 0, Either voltage or current leading
  — L–C circuit
- $i = i_0 \sin(\omega t \pm \phi)$
  $Z = \sqrt{R^2 + (X_c - X_L)^2}$
  $X = X_c - X_L$
  $= \left(\frac{1}{\omega c} - \omega L\right)$
  $\tan\phi = \left(\frac{1}{\omega c} - \omega L\right)/R$, $i_0 = \frac{E_0}{Z}$
  — L–C–R circuit
- $i = i_0 \sin(\omega t + \phi)$
  $Z = \sqrt{R^2 + X_L^2}$
  $X_L = \omega L$
  $\tan\phi = \omega L/R$
  $i_0 = E_0/Z$
  - power factor = R/Z
  - voltage is leading
  — L–R circuit

MIND MAP : LEARNING MADE SIMPLE CHAPTER - 8



MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 9

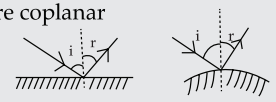
Ray Optics & Optical Instruments

Sign Conversion

- Pole is taken as origin
- Principle axis as the X-axis
- All distance measured from origin (or pole).
- All distance measured in the direction of incident ray is taken + ve.
- All distance measured in the direction opposite to the incident ray is taken - ve.

Reflection of Light

- $\angle i = \angle r$
- Incident ray reflected ray and normal to the reflecting surface are coplanar

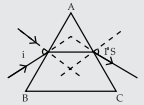


Prism

Angle of deviation $\delta = i - i' - A$

$\delta_{minimum} = 2i - A \left[i = i' \right] \rightarrow$

$\delta_{minimum} = (\mu - 1)A$, if A is small



Scattering of Light

Light scattered i.e. redirected in different paths when interacts with particle matters e.g. sunset, sunrise, colours, blue colour of sky

Spherical Mirror

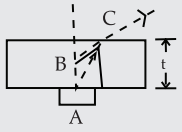
$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} = \frac{1}{f}$$

Lateral Magnification $= \frac{h_2}{h_1} = -\frac{v}{u}$

Refraction of light

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$$

$$\mu = \frac{real\ depth}{apparent\ depth}$$



$\Delta t = \left(1 - \frac{1}{\mu}\right) t$ = image shift

Optical Instruments

Microscope

$M = 1 + \frac{D}{f}$ [image at near point]

$M = D / f$ [image at infinite]

Compound Microscope

$M = \frac{v}{u}\left[\frac{D}{f_e}\right]$ [normal adjustment]

$M = \frac{v}{u}\left(1 + \frac{D}{f_e}\right) = -\frac{1}{f_o}\left(1 + \frac{D}{f_e}\right)$

For final image at least distance

Telescope

$M = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$ [image at near point]

$M = -\frac{fo}{fe}$ [image at infinite]

Total internal refection

When ray passes from optically denser to rarer medium.
If incident angle (i) further increased than $(\theta)_C$ critical angle entire light is then reflected back to the denser medium again is called T.I.R. It is used in optical fibre.

Critical Angle

- Incident angle (θ_c) for which angle of refraction is 90°

i.e. $\sin\theta_c = 1/\mu$

$\theta_c = \sin^{-1}\left(\frac{1}{\mu}\right)$

When ray passes from optically denser to rarer medium.

Refraction on Spherical surface

μ_1 (μ_2

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

Lateral Magnification

$m = \frac{h_2}{h_1} = \frac{\mu_1 v}{\mu_2 u}$

$= \frac{R - v}{R - u}$

Lens Maker's Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{\mu_2 - \mu_1}{\mu_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \left[If\ \mu_2 = \mu, \mu_1 = 1\ (air)\right.$$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ (lens formula)

Lateral Magnification $= \frac{h_2}{h_1} = \frac{v}{u}$

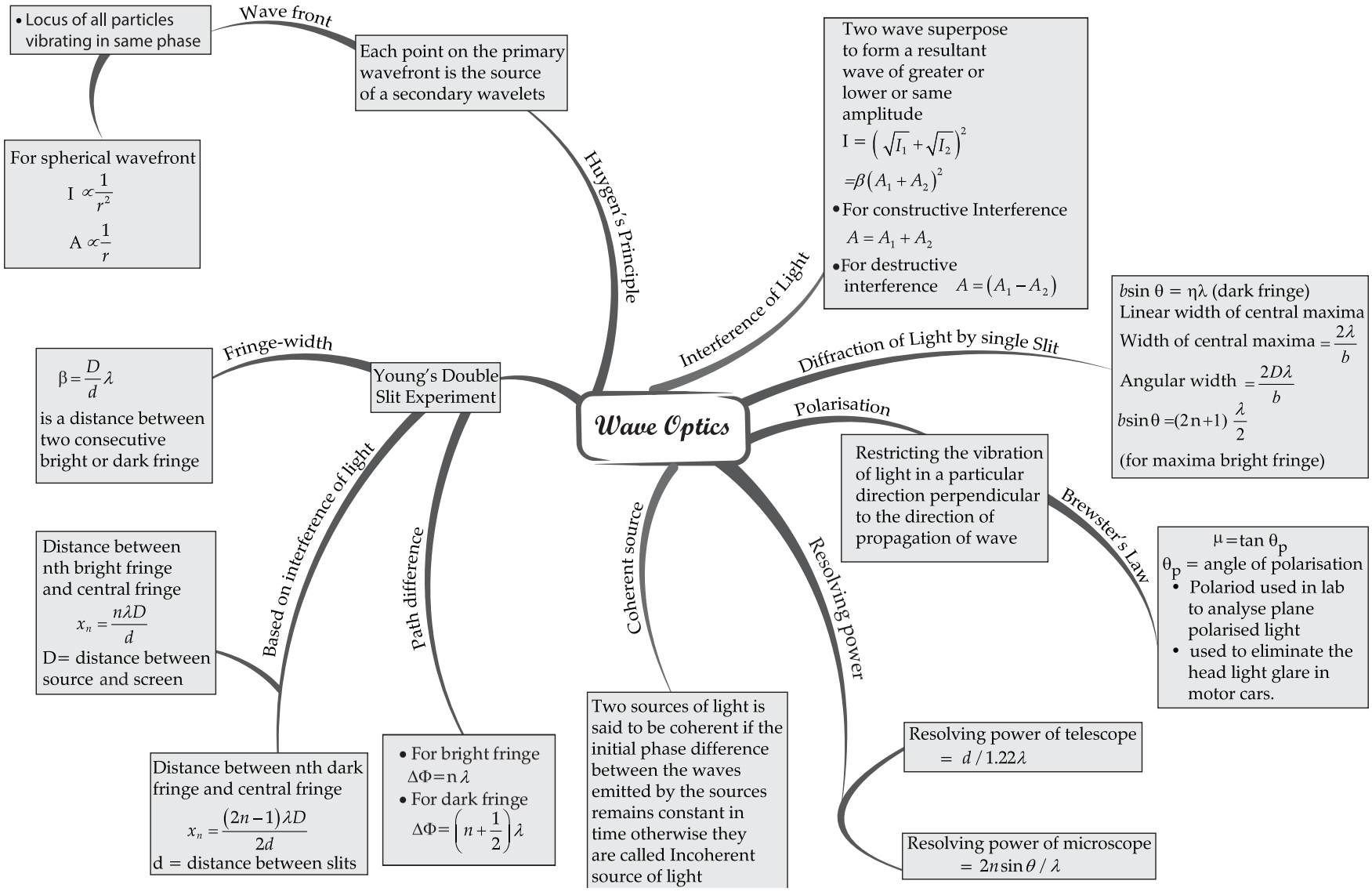
Power of a lens

$P = \frac{1}{f}$

[For combination of lens]

$P = \frac{1}{f_1} + \frac{1}{f_2}$

MIND MAP : LEARNING MADE SIMPLE CHAPTER - 10



MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 11

Dual Nature of Radiation & Matter

Dual Nature of Radiation

- Light has both wave character as well as particle
- Interference explained by wave nature
- When light of sufficiently low wavelength, it behaves as a particle
- Light particle having definite energy and definite linear momentum called "photon" or Energy packets.

Energy of each photon = $h\nu = hc/\lambda$

Momentum of each photon = $h/\lambda = E/c$

Contribution

Einstein, after an average academic career put forward quantum theory of light in 1905 while working as a grade III technical officer in a patent office

Matter waves

All matter can exhibit wave–like behaviour e.g. beam of electrons can be diffracted like water wave

de - Broglie relation

$\lambda = h/p$

λ = wavelength associated with particle or de-Broglie wavelength

p = momentum

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2\,m\,K_{max}}}$$

Davisson–Germer Experiment

A beam of electrons emitted by electron gun is made to fall on nickel crystal cut along cubical axis at a particular angle .Scattered beam of electrons is received by detector.

Results : λ = de Broglie's wavelength

$= h/p$

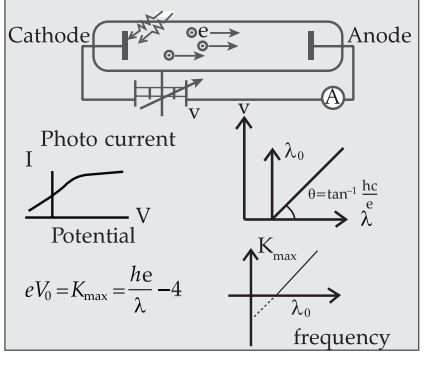
$= 1227/\sqrt{v}$ nm

$= 1227/\sqrt{54}$ nm

$= 0.167$ nm $= 1.67$ A°

This experiment verifies the wave nature of electrons & relation with de-Broglie wavelength.

Hertz and Lenard's Observation



Photoelectric Effect

- When light of sufficiently small wavelength is incident on metal surface, electrons are ejected from the metal, the phenomenon is called photoelectric effect.
- Ejected electrons are called photoelectrons
- Minimum energy equal to work function (ϕ) must be given to an electron so as to bring it out of the metal

Einstein's Photoelectric equation

- $K_{max} = E - \phi = eV_0$

  $= \frac{hc}{\lambda} - \phi$, V_0 = stopping potential

  K_{max} = maximum kinetic energy of elected electrons

Hore, $\lambda_0 = hc/\phi$

λ_0 = Threshold Wavelength

$\lambda_0 = c/\lambda_0 = \phi/h$

λ_0 = Threshold frequency

$K_{max} = \lambda(\nu - \nu_0)$

- If $\lambda = \lambda_0 = hc/\phi$

  $K_{max} = 0$, *i.e.*

  Electron may just come out.
- If $\lambda > \lambda_0$

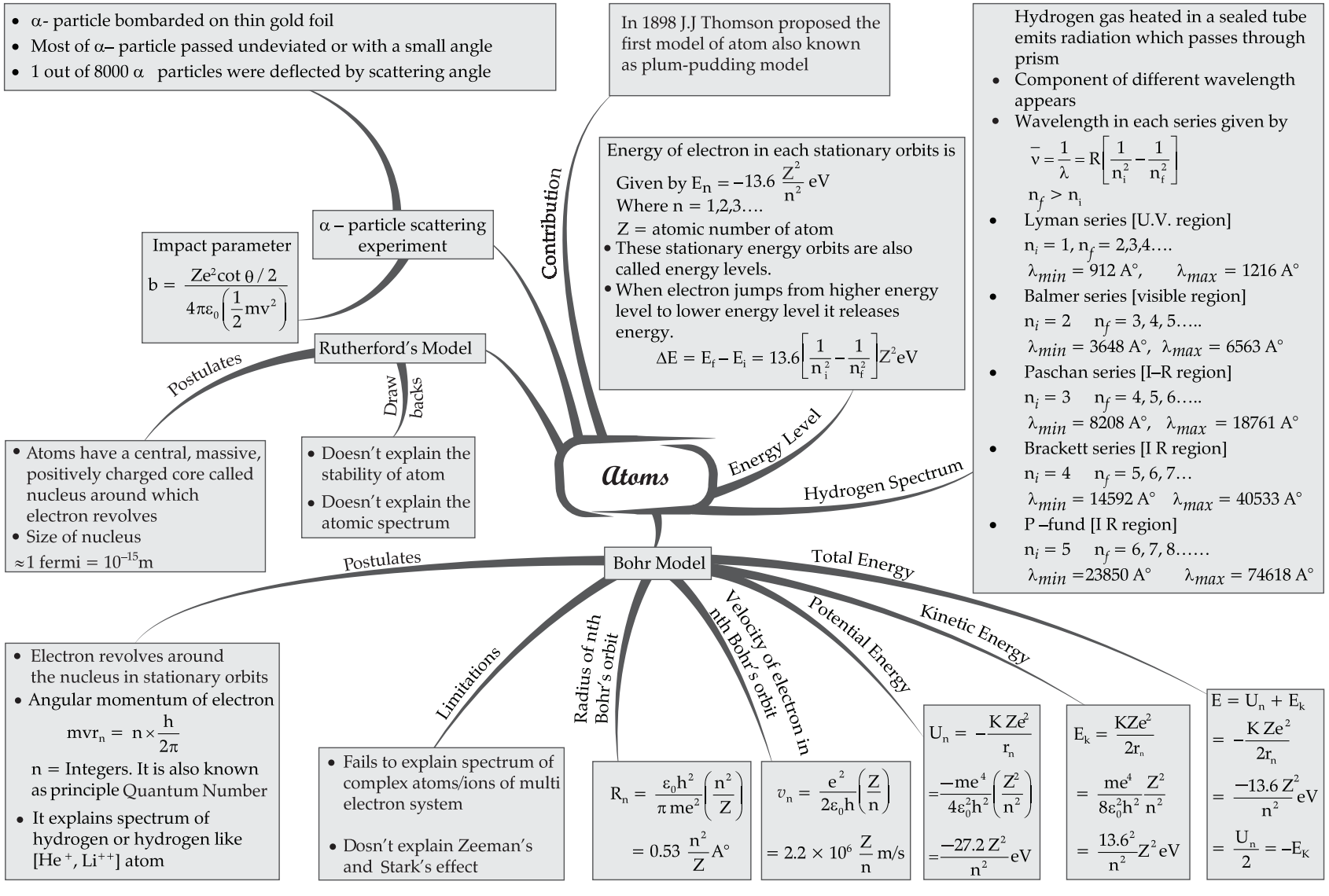
  i.e. $E < \phi$

  no electron will come out
- If $\lambda \le \lambda_0$.

  Photoelectric effect takes place this λ_0
- λ_0 = depends on metal used

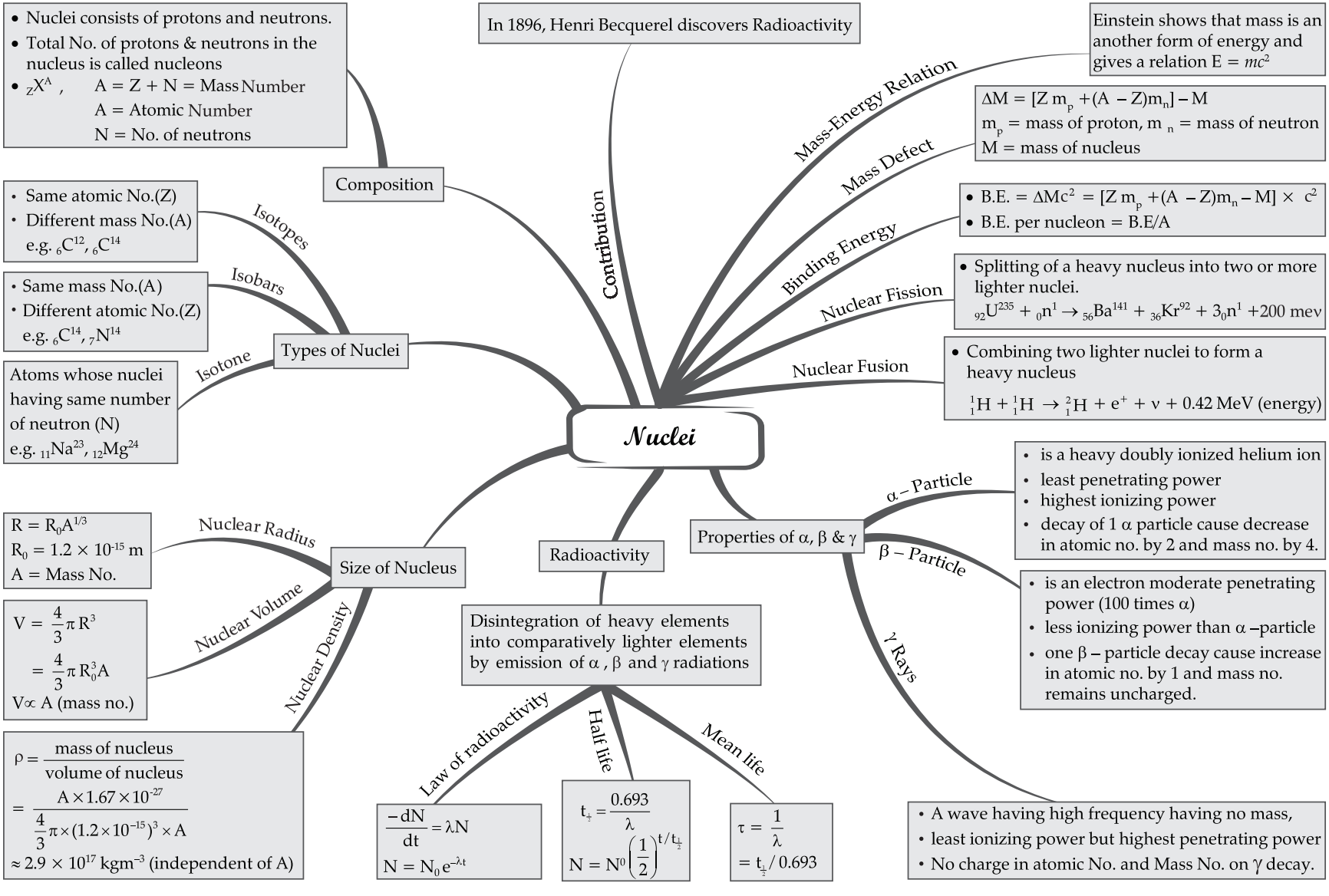
MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 12



MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 13



MIND MAP : LEARNING MADE SIMPLE CHAPTER - 14

Semiconductor Electronics: Materials, Devices and Simple Circuits

Energy Band

A range of energies associated with the quantum states of electrons in a crystalline solid.

- **Valence Band**: The energy band which are completely filled with electron at OK.
- **Conduction Band**: The bands with higher energy above the valence band (C.B. / E_g = Band gap / V.B.)
- **Band Gap or Forbidden Gap**: The difference between the highest energy in a valence band and lowest energy in the next higher band.

Contribution

In 1927, Walter Heitler and Fritz London discover bands

Classification of metals, Semiconductor & Insulator on the basis of bond theory

- **Conductors**
  - → Eg = 0 or least
  - → Density of charge carriers is very high (For electrons e) $= 9 \times 10^{23}\ m^{-3}$ for For Hole (h)
  - $= 0\ m^{-3}$ (in case of copper)
  - C.B. / V.B. $E_g = 0$
- **Semiconductors**
  - → Eg = 0.72 eV for Ge
  - ≈ 1.1ev for Si
  - $0 < E_g <$ Insulator
  - → very low (e)
    - For– n–type $= 7 \times 10^{15} m^{-3}$ for *Si* doped with P (C.B / E_g / V.B)
    - For– p–type $= 1 \times 10^{9} m^{-3}$ for Si doped with Al
- **Insulators**
  - → Eg = very high ≈ 7eV for diamond
  - → negligible ≈ 0
  - (C.B / E_g >> / V.B)

Intrinsic Semiconductor

- Pure semiconductor
- Charge carrier concentration $n_i = n_e = n_h$
- $I = n_e + n_h$
- e.g. Si (Silicon)

Extrinsic Semiconductor

Impure or doped semiconductor

- **N–Type semiconductor**
  - Si or Ge doped with pentavalent (P,As, Sb) element
  - Electrons are majority carriers
  - Holes are minority carriers
- **P–Type Semiconductor**
  - Si or Ge doped with trivalent (B, Al) element
  - Electrons are minority carriers
  - Holes are majority carriers

Semiconductor Diode

- **P.N junction Diode**: Device made by a close contact of N– type and P–type semiconductors
- **Forward Biased**
  - +ve terminal to P-side
  - –ve to N–side
  - diffusion current increases
  - depletion layer reduced
- **Reverse Biased**
  - –ve terminal to P– side
  - +ve terminal to N–Side
  - diffusion current decreases
  - depletion layer increases
- **Light Emitting Diode/LED**: used in T.V. or electronic gadgets.
- **Photo Diode**: is a P.N. junction whose function is controlled by the light allowed to fall on it.
- **Zener Diode**: used as a voltage regulator

Transistor

is a three terminal device

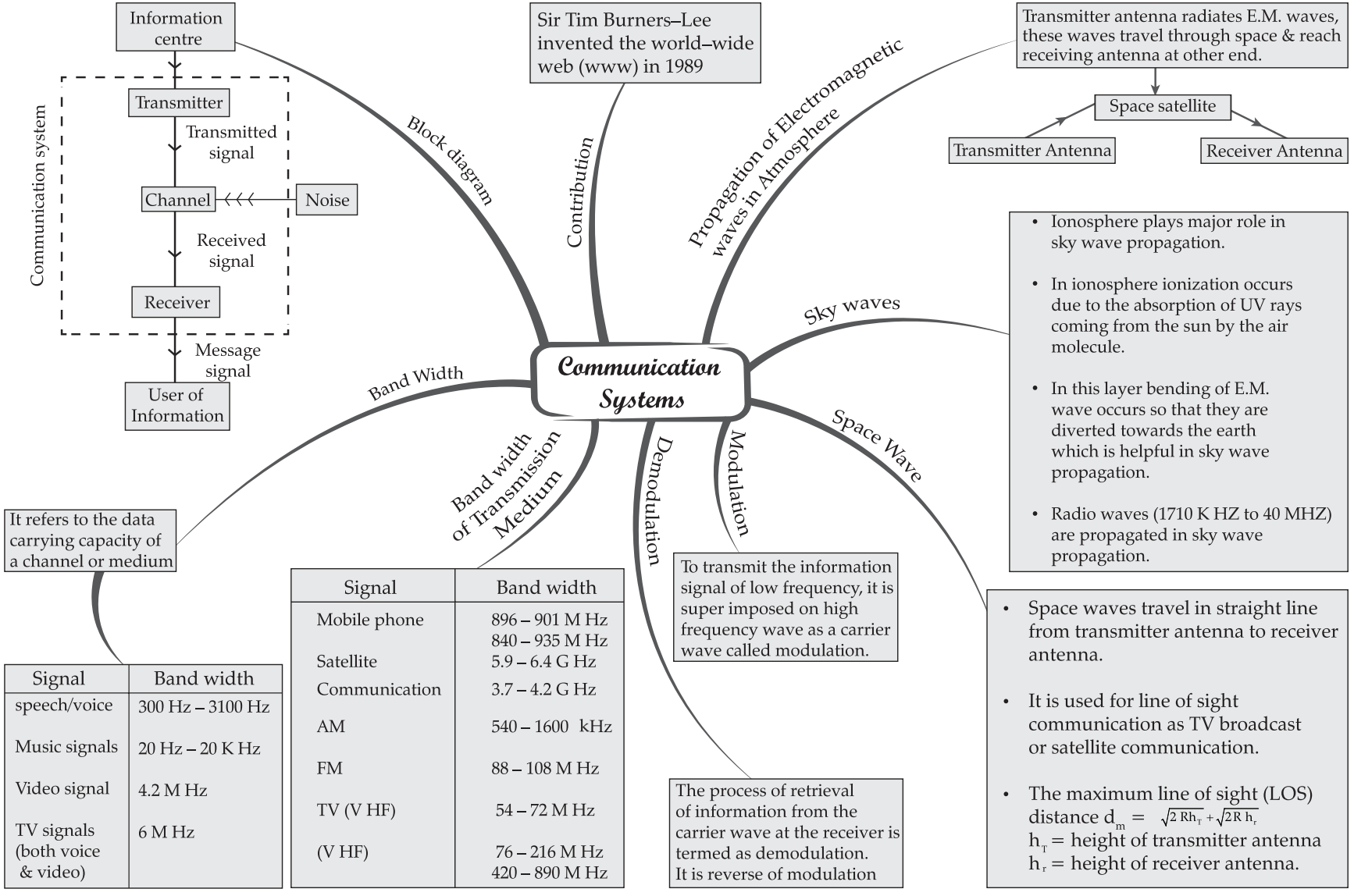
- **n–p–n transistor**
  - $I_E = I_B + I_c$
  - $\alpha = I_c / I_E$
  - $\beta = I_c / I_B$
  - $\beta = \frac{\alpha}{1-\alpha}$
  - E base C: p n p — p-n-p transistor
  - E base C: n p n — n-p-n transistor
- **p–n–p transistor**
  - In normal operation emitter, base junction is always forward biased
  - E base C: p n p — p-n-p transistor

Logic Gates

- 1-Or Gate: A, B → Y=A+B
- 2-AND Gate: A, B → Y=A.B
- 3-NOT Gate: A → $Y=\overline{A}$
- 4-NOR Gate: A, B → $Y=\overline{A+B}$
- 5-NAND Gate: A, B → $Y=\overline{A.B}$

MIND MAP : LEARNING MADE SIMPLE

CHAPTER - 15



Signal	Band width
speech/voice	300 Hz – 3100 Hz
Music signals	20 Hz – 20 K Hz
Video signal	4.2 M Hz
TV signals (both voice & video)	6 M Hz

Signal	Band width
Mobile phone	896 – 901 M Hz 840 – 935 M Hz
Satellite Communication	5.9 – 6.4 G Hz 3.7 – 4.2 G Hz
AM	540 – 1600 kHz
FM	88 – 108 M Hz
TV (V HF)	54 – 72 M Hz
(V HF)	76 – 216 M Hz 420 – 890 M Hz

WRITING NOTES

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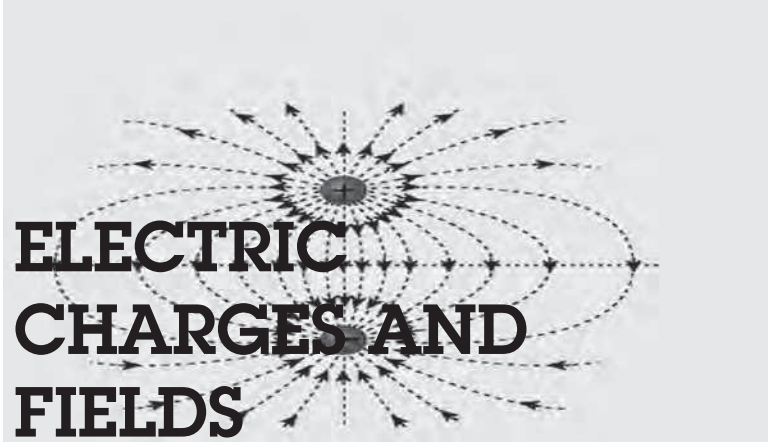
CHAPTER

1 ELECTRIC CHARGES AND FIELDS



Chapter Objectives

This chapter will help you understand :

- ***Electric charge and force :*** *Introduction; Electric charges; Conductors and insulators; Charging by induction; Basic properties of electric charge; Coulomb's Law; and Forces between multiple charges.*
- ***Electric field :*** *Introduction; Electric field; Electric field lines; Electric flux; Electric dipole; Dipole in a uniform external electric field; Continuous charge distribution; Gauss's Law and Application of Gauss's Law.*

TOPIC - 1
Electric Charge and Force P. 01

TOPIC - 2
Electric Field P. 10

TOPIC-1
Electric Charge and Force

Quick Review

- Electrostatics deals with the study of forces, fields and potentials arising from static charges. The name electricity is coined from the Greek word ***electron*** meaning *amber*.
- The **triboelectric effect** (also known as tribo-electric charging) is a type of contact electrification on which certain materials become electrically charged after they come into frictional contact with a different material.
- Coulomb's law should be used for point charges in vacuum at rest. It is not valid for charges in motion.
- The electrostatic force acts along the line joining the two charges. It obeys Newton's third law of motion.
- Coulomb's force is not affected by the presence of other charges in the neighbourhood; hence, the principle of superposition is valid.
- When a body is charged, the charge is distributed equally over the body. The distribution of charge may be one, two or three dimensional. Depending upon that it can have charge.
  - **Linear charge density :** When a charge is distributed along a line (straight or curved), the charge distribution is called linear charge distribution and the charge per unit length is called linear charge density, q/L.
  - **Surface charge density :** When a charge is distributed over a surface (plane or curved), the charge distribution is called surface charge distribution and the charge per unit area is called surface charge density, q/A.
  - **Volume charge density :** When a charge is distributed over the entire volume of the body, the charge distribution is called volume charge distribution and charge per unit volume is called volume charge density, q/V.

TIPS...

- *Study the Coulomb's law.*
- *Draw diagrams of the charges in the solution.*
- *Learn formulae and values of constants.*

TRICKS...

- *Sum all the x-components, getting the x-component of the resultant electric force.*
- *Sum all the y-components, getting the y-component of the resultant electric force.*
- *Use the Pythagorean Theorem and trigonometry to find the magnitude and direction of the resultant force if desired.*

Know the Links

- *http://physics.bu.edu/~duffy/PY106/Electricfield.html*
- *https://www.toppr.com/bytes/electricity-and-magnetism/*
- *https://www.electrical4u.com/electric-potential/*

Multiple Choice Questions (1 mark each)

Q. 1. Plastic rod rubbed with fur and glass rod rubbed with silk

(a) repel each other. **(b) mix up with each other.**
(c) attract each other. **(d) None of the above**

[OB Value Add.]

Ans. Correct option : (c)

Explanation : Rubbing a rod with certain materials will cause the rod to become charged. If a plastic rod rubbed with fur becomes negatively charged and a glass rod rubbed with silk becomes positively charged.

Q. 2. Electric charge between two bodies can be produced by

(a) sticking. **(b) rubbing.**
(c) oiling. **(d) passing AC current.**

[OB Value Add.]

Ans. Correct option : (b)

Explanation : The tribo-electric effect is a type of contact electrification on which certain materials become electrically charged after they come into frictional contact with a different material.

Q. 3. Electric charges under action of electric forces is called

(a) electrostatic. **(b) electric flux.**
(c) electric field. **(d) electric field lines.**

[OB Value Add.]

Ans. Correct option : (a)

Explanation : Coulomb force, also called electrostatic force or Coulomb interaction, attraction or repulsion of particles or objects because of their electric charge.

Q. 4. Law stating that "force is directly proportional to product of charges and inversely proportional to square of separation between them" is called

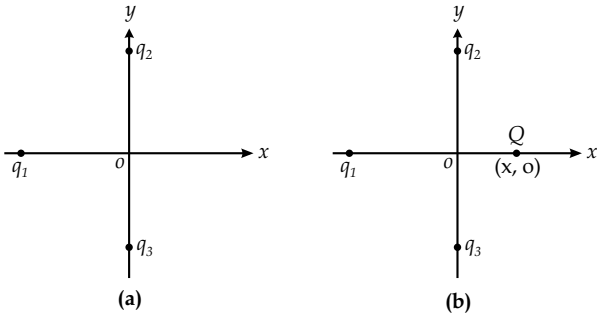
(a) Newton's law. **(b) Coulombs law.**
(c) Gauss's law. **(d) Ohm's law.**

[OB Value Add.]

Ans. Correct option : (b)

Explanation : Coulomb's law states that : The magnitude of the electrostatic force of attraction or repulsion between two-point charges is directly proportional to the product of the magnitudes of charges and inversely proportional to the square of the distances between them.

Q. 5. In given figure, two positive charges q_2 and q_3 fixed along the y axis, exert a net electric force in the + x direction on a charge q_1 fixed along the x axis. If a positive charge Q is added at (x, 0), the force on q_1



(a) shall increase along the positive x-axis.
(b) shall decrease along the positive x-axis.
(c) shall point along the negative x-axis.
(d) shall increase but the direction changes because of the intersection of Q with q_2 and q_3.

[NCERT Exemp. Q. 1.1, Page 1]

Ans. Correct option : (a)

Explanation : Net force on charge q_1, by other charges q_2 and q_3 is along the + x-direction, so nature of force between q_1 and q_2 and q_1 and q_3 is attractive. This is possible when charge q_1 is negative. Now, if a positive charge Q is placed at (x, 0), then, the force on q_1 shall increase. The direction will be along positive x-axis.

Tick Two or More Options

Q. 6. The magnitude of electric force, F is

(a) directly proportional to the multiplication of both charges.
(b) directly proportional to the distance between both charges.
(c) inversely proportional to the square of the distance between both charges.
(d) constant. **[OB Value Add.]**

Ans. Correct options : (a) and (c)

Explanation : The magnitude of the electric force F is directly proportional to the amount of one electric charge, q_1, multiplied by the other, q_2, and inversely proportional to the square of the distance 'r' between their centres. Expressed in the form of an equation, this relation, called Coulomb's law, may be written by including the proportionality factor k as $F = \frac{kq_1q_2}{r^2}$.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. A polythene piece rubbed with wool is found to have a negative charge of 3×10^{-7} C.

(a) Estimate the number of electrons transferred (from which to which)?
(b) Is there a transfer of mass from wool to polythene?

[NCERT Ex. Q. 11, Page 46]

Ans. (a) As the wool is positively charged and the polythene is negatively charged, so we can say that

few number of electrons are transferred from wool to polythene.

Charge on the polythene, $q = -3 \times 10^{-7}$ C

Amount of charge on an electron, $e = -1.6 \times 10^{-19}$ C

Let number of electrons transferred from wool to polythene be n.

So, by using the given equation we can calculate the value of n,

$$q = ne$$

$$\Rightarrow n = \frac{q}{e} = \frac{-3 \times 10^{-7}}{-1.6 \times 10^{-19}} = 1.87 \times 10^{12}$$

Therefore, the number of electrons transferred from wool to polythene is 1.87×10^{12} **[1]**

(b) Yes,

Mass is also transfer as electron is transferred from wool to polythene and electrons have some masses.

Mass of one electron, $m_e = 9.1 \times 10^{-31}$ kg

Total mass transferred, $m = m_e \times n$

$= 9.1 \times 10^{-31} \times 1.87 \times 10^{12}$

$= 1.706 \times 10^{-18}$ kg

From above result, the mass transferred is too low that it can be neglected. **[1]**

Q. 2. When a glass rod is rubbed with a silk cloth, charges appear on both. A similar phenomenon is observed with many other pairs of bodies. Explain how this observation is consistent with the law of conservation of charge.

[NCERT Ex. Q. 1.5, Page 46]

Ans. When we rub two bodies together then a charge is developed on both of the bodies and the developed charge over the bodies are equal but opposite in nature. This phenomenon of getting a charge is called as charge by friction. **[1]**

So, the net charge on both of the bodies will be zero because an equal amount of charge is also repelling it. When we rubbed the glass rod with the silk cloth, there will be charge with opposite magnitude is generated over there.

This phenomenon is in consistence and follows the law of conservation of charge. A similar phenomenon is observed with many other pairs of bodies. **[1]**

Q. 3. Check that the ratio $\frac{ke^2}{Gm_em_p}$ is dimensionless. Look up a Table of Physical Constants and determine the value of this ratio. What does the ratio signify? [NCERT Ex. Q. 1.3, Page 46]

Ans. The ratio is given as : $\frac{ke^2}{Gm_em_p}$

Where,

G = Gravitational constant in Nm^2kg^{-2}

m_e and m_p = Masses of electron and proton in kg

e = Electric charge in coulomb, C

$k = \frac{1}{4\pi\varepsilon_0}$ in Nm^2C^{-2}

Therefore, unit of given ratio will be,

$$\frac{ke^2}{Gm_em_p} = \frac{[Nm^2C^{-2}][C^2]}{[Nm^2kg^{-2}][kg][kg]}$$

$$= M^0L^0T^0$$ **[1]**

From the table :

$e = 1.6 \times 10^{-19}$ C

$G = 6.67 \times 10^{-11}$ Nm^2kg^{-2}

$m_e = 9.1 \times 10^{-31}$ kg

$m_p = 1.66 \times 10^{-27}$ kg

So, the numerical value of the given ratio will be:

$$\Rightarrow \frac{ke^2}{Gm_em_p} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.67 \times 10^{-27}}$$

$$\approx 2.3 \times 10^{39}$$

So, the value of this ratio is 2.3×10^{39}.

That means the above ratio is the ratio of the electric force to the gravitational force between a proton and an electron when the distance between them is constant. **[1]**

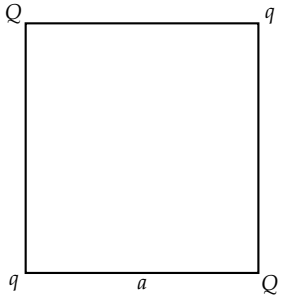
Q. 4. (a) Explain the meaning of the statement 'electric charge of a body is quantised'.

(b) Why can one ignore quantisation of electric charge when dealing with macroscopic, *i.e.*, large scale charges? [NCERT Ex. Q. 1.4, Page 46]

Ans. **(a)** Electric charge of any body is quantised meaning only integral (1, 2, …n) number of electrons can be transferred from a body to another. Charges do not get transferred in fractions forms. So, the total charge possessed by a body is only in integral multiples of electric charge. **[1]**

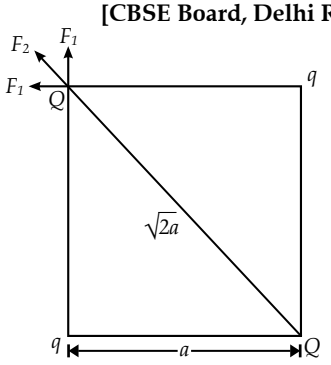
(b) In the case of large scale, the charge which are used over there are comparatively too huge to the magnitude of the electric charge. Hence, for the macroscopic level the quantisation of charge is of no use. Therefore, it is ignored and the electric charge is considered to be continuous. **[1]**

Q. 5. Four-point charges Q, q, Q and q are placed at the corners of a square of side 'a' as shown in the figure. Find the resultant electric force on a charge Q.



[CBSE Board, Delhi Region, 2018]

Ans.



F_1 = Force acts at Q by q

F_2 = Force acts at Q by Q

$F_1 = \frac{k.q.Q}{a^2}$, and

$F_2 = \frac{k.Q.Q}{(\sqrt{2}a)^2}$ **[1]**

Solving both

$F_1 = \frac{k.q.Q}{a^2}$, $F_2 = \frac{k.Q^2}{2a^2}$

Resultant force will be:

$F_{net} = \sqrt{2}F_1 + F_2$

$= \sqrt{2}\frac{k.q.Q}{a^2} + \frac{k.Q^2}{2a^2}$

[Resultant F of F_1 and F_1 will be $\sqrt{2}F_1$ because F_1 and F_1 are perpendicular.]

So, the resultant electric force on a charge Q will be $\sqrt{2}\frac{k.q.Q}{a^2} + \frac{k.Q^2}{2a^2}$. **[1]**

Q. 6. What is the force between two small charged spheres having charges of 2×10^{-7} C and 3×10^{-7} C placed 30 cm apart in air?

[NCERT Ex. Q. 1.1, Page 46]

Ans. Given that,

Charge on first sphere, $q_1 = 2 \times 10^{-7}$ C

Charge on second sphere, $q_2 = 3 \times 10^{-7}$ C

Distance between the two charges, $r = 30$ cm $= 0.3$ m

Electrostatic force between the spheres is given by :

$F = \frac{1}{4\pi\varepsilon_0}.\frac{q_1 q_2}{r^2}$

Where,

ε_0 = Permittivity of free space

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ N

Then force,

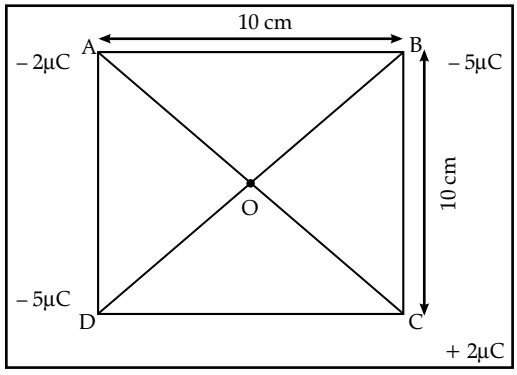
$F = 6 \times 10^9$ N

Since, the charges are having same nature; therefore, force between them will be repulsive. **[1]**

Q. 7. Four-point charges $q_A = 2\ \mu C$, $q_B = -5\ \mu C$, $q_C = 2\ \mu C$ and $q_D = -5\ \mu C$ are located at the corners of a square $ABCD$ of side 10 cm. What is the force on a charge of $1\ \mu C$ placed at the centre of the square?

[NCERT Ex. Q. 1.6, Page 46]

Ans.



In given square,

Sides : $AB = BC = CD = DA = 10$ cm

Diagonals : $AC = BD = 10\sqrt{2}$ cm

$AO = OC = DO = OB = 5\sqrt{2}$ cm

At the centre point 'O', placed a charge of 1μC.

So, for the above cases :

The repulsive force between the corner A and the centre O is same in magnitude with the repulsive force by the corner C to the centre O, but these forces are opposite in direction. So, these forces will cancel each other and from A and C no forces are applied on the centre O. **[1]**

Similarly, from the corner C the attractive force is applying on to the centre O and another force with the same magnitude is applying on the centre O, also these two forces are opposite in direction.so, it is also opposing each other.

Therefore, the net force applying in the centre is zero. Because all the forces are cancelled by each other. **[1]**

Q. 8. The electrostatic force on a small sphere of charge 0.4 μC due to another small sphere of charge –0.8 μC in air is 0.2 N.

(a) What is the distance between the two spheres?

(b) What is the force on the second sphere due to the first? **[NCERT Ex. Q. 1.2, Page 46]**

Ans. **(a)** Charge on first sphere, $q = 0.4\ \mu C = 0.4 \times 10^{-6}$ C

Charge on second sphere, $q = -0.8\ \mu C = -0.8 \times 10^{-6}$ C

Electrostatic force on the first sphere, $F = 0.2$ N

Electrostatic force between the spheres is given by :

$F = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1 q_2}{r^2}$

Where,

ε_0 = Permittivity of free space

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \text{Nm}^2\text{C}^{-2}$

$r^2 = \frac{1}{4\pi\varepsilon_0}.\frac{q_1 q_2}{F}$

$= \frac{0.4 \times 10^{-6} \times 0.8 \times 10^{-6} \times 9 \times 10^9}{0.2}$

$= 144 \times 10^{-4}\ \text{m}^2$

$\therefore\ r = \sqrt{144 \times 10^{-4}}$

$= 12 \times 10^{-2}$

$= 0.12$ m

Distance between the two spheres = 0.12 m **[1]**

(b) Both spheres attract each other with the same force. So, the force on the second sphere due to first sphere will be 0.2 N. **[1]**

Q. 9. Suppose the spheres A and B in Q.5 (SATQ) have identical sizes. A third sphere of the same size but uncharged is brought in contact with the first, then brought in contact with the second, and finally removed from both. What is the new force of repulsion between A and B?

[NCERT Ex. Q. 1.13, Page 47]

Ans. Given that,
Distance between the spheres, A and B, $r = 0.5$ m
Initially, the charge on each of sphere,
$q = 1.3 \times 10^{-7}$ C
When the sphere A is touched with an uncharged sphere C, then half of the charge will be transferred to the sphere C. Hence the charge on both the spheres A and C will be $\frac{q}{2}$. **[1]**

After that when sphere C with charge $\frac{q}{2}$ is brought in touch with sphere B with charge q, then charge on each of the sphere will be divided in two equal parts,

$$\frac{1}{2}\left(q+\frac{q}{2}\right)=\frac{3q}{4}$$

So, the charge on each of the spheres, C and B, is $\frac{3q}{4}$
Force of repulsion between sphere A and B is :

$$F = \frac{1}{4\pi\varepsilon_0} \times \frac{q_A q_B}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{\frac{q}{2} \times \frac{3q}{4}}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{3q^2}{8r^2}$$

$$= \frac{9 \times 10^9 \times 3 \times (6.5 \times 10^{-7})^2}{8 \times (0.5)^2}$$

$$= 5.703 \times 10^{-3} \text{ N}$$

So, the force of attraction between the two speheres is 5.703×10^{-3} N. **[1]**

Short Answer Type Questions **(3 marks each)**

Q. 1. A paisa coin is made up of Al-Mg alloy and weighs 0.75g. It has a square shape and its diagonal measures 17 mm. It is electrically neutral and contains equal amounts of positive and negative charges.
Treating the paisa coins made up of only Al, find the magnitude of equal number of positive and negative charges. What conclusion do you draw from this magnitude?
[NCERT Exemp. Q. 20, Page 6]

Ans. Given that,
The coin is made of aluminium
Diagonal of the square coin = 17 mm
Atomic mass of Al = 27 g
Atomic number of Al = 13
As we know,
One molar mass of substance has 6.023×10^{23} number of atoms. **[1]**
27 g Al contains 6.023×10^{23} atoms
1 g of Al contains = $\frac{6.023 \times 10^{23}}{27}$ atoms
0.75 g of Al contains = $\frac{[1.023 \times 10^{23} \times 0.75]}{27}$
$= 1.67 \times 10^{23}$ atoms
And we know that,
One atom of Al contains 13 electrons 13 protons. **[1]**
1.67×10^{22} atom of Al contains = $13 \times 1.67 \times 10^{22}$
$= 21.71 \times 10^{22}$ electron
Hence number of proton = 21.71×10^{22}
The charge on one electron = 1.6×10^{-19} Coulomb
Charge on 21.71×10^{22} electrons
$= 1.6 \times 10^{-19} \times 21.71 \times 10^{22}$
$= 34.736 \times 10^3$ C
So, the amount of positive and negative charge on the coin = 34.736×10^3 C
q = 34.8 kC of ±ve charge. This is a very large amount of charge.
This concludes that all the coins contain a very large amount of positive and negative charges. **[1]**

Q. 2. Consider a coin of above question It is electrically neutral and contains equal amounts of positive and negative charge of magnitude 34.8 kC. Suppose that these equal charges were concentrated in two-point charges separated by

(i) 1 cm ($\sim \frac{1}{2}$ x diagonal of the one paisa coin),

(ii) 100 m ($\sim$ length of a long building), and

(iii) 10^6 m (radius of the earth).

Find the force on each such point charge in each of the three cases. What do you conclude from these results? [NCERT Exemp. Q. 21, Page 6]

Ans. It is clear that charges are equal and opposite. So, by Coulomb's law, force of attraction between charges,

$$F = \frac{1}{4\pi\varepsilon_0} \times \frac{q_1 q_2}{r^2}$$

$$q_1 = q_2 = 34.8 \text{ kC}$$

$$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$F = \frac{34.8 \times 10^3 \times 34.8 \times 10^3 \times 9 \times 10^9}{r^2}$$

$$= \frac{34.8 \times 34.8 \times 9 \times 10^{15}}{r^2}$$

$$= \frac{10899.36 \times 10^{15}}{r^2} \text{N}$$

$$\cong \frac{1.1 \times 10^{19}}{r^2} \text{N}$$

$$\therefore F = \frac{1.1 \times 10^{19}}{r^2} \text{N}$$

(i) Given that,
r_1 = 1 cm = 0.01 m

$$F_1 = \frac{1.1 \times 10^{19}}{0.01 \times 0.01} = 1.1 \times 10^{23}$$

$= 1.1 \times 10^{23}$ N towards the charges **[1]**

(ii) Given that,

$r_2 = 100$ m

$\therefore F = \frac{1.1\times10^{19}}{100\times100} = 1.1\times10^{15}$

$= 1.1\times10^{15}$ N towards the charges. **[1]**

(iii) Given that,

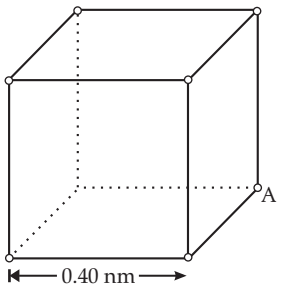
$r_3 = 106$ m

$\therefore F_3 = \frac{1.1\times10^{19}}{10^6\times10^6} = 1.1\times10^{7}$

$= 1.1\times10^{7}$ N towards the charges.

This electrostatic force varies from order 10^7 N to 10^{23} N. The minimum force 10^7 is equivalent to the force of attraction between the earth and 1 million kg body which is too much high. So electrostatic force is so many times larger than the gravitational force. **[1]**

Q. 3. Figure represents a crystal unit of caesium chloride, CsCl. The caesium atoms, represented by open circles are situated at the corners of a cube of side 0.40nm, whereas a Cl atom is situated at the centre of the cube. The Cs atoms are deficient in one electron while the Cl atom carries an excess electron.



(i) What is the net electric field on the Cl atom due to eight Cs atoms?

(ii) Suppose that the Cs atom at the corner *A* is missing. What is the net force now on the Cl atom due to seven remaining Cs atoms?

[NCERT Exemp. Q. 22, Page 6]

Ans. **(i)** It is clear from the given figure that Cl^-atom is at the centre whereas Cs^+ atoms are at eight corners at equal distance. So, by symmetry net force on Cl^- atom due to other Cs^+ atoms will be zero. So by the net force on Cl^- due to Cs^+ ion will be zero, as atoms of Cs^+ attracts the Cl^- equally in opposite direction with pairs diagonally, *i.e.*, (*B*, *H*), (*C*, *E*), (*D*, *F*). **[1½]**

(ii) In the given question, removing Cs atom at the corner *A* is equivalent to adding a singly charged negative *Cs* ion at point *A*. so,

Net force $= \frac{q^2}{(4\pi\varepsilon_0 r^2)}$

Where, q= charge on electron, r = distance between *Cl* and *Cs* atoms

Applying Pythagoras theorem, we have

$r = \sqrt{(0.02)^2 + (0.20)^2 + (0.20)^2}\times10^{-19}$ m

$= 0.346\times10^{-9}$

Now,

Net force $= \frac{1}{4\pi\varepsilon_0 r^2}$

$= \frac{9\times10^9\times(1.6\times10^{-19})^2}{(0.346\times10^{-9})^2}$

$= 1.92\times10^{-9}$ N

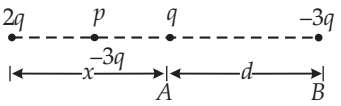
The direction is the force is from *A* to Cl^-.

Force on Cl^- ion,

$F = 1.92\times10^{-9}$ N towards Cl^- ion. **[1½]**

Q. 4. Two charges q and $-3q$ are placed fixed on x-axis separated by distance 'd'. Where should a third charge $2q$ be placed such that it will not experience any force? [NCERT Exemp. Q. 23, Page 6]

Ans. Where, let us keep the charge $2q$ at a distance r from *A*.



So, charge $2q$ will not experience any force.

When force of repulsion on it due to q is balanced by force of attraction on it due to $-3q$, at *B*, where $AB = d$.

So that,

Force of attraction by $-3q$ = Force of repulsion by q **[1½]**

$$\Rightarrow \quad \frac{2q\times q}{4\pi\varepsilon_0 x^2} = \frac{2q\times 3q}{4\pi\varepsilon_0 (x+d)^2}$$

$$\Rightarrow \quad (x+d)^2 = 3x^2$$

$$\Rightarrow x^2 + d^2 + 2xd = 3x^2$$

$$\Rightarrow 2x^2 - 2dx - d^2 = 0$$

$$\Rightarrow \quad x = \frac{d}{2} \pm \frac{\sqrt{3}d}{2}$$

[Negative (–) sign be between q and $-3q$ and hence is unadaptable.]

$x = \frac{d}{2} + \frac{\sqrt{3}d}{2}$

$= \frac{d}{2}(1+\sqrt{3})$ to the left of q **[1½]**

So, x will be : $x = \frac{d}{2} + \frac{\sqrt{3}d}{2}$ to the left of q.

Q. 5. (a) Two insulated charged copper spheres *A* and *B* have their centres separated by a distance of 50 cm. What is the mutual force of electrostatic repulsion if the charge on each is 6.5×10^{-7} C? The radii of *A* and *B* are negligible compared to the distance of separation.

(b) What is the force of repulsion if each sphere is charged double the above amount, and the distance between them is halved?

[NCERT Ex. Q. 12, Page 46]

Ans. **(a)** Charge on sphere *A*, $q_A = 6.5\times10^{-7}$ *C*

Charge on sphere *B*, $q_B = 6.5\times10^{-7}$C

Distance between the spheres, r = 50 cm = 0.5 m

Force of repulsion between the two spheres,

$F = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_A q_B}{r^2}$

Where ε_0= Permittivity of free space and

$\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ \text{Nm}^2\text{C}^{-2}$

So,

$F = \frac{9\times10^9\times(6.5\times10^{-7})^2}{(0.5)^2}$

$= 1.52\times10^{-2}\ \text{N}$

Thus, the force between two spheres is 1.52×10^{-2} N. **[1½]**

(b) After twice (doubling) the charge,

Charge on sphere A, $q_A = 1.3\times10^{-6}$ C

Charge on sphere B, $q_B = 1.3\times10^{-6}$ C

The distance between the spheres is halved.

$\therefore r = \frac{0.5}{2} = 0.25$ m

Force of repulsion between the two spheres,

$F = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_A q_B}{r^2}$

$= \frac{9\times10^9\times1.3\times10^{-6}\times1.3\times10^{-6}}{(0.25)^2}$

$= 16\times1.52\times10^{-2}$

$= 0.243$ N

So, the force between the two spheres is 0.243 N. **[1½]**

Long Answer Type Questions **(5 marks each)**

Q. 1. A particle of mass m and charge $(-q)$ enters the region between the two charged plates initially moving along x-axis with speed v_x. The length of plate is L and an uniform electric field E is maintained between the plates. Show that the vertical deflection of the particle at the far edge of the plate is $qEL^2/(2m\,v_x^2)$. Compare this motion with motion of a projectile in gravitational field discussed in Section 4.10 of Class XI Textbook of Physics. [NCERT Add. Ex. Q. 1.33, Page 50]

Ans. Given that,

Charge on a particle of mass $m = -q$

Velocity of the particle $= v_x$

Length of the plates $= L$

Magnitude of the uniform electric field between the plates $= E$

Mechanical force, F = Mass (m) × Acceleration (a)

$\Rightarrow a = \frac{F}{m}$

$\Rightarrow a = \frac{qE}{m}$ ...(1)

[As electric force, $F = qE$] **[1]**

Time taken by the particle to cross the field of length L is given by,

$t = \frac{\text{Length of the plate}}{\text{Velocity of the particle}}$

$= \frac{L}{v_x}$ ...(2) **[1]**

For vertical direction,

Initial velocity, $u = 0$

$s = ut + \frac{1}{2}at^2$

$\Rightarrow s = 0 + \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{L}{v_x}\right)^2$

$\Rightarrow s = \frac{qEL^2}{2mv_x^2}$

[From equations 1 and 2] **[2]**

So, the vertical deflection of the particle at the far edge of the plate is $\frac{qEL^2}{2mv_x^2}$.

This is similar to the motion of horizontal projectiles under gravity. **[1]**

Q. 2. Suppose that the particle in Exercise in 1.33 is an electron projected with velocity $v_x = 2.0\times10^6\ \text{ms}^{-1}$. If E between the plates separated by 0.5 cm is 9.1×10^2 N/C, where will the electron strike the upper plate? ($|e| = 1.6\times10^{-19}$ C, $m_e = 9.1\times10^{-31}$ kg.) [NCERT Add. Ex. Q. 1.34, Page 50]

Ans. Velocity of the particle, $v_x = 2.0\times10^6$ m/s

Separation of the two plates, $s = 0.5$ cm $= 0.005$ m

Electric field between the two plates, $E = 9.1\times10^2$ N/C

Charge on an electron, $q = 1.6\times10^{-19}$ C

Mass of an electron, $m_e = 9.1\times10^{-31}$ kg

Let assume that the electron strike the upper plate at the end of plate L, and the deflection due to this is s. so,

$s = \frac{q\text{EL}^2}{2mv_x^2}$

$\Rightarrow L = \sqrt{\frac{2\text{s}mv_x^2}{q\text{E}}}$ **[2½]**

$= \sqrt{\frac{2\times0.005\times9.1\times10^{-31}\times2\times2\times10^{12}}{1.6\times10^{-19}\times9.1\times10^2}}$

$= \sqrt{0.00025}$

$= 0.016$ m

$= 1.6$ cm

So, the electron will strike the upper plate after travelling 1.6 cm. **[2½]**

Q. 3. An oil drop of 12 excess electrons is held stationary under a constant electric field of $2.55\times10^4\ \text{NC}^{-1}$ in Millikan's oil drop experiment. The density of the oil is 1.26 g cm^{-3}. Estimate the radius of the drop. ($g = 9.81\ \text{m s}^{-2}$; $e = 1.60\times10^{-19}$ C). [NCERT Add. Ex. Q. 1.25, Page 50, Page 48]

Ans. Given that,

Excess electrons on an oil drop, $n = 12$

Electric field intensity, $E = 2.55 \times 10^4\ NC^{-1}$
Density of oil, $\rho = 1.26\ gm/cm^3 = 1.26 \times 10^3\ kg/m^3$
Acceleration due to gravity, $g = 9.81\ ms^{-2}$
Charge on an electron, $e = 1.6 \times 10^{-19}\ C$
Let assume the radius of the oil drop $= r$
Force, F due to electric field, E is equal to the weight of the oil drop, W :

$$F = W$$
$$Eq = mg$$

And, $q = ne$

So, $Eq = \frac{4}{3}\pi r^3 \times \rho \times g$ **[2]**

Where,
q = Net charge on the oil drop $= ne$
m = Mass of the oil drop
= Volume of the oil drop × Density of oil

$$= \frac{4}{3}\pi r^3 \times \rho \quad \textbf{[2]}$$

$$r = \sqrt[3]{\frac{3Ene}{4\pi\rho g}}$$
$$= \sqrt[3]{\frac{3 \times 2.55 \times 10^4 \times 12 \times 1.6 \times 10^{-19}}{4 \times 3.14 \times 1.26 \times 10^3 \times 9.81}}$$
$$= \sqrt[3]{946.09 \times 10^{-21}}\ m$$
$$= 9.82 \times 10^{-7} m$$
$$= 9.82 \times 10^{-4}\ mm$$

So, the radius of the oil drop is 9.82×10^{-4} mm. **[1]**

Q. 4. It is now believed that protons and neutrons (which constitute nuclei of ordinary matter) are themselves built out of more elementary units called quarks. A proton and a neutron consist of three quarks each. Two types of quarks, the so-called 'up' quark (denoted by *u*) of charge + $\left(\frac{2}{3}\right)$ *e*, and the 'down' quark (denoted by *d*) of charge $\left(\frac{-1}{3}\right)$ e, together with electrons build up ordinary matter. (Quarks of other types have also been found which give rise to different unusual varieties of matter.) Suggest a possible quark composition of a proton and neutron.

[NCERT Add. Ex. Q. 1.31, Page 49]

Ans. A proton has three quarks. Let there be n up quarks in a proton, each having a charge of $\left(+\frac{2}{3}\right)$ e.

Charge due to n up quarks $= \left(\frac{2}{3}e\right) n$

Number of down quarks in a proton $= 3 - n$

Each down quark has a charge of $-\frac{1}{3}$e

Charge due to $(3 - n)$ down quarks $= \left(-\frac{1}{3}e\right)(3-n)$

Total charge on a proton $= +e$ **[1]**

So,

$$e = \left(\frac{2}{3}e\right) n + \left(-\frac{1}{3}e\right)(3-n)$$

$$\Rightarrow e = \frac{2ne}{3} - e + \frac{ne}{3}$$

$\Rightarrow 2e = ne \Rightarrow n = 2$ **[1]**

Number of up quarks in a proton, $n = 2$
Number of down quarks in a proton $= 3 - n = 3 - 2 = 1$
Thus, a proton can be represented as '*uud*'. **[1]**
A neutron also has three quarks. Let there be n up quarks in a neutron.

Charge on a neutron due to n up quarks $= \left(+\frac{2}{3}e\right) n$

Number of down quarks is $(3 - n)$, and each having a charge of $-\frac{1}{3}e$.

Charge on a neutron due to $(3 - n)$ down quarks $= \left(-\frac{1}{3}e\right)(3-n)$

Total charge on a neutron = 0
So,

$$0 = \left(\frac{2}{3}e\right) n + \left(-\frac{1}{3}e\right)(3-n) \quad \textbf{[1]}$$

$$\Rightarrow 0 = \frac{2ne}{3} - e + \frac{ne}{3}$$

$\Rightarrow e = ne \Rightarrow n = 1$
Number of up quarks in a neutron, $n = 1$
Number of down quarks in a neutron $= 3 - n = 2$
So, a neutron can be represented as '*udd*'. **[1]**

Q. 5. There is another useful system of units, besides the SI/mks, a system called the cgs (centimetre-gram-second) system. In this system Coulomb's law is given by : $F = \frac{Qq}{r^2}\hat{r}$ where the distance r is measured in cm (= 10^{-2} m), F in dynes (= 10^{-5} N) and the charges in electrostatic units (esu units), where 1 esu unit of charge = $\frac{1}{|3|} \times 10^{-9}$ C. The number $|3|$ actually arises from the speed of light in vacuum which is now taken to be exactly given by $c = 2.99792458 \times 10^8$ m/s. An approximate value of c then is $c = |3| \times 10^8$ m/s.

(i) Show that the coulomb law in CGS units yields. 1 esu of charge = 1 (dyne)$^{1/2}$ cm. Obtain the dimensions of units of charge in terms of mass M, length L and time T. Show that it is given in terms of fractional powers of M and L.

(ii) Write 1 esu of charge = x C, where x is a dimensionless number. Show that this gives

$$\frac{1}{4\pi\varepsilon_0} = \frac{10^{-9}}{x^2}\frac{N.m^2}{C^2}$$

With $x = \frac{1}{3} \times 10^9$ we have,

$$\frac{1}{4\pi\varepsilon_0} = 3^2 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

$$\text{or, } \frac{1}{4\pi\varepsilon_0} = (2.99792458)^2 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \text{(exactly)}$$

[NCERT Exemp. Q. 1.29, Page 8]

Ans. **(i)** From the relation,

$$F = \frac{Qq}{r^2} = 1 \text{ dyne} = \frac{[1 \text{ esu of charge}]^2}{[1 \text{ cm}]^2}$$

So, 1 esu of charge

$$= (1 \text{ dyne})^{\frac{1}{2}} \times 1 \text{ cm} = F^{\frac{1}{2}} \cdot L = [\text{MLT}^{-2}]^{\frac{1}{2}} L$$

So, 1 esu of charge = $\text{M}^{\frac{1}{2}}\text{L}^{\frac{3}{2}}\text{T}^{-1}$

Thus, esu of charge is represented in terms of fractional powers $\frac{1}{2}$ of M and $\frac{3}{2}$ of L. **[2½]**

(ii) Let 1 esu of charge = x C. where x is a dimensionless number Coulomb force on two charges, each of magnitude 1 esu separated by 1 cm is dyne = 10^{-5} N. This situation is equivalent to two charges of magnitude x C separated by 10^{-2} m.

$$\therefore F = \frac{1}{4\pi\varepsilon_0} \frac{x^2}{(10^{-2})^2} = 1 \text{ dyne} = 10^{-5} \text{N}$$

Put the value of constant:

$$\therefore \frac{1}{4\pi\varepsilon_0} = \frac{10^{-9}}{x^2} \frac{\text{Nm}^2}{C^2}$$

Taking, $x = \frac{1}{|3| \times 10^9}$

We get,

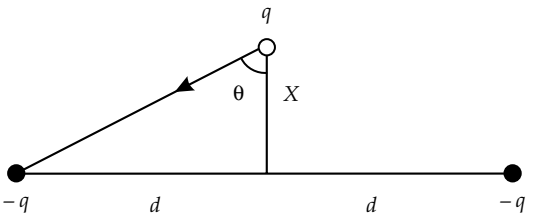
$$\frac{1}{4\pi\varepsilon_0} = |3| \times 10^9 \times 10^{18} \frac{Nm^2}{C^2}$$

$$= 9 \times 10^9 \frac{Nm^2}{C^2}$$

If $|3| \to 2.99792458$,

we get, $\frac{1}{4\pi\varepsilon_0} = 8.98755 \times 10^9 \text{ Nm}^2\text{C}^{-2}$. **[2½]**

Q. 6. Two charges $-q$ each are fixed separated by distance $2d$. A third charge q of mass m placed at the mid-point is displaced slightly by x ($x << d$) perpendicular to the line joining the two-fixed charged as shown in Figure. Show that q will perform simple harmonic oscillation of time period. $T = [(8\pi^3\varepsilon_0 md^3) / q^2]^{\frac{1}{2}}$



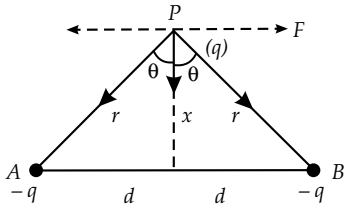
[NCERT Exemp. Q. 1.30, Page 9]

Ans. From the figure,

Given that,

Two charge $-q$ at A and B

$AB = AO + OB = 2d$



x = Small distance perpendicular to O.

i.e., $x << d$ mass of charge q is. So, force of attraction at P towards A and B are each

$$F = \frac{q(q)}{4\pi\varepsilon_0 r^2}$$

where $AP = BP = r$

Horizontal components of these forces F_n are cancel out. Vertical components along PO are added up. **[2½]**

If $\angle APO = \theta$, the net force on q along PO is

$F' = 2 \text{ F} \cos\theta$

$$= \frac{2q^2}{4\pi\varepsilon_0 r^2} \left(\frac{x}{r}\right)$$

$$= \frac{2q^2 x}{4\pi\varepsilon_0 (d^2 + x^2)^{\frac{3}{2}}}$$

When, $x << d$, $F' = \frac{2q^2 x}{4\pi\varepsilon_0 d^3} = Kx$

where, $K = \frac{2q^2}{4\pi\varepsilon_0 d^3}$

$\Rightarrow F \propto x$

So that, force on charged components is proportional to its displacement from the centre O and it is directed towards O.

So it is clear that, the motion of charge g would be simple harmonic, where

$$\omega = \sqrt{\frac{K}{m}}$$

and $T = \frac{2\pi}{\omega}$

$$= 2\pi \sqrt{\frac{m}{K}}$$

$$= 2\pi \sqrt{\frac{m.4\pi\varepsilon_0 d^3}{2q^2}}$$

$$= \left[\frac{8\pi^3\varepsilon_0 md^3}{q^2}\right]^{\frac{1}{2}}$$

[2½]

TOPIC-2
Electric Field

Quick Review

- When there is more than one charge in a region, the electric field lines will not be straight lines. The electric field lines will curve in response to the different charges. In every case, though, the field is highest where the field lines are close together and decreases as the lines get further apart.
- Electric field lines of force have a tendency to get separated from each other in the direction perpendicular to their lengths. They repel if they are of like charges.
- These lines of force are like elastic string, they come to contract in length, that is, attract each other with respect to the opposite charges.
- Closeness of lines of forces symbolises more strength of electric field and vice versa.
- Parallel lines indicate uniform field.
- Two lines of forces never intersect each other.
- Lines of force never pass through a conductor, that is, field inside a conductor is always zero.
- The tangential direction at any point on the lines of forces indicates the direction of the force acting on the positive charge at that point.
- Gauss' Law is a powerful method of calculating electric fields.
- The permittivity of free space : It is a constant related to the constant k that appears in Coulomb's law. The relationship between the two is this :

$$k = \frac{1}{4\pi\varepsilon_0}$$

or, equivalently,

$$\varepsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12}\ \mathrm{C^2/Nm^2}$$

TIPS...
- *Study the properties of Electric field.*
- *Understand the Gauss's Law.*
- *Study the application of Gauss's law.*

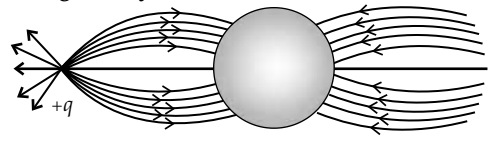
TRICKS...
- *Practice a lot in Gauss's law.*
- *Make flow chart to revise quickly.*

Know the Links

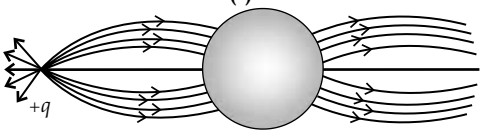
- *https://www.electrical4u.com/what-is-electric-field/*
- *https://www.electrical4u.com/electric-potential/*

Multiple Choice Questions **(1 mark each)**

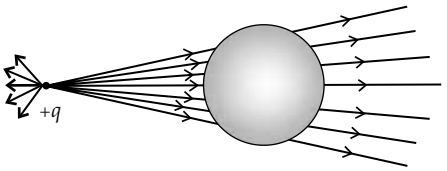
Q. 1. A point positive charge is brought near an isolated conducting sphere in Figure. The electric field is best given by :



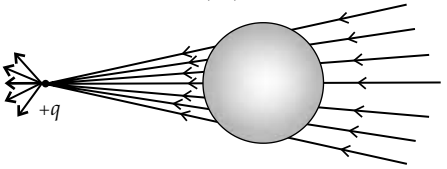
(i)



(ii)



(iii)



(iv)

(a) Fig (i) **(b) Fig (iii)**
(c) Fig (ii) **(d) Fig (iv)**

[NCERT Exemp. Q. 1.2, Page 2]

Ans. Correct option : (a)

Explanation : As given charge is + q and lines of forces in positive charge must be outwards from positive charge q. Now, as the positive charge is kept near an isolated conducting sphere, due to induction left part of sphere gets accumulated negative charge and right part positive, and lines of force from right part of sphere must emerge outwards normally.

So, verifies the answer (a).

As lines of forces are not perpendicular to the surface of sphere, so options (iii) and (iv) are not true again.

Q. 2. A point charge +q, is placed at a distance d from an isolated conducting plane. The field at a point P on the other side of the plane is

(a) directed perpendicular to the plane and away from the plane.

(b) directed perpendicular to the plane but towards the plane.

(c) directed radially away from the point charge.

(d) directed radially towards the point charge.

[NCERT Exemp. Q. 1.6, Page 3]

Ans. Correct option : (a)

Explanation : When you place a positive charge near a conducting plane, then electric field lines from positive charges will enter into the conducting plane (from the side where positive charge is kept) and emerge from opposite side of the plane.

In both cases, the direction of electric field lines will always be perpendicular to the surface of the plane.

Q. 3. A hemisphere is uniformly charged positively. The electric field at a point on a diameter away from the centre is directed

(a) perpendicular to the diameter.

(b) parallel to the diameter.

(c) at an angle tilted towards the diameter.

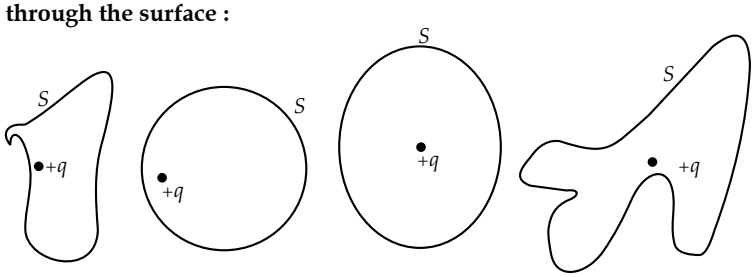
(d) at an angle tilted away from the diameter.

[NCERT Exemp. Q. 1.7, Page 3]

Ans. Correct option : (a)

Explanation : As the side or diameter of hemi-sphere is plane surface, and whole hemi-sphere is charged with positive charge so, the electric field line of forces emerging outward will be perpendicular to the plane surface or diameter.

Q. 4. The electric flux through the surface :



(i) (ii) (iii) (iv)

(a) in Figure (iv) is the largest.

(b) in Figure (iii) is the least.

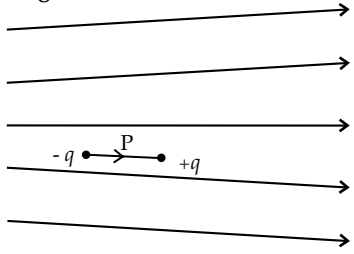
(c) in Figure (ii) is same as Figure (iii) but is smaller than Figure (iv).

(d) is the same for all the figures. **[NCERT Exemp. Q. 1.3, Page 2]**

Ans. Correct option : (d)

Explanation : Electric flux, through the closed surface (or space) depends only on the charge enclosed inside the surface. Here, charges inside all figures are same. So, electric flux will remain same.

Q. 5. Figure shows electric field lines in which an electric dipole p is placed as shown. Which of the following statements is correct?



(a) The dipole will not experience any force.

(b) The dipole will experience a force towards right.

(c) The dipole will experience a force towards left.

(d) The dipole will experience a force upwards.

[NCERT Exemp. Q. 1.5, Page 3]

Ans. *Correct op*tion : (c)

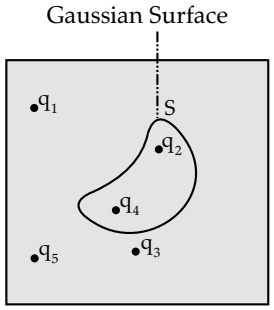
Explanation : We know electric field emerges radially outward from positive point charge.

In the figure given above, space between field lines is increasing (or density of electric field line is decreasing). In other words, the electric force is decreasing while moving from left to right.

Thus, the force on charge – q is greater than the force on charge + q in turn dipole will experience a force towards left direction.

Q. 6. Five charges q_1, q_2, q_3, q_4, and q_5 are fixed at their positions as shown in Figure. S is a Gaussian surface. The Gauss's law is given by:

$$\oint_S \mathbf{E}.\mathbf{ds} = \frac{q}{\varepsilon_0}$$



Which of the following statements is correct?

(a) E on the LHS of the above equation will have a contribution from q_1, q_5 and q_3 while q on the RHS will have a contribution from q_2 and q_4 only.

(b) E on the LHS of the above equation will have a contribution from all charges while q on the RHS will have a contribution from q_2 and q_4 only.

(c) E on the LHS of the above equation will have a contribution from all charges while q on the RHS will have a contribution from q_1, q_3 and q_5 only.

(d) Both E on the LHS and q on the RHS will have contributions from q_2 and q_4 only.

[NCERT Exemp. Q. 1.4, Page 3]

Ans. ***Correct option :*** (b)

Explanation : As all charges are positive (or of same signs) so electric field lines on R.H.S. of Gaussian surface will be due to q_2 and q_4 only.

On L.H.S. of Gaussian surface, the electric field lines on 'E' will be due to q_1,q_2, q_3,q_4 and q_5. So, answer (*b*) is verified.

Tick Two or More Options

Q. 7. The Electric field at a point is

(a) always continuous.

(b) continuous if there is no charge at that point.

(c) discontinuous only if there is a negative charge at that point.

(d) discontinuous if there is a charge at that point.

[NCERT Exemp. Q. 1.9, Page 4]

Ans. Correct options : (b) and (d)

Explanation : Either positive or negative charges will interact the lines of electric field so makes the electric field discontinuous.

If there is no other charge inside the electric field then the lines will not be affected. So electric field becomes continuous. So, answers (b) and (d) are verified.

Q. 8. Consider a region inside which there are various types of charges but the total charge is zero. At points outside the region

(a) the electric field is necessarily zero.

(b) the electric field is due to the dipole moment of the charge distribution only.

(c) the dominant electric field is $\propto \frac{1}{r^3}$, for large r, where r is the distance from a origin in this region.

(d) the work done to move a charged particle along a closed path, away from the region, will be zero.

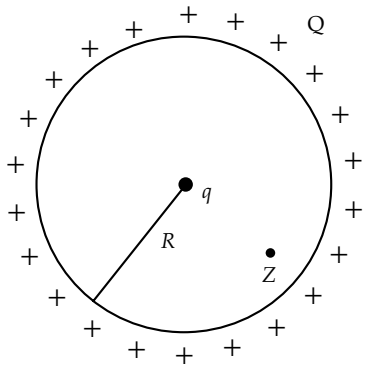
[NCERT Exemp. Q. 1.11, Page 4]

Ans. Correct options : (c) and (d)

Explanation : Although net charge in a dipole is zero but its electric field is proportional to $\frac{1}{r^3}$

Work done again electric field is conservative, so net work done in a closed loop is always zero. So, answer (c) and (d) are verified.

Q. 9. A positive charge Q is uniformly distributed along a circular ring of radius R. A small test charge q is placed at the centre of the ring. Then :



(a) If $q > 0$ and is displaced away from the centre in the plane of the ring, it will be pushed back towards the centre.

(b) If $q < 0$ and is displaced away from the centre in the plane of the ring, it will never return to the centre and will continue moving till it hits the ring.

(c) If $q < 0$, it will perform SHM for small displacement along the axis.

(d) q at the centre of the ring is in an unstable equilibrium within the plane of the ring for $q > 0$.

[NCERT Exemp. Q. 1.13, Page 5]

Ans. Correct options : (a), (b), (c) and (d)

Explanation : For '*a*', charge is uniformly distributed along the ring. It is not sphere in which charge is only outside. So, positive charge of ring will interact equally a charge placed at centre of ring but will be in unstable equilibrium.

For '*c*', if '*q*' is displaced slightly (or small), it will perform SHM and stops at centre. Options (a) and (b) are verified in a similar way.

Q. 10. If $\oint_s$ E. dS = 0 over a surface, then

(a) the electric field inside the surface and on it is zero.

(b) the electric field inside the surface is necessarily uniform.

(c) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

(d) all charges must necessarily be outside the surface.
[NCERT Exemp. Q. 1.8, Page 4]

Ans. Correct options : (c) and (d)

Explanation : Flux in Gaussian surface is zero. So, the net charge inside the closed surface either is zero or charges are outside the surface. If charge or charges are outside the Gaussian surface, then entering leaving lines of electric field will be equal so net flux (lines of electric filed) is zero verifies answers (c) and (d).

Q. 11. If there were only one type of charge in the universe, then

(a) $\oint_s E.dS \neq 0$ **on any surface.**

(b) $\oint_s E.dS = 0$ **if the charge is outside the surface.**

(c) $\oint_s E.dS$ **could not be defined.**

(d) $\oint_s E.dS = \frac{q}{\varepsilon_0}$ **if charges of magnitude** q **were inside the surface.**

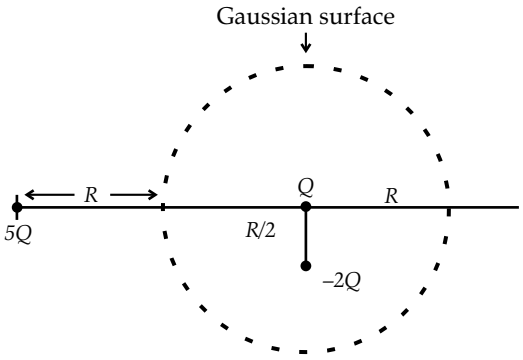
[NCERT Exemp. Q. 1.10, Page 4]

Ans. Correct options : (b) and (d)

Explanation : If a charge q is enclosed inside Gaussian surface then (d) is true. If Gaussian surface (or space) is outside the charge and (c) and (a) are not true.

So, answers (b) and (d) are true.

Q. 12. Refer to the arrangement of charges in Figure and a Gaussian surface of radius R **with** Q **at the centre. Then**



(a) total flux through the surface of the sphere is $\frac{-Q}{\varepsilon_0}$.

(b) field on the surface of the sphere is $\frac{-Q}{4\pi\varepsilon_0 R^2}$

(c) flux through the surface of sphere due to $5Q$ **is zero.**

(d) field on the surface of sphere due to $-2Q$ **is same everywhere.** **[NCERT Exemp. Q. 1.12, Page 4]**

Ans. Correct options : (a) and (c)

Explanation :

(a) It is true by Gauss's law. Here net charge = $-2Q + Q = -Q$.

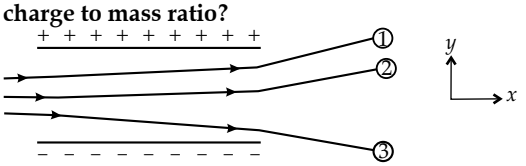
(b) It is the electric field on a conducting sphere. There is no any conducting sphere but only surface or spherical space is there.

(c) $5Q$ charge outside the Gaussian surface will not contribute to electric flux in Gaussian surface (or space).

(d) For the distance of Gaussian surface from $-2Q$ is different, so field will not be same on the surface.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Figure shows tracks of three charged particles in a uniform electrostatic field. Give the signs of the three charges. Which particle has the highest charge to mass ratio?



[NCERT Ex. Q. 1.14, Page 47]

Ans. As we know that the opposite charges attract each other and same charges repel each other. So, it can be observed that particles 1 and 2 both move towards the positively charged plate and repel away from the negatively charged plate. It means these two particles are negatively charged. And for particle 3, it can also be observed that it moves towards the negatively charged plate and repels away from the positively charged plate. So, particle 3 is positively charged. **[1]**

The charge to mass ratio (e/m) is directly proportional to the displacement or amount of deflection at a given velocity. Since the deflection of particle 3 is the maximum as compare to 1 and 2 particles. Therefore, it will have higher charge to mass ratio. **[1]**

Q. 2. A conducting sphere of radius 10 cm has an unknown charge. If the electric field 20 cm from the centre of the sphere is 1.5×10^3 **N/C and points radially inward, what is the net charge on the sphere?** **[NCERT Ex. Q. 1.21, Page 48]**

Ans. As we know,

Electric field intensity (E) at a distance (d) from the centre of a sphere containing net charge 'q' is given by the relation,

$$E = \frac{q}{4\pi\varepsilon_0 d^2}$$

Where, q = Net charge = 1.5×10^3 N/C

d = Distance from the centre = 20 cm = 0.2 m

ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

$$q = E(4\pi\varepsilon_0)d^2 \quad \textbf{[1]}$$

$$= \frac{1.5 \times 10^3 \times (0.2)^2}{9 \times 10^9}$$

$= 6.67 \times 10^{-9}$ C
$= 6.67$ nC
So, the net charge on the sphere is 6.67 nC. **[1]**

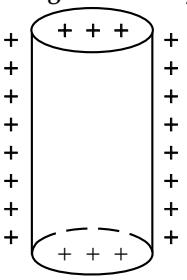
Q. 3. (a) An electrostatic field line is a continuous curve. That is, a field line cannot have sudden breaks. Why not?

(b) Explain why two field lines never cross each other at any point? [NCERT Ex. Q. 1.7, Page 46]

Ans. **(a)** An electrostatic field line is a continuous curve because when a charge enters in electrostatic field, charge experiences a continuous force. The charge moves continuously so that the field line cannot contain sudden breaks and charge cannot jump from one point to the other suddenly. Hence an electrostatic field line is a continuous curve. **[1]**

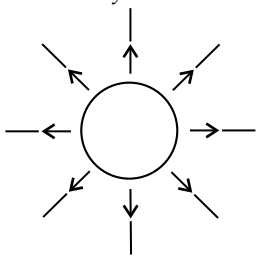
(b) If two field lines cross each other at a point, then electric field intensity will show two directions at that point, which is impossible. Hence, two field lines never cross each other. **[1]**

Q. 4. Sketch the electric field lines for a uniformly charged hollow cylinder shown in Figure.

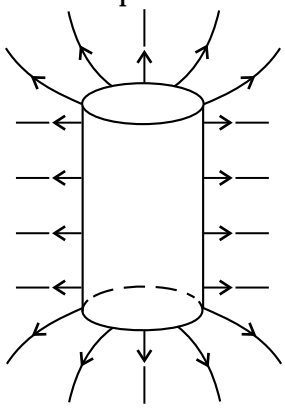


[NCERT Exemp. Q. 18, Page 5]

Ans. As there is no charge inside the hollow cylinders, so no negative charge will be induced. Only positive charge is spreading uniformly on the surface and lines of force emerge away perpendicularly from the surface to infinity.



Top view



Side view

[1]

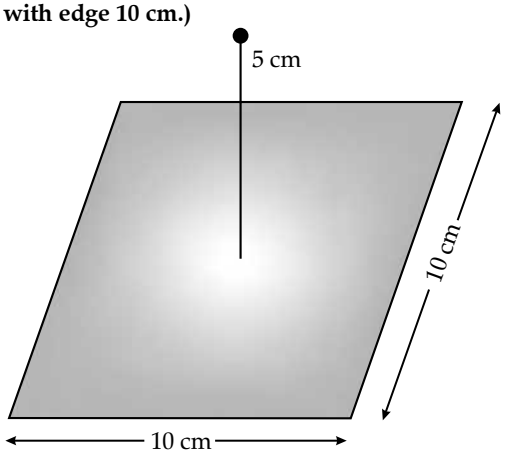
Q. 5. The dimensions of an atom are of the order of an Angstrom. Thus, there must be large electric fields between the protons and electrons. Why, then is the electrostatic field inside a conductor zero? [NCERT Exemp. Q. 16, Page 5]

Ans. In an atom, number of electrons and protons are equal and the electric fields bind the atoms to neutral entity. Electrostatic fields are caused by excess charges. **[½]**
But there can be no excess charge on the inter surface of an isolated conductor. So, the electrostatic fields inside a conductor are zero despite the fact that the dimensions of an atom are of the order of an Angstrom. **[½]**

Q. 6. Define Electric Flux. Write its SI unit.
[CBSE Board, Foreign Scheme, 2016]

Ans. Electric flux is a number of electric field lines passing through a given area. Its SI unit is : Nm^2C^{-1}. **[1]**

Q. 7. A point charge +10 μC is at distance 5 cm directly above the centre of a square of side 10 cm, as shown in Figure. What is the magnitude of the electric flux through the square?
(Hint : Think of the square as one face of a cube with edge 10 cm.)



[NCERT Ex. Q. 1.18, Page 47]

Ans. The square can be considered as one face of a cube of edge 10 cm with a centre where charge 'q' is placed. According to Gauss's theorem for a cube, total electric flux is through all its six faces.

$$\phi_{Total} = \frac{q}{\varepsilon_0}$$

So, electric flux through one face of the cube, that is, through the square,

$$\phi = \frac{\phi_{Total}}{6}$$

$$= \frac{1}{6} \times \frac{q}{\varepsilon_0}$$ **[1]**

Where, ε_0 = Permittivity of free space
$= 8.854 \times 10^{-12}\ N^{-1}C^2\,m^{-2}$
$q = 10\ \mu C = 10 \times 10^{-6}$ C

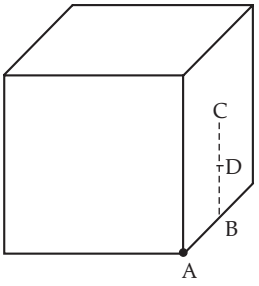
$$\phi = \frac{1}{6} \times \frac{10 \times 10^{-6}}{8.854 \times 10^{-12}}$$

$= 1.88 \times 10^5\ Nm^2C^{-1}$
So, electric flux through the square is 1.88×10^5 Nm^2C^{-1}. **[1]**

Q. 8. What is the net flux of the uniform electric field of (Q. 6, SATQ) through a cube of side 20 cm oriented so that its faces are parallel to the coordinate planes? [NCERT Ex. Q. 1.16, Page 47]

Ans. All the faces of a cube are parallel to the coordinate axes. So, the number of field lines entering the cube is equal to the number of field lines emerging out of the cube. As a result, net flux through the cube is zero. **[1]**

Q. 9. What will be the total flux through the faces of the cube in figure with side of length a if *a* charge *q* is placed at :



(a) A : a corner of the cube.
(b) B : mid-point of an edge of the cube.
(c) C : centre of a face of the cube.
(d) D : mid-point of B and C.
[NCERT Exemp. Q. 19, Page 6]

Ans. **(a)** If a charge is placed at corner A of the cube then only $\frac{1}{8}^{\text{th}}$ portion of the charge lies inside the Gaussian surface. So, total flux through the faces of the given cube = $\frac{q}{8\varepsilon_0}$. **[½]**

(b) If a charge is placed at point B of the cube then only $\frac{1}{4}^{\text{th}}$ portion of the charge lies inside the Gaussian surface. So, total flux through the faces of the given cube = $\frac{q}{4\varepsilon_0}$. **[½]**

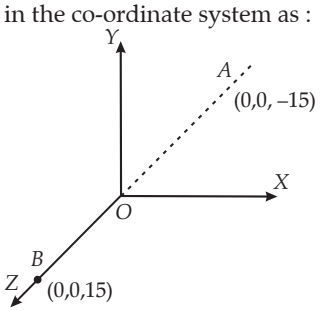
(c) If a charge is placed at point *C* of the cube then only $\frac{1}{2}$ portion of the charge lies inside the Gaussian surface. So, total flux through the faces of the given cube = $\frac{q}{2\varepsilon_0}$. **[½]**

only

(d) If a charge is placed at point *D* of the cube then $\frac{1}{2}$ portion of the charge lies inside the Gaussian surface. So, total flux through the faces of the given cube = $\frac{q}{2\varepsilon_0}$. **[½]**

Q. 10. A system has two charges $q_A = 2.5 \times 10^{-7}$ C and $q_B = -2.5 \times 10^{-7}$ C located at points A : (0, 0, –15 cm) and B : (0, 0, +15 cm), respectively. What are the total charge and electric dipole moment of the system? [NCERT Ex. Q. 1.9, Page 46]

Ans. The charges are located at the given points as shown in the co-ordinate system as :



At point *A*, total charge amount, $q_A = 2.5 \times 10^{-7}$C

At point *B*, total charge amount, $q_B = -2.5 \times 10^{-7}$C

Total charge of the system is,

$q_A + q_B = 2.5 \times 10^{-7}\text{C} - 2.5 \times 10^{-7} = 0$

Distance between two charges at points *A* and *B*,

$d = 15 + 15 = 30$ cm $= 0.3$ m

Electric dipole moment of the system is given by,

$p = q_A \times d = q_B \times d$

$= 2.5 \times 10^{-7} \times 0.3$

$= 7.5 \times 10^{-8}$C

So, the electric dipole moment of the system is $= 7.5 \times 10^{-8}$m along negative z-axis. **[2]**

Q. 11. A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge Q is placed at the centre of the spherical cavity. What will be surface charge density on (i) the inner surface, and (ii) the outer surface? [NCERT Exemp. Q. 15, Page 5]

Ans. When positive charge is placed at the centre of the spherical cavity then an equal amount of negative charge ($-Q$) appears on inner surface of the sphere due to induction.

This charge is distributed uniformly on the inner surface. So, an equal amount of positive charge ($+Q$) also appears on outer surface of the sphere.

Now surface charge density on inner surface =

$$\frac{\text{Total charge on inner surface}}{\text{Area of the inner surface}} = \frac{-Q}{(4\pi R_1^2)}$$ **[1]**

Surface charge density on outer surface =

$$\frac{\text{Total charge on outer surface}}{\text{Area of the outer surface}} = \frac{+Q}{(4\pi R_2^2)}$$ **[1]**

Q. 12. How does the electric flux due to a point charge enclosed by a spherical Gaussian surface get affected when its radius is increased?
[CBSE Board, Delhi Region, 2016]

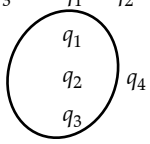
Ans. There will be no effect on electric flux, when the radius of Gaussian surface is increased because the charge enclosed by the Gaussian surface remains the same. **[1]**

Q. 13. If the total charge enclosed by a surface is zero, does it imply that the electric field everywhere on the surface is zero? Conversely, if the electric field everywhere on a surface is zero, does it imply that net charge inside is zero.
[NCERT Exemp. Q. 17, Page 5]

Ans. According to Gauss's law of electrostatics,

$$\phi = \oint_s E.dS = \frac{q}{\varepsilon_0}$$

Now, Consider a Gaussian surface enclosing charges q_1, q_2, q_3 and $q_1 + q_2 + q_3 = 0$ **[1]**



But the term q enclosed on the R.H.S of the equation includes the sum of all charges enclosed by the surface called gaussian surface which is equal to zero. But, on L.H.S., it will be not zero due to q_4. So, first statement is not correct. And,

If electric field on Gaussian surface is zero, then, in above case, it is possible only when $q_4 = 0$, *i.e.*, if everywhere, on Gaussian surface, electric field is zero then net charge will be zero. **[1]**

Q. 14. An arbitrary surface encloses a dipole. What is the electric flux through this surface?

[NCERT Exemp. Q. 14, Page 5]

Ans. By Gauss' theorem of electrostatics, the flux crossing a closed surface is equal to the net charge enclosed by the surface divided by ε_0.

$$\phi = \oint_s E.dS = \frac{q}{\varepsilon_0}$$ **[1]**

Net charge inside the Gaussian surface due to a dipole, $+q - q = 0$

$$\phi = \oint_s E.dS = \frac{q}{\varepsilon_0} = 0$$

Since the net charge of a dipole is zero, the total flux crossing the surface is also zero. **[1]**

Short Answer Type Questions **(3 marks each)**

Q. 1. Two-point charges $q_A = 3\ \mu C$ and $q_B = -3\ \mu C$ are located 20 cm apart in vacuum.

(a) What is the electric field at the midpoint *O* of the line *AB* joining the two charges?

(b) If a negative test charge of magnitude 1.5 × 10^{-9} C is placed at this point, what is the force experienced by the test charge?

[NCERT Ex. Q. 1.8, Page 46]

Ans. (a) Given that,

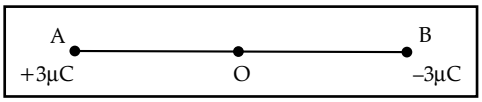
$q_A = 3\ \mu C = 3 \times 10^{-6} C$

$q_B = -3\ \mu C = -3 \times 10^{-6} C$

Distance $AB = 20$ cm $= 0.2$ m

In this situation, point, *O* is the mid-point of line *AB*. So :

$$OB = OA = \frac{0.2}{2} = 0.1 \text{ m}$$



Electric field at point O caused by $+3\ \mu C$ charge,

$$E_1 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{3\times10^{-6}}{(OA)^2} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{3\times10^{-6}}{(10\times10^{-2})^2} NC^{-1} \text{ along } OB$$

Where, ε_0 = Permittivity of free space and $\frac{1}{4\pi\varepsilon_0} = 9\times10^9 Nm^2C^{-2}$

Therefore,

Electric field at point *O* caused by $-3\ \mu C$ charge,

$$E_2 = \left| \frac{1}{4\pi\varepsilon_0} \cdot \frac{-3\times10^{-6}}{(OB)^2} \right|$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{3\times10^{-6}}{(10\times10^{-2})^2} \text{NC}^{-1} \text{ along } OB$$

$$\therefore E_1 + E_2 = 2 \times \frac{1}{4\pi\varepsilon_0} \cdot \frac{3\times10^{-6}}{(10\times10^{-2})^2} NC^{-1} \text{ along } OB$$

[Since the magnitude of E_1 and E_2 are equal and in the same direction.]

$$\therefore E = 2\times9\times10^9 \times \frac{3\times10^{-6}}{(10\times10^{-2})^2} \text{NC}^{-1} \text{ along } OB$$

$$= 5.4\times10^6 \text{ NC}^{-1}$$

So, the electric field at mid-point *O* is $5.4 \times 10^6\ NC^{-1}$ along *OB*.

The direction of electric field is always positive to negative so that direction of electric field will in direction of *OB*. **[1½]**

(b) $q = 1.5 \times 10^{-9}$ C

Let Force experienced by the test charge = *F*

As we know the formula of electrostatic force,

$F = qE$

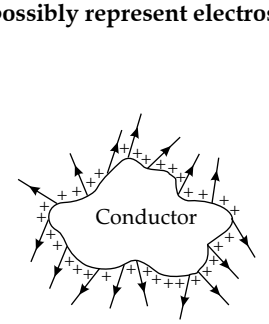
Put the values in this formula we get

$F = 1.5 \times 10^{-9} \times 5.4 \times 10^6 = 8.1 \times 10^{-3}$ N

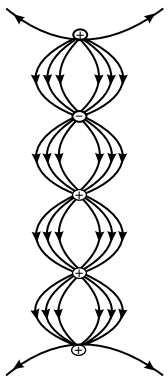
Here placed charge is negative so negative charge at point B will repel while positive change at point A will attract. Hence the direction of force will along *OA*.

So, the force experienced by the test charge is 8.1×10^{-3} N along the direction of *OA*. **[1½]**

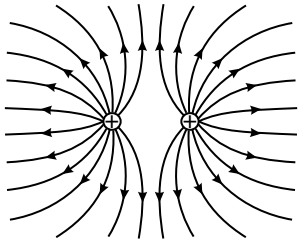
Q. 2. Which among the curves shown in Figure cannot possibly represent electrostatic field lines?



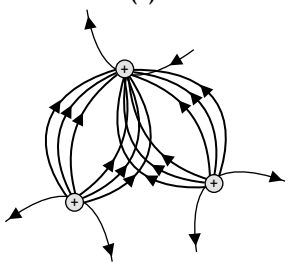
(a)



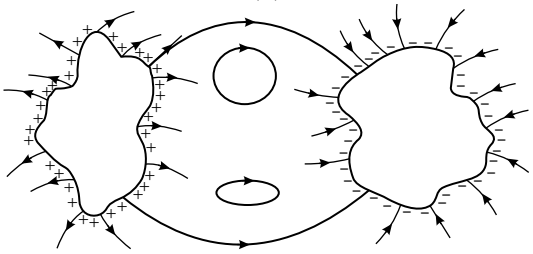
(b)



(c)



(d)



(e)

[NCERT Add. Ex. Q. 1.26, Page 48]

Ans. **(a)** The field lines shown in (a) do not represent electrostatic field lines because field lines must be normal to the surface of the conductor. **[½]**

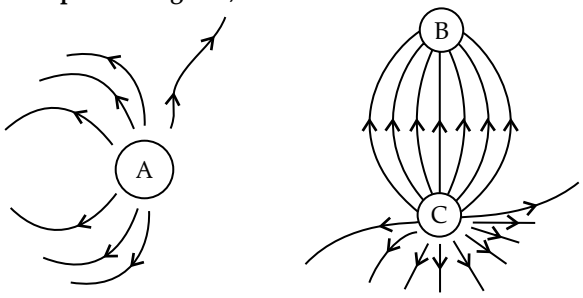
(b) The field lines shown in (b) do not represent electrostatic field lines because the field lines cannot emerge out from a negative charge and cannot terminate at a positive charge. **[½]**

(c) The field lines showed in (c) represent electrostatic field lines. This is because the field lines emerge from the positive charges and repel each other. **[½]**

(d) The field lines showed in (d) do not represent electrostatic field lines because the field lines should not intersect each other. **[½]**

(e) The field lines showed in (e) do not represent electrostatic field lines because closed loops are not formed in the area between the field lines. **[1]**

Q. 3. Figure shows the electric field lines around three-point charges A, B and C.



(a) Which charges are positive?

(b) Which charge has the largest magnitude? Why?

(c) In which region or regions of the picture could the electric field be zero? Justify your answer.

(d) (i) near A, (ii) near B, (iii) near C, (iv) nowhere.

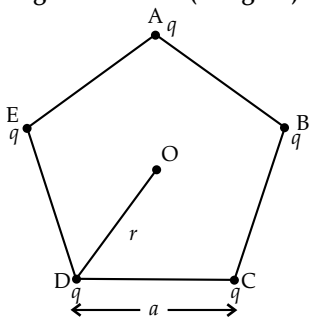
[NCERT Exemp. Q. 1.24, Page 7]

Ans. **(a)** Electric lines of forces diverge out from positive charge. So, A and C are positive charges. **[1]**

(b) As the density of electric lines forces from charge increases, the intensity of electric field or magnitude of charge will increase. Hence, maximum numbers of field lines are associated with C. **[1]**

(c) Point between two like charges where electrostatic force is zero is called neutral point. So, the neutral point lies between A and C only. Now the position of neutral point depends on the strength of the forces of charges. Here, more number of electric lines of forces shows higher strength of charge C than A. So, neutral point lies near A. As the magnitude of charge C is larger than A so lines of force of C will be stronger than of A. So, the neutral point lies somewhere near A. So, option (a) is the correct answer. **[1]**

Q. 4. Five charges, q each are placed at the corners of a regular pentagon of side 'a' (in figure).



(a) (i) What will be the electric field at O, the centre of the pentagon?

(ii) What will be the electric field at O if the charge from one of the corners (say A) is removed?

(iii) What will be the electric field at O if the charge q at A is replaced by $-q$?

(b) How would your answer to (a) be affected if pentagon is replaced by n-sided regular polygon with charge q at each of its corners?

[NCERT Exemp. Q. 1.25, Page 7]

Ans. **(a) (i)** The point O is equidistant from all the charges at the end point of pentagon. So, due to symmetry, the forces due to all the charges are cancel out each other. So, electric field at O is zero.

(ii) When charge q is removed a negative charge will develop at A giving electric field

$$E = \frac{(q \times 1)}{4\pi\varepsilon_0 r^2} \text{ along } OA.$$

(iii) If charge q at A is replaced by $-q$, then two negative charges $-2q$ will develop there. Thus, the value of electric field,

$$E = \frac{2q}{4\pi\varepsilon_0 r^2} \text{ along } OA. \quad \textbf{[1½]}$$

(b) Pentagon is replaced by the n-sided polygon :
When pentagon is replaced by n-sided regular polygon with charge q at each of its corners, the electric field at O would continue to be zero as

symmetry of the charges is due to the regularity of the polygon. It doesn't depend on the number of sides or the number of charges. **[1½]**

Q. 5. Careful measurement of the electric field at the surface of a black box indicates that the net outward flux through the surface of the box is 8.0×10^3 Nm^2/C.

(a) What is the net charge inside the box?

(b) If the net outward flux through the surface of the box were zero, could you conclude that there were no charges inside the box? Why or Why not?

[NCERT Ex. Q. 1.17, Page 47]

Ans. **(a)** Net outward flux through the surface of the box, $\phi = 8.0 \times 10^3$ Nm^2/C

For a body containing net charge q, flux is given by the relation,

$$\phi = \frac{q}{\varepsilon_0}$$

Where,

ε_0 = Permittivity of free space

$= 8.854 \times 10^{-12}\ N^{-1}C^2m^{-2}$

$q = \varepsilon_0\phi$

$= 8.854 \times 10^{-12} \times 8.0 \times 10^3$

$= 7.08 \times 10^{-8}$

$= 0.07\ \mu C$

So, the net charge inside the box is 0.07 μC. **[1½]**

(b) No, net flux emerging out through a body depends on the net charge contained in the body. If net flux is zero, then it can be inferred that net charge inside the body is zero. The body may have equal amount of positive and negative charges. **[1½]**

Q. 6. Consider a uniform electric field $E = 3 \times 10^3\ \hat{i}$ N/C.

(a) What is the flux of this field through a square of 10 cm on a side whose plane is parallel to the *yz* plane?

(b) What is the flux through the same square if the normal to its plane makes a 60° angle with the x-axis? [NCERT Ex. Q. 1.15, Page 47]

Ans. **(a)** Electric field intensity, $= 3 \times 10^3\ \hat{i}$ N/C

Magnitude of electric field intensity,

$|\vec{E}| = 3 \times 10^3$ N/C

Side of the square, $s = 10$ cm $= 0.1$ m

Area of the square, $A = s^2 = 0.01$ m^2

The plane of the square is parallel to the y-z plane. Hence, angle between the unit vector normal to the plane and electric field, $\theta = 0$.

Flux (ϕ) through the plane is given by the relation,

$\phi = |\vec{E}|$ A Cosθ

$= 3 \times 10^3 \times 0.01 \times \cos 0°$

$= 30$ Nm^2/C **[1½]**

(b) Plane makes an angle of 60° with the x-axis. Hence, $\theta = 60°$

Flux, $\phi = |\vec{E}|$ A cosθ

$= 3 \times 10^3 \times 0.01 \times \cos 60°$

$= 15$ N m^2/C **[1½]**

Q. 7. In a certain region of space, electric field is along the z-direction throughout. The magnitude of electric field is, however, not constant but increases uniformly along the positive z-direction, at the rate of 10^5 NC^{-1} per metre. What are the force and torque experienced by a system having a total dipole moment equal to 10^{-7}cm in the negative z-direction? [NCERT Add. Ex. Q. 1.27, Page 49]

Ans. Given that,

Dipole moment of the system,

$p = q \times dl$

$= -10^{-7}$ cm

Rate of increase of electric field per unit length,

$\frac{dE}{dl} = 10^5$ NC^{-1}/m **[1]**

As we know that, the force, F experienced by the system is given by,

$F = qE$

$$F = q\frac{dE}{dl} \times dl$$

$$= p \times \frac{dE}{dl}$$ **[1]**

Put the values

$= -10^{-7} \times 10^{+5}$

$= -10^{-2}$ N

The force is -10^{-2} N in the negative z-direction which is opposite to the direction of electric field.

So, the angle between electric field and dipole moment will be 180°.

Torque (τ) is given by the relation, $\tau = pE \sin 180° = 0$

So, the torque experienced by the system is zero. **[1]**

Q. 8. An electric dipole with dipole moment 4×10^{-9} C m is aligned at 30° with the direction of a uniform electric field of magnitude 5×10^4 NC^{-1}. Calculate the magnitude of the torque acting on the dipole.

[NCERT Ex. Q. 1.10, Page 46]

Ans. Given that,

Electric dipole moment, $p = 4 \times 10^{-9}$ Cm

Angle made by p with a uniform electric field, $\theta = 30°$

Electric field, $E = 5 \times 10^4$ NC^{-1}

As we know that,

Torque acting on the dipole is given by the relation,

$\tau = pE \sin\theta$ **[1½]**

$= 4 \times 10^{-9} \times 5 \times 10^4 \times \sin 30°$

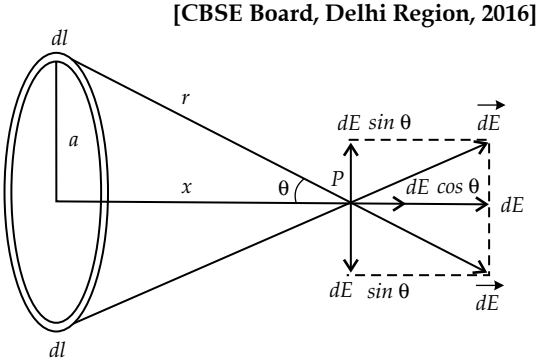
$= 20 \times 10^{-5} \times \frac{1}{2} = 10^{-4}$ Nm

So, the magnitude of the torque acting on the dipole is 10^{-4} Nm. **[1½]**

Q. 9. A charge is distributed uniformly over a ring of radius '*a*'. Obtain an expression for the electric intensity E at a point on the axis of the ring. Hence show that for point's at large distances from the ring, it behaves like a point charge.

[CBSE Board, Delhi Region, 2016]

Ans.



[1]

Let assume that the ring is placed with its plane perpendicular to the x-axis, as shown in the above figure. Consider a small element *dl* of the ring.

So the charge *dq* on the element *dl* is :

As shown in the figure, the field $d\vec{E}$ has two components :

(i) the axial point $dE\cos\theta$

(ii) the perpendicular component $dE\sin\theta$ **[1]**

Since the perpendicular components of any two diametrically opposite elements are equal and opposite, they all cancel out in pairs. Only the axial components will add up to produce the resultant field $\vec{E}$ at point *P*, which is given by,

$$E = \int_0^{2\pi a} dE\cos\theta$$

$$E = \int_0^{2\pi a} \frac{kq}{2\pi a}\frac{dl}{r^2}\frac{x}{r} = \frac{kqx}{2\pi a}\frac{1}{r^3}\int_0^{2\pi a} dl \qquad [as\ \cos\theta = \frac{x}{r}]$$

$$E = \frac{kqx}{2\pi a}\frac{1}{r^3}[l]_0^{2\pi a}$$

$$= \frac{kqx}{2\pi a}\frac{1}{(x^2+a^2)^{\frac{3}{2}}}2\pi a \qquad [as\ r^2 = x^2 + a^2]$$

$$\text{or } E = \frac{kqx}{(x^2+a^2)^{\frac{3}{2}}} = \frac{1}{4\pi\varepsilon_0}\frac{qx}{(x^2+a^2)^{\frac{3}{2}}}$$

Special case : For points at large distances from the ring $x >> a$.

$$E = \frac{kq}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2}$$

This is the same as the field due to a point charge, indicating that for far off axial points, the charged ring behaves as a point charge. **[1]**

Q. 10. A uniformly charged conducting sphere of 2.4 m diameter has a surface charge density of 80.0 μC/m^2.

(a) Find the charge on the sphere.

(b) What is the total electric flux leaving the surface of the sphere? [NCERT Ex. Q. 1.22, Page 48]

Ans. **(a)** Given that,

Diameter of the sphere, $d = 2.4$ m

Radius of the sphere, $r = 1.2$ m

Surface charge density, $\sigma = 80.0\ \mu C/m^2$

$= 80 \times 10^{-6}\ C/m^2$

Total charge on the surface of the sphere,

Q = Charge density × Surface area

$= \sigma \times 4\pi r^2$

$= 80 \times 10^{-6} \times 4 \times 3.14 \times (1.2)^2$

$= 1.447 \times 10^{-3}$ C

Therefore, the charge on the sphere is 1.447×10^{-3} C. **[1½]**

(b) Total electric flux (ϕ_{Total}) leaving out the surface of a sphere containing net charge Q is given by the relation,

$$\phi_{Total} = \frac{Q}{\varepsilon_0}$$

Where, ε_0 = Permittivity of free space

$\varepsilon_0 = 8.854 \times 10^{-12}\ N^{-1}C^2m^{-2}$

$Q = 1.447 \times 10^{-3}$ C

$$\phi_{Total} = \frac{1.44\times10^{-3}}{8.854\times10^{-12}}$$

$= 1.63 \times 10^8\ NC^{-1}m^2$

So, the total electric flux leaving the surface of the sphere is $1.63 \times 10^8\ NC^{-1}m^2$. **[1½]**

Q. 11. An infinite line charge produces a field of 9 × 10^4 N/C at a distance of 2 cm. Calculate the linear charge density. [NCERT Ex. Q. 1.23, Page 48]

Ans. Electric field produced by the infinite line charges at a distance d having linear charge density λ is, E

$$E = \frac{\lambda}{2\pi\varepsilon_0 d}$$

$$\lambda = 2\pi\varepsilon_0 dE$$ **[1½]**

Where,

$d = 2$ cm $= 0.02$ m

$E = 9 \times 10^4$ N/C

ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ Nm^2C^{-2}$$

$$\lambda = \frac{0.02\times9\times10^4}{2\times9\times10^9} = 10^{-7} C/m$$

Thus, the linear charge density is 10^{-7}C/m. **[1½]**

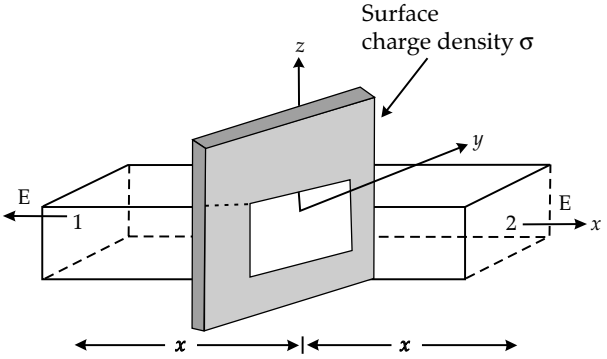
Q. 12. (a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density σ.

[CBSE Board, Delhi Region, 2017]

Ans. Field due to a uniformly charged infinite plane sheet :

Let σ be the uniform surface charge density of an infinite plane sheet.

Let's take the Gaussian surface to be a rectangular parallelepiped of cross sectional area A, as shown.



[1]

As seen from the figure, only the two faces 1 and 2 will contribute to the flux; electric field lines are parallel to the other faces and they, therefore, do not contribute to the total flux.

The unit vector normal to surface 1 is in $-x$ direction while the unit vector normal to surface 2 is in the $+x$ direction.

Therefore, fluxes $E.\Delta S$ through both the surfaces are equal and add up. Therefore, the net flux through the Gaussian surface is $2EA$. The charge enclosed by the closed surface is σA. **[1]**

So, by Gauss's law,

$$2EA=\frac{\sigma A}{\varepsilon_0}$$

or, $E=\frac{\sigma}{2\varepsilon_0}$

Vectorically,

$$E=\frac{\sigma}{2\varepsilon_0}\hat{n}$$

Where $\hat{n}$ is a unit vector normal to plane and going away from it.

E is directed away from the plate if σ is positive and toward the plate if σ is negative. **[1]**

Q. 13. Find the electric field intensity due to a uniformly charged spherical shell at a point

(i) outside the shell and

(ii) inside the shell.

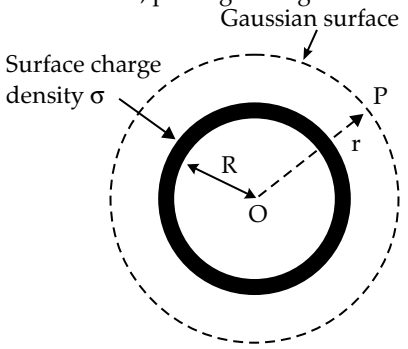
Plot the graph of electric field with distance from the centre of the shell.

[CBSE Board, Delhi Region, 2016]

Ans. (i) Field outside the shell :

Consider a point *P* outside the shell with radius vector r.

Let's take Gaussian surface to be a sphere of radius *r* and with centre *O*, passing through *P*.



The electric field at each point of the Gaussian surface, therefore, has the same magnitude *E* and is along the radius vector at each point.

Thus, *E* and Δ*S* at every point are parallel and the flux through each element is *E*Δ*S*. Summing over all Δ*S*, the flux through the Gaussian surface is,

$= E \times 4\pi r^2$.

The charge enclosed is,

$= \sigma \times 4\pi R^2$ [By Gauss's Law]

$$E\times 4\pi r^2=\frac{\sigma}{\varepsilon_0}4\pi R^2$$

Or, $E=\frac{\sigma R^2}{\varepsilon_0 r^2}=\frac{q}{4\pi\varepsilon_0 r^2}$

Where $q=4\pi R^2\sigma$ is the total charge on the spherical shell.

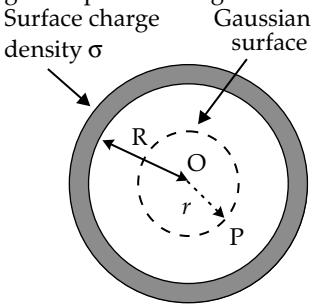
Vectorially,

$$E=\frac{q}{4\pi\varepsilon_0 r^2}\hat{r}$$

The electric field is directed outward if $q > 0$ and inward if $q < 0$. This, however, is exactly the field produced by a charge '*q*' placed at the centre O. Thus, for points outside the shell, the field due to a uniformly charged shell is as if the entire charge of the shell is concentrated at its centre. **[1½]**

(ii) Field inside the shell :

Let the point *P* is inside the shell. The Gaussian surface is again a sphere through P centred at *O*.



The flux through the Gaussian surface,

$= E \times 4\pi r^2$

However, in this case, the Gaussian surface encloses no charge.

Gauss's law then gives,

$= E \times 4\pi r^2$

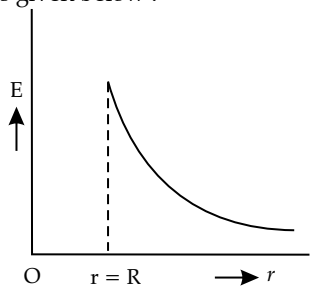
$= 0$

That means, $E = 0$ $(r < R)$

So, the field due to a uniformly charged thin shell is zero at all points inside the shell.

The variation of the electric field intensity $E(r)$ with distance, *r* from the centre for shell

$0 \le r \le \infty$ is given below :



[1½]

Q. 14. A point charge of 2.0 μC is at the centre of a cubic Gaussian surface 9.0 cm on edge. What is the net electric flux through the surface?

[NCERT Ex. Q. 1.19, Page 48]

Ans. As we know

Net electric flux (ϕ_{Net}) through the cubic surface is given by,

$$\phi_{Net}=\frac{q}{\varepsilon_0}$$ **[1]**

Where,

ε_0 = Permittivity of free space

$= 8.854 \times 10^{-12}\ N^{-1}C^2m^{-2}$

q = Net charge contained inside the cube

$= 2.0\ \mu C = 2 \times 10^{-6}\ C$

$$\phi_{Net}=\frac{2\times10^{-6}}{8.854\times10^{-12}}$$ **[1]**

$= 2.26 \times 10^5\ Nm^2C^{-1}$

The net electric flux through the surface is $2.26 \times 10^5\ Nm^2C^{-1}$. **[1]**

Q. 15. A point charge causes an electric flux of –1.0 × 10^3 Nm^2/C to pass through a spherical Gaussian surface of 10.0 cm radius centred on the charge.

(a) If the radius of the Gaussian surface were doubled, how much flux would pass through the surface?

(b) What is the value of the point charge?

[NCERT Ex. Q. 1.20, Page 48]

Ans. **(a)** Given that,

Electric flux, $\phi = -1.0 \times 10\ Nm^2C^{-1}$

Radius of the Gaussian surface, $r = 10.0$ cm

Electric flux emerging out through a surface depends on the net charge enclosed inside the body and it does not depend on the size of the body. If the radius of the Gaussian surface is doubled, then the flux passing through the surface remains the same that is $-1.0 \times 10^3\ Nm^2C^{-1}$. **[1½]**

(b) Electric flux is,

$$\phi = \frac{q}{\varepsilon_0}$$

Where,

q = Net charge enclosed by the spherical surface

ε_0 = Permittivity of free space = $8.854 \times 10^{-12}\ N^{-1}C^2m^{-2}$

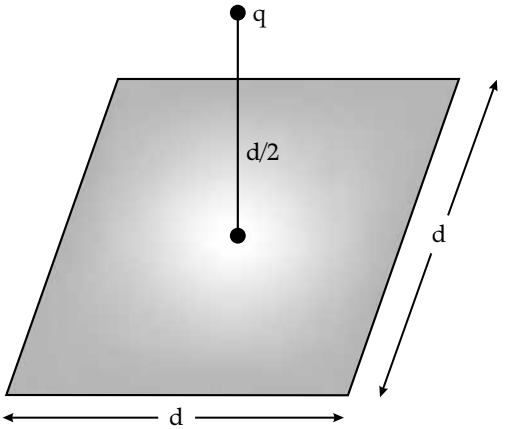
And we know that,

$$q = \phi\varepsilon_0$$
$$= -1.0 \times 10^3 \times 8.854 \times 10^{-12}$$
$$= -8.854 \times 10^{-9}\ C$$
$$= -8.854\ nC$$

So, the value of the point charge is -8.854 nC. **[1½]**

Long Answer Type Questions (5 marks each)

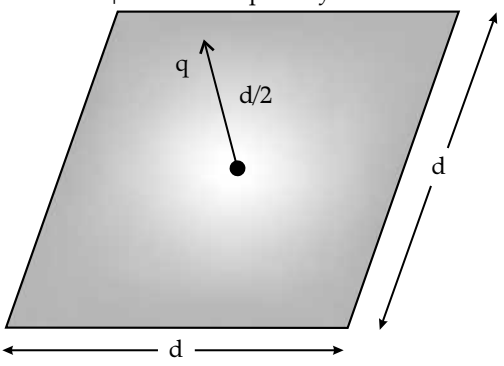
Q. 1. (a) Define electric flux. Is it a scalar or a vector quantity?



A point charges q is at a distance of d/2 directly above the centre of a square of side a, as shown in the figure. Use Gauss' law to obtain the expression for the electric flux through the square.

(b) If the point charge is now moved to a distance 'd' from the centre of the square and the side of the square is doubled, explain how the electric flux will be affected. [CBSE Board, Delhi Region, 2018]

Ans. **(a) Electric flux :** The number of electric field lines passing through any area normally is known as electric flux ϕ. It is scalar quantity.



Making square of the cube and charge becomes at centre.

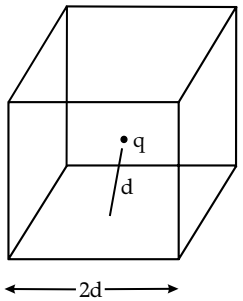
$$\phi = \frac{q_{in}}{\varepsilon_0}$$

$$\phi = \frac{q}{\varepsilon_0}$$

From only one square of cube

$$\phi = \frac{q}{6\varepsilon_0}$$ **[2½]**

(b) If we moved point charge 'to a distance 'd' from the center



$$\phi = \frac{q}{\varepsilon_0}$$

$$\phi = \frac{q}{6\varepsilon_0}$$

That means the flux will remain same in both cases. **[2½]**

Q. 2. (a) Derive an expression for the electric field E due to a dipole of length '$2a$' at a point distant r from the centre of the dipole on the axial line.

(b) Draw a graph of E versus r for $r >> a$.

(c) If this dipole were kept in a uniform external electric field E_0, diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expressions for the torque acting on the dipole in both the cases.

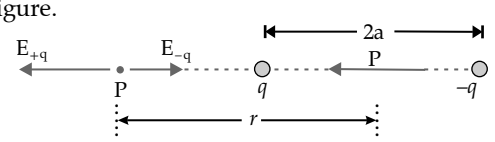
[CBSE Board, Delhi Region, 2017]

Ans. **(a) The field of an electric dipole :**

The electric field of the pair of charges ($-q$ and q) at any point in space can be found out from Coulomb's law and the superposition principle.

For points on the axis :

Let the point P be at distance r from the centre of the dipole on the side of the charge q, as shown in Figure.



Then,

$$E_{-q} = -\frac{q}{4\pi\varepsilon_0(r+a)^2}\hat{p}$$

where $\hat{p}$ is the unit vector along the dipole axis (from $-q$ to q). Also

$$E_{+q} = -\frac{q}{4\pi\varepsilon_0(r-a)^2}\hat{p}$$

The total field at P is

$$E = E_{+q} + E_{-q}$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right]\hat{p}$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{4ar}{(r^2-a^2)^2}\hat{p}$$

For $r >> a$

$$E = \frac{4qa}{4\pi\varepsilon_0 r^3}\hat{p} \qquad (r >> a)$$

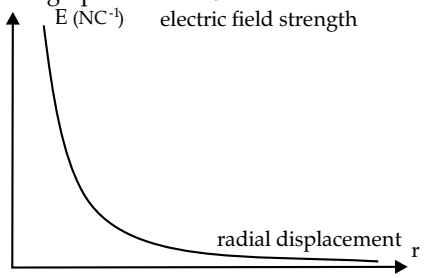
[2½]

(b) When $r > > a$ then,

$$|E_p| = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{r^3}$$

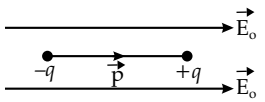
$$|E_p| \propto \frac{1}{r^3}$$

So the graph will be like below :



[1]

(c) For stable equilibrium, the angle between p and E is 0°.



For unstable equilibrium, the angle between $\vec{p}$ and E is 180°

Torque in both the cases is given by,

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$\Rightarrow |\vec{\tau}| = pE\sin\theta$$

In first case,

$$|\vec{\tau}| = pE\sin 0^\circ = 0 \qquad (\theta = 0^\circ)$$

In second case,

$$|\vec{\tau}| = pE\sin 180^\circ = 0 \qquad (\theta = 180^\circ)$$

[1½]

Q. 3. Two large, thin metal plates are parallel and close to each other. On their inner faces, the plates have surface charge densities of opposite signs and of magnitude 17.0×10^{-22} C/m^2. What is E :

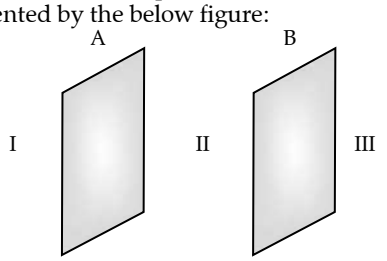
(a) in the outer region of the first plate,

(b) in the outer region of the second plate, and

(c) between the plates?

[NCERT Ex. Q. 1.24 ,Page 48]

Ans. From the above question the condition is represented by the below figure:



As, A and B are two parallel plates close to each other. The outer region of plate A is labelled as I and outer region of plate B is labelled as III, and the region between the plates, A and B, is labelled as II. So,

Charge density of plate A, $\sigma = 17.0 \times 10^{-22}$ C/m^2

[1½]

Charge density of plate B, $\sigma = -17.0 \times 10^{-22}$ C/m^2

[1½]

In the regions, I and III, electric field E is zero. This is because charge is not enclosed by the respective plates.

Electric field E in region II is given by the relation,

$$E = \frac{\sigma}{\varepsilon_0}$$

Where,

ε_0 = Permittivity of free space

$= 8.854 \times 10^{-12}\ N^{-1}C^2m^{-2}$

$$E = \frac{17.0 \times 10^{-22}}{8.854 \times 10^{-12}}$$

$= 1.92 \times 10^{-10}$ N/C

So, the electric field between the plates is 1.92×10^{-10} N/C. **[2]**

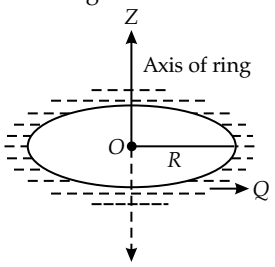
Q. 4. Total charge $-Q$ is uniformly spread along length of a ring of radius R. A small test charge $+q$ of mass m is kept at the centre of the ring and is given a gentle push along the axis of the ring.

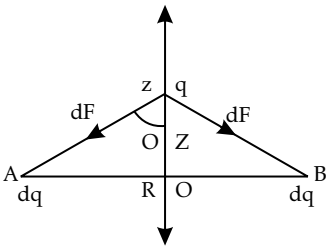
(a) Show that the particle executes a simple harmonic oscillation.

(b) Obtain its time period.

[NCERT Exemp. Q. 1.31, Page 9]

Ans. Slight push on q along the axis of the ring gives rise to the situation shown in Figure. A and B are two points on the ring at the end of a diameter.





[1]

Plane of the ring

Let the charge q is displaced slightly by z ($z << R$) along the axis of ring. Let force on the charge q will be towards O. The motion of charge q to be simple harmonic, if the force on charge q must be proportional to z and is directed towards O.

Electric field at axis of the ring at a distance z from the centre of ring

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Qz}{(R^2+z^2)^{\frac{3}{2}}}; \qquad \text{towards O}$$

Net force on the charge $F_{net} = qE$

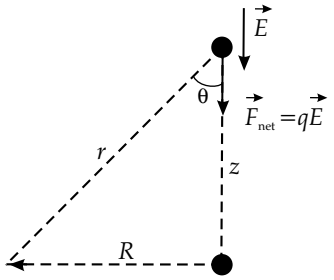
$$F_{net} = \frac{1}{4\pi\varepsilon_0}\frac{qQz}{(R^2+z^2)^{\frac{3}{2}}}$$

$$\Rightarrow \qquad F_{net} = \frac{1}{4\pi\varepsilon_0}\frac{qQz}{R^3\left(1+\frac{z^2}{R^2}\right)^{\frac{3}{2}}}$$

As $z << R$ then, $F_{net} = \frac{1}{4\pi\varepsilon_0}\frac{qQz}{R^3}$

or, $\overrightarrow{F_{net}} = -K\vec{z}$

where $K = \frac{Qq}{4\pi\varepsilon_0 R^3} = \text{constant}$ **[1½]**



[1]

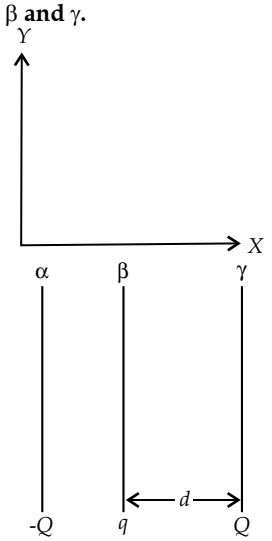
Clearly, force on q is proportional to negative of its displacement. Therefore, motion, of q is simple harmonic.

$$\omega = \sqrt{\frac{K}{m}} \text{ and } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{K}}$$

$$T = 2\pi\sqrt{\frac{m4\pi\varepsilon_0 R^3}{Qq}}$$

$$T = 2\pi\sqrt{\frac{4\pi\varepsilon_0 mR^3}{Qq}}$$ **[1½]**

Q. 5. Two fixed, identical conducting plates (α and β), each of surface area S are charged to $-Q$ and q, respectively, where $Q > q > 0$. A third identical plate (γ), free to move is located on the other side of the plate with charge q at a distance d (in Figure). The third plate is released and collides with the plate β. Assume the collision is elastic and the time of collision is sufficient to redistribute charge amongst β and γ.



(a) Find the electric field acting on the plate γ before collision.

(b) Find the charges on β and γ after the collision.

(c) Find the velocity of the plate γ after the collision and at a distance d from the plate β.

[NCERT Exemp. Q. 1.28, Page 8]

Ans. **(a)** Net electric field at plate y before collision is equal to the sum of electric field at plate γ due to plate α and β

The electric field at plate γ due to plate α is $E_1 = \frac{-Q}{S(2\varepsilon_0)}$ to the left.

The electric field at plate γ due to plate β is $E_2 = \frac{q}{S(2\varepsilon_0)}$ to the right.

So, the net electric field at plate y before collision,

$$E = E_1 + E_2 = \frac{q-Q}{S(2\varepsilon_0)} \text{ to the left, if } Q > q. \qquad \textbf{[1½]}$$

(b) During collision, plates β and γ are together. Their potentials become same.

Suppose charge on plate β is q_1, and charge on plate γ is q_2. At any point O. in between the two plates, the electric field must be zero.

Electric field at O due to plate $\alpha = \frac{-Q}{S(2\varepsilon_0)}$ to the left

Electric field at O due to plate $\beta = \frac{-q_1}{S(2\varepsilon_0)}$ to the right

Electric field at O due to plate $\gamma = \frac{-q_2}{S(2\varepsilon_0)}$ to the left

As the electric field at O is zero, therefore

$$\frac{Q+q_2}{S(2\varepsilon_0)} = \frac{q_1}{S(2\varepsilon_0)}$$

$$\therefore Q + q_2 = q_1$$

$$Q = q_1 - q_2 \qquad \text{...(i)}$$

As there is no loss of charge on collision,

$$Q + q = q^1 + q^2 \qquad \text{...(ii)}$$

On solving Equations (i) and (ii), we get

$q_1 = \left(Q + \frac{q}{2}\right) =$ Charge on plate β

$q_2 = \left(\frac{q}{2}\right) =$ Charge on plate γ

[1½]

(c) After collision, at a distance d from plate β.
Let the velocity of plate γ be v. After the collision, electric field at plate γ is

$$E_2 = \frac{-Q}{2\varepsilon_0 S} + \frac{(Q + \frac{q}{2})}{2\varepsilon_0 S} = \frac{\frac{q}{2}}{2\varepsilon_0 S} \text{ to the right}$$

Just before collision, electric field at plate γ is

$$E_1 = \frac{Q-q}{2\varepsilon_0 S}$$

If F; is force on plate γ before collision, then

$$F_1 = E_1 Q = \frac{(Q-q)Q}{2\varepsilon_0 s}$$

Similarly, Force F_2 on plate r after collision,

$$F_2 = E_2,\ \frac{q}{2} = \frac{(q/2)^2}{2\varepsilon_0 S}$$

Total work done by the electric field in round trip movement of plate γ

$$w = (F_1 + F_2)d$$

$$= \frac{[(Q-q)Q + (\frac{q}{2})^2]d}{2\varepsilon_0 S} = \frac{(Q - \frac{q}{2})^2 d}{2\varepsilon_0 S}$$

If m is mass of plate γ, the K.E. gained by plate

$$\gamma = \frac{1}{2}mv^2$$

According to work-energy principle,

$$\frac{1}{2}mv^2 = \mathrm{W} = \frac{\left(Q - \frac{q}{2}\right)^2 d}{2\varepsilon_0 S}$$

$$v^2 = \frac{2d\left(Q - \frac{q}{2}\right)^2}{2\varepsilon_0 S m}$$

$$v = \left(Q - \frac{q}{2}\right)\left(\frac{d}{m\varepsilon_0 S}\right)^{\frac{1}{2}}$$

[2]

Q. 6. (a) Consider an arbitrary electrostatic field configuration. A small test charge is placed at a null point (*i.e.*, where $E = 0$) of the configuration. Show that the equilibrium of the test charge is necessarily unstable.

(b) Verify this result for the simple configuration of two charges of the same magnitude and sign placed a certain distance apart.

[NCERT Add. Ex. Q. 1.32, Page 50]

Ans. (a) Let the equilibrium of the test charge be stable. Then, if a test charge is in equilibrium and displaced from its position in any direction, then it experiences a restoring force towards a null point, where the electric field is zero. All the field lines near the null point are directed inwards towards the null point. There is a net inward flux of electric field through a closed surface around the null point.
According to Gauss's law, the flux of electric field through a surface, which is not enclosing any charge, is zero. Hence, the equilibrium of the test charge cannot be stable. **[2½]**

(b) The two charges of same magnitude and same sign are placed at a certain distance. The mid-point of the joining line of the charges is the null point. When a test charged is displaced along the line, it experiences a restoring force. If it is displaced normal to the joining line, then the net force takes it away from the null point. So, the charge is unstable because stability of equilibrium requires restoring force in all directions. **[2½]**

Q. 7. In 1959, Lyttleton and Bondi suggested that the expansion of the Universe could be explained if matter carried a net charge. Suppose that the Universe is made up of hydrogen atoms with a number density *N*, which is maintained a constant. Let the charge on the proton be :

$e_p = -(1 + y)e$ where 'e' is the electronic charge.

(a) Find the critical value of y such that expansion may start.

(b) Show that the velocity of expansion is proportional to the distance from the centre.

[NCERT Exemp. Q. 1.26, Page 7]

Ans. (a) Let us assume that the universe is a perfect sphere of radius 'R' and its constituent hydrogen atoms which is distributed uniformly in the sphere. As we know that the hydrogen atom contains one proton and one electron, charge on each hydrogen atom.

$$e_H = e_p + e$$

$$= -(1 + y)e + e$$

$$= -ye = |ye|$$

If E is electric field intensity at distance R, on the surface of the sphere, then according to Gauss' theorem,

$$\oint E.dS = \frac{q}{\varepsilon_0} \qquad i.e., E(4\pi R^2) = \frac{4}{3}\frac{\pi R^3 N|ye|}{\varepsilon_0}$$

$$E = \frac{1}{3}\frac{N|ye|R}{\varepsilon_0} \qquad \text{...(i)}$$

Now, suppose, mass of each hydrogen atom $\simeq m_p$ = Mass of a proton,
G_R = Gravitational field at distance R on the sphere. Then,

$$-4\pi R^2 G_R = 4\pi G m_p (\frac{4}{3}\pi R^3)N$$

$$\Rightarrow \quad G_R = \frac{-4}{3}\pi G m_p N R \qquad \text{...(ii)}$$

Gravitational force on this atom is

$$F_G = m_P \times G_R = \frac{-4\pi}{3} G m_p{}^2 N R \qquad \text{...(iii)}$$

Coulomb force on hydrogen atom at R is

$$F_C = (ye)E = \frac{1}{3}\frac{Ny^2e^2R}{\varepsilon_0} \qquad \text{[From equation (i)]}$$

Now, to start expansion $F_C > F_G$ and critical value of γ to start expansion would be when

$$F_C = F_G$$

$$\Rightarrow \frac{1}{3}\frac{Ny^2e^2R}{\varepsilon_0} = \frac{4\pi}{3}Gm_p^2NR$$

$$\Rightarrow \quad y^2 = (4\pi\varepsilon_0)G\left(\frac{m_p}{e}\right)^2$$

$$= \frac{1}{9\times10^9}\times(6.67\times10^{-11})\left(\frac{(1.66\times10^{-27})^2}{(1.6\times10^{-19})^2}\right) = 79.8\times10^{-38}$$

$$\Rightarrow y = \sqrt{79.8\times10^{-38}} = 8.9\times10^{-19} \approx 10^{-18}$$

Thus, 10^{-18} is the required critical value of y corresponding to which expansion of universe would start. **[2½]**

(b) Net force experience by the hydrogen atom is given by

$$F = F_C - F_G = \frac{1}{3}\frac{Ny^2e^2R}{\varepsilon_0} - \frac{4\pi}{3}Gm_p^2NR$$

If acceleration of hydrogen atom is representing by $\frac{d^2R}{dt^2}$, then

$$m_p\frac{d^2R}{dt^2} = F = \frac{1}{3}\frac{Ny^2e^2R}{\varepsilon_0} - \frac{4\pi}{3}Gm_p^2NR$$

$$= \left(\frac{1}{3}\frac{Ny^2e^2}{\varepsilon_0} - \frac{4\pi}{3}Gm_p^2N\right)R$$

$$\therefore \frac{d^2R}{dt^2} = \frac{1}{m_p}\left[\frac{1}{3}\frac{Ny^2e^2}{\varepsilon_0} - \frac{4\pi}{3}Gm_p^2N\right]R = \alpha^2R \quad ...(iv)$$

$$where,\ \alpha^2 = \frac{1}{m_p}\left[\frac{1}{3}\frac{Ny^2e^2}{\varepsilon_0} - \frac{4\pi}{3}Gm_p^2N\right]$$

The general solution of equation (iv) is given by $R = Ae^{\alpha t} + Be^{-\alpha t}$ We are looking for expansion, here, so, $B = 0$ and $R = Ae^{\alpha t}$.

$$\text{Velocity of expansion, } v = \frac{dR}{dt} = Ae^{\alpha t}(\alpha) = \alpha Ae^{\alpha t} = \alpha R$$

So, $v \propto R$, *i.e.*, velocity of expansion is proportional to the distance from the centre. **[2½]**

Q. 8. Consider a sphere of radius *R* with charge density distributed as

$\rho(r) = kr$ for $r \le R$

$= 0$ for $r > R$.

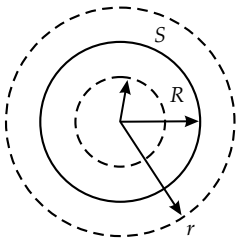
(a) Find the electric field at all points *r*.

(b) Suppose the total charge on the sphere is $2e$ where e is the electron charge. Where can two protons be embedded such that the force on each of them is zero. Assume that the introduction of the proton does not alter the negative charge distribution.

[NCERT Exemp. Q. 1.27, Page 8]

Ans. **(a)** Let us consider a sphere 'S' of radius '*R*' and two hypothetic sphere of radius $r < R$ and $r > R$.

Now, for point $r < R$. electric field intensity will be given by.



$$\oint E.dS = \frac{1}{\varepsilon_0}\int \rho dV$$

$$[\text{For } dV,\ V = \frac{4}{3}\pi r^3 \Rightarrow dV = 3\times\frac{4}{3}\pi r^2dr = 4\pi r^2dr]$$

$$\Rightarrow \oint E.dS = \frac{1}{\varepsilon_0}4\pi k\int_0^r r^3dr \qquad [\because p(r) = kr]$$

$$\Rightarrow (E)4\pi r^2 = \frac{4\pi k}{\varepsilon_0}\frac{r^4}{4}$$

$$\Rightarrow \quad E = \frac{1}{4\varepsilon_0}kr^2$$

Here, charge density is positive

So, direction of E is radially outwards.

For points $r > R$, electric field intensity will be given by,

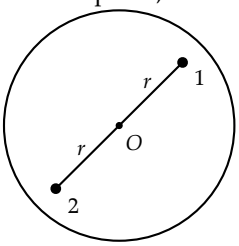
$$\oint E.dS = \frac{1}{\varepsilon_0}\int \rho dV$$

$$\Rightarrow E(4\pi r^2) = \frac{4\pi k}{\varepsilon_0}\int_0^R r^3dr = \frac{4\pi k}{\varepsilon_0}\frac{R^4}{4}$$

$$\Rightarrow \quad E = \frac{k}{4\varepsilon_0}\frac{R^4}{r^2}$$

Charge density is again positive. So, the direction of E is radially outward. **[2½]**

(b) The two protons must be on the opposite sides of the centre along a diameter following the rule of symmetry. This can be shown by the figure given below. Charge on the sphere,



$$q = \int_0^R \rho dV = \int_0^R (kr)4\pi r^2dr$$

$$q = 4\pi k\frac{R^4}{4} = 2e$$

$$\therefore\ k = \frac{2e}{\pi R^4}$$

If protons 1 and 2 are embedded at distance r from the centre of the sphere as shown, then attractive force on proton 1 due to charge distribution is,

$$F_1 = -eE = \frac{-ekr^2}{4\varepsilon_0}$$

Repulsive force on proton 1 due to proton 2 is,

$$F_2 = \frac{e^2}{4\pi\varepsilon_0(2r)^2}$$

Net force on proton 1,

$F = F_1 + F_2$

$$F = \frac{-ekr^2}{4\varepsilon_0} + \frac{e^2}{16\pi\varepsilon_0 r^4}$$

So,

$$F = \left[\frac{-er^2}{4\varepsilon_0}\frac{2e}{\pi R^4} + \frac{e^2}{16\pi\varepsilon_0 r^4}\right]$$

Thus, net force on proton 1 will be zero, when

$$\frac{er^2 2e}{4\varepsilon_0 \pi R^4} = \frac{e^2}{16\pi\varepsilon_0 r^2}$$

$$\Rightarrow \quad r^4 = \frac{R^4}{8}$$

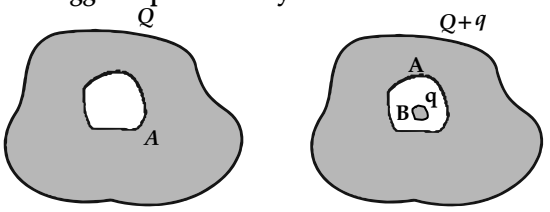
$$\Rightarrow \quad r = \frac{R}{(8)^{\frac{1}{4}}}$$

This is the distance of each of the two protons from the centre of the sphere. **[2½]**

Q. 9. (a) A conductor A with a cavity as shown in Figure (a), is given a charge *Q*. Show that the entire charge must appear on the outer surface of the conductor.

(b) Another conductor B with charge *q* is inserted into the cavity keeping B insulated from A. Show that the total charge on the outside surface of A is *Q* + *q* in Figure (b).

(c) A sensitive instrument is to be shielded from the strong electrostatic fields in its environment. Suggest a possible way.



(a) **(b)**

[NCERT Add. Ex. Q. 1.28, Page 49]

Ans. **(a)** Let us consider a Gaussian surface that is lying wholly within a conductor and enclosing the cavity. The electric field intensity 'E' inside the charged conductor is zero.

Let '*q*' is the charge inside the conductor and ε_0 is the permittivity of free space.

According to Gauss's law,

$$\text{Flux, } \oint \vec{E}\cdot ds = \frac{q}{\varepsilon_0}$$

But,

$E = 0$

$q/\varepsilon_0 = 0$

$\because \varepsilon_0 \neq 0$

$\therefore q = 0$

That means, charge inside the conductor is zero. The entire charge *Q* appears on the outer surface of the conductor. **[2]**

(b) The outer surface of conductor *A* has a charge of amount *Q*. Another conductor *B* having charge $+q$ is kept inside conductor *A* and it is insulated from *A*. Hence, a charge of amount $-q$ will be induced in the inner surface of conductor *A* and $+q$ is induced on the outer surface of conductor *A*. Therefore, total charge on the outer surface of conductor A is $Q + q$. **[1½]**

(c) A sensitive instrument can be shielded from the strong electrostatic field in its environment by enclosing it fully inside a metallic surface. A closed metallic body acts as an electrostatic shield. **[1½]**

Q. 10. (a) Use Gauss' law to derive the expression for the electric field, due to a straight uniformly charged infinite line of charge density λ C/m.

(b) Draw a graph to show the variation of *E* with perpendicular distance '*r*' from the line of charge.

[CBSE Board, Delhi Region, 2018]

Ans. **(a)** $\phi = \oint \vec{E}.d\vec{a}$

$$\phi = \int \vec{E}.d\vec{a}_1 + \int \vec{E}.d\vec{a}_2 + \int \vec{E}.d\vec{a}_3$$

$$\phi = \int \vec{E}.d\vec{a}_1 \cos 90^o + \int \vec{E}.d\vec{a}_2 \cos 0^o + \int \vec{E}.d\vec{a}_3 \cos 90^o$$

$$\phi = \int E.da_2$$

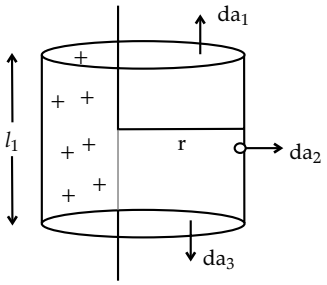
$$\phi = E.2\pi r\ell_1$$

But, $\phi = \frac{\sum q_{in}}{\varepsilon_0} = \frac{\lambda\ell_1}{\varepsilon_0}$

Equate them

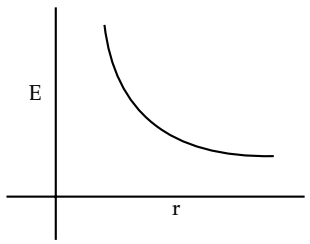
$$\frac{\lambda\ell_1}{\varepsilon_0} = E.2\pi r\ell_1$$

$$E = \frac{\lambda}{2\pi r\varepsilon_0}$$



[2½]

(b)



$$E = \frac{\lambda}{2\pi r\varepsilon_0}$$

$$E \propto \frac{1}{r}$$

[2½]

Q. 11. A hollow charged conductor has a tiny hole cut into its surface. Show that the electric field in the hole is $\left(\frac{\alpha}{2\varepsilon_0}\right)$, where is the unit vector in the

outward normal direction, and σ is the surface charge density near the hole.

[NCERT Add. Ex. Q. 1.29, Page 49]

Ans. Let us consider a conductor with a cavity or a hole. Electric field inside the cavity is zero.

Let E is the electric field just outside the conductor, q is the electric charge, σ is the charge density and ε_0 is the permittivity of free space.

Charge $q = \sigma \times ds$

According to Gauss's law, flux, $\phi = E.\,ds = \frac{q}{\varepsilon_0}$

$$\Rightarrow E.\,ds = \sigma \times \frac{ds}{\varepsilon_0}$$

$$\therefore E = \frac{\sigma}{\varepsilon_0 \hat{n}} \qquad \textbf{[2½]}$$

Thus, the electric field just outside the conductor is $\frac{\sigma}{\varepsilon_0 \hat{n}}$. This field is a superposition of field due to the cavity E' and the field due to the rest of the charged conductor E'. These fields are equal and opposite inside the conductor and equal in magnitude and direction outside the conductor.

$\therefore E' + E' = E$

$$\Rightarrow \quad E' = \frac{E}{2} = \frac{\sigma}{\varepsilon_0 \hat{n}}$$

Thus, the field due to the rest of the conductor will be $\frac{\sigma}{\varepsilon_0 \hat{n}}$. **[2½]**

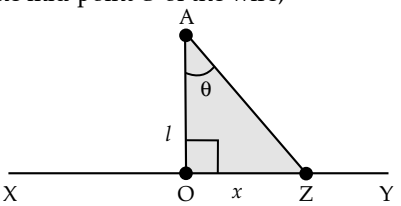
Q. 12. Obtain the formula for the electric field due to a long thin wire of uniform linear charge density λ without using Gauss's law.
[Hint : Use Coulomb's law directly and evaluate the necessary integral.]

[NCERT Add. Ex. Q. 1.30, Page 49]

Ans. Take a long thin wire XY of uniform linear charge density λ.

X Y

Consider a point A at a perpendicular distance l from the mid-point O of the wire,



Let E, be the electric field at point A due to the wire, XY.

Consider a small length element dx on the wire section with OZ = x

Let q be the charge on this piece.

$\therefore q = \lambda dx$

Electric field due to the piece,

However,

$$dE = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda dx}{(AZ)^2}$$

$$AZ = \sqrt{l^2 + x^2}$$

$$\therefore dE = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda dx}{(l^2 + x^2)} \qquad \textbf{[1½]}$$

The electric field is resolved into two rectangular components.

Perpendicular component = $dE \cos\theta$

Parallel component = $dE \sin\theta$

When the whole wire is considered, the component $dE \sin\theta$ is cancelled. Only the perpendicular component $dE \cos\theta$ affects point A.

So, effective electric field at point A due to the element dx is dE_1.

$$\therefore dE_1 = \frac{\lambda\, dx \cos\theta}{4\pi\varepsilon_0 \left(x^2 + l^2\right)} \qquad \ldots(1)$$

In ΔAZO, $\tan\theta = \frac{x}{l} \Rightarrow x = l \,.\, \tan\theta$ …(2)

On differentiating equation (2), we obtain,

$$\frac{dx}{d\theta} = l\, x \sec^2\theta \Rightarrow dx = l\, x \sec^2\theta d\theta \qquad \ldots(3)$$

From equation (2), we have,

$x^2 + l^2 = l^2 \sec^2\theta$ ...(4)

Putting equations (3) and (4) in equation (1), we obtain

$$dE_1 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda l \sec^2\theta d\theta . \cos\theta}{l^2 \sec^2\theta}$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda \cos\theta d\theta}{l} \qquad \ldots(5)$$

[1½]

The wire is so long that θ tends from $-\pi/2$ to $\pi/2$.

By integrating equation (5), we obtain the value of field E_1 as,

$$E_1 = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4\pi\varepsilon_0} \frac{\lambda}{l} \cos\theta \, d\theta$$

$$E_1 = \frac{1}{4\pi\varepsilon_0} \frac{\lambda}{l} \left[\sin\theta\right]_{-\pi/2}^{\pi/2}$$

$$\Rightarrow E_1 = \frac{1}{4\pi\varepsilon_0} \frac{\lambda}{l} \times 2$$

$$\Rightarrow E_1 = \frac{\lambda}{2\pi\varepsilon_0 l}$$

So, the electric field due to long wire is $\frac{\lambda}{\sqrt{2}\pi\varepsilon_0 l}$ **[2]**

Some Commonly Made Errors

- Students confuse to differentiate the Coulomb's Law and Gauss's Law.
- Students do not study the Applications of Laws.

- Do not ignore to study Electric Field Intensity.
- Generally, students get confused with the term 'Electric Flux' and 'Electric Field Intensity'.
- Try to solve numerical by making suitable diagram, do not skip the diagrams.

EXPERT ADVICE

- ☞ *Apply method of elimination for MCQ questions. Look at the answers before even starting to solve the questions and see if you can eliminate any answer. When calculating answer for a big question you may not need to solve the problem fully in order to choose the right answer because you could eliminate the other three.*
- ☞ *Convert all units to SI, with charges in coulombs and distances in meters, so as to be consistent with the SI value of the Coulomb constant.*
- ☞ *Learn the values of universal constants.*
- ☞ *Draw suitable diagram to explain electric field due to different-different conductors.*
- ☞ *Don't skip any important topics, like of Gauss's Law and its application, Coulombs Law, Electric Field Intensity.*

OSWAAL LEARNING TOOLS

For Suggested Online Videos

https://youtu.be/_9Xo-7oMMxw **Or Scan the Code**	*https://youtu.be/hm-c2V_yx8k* **Or Scan the Code**
https://youtu.be/X10kVPTfqjA **Or Scan the Code**	*https://goo.gl/M2ii3j* **Or Scan the Code**

CHAPTER

2 ELECTROSTATIC POTENTIAL AND CAPACITANCE

Chapter Objectives

This chapter will help you understand :

- ***Electrostatic potential :*** *Introduction; Electrostatic potential- due to point charge, electric dipole and system of charges; Equipotential surfaces; Relation between Electric field and Potential; Potential energy of a system of charges; Potential energy in an external field; Potential energy of a single charge; Potential energy of two charges in an external field; and Potential energy of a dipole in an external field.*
- ***Electrostatic capacitance :*** *Electrostatics of conductors; Dielectrics and Polarisation; Capacitors and Capacitance; The parallel plate capacitor; Effects of dielectric on capacitors; Energy stored in a capacitor; Van de Graff generator.*

TOPIC - 1
Electrostatic Potential P. 29

TOPIC - 2
Electrostatic Capacitance P. 43

TOPIC-1
Electrostatic Potential

Quick Review

- The electric potential is another useful term in Physics. It provides an alternative to the electric field in electrostatics problems.
- The potential is easier to understand because it is a single and scalar number, instead of a vector.
- The difference in potential between two places or points measures the degree to which charges are influenced to move from one place to another.
- For defining electric potential, we take the reference of zero potential at infinity.
- The electric field lines are always perpendicular to every point of equipotential surface.
- The work done in moving a charge between any two points on equipotential surface is always zero.

TIPS...

- *Study the electric potential due to a Point charge, a Dipole and system of charges.*
- *Draw diagrams of the charges in the solution.*
- *Learn Formulae and values of constants.*

TRICKS...

- *Understand the concept- after earthing a positively charged conductor electrons flow from earth to conductor and if a negatively charged conductor is earthed then electrons flows from conductor to earth.*
- *When a charged spherical conductor is placed inside a hollow insulated conductor and connected through a fine conducting wire the charge will be completely transferred from inner conductor to outer conductor.*

Know the Links

- ☞ *http://school.eckovation.com/short-notes-electric-potential-potential-energy/*
- ☞ *http://teacher.nsrl.rochester.edu/phy122/Lecture_Notes/Chapter25/Chapter25.html*
- ☞ *https://www.quora.com/What-is-the-difference-between-electrical-potential-and-electrostatic-potential*

Multiple Choice Questions (1 mark each)

Q. 1. The electrostatic potential on the surface of a charged conducting sphere is 100 V.
Two statements are made in this regard :
S_1 : At any point inside the sphere, electric intensity is zero.
S_2 : At any point inside the sphere, the electrostatic potential is 100 V.
Which of the following is a correct statement?

(a) S_1 is true, but S_2 is false.
(b) Both S_1 and S_2 are false.

(c) S_1 is true, S_2 is also true and S_1 is the cause of S_2.
(d) S_1 is true, S_2 is also true but the statements are independent. [NCERT Exemp. Q. 2.4, Page 11]

Ans. Correct option : (c)

Explanation : The relation between electric field intensity E and potential (V) is,

$$= -\frac{dV}{dr}$$

Where, Electric field intensity, $E = 0$ inside the sphere

So that, $\frac{dV}{dr}=0$

This means that V = constant. So, if $E = 0$ inside charged sphere, the potential is constant or $V = 100$ everywhere inside the sphere and it verifies the shielding effect also. So, it verifies the option (c).

Q. 2. Equipotential at a great distance from a collection of charges whose total sum is not zero are approximately
(a) spheres. (b) planes.
(c) paraboloids (d) ellipsoids.
[NCERT Exemp. Q. 2.5, Page 11]

Ans. Correct option : (a)

Explanation : For equipotential surface, these surfaces are perpendicular to the field lines. So there must be electric field, which cannot be without charge.

So the algebraic sum of all charges must not be zero. Equipotential surface at a great distance means that space of charge is negligible as compared to distance. So the collection of charges is considered as a point charge.

Electric potential due to point charge is,

$$V = k_e \frac{q}{r}$$

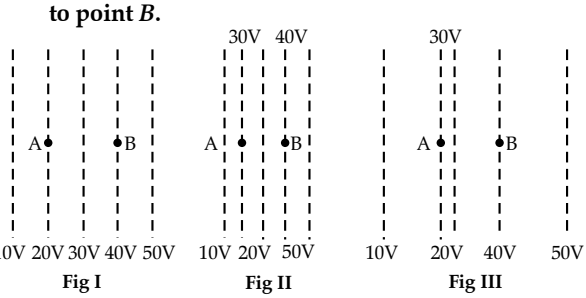
which explains that electric potentials due to point charge is same for all equidistant points. The locus of these equidistant points, which are at same potential, forms spherical surface.

Q. 3. A positively charged particle is released from rest in an uniform electric field. The electric potential energy of the charge
(a) remains a constant because the electric field is uniform.
(b) increases because the charge moves along the electric field.
(c) decreases because the charge moves along the electric field.
(d) decreases because the charge moves opposite to the electric field. [NCERT Exemp. Q. 2.2, Page 10]

Ans. Correct option : (c)

Explanation : As we know that, an equipotential surface is always perpendicular to the direction of electric field. Positive charge experiences the force in the direction of electric field. When a positive charge is released from rest in uniform electric field, its velocity increases in the direction of electric field. So K.E. increases, and the P.E. decreases due to law of conservation of energy.

Q. 4. Figure shows some equipotential lines distributed in space. A charged object is moved from point A to point B.



(a) The work done in Figure (i) is the greatest.
(b) The work done in Figure (ii) is least.
(c) The work done is the same in Figure (i), Figure (ii) and Figure (iii).
(d) The work done in Figure (iii) is greater than Figure (ii), but equal to that in Figure (i).
[NCERT Exemp. Q. 2.3, Page 11]

Ans. Correct option : (c)

Explanation : The work done by the electrostatic force is given by $W_{12} = q\,(V_2 - V_1)$

As the potential difference between A and B in all three figures are equal, 20 V, so work done by any charge in moving from A to B surface will be equal.

Tick Two or More Options

Q. 5. In a region of constant potential,
(a) the electric field is uniform.
(b) the electric field is zero.
(c) there can be no charge inside the region.
(d) the electric field shall necessarily change if a charge is placed outside the region.
[NCERT Exemp. Q. 2.10, Page 12]

Ans. Correct options : (b) and (c)

Explanation :

Constant potential

$\Rightarrow dV = 0$ so by relation $E = -\frac{dV}{dR}$, $E = 0$

That means, the electric field is not uniform discards answer (a) and agrees with answer (b).

As potential may be outside the charge also, so there can be no charge inside the regions of constant potential. It verifies answer (c).

If a charge is placed in outside region, potential difference in region will not be changed or electric field will not be changed. It makes answer (d) false.

Q. 6. If a conductor has a potential $V \neq 0$ and there are no charges anywhere else outside, then
(a) there must be charges on the surface or inside itself.
(b) there cannot be any charge in the body of the conductor.
(c) there must be charges only on the surface.
(d) there must be charges inside the surface.
[NCERT Exemp. Q. 2.12, Page 13]

Ans. **Correct options :** (a) and (b)
Explanation : As we know that, the excess charge can reside only on the surface of a conductor and inside net positive and negative charge is zero. Any charge can reside inside the hollow shell or body. So verifies answer (a) and discards answer (c).
And inside the solid material of conducting body there is no charge, it comes to outer surface. So verifies answer (b) and discards answer (d).

Q. 7. Consider a uniform electric field in the Z direction. The potential is a constant
(a) in all space.
(b) for any x for a given z.
(c) for any y for a given z.
(d) on the x-y plane for a given z.
[NCERT Exemp. Q. 2.7, Page 12]

Ans. **Correct options :** (b), (c) and (d)
Explanation : As we know that equipotential surfaces are perpendicular to the direction electric field lines. Here electric field is in + Z-direction.
The electric field in direction suggests that equipotential surfaces are in x-y plane. Therefore, the potential is a constant for any x for a given z, for any y for a given z and, on the x-y plane for a given z.
So, answers (b), (c) and (d) are verified respectively.

Q. 8. Equipotential surfaces
(a) are closer in regions of large electric fields compared to regions of lower electric fields.
(b) will be more crowded near sharp edges of a conductor.
(c) will be more crowded near regions of large charge densities.
(d) will always be equally spaced.
[NCERT Exemp. Q. 2.8, Page 12]

Ans. **Correct options :** (a), (b) and (c)
Explanation : Equipotential surfaces are closer in regions of large electric fields because electric field intensity is inversely proportional to the separation between equipotential surfaces.
As the electric field intensities is large near sharp edges of a charged conductor or near the regions of large charge densities. Therefore, numbers of equipotential surfaces are closer to such places or in other words they are more crowded.

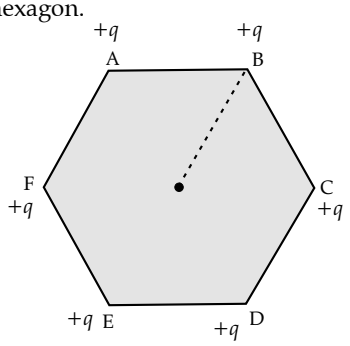
Q. 9. The work done to move a charge along an equipotential from A to B
(a) cannot be defined as $-\int_A^B E.dl$
(b) must be defined as $-\int_A^B E.dl$
(c) is zero.
(d) can have a non-zero value.
[NCERT Exemp. Q. 2.9, Page 12]

Ans. **Correct options :** (b) and (c)
Explanation : Work done to move a charge q from A to B along an equipotential surface,
$W = q\,(V_A - V_B)$
$= q \int_A^B dV = -\int_A^B E.dl$
Also, for an equipotential surface,
$V_A - V_B$ = Constant
$W = 0$

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. A regular hexagon of side 10 cm has a charge 5 μC at each of its vertices. Calculate the potential at the centre of the hexagon.
[NCERT Ex. Q. 2.2, Page 10]

Ans. According to the above conditions, the figure shows six equal amounts of charges, q, at the vertices of a regular hexagon.



Given that,
$q = 5\,\mu C = 5 \times 10^{-6}$ C
Side of the hexagon, $l = AB = BC = CD = DE = EF = FA = 10$ cm
Distance of each vertex from centre O, $d = 10$ cm
Electric potential at point O,

Where, $V = \dfrac{6 \times q}{4\pi\varepsilon_0 d}$ **[1]**

ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ NC}^{-2}\text{m}^2$$

So that $V = \dfrac{6 \times 9 \times 10^9 \times 5 \times 10^{-6}}{0.1}$

$= 2.7 \times 10^6$ V

So, the potential at the centre of the hexagon is 2.7 $\times 10^6$ V. **[1]**

Q. 2. Can there be a potential difference between two adjacent conductors carrying the same charge?
[NCERT Exemp. Q. 2.16, Page 14]

Ans. If in two conductors flowing current is same, then both may be considered in series. So
Ohm's law becomes $V \propto R$ **[1]**
So, if the resistances (which is the function of ρ, l and A) are different then potential difference will be different.
So, there can be potential difference between two adjacent conductors carrying the same charge or current if either their length or area of cross-section (A) and resistivity are different. **[1]**

Q. 3. Can the potential function have a maximum or minimum in free space?
[NCERT Exemp. Q. 2.17, Page 14]

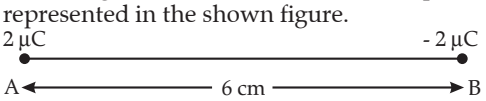
Ans. No, the potential function cannot have a maximum or minimum in free space. **[1]**
In the absence of free space or atmosphere, the phenomenon of electric field or potential leakage cannot be prevented. Hence, the potential function does not have maximum or minimum in free space. **[1]**

Q. 4. Do free electrons travel to region of higher potential or lower potential?
[NCERT Exemp. Q. 2.15, Page 14]

Ans. In a uniform electric field, free electrons move opposite to the direction of electric field. **[1]**
As the electric field is always direct from higher potential to lower travel so, electrons move from lower potential to higher potential. **[1]**

Q. 5. Two charges 2 μC and -2 μC are placed at points *A* and *B* 6 cm apart.
(a) Identify an equipotential surface of the system.
(b) What is the direction of the electric field at every point on this surface? [NCERT Ex. Q. 2.3, Page 11]

Ans. According to the condition of the question is represented in the shown figure.

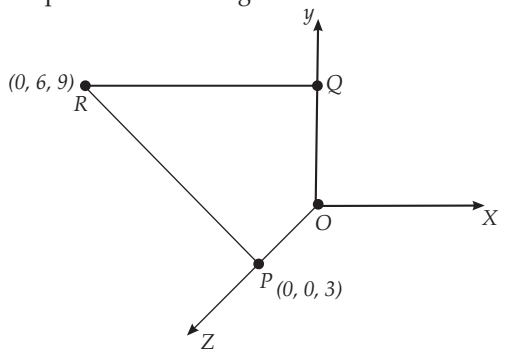


(a) An equipotential surface is the plane on which total potential is zero everywhere. This plane is normal to line *AB*. The plane is located at the mid-point of line *AB* because the magnitude of charges is the same. **[1]**

(b) The direction of the electric field at every point on this surface is normal to the plane in the direction of *AB*. **[1]**

Q. 6. A charge of 8 mC is located at the origin. Calculate the work done in taking a small charge of -2×10^{-9} C from a point *P* (0, 0, 3 cm) to a point *Q* (0, 4 cm, 0), via a point *R* (0, 6 cm, 9 cm).
[NCERT Add. Ex. Q. 2.12, Page 13]

Ans. Given that,
Charge located at the origin, $q = 8$ mC$= 8 \times 10^{-3}$ C
$q_1 = -2 \times 10^{-9}$ C
Condition according to the question - all the points are represented in the figure shown below.



Point *P* is at a distance, $d_1 = 3$ cm, from the origin along z-axis.
Point *Q* is at a distance, $d_2 = 4$ cm, from the origin along y-axis
Potential at point *P*,

$$V_1 = \frac{q}{4\pi\varepsilon_0 \times d_1}$$

Potential at point *Q*,

$$V_2 = \frac{q}{4\pi\varepsilon_0 \times d_2}$$

[1]

As we know the work done (W) by the electrostatic force is independent of the path is given by,
So that, $W = q_1[V_2 - V_1]$

$$= q_1\left[\frac{q}{4\pi\varepsilon_0 d_2} - \frac{q}{4\pi\varepsilon_0 d_1}\right]$$

$$= \frac{qq_1}{4\pi\varepsilon_0}\left[\frac{1}{d_2} - \frac{1}{d_1}\right] \quad \ldots(i)$$

Where, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^{2}C^{-2}
So, work done,

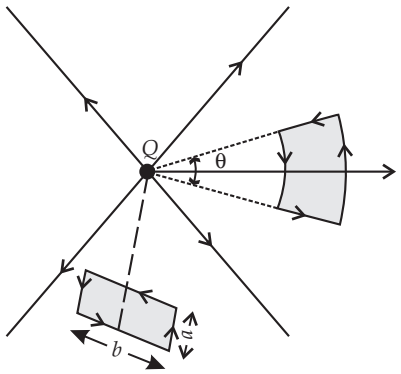
$$W = 9 \times 10^9 \times 8 \times 10^{-3} \times (-2 \times 10^{-9})\left[\frac{1}{0.04} - \frac{1}{0.03}\right]$$

$$= -144 \times 10^{-3} \times \left(\frac{-25}{3}\right)$$

$$= 1.2 \text{ J}$$

So, work done during the process is 1.2 J. **[1]**

Q. 7. A test charge *q* is made to move in the electric field of a point charge *Q* along two different closed paths in Figure. First path has sections along and perpendicular to lines of electric field. Second path is a rectangular loop of the same area as the first loop. How does the work done compare in the two cases?

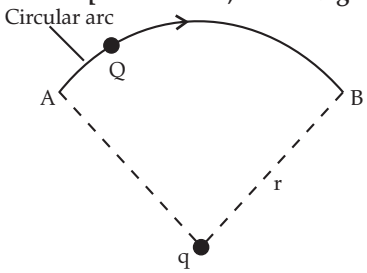


[NCERT Exemp. Q. 2.18, Page 14]

Ans. The electrostatic work done is conservative. **[1]**
So, work done in closed loop is always zero, it does not depend on the nature of closed path. **[1]**

Q. 8. What is the amount of work done in moving a point charge *Q* around a circular arc of radius '*r*' at the centre of which another point charge '*q*' is located? [CBSE Board, Delhi Region, 2016]

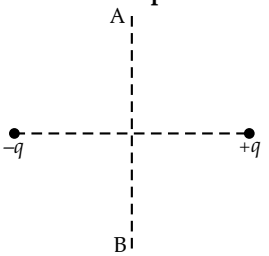
Ans.



A Charge particle of charge *Q* moving on circular arc where at the centre the charge *q* is located.

Work done by electrostatic force on moving charge Q,
$W_{field} = -\Delta U$
(Where ΔU is change in potential energy)
$U_A = qV_A$
$U_B = qV_B$
Here $V_A = V_B = \frac{kq}{r}$
$W_{field} = -(U_B - U_A)$ [So $U_A - U_B$]
$W_{field} = 0$ **[1]**

Q. 9. A charge 'q' is moved from a point A above a dipole of dipole moment 'p' to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process.



[CBSE Board, All India Region, 2016]

Ans. The net force on the charge q is directed parallel to the dipole and hence perpendicular to its motion along AB.
So that, Work done = 0 **[1]**

Q. 10. Find the work done in bringing a charge q from perpendicular distance r_1 to r_2 ($r_2 > r_1$).
[CBSE Board, Delhi Region, 2018]

Ans. Work done to moving from r_1 to r_2,

$$dV = \int E.dr$$

$$dV = \int_{r_1}^{r_2} \frac{\lambda}{2\pi\varepsilon_0 r} dr$$

$$dV = \frac{\lambda}{2\pi\varepsilon_0} \int_{r_1}^{r_2} \frac{1}{r} dr$$

$$= \frac{\lambda}{2\pi\varepsilon_0} \left[\log \frac{r_2}{r_1} \right]$$

$$W = qV$$

$$W = q \left[\frac{\lambda}{2\pi\varepsilon_0} \log \left(\frac{r_2}{r_1} \right) \right]$$

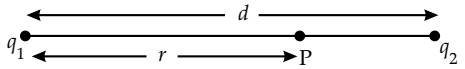
[2]

Short Answer Type Questions

(3 or 4 marks each)

Q. 1. Two charges 5×10^{-8} C and -3×10^{-8} C are located 16 cm apart. At what point(s) on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero.
[NCERT Ex. Q. 2.1, Page 10]

Ans. Given that,
$q_1 = 5 \times 10^{-8}$ C
$q_2 = -3 \times 10^{-8}$ C
$d = 16$ cm $= 0.16$ m



Now, let a point P on the line joining the two charges,
r = Distance of point P from charge q_1
Let the electric potential, V at point P be zero.
Potential at point P is the sum of potentials caused by charges q_1 and q_2 respectively.

$$\therefore V = \frac{q_1}{4\pi\varepsilon_0 r} + \frac{q_2}{4\pi\varepsilon_0 (d-r)} \quad \ldots(i)$$

Where,
ε_0 = Permittivity of free space
For $V = 0$, equation (i) becomes

$$\frac{q_1}{4\pi\varepsilon_0 r} = -\frac{q_2}{4\pi\varepsilon_0 (d-r)}$$

$$\frac{q_1}{r} = \frac{-q_2}{d-r}$$

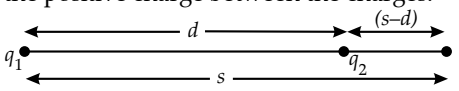
$$\frac{5\times10^{-8}}{r} = -\frac{(-3\times10^{-8})}{(0.16-r)}$$

$$\frac{0.16}{r} - 1 = \frac{3}{5}$$

$$\frac{0.16}{r} = \frac{8}{5}$$

$$r = 0.1 \text{ m} = 10 \text{ cm}$$

So, the potential is zero at a distance of 10 cm from the positive charge between the charges. **[1½]**



Let's assume a point P is outside the system of two charges at a distance s from the negative charge, where potential is zero, as shown in the figure shown above.
For above arrangement, potential is given by,

$$V = \frac{q_1}{4\pi\varepsilon_0 s} + \frac{q_2}{4\pi\varepsilon_0 (s-d)} \quad \ldots(ii)$$

For $V = 0$, equation (ii) reduces to

$$\frac{q_1}{4\pi\varepsilon_0 s} = -\frac{q_2}{4\pi\varepsilon_0 (s-d)}$$

$$\frac{q_1}{s} = \frac{-q_2}{s-d}$$

$$\frac{5\times10^{-8}}{s} = -\frac{(-3\times10^{-8})}{(s-0.16)}$$

$$1 - \frac{0.16}{s} = \frac{3}{5}$$

$$\therefore \frac{0.16}{s} = \frac{2}{5}$$

So that, $s = 0.4\ m = 40$ cm

That is, the required point is 40 cm away from q_1 and (40 – 16) = 24 cm from q_2 such that, potential is zero. **[1½]**

Q. 2. Two charges $-q$ and $+q$ are located at points (0, 0, $-a$) and (0, 0, a), respectively.

(a) What is the electrostatic potential at the points (0, 0, z) and (x, y, 0)?

(b) Obtain the dependence of potential on the distance r of a point from the origin when $\frac{r}{a} >> 1$.

(c) How much work is done in moving a small test charge from the point (5, 0, 0) to (−7, 0, 0) along the x-axis? Does the answer change if the path of the test charge between the same points is not along the x-axis? [NCERT Add. Ex. Q. 2.21, Page 14]

Ans. **(a)** Zero at the point (x, y, 0)

Charge $-q$ is located at (0, 0, $-a$) and charge $+q$ is located at (0, 0, a).

So, they form a dipole. Point (0, 0, z) is on the axis of this dipole and point (x, y, 0) is normal to the axis of the dipole.

So, electrostatic potential at point (x, y, 0) is zero.

Electrostatic potential at point (0, 0, z) is given by,

$$V=\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{z-a}\right)+\frac{1}{4\pi\varepsilon_0}\left(-\frac{q}{z+a}\right)$$

$$=\frac{q(z+a-z+a)}{4\pi\varepsilon_0(z^2-a^2)}$$

$$=\frac{2qa}{4\pi\varepsilon_0(z^2-a^2)}=\frac{p}{4\pi\varepsilon_0(z^2-a^2)}$$

Where,

ε_0 = Permittivity of free space

p = Dipole moment of the system of two charges = $2qa$ **[1]**

(b) Distance 'r' is much greater than half of the distance between the two charges.

So, the potential (V) at a distance r is inversely proportional to square of the distance *i.e.*,

$$V \propto \frac{1}{r^2}$$ **[1]**

(c) Zero,

The answer does not change if the path of the test is not along the x-axis.

A test charge is moved from point (5, 0, 0) to point (−7, 0, 0) along the x-axis. So,

Electrostatic potential (V_1) at point (5, 0, 0) is given by,

$$V_1=\frac{-q}{4\pi\varepsilon_0}\frac{1}{\sqrt{(5-0)^2+(-a)^2}}+\frac{q}{4\pi\varepsilon_0}\frac{1}{\sqrt{(5-0)^2+a^2}}$$

$$=\frac{-q}{4\pi\varepsilon_0\sqrt{25^2+a^2}}+\frac{q}{4\pi\varepsilon_0\sqrt{25+a^2}}$$

$$=0$$

Electrostatic potential, V_2, at point (−7, 0, 0) is given by,

$$V_2=\frac{-q}{4\pi\varepsilon_0}\frac{1}{\sqrt{(-7)^2+(-a)^2}}+\frac{q}{4\pi\varepsilon_0}\frac{1}{\sqrt{(-7)^2+(a)^2}}$$

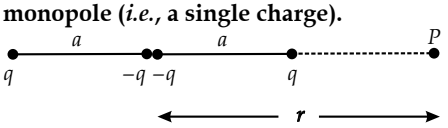
$$=\frac{-q}{4\pi\varepsilon_0\sqrt{49+a^2}}+\frac{q}{4\pi\varepsilon_0}\frac{1}{\sqrt{49+a^2}}$$

$$=0$$

work done, $W = q_0 (V_2 - V_1) = 0$. So, no work is done in moving a small test charge from point (5, 0, 0) to point (−7, 0, 0) along the x-axis.

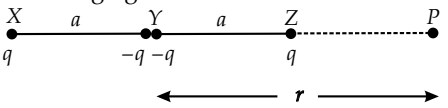
The answer will not change because work done by the electrostatic field in moving a test charge between the two points is independent of the path connecting the two points. **[1]**

Q. 3. Figure shows a charge array known as an electric quadrupole. For a point on the axis of the quadrupole, obtain the dependence of potential on r for $\frac{r}{a} >> 1$, and contrast your results with that due to an electric dipole, and an electric monopole (*i.e.*, a single charge).



[NCERT Add. Ex. Q. 2.22, Page 14]

Ans. Four charges of same magnitude are placed at points X, Y, Y, and Z respectively, as shown in the following figure.



A point is located at P, which is r distance away from point Y. The system of charges forms an electric quadrupole.

It can be considered that the system of the electric quadrupole has three charges.

Charge $+q$ placed at point X

Charge $-2q$ placed at point Y

Charge $+q$ placed at point Z

$XY = YZ = a$

$YP = r$

$PX = r + a$

$PZ = r - a$

So, the Electrostatic potential caused by the system of three charges at point P is given by,

$$V=\frac{1}{4\pi\varepsilon_0}\left[\frac{q}{XP}-\frac{2q}{YP}+\frac{q}{ZP}\right]$$

$$=\frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r+a}-\frac{2q}{r}+\frac{q}{r-a}\right]$$

$$=\frac{q}{4\pi\varepsilon_0}\left[\frac{r(r-a)-2(r+a)(r-a)+r(r+a)}{r(r+a)(r-a)}\right]$$

$$=\frac{q}{4\pi\varepsilon_0}\left[\frac{r^2-ra-2r^2+2a^2+r^2+ra}{r(r^2-a^2)}\right]$$

$$=\frac{q}{4\pi\varepsilon_0}\left[\frac{2a^2}{r(r^2-a^2)}\right]=\frac{2qa^2}{4\pi\varepsilon_0 r^3\left(1-\frac{a^2}{r^2}\right)}$$ **[1½]**

Since, $\frac{r}{a} >> 1$,

$$\therefore \frac{a}{r} << 1$$

$$\therefore V=\frac{2qa^2}{4\pi\varepsilon_0 r^3}$$

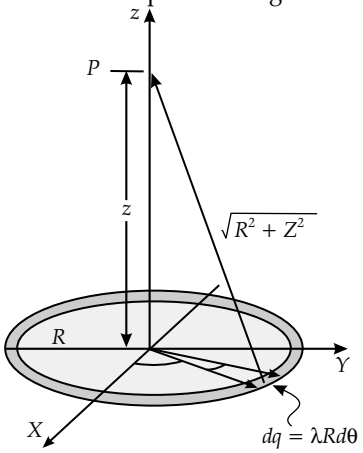
$\frac{a^2}{r^2}$ is taken as negligible. It can be inferred that potential, $V \propto \frac{1}{r^3}$

However, it is known that for a dipole, $V \propto \frac{1}{r^2}$

And, for a monopole, $V \propto \frac{1}{r}$ **[1½]**

Q. 4. Calculate potential on the axis of a ring due to charge Q uniformly distributed along the ring of radius R. [NCERT Exemp. Q. 2.23, Page 14]

Ans. Let us consider a ring of radius R having charge $+Q$ distributed uniformly. Also a point P at distance z on its axis passes through centre O and it is perpendicular to the plane of ring.



[1]

Linear charge density :

$\lambda = Q/2\pi R$

A small element of charge is the product of the linear charge density and the small arc length :

$dQ = \lambda R d\theta$

$$dV = \frac{dQ}{4\pi\varepsilon_0 r}$$

$$= \frac{\lambda R d\theta}{4\pi\varepsilon_0 \sqrt{R^2+z^2}}$$

So,

$$V = \frac{\lambda R}{4\pi\varepsilon_0\sqrt{R^2+z^2}} \int_0^{2\pi} d\theta = \frac{\lambda 2\pi R}{4\pi\varepsilon_0\sqrt{R^2+z^2}}$$

$$= \frac{Q}{4\pi\varepsilon_0\sqrt{R^2+Z^2}}$$

So, the potential on the axis of a ring due to charge Q uniformly distributed along the ring of radius R will be :

$$V = \frac{Q}{4\pi\varepsilon_0\sqrt{R^2+z^2}}$$

[2]

Q. 5. Prove that a closed equipotential surface with no charge within itself must enclose an equipotential volume. [NCERT Exemp. Q. 2.19, Page 14]

Ans. Let us assume that inside the enclosed equipotential surface, potential is not same and let the potential just inside the equipotential surface is different to that on the equipotential surface, causing in a potential gradient,

$= \frac{dV}{dr}$

So electric field will exist inside surface which is equal to, $E = -\frac{dV}{dr}$ **[1½]**

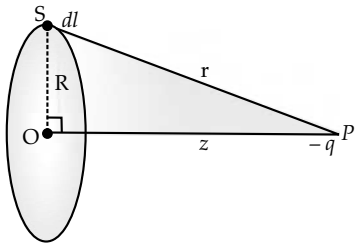
The field lines pointing inwards or outwards from the surface are perpendicular to equipotential surfaces or the field lines cannot be on the equipotential surface.

The field lines can be on the equipotential surface if field lines can originate from the charge inside, which contradicts the original assumption. Thus, the entire volume inside equipotential surface has no charge. **[1½]**

Q. 6. Calculate potential energy of a point charge $-q$ placed along the axis due to a charge $+Q$ uniformly distributed along a ring of radius R. Sketch P.E. as a function of axial distance z from the centre of the ring. Looking at graph, can you see what would happen if $-q$ is displaced slightly from the centre of the ring (along the axis)?

[NCERT Exemp. Q. 2.22, Page 14]

Ans. Let us assume a ring of radius R having charge $+Q$ distributed uniformly over the ring. Also, a point P at distance z on its axis passes through centre O and is perpendicular to the plane of ring.



Again, assume an element of ring at S of length dl having charge dq and SP is equal to r.

Then potential energy due to element dl at P.

If dq is charge on element dl of ring, then

$$dV = \frac{-kdq}{r}$$

Where,

$k = \frac{1}{4\pi\varepsilon_0}$ and as Q is positive charge so potential due to dq charge will be negative.

Charge on $2\pi R$ length of ring $= Q$

Charge on dl length of ring $dq = \frac{Q}{2\pi R} dl$

So potential due to element dl at P

$$dV = \frac{-k.Qdl}{2\pi Rr}$$

so, $dW = dV.q$ and $r = \sqrt{R^2+z^2}$

so, $dW = \frac{-kQqdl}{2\pi R\sqrt{R^2+z^2}}$

Integrating both sides, over a ring, we have,

$$\int_0^W dW = -\int_0^{2\pi R} \frac{kqQdl}{2\pi R\sqrt{R^2+z^2}}$$

$$W = -\frac{kqQ2\pi R}{2\pi R R\sqrt{1+\frac{z^2}{R^2}}} = \frac{kqQ}{R\sqrt{1+\frac{z^2}{R^2}}}$$

This work done converts into P.E. at P, so

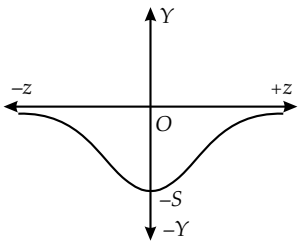
$$U = \frac{-Qq}{4\pi\varepsilon_0 R\sqrt{1+\frac{z^2}{R^2}}}$$ **[2]**

Let $\frac{Qq}{4\pi\varepsilon_0 R} = S$ (a new constant)

$$U = \frac{-S}{\left[1+\frac{z^2}{R^2}\right]^{\frac{1}{2}}}$$

At $z = -\infty$ or $+\infty$

$$U_2 = \frac{-S}{\left[1+\frac{z^2}{R^2}\right]^{\frac{1}{2}}} = 0$$



At $z = 0$, $U = -S$ **[1]**

Q. 7. Define an equipotential surface. Draw equipotential surfaces:

(i) in the case of a single point charge and

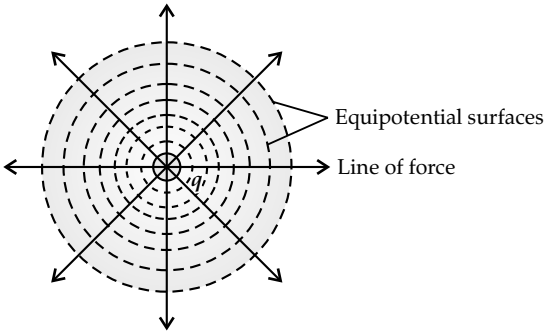
(ii) in a constant electric field in Z-direction. Why the equipotential surfaces about a single charge are not equidistant?

(iii) Can electric field exist tangential to an equipotential surface? Give reason.

[CBSE Board, O.D, I, II, III, 2016]

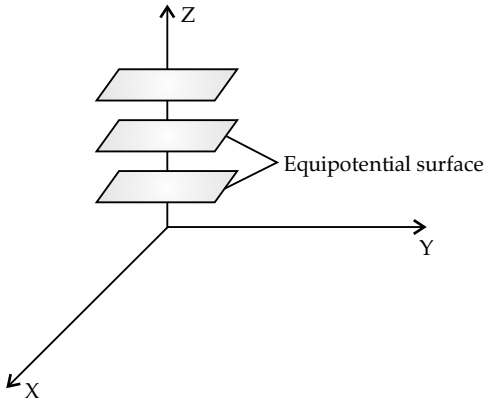
Ans. **(i)** Equipotential surface is a surface which has equal potential at every point on it.

Equipotential surfaces due to single point charge are concentric sphere having charge at the centre.



In constant electric filed along z-direction, the perpendicular distance between equipotential surfaces remains same. **[1]**

(ii)



For single charge, equipotential surface will be series of concentric spherical shells with charge at a centre,

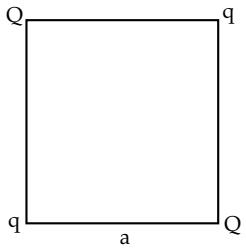
$$dr \propto \frac{1}{E}$$

The separation dr between equipotential surface will go on increasing with the decrease in electric field. **[1]**

(iii) No, because if the surface is not equipotential then it would mean that there is tangential component of electric field along surface.

This component will result in motion of electrons, but since we have static fields, this is not possible. **[1]**

Q. 8. Four-point charges Q, q, Q and q are placed at the corners of a square of side 'a' as shown in the figure.



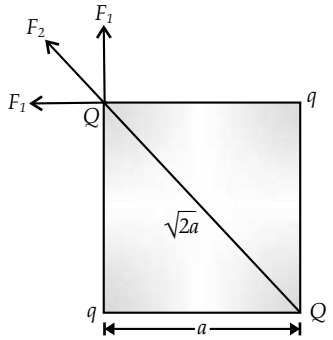
Find the

(a) resultant electric force on a charge Q, and

(b) potential energy of this system.

[CBSE Board, Delhi Region, 2018]

Ans. (a)



$F_1 \rightarrow$ force act at Q by q

$F_2 \rightarrow$ force act at Q by Q

$$F_1 = \frac{k.q.Q}{a^2},\ F_2 = \frac{k.Q.Q}{\left(\sqrt{2}a\right)^2}$$

$$F_1 = \frac{k.q.Q}{a^2},\ F_2 = \frac{k.Q^2}{2a^2}$$

$$F_{net} = \sqrt{2}F_1 + F_2$$

$$= \sqrt{2}\frac{k.Q.q}{a^2} + \frac{k.Q^2}{2a^2}$$

[F of F_1 and F_2 will be $\sqrt{2}F_1$ because F_1 and F_2 perpendicular]

[1½]

(b) The potential energy of system of n charges is

$$U = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\left[\sum_{j(j>i)}^{n}\frac{q_i q_j}{r_{ij}}\right]$$

Therefore for four (04) charges system we have–

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_1q_4}{r_{14}} + \frac{q_2q_3}{r_{23}} + \frac{q_2q_4}{r_{24}} + \frac{q_3q_4}{r_{34}}\right]$$

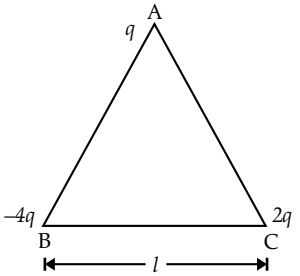
For the given system of charges we have

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{Qq}{a} + \frac{Q^2}{a\sqrt{2}} + \frac{Qq}{a} + \frac{qQ}{a} + \frac{q^2}{a\sqrt{2}} + \frac{Qq}{a}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{4Qq}{a} + \frac{q^2}{a\sqrt{2}} + \frac{Q^2}{a\sqrt{2}}\right] \quad \textbf{1}$$

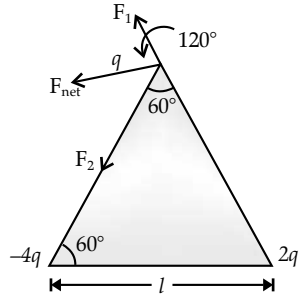
$$U = \frac{1}{4\pi\varepsilon_0\, a}\left[\frac{4Qq}{1} + \frac{q^2}{\sqrt{2}} + \frac{Q^2}{\sqrt{2}}\right]$$

Q. 9. Three-point charges q, $-4q$ and $2q$ are placed at the vertices of an equilateral triangle ABC of side 'l' a shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge q.



[CBSE Board, Delhi Region, 2018]

Ans. **(a)** The resultant force on charge q at A is



$$F_q = \sqrt{F_C^2 + F_B^2 + 2F_BF_C\cos\theta}$$

Now,

$$\left|\overrightarrow{F_c}\right| = \frac{k\cdot q\cdot 2q}{l^2} = \frac{2kq^2}{l^2} = F_c$$

$$\overrightarrow{F_B} = \frac{kq(-4q)}{l^2} = -\frac{4kq^2}{l^2} \Rightarrow F_B = \frac{4kq^2}{l^2}$$

Now from eqn. (1) and (2)

$$F_B = 2F_c$$

and angle between F_B and F_c is 120°, so force on q is **2**

$$F_q = \sqrt{F_c^2 + (2F_c)^2 + 2\cdot(2F_c)\cdot F_c\cdot\cos 120^\circ}$$

$$= \sqrt{5F_c^2 + 4F_c^2\left(-\frac{1}{2}\right)} = \sqrt{5F_c^2 - 2F_c^2} = F_c\sqrt{3}$$

$$\boxed{F_q = 2\sqrt{3}\frac{kq^2}{l^2}}\ \left[\because \text{from eqn (1), } F_c = \frac{2kq^2}{l^2}\right] \quad \textbf{[3]}$$

Q. 10. Find out the amount of the work done to separate the charges at infinite distance.

[CBSE Board, Delhi Region, 2018]

Ans. Work done required to separate the charges at infinity equal to electric potential energy of system. So

$$W_1 = \frac{k.q.2q}{l}, W_2 = \frac{k.(-4q).q}{l}, W_3 = \frac{k.(-4q)(2q)}{l}$$

$$W_1 = \frac{2kq^2}{l}, W_2 = \frac{-4k.q^2}{l}, W_3 = \frac{-8k.q^2}{l}$$

$$W = W_1 + W_2 + W_3$$

$$W = \frac{2kq^2}{l} - \frac{4k.q^2}{l} - \frac{8k.q^2}{l}$$

$$W = \frac{-10k.q^2}{l}$$

[3]

Q. 11. A 600 pF capacitor is charged by a 200 V supply. It is then disconnected from the supply and is connected to another uncharged 600 pF capacitor. How much electrostatic energy is lost in the process? [NCERT Ex. Q. 2.11, Page 88]

Ans. Capacitance of the capacitor, C = 600 pF

Potential difference, V = 200 V

Electrostatic energy stored in the capacitor is given by,

$$E = \frac{1}{2}CV^2$$

$$= \frac{1}{2}\times 600\times 10^{-12}\times 200^2$$

$$= 1.2\times 10^{-5}\ \text{J}$$

If supply is disconnected from the capacitor and another capacitor of capacitance C = 600 pF is connected to it, then equivalent capacitance (C') of the combination is given by,

$$\frac{1}{C'} = \frac{1}{C} + \frac{1}{C}$$

$$= \frac{1}{600} + \frac{1}{600} = \frac{2}{600} = \frac{1}{300}$$

New electrostatic energy can be calculated as

$$E' = \frac{1}{2} \times C' \times V^2$$
$$= \frac{1}{2} \times 300 \times 200^2$$
$$= 0.6 \times 10^{-5} \text{ J}$$

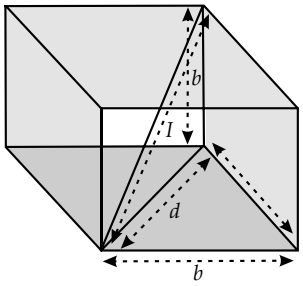
Loss in electrostatic energy = $E - E'$
$= 1.2 \times 10^{-5} - 0.6 \times 10^{-5}$
$= 0.6 \times 10^{-5}$
$= 6 \times 10^{-6}$ J

Therefore, the electrostatic energy lost in the process is 6×10^{-6} J. **[3]**

Long Answer Type Questions **(5 marks each)**

Q. 1. A cube of side b has a charge q at each of its vertices. Determine the potential and electric field due to this charge array at the centre of the cube. [NCERT Add. Ex. Q. 2.13, Page 13]

Ans. Given that,
Length of the side of a cube = b
Charge at each of its vertices = q
A cube of side b as shown in the below figure :



[1]

Where,
d = Diagonal of one of the six faces of the cube
$d = \sqrt{b^2 + b^2} = \sqrt{2b^2}$
$d = b\sqrt{2}$
l = Length of the diagonal of the cube
$l = \sqrt{d^2 + b^2}$
$l = \sqrt{(\sqrt{2}b)^2 + b^2} = \sqrt{2b^2 + b^2} = \sqrt{3b^2}$
$l = b\sqrt{3}$

$r = \frac{l}{2} = \frac{b\sqrt{3}}{2}$ is the distance between the centre of the cube and one of the eight vertices. **[2]**

The electric potential (V) at the centre of the cube is due to the presence of eight charges at the vertices.

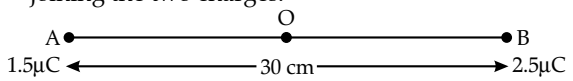
$$V = \frac{8q}{4\pi\varepsilon_0}$$
$$= \frac{8q}{4\pi\varepsilon_0\left(b\frac{\sqrt{3}}{2}\right)}$$
$$= \frac{4q}{\sqrt{3}\pi\varepsilon_0 b}$$

So, the potential at the centre of the cube is $\frac{4q}{\sqrt{3}\pi\varepsilon_0 b}$.

The electric field at the centre of the cube, due to the eight charges, gets cancelled. This is because the charges are distributed symmetrically with respect to the centre of the cube. So, the electric field is zero at the centre. **[2]**

Q. 2. Two tiny spheres carrying charges 1.5 μC and 2.5 μC are located 30 cm apart. Find the potential and electric field :
(a) at the mid-point of the line joining the two charges, and
(b) at a point 10 cm from this midpoint in a plane normal to the line and passing through the mid-point. [NCERT Add. Ex. Q. 2.14, Page 13]

Ans. Given that,
Magnitude of charge located at A, $q_1 = 1.5\ \mu$C
Magnitude of charge located at B, $q_2 = 2.5\ \mu$C
Distance between the two charges, $d = 30$ cm $= 0.3$ m
According to the condition in the question, two charges placed at points A and B are represented in the below figure. O is the mid-point of the line joining the two charges.



(a) Let V_1 and E_1 are the electric potential and electric field respectively at O.
V_1 = Potential due to charge at A + Potential due to charge at B

$$V_1 = \frac{q_1}{4\pi\varepsilon_0\left(\frac{d}{2}\right)} + \frac{q_2}{4\pi\varepsilon_0\left(\frac{d}{2}\right)} = \frac{1}{4\pi\varepsilon_0\left(\frac{d}{2}\right)}(q_1 + q_2)$$

Where,
ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ NC}^{-2}\text{m}^2$$

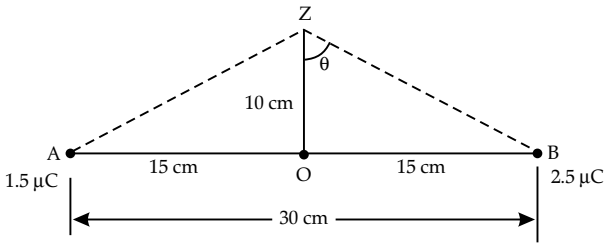
So, $V_1 = \frac{9 \times 10^9 \times 10^{-6}}{\left(\frac{0.30}{2}\right)}(2.5 + 1.5) = 2.4 \times 10^5$ V

E_1 = Electric field due to q_2 − Electric field due to q_1

$$= \frac{q_2}{4\pi\varepsilon_0\left(\frac{d}{2}\right)^2} - \frac{q_1}{4\pi\varepsilon_0\left(\frac{d}{2}\right)^2}$$
$$= \frac{9 \times 10^9}{\left(\frac{0.30}{2}\right)^2} \times 10^6 \times (2.5 - 1.5)$$
$$= 4 \times 10^5 \text{ Vm}^{-1}$$

So, the potential at mid-point is 2.4×10^5 V and the electric field at mid-point is 4×10^5 V m^{-1}. The field is directed from the larger charge to the smaller charge. **[2½]**

(b) Now, consider a point Z such that normal distance OZ = 10 cm = 0.1 m, as shown in the following figure.



V_2 and E_2 are the electric potential and electric field respectively at Z.

It can be observed from the figure that distance,

$$BZ=AZ=\sqrt{(0.1)^2+(0.15)^2}=0.18 \text{ m}$$

V_2= Electric potential due to A + Electric Potential due to B

$$=\frac{q_1}{4\pi\varepsilon_0(AZ)}+\frac{q_2}{4\pi\varepsilon_0(BZ)}$$

$$=\frac{9\times10^9\times10^{-6}}{0.18}(1.5+2.5)$$

$$=2\times10^5$$

Electric field due to q_1 at Z,

$$E_A=\frac{q_1}{4\pi\varepsilon_0(AZ)^2}$$

$$=\frac{9\times10^9\times1.5\times10^{-6}}{(0.18)^2}$$

$$=0.416\times10^6 \text{ V/m}$$

Electric field due to q_2 at Z,

$$E_B=\frac{q_2}{4\pi\varepsilon_0(BZ)^2}$$

$$=\frac{9\times10^9\times2.5\times10^{-6}}{(0.18)^2}$$

$$=0.69\times10^6 \text{ Vm}^{-1}$$

The resultant field intensity at Z,

$$E=\sqrt{E_A^2+E_B^2+2E_AE_B\cos2\theta}$$

Where, 2θ is the angle, $\angle AZB$

From the figure, we obtain

$$\cos\theta=\frac{0.10}{0.18}=\frac{5}{9}=0.5556$$

$$\theta=\cos^{-1}0.5526=56.25$$

$$\therefore 2\theta=112.5^\circ$$

$$\cos2\theta=-0.38$$

$$E=\sqrt{(0.416\times10^6)\times(0.69\times10^6)+2\times0.416\times0.69\times10^{12}\times(-0.38)}$$

$$=6.6\times10^5 \text{ Vm}^{-1}$$

So, the potential at a point 10 cm (perpendicular to the mid-point) is 2.0×10^5 V and electric field is 6.6×10^5 V m^{-1}. **[2½]**

Q. 3. Two charges q_1 and q_2 are placed at (0, 0, d) and (0, 0, $-d$) respectively. Find locus of points where the potential a zero.

[NCERT Exemp. Q. 2.32, Page 15]

Ans. Let the potential at any point $P(x, y, z)$ is zero then,

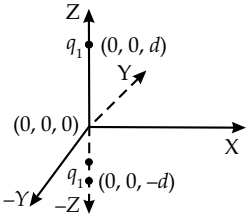
$V_1 + V_2 = 0$

$$\frac{kq_1}{\sqrt{x^2+y^2+(z-d)^2}}+\frac{kq_2}{\sqrt{x^2+y^2+(z+d)^2}}=0$$

$$\frac{q_1}{\sqrt{x^2+y^2+(z-d)^2}}+\frac{q_2}{\sqrt{x^2+y^2+(z+d)^2}}=0$$

$$\frac{q_1}{q_2}=\frac{-\sqrt{x^2+y^2+(z-d)^2}}{\sqrt{x^2+y^2+(z+d)^2}}$$

$$\frac{q_1^2}{q_2^2}=\frac{x^2+y^2+z^2+d^2-2zd}{x^2+y^2+z^2+d^2+2zd}$$



[2]

Componendo and dividendo of

$$\frac{x}{a}=\frac{y}{b} \text{ is } \frac{x+a}{x-a}=\frac{y+b}{y-b}$$

Then componendo and dividendo of

$$\frac{\left(\frac{q_1}{q_2}\right)^2}{1}=\frac{x^2+y^2+z^2+d^2-2zd}{x^2+y^2+z^2+d^2+2zd}$$

$$\frac{\left(\frac{q_1}{q_2}\right)^2+1}{\left(\frac{q_1}{q_2}\right)^2-1}=\frac{x^2+y^2+z^2+d^2-2dz+(x^2+y^2+z^2+d^2+2dz)}{x^2+y^2+z^2+d^2-2dz-(x^2+y^2+z^2+d^2+2dz)}$$

$$\left[\frac{\left(\frac{q_1}{q_2}\right)^2+1}{\left(\frac{q_1}{q_2}\right)^2-1}\right]=\frac{2(x^2+y^2+z^2+d^2)}{-4dz}$$

$$x^2+y^2+z^2+d^2=-2dz\left[\left[\frac{\left(\frac{q_1}{q_2}\right)^2+1}{\left(\frac{q_1}{q_2}\right)^2-1}\right]\right]$$

$$x^2+y^2+z^2+2d\left[\frac{\left(\frac{q_1}{q_2}\right)^2+1}{\left(\frac{q_1}{q_2}\right)^2-1}\right]z+d^2=0$$

$$x^2+y^2+z^2+2d\left[\frac{q_1^2+q_2^2}{q_1^2-q_2^2}\right]z+d^2=0$$

This is the equation of sphere required point is on z axis $a = 0$, $b = 0$ and

$$z=2d\left[\frac{q_1^2+q_2^2}{q_1^2-q_2^2}\right]$$

$$\left(0,0,-2d\left[\frac{q_1^2+q_2^2}{q_1^2-q_2^2}\right]\right)$$

[3]

If $q_1 = -q_2$, then $z = 0$

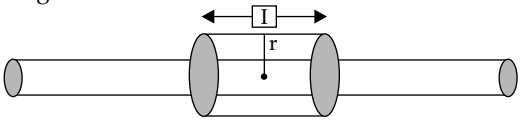
Therefore, locus of points where potential is zero is the plane through mid point of the two charges.

Q. 4. Find the equation of the equipotential for an infinite cylinder of radius r_0, carrying charge of linear density λ.

[NCERT Exemp. Q. 2.24, Page 14]

Ans. Potential at distance *r* :

Potential at distance '*r*' from the line consider the electric field. We know that from symmetry the field lines must be radially outward. So, draw a cylindrical Gaussian surface of radius '*r*' and length '*l*'.



Then

$$\oint E.ds = \frac{1}{\varepsilon_0}\lambda l$$

[1]

$$\text{Or, } E_r 2\pi rl = \frac{1}{\varepsilon_0}\lambda l$$

$$\Rightarrow E_r = \frac{\lambda}{2\pi\varepsilon_0 r}$$

So, if r_0 is the radius,

$$V(r) - V(r_0) = -\int_{r_0}^{r} \mathrm{E}.dl = \frac{\lambda}{2\pi\varepsilon_0} ln\frac{r_0}{r}$$

[1]

For a given V,

$$\ln\frac{r}{r_0} = -\frac{2\pi\varepsilon_o}{\lambda}[V(r) - V(r_0)]$$

$$r = r_0 e^{\left(\frac{-2\pi\varepsilon_0[V(r)-V(r_0)]}{\lambda}\right)}$$

The equipotential surfaces are cylindrical surface of radius

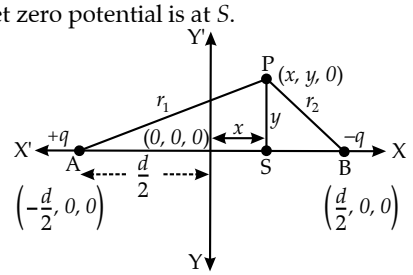
$$\Rightarrow r = r_0 e^{\frac{-2\pi\varepsilon_0 V(r_0)}{\lambda} + \frac{2\pi\varepsilon_0 V(r)}{\lambda}}$$

[3]

Q. 5. Two-point charges of magnitude $+q$ and $-q$ are placed at $(-\frac{d}{2}, 0, 0)$ and $(\frac{d}{2}, 0, 0)$, respectively. Find the equation of the equipotential surface where the potential is zero.

[NCERT Exemp. Q. 2.25, Page 15]

Ans. The potential due to charge $+q$ and $-q$ will be zero in between the line joining the two charges $+q$ and $-q$. Let zero potential is at *S*.



Then equipotential surface will pass through *S* and perpendicular to line joining two chargers or *AB*.

Then,

$r^2_1 = AS^2 + SP^2$

$$= \left(x + \frac{d}{2}\right)^2 + y^2$$

$$r_1 = \sqrt{\left(x + \frac{d}{2}\right)^2 + y^2}$$

Similarly, $r_2 = \sqrt{\left(x - \frac{d}{2}\right)^2 + y^2}$ **[2½]**

So, net potential at $P = 0$

$$\frac{kq}{r_1} + \frac{k(-q)}{r_2} = 0 \text{ where } k = \frac{1}{4\pi\varepsilon_0}$$

$$\Rightarrow kq\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = 0 \qquad [\because kq \neq 0]$$

$$\Rightarrow \frac{1}{r_1} - \frac{1}{r_2} = 0$$

$$\Rightarrow \frac{1}{r_1} = \frac{1}{r_2} \Rightarrow r_1 = r_2$$

$$\Rightarrow \left(x + \frac{d}{2}\right)^2 + y^2 = \left(x - \frac{d}{2}\right)^2 + y^2$$

$$\Rightarrow \left(x + \frac{d}{2}\right)^2 = \left(x - \frac{d}{2}\right)^2$$

$$\Rightarrow x^2 + \frac{d^2}{4} + dx = x^2 + \frac{d^2}{2} - dx$$

$$2dx = 0$$

$$2d \neq 0$$

$$\therefore x = 0$$

Its mean, equipotential surface will be perpendicular to X-axis passing through $x = 0$, that is, origin in *Y*-*Z* plane.

So, the equation is that of a plane $x = 0$ **[2½]**

Q. 6. (a) In a quark model of elementary particles, a neutron is made of one up quarks [charge (2/3) *e*] and two down quarks [charges –(1/3) *e*]. Assume that they have a triangle configuration with side length of the order of 10^{-15} m. Calculate electrostatic potential energy of neutron and compare it with its mass 939 MeV.

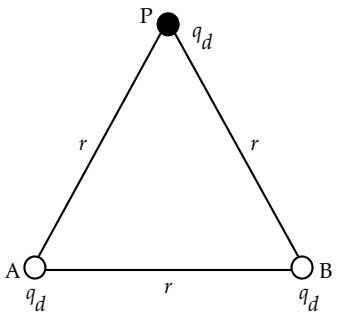
(b) Repeat above exercise for a proton which is made of two up and one down quark.

[NCERT Exemp. Q. 2.28, Page 15]

Ans.

(a) $q_d = -\frac{1}{3}e$ [Charge on down quark]

$q_u = +\frac{2}{3}e$ [Charge on up quark]



Potential energy $U=\frac{kq_1q_2}{r}$

Where,

$k=\frac{1}{4\pi\varepsilon_0}$

So,

$U=\frac{kq_1q_2}{r}+\frac{kq_1q_3}{r}+\frac{kq_2q_3}{r}$

$\therefore U_n=\frac{1}{4\pi\varepsilon_0}\frac{(-q_d)(-q_d)}{r}+\frac{(-q_d)q_u}{4\pi\varepsilon_0}+\frac{q_u(-q_d)}{4\pi\varepsilon_0 r}$

$=\frac{q_d}{4\pi\varepsilon_0 r}[+q_d-q_u-q_u]$ [Taking sign of charge]

$=\frac{q_d}{4\pi\varepsilon_0 r}[q_d-2q_u]=\frac{9\times10^9\times\frac{1}{3}e}{10^{-15}}\left[\frac{1}{2}e-2.\frac{2}{3}e\right]$

[Nature sign of charge taken already]

$=\frac{9\times10^9\times e}{3\times10^{-15}}.\frac{e}{3}[1-4]$ Joule

$=\frac{-3\times9\times10^9\times1.6\times10^{-19}}{9\times10^{-15}}e$ Joule

$=-4.8\times10^{9-19+15}eV=-4.8\times10^5eV=-0.48\times10^6eV$

$U=-0.48$ MeV

So, charges inside neutron [$1q_u$ and $2q_d$] are attracted by energy of 0.48 MeV.

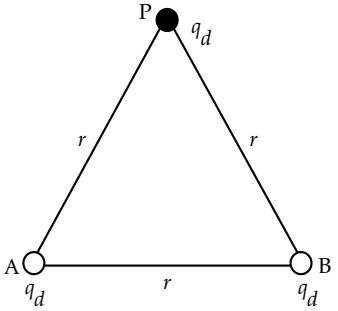
Energy released by a neutron when converted into energy is m_nC^2 939 MeV.

Required ratio will be

$=\frac{1-0.481\text{MeV}}{939\text{MeV}}=0.0005111=5.11\times10^{-4}$

[2½]

(b) P.E. of proton consists of 2 up and 1 down quark



(A proton)

$r=10^{-15}$ m

$q_d=-\frac{1}{3}e, q_u=\frac{2}{3}e$

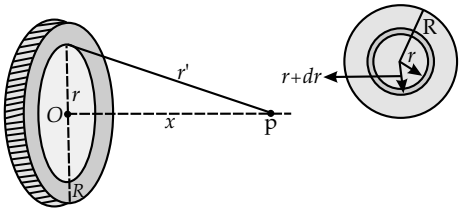
$U_p=\frac{1}{4\pi\varepsilon_0}\frac{q_u\times q_u}{r}+\frac{q_u(-q_d)}{4\pi\varepsilon_0 r}+\frac{q_u(-q_d)}{4\pi\varepsilon_0 r}$

$=\frac{q_u}{4\pi\varepsilon_0 r}[q_u-q_d-q_d]$

$=\frac{q_u}{4\pi\varepsilon_0 r}[q_u-2q_d]=\frac{9\times10^9}{10^{-15}}\frac{2}{3}e\left[\frac{2}{3}e-2.\frac{1}{3}e\right]=0$ **[2½]**

Q. 7. Calculate potential on the axis of a disc of radius R due to a charge Q uniformly distributed on its surface. [NCERT Exemp. Q. 2.31, Page 15]

Ans. Consider a point *P* on the axis perpendicular to the plane of disc and at distance *x* from the centre *O* of disc as shown in the figure.



Now consider a ring of radius *r* of thickness *dr* on disc of radius R, as shown in figure, Again let the charge on the ring is *dq* then potential *dV* due to ring at P, will be,

$dV=\frac{kdq}{r'}$ [$\because r'=\sqrt{r^2+x^2}$]

dq is the charge on the ring= σ .area of the ring

$=\sigma.\left[\pi(r+dr)^2-\pi r^2\right]$

$dq=\sigma.\pi\left[r^2+dr^2+2rdr-r^2\right]$

Because, *dr* is small therefore dr^2 is negligible.

$\therefore dq=\sigma\pi(2rdr)=2\pi r\sigma.dr$

$\therefore dV=\frac{k.2\pi r\sigma dr}{\sqrt{(r^2+x^2)}}$ **[1]**

So the potential to charged disc

$\int_0^V dV=\int_0^R\frac{k2\pi r\sigma\,dr}{\sqrt{r^2+x^2}}$ **[2]**

$V=k.2\pi\sigma.\int_0^R\frac{rdr}{(r^2+x^2)^{1/2}}$

$=k.2\pi\sigma.\int_0^R r.(r^2+x^2)^{\frac{1}{2}}dr$

$=2\pi k\sigma\left[(R^2+x^2)^{\frac{1}{2}}-x\right]$

$=\frac{2\pi\sigma}{4\pi\varepsilon_0}\left[(R^2+x^2)^{\frac{1}{2}}-x\right]$

[As we know that $\pi R^2\sigma = Q$ (charge on disc), So, $\sigma=\frac{Q}{\pi R^2}$]

$=\frac{2\pi R^2\sigma}{4\pi\varepsilon_0 R^2}\left[\sqrt{R^2+x^2}-x\right]$

$V=\frac{2Q}{4\pi\varepsilon_0 R^2}\left[\sqrt{R^2+x^2}-x\right]$

[2]

Q. 8. Two charges *–q* each are separated by distance 2*d*. A third charge *+q* is kept at mid-point O. Find potential energy of *+q* as a function of small distance x from *O* due to *– q* charges. Sketch P.E. v/s *x* and convince yourself that the charge at *O* is in an unstable equilibrium.

[NCERT Exemp. Q. 2.33, Page 15]

Ans. Let equilibrium of *+q* is at *P* at a distance *x* from mid-point of line joining two charges.

Force FA on $+q$ is towards left side and force FB is towards right side, so for equilibrium of $+q$ at P,

$$F_A = F_B$$

$$\frac{-kq^2}{(d-x)^2} = \frac{-kq^2}{(d+x)^2}$$

$$\therefore (d-x)^2 = (d+x)^2$$

$$d - x = d + x$$

$$-2x = 0$$

$$x = 0$$ **[1]**

So, Equilibrium position of charge $+q$ between two $-q$ charges, is at mid-point (O) of line joining the two charges ($-q$) and ($-q$).

Now the potential energy of $+q$ as a function of small distance x from balance condition (O) towards any of ($-q$) charge.

So, let new position of charge ($+q$) from a small distance x from (O)

$$U = \frac{k.(q)(-q)}{(d-x)} + \frac{k(q)(-q)}{(d+x)}$$

$$\left(\text{As we know that, } U = \frac{kq_1q_2}{r_1}\right)$$

$$= -kq^2\left[\frac{1}{(d-x)} + \frac{1}{(d+x)}\right]$$

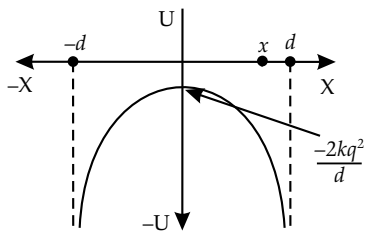
$$= -kq^2\left[\frac{d+x+d-x}{(d-x)(d+x)}\right]$$

$$= -kq^2\left[\frac{2d}{d^2-x^2}\right]$$

$$U = \frac{-q^2}{4\pi\varepsilon_0}.\frac{2d}{(d^2-x^2)}$$ **[2]**

So, U is the P.E. as a function of x

x	U
0	$-\frac{2k}{d}q^2$
$\frac{d}{2}$	$\frac{4}{3}\left(\frac{-2kq^2}{d}\right)$
$-\frac{d}{2}$	$\frac{4}{3}\left(\frac{-2kq^2}{d}\right)$
$+d$	$-\infty$
d	$-\infty$



[2]

Q. 9. In a hydrogen atom, the electron and proton are bound at a distance of about 0.53 Å :

(a) Estimate the potential energy of the system in eV, taking the zero of the potential energy at infinite separation of the electron from proton.

(b) What is the minimum work required to free the electron, given that its kinetic energy in the orbit is half the magnitude of potential energy obtained in (a)?

(c) What are the answers to (a) and (b) above if the zero of potential energy is taken at 1.06 Å separation?

[NCERT Add. Ex. Q. 2.18, Page 14]

Ans. Given that,

Distance between electron-proton of a hydrogen atom, $d = 0.53$ Å

Charge on an electron, $q_1 = -1.6 \times 10^{-19}$ C

Charge on a proton, $q_2 = +1.6 \times 10^{-19}$ C

(a) Potential at infinity is zero.

Potential energy of the system, P :

p = Potential energy at infinity − Potential energy at distance, d

$$= 0 - \frac{q_1q_2}{4\pi\varepsilon_0 d}$$

Where,

ε_0 is the permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9\times10^9 \text{Nm}^2\text{C}^{-2}$$

$\therefore$ Potential energy

$$= 0 - \frac{9\times10^9\times(1.6\times10^{-19})^2}{0.53\times10^{10}} = -43.7\times10^{-19} \text{ J}$$

Since, 1.6×10^{-19} J = 1 eV,

$\therefore$ Potential energy = -43.7×10^{-19} J

$$= \frac{-43.7\times10^{-19}}{1.6\times10^{-19}} = -27.2 \text{ eV}$$

So, the potential energy of the system is -27.2 eV. **[1½]**

(b) Kinetic energy is half of the magnitude of potential energy.

$$\text{Kinetic energy} = \frac{1}{2}\times(-27.2) = 13.6 \text{ eV}$$

Total energy = 13.6 − 27.2 = –13.6 eV

Therefore, the minimum work required to free the electron is 13.6 eV. **[1½]**

(c) When zero of potential energy is taken, $d_1 = 1.06$ Å

Potential energy of the system = Potential energy at d_1 − Potential energy at d

$$= \frac{q_1q_2}{4\pi\varepsilon_0 d_1} - 27.2 \text{ eV}$$

$$= \frac{9\times10^9\times(1.6\times10^{-19})^2}{1.06\times10^{-10}} - 27.2 \text{ eV}$$

$$= 21.73\times10^{-19} \text{ J} - 27.2 \text{ eV}$$

$$= \frac{21.73\times10^{-19}\text{J}}{1.6\times10^{-19}}\text{eV} - 27.2 \text{ eV}$$

$$= 13.58 \text{ eV} - 27.2 \text{ eV}$$

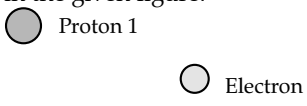
$$= -13.6 \text{ eV}$$

[2]

Q. 10. If one of the two electrons of a H_2 molecule is removed, we get a hydrogen molecular ion H_2^+. In the ground state of an, H_2^+ the two protons are separated by roughly 1.5 Å, and the electron is roughly 1 Å from each proton. Determine the potential energy of the system. Specify your choice of the zero of potential energy.

[NCERT Add. Ex. Q. 2.19, Page 14]

Ans. The system of two protons and one electron is represented in the given figure.



Charge on proton 1, $q_1 = 1.6 \times 10^{-19}$ C
Charge on proton 2, $q_2 = 1.6 \times 10^{-19}$ C
Charge on electron 3, $q_3 = -1.6 \times 10^{-19}$ C
Distance between protons 1 and 2, $d_1 = 1.5 \times 10^{-10}$ m
Distance between proton 1 and electron, $d_2 = 1 \times 10^{-10}$ m
Distance between proton 2 and electron, $d_3 = 1 \times 10^{-10}$ m
The potential energy at infinity is zero.
Potential energy of the system,

$$V = \frac{q_1 q_2}{4\pi\varepsilon_0 d_1} + \frac{q_2 q_3}{4\pi\varepsilon_0 d_2} + \frac{q_3 q_1}{4\pi\varepsilon_0 d_3}$$ **[2½]**

Substituting $\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ \text{Nm}^2\text{C}^{-2}$, we obtain

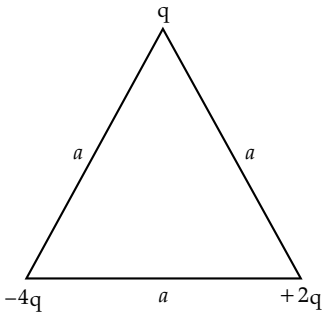
$$V = \frac{9\times10^9\times10^{-19}\times10^{-19}}{10^{-10}}\left[\frac{(1.6)^2}{1.5} + -(1.6)^2 + -(1.6)^2\right]$$

$$= -30.7\times10^{-19}\ \text{J}$$

$$= -19.2\ \text{eV}$$

So, the potential energy of the system is -19.2 eV. **[2½]**

Q. 11. Obtained an expression for the work done to dissociate the system of three charges placed at the vertices of an equilateral triangle of side 'a' as shown below.



[CBSE Board, Delhi Region, 2016]

Ans. Total electrostatic potential energy of system

$$U = U_{12} + U_{23} + U_{31}$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{q(-4q)}{a} + \frac{(-4q)(2q)}{a} + \frac{q(2q)}{a}\right]$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{10q^2}{a}$$

Work done to dissociate the system

$$W = -U$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{10q^2}{a}$$ **[5]**

TOPIC-2
Electrostatic Capacitance

Quick Review

- Two plates of unequal area can also form a capacitor, but effective overlapping area is considered.
- Capacitance of a parallel plate capacitor doesn't depend upon the charge given, potential raised or nature of metals and thickness of plates.
- The distance between the plates is kept small to avoid fringing or edge effect (non-uniformity of the field) at the boundaries of the plates.
- A spherical capacitor behaves as a parallel plate capacitor if it's spherical surfaces have large radii and are close to each other.
- The capacity of a conductor is defined as the ratio between the charge of the conductor to its potential : C = Q/V
- Capacitance is the electrical property of a capacitor and it is the measure of a capacitors ability to store an electrical charge onto its two plates.
- The dielectric provides mechanical support between the two plates allowing the plates to be closer together without touching.
- The unit of capacitance is the Farad (abbreviated to F) named after the British physicist Michael Faraday.
- Units in :
  - S.I. - farad (coulomb/volt)
  - C.G.S. - stat farad (stat-coulomb/stat-volt)
- Just like the Resistor, the Capacitor, sometimes referred to as a condenser, is a simple passive device that is used to "store electricity" on its plates.

TIPS...
- *Study the dielectrics and electric polarisation.*
- *Understand the combination of capacitors and capacitance.*
- *Study the energy stored in a capacitance.*

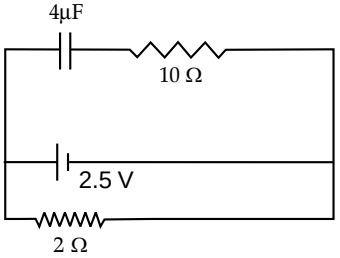
TRICKS...
- *In series, charge remains the same and potential across the capacitors may be different.*
- *In parallel, potential difference across each capacitor remains the same and changes may be different.*

Know the Links

- *http://school.eckovation.com/short-notes-electric-potential-potential-energy/*
- *http://physicscatalyst.com/article/solve-electric-force-field-problems/*
- *https://www.askiitians.com/revision-notes/physics/electrostatic-potential-and-capacitance/*

Multiple Choice Questions (1 mark each)

Q. 1. A capacitor of 4 μF is connected as shown in the circuit Figure. The internal resistance of the battery is 0.5Ω. The amount of charge on the capacitor plates will be :



(a) 0 μC **(b)** 4 μC

(c) 16 μC **(d)** 8 μC

[NCERT Exemp. Q. 2.1, Page 10]

Ans. Correct option : (d)

Explanation : As capacitor offer infinite resistance for DC circuit. So current from cell will not flow across branch of 4 μF and 10 Ω. So current will flow across 2 ohm branch.

So current flows through across 2 Ω resistance from left to right is,

$$I=\frac{V}{(R+r)}$$

$$=\frac{2.5\text{ V}}{(2+0.5)}$$

$$=1\text{ A}$$

So Potential Difference (PD) across 2 Ω resistance $V = RI = 2\times1 = 2$ Volt.

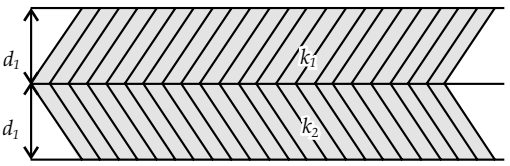
As battery, capacitor and 2 branches are in parallel. So PD will remain same across all three branches.

As current does not flow through capacitor branch so no potential drop will be across 10 Ω.

So PD across 4 F capacitor = 2 Volt

$Q = CV = 2\mu\text{F}\times 2\text{V} = 8\ \mu\text{C}$

Q. 2. A parallel plate capacitor is made of two dielectric blocks in series. One of the blocks has thickness d_1 and dielectric constant k_1 and the other has thickness d_2 and dielectric constant k_2 as shown in Figure. This arrangement can be thought as a dielectric slab of thickness d (= d_1+d_2) and effective dielectric constant k. The k is :



(a) $\frac{k_1d_1+k_2d_2}{d_1+d_2}$ **(b)** $\frac{k_1d_1+k_2d_2}{k_1+k_2}$

(c) $\frac{k_1k_2(d_1+d_2)}{(k_1d_1+k_2d_2)}$ **(d)** $\frac{2k_1k_2}{k_1+k_2}$

[NCERT Exemp. Q. 2.6, Page 12]

Ans. Correct option : (c)

Explanation : Capacitance of a parallel plate capacitor filled with dielectric of constant k_1 and thickness d_1 is,

$$C_1=\frac{k_1\varepsilon_0A}{d_1}$$

Similarly, for other capacitance of a parallel plate capacitor filled with dielectric of constant k_2 and thickness d_2 is,

$$C_2=\frac{k_2\varepsilon_0A}{d_2}$$

Both capacitors are in series so equivalent capacitance C is related as :

$$\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{d_1}{k_1\varepsilon_0A}+\frac{d_2}{k_2\varepsilon_0A}$$

$$=\frac{1}{\varepsilon_0A}\left[\frac{k_2d_1+k_1d_2}{k_1k_2}\right]$$

So,

$$C=\frac{k_1k_2\varepsilon_0A}{(k_1d_2+k_2d_1)} \quad \ldots(\text{I})$$

$$C'=\frac{k\varepsilon_0A}{d} \quad \ldots(\text{II})$$

Where $d = (d_1+d_2)$

So, multiply the numerator and denominator of eqn. (I) with (d_1+d_2),

$$C=\frac{k_1k_2\varepsilon_0A}{(k_1d_2+k_2d_1)}\cdot\frac{(d_1+d_2)}{(d_1+d_2)}=\frac{k_1k_2}{(k_1d_2+k_2d_1)}\cdot\frac{\varepsilon_0A}{(d_1+d_2)} \quad \ldots(\text{III})$$

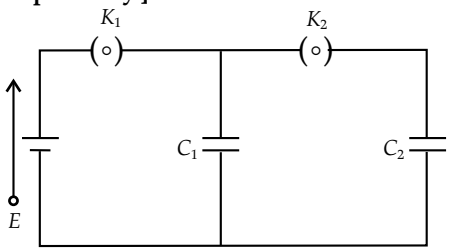
Comparing eqns. (II) and (III), the dielectric constant of new capacitor is :

$$k=\frac{k_1k_2(d_1+d_2)}{(k_1d_2+k_2d_1)}$$

Tick Two or More Options

Q. 3. In the circuit shown in Figure, initially key K_1 is closed and key :

K_2 is open. Then K_1 is opened and K_2 is closed (order is important). [Take Q_1' and Q_2' as charges on C_1 and C_2 and V_1 and V_2 as voltage respectively.]



Then

(a) charge on C_1 gets redistributed such that $V_1 = V_2$

(b) charge on C_1 gets redistributed such that $Q_1' = Q_2'$

(c) charge on C_1 gets redistributed such that $C_1V_1 + C_2V_2 = C_1E$

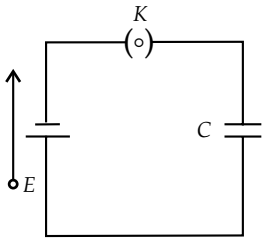
(d) charge on C_1 gets redistributed such that $Q_1' + Q_2' = Q$ [NCERT Exemp. Q. 2.11, Page 12]

Ans. Correct options : (a) and (d)

Explanation : When key k_1 is closed and key k_2 is open, the capacitor C_1 is charged by cell and when k_1 is opened and k_2 is closed, the charge stored by capacitor C_1 gets redistributed between C_1 and C_2. So, the charge on C_1 gets redistributed such that $Q_1' + Q_2' = Q$

As C_1 and C_2 both are in parallel combination, so their potential will be equal, *i.e.*, $V_1 = V_2$. It verifies the answer (a).

Q. 4. A parallel plate capacitor is connected to a battery as shown in Figure. Consider two situations :



A : Key K is kept closed and plates of capacitors are moved apart using insulating handle.

B : Key K is opened and plates of capacitors are moved apart using insulating handle.

Choose the correct option(s).

(a) In A : Q remains same but C changes.

(b) In B : V remains same but C changes.

(c) In A : V remains same and hence Q changes.

(d) In B : Q remains same and hence V changes.

[NCERT Exemp. Q. 2.13, Page 13]

Ans. Correct options : (c) and (d)

Explanation :

(i) In case A : when the space between the plates of capacitor increases, the capacitance decreases by relation,

$$C=\frac{K\varepsilon_0 A}{d}$$

But battery remains same, *i.e.*, potential difference across plate remains 'V' same. So by $Q = CV$ relation, Q also decreases verifies answer (c) and discards answer (a).

(ii) In case B : K is open, and capacitance decreases by moving apart plates of capacitor, so by relation $Q = CV$, here K is open, so charge Q remains same in turn V will increase on decreasing C. Hence answer (d) is verified.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Consider two conducting spheres of radii R_1 and R_2 with $R_1 > R_2$. If the two are at the same potential, the larger sphere has more charge than the smaller sphere. State whether the charge density of the smaller sphere is more or less than that of the larger one.

[NCERT Exemp. Q. 2.14, Page 13]

Ans. Let Q_1 and Q_2 be the charges on radii R_1 and R_2

And, we know that

$$V_1=\frac{1}{4\pi\varepsilon}\frac{Q_1}{R_1} \text{ and } V_2=\frac{1}{4\pi\varepsilon}\frac{Q_2}{R_2}$$

As $V_1 = V_2$, so :

$$\Rightarrow \quad \frac{1}{4\pi\varepsilon}\frac{Q_1}{R_1}=\frac{1}{4\pi\varepsilon}\frac{Q_2}{R_2} \qquad \textbf{[1]}$$

$$\Rightarrow \quad \frac{Q_1}{R_1}=\frac{Q_2}{R_2}$$

$$\Rightarrow \quad \frac{Q_1}{Q_2}=\frac{R_1}{R_2}$$

$$\Rightarrow \quad \frac{\sigma_1\times 4\pi(R_1)^2}{\sigma_2\times 4\pi(R_2)^2}=\frac{R_1}{R_2}$$

$$\Rightarrow \quad \frac{\sigma_1}{\sigma_2}=\frac{R_2}{R_1}$$

And, it is given that $R_1 > R_2$

It means, $\sigma_1 < \sigma_2$

So, the charge density of smaller sphere is more than that of larger sphere. **[1]**

Q. 2. Three capacitors each of capacitance 9 pF are connected in series.

(a) What is the total capacitance of the combination?

(b) What is the potential difference across each capacitor if the combination is connected to a 120 V supply? [NCERT Ex. Q. 2.1, Page 10]

Ans. **(a)** Given that,

Capacitance of each of the three capacitors, $C = 9$ pF

Equivalent capacitance (C') of the combination of the capacitors is,

$$\frac{1}{C'}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}$$

$$=\frac{1}{9}+\frac{1}{9}+\frac{1}{9}=\frac{3}{9}=\frac{1}{3}$$

$\therefore C' = 3\,\mu$F

So, total capacitance of the combination is 3 μF. **[1]**

(b) Given that,

Supply voltage, $V = 100$ V

Potential difference (V') across each capacitor is equal to one-third of the supply voltage.

$$\therefore V'=\frac{V}{3}=\frac{120}{3}=40 \text{ V}$$

So, the potential difference across each capacitor is 40 V. **[1]**

Q. 3. A 12 pF capacitor is connected to a 50 V battery. How much electrostatic energy is stored in the capacitor? [NCERT Ex. Q. 2.10, Page 12]

Ans. Given that,

Capacitor of the capacitance, C = 12 pF = 12×10^{-12} F

Potential difference, $V = 50$ V

As we know the, electrostatic energy stored in the capacitor is,

$$E=\frac{1}{2}CV^2$$ **[1]**

$$=\frac{1}{2}\times 12\times 10^{-12}\times (50)^2$$

$$=1.5\times 10^{-8} \text{ J}$$

So, the electrostatic energy stored in the capacitor is 1.5×10^{-8} J. **[1]**

Short Answer Type Questions (3 or 4 marks each)

Q. 1. Two charged conducting spheres of radii a and b are connected to each other by a wire. What is the ratio of electric fields at the surfaces of the two spheres? Use the result obtained to explain why charge density on the sharp and pointed ends of a conductor is higher than on its flatter portions. [NCERT Add. Ex. Q. 2.20, Page 14]

Ans. Assume 'a' be the radius of a sphere A, Q_A be the charge on the sphere, and C_A be the capacitance of the sphere.

Let b be the radius of a sphere B, Q_B be the charge on the sphere, and C_B be the capacitance of the sphere. Since the two spheres are connected to a wire, their potential (V) will become equal.

Now, Let E_A be the electric field of sphere A and E_B be the electric field of sphere B.

So, their ratio,

$$\frac{E_A}{E_B}=\frac{Q_A}{4\pi\varepsilon_0\times a^2}\times\frac{b^2\times 4\pi\varepsilon_0}{Q_B}$$

$$\frac{E_A}{E_B}=\frac{Q_A}{Q_B}\times\frac{b^2}{a^2} \quad \ldots(1)$$

However, $\frac{Q_A}{Q_B}=\frac{C_AV}{C_BV}$ **[1½]**

And, $\frac{C_A}{C_B}=\frac{a}{b}$

$$\therefore \frac{Q_A}{Q_B}=\frac{a}{b} \quad \ldots(2)$$

Putting the value of (2) in (1), we obtain

$$\therefore \frac{E_A}{E_B}=\frac{a}{b}\frac{b^2}{a^2}=\frac{b}{a}$$

So, the ratio of electric fields at the surface is $\frac{b}{a}$. A sharp and pointed end can be treated as a sphere of very small radius and a flat portion behaves as a sphere of much larger radius. So that, charge density on sharp and pointed ends of a conductor is much higher than on its flatter portions. **[1½]**

Q. 2. A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q.

(a) A charge q is placed at the centre of the shell. What is the surface charge density on the inner and outer surfaces of the shell?

(b) Is the electric field inside a cavity (with no charge) zero, even if the shell is not spherical, but has any irregular shape? Explain.

[NCERT Add. Ex. Q. 2.15, Page 14]

Ans. **(a)** Given that,

Charge placed at the centre of a shell is $+q$. And, a charge of magnitude $-q$ will be induced to the inner surface of the shell. So, total charge on the inner surface of the shell is $-q$.

As we know the surface charge density at the inner surface of the shell is given by the relation,

$$\sigma_1=\frac{\text{Total charge}}{\text{Inner surface area}}=\frac{-q}{4\pi r_1^2} \quad \ldots(i)$$

A charge of $+q$ is induced on the outer surface of the shell. A charge of magnitude Q is placed on the outer surface of the shell. Therefore, total charge on the outer surface of the shell is $Q + q$. Surface charge density at the outer surface of the shell,

$$\sigma_2=\frac{\text{Total charge}}{\text{Outer surface area}}=\frac{Q+q}{4\pi r_2^2} \quad \ldots\ldots(ii)$$ **[1½]**

(b) Yes, The electric field intensity inside a cavity is zero, even if the shell is not spherical and has any irregular shape. Take a closed loop such that a part of it is inside the cavity along a field line while the rest is inside the conductor. Net work done by the field in carrying a test charge over a closed loop is zero because the field inside the conductor is zero. Hence, electric field is zero, whatever is the shape. **[1½]**

Q. 3. (a) Show that the normal component of electrostatic field has a discontinuity from one side of a charged surface to another given by

$$(E_2 - E_1).\hat{n} = \frac{\sigma}{\varepsilon_0}$$

Where $\hat{n}$ is a unit vector normal to the surface at a point and σ is the surface charge density at that point. (The direction of $\hat{n}$ is from side 1 to side 2.) Hence show that just outside a conductor, the electric field is $\sigma\hat{n}/\varepsilon_0$

(b) Show that the tangential component of electrostatic field is continuous from one side of a charged surface to another.

[Hint : For (a), use Gauss's law. For, (b) use the fact that work done by electrostatic field on a closed loop is zero.] [NCERT Add. Ex. Q. 2.16, Page 14]

Ans. **(a)** If the electric field on one side of a charged body is E_1 and electric field on the other side of the same body is E_2 and If infinite plane charged body has a uniform thickness, then electric field due to one surface of the charged body is,

$$\overrightarrow{E_1} = -\frac{\sigma}{2\varepsilon_0}\hat{n} \quad ...(i)$$

Where,

$\hat{n}$ = Unit vector normal to the surface at a point

σ = Surface charge density at that point

Electric field due to the other surface of the charged body,

$$\overrightarrow{E_2} = +\frac{\sigma}{2\varepsilon_0}\hat{n} \quad ...(ii)$$

Electric field at any point due to the two surfaces,

$$\overrightarrow{E_2} - \overrightarrow{E_1} = \frac{\sigma}{2\varepsilon_0}\hat{n} + \frac{\sigma}{2\varepsilon_0}\hat{n} = \frac{\sigma}{\varepsilon_0}\hat{n}$$

$$(\overrightarrow{E_2} - \overrightarrow{E_1}).\hat{n} = \frac{\sigma}{\varepsilon_0}\hat{n} \quad ...(iii)$$ **[1½]**

Since inside a closed conductor, $\overrightarrow{E_1} = 0$,

$$\therefore \vec{E}_1 = \overrightarrow{E_2} = -\frac{\sigma}{2\varepsilon_0}\hat{n}$$

So, the electric field just outside the conductor is $\frac{\sigma}{\varepsilon_0}\hat{n}$

(b) When a charged particle is moved from one point to the other on a closed loop, the work done by the electrostatic field will be zero.

So, the tangential component of electrostatic field is continuous from one side of a charged surface to the other. **[1½]**

Q. 4. A long-charged cylinder of linear charged density λ is surrounded by a hollow co-axial conducting cylinder. What is the electric field in the space between the two cylinders?

[NCERT Add. Ex. Q. 2.17, Page 14]

Ans. Given that,

Charge density of the long-charged cylinder of length L and radius r is λ. Another cylinder of same length surrounds the pervious cylinder. The radius of this cylinder is R.

Let E be the electric field produced in the space between the two cylinders. Electric flux through the Gaussian surface is given by Gauss's theorem as,

$$\phi = E(2\pi d)L$$

Where,

d = Distance of a point from the common axis of the cylinders Let q be the total charge on the cylinder. It is given by,

$$\therefore \phi = E(2\pi dL) = \frac{q}{\varepsilon_0}$$ **[1]**

Where,

q = Charge on the inner sphere of the outer cylinder

ε_0 = Permittivity of free space

So, the electric field in the space between the two cylinders is

$$E(2\pi dL) = \frac{\lambda L}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0 d}$$ **[2]**

Therefore, the electric field in the space between the two cylinders is $\frac{\lambda}{2\pi\varepsilon_0 d}$

Q. 5. A spherical conductor of radius 12 cm has a charge of 1.6×10^{-7}C distributed uniformly on its surface. What is the electric field

(a) inside the sphere.

(b) Just outside the sphere

(c) at a point 18 cm from the centre of the sphere

[NCERT Ex. Q. 2.4, Page 11]

Ans. Given that,

Radius of the spherical conductor, r = 12 cm = 0.12 m

Charge is uniformly distributed over the conductor, $q = 1.6 \times 10^{-7}$ C

(a) As we know that,

An Electric field inside a spherical conductor is zero.

This is because if there is field inside the conductor, then charges will move to neutralise it. **[1]**

(b) Electric field E just outside the conductor is,

$$E = \frac{q}{4\pi\varepsilon_0 r^2}$$

Where,

ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9\times10^9 \text{ Nm}^2\text{C}^{-2}$$

$$\text{so, } E = \frac{1.6\times10^{-7}\times9\times10^9}{(0.12)^2} = 10^5 \text{ NC}^{-1}$$

So, the electric field just outside the sphere is 10^5 NC^{-1}. **[1]**

(c) Let the electric field at a point 18 m from the centre of the sphere = E_1

Distance of the point from the centre, d = 18 cm = 0.18 m

$$E_1 = \frac{q}{4\pi\varepsilon_0 d^2}$$

$$=\frac{9\times10^{9}\times1.6\times10^{-7}}{(18\times10^{-2})^{2}}$$

$=4.4\times10^{4}$ N/C

So, the electric field at a point 18 cm from the centre of the sphere is $4.4 \times 10^{4}\,NC^{-1}$. **[1]**

Q. 6. Three capacitors of capacitances 2 pF, 3 pF and 4 pF are connected in parallel.

(a) What is the total capacitance of the combination?

(b) Determine the charge on each capacitor if the combination is connected to a 100 V supply.

[NCERT Ex. Q. 2.7, Page 12]

Ans. **(a)** Given that,

$C_1 = 2$ pF

$C_2 = 3$ pF

$C_3 = 4$ pF

For the parallel combination of the capacitors, equivalent capacitor C′ is the algebraic sum,

$C' = 2+3+4 = 9$ pF

So, total capacitance of the combination is 9 pF. **[1½]**

(b) Given that,

Supply voltage, $V = 100$ V

The voltage through all the three capacitors is same $= V = 100$ V

Charge on a capacitor of capacitance C and potential difference V is given by the relation,

$q = VC$ …(i)

For $C = 2$ pF,

Charge $= VC = 100\times2 = 200$ pC $= 2\times10^{-10}$ C

For C = 3 pF,

Charge $= VC = 100\times3 = 300$ pC $= 3\times10^{-10}$ C

For $C = 4$ pF,

Charge = VC $= 100\times4 = 200$ pC $= 4\times10^{-10}$ C **[1½]**

Q. 7. Explain what would happen if in the capacitor given in Q. 10 (SATQ), a 3 mm thick mica sheet (of dielectric constant = 6) were inserted between the plates,

(a) While the voltage supply remained connected.

(b) After the supply was disconnected.

[NCERT Ex. Q. 2.1, Page 10]

Ans. **(a)** Given that,

Dielectric constant of the mica sheet, $K = 6$

Initial capacitance, $C = 1.771 \times 10^{-11}$ F

Supply voltage, $V = 100$ V

New capacitance, $C' = KC = 6\times1.771\times10^{-11}$

$= 10.6 \times 10^{-11}$ F

New charge, $q' = C'V' = 100\times10.6\times10^{-11}$

$= 10.6 \times 10^{-9}$ C

As the capacitor is connected to supply, the potential across the plates remains 100 V. **[1½]**

(b) Dielectric constant, $K = 6$

Initial capacitance, $C = 1.771 \times 10^{-11}$ F

New capacitance, $C' = KC = 6\times1.771\times10^{-11} = 106$ pF

So, if supply voltage is removed, then there will be no effect on the amount of charge on the plates.

Charge $= 1.771 \times 10^{-9}$ C

Potential across the plates is given by,

$$\therefore V'=\frac{q}{C'}$$

$$=\frac{1.771\times10^{-9}}{106\times10^{-12}}=16.7\text{ V}$$

[1½]

Q. 8. A parallel plate capacitor with air between the plates has a capacitance of 8 pF (1 pF = 10^{-12} F). What will be the capacitance if the distance between the plates is reduced by half, and the space between them is filled with a substance of dielectric constant 6? [NCERT Ex. Q. 2.5, Page 11]

Ans. Given that,

Capacitance between the parallel plates of the capacitor, $C = 8$ pF

Initially, the distance between the parallel plates $= d$

Dielectric constant of air, $K = 1$

As we know the capacitance, C is,

$$C=\frac{K\varepsilon_0 A}{d}$$

$$=\frac{\varepsilon_0 A}{d} \quad \text{…(i)}$$

[1]

Where,

A = Area of each plate

ε_0 = Permittivity of free space

Now, If distance between the plates is reduced to half, then new distance, $d' = \frac{d}{2}$

Dielectric constant of the substance filled in between the plates, $k' = 6$

So, the new capacitance of the capacitor will be :

$$C'=\frac{K'\varepsilon_0 A}{d}=\frac{6\varepsilon_0 A}{\frac{d}{2}} \quad \text{...(ii)}$$

[1]

Now, taking ratios of equations (i) and (ii),

$C'=2\times6C$

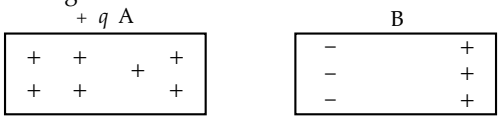
$=12 \quad =12\times8=96$ pF

So, the capacitance between the plates is 96 pF. **[1]**

Q. 9. Prove that, if an insulated, uncharged conductor is placed near a charged conductor and no other conductors are present, the uncharged body must be intermediate in potential between that of the charged body and that of infinity.

[NCERT Exemp. Q. 2.21, Page 14]

Ans. Consider a charged body (A) (say with positive charge) and an insulated uncharged conductor (B) is placed near the charged conductor (A) as shown in the figure :



As,

$$V=\frac{kq}{r}$$

[1]

where k and q are constants,

So $V\propto\frac{1}{r}$ or at infinity, $V \to 0$

Uncharged conductor is between charged conductor and infinity, so potential decreases from body A to infinity. So the potential of uncharged body varies between potential of A and infinity. **[2]**

Q. 10. In a parallel plate capacitor with air between the plates, each plate has an area of 6×10^{-3} m^2 and the distance between the plates is 3 mm. Calculate the capacitance of the capacitor. If this capacitor is

connected to a 100 V supply, what is the charge on each plate of the capacitor?

[NCERT Ex. Q. 2.8, Page 12]

Ans. Given that,

Area of each plate of the parallel plate capacitor, $A = 6 \times 10^{-3}\ m^2$

Distance between the plates, $d = 3\ mm = 3 \times 10^{-3}\ m$

Supply voltage, $V = 100\ V$

As we know the capacitance C of a parallel plate capacitor is given by,

$$C = \frac{\varepsilon_0 A}{d}$$ **[1]**

Where,

ε_0 = Permittivity of free space

$= 8.854 \times 10^{-12}\ N^{-1}m^{-2}C^2$

$$C = \frac{[8.854 \times 10^{-12} \times 6 \times 10^{-3}]}{3 \times 10^{-3}}$$

$$= 17.71 \times 10^{-12}\ F$$

$$V = \frac{q}{C}$$

$$\therefore q = CV$$

$$= 100 \times 17.71 \times 10^{-12} = 1.771 \times 10^{-9}\ C$$

So, the capacitance of the capacitor is 17.71 pF and charge on each plate is $1.771 \times 10^{-9}\ C$. **[2]**

Q. 11. A capacitor has some dielectric between its plates, and the capacitor is connected to a DC source. The battery is now disconnected and then the dielectric is removed. State whether the capacitance, the energy stored in it, electric field, charge stored and the voltage will increase, decrease or remain constant. [NCERT Exemp. Q. 2.20, Page 14]

Ans. The capacitance of the parallel plate capacitor, filled with dielectric medium of dielectric constant 'K' is given as-

$$C = \frac{K\varepsilon_0 A}{d}$$ **[1]**

As 'K' is positive and more than one, so by removing dielectric slab, and keeping A and d constant, capacitance of capacitor will decrease. When battery and dielectric slab from capacitor is removed, the charge remains same as it was when battery connected earlier.

As the energy stored in an isolated charge capacitor is $E = \frac{q^2}{2C}$

When capacitance C is decreased by removing dielectric slab but q remains same, so the energy stored in capacitor will increase. We know that,

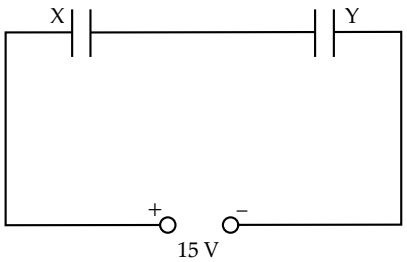
$$V = \frac{q}{C}$$

where q is same and C is decreased so potential will increase. Potential will increase.

As, $E = \frac{V}{d}$

Distance between plates of capacitor is same and potential is increased as discussed above, so electric field between the plates of capacitor will increase. **[2]**

Q. 12. Two parallel plate capacitors X and Y have the same area of plates and same separation between them. X has air between the plates while Y contains a dielectric medium of ε, = 4



(i) Calculate capacitance of each capacitor if equivalent capacitance of the combination is 4 μF.

(ii) Calculate the potential difference between the plates of X and Y.

(iii) Estimate the ratio of electrostatic energy stored in X and Y. [CBSE Board, Delhi Region, 2016; CBSE Board, Foreign Scheme, 2016]

Ans. **(i)** Capacitance of a parallel plate capacitor is given by,

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

Capacitor of plate Y is,

$$C_Y = \frac{\varepsilon_0 4A}{d}$$

Thus,

$$C_Y = 4C_X$$

$$\Rightarrow C_{eq} = \frac{C_X C_Y}{C_X + C_Y} = 4\mu F$$

$$\Rightarrow \frac{C_X(4C_X)}{5C_X} = 4\mu F$$

$$\Rightarrow C_X = 5\mu F$$

$$\Rightarrow C_Y = 20\mu F$$ **[1]**

(ii) Potential Difference between plates X and Y can be calculated as,

$$Q = CV$$

$$\Rightarrow \quad C_X V_X = C_Y V_Y$$

$$\Rightarrow \quad \frac{V_X}{V_Y} = \frac{C_Y}{C_X} = 4$$

$$\Rightarrow \quad V_X = 4V_Y$$

$$\text{Also, } V_Y + V_X = 15$$

$$\Rightarrow \quad V_Y = 3V$$

$$\Rightarrow \quad V_X = 12V$$ **[1]**

(iii) The ratio of electrostatic energies can be calculated as:

$$E = \frac{Q^2}{2C}$$

$$\Rightarrow \frac{E_X}{E_Y} = \frac{C_Y}{C_X} = 4$$

$$\Rightarrow \frac{E_X}{E_Y} = \frac{4}{1}$$

$$\Rightarrow E_X : E_Y :: 4 : 1$$ **[1]**

Q. 13. Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1: 2 so that the energy stored in the two cases becomes the same.

[CBSE Board, All India Region, 2016]

Ans. Given,

The ratio of the two capacitance is $C_2 = 2C_1$

When the capacitor are connected in parallel the total capacitance will be

$C_p = C_1 + C_2 = 3C_1$

Energy stored in the capacitor:

$$E = \frac{1}{2}C_pV_p^2$$

$$= \frac{3C_1V_p^2}{2}$$

When the capacitor are connected in series the total capacitance will be

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_s = \frac{2C_1}{3}$$

Energy stored in the capacitor :

$$E = \frac{1}{2}C_sV_s^2$$

$$= \frac{C_1V_s^2}{3}$$

According to the question

$$\frac{3C_1V_p^2}{2} = \frac{C_1V_s^2}{3}$$

$$\frac{V_p}{V_s} = \frac{\sqrt{2}}{3}$$ **[3]**

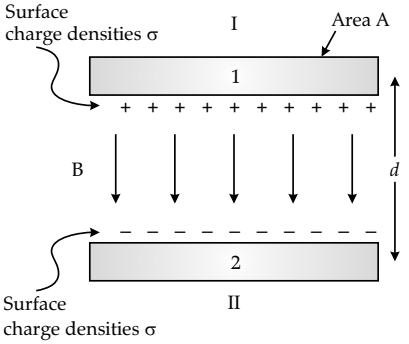
Q. 14. (a) If two similar large plates; each of area A having surface charge densities +σ and –σ are separated by a distance d in air, find the expressions for field at points between the two plates and on outer side of the plates. Specify the direction of the field in each case.

(b) the potential difference between the plates.

(c) the capacitance of the capacitor so formed.

[CBSE Board, All India Region, 2016]

Ans. (a) We are given two similar large plates separated by a small distance (d) and having area (A)



Surface charge density of plate 1, $\sigma = Q/A$, and that of plate 2 is $-\sigma$.

Electric field in different regions:

Outer region I:

$$E = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$

Outer region II:

$$E = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$

In the inner region between plates 1 and 2, the electric fields due to the two charged plates add up. So,

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$$

The direction of electric field is from the positive to the negative plate.

(b) For uniform electric field, potential difference is simply the electric field multiplied by the distance between the plates, *i.e.*

$$V = Ed = \frac{Qd}{\varepsilon_0 A}$$

(c) Capacitance C of the parallel plate capacitor,

$$C = \frac{Q}{V} = \frac{E_0A}{d}$$

We know that the potential of the metallic sphere is given by, where "r" is the radius of the sphere.

Now, the potential of the metallic sphere of radius R is given by, $V = \frac{Q}{4\pi\varepsilon_0 r}$

where "r" is the radius of the sphere.

Now, the potential of the metallic sphere of radius R is given by.

$$V_R = \frac{Q}{4\pi\varepsilon_0 r}$$

$$V_R = \frac{\sigma(4\pi R^2)}{4.4\pi\varepsilon_0 R}$$

$$V_R = \frac{\sigma R}{\varepsilon_0} \quad ...(i)$$

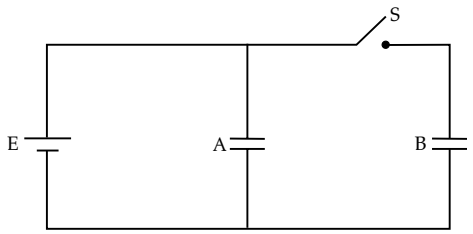
Similarly, the potential of the metallic sphere of radius $2R$ is given by

$$V_{2R} = \frac{\sigma^2 R}{\varepsilon_0} \quad ...(ii)$$

From the relation (i) and (ii) we know that $V_{2R} > V_R$.

So, when both the spheres are connected the charge flow from the sphere of radius of $2R$ to R. **[3]**

Q. 15. Two identical parallel plate capacitors *A* and *B* are connected to a battery of *V* volts with the switch *S* closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant K. Find the ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric.



[CBSE Board, All India Region, 2017]

Ans. Two capacitors are connected in parallel. Hence, the potential on each of them remains the same.
So, the charge on each is,
$Q_A = CV = Q_B$
So that, the energy stored in the system is,

$$U_{\text{initial}} = \frac{1}{2}CV^2 + \frac{1}{2}CV^2 = CV^2$$

When a dielectric slab is introduced, the capacitance changes to KC.
As the switch is open, only voltage across A remains the same. The voltage across B changes to $V' = \frac{Q}{C'} = \frac{Q}{KC} = \frac{V}{K}$.
Hence, new energy stored in the system is

$$U_{\text{final}} = \frac{1}{2}KCV^2 + \frac{1}{2}KC\frac{V^2}{K^2}$$

$$\therefore U_{\text{final}} = \frac{1}{2}KCV^2 + \frac{1}{2}\frac{CV^2}{K} = \frac{1}{2}CV^2\left(K + \frac{1}{K}\right)$$

$$\therefore \frac{U_{\text{initial}}}{U_{\text{final}}} = \frac{1}{K + \frac{1}{K}} = \frac{K}{K^2+1}$$

[3]

Long Answer Type Questions (5 marks each)

Q. 1. Two metal spheres, one of radius R and the other of radius $2R$, both have same surface charge density σ. They are brought in contact and separated. What will be new surface charge densities on them? [NCERT Exemp. Q. 2.29, Page 15]

Ans. Let surface charge density of both the spheres are σ and their charges are q_1 and q_2.

$$\therefore q_1 = \sigma.A_1 = \sigma.4\pi R^2$$

$$q_2 = \sigma.A_2 = \sigma.4\pi(2R)^2 = \sigma.4\pi R^2.4 = 4q_1$$

Both charged spheres are kept in contact, so charge flows between them and their potential becomes equal, let the charges on them now become q_1' and q_2'.

So, $V_1 = V_2$ $\left(\because V = \frac{kq}{r}\right)$

So, $\frac{kq_1'}{R} = \frac{kq_2'}{(2R)}$ $\left[\because k = \frac{1}{4\pi\varepsilon_0}\right]$

Where q_1' and q_2' are the charges on spheres after redistribution of charges **[2½]**

$$\frac{q_1'}{R} = \frac{q_2'}{2R}$$

$$\therefore q_2' = 2q_1' \quad \ldots(\text{I})$$

By law of conservation of charges

$$q_1 + q_2 = q_1' + q_2'$$

$$q_1 + 4q_1 = q_1' + 2q_1' \quad \text{(From I)}$$

$$5q_1 = 3q_1'$$

$$q_1' = \frac{20}{3}\pi R^2\sigma$$

$$\sigma_1 = \frac{q_1'}{A_1} = \frac{\frac{20}{3}\pi R^2\sigma}{4\pi R^2} = \frac{5}{3}\sigma$$

$$\sigma_2 = \frac{q_2'}{A_2} = \frac{2.\frac{20}{3}\pi R^2\sigma}{4\pi.4R^2} = \frac{5}{6}\sigma$$

So the values will be

$$\sigma_1 = \frac{5}{3}\sigma \text{ and } \sigma_2 = \frac{5}{6}\sigma$$

[2½]

Q. 2. An electrical technician requires a capacitance of $2\ \mu$F in a circuit across a potential difference of 1 kV. A large number of $1\ \mu$F capacitors are available to him each of which can withstand a potential difference of not more than 400 V. Suggest a possible arrangement that requires the minimum number of capacitors.
[NCERT Add. Ex. Q. 2.23, Page 14]

Ans. Given that,
Total required capacitance, $C = 2\ \mu$F
Potential difference, $V = 1$ kV $= 1000$ V
Capacitance of each capacitor, $C_1 = 1\ \mu$F
Each capacitor can withstand a potential difference, $V_1 = 400$ V
Let assume a number of capacitors are connected in series and these series circuits are connected in parallel (row) to each other. The potential difference across each row must be 1000 V and potential difference across each capacitor must be 400 V.
So, number of capacitors in each row is,

$$\frac{1000}{400} = 2.5$$

[1]

So, there are three capacitors in each row.

$$\text{Capacitance of each row} = \frac{1}{1+1+1} = \frac{1}{3}\mu\text{F}$$

Let there are n rows, each having three capacitors, which are connected in parallel series.
So, resultant capacitance of the circuit is,

$$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \ldots\ldots\ldots\ldots\text{n terms}$$

$$= \frac{n}{3}$$

[1]

However, capacitance of the circuit is given as $2\ \mu$F.

$$\therefore \frac{n}{3} = 2$$

$n = 6$

So, 6 rows of three capacitors are present in the circuit. A minimum of 6×3, that is, 18 capacitors are required for the given arrangement. **[3]**

Q. 3. What is the area of the plates of a 2 F parallel plate capacitor, given that the separation between the plates is 0.5 cm? [You will realize from your answer why ordinary capacitors are in the range

of μF or less. However, electrolytic capacitors do have a much larger capacitance (0.1 F) because of very minute separation between the conductors.]

[NCERT Add. Ex. 2.24, Page 14]

Ans. Given that,

Capacitance of a parallel capacitor, $V = 2$ F

Distance between the two plates, $d = 0.5$ cm $= 0.5 \times 10^{-2}$ m

The relation gives capacitance of a parallel plate capacitor,

$$C=\frac{\varepsilon_0 A}{d}$$

$$A=\frac{Cd}{\varepsilon_0}$$ **[2]**

Where,

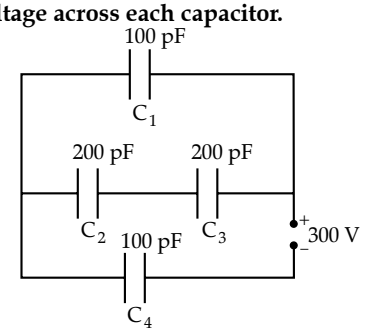
ε_0= Permittivity of free space

$= 8.85 \times 10^{-12}\ C^2N^{-1}m^{-2}$

$$\therefore A=\frac{2\times0.5\times10^{-2}}{8.85\times10^{-12}}=1130\ km^2$$

So, the area of the plates is too large. To avoid this situation, the capacitance is taken in the range of μF. **[3]**

Q. 4. Obtain the equivalent capacitance of the network in Figure, For a 300 V supply, determine the charge and voltage across each capacitor.



[NCERT Add. Ex. Q. 2.25, Page 15]

Ans. Given that,

Capacitance of capacitor C_1 is 100 pF.

Capacitance of capacitor C_2 is 200 pF.

Capacitance of capacitor C_3 is 200 pF.

Capacitance of capacitor C_4 is 100 pF.

Supply potential, $V = 300$ V

Capacitors C_2 and C_3 are connected in series. Let their equivalent capacitance be C'

$$\therefore \frac{1}{C'}=\frac{1}{200}+\frac{1}{200}=\frac{2}{200}$$

$$C'=100\ pF$$

Capacitors C_1 and C' are in parallel. Let their equivalent capacitance be C''

$$\therefore C''=C'+C_1$$

$$=100+100=200\ pF$$ **[1]**

C'' and C_4 are connected in series. Let their equivalent capacitance be C.

$$\therefore \frac{1}{C}=\frac{1}{C''}+\frac{1}{C_4}$$

$$=\frac{1}{200}+\frac{1}{100}=\frac{2+1}{200}$$

$$C=\frac{200}{3}\ pF$$

So, the equivalent capacitance of the circuit is $\frac{200}{3}$ pF

Potential difference across $C'' = V''$ Potential difference across $C_4 = V_4$

$$\therefore V''+V_4=300\ V$$ **[1]**

Charge on C_4 is given by,

$$Q_4= CV$$

$$=\frac{200}{3}\times10^{-12}\times300$$

$$=2\times10^{-8}\ C$$

$$\therefore V_4=\frac{Q_4}{C_4}$$

$$=\frac{2\times10^{-8}}{100\times10^{-12}}=200\ V$$

$\therefore$ Voltage across C_1 is given by,

$V_1 = V - V_4$

$= 300 - 200 = 100$ V

Hence, potential difference, V_1, across C_1 is 100 V.

Charge on C_1 is given by,

$$Q_1=C_1V_1$$

$$= 100 \times 10^{-12} \times 100 = 10^{-8}\ C$$ **[1]**

C_2 and C_3 having same capacitances have a potential difference of 100 V together. Since C_2 and C_3 are in series, the potential difference across C_2 and C_3 is given by,

$V_2 = V_3 = 50$ V

So, the charge on C_2 is given by,

$$Q_2=C_2V_2$$

$$=200 \times 10^{-12}\times50 = 10^{-8}\ C$$

And charge on C_3 is given by,

$$Q_3=C_3V_3$$

$$=200\times10^{-12}\times50= 10^{-8}\ C$$

So, the equivalent capacitance of the given circuit is $\frac{200}{3}$ pF with charges,

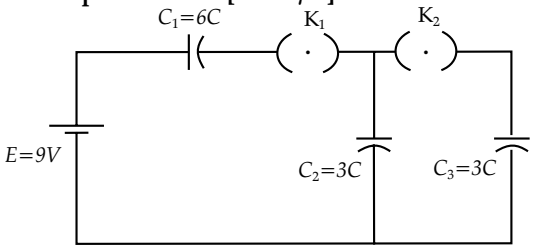
$Q_1 = 10^{-8}$ C, $V_1 = 100$ V

$Q_2 = 10^{-8}$ C, $V_2 = 50$ V

$Q_3 = 10^{-8}$ C, $V_3 = 50$ V

$Q_4 = 2\times10^{-8}$, $V_4 = 200$ V **[2]**

Q. 5. In the circuit shown in Figure, initially K_1 is closed and K_2 is open. What are the charges on each capacitor. Then K_1 was opened and K_2 was closed (order is important), what will be the charge on each capacitor now? [$C = 1\ \mu$F]



[NCERT Exemp. Q. 2.30, Page 15]

Ans. When K_2 is open and K_1 is closed the capacitors C_1 and C_2 will charge and potential develops across them, that is, V_1 and V_2 respectively which will be equal to the potential of battery 9 V.

$\therefore V_1+V_2=9$ …(I)

[1]

$\because \quad V=\frac{q}{C}$

or $\quad V\alpha\frac{1}{C}$

So that, $\frac{V_1}{V_2}=\frac{C_2}{C_1}$

$\frac{V_1}{V_2}=\frac{3C}{6C}$

$3V_2 = 6V_1$

$V_2 = 2V_1$ …(II) **[1]**

From equations (I) and (II)

$V_1 + 2V_1 = 9$

$3V_1 = 9$

$V_1 = 3$ Volt

$V_2 = 2 \times 3$ Volt $= 6$ Volt

$\therefore q_1 = C_1V_1 = 6C \times 3 = 18$ C

[From (II) $C = 1\mu$F] **[1]**

$=18\times1\ \mu\text{F} = 18\ \mu\text{C}$

$q_2=C_2V_2=3C\times6$

$= 3\times1\ \mu\text{F} \times6 = 18\ \mu\text{C}$

So, charges in each capacitor will be : $q_1=q_2=18\ \mu$C

When k_1 is open and k_2 is closed then charge q_2 will be distributed among C_2 and C_3. Let it be q_2 and q_3.

$\therefore q_2=q_2+q_3$

As C_2 and C_3 are now in parallel combination so them potentials remains same (V)

$\therefore q_2=C_2V+C_3V$

$18\ \mu\text{C}=3\times1\ \mu\text{F}\times V+3\times1\ \mu\text{F}\times V$

$18 = 6$ V

$V= 3$ Volt

So potential on C_2 and C_3 capacitors is 3 Volt each.

$q_2'=C_2V = 3\times1\ \mu\text{F} \times3$ Volt $= 9\ \mu$C

$q_3=C_3V = 3\times1\ \mu\text{F} \times3$ Volt $= 9\ \mu$C **[2]**

Q. 6. A parallel plate capacitor is filled by a dielectric whose relative permittivity varies with the applied voltage (U) as $\varepsilon = \alpha U$ where $\alpha = 2V^{-1}$. A similar capacitor with no dielectric is charged to $U_0 = 78$ V. It is then connected to the uncharged capacitor with the dielectric. Find the final voltage on the capacitors.

[NCERT Exemp. Q. 2.26, Page 15]

Ans. Let us assume that the final voltage be U :

If C is the capacitance of the capacitor without the dielectric, then the charge on the capacitor will be, $Q_1 = CU$

The capacitor with the dielectric has a capacitance εC. So, the charge on the capacitor will be, $Q_2 = \varepsilon U = \alpha CU^2$

The initial charge on the capacitor that was charged is $Q_0 = CU_0$ **[2½]**

From the conservation of charges,

$Q_0 = Q_1 + Q_2$

Or, $CU_0 = CU + \alpha CU^2$

$\Rightarrow \alpha U^2 + U - U_0 = 0$

$\therefore \quad U=\frac{-1\pm\sqrt{1+4\alpha U_0}}{2\alpha}$

As, $\alpha = 2V^{-1}$ and $U_0 = 78$ V

$=\frac{-1\pm\sqrt{1+624}}{4}$

$=\frac{-1\pm\sqrt{625}}{4}$ volts

As U is positive

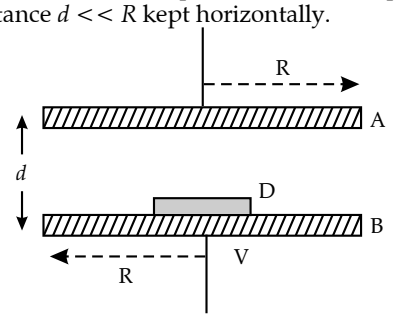
$U=\frac{\sqrt{625}-1}{4}=\frac{24}{4}=6$ V

So, final potential on both the capacitors becomes 6 volts. **[2½]**

Q. 7. A capacitor is made of two circular plates of radius R each, separated by a distance $d << R$. The capacitor is connected to a constant voltage. A thin conducting disc of radius $r << R$ and thickness $t << r$ is placed at a centre of the bottom plate. Find the minimum voltage required to lift the disc if the mass of the disc is m.

[NCERT Exemp. Q. 2.27, Page 15]

Ans. Let A and B are circular plates of radius R separated by distance $d << R$ kept horizontally.



A thin conducting disc D of radius $r << R$ of thickness 't' is placed concentrically on lower plate B as shown in figure.

Let plate A and B charged with potential V.

The magnitude of electric field E between plates of capacitor

$E=\frac{V}{d}$ $\qquad \left[\because E=\frac{-dV}{dR}\right]$

Consider Gaussian surface along circular disc D.

By Gauss's law, $\oint \text{E.ds}=\frac{q'}{\varepsilon_0}$

$\frac{V}{d}.\pi r^2=\frac{q'}{\varepsilon_0}$ **[2]**

q' is the charge conducted by plate B to disc D during charging. Nature of charge on plate B and disc will be same so repulsive force acts between B and D.

So, the charge on disc $D=q'=\frac{V}{d}.\pi r^2\varepsilon_0$

Electrostatic repulsive force acting on disc in upward direction

$F=q'E=\frac{V}{d}.\pi r^2\varepsilon_0.\frac{V}{d}=\frac{V^2}{d^2}\pi r^2\varepsilon_0$

This repulsive force will be balanced by weight mg of disc D.

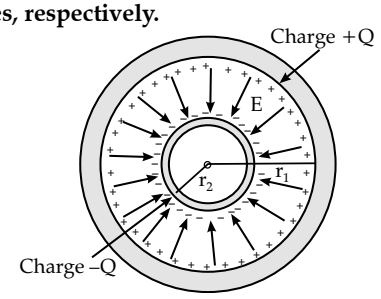
$$\varepsilon_0 \frac{V^2}{d^2}\pi r^2 = mg$$

$$\Rightarrow \quad V = \sqrt{\frac{mgd^2}{\pi\varepsilon_0 r^2}} \qquad \textbf{[3]}$$

Q. 8. A spherical capacitor consists of two concentric spherical conductors, held in position by suitable insulating supports (see Figure). Show that the capacitance of a spherical capacitor is given by

$$C = \frac{4\pi\varepsilon_0 r_1 r_2}{r_1 - r_2}$$

where r_1 and r_2 are the radii of outer and inner spheres, respectively.



[NCERT Add. Ex. Q. 2.29, Page 15]

Ans. Given that,

Radius of the outer shell = r_1

Radius of the inner shell = r_2

The inner surface of the outer shell has charge $+Q$.

The outer surface of the inner shell has induced charge $-Q$.

As we know the potential difference between the two shells is given by,

$$V = \frac{Q}{4\pi\varepsilon_0 r_2} - \frac{Q}{4\pi\varepsilon_0 r_1} \qquad \textbf{[2½]}$$

Where,

ε_0 = Permittivity of free space

$$V = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$$

$$= \frac{Q(r_1 - r_2)}{4\pi\varepsilon_0 r_1 r_2}$$

Capacitance of the given system is given by,

$$C = \frac{\text{Charge } (Q)}{\text{Potential difference } (V)}$$

$$= \frac{4\pi\varepsilon_0 r_1 r_2}{r_1 - r_2} \qquad \textbf{[2½]}$$

Hence proved.

Q. 9. A spherical capacitor has an inner sphere of radius 12 cm and an outer sphere of radius 13 cm. The outer sphere is earthed, and the inner sphere is given a charge of 2.5 μC. The space between the concentric spheres is filled with a liquid of dielectric constant 32.

(a) Determine the capacitance of the capacitor.

(b) What is the potential of the inner sphere?

(c) Compare the capacitance of this capacitor with that of an isolated sphere of radius 12 cm. Explain why the latter is much smaller.

[NCERT Add. Ex. Q. 2.30, Page 15]

Ans. Given that,

Radius of the inner sphere, r_2 = 12 cm = 0.12 m

Radius of the outer sphere, r_1 = 13 cm = 0.13 m

Charge on the inner sphere,

$Q = 2.5\,\mu\text{C} = 2.5 \times 10^{-6}$

Dielectric constant of a liquid, $\varepsilon_r = 32$

(a) As we know the capacitance of the capacitor is,

$$C = \frac{4\pi\varepsilon_0\varepsilon_r r_1 r_2}{r_1 - r_2}$$

Where,

ε_0 = Permittivity of free space

$= 8.85 \times 10^{-12}\ \text{C}^2\text{N}^{-1}\text{m}^{-2}$

$$\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ \text{Nm}^2\text{C}^{-2}$$

$$\therefore \quad C = \frac{32\times0.12\times0.13}{9\times10^9\times(0.13-0.12)}$$

$$\approx 5.5\times10^{-9}\ \text{F}$$

So, the capacitance of the capacitor is approximately 5.5×10^{-9} F. **[2]**

(b) Potential of the inner sphere is given by,

$$V = \frac{q}{C}$$

$$= \frac{2.5\times10^{-6}}{5.5\times10^{-9}} = 4.5\times10^2\ \text{V}$$

So, the potential of the inner sphere is 4.5×10^2 V. **[1]**

(c) Radius of an isolated sphere, $r = 12 \times 10^{-2}$ m

As we know the capacitance of the sphere is,

$$C' = 4\pi\varepsilon_0 r$$

$$= 4\pi\times8.85\times10^{-12}\times12\times10^{-2} = 1.33\times10^{-11}\ \text{F}$$

The capacitance of the isolated sphere is less in comparison to the concentric spheres because the outer sphere of the concentric spheres is earthed.

So, the potential difference is less, and the capacitance is more than the isolated sphere. **[2]**

Q. 10. Answer carefully :

(a) Two large conducting spheres carrying charges Q_1 and Q_2 are brought close to each other. Is the magnitude of electrostatic force between them exactly given by $Q_1Q_2/4\pi\varepsilon_0 r^2$, where r is the distance between their centres?

(b) If Coulomb's law involved $1/r^3$ dependence (instead of $1/r^2$), would Gauss's law be still true?

(c) A small test charge is released at rest at a point in an electrostatic field configuration. Will it travel along the field line passing through that point?

(d) What is the work done by the field of a nucleus in a complete circular orbit of the electron? What if the orbit is elliptical?

(e) We know that electric field is discontinuous across the surface of a charged conductor. Is electric potential also discontinuous there?

(f) What meaning would you give to the capacitance of a single conductor?

(g) Guess a possible reason why water has a much greater dielectric constant (= 80) than say, mica (= 6). **[NCERT Add. Ex. Q. 2.31, Page 91]**

Ans. **(a)** The force between two conducting spheres is not exactly given by the expression,

$$=\frac{Q_1Q_2}{4\pi\varepsilon_0 r^2},$$

because there is a non-uniform charge distribution on the spheres. **[½]**

(b) Gauss's law will not be true, if Coulomb's law involved $\frac{1}{r^3}$ dependence, instead of $\frac{1}{r^2}$, on r. **[½]**

(c) Yes,
If a small test charge is released at rest at a point in an electrostatic field configuration, then it will travel along the field lines passing through the point, only if the field lines are straight. This is because the field lines give the direction of acceleration and not of velocity. **[½]**

(d) Whenever the electron completes an orbit, either circular or elliptical, the work done by the field of a nucleus is zero. **[1]**

(e) No,
Electric field is discontinuous across the surface of a charged conductor. However, electric potential is continuous. **[½]**

(f) The capacitance of a single conductor is considered as a parallel plate capacitor with one of its two plates at infinity. **[1]**

(g) Water has an unsymmetrical space as compared to mica. Since it has a permanent dipole moment, it has a greater dielectric constant than mica. **[1]**

Q. 11. A cylindrical capacitor has two co-axial cylinders of length 15 cm and radii 1.5 cm and 1.4 cm. The outer cylinder is earthed, and the inner cylinder is given a charge of 3.5 μC. Determine the capacitance of the system and the potential of the inner cylinder. Neglect end effects (*i.e.*, bending of field lines at the ends).

[NCERT Add. Ex. Q. 2.32, Page 91]

Ans. Given that,
Length of a co-axial cylinder, l = 15 cm = 0.15 m
Radius of outer cylinder, r_1 = 1.5 cm = 0.015 m
Radius of inner cylinder, r_2 = 1.4 cm = 0.014 m
Charge on the inner cylinder, q = 3.5 μC = 3.5 $\times 10^{-6}$ C
As we know that,
the capacitance of a co-axial cylinder of radii r_1 and r_2 is given by the relation,
Capacitance of a co-axial cylinder of radii r_1 and r_2 is given by the relation,

$$C=\frac{2\pi\varepsilon_0 l}{\log\frac{r_1}{r_2}}$$ **[2]**

Where,
ε_0 = Permittivity of free space = 8.85 x 10^{-12} C^2N^{-1} m^{-2}

$$\therefore C=\frac{2\pi\times8.85\times10^{-12}\times0.15}{2.3026\log_{10}\left(\frac{0.15}{0.14}\right)}$$

$$=\frac{2\pi\times8.85\times10^{-12}\times0.15}{2.3026\times0.0299}=1.2\times10^{-10}\text{ F}$$ **[2]**

Potential difference of the inner cylinder is given by,

$$V=\frac{q}{C}$$

$$=\frac{3.5\times10^{-6}}{1.2\times10^{-10}}=2.92\times10^{4}\text{ V}$$ **[1]**

Q. 12. A parallel plate capacitor is to be designed with a voltage rating 1 kV, using a material of dielectric constant 3 and dielectric strength about 10^7 Vm^{-1}. (Dielectric strength is the maximum electric field a material can tolerate without breakdown, *i.e.*, without starting to conduct electricity through partial ionization.) For safety, we should like the field never to exceed, say 10% of the dielectric strength. What minimum area of the plates is required to have a capacitance of 50 pF?

[NCERT Add. Ex. Q. 2.33, Page 91]

Ans. Given that,
Potential rating of a parallel plate capacitor,
V = 1 kV = 1000 V
Dielectric constant of a material, ε_r = 3
Dielectric strength = 10^7 V/m
For safety, the field intensity never exceeds 10% of the dielectric strength.
So, electric field intensity, E = 10% of 10^7 = 10^6 V/m
Capacitance of the parallel plate capacitor, C = 50 pF = 50 $\times$ 10^{-12} F
Distance between the plates is given by,

$$d=\frac{V}{E}$$

$$\frac{1000}{10^6}=10^{-3}\text{ m}$$

Capacitance is given by the relation,

$$C=\frac{\varepsilon_0\varepsilon_r A}{d}$$ **[2]**

Where,
A = Area of each plate
ε_0 = Permittivity of free space
= 8.85 $\times$ 10^{-12} $N^{-1}C^2m^{-2}$

$$\therefore A=\frac{Cd}{\varepsilon_0\varepsilon_r}$$

$$=\frac{50\times10^{-12}\times10^{-13}}{8.85\times10^{-12}\times3}\approx19\text{ cm}$$

So, the area of each plate is about 19 cm^2. **[3]**

Q. 13. Describe schematically the equipotential surfaces corresponding to -

(a) a constant electric field in the z-direction,

(b) a field that uniformly increases in magnitude but remains in a constant (say, *z*) direction,

(c) a single positive charge at the origin, and

(d) a uniform grid consisting of long equally spaced parallel charged wires in a plane.

[NCERT Add. Ex. Q. 2.34, Page 91]

Ans. **(a)** Equidistant planes parallel to the *x*-*y* plane are the equipotential surfaces. **[1]**

(b) Planes parallel to the *x-y* plane are the equipotential surfaces with the exception that when the planes get closer, the field increases. **[2]**

(c) Concentric spheres centered at the origin are equipotential surfaces. **[1]**

(d) A periodically varying shape near the given grid is the equipotential surface. This shape gradually reaches the shape of planes parallel to the grid at a larger distance. **[1]**

Q. 14. In a Van de Graaff type generator a spherical metal shell is to be a 15×10^6 V electrode. The dielectric strength of the gas surrounding the electrode is 5×10^7 Vm^{-1}. What is the minimum radius of the spherical shell required? (You will learn from this exercise why one cannot build an electrostatic generator using a very small shell which requires a small charge to acquire a high potential.)

[NCERT Add. Ex. Q. 2.35, Page 92]

Ans. Given that,

Potential difference, $V = 15 \times 10^6$ V

Dielectric strength of the surrounding gas

$$= 5 \times 10^7 \frac{V}{m}$$

Electric field intensity, E = Dielectric strength

$$= 5 \times 10^7 \frac{V}{m}$$

Minimum radius of the spherical shell required for the purpose is given by,

$$r = \frac{V}{E}$$

$$= \frac{15 \times 10^6}{5 \times 10^7} = 0.3 \text{ m} = 30 \text{ cm}$$

So, the minimum radius of the spherical shell required is 30 cm. **[5]**

Q. 15. A small sphere of radius r_1 and charge q_1 is enclosed by a spherical shell of radius r_2 and charge q_2. Show that if q_1 is positive, charge will necessarily flow from the sphere to the shell (when the two are connected by a wire) no matter what the charge q_2 on the shell is.

[NCERT Add. Ex. Q. 2.36, Page 92]

Ans. As we know the Gauss's Law,

According to this law, the electric field between a sphere and a shell is determined by the charge q_1 on a small sphere. **[2½]**

So, the potential difference, V, between the sphere and the shell is independent of charge q_2.

For positive charge q_1, potential difference V is always positive. **[2½]**

Q. 16. Answer the following :

(a) The top of the atmosphere is at about 400 kV with respect to the surface of the earth, corresponding to an electric field that decreases with altitude. Near the surface of the earth, the field is about 100 V m^{-1}. Why then do we not get an electric shock as we step out of our house into the open? (Assume the house to be a steel cage so there is no field inside)

(b) A man fixes outside his house one evening a two-metre-high insulating slab carrying on its top a large aluminium sheet of area 1 m^2. Will he get an electric shock if he touches the metal sheet next morning?

(c) The discharging current in the atmosphere due to the small conductivity of air is known to be 1800 A on an average over the globe. Why then does the atmosphere not discharge itself completely in due course and become electrically neutral? In other words, what keeps the atmosphere charged?

(d) What are the forms of energy into which the electrical energy of the atmosphere is dissipated during a lightning? (Hint : The earth has an electric field of about 100 Vm^{-1} at its surface in the downward direction, corresponding to a surface charge density = -10^{-9} C m^{-2}. Due to the slight conductivity of the atmosphere up to about 50 km (beyond which it is good conductor), about + 1800 C is pumped every second into the earth as a whole. The earth, however, does not get discharged since thunderstorms and lightning occurring continually all over the globe pump an equal amount of negative charge on the earth.)

[NCERT Add. Ex. Q. 2.37, Page 92]

Ans. **(a)** We do not get an electric shock as we step out of our house because the original equipotential surfaces of open air changes, keeping our body and the ground at the same potential. **[1]**

(b) Yes, the man will get an electric shock if he touches the metal slab next morning. The steady discharging current in the atmosphere charges up the aluminium sheet. As a result, its voltage rises gradually. The raise in the voltage depends on the capacitance of the capacitor formed by the aluminium slab and the ground. **[1½]**

(c) The occurrence of thunderstorms and lightning charges the atmosphere continuously. Hence, even with the presence of discharging current of 1800 A, the atmosphere is not discharged completely. The two opposing currents are in equilibrium and the atmosphere remains electrically neutral. **[1½]**

(d) During lightning and thunder storm, light energy, heat energy, and sound energy are dissipated in the atmosphere. **[1]**

Q. 17. The plates of a parallel plate capacitor have an area of 90 cm^2 each and are separated by 2.5 mm. The capacitor is charged by connecting it to a 400 V supply.

(a) How much electrostatic energy is stored by the capacitor?

(b) View this energy as stored in the electrostatic field between the plates, and obtain the energy per unit volume u. Hence arrive at a relation between u and the magnitude of electric field *E* between the plates. **[NCERT Add. Ex. Q. 2.26, Page 90]**

Ans. Given that,

Area of the plates of a parallel plate capacitor, $A = 90 \text{ cm}^2 = 90 \times 10^{-4} \text{ m}^2$

Distance between the plates, $d = 2.5 \text{ mm} = 2.5 \times 10^{-3}$ m

Potential difference across the plates, $V = 400$V

(a) The Capacitance of capacitor is

$$C=\frac{\varepsilon_0 A}{d}$$

As we know the electrostatic energy stored in the capacitor is

$$E_1=\frac{1}{2}CV^2$$

$$=\frac{1}{2}\frac{\varepsilon_0 A}{d}V^2$$

Where,

ε_0 = Permittivity of free space

$= 8.85 \times 10^{-12}\, C^2N^{-1}m^{-2}$

$$=\frac{1\times8.85\times10^{-12}\times90\times10^{-4}\times(400)^2}{2\times2.5\times10}=2.55\times10 \quad \text{J}$$

So, the electrostatic energy stored by the capacitor is 2.55×10^{-6} J **[2½]**

(b) Volume of the given capacitor,

$V' = A \times d$

$= 90 \times 10^{-4} \times 25 \times 10^{-3}$

$= 2.25 \times 10^{-4}\ m^3$

So, the energy stored in the capacitor per unit volume is

$$u=\frac{E_1}{V'}$$

$$=\frac{2.55\times10^{-6}}{2.25\times10^{-4}}=0.113\ \text{Jm}^{-3}$$

Again, $u=\frac{E_1}{V'}$

$$=\frac{\frac{1}{2}CV^2}{Ad}=\frac{\frac{\varepsilon_0 A}{2d}V^2}{Ad}=\frac{1}{2}\varepsilon_0\left(\frac{V}{d}\right)^2$$

where,

$\frac{V}{d}$ = Electric intensity = E

$$\therefore u=\frac{1}{2}\varepsilon_0 E^2$$

[2½]

Q. 18. A 4 μF capacitor is charged by a 200 V supply. It is then disconnected from the supply and is connected to another uncharged 2 μF capacitor. How much electrostatic energy of the first capacitor is lost in the form of heat and electromagnetic radiation? [NCERT Add. Ex. Q. 2.27, Page 90]

Ans. Given that,

Capacitance of a charged capacitor,

$C_1 = 4\mu F = 4\times10^{-6}$ F

Supply voltage, V_1 = 200 V

As we know that,

Electrostatic energy stored in C_1 is given by,

$$E_1=\frac{1}{2}C_1V_1^2$$

$$=\frac{1}{2}\times4\times10^{-6}\times(200)^2$$

$$=8\times10^{-2}\ \text{J}$$

Capacitance of an uncharged capacitor,

$C_2 = 2\mu F = 2\times10^{-6}$ F

When C_2 is connected to the circuit, the potential acquired by it is V_2. **[2½]**

According to the conservation of charge, initial charge on capacitor C_1 is equal to the final charge on capacitors, C_1 and C_2.

$$V_2(C_1+C_2)=C_1V_1$$

$$V_2\times(4+2)\times10^{-6}=4\times10^{-6}\times200$$

$$V_2=\frac{400}{3}V$$

Electrostatic energy for the combination of two capacitors is,

$$E_2=\frac{1}{2}(C_1+C_2)V_2^2$$

$$=\frac{1}{2}(2+4)\times10^{-6}\times\left(\frac{400}{3}\right)^2$$

$$=5.33\times10^{-2}\ \text{J}$$

So, amount of electrostatic energy lost by capacitor C_1

$= E_1 - E_2$

$= 0.08 - 0.0533 = 0.0267$

$= 2.67 \times 10^{-2}$ J **[2½]**

Q. 19. Show that the force on each plate of a parallel plate capacitor has a magnitude equal to *QE*, where *Q* is the charge on the capacitor, and *E* is the magnitude of electric field between the plates. Explain the origin of the factor $\frac{1}{2}$.

[NCERT Add. Ex. Q. 2.28, Page 90]

Ans. Let assume that F be the force applied to separate the plates of a parallel plate capacitor by a distance of x.

So, the work done by the force to do so = $F\Delta x$

As a result, the potential energy of the capacitor increases by an amount given as, $uA\Delta x$.

Where,

u = Energy density

A = Area of each plate

d = Distance between the plates

V = Potential difference across the plates

The work done will be equal to the increase in the potential energy, *i.e.*,

$F\Delta x = uA\Delta x$

$$F=uA=\left(\frac{1}{2}\varepsilon_0 E^2\right)A$$ **[2]**

As we know that an electric intensity is given by,

$$E=\frac{V}{d}$$

$$\therefore F=\frac{1}{2}\varepsilon_0\left(\frac{V}{d}\right)EA=\frac{1}{2}\left(\varepsilon_0 A\frac{V}{d}\right)E$$

Capacitance of any capacitor,

$$C=\frac{\varepsilon_0 A}{d}$$

$$\therefore F=\frac{1}{2}(CV)E$$

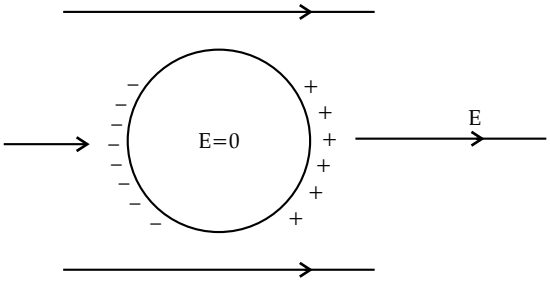
Charge on the capacitor is, $Q = CV$

$$\therefore F=\frac{1}{2}QE$$ **[2]**

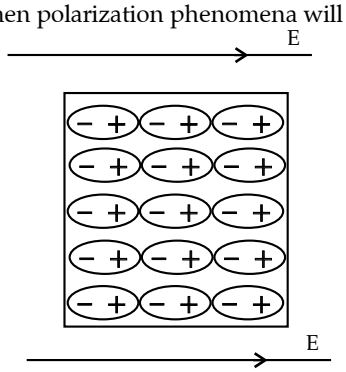
The physical origin of the factor, $\frac{1}{2}$, in the force formula lies in the fact that just outside the conductor, field is E and inside it is zero. Hence, it is the average value, $\frac{E}{2}$, of the field that contributes to the force. **[1]**

Q. 20. Distinguish, with the help of a suitable diagram, the difference in the behaviour of a conductor and a dielectric placed in an external electric field. How does polarised dielectric modify the original external field? [CBSE Board, Delhi Region, 2016]

Ans. When conductor is placed in an external electric field then induction phenomena occur, due to which induced charge get develop on the conductor surface.



When dielectric is placed in an external electric field then polarization phenomena will occur.



[5]

Q. 21. A capacitor of capacitance C is charged fully by connecting it to a battery of emf E. It is then disconnected from the battery. If the separation between the plates of the capacitor is now doubled, how will the following change?

(i) charge stored by the capacitor.

(ii) field strength between the plates.

(iii) energy stored by the capacitor.

Justify your answer in each case.

[CBSE Board, Delhi Region, 2016]

Ans. After disconnected from battery and doubling the separation between two plates

(i) Charge on capacitor remains same, *i.e.*,

$$CV = C'V'$$

$$\Rightarrow CV = \left(\frac{C}{2}\right)V'$$

$$\Rightarrow \quad V' = 2V$$

(ii) Electric field between the plates

$$E' = \frac{V'}{d'} = \frac{2V}{2d}$$

$$E' = \frac{V}{d} = E$$

$\Rightarrow$ Electric field between the two plates remains same.

(iii) Energy stored in capacitor when connected from battery

$$U_1 = \frac{q^2}{2C}$$

Now, energy stored in capacitor after disconnection from battery

$$U_2 = \frac{q^2}{(2C')} = \frac{q^2}{2 \times \left(\frac{C}{2}\right)} = \frac{q^2}{C}$$

$$\Rightarrow U_2 = 2\left(\frac{q^2}{2C}\right) = 2U_1$$

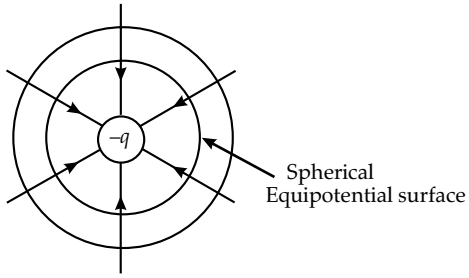
$$\Rightarrow U_2 = 2U_1$$

Energy stored in capacitor gets doubled to its initial value. **[5]**

Q. 22. Explain why, for any charge configuration, the equipotential surface through a point is normal to the electric field at that point. Draw a sketch of equipotential surfaces due to a single charge (–q), depicting the electric field lines due to the charge.

[CBSE Board, Delhi Region, 2016]

Ans. If the field were not normal to the equipotential surface, it would have non-zero component along the surface. To move a unit test charge against the direction of the component of the field, work would have to be done. But this is in contradiction to the definition of an equipotential surface: there is no potential difference between any two points on the surface and no work is required to move a test charge on the surface. **[4]**

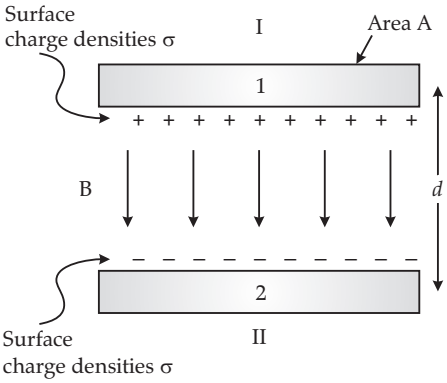


[1]

Q. 23. Define the capacitance of a capacitor. Obtain the expression for the capacitance of a parallel plate capacitor in vacuum in terms of plate area A and separation d between the plates.

[CBSE Board, Foreign Scheme, 2017]

Ans. Capacitance of a capacitor is defined as the ratio of the charge stored on any of the plates of capacitor to the potential between the plates. **[2]**



Electric field between the plates of capacitor

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0}$$

$$V = Ed = \frac{Qd}{A\varepsilon_0}$$

$$\text{Capacitance, } C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d} \qquad \textbf{[3]}$$

Q. 24. A slab of material of dielectric constant K has the same area as the plates of a parallel plate capacitor but has a thickness $\frac{3d}{4}$. Find the ratio of the capacitance with dielectric inside it to its capacitance without the dielectric.

[CBSE Board, Foreign Scheme, 2017]

Ans. Capacitance having a dielectric of thickness 't' is given by,

$$C = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}}$$

When the thickness of the plates, $t = \frac{3d}{4}$, then,

$$C = \frac{\varepsilon_0 A}{d - \frac{3d}{4} + \frac{3d}{4K}}$$

$$= \frac{4\varepsilon_0 A}{d\left(\frac{K+3}{K}\right)}$$

$$= \left(\frac{4K}{K+3}\right)\left(\frac{\varepsilon_0 A}{d}\right)$$

$$= \frac{4K}{K+3} C_0 \qquad \textbf{[5]}$$

Some Commonly Made Errors

- Students confuse to differentiate the combination of capacitors with the combination of resistors.
- Students do not study the effect of dielectric on the Capacitance of a Capacitor.
- Do not ignore to study 'Polarization and polar atom'.
- Generally, students get confused with the term 'electric potential differences' and 'equipotential surfaces'.
- Try to solve numerical by making suitable diagram, do not skip the diagrams.

EXPERT ADVICE

☞ *Apply method of elimination for MCQ questions. Look at the answers before even starting to solve the questions and see if you can eliminate any answer. When calculating answer for a big question you may not need to solve the problem fully in order to choose the right answer because you could eliminate the other three.*

☞ *Convert all units to SI, with charges in coulombs and distances in meters, so as to be consistent with the SI value of the Coulomb constant.*

OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/MnpJocNB6mc
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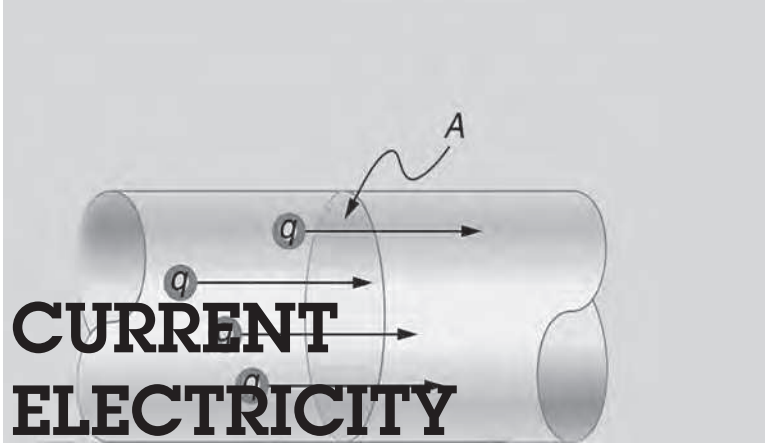
CHAPTER

3 CURRENT ELECTRICITY



Chapter Objectives

This chapter will help you understand :

- ***Electric Current and Resistance :*** *Introduction; Electric current; Resistance, Resistivity and current density; Ohm's law and its limitations; Colour coding of resistors; Temperature dependence of resistivity; Combination of electric resistances; Drift velocity and mobility of electrons; Electrical energy and power.*
- ***Electric Cell and Bridges :*** *Introduction; Electric cell, EMF and internal resistance; Combination of electric cells; Kirchhoff's Law; Wheatstone Bridge and Meter Bridge; Potentiometer.*

TOPIC - 1
Electric Current and Resistance **P. 60**

TOPIC - 2
Electric Cell and Bridges **P. 71**

TOPIC-1
Electric Current and Resistance

Quick Review

- Electricity travels at the speed of light - more than 1,86,000 miles per second.
- A spark of static electricity can measure up to 3000 volts.
- Lightning is a discharge of electricity in the atmosphere. Lightning bolts can travel at around 1,30,000 miles per hour and reach nearly 54000 °F in temperature.
- **Benjamin Franklin** carried out extensive electricity research in the 18th century, inventing the lightning rod amongst his many discoveries. In the event of a lightning strike, the lighting rod conducts the strike through a grounded wire, protecting the building.
- Electrons can be motivated to flow through a conductor by the same force manifested in static electricity.
- Voltage is the measure of specific potential energy (potential energy per unit charge) between two locations. In layman's terms, it is the measure of "push" available to motivate electrons.

TIPS...

- *Study the ohm's law and its limitations.*
- *Draw required diagram of the circuit in the solution.*
- *The total resistance in the series combination is more than the greatest resistance in the circuit.*
- *The total resistance in the parallel combination is less than the least resistance in the circuit.*

TRICKS...

- *Total resistance R of n resistors connected in series is given by-*

  $R = R_1 + R_2 + \ldots + R_n$

- *Total resistance R of n resistors connected in parallel is given by-*

  $\frac{I}{R} = \frac{I}{R_1} + \frac{I}{R_2} + \ldots + \frac{I}{R_n}$

Know the Links

- *https://www.jagranjosh.com/articles/cbse-class-12th-physics-notes-current-electricity-part-ii-1469528395-1*
- *http://www.learncbse.in/current-electricity-cbse-notes-class-12-physics/*
- *http://www.berwickelectric.com/the-electrical-blog/bid/71719/10-Fun-Facts-about-Electricity*

Multiple Choice Questions (1 mark each)

Q. 1. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of j (current density) changes in an exact manner, while the current I remain unaffected. The agent that is essentially responsible for is

(a) source of emf.

(b) electric field produced by charges accumulated on the surface of wire.

(c) the charges just behind a given segment of wire which push them just the right way by repulsion.

(d) the charges ahead.

[NCERT Exemp. Q. 3.1, Page 16]

Ans. Correct option : (b)

Explanation : As we know that current density (J) depends on

(i) conductivity [$\sigma = 1/\rho = 1/R.A$]

(ii) Electric field [$J = \sigma E$]

(iii) current and length and area of cross-section

But in the given options only b, that is, electric field can be varied by the charges accumulated on the surface of wire.

Q. 2. Which of the following characteristics of electrons determines the current in a conductor?

(a) Drift velocity alone

(b) Thermal velocity alone

(c) Both drift velocity and thermal velocity

(d) Neither drift nor thermal velocity.

[NCERT Exemp. Q. 3.6, Page 17]

Ans. Correct option : (a)

Explanation : As we know that,

$I = Anev_d$

So current,

$I \alpha v_d$

And, current (I) also depend on n, the number of free electrons which increases on increasing temperature which makes more collision between electrons increases resistance or decrease current.

Q. 3. A metal rod of length 10 cm and a rectangular cross-section of 1 cm $\times \frac{1}{2}$ cm is connected to a battery across opposite faces. The resistance will be

(a) maximum when the battery is connected across 1 cm $\times \frac{1}{2}$ cm faces.

(b) maximum when the battery is connected across 10 cm $\times$ 1 cm faces.

(c) maximum when the battery is connected across 10 cm $\times \frac{1}{2}$ cm faces.

(d) same irrespective of the three faces.

[NCERT Exemp. Q. 3.5, Page 17]

Ans. Correct option : (a)

Explanation : As we know that,

$$R = \rho\left(\frac{l}{A}\right)$$

The maximum resistance will be achieved when the value of $\frac{l}{A}$, is maximum, so that 'A' must be minimum and it is minimum when area of cross section is 1 cm $\times \frac{1}{2}$ cm.

Tick Two or More Options

Q. 4. Temperature dependence of resistivity $\rho(T)$ of semiconductors, insulators and metals is significantly based on the following factors

(a) number of charge carriers can change with temperature T.

(b) time interval between two successive collisions can depend on T.

(c) length of material can be a function of T.

(d) mass of carriers is a function of T.

[NCERT Exemp. Q. 3.9, Page 18]

Ans. Correct option : (a) and (b)

Explanation : As we know that resistivity (ρ) depends on-

(i) mass of charge-carrier (m),

(ii) relaxation time (τ).

And the length and mass cannot be function of T as the mass of a body is constant anywhere. So, it discards option (d) and length of body changes negligibly with temperature discards option (c).

As τ decreases on increasing T due to rise in speed of change-carriers and n increases on increasing temperature. So, it will affect the ρ or ρ is function of T verifies options (a) and (b).

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Define mobility of a charge carrier. What is its relation with relaxation time?

[CBSE Board, Delhi Region, 2016]

Ans. Mobility of a charge carriers is defined as the drift velocity of the charge carrier per unit electric field is-

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$ **[1]**

Q. 2. When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is 2.5 $\times$ 10^{-4} m/s. If the electron density in the wire is 8 $\times$ 10^{28} m^{-3}, calculate the resistivity of the material of wire.

[CBSE Board, Delhi Region, 2016]

Ans. Given that,

$V = 5V$

$l = 0.1$ m
$v_d = 2.5 \times 10^{-4}$ m/s
$n = 8 \times 10^{28}$
As we know that,
$E = \rho J$
$E = \frac{V}{l}$ and $J = nev_d$ **[1]**

Put the values,
$\frac{V}{l} = \rho nev_d$
$\rho = \frac{V}{\ln ev_d}$
$= \frac{5}{0.1 \times 8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4}}$
$= 1.56 \times 10^{-5}$ ohm – m
[1]

Q. 3. For wiring in the home, one uses Cu wires or Al wires. What considerations are involved in this?
[NCERT Exemp. Q. 3.16, Page 19]

Ans. As the metals have low resistivity so, metals have low resistance. The cost of metals used in electric circuits decreases from Ag, Cu, Al, Fe (steel). But Ag is costly, so Cu or Al wires are used in wiring. **[1]**

Q. 4. The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.4 Ω, what is the maximum current that can be drawn from the battery? [NCERT Ex. Q. 3.1, Page 16]

Ans. Given that,
Emf of the battery, $E = 12$ V
Internal resistance of the battery, $r = 0.4\ \Omega$
Let maximum current drawn from the battery $= I$
According to Ohm's law,
$E = IR$
$I = \frac{E}{R}$ **[1]**

Put the values,
$I = \frac{12}{0.4}$
$I = 30$ ampere
The maximum current drawn from the given battery is 30 A. **[1]**

Q. 5. A negligibly small current is passed through a wire of length 15 m and uniform cross-section 6.0×10^{-7} m², and its resistance is measured to be 5.0 Ω. What is the resistivity of the material at the temperature of the experiment?
[NCERT Ex. Q. 3.6, Page 17]

Ans. Given that,
Length of the wire, $L = 15$ m
Area of cross-section of the wire, $A = 6.0 \times 10^{-7}$ m²
Resistance of the material of the wire, $R = 5.0\ \Omega$
Let the resistivity of the material of the wire is ρ
So, the resistance is related with the resistivity as,
$R = \rho \frac{L}{A}$
So, $\rho = \frac{RA}{L}$ **[1]**

Put the values,
$= \frac{(5 \times 6 \times 10^{-7})}{15}$
$= 2 \times 10^{-7}\ \Omega$ m
So, the resistivity of the material is $2 \times 10^{-7}\ \Omega$ m. **[1]**

Q. 6. A silver wire has a resistance of 2.1 Ω at 27.5 °C, and a resistance of 2.7 Ω at 100 °C. Determine the temperature coefficient of resistivity of silver.
[NCERT Ex. Q. 3.7, Page 18]

Ans. Given that,
Temperature, $T_1 = 27.5$°C
Resistance of the silver wire at T_1, $R_1 = 2.1\ \Omega$
Temperature, $T_2 = 100$°C
Resistance of the silver wire at T_2, $R_2 = 2.7\ \Omega$
Let the temperature coefficient of silver is α
As we know that- the temperature coefficient is related with temperature and resistance as,
$\alpha = \frac{(R_2 - R_1)}{R_1(T_2 - T_1)}$ **[1]**
$= \frac{(2.7 - 2.1)}{2.1(100 - 27.5)} = 0.0039\ °C^{-1}$

So, the temperature coefficient of silver is $0.0039° C^{-1}$. **[1]**

Q. 7. Why are alloys used for making standard resistance coils? [NCERT Exemp. Q. 3.17, Page 19]

Ans. Dependence of resistance on change of temperature, humidity, pressure, etc. must be negligible. Alloys has small value of temperature coefficient and are not affected by moisture, etc. **[1]**
Alloys has higher resistivity in turn the higher resistance so need smaller length to make coils which decrease the effect of inductance. Due to these reasons alloys are used to make standard resistance coils. **[1]**

Q. 8. Nichrome and copper wires of same length and same radius are connected in series. Current *I* is passed through them. Which wire gets heated up more? Justify your answer.
[CBSE Board, Delhi Region, 2017]

Ans. Heat dissipate in a wire is given by :
$H = I^2Rt$ **[½]**
$H = I^2 \frac{\rho l}{A} \times t$ $\left[R = \frac{\rho l}{A}\right]$
For same current I, length l, and area A
The heat dissipation, H depends upon only on ρ
$H \propto \rho$
$\rho_{\text{nichrome}} > \rho_{\text{copper}}$
So,
$H_{\text{nichrome}} > H_{\text{copper}}$ **[½]**

Q. 9. Is the motion of a charge across junction momentum conserving? Why or why not?
[NCERT Exemp. Q. 3.12, Page 19]

Ans. In the circuit, when an electron approaches a junction, it faces a uniform electric field E (which keep the drift velocity fixed.) as drift velocity (v_d) is directly proportional to electric field (E). That is why there is accumulation of charges on the surface of wires at the junction. **[1]**
These produce additional electric fields. These fields alter the direction of momentum. Thus, the motion of a charge across junction is not momentum conserving. **[1]**

Q. 10. The relaxation time τ is nearly independent of applied electric field *E* whereas it changes significantly with temperature T. First fact is (in part) responsible for Ohm's law whereas the second fact leads to variation of ρ with temperature. Elaborate why?
[NCERT Exemp. Q. 3.13, Page 19]

Ans. As we know that as the drift velocity increases, the relaxation time (τ) (average time between successive collision) decreases which increases the ρ by formula :

$$\rho = \frac{m}{ne^2 r}$$ **[1]**

The drift velocity (v_d) changes of the order of 1 mm on increasing electric field, whereas the drift velocity increases of the order of 10^2 m/s when the number of free electrons (n) increases on increasing temperature (T).

So, due to increase in v_d the relaxation time (τ) considerably decreases in metal or conductor. **[1]**

Q. 11. Power *P* is to be delivered to a device via transmission cables having resistance RC. If *V* is the voltage across *R* and *I* the current through it, find the power wasted and how can it be reduced.
[NCERT Exemp. Q. 3.18, Page 19]

Ans. As we know that power,

$P = VI$

So, to transmit a constant power P through transmission cable there are two ways :

(i) If a constant power (P) is transmitted at low voltage (V) and high current (I). In this method high current will produce higher heat by $H = I^2R$ and power loss through cable is higher. **[½]**

(ii) If a constant or same power be transmitted at high voltage (V) and low current. It gives lower loss of power as heat. But need thicker insulation during transmission. **[½]**

So, to transmit high power at long distance, we use low current and high (132 kV) voltage to minimise heat losses through towers and thick (long) insulator.

To transmit power supply at short distance, we can transmit power at low 440V, 220V, 11kV with higher current. **[1]**

Short Answer Type Questions (3 marks each)

Q. 1. Two electric bulbs *P* and *Q* have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation of these bulbs.
[CBSE Board, Delhi Region, 2018]

Ans. For series combination, power dissipated by a bulb is directly proportional to its resistance. **[1½]**

$$P \propto R$$

$$\frac{P_1}{P_2} = \frac{R_1}{R_2} = \frac{1}{2}$$ **[1½]**

Q. 2. (a) Define the term 'conductivity' of a metallic wire. Write its SI unit.

(b) Using the concept of free electrons in a conductor, derive the expression for the conductivity of a wire in terms of number density and relaxation time. Hence, obtain the relation between current density and the applied electric field *E*.
[CBSE Board, Delhi Region, 2018]

Ans. **(a)** Conductivity **:**

The reciprocal of the resistivity of a material is called its conductivity and is denoted by σ.

So,

$$\text{Conductivity}(\sigma) = \frac{1}{\text{Resistivity}(\rho)}$$

The *SI* unit of conductivity is, ohm^{-1} m^{-1} or mho metre^{-1} **[1]**

(b) When a potential difference V is applied across a conductor of length l, the drift velocity in term of v is given by :

$$v_d = \frac{eE}{m}\tau = \frac{eV}{ml}\tau$$

If area of cross-section A and n number of electron per unit volume the current through

$$I = neAv_d = neA\left(\frac{eE}{m}\tau\right) \quad [v_d = \frac{eE}{ml}\tau]$$

$$I = \frac{ne^2E\tau}{m}$$

$$I = \frac{Ane^2V}{ml}\tau \qquad \left[\because \frac{V}{l} = E\right]$$

$$\frac{V}{R} = \frac{Ane^2V}{ml}\tau \qquad [\because V = IR]$$

$$R = \frac{ml}{Ane^2\tau}$$

$$\frac{\rho l}{A} = \frac{ml}{Ane^2\tau} \qquad \left[\because R = \frac{\rho l}{A}\right]$$

$$\rho = \frac{m}{ne^2\tau}$$

So,

$$\sigma = \frac{1}{\frac{m}{ne^2\tau}}$$

$$\sigma = \frac{ne^2\tau}{m}$$ [1]

According to Ohm's law :

$$V = IR$$

$$V = I\frac{\rho l}{A} \qquad \left[\because j = \frac{I}{A}\right]$$

$$\frac{V}{l} = j \times \rho$$

$$E = j \times \rho$$

$$\sigma \times E = j \qquad \left[\because \frac{1}{\rho} = \sigma\right]$$

[1]

Q. 3. Choose the correct alternative :

(a) Alloys of metals usually have (greater/less) resistivity than that of their constituent metals.

(b) Alloys usually have much (lower/higher) temperature coefficients of resistance than pure metals.

(c) The resistivity of the alloy manganin is nearly independent of/increases rapidly with increase of temperature.

(d) The resistivity of a typical insulator (*e.g.*, amber) is greater than that of a metal by a factor of the order of ($10^{22}/10^{3}$). [NCERT Add. Ex. Q. 3.19, Page 19]

Ans. **(a)** Alloys of metals usually have greater resistivity than that of their constituent metals. **[½]**

(b) Alloys usually have lower temperature coefficients of resistance than pure metals. **[½]**

(c) The resistivity of the alloy, manganin, is nearly independent of increase of temperature. **[1]**

(d) The resistivity of a typical insulator is greater than that of a metal by a factor of the order of 10^{22}. **[1]**

Q. 4. The number density of free electrons in a copper conductor estimated is 8.5×10^{28} m^{-3}. How long does an electron take to drift from one end of a wire 3.0 m long to its other end? The area of cross-section of the wire is 2.0×10^{-6} m^2 and it is carrying a current of 3.0 A. [NCERT Ex. Q. 3.13, Page 19]

Ans. Given that,

Number density of free electrons in a copper conductor, $n = 8.5\times10^{28}$ m^{-3}

Length of the copper wire, $l = 3.0$ m

Area of cross-section of the wire, $A = 2.0\times10^{-6}$ m^2

Current carried by the wire, $I = 3.0$ A,

And we know that,

$I = nAev_d$ **[1]**

Where,

e = Electric charge $= 1.6\times10^{-19}\,C$

v_d = Drift velocity

$$v_d = \frac{\text{Length of the wire}(l)}{\text{Time taken to cover } l(t)}$$ **[1]**

$$I = nAe\frac{l}{t}$$

$$t = \frac{neAl}{I}$$

$$= \frac{3\times8.5\times10^{28}\times2\times10^{-6}\times1.6\times10^{-19}}{3}$$

$$= 2.7\times10^{4}\ s$$

So, the time taken by an electron to drift from one end of the wire to the other is 2.7×10^4 s. **[1]**

Q. 5. Two conductors are made of the same material and have the same length. Conductor A is a solid wire of diameter 1 mm. Conductor B is a hollow tube of outer diameter 2 mm and inner diameter 1 mm. Find the ratio of resistance R_A to R_B.

[NCERT Exemp. Q. 3.26, Page 21]

Ans.

Conductor A (solid wire R_A)	**Conductor B (hollow tube R_B)**
$l_1 = 1$	$l_2 = 1$ (Given)
$A_1 = \pi r_1^2$	$A_2 = \pi r_2^2 - \pi r_1^2$
$r_1 = \frac{1}{2}mm = 0.5\times10^{-3}m$	$r_2 = \frac{2}{2}mm = 1\times10^{-3}m$
$\rho_1 = \rho$	$\rho_2 = \rho$

$$\Rightarrow \frac{R_A}{R_B} = \frac{\frac{\rho_1 l_1}{A_1}}{\frac{\rho_2 l_2}{A_2}}$$

$$\Rightarrow \frac{R_A}{R_B} = \frac{\rho_1 l_1}{A_1}\times\frac{A_2}{\rho_2 l_2}$$

$$\Rightarrow \frac{R_A}{R_B} = \frac{A_2}{A_1}$$

$$\therefore \frac{R_A}{R_B} = \frac{A_2}{A_1} = \frac{\pi r_2^2 - \pi r_1^2}{\pi r_1^2} = \frac{\pi(r_2^2 - r_1^2)}{\pi r_1^2} = \left[\frac{r_2}{r_1}\right]^2 - 1$$

$$\Rightarrow \left[\frac{1\times10^{-3}}{0.5\times10^{-3}}\right] - 1 = (2)^2 - 1 = 4 - 1 = \frac{3}{1}$$

$$\therefore \frac{R_A}{R_B} = \frac{3}{1}$$ **[3]**

Q. 6. What conclusion can you draw from the following observations on a resistor made of alloy manganin?

Current A	**Voltage V**	**Current A**	**Voltage V**
0.2	3.94	3.0	59.2
0.4	7.87	4.0	78.8
0.6	11.8	5.0	98.6
0.8	15.7	6.0	118.5
1.0	19.7	7.0	138.2
2.0	39.4	8.0	158.0

[NCERT Add. Ex. Q. 3.17, Page 19]

Ans. It is clear from the given table that the ratio of voltage with current is a constant, which is equal to 19.7 so the, manganin is an ohmic conductor, that is, the alloy follows the **Ohm's law.** **[1½]**

According to this law, the ratio of voltage with current gives the resistance of the conductor.

So, the resistance of manganin will be 19.7 Ω. **[1½]**

Q. 7. Answer the following questions :

(a) A steady current flow in a metallic conductor of non-uniform cross-section. Which of these quantities is constant along the conductor : current, current density, electric field, drift speed?

(b) Is Ohm's law universally applicable for all conducting elements? If not, give examples of elements which do not obey Ohm's law.

(c) A low voltage supply from which one needs high currents must have very low internal resistance. Why?

(d) A high tension (*HT*) supply of, say, 6 kV must have a very large internal resistance. Why?

[NCERT Add. Ex. Q. 3.18, Page 19]

Ans. **(a)** When a steady current flow in a metallic conductor of non-uniform cross-section, the current flowing through the conductor is constant. Current density, electric field, and drift speed are inversely proportional to the area of cross constant. **[½]**

(b) No, Ohm's law is not universally applicable for all conducting elements. Vacuum diode semi-conductor is a non-ohmic conductor. Ohm's law is not valid for it. **[½]**

(c) According to Ohm's law,

$$V = IR$$

Where,
Voltage, V is directly proportional to current, I
Resistance, R is the internal resistance of the source.
So, if V is low, then R must be very low, so that high current can be drawn from the source. **[1]**

(d) In order to prohibit the current from exceeding the safety limit, a high-tension supply must have a very large internal resistance. If the internal resistance is not large, then the current drawn can exceed the safety limits in case of a short circuit. **[1]**

Q. 8. At room temperature (27.0 °C) the resistance of a heating element is 100 Ω. What is the temperature of the element if the resistance is found to be 117 Ω, given that the temperature coefficient of the material of the resistor is 1.70×10^{-4} °C^{-1}.

[NCERT Ex. Q. 3.5, Page 17]

Ans. Given that,
Room temperature, $T = 27$°C
Resistance of the heating element at T, $R = 100\ \Omega$
Let T_1 is the increased temperature of the filament.
Resistance of the heating element at T_1, $R_1 = 117\ \Omega$
Temperature co-efficient of the material of the filament,
$\alpha = 1.70 \times 10^{-4}$ °C^{-1}
α is given by the relation,

$$\alpha = \frac{(R_1 - R)}{R(T_1 - T)}$$

$$(T_1 - T) = \frac{(R_1 - R)}{R\alpha}$$ **[2]**

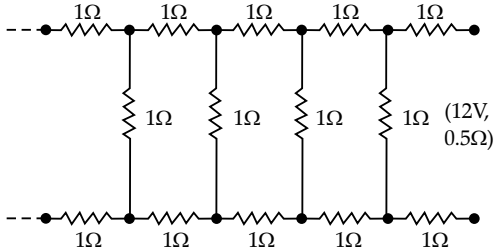
Put the values,

$$(T_1 - 27) = \frac{(117 - 100)}{100(1.7\times10^{-4})}$$

$$T_1 - 27 = 1000$$
$$T_1 = 1027\ °C$$

so, at 1027 °C, the resistance of the element is 117 Ω. **[1]**

Q. 9. Determine the current drawn from a 12V supply with internal resistance 0.5 Ω by the infinite network shown in Figure. Each resistor has 1 Ω resistance.



[NCERT Add. Ex. Q. 3.21, Page 20]

Ans. Given that,
The resistance of each resistor connected in the given circuit, $R = 1\ \Omega$
Internal resistance, $r = 0.5\ \Omega$
Voltage, $V = 12$ volt
Let the equivalent resistance of the given circuit = R'
The given network is infinite.
So, the equivalent resistance is given by the relation,

$$\therefore \quad R' = 2 + \frac{R'}{(R'+1)}$$

$$(R')^2 - 2R' - 2 = 0$$

$$R' = \frac{2 \pm \sqrt{4+8}}{2}$$

$$= \frac{2 \pm \sqrt{12}}{2} = 1 \pm \sqrt{3}$$ **[1½]**

Negative value of R′ cannot be accepted. So, equivalent resistance will be

$$R' = \left(1 + \sqrt{3}\right) = 1 + 1.73 = 2.73\ \Omega$$

And it is given that the internal resistance of the circuit, $r = 0.5\ \Omega$
So, total resistance of the given circuit = 2.73 + 0.5
= 3.23 Ω
Supply voltage, $V = 12$ V
According to Ohm's Law,
Current drawn from the source is given by the ratio,
=3.72 A. **[1½]**

Q. 10. First a set of 'n' equal resistors of R each are connected in series to a battery of emf E and internal resistance R. A current I is observed to flow. Then the n resistors are connected in parallel to the same battery. It is observed that the current is increased 10 times. What is 'n'?

[NCERT Exemp. Q. 3.22, Page 20]

Ans. When n resistance of each $R\ \Omega$ are connected in series and parallel then
For Series combination :
$R_S = R+R+R+\ldots.\ n$ times
$R_S = nR$
For Parallel combination :

$$\frac{1}{R_p} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} \ldots\ldots\ldots n \text{ times}$$

$$\frac{1}{R_p} = \frac{n}{R}$$

$$R_p = \frac{R}{n}$$ **[1½]**

When n resistors are connected in series connected with battery of emf E then current (I) flows. So,

$$\frac{E}{(r+nR)} = I \quad \ldots(i)$$

Now n resistance are connected in parallel combination then the current in circuit increased to 10 times of I

$$\therefore \quad \frac{E}{R + \frac{R}{n}} = 10I$$

$$\frac{E}{R + \frac{R}{n}} = \frac{10E}{R+nR}$$

$$\frac{1}{R(1+\frac{1}{n})} = \frac{10}{R(1+n)}$$

$$10\left(1+\frac{1}{n}\right) = 1+n$$

$$10 + \frac{10}{n} - 1 - n = 0$$

$$-n + \frac{10}{n} + 9 = 0$$

Rearrange the above equation :

$n^2 - 9n - 10 = 0$

solving this quadratic equation :

$(n+1)(n-10) = 0$

or $n = -1$, is not possible

so, $n = 10$

So that, there are 10 resistors in combination. **[1½]**

Q. 11. (a) Three resistors 1 Ω, 2 Ω, and 3 Ω are combined in series. What is the total resistance of the combination?

(b) If the combination is connected to a battery of emf 12 V and negligible internal resistance, obtain the potential drop across each resistor.

[NCERT Ex. Q. 3.3, Page 127]

Ans. **(a)** Given that

$r_1 = 1\ \Omega$

$r_2 = 2\ \Omega$

$r_3 = 3\ \Omega$

All resistors are connected in series combination.

So, the total resistance, R of the above series combination can be calculated by the algebraic sum of individual resistances, $R = r_1 + r_2 + r_3$

Now, total resistance is given by, $R = r_1 + r_2 + r_3$

$= 1\ \Omega + 2\ \Omega + 3\ \Omega = 6\ \Omega$

Total resistance of the circuit is 6 Ω. **[1½]**

(b) Given that,

Emf of the battery, $E = 12$ V

Total resistance of the circuit, $R = 6\ \Omega$

Let the current flowing through the circuit = I

So, according to the Ohm's law,

$$I = \frac{E}{R}$$

$$= \frac{12}{6} = 2\ A$$

Potential drop across 1 Ω resistor = V_1

From Ohm's law, the value of V_1,

$V_1 = 2 \times 1 = 2$ V … (i)

Potential drop across 2 Ω resistor = V_2

Again, from Ohm's law, the value of V_2,

$V_2 = 2 \times 2 = 4$ V …(ii)

Potential drop across 3 Ω resistor = V_3

Again, from Ohm's law, the value of V_3,

$V_3 = 2 \times 3 = 6$ V …(iii)

So, the potential drop across 1 Ω, 2 Ω, and 3 Ω resistors are 2 V, 4 V, and 6 V, respectively. **[1½]**

Q. 12. (a) Three resistors 2 Ω, 4 Ω and 5 Ω are combined in parallel. What is the total resistance of the combination?

(b) If the combination is connected to a battery of emf 20 V and negligible internal resistance, determine the current through each resistor, and the total current drawn from the battery.

[NCERT Ex. Q. 3.4, Page 127]

Ans. Given that,

$R_1 = 2\ \Omega$,

$R_2 = 4\ \Omega$, and

$R_3 = 5\ \Omega$

All resistances are combined in parallel combination.

(a) There are three resistors of resistances and all are connected in parallel. so, total resistance (R) of the combination is given by,

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{5} = \frac{(10+5+4)}{20}$$

$$= \frac{19}{20}$$

So,

$$R = \frac{20}{19}\ \Omega$$

So, the total resistance of the combination is $\frac{20}{19}$ ohm. **[1½]**

(b) Current (I_1) flowing through resistor R_2 is given by

$$I_1 = \frac{V}{R_1} = \frac{20}{2} = 10\ A$$

Current (I_2) flowing through resistor R_2 is given by,

$$I_2 = \frac{v}{R_2} = \frac{20}{4} = 5\ A$$

Current (I_3) flowing through resistor R_3 is given by,

$$I_3 = \frac{V}{R_3} = \frac{20}{5} = 4A$$

Total current, $I = I_1 + I_2 + I_3 = 10 + 5 + 4 = 19$ A

so, the current through each resister is 10 A, 5 A, and 4 A respectively and the total current is 19 A. **[1½]**

Q. 13. Let there be n resistors R_1 .....R_n with R_{max} = max (R_1..... R_n) and R_{min} = min (R_1 .... R_n). Show that when they are connected in parallel, the resultant resistance $R_P < R_{min}$ and when they are connected in series, the resultant resistance $R_S > R_{max}$. Interpret the result physically.

[NCERT Exemp. Q. 3.23, Page 20]

Ans. Let R_{min} and R_{max} are the minimum and maximum resistances among all resistance :

$R_1, R_2 \ldots R_n$.

When resistors are connected in parallel then equivalent resistance

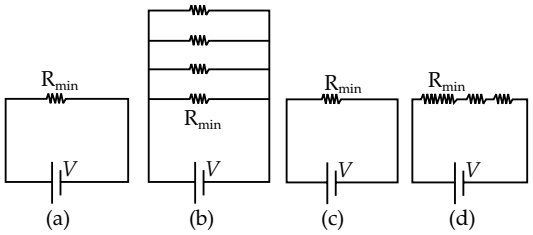
$$\left(\frac{1}{R_p}\right) = \left(\frac{1}{R_1}\right) + \ldots + \left(\frac{1}{R_n}\right)$$

Now, $$\frac{R_{min}}{R_P} = \left(\frac{R_{min}}{R_1}\right) + \left(\frac{R_{min}}{R_2}\right) + \ldots + \left(\frac{R_{min}}{R_n}\right) > 1$$

and $R_S = R_1 + \ldots\ldots + R_n \geq R_{max}$.

In Figure (b),

R_{min} provides an equivalent route as in Figure (a) for current.
But, in addition there are (n – 1) routes by the remaining (n – 1) resistors.
Current in Figure (b) > current in Figure (a).
Effective Resistance in Figure (b) < R_{min}.
Second circuit evidently affords a greater resistance.
You can use Figure (c) and (d) and prove $R_s > R_{max}$. **[1½]**



[1½]

Long Answer Type Questions **(5 marks each)**

Q. 1. The earth's surface has a negative surface charge density of 10^{-9} C m^{-2}. The potential difference of 400 kV between the top of the atmosphere and the surface results (due to the low conductivity of the lower atmosphere) in a current of only 1800 A over the entire globe. If there were no mechanism of sustaining atmospheric electric field, how much time (roughly) would be required to neutralise the earth's surface?
(This never happens in practice because there is a mechanism to replenish electric charges, namely the continual thunderstorms and lightning in different parts of the globe). (Radius of earth = 6.37×10^6 m.)
[NCERT Add. Ex. Q. 3.14, Page 128]

Ans. Given that,
Surface charge density of the earth, $\sigma = 10^{-9}$ Cm^{-2}
Current over the entire globe, I = 1,800 A
Radius of the earth, $r = 6.37 \times 10^6$ m
Now, calculate the surface area of the earth, $A = 4\pi r^2$
$= 4\pi \times (6.37 \times 10^6)^2$
$= 5.09 \times 10^{14}$ m^2
Charge on the earth surface, $q = \sigma \times A$
$= 10^{-9} \times 5.09 \times 10^{14}$
$= 5.09 \times 10^5$ C **[2½]**
Time taken to neutralise the earth's surface = t
Now the current, I will be :

$$I = \frac{q}{t}$$

$$t = \frac{q}{I} = \frac{5.09 \times 10^5}{1{,}800} = 282.77\ s$$

so, the time taken to neutralise the earth's surface is 282.77 s. **[2½]**

Q. 2. A heating element using nichrome connected to a 230 V supply draws an initial current of 3.2 A which settle after a few seconds to a steady temperature of the heating element if the room temperature is 27.0°C? Temperature coefficient of resistance of nichrome averaged over the temperature range involved is 1.70×10^{-4} °C^{-1}.
[NCERT Ex. Q. 3.8, Page 127]

Ans. Given that,
Supply voltage, V = 230 V
Initial current drawn, I_1 = 3.2 A
Initial resistance = R_1,
According to ohm's law,

$$R_1 = \frac{V}{I} \quad \textbf{[1]}$$

$$= \frac{230}{3.2}$$

$$= 71.87\ \Omega$$

Steady state value of the current, I_2 = 2.8 A **[1]**
Resistance at the steady state, R_2,

$$= \frac{230}{2.8} = 82.14\ \Omega$$

Temperature co-efficient of nichrome, $\alpha = 1.70 \times 10^{-4}$ °C^{-1}
Initial temperature of nichrome, T_1 = 27.0°C **[1]**
Let assume the steady state temperature reached by nichrome = T_2
T_2 can be obtained by the relation for α,

$$\alpha = \frac{(R_2 - R_1)}{R_1(T_2 - T_1)} \quad \textbf{[1]}$$

$$(T_2 - 27°C) = \frac{(82.14 - 71.87)}{71.87 \times 1.7 \times 10^{-4}}$$

$$T_2 - 27 = 840.5$$

$$T_2 = 840.5 + 27 = 867.5\ °C$$

So, the steady temperature of the heating element is 867.5°C. **[1]**

Q. 3. Two wires of equal length, one of aluminium and the other of copper have the same resistance. Which of the two wires is lighter? Hence explain why aluminium wires are preferred for overhead power cables.
($\rho_{Al} = 2.63 \times 10^{-8}\ \Omega$ m, $\rho_{Cu} = 1.72 \times 10^{-8}\ \Omega$ m, Relative density of Al = 2.7, of Cu = 8.9.)
[NCERT Add. Ex. Q. 3.16, Page 129]

Ans. Given that,
Resistivity of aluminium, $\rho_{Al} = 2.63 \times 10^{-8}\ \Omega$ m
Relative density of aluminium, d_1 = 2.7
Resistivity of copper, $\rho_{Cu} = 1.72 \times 10^{-8}\ \Omega$ m
Relative density of copper, d_2 = 8.9
Let assume,
l_1 be the length of aluminium wire and m_1 be its mass
R_1, Resistance of the aluminium wire

A_1, Area of cross-section of the aluminium wire
l_2 be the length of copper wire and m_2 be its mass
R_2, Resistance of the copper wire
A_2, Area of cross-section of the copper wire
The two relations can be written as :

$$R_1 = \rho_1 \frac{l_1}{A_1} \quad ...(i)$$

$$R_2 = \rho_2 \frac{l_2}{A_2} \quad ...(ii)$$ **[1½]**

And it is given that, $R_1 = R_2$
So,

And,

$$l_1 = l_2$$

$$\frac{\rho_1}{A_1} = \frac{\rho_2}{A_2}$$

$$\frac{A_2}{A_1} = \frac{\rho_2}{\rho_1}$$

or

$$\frac{A_1}{A_2} = \frac{\rho_1}{\rho_2} = \frac{2.63 \times 10^{-8}}{1.72 \times 10^{-8}} = \frac{2.63}{1.72}$$ **[1½]**

Mass of the aluminium wire,
m_1 = Volume × Density
$= A_1 l_1 \times d_1 = A_1 l_1 d_1$ …(iii)
Mass of the copper wire,
m_2 = Volume × Density
$= A_2 l_2 \times d_2 = A_2 l_2 d_2$ …(iv)
Dividing equation (iii) by equation (iv), we get

$$\frac{m_1}{m_2} = \frac{A_1 l_1 d_1}{A_2 l_2 d_2}$$

$$For\ l_1 = l_2$$

$$\frac{m_1}{m_2} = \frac{A_1 d_1}{A_2 d_2}$$

$$For\ \frac{A_1}{A_2} = \frac{2.63}{1.72}$$

$$\frac{m_1}{m_2} = \frac{2.63}{1.72} \times \frac{2.7}{8.9} = 0.46$$

It is clear from this ratio that m_1 is less than m_2. That means, aluminium is lighter than copper.

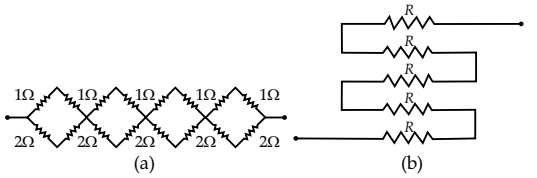
Since, aluminium is lighter, it is preferred for overhead power cables over copper. **[2]**

Q. 4. (a) Given n resistors each of resistance *R*, how will you combine them to get the (i) maximum (ii) minimum effective resistance?

What is the ratio of the maximum to minimum resistance?

(b) Given the resistances of 1 Ω, 2 Ω, 3 Ω, how will be combine them to get an equivalent resistance of (i) $\left(\frac{11}{3}\right)$ Ω (ii) $\left(\frac{11}{5}\right)$ Ω, (iii) 6 Ω, (iv) $\left(\frac{6}{11}\right)$ Ω?

(c) Determine the equivalent resistance of networks shown in Figure.



[NCERT Add. Ex. Q. 3.20, Page 129]

Ans. **(a)** Given that,
Total number of resistors = n
Resistance of each resistor = R

1. If n resistors are connected in series, effective resistance R_1 is the maximum, given by the product nR.
so, the maximum resistance of the combination, $R_1 = nR$ **[1]**

2. If n resistors are connected in parallel, the effective resistance (R_2) is the minimum, given by the ratio.
So, minimum resistance of the combination, $R_2 = \frac{R}{n}$ **[1]**

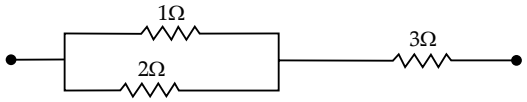
3. The ratio of the maximum to the minimum resistance is,

$$\frac{R_2}{R_1} = \frac{nR}{(R/n)} = n^2$$ **[1]**

(b) The resistance of the given resistors is,
$R_1 = 1\ \Omega, R_2 = 2\ \Omega, R_3 = 3\ \Omega$

Equivalent resistance, $R' = \frac{11}{3}\ \Omega$

Consider the following combination of the resistors

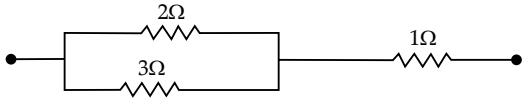


Equivalent resistance of the circuit is given by,

$$R' = \frac{(2 \times 1)}{(2+1)} + 3 = \frac{2}{3} + 3 = \frac{11}{3}\ \Omega$$

Equivalent resistance, $R' = \frac{11}{5}\ \Omega$

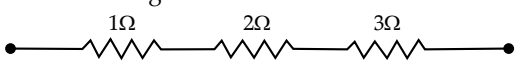
Consider the following combination of the resistors



Equivalent resistance of the circuit is given by,

$$R' = \frac{(2 \times 3)}{(2+3)} + 1 = \frac{6}{5} + 1 = \frac{11}{5}\ \Omega$$

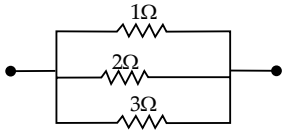
(iv) Equivalent resistance, $R' = 6\ \Omega$
Consider the series combination of the resistors, as shown in the given circuit.



Equivalent resistance of the circuit is given by the sum,
$R' = 1 + 2 + 3 = 6\ \Omega$

(v) Equivalent resistance, $R' = \frac{6}{11}\ \Omega$

Consider the series combination of the resistors, as shown in the given circuit.



Equivalent resistance of the circuit is given by,

$$R' = \frac{(1\times 2\times 3)}{(1\times 2)+(2\times 3)+(3\times 1)} = \frac{6}{11}\ \Omega$$ **[1]**

(c) The equivalent resistance of networks as shown in Figure (a)

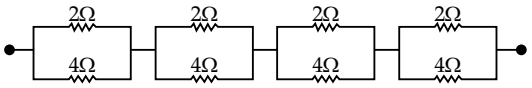
It can be observed from the given circuit that in the first small loop, two resistors of resistance 1 Ω each are connected in series.

Hence, their equivalent resistance = (1+1) = 2 Ω

It can also be observed that two resistors of resistance 2 Ω each are connected in series.

Hence, their equivalent resistance = (2 + 2) = 4 Ω

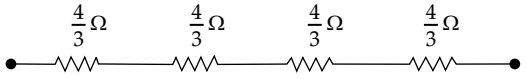
Therefore, the circuit can be redrawn as



It can be observed that 2 Ω and 4 Ω resistors are connected in parallel in all the four loops. So, equivalent resistance (R') of each loop is given by,

$$R' = \frac{(2\times 4)}{(2+4)} = \frac{8}{6} = \frac{4}{3}\ \Omega$$

The circuit reduces to-



All four resistors are connected in series.

So, the equivalent resistance of the circuit will be

$$= \frac{4}{3}R \times 4 = \frac{16}{3}R$$

Figure (b)

It can be observed from the given circuit that five resistors of resistance R each are connected in series.

So equivalent resistance = $R + R + R + R + R = 5R$ **[1]**

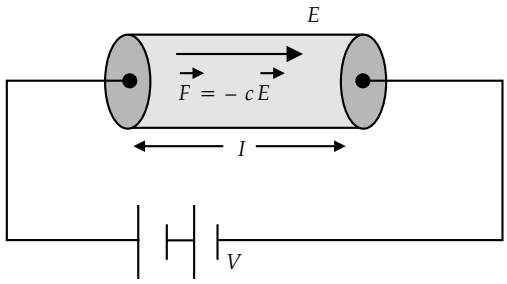
Q. 5. (i) Derive an expression for drift velocity of free electrons.

(ii) How does drift velocity of electrons in a metallic conductor vary with increase in temperature? Explain. [CBSE Board, All India Region, 2016]

Ans. (i) As we know that the free electrons are in continuous random motion. They collide each other and they undergo change in random direction at each collision and the thermal velocities are randomly distributed in all directions.

$\therefore$ Average thermal velocity,

$U = (u_1 + u_2 + \ldots\ldots + u_n)/n$ is zero …(i)



The electric field E exerts an electrostatic force '$-Ee$'. So, the acceleration of each electron is,

$$|\bar{a}| = \frac{-e\bar{E}}{m}$$ ...(2)

$m \rightarrow$ Mass of an electron

$e \rightarrow$ Charge on an electron

Drift velocity,

$$\bar{v}_d = \frac{\bar{v}_1 + \bar{v}_2 + \ldots\ldots\ldots + \bar{v}_5}{n}$$

$$\bar{v}_d = \frac{(\bar{u}_1 + \bar{a}\tau_1) + (\bar{u}_2 + \bar{a}\tau_2) + \ldots\ldots + (\bar{u}_n + \bar{a}\tau_n)}{n}$$

Where,

$\bar{u}_1, \bar{u}_2 \rightarrow$ Thermal velocities of the electrons

$\bar{a}\tau_1, \bar{a}\tau_2 \rightarrow$ Velocity acquired by electrons

$\tau_1, \tau_2 \rightarrow$ Time elapsed after the collision

Now solving the above equation :

$$\bar{v}_d = \frac{(\bar{u}_2 + \bar{u}_2 + \ldots\ldots + \bar{u}_n)}{n} + \frac{\bar{a}(\tau_1 + \tau_2 + \ldots\ldots + \tau_n)}{n}$$

Since $\frac{(\bar{u}_2 + \bar{u}_2 + \ldots\ldots + \bar{u}_n)}{n} = 0$

$\therefore \bar{v}_a = a\tau$

Where, $\tau = \frac{\tau_1 + \tau_2 + \tau_3 + \ldots\ldots + \tau_n}{n}$ is the average time elapsed

Substituting from equation (2)

$$\bar{v}_a = \frac{-e\bar{E}}{m}\tau$$ ...(2)

An expression for drift velocity of free electrons is

$$\bar{v}_a = \frac{-e\bar{E}}{m}\tau$$ **[2½]**

(ii) Drift velocity of electrons in a metallic conductor decreases with increase in temperature.

As we know that,

If, we increase the temperature of the metallic conductor the collision between the electrons and ions increases, which results in the decrease in the relaxation time.

so, from the equation (4)

$v_d \propto \tau$

Hence, the drift velocity decreases. **[2½]**

Q. 6. A room has *AC* run for 5 hours a day at a voltage of 220 V. The wiring of the room consists of Cu of 1 mm radius and a length of 10 m. Power

consumption per day is 10 commercial units. What fraction of it goes in the joule heating in wires? What would happen if the wiring is made of aluminium of the same dimensions?

[ρ_{cu} = 1.7 × 10^{-8} Ωm, ρ_{Al} = 2.7 × 10^{-8} Ωm]

[NCERT Exemp. Q. 3.29, Page 21]

Ans. Total energy consumed in 5 hrs a day by *AC* and wiring =10 kWh

Energy consumed in 1 hr by *AC* and wiring =2 kWh

So, total power of *AC* and wire = 2000 W

$P = VI$

$\Rightarrow I = \frac{P}{V} = \frac{2000}{220} \cong 9.0\,A$

Let P_0 is power of wiring then,

$P_0 = I^2 R_w$ [R_w = resistance of wiring]

$\Rightarrow \frac{9\times9\times1.7\times10^{-8}\times10}{3.14\times1\times10^{-3}\times1\times10^{-3}} = \frac{81\times17\times10^{-2}}{3.14}$

$\Rightarrow \frac{1377}{3.14}\times10^{-2} = 4.38 = 4.4$ watt

So, loss of energy in wiring $\cong$ 4.4 J/sec

The fractional loss due to heating of wires-

$\Rightarrow \frac{4.4}{2000}\times100 = 0.22\%$ **[2]**

$\Rightarrow \frac{P_{Al}(\text{wiring})}{P_{cu}(\text{wiring})} = \frac{I^2R_{Al}}{I^2R_{cu}}$

$\Rightarrow = \frac{\rho_{Al}\frac{l_{Al}}{A_{Al}}}{\rho_{cu}\frac{l_{cu}}{A_{cu}}}$

$as\ I_{Al} = I_{cu}\ and\ A_{Al} = A_{cu}$ **[1]**

$\frac{P_{Al}}{P_{cu}} = \frac{\rho_{Al}}{\rho_{cu}}$

or

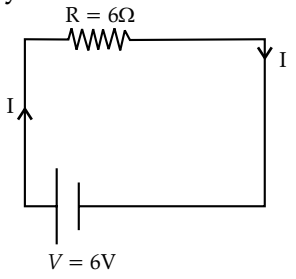
$P_{Al} = \frac{2.7\times10^{-8}}{1.7\times10^{-8}}\times4.4\,\text{watt}$

So, power loss in Al wiring will be 7 watt.

The fractional loss due to Al wiring

$= \frac{7}{2000}\times100\% = 0.35\%$ **[2]**

Q. 7. (a) Consider circuit in Figure. How much energy is absorbed by electrons from the initial state of no current (ignore thermal motion) to the state of drift velocity?



(b) Electrons give up energy at the rate of RI^2 per second to the thermal energy. What time scale would one associate with energy in problem (a)?

n = no of electron/volume = $10^{29}/m^3$,

length of circuit = 10 cm,

cross-section = A = 1 mm^2

[NCERT Exemp. Q. 3.31, Page 21]

Ans. Given that,

A = 1 mm^2 = 10^{-6} m^2

R = 6 ohm

$n = \frac{10^{29}}{m^3}$

V = 6 V

$e = 1.6\times10^{-19}$ C

According to ohm's law-

$I = \frac{V}{R}$

$= \frac{6}{6} = 1$ amp.

$m_e = 9.1\times10^{-31}$ kg

$L = 10\text{ cm} = 10^{-1}$ m

(a) We know that,

$I = Anev_d$

So $v_d = \frac{I}{Ane}$

$= \frac{1}{10^{-6}} \times 10^{29} \times 1.6 \times 10^{-19}$ m/s

$v_d = \frac{10^{-4}}{1.6}$ m/s

K.E = ½ mv_d^2 per electron

Number of electrons (free) in wire = n (volume of wire) = $n \times Al$

$\text{kinetic energy} = \frac{1}{2}mv_d^2 Anl$

$K.E = \frac{1}{2}\times9.1\times10^{-31}\times\frac{10^{-4}\times10^{-4}}{1.6\times1.6}\times10^{-6}\times10^{29}\times10^{-1}$

$= \frac{9.1}{2\times2.56}\times10^{-31-8-7+29} = 1.78\times10^{-46+29}$

$= 1.78\times10^{-17}\,J$

So, to start flow of current I, the electrons will take energy from cell

= KE of all electrons = 1.78× 10^{-17}J **[2½]**

(b) Loss of energy during current flowing =I^2R.

P=1×1×6=6 Joule per second

As we know that the energy = P.t

Or,

$t = \frac{E}{P}$

$= 1.78 \times \frac{10^{-17}}{6}$

$\cong 0.29\times10^{-17}$

$\cong 0.3\times10^{-17}$

$= 3\times10^{-18}$ sec **[2½]**

TOPIC-2
Electric Cell and Bridges

Quick Review

- In the year 1845, **Gustav Kirchhoff** (German physicist) introduces a set of laws which deal with current and voltage in the electrical circuits. The Kirchhoff's Laws are generally named **as KCL (Kirchhoff's Current Law)** and **KVL (Kirchhoff's Voltage Law).**
- The best-known bridge circuit is the Wheatstone bridge; the term was invented by **"Samuel Hunter Christie"** and popularized by **"Charles Wheatstone".** A bridge circuit is mainly used to measure **resistance**.
- The Wheatstone bridge circuit has also been widespread to calculate impedance in *AC* circuits, and also to calculate inductance, resistance, capacitance, and dissipation factor individually.
- *AC* Bridges consist of a source, balance detector and four arms.
- In *AC* bridges, all the four arms consist of impedance. The *AC* bridges are formed by replacing the *DC* battery with an *AC* source and galvanometer by detector of Wheatstone bridge.

TIPS...

- *Circuit : A DC circuit is a closed loop conducting lane in which an electrical current flow.*
- *Path : A single lane is used to connect the sources or elements.*
- *Node : A node is a connection in a circuit where multiple elements are connected together, and it is denoted with a dot.*
- *Branch : A branch is a single or collection of elements which are connected between two nodes like resistors or a source.*
- *Loop : A loop in a circuit is a closed path, where no circuit element or node is met more than once.*
- *Mesh : A mesh doesn't contain any closed path, but it is a single open loop, and it does not contain any components inside a mesh.*

TRICKS...

- *When identical cells are connected in parallel the total emf = E, the emf of any one of the cells. Here, the sum of the reciprocals of the individual internal resistances is equal to the reciprocal of the total internal resistance.*

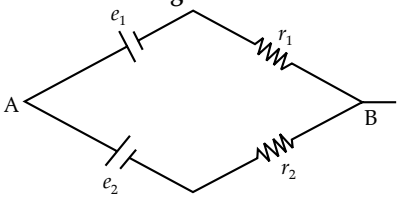
Know the Links

- *https://www.electrical4u.com/maxwell-bridge-inductance-capacitance-bridge/*
- *https://www.vidyakul.com/ncert-solutions-class-12-physics-chapter-3-Current-Electricity/*
- *http://www.bbc.co.uk/schools/gcsebitesize/science/add_aqa/electricity/circuitsrev4.shtml*

Multiple Choice Questions

(1 mark each)

Q. 1. Two batteries of emf ε_1 and ε_2 ($\varepsilon_2 > \varepsilon_1$) and internal resistances r_1 and r_2 respectively are connected in parallel as shown in figure :



(a) The equivalent emf ε_{eq} of the two cells is between ε_1 and ε_2, *i.e.* $\varepsilon_1 < \varepsilon_{eq} < \varepsilon_2$.
(b) The equivalent emf ε_{eq} is smaller than ε_1.
(c) The ε_{eq} is given by $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$ always.
(d) ε_{eq} is independent of internal resistances r_1 and r_2.
[NCERT Exemp. Q. 3.2, Page 16]

Ans. Correct option : (a)
Explanation : As we know that the equivalent emf in parallel combination of cells is :

$$\varepsilon_{eq} = \frac{(\varepsilon_1 r_2 + \varepsilon_2 r_1)}{(r_1 + r_2)}$$

so, it is clear that part '*c*' and '*d*' are incorrect by formula. According to this formula only option (a), is correct.

Q. 2. A resistance R is to be measured using a meter bridge. Student chooses the standard resistance S to be 100 Ω. He finds the null point at l_1 = 2.9 cm. He is told to attempt to improve the accuracy. Which of the following is a useful way?
(a) He should measure l_1 more accurately.
(b) He should change *S* to 1000 Ω and repeat the experiment.
(c) He should change *S* to 3 Ω and repeat the experiment.
(d) He should give up hope of a more accurate measurement with a meter bridge.
[NCERT Exemp. Q. 3.3, Page 17]

Ans. **Correct option :** (c)

Explanation : to calculate resistance, R -

$$R = S\left[\frac{l_1}{(100-l_1)}\right]$$

$$= 100\left[\frac{2.9}{97.1}\right]$$

$= 2.98\ \Omega$

So to get balance point near to 50 cm (middle) we have to take S=3 Ω, as here R :S =2.9 : 97.1 implies that S is nearly 33 times to R.

In order to make ratio R and S=1 :1, we must take the resistance S =3 Ω, which verifies options (c).

Q. 3. Two cells of emf's approximately 5V and 10V are to be accurately compared using a potentiometer of length 400 cm :

(a) The battery that runs the potentiometer should have voltage of 8V.

(b) The battery of potentiometer can have a voltage of 15 V and R adjusted so that the potential drop across the wire slightly exceeds 10 V.

(c) The first portion of 50 cm of wire itself should have a potential drop of 10 V.

(d) Potentiometer is usually used for comparing resistances and not voltages.

[NCERT Exemp. Q. 3.4, Page 17]

Ans. **Correct option :** (b)

Explanation : Given that,

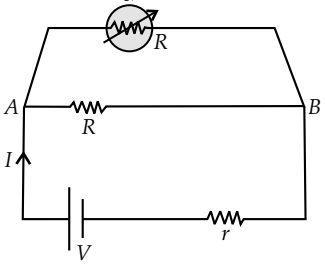
emf of primary cells are 5 V and 10 V

The potential drop across potentiometer wire must be slightly more than that larger emf 10 V.

So, the battery should be of 15V and about 4V potential is dropped by using rheostat or resistances. So option (b) is correct.

Tick Two or More Options

Q. 4. Consider a simple circuit shown in Figure stands for a variable resistance R'. R' can vary from R_0 to infinity. r is internal resistance of the battery ($r << R << R_0$).



(a) Potential drop across AB is nearly constant as R' is varied.

(b) Current through R' Ω is nearly a constant as R' Ω is varied.

(c) Current, I depend sensitively on R'.

(d) $\frac{I \geq V}{(r+R)}$ always. [NCERT Exemp. Q. 3.8, Page 18]

Ans. **Correct option :** (a) and (d)

Explanation : As it is given that-

$r << R << R_0 < R'$ from question,

$R' > R$ where, R' and R are in parallel combination, so the equivalent resistance will be always less than R.

So, $\frac{I \geq V}{(r+R)}$

And potential across AB will remain nearly constant.

Q. 5. Kirchhoff 's junction rule is a reflection of :

(a) conservation of current density vector.

(b) conservation of charge.

(c) the fact that the momentum with which a charged particle approaches a junction is unchanged (as a vector) as the charged particle leaves the junction.

(d) the fact that there is no accumulation of charges at a junction. [NCERT Exemp. Q. 3.7, Page 18]

Ans. **Correct option :** (b) and (d)

Explanation : As we know the junction rule, algebraic sum of current or charge flowing per unit time towards a junction in an electric network is zero, that verifies, law of conservation of charges so option (b) is correct and no any charges accumulate at junction as the sum of entering and outgoing charge are equal, at any time interval. It verifies answer (d).

Q. 6. The measurement of an unknown resistance R is to be carried out using Wheat stone bridge. Two students perform an experiment in two ways. The first students take R_2 = 10 Ω and R_1 = 5 Ω. The other student takes R_2 = 1000 Ω and R_1= 500 Ω. In the standard arm, both take R_3 = 5 Ω. Both find $R = (R_2/R_1)R_3$ = 10 Ω within errors.

(a) The errors of measurement of the two students are the same.

(b) Errors of measurement do depend on the accuracy with which R_2 and R_1 can be measured.

(c) If the student uses large values of R_2 and R_1, the currents through the arms will be feeble. This will make determination of null point accurately more difficult.

(d) Wheatstone bridge is a very accurate instrument and has no errors of measurement.

[NCERT Exemp. Q. 3.10, Page 18]

Ans. **Correct option :** (b) and (c)

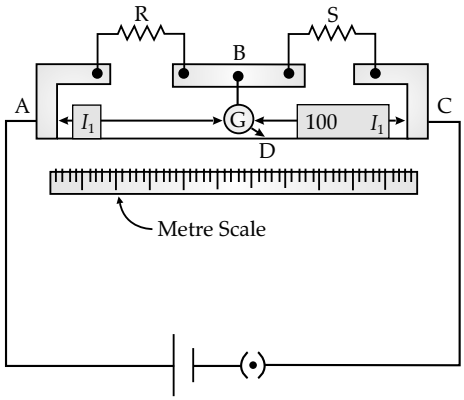
Explanation : As the ratio of R_2/R_3 and standard resistance are same, so value of unknown resistance for both students are 10.

So, we can say the Wheatstone Bridge is most sensitive.

The results of both students depend on the accuracy of resistances used. So, option (b) is verified.

When R_1 and R_2 is larger, the current through galvanometer becomes weak. It will make difficult to find out null point more accurately. So, option (c) is verified.

Q. 7. In a meter bridge the point D is a neutral point in figure :



(a) The meter bridge can have no other neutral point for this set of resistances.

(b) When the jockey contacts a point on meter wire left of *D*, current flows to *B* from the wire.

(c) When the jockey contacts a point on the meter wire to the right of *D*, current flows from *B* to the wire through galvanometer.

(d) When *R* is increased, the neutral point shifts to left. [NCERT Exemp. Q. 3.11, Page 19]

Ans. Correct option : (a) and (c)

Explanation : When jockey is at *D*, the current does not flow through galvanometer. So, the potentials at *B* and *D* are equal. Potentials at different points on the wire are different.

The point *D* is unique to get null point verifies the option (a).

When jockey is shifted to right of *D* on wire the potential in wire towards right side becomes smaller or $V_B > V_D$ so current flows from *B* to *D* in wire verify the option (c) and discards option (b).

When *R* is increased potential drop across *R* increases. So, potential at *B* increases to get null point. Jockey must move to right.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. A cell of emf E and internal resistance r is connected across an external resistance *R*. Plot a graph showing the variation of *P.D.* across *R*, verses *R*. [NCERT Exemp. Q. 3.21, Page 20]

Ans. When the cell of emf, *E* and internal resistance, *r* is connected across an external resistance, *R* the relationship between the voltage across, *R* is given by

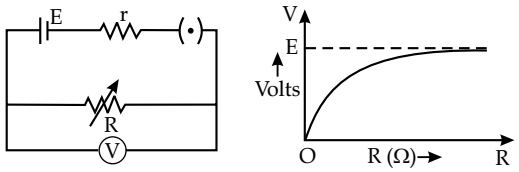
$$V = \frac{E}{[1+(r/R)]}$$

From the above relation,

Now,

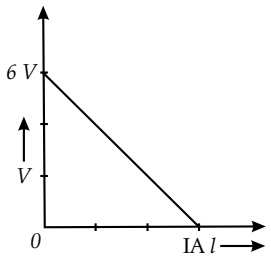
as $R = 0 \Rightarrow V = 0$ & $R = \infty \Rightarrow V = E$ **[1]**

So, this variation is shown in the figure given below.



[1]

Q. 2. The plot of the variation of potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell?



[CBSE Board, All India Regions, 2016]

Ans. Let *E* and *r* be the emf and internal resistance of all cells, respectively and *e* and *r'* be the emf and internal resistance of each cell, respectively.

As we know,

According to the definition of the terminal potential difference, $V = E - Ir$ **[½]**

Voltage across cell combination :

$V = E - IR$

when current,

$I = 0$

$\Rightarrow V = E$

From graph, when $I = 0, V = 0$

$\Rightarrow$ emf $E = 6V$

As $I = 1A$ at $V = 0$ (from graph)

so, $V = E - Ir$

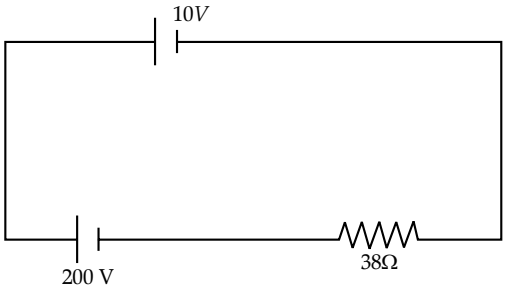
$\Rightarrow$ $0 = 6 - 1r \Rightarrow -6 = -1r$

so, $r = 6\,\text{ohm}$

As per the question the cells are connected in the series, so we can write.

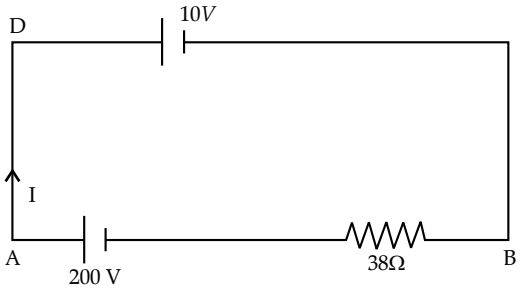
$$r' = \frac{r}{3} = 2\,\text{ohm}$$ **[½]**

Q. 3. A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance 38 Ω as shown in the figure. Find the value of current in the circuit.



[CBSE Board, Delhi Region, 2018]

Ans. Applying Kirchhoff's law in *ABCDA*-



$10 - 200 + 38i = 0$

$38i = 190$

$i = 190/38 = 5A$

So, the value of current in the circuit will be 5 A. **[2]**

Q. 4. What are the advantages of the null-point method in a Wheatstone bridge? What additional measurements would be required to calculate $R_{unknown}$ by any other method?

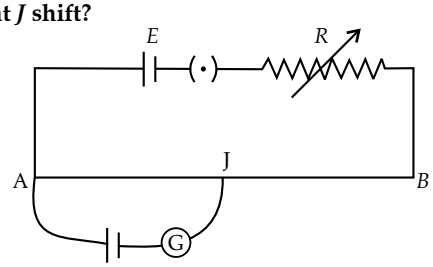
[NCERT Exemp. Q. 3.14, Page 19]

Ans. The advantage of null point method in a Wheatstone bridge is that the resistance of galvanometer does not affect the balance point and there is no need to determine current in resistances and galvanometer and the internal resistance of a galvanometer. **[1]**

$R_{unknown}$ can be calculated by applying Kirchhoff's law to the circuit.

We would need additional accurate measurement of all the currents in resistances and galvanometer and internal resistance of the galvanometer. **[1]**

Q. 5. *AB* is a potentiometer wire (Figure). If the value of R is increased, in which direction will the balance point *J* shift?



[NCERT Exemp. Q. 3.19, Page 19]

Ans. If R is increased current in main circuit will decrease (by $V = IR$) as the potential (E) is constant. So, in turn potential difference across AB will decrease (by $V = IR$). **[1]**

As R of AB is constant so potential gradient,

$$k = \frac{V}{AB}$$

will decrease. So, to balance potential across AB equal to potential of secondary circuit (E′) the length AJ' must be larger than earlier AJ. So the point J shifts towards B. **[1]**

Q. 6. In a potentiometer arrangement, a cell of emf 1.25 V gives a balance point at 35.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 63.0 cm, what is the emf of the second cell? [NCERT Ex. Q. 3.12, Page 128]

Ans. Given that,

emf of the cell, $E_1 = 1.25$ V

Balance point of the potentiometer, $l_1 = 35$ cm

New balance point of the potentiometer, $l_2 = 63$ cm

If the cell is replaced by another cell of emf, E_2 then,

According to the balance condition-

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$E_2 = E_1 \times \frac{l_2}{l_1}$$

$$= 1.25 \times \frac{63}{35} = 2.25V$$

Therefore, emf of the second cell is 2.25V. **[2]**

Q. 7. What is the advantage of using thick metallic strips to join wires in a potentiometer?

[NCERT Exemp. Q. 3.15, Page 19]

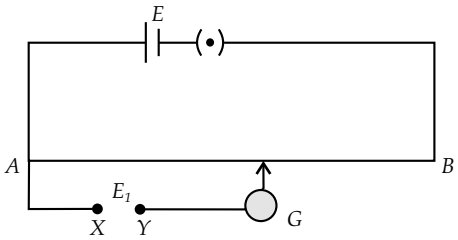
Ans. The metal strips have low resistance and need not be counted in the potentiometer length l_1 of the null point. **[1]**

One measures only their lengths along the straight segments (of lengths 1 meter each). This is easily done with the help of centimetre rulings or metre ruler and leads to accurate measurements. **[1]**

Q. 8. While doing an experiment with potentiometer (Figure) it was found that the deflection is one sided and (i) the deflection decreased while moving from one end A of the wire to the end B, (ii) the deflection increased. while the jockey was moved towards the end B.

(i) Which terminal + or – ve of the cell E_1, is connected at X in case (i) and how is E_1 related to E ?

(ii) Which terminal of the cell E_1 is connected at X in case (ii)?

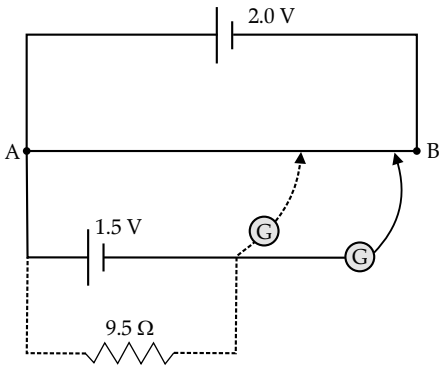


[NCERT Exemp. Q. 3.20, Page 20]

Ans. **(i)** Deflection in galvanometer is one sided and the deflection decreases while moving from end A to end B, this is possible only when positive terminal of the cell E_1, is connected at X and $E_1 > E$. **[1]**

(ii) The deflection in galvanometer is one sided and the deflection increased, while moving from one end A of the wire to the end S, this is possible only when negative terminal of the cell E_1 is connected at X. **[1]**

Q. 9. Figure shows a 2.0 V potentiometer used for the determination of internal resistance of a 1.5 V cell. The balance point of the cell in open circuit is 76.3 cm. When a resistor of 9.5 Ω is used in the external circuit of the cell, the balance point shifts to 64.8 cm length of the potentiometer wire. Determine the internal resistance of the cell.



[NCERT Add. Ex. Q. 3.23, Page 130]

Ans. Given that,

Balance point of the cell in open circuit, $l_1 = 76.3$ cm
An external resistance (R) is connected to the circuit with $R = 9.5\ \Omega$
New balance point of the circuit, $l_2 = 64.8$ cm
Let assume that,
Internal resistance of the cell = r
Current flowing through the circuit = I **[1]**
Then, the relation connecting resistance and emf is,

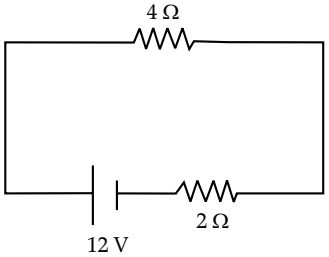
$$r = \frac{l_1 - l_2}{l_2} R$$

$$= \frac{76.3 - 64.8}{64.8} \times 9.5 = 1.68 \text{ohm}$$

So, the internal resistance of the cell is 1.68 Ω. **[1]**

Short Answer Type Questions **(3 marks each)**

Q. 1. A battery of emf 12 V and internal resistance 2 Ω is connected to a 4 Ω resistor as shown in the figure.

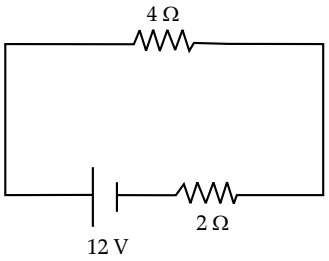


(a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.

(b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit?

[CBSE Board, All India Region, 2016]

Ans. (a) Voltmeter across cell :



$$I = \frac{E}{R+r} = \frac{12}{2+4} = 2\text{A}$$

$$V_1 = E - Ir = 12 - 2 \times 2 = 8\text{V}$$

Voltmeter across resistor :

$$V_2 = \frac{ER}{R+r} = \frac{12 \times 4}{4+2} = 8\text{V}$$

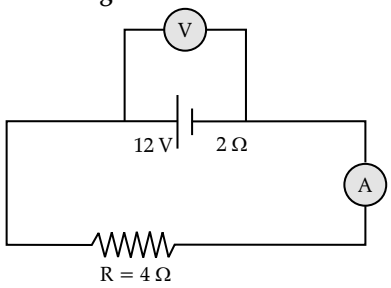
$$So,\ V_1 = V_2$$

So, from the above relation we can see that when voltmeter placed across the cell and across the resistor, gives the same reading. **[2]**

(b) Voltmeter connects in parallel because it measures the difference in voltage potential between two points. Ammeter connects in series because it needs to measure the total amount of current flow. **[1]**

Q. 2. (a) The potential difference applied across a given resistor is altered so that the heat product per second increases by a factor of 9. By what factor does the applied potential difference change?

(b) In the figure shown, an ammeter A and a resistor of 4 Ω are connected to the terminals of the source. The emf of the source is 12 V having an internal resistance of 2 Ω. Calculate the voltmeter and ammeter reading.



[CBSE Board, Delhi Region, 2017]

Ans. (a) Let the heat dissipated per unit time

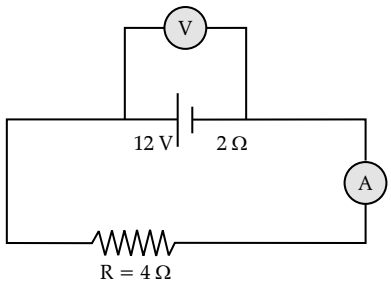
$$H = \frac{V^2}{R} = \frac{(12)^2}{6} = 24\ J/\text{sec}$$

The new heat dissipated per unit time (H′) = $H \times 9$
$= 24 \times 9 = 216\ J$/sec
Let assume the new voltage be V'

$$\frac{(V)^2}{R} = 216$$

Put the values
$(V')^2 = 216 \times 6$
or
$V' = 36$ volt **[1½]**

(b)



Given that,
emf (E) = 12 V
internal resistance (r) = 2 Ω
external resistance (R) = 4 Ω
then current,

$$I = \frac{E}{(R+r)}$$

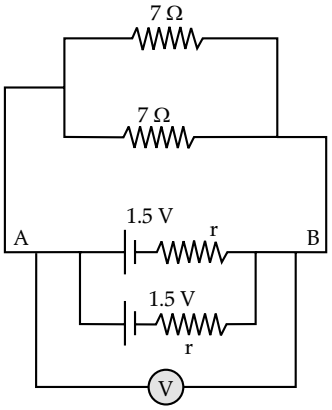
$$= \frac{12}{(4+2)} = 2\text{ A}$$

So, reading of ammeter will be 2 A
As we know that the,
$V = E - Ir$
$V = 12 - 2 \times 2$
$V = 12 - 4 = 8$ V
So, the reading in the voltmeter will be 8 V. **[1½]**

Q. 3. Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of 7 Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V. Calculate the internal resistance of each cell.

[CBSE Board, Delhi Region, 2016]

Ans.



E_{eq} of cells across AB $E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$

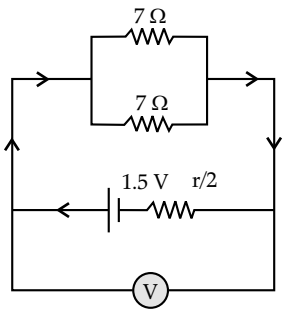
$$E_1 = E_2 = 1.5V$$

$$E_{eq} = \frac{1.5r + 1.5r}{2r}$$

$$E_{eq} = 1.5\,\text{volt}$$

Equivalent resistance, $r_{eq} = \frac{r^2}{2r} = \frac{r}{2}$

So, the circuit diagram will be-



[1½]

$$i = \frac{1.5}{\frac{7}{2} + \frac{r}{2}}$$

$$\Rightarrow \quad i = \frac{3}{7+r}$$

Reading of voltmeter-

$$v = ir$$

$$1.4 = i\left(\frac{7}{2}\right)$$

$$\left(\frac{3}{7+r}\right)\frac{7}{2} = 1.4$$

$$\Rightarrow 7 + r = 7.5 = 0.5 \text{ ohm}$$

[1½]

Q. 4. A storage battery of emf 8.0 V and internal resistance 0.5 Ω is being charged by a 120 V dc supply using a series resistor of 15.5 Ω. What is the terminal voltage of the battery during charging? What is the purpose of having a series resistor in the charging circuit?

[NCERT Ex. Q. 3.11, Page 128]

Ans. Given that,
emf of the storage battery, E = 8.0 V
Internal resistance of the battery, r = 0.5 Ω
DC supply voltage, V = 120 V
Resistance of the resistor, R = 15.5 Ω
Let assume the effective voltage in the circuit = V_1
R is connected to the storage battery in series.
So, it can be
$V_1 = V - E$
$V_1 = 120 - 8 = 112$ V
Current flowing in the circuit = I
According to Ohm's law,

$$I = \frac{V'}{R+r}$$

$$= \frac{112}{15.5+5} = \frac{112}{16} = 7A$$

[1½]

Voltage across resistor R given by the product,
$IR = 7 \times 15.5 = 108.5$ V
DC supply voltage = Terminal voltage of battery + Voltage drop across R
Terminal voltage of battery = 120 – 108.5 = 11.5 V
A series resistor in a charging circuit limits the current drawn from the external source. So, the current will be extremely high in its absence. This is very dangerous. **[1.5]**

Q. 5. A battery of emf 10 V and internal resistance 3 Ω is connected to a resistor. If the current in the circuit is 0.5 A, what is the resistance of the resistor? What is the terminal voltage of the battery when the circuit is closed?

[NCERT Ex. Q. 3.2, Page 127]

Ans. Given that,
Emf of the battery, E = 10 V

Internal resistance of the battery, $r = 0.4\ \Omega$
Current in the circuit, $I = 0.5$ A
Let assume the resistance of the resistor = R
Now, using Ohm's law,

$$I = \frac{E}{(R+r)} \quad \textbf{[1]}$$

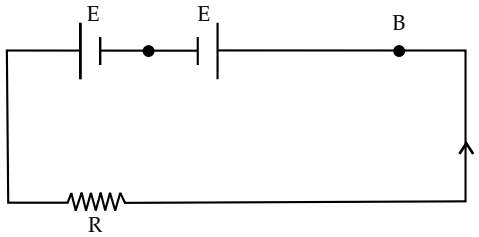
$$(R + r) = \frac{E}{I}$$

Put the values :

$$(R + r) = \frac{10}{0.5} = 20\ \Omega$$

So, $R = 20 - 3 = 17\ \Omega$
Terminal voltage of the resistor = V **[1]**
According to Ohm's law,
$V = IR$
$= 0.5 \times 17 = 8.5$ V
So, the resistance of the resistor is 17 Ω and the terminal voltage is 8.5 V. **[1]**

Q. 6. Two cells of same emf E but internal resistance r_1 and r_2 are connected in series to an external resistor R (Figure). What should be the value of R so that the potential difference across the terminals of the first cell becomes zero.



[NCERT Exemp. Q. 3.25, Page 20]

Ans. Current ***I*** flowing in the circuit,

$$I = \frac{E+E}{R+r_1+r_2}$$

$$V_1 = E - Ir_1 = E - \frac{2E.r_1}{R+r_1+r} \quad \textbf{[1]}$$

The net potential difference across 1^{st} *cell* $v_1 = 0 (given)$

$$\therefore \quad E - \frac{2E.r_1}{R+r_1+r} = 0$$

after solving the above equation:

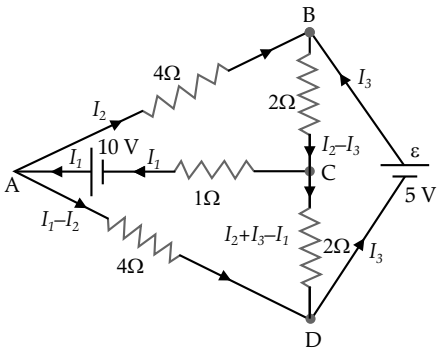
$$\frac{2.r_1}{R+r_1+r} = \frac{1}{1}$$

or

$$2r_1 = r_1 + r_2 + R$$
$$r_1 - r_2 = R \quad \textbf{[1]}$$

So, this above equation is the desired condition for the potential difference across 1^{st} cell to be zero. **[1]**

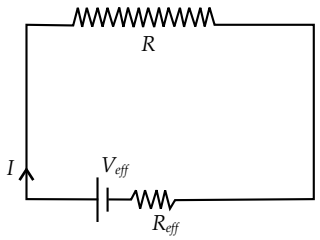
Q. 7. Suppose there is a circuit consisting of only resistances and batteries. Suppose one is to double (or increase it to *n*-times) all voltages and all resistances. Show that currents are unaltered. Do this for the circuit given in the Figure



[NCERT Exemp. Q. 3.27, Page 21]

Ans. After rearranging the circuit, let assume R_{eff} be the internal resistance of the battery, the effective external resistance is R and the effective voltage of the battery is V_{eff}.

Then the circuit will be :



The current through R is given by ohm's law,

$$I = \frac{V_{eff}}{(R_{eff} + R)} \quad \textbf{[1]}$$

If all the resistances and the effective voltage are increased n-times, then we

$V_{eff\ new} = nV_{eff}$
and $R_{eff\ new} = n\ R_{eff}$
also $R_{new} = n\ R$ **[1]**

Then, the new current is given by :

$$I' = \frac{nV_{eff}}{(nR_{eff} + nR)} = \frac{\left(V_{eff}\right)}{(R_{efff} + R)} = I$$

As the current remains same if the R, E and r of a circuits is increased by n times, that is, nR, nE, nr. So, the current remains same. **[1]**

Q. 8. (i) Plot a graph showing variation of voltage vs the current drawn from the cell.

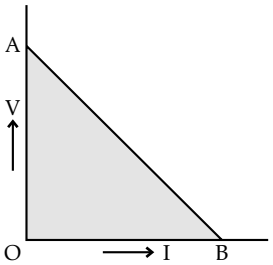
How can one get information from this plot about the emf of the cell and its internal resistance?

(ii) Two cells of emf's E_1 and E_2 and internal resistance r_1 and r_2 are connected in parallel.

Obtain the expression for the emf and internal resistance of a single equivalent cell that can replace this combination?

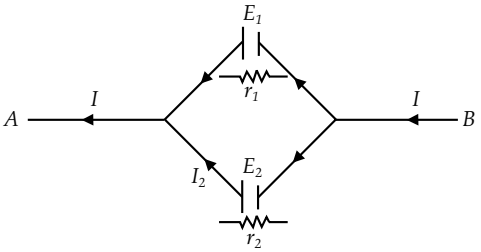
[CBSE Board, Foreign Scheme, 2016]

Ans. (i) The relation between V and I is given by :
$V = E - Ir$
Thus, the graph between V and I is shown as below :



emf is given by the intercept on the Y- axis.
Internal resistance is given by the slope of the line in the above graph. **[1½]**

(ii) Consider the circuit :



Where, E_1 and E_2 are the emf of two cells.
r_1 and r_2 are the internal resistance of cell,
I_1 and I_2 current due to two cell,
Terminal potential difference across the first cell is given by,

$V = E_1 - I_1 r_1$

So,

$$I_1 = \frac{(E_1 - V)}{r_1}$$

Let E be the effective emf and r the resultant internal resistance.
Now, consider I as the current flowing through the cell.
So,

$$I = I_1 + I_2$$

$$\Rightarrow \quad I = \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$$

$$\Rightarrow \quad I = \frac{r_2(E_1 - V) + r_1(E_2 - V)}{r_1 r_2}$$

$$\Rightarrow \quad Ir_1r_2 = E_1r_2 + E_2r_1 - (r_1 + r_2)V$$

$$\Rightarrow \quad V = \frac{E_1r_2 + E_2r_1}{r_1 + r_2} - \frac{Ir_1r_2}{r_1 + r_2}$$

Now, comparing the equation with

$$V = E - Ir$$

then we get

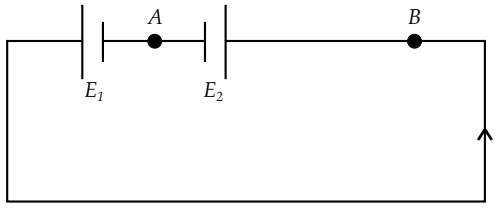
$$E = \frac{E_1r_2 + E_2r_1}{r_1 + r_2}$$

and

$$\text{Internal resistance, } r = \frac{r_1r_2}{r_1 + r_2}$$

[1½]

Q. 9. The circuit in Figure shows two cells connected in opposition to each other. Cell E_1 is of emf 6V and internal resistance 2 Ω; the cell E_2 is of emf 4V and internal resistance 8 Ω. Find the potential difference between the points *A* and *B*.

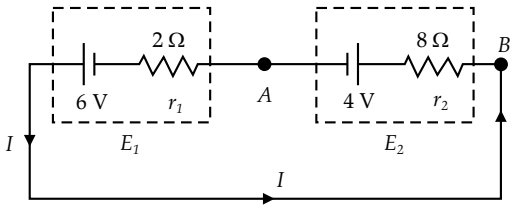


[NCERT Exemp. Q. 3.24, Page 20]

Ans. The above figure can be redrawn as given here. The direction of current in circuit will be as shown in the figure. So, point *B* is at higher potential than *A*.

$V_B > V_A$

Current (I) in circuit, $I = \frac{E_1 + E_2}{r_1 + r_2} = \frac{(6-4)\ V}{(2+8)\ \Omega} = 0.2\ A$ **[1]**



[1]

For positive potential A is near to positive terminal of E_2 so has +4 V. So, potential across
E_1 and E_2 :

$E_1 = V - Ir_1 = 6 - 0.2 \times 2 = 6 - 0.4 = 5.6$ V

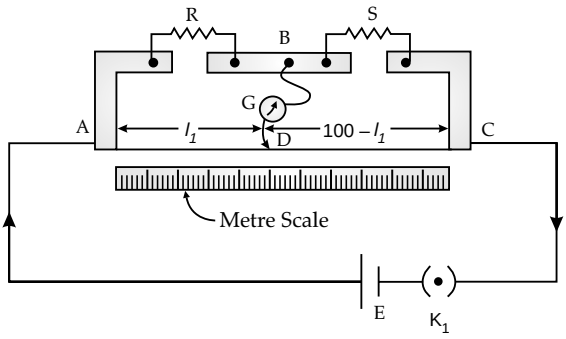
$E_2 = V + Ir_2 = 4 + 0.2 \times 8 = 4 + 1.6 = 5.6$ V

So, potential between *A* and *B* will be $E_2 = 5.6$ Volt.

As current is flowing from *B* to *A*.

So, potential at *B* is larger than *A*. **[1]**

Q. 10. (a) In a metre bridge in Figure, the balance point is found to be at 39.5 cm from the end *A*, when the resistor Y is of 12.5 Ω.



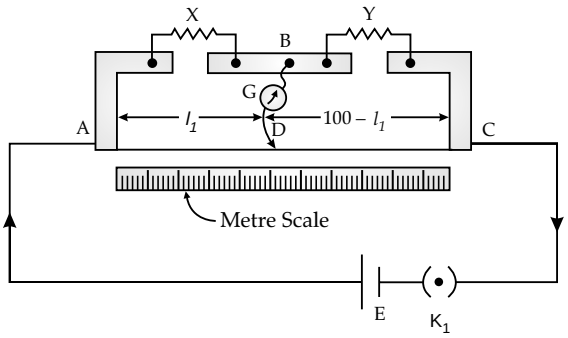
Determine the resistance of *X*. Why are the connections between resistors in a Wheatstone or meter bridge made of thick copper strips?

(b) Determine the balance point of the bridge above if *X* and *Y* are interchanged.

(c) What happens if the galvanometer and cell are interchanged at the balance point of the bridge? Would the galvanometer show any current?

[NCERT Ex. Q. 3.10, Page 128]

Ans. A metre bridge with resistors *X* and *Y* is represented in the given figure.



(a) Given that,
Balance point from end A, $l_1 = 39.5$ cm
Resistance of the resistor $Y = 12.5\ \Omega$
Condition for the balance is given as,

$$\frac{X}{Y} = \frac{100 - l_1}{l_1}$$

$$X = \frac{100 - 39.5}{39.5} \times 12.5 = 8.2\ \Omega$$

So, the resistance of resistor X is 8.2 Ω. **[1]**

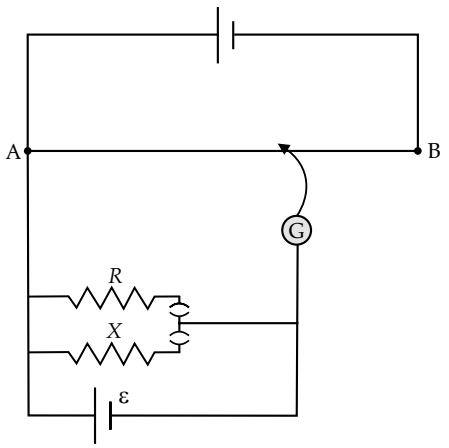
(b) The connection between resistors in a Wheatstone or metre bridge is made of thick copper strips to minimise the resistance, which is not taken into consideration in the bridge formula.

If X and Y are interchanged, then l_1 and $(100 - l_1)$ get interchanged.

The balance point of the bridge will be $(100 - l_1)$ from A.
$100 - l_1 = 100 - 39.5 = 60.5$ cm
So, the balance point is 60.5 cm from A. **[1]**

(c) When the galvanometer and cell are interchanged at the balance point of the bridge, the galvanometer will show no deflection. Hence, no current would flow through the galvanometer. **[1]**

Q. 11. Figure shows a potentiometer circuit for comparison of two resistances. The balance point with a standard resistor $R = 10.0\ \Omega$ is found to be 58.3 cm, while that with the unknown resistance X is 68.5 cm. Determine the value of X. What might you do if you failed to find a balance point with the given cell of emf ε?



[OB Value Add.]

Ans. Given that,
Resistance of the standard resistor, $R = 10.0\ \Omega$
Balance point for this resistance, $l_1 = 58.3$ cm
Current in the potentiometer wire = i
so, potential drop across R,
$E_1 = iR$
Resistance of the unknown resistor = X
Balance point for this resistor, $l_2 = 68.5$ cm
So, potential drop across X, $E_2 = iX$ **[1]**
The relation connecting emf and balance point is,

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{iR}{iX} = \frac{l_1}{l_2}$$

$$X = \frac{l_1}{l_2} \times R$$

$$= \frac{68.5}{58.3} \times 10 = 11.749\ \Omega$$

so, the value of the unknown resistance, X is 11.75 Ω. **[1]**
If we fail to find a balance point with the given cell of emf, ε, then the potential drop across R and X must be reduced by putting a resistance in series with it. Only if the potential drop across R or X is smaller than the potential drop across the potentiometer wire AB, a balance point is obtained. **[1]**

Long Answer Type Questions **(5 marks each)**

Q. 1. (a) Six lead-acid type of secondary cells each of emf 2.0 V and internal resistance 0.015 Ω are joined in series to provide a supply to a resistance of 8.5 Ω. What are the current drawn from the supply and its terminal voltage?

(b) A secondary cell after long use has an emf of 1.9 V and a large internal resistance of 380 Ω. What maximum current can be drawn from the cell? Could the cell drive the starting motor of a car?

[NCERT Add. Ex. Q. 3.15, Page 129]

Ans. (a) Given that,
Number of secondary cells, $n = 6$
emf of each secondary cell, $E = 2.0$ V
Internal resistance of each cell, $r = 0.015\ \Omega$
series resistor is connected to the combination of cells.
Resistance of the resistor, $R = 8.5\ \Omega$
Current drawn from the supply = I, which is given by the relation

$$I = \frac{nE}{R + nr}$$

$$= \frac{6 \times 2}{8.5 + 6 \times 0.0015}$$

$$= \frac{12}{8.59} = 1.39\ A$$

[2]

So, the terminal voltage, $V = IR = 1.39 \times 8.5 = 11.87$ A
Hence, the current drawn from the supply is 1.39 A and terminal voltage is 11.87 A. **[½]**

(b) Given that,
After a long use, emf of the secondary cell, $E = 1.9$ V
Internal resistance of the cell, $r = 380\ \Omega$
so, maximum current,

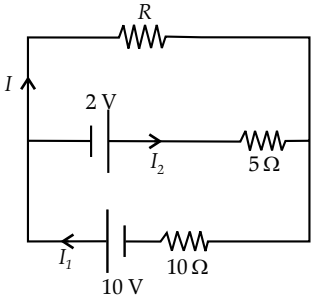
$$I_{max} = \frac{E}{r}$$

$$= \frac{1.9}{380} = 0.005\,A$$

[1]

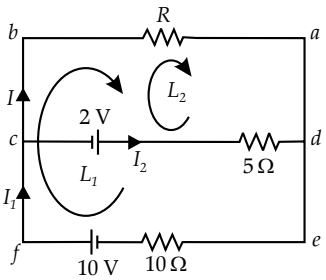
The maximum current drawn from the cell is 0.005 A.
So, a large current is required to start the motor of a car, the cell cannot be used to start a motor. **[1½]**

Q. 2. Two cells of voltage 10 V and 2 V and internal resistances 10 Ω and 5 Ω respectively, are connected in parallel with the positive end of 10 V battery connected to negative pole of 2V battery (Figure). Find the effective voltage and effective resistance of the combination.



[NCERT Exemp. Q. 3.28, Page 21]

Ans. Applying Kirchhoff's law at junction c,

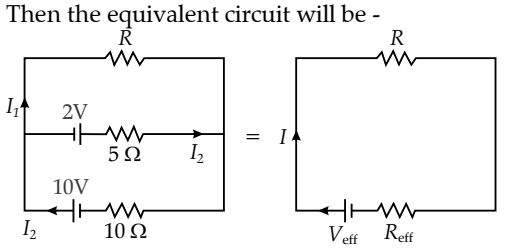


$I_1 = I + I_2$
Applying Kirchhoff's voltage law in loop L_1 (*efbae*) outer loop :
$10 - IR - 10I_1 = 0$
$10 = IR + 10I_1$ ...(i)
Applying Kirchhoff's law in loop L_2 (*cbadc*)-
$-2 - IR + 5I_2 = 0$
$2 = 5I_2 - RI$ **[1]**
And
$I_1 = I + I_2$ Then,
$I_2 = I_1 - I$
Rewrite the above equation :
$2 = 5\,(I_1 - I) - R_1$
Or, $4 = 10\,I_1 - 10\,I - 2RI$ **[1]**
Subtracting equation (ii) from (i), we get :
$6 = 3RI + 10\,I$
$2 = I\,(R + \frac{10}{3})$

Also, the external resistance is R.
According to ohm's law :
$V = I(R + R_{eff})$ **[1]**
On comparing,
we have $V = 2$V and effective internal resistance
$(R_{eff}) = \left(\frac{10}{3}\right)\Omega$

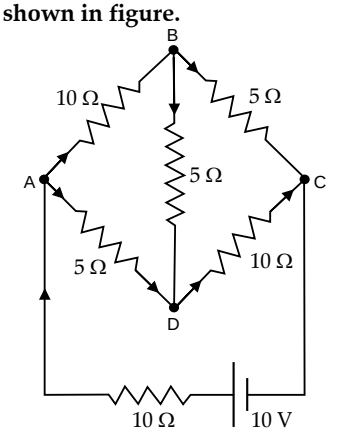
Since, the equivalent internal resistance R_{eff} of two cells will be $\left(\frac{10}{3}\right)\Omega$

Being the parallel combination of 5 Ω and 10 Ω. **[1]**
Then the equivalent circuit will be -



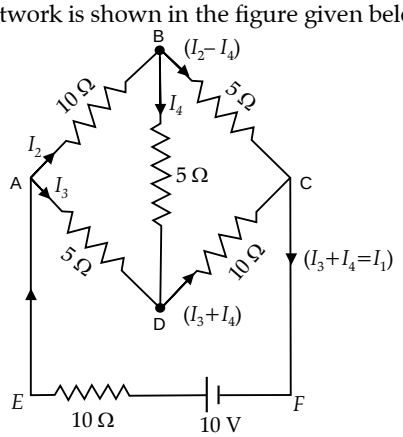
[1]

Q. 3. Determine the current in each branch of the network shown in figure.



[NCERT Ex. Q. 3.9, Page 128]

Ans. The current flowing through various branches of the network is shown in the figure given below :



I_1 = Current flowing through the outer circuit
I_2 = Current flowing through branch *AB*
I_3 = Current flowing through branch *AD*
$I_2 - I_4$ = Current flowing through branch *BC*
$I_3 + I_4$ = Current flowing through branch *CD*
I_4 = Current flowing through branch *BD*
For the closed circuit ABDA, potential is zero, that is,

$10I_2 + 5I_4 - 5I_3 = 0$

$2I_2 + I_4 - I_3 = 0$

$I_3 = 2I_2 + I_4$ …(1)

[1]

For the closed circuit BCDB, potential is zero, that is,

$5(I_2 - I_4) - 10(I_3 + I_4) - 5I_4 = 0$

$5I_2 + 5I_4 - 10I_3 - 10I_4 - 5I_4 = 0$

$5I_2 - 10I_3 - 20I_4 = 0$

$I_2 = 2I_3 + 4I_4$ …(2)

[1]

For the closed circuit *ABCFEA*, potential is zero, that is,

$-10 + 10(I_1) + 10(I_2) + 5(I_2 - I_4) = 0$

$10 = 15I_2 + 10I_1 - 5I_4$

$3I_2 + 2I_1 - I_4 = 2$ …(3)

[1]

From equations (1) and (2), we obtain

$I_3 = 2(2I_3 + 4I_4) + I_4$

$I_3 = 4I_3 + 8I_4 + I_4$

$3I_3 = 9I_4$

$3I_4 = I_3$ …(4)

Putting equation (4) in equation (1), we obtain

$I_3 = 2I_2 + I_4$

$-4I_4 = 2I_2$

$I_2 = -2I_4$ …(5)

It is evident from the given figure that,

$I_1 = I_3 + I_2$ …(6)

Putting equation (6) in equation (1), we obtain

$3I_2 + 2(I_3 + I_2) - I_4 = 2$

$5I_2 + 2I_3 - I_4 = 2$ … (7)

Putting equations (4) and (5) in equation (7), we obtain

$5(-2I_4) + 2(-3I_4) - I_4 = 2$

$10I_4 - 6I_4 - I_4 = 2$

$17I_4 = -2$

$I_4 = -2/17$ A

Equation (4) reduces to

$I_3 = -3(I_4)$

$= -3\left(\frac{-2}{17}\right) = \frac{6}{17}$ A

$I_2 = -2(I_4)$

$= \left(\frac{-2}{17}\right) = \frac{4}{17}$ A

$I_2 - I_4 = \frac{4}{17} - \left(\frac{-2}{17}\right) = \frac{6}{17}$ A

$I_3 + I_4 = \frac{6}{17} + \left(\frac{-2}{17}\right) = \frac{4}{17}$ A

$I_1 = I_3 + I_2$

$= \frac{6}{17} + \frac{4}{17} = \frac{10}{17}$ A **[1]**

Therefore, current

In branch AB $= \frac{4}{17}$ A

In branch BC $= \frac{6}{17}$ A

In branch CD $= \frac{-4}{17}$ A

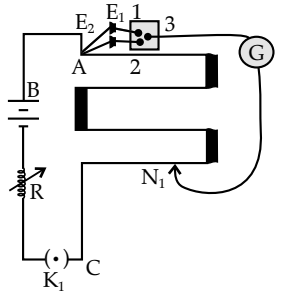
In branch AD $= \frac{6}{17}$ A

In branch BD $= \left(\frac{-2}{17}\right)$ A

Total current $= \frac{4}{17} + \frac{6}{17} + \left(\frac{-4}{17}\right) + \frac{6}{17} + \left(\frac{-2}{17}\right)$

$= 10/17$ A **[1]**

Q. 4. In an experiment with a potentiometer, V_B = 10V. R is adjusted to be 50 Ω (Figure). A student wanting to measure voltage E_1 of a battery (approx. 8 V) finds no null point possible. He then diminishes R to 10 Ω and is able to locate the null point on the last (4^{th}) segment of the potentiometer. Find the resistance of the potentiometer wire and potential drop per unit length across the wire in the second case.



[NCERT Exemp. Q. 3.30, Page 21]

Ans. Let R′ be the resistance of potentiometer wire.

Variable resistance, R= 50 Ω

I is the current in primary circuit which is at E_B = 10 V.

$\therefore \quad I = \frac{V_B}{R + R'}$

$\Rightarrow \frac{10}{50 + R} = I$ (in primary circuit) ...(i) **[1]**

Potential difference across the wire of potentiometer- $V' = IR'$

From eq. I

$\therefore V' = \frac{10R'}{R + 50}$ ...(ii)

As with R = 50 Ω resistance, null point cannot be obtained by 8 Volt.

So, V' < 8 Volt.

$\frac{10R'}{50 + R'} < 8$ (No balance point)

[1]

As $50 + R'$ is positive, so we can multiply above equation by positive number and we get

$10R' < 400 + 8R'$

$2R' < 400$

$R' < 200$ ...(iii)

Similarly, null point obtained by $R = 10\ \Omega$. Then $V'' > 8$ at balance point. So, it is possible when

$\frac{10R'}{10+R'} > 8$ (as from I, $R = 10$)

Similarly, multiply above equation by positive number $10 + R'$ to both sides

$10R' > 80 + 8R'$

$2R' > 80$

$R' > 40$ ...(iv)

As the null point is obtained on 4th segment or at ¾ of total length So, at ¾ R' (No balance point)

or $\frac{10 \times \frac{3}{4}R'}{10+R'} < 8$ (at balance point) **[1]**

So, $7.5R' < 80 + 8R'$

$-0.5R' < 80$

$-R' < 160$

$R' > -160$

R' can never be negative so, –160 Ω is considered 160 Ω

So $160 < R' < 200$ ...(v)

Any R' between 160 Ω and 200 Ω will achieve null point. Since the null point is on last 4th segment of potentiometer wire, so the potential drop across 400 cm < 8 Volt.

So, $K \times 400$ cm > 8V (At balance point) **[1]**

$K > \frac{8}{400}$ volt/cm

$K > \frac{8}{4}$ volt/m

$K > 2$ volt/m

As balance point is at 4th wire, so no balance point at 3 m,

$K.3 < 8$ (No balance point)

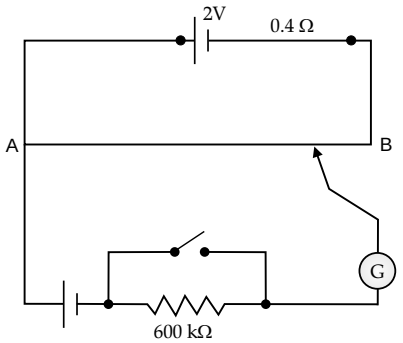
$K < \frac{8}{3}$ volt/m

$K < 2\frac{2}{3}$ volt/m

So,

$2\frac{2}{3}$ volt/m $> K > 2$ volt/m **[1]**

Q. 5. Figure shows a potentiometer with a cell of 2.0 V and internal resistance 0.40 Ω maintaining a potential drop across the resistor wire *AB*. *A* standard cell which maintains a constant emf of 1.02 V (for very moderate currents up to a few mA) gives a balance point at 67.3 cm length of the wire. To ensure very low currents drawn from the standard cell, a very high resistance of 600 kΩ is put in series with it, which is shorted close to the balance point. The standard cell is then replaced by a cell of unknown emf ε and the balance point found similarly, turns out to be at 82.3 cm length of the wire.



(a) What is the value ε?

(b) What purpose does the high resistance of 600 kΩ have?

(c) Is the balance point affected by this high resistance?

(d) Is the balance point affected by the internal resistance of the driver cell?

(e) Would the method work in the above situation if the driver cell of the potentiometer had an emf of 1.0 V instead of 2.0 V?

(f) Would the circuit work well for determining an extremely small emf, say of the order of a few *mV* (such as the typical emf of a thermo-couple)? If not, how will you modify the circuit?

[NCERT Add. Ex. Q. 3.22, Page 130]

Ans. **(a)** Constant emf of the given standard cell, E_1 = 1.02 V

Balance point on the wire, l_1 = 67.3 cm

A cell of unknown emf, ε replaced the standard cell.

now, the new balance point on the wire, l = 82.3 cm

The relation connecting emf and balance point is,

$\frac{E_1}{l_1} = \frac{\varepsilon}{l}$

$\varepsilon = \frac{l}{l_1} \times E_1$

$= \frac{82.3}{67.3} \times 1.02 = 1.247 V$

The value of unknown emf is 1.247 V. **[1]**

(b) The purpose of using the high resistance of 600 kΩ is to reduce the current through the galvanometer when the movable contact is far from the balance point. **[1]**

(c) The balance point is not affected by the presence of high resistance. **[½]**

(d) The point is not affected by the internal resistance of the driver cell. **[½]**

(e) The method would not work if the driver cell of the potentiometer had an emf of 1.0 V instead of 2.0 V. This is because if the emf of the driver cell of the potentiometer is less than the emf of the other cell, then there would be no balance point on the wire. **[1]**

(f) The circuit would not work well for determining an extremely small emf. As the circuit would be unstable, the balance point would be close to end *A*. Hence, there would be a large percentage of error.

The given circuit can be modified if a series resistance is connected with the wire *AB*. So, the potential drop across *AB* is slightly greater than the emf measured.

So, the percentage error would be small. **[1]**

Some Commonly Made Errors

- Students confuse to differentiate the electrical bridges.
- Try to solve numerical by making suitable diagram, do not skip the diagrams.
- The first and most straightforward network analysis technique is called the Branch Current Method. In this method, we assume directions of currents in a network, then write equations describing their relationships to each other through Kirchhoff's and Ohm's Laws. Once we have one equation for every unknown current, we can solve the simultaneous equations and determine all currents, and therefore, all voltage drops in the network.

EXPERT ADVICE

- ☞ *Apply method of elimination for MCQ questions. Look at the answers before even starting to solve and see if you can eliminate any answer. When calculating answer for a big question you may not need to solve the problem fully in order to choose the right answer because you could eliminate the other 3.*
- ☞ *Mathematically, analyse the circuit, solving for all values of voltage, current, etc.*
- ☞ *Carefully measure those quantities, to verify the accuracy of your analysis.*
- ☞ *If there are any substantial errors (greater than a few percent), carefully check your circuit's construction against the diagram, then carefully re-calculate the values and re-measure.*

OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/zfJiJkxM02Y

Or Scan the Code

https://youtu.be/ZDoyIghUI44

Or Scan the Code

https://youtu.be/n__cezo9R1U

Or Scan the Code

https://youtu.be/TS5ihKbOOOs

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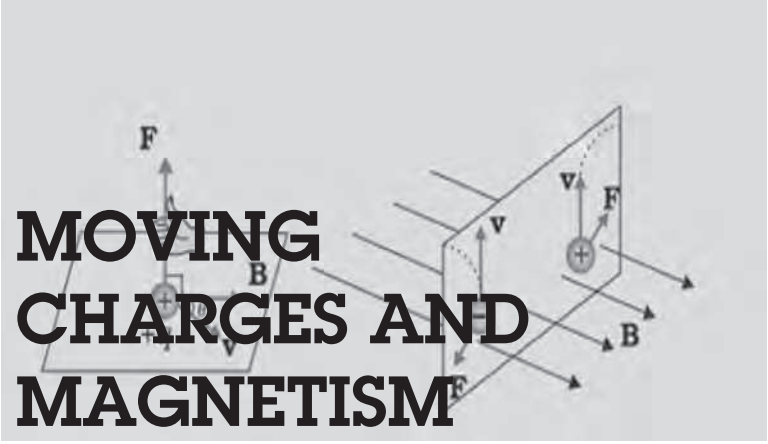
CHAPTER

4 MOVING CHARGES AND MAGNETISM



Chapter Objectives

This chapter will help you understand :

- ***Moving Charges :*** *Magnetic force on current carrying wire; Motion in a magnetic field; Velocity selector; Cyclotron; Magnetic field on axis of circular current carrying loop; Ampere's circuital law.*
- ***Magnetism :*** *Force between two parallel current carrying wires; Torque on current carrying loop in magnetic field; Moving coil galvanometer; The solenoid and the toroid.*

TOPIC - 1
Moving Charges P. 84

TOPIC - 2
Magnetism P. 97

TOPIC-1
Moving Charges

Quick Review

- Electrons cannot move along with bands which are completely full, and bands which are completely empty have no electrons to move. Hence, conduction can only take place in a band (from atom to atom at a given level) if the band is partly empty. In general, the random thermal energy inside a solid will give some electrons enough energy to jump up to the mostly empty conduction band.
- When an electron moves from conduction band to valance band there is an empty identity left. This is known as Hole.
- When drawing diagrams that shows what happens to electrons in a material or electronic device, it is simpler to ignore the individual atoms and just draw some bands in which electrons must stay.

Know the Links

- *http://www.learncbse.in/ncert-exemplar-problems-class-12-physics-moving-charges-magnetism/*
- *https://edurev.in/studytube/NCERT-Solutions--Moving-Charges-And-Magnetism--Class-12--Physics/0ff81511-78f0-4830-aceb-362b1824ffdd*

TIPS...

- *Direct derivations from topics like magnetic force on axis of circular coil, ampere's circuital law, forces between two current carrying wires and torque on current carrying loop in magnetic field are asked.*
- *Numerical from mentioned topics are most of the time simple, based on derived relations or on motion in magnetic field, torque on current carrying loop placed in magnetic field, velocity selector are asked.*

TRICKS...

- *Electrostatic field lines originate at a positive charge and terminate at a negative charge or fade at infinity. Magnetic field lines always form closed loops.*
- *Ampere's Circuital law is not independent of the Biot-Savart law. It can be derived from the Biot-Savart law. Its relationship to the Biot-Savart law is similar to the relationship between Gauss's law and Coulomb's law.*
- *The resolution of this paradox implies that electricity and magnetism are linked phenomena (electromagnetism) and that the Lorentz force expression does not imply a universal preferred frame of reference in nature.*

Multiple Choice Questions (1 Mark each)

Q. 1. Two charged particles traverse identical helical paths in a completely opposite sense in a uniform magnetic field $B = B_0\hat{k}$.

(a) They have equal z-components of momenta.

(b) They must have equal charges.

(c) They necessarily represent a particle-antiparticle pair.

(d) The charge to mass ratio satisfy : $\left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$

[NCERT Exemp. Q. 4.1, Page 22]

Ans. Correct option : (d)

Explanation : When charge/mass ratio of these two particles is same and charges on them are of opposite nature then the charged particles will traverse identical helical paths in a completely opposite sense. Therefore, option (d) is correct. $\left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$

Q. 2. Biot-Savart law indicates that the moving electrons (velocity v) produce a magnetic field B such that

(a) $B \perp$ v.

(b) B || v.

(c) it obeys inverse cube law.

(d) it is along the line joining the electron and point of observation. [NCERT Exemp. Q. 4.2, Page 22]

Ans. Correct option : (a)

Explanation : In Biot-Savart's law, magnetic field $B|| \; idl \times r$ and idl due to flow of electron is in opposite direction of v and by direction of cross product of two vectors, B.

$$dB = \frac{I.dl\sin\theta}{r^2}$$

$$or\, dB = \frac{I \times dl}{r}$$

According to Biot-Savart law, if magnetic field is not perpendicular to the motion of charge then it will not move in helical path, which is not possible for motion of a charge in magnetic field. So, the magnetic field is perpendicular to the direction of flow of charge verifies answer '*a*'.

Q. 3. A current carrying circular loop of radius R is placed in the x-y plane with centre at the origin. Half of the loop with $x > 0$ is now bent so that it now lies in the y-z plane.

(a) The magnitude of magnetic moment now diminishes.

(b) The magnetic moment does not change.

(c) The magnitude of B at (0.0.z), $z >> R$ increases.

(d) The magnitude of B at (0.0.z), $z >> R$ is unchanged.

[NCERT Exemp. Q. 4.3, Page 23]

Ans. Correct option : (a)

Explanation : For a circular loop of radius R, carrying current I in x-y plane, the magnetic moment $M = I \times \pi R_2$.

It acts perpendicular to the loop along z-direction.

When half of the current loop is bent in y-z plane, then magnetic moment due to half current loop is x-y plane, $M_1 = I(\pi R_2/2)$ acting along z-direction.

Magnetic moment due to half current loop in y-z plane, $M_2 = I(\pi R_2/2)$ along x-direction.

Net magnetic moment due to entire bent current loop,

$$M_{net} = \sqrt{M_1^2 + M_2^2}$$

$$= \sqrt{2}\,\frac{I\pi R^2}{2}$$

$$= \frac{M}{\sqrt{2}}$$

Therefore, $M_{net} < M$ or M diminishes.

Q. 4. A circular current loop of magnetic moment M is in an arbitrary orientation in an external magnetic field B. The work done to rotate the loop by 30° about an axis perpendicular to its plane is

(a) MB. **(b) $\frac{\sqrt{3}MB}{2}$.**

(c) $\frac{MB}{2}$. **(d) zero.**

[NCERT Exemp. Q. 4.6, Page 23]

Ans. Correct option : (d)

Explanation : The work done to rotate the loop in magnetic field $W = MB(\cos\theta_1 - \cos\theta_2)$.

When current carrying coil is rotated then there will be no change in angle between magnetic moment and magnetic field.

Here, $\theta_1 = \theta_2 = \alpha$

$\Rightarrow W = MB(\cos\alpha - \cos\alpha) = 0$.

Tick Two or More Options

Q. 5. Consider a wire carrying a steady current, I placed in a uniform magnetic field B perpendicular to its length. Consider the charges inside the wire. It is known that magnetic forces do no work. This implies that,

(a) motion of charges inside the conductor is unaffected by B since they do not absorb energy.

(b) some charges inside the wire move to the surface as a result of B.

(c) if the wire moves under the influence of B, no work is done by the force.

(d) if the wire moves under the influence of B, no work is done by the magnetic force on the ions, assumed fixed within the wire.

[NCERT Exemp. Q. 4.8, Page 24, Page 24]

Ans. Correct options : (b) and (d)

Explanation : Magnetic force on a conductor of length l carrying a current I placed in a uniform

magnetic field B is given by, $F = I\,(l \times B)$ or $|F| = I\,|l||\,B|\sin\theta$.

The direction of force is given by Fleming's left Hand rule and F is perpendicular to the direction of magnetic field B, Therefore, work done by the magnetic force on the ions is zero.

Q. 6. Two identical current carrying coaxial loops, carry current I in an opposite sense. A simple amperian loop passes through both of them once. Calling the loop as *C*,

(a) $\oint \boldsymbol{B.dl} = \mp 2\mu_o I.$

(b) the value of $\oint \boldsymbol{B.dl}$ is independent of sense of *C*.

(c) there may be a point on *C* where *B* and *dl* are perpendicular.

(d) *B* vanishes everywhere on *C*.

[NCERT Exemp. Q. 4.9, Page 24]

Ans. Correct options : (b) and (c)

Explanation : Applying the Ampere's circuital law, we have $\oint_c \vec{B}.\vec{dl} = \mu_0(I - I) = 0$. Hence, option (*a*) is wrong.

Also, $\oint_c \vec{B}.\vec{dl} = 0$, so, $\oint_c B.dl = 0$ is independent of sense of C. Therefore, option (b) is correct.

Again there will be a point of loop C, lying at the axis of two loops A and B, hence, $\vec{B}$ and $\vec{dl}$ are perpendicular to each other. So, option (c) is correct.

Here, option (d) is wrong because value of $\vec{B}$ does not vanish on various points of C.

Q. 7. A cubical region of space is filled with some uniform electric and magnetic fields. An electron enters the cube across one of its faces with velocity v and a positron enters via opposite face with velocity - v. At this instant,

(a) the electric forces on both the particles cause identical accelerations.

(b) the magnetic forces on both the particles cause equal accelerations.

(c) both particles gain or lose energy at the same rate.

(d) the motion of the centre of mass (*CM*) is determined by *B* alone.

[NCERT Exemp. Q. 4.10, Page 24]

Ans. Correct options : (b), (c), and (d)

Explanation : Here, magnitude of acceleration of electron and positron is same but direction is different. Hence, option (*a*) is not correct.

Here,

Magnetic force on electron, $\vec{F} = -e(\vec{v} \times \vec{B})$

Magnetic force on positron, $\vec{F'} = e(\vec{v} \times \vec{B})$

Acceleration of electron, $\vec{a} = \dfrac{\vec{F}}{m} = -e\dfrac{\left(\vec{v} \times \vec{B}\right)}{m}$

Acceleration of positron, $\vec{a'} = \dfrac{e\left(\vec{v} \times \vec{B}\right)}{m}$

Now,

Net magnetic force on electron positron pair $= -2e\left(\vec{v} \times \vec{B}\right)$

Net electric force on electron-positron pair = 0.

Q. 8. A charged particle would continue to move with a constant velocity in a region wherein,

(a) $E = 0, B \neq 0$. **(b) $E \neq 0, B \neq 0$.**

(c) $E \neq 0, B = 0$. **(d) $E = 0, B = 0$.**

[NCERT Exemp. Q. 4.11, Page 24]

Ans. Correct options : (a), (b), and (d)

Explanation : Given that,

Force on charged particle due to magnetic field, $F_m = q(v \times B)$

Force on charged particle due to electric field $F_E = qE$.

Now, $F_E = 0$ if $E = 0$

Also, $F_m = 0$ if $\sin\theta = 0$ or $\theta° = 0°$ or $180°$.

It may be also possible that resultant of $q\,E + q\,(v \times B)$ is equal to zero.

Therefore, options (a), (b), and (d) are correct.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. What can be the causes of helical motion of a charged particle?

[CBSE Board, Delhi Region, 2016]

Ans. A particle moves along a curved path if it moves in the presence of a magnetic field. Now, when there is an angular projection, that is, when charge particle is projected in a uniform magnetic field in such a way that angle between velocity and magnetic field is neither 90° nor 0°, or 180° then charge particle move on helical path. **[½]**

The vertical component (v sin θ) will be perpendicular to the field and hence, will rotate the particle in a circular path. However, the horizontal component (v cos θ) will move the particle in a straight line path. So, the cumulative motion of the particle is a helix. **[½]**

Q. 2. The wires which connect the battery of an automobile to its starting motor carry a current of 300 A (for a short time). What is the force per unit length between the wires if they are 70 cm long and 1.5 cm apart? Is the force attractive or repulsive? [NCERT Add. Ex. Q. 4.22, Page 171]

Ans. Current in both wires, $I = 300$ A

Distance between the wires, $r = 1.5$ cm $= 0.015$ m

Length of the two wires, $l = 70$ cm $= 0.7$ m

Force between the two wires is given by the relation,

$$F = \frac{\mu_0 I^2}{2\pi r}$$ **[½]**

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7}$ TmA^{-1}

$$\therefore\ F = \frac{4\pi \times 10^{-7} \times (300)^2}{2\pi \times 0.015}$$

$$= 1.2 \text{ N/m}$$

[1]

Since, the direction of the current in the wires is opposite, a repulsive force exists between them. [½]

Q. 3. What is the magnitude of magnetic force per unit length on a wire carrying a current of 8 A and making an angle of 30° with the direction of a uniform magnetic field of 0.15 T?
[NCERT Ex. Q. 4.5, Page 169]

Ans. Given that,
Current in the wire, $I = 8$ A
Magnitude of the uniform magnetic field, $B = 0.15$ T
Angle between the wire and magnetic field, $\theta = 30°$
Magnetic force per unit length on the wire is given as : $F = BI\ sin\theta$
$= 0.15 \times 8 \times 1 \times \sin 30°$
$= 0.6\ \text{Nm}^{-1}$
Hence, the magnetic force per unit length on the wire is $0.6\ \text{Nm}^{-1}$. **[2]**

Q. 4. A **3.0 cm wire carrying a current of 10 A is placed inside a solenoid perpendicular to its axis. The magnetic field inside the solenoid is given to be 0.27 T. What is the magnetic force on the wire?**
[NCERT Ex. Q. 4.6, Page 169]

Ans. Given that,
Length of the wire, $l = 3$ cm $= 0.03$ m
Current flowing in the wire, $I = 10$ A
Magnetic field, $B = 0.27$ T
Angle between the current and magnetic field, $\theta = 90°$ **[1]**
Magnetic force exerted on the wire is given as :
$F = BIlsin\theta$
$= 0.27 \times 10 \times 0.03 \sin 90°$
$= 8.1 \times 10^{-2}$ N
Hence, the magnetic force on the wire is 8.1×10^{-2} *N*. The direction of the force can be obtained from Fleming's left hand rule. **[1]**

Q. 5. A circular coil of wire consisting of 100 turns, each of radius 8.0 cm carries a current of 0.40 A. What is the magnitude of the magnetic field *B* at the centre of the coil? [NCERT Ex. Q. 4.1, Page 169]

Ans. Given that,
Number of turns on the circular coil, $n = 100$
Radius of each turn, $r = 8.0$ cm $= 0.08$ m
Current flowing in the coil, $I = 0.4$ A
Magnitude of the magnetic field at the centre of the coil is given by the relation,

$$\left|\vec{B}\right| = \frac{\mu_0}{4\pi}\frac{2\pi nI}{r}$$ **[1]**

Where,
μ_0 = Permeability of free space
$= 4\pi \times 10^{-7}$ T m A^{-1}

$$\left|\vec{B}\right| = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2\pi \times 100 \times 0.4}{r}$$
$$= 3.14 \times 10^{-4}\ T$$ **[1]**

Hence, the magnitude of the magnetic field is $3.14 \times 10^{-4}\ T$.

Q. 6. A long straight wire carries a current of 35 A. What is the magnitude of the field *B* at a point 20 cm from the wire? [NCERT Ex. Q. 4.2, Page 169]

Ans. Given that,
Current in the wire, $I = 35$ A
Distance of a point from the wire, $r = 20$ cm $= 0.2$ m
Magnitude of the magnetic field at this point is given as :

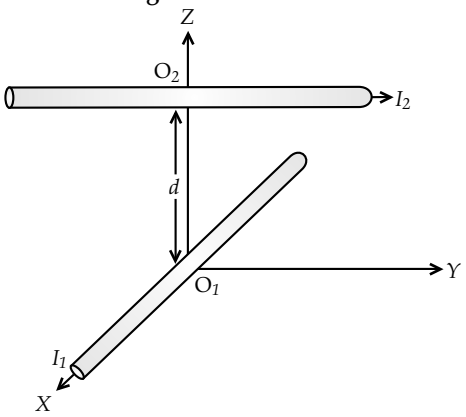
$$\left|\vec{B}\right| = \frac{\mu_0}{4\pi}\frac{2I}{r}$$ **[1]**

Where,
μ_0 = Permeability of free space
$= 4\pi \times 10^{-7}$ TmA^{-1}

$$\left|\vec{B}\right| = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2 \times 35}{0.2}$$
$$= 3.5 \times 10^{-5}\ T$$ **[1]**

Hence, the magnitude of the magnetic field at a point 20 cm from the wire is 3.5×10^{-5} T.

Q. 7. Two long wires carrying current I_1 and I_2 are arranged as shown in Figure. The one carrying current I_1 is along is the *x*-axis. The other carrying current I_2 is along a line parallel to the *y*-axis given by $x = 0$ and $z = d$. Find the force exerted at O_2 because of the wire along the *x*-axis.



[NCERT Exemp. Q. 4.16, Page 25]

Ans. At O_2, the magnetic field due to I_1 is along the *y*-axis. The second wire is along the *y*-axis so that, the angle between I_2 and B_1 is zero and hence, the force is zero. **[1]**

Q. 8. Show that a force that does no work must be a velocity dependent force.
[NCERT Exemp. Q. 4.13, Page 25]

Ans. Let no work is done by a force, so we have
$W = F\ xdl = 0$
As $dl = v\ dt$ and dt is not equal to zero
$F.v\ dt = 0$
$F.v = 0$
F must be velocity dependent which implies that angle between F and v is 90°.
If direction of v changes then direction of F will also change, so that above condition is satisfied. **[2]**

Q. 9. The magnetic force depends on v which depends on the inertial frame of reference. Does then the magnetic force differ from inertial frame to frame? Is it reasonable that the net acceleration has a different value in different frames of reference?
[NCERT Exemp. Q. 4.14, Page 25]

Ans. Magnetic force is frame dependent and it will differ from frame to frame. Net acceleration arising from this is however frame independent (non-relativistic physics) for inertial frames. **[1]**

Q. 10. A proton and an electron travelling along parallel paths enter a region of uniform magnetic field, acting perpendicular to their paths. Which of them will move in a circular path with higher frequency? [CBSE Board, Delhi Region, 2018]

Ans. From Lorentz formula we know that an electron and a proton will experience a force

$F = q\vec{v} \times \vec{B}$

Electron and proton have equal charge but of opposite polarity so they will move in a circle in opposite directions, magnetic force will provide the required centripetal force

$$\frac{mv^2}{r} = qvB\sin 90^o$$

$$\frac{mv^2}{r} = qvB$$

Therefore,

$$v = \frac{qBr}{m}$$

Time period of revolution is

$$T = \frac{2\pi r}{\omega}$$

$$T = \frac{2\pi m}{qB}$$

Which gives frequency of revolution as

$$v = \frac{qB}{2\pi m}$$

We can see that frequency of revolution is inversely proportional to mass and mass of proton is greater than the electron, so an electron will move in circular path with greater frequency. As we know charge particle enters into perpendicular magnetic field it traces circular path. **[1]**

Short Answer Type Questions

(3 or 4 marks each)

Q. 1. A circular coil of 20 turns and radius 10 cm is placed in a uniform magnetic field of 0.10 T normal to the plane of the coil. If the current in the coil is 5.0 A, what is the

(a) total torque on the coil,

(b) total force on the coil,

(c) average force on each electron in the coil due to the magnetic field?

(The coil is made of copper wire of cross-sectional area 10^{-5} m^2, and the free electron density in copper is given to be about 10^{29} m^{-3}.)

[NCERT Add. Ex. Q. 4.25, Page 172]

Ans. Number of turns on the circular coil, $n = 20$

Radius of the coil, $r = 10$ cm $= 0.1$ m

Magnetic field strength, $B = 0.10$ T

Current in the coil, $I = 5.0$ A

(a) The total torque on the coil is zero because the angle between force and the normal to the loop, is zero.

so, $\tau = BINA \sin\theta$ is zero. **[½]**

(b) The total force on the coil is zero because the field is uniform. **[½]**

(c) Cross-sectional area of copper coil, $A = 10^{-5}$ m^2

Number of free electrons per cubic meter in copper,

$$n = \frac{10^{29}}{m^3}$$

Charge on the electron, $e = 1.6 \times 10^{-19}$ C

Magnetic force, $F = Bev_d$ **[1]**

Where,

v_d = Drift velocity of electrons

$$= \frac{I}{neA}$$

$$\therefore F = \frac{Bel}{neA}$$

$$= \frac{0.10 \times 5.0}{10^{29} \times 10^{-5}} = 5 \times 10^{-25}\ N$$

Hence, the average force on each electron is 5×10^{-25} N. **[1]**

Q. 2. Answer the following questions :

(a) A magnetic field that varies in magnitude from point to point but has a constant direction (east to west) is set up in a chamber. A charged particle enters the chamber and travels undeflected along a straight path with constant speed. What can you say about the initial velocity of the particle?

(b) A charged particle enters an environment of a strong and non-uniform magnetic field varying from point to point both in magnitude and direction, and comes out of it following a complicated trajectory. Would its final speed equal the initial speed if it suffered no collisions with the environment?

(c) An electron travelling west to east enters a chamber having a uniform electrostatic field in north to south direction. Specify the direction in which a uniform magnetic field should be set up to prevent the electron from deflecting from its straight line path.

[NCERT Add. Ex. Q. 4.18, Page 170]

Ans. **(a)** The initial velocity of the particle is either parallel or anti-parallel to the magnetic field. Hence, it travels along a straight path without suffering any deflection in the field. **[1]**

(b) Yes, the final speed of the charged particle will be equal to its initial speed. This is because magnetic force can change the direction of velocity, but not its magnitude. **[1]**

(c) An electron travelling from West to East enters a chamber having a uniform electrostatic field in the North-South direction. This moving electron can remain undeflected if the electric force acting on

it is equal and opposite of electric force. Electric force is directed towards the South. According to Fleming's left hand rule, magnetic field should be applied in a vertically downward direction. **[1]**

Q. 3. (a) A circular coil of 30 turns and radius 8.0 cm carrying a current of 6.0 A is suspended vertically in a uniform horizontal magnetic field of magnitude 1.0 T. The field lines make an angle of 60° with the normal of the coil. Calculate the magnitude of the counter torque that must be applied to prevent the coil from turning.

(b) Would your answer change, if the circular coil in (a) were replaced by a planar coil of some irregular shape that encloses the same area? (All other particulars are also unaltered.)

[NCERT Ex. Q. 4.13, Page 169]

Ans. Number of turns on the circular coil, $n = 30$

Radius of the coil, $r = 8.0$ cm $= 0.08$ m

Area of the coil $\pi r^2 = \pi (0.08)^2 = 0.0201$ m^2

Current flowing in the coil, $I = 6.0$ A

Magnetic field strength, $B = 1$ T

Angle between the field lines and normal with the coil surface, $\theta = 60°$ **[1]**

The coil experiences a torque in the magnetic field. Hence, it turns. The counter torque applied to prevent the coil from turning is given by the relation,

$\tau = n\,IBA\,sin\theta$

$= 30 \times 6 \times 1 \times 0.0201 \times \sin 60°$

$= 3.133$ N m **[1]**

(b) It can be inferred from relation $\tau = n\,IBA sin\theta$ that the magnitude of the applied torque is not dependent on the shape of the coil. It depends on the area of the coil. Hence, the answer would not change if the circular coil in the above case is replaced by a planar coil of some irregular shape that encloses the same area. **[1]**

Q. 4. A long straight wire in the horizontal plane carries a current of 50 A in north to south direction. Give the magnitude and direction of *B* at a point 2.5 m east of the wire. [NCERT Ex. Q. 4.3, Page 169]

Ans. Current in the wire, $I = 50$ A

A point is 2.5 m away from the East of the wire.

∴ Magnitude of the distance of the point from the wire, $r = 2.5$ m.

Magnitude of the magnetic field at that point is given by the relation,

$$\left|\vec{B}\right| = \frac{\mu_0}{4\pi}\frac{2I}{r}$$ **[1]**

Where,

μ_0 = Permeability of free space

$= 4\pi \times 10^{-7}$ T m A^{-1}

$$\left|\vec{B}\right| = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2 \times 50}{2.5} = 4 \times 10^{-6} \text{ T}$$ **[1]**

The point is located normal to the wire length at a distance of 2.5 m. The direction of the current in the wire is vertically downward. Hence, according to the Maxwell's right hand thumb rule, the direction of the magnetic field at the given point is vertically upward. **[1]**

Q. 5. A horizontal overhead power line carries a current of 90 A in east to west direction. What is the magnitude and direction of the magnetic field due to the current 1.5 m below the line?

[NCERT Ex. Q. 4.4, Page 169]

Ans. Current in the power line, $I = 90$ A

Point is located below the power line at distance, $r = 1.5$ m

Hence, magnetic field at that point is given by the relation,

$$\left|\vec{B}\right| = \frac{\mu_0}{4\pi}\frac{2I}{r}$$ **[1]**

Where,

μ_0 = Permeability of free space

$= 4\pi \times 10^{-7}$ TmA^{-1}

$$\left|\vec{B}\right| = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2 \times 90}{1.5}$$

$= 1.2 \times 10^{-5}$ T **[1]**

The current is flowing from East to West. The point is below the power line. Hence, according to Maxwell's right hand thumb rule, the direction of the magnetic field is towards the South. **[1]**

Q. 6. Two long and parallel straight wires *A* and *B* carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire *A*.

[NCERT Ex. Q. 4.7, Page 169]

Ans. Current flowing in wire A, $I_A = 8.0$ A

Current flowing in wire B, $I_B = 5.0$ A

Distance between the two wires, $r = 4.0$ cm $= 0.04$ m

Length of a section of wire A, $L = 10$ cm $= 0.1$ m

Force exerted on length L due to the magnetic field is given as :

$$F = \frac{\mu_0 I_A I_B L}{2\pi r}$$ **[1]**

Where, μ_0 = Permeability of free space $= 4\pi \times 10^{-7}$ T m A^{-1}

$$F = \frac{4\pi \times 10^{-7} \times 8 \times 5 \times 0.1}{2\pi \times 0.04} = 2 \times 10^{-5} N$$ **[1]**

The magnitude of force is 2×10^{-5} N. This is an attractive force normal to A towards B because the direction of the currents in the wires is the same. **[1]**

Q. 7. A current carrying loop consists of 3 identical quarter circles of radius *R*, lying in the positive quadrants of the *x-y*, *y-z*, and *z-x* planes with their centres at the origin, joined together. Find the direction and magnitude of B at the origin.

[NCERT Exemp. Q. 4.17, Page 25]

Ans. Magnetic field at origin due to the quarter circle lying on the x - y plane is

$$\vec{B}_{xy} = \frac{\mu_0 I\left(\frac{\pi}{2}\right)}{4\pi R}\hat{k} = \left(\frac{\mu_0}{4}\right)\frac{I}{2R}\hat{k}$$

Similarly, Magnetic field at origin due to the quarter circle lying on the y - z plane is

$$\vec{B}_{yz} = \left(\frac{\mu_0}{4}\right)\frac{I}{2R}\hat{i}$$

Similarly, Magnetic field at origin due to the quarter circle lying on the z - x plane is

$$\vec{B}_{zx} = \left(\frac{\mu_0}{4}\right)\frac{I}{2R}\hat{j}$$

$$\vec{B} = \vec{B}_{xy} + \vec{B}_{yz} + \vec{B}_{zx}$$

$$\Rightarrow \vec{B} = \left(\frac{\mu_0}{4\pi}\frac{I}{R}\frac{\pi}{2}\right)\hat{k} + \left(\frac{\mu_0}{4\pi}\frac{I}{R}\frac{\pi}{2}\right)\hat{i} + \left(\frac{\mu_0}{4\pi}\frac{I}{R}\frac{\pi}{2}\right)\hat{j}$$

$$\Rightarrow \vec{B} = \left(\frac{\mu_0}{8}\frac{I}{R}\right)(\hat{i} + \hat{j} + \hat{k}). \quad \textbf{[3]}$$

Q. 8. A charged particle of charge *e* and mass *m* is moving in an electric field *E* and magnetic field *B*. Construct dimensionless quantities and quantities of dimension [T]$^{-1}$.

[NCERT Exemp. Q. 4.18, Page 25]

Ans. If a charged particle is moving in electric and magnetic field, we cannot construct any dimensionless quantity with these physical quantities. For a charged particle moving perpendicular to the magnetic field, the magnetic Lorentz forces provides necessary centripetal force for revolution. **[1]**

$$\frac{mv^2}{R} = evB$$

On simplifying the terms, we have

$$\therefore \frac{eB}{m} = \frac{v}{R} = \omega$$

We know that

$$B = \frac{F}{ev} = \frac{\left[MLT^{-2}\right]}{\left[AT\right]\left[LT^{1}\right]} = \left[MA^{-1}T^{-2}\right]$$

and then dimensional formula of angular frequency

$$\therefore [\omega] = \left[\frac{eB}{m}\right] = \left[\frac{v}{R}\right]$$

$$[\omega] = \frac{\left[AT\right]\left[AM^{-1}T^{-2}\right]}{M} = [\mathrm{T}^{-1}]$$

[2]

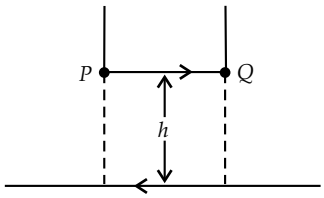
No dimensionless quantity can be constructed using given quantities. **[1]**

Q. 9. An electron enters with a velocity v = v $\hat{i}$ into a cubical region (faces parallel to coordinate planes) in which there are uniform electric and magnetic fields. The orbit of the electron is found to spiral down inside the cube in plane parallel to the *x-y* plane. Suggest a configuration of fields *E* and *B* that can lead to it.

[NCERT Exemp. Q. 4.19, Page 25]

Ans. In the given question, orbit of electron is spiral down inside the cube in a plane parallel to x-y plane, so, direction of magnetic field is along $+z$ direction. Therefore, $B = B_o\hat{k}$. Also, electric field will be along positive x-axis, that is, $E = E_o\hat{i}$, where $E_o > 0$. **[3]**

Q. 10. A long straight wire carrying current of 25 A rests on a table as shown in Figure. Another wire *PQ* of length 1 m, mass 2.5 g carries the same current but in the opposite direction. The wire *PQ* is free to slide up and down. To what height will *PQ* rise?



[NCERT Exemp. Q. 4.22, Page 26]

Ans. In equilibrium state,

$$mg = \frac{\mu_0}{4\pi}\frac{2I_1I_2}{h}l$$

[1]

According to the question,

$$h = \frac{\mu_0}{4\pi}\frac{2I_1I_2}{mg}l$$

$$h = \frac{\mu_0}{4\pi}\frac{10^{-7}\times 2\times 25\times 25\times 1}{(2.5\times 10^{-3})\times 9.8} = 51\times 10^{-4} = 0.51 \text{ cm}.$$

[2]

Q. 11. State Ampere's circuital law. Use this law to find magnetic field due to straight infinite current carrying wire. How are the magnetic field lines different from the electrostatic field lines?

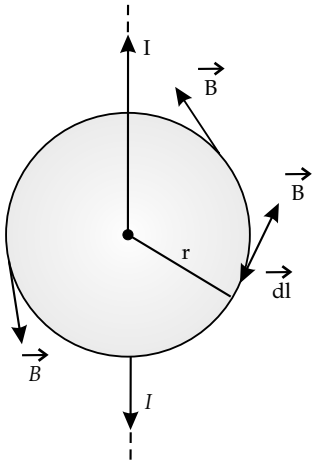
[CBSE Board, Delhi Region/All India Region, 2016]

Ans. Ampere's circuital law states that the line integral of the magnetic field B around any closed circuit is equal to $\mu_{o=}$ (permeability constant) times the total current I passing through the circuit

$\int B.dl = \mu_0 I$

Magnetic field due to straight infinite current carrying wire :

The figure below shows a straight infinite current carrying wire around which a circular loop of radius r is shown.



[½]

The field lines are circular, and hence, the field B at any point of the circular loop is directed along the tangent to the circle at that point.

The magnitude of field B is the same at every point of the circular loop.

Therefore, we get

$\int B\cdot dl = \int B\cdot dl\cos\theta = \int B\cdot dl\cos 0 = B\int dl = B(2\pi r)$

Now, from Ampere's circuital law, the above equation is

$$\therefore B = \frac{\mu_o I}{2\pi r}$$

[½]

Difference between magnetic field lines and electrostatic field lines :

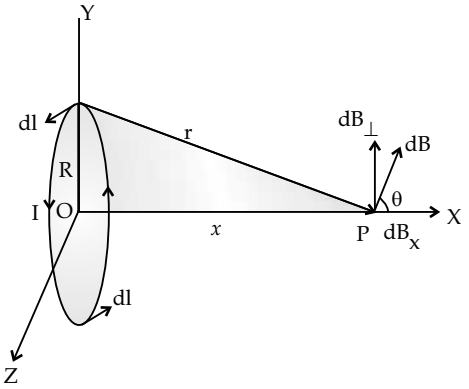
Magnetic field lines form closed loops, whereas electrostatic field lines do not. The magnetic field lines are produced by moving charges, whereas the electrostatic field lines are produced by static positive or negative charges. **[2]**

Q. 12. Use Biot-Savart law to derive the expression for the magnetic field on the axis of a current carrying circular loop of radius *R*.

Draw the magnetic field lines due to a circular wire carrying current *I*.

[CBSE Board, All India Region, 2016]

Ans.



[½]

I = Current in the loop

R = Radius of the loop

X-axis = Axis of the loop

x = Distance between O and P

dl = Conducting element of the loop

According to the Biot–Savart law, the magnetic field at P is

$$dB = \frac{\mu_0 I}{4\pi}\frac{|dI \times r|}{r^3}$$

$$R^2 = x^2 + R^2$$

$|dI \times r| = rdI$ (Because they are perpendicular)

$$dB = \frac{\mu_0}{4\pi}\frac{IdI}{(x^2 + R^2)}$$

dB has two components : dB_x and $dB_\perp$.

$dB \perp$ is cancelled out and only the x-component remains.

Therefore, $dB_x = dB\cos\theta$ **[½]**

$$\cos\theta = \frac{R}{\left(x^2 + R^2\right)^{1/2}}$$

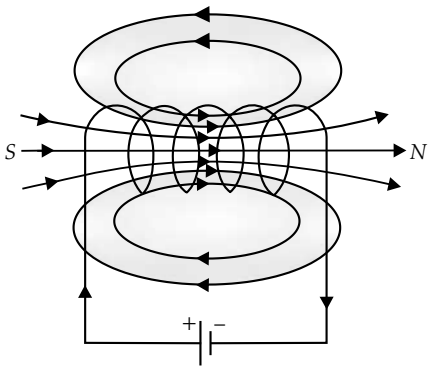
$$\therefore\ dB_x = \frac{\mu_0 IdI}{4\pi}\frac{R}{(x^2 + R^2)^{3/2}}$$

Summation of dl over the loop is given by $2\pi R$.

$$\therefore B = B = B_x\hat{i} = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}\hat{i}$$

[1]

Magnetic field lines due to circular wire carrying current I.



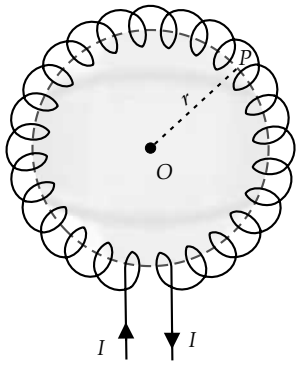
[1]

Q. 13. An iron ring of relative permeability μ_r has windings of insulated copper wire of *n* turns per metre.

When the current in the windings is *I*, find the expression for the magnetic field in the ring. [2]

[CBSE Board, Delhi Region, 2018]

Ans.



[2]

Applying Ampere's Law

$$\oint \vec{B}.d\vec{\ell} = \mu_0 \Sigma i$$

$$\oint B.d\ell\cos 0° = \mu_0 \Sigma i$$

$$\oint d\ell = \mu_0 \Sigma i \qquad [\cos\ 0° = 1]$$

$$B.2\pi r = \mu_0[n.2\pi r.i]$$

$$B = \mu_0 ni \qquad \mu_r = \frac{\mu_m}{\mu_0}$$

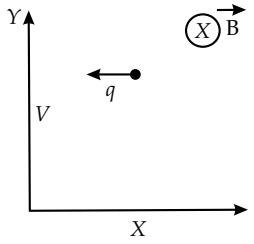
$B = \mu_r.\mu_0 ni$ $\quad \mu_m$(permeability of iron) $= \mu_r.\mu_0$

$\boxed{B = \mu_r.\mu_0 ni}$

Q. 14. (a) A point charge *q* moving with speed *v* enters a uniform magnetic field *B* that is acting into the plane of the paper as shown. What is the path followed by the charge *q* and in which plane does it move?

(b) How does the path followed by the charge get affected if its velocity has a component parallel to $\vec{B}$?

(c) If an electric field $\vec{E}$ is also applied such that the particle continues moving along the original straight line path, what should be the magnitude and direction of the electric field $\vec{E}$?



[CBSE Board, Foreign Scheme, 2016]

Ans. **(a)** When a charged particle having charge q moves inside a magnetic field $\vec{B}$ with velocity v, it experiences a force $\vec{F} = q(\vec{v} \times \vec{B})$.
When $\vec{v}$ is perpendicular to $\vec{B}$, the force $\vec{F}$ on the charged particle acts as the centripetal force and makes it move along a circular path.
The point charge travels in the plane perpendicular to both $\vec{v}$ and $\vec{B}$. **[1]**

(b) If a component of velocity of the charge particle is parallel to the direction of the force of the electric field, then the force experienced due to that component will be zero, because
$F = qvB\sin 0° = 0$, and particle will move in a straight line. Also, the force experienced by the component perpendicular to $\vec{B}$ moves the particle in a circular path. The combined effect of both the components will move the particle in a helical path. **[1]**

(c) The direction of the magnetic force is along negative Y-axis and so, the direction of electric force should be along the positive Y-axis to counter balance the magnetic force and charge particle will move in the straight line path.
Therefore, the direction of electric field is along the positive Y-axis and its magnitude is given by $E = vB$. **[1]**

Long Answer Type Questions **(5 marks each)**

Q. 1. A uniform magnetic field of 1.5 T exists in a cylindrical region of radius 10.0 cm, its direction parallel to the axis along east to west. A wire carrying current of 7.0 A in the north to south direction passes through this region. What is the magnitude and direction of the force on the wire if,
(a) the wire intersects the axis,
(b) the wire is turned from *N-S* to northeast-northwest direction,
(c) the wire in the *N-S* direction is lowered from the axis by a distance of 6.0 cm?
[NCERT Add. Ex. Q. 4.23, Page 171]

Ans. Magnetic field strength, B = 1.5 T
Radius of the cylindrical region, r = 10 cm = 0.1 m
Current in the wire passing through the cylindrical region, I = 7 A

(a) If the wire intersects the axis, then the length of the wire is the diameter of the cylindrical region.
Thus, $l = 2r = 0.2$ m
Angle between magnetic field and current, θ = 90°
Magnetic force acting on the wire is given by the relation,
$F = BIl \sin\theta$
$= 1.5 \times 7 \times 0.2 \times \sin 90°$
$= 2.1\ N$
Hence, a force of 2.1 N acts on the wire in a vertically downward direction. **[1]**

(b) New length of the wire after turning it to the Northeast-Northwest direction can be given as :
$$l_1 = \frac{l}{\sin\theta}$$
Angle between magnetic field and current, θ = 45°
Force on the wire,
$F = BIl_1 \sin\theta$
$= BIl$
$= 1.5 \times 7 \times 0.2$
$= 2.1\ N$
Hence, a force of 2.1 N acts vertically downward on the wire. This is independent of angle θ because l sin θ is fixed. **[2]**

(c) The wire is lowered from the axis by distance, d = 6.0 cm
Let l_2 be the new length of the wire.
$$\therefore \quad \left(\frac{l_2}{2}\right)^2 = 4(d + r)$$
$$= 4(10 + 6) = 4(16)$$
$$\therefore \quad l_2 = 8 \times 2 = 16 \text{ cm} = 0.16 \text{ m}$$
Magnetic force exerted on the wire,
$F_2 = BIl_2$
$= 1.5 \times 7 \times 0.16$
$= 1.68\ N$
Hence, a force of 1.68 N acts in a vertically downward direction on the wire. **[2]**

Q. 2. A magnetic field set up using Helmholtz coils is uniform in a small region and has a magnitude of 0.75 T. In the same region, a uniform electrostatic field is maintained in a direction normal to the common axis of the coils. A narrow beam of (single species) charged particles all accelerated through 15 kV enters this region in a direction perpendicular to both the axis of the coils and the electrostatic field. If the beam remains undeflected when the electrostatic field is 9.0 × 10^{-5} V m^{-1}, make a simple guess as to what the beam contains. Why is the answer not unique?
[NCERT Add. Ex. Q. 4.20, Page 171]

Ans. Magnetic field, B = 0.75 T
Accelerating voltage, V = 15 kV = 15 × 10^3 V
Electrostatic field, $E = 9 \times 10^5\ \text{Vm}^{-1}$
Mass of the electron = m
Charge of the electron = e
Velocity of the electron = v
Kinetic energy of the electron = eV

$$\frac{1}{2}mv^2 = eV$$ **[1]**

$$\therefore \frac{e}{m} = \frac{v^2}{2V} \quad ...(1)$$

Since, the particle remains undeflected by electric and magnetic fields, we can infer that the electric field is balancing the magnetic field.

$$\therefore eE = evB$$

$$v = \frac{E}{B} \quad ...(2)$$ **[1]**

Putting equation (2) in equation (1), we get

$$\frac{e}{m} = \frac{1}{2}\frac{\left(\frac{E}{B}\right)^2}{V} = \frac{E^2}{2VB^2}$$

$$= \frac{\left(9.2\times10^5\right)^2}{2\times15000\times\left(0.75\right)^2} = 4.8\times10^7 \text{ C/kg}$$ **[2]**

This value of specific charge e/m is equal to the value of deuteron or deuterium ions.

This is not a unique answer. Other possible answers are He^{++}, Li^{++}, etc. **[1]**

Q. 3. An electron emitted by a heated cathode and accelerated through a potential difference of 2.0 kV, enters a region with uniform magnetic field of 0.15 T. Determine the trajectory of the electron if the field

(a) is transverse to its initial velocity,

(b) makes an angle of 30° with the initial velocity.

[NCERT Add. Ex. Q. 4.19, Page 171]

Ans. Magnetic field strength, $B = 0.15$ T

Charge on the electron, $e = 1.6 \times 10^{-19}$ C

Mass of the electron, $m = 9.1 \times 10^{-31}$ kg

Potential difference, $V = 2.0$ kV $= 2 \times 10^3$ V

Thus, kinetic energy of the electron $= eV$

$$\Rightarrow eV = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2eV}{m}} \quad ...(i)$$ **[1]**

Where,

v = velocity of the electron

(a) Magnetic force on the electron provides the required centripetal force of the electron.

Hence, the electron traces a circular path of radius r.

Magnetic force on the electron is given by the relation, Bev.

Centripetal force,

$$= \frac{mv^2}{r}$$

$$\therefore Bev = \frac{mv^2}{r}$$

$$r = \frac{mv}{Be} \quad ...(2)$$

From equations (1) and (2), we get

$$r = \frac{m}{Be}\left[\frac{2eV}{m}\right]^{\frac{1}{2}}$$

$$= \frac{9.1\times10^{-31}}{0.15\times1.6\times10^{-19}}\times\left(\frac{2\times1.6\times10^{-19}\times2\times10^3}{9\times10^{-31}}\right)^{\frac{1}{2}}$$

$$= 100.55\times10^{-5}$$

$$= 1.01\times10^{-3} \text{ m}$$

$$= 1 \text{ mm}$$ **[2]**

Hence, the electron has a circular trajectory of radius 1.0 mm normal to the magnetic field.

(b) When the field makes an angle θ of 30° with initial velocity, the initial velocity will be, $v_1 = v \cos\theta$

From equation (2), we can write the expression for new radius as :

$$r_1 = \frac{mv_1}{Be}$$

$$= \frac{mv\sin\theta}{Be}$$

$$= \frac{9.1\times10^{-31}}{0.15\times1.6\times10^{-19}}\times\left[\frac{2\times1.6\times10^{-19}\times2\times10^3}{9\times10^{-31}}\right]^{\frac{1}{2}}\times\sin30°$$

$$= 0.5\times10^{-3} \text{ m}$$

$$= 0.5 \text{ mm}$$ **[2]**

Hence, the electron has a helical trajectory of radius 0.5 mm along the magnetic field direction.

Q. 4. For a circular coil of radius R and N turns carrying current I, the magnitude of the magnetic field at a point on its axis at a distance x from its centre is given by,

$$B = \frac{\mu_0 IR^2N}{2(x^2+R^2)^{\frac{3}{2}}}$$

(a) Show that this reduces to the familiar result for field at the centre of the coil.

(b) Consider two parallel co-axial circular coils of equal radius R, and number of turns N, carrying equal currents in the same direction, and separated by a distance R. Show that the field on the axis around the mid- point between the coils is uniform over a distance that is small as compared to R, and is given by,

$$B = 0.72\frac{\mu_0 NI}{R}$$ **, approximately.**

[Such an arrangement to produce a nearly uniform magnetic field over a small region is known as Helmholtz coils.]

[NCERT Add. Ex. Q. 4.16, Page 170]

Ans. Radius of circular coil $= R$

Number of turns on the coil $= N$

Current in the coil $= I$

Magnetic field at a point on its axis at distance x is given by the relation,

$$B = \frac{\mu_0 IR^2N}{2(x^2+R^2)^{\frac{3}{2}}}$$ **[½]**

Where, μ_0 = Permeability of free space $= 4\pi\times10^{-7}$ T m A^{-1}

(a) If the magnetic field at the centre of the coil is considered, then $x = 0$.

$$\therefore\ B=\frac{\mu_0 IR^2N}{2R^3}=\frac{\mu_0 IN}{2R}$$

This is the familiar result for magnetic field at the centre of the coil. **[½]**

(b) Radii of two parallel co-axial circular coils = R
Number of turns on each coil = N
Current in both coils = I
Distance between both the coils = R
Let us consider point Q at distance d from the centre.
Then, one coil is at a distance of $\frac{R}{2}+d$ from point Q.

Magnetic field at a point Q is given as :

$$\therefore\ B_1=\frac{\mu_0 NIR^2}{2\left[\left(\frac{R}{2}+d\right)^2+R^2\right]^{\frac{3}{2}}}$$ **[½]**

Also, the other coil is at a distance of $\frac{R}{2}-d$ from point Q.
Magnetic field due to this coil is given as :

$$\therefore\ B_2=\frac{\mu_0 NIR^2}{2\left[\left(\frac{R}{2}-d\right)^2+R^2\right]^{\frac{3}{2}}}$$ **[½]**

Total magnetic field,

$$B=B_1+B_2$$

$$=\frac{\mu_2 IR^2}{2}\left[\begin{array}{l}\left\{\left(\left(\frac{R}{2}-d\right)^2+R^2\right)\right\}^{-\frac{3}{2}}\\ +\left\{\left(\frac{R}{2}+d\right)^2+R^2\right\}^{-\frac{3}{2}}\end{array}\right]$$

$$=\frac{\mu_2 IR^2}{2}\left[\begin{array}{l}\left(\frac{5R^2}{4}+d^2-Rd\right)^{-\frac{3}{2}}\\ +\left(\frac{5R^2}{4}+d^2+Rd\right)^{-\frac{3}{2}}\end{array}\right]$$

$$=\frac{\mu_2 IR^2}{2}\times\left(\frac{5R^2}{4}\right)^{-\frac{3}{2}}\left[\begin{array}{l}\left(1+\frac{4}{5}\frac{d^2}{R^2}-\frac{4}{5}\frac{d}{R}\right)^{-\frac{3}{2}}\\ +\left(1+\frac{4}{5}\frac{d^2}{R^2}+\frac{4}{5}\frac{d}{R}\right)^{-\frac{3}{2}}\end{array}\right]$$ **[1]**

For $d << R$, neglecting the factor $\frac{d^2}{R^2}$, we get :

$$\approx\frac{\mu_0 IR^2}{2}\times\left(\frac{5R^2}{4}\right)^{-\frac{3}{2}}\times\left[\left(1-\frac{4d}{5R}\right)^{-\frac{3}{2}}+\left(1+\frac{4d}{5R}\right)^{-\frac{3}{2}}\right]$$

$$\approx\frac{\mu_0 IR^2N}{2R^3}\times\left(\frac{4}{5}\right)^{\frac{3}{2}}\left[1-\frac{6d}{5R}+1+\frac{6d}{5R}\right]$$

$$B=\left(\frac{4}{5}\right)^{\frac{3}{2}}\frac{\mu_0 IN}{R}=0.72\left(\frac{\mu_0 IN}{R}\right)$$ **[2]**

Hence, it is proved that the field on the axis around the mid-point between the coils is uniform.

Q. 5. Two concentric circular coils X and Y of radii 16 cm and 10 cm, respectively, lie in the same vertical plane containing the north to south direction. Coil X has 20 turns and carries a current of 16 A; coil Y has 25 turns and carries a current of 18 A. The sense of the current in X is anticlockwise, and clockwise in Y, for an observer looking at the coils facing west. Give the magnitude and direction of the net magnetic field due to the coils at their centre. [NCERT Add. Ex. Q. 4.14, Page 170]

Ans. Radius of coil X, r_1 = 16 cm = 0.16 m
Radius of coil Y, r_2 = 10 cm = 0.1 m
Number of turns of on coil X, n_1 = 20
Number of turns of on coil Y, n_2 = 25
Current in coil X, I_1 = 16 A
Current in coil Y, I_2 = 18 A
Magnetic field due to coil X at their centre is given by the relation,

$$B_1=\frac{\mu_0 n_1 I_1}{2r_1}$$ **[½]**

Where, μ_0 = Permeability of free space = $4\pi\times10^{-7}$ T m A^{-1}

$$B_1=\frac{4\pi\times10^{-7}\times20\times16}{2\times0.16}=4\pi\times10^{-4}\,T$$ (towards East) **[1½]**

Magnetic field due to coil Y at their centre is given by the relation,

$$B_2=\frac{\mu_0 n_2 I_2}{2r_2}$$ **[½]**

Where, μ_0 = Permeability of free space = $4\pi\times10^{-7}$ T m A^{-1}

$$B_1=\frac{4\pi\times10^{-7}\times25\times18}{2\times0.10}=9\pi\times10^{-4}\,T$$ (Towards West) **[½]**

Hence, net magnetic field can be obtained as :

$$B=B_2-B_1=9\pi\times10^{-4}\,T-4\pi\times10^{-4}\,T$$
$$=5\pi\times10^{-4}\,T$$
$$=5\times3.14\times10^{-4}\,T$$
$$=1.57\times10^{-3}\,T$$ (Towards West) **[2]**

Q. 6. In a chamber, a uniform magnetic field of 6.5 G (1 G = 10^{-4} T) is maintained. An electron is shot into the field with a speed of 4.8 × 10^6 m s^{-1} normal to the field. Explain why the path of the electron is a circle. Determine the radius of the circular orbit. (e =1.6 × 10^{-19} C, m_e = 9.1×10^{-31} kg)
[NCERT Ex. Q. 4.11, Page 169]

Ans. Magnetic field strength, B = 6.5 G = 6.5 × 10^{-4} T
Speed of the electron, v = 4.8 × 10^6 m/s
Charge on the electron, e = 1.6 × 10^{-19} C
Mass of the electron, m_e = 9.1 × 10^{-31} kg
Angle between the shot electron and magnetic field, θ = 90°
Magnetic force exerted on the electron in the magnetic field is given as : $F = evB\ sin\theta$
This force provides centripetal force to the moving electron. Hence, the electron starts moving in a circular path of radius r.

Hence, centripetal force exerted on the electron,

$$F_e = \frac{mv^2}{r}$$ **[1]**

In equilibrium, the centripetal force exerted on the electron is equal to the magnetic force, that is,

$$F_e = F$$

$$\Rightarrow \quad \frac{mv^2}{r} = evB \sin\theta$$

$$\Rightarrow \quad r = \frac{mv}{eB\sin\theta}$$ **[2]**

So,

$$r = \frac{9.1\times10^{-31}\times4.8\times10^{6}}{6.5\times10^{-4}\times1.6\times10^{-19}\times\sin 90^\circ} = 4.2\times10^{-2}\text{m}$$

$$= 4.2 \text{ m}$$ **[2]**

Hence, the radius of the circular orbit of the electron is 4.2 cm.

Q. 7. In the question 6, obtain the frequency of revolution of the electron in its circular orbit. Does the answer depend on the speed of the electron? Explain. [NCERT Ex. Q. 4.12, Page 169]

Ans. Magnetic field strength, $B = 6.5 \times 10^{-4}$ T

Charge of the electron, $e = 1.6 \times 10^{-19}$ C

Mass of the electron, $m_e = 9.1 \times 10^{-31}$ kg

Velocity of the electron, v = 4.8×10^6 m/s

Radius of the orbit, $r = 4.2$ cm = 0.042 m

Frequency of revolution of the electron = ν

Angular frequency of the electron = $\omega = 2\pi\nu$

Velocity of the electron is related to the angular frequency as : $v = r\omega$

In the circular orbit, the magnetic force on the electron is balanced by the centripetal force.

Hence, we can write :

$$\frac{mv^2}{r} = evB$$ **[1]**

$$\Rightarrow eB = \frac{mv}{r} = \frac{m(r\omega)}{r} = \frac{m(r\cdot 2\pi v)}{r}$$

$$\Rightarrow v = \frac{Be}{2\pi m}$$ **[1]**

This expression for frequency is independent of the speed of the electron. On substituting the known values in this expression, we get the frequency as :

$$v = \frac{6.5\times10^{-4}\times1.6\times10^{-19}}{2\times3.14\times9.1\times10^{-31}} = 1.82\times16^{6}\ Hz \approx 18\ MHz$$ **[2]**

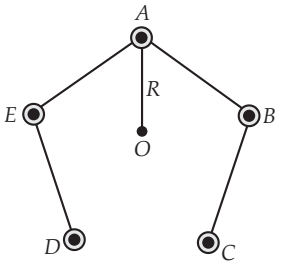
Hence, the frequency of the electron is around 18 MHz and is independent of the speed of the electron. **[1]**

Q. 8. Five long wires *A*, *B*, *C*, *D*, and E, each carrying current *I* are arranged to form edges of a pentagonal prism as shown in the Figure. Each carries current out of the plane of paper.

(a) What will be magnetic induction at a point on the axis *O*? Axis is at a distance R from each wire.

(b) What will be the field if current in one of the wires (say *A*) is switched off?

(c) What if current in one of the wire (say *A*) is reversed?



[NCERT Exemp. Q. 4.29, Page 27]

Ans. **(a)** Suppose the five wires *A*, *B*, *C*, *D*, and *E* be perpendicular to the plane of paper at locations as shown in figure.

Thus, magnetic field induction due to five wires will be represented by various sides of a closed pentagon in one order, lying in the plane of paper. So, its value is zero. **[1]**

(b) Since, the vector sum of magnetic field produced by each wire at *O* is equal to 0. Therefore, magnetic induction produced by one current carrying wire is equal in magnitude of resultant of four wires and opposite in direction.

Therefore, the field if current in one of the wires (say A) is switched off is $\frac{\mu_0 I}{2\pi R}$ perpendicular to *AO* towards left. **[2]**

(c) If current in wire *A* is reversed, then total magnetic field induction at *O* = Magnetic field induction due to wire *A* + magnetic field induction due to wires *B*, *C*, *D*, and *E*.

$$= \frac{\mu_0 2I}{\pi R} + \frac{\mu_0 2I}{\pi R}$$

$\frac{\mu_0 I}{\pi R}$ perpendicular to *AO* towards left. **[2]**

Q. 9. An electron and a positron are released from (0, 0, 0) and (0, 0, 1.5R), respectively, in a uniform magnetic field $B = B\,\hat{i}$, each with an equal momentum of magnitude $p = e\,BR$. Under what conditions on the direction of momentum will the orbits be non-intersecting circles?

[NCERT Exemp. Q. 4.25, Page 26]

Ans. Here, *B* is along the x axis, for a circular orbit the momenta of the two particles are in the *y* - *z* plane. Let p_1 and p_2 be the momentum of the electron and positron, respectively. Both of them define a circle of radius *R*. They shall define circles of opposite sense. **[1]**

Suppose p_1 makes an angle θ with the $y-axis$, p_2 must make the same angle. The centres of the respective circles must be perpendicular to the momenta and at a distance *R*. Let the centre of the electron be at C_e and of the positron at C_p.

The coordinates of C_e are

$C_e \equiv (0, -R\sin\theta, R\cos\theta)$

The coordinates of C_p are

$$C_p \equiv \left[0, -R\sin\theta, \left(\frac{3}{2}\right)R - R\cos\theta\right]$$

The circles of the two shall not overlap, if the distances between the two centres are greater than 2*R*. **[2]**

Let d be the distance between C_p and C_e.

$$d^2 = \left[R\sin\theta - \left(-R\sin\theta\right)\right]^2 + \left[R\cos\theta - \left(\frac{3}{2}R - R\cos\theta\right)\right]^2$$

Then, $d^2 = \left(2R\sin\theta\right)^2 + \left(-\frac{3}{2}R - 2R\cos\theta\right)^2$

$$= 4R^2\sin^2\theta + \frac{9}{4}R^2 - 6R^2\cos\theta + 4R^2\cos^2\theta$$

$$= 4R^2 + \frac{9}{4}R^2 - 6R^2\cos\theta$$ **[1]**

Since, d has to be greater than $2R$

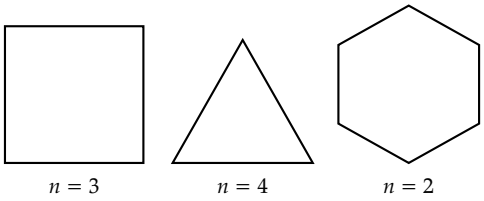
$$d^2 > 4R^2$$

$$\Rightarrow 4R^2 + \frac{9}{4}R^2 - 6R^2\cos\theta > 4R^2$$

$$\Rightarrow \frac{9}{4} > 6\cos\theta \text{ or } \cos\theta < \frac{3}{8}$$ **[1]**

Q. 10. A uniform conducting wire of length 12*a* and resistance *R* is wound up as a current carrying coil in the shape of (i) an equilateral triangle of side *a*; (ii) a square of sides a and, (iii) a regular hexagon of sides a. The coil is connected to a voltage source *V*. Find the magnetic moment of the coils in each case. [NCERT Exemp. Q. 4.26, Page 26]

Ans.



[1]

Area of equilateral triangle: $A = \frac{\sqrt{3}}{4}a^2$

Area of a square $A = a^2$

Area of regular hexagon $A = \frac{3\sqrt{3}}{2}a^2$ **[1]**

Current I is same for all

Magnetic moment $m = nIA$

Here, n is number of turns.

(i) For equilateral triangle :

$$\therefore m = nIA = 4I\left(\frac{\sqrt{3}}{4}a^2\right) = Ia^2\sqrt{3}$$ **[1]**

(ii) For square of side a

$$\therefore m = nIA = 3Ia^2$$ **[1]**

(iii) For a regular hexagon of sides a,

$$\therefore m = nIA = 2\times I\times\left(\frac{3\sqrt{3}}{2}\right)^2 = 3\sqrt{3}a^2I.$$ **[1]**

Q. 11. Consider a circular current-carrying loop of radius *R* in the *x-y* plane with centre at origin. Consider the line integral $\Im(L) = \left|\int_{-L}^{L}\mathbf{B}.\mathbf{dl}\right|$ taken along z-axis.

(a) Show that $\Im$ (L) monotonically increases with *L*.

(b) Use an appropriate Amperian loop to show that $\Im(\infty) = \mu_0 I$, where *I* is the current in the wire.

(c) Verify directly the above result.

(d) Suppose we replace the circular coil by a square coil of sides *R* carrying the same current *I*. what can you say about $\Im$ (L) and $\Im(\infty)$?

[NCERT Exemp. Q. 4.27, Page 27]

Ans. **(a)** B(z) point in the same direction of Z-axis and hence $\Im$ (L) is monotonical function of L as B and dl are along the same direction. So

$B\cdot dl = Bdl\cos\theta = Bdl\cos 0^0 = Bdl.$ **[½]**

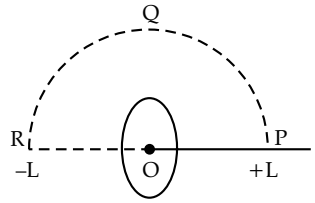
(b) $\Im$ (L) + Contribution from large distance on contour $C = \mu_0 I$.

$\therefore$ As $L\to\infty$

Contribution from large distance $\to 0\cdot\left(As\ B\propto\frac{1}{r^3}\right)$

$\therefore \Im(\infty) = \mu_0 I.$ **[½]**

(c) The magnetic field due to circular current-carrying loop of radius R in X-Y plane with centre at origin at any point lying at distance a from origin. Consider a loop of current- carrying conductor placed in X-Y plane. A point P is in $+Z$ direction at distance z, that is, $OP = z$.



[1]

Again consider an element dz on loop of conductor as shown in figure

Let angle between R and QP = θ then magnetic field at P due to loop is

$B_z = \frac{\mu_0 IR^2}{2(z^2+R^2)^{3/2}}$ [Integrating both sides *with respect to* z and z can vary from $-d$ to $+d$]

$$\int_{-\infty}^{\infty} B_z dz = \int_{-\infty}^{\infty}\frac{\mu_0 IR^2 dz}{2(z^2+R^2)^{3/2}}$$

$$\tan\theta = \frac{z}{R}$$

$$z = R\tan\theta$$

Differentiating both sides.

$$dz = R\cdot\sec^2\theta\cdot d\theta$$

$$\cos\theta = \frac{R}{\sqrt{z^2+R^2}}$$

$$\cos^2\theta = \frac{R^2}{z^2+R^2}$$

$$\int_{-\infty}^{\infty} B_z dz = \frac{\mu_0 I}{2}\int_{-\infty}^{+\infty}\frac{R^2}{(z^2+R^2)}\frac{dz}{\sqrt{z^2+R^2}}$$

$$= \frac{\mu_0 I}{2}\int_{-\pi/2}^{+\pi/2}\cos^2\theta\cdot\frac{R\sec^2\theta\cdot d\theta}{\sqrt{z^2+R^2}}$$

$$\left[\because \cos^2\theta = \frac{R^2}{z^2+R^2}\right]$$

$$= \frac{\mu_0 I}{2}\int_{-\pi/2}^{+\pi/2}\frac{Rd\theta}{\sqrt{z^2+R^2}} = \frac{\mu_0 I}{2}\int_{-\pi/2}^{\pi/2}\cos\theta d\theta$$

$$= \frac{\mu_0 I}{2}[\sin\theta]_{-\pi/2}^{\pi/2} = \frac{\mu_0 I}{2}\left[\sin\frac{\pi}{2} - \sin\frac{-\pi}{2}\right]$$

$$= \frac{\mu_0 I}{2}[1+1] = \mu_0 I = \int_{-\infty}^{+\infty} B_z d\hat{z} = \mu_0 I \quad \textbf{[2]}$$

(d) Because the area of square loop is smaller than the area of circular loop, for the same length of conducting wires, hence,

loop $B(z)_{square\ loop}$ <loop $B(z)_{circular\ loop}$

$\Im_s(L)_{sq.} = \Im_s(L)_{circular}$ [$\therefore$ length of conducting wire are equal]

By using arguments as in (b) part, B_z does not depend on length of wire

$\therefore \Im(\infty)_{sq.\ loop} = \Im(\infty)_{circular\ loop}$

Magnetic field due to circular or square loops remain same, that is,

$B(z)_{circular\ loop} = B(z)_{square\ loop} = \mu_0 I$

So, $\Im(\infty)_{sq.\ loop} = \Im(\infty)_{circular\ loop} = \mu_0 I$

Hence, proved. [1]

TOPIC-2
Magnetism

Quick Review

- Magnetic resonance imaging (MRI) machines use magnets, and they generate stronger fields than the Earth. In fact, it is about 60,000 times stronger than the Earth's magnetic field.
- The most powerful magnet in the universe is actually a star called a magnetar. These are stars that have died off and had a supernova explosion.
- Strong rare earth magnets can turn some metals into magnets. Ferromagnetic materials like iron can be magnetised with a strong permanent magnet.
- Magnets are ancient. Well, today's man-made magnets may not be so ancient, but the Chinese are said to have used lodestone, a natural magnet as far back as date. In fact, ancient mariners are said to have used lodestones to help them navigate.
- There are magnetic hills, said to pull cars and other large magnetic objects towards certain locations. However, researchers have found out that these are not really a magnetic anomaly as much as a topographical illusion.
- Magnets have been used to study bee communication patterns, migratory cycles and several other animal behaviors. This is because many animals can sense magnetic fields. For instance, some sharks are repelled by them and birds and turtles navigate by them.

TIPS...

- *Study the basic concept of electric charge and electric particles.*
- *Understand the Fleming's left rule and right-hand rule.*
- *Study about the law's given in chapter for proper numerical solution.*

TRICKS...

- *Magnetics is all about understanding a few basic laws and then using integration to find out the magnetic field produced by a current distribution. All questions will require you to use Biot-Savart's Law or Ampere's Law*
- *The uniqueness theorem states that given a charge distribution in a region and boundary conditions (i.e. values of potential at the boundaries) there exists a unique solution to the potential in Magnetization region.*

Know the Links

- ☞ *https://www.apexmagnets.com/news-how-tos/8-strange-facts-about-magnets-and-magnetism/*
- ☞ *https://study.com/academy/lesson/how-magnetic-forces-affect-moving-charges.html*
- ☞ *https://www.quora.com/How-do-I-study-moving-charges-and-magnetism-in-a-day-I-am-confused-and-referring-to-all-kinds-of-books-Which-is-the-best-for-the-boards-the-NCERT-or-SL-Arora*

Multiple Choice Questions (1 mark each)

Q. 1. An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?

(a) The electron will be accelerated along the axis.

(b) The electron path will be circular about the axis.

(c) The electron will experience a force at 45° to the axis and hence execute a helical path.

(d) The electron will continue to move with uniform velocity along the axis of the solenoid.

[NCERT Exemp. Q. 4.4, Page 23]

Ans. **Correct option :** (d)
Explanation : The magnetic field inside the long current carrying solenoid is uniform. Therefore, magnitude of force on the electron of charge ($-e$) is given by $F = -ev\text{B}\sin\theta = 0$ ($\theta = 0°$) as magnetic field and velocity are parallel. The electron will continue to move with uniform velocity along the axis of the solenoid.

Q. 2. In a cyclotron, a charged particle
(a) undergoes acceleration all the time.
(b) speeds up between the dees because of the magnetic field.
(c) speeds up in a dee.
(d) slows down within a dee and speeds up between dees. **[NCERT Exemp. Q. 4.5, Page 23]**

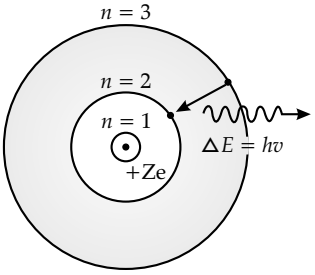
Ans. **Correct option :** (a)
Explanation : In a cyclotron, a charged particle describes the circular path inside the dees and is accelerated while going from one dee to another due to the electric field. While moving in the circular path, charged particles have centripetal acceleration which is provided by the magnetic force due to the magnetic field. A charged particle undergoes acceleration all the time, inside the cyclotron.

Tick Two or More Options

Q. 3. The gyro-magnetic ratio of an electron in an H-atom, according to Bohr model, is
(a) independent of which orbit it is in.
(b) negative.
(c) positive.
(d) increases with the quantum number n.
[NCERT Exemp. Q. 4.7, Page 23]

Correct options : (a) and (b)
Explanation : We define the magnetic moment of the current loop as,

m or $\mu_l = IA$ ...(i)



The electron performs uniform circular motion around a stationary heavy nucleus of charge $+Ze$. This constitutes a current I,

$$I = \frac{e}{T}$$

and T is the time period of revolution. Let r be orbital radius of the electron, and v be the orbital speed. Then,

$$T = \frac{2\pi r}{v}$$

$$I = \frac{ev}{2\pi r} \text{ and } A = \pi r^2$$

So, by substituting this in equation (i), we get

$$\mu_l = \frac{evr}{2}$$

And by multiplying-dividing this equation with m_e, we get

$$\mu_l = \frac{-e}{2m_e}L$$

The gyro-magnetic ratio of an electron in an H-atom is equal to the ratio of the magnetic moment and the angular momentum of the electron. Both are related as,

$$\mu_l = -\left(\frac{e}{2m_e}\right)L$$

The negative sign indicates that the angular momentum of the electron is opposite in direction to the magnetic moment.

Very Short Answer Type Questions (1 or 2 mark each)

Q. 1. Verify that the cyclotron frequency $\omega = \frac{eB}{m}$ has the correct dimensions of $[\text{T}]^{-1}$
[NCERT Exemp. Q. 4.12, Page 25]

Ans. For a charge particle moving perpendicular to the magnetic field :

$$\frac{(mv^2)}{R} = qvB$$

$$\Rightarrow \frac{qB}{m} = \frac{v}{R} = \omega$$

$$\Rightarrow = \omega = \frac{qB}{m}$$

$$= \frac{v}{\text{R}} = [\text{T}]^{-1}. \quad \textbf{[1]}$$

Q. 2. Describe the motion of a charged particle in a cyclotron if the frequency of the radio frequency (R_f) field were doubled.
[NCERT Exemp. Q. 4.15, Page 25]

Ans. Particle will accelerate and decelerate alternatively. So, the radius of path in the Dee's will remain unchanged. **[1]**

Short Answer Type Questions (3 or 4 marks each)

Q. 1. A toroid has a core (non-ferromagnetic) of inner radius 25 cm and outer radius 26 cm, around which 3500 turns of a wire are wound. If the current in the wire is 11 A, what is the magnetic field (a) outside the toroid, (b) inside the core of the toroid, and (c) in the empty space surrounded by the toroid. [NCERT Add. Ex. Q. 4.17, Page 170]

Ans. Given that,

Inner radius of the toroid, $r_1 = 25$ cm $= 0.25$ m

Outer radius of the toroid, $r_2 = 26$ cm $= 0.26$ m

Number of turns on the coil, $N = 3500$

Current in the coil, $I = 11$ A

(a) Magnetic field outside a toroid is zero. It is non-zero only inside the core of a toroid. **[½]**

(b) Magnetic field inside the core of a toroid is given by the relation,

$$B = \frac{\mu_0 NI}{l}$$ **[½]**

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7}\ \text{TmA}^{-1}$

$$l = \text{length of toroid}$$

$$= 2\pi\left[\frac{r_1 + r_2}{2}\right]$$

$$= \pi(0.25 + 0.26)$$

$$= 0.51\pi$$

$$\therefore B = \frac{4\pi \times 10^{-7} \times 3500 \times 11}{0.51\pi}$$

$$\approx 3.0 \times 10^{-2}\ \text{T}$$ **[1]**

(c) Magnetic field in the empty space surrounded by the toroid is zero. **[1]**

Q. 2. A straight horizontal conducting rod of length 0.45 m and mass 60 g is suspended by two vertical wires at its ends. A current of 5.0 A is set up in the rod through the wires. (a) What magnetic field should be set up normal to the conductor in order that the tension in the wires is zero? (b) What will be the total tension in the wires if the direction of current is reversed keeping the magnetic field same as before? (Ignore the mass of the wires.) g = 9.8 m /s^{-2}. [NCERT Add. Ex. Q. 4.21, Page 171]

Ans. Length of the rod, $l = 0.45$ m

Mass suspended by the wires, $m = 60\ \text{g} = 60 \times 10^{-3}$ kg

Acceleration due to gravity, $g = 9.8\ \text{m/s}^2$

Current in the rod flowing through the wire, $I = 5$ A

(a) Magnetic field B is equal and opposite to the weight of the wire, that is,

$$BIl = mg$$

$$\therefore B = \frac{mg}{Il}$$

$$= \frac{60 \times 10^{-3} \times 9.8}{5 \times 0.45} = 0.26\ \text{T}$$ **[1]**

A horizontal magnetic field of 0.26 T normal to the length of the conductor should be set up in order to get zero tension in the wire. The magnetic field should be such that Fleming's left hand rule gives an upward magnetic force. **[1]**

(b) If the direction of the current is revered, then the force due to magnetic field and the weight of the wire acts in a vertically downward direction.

Total tension in the wire $= BIl + mg$

$$= 0.26 \times 5 \times 0.45 + \left(60 \times 10^{-3}\right) \times 9.8$$

$$= 1.176\ \text{N}$$ **[1]**

Q. 3. A magnetic field of 100 G (1 G = 10^{-4} T) is required which is uniform in a region of linear dimension about 10 cm and area of cross-section about 10^{-3} m^2. The maximum current carrying capacity of a given coil of wire is 15 A and the number of turns per unit length that can be wound round a core is at most 1000 turns m^{-1}. Suggest some appropriate design particulars of a solenoid for the required purpose. Assume the core is not ferromagnetic [NCERT Add. Ex. Q. 4.15, Page 170]

Ans. Magnetic field strength, $B = 100\ \text{G} = 100 \times 10^{-4}$ T

Number of turns per unit length, $n = 1000\ \text{turns m}^{-1}$

Current flowing in the coil, $I = 15$ A

Permeability of free space, $\mu_0 = 4\pi \times 10^{-7}\ \text{T m A}^{-1}$

Magnetic field is given by the relation,

$$B = \mu_0 nI$$

$$\Rightarrow nI = \frac{B}{\mu_0} = \frac{100 \times 10^{-4}}{4\pi \times 10^{-7}} = 7957.74 \approx 8000\ A\text{m}^{-1}$$ **[2]**

If the length of the coil is taken as 50 cm, radius 4 cm, number of turns 400, and current 10 A, then these values are not unique for the given purpose. There is always a possibility of some adjustments with limits. **[1]**

Q. 4. A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, estimate the magnitude of B inside the solenoid near its centre. [NCERT Ex. Q. 4.8, Page 169]

Ans. Length of the solenoid, $l = 80$ cm $= 0.8$ m

There are five layers of windings of 400 turns each on the solenoid.

$\therefore$ Total number of turns on the solenoid, $N = 5 \times 400 = 2000$

Diameter of the solenoid, $D = 1.8$ cm $= 0.018$ m

Current carried by the solenoid, $I = 8.0$ A

Magnetic field inside the solenoid near its centre is given by the relation,

by the relation, $B = \frac{\mu_0 NI}{l}$ **[1]**

Where, μ_0 = Permeability of free space $= 4\pi \times 10^{-7}$ TmA^{-1}

$$B = \frac{4\pi \times 10^{-7} \times 2000 \times 8}{0.8}$$

$$= 2.512 \times 10^{-2}\ T$$

Hence, the magnitude of the magnetic field inside the solenoid near its centre is 2.512×10^{-2} T. **[2]**

Q. 5. A square coil of side 10 cm consists of 20 turns and carries a current of 12 A. The coil is suspended vertically and the normal to the plane of the coil makes an angle of 30° with the direction of a uniform horizontal magnetic field of magnitude 0.80 T. What is the magnitude of torque experienced by the coil? [NCERT Ex. Q. 4.9, Page 169]

Ans. Length of a side of the square coil, $l = 10$ cm $= 0.1$ m
Current flowing in the coil, $I = 12$ A
Number of turns on the coil, $n = 20$
Angle made by the plane of the coil with magnetic field, $\theta = 30°$
Strength of magnetic field, $B = 0.80$ T
Magnitude of the magnetic torque experienced by the coil in the magnetic field is given by the relation,
$\tau = n\,BIA\,sin\theta$ **[1]**
Where,
A = Area of the square coil
$= l \times l = 0.1 \times 0.1 = 0.01$ m^2
So, $\tau = 20 \times 0.8 \times 12 \times 0.01 \times \sin 30°$ **[2]**
$= 0.96$ N m
Hence, the magnitude of the torque experienced by the coil is 0.96 N m. **[1]**

Q. 6. Do magnetic forces obey Newton's third law? Verify for two current elements $dl_1 = dl\,\hat{i}$ located at the origin and $dl_2 = dl\,\hat{j}$ located at (0, *R*, 0). Both carry current *I*. [NCERT Exemp. Q. 4.20, Page 25]

Ans. Magnetic field due to current element, Idl_1 at a point on element dl_2,

$$\vec{B} = \frac{\mu_o}{4\pi}\frac{Idl}{R^2}\hat{k}$$

[1]

Force due to dl_2 is given by, $\vec{F} = (Idl)\,\hat{j} \times \vec{B}$ **[1]**

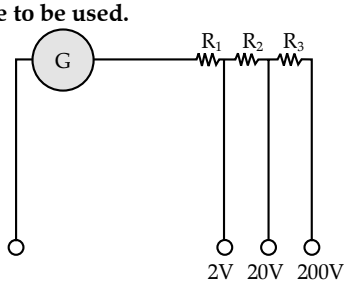
$$\vec{F} = (Idl)\,\hat{j} \times \left(\frac{\mu_0}{4\pi}\frac{Idl}{R^2}\right)\hat{k}$$

$$\Rightarrow \vec{F} = \left(\frac{\mu_0}{4\pi}\frac{I^2dl^2}{R^2}\right)\hat{i}$$

[1]

Force due to dl_2 on dl_1 is zero and force due to dl_1 on dl_2 is non-zero. It appears Newton's third law is not obeyed here. **[1]**

Q. 7. A multirange voltmeter can be constructed by using a galvanometer circuit as shown in Fig. 4.2. We want to construct a voltmeter that can measure 2V, 20V and 200V using a galvanometer of resistance 10Ω and that produces maximum deflection for current of 1 *mA*. Find R_1, R_2 and R_3 that have to be used.



[NCERT Exemp. Q. 4.21, Page 26]

Ans. In the given question,
$G = 10\ \Omega$, $i_G = 1$ mA $= 10^{-3}$ A
For 2 V range,
$i_G(G + R_1) = 2$
$\Rightarrow i_G = 2\ \text{k}\Omega$ **[1]**
For 20 V range,
$i_G(G + R_1 + R_2) = 20$
$\Rightarrow R_2 = 18\ \text{k}\Omega$ **[1]**
For 200 V range,
$i_G(G + R_1 + R_2 + R_3) = 200$
$\Rightarrow R_3 = 180\ \text{k}\Omega$. **[1]**

Q. 8. A galvanometer coil has a resistance of 12 Ω and the metre shows full scale deflection for a current of 3 mA. How will you convert the metre into a voltmeter of range 0 to 18 V? [NCERT Add. Ex. Q. 4.27, Page 172]

Ans. Resistance of the galvanometer coil, $G = 12\ \Omega$
Current for which there is full scale deflection, $I_s = 3$ mA $= 3\times10^{-3}$ A range of the voltmeter is 0, which needs to be converted to 18 V.
$V = 18$ V **[1]**
Let a resistor of resistance R be connected in series with the galvanometer to convert it into a voltmeter. This resistance is given as :

$$R = \frac{V}{I_g} - G$$

$$= \frac{18}{3\times10^{-3}} - 12 = 6000 - 12 = 5988\ \Omega$$

Hence, a resistor of resistance 5988Ω is to be connected in series with the galvanometer. **[2]**

Q. 9. A galvanometer coil has a resistance of 15 Ω and the metre shows full scale deflection for a current of 4 mA. How will you convert the metre into an ammeter of range 0 to 6 A? [NCERT Add. Ex. Q. 4.28, Page 172]

Ans. Resistance of the galvanometer coil, $G = 15\ \Omega$
Current for which the galvanometer shows full scale deflection,
$= 4$ mA $= 4 \times 10^{-3}$ A
Range of the ammeter is 0, which needs to be converted to 6 A.
Current, $I = 6$ A
A shunt resistor of resistance S is to be connected in parallel with the galvanometer to convert it into an ammeter. The value of S is given as :

$$S = \frac{I_g G}{I - I_g}$$ **[1]**

$$= \frac{4\times10^{-3}\times15}{6 - 4\times10^{-3}}$$

$$S = \frac{6\times10^{-2}}{6 - 0.004} = \frac{0.06}{5.996}$$

$$\approx 0.01\ \Omega = 10\ \text{m}\Omega$$

Hence, a shunt resistor 10 mΩ is to be connected in parallel with the galvanometer. **[2]**

Q. 10. Seema's uncle was advised by his doctor to have an *MRI* (Magnetic Resonance Imaging) scan of his brain. Her uncle felt it to be expensive and wanted to postpone it. When Seema learnt about this, she took the help of her family and also approached

the doctor, who also offered a substantial discount. She then convinced her uncle to undergo the test to enable the doctor to know the condition of his brain. The information thus obtained greatly helped the doctor to treat him properly. Based on the above paragraph, answer the following questions :

(a) What according to you are the values displayed by Seema, her family and the doctor?

(b) What could be the possible reason for *MRI* test to be so expensive?

(c) Assuming that *MRI* test was performed using a magnetic field of 0.1 T., find the minimum and maximum values of the force that the magnetic field could exert on a proton (charge = 1.6×10^{-19} *C*) moving with a speed of 10^4 m/s.

[CBSE Board, Delhi Region/All India Region, 2016]

Ans. **(a)** Depend on student what he / she think about value of Seema. **[1]**

(b) Availability of MRI test is very less. Only in big Hospital, it is available so due to monopoly of big hospital they are just charging huge amount. **[1]**

(c) Force experienced by proton $\vec{F} = q\ (\vec{v} \times \vec{B})$

$|\vec{F}| = qvB \sin\theta$

Maximum force $|\vec{F}|_{max} = qvB$

$\Rightarrow |\vec{F}|_{max} = 1.6 \times 10^{-19} \times 10^4 \times 0.1$

$|\vec{F}|_{max} = 1.6 \times 10^{-16}$ N

Minimum force $|\vec{F}|_{min}$ = Zero **[1]**

Q. 11. State the principle of a cyclotron. Show that the time period of revolution of particles in a cyclotron is independent of their speeds. Why is this property necessary for the operation of a cyclotron? [CBSE Board, Delhi Region, 2016]

Ans. Principle. The frequency of revolution of the charges particle in magnetic field is independent of its energy. If particle is performing circular motion due to magnetic force, then centripetal force = Magnetic force

$$\Rightarrow \quad \frac{mv^2}{r} = qv\ B \sin 90°$$

$$\Rightarrow \quad r = \frac{mv}{qB}$$

$$\therefore \text{Time period } = \frac{2\pi r}{v}$$

$$T = \frac{2\pi}{v} \cdot \frac{mv}{qB} = \frac{2\pi m}{qB}$$ **[1]**

The time period of revolution of particles in a cyclotron is independent of their speeds **[1]**

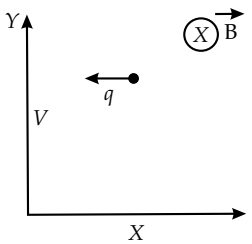
This property is necessary for the operation of a cyclotron to attain the resonance condition ($v_a = v_c$) where v_a is the frequency of the applied voltage and v_c is cyclotron frequency. **[1]**

Long Answer Type Questions **(5 marks each)**

Q. 1. A solenoid 60 cm long and of radius 4.0 cm has 3 layers of windings of 300 turns each. A 2.0 cm long wire of mass 2.5 g lies inside the solenoid (near its centre) normal to its axis; both the wire and the axis of the solenoid are in the horizontal plane. The wire is connected through two leads parallel to the axis of the solenoid to an external battery which supplies a current of 6.0 A in the wire. What value of current (with appropriate sense of circulation) in the windings of the solenoid can support the weight of the wire? (g = 9.8 m/s^{-2})

[NCERT Add. Ex. Q. 4.26, Page 172]

Ans.



Length of the solenoid, L = 60 cm = 0.6 m

Radius of the solenoid, r = 4.0 cm = 0.04 m

It is given that there are 3 layers of windings of 300 turns each.

Total number of turns, $n = 3 \times 300 = 900$ Length of the wire, l = 2 cm = 0.02 m

Mass of the wire, m = 2.5 g = 2.5×10^{-3} kg

Current flowing through the wire, i = 6 A

Acceleration due to gravity, g = 9.8 m/s^2

Magnetic field produced inside the solenoid,

$$B = \frac{\mu_0 nI}{L}$$ **[1]**

Where,

μ_o = Permeability of free space = $4\pi \times 10^{-7}$ T m A^{-1}

I = Current flowing through the windings of the solenoid

Magnetic force is given by the relation,

$F = Bil$

$$= \frac{\mu_0 nI}{L} il$$ **[2]**

Also, the force on the wire is equal to the weight of the wire.

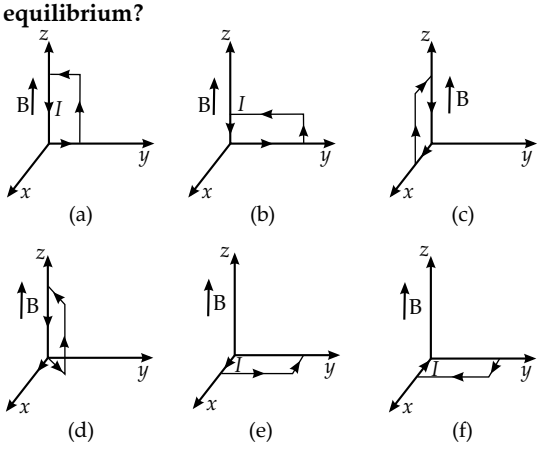
$$\therefore mg = \frac{\mu_0 nIil}{L}$$

$$I = \frac{mgL}{\mu_0 nil}$$

$$= \frac{2.5 \times 10^{-3} \times 9.8 \times 0.6}{4\pi \times 10^{-7} \times 900 \times 0.02 \times 6} = 108 \text{ A}$$ **[2]**

Hence, the current flowing through the solenoid is 108 A.

Q. 2. A uniform magnetic field of 3000 G is established along the positive z-direction. A rectangular loop of sides 10 cm and 5 cm carries a current of 12 A. What is the torque on the loop in the different cases shown in the Figure? What is the force on each case? Which case corresponds to stable equilibrium?



[NCERT Add. Ex. Q. 4.24, Page 171]

Ans. Magnetic field strength, $B = 3000\ G = 3000 \times 10^{-4}$ $T = 0.3\ T$

Length of the rectangular loop, $l = 10$ cm

Width of the rectangular loop, $b = 5$ cm

Area of the loop,

$A = l \times b = 10 \times 5 = 50\ cm^2 = 50 \times 10^{-4}\ m^2$

Current in the loop, $I = 12$ A

Now, taking the anti-clockwise direction of the current as positive and vice-versa :

(a) Torque, $\vec{\tau} = I\vec{A} \times \vec{B}$

From the given figure, it can be observed that A is normal to the y-z plane and B is directed along the z-axis.

$$\tau = 12 \times \left(50 \times 10^{-4}\right)\hat{i} \times 0.3\hat{k}$$

$$= -1.8 \times 10^{-2}\ \hat{j}\ \text{N m}$$

The torque is -1.8×10^{-2} Nm along the negative y-direction. The force on the loop is zero because the angle between A and B is zero. **[1]**

(b) This case is similar to case (a). Hence, the answer is the same as (a). **[½]**

(c) Torque, $\vec{\tau} = I\vec{A} \times \vec{B}$

From the given figure, it can be observed that A is normal to the x-z plane and B is directed along the z-axis.

$$\therefore \tau = -12 \times \left(50 \times 10^{-4}\right)\hat{j} \times 0.3\hat{k}$$

$$= -1.8 \times 10^{-2}\ \hat{i}\ \text{Nm}$$

[1]

The torque is 1.8×10^{-2} Nm along the negative x direction and the force is zero.

(d) Magnitude of torque is given as :

$$|\tau| = IAB$$

$$= 12 \times 50 \times 10^{-4} \times 0.3$$

$$= 1.8 \times 10^{-2}\ \text{Nm}$$

The torque is 1.8×10^{-2} Nm at an angle of 240° with positive x direction. The force is zero. **[½]**

(e) Torque, $\vec{\tau} = I\vec{A} \times \vec{B}$

$$= \left(50 \times 10^{-4} \times 12\right)\hat{k} \times 0.3\hat{k}$$

$$= 0$$ **[1]**

Hence, the torque is zero. The force is also zero.

(f) Torque, $\vec{\tau} = I\vec{A} \times \vec{B}$

$$= \left(50 \times 10^{-4} \times 12\right)\hat{k} \times 0.3\hat{k}$$

$$= 0$$

Hence, the torque is zero. The force is also zero. In case (e), the direction of $\vec{A}$ and $\vec{B}$ is the same and the angle between them is zero. If displaced, they come back to an equilibrium. Hence, its equilibrium is stable. Whereas, in case (f), the direction of I_A and B is opposite. The angle between them is 180°. If disturbed, it does not come back to its original position. Hence, its equilibrium is unstable. **[1]**

Q. 3. Two moving coil meters, M_1 and M_2 have the following particulars :

$R_1 = 10\ \Omega$, $N_1 = 30$,

$A_1 = 3.6 \times 10^{-3}\ m^2$, $B_1 = 0.25$ T

$R_2 = 14\ \Omega$, $N_2 = 42$,

$A_2 = 1.8 \times 10^{-3}\ m^2$, $B_2 = 0.50$ T

(The spring constants are identical for the two meters). Determine the ratio of

(a) current sensitivity and

(b) voltage sensitivity of M_2 and M_1.

[NCERT Ex. Q. 4.10, Page 169]

Ans. **For moving coil meter M_1 :**

Resistance, $R_1 = 10\ \Omega$

Number of turns, $N_1 = 30$

Area of cross-section, $A_1 = 3.6 \times 10^{-3}\ m^2$

Magnetic field strength, $B_1 = 0.25$ T

Spring constant $K_1 = K$

For moving coil meter M_2 :

Resistance, $R_2 = 14\ \Omega$

Number of turns, $N_2 = 42$

Area of cross-section, $A_2 = 1.8 \times 10^{-3}\ m^2$

Magnetic field strength, $B_2 = 0.50$ T

Spring constant, $K_2 = K$

(i) Current sensitivity of M_1 is given as :

$$I_{s1} = \frac{N_1 B_1 A_1}{K_1}$$

And, current sensitivity of M_2 is given as :

$$I_{s2} = \frac{N_2 B_2 A_2}{K_2}$$

$$\therefore \text{Ratio } \frac{I_{s2}}{I_{s1}} = \frac{N_2 B_2 A_2 k_1}{N_1 B_1 A_1 k_2}$$

$$= \frac{42 \times 0.5 \times 1.8 \times 10^{-3} \times K}{K \times 30 \times 0.25 \times 3.6 \times 10^{-3}} = 1.4$$

Hence, the ratio of current sensitivity of M_2 to M_1 is 7:5. **[2½]**

(ii) Voltage sensitivity for M_2 is given as :

$$V_{s2} = \frac{N_2 B_2 A_2}{K_2 R_2}$$

And, voltage sensitivity for M_1 is given as :

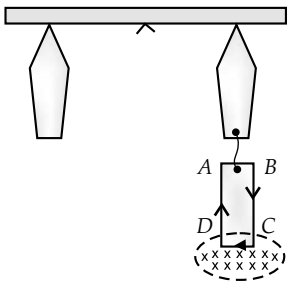
$$V_{s1} = \frac{N_1 B_1 A_1}{K_1 R_1}$$

$$\therefore \text{Ratio } \frac{V_{s2}}{V_{s1}} = \frac{N_2 B_2 A_2 R_1 k_1}{N_1 B_1 A_1 R_2 k_2}$$

$$= \frac{42 \times 0.5 \times 1.8 \times 10^{-3} \times 10 \times K}{K \times 14 \times 30 \times 0.25 \times 3.6 \times 10^{-3}} = 1$$

Hence, the ratio of voltage sensitivity M_2 to M_1 is 1. **[2½]**

Q. 4. A 100 turn rectangular coil *ABCD* (in *XY* plane) is hung from one arm of a balance figure. A mass 500 g is added to the other arm to balance the weight of the coil. A current 4.9 A passes through the coil and a constant magnetic field of 0.2 T acting inward (in *xz* plane) is switched on such that only arm *CD* of length 1 cm lies in the field. How much additional mass '*m*' must be added to regain the balance?



[NCERT Exemp. Q. 4.23, Page 26]

Ans. When the magnetic field is off $\Sigma \tau = 0$ **[½]**

Now, $Mgl = W_{coil}\, l$

$500\ g\ l = W_{coil}\, l$

$W_{coil} = 500 \times 9.8$ N **[1]**

When the magnetic field is switched on, taking moment of force about mid-point, we have

$Mgl + mgl = W_{coil}\, l + (IBL \sin 90^\circ)\, l$ **[1]**

$mgl = BIL\, l$ **[½]**

$$m = \frac{(BIL)}{g}$$

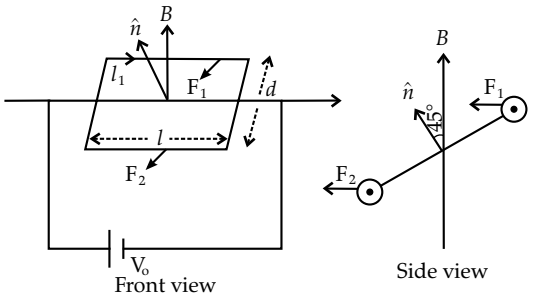
$$= \frac{0.2 \times 4.9 \times 1 \times 10^{-2}}{9.8}$$ **[2]**

$= 10^{-3}$ kg = 1 g. Therefore, 1 g of additional mass must be added to regain the balance.

Q. 5. A rectangular conducting loop consists of two wires on two opposite sides of length *l* joined together by rods of length *d*. The wires are each of the same material but with cross-sections differing by a factor of 2. The thicker wire has a resistance *R* and the rods are of low resistance, which in turn are connected to a constant voltage source *V*. The loop is placed in uniform a magnetic field *B* at 45° to its plane. Find τ, the torque exerted by the magnetic field on the loop about an axis through the centres of rods.

[NCERT Exemp. Q. 4.24, Page 26]

Ans.



[2]

Torque on the first wire is given by,

$$F_1 = i_1 lB = \frac{V_o}{2R} lB,$$

Torque due to for a F_1

$$\tau_1 = F_1 \times \frac{d}{2} \sin 45°$$

$$\tau_1 = \frac{d}{2\sqrt{2}} F_1 = \frac{V_o l d B}{4\sqrt{2} R}$$

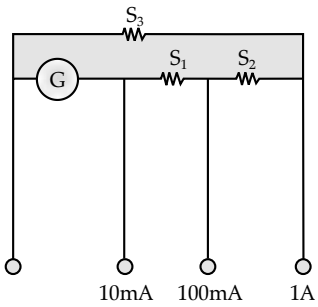
Torque on the second wire is given by,

$$F_2 = i_2 lB = \frac{V_o}{R} lB, \tau_2 = F_2 \times \frac{d}{2} \sin 45° = \frac{d}{2\sqrt{2}} F_2 = \frac{V_o l d B}{2\sqrt{2} R}$$

Net torque, $\tau = \tau_1 - \tau_2$

$$\Rightarrow \tau = \frac{1}{4\sqrt{2}} \frac{V_o A B}{R}$$ **[3]**

Q. 6. A multi-range current meter can be constructed by using a galvanometer circuit as shown in Figure. We want a current meter that can measure 10 mA, 100 mA, and 1 A using a galvanometer of resistance 10 Ω and that produces maximum deflection for current of 1 mA. Find S_1, S_2, and S_3 that have to be used.



[NCERT Exemp. Q. 4.28, Page 27]

Ans. $i_G.G = (i_1 - i_G)(S_1 + S_2 + S_3)$ for i_1 = 10 mA **[1]**

$i_G (G + S_1) = (i_2 - i_G)(S_2 + S_3)$ for i_2 =100 mA **[1]**

and $i_G (G + S_1 + S_2) = (i_3 - i_G)(S_2 + S_3)$ for i_3 = 1 A **[1]**

gives S_1 = 1 ohm,

S_2 = 0.1 ohm and

S_3=0.01 ohm. **[2]**

Q. 7. Answer the following

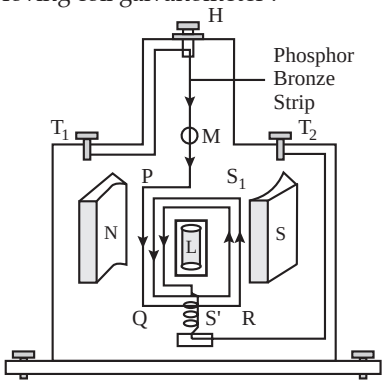
(a) With the help of a neat and labelled diagram, explain the principle and working of a moving coil galvanometer.

(b) What is the function of uniform radial field and how is it produced?

(c) Define current sensitivity of a galvanometer. How is current sensitivity increased?

[CBSE Board, Foreign Scheme, 2016]

Ans. **(a)** Moving coil galvanometer :



[1½]

Principle : The underlying principle of moving coil galvanometer is that a current carrying coil, placed in a uniform magnetic field, experiences torque.

Consider a rectangular coil for which no. of turns = N

Area of cross-section is $A = lb$

Intensity of the uniform magnetic field = B

Current through the coil = I

Therefore,

Deflecting torque is given by,

$BIL \bullet b = \frac{B}{A}$ **[½]**

For N number of turns,

$\tau = \frac{NB}{A}$ **[½]**

Restoring torque in the spring = $k\theta$ **[½]**

Therefore,

$NBIA = k\theta$ **[½]**

This implies,

$I = \frac{k}{NBA}\ \theta$

That is,

I θ

(b) **(i)** The soft iron coil in a galvanometer will make the field radial. Also, it increases the strength of the magnetic field. **[½]**

(ii) Current sensitivity in the galvanometer is given by, $\frac{\theta}{I} = \frac{NBA}{k}$

Voltage sensitivity in the galvanometer is given by,

$\frac{\theta}{V} = \frac{\theta}{IR} = \frac{nBA}{k}\ \ \frac{1}{R}$

The above two equations imply that increasing the current sensitivity may not necessarily increase the voltage sensitivity. **[1]**

Some Commonly Made Errors

- Students get confused to determine the direction of electric current and force exerted on the conductor.
- Generally, students do not draw diagrams. Try to explain your answers by making suitable diagram.

EXPERT ADVICE

- *Generally, students confuse between Ampere's circuital law and Biot-Savart Law. Ampere's Circuital law is not independent of the Biot-Savart law. It can be derived from the Biot-Savart law.*
- *It can often be very confusing getting the direction correct in complicated systems in both electricity and magnetism. For overcoming this problem, getting the direction right, the actual current will be in the direction opposite to which you have assumed with the same magnitude.*

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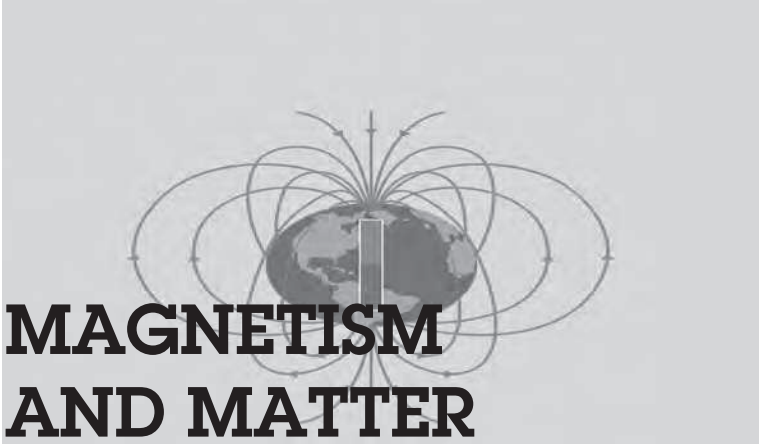
CHAPTER

5 MAGNETISM AND MATTER



Chapter Objectives

This chapter will help you understand :

- ***Magnetism and matter :*** *Introduction; Bar magnets; Magnetism and Gauss's Law; The Earth magnetism; Magnetic properties; Permanent magnets and Electromagnets.*

Quick Review

- The magnetism is associated with magnetic fields, which arises due to the motion of electric charges. This motion can take many forms. It can be an electric current in a conductor or charged particles moving through space, or it can be the motion of an electron in an atomic orbital.
- Magnetism is also associated with elementary particles, such as the electron that have a property called spin.
- The compass in real sense was created by a Chinese geomantic omen master in the late Tang Dynasty, who originally used it for divination.
- Magnetic monopole, hypothetical particle with a magnetic charge, a property analogous to an electric charge.
- In 1845, Faraday became the first to classify substances as either diamagnetic or para-magnetic.
- The Curie temperature is named for the French physicist Pierre Curie (1859–1906) who discovered the laws that relate some magnetic properties to change in temperature in 1895.

TIPS...

- *Study the bar magnets as a solenoid and Earth's magnetism.*
- *Study the concept of Para-magnetism, Diamagnetism and Ferro-magnetism.*
- *Study the Gauss's law for magnetic fields.*
- *Understand the diagrams to explain magnetic lines.*

TRICKS...

- *For all formulae involving the length of the magnet, denoted by l, the symbol 'l' always stands for half the length of the magnet. Complete length would be '2l'.*

- When a material makes the transition from the normal to super-conducting state, it actively excludes magnetic fields from its interior, this is called the **Meissner effect**.
- Maxwell's equations are a set of four differential equations that form the theoretical basis for describing classical electromagnetism :
  - **Gauss's law for electrostatics :** Electric charges produce an electric field. The electric flux across a closed surface is proportional to the charge enclosed.

$$\oint E.dA = \frac{q\text{enclosed}}{\varepsilon_0}$$

  - **Gauss's law for magnetism :** There are no magnetic monopoles. The magnetic flux and Faraday's law quantitative across a closed surface is zero.

$$\oint B.dA = 0$$

  - **Faraday's law :** Time-varying magnetic fields produce an electric field.

$$\oint E.dl = \frac{-d\phi_B}{dt}$$

  - **Ampère's law :** Steady currents and time-varying electric fields (the latter due to Maxwell's correction) produce a magnetic field.

$$\oint B.dl = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_e}{dt}$$

❖ The **Nobel Prize in Physics 2007** was awarded jointly to **Albert Fert** and **Peter Grünberg** for the discovery of **"Giant Magnetoresistance"**.

Know the Links

☞ *https://www.livescience.com/38059-magnetism.html*
☞ *https://www.britannica.com/science/magnetism*
☞ *https://brilliant.org/wiki/maxwells-equations/*

Multiple Choice Questions (1 mark each)

Q. 1. A toroid of n turns, mean radius R and cross-sectional radius a carries current I. It is placed on a horizontal table taken as x-y plane. Its magnetic moment m

(a) is non-zero and points in the z-direction by symmetry.

(b) points along the axis of the toroid ($m = m\phi$).

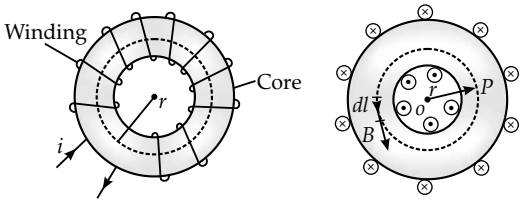
(c) is zero, otherwise there would be a field falling as $\frac{1}{r^3}$ at large distances outside the toroid.

(d) is pointing radially outwards.

[NCERT Exemp. Q. 5.1, Page 28]

Ans. Correct option : (c)

Explanation : As we know that a toroid can be considered as a ring shaped closed solenoid. So that it is like an endless cylindrical solenoid.



So, the magnetic field is only confined inside the body of a toroid in the form of concentric magnetic lines of force.

For any point inside the empty space surrounded by toroid and outside the toroid, the magnetic field B is zero because the net current enclosed in these spaces is zero. So that, the magnetic moment of toroid is zero. In general, if we take r as a long distance outside the toroid, the $m \propto \frac{1}{r^3}$ but this case is not possible here.

Q. 2. Consider the two idealized systems : (i) a parallel plate capacitor with large plates and small separation and (ii) a long solenoid of length $L \gg R$, radius of cross-section. In (i) E is ideally treated as a constant between plates and zero outside. In (ii) magnetic field is constant inside the solenoid and zero outside. These idealized assumptions, however, contradict fundamental laws as below :

(a) case (i) contradicts Gauss's law for electrostatic fields.

(b) case (ii) contradicts Gauss's law for magnetic fields.

(c) case (i) agrees with $\oint_s E.dl = 0$

(d) case (ii) contradicts $\oint H.dl = I_{en}$

[NCERT Exemp. Q. 5.4, Page 29]

Ans. Correct option : (b)

Explanation : According to Gauss's law of electrostatic field,

$$\oint E.ds = \frac{q}{\varepsilon_0}$$

So it does not contradict for electrostatic field as the electric field lines do not form continuous path.

According to Gauss's law of magnetic field,

$$\oint B.ds = 0$$

It is clear that it contradicts for magnetic field because there is magnetic field inside the solenoid, and no field outside the solenoid carrying current, but the magnetic field lines form the closed paths.

Q. 3. The magnetic field of Earth can be modelled by that of a point dipole placed at the centre of the Earth. The dipole axis makes an angle of 11.3° with the axis of Earth. At Mumbai, declination is nearly zero. Then,

(a) the declination varies between 11.3° W to 11.3° E.

(b) the least declination is 0°.

(c) the plane defined by dipole axis and Earth axis passes through Greenwich.

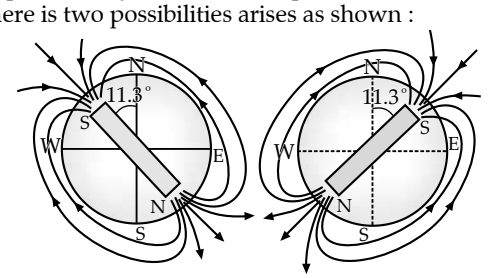
(d) declination averaged over Earth must be always negative. **[NCERT Exemp. Q. 5.2, Page 28]**

Ans. Correct option : (a)

Explanation : The magnetic field lines of the Earth resemble that of a hypothetical magnetic dipole located at the centre of the Earth.

The axis of the dipole does not coincide with the axis of rotation of the Earth and it is tilted at some angle (angle of declination).

In this situation, the angle of declination is approximately 11.3° with respect to the later. So, there is two possibilities arises as shown :



So that the declination varies between 11.3° W to 11.3° E.

Q. 4. A paramagnetic sample shows a net magnetization of 8 Am^{-1} when placed in an external magnetic field of 0.6 T at a temperature of 4 K. When the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K, the magnetization will be

(a) $\frac{32}{3}$ Am^{-1} **(b)** $\frac{2}{3}$ Am^{-1}

(c) 6 Am^{-1} **(d)** 2.4 Am^{-1}

[NCERT Exemp. Q. 5.5, Page 29]

Ans. Correct option : (b)

Explanation : According to the Curie's law of magnetisation, the magnetic susceptibility for a substance is directly proportional to magnetic field induction (B) and inversely proportional to the absolute temperature (T).

$$I \propto \frac{B}{T}$$

So that,

$$\frac{I_2}{I_1} = \frac{B_2 T_1}{B_1 T_2}$$

$I_1 = 8\ Am^{-1}$

$B_1 = 0.6\ T$

$T_1 = 4\ k$

$I_2 = ?$

$B_2 = 0.2\ T$

$T_2 = 16\ k$

Put the values,

$$I_2 = \frac{B_2 T_1}{B_1 T_2} I_1 = \frac{0.2 \times 8 \times 4}{0.6 \times 16} = \frac{2}{3}\ Am^{-1}$$

Q. 5. In a permanent magnet at room temperature

(a) magnetic moment of each molecule is zero.

(b) the individual molecules have non-zero magnetic moment which are all perfectly aligned.

(c) domains are partially aligned.

(d) domains are all perfectly aligned.

[NCERT Exemp. Q. 5.3, Page 29]

Ans. Correct option : (c)

Explanation : At room temperature permanent magnet behaves like a ferromagnetic substance. So that,

When there is no magnetic field in ferromagnetic substance domains are randomly spread.

So resultant magnetic moment is about zero.

But when this substance is placed in strong magnetic field, some domains aligned in external field permanently even in absence of external magnetic field.

So domains partially aligned permanently verify the option (c).

Tick Two or More Options

Q. 6. The primary origin(s) of magnetism lies in

(a) atomic currents.

(b) Pauli exclusion principle.

(c) polar nature of molecules.

(d) intrinsic spin of electron.

[NCERT Exemp. Q. 5.7, Page 30]

Ans. Correct options : (a) and (d)

Explanation : Motion of charge particle produces magnetism and nature of magnetism depends on the nation of motion of charge particle.

In atom, electrons revolve and spin around the nucleus which in turn produces current and due to magnetic effect of current, magnetism produces in the material.

Q. 7. Essential difference between electrostatic shielding by a conducting shell and magneto statics shielding is due to

(a) electrostatic field lines can end on charges and conductors have free charges.

(b) lines of B can also end but conductors cannot end them.

(c) lines of B cannot end on any material and perfect shielding is not possible.

(d) shells of high permeability materials can be used to divert lines of B from the interior region.

[NCERT Exemp. Q. 5.9, Page 30]

Ans. Correct options : (a), (c) and (d)

Explanation : Conductors have free charge particles so the lines of force can be stopped by conductors that give shielding effect.

As non-existence of monopole magnetic field lines cannot be stopped or shield.

Magnetic field lines are affected by magnetic material so it can be repelled by using a high permeability magnetic material to get the region of no magnetic field.

Q. 8. A long solenoid has 1,000 turns per metre and carries a current of 1 A. It has a soft iron core of μ_r = 1,000. The core is heated beyond the Curie temperature, T_c.

(a) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically.

(b) The H and B fields in the solenoid are nearly unchanged.

(c) The magnetisation in the core reverses direction.

(d) The magnetisation in the core diminishes by a factor of about 10^8.

[NCERT Exemp. Q. 5.8, Page 30]

Ans. Correct options : (a) and (d)

Explanation : As we know that the, the magnetic field intensity,

$H = nI$

where

n = Number of turns per metre of a solenoid and

I = Current

$B = \mu_0 \mu_r I$

Also, at normal temperature, a solenoid behaves as a ferro-magnetic substance and at the temperature beyond the Curie temperature, it behaves as a para-magnetic substance.

As we know, for solenoid $H = nI$

$\Rightarrow \quad H = 1,000 \times 1 = 1,000\ Am$

So that, H is a constant, so it is nearly unchanged because of heating n and I remain constant.

But

$$B = \mu_0 \mu_r nI$$
$$= \mu_0 nI \mu_r$$
$$= k \text{ (constant) } \mu_r$$

But there is a large decrease in the susceptibility of the core on heating it beyond critical temperature; hence magnetic field will decrease drastically. Now, for magnetisation in the core, when temperature of the iron core of a solenoid is raised beyond Curie temperature, then it behaves as a para-magnetic material, where

$$(\chi_m)_{\text{Fero}} \approx 10^3$$

and $$(\chi_m)_{\text{Para}} \approx 10^{-5}$$

$$\Rightarrow \frac{(\chi_m)_{\text{Fero}}}{(\chi_m)_{\text{Para}}} = \frac{10^3}{10^{-5}} = 10^8$$

Q. 9. Let the magnetic field on earth be modelled by that of a point magnetic dipole at the centre of Earth. The angle of dip at a point on the geographical equator

(a) is always zero.

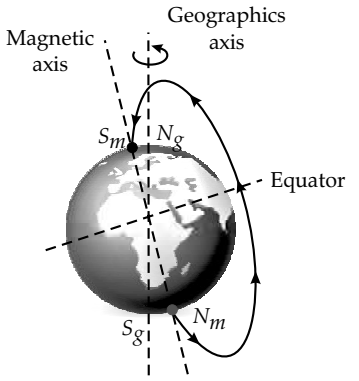
(b) can be zero at specific points.

(c) can be positive or negative.

(d) is bounded. [NCERT Exemp. Q. 5.10, Page 30]

Ans. Correct options : (b), (c) and (d)

Explanation : Angle of inclination or dip is the angle between the direction of intensity of total magnetic field of the Earth and a horizontal line in the magnetic meridian.



If the total magnetic field of the Earth is modelled by a point magnetic dipole at the centre, then it is in the same plane of geographical equator, thus the angle of dip at a point on the geographical equator is bounded in a range from positive to negative values.

Q. 10. S is the surface of a lump of magnetic material.

(a) Lines of B are necessarily continuous across S.

(b) Some lines of B must be discontinuous across S.

(c) Lines of H are necessarily continuous across S.

(d) Lines of H cannot all be continuous across S.

[NCERT Exemp. Q. 5.6, Page 29]

Correct options : (a) and (d)

Explanation : Magnetic field lines for magnetic induction (B) form continuous lines, so lines of B are necessarily continuous across S.

And, also magnetic intensity (H) to magnetise varies for inside and outside the lump. So, line of H cannot all be continuous across S.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude equal to 4.5 × 10^{-2} J. What is the magnitude of magnetic moment of the magnet? [NCERT Ex. Q. 5.3, Page 200]

Ans. Given that,

$\theta = 30°$

$B = 0.25$ T

$\tau = 4.5 \times 10^{-2}$ J

As we know that torque is :

$\tau = m B \sin\theta$ **[½]**

Put the values,

$4.5 \times 10^{-2} = m\ (0.25) \sin 30°$

So that magnetic moment 0.36 J/T. **[½]**

Q. 2. A proton has spin and magnetic moment just like an electron. Why then its effect is neglected in magnetism of materials?

[NCERT Exemp. Q. 5.11, Page 30]

Ans. As we know that the magnetic moment of electron or proton is inversely proportional to the mass of electron or proton.

If m_e is the mass of the electron,

Then magnetic moment of electron, $\mu_e \propto 1/m_e$

If m_p is the mass of the proton,

then magnetic moment of the proton, $\mu_p \propto 1/m_p$

$$\frac{\mu_e}{\mu_p} = \frac{\frac{1}{m_e}}{\frac{1}{m_p}}$$

$$\frac{\mu_e}{\mu_p} = \frac{m_p}{m_e}$$

But, $m_e << m_p$

So that, $\mu_e >> \mu_p$ **[1½]**

Since the mass of a proton is approximately 2,000 times larger than the mass of an electron, the magnetic moment of electron will be 2,000 times larger than the magnetic moment of a proton.

So that its effect is neglected. **[½]**

Q. 3. A ball of superconducting material is dipped in liquid nitrogen and placed near a bar magnet.

(i) In which direction will it move?

(ii) What will be the direction of its magnetic moment?

[NCERT Exemp. Q. 5.15, Page 31]

Ans. The magnetic nature of super-conducting material and nitrogen are diamagnetic in nature. If a ball of super-conducting material is dipped in liquid nitrogen then it behaves as diamagnetic material. **[1]**

When it is placed near a bar magnet, it will be feebly magnetised opposite to the direction of magnetising field, therefore,

(i) It will repel away from the magnet.

(ii) The direction of the magnetic moment will be opposite to the direction of the magnetic field of the bar magnet. That means its direction is from left to right. **[1]**

Q. 4. 5.5 A closely wound solenoid of 800 turns and area of cross section 2.5 × 10^{-4} m^2 carries a current of 3.0 A. Explain the sense in which the solenoid acts like a bar magnet. What is its associated magnetic moment? [NCERT Ex. Q. 5.5, Page 201]

Ans. Given that,

Number of turns in the solenoid, $n = 800$

Area of cross-section, $A = 2.5 \times 10^{-4}$ m^2

Current in the solenoid, I = 3.0 A

A current-carrying solenoid behaves as a bar magnet because a magnetic field develops along its axis, *i.e.*, along its length. The magnetic moment associated with the given current-carrying solenoid is :

$M = n\,I\,A$ **[1½]**

$= 800 \times 3 \times 2.5 \times 10^{-4}$

$= 0.6\,JT^{-1}$ **[½]**

Q. 5. If the solenoid in Exercise 5.5 (Q. 4 VSATQ) is free to turn about the vertical direction and a uniform horizontal magnetic field of 0.25 T is applied, what is the magnitude of torque on the solenoid when its axis makes an angle of 30° with the direction of applied field? [NCERT Ex. Q. 5.6, Page 201]

Ans. Given that,

Magnetic field strength, $B = 0.25$ T

Magnetic moment, $M = 0.6$ Nm T^{-1}

The angle θ, between the axis of the solenoid and the direction of the applied field is 30°.

So, the torque acting on the solenoid is :

$\tau = MB\sin\theta$

$= 0.6 \times 0.25 \sin 30^\circ$

$= 7.5 \times 10^{-2}$ J **[2]**

Q. 6. A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip pointing down at 22° with the horizontal. The horizontal component of the Earth's magnetic field at the place is known to be 0.35 G. Determine the magnitude of the Earth's magnetic field at the place. [NCERT Ex. Q. 5.10, Page 201]

Ans. Given that,

Horizontal component of Earth's magnetic field, $B_H = 0.35$ G

Angle made by the needle with the horizontal plane = Angle of dip, $\delta = 22°$

Earth's magnetic field strength = B

As we know the relation between B and B_H as :

$B_H = B\cos\theta$

$\therefore\ B = \dfrac{B_H}{\cos\ \delta}$

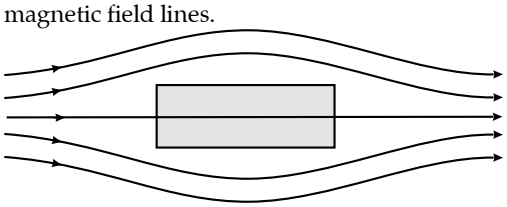
$= \dfrac{0.35}{\cos\ 22^\circ} = 0.377$ G

So, the strength of Earth's magnetic field at the given location is 0.377 G. **[2]**

Q. 7. Depict the behaviour of magnetic field lines in the presence of a diamagnetic material.

[CBSE Board, All India Region, 2016]

Ans. Magnetic permeability of diamagnetic substance is less than air, so it does not allow lines to pass through it. Thus, diamagnetic substances expel magnetic field lines.



[1]

Q. 8. Explain quantitatively the order of magnitude difference between the diamagnetic susceptibility of N_2 (~5 × 10^{-9}) and Cu (~10^{-5}). (at STP)

[NCERT Exemp. Q. 5.13, Page 30]

Ans. Density of nitrogen

$$\rho_{N_2} = \frac{28\text{ g}}{22.4\text{ L}} = \frac{28\text{ g}}{22400\text{ cc}}$$

Also, density of copper

$$\rho_{Cu} = \frac{8\text{ g}}{22.4\text{ L}} = \frac{8\text{ g}}{22400\text{ cc}}$$

So, ratio of both densities

$$\frac{\rho_{N_2}}{\rho_{Cu}} = \frac{28}{22400} \times \frac{1}{8} = 16 \times 10^{-4}$$

Also given

$$\frac{\chi_{N_2}}{\chi_{Cu}} = \frac{5 \times 10^{-9}}{10^{-5}} = 5 \times 10^{-4}$$

We know that,

$$\chi = \frac{\text{Magnetisation}(I)}{\text{Magnetic intensity}\ (H)}$$

$$= \frac{\text{Magnetic moment}\ (M)/\text{Volume}\ (V)}{H}$$

$$= \frac{M}{HV}$$

$$= \frac{M}{H(\text{Mass/Density})}$$

$$= \frac{M\rho}{Hm}$$

$$\chi \propto \rho \qquad \left(\because \frac{M}{Hm} = \text{constant}\right)$$

So that,

$$\frac{\chi_{N_2}}{\chi_{Cu}} = \frac{\rho_{N_2}}{\rho_{Cu}} = 1.6 \times 10^{-4}$$

So, the major difference between in susceptibility of N_2 and Cu is due to their density. Hence we can

say that magnitude difference or major difference between the diamagnetic susceptibility of N_2 and Cu is 1.6×10^{-4}. **[2]**

Q. 9. From molecular view point, discuss the temperature dependence of susceptibility for diamagnetism, para-magnetism and ferro-magnetism. [NCERT Exemp. Q. 5.14, Page 31]

Ans. Diamagnetism is due to orbital motion of electrons developing magnetic moments opposite to applied field and hence is not much affected by temperature. **[1]**

Para-magnetism and ferro-magnetism is due to alignments of atomic magnetic moments in the direction of the applied field. As temperature is raised, this alignment is disturbed so that susceptibilities of both decrease as temperature increases. **[1]**

Q. 10. Write two properties of a material suitable for making (a) *a* permanent magnet, and (b) an electromagnet.

[CBSE Board, All India Region, 2016]

Ans. Two properties of material used for making permanent magnets are :

(a) high coercivity

(b) high retentivity and high hysteresis loss **[1]**

Two properties of material used for making electromagnets are :

(a) high permeability

(b) low coercivity **[1]**

Q. 11. A permanent magnet in the shape of a thin cylinder of length 10 cm has $M = 10^6$ A/m. Calculate the magnetisation current I_M.

[NCERT Exemp. Q. 5.12, Page 30]

Ans. Given that,

Intensity of magnetisation, $M = 10^6$ A/m.

Length of cylinder, $l = 10$ cm $= 10 \times 10^{-2}$ m $= 0.1$ m

Magnetisation current $= I_M$

As we know that the intensity of magnetisation,

$$M = \frac{I_M}{l}$$

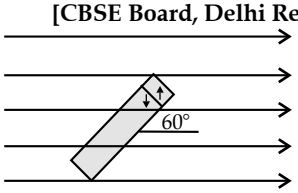
$\Rightarrow I_M = M \times l = 10^6 \times 0.1 = 10^5$A

So that the magnetisation current I_M will be 10^5 A. **[2]**

Short Answer Type Questions **(3 marks each)**

Q. 1. A bar magnet of magnetic moment 6 J/T is aligned at 60° with a uniform external magnetic field of 0.44 T. Calculate (a) the work done in turning the magnet to align its magnetic moment (i) normal to the magnetic field, (ii) opposite to the magnetic field, and (b) the torque on the magnet in the final orientation in case (ii). [CBSE Board, Delhi Region, 2018]

Ans.



$B = 0.44$ T(magnetic field)

$\mu = 6$ J/T (Magnetic moment)

$\Delta W = \mu B[\cos\theta_1 - \cos\theta_2]$

(a) **(i)** Magnetic moment align normal to magnetic field

$\theta_1 = 60^\circ, \theta_2 = 90^\circ$

$\Delta W = 6 \times 0.44\ [\cos 60^\circ - \cos 90°]$

$= 6 \times 0.44 \times \left[\frac{1}{2} - 0\right]$

$= 3 \times 0.44$

$\Delta W = 1.32$ J

(ii) Opposite to magnetic field $\theta_1 = 60°, \theta_2 = 180°$

$\Delta W = 6 \times 0.44[\cos 60^\circ - \cos 180°]$

$\Delta W = 6 \times 0.44 \times \left[\frac{1}{2} - (-1)\right]$

$\Delta W = 6 \times 0.44\left[\frac{3}{2}\right]$

$= 3 \times 0.44 \times 3$

$\Delta W = 3.96$ J **[2]**

(b) Torque acting on magnet placed in magnetic field

$\tau = \mu B \sin\theta$

$\tau = \mu B \sin 180° = 0$ **[1]**

Q. 2. A short bar magnet has a magnetic moment of 0.48 JT^{-1}. Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on (a) the axis, (b) the equatorial lines (normal bisector) of the magnet.

[NCERT Ex. Q. 5.12, Page 201]

Ans. Given that,

Magnetic moment of the bar magnet, $M = 0.48\ JT^{-1}$

(a) Distance, $d = 10$ cm $= 0.1$ m

The magnetic field at distance d, from the centre of the magnet on the axis is given by the relation :

$$B = \frac{\mu_0}{4\pi}\frac{2M}{d^3}$$

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7}$ Tm A^{-1}

$$\therefore B = \frac{4\pi \times 10^{-7} \times 2 \times 0.48}{4\pi \times (0.1)^3}$$

$$= 0.96 \times 10^{-4}\text{T} = 0.96\text{ G}$$

The magnetic field is along the S–N direction. **[1½]**

(b) The magnetic field at a distance of 10 cm (*i.e.*, $d = 0.1$ m) on the equatorial line of the magnetic is given as :

$$B = \frac{\mu_0 \times M}{4\pi \times d^3}$$

$$= \frac{4\pi \times 10^{-7} \times 0.48}{4 \times (0.1)^3}$$

$$= 0.48 \text{ G}$$

The magnetic field is along the N–S direction. **[1½]**

Q. 3. A short bar magnet placed in a horizontal plane has its axis aligned along the magnetic north-south direction. Null points are found on the axis of the magnet at 14 cm from the centre of the magnet. The Earth's magnetic field at the place is 0.36 G and the angle of dip is zero. What is the total magnetic field on the normal bisector of the magnet at the same distance as the null-point (*i.e.*, 14 cm) from the centre of the magnet? (At null points, field due to a magnet is equal and opposite to the horizontal component of Earth's magnetic field.) [NCERT Ex. Q. 5.13, Page 201]

Ans. Given that,

Earth's magnetic field at the given place, $H = 0.36$ G

The magnetic field at a distance d, on the axis of the magnet is given as :

$$B_1 = \frac{\mu_0}{4\pi}\frac{2M}{d^3} = H \quad \ldots(i)$$

Where,

μ_0 = Permeability of free space

M = Magnetic moment

The magnetic field at the same distance d, on the equatorial line of the magnet is given as :

$$B_2 = \frac{\mu_0 M}{4\pi d^3} = \frac{H}{2}$$ [Using equation (i)]

Total magnetic field, $B = B_1 + B_2$

$$= H + \frac{H}{2}$$

$0.36 + 0.18 = 0.54$ G

So that, the magnetic field is 0.54 G in the direction of Earth's magnetic field. **[3]**

Q. 4. If the bar magnet in exercise 5.13 is turned around by 180°, where will the new null points be located? [NCERT Ex. Q. 5.14, Page 201]

Ans. The magnetic field on the axis of the magnet at a distance, $d_1 = 14$ cm, can be written as :

$$B_1 = \frac{\mu_0 2M}{4\pi (d_1)^3} = H \quad \ldots(1)$$

Where,

M = Magnetic moment

μ_0 = Permeability of free space

H = Horizontal component of the magnetic field at d_1

If the bar magnet is turned through 180°, then the neutral point will lie on the equatorial line.

So that, the magnetic field at a distance, d_2, on the equatorial line of the magnet can be written as :

$$B_2 = \frac{\mu_0 M}{4\pi (d_2)^3} = H \quad \ldots(2)$$

[1]

Equating equations (1) and (2), we get :

$$\frac{2}{(d_1)^3} = \frac{1}{(d_2)^3}$$

$$\left(\frac{d_2}{d_1}\right)^3 = \frac{1}{2}$$

$$\therefore \quad d_2 = d_1 \times \left(\frac{1}{2}\right)^{\frac{1}{3}}$$

$$= 14 \times 0.794 = 11.1 \text{ cm}$$

The new null points will be located 11.1 cm on the normal bisector. **[2]**

Q. 5. A short bar magnet of magnetic moment 5.25×10^{-2} JT^{-1} is placed with its axis perpendicular to the Earth's field direction. At what distance from the centre of the magnet, the resultant field is inclined at 45° with Earth's field on (a) its normal bisector and (b) its axis. Magnitude of the Earth's field at the place is given to be 0.42 G. Ignore the length of the magnet in comparison to the distances involved. [NCERT Ex. Q. 5.15, Page 202]

Ans. Given that,

Magnetic moment of the bar magnet,

$M = 5.25 \times 10^{-2}$ JT^{-1}

Magnitude of Earth's magnetic field at a place,

$H = 0.42$ G $= 0.42 \times 10^{-4}$ T

(a) The magnetic field at a distance R from the centre of the magnet on the normal bisector is given by the relation :

$$B = \frac{\mu_0 M}{4\pi R^3}$$

Where,

μ_0 = Permeability of free space = $4\pi \times 10^{-7}$ TmA^{-1}

When the resultant field is inclined at 45° with Earth's field, $B = H$

$$\therefore \frac{\mu_0 M}{4\pi R^3} = H = 0.42 \times 10^{-4}$$

$$R^3 = \frac{\mu_0 M}{0.42 \times 10^{-4} \times 4\pi}$$

$$= \frac{4\pi \times 10^{-7} \times 5.25 \times 10^{-2}}{4\pi \times 0.42 \times 10^{-4}}$$

$$= 12.5 \times 10^{-5}$$

$\therefore R = 0.05$ m $= 5$ cm **[1½]**

(b) The magnetic field at a distance R' from the centre of the magnet on its axis is given as :

$$B' = \frac{\mu_0 2M}{4\pi R^3}$$

The resultant field is inclined at 45° with Earth's field.

$$\therefore \quad B' = H$$

$$\frac{\mu_0 2M}{4\pi (R')^3} = H$$

$$(R')^3 = \frac{\mu_0 2M}{4\pi \times H}$$

$$= \frac{4\pi \times 10^{-7} \times 2 \times 5.25 \times 10^{-2}}{4\pi \times 0.42 \times 10^{-4}}$$

$$= 25 \times 10^{-5}$$

$\therefore R' = 0.063$ m $= 6.3$ cm **[1½]**

Q. 6. A bar magnet of magnetic moment 1.5 JT^{-1} lies aligned with the direction of a uniform magnetic field of 0.22 T.

(a) What is the amount of work required by an external torque to turn the magnet so as to align its magnetic moment :

(i) normal to the field direction

(ii) opposite to the field direction?

(b) What is the torque on the magnet in cases (i) and (ii)? [NCERT Ex. Q. 5.7, Page 201]

Ans. **(a)** Given that,

Magnetic moment, $M = 1.5\ JT^{-1}$

Magnetic field strength, $B = 0.22$ T

(i) Initial angle between the axis and the magnetic field, $\theta_1 = 0°$

Final angle between the axis and the magnetic field, $\theta_2 = 90°$

The work required to make the magnetic moment normal to the direction of magnetic field is given as :

$$W = -MB(\cos\theta_2 - \cos\theta_1)$$
$$= -1.5 \times 0.22(\cos 90° - \cos 0°)$$
$$= -0.33(0 - 1)$$
$$= 0.33\ J$$

(ii) Initial angle between the axis and the magnetic field, $\theta_1 = 0°$

Final angle between the axis and the magnetic field, $\theta_2 = 180°$

The work required to make the magnetic moment opposite to the direction of magnetic field is given as :

$$W = -MB(\cos\theta_2 - \cos\theta_1)$$
$$= -1.5 \times 0.22(\cos 180° - \cos 0°)$$
$$= -0.33(-1 - 1)$$
$$= 0.66\ J$$ **[1½]**

(b) For case (i) : $\theta = \theta_2 = 90°$

$\therefore$ Torque, $\tau = MB\sin\theta$

$= 1.5 \times 0.22 \sin 90°$

$= 0.33$ J

For case (ii) : $\theta = \theta_2 = 180°$

$\therefore$ Torque, $\tau = MB\sin\theta$

$= MB\sin 180° = 0$ J **[1½]**

Q. 7. A closely wound solenoid of 2,000 turns and area of cross-section $1.6 \times 10^{-4}\ m^2$, carrying a current of 4.0 A, is suspended through its centre allowing it to turn in a horizontal plane.

(a) What is the magnetic moment associated with the solenoid?

(b) What is the force and torque on the solenoid if a uniform horizontal magnetic field of 7.5×10^{-2} T is set up at an angle of 30° with the axis of the solenoid? [NCERT Ex. Q. 5.8, Page 201]

Ans. Given that,

Number of turns on the solenoid, $n = 2{,}000$

Area of cross-section of the solenoid, $A = 1.6 \times 10^{-4}\ m^2$

Current in the solenoid, $I = 4$ A

(a) The magnetic moment along the axis of the solenoid is calculated as :

$M = nAI$

$= 2{,}000 \times 1.6 \times 10^{-4} \times 4$

$= 1.28\ Am^2$ **[1½]**

(b) Magnetic field, $B = 7.5 \times 10^{-2}$ T

Angle between the magnetic field and the axis of the solenoid, $\theta = 30°$

Torque will be,

$\tau = MB\sin\theta$

$= 1.28 \times 7.5 \times 10^{-2} \sin 30°$

$= 4.8 \times 10^{-2}$ Nm

So that, the magnetic field is uniform, the force on the solenoid is zero. The torque on the solenoid is 4.8×10^{-2} Nm. **[1½]**

Q. 8. A short bar magnet of magnetic moment $m = 0.32\ JT^{-1}$ is placed in a uniform magnetic field of 0.15 T. If the bar is free to rotate in the plane of the field, which orientation would correspond to its (a) stable, and (b) unstable equilibrium? What is the potential energy of the magnet in each case? [NCERT Ex. Q. 5.4, Page 200]

Ans. Given that,

$m = 0.32$ J/T

$B = 0.15$ T

The bar is free to rotate in the plane of the field. It will be in stable position in which the net torque is zero.

It means the field is uniform, there will be no net force. The angle between the field and the moment should be either zero or 180°. This means that it can be parallel or anti-parallel to field.

(a) For stable equilibrium, magnetic moment m should be parallel to applied magnetic field.

(b) For unstable equilibrium, magnetic moment should be anti-parallel to applied magnetic field. **[1]**

Now, the potential energy of the magnet in each case will be,

For stable equilibrium

Potential energy, $U = -mB$

Putting the values

$U = -0.32 \times 0.15$

$U = -0.048$ J

For unstable equilibrium

$U = mB$

Putting the values

$U = 0.32 \times 0.15$

$U = +0.048$ J **[2]**

Q. 9. A bar magnet of magnetic moment m and moment of inertia I (about centre, perpendicular to length) is cut into two equal pieces, perpendicular to length. Let T be the period of oscillations of the original magnet about an axis through the mid-point, perpendicular to length, in a magnetic field B. What would be the similar period T' for each piece? [NCERT Exemp. Q. 5.19, Page 31]

Ans. As we know that the, time period due to the oscillation is given by,

$$T=2\pi\sqrt{\frac{I}{MB}}$$

Where,

M = Magnetic moment of the magnet

B = Uniform magnetic field

If the magnet of magnetic field M is cut into n pieces in equal proportion then the magnetic moment M of all equal parts is

$nM' = M$

Magnetic dipole moment for two parts of magnet is, $M' = \frac{M}{2}$

So that the magnetic moment of each parts of magnet is,

$$M' = \frac{M}{2}$$

And we know that the, moment of inertia, $I = \frac{ml^2}{12}$

When magnet is cut into two equal parts, perpendicular to length, then moment of inertia of each part of magnet about an axis perpendicular to length passing through its centre is,

$$I' = \frac{m'l'^2}{12}$$

$$I' = \frac{m}{2}\frac{(l/2)^2}{12} \quad \left(\because l' = \frac{l}{2}\right)$$

$$I' = \frac{ml^2}{12} \times \frac{I}{8} = \frac{I}{8} \quad \left[\because I = \frac{ml^2}{12}\right]$$

So, $I' = \frac{I}{8}$

Magnetic dipole moment

$$M' = \frac{M}{2}$$

Its time period of oscillation is

$$T' = 2\pi\sqrt{\frac{I'}{M'B}}$$

$$= 2\pi\sqrt{\frac{\frac{I}{8}}{\left(\frac{M}{2}\right)B}}$$

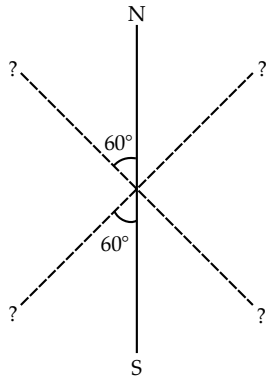
$$= \frac{2\pi}{2}\sqrt{\frac{I}{MB}}$$

So,

$$T' = \frac{T}{2}$$

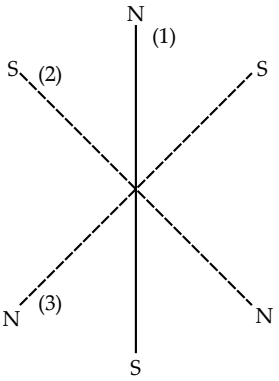
[3]

Q. 10. Three identical bar magnets are riveted together at centre in the same plane as shown in Figure. This system is placed at rest in a slowly varying magnetic field. It is found that the system of magnets does not show any motion. The north-south poles of one magnet are shown in the Figure. Determine the poles of the remaining two.



[NCERT Exemp. Q. 5.17, Page 31]

Ans. The system will be in stable equilibrium, if the net force on the system is zero and net torque on the system is also zero which is possible only when the poles of the remaining two magnets are as shown as below.



The north pole of the magnet one is equally attracted by the south pole of magnet two and three magnets placed at the equal distance. **[3]**

Q. 11. Suppose we want to verify the analogy between electrostatic and magneto statics by an explicit experiment. Consider the motion of

(i) electric dipole p in an electrostatic field E and

(ii) magnetic dipole M in a magnetic field B. Write down a set of conditions on E, B, p, M so that the two motions are verified to be identical. (Assume identical initial conditions.)

[NCERT Exemp. Q. 5.18, Page 31]

Ans. Let consider the angle between p and E is θ.

Torque on electric dipole of moment p in electric field (E)

Then,

$\tau = pE \sin\theta$.

And the angle between M and B is also θ. Torque on magnetic dipole moment M in magnetic field B will be,

$\tau' = MB \sin\theta$

Two motions will be identical, if

$pE \sin\theta = MB \sin\theta$

$pE = MB$ …(i)

As we know that, the relation between E and B is

$E = cB$

By, putting the value of E in *Eq.* (i),

$pcB = M$B

So, $p = \frac{M}{c}$ **[3]**

Q. 12. A circular coil of 16 turns and radius 10 cm carrying a current of 0.75 A rests with its plane normal to an external field of magnitude 5.0×10^{-2} T. The coil is free to turn about an axis in its plane perpendicular to the field direction. When the coil is turned slightly and released, it oscillates about its stable equilibrium with a frequency of 2.0 s^{-1}. What is the moment of inertia of the coil about its axis of rotation? [NCERT Ex. Q. 5.9, Page 201]

Ans. Given that,

Number of turns in the circular coil, $N = 16$

Radius of the coil, $r = 10$ cm $= 0.1$ m

Cross-section of the coil, $A = \pi r^2 = \pi \times (0.1)^2$ m^2

Current in the coil, $I = 0.75$ A

Magnetic field strength, $B = 5.0 \times 10^{-2}$ T

Frequency of oscillations of the coil, $\nu = 2.0\ s^{-1}$

Magnetic moment, $M = NIA = NI\pi r^2$

$= 16 \times 0.75 \times \pi \times (0.1)^2$

$= 0.377\ JT^{-1}$

Frequency is given by the relation :

$$\nu = \frac{1}{2\pi}\sqrt{\frac{MB}{I}}$$

Where

I = Moment of inertia of the coil

As we know that,

$$I = \frac{MB}{4\pi^2\nu^2}$$

$$= \frac{0.377 \times 5 \times 10^{-2}}{4\pi^2 \times (2)^2}$$

$$= 1.19 \times 10^{-4}\ \text{kg m}^2$$

So that, the moment of inertia of the coil about its axis of rotation is 1.19×10^{-4} kg m^2. **[3]**

Q. 13. A magnetic dipole is under the influence of two magnetic fields. The angle between the field directions is 60°, and one of the fields has a magnitude of 1.2×10^{-2} T. If the dipole comes to stable equilibrium at an angle of 15° with this field, what is the magnitude of the other field? [NCERT Add. Ex. Q. 5.21, Page 203]

Ans. Given that,

Magnitude of one of the magnetic fields, $B_1 = 1.2 \times 10^{-2}$ T

Angle between the two fields, $\theta = 60°$

At stable equilibrium, the angle between the dipole and field B_1, $\theta_1 = 15°$

Angle between the dipole and field B_2, $\theta_2 = \theta - \theta_1$

$= 60° - 15° = 45°$

Let the magnitude of the other magnetic field is B_2.

At rotational equilibrium, the torques between both the fields must balance each other.

Torque due to field B_1 = Torque due to field B_2

$$MB_1 \sin\theta_1 = MB_2 \sin\theta_2 \quad \textbf{[1]}$$

Where,

M = Magnetic moment of the dipole

$$\therefore \quad B_2 = \frac{B_1 \sin\theta_1}{\sin\theta_2}$$

$$= \frac{1.2 \times 10^{-2} \times \sin 15°}{\sin 45°}$$

$$= 4.39 \times 10^{-3}\ \text{T}$$

So that, the magnitude of the other magnetic field is 4.39×10^{-3} T. **[2]**

Q. 14. A long straight horizontal cable carries a current of 2.5 A in the direction 10° south of west to 10° north of east. The magnetic meridian of the place happens to be 10° west of the geographic meridian. The Earth's magnetic field at the location is 0.33 G, and the angle of dip is zero. Locate the line of neutral points (ignore the thickness of the cable). (At neutral points, magnetic field due to a current-carrying cable is equal and opposite to the horizontal component of Earth's magnetic field.) [NCERT Add. Ex. Q. 5.18, Page 202]

Ans. Given that,

Current in the wire, $I = 2.5$ A

Angle of dip at the given location on Earth, $= 0°$

Earth's magnetic field, $H = 0.33$ G $= 0.33 \times 10^{-4}$ T

The horizontal component of Earth's magnetic field is :

$$H_H = H\cos\delta$$

$$= 0.33 \times 10^{-4} \times \cos 0°$$

$$= 0.33 \times 10^{-4}\ \text{T} \quad \textbf{[1]}$$

The magnetic field at the neutral point at a distance R from the cable is :

$$H_H = \frac{\mu_0 I}{2\pi R}$$

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7}$ Tm A^{-1}

As we know that,

$$R = \frac{\mu_0 I}{2\pi H_H}$$

$$= \frac{4\pi \times 10^{-7} \times 2.5}{2\pi \times 0.33 \times 10^{-4}}$$

$$= 15.15 \times 10^{-3}\ \text{m}$$

$$= 1.51\ \text{cm}$$

So that, a set of neutral points parallel to and above the cable are located at a normal distance of 1.51 cm. **[2]**

Q. 15. A telephone cable at a place has four long straight horizontal wires carrying a current of 1.0 A in the same direction east to west. The Earth's magnetic field at the place is 0.39 G, and the angle of dip is 35°. The magnetic declination is nearly zero. What are the resultant magnetic fields at points 4.0 cm below the cable? [NCERT Add. Ex. Q. 5.19, Page 202]

Ans. Given that,

Number of horizontal wires in the telephone cable, $n = 4$

Current in each wire, $I = 1.0$ A

Earth's magnetic field at a location, $H = 0.39\ G = 0.39 \times 10^{-4}$ T

Angle of dip at the location, $\delta = 35°$

Angle of declination, $\theta \sim 0°$

For a point 4 cm below the cable :
Distance, r = 4 cm = 0.04 m
The horizontal component of Earth's magnetic field is :
$H_H = H\cos\delta - B$
Where,
B = Magnetic field at 4 cm due to current I in the four wires

$$B = 4\times\frac{\mu_0 I}{2\pi r}$$

μ_0 = Permeability of free space = $4\pi \times 10^{-7}$ Tm A^{-1}

$$\therefore B = 4\times\frac{4\pi\times10^{-7}\times1}{2\pi\times0.04}$$

$= 0.2 \times 10^{-4}$ T = 0.2 G
$\therefore H_H = 0.39\cos 35° - 0.2$
$= 0.39 \times 0.819 - 0.2 \approx 0.12$ G
The vertical component of Earth's magnetic field is :
$H_v = H\sin\delta$
$= 0.39 \sin 35°$
$= 0.22$ G
The angle made by the field with its horizontal component is :

$$\theta = \tan^{-1}\frac{H_v}{H_H}$$

$$= \tan^{-1}\frac{0.22}{0.12} = 61.39°$$

The resultant field at the point is given as :

$$H_1 = \sqrt{(H_v)^2 + (H_H)^2}$$

$$= \sqrt{(0.22)^2 + (0.12)^2} = 0.25 \text{ G}$$ **[1½]**

For a point 4 cm above the cable :
Horizontal component of Earth's magnetic field :
$H_H = H\cos\delta + B$
$= 0.39\cos 35° + 0.2 = 0.52$ G
Vertical component of Earth's magnetic field :
$H_v = H\sin\delta$
$= 0.39 \sin 35° = 0.22$ G

$$\text{Angle, } \theta = \tan^{-1}\frac{H_v}{H_H} = \tan^{-1}\frac{0.22}{0.52} = 22.9°$$

And resultant field :

$$H_2 = \sqrt{(H_v)^2 + (H_H)^2}$$

$$= \sqrt{(0.22)^2 + (0.52)^2}$$

$= 0.56$ T **[1½]**

Q. 16. A compass needle free to turn in a horizontal plane is placed at the centre of circular coil of 30 turns and radius 12 cm. The coil is in a vertical plane making an angle of 45° with the magnetic meridian. When the current in the coil is 0.35 A, the needle points west to east.

(a) Determine the horizontal component of the Earth's magnetic field at the location.

(b) The current in the coil is reversed, and the coil is rotated about its vertical axis by an angle of 90° in the anticlockwise sense looking from above. Predict the direction of the needle. Take the magnetic declination at the places to be zero.

[NCERT Add. Ex. Q. 5.20, Page 203]

Ans. Given that,
Number of turns in the circular coil, N = 30
Radius of the circular coil, r = 12 cm = 0.12 m
Current in the coil, I = 0.35 A
Angle of dip, δ = 45°

(a) The magnetic field due to current I, at a distance r, is :

$$B = \frac{\mu_0 2\pi NI}{4\pi r}$$

Where,
μ_0 = Permeability of free space = $4\pi \times 10^{-7}$ TmA^{-1}

$$\therefore B = \frac{4\pi\times10^{-7}\times2\pi\times30\times0.35}{4\pi\times0.12}$$

$= 5.49 \times 10^{-5}$ T
And, the compass needle points from West to East. So that, the horizontal component of Earth's magnetic field is :
$B_H = B\sin\delta$
$= 5.49 \times 10^{-5} \sin 45°$
$= 3.88 \times 10^{-5}$ T
$= 0.388$ G **[1½]**

(b) When the current in the coil is reversed and the coil is rotated about its vertical axis by an angle of 90°, then the needle will also reverse its original direction. So, in this case, the needle will point from East to West. **[1½]**

Q. 17. At a certain location in Africa, a compass points 12° west of the geographic north. The north tip of the magnetic needle of a dip circle placed in the plane of magnetic meridian points 60° above the horizontal. The horizontal component of the Earth's field is measured to be 0.16 G. Specify the direction and magnitude of the Earth's field at the location. [NCERT Ex. Q. 5.11, Page 201]

Ans. Given that,
Angle of declination, θ = 12°
Angle of dip, δ = 60°
Horizontal component of Earth's magnetic field, B_H = 0.16 G
Earth's magnetic field at the given location = B
As we know the relation between B and B_H as :

$$B_H = B\cos\delta$$

$$\therefore B = \frac{B_H}{\cos\delta}$$

$$= \frac{0.16}{\cos 60°}$$

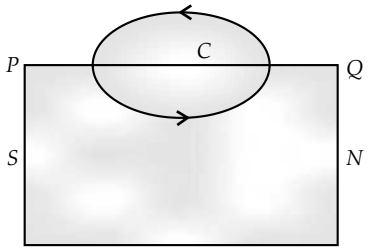
$= 0.32$ G

Earth's magnetic field lies in the vertical plane, 12° west of the geographic meridian, making an angle of 60° (upwards) with the horizontal direction. Its magnitude is 0.32 G. **[3]**

Q. 18. Use (i) the Ampere's law for H and (ii) continuity of lines of B, to conclude that inside a bar magnet, (a) lines of H run from the N pole to S pole, while (b) lines of B must run from the S pole to N pole.

[NCERT Exemp. Q. 5.20, Page 31]

Ans.



Let the magnetic field line of B through the bar magnet must be in a closed loop as shown above in figure.

Consider C is an Amperian loop

So, $\int_Q^P H.dl = \int_Q^P \frac{B}{\mu_0}.dl$

As the angle between B and dl is less than 90° inside the bar magnet. So, cos θ > 0 so that it is positive.

$\int_Q^P H.dl = \int_Q^P \frac{B}{\mu_0}.dl > 0$

So that, the lines of B must run from South Pole (S) to north pole (N) inside the bar magnet. By Ampere's law,

$\oint_{PQP} H.dl = 0$

So, $\Rightarrow \oint_{PQP} H.dl = \int_P^Q H.dl + \int_Q^P H.dl = 0$

As $\int_Q^P H.dl > 0$ (Inside the magnet)

So, $\int_P^Q H.dl > 0$ (Outside the magnet)

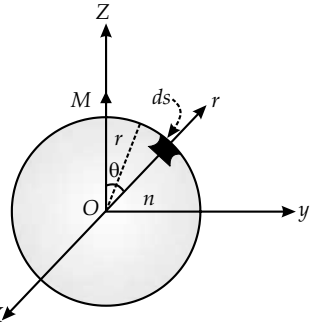
So that, it will be so angle between H and dl is more than 90° so the cos θ is negative.

That means the line of H must run from N-pole to S-pole inside the bar magnet. **[3]**

Q. 19. Verify the Gauss's law for magnetic field of a point dipole of dipole moment m at the origin for the surface which is a sphere of radius R.

[NCERT Exemp. Q. 5.16, Page 31]

Ans. The magnetic field of a point dipole of dipole moment m at the origin for the surface which is a sphere of radius R is shown below :



[1]

We have to prove that : $\oint B.dS = 0$.

This is called Gauss's law in magnetism.

Now, magnetic moment of dipole at origin O is along z-axis.

Let P be a point at distance r from O and OP makes an angle θ with z-axis.

Component of M along $OP = M \cos \theta$.

Now, the magnetic field induction at P due to dipole of moment $M \cos \theta$ is

$B = \frac{\mu_0}{4\pi}\frac{2M\cos\theta}{r^3}\hat{r}$

From the diagram, r is the radius of sphere with centre at O lying in yz – plane.

Take an elementary area, dS of the surface at P. Then,

$dS = r(r\sin\theta\, d\theta)\hat{r} = r^2 \sin\theta\, d\theta\, \hat{r}$

$$\oint B.dS = \oint \frac{\mu_0}{4\pi}\frac{2M\cos\theta}{r^3}\hat{r}(r^2\sin\theta\, d\theta\, \hat{r})$$

$$= \frac{\mu_0}{4\pi}\frac{M}{r}\int_0^{2\pi} 2\sin\theta\cdot\cos\theta\, d\theta$$

$$= \frac{\mu_0}{4\pi}\frac{M}{r}\int_0^{2\pi} \sin 2\theta\, d\theta$$

$$= -\frac{\mu_0}{4\pi}\frac{M}{2r}\left(\frac{\cos 2\theta}{2}\right)_0^{2\pi}$$

$$= -\frac{\mu_0}{4\pi}\frac{M}{2r}[\cos 4\pi - \cos 0]$$

$$= -\frac{\mu_0}{4\pi}\frac{M}{2r}[1-1]$$

$$= 0$$

Thus proved **[2]**

Q. 20. A mono-energetic (18 keV) electron beam initially in the horizontal direction is subjected to a horizontal magnetic field of 0.04 G normal to the initial direction. Estimate the up or down deflection of the beam over a distance of 30 cm (m_e = 9.11 × 10^{-31} Kg).

[Note : Data in this exercise are so chosen that the answer will give you an idea of the effect of Earth's magnetic field on the motion of the electron beam from the electron gun to the screen in a TV set.]

[NCERT Add. Ex. Q. 5.22, Page 203]

Ans. Given that,

Energy of an electron beam, $E = 18$ keV = 18×10^3 eV

Charge on an electron, $e = 1.6 \times 10^{-19}$ C

$E = 18 \times 10^3 \times 1.6 \times 10^{-19}$ J

Magnetic field, $B = 0.04$ G

Mass of an electron, $m_e = 9.11 \times 10^{-31}$ kg

Distance up to which the electron beam travels, d = 30 cm = 0.3 m

As we know that the kinetic energy of the electron beam is :

$E = \frac{1}{2}mv^2$

So that,

$$v = \sqrt{\frac{2E}{m}}$$

$$= \sqrt{\frac{2\times 18\times 10^3 \times 1.6\times 10^{-19}\times 10^{-15}}{9.11\times 10^{-31}}}$$

$$= 0.795\times 10^8 \text{ m/s}$$

[1½]

The electron beam deflects along a circular path of radius, r and the force due to the magnetic field balances the centripetal force of the path.

$$Bev = \frac{mv^2}{r}$$

$$\therefore \quad r = \frac{mv}{Be}$$

$$= \frac{9.11 \times 10^{-31} \times 0.795 \times 10^{8}}{0.4 \times 10^{-4} \times 1.6 \times 10^{-19}}$$

$$= 11.3 \text{ m}$$

Let the up and down deflection of the electron beam be $x = r(1 - \cos\theta)$

Where,

θ = Angle of declination

$$\sin\theta = \frac{d}{r}$$

$$= \frac{0.3}{11.3}$$

$$\theta = \sin^{-1}\frac{0.3}{11.3} = 1.521°$$

And $x = 11.3\,(1 - \cos 1.521°)$

$= 0.0039$ m $= 3.9$ mm

So that, the up and down deflection of the beam is 3.9 mm. **[1½]**

Q. 21. A sample of paramagnetic salt contains 2.0 × 10^{24} atomic dipoles each of dipole moment 1.5 × 10^{-23} JT^{-1}. The sample is placed under a homogeneous magnetic field of 0.64 T, and cooled to a temperature of 4.2 K. The degree of magnetic saturation achieved is equal to 15%. What is the total dipole moment of the sample for a magnetic field of 0.98 T and a temperature of 2.8 K? (Assume Curie's law). [NCERT Add. Ex. Q. 5.23, Page 203]

Ans. Given that,

Number of atomic dipoles, $n = 2.0 \times 10^{24}$

Dipole moment of each atomic dipole, $M = 1.5 \times 10^{-23}$ JT^{-1}

When the magnetic field, $B_1 = 0.64$ T

Sample is cooled to a temperature, $T_1 = 4.2°$ K

As we know that,

The total dipole moment of the atomic dipole, M_{tot}

$= n \times M$

$= 2 \times 10^{24} \times 1.5 \times 10^{-23}$

$= 30$ JT^{-1}

Magnetic saturation is achieved at 15%.

So effective dipole moment,

$$M_1 = \frac{15}{100} \times 30 = 4.5 \text{ JT}^{-1}$$ **[1½]**

When the magnetic field, $B_2 = 0.98$ T

Temperature, $T_2 = 2.8°$K

Let its total dipole moment= M_2

According to Curie's law, the ratio of two magnetic dipoles will be,

$$\frac{M_2}{M_1} = \frac{B_2}{B_1} \times \frac{T_1}{T_2}$$

$$\therefore M_2 = \frac{B_2 T_1 M_1}{B_1 T_2}$$

$$= \frac{0.98 \times 4.2 \times 4.5}{2.8 \times 0.64}$$

$$= 10.336 \text{ J T}^{-1}$$

So that, 10.335 JT^{-1} is the total dipole moment of the sample for a magnetic field of 0.98 T and a temperature of 2.8 K. **[1½]**

Q. 22. A Rowland ring of mean radius 15 cm has 3,500 turns of wire wound on a ferromagnetic core of relative permeability 800. What is the magnetic field B in the core for a magnetising current of 1.2 A? [NCERT Add. Ex. Q. 5.24, Page 203]

Ans. Given that,

Mean radius of a Rowland ring, $r = 15$ cm $= 0.15$ m

Number of turns on a ferro-magnetic core,

$N = 3{,}500$

Relative permeability of the core material, $\mu_r = 800$

Magnetising current, $I = 1.2$ A

The magnetic field is given by the relation :

$$A = \frac{\mu_r \mu_0 IN}{2\pi r}$$ **[1]**

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7}$ Tm A^{-1}

$$B = \frac{800 \times 4\pi \times 10^{-7} \times 1.2 \times 3500}{2\pi \times 0.15} = 4.48 \text{ T}$$

So that, the magnetic field in the core is 4.48 T. **[2]**

Q. 23. The magnetic moment vectors μ_s and μ_l associated with the intrinsic spin angular momentum S and orbital angular momentum l, respectively, of an electron are predicted by quantum theory (and verified experimentally to a high accuracy) to be given by :

$$\mu_s = -\frac{e}{m} S,$$

$$\mu_l = -\frac{e}{2m} l$$

Which of these relations is in accordance with the result expected classically? Outline the derivation of the classical result.

[NCERT Add. Ex. Q. 5.25, Page 203]

Ans. The magnetic moment associated with the orbital angular momentum is valid with the classical mechanics.

As we know that, the magnetic moment associated with the orbital angular (l) momentum is given for current i and area of cross-section A, is

Magnetic moment,

$$\mu_i = \text{IA} = \left(\frac{e}{\text{T}}\right)\pi r^2 \qquad \ldots(1)$$

Where,

e =Charge of the electron

r =Radius of the circular orbit

T=Time taken to complete one rotation around the circular orbit of radius r

So, the angular momentum,

$$I = mvr = mr\left(\frac{2\pi r}{T}\right) \qquad \ldots(2)$$

Where,
m = Mass of the electron
v = Velocity of the electron **[1]**
Dividing equation (1) by equation (2), we get :

$$\frac{\mu_l}{l} = \frac{-e}{2m}$$

So, it is clear that, from the above given two relations,

$$\frac{\mu_l}{l} = \frac{-e}{2m}$$

This is in accordance with classical physics. **[2]**

Q. 24. In what way is the behaviour of a diamagnetic material different from that of a paramagnetic, when kept in an external magnetic field?
[CBSE Board, Foreign Scheme, 2016]

Ans. Difference between :

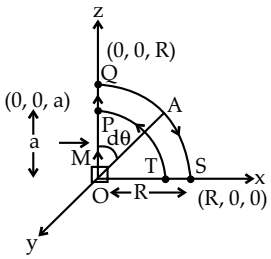
S. No.	Diamagnetic	Para-magnetic
(i)	Diamagnetic substance would move towards the weaker region of the magnetic field.	Para-magnetic substance moves towards the strong region of the magnetic field.
(ii)	Diamagnetic substance is repelled by a magnet.	Para-magnetic substance moves towards the magnet.
(iii)	Diamagnetic substance gets aligned perpendicular to the field.	Para-magnetic substance gets aligned along the magnetic field.

[3]

Long Answer Type Questions **(5 marks each)**

Q. 1. Verify the Ampere's law for magnetic field of a point dipole of dipole moment $m = m\hat{k}$. Take C as the closed curve running clockwise along
(i) the z-axis from $z = a > 0$ to $z = R$;
(ii) along the quarter circle of radius R and centre at the origin, in the first quadrant of x-z plane;
(iii) along the x-axis from $x = R$ to $x = a$, and
(iv) along the quarter circle of radius, a and centre at the origin in the first quadrant of x-z plane.
[NCERT Exemp. Q. 5.21, Page 31]

Ans. Let us consider a figure according to the given situation,



In this figure from P to Q, every point on the z-axis lies at the axial line of magnetic dipole of moment M.
Magnetic field induction, B at a point (0, 0, z) distance z from the magnetic dipole of magnetic moment is,

$$|B| = \frac{\mu_0}{4\pi}\frac{2|M|}{z^3} = \frac{\mu_0 M}{2\pi z^3}$$

(i) By Ampere's law, along z-axis from P to Q

$$\int_P^Q B.dl = \int_P^Q B.dl\cos 0° = \int_a^R B.dz$$

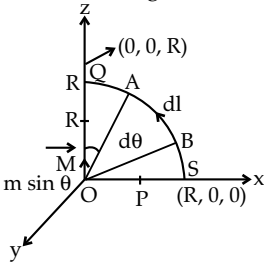
$$= \int_a^R \frac{\mu_0}{2\pi}\frac{M}{z^3}dz$$

$$= \frac{\mu_0 M}{2\pi}\left(\frac{-1}{2}\right)\left(\frac{1}{R^2} - \frac{1}{a^2}\right)$$

$$= \frac{\mu_0 M}{4\pi}\left(\frac{1}{a^2} - \frac{1}{R^2}\right)$$

[1]

(ii) By Ampere's law along the quarter circle QS of radius R as shown in the figure,



The point A can be considered on the equatorial line of the magnetic dipole of moment M sinθ.
So, magnetic field at point A on the circular arc is,

$$B = \frac{\mu_0}{4\pi}\frac{M\sin\theta}{R^3}; \quad dl = Rd\theta$$

$$\therefore\ d\theta = \frac{dl}{R}$$

By Ampere's law,

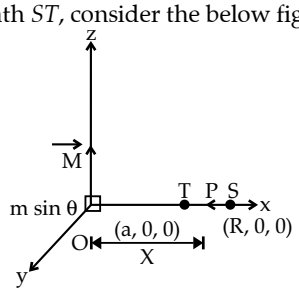
$$\int B.dl = \int B.dl\cos\theta = \int_0^{\pi/2} \frac{\mu_0}{4\pi}\frac{M\sin\theta}{R^3}Rd\theta$$

$$\text{Circular arc} = \frac{\mu_0}{4\pi}\frac{M}{R}(-\cos\theta)_0^{\pi/2}$$

$$= \frac{\mu_0}{4\pi}\frac{M}{R^2}$$

[1]

(iii) By Ampere's law along x-axis from ($X = R$ to $X = a$) over the path ST, consider the below figure,



In the above figure, every point lies on the equatorial line of magnetic dipole.

Magnetic field induction at a point [(P) at a)] distance x from the dipole is,

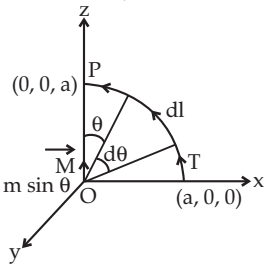
$$B = \frac{\mu_0}{4\pi}\frac{M}{x^3}$$

If angle between ($-M$) and dl is 90°.

$$\int_S^T B.dl = \int_R^a -\frac{\mu_0 M}{4\pi x^3}.dl = 0$$

So, **[1½]**

(iv) By Ampere's law along the quarter circular arc TP of radius a shown below,



From case (ii), we have the line integral of B along the quarter circle TP of radius a is circular arc TP

$$\int B.dl = \int_{\pi/2}^{0} \frac{\mu_0}{4\pi}\frac{M\sin\theta}{a^3}ad\theta$$

$$= \frac{\mu_0}{4\pi}\frac{M}{a^2}\int_{\pi/2}^{0}\sin\theta d\theta$$

$$= \frac{\mu_0}{4\pi}\frac{M}{a^2}[-\cos\theta]_{\pi/2}^{0} = \frac{\mu_0 M}{4\pi a^2}[-1+0]$$

$$= \frac{-\mu_0}{4\pi}\frac{M}{a^2}$$

$$\oint_{PQST} B.dl = \int_P^Q B.dl + \int_Q^S B.dl + \int_S^T B.dl + \int_T^P B.dl$$

$$= \frac{\mu_0 M}{4}\left[\frac{1}{a^2} - \frac{1}{R^2}\right] + \frac{\mu_0}{4\pi}\frac{M}{R^2} + 0 + \left(-\frac{\mu_0}{4\pi}\frac{M}{a^2}\right)$$

$$= 0$$

So that, $\oint_{PQST} B.dl = 0$ **[1½]**

Q. 2. Assume the dipole model for Earth's magnetic field B which is given by B_V = vertical component of magnetic field = $\frac{\mu_0}{4\pi}\frac{2m\cos\theta}{r^3}$

B_H = Horizontal component of magnetic field = $\frac{\mu_0}{4\pi}\frac{\sin\theta\, m}{r^3}$

θ = 90° – latitude as measured from magnetic equator.

Find loci of points for which

(a) $|B|$ is minimum;

(b) dip angle is zero; and

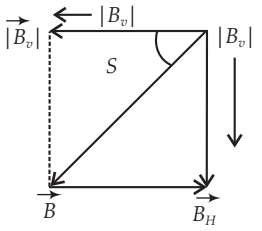
(c) dip angle is ± 45°.

[NCERT Exemp. Q. 5.23, Page 32]

Ans. (a) As given that,

$$B_V = \frac{\mu_0}{4\pi}\frac{2m\cos\theta}{r^3} \quad \ldots(i)$$

$$B_H = \frac{\mu_0}{4\pi}\frac{\sin\theta m}{r^3} \quad \ldots(ii)$$



[1]

Squaring both the equations and adding, we get

$$B^2 = B_V^2 + B_H^2$$

$$= \left(\frac{\mu_0}{4\pi}\right)\frac{m^2}{r^6}[4\cos^2\theta + \sin^2\theta]$$

$$B = \sqrt{B_V^2 + B_H^2}$$

$$= \frac{\mu_0}{4\pi}\frac{m}{r^3}[3\cos^2\theta + 1]^{1/2} \quad \ldots(iii)$$

From *Eq.* (iii), the value of B is minimum,

If $\cos\theta = \frac{\pi}{2}$

$\theta = \frac{\pi}{2}$. So, the magnetic equator is the locus.

(b) For angle of dip δ is

$$\tan\delta = \frac{B_V}{B_H} = \frac{\frac{\mu_0}{4\pi}\cdot\frac{2m\cos\theta}{r^3}}{\frac{\mu_0}{4\pi}\cdot\frac{\sin\theta.m}{r^3}}$$

$= 2\cot\theta$ …(iv)

$\tan\delta = 2\cot\theta$

For dip angle is zero, so, that δ = 0,

$$\cot\theta = 0, \theta = \frac{\pi}{2}$$

So locus is magnetic equator. **[2]**

(c) $\tan\delta = \frac{B_V}{B_H}$

If, (δ = ±45°)

or, $\frac{B_V}{B_H} = \tan(\pm 45°)$, $\frac{B_V}{B_H} = \tan 45°$

$\frac{B_V}{B_H} = 1, \boxed{B_V = B_H}$

$2\cot\theta = 1$ from *Eq.* (iv)

$\cot\theta = \frac{1}{2}$

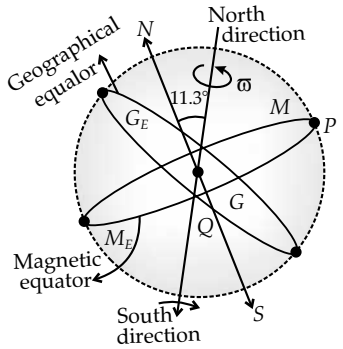
$\tan\theta = 2$

So, $\theta = \tan^{-1}(2)$

So that, $\theta = \tan^{-1}(2)$ is the locus. **[2]**

Q. 3. Consider the plane S formed by the dipole axis and the axis of Earth. Let P be point on the magnetic equator and in S. Let Q be the point of intersection of the geographical and magnetic equators. Obtain the declination and dip angles at P and Q. [NCERT Exemp. Q. 5.24, Page 32]

Ans. The declination is zero as P is in the plane S, needle is in the North direction .



[2½]

P is lies on the magnetic equator, so the angle of dip = 0, so that the value of dip at equator is zero.

Q is also on the magnetic equator, so that the angle of dip is also zero.

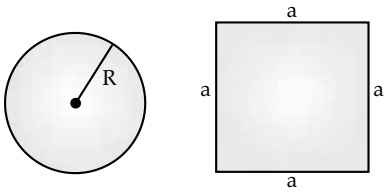
As the angle between axes of rotation of Earth titled on its magnetic axis by 11.3°, so the declination will be 11.3° between P and Q. **[2½]**

Q. 4. There are two current carrying planar coils made each from identical wires of length L. C_1 is circular (radius R) and C_2 is square (side a). They are so constructed that they have same frequency of oscillation when they are placed in the same uniform B and carry the same current. Find a in terms of R. [NCERT Exemp. Q. 5.25, Page 32]

Ans. As given that C_1 is circular coil of radius R, length L, and number of turns per unit length

$$n_1 = \frac{L}{2\pi R}$$

C_2 is square of side a and perimeter L, number of turns per unit length $n_2 = \frac{L}{4a}$



So magnetic moment of C_1

$$m_1 = n_1 I A_1 = \frac{L.I.\pi R^2}{2\pi R}$$

And magnetic moment of C_2

$$m_2 = n_2 I A_2 = \frac{L}{4a}.I.a^2$$

$$m_1 = \frac{LIR}{2} \quad \ldots(i)$$

[1]

$$m_2 = \frac{LIa}{4} \quad \ldots(ii)$$

Moment of inertia of C_1

$$I_1 = \frac{MR^2}{2} \quad \ldots(iii)$$

Moment of inertia of C_2

$$\Rightarrow I_2 = \frac{Ma^2}{12} \quad \ldots(iv)$$

As the frequencies for both coil, are given by

$f_1 = f_2$

$$\left(\frac{2\pi}{T_1}\right) = \left(\frac{2\pi}{T_2}\right), (T_1 = T_2) \quad \ldots(v)$$

So, time period of C_1

$$\Rightarrow T_1 = 2\pi\sqrt{\frac{I_1}{m_1 B}}$$

And time period of C_2

$$\Rightarrow T_2 = 2\pi\sqrt{\frac{I_2}{m_2 B}}$$

[2]

From (v) : $T_1 = T_2$

$$2\pi\sqrt{\frac{I_1}{m_1 B}} = 2\pi\sqrt{\frac{I_2}{m_2 B}}$$

$$\frac{I_1}{m_1} = \frac{I_2}{m_2}$$

$$\frac{m_2}{m_1} = \frac{I_2}{I_1}$$

or $\quad \ldots(vi)$

Putting the values by *Eqs.* (i), (ii), (iii) and (iv) in (vi)

So, $\frac{LIa.2}{4 \times LIR} = \frac{Ma^2.2}{12.MR^2}$

$$\frac{a}{2R} = \frac{a^2}{6R^2}$$

$$a = 3\text{R}$$

[2]

Q. 5. Answer the following :

(a) A vector needs three quantities for its specification. Name the three independent quantities conventionally used to specify the earth's magnetic field.

(b) The angle of dip at a location in southern India is about 18°. Would you expect a greater or smaller dip angle in Britain?

(c) If you made a map of magnetic field lines at Melbourne in Australia, would the lines seem to go into the ground or come out of the ground?

(d) In which direction would a compass free to move in the vertical plane point to, if located right on the geomagnetic north or south pole?

(e) The earth's field, it is claimed, roughly approximates the field due to a dipole of magnetic moment 8×10^{22} J T^{-1} located at its centre. Check the order of magnitude of this number in some way.

(f) Geologists claim that besides the main magnetic *N-S* poles, there are several local poles on the earth's surface oriented in different directions. How is such a thing possible at all?

[NCERT Ex. Q. 5.1, Page 200]

Ans. (a) The three independent quantities conventionally used for specifying Earth's magnetic field are :

1. Magnetic declination,

2. Angle of dip,

3. Horizontal component of Earth's magnetic field **[1]**

(b) The angle of dip at a point depends on how far the point is located with respect to the North Pole or the South Pole. So, as the location of Britain on the globe is near to the magnetic North Pole, the angle of dip would be greater in Britain (~70°) than in southern India. **[½]**

(c) It is assumed that a huge bar magnet is dipped inside Earth with its North Pole near the geographic South Pole and its south pole near the geographic North Pole.
Magnetic field lines emanate from a magnetic North Pole and terminate at a magnetic South Pole. So that, in a map depicting Earth's magnetic field lines, the field lines at Melbourne, Australia would seem to leave the ground. **[1]**

(d) If a compass is located on the geomagnetic North Pole or the South Pole, then the compass will be free to move in the horizontal plane while Earth's field is exactly vertical to the magnetic poles. In such a case, the compass can point in any direction. **[1]**

(e) Magnetic moment, $M = 8\times10^{22}\,\text{JT}^{-1}$
Radius of Earth, $r = 6.4\times10^{6}$ m
As we know that the,

Magnetic field strength, $B = \dfrac{\mu_0 M}{4\pi r^3}$

Where,
μ_0 = Permeability of free space
$= 4\pi\times10^{-7}\,\text{TmA}^{-1}$
So,

$$B = \frac{4\pi\times10^{-7}\times10^{22}}{4\pi\times(6.4\times10^{6})^3} = 0.3\,\text{G}$$

This quantity is of the order of magnitude of the observed field on Earth. **[1]**

(f) Yes, there are several local poles on Earth's surface oriented in different directions. *A* magnetised mineral deposit is an example of a local N-S pole. **[½]**

Q. 6. Answer the following :

(a) The earth's magnetic field varies from point to point in space. Does it likewise change with time? Provided that this is true, on what time scale does it change apparently?

(b) Knowing that the Earth's core contains iron, geologists do not consider it as a source of the Earth's magnetism. Why?

(c) The charged currents in the external conducting regions of the Earth's core are assumed to be responsible for Earth's magnetism. What could be the source of energy that can sustain these currents?

(d) The Earth may have even switched the course of its field a few times amid its history of 4 to 5 billion years. By what method can geologists think about the Earth's field in such distant past?

(e) The Earth's field departs from its dipole shape considerably at large distances (greater than about 30,000 km). What parameters may be considered and held responsible for this distortion?

(f) Interstellar space has an extremely weak magnetic field of the order of 10^{-12} T. Can such a weak field be of any significant consequence? Explain.

[NCERT Ex. Q. 5.2, Page 200]

Ans. **(a)** Earth's magnetic field changes with time. It takes a couple of hundred years to change by an obvious amount. The variation in Earth's magnetic field with the time can't be ignored. **[½]**

(b) Iron in the Earth's core cannot be considered as sources of Earth's magnetism as this iron is in its molten form (non-ferro-magnetic). **[½]**

(c) The radio-activity in the Earth's interior is the source of energy that sustains the currents in the outer conducting regions of Earth's core. These charged currents are considered to be responsible for Earth's magnetism. **[1]**

(d) Earth reversed the direction of its field several times during its history of 4–5 billion years. These magnetic fields got weakly recorded in rocks during their solidification. One can get clues about the geomagnetic history from the analysis of this rock magnetism. **[1]**

(e) Earth's field departs from its dipole shape substantially at large distances (>30,000 km) because of the presence of the ionosphere. In this region, Earth's field gets modified because of the field of single ions. While in motion, these ions produce the magnetic field associated with them. **[1]**

(f) An extremely weak magnetic field can bend charged particles moving in a circle. This may not be noticeable for a large radius path. With reference to the gigantic interstellar space, the deflection can affect the passage of charged particles. **[1]**

Q. 7. What are the dimensions of χ, the magnetic susceptibility? Consider an *H*-atom. Guess an expression for χ, up to a constant by constructing a quantity of dimensions of χ, out of parameters of the atom : *e*, *m*, *v*, *R* and μ_0. Here, m is the electronic mass, *v* is electronic velocity, *R* is Bohr radius. Estimate the number so obtained and compare with the value of $\chi \sim 10^{-5}$ for many solid materials.

[NCERT Exemp. Q. 5.22, Page 32]

Ans. Magnetic susceptibility is a measure of how a magnetic material responds to an external field,

So, $\chi_m = \dfrac{(\text{Intensity of magnetisation})}{(\text{Magnetising force})}$

As M and *H*, both have same units and dimensions. Thus, χ has no dimension. Here, χ is to be related with *e*, *m*, v, *R* and μ_0.
As we know that, the dimensions of μ_0 is
$= [MLQ^{-2}]$
By Biot-Savart's law,

$$dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$

So, $\mu_0 = \dfrac{4\pi^2 dB}{Idl\sin\theta} = \dfrac{4\pi r^2}{Idl\sin\theta}\times\dfrac{F}{qv\sin\theta}$

$$\left[\because dB = \frac{F}{qv\sin\theta}\right],$$

So, dimension of μ_0

$$= \frac{L^2\times[MLT^{-2}]}{[QT^{-1}][L]\times1\times[Q][LT^{-1}]\times1}$$
$$= [MLQ^{-2}]$$

$$\therefore F = MLT^{-2}, I = QT^{-1}, V = LT^{-1}, \theta = 90^\circ$$

where *Q* is the dimension of charge.

χ depends on magnetic moment induced when H is turned on and M and H both have same unit. **[2½]**

So, χ is dimensionless, it should have no involvement of charge Q in its dimensional formula. It will be so if μ_0 and e together should have the value $\mu_0 e^2$, as e has the dimensions of charge.

Consider, dimension of $\chi = \mu_0 e^2 m^a v^b R^c$ …(i)

where a, b and c are the power of m, v and R respectively, such that relation (i) is satisfied. Dimensional equation of (i) is

$$[M^0L^0T^0Q^0] = [MLQ^{-2}] \times [Q^2][M^a] \times [LT^{-1}]^b \times [L]^c$$
$$= [M^{1+a} + L^{1+b+c}T^{-b}Q^0]$$

Equating the power of M, L and T, we have,

$0 = 1 + a$

So, $a = -1$

$0 = -b$

So, $b = 0$ $0 = 1 + 0 + c$

$0 = 1 + b + c$

So, $c = -1$

By putting values of a, b, c in *Eq.* (i), we get

$$\chi = \mu_0 e^2 m^{-1} v^2 R^{-1}$$

$$\chi = \frac{\mu_0 e^2}{mR} \quad \text{…(ii)}$$

where,

$\mu_0 = 4\pi \times 10^{-7}\ \text{TmA}^{-1}$

$e = 1.6 \times 10^{-19}\ \text{C}$

$m = 9.1 \times 10^{-31}\ \text{kg}$

$R = 10^{-10}\ \text{m}$

$$\chi = \frac{(4\pi \times 10^{-7}) \times (1.6 \times 10^{-19})^2}{(9.1 \times 10^{-31}) \times (10^{-10})}$$
$$= 3.533 \times 10^{-4} \quad \textbf{[2½]}$$

Q. 8. Answer the following questions :

(a) Why does a paramagnetic sample display greater magnetisation (for the same magnetising field) when cooled?

(b) Why is diamagnetism, in contrast, almost independent of temperature?

(c) If a toroid uses bismuth for its core, will the field in the core be (slightly) greater or (slightly) less than when the core is empty?

(d) Is the permeability of a ferromagnetic material independent of the magnetic field? If not, is it more for lower or higher fields?

(e) Magnetic field lines are always nearly normal to the surface of a ferro-magnet at every point. (This fact is analogous to the static electric field lines being normal to the surface of a conductor at every point.) Why?

(f) Would the maximum possible magnetisation of a paramagnetic sample be of the same order of magnitude as the magnetisation of a ferro-magnet? [NCERT Add. Ex. Q. 5.16, Page 202]

Ans. **(a)** Owing to the random thermal motion of molecules, the alignments of dipoles get disrupted at high temperatures. On cooling, this disruption is reduced. So that, a paramagnetic sample displays greater magnetisation when it is cooled. **[1]**

(b) The induced dipole moment in a diamagnetic substance is always opposite to the magnetising field. So that, the internal motion of the atoms (which is related to the temperature) does not affect the diamagnetism of a material. **[1]**

(c) Bismuth is a diamagnetic substance. So that, a toroid with a bismuth core has a magnetic field slightly greater than a toroid whose core is empty. **[½]**

(d) The permeability of ferromagnetic materials is not independent of the applied magnetic field. As observed from the hysterisis curve, it is greater for a lower field and vice-versa. **[½]**

(e) The permeability of a ferro-magnetic material is not less than one. It is always greater than one. So that, magnetic field lines are always nearly normal to the surface of such materials at every point. **[1]**

(f) The maximum possible magnetisation of a paramagnetic sample can be of the same order of magnitude as the magnetisation of a ferro-magnet. This requires high magnetising fields for saturation. **[1]**

Q. 9. Answer the following questions :

(a) Explain qualitatively on the basis of domain picture the irreversibility in the magnetisation curve of a ferro-magnet.

(b) The hysteresis loop of a soft iron piece has a much smaller area than that of a carbon steel piece. If the material is to go through repeated cycles of magnetisation, which piece will dissipate greater heat energy?

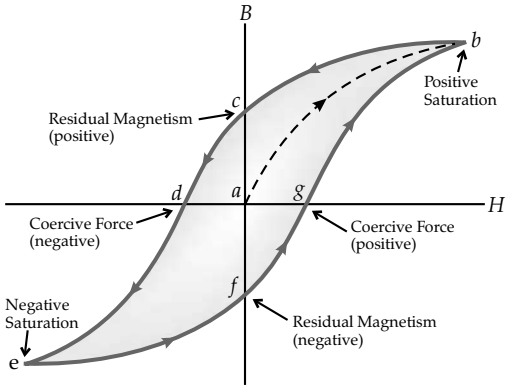
(c) 'A system displaying a hysteresis loop such as a ferro-magnet, is a device for storing memory?' Explain the meaning of this statement.

(d) What kind of ferromagnetic material is used for coating magnetic tapes in a cassette player, or for building 'memory stores' in a modern computer?

(e) A certain region of space is to be shielded from magnetic fields. Suggest a method.

[NCERT Add. Ex. Q. 5.17, Page 202]

Ans. The hysteresis curve (B-H curve) of a ferromagnetic material is :



[1]

(a) It can be observed from the given curve that magnetisation persists even when the external field is removed. This reflects the irreversibility of a ferro-magnet. **[½]**

(b) The dissipated heat energy is directly proportional to the area of a hysteresis loop. A carbon steel

piece has a greater hysteresis curve area. Hence, it dissipates greater heat energy. **[½]**

(c) The value of magnetisation is memory or record of hysteresis loop cycles of magnetisation. These bits of information correspond to the cycle of magnetisation. Hysteresis loops can be used for storing information. **[1]**

(d) Ceramic is used for coating magnetic tapes in cassette players and for building memory stores in modern computers. **[1]**

(e) A certain region of space can be shielded from magnetic fields if it is surrounded by soft iron rings. In such arrangements, the magnetic lines are drawn out of the region. **[1]**

Some Commonly Made Errors

- Students confuse to differentiate the Gauss's law for magnetic field and Electric field.
- Students do not study the concept of Para-magnetism, Diamagnetism and Ferro-magnetism substances.
- Don't ignore to study permanent magnets and electromagnets.
- Generally, students get confused with the term 'dipole in a uniform magnetic field' with 'dipole in an electric field'. So understand the concept of dipole.
- Try to solve numerical by making suitable diagram, do not skip the diagrams.

EXPERT ADVICE

- ☞ *In physics, drawing a picture can really make things easier. You just need to draw a rough idea of the situation given in the question. You can use arrows to understand and visualise the directions given in the question.*
- ☞ *Memorize definitions and laws, especially Gauss's Law, the Earth magnetism and bar magnets.*
- ☞ *Make a separate list of all formulae along with what each variable represents and derivations.*
- ☞ *Convert all units to SI, with charges in Coulombs and distances in meters, magnetic field intensity in Tesla.*
- ☞ *Learn the values of universal constants.*
- ☞ *Draw suitable diagram to explain the answer.*

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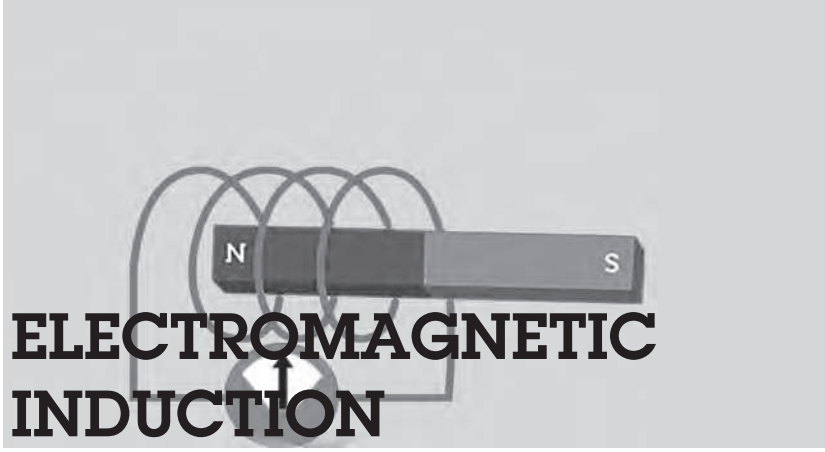
CHAPTER

6 ELECTROMAGNETIC INDUCTION



Chapter Objectives

This chapter will help you understand :

- ***Electromagnetic induction :*** *Introduction, The experiments of Faraday and Henry; Magnetic flux; Faraday's Law of induction; Lenz's Law and Conservation of energy; Motional electromotive force; Energy consideration; Eddy currents; Inductance; and AC generator.*

Quick Review

- The term 'electromagnetic induction' refers to the generation of an electric current by passing a metal wire through a magnetic field. The discovery of electromagnetic induction in 1831 was preceded a decade earlier by a related discovery by Danish physicist Hans Christian Oersted (1777–1851).
- Oersted showed that an electric current produces a magnetic field. That is, if you place a simple magnetic compass near any of the electrical wires in your home that is carrying a current, you can detect a magnetic field around the wires. If an electric current can produce a magnetic field, physicists reasoned, perhaps the reverse effect could be observed as well. So they set out to generate an electric current from a magnetic field.
- Electromagnetic effect was first observed in 1831 by English physicist Michael Faraday (1791–1867) and shortly thereafter by American physicist Joseph Henry (1797–1878).
- The word "electromagnetism" in physics is used to describe one of the fundamental forces of nature. This force is between sub-atomic particles such as protons and electrons. It helps to hold matter together. Electromagnetism is also used to describe how a magnetic field is created by the flowing of electric current.

TIPS...

- *Make proper use of Kirchhoff's Laws.*
- *It can often be very confusing getting the direction correct in complicated systems in both electricity and magnetism.*
- *Try to get a perspective from Energy Conservation.*
- *The uniqueness theorem states that given a charge distribution in a region and boundary conditions*
- *Understand the concept of induced EMF properly.*

TRICKS...

- *EM induction has three parameters are always involved : Magnet/Electromagnet/Magnetic field; a wire wound coil/Inductor; Movement of coil with respect to field or vice-versa.*
- *The Units of all the parameters must belong to any one standard unit system, i.e., SI or CGS.*
- *Facts of Fleming's right hand rule.*
- *Magnetics is all about understanding a few basic laws and then using integration to find out the magnetic field produced by a current distribution.*
- *All questions will require you to use Biot-Savart's Law or Ampere's Law and then calculate the magnetic field.*

Know the Links

- *https://www.electronics-tutorials.ws/electromagnetism/electromagnetic-induction.html*
- *https://www.techopedia.com/definition/19997/electromagnetic-induction*
- *https://www.nde-ed.org/EducationResources/HighSchool/.../electroinduction.htm*
- *https://www.allaboutcircuits.com › ... › Magnetism and Electromagnetism*

Multiple Choice Questions

(1 mark each)

Q. 1. A square of side L meters lies in the *x-y* plane in a region where the magnetic field is given by $B = B_0 (2\hat{i} + 3\hat{j} + 4\hat{k})$ Tesla, where B_0 is constant. The magnitude of flux passing through the square is

(a) $2B_0L^2$**Wb.** **(b)** $3B_0L^2$**Wb.**
(c) $4B_0L^2$**Wb.** **(d)** $\sqrt{29}B_0L^2$**Wb**

[NCERT Exemp. Q. 6.1, Page 33]

Ans. Correct option : (c)

Explanation : Magnetic flux is defined as the total number of magnetic lines of force passing normally through an area placed in a magnetic field and is equal to the magnetic flux linked with that area.

Square lies in *X*-*Y* plane in $\vec{B}$ so $\vec{A} = L^2\hat{k}$

$Q = B.A$

$= B_0(2\hat{i} + 3\hat{j} + 4\hat{k}).(L^2\hat{k})$

$= B_0[2\times\hat{i}\hat{k} + 3\times\hat{j}\hat{k} + 4\times\hat{k}.\hat{k}]$

$= B_0L^2[0 + 0 + 4]$

$= 4B_0L^2Wb.$

Q. 2. A loop, made of straight edges has six corners at *A*(0, 0, 0), *B*(*L*, 0, 0) *C*(*L*, *L*, 0), *D*(0, *L*, 0), *E*(0, *L*, *L*) and *F*(0, 0, *L*). A magnetic field $B=B_0$ ($i\hat{}$ + $k\hat{}$) Tesla is present in the region. The flux passing through the loop *ABCDEFA* (in that order) is

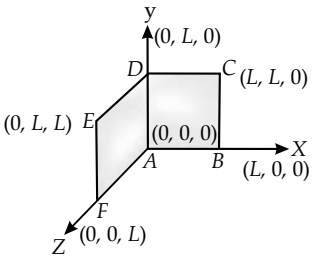
(a) B_0L^2**Wb.** **(b)** $2B_0L^2$**Wb.**

(c) $\sqrt{2}B_0L^2$**Wb.** **(d)** $4B_0L^2$**Wb.**

[NCERT Exemp. Q. 6.2, Page 33]

Ans. Correct option : (b)

Explanation : The loop can be considered in two planes :



(i) Plane of *ABCDA* is in *X*-*Y* plane. So its vector $\vec{A}$ is in *Z*-direction. Hence,

$A_1 = |A|\hat{k} = L^2\hat{k}$

(ii) Plane of *DEFAD* is in *Y*-*Z* plane

So $A_2 = |A|\hat{i} = L^2\hat{i}$

$\therefore \quad A = A_1 + A_2 = L^2(\hat{i} + \hat{k})$

$B = B_0(\hat{i} + \hat{k})$

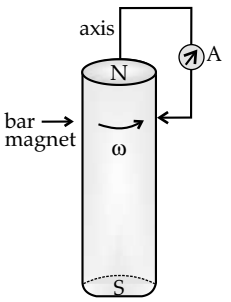
So, $Q = B.A = B_0(\hat{i} + \hat{k}).L^2(\hat{i} + \hat{k}) = B_0L^2$

$[\hat{i}.\hat{i}+\hat{i}.\hat{k}+\hat{k}.\hat{i}+\hat{k}.\hat{k}]$

$= B_0L^2[1 + 0 + 0 + 1] \qquad (\because \cos 90° = 0)$

$= 2B_0L^2Wb$

Q. 3. A cylindrical bar magnet is rotated about its axis in the figure. A wire is connected from the axis and is made to touch the cylindrical surface through a contact. Then



(a) a direct current flows in the ammeter *A*.

(b) no current flows through the ammeter *A*.

(c) an alternating sinusoidal current flows through the ammeter A with a time period $2\pi/\omega$.

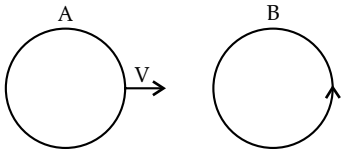
(d) a time varying non-sinusoidal current flows through the ammeter A. [NCERT Exemp. Q. 6.3, Page 34]

Ans. Correct option : (b)

Explanation : The phenomenon of electromagnetic induction is used in this problem. Whenever the number of magnetic lines of force (magnetic flux) passing through a circuit changes (or a moving conductor cuts the magnetic flux) an emf is produced in the circuit (or emf induces across the ends of the conductor) is called induced emf. The induced emf persists only as long as there is a change or cutting of flux.

When cylindrical bar magnet is rotated about its axis, no change in flux linked with the circuit takes place, consequently no emf induces and hence, no current flows through the ammeter *A*. Hence the ammeter shows no deflection.

Q. 4. There are two coils *A* and *B* as shown in figure. A current starts flowing in *B* as shown, when *A* is moved towards *B* and stops when *A* stops moving. The current in *A* is counter clockwise. *B* is kept stationary when *A* moves. We can infer that



(a) there is a constant current in the clockwise direction in *A*.

(b) there is a varying current in *A*.

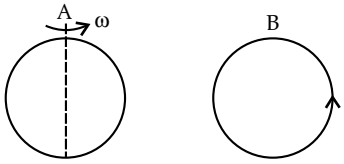
(c) there is no current in *A*.

(d) there is a constant current in the counter clockwise direction in *A*. [NCERT Exemp. Q. 6.4, Page 34]

Ans. Correct option : (d)

Explanation : When coil *A* moves towards coil *B* with constant velocity, so rate of change of magnetic flux due to coil *B* in coil *A* will be constant that gives constant current in coil *A* in same direction as in coil *B* by Lenz's law.

Q. 5. Same as problem 4 except the coil *A* is made to rotate about a vertical axis figure. No current flows in *B* if *A* is at rest. The current in coil *A*, when the current in *B* (at t = 0) is counter-clockwise and the coil *A* is as shown at this instant, *t* = 0, is



(a) constant current clockwise.

(b) varying current clockwise.

(c) varying current counter-clockwise.

(d) constant current counter-clockwise.

[NCERT Exemp. Q. 6.5, Page 34]

Ans. Correct option : (a)

Explanation : In this case the direction of the induced electromotive force/induced current is determined by the Lenz's law. According to the Lenz's law the direction of induced emf or current in a cir-

cuit is such as to oppose the cause that produces it. This law is based upon law of conservation of energy.

When the current in coil B (at $t = 0$) is counter-clockwise and the coil A is considered above it. The counter-clockwise flow of the current in coil B is equivalent to North Pole of magnet and magnetic field lines are eliminating upward to coil A. When coil A starts rotating at $t = 0$, the current in coil A is constant along clockwise direction by Lenz's rule.

Q. 6. The self-inductance L of a solenoid of length l and area of cross-section A, with a fixed number of turns N increases as

(a) l and A increase.

(b) l decreases and A increases.

(c) l increases and A decreases.

(d) both l and A decrease.

[NCERT Exemp. Q. 6.6, Page 34]

Ans. Correct option : (b)

Explanation : As we know that,

$$L = \mu_r \mu_0 \frac{N^2 A}{l}$$

As L is constant for a coil,

$$L \propto A \text{ and } L \propto \frac{1}{l}$$

As μ_r and N are constant here so, to increase L for a coil, area A must be increased and l must be decreased so verify Answer (b).

Important point : The self and mutual inductance of capacitance and resistance depend on the geometry of the devices as well as permittivity/ permeability of the medium.

Tick Two or More Options

Q. 7. A metal plate is getting heated. It can be because

(a) a direct current is passing through the plate.

(b) it is placed in a time varying magnetic field.

(c) it is placed in a space varying magnetic field, but does not vary with time.

(d) a current (either direct or alternating) is passing through the plate.

[NCERT Exemp. Q. 6.7, Page 34]

Ans. Correct options : (a), (b) and (d)

Explanation : A plate or conductor is heated by heating effect of current ($H=I^2Rt$). It verifies answers (a) and (d).

When time-dependent magnetic field varies across the metal plate due to production of eddy currents, heating of plate takes place verifying Answer (b).

Q. 8. An *e.m.f.* is produced in a coil, which is not connected to an external voltage source. This can be due to

(a) the coil being in a time varying magnetic field.

(b) the coil moving in a time varying magnetic field.

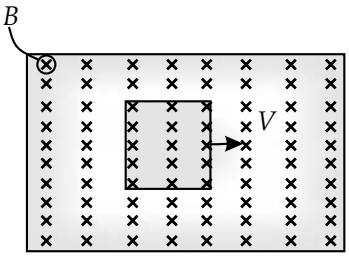
(c) the coil moving in a constant magnetic field.

(d) the coil is stationary in external spatially varying magnetic field, which does not change with time.

[NCERT Exemp. Q. 6.8, Page 35]

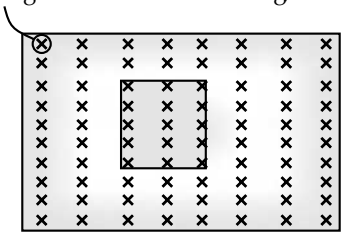
Ans. Correct options : (a), (b) and (c)

Explanation : As we know whenever the number of magnetic lines of force (magnetic flux) passing through a circuit changes, an emf is produced in the circuit called induced emf. The induced emf persists only as long as there is a change or cutting of flux.



In this problem, magnetic flux linked with the isolated coil changes when the coil is placed in the region of a time-varying magnetic field, the coil moving in a constant magnetic field or in time-varying magnetic field.

B (is varying with time or increasing/decreasing)



Q. 9. The mutual inductance M_{12} of coil 1 with respect to coil 2

(a) increases when they are brought nearer.

(b) depends on the current passing through the coils.

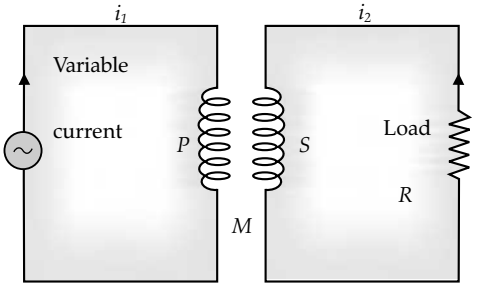
(c) increases when one of them is rotated about an axis.

(d) is the same as M_{21} of coil 2 with respect to coil 1.

[NCERT Exemp. Q. 6.9, Page 35]

Ans. Correct options : (a) and (d)

Explanation : **Mutual induction :** Whenever the current passing through a coil or circuit changes, the magnetic flux linked with a neighbouring coil or circuit will also change. Hence an emf will be induced in the neighbouring coil or circuit. This phenomenon is called 'mutual induction'.



According to Faraday's second law emf induces in secondary coil,

$e_2 = -N_2 \frac{d\phi_2}{dt}$ and $e_2 = -M\frac{di_1}{dt}$

As we know that, $M_{12} = M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l$ where $r_1 l$ is the common area of cross-section of coil so M_{12} does not depend on passing current and rotation. l is also common length. So answers (a) and (d) are verified.

Q. 10. A circular coil expands radially in a region of magnetic field and no electromotive force is produced in the coil. This can be because

(a) the magnetic field is constant.

(b) the magnetic field is in the same plane as the circular coil and it may or may not vary.

(c) the magnetic field has a perpendicular (to the plane of the coil) component whose magnitude is decreasing suitably.

(d) there is a constant magnetic field in the perpendicular (to the plane of the coil) direction.

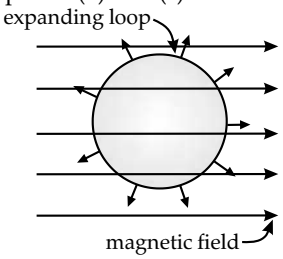
[NCERT Exemp. Q. 6.10, Page 35]

Ans. Correct options : (b) and (c)

Explanation : When *a* circular coil expands radially in a region of magnetic field, induced emf developed is $e = Blv = B \times$ rate of change of area.

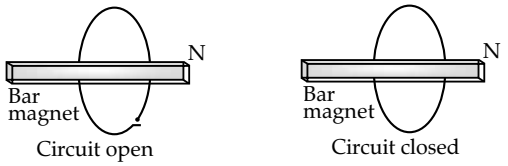
It is given that the magnetic field B is in a plane perpendicular to the plane of the circular coil. Since e=0, magnetic field should be in the plane of circular coil so that its component, which is perpendicular to plane of coil, is zero. If the magnetic field decreases with the magnetic flux linked with the coil, then $e = \frac{d\phi}{dt} = 0$.

Hence, the options (b) and (c) are correct.



Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Consider a magnet surrounded by a wire with an on/off switch *S* in the figure. If the switch is thrown from the off position (open circuit) to the on position (closed circuit), will a current flow in the circuit? Explain.



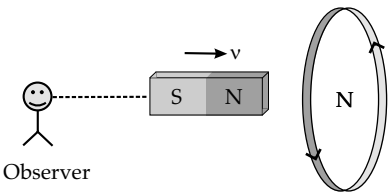
[NCERT Exemp. Q. 6.11, Page 35]

Ans. As we know that induced emf or induced current can flow when there is change in magnetic flux with respect to time and magnetic flux $\phi = BA\cos\theta$. As there is no any change B, *i.e.*, magnet A, *i.e.*, area of circuit and no change in angle between $\vec{B}$ and $\vec{A}$. So, no induced current will flow. **[1]**

Q. 2. A wire in the form of a tightly wound solenoid is connected to a *DC* source, and carries a current. If the coil is stretched so that there are gaps between successive elements of the spiral coil, will the current increase or decrease? Explain.

[NCERT Exemp. Q. 6.12, Page 36]

Ans. When the coil is stretched the magnet flux with leak through the gaps between two turns. According to Lenz's law, the emf is produced in nearby coil which will oppose the increase or decrease in flux in coil. So, on stretching the coil, the current will increase. When the coil is stretched then n (numbers of turns per unit length) will decrease. So L or reactance decreases which makes current increase. **[1½]**

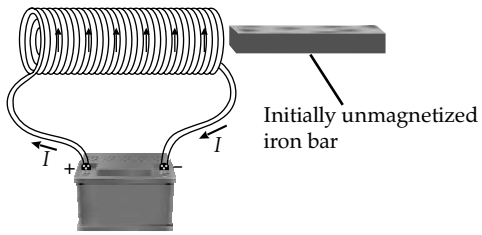


[½]

Q. 3. A solenoid is connected to a battery so that a steady current flows through it. If an iron core is inserted into the solenoid, will the current increase or decrease? Explain.

[NCERT Exemp. Q. 6.13, Page 36]

Ans. This problem is based on Lenz's law and according to this law, the direction of induced emf or current in a circuit is such as to oppose the cause that produces it. **[½]**

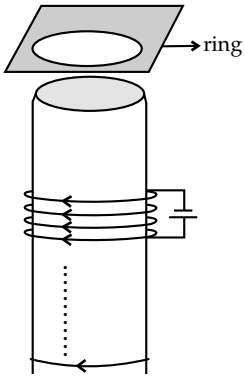


[½]

When the iron core is inserted in the current carrying solenoid, the magnetic field increases due to the magnetisation of iron core and hence the flux increases. So, the emf induced in the coil must oppose this increase in flux, so the current induced in the coil in such a direction that it will oppose the increasing magnetic field which can be done by making decrease in current. So, the current will decrease. **[1]**

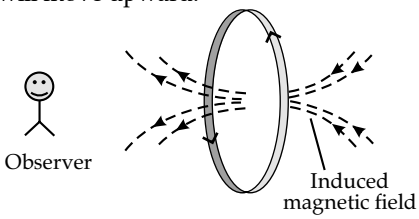
Q. 4. Consider a metal ring kept on top of a fixed solenoid (say on a cardboard) figure. The centre of the ring coincides with the axis of the solenoid.

If the current is suddenly switched on, the metal ring jumps up. Explain.



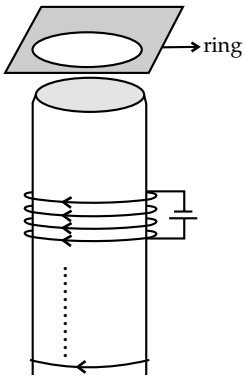
[NCERT Exemp. Q. 6.14, Page 36]

Ans. Initially, no flux was passes through the metal ring. When the current is switched on, flux passes through the ring. According to Lenz's law this increase will be resisted and this can happen if the ring moves away from the solenoid. If the current in the solenoid is as shown, the flux (downward) increases and this will cause a counter clockwise current (as seen from the top of the ring). As the flow of current is in the opposite direction to that in the solenoid, they will repel each other and the ring will move upward. **[1½]**



[½]

Q. 5. Consider a metal ring kept (supported by a cardboard) on top of a fixed solenoid carrying a current *I* figure. The centre of the ring coincides with the axis of the solenoid. If the current in the solenoid is switched off, what will happen to the ring?



[NCERT Exemp. Q. 6.15, Page 36]

Ans. This problem is based on Lenz's law and according to this law, the direction of induced emf or current in a circuit is such as to oppose the cause that produces it. When the switch is opened current in the circuit of solenoid stops flowing. Initially, there is some magnetic flux linked with the solenoid and now if current in the circuit stops, the magnetic flux falls to zero or we can say that magnetic flux linked through the ring decreases. According to Lenz's law, this decrease in flux will be opposed and the ring experiences downward force towards the solenoid. **[1]**

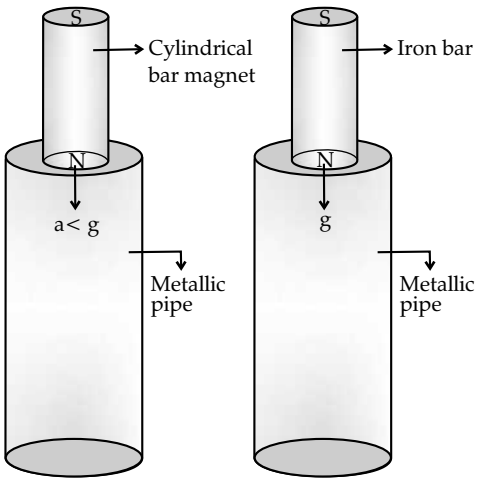
This happens because the current 'I" will decreases and cause a clockwise current (as seen from the top of the ring in figure) to increase the decreasing flux. This can be done if the direction of induced magnetic field is same as that of solenoid. This makes the opposite sense of flow of current in the ring (when viewed from the bottom of the ring) and solenoid forming opposite magnetic pole in front of each other.

Hence, they will attract each other but as ring is placed at the cardboard it could not be able to move downwards. **[1]**

Q. 6. Consider a metallic pipe with an inner radius of 1 cm. If a cylindrical bar magnet of radius 0.8 cm is dropped through the pipe, it takes more time to come down than it takes for a similar un-magnetized cylindrical iron bar dropped through the metallic pipe. Explain. [NCERT Exemp. Q. 6.16, Page 36]

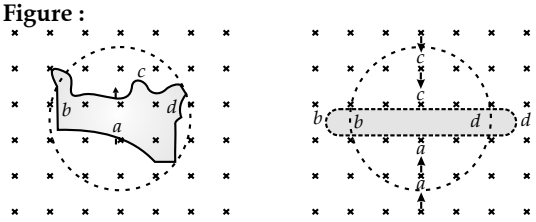
Ans. When a cylindrical bar magnet of radius 0.8 cm is dropped inside a hollow cylindrical metallic pipe of inner radius 1 cm the magnetic flux across the pipe changes, and due to induction, eddy current will produce. **[½]**

The direction of eddy current will be in such a way so that it can oppose the cause, *i.e.*, motion of magnet. So upward force acts on bar magnet and decreases the acceleration, which in turn takes more time to fall magnet through the pipe, as compared to simple metallic piece in place of magnet. **[1]**



[½]

Q. 7. Use Lenz's law to determine the direction of induced current in the situations described by Figure :



(a) A wire of irregular shape turning into a circular shape;

(b) A circular loop being deformed into a narrow straight wire. [NCERT Ex. Q. 6.2, Page 230]

Ans. According to Lenz's law, the direction of the induced emf is such that it tends to produce a current that opposes the change in the magnetic flux that produces in it.

(a) When the shape of the wire changes, the flux piercing through the unit surface area increases (or the number of magnetic fields line passing through the loop increases). As a result, the induced current produces an opposing flux. Hence, the induced current flows along *c*, *b*, *a* and *d*. **[1]**

(b) When the shape of a circular loop is deformed into a narrow straight wire, the flux piercing the surface decreases. Hence, the induced current flows along *a*, *b*, *c* and *d*. **[1]**

Q. 8. A 1.0 m long metallic rod is rotated with an angular frequency of 400 radsec^{-1} about an axis normal to the rod passing through its one end. The other end of the rod is in contact with a circular metallic ring. A constant and uniform magnetic field of 0.5 T parallel to the axis exists everywhere. Calculate the emf developed between the centre and the ring. [NCERT Ex. Q. 6.5, Page 230]

Ans. Length of the rod, $l = 1$ m
Angular frequency, $\omega = 400$ rad/s
Magnetic field strength, $B = 0.5$ T
One end of the rod has zero linear velocity, while the other end has a linear velocity of $l\omega$. **[½]**

Average linear velocity of the rod, $v = \frac{l\omega + 0}{2} = \frac{l\omega}{2}$ **[½]**

Emf developed between the centre and the ring,

$$e = Blv = \frac{Bl^2\omega}{2} = \frac{0.5 \times (1)^2 \times 400}{2} = 100\ V$$

Hence, the emf developed between the centre and the ring is 100 V. **[1]**

Q. 9. A horizontal straight wire 10 m long extending from east to west is falling with a speed of 5.0 m/sec, at right angles to the horizontal component of the earth's magnetic field, 0.30×10^{-4} Wbm^{-2}.

(a) What is the instantaneous value of the emf induced in the wire?

(b) What is the direction of the emf?

(c) Which end of the wire is at the higher electrical potential? [NCERT Ex. Q. 6.7, Page 230]

Ans. Given that,
Length of the wire, $l = 10$ m
Falling speed of the wire, v = 5.0 m/s
Magnetic field strength, $B = 0.3 \times 10^{-4}\ Wbm^{-2}$

(a) Emf induced in the wire,

$$e = Blv = 0.3 \times 10^{-4} \times 5 \times 10 = 1.5 \times 10^{-3}\ V$$ **[1]**

(b) Using Fleming's right hand rule, it can be inferred that the direction of the induced emf is from West to East end of the falling wire. **[½]**

(c) The eastern end of the wire is at a higher potential. **[½]**

Q. 10. Current in a circuit falls from 5.0 A to 0.0 A in 0.1 s. If an average emf of 200 V induced, give an estimate of the self-inductance of the circuit. [NCERT Ex. Q. 6.8, Page 230]

Ans. Given that,
Initial current, $I_1 = 5.0$ A
Final current, $I_2 = 0.0$ A

Change in current, $dl = l_1 - l_2 = 5A$ **[½]**
Time taken for the change, $t = 0.1$ s
Average emf, $e = 200$ V
For self-inductance (L) of the coil, we have the relation for average emf as :

$$e = L\left(\frac{di}{dt}\right)$$

$$L = \frac{e}{\left(\frac{di}{dt}\right)} = \frac{200}{\frac{5}{0.1}} = 4H$$

Hence, the self-induction of the coil is 4 H. **[1½]**

Q. 11. A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 A to 20 A in 0.5 s, what is the change of flux linkage with the other coil? [NCERT Ex. Q. 6.9, Page 230]

Ans. Mutual inductance of a pair of coils, μ = 1.5 H
Initial current, $I_1 = 0$ A
Final current $I_2 = 20$ A
Change in current, $dI = I_2 - I_1 = 20 - 0 = 20\ A$…(1)
Time taken for the change, dt = 0.5 s
Induced emf, $e = \frac{d\phi}{dt}$ …(2) **[1]**

Where $d\phi$ is the change in the flux linkages with the coil.
Emf is related with mutual inductance as :
Equating (1) and (2), we get

$$\frac{d\phi}{dt} = \mu\frac{dI}{dt}$$

$$d\phi = 1.5 \times (20) = 30\,\text{Wb}$$

Hence, the change in the flux linkage is 30 Wb. **[1]**

Q. 12. A jet plane is travelling towards west at a speed of 1,800 km/h. What is the voltage difference developed between the ends of the wing having a span of 25 m, if the Earth's magnetic field at the location has a magnitude of 5×10^{-4} T and the dip angle is 30°. [NCERT Ex. Q. 6.10, Page 230]

Ans. Speed of the jet plane, v = 1,800 km/h = 500 m/s
Wing span of jet plane, $l = 25$ m
Earth's magnetic field strength, $B = 5 \times 10^{-4}$ T
Angle of dip, $\delta = 30°$
Vertical component of Earth's magnetic field,

$$B_V = B\sin\delta = 5 \times 10^{-4} \sin 30° = 2.5 \times 10^{-4}\ \text{T}$$ **[½]**

Voltage difference between the ends of the wing can be calculated as :

$$e = (B_V) \times l \times v = 2.5 \times 10^{-4} \times 25 \times 500 = 3.125\ V$$ **[½]**

Hence, the voltage difference developed between the ends of the wings is 3.125 V. **[1]**

Q. 13. An air-cored solenoid with length 30 cm, area of cross-section 25 cm^2 and number of turns 500, carries a current of 2.5 A. The current is suddenly switched off in a brief time of 10^{-3} s. How much is the average back emf induced across the ends of

the open switch in the circuit? Ignore the variation in magnetic field near the ends of the solenoid.

[NCERT Add. Ex. Q. 6.15, Page 232]

Ans. Length of the solenoid, $l = 30$ cm $= 0.3$ m
Area of cross-section, $A = 25\ cm^2 = 25 \times 10^{-4}\ m^2$
Number of turns on the solenoid, $N = 500$
Current in the solenoid, $I = 2.5$ A
Current flows for time, $t = 10^{-3}$ s

Average back emf, $e = \frac{d\phi}{dt}$ …(1) **[½]**

Where,

$d\phi$ = Change in flux $= NAB$…(2) **[½]**

Where,

B = Magnetic field strength $= \mu_0 \frac{Nl}{l}$ …(3) **[½]**

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7}\ TmA^{-1}$

Using equations (2) and (3) in equation (1), we get

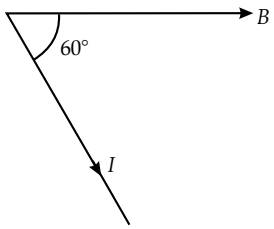
$$e = \frac{\mu_0 N^2 lA}{lt} = \frac{4\pi \times 10^{-7} \times (500)^2 \times 2.5 \times 25 \times 10^{-4}}{0.3 \times 10^{-3}} = 6.5\ V$$

Hence, the average back emf induced in the solenoid is 6.5 V. **[½]**

Q. 14. At a place, the horizontal component of earth's magnetic field is *B* and angle of dip is 60°. What is the value of horizontal component of the earth's magnetic field at equator?

[CBSE Board, Delhi Region, 2017]

Ans.



[1]

I is the total magnetic field.

Now, $I \cos 60° = B$

$$\Rightarrow I = \frac{B}{\cos 60°} = \frac{B}{\frac{1}{2}} = 2B$$

At equator, dip angle is 0°.

$\therefore B_H = I \cos 0° = I = 2B.$ **[1]**

Q. 15. Give one example of use of eddy currents.

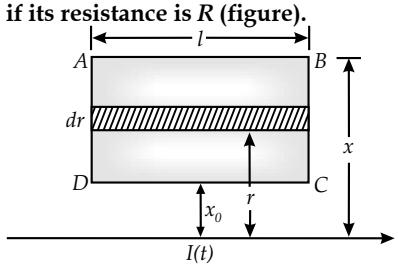
[CBSE Board, Foreign Scheme, 2016]

Ans. When an alternating magnetic field is applied to a magnetic material, an emf is induced in the material itself, according to Faraday's law of electromagnetic induction.

Since the magnetic material is a conducting material, therefore these emf's circulates current within the body of the material. These circulating current are called Eddy Current. There are various uses of Eddy Current. **[1]**

Example : Electromagnetic damping; Induction furnace; Magnetic brakes; Induction Motor; Speed meters and Energy meters. **[1]**

Q. 16. Consider an infinitely long wire carrying a current *I*(*t*), with $\frac{dI}{dt} = \lambda$ = constant. Find the current produced in the rectangular loop of wire *ABCD* if its resistance is *R* (figure).



[NCERT Exemp. Q. 6.25, Page 38]

Ans. Consider a strip of width dr and length l inside a rectangle at distance r from the surface of current carrying conductor. The magnetic field at strip due to current carrying wire is given by at a distance r from the wire,

Field $B(r) = \frac{\mu_0 I}{2\pi r}$ (out of paper)

So, total flux through the loop is,

$$\phi_m = \vec{B} \cdot \vec{A} = \frac{\mu_0 I}{2\pi} l \int_{x_0}^{x} \frac{dr}{r} = \frac{\mu_0 Il}{2\pi} \ln \frac{x}{x_0}$$ …(i) **[1]**

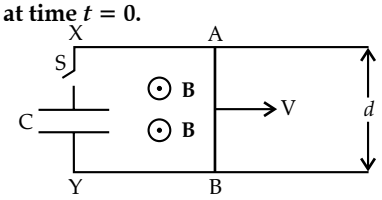
The emf induced can be obtained by differentiating the equation (i) with respect to t and then applying Ohm's law,

$$\cdot I = \frac{\varepsilon}{R} \text{ and } |\varepsilon| = \frac{d\phi}{dt}$$

We have, induced current

$$I = \frac{I}{R}\frac{d\phi}{dt} = \frac{\mu_0 l}{2\pi R}\frac{\lambda}{R} \ln \frac{x}{x_0} \qquad \left(\because \frac{dl}{dt} = \lambda\right)$$ **[1]**

Q. 17. Find the current in the sliding rod *AB* (resistance = *R*) for the arrangement shown in figure. $\vec{B}$ is constant and is out of the paper. Parallel wires have no resistance. $\vec{v}$ is constant. Switch *S* is closed at time *t* = 0.



[NCERT Exemp. Q. 6.29, Page 39]

Ans. If Q (t) is charge on the capacitor (remember that current flows from A to B)

$$I = \frac{vBd}{R} - \frac{Q}{RC}$$

$$\Rightarrow \frac{Q}{RC} + \frac{dQ}{dt} = \frac{vBd}{R}$$ **[1]**

This is the linear differential equation. On solving, we get, $Q = vBdC + Ae^{-t/RC}$

where A is an constant, At $t = 0$, $Q = 0$ $\therefore A = -$v Bdc $\therefore$ $Q = vBdC\,(1 - e^{-t/RC})$

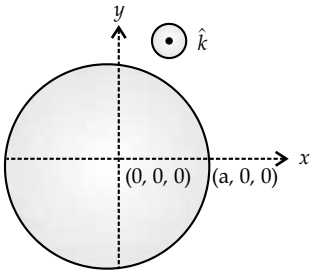
Differentiating, w.r.to 't'

$$I = \frac{dQ}{dt} = vBdc \times \frac{1}{Rc} e^{-t/RC}$$

$$I = \frac{vBd}{R} e^{-t/RC}$$ **[1]**

Short Answer Type Questions (3 or 4 marks each)

Q. 1. A magnetic field in a certain region is given by $B = B_0 \cos(\omega t)\,\hat{k}$ and a coil of radius 'a' with resistance R is placed in the x-y plane with its centre at the origin in the magnetic field see figure. Find the magnitude and the direction of the current at (a, 0, 0) at $t = \frac{\pi}{2\omega}$, $t = \frac{\pi}{\omega}$ and $t = \frac{3\pi}{2\omega}$.



[NCERT Exemp. Q. 6.17, Page 36]

Ans. The direction of magnetic field is along Z-axis. So the magnetic flux passing through circular region of coil of radius a placed in X-Y plane,

$\phi = \vec{B}.\vec{A} = BA\cos\theta$ **[½]**

The direction area vector A of coil is along Z-direction and B also in Z-direction of $\theta = 0°$.

$\phi = (B_0\cos\omega t)\cdot(\pi a^2)\cos 0° = B_0\pi a^2\cos\omega t$ **[½]**

By Faraday's law of electromagnetic induction,

$\varepsilon = \frac{d\phi}{dt} = B_0\pi a^2 sin\omega t.\omega$

$\varepsilon = B_0\pi a^2\omega sin\omega t$

This causes flow of induced current, which is given by

$I = \frac{B_0(\pi a^2)\omega\sin\omega t}{R}$

Now, the value of current at different instants,

(i) $t = \frac{\pi}{2\omega}$

$I = \frac{B_0(\pi a^2)\omega}{R}$ along $\hat{j}$

Because $\sin\omega t = \sin\left(\omega\frac{\pi}{2\omega}\right) = \sin\frac{\pi}{2} = 1$ **[1]**

(ii) $t = \frac{\pi}{\omega}$, $I = \frac{B(\pi a^2)\omega}{R} = 0$

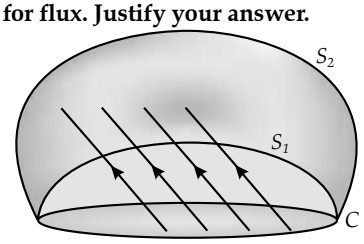
Because, $\sin\omega t = \sin\left(\omega\frac{\pi}{\omega}\right) = \sin\pi = 0$ **[1]**

(iii) $t = \frac{2}{3}\frac{\pi}{\omega}$

$I = \frac{B(\pi a^2)\omega}{R}$ along $-\hat{j}$

$\sin\omega t = \sin\left(\omega\cdot\frac{3\pi}{2\omega}\right) = \sin\frac{3\pi}{2} = -1$ **[1]**

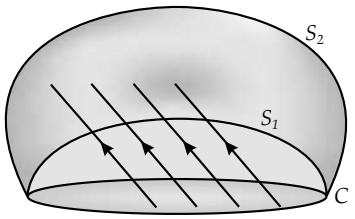
Q. 2. Consider a closed loop C in a magnetic field (figure). The flux passing through the loop is defined by choosing a surface whose edge coincides with the loop and using the formula $\Phi = B_1dA_1, B_2dA_2$ …. Now, if we choose two different surfaces S_1 and S_2 having C as their edge, would we get the same answer for flux. Justify your answer.



[NCERT Exemp. Q. 6.18, Page 36]

Ans. We would get the same answer for magnetic flux. Let us discuss its reason in detail. The magnetic flux linked with the surface can be considered as the number of magnetic field lines passing through the surface. So, let $d\Phi = B.\,dA$ represents magnetic lines in an area A to B. **[1]**

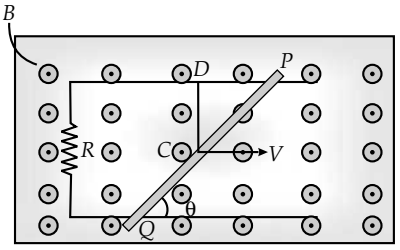
Magnetic field cannot end or start in space, this property of magnetic field lines is based upon the concept of continuity. Therefore, the number of lines passing through surface S_1 must be the same as the number of lines passing through the surface S_2. Therefore, in both the cases we get the same magnetic flux. **[1]**



[½]

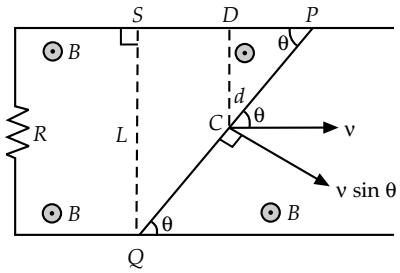
Important point : Magnetic field lines can neither be originated, nor be destroyed in space. This property is based on the concept of continuity. **[½]**

Q. 3. Find the current in the wire for the configuration shown in figure. Wire PQ has negligible resistance. $\vec{B}$., the magnetic field is coming out of the paper. θ is a fixed angle made by PQ travelling smoothly over two conducting parallel wires separated by a distance d.



[NCERT Exemp. Q. 6.19, Page 36]

Ans. The motional electric field E due to the motion along CD is $E = vB$. **[½]**



[1]

The direction of E will be perpendicular to both v and B ($\therefore F=q \times B$). **[½]**

F is force on free charge particle of PQ.

So the motional emf $\varepsilon=E$ along $PQ\times$ Effective length PQ.

Electric field along $PQ=v\times B=vB\sin\theta$. **[½]**

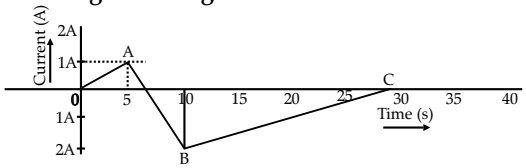
$$\varepsilon = (vB\ \sin\theta)\left(\frac{d}{\sin\theta}\right)$$

$$e = vBd$$ **[½]**

$$\therefore \text{Induced current} = \frac{\varepsilon}{R} = \frac{vBd}{R}$$

It is independent of q. **[1]**

Q. 4. A (current vs. time) graph of the current passing through a solenoid is shown in figure. For which time is the back electromotive force (u) a maximum. If the back e.m.f. at t = 3 s is e, find the back e.m.f. at t = 7 s, 15 s and 40 s. *OA*, *AB* and *BC* are straight line segments.



[NCERT Exemp. Q. 6.20, Page 37]

Ans. The maximum back electromotive force (u) will be maximum when there is maximum rate of change of magnetic flux which is directly proportional to the rate of change of current.

Maximum change or rate of current will be where (I-t) graph makes maximum angle with time axis which is in part AB. **[1]**

So the maximum back emf will occur between 5 s and 10 s. As the back emf at t =3 s, it is 'e'. (Given)

Rate of change of current at t =3 s = slope of OA graph with time axis. So, rate of change of current at 3 $s = \frac{1}{5} A/s$. **[1]**

So back electromotive force at t = 3 $s = L \times \frac{1}{5} = \frac{L}{5} = e$ (Given)

$\because e = L.\frac{dl}{dt}$ and L = Constant for solenoid. **[½]**

Similarly, back emf u_1 between 5 s and 10 s,

$$u_1 = L.\left(\frac{-3}{5}\right) = -3\frac{L}{5} = -3e$$ **[½]**

Back emf between 10 s and 30 s,

$$u_2 = L\frac{[0-(-2)]}{30-10} = \frac{+2L}{20} = \frac{+1}{2}\frac{L}{5}$$

$$u_2 = +\frac{1}{2}e$$ **[½]**

So back emf at 7 s = $-3e$

Back emf at 15 s = $+\frac{1}{2}e$

At 40 s graph is along the time axis, *i.e.*, its slope with time axis is zero.

So, $\frac{dI}{dt} = 0$.

Or back emf at 40 s = 0 **[½]**

Q. 5. There are two coils *A* and *B* separated by some distance. If a current of 2 A flows through *A*, a magnetic flux of 10^{-2} Wb passes through *B* (no current through *B*). If no current passes through A and a current of 1 A passes through *B*, what is the flux through *A*?

[NCERT Exemp. Q. 6.21, Page 37]

Ans. Let I_A current is passing through coil A having mutual inductance M_{AB} with respect to coil B.

N_A=number of turns in coil A

N_B= number of turns in coil B

ϕ_A= flux linked with coil A due to coil B

ϕ_B= flux linked with coil B due to coil A

M_{BA}=Mutual inductance of coil B with respect to coil A

Then,

Total flux through $B=M_2\,\phi_2=M_{BA}I_1$ **[1]**

$10^{-2}=M_{BA}\times 2$

$$M_{BA} = \frac{10^{-2}}{2} = 5 \text{ mH}$$

Now total flux through $A=M_A\phi_A=M_{AB}I_2$ [$\therefore M_{AB}=M_{BA}$]

=5 mH × 1 A= 5 mWb **[3]**

Q. 6. Predict the direction of induced current in the situations described by the following figures (a) to Figure (f).

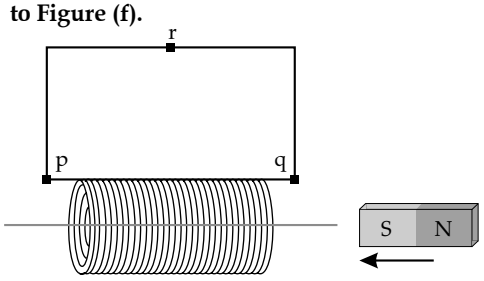


Figure (a)

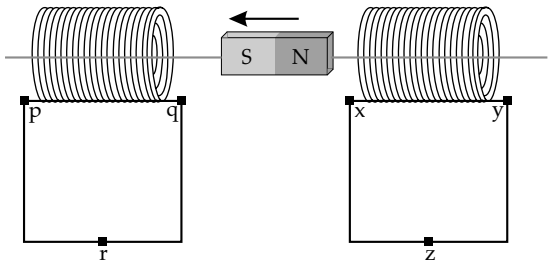
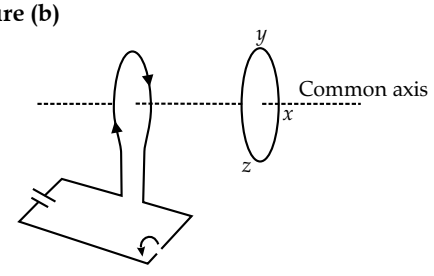
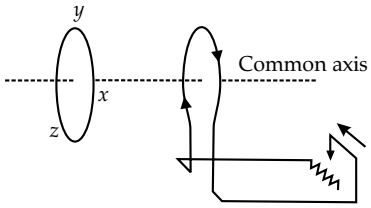


Figure (b)



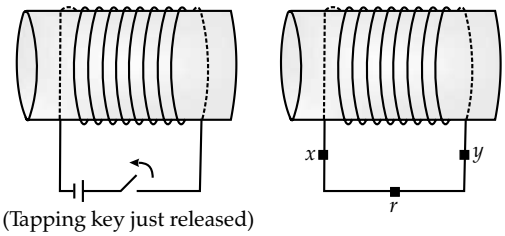
(Tapping key just closed)

Figure (c)



(Rheostat setting being charged)

Figure (d)



(Tapping key just released)

Figure (e)

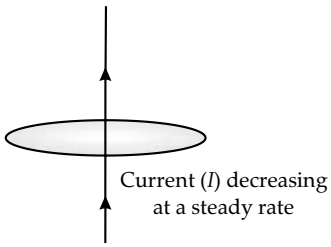
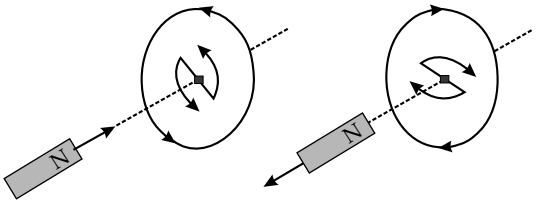


Figure (f) **[NCERT Ex. Q. 6.1, Page 229]**

Ans. The direction of the induced current in a closed loop is given by Lenz's law. The given pairs of figures show the direction of the induced current when the North Pole of a bar magnet is moved towards and away from a closed loop respectively. Using Lenz's rule, the direction of induced current in the given situations can be predicted as follows : **[½]**



[½]

(a) As the S-pole of the magnet is approaching the coil, the current produced at the end q will be clockwise. Hence the direction of the induced current is along $qrpq$. **[½]**

(b) The induced current in the left coil will flow to oppose the incoming S-pole, hence, in clockwise direction along prqp in outer circuit and in the right coil as N-pole is going away from it, hence it will induce S-pole hence the direction of the induced current in left coil is from $yzxy$. **[½]**

(c) As the current is in clockwise direction due to which S-pole is produced and in order to oppose this, S-pole is produced and the direction of the induced current is along $yzxy$. **[½]**

(d) As current in the right circuit, flowing in anti-clockwise direction is decreased and in order to oppose this cause the direction of the induced current is along zyxz. **[½]**

(e) As the key is released in left circuit, the N-polarity decreases, hence to oppose it, S-polarity in right coil should increase hence, the direction of the induced current is along $xryx$. **[½]**

(f) No current is induced since the field lines are lying in the plane of the closed loop. **[½]**

Q. 7. A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1 s, what is the induced emf in the loop while the current is changing?

[NCERT Ex. Q. 6.3, Page 230]

Ans. Given that,

Number of turns on the solenoid = 15 turns/cm = 1,500 turns/m

Number of turns per unit length, n = 1,500 turns

The solenoid has a small loop of area, $A = 2.0\ \text{cm}^2 = 2\times10^{-4}\ \text{m}^2$

Current carried by the solenoid changes from 2 A to 4 A.

$\therefore$ Change in current in the solenoid, di = 4 – 2 = 2 A **[½]**

Change in time, dt = 0.1 s

Induced *emf* in the solenoid is given by Faraday's law as :

$$e = \frac{d\phi}{dt} \quad \text{...(i)} \qquad \textbf{[½]}$$

Where,

ϕ = Induced flux through the small loop

$= BA$ ...(ii) **[½]**

B = Magnetic field

$= \mu_0 ni$ ...(iii) **[½]**

μ_0 = Permeability of free space

$= 4\pi\times10^{-7}\ Hm^{-1}$

Hence, equation (i) reduces to :

$$e = \frac{d}{dt}\left(\frac{di}{dt}\right) = A\mu_0 n\times\left(\frac{di}{dt}\right)$$

$$= 2\times10^{-4}\times4\pi\times10^{-7}\times1{,}500\times\frac{2}{0.1}$$

$$= 7.54\times10^{-6}\ \text{V}$$

Hence, the induced voltage in the loop is 7.54×10^{-6} V. **[1]**

Q. 8. A rectangular wire loop of sides 8 cm and 2 cm with a small cut is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the loop. What is the emf developed across the cut if the velocity of the loop is 1 cm/sec in a direction normal to the (a) longer side, (b) shorter side of the loop? For how long does the induced voltage last in each case?

[NCERT Ex. Q. 6.4, Page 230]

Ans. Length of the rectangular wire, $l = 8$ cm $= 0.08$ m
Width of the rectangular wire, $b = 2$ cm $= 0.02$ m
Hence, area of the rectangular loop,
$A = lb = 0.08 \times 0.02 = 16 \times 10^{-4}\ m^2$ **[1]**
Magnetic field strength, $B = 0.3$ T
Velocity of the loop, v = 1 cm/s = 0.01 m/s

(a) Emf developed in the loop is given as :

$e = Blv = 0.3 \times 0.08 \times 0.01 = 2.4 \times 10^{-4}$ V **[½]**

Time taken to travel along the width,

$$t = \frac{\text{Distance travelled}}{\text{Velocity}} = \frac{b}{v} = \frac{0.02}{0.01} = 2\ s$$ **[½]**

Hence, the induced voltage is $2.4 \times 10^{-4}\ V$ which lasts for 2 s.

(b) Emf developed, $e = Bbv = 0.3 \times 0.02 \times 0.01 = 10^{-4}\ V$ **[1]**

Time taken to travel along the length,

$$t = \frac{\text{Distance travelled}}{\text{Velocity}} = \frac{1}{v} = \frac{0.08}{0.01} = 8\ s$$

Hence, the induced voltage is $0.6 \times 10^{-4}\ V$ which lasts for 8 s. **[1]**

Q. 9. A circular coil of radius 8.0 cm and 20 turns is rotated about its vertical diameter with an angular speed of 50 rad s^{-1} in a uniform horizontal magnetic field of magnitude 3.0×10^{-2} T. Obtain the maximum and average emf induced in the coil. If the coil forms a closed loop of resistance 10 Ω, calculate the maximum value of current in the coil. Calculate the average power loss due to Joule heating. Where does this power come from?

[NCERT Ex. Q. 6.6, Page 230]

Ans. As we know that from given data,
Max induced *emf* = 0.603 V
Average induced *emf* = 0 V
Max current in the *coil* = 0.0603 A
Average power *loss* = 0.018 W
(Power comes from the external rotor)
Radius of the circular coil, $r = 8$ cm $= 0.08$ m
Area of the coil, $A = \pi r^2 = \pi \times (0.08)^2\ m^2$ **[1]**
Number of turns on the coil, $N = 20$
Angular speed, $\omega = 50$ rad/s
Magnetic field strength, $B = 3 \times 10^{-2}$ T
Resistance of the loop, $R = 10\ \Omega$
Maximum induced *emf* is given as :

$e = N\omega AB = 20 \times 50 \times \pi \times (0.08)^2 \times 3 \times 10^{-2} = 0.603$ V **[1]**

The maximum emf induced in the coil is 0.603 V.
Over a full cycle, the average emf induced in the coil is zero.
Maximum current is given as :

$$l = \frac{e}{R} = \frac{0.603}{10} = 0.0603\ A$$ **[½]**

Average power loss due to joule heating :

$$P = \frac{el}{2} = \frac{0.603 \times 0.0603}{2} = 0.018\ W$$ **[½]**

The current induced in the coil produces a torque opposing the rotation of the coil. The rotor is an external agent. It must supply a torque to counter this torque in order to keep the coil rotating uniformly. Hence, dissipated power comes from the external rotor. **[1]**

Q. 10. Suppose the loop in Exercise 6.4 is stationary but the current feeding the electromagnet that produces the magnetic field is gradually reduced so that the field decreases from its initial value of 0.3 T at the rate of 0.02 Ts^{-1}. If the cut is joined and the loop has a resistance of 1.6 Ω. How much power is dissipated by the loop as heat? What is the source of this power?

[NCERT Add. Ex. Q. 6.11, Page 231]

Ans. Sides of the rectangular loop are 8 cm and 2 cm.
Hence, area of the rectangular wire loop,
A = Length × Width $= 8 \times 2 = 16\ cm^2 = 16 \times 10^{-4}\ m^2$
Initial value of the magnetic field, $B' = 0.3$ T
Rate of decrease of the magnetic field,

$$\frac{dB}{dt} = 0.02\ \text{T/s}$$ **[½]**

Emf developed in the loop is given as :

$$e = \frac{d\phi}{dt}$$ **[½]**

Where,
dϕ = Change in flux through the loop area = AB

$$\therefore e = \frac{d(AB)}{dt} = \frac{AdB}{dt} = 16 \times 10^{-4} \times 0.02 = 0.32 \times 10^{-4}\ \text{V}$$

Resistance of the loop, $R = 1.6\ \Omega$
The current induced in the loop is given as :

$$i = \frac{e}{R} = \frac{0.32 \times 10^{-4}}{1.6} = 2 \times 10^{-5}\ A$$ **[2]**

Power dissipated in the loop in the form of heat is given as :

$P = i^2R = (2 \times 10^{-5})^2 \times 1.6 = 6.4 \times 10^{-10}$ W

The source of this heat loss is an external agent, which is responsible for changing the magnetic field with time. **[1]**

Q. 11. It is desired to measure the magnitude of field between the poles of a powerful loudspeaker magnet. A small flat search coil of area 2 cm^2 with 25 closely wound turns, is positioned normal to the field direction, and then quickly snatched out of the field region. Equivalently, one can give it a quick 90° turn to bring its plane parallel to the field direction). The total charge flown in the coil (measured by a ballistic galvanometer connected to coil) is 7.5 mC. The combined resistance of the coil and the galvanometer is 0.50 Ω. Estimate the field strength of magnet.

[NCERT Add. Ex. Q. 6.13, Page 231]

Ans. As we know that given in the question,
Area of the small flat search coil, $A = 2\ cm^2 = 2 \times 10^{-4}\ m^2$
Number of turns on the coil, $N = 25$
Total charge flowing in the coil, $Q = 7.5$ mC $= 7.5 \times 10^{-3}$ C
Total resistance of the coil and galvanometer, $R = 0.50\ \Omega$
Induced current in the coil,

$$I = \frac{\text{Induced emf(e)}}{R} \quad \ldots(1)$$

Induced emf is given as :

$$e = -N\frac{d\phi}{dt} \quad \ldots(2)\ \textbf{[1]}$$

Where,

$d\phi$ = Charge in flux

Combining equations (1) and (2), we get

$$I = -\frac{N\frac{d\phi}{dt}}{R}$$

$$Idt = -\frac{N}{R}d\phi \quad \ldots(3)\ [½]$$

Initial flux through the coil, $\phi_i = BA$

Where,

B = Magnetic field strength

Final flux through the coil, $\phi_f = 0$

Integrating equation (3) on both sides, we have

$$\int Idt = -\frac{N}{R}\int_{\phi_i}^{\phi_f} d\phi$$

But total charge, $Q = \int Idt$ [½]

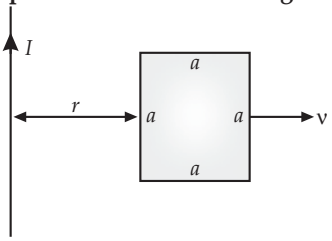
$$\therefore\ Q = \frac{-N}{R}(\phi_f - \phi_i) = \frac{-N}{R}(0 - \phi_i) = +\frac{N\phi_i}{R}$$

$$Q = \frac{NBA}{R}$$

$$\therefore\ B = \frac{QR}{NA} = \frac{7.5\times10^{-3}\times0.5}{25\times2\times10^{-4}} = 0.75\ \text{T}$$

Hence, the field strength of the magnet is 0.75 T. **[2]**

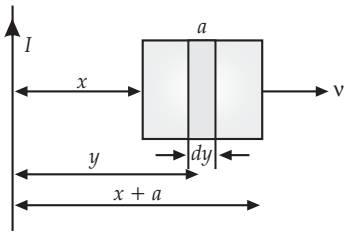
Q. 12. (a) Obtain an expression for the mutual inductance between a long straight wire and a square loop of side a as shown in Figure.



(b) Now assume that the straight wire carries a current of 50 A and the loop is moved to the right with a constant velocity, v = 10 m/s. Calculate the induced emf in the loop at the instant when x = 0.2 m. Take a = 0.1 m and assume that the loop has a large resistance.

[NCERT Add. Ex. Q. 6.16, Page 232]

Ans.



[1]

(a) Take a small element dy in the loop at a distance y from the long straight wire (as shown in the given figure).

Magnetic flux associated with element dy, $d\phi = BdA$ **[½]**

Where,

dA = Area of element $dy = a\ dy$

B = Magnetic field at distance $y = \frac{\mu_0 I}{2\pi y}$ [½]

I = Current in the wire

μ_0 = Permeability of free space = $4\pi\times10^{-7}\ TmA^{-1}$

$$\therefore d\phi = \frac{\mu_0 Ia}{2\pi}\frac{dy}{y}$$

$$\phi = \frac{\mu_0 Ia}{2\pi}\int\frac{dy}{y}$$ [½]

y tends from x to $a + x$

$$\therefore \phi = \frac{\mu_0 Ia}{2\pi}\int_x^{a+x}\frac{dy}{y} = \frac{\mu_0 Ia}{2\pi}[\log_e y]_x^{a+x} = \frac{\mu_0 Ia}{2\pi}\log_e\left(\frac{a+x}{x}\right)$$ [½]

For mutual inductance M, the flux is given as :

$$\phi = MI$$

$$\therefore MI = \frac{\mu_0 Ia}{2\pi}\log\left(\frac{a}{x}+1\right)$$

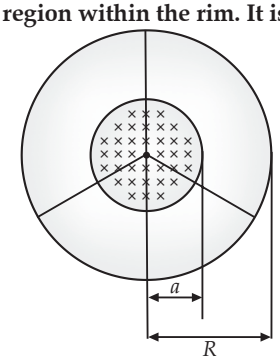
$$M = \frac{\mu_0 a}{2\pi}a\ \log_e\left(1+\frac{a}{x}\right)$$ [½]

(b) Emf induced in the loop, $e = B'av = \frac{\mu_0 Iav}{2\pi x}$ (Given),

I = 50 A, x = 0.2 m, a = 0.1 m, v = 10 m/s

$$e = \frac{4\pi\times10^{-7}\times50\times0.1\times10}{2\pi\times0.2} = 5\times10^{-5}\ \text{V}$$ [½]

Q. 13. A line charge λ per unit length is lodged uniformly onto the rim of a wheel of mass M and radius R. The wheel has light non-conducting spokes and is free to rotate without friction about its axis in figure. A uniform magnetic field extends over a circular region within the rim. It is given by,



$\boldsymbol{B = -B_0\mathbf{k}}$ $\quad (r \le a;\ a < R) = 0$ **(otherwise)**

What is the angular velocity of the wheel after the field is suddenly switched off?

[NCERT Add. Ex. Q. 6.17, Page 232]

Ans. Line charge per unit length,

$$\lambda = \frac{\text{Total charge}}{\text{Length}} = \frac{Q}{2\pi r}$$ **[1]**

Where,

r = Distance of the point within the wheel

Mass of the wheel = M

Radius of the wheel = R

Magnetic field, $B = -\vec{B} = B_0\hat{k}$

At distance r, the magnetic force is balanced by the centripetal force,

$$BQv = \frac{Mv^2}{r}$$ [½]

Where

v = linear velocity of the wheel

$$B2\pi r\lambda = \frac{Mv}{r}$$

$$v = \frac{B2\pi\lambda r^2}{M}$$

Angular velocity, $\omega = \frac{v}{R} = \frac{B2\pi r^2\lambda}{MR}$

For $r \le a$ and $a < R$

we get,

$$\omega = -\frac{2\pi B_0 a^2 \lambda}{MR}\hat{k}$$

[1]

Here, negative sign shows that ω is along negative z-axis or along the direction of B. **[½]**

Q. 14. Obtain the expression for the mutual inductance of two long co-axial solenoids S_1 and S_2 wound one over the other, each of length L and radii r_1 and r_2 and n_1 and n_2, number of turns per unit length, when a current I is set up in the outer solenoid S_2.

[CBSE Board, Delhi Region, 2017]

Ans. Magnetic field B formed along the axis of outer solenoid,

$B = \mu_0 n_2 I$.

Flux linked with inner solenoid S_1 per turn.

$= B.A = \pi r_{21}^2 B = \pi r_{21}^2 \times \mu_0 n_2 I$

Flux linked with whole of inner solenoid having total no of turns n_1L,

$\phi = \pi r_1^2 \times \mu_0 n_2 I \times n_1 L$

Now, $\phi = MI$ (M is mutual inductance)

So, $MI = \mu_0 \pi r_1^2 n_1 n_2 LI$

$M = \mu_0 \pi r_1^2 n_1 n_2 L$

[4]

Q. 15. Define the term 'self-inductance' and write its *S.I.* unit. [CBSE Board, Delhi Region, 2017]

Ans. Self-inductance is the property of a coil by virtue of which, the coil opposes any change in the strength of current flowing through its by inducing an emf in itself. We know that ϕ is directly proportional to current. **[1]**

Hence,

$$e = \frac{d\phi}{dt} = \frac{d}{dt}(LI)$$

or

$$e = -L\frac{di}{dt}$$

So, we get $L = \frac{-e}{(di/dt)}$. **[1]**

The *SI* unit of self-inductance is Henry. Therefore, self-inductance of a coil is said to be 1 henry when a current change at the rate of 1 amp/sec through the coil induces an emf of 1 volt in the coil. As 1 henry = 1 volt/(1 amp/sec) = 1 volt-sec/amp. **[1]**

Q. 16. A circular coil of cross-sectional area 200 cm^2 and 20 turns is rotated about the vertical diameter with angular speed of 50 rad s^{-1} in a uniform magnetic field of magnitude 3.0 x 10^{-2} T. Calculate the maximum value of the emf in the coil.

[CBSE Board, Delhi Region, 2017]

Ans. Given that,

Area = 200 cm^2 = 2 x 10^{-2} m^2

Total number of turns, N = 20

Angular speed, ω = 50 $rads^{-1}$

Magnetic field = 3 x 10^{-2} T

Maximum value of emf, $e_{max} = n\,AB\omega$ **[1]**

= 20 x 200 x 10^{-4} x 3 x 10^{-2} x 50 V

= 4 x 3 x 5 x 10^{-2} V

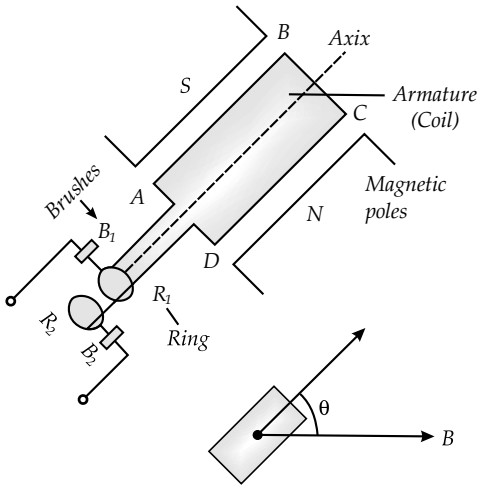
= 60 x 10^{-2} V

= 0.6 V **[2]**

Q. 17. Draw a labelled diagram of AC generator. Derive the expression for the instantaneous value of the emf induced in the coil.

[CBSE Board, Delhi Region, 2017]

Ans. The labelled diagram of AC generator is as below :



[1]

Labelled Diagram of an AC Generator :

Let at any moment t the perpendicular vector to the plane of coil makes angle θ with direction of magnetic field (B). So, flux passing through the coil,

$\phi = nAB\cos\theta$.

And $\theta = \omega t$

[ω is angular velocity of the coil.]

So, $\phi = nAB\cos\omega t$ **[1]**

If e is the instantaneous induced emf produced in the coil, then maximum value or peak value of e is attained when $\sin\omega t = \pm 1$

$\therefore e_{max} = n\omega AB$

So, $e = e_{max}\sin\omega t$ **[½]**

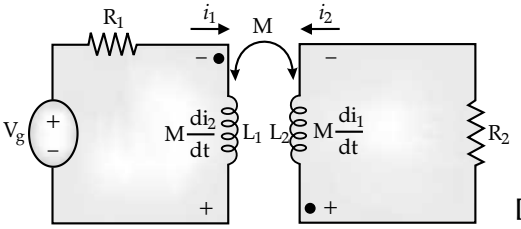
Since, the value of the sine function varies between +1 and −1. So, the polarity of emf changes with time and also, output voltage is sinusoidal in nature. **[½]**

Q. 18. Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

[CBSE Board, All India Region, 2017]

Ans. **Mutual Inductance** : The phenomenon according to which an opposing emf is produced as result of change in current or magnetic flux linked with a neighbouring coil. **[½]**

Mutual inductance of two long solenoids :



[½]

Two long solenoids of same length l let n_1 be the number of turns per unit length of S_1, n_2 be the number of turns per unit length of S_2, I, be current passed through S_1 and ϕ_{21} be flux linked with S_2 due to current flowing in S_1

$\phi_{21} \propto I_1$ or $\phi_{21} = M_{21} I_1$ [½]

where M_{21} = co-efficient of mutual induction of two solenoid

When current is passed through S_1, an emf is induced in solenoid S_2. Magnetic field produced inside S_1 on passing current,

$B_1 = \mu_0 n_1 I_1$

Magnetic flux linked with each turn of the solenoid S_2 will be equal to B_1 times the area of cross-section of solenoid S_1

So, magnetic flux linked with each turn of the solenoid $S_2 = B_1, A$

Therefore, total magnetic flux linked with solenoid S_2 will be

$$\phi_{21} = B_1 A \times n_2 \ell = \mu_0 n_1 I_1 \times A \times n_2 \ell = \mu_0 n_1 n_2 A\, I_1 \ell$$

$$M_{21} = \mu_0 n_1 n_2 A \ell \quad \ldots(i)$$

Similarly, the mutual inductance between the two solenoids, when current is passing through S_2 and induced emf is produced in solenoid S_1 and is given by

$$M_{12} = \mu_0 n_1 n_2 A \ell$$

$M_{12} = M_{21} = M$ (say) [½]

Hence co-efficient of mutual induction between the two long solenoid,

$$M = \mu_0 n_1 n_2 A \ell$$

We can write equation (i) as

$$M = \mu_0 \left(\frac{N_1}{\ell}\right)\left(\frac{N_2}{\ell}\right) \pi r_1^2 \times \ell = \frac{\mu_0 N_1 N_2 A}{\ell}$$

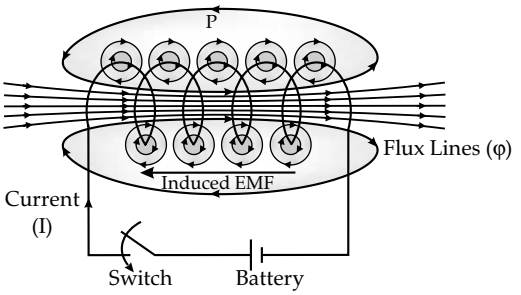
If core of any other magnetic material μ is placed then,

$$M = \frac{\mu_0 \mu_r N_1 N_2 A}{\ell}$$ [1]

Q. 19. Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor *L* connected across a source of emf.

[CBSE Board, All India Region, 2017]

Ans. Self-inductance : Self-inductance or in other words inductance of the coil is defined as the property of the coil due to which it opposes the change of current flowing through it. Inductance is attained by a coil due to the self-induced emf produced in the coil itself by changing the current flowing through it. [½]



[½]

The self-induction opposes the growth of the current and when the current is switched off, the self-induction opposes the decay of the current; so, self-induction is also called the inertia of electricity. [½]

Self-inductance of long solenoid : A long solenoid is one whose length is very large as compared to its area of cross-section.

Magnetic field (B) at any point inside $B = \frac{\mu_0 NI}{\ell}$ …(i)

Magnetic flux through each two of the solenoid $\phi = B \times A$ of the each turns $\phi = \left(\mu_0 \frac{N}{\ell} I\right) A$

Where A = area of each turn of the solenoid

Total magnetic flux linked with the solenoid = flux through each turn × total number of turns

$$N\phi = \mu_0 \frac{N}{\ell} IA \times N \quad \ldots(ii)$$

If L is co-efficient of self-inductance of the solenoid then,

$$\therefore N\phi = LI \quad \ldots(iii)$$

From (ii) and (iii), we get

$$LI = \mu_0 \frac{N}{\ell} IA \times N$$ [½]

or $L = \frac{\mu_0 N^2 A}{\ell}$ …(iv)

The magnitude of emf is given by

Let or $e = L\frac{dI}{dt}$

Multiplying (I) to both sides, we get

$eIdt = LIdI$ …(v)

But $I = \frac{dq}{dt}$ or $Idt = dq$

Also, work done (dw) = Voltage (e) × Charge (dq)

Or $dw = e \times dq = eIdt$ …(vi)

So from (v) and (vi), we have

$dw = LIdI$ …(vii)

Total work done in increasing current from zero to I_0, we have

By integrating both sides of equation (vii), we get

$$\int_0^w dw = \int_0^{I_0} LIdI$$

$$w = \frac{1}{2} LI_0^2$$

This work done through inductor is stored as the potential energy (u) in the magnetic field of inductor,

$$u = \frac{1}{2} LI_0^2$$

[1]

Q. 20. (a) State the condition under which a charged particle moving with velocity v goes undeflected in a magnetic field B.

(b) An electron, after being accelerated through a potential difference of 10^4V, enters a uniform magnetic field of 0.04 T, perpendicular to its direction of motion. Calculate the radius of curvature of its trajectory.

[CBSE Board, Foreign Scheme, 2017]

Ans. (a) The electron will move undeflected in the presence of crossed electric and magnetic field if the electric force on the electron is balanced by the magnetic force acting on the electron, **[½]**

Mathematically $qE = qvB$ **[½]**

Where q = Charge, E = Electric field, B = Magnetic field, v = Velocity of the electron.

In the given problem $q = e$ (Charge on electron)

$eE = evB$, so,

$$v = \frac{E}{B}$$

(b) We know that,

Kinetic energy = eV

The magnetic force equals to centrifugal force,

$$eVB = \frac{mv^2}{r} \quad \textbf{[1]}$$

$$r = \frac{mv}{Bq} \quad ...(i)$$

As we know that the kinetic energ y,

$$\frac{1}{2}mv^2 = eV$$

$$v^2 = \frac{2eV}{m}$$

$$v = \sqrt{\frac{2eV}{m}} \quad ...(ii)$$

using equation (i), $r = \frac{mv}{eB}$

$$r = \frac{m}{eB}\sqrt{\frac{2eV}{m}}$$

$$r = \frac{\sqrt{2qmV}}{eB}$$

$$r = \frac{1}{B}\sqrt{\frac{2mV}{e}}$$

Formula for radius of the path,

$$r = \frac{1}{B}\sqrt{\frac{2mV}{e}} = \frac{1}{0.04}\sqrt{\frac{2\times 9.1\times 10^{-31}\times 10^4}{1.6\times 10^{-19}}} = 8.43\times 10^{-3} \text{ m} \quad \textbf{[1]}$$

Q. 21. (i) Define self-inductance. Write its *SI* units.

(ii) A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1 s, what is the induced emf in the loop while the current is changing? [CBSE Board, Foreign Scheme, 2016]

Ans. (i) Self-inductance is the property of a coil by virtue of which, the coil opposes any change in the strength of current flowing through it by inducing an emf in itself. We know that ϕ is directly proportional to current. Hence, **[½]**

$$\phi = \frac{d\phi}{dt} = \frac{d}{dt}(LI)$$

or

$$e = -L\frac{di}{dt}$$

So, we get L= $\frac{-e}{\left(\frac{di}{dt}\right)}$. **[½]**

The SI unit of self-inductance is Henry. Therefore, self-inductance of a coil is said to be 1 henry when a current change at the rate of 1 amp/sec through the coil induces an emf of 1 volt in the coil. As 1 henry = 1 volt/(1 amp/sec)=1 volt-sec/amp. **[½]**

(ii) Number of turns on the solenoid = 15 turns/cm = 1,500 turns/m

Number of turns per unit length, n = 1,500 turns

The solenoid has a small loop of area, A = 2.0 cm^2 = 2×10^{-4} m^2

Current carried by the solenoid changes from 2 A to 4 A.

Change in current in the solenoid, di = 4 − 2 = 2 A

Change in time, dt = 0.1 s

Induced emf in the solenoid is given by Faraday's law as :

$$e = \frac{d\phi}{dt} \quad \text{…(i)}$$

Where,

ϕ = Induced flux through the small loop = BA …(ii)

B = Magnetic field = $\mu_0 ni$

μ_0 = Permeability of free space = $4\pi\times 10^{-7}$ H/m

Hence, equation (i) reduces to :

$$e = \frac{d}{dt}(BA) = A\mu_0 n\times\left(\frac{di}{dt}\right)$$

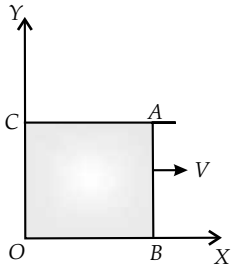
$$= 2\times 10^{-4}\times 4\pi\times 10^{-7}\times 1{,}500\times\frac{2}{0.1}$$

$$= 7.54\times 10^{-6} \text{ V}$$

Hence, the induced voltage in the loop is 7.54×10^{-6} V. **[1½]**

Long Answer Type Questions **(5 marks each)**

Q. 1. A magnetic field $B = B_0 \sin(\omega t)\ \hat{k}$ covers a large region where a wire AB slides smoothly over two parallel conductors separated by a distance d (figure). The wires are in the x-y plane. The wire AB (of length d) has resistance R and the parallel wires have negligible resistance. If AB is moving with velocity v, what is the current in the circuit? What is the force needed to keep the wire moving at constant velocity?



[NCERT Exemp. Q. 6.22, Page 37]

Ans. In the given figure, *CA* and *OB* are long parallel conducting wires, connected by *d* and conductor *CO*. The resistance of *ACOB* is negligible. **[1]**

Let wire *AB* at $t = 0$ is at $x = 0$, *i.e.*, on *Y*-axis. Now *AB* moves with velocity $v\hat{i}$. **[½]**

Let at any time *t*,

position of conductor *AB* is $x(t) = v\hat{i}t$ **[½]**

Motional emf across AB

$e_1 = Blv = (B_0 \sin \omega t)vd(-\hat{j})$

And emf due to change in field (along *OBAC*)

$e_2 = -\dfrac{d(\phi_B)}{dt}$

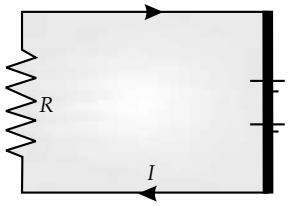
$\phi_B = (B_0 \sin\omega t)\big(x(t)d\big)$ [where, area $A = xd$]

$e_2 = -B_0\omega \cos\omega t x(t)d$

Total emf in the circuit = emf due to change in field (along *OBAC*) + the motional emf across *AB*

$e_1 + e_2 = -B_0 d[\omega x \cos(\omega t) + v \sin(\omega t)]$ **[1½]**

The equivalent electrical diagram is shown in the diagram below.



[½]

Electric current in clockwise direction is given by

$$= \frac{B_0 d}{R}(\omega x \cos\omega t + v \sin\omega t)$$

The force acting on the conductor is given by $F = ilB \sin 90° = ilB$

Substituting the values,

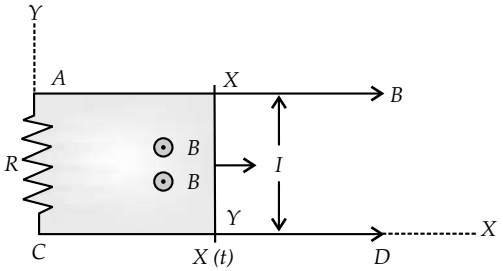
$\overline{F_m} = \dfrac{B_0 d}{R}(\omega x \cos\omega t + v \sin\omega t)(d)(B_0 \sin\omega t)(-\hat{i})$

External force needed on wise is along positive *x*-axis to keep moving it with constant velocity is given by,

$\overline{F}_{ext} = \dfrac{B_0^2 d^2}{R}(\omega x \cos\omega t + v \sin\omega t)\sin\omega t(\hat{i})$

This is required expression for force. **[1]**

Q. 2. A conducting wire *XY* of mass *m* and negligible resistance slides smoothly on two parallel conducting wires as shown in figure. The closed circuit has a resistance *R* due to *AC*. *AB* and *CD* are perfect conductors. There is a magnetic field $B = B(t)\hat{k}$.



(i) Write down equation for the acceleration of the wire *XY*.

(ii) If *B* is independent of time, obtain v(*t*), assuming $v(0) = u_0$.

(iii) For (ii), show that the decrease in kinetic energy of *XY* equals the heat lost in *R*.

[NCERT Exemp. Q. 6.23, Page 37]

Ans. **(i)** Let wire *XY* at $t=0$ is at $x=0$

And at $t = t$ is $x = x(t)$

Magnetic flux is a function of time $\phi(t) = B(t) \times A$

$\therefore \quad \phi(t) = B(t)[l \cdot x(t)]$

$$\varepsilon = -\frac{d\phi(t)}{dt} = -\frac{dB(t)}{dt} l \cdot x(t) - B(t) l \cdot \frac{dx(t)}{dt}$$

$$\varepsilon = \frac{-dB(t)}{dt} l \cdot x(t) - B(t) l v(t)$$ **[1]**

The direction of induced current by Fleming's Right or by Lenz's law is in clockwise direction in loop XY < AX.

$$I = \frac{\varepsilon}{R} = \frac{-l}{R}\left[x(t)\frac{dB(t)}{dt} + B(t)v(t)\right]$$ …(I)

The force acting on the conductor is $F = B(t)\, I\, l \sin 90°$

$F = B(t) I.l$

$$F = \frac{B(t) l \varepsilon}{R} = \frac{-B(t) l^2}{R}\left[\frac{-dB(t)}{dt}.x(t) - B(t).v(t)\right]$$

$$\frac{md^2x}{dt^2} = \frac{-B(t)l^2}{R}\left[x(t)\frac{dB(t)}{dt} + B(t)v(t)\right]$$

Or $\dfrac{d^2x}{dt^2} = \dfrac{-l^2}{mR}B(t)\left[x(t)\dfrac{dB(t)}{dt} + B(t).v(t)\right]$ ...(II) **[1]**

(ii) Now B is independent of time *i.e. B* does not change with time or it is constant.

$\therefore \dfrac{dB}{dt} = 0, \mathrm{B}(t) = \mathrm{B}$ and $v(t) = v$ …(III)

Put (III) in (II), we get,

$$\frac{d^2x}{dt^2} = \frac{-l^2}{mR}[0 + Bv]$$

$$\frac{d^2x}{dt^2} + \frac{B^2l^2}{mR}\frac{dx}{dt} = 0$$

$$\frac{dv}{dt} + \frac{B^2l^2}{mR}v = 0$$

Integrating using variable separable from differential equation, we have

$$v = A\exp\left(\frac{-l^2B^2t}{mR}\right)$$ …(IV)

At $t = 0$, $v = u$

$$\therefore v(t) = u\exp\left(\frac{-l^2B^2t}{mR}\right)$$ **[1]**

(iii) Heat lost per second in (ii) where $\frac{dB}{dt}=0$

$H = I^2R$

Magnitude of current from equation I in (i) part

$$I=\frac{Blv}{R}=\frac{-l}{R}[0+Bv] \qquad \left[\because \frac{dB}{dt}=0\right]$$

Heat produced per second $H = I^2R$

$$\therefore H=\frac{B^2l^2v^2}{R^2}\cdot R$$

$$H=\frac{B^2l^2}{R}u^2\exp^{\left(\frac{-2l^2B^2t}{mR}\right)}$$ [v from eqn. (IV) in (iii) part]

[1]

$$\text{Power lost}=\int_0^t I^2R\,dt=\frac{B^2l^2u^2}{R}\int_0^t e^{\frac{-2l^2B^2t}{mR}}dt$$

$$\left[\because v^2=u^2\exp^{\frac{-2l^2B^2t}{mR}}\right]$$

$$\text{Power lost}=\frac{B^2l^2u^2}{R}\frac{mR}{2l^2B^2}[1-e^{\left(\frac{-2l^2B^2t}{mr}\right)}]$$

$$=\frac{mu^2}{2}\left[1-e^{\frac{-2l^2B^2t}{mR}}\right]=\frac{mu^2}{2}-\frac{m}{2}u^2e^{\frac{-2l^2B^2t}{mR}}$$

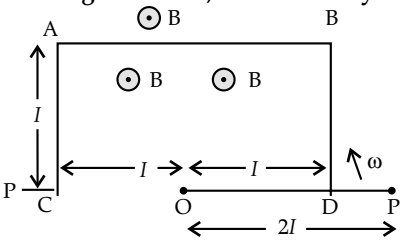
$$=\left[\frac{mu^2}{2}-\frac{mv^2(t)}{2}\right]$$

= Initial *K.E.* – Final *K.E.*

Power lost= Decrease in kinetic energy

This proves that decrease in *K.E.* of *XY* is equal to the heat lost in *R*. **[1]**

Q. 3. *ODBAC* is a fixed rectangular conductor of negligible resistance (*CO* is not connected) and *OP* is a conductor which rotates clockwise with an angular velocity ω figure. The entire system is in a uniform magnetic field *B* whose direction is along the normal to the surface of the rectangular conductor *ABDC*. The conductor *OP* is in electric contact with *ABDC*. The rotating conductor has a resistance of λ per unit length. Find the current in the rotating conductor, as it rotates by 180°.



[NCERT Exemp. Q. 6.24, Page 38]

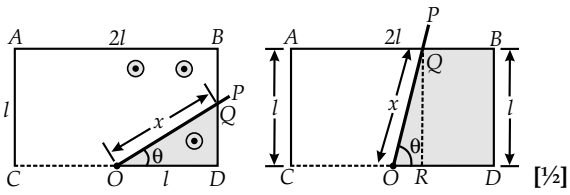
Ans. When the conductor *OP* is rotated, then the rate of change of area and hence the rate of change of flux can be considered uniform form

$$0<\theta<\frac{\pi}{4};\frac{\pi}{4}<\theta<\frac{3\pi}{4} \text{ and } \frac{3\pi}{4}<\theta<\pi.$$

(i) Let us first assume the position of rotating conductor at time interval,

$t = 0$ to $t=\frac{\pi}{4\omega}$ (or T/8)

[1]



[½]

The rod *OP* will make contact with the side *BD*. Let the length *OQ* of the contact after some time interval *t* such that $0<t<\frac{\pi}{4\omega}$ or $0<t<\frac{T}{8}$ be *x*.

$$\phi_m=BA=B\left(\frac{1}{2}\times QD\times OD\right)=B\left(\frac{1}{2}\times l\tan\theta\times l\right)$$

$$\Rightarrow \phi_m=\frac{1}{2}Bl^2\tan\theta, \text{ where } \theta=\omega t$$

By applying Faraday's law of *EMI*,

Thus, the magnitude of the emf induced is

$$|\varepsilon|=\left|\frac{d\phi}{dt}\right|=\frac{1}{2}Bl^2\omega\sec^2\omega t$$

The current induced in the circuit will be $I=\frac{\varepsilon}{R}$ where, *R* is the resistance of the rod in contact.

where, $R\propto\lambda$

$$R=\lambda x=\frac{\lambda l}{\cos\omega t}$$

$$\therefore I=\frac{1}{2}\frac{Bl^2\omega}{\lambda l}\sec^2\omega t\cos\omega t=\frac{Bl\omega}{2\lambda\cos\omega t}$$ **[1]**

(ii) Now let the rod *OP* will make contact with the side *AB*. And the length of *OQ* of the contact after some time interval *t* such that $\frac{\pi}{4\omega}<t<\frac{3\pi}{4\omega}$ or $\frac{T}{8}<t<\frac{3T}{8}$ be *x*. The flux through the area *OQBD* is

$$\phi_m=\left(l^2+\frac{1}{2}\frac{l^2}{\tan\theta}\right)B$$

Where, $\theta=\omega t$

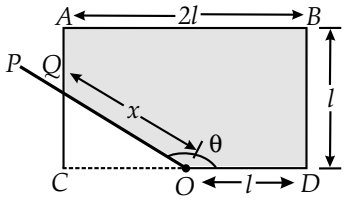
Thus, the magnitude of emf induced in the loop is

$$|\varepsilon|=\left|\frac{d\phi}{dt}\right|=\frac{Bl^2\omega\sec^2\omega t}{2\tan^2\omega t}$$

The current induced in the circuit is

$$I=\frac{\varepsilon}{R}=\frac{\varepsilon}{\lambda x}=\frac{\varepsilon\sin\omega t}{\lambda l}=\frac{1}{2}\frac{Bl\omega}{\lambda\sin\omega t}$$ **[1]**

(iii) Similarly, for time interval $\frac{3\pi}{4\omega}<t<\frac{\pi}{\omega}$ or $\frac{3T}{8}<t<\frac{T}{2}$, the rod will be in touch with *AC*.



[½]

The flux through *OQABD* is given by

$$\phi_m=\left(2l^2-\frac{l^2}{2\tan\omega t}\right)B$$

And the magnitude of emf generated in loop is given by

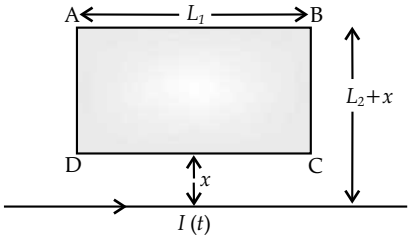
$$\varepsilon = \frac{d\phi}{dt} = \frac{B\omega l^2 \sec^2 \omega t}{2\tan^2 \omega t}$$

$$I = \frac{\varepsilon}{R} = \frac{\varepsilon}{\lambda x} = \frac{1}{2}\frac{Bl\omega}{\lambda \sin \omega t}$$

These are the required expressions. **[1]**

Q. 4. A rectangular loop of wire *ABCD* is kept close to an infinitely long wire carrying a current $I(t) = I_0\left(\frac{I-t}{T}\right)$ for $0 \le t \le T$ and $I(0) = 0$ for $t > T$(figure).

Find the total charge passing through a given point in the loop, in time *T*. The resistance of the loop is *R*.



[NCERT Exemp. Q. 6.26, Page 38]

Ans. To find the charge that passes through the circuit, first we have to find the relation between instantaneous current and instantaneous magnetic flux linked with it. The emf induced can be obtained by differentiating the expression of magnetic flux linked with respect to '*t*' and then applying Ohm's law. **[1]**

We get

$$I = \frac{e}{R} = \frac{1}{R}\frac{d\phi}{dt}$$

According to the problem electric current is given as a function of time.

$$I(t) = \frac{dQ}{dt} \text{ or } \frac{dQ}{dt} = \frac{1}{R}\frac{d\phi}{dt}$$

Integrating the variable separately in the form of differential equation for finding the charge Q that passed in time t, we have

$$Q(t_1) - Q(t_2) = \frac{1}{R}[\phi(t_1) - \phi(t_2)] \quad \textbf{[1]}$$

$$\phi(t_1) = L_1 \frac{\mu_0}{2\pi}\int_x^{L_2+2x} \frac{dx'}{x'} I(t_1)$$

$$= \frac{\mu_0 L_1}{2\pi} I(t_1) \ln \frac{L_2 + x}{x} \quad \textbf{[1]}$$

Therefore, the magnitude of charge is

$$Q = \frac{1}{R}[\phi(T) - \phi(0)]$$

$$Q = \frac{\mu_0 L_1}{2\pi} \ln \frac{L_2 + x}{x}[I(T) - I(0)] \quad \textbf{[1]}$$

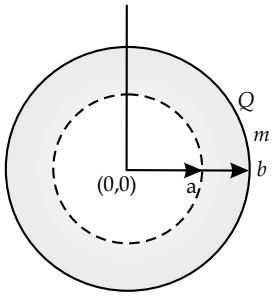
Now $I(T) = 0$ and $I(0) = 1$.

$$\therefore \quad Q = \frac{\mu_0 L_1}{2\pi} I_0 \ln\left(\frac{L_2 + x}{x}\right) \quad \textbf{[1]}$$

Q. 5. A magnetic field *B* is confined to *a* region $r \le a$ and points out of the paper (the *z*-axis), $r = 0$ being the centre of the circular region. A charged ring (charge = *Q*) of radius *b*, $b > a$ and mass *m* lies in the *x-y* plane with its centre at the origin. The ring is free to rotate and is at rest. The magnetic field is brought to zero in time Δt. Find the angular velocity ω of the ring after the field vanishes.

[NCERT Exemp. Q. 6.27, Page 38]

Ans. As the magnetic field is reduced to zero in Δt, so magnetic flux linked with the ring reduces from maximum to zero. Change of magnetic flux across the conducting ring induces *emf*. The induced *emf* causes the electric field across the ring. **[1]**



[½]

By the relations between electric field and potential we get,

The induced *emf* = Electric field $E \times (2\pi b)$ …(i)

$(\because V = E \times d)$

By Faraday's law of EMI,

$$|\varepsilon| = \frac{d\phi}{dt} = A\frac{dB}{dt}$$

$$|\varepsilon| = \frac{B\pi a^2}{\Delta t} \quad \textbf{…(ii)}$$

From Eqs. (i) and (ii), we have

$$2\pi bE = \varepsilon = \frac{B\pi a^2}{\Delta t} \quad \textbf{[1]}$$

As we know the electric force experienced by the charged ring, $F_e = QE$ This force try to rotate the coil, and the torque is given by

Torque = $b \times$ Force

$$\tau = QEb = Q\left[\frac{B\pi a^2}{2\pi b \Delta t}\right] b = Q\frac{Ba^2}{2\Delta t} \quad \textbf{[1]}$$

If ΔL is the change in angular momentum,

$$\Delta L = \text{Torque} \times \Delta t = Q\frac{Ba^2}{2}$$

Since, initial angular momentum = 0

And Torque $\times \Delta t$ = Change in angular momentum

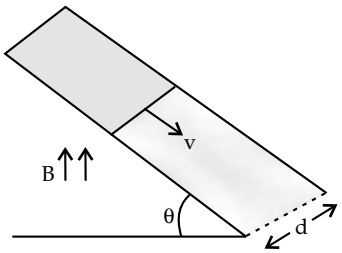
Final angular momentum $= mb^2\omega = \frac{QBa^2}{2}$ **[1]**

Where, $mb^2 = I$ (moment of inertia of ring)

$$\omega = \frac{QBa^2}{2mb^2}$$

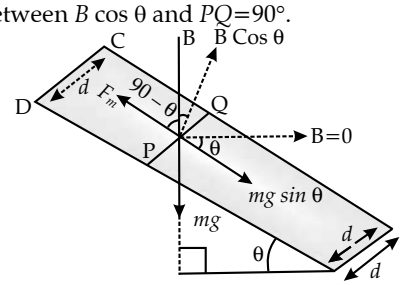
On rearranging the terms, we have the required expression of angular speed. **[½]**

Q. 6. A rod of mass *m* and resistance R slides smoothly over two parallel perfectly conducting wires kept sloping at an angle θ with respect to the horizontal given figure. The circuit is closed through a perfect conductor at the top. There is a constant magnetic field *B* along the vertical direction. If the rod is initially at rest, find the velocity of the rod as a function of time.



[NCERT Exemp. Q. 6.28, Page 39]

Ans. From free body diagram the component of B perpendicular to wire PQ or line F_m is $B\cos\theta$: Angle between $B\cos\theta$ and PQ=90°. **[1]**



[1]

Applying Newton's law of motion, we have

$$m\frac{d^2x}{dt^2} = mg\sin\theta - \frac{B\cos\theta d}{R}\left(\frac{dx}{dt}\right) \times (Bd)\cos\theta$$

$$\frac{dv}{dt} = g\sin\theta - \frac{B^2d^2}{mR}(\cos\theta)^2 v$$

$$\frac{dv}{dt} + \frac{B^2d^2}{mR}(\cos\theta)^2 v = g\sin\theta \quad \textbf{[1]}$$

On solving, we get this linear differential equation, we have

$$v = \frac{g\sin\theta}{\frac{B^2d^2\cos^2\theta}{mR}} + A\exp\left[-\frac{B^2d^2}{mR}\left(\cos^2\theta\right)t\right]$$

Here, A is a constant to be determined by initial conditions.

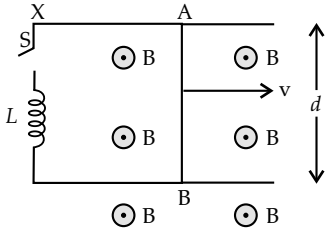
Final answer will be

$$v = \frac{mgR\ \sin\theta}{B^2d^2\cos^2\theta}\left[1 - \exp\left\{-\frac{B^2d^2}{mR}\left(\cos^2\theta\right)t\right\}\right] \quad \textbf{[1]}$$

Let us consider a new constant $\frac{mR}{B^2.d^2.\cos^2\theta} = \alpha$

$$\boxed{v = \alpha g\sin\theta\left(1 - e^{\frac{-t}{a}}\right)} \quad \textbf{[1]}$$

Q. 7. Find the current in the sliding rod AB (resistance = R) for the arrangement shown in figure. $\vec{B}$ is constant and is out of the paper. Parallel wires have no resistance. $\vec{v}$ is constant. Switch S is closed at time $t = 0$.



[NCERT Exemp. Q. 6.30, Page 39]

Ans. Conductor AB moves towards right with speed v and magnetic $\vec{B}$ perpendicularly upwards so angle between B and v is 90° and an induced emf flows in loop $AXYB$. **[1]**

$$\varepsilon = \frac{d}{dt}(\vec{B}.\vec{A}) = \frac{d}{dt}BA\cos 0 = \frac{d}{dt}B(d\times v\times t) = Bvd$$

$\because$ Angle between $\vec{B}$ and $\vec{A}$ is 0°.

$\therefore\ \varepsilon = \mathrm{B}.v.d$

At $t = 0$, current starts increasing in loop along with inductor L due to potential difference **[1]**

Applying Kirchhoff's voltage rule, we have

$$-L\frac{dI}{dt} + vBd = IR$$

$$\Rightarrow L\frac{dI}{dt} + IR = vBd \quad \textbf{[1]}$$

Solving this linear differential equation, we have

$$I = \frac{vBd}{R} + Ae^{-Rt/2}$$

At $t = 0, I = 0$

$$\Rightarrow\ A = -\frac{vBd}{R} \Rightarrow I = \frac{vBd}{R}(1 - e^{-Rt/L})$$

This is the required current expression. **[2]**

Q. 8. A metallic ring of mass m and radius 1 (ring being horizontal) is falling under gravity in a region having a magnetic field. If z is the vertical direction, the z-component of magnetic field is $B_z = B_0 (1+\lambda z)$. If R is the resistance of the ring and if the ring falls with a velocity v, find the energy lost in the resistance. If the ring has reached a constant velocity, use the conservation of energy to determine v in terms of m, B, λ and acceleration due to gravity g. [NCERT Exemp. Q. 6.31, Page 39]

Ans. Magnetic flux across the ring of mass 'm' and radius 'l' falling under gravity in a region of having magnetic field $B_z = B_0(1+\lambda z)$ is **[1]**

$$\phi = B_z.\vec{A} = B_0(1+\lambda z).\pi l^2.$$

The angle between $\vec{B}$ and $\vec{A}$ is 0°

$$\varepsilon = \frac{d}{dt}[B_0(1+\lambda z)\pi]^2 \quad \textbf{[1]}$$

$$IR = (B_0\pi l^2)\left[0 + \lambda\frac{dz}{dt}\right]$$

$$I = \frac{B_0\pi\lambda l^2}{R}\frac{dz}{dt} = \frac{B_0\pi\lambda l^2}{R}v \quad \textbf{[1]}$$

Put the value of I in Energy loss formula, I^2R

$$\text{Energy lost, } = \frac{B_0^2\pi^2\lambda^2 l^4}{R^2}v^2.R$$

$$\text{Energy lost} = \frac{B_0^2\pi^2\lambda^2 l^4 v^2}{R}$$

The energy must come from decrease in P.E.

$$= mg\frac{dz}{dt} = mgv \quad \textbf{[1]}$$

$$\therefore mgv = \frac{B_0^2\pi^2\lambda^2 v^2 l^4}{R}$$

$$v = \frac{mgR}{B_0^2\pi^2\lambda^2 l^4} \text{ or } v = \frac{mgR}{(\pi l^2\lambda B_0)^2} \quad \textbf{[1]}$$

Q. 9. A long solenoid '*S*' has *n* turns per meter, with diameter '*a*'. At the centre of this coil we place a smaller coil of '*N*' turns and diameter '*b*' (where $b < a$). If the current in the solenoid increases linearly, with time, what is the induced *emf* appearing in the smaller coil. Plot graph showing nature of variation in *emf*, if current varies as a function of $mt^2 + C$.

[NCERT Exemp. Q. 6.32, Page 39]

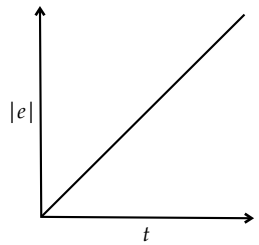
Ans. Varying magnetic field $B(t)$ in solenoid is

$B_1(t) = \mu_0 nI(t)$ **[1]**

This varying magnetic field changes flux in the smaller coil.

Magnetic flux in second coil,

$\phi_2 = B_1(t).A = \mu_0 nI(t).\pi b^2$ **[1]**



[½]

Magnetic field due to a solenoid $S, B = \mu_0 nl$

Magnetic flux in smaller coil $\phi = NBA$ where $A = \pi b^2$

Applying Faraday's law of EMI, we have

$$\text{So, } e = \frac{-d\phi}{dt} = \frac{-d}{dt}(NBA) = -N\pi b^2 \frac{d(B)}{dt} = -N\pi b^2 \mu_0 n \frac{dl}{dt}$$

$$= -Nn\pi\mu_0 b^2 \frac{d}{dt}(mt^2 + C) = -\mu_0 Nn\pi b^2 2mt$$

$$e = -\mu_0 Nn\pi b^2 2mt$$

[2]

Negative sign (−) signifies opposite nature of induced emf. The magnitude of emf varies with time as shown in the figure. **[½]**

Q. 10. A square loop of side 12 cm with its sides parallel to *X* and *Y* axes is moved with a velocity of 8 cm s^{-1} in the positive *x*-direction in an environment containing *a* magnetic field in the positive *z*-direction. The field is neither uniform in space nor constant in time. It has a gradient of 10^{-3} T cm^{-1} along the negative *x*-direction (that is it increases by 10^{-3} T cm^{-1} as one moves in the negative *x*-direction), and it is decreasing in time at the rate of 10^{-3} T s^{-1}. Determine the direction and magnitude of the induced current in the loop if its resistance is 4.50 Ω.

[NCERT Add. Ex. Q. 6.12, Page 37]

Ans. Side of the square loop, s = 12 cm = 0.12 m

Area of the square loop, $A = 0.12 \times 0.12 = 0.0144\ m^2$ **[1]**

Velocity of the loop, v = 8 cm/s = 0.08 m/s

Gradient of the magnetic field along negative *x*-direction,

$\frac{dB}{dx} = 10^{-3}\ \text{Tcm}^{-1} = 10^{-1}\ \text{Tm}^{-1}$

And, rate of decrease of the magnetic field,

$\frac{dB}{dx} = 10^{-3}\ \text{Ts}^{-1}$

Resistance of the loop, $R = 4.5\ \text{m}\Omega = 4.5 \times 10^{-3}\ \Omega$ **[1]**

Rate of change of the magnetic flux due to the motion of the loop in a non-uniform magnetic field is given as :

$$\frac{d\phi}{dt} = A \times \frac{dB}{dx} \times v$$

$$= 144 \times 10^{-4}\ \text{m}^2 \times 10^{-1} \times 0.08$$

$$= 11.52 \times 10^{-5}\ \text{Tm}^2\text{s}^{-1}$$ **[1]**

Rate of change of the flux due to explicit time variation in field B is given as :

$$\frac{d\phi}{dt} = A \times \frac{dB}{dx}$$

$$= 144 \times 10^{-4} \times 10^{-3}$$

$$= 1.44 \times 10^{-5}\ \text{Tm}^2\text{s}^{-1}$$ **[1]**

Since the rate of change of the flux is the induced *emf*, the total induced *emf* in the loop can be calculated as :

$$e = 1.44 \times 10^{-5} + 11.52 \times 10^{-5}$$

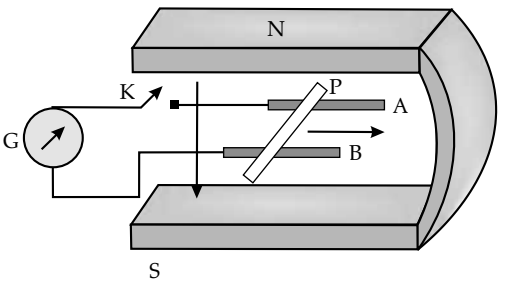
$$= 12.96 \times 10^{-5}\ \text{V}$$

$$\therefore \text{Induced current, } i = \frac{e}{R} = \frac{12.96 \times 10^{-5}}{1.5 \times 10^{-3}} = 2.88 \times 10^{-2} A$$

Hence, the direction of the induced current is such that there is an increase in the flux through the loop along positive *z*-direction. **[1]**

Q. 11. Figure shows a metal rod *PQ* resting on the smooth rails *AB* and positioned between the poles of a permanent magnet. The rails, the rod, and the magnetic field are in three mutual perpendicular directions. A galvanometer *G* connects the rails through a switch *K*. Length of the rod = 15 cm, *B* = 0.50 T, resistance of the closed loop containing the rod = 9.0 mΩ. Assume the field to be uniform.

(a) Suppose *K* is open and the rod is moved with a speed 12 cm s^{-1} of in the direction shown. Give the polarity and magnitude of the induced emf.



(b) Is there an excess charge built up at the ends of the rods when *K* is open? What if *K* is closed?

(c) With *K* open and the rod moving uniformly, there is no net force on the electrons in the rod *PQ* even though they do experience magnetic force due to the motion of the rod. Explain.

(d) What is the retarding force on the rod when *K* is closed?

(e) How much power is required (by an external agent) to keep the rod moving at the same speed (= 12 cms^{-1}) when *K* is closed? How much power is required when *K* is open?

(f) How much power is dissipated as heat in the closed circuit? What is the source of this power?

(g) What is the induced emf in the moving rod if the magnetic field is parallel to the rails instead of being perpendicular?

[NCERT Add. Ex. Q. 6.14, Page 231]

Ans. Length of the rod, $l = 15$ cm $= 0.15$ m
Magnetic field strength, $B = 0.50$ T
Resistance of the closed loop, $R = 9\ \text{m}\ \Omega = 9\times10^{-3}\ \Omega$
Speed of the rod, v = 12 cm/s = 0.12 m/s

(a) Induced emf is given as :

$e = Bvl$
$= 0.5 \times 0.12 \times 0.15$
$= 9 \times 10^{-3}$ V
$= 9$ mV

The polarity of the induced emf is such that end P shows positive while end Q shows negative ends. **[½]**

(b) Yes; when key K is closed, excess charge is maintained by the continuous flow of current. When key K is open, there is excess charge built up at both ends of the rods. When key K is closed, excess charge is maintained by the continuous flow of current, *i.e.*

$$i = \frac{e}{R} = \frac{9\times10^{-3}}{9\times10^{-3}} = 1 \text{ Amp.}$$

[1]

(c) Magnetic force is cancelled by the electric force set-up due to the excess charge of opposite nature at both ends of the rod. There is no net force on the electrons in rod PQ when key K is open and the rod is moving uniformly. This is because magnetic force is cancelled by the electric force set-up due to the excess charge of opposite nature at both ends of the rods. **[½]**

(d) Retarding force exerted on the rod, $F = IBl$
Where,
I = Current flowing through the rod,

$$= \frac{e}{R} = \frac{9\times10^{-3}}{9\times10^{-3}} = 1\ A$$

$$\therefore F = 1\times0.5\times0.15 = 75\times10^{-3}\ N$$

[½]

(e) 9 mW; no power is expended when key K is open.
Speed of the rod, $v = 12$ cm/s $= 0.12$ m/s
Hence, power is given as :

$P = Fv$
$= 75\times10^{-3}\times0.12$
$= 9\times10^{-3}\ W$
$= 9$ mW

[1]

When key K is open, no power is expended.

(f) 9 mW power is provided by an external agent.
Power dissipated as heat $= I^2R = (1)^2\times9\times10^{-3} = 9$ mW
The source of this power is an external agent. **[1]**

(g) Zero in this case, no emf is induced in the coil because the motion of the rod does not cut across the field lines. **[½]**

Q. 12. (a) State the principle of an ac generator and explain its working with the help of a labelled diagram. Obtain the expression for the emf induced in a coil having N turns each of cross-sectional area A, rotating with a constant angular speed 'ω' in a magnetic field $\vec{B}$ directed perpendicular to the axis of rotation.

(b) An aeroplane is flying horizontally form west to east with a velocity of 900 km/hour. Calculate the potential difference developed between the ends of its wings having a span of 20 m. The horizontal component of the Earth's magnetic field is 5×10^{-4} T and the angle of dip is 30°.

[CBSE Board, Delhi Region, 2018]

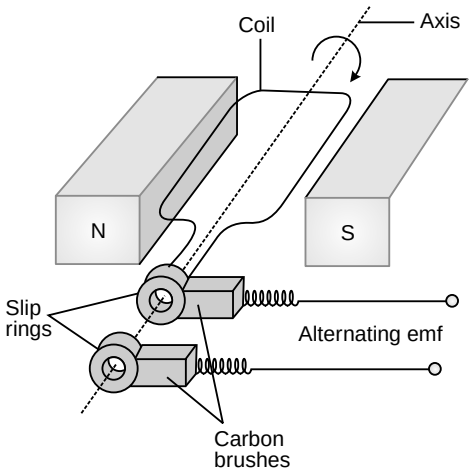
Ans. (a) AC Generator : Consider a coil $PQRS$ free to rotate in a uniform magnetic field B. The axis of rotation of the coil is perpendicular to the field B. The flux through the coil will be $\phi = \phi_m \cos\omega t$. **[½]**

$$N\phi = NBA\cos\theta$$
$$N\phi = NBA\cos(\omega t)$$
$$N\frac{d\phi}{dt} = NBA\frac{d}{dt}\cos(\omega t) \quad \text{[Differentiate]}$$
$$N\frac{d\phi}{dt} = -NBA\omega\sin(\omega t) \quad \left(\varepsilon = -N\frac{d\phi}{dt}\right)$$

[1]

$$-\varepsilon = -NBA\omega\sin(\omega t)$$
$$\varepsilon_0 = NBA\omega$$
$$\varepsilon = \varepsilon_0\sin(\omega t)$$

[½]



[1]

The AC generator work on change in flux induce emf in coil on [EMI]

(b) $V = 900\times\dfrac{1{,}000}{3{,}600} = 250$ m/s

$l = 20$ m

When aeroplane is going in West to East, its wings cross-vertical component of the Earth's magnetic field,

$B = 5\times10^{-4}$ Tesla
$\theta = 30°$
$B\sin\theta = B_\perp$
$B_\perp = 5\times10^{-4}\times\sin30° = 2.5\times10^{-14}$
$\varepsilon = B\ell v$
$= 2.5\times10^{-4}\times20\times250$
$= 50\times25\times10^{-3}$
$= 125\times10^{-2}$
$= 1.25$ Volt

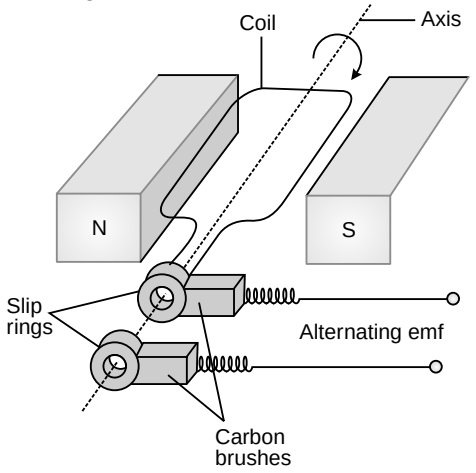
[2]

Q. 13. (a) Draw a labelled diagram of an ac generator. Obtain the expression for the emf induced in the rotating coil of N turns each of cross-sectional area A, in the presence of a magnetic field $\vec{B}$

(b) A horizontal conducting rod 10 m long extending from east to west is falling with a speed 5.0 ms^{-1} at right angles to the horizontal component of the Earth's magnetic field, 0.3 × 10^{-4} Wb m^{-2}. Find the instantaneous value of the emf induced in the rod.

[CBSE Board, All India Region, 2017]

Ans. **(a)** AC generator :



[1]

Let at any instant total magnetic flux link with the armature will is given *G*.

(Where, θ= ωt is the angle made by area vector of coil with magnetic field.)

$$\phi = \text{NBA}\cos\theta = \text{NBA}\cos\omega t$$

$$\therefore \frac{d\phi}{dt} = -\text{NBA}\ \omega\ \sin\omega t$$

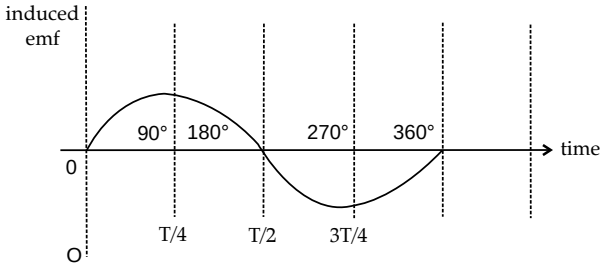
$$-\frac{d\phi}{dt} = \text{NBA}\ \omega\ \sin\omega t$$

By Faraday's law of EMI, $e = -\frac{d\phi}{dt}$

Induced emf in coil will be given by,

$e = NBA\ \omega \sin\omega t = e_0 \sin\omega t$

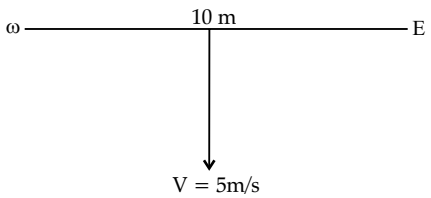
Where, $e_0 = NBA$ and $\omega =$ Peak value of induced emf **[1]**



[½]

The mechanical energy spent in rotating the coil in magnetic field appears in form of electrical energy.

(b)



[1]

Given velocity of straight rod v = 5 m/s horizontal component of the Earth's magnetic field,

$B = 0.30 \times 10^{-4}$ Wb/m^2, length of wire $\ell = 10$ m

So the emf induced in the wire is given by $e = B\ell v \sin\theta$ $(\theta = 90°)$

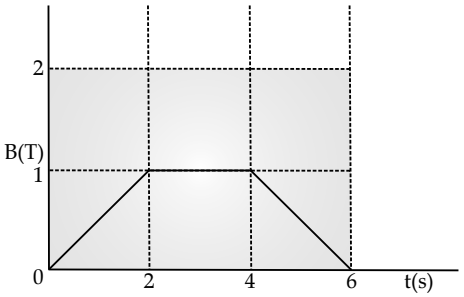
$e = 0.30 \times 10^{-4} \times 10 \times 5$ (Wire is falling at right angle to the Earth horizontal magnetic field component)

$e = 1.5 \times 10^{-3}$ volt

Air moves from west to east (According to Fleming's right hand rule) **[1½]**

Q. 14. (a) State Faraday's law of electromagnetic induction.

(b) The magnetic field through a circular loop of wire 12 cm in radius and 8·5 Ω resistance, changes with time as shown in the figure. The magnetic field is perpendicular to the plane of the loop. Calculate the induced current in the loop and plot it as a function of time.



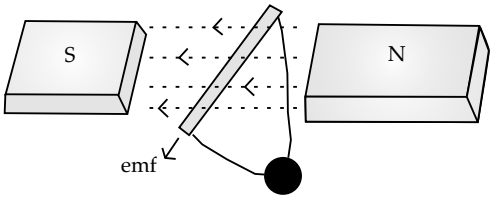
(c) Show that Lenz's law is a consequence of conservation of energy.

[CBSE Board, Foreign Scheme, 2017]

Ans. **(a)** "The phenomenon whereby an emf and hence current is induced in any conductor which is cut across or is cut by a magnetic flux is known as electromagnetic induction". **[½]**

Faraday's First Law It states as follows :

"Whenever the magnetic flux linked with a circuit changes, an emf is always induced in it". Or "Whenever a conductor cuts magnetic flux, an emf is induced in that conductor". **[½]**



[½]

Faraday's Second Law It states as follows : "The magnitude of induced emf is equal to the rate of change of flux-linkages".

Mathematically, $e = -N\frac{d\phi}{dt}$ volts

Where e = Induced emf, $\frac{d\phi}{dt}$ = Rate of change of flux and N = Number of turns of the coil.

[Usually, a minus sign is given to the right-hand side expression to signify the fact that the induced emf sets up current in such a direction that magnetic effect produced by it opposes the very cause producing it.] **[½]**

(b) We know that, $\phi = BA\cos\theta$
But $\theta = 0°$ (fields and area vector in same direction)
So, $\phi = BA$
We also know that,

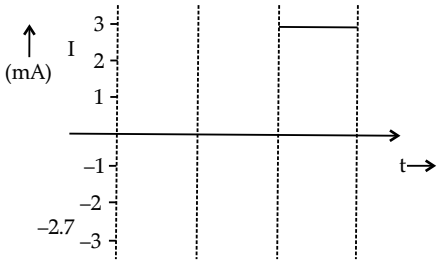
$$\varepsilon = \frac{-d\phi}{dt}$$

$\varepsilon = -0.023$ V,

$$I = \frac{\varepsilon}{R} = -2.7 \text{ mA for } (0<t<2 \text{ s})$$

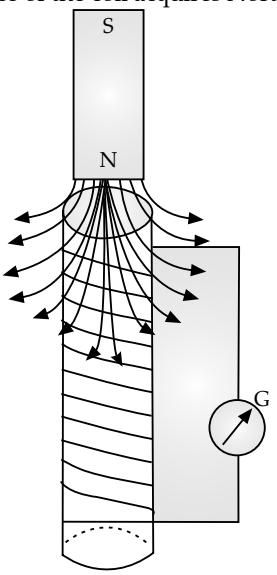
Time/voltage and current)	0<t<2 s	2<t<4 s	4<t<6 s
ε(v)	−0.023	0	+0.023
I(A)I	−2.7	0	+2.7

[½]



[½]

(c) **Lenz's Law** : Copper coils are wound on a cylindrical cardboard and the two ends of the coil are connected to a sensitive galvanometer. A magnet is moved towards the coil (Figure). The upper face of the coil acquires North polarity.

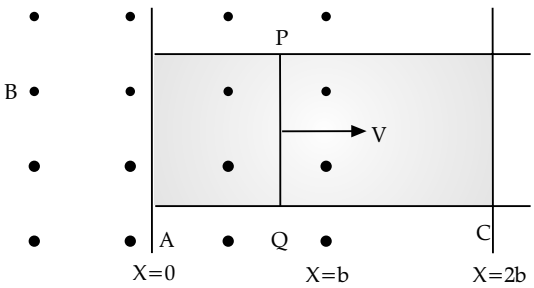


[½]

Consequently, work has to be done to move the magnet further against the force of repulsion.
When we withdraw the magnet away from the coil, its upper face acquires South polarity. Now the work done is against the force of attraction. When the magnet is moved, the number of magnetic lines of force linking the coil changes, which causes an induced current to flow through the coil. The direction of the induced current, according to Lenz's law is always to oppose the motion of the magnet. The work done in moving the magnet is converted into electrical energy. This energy is dissipated as heat energy in the coil. If on the contrary, the direction of the current helps the motion of the magnet, it would start moving faster increasing the change of magnetic flux linking the coil.
This results in the increase of induced current. Hence kinetic energy and electrical energy would be produced without any external work being done, but this is impossible. Therefore, the induced current always flows in such a direction to oppose the cause. Thus it is proved that Lenz's law is the consequence of conservation of energy. **[1½]**

Q. 15. (a) When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects. Identify the phenomenon causing this deflection and write the factors on which the amount and direction of the deflection depends. State the laws describing this phenomenon.

(b) Sketch the change in flux, emf and force when a conducting rod *PQ* of resistance *R* and length *l* moves freely to and fro between *A* and *C* with speed v on a rectangular conductor placed in uniform magnetic field as shown in the figure.



[CBSE Board, Delhi Region, 2016]

Ans. **(a)** When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects because of the phenomenon of electromagnetic induction. **[½]**
Factors on which the amount and direction of deflection depend are :

(i) Speed of the magnet

(ii) Polarity of the magnet **[½]**

Laws describing electromagnetic induction are :

(1) **Faraday's law of electromagnetic induction :** The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit. **[½]**

$$e = -\frac{d\phi}{dt}$$

[½]

(2) **Lenz's law :** The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it. **[½]**

(b) The flux enclosed by the rod is,

$$\phi = Bl_x \quad \text{for } 0 \le x \le b$$
$$= Bl_b \quad \text{for } b \le x \le 2b$$

Now, the magnitude of induced emf is,

$$\varepsilon = -\frac{d\phi}{dt} = -Bl\frac{dx}{dt} = -Blv \qquad \text{for } 0 \le x \le b$$

$$= -Bl\frac{db}{dt} = 0 \qquad \text{for } b \le x \le 2b$$

Now, the magnitude of induced current when induced emf is non-zero is $\varepsilon = IR$

$$\therefore I = \frac{\varepsilon}{R} = \frac{Blv}{R}$$

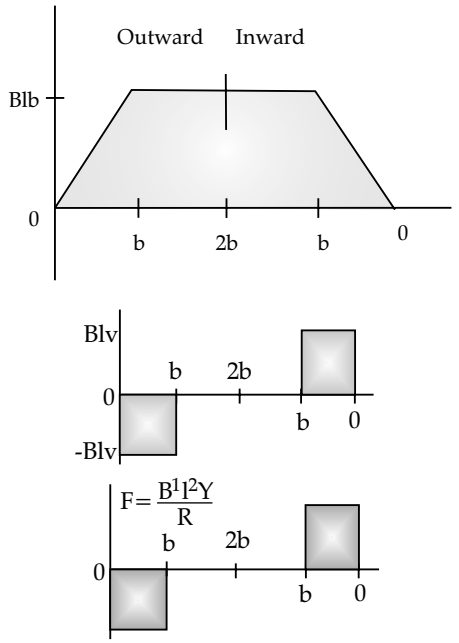
The force required to keep the conductor in motion is $F = BIl$

$$\therefore F = B\frac{Blv}{R}l = \frac{B^2l^2v}{R}$$

$$\therefore F = \frac{B^2l^2v}{R} \qquad \text{for } b \le x < b$$

$$= 0 \qquad \text{for } b \le x \le 2b$$

Therefore, the variation of flux, emf and force is **[1]**



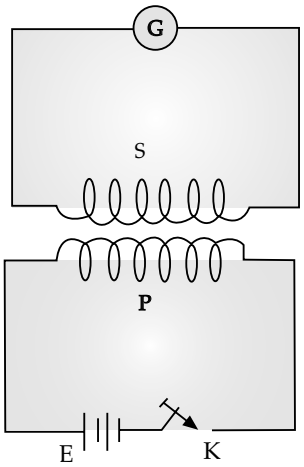
[2½]

Q. 16. (a) Explain the meaning of the term mutual inductance. Consider two concentric circular coils, one of radius r_1 and the other of radius r_2 ($r_1 < r_2$) placed coaxially with centres coinciding with each other. Obtain the expression for the mutual inductance of the arrangement.

(b) A rectangular coil of area A, having number of turns N is rotated at 'f' revolutions per second in a uniform magnetic field B, the field being perpendicular to the coil. Prove that the maximum emf induced in the coil is $2\pi f NBA$.

[CBSE Board, All India Region, 2016]

Ans. **Mutual inductance :** The phenomenon according to which an opposing emf is produced in a coil as a result of change in current; hence, the magnetic flux linked with a neighbouring coil is called m**utual induction.** **[½]**



[½]

Coefficient of mutual induction : Consider two coils P and S. Suppose that a current I is flowing through the coil P at any instant, *i.e.*,

$\Phi \propto I$

$\Phi = MI$ … (i)

Where, M is called coefficient of mutual induction **[½]**

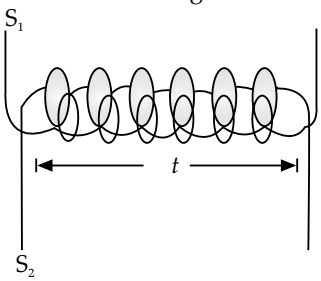
If 'e' is the induced emf produced in the S-coil, then

$$e = -\frac{d\phi}{dt} = \frac{d}{dt}(MI)$$

$$= -M\frac{dI}{dt}$$

[½]

Mutual inductance of two concentric coils, one of radius r_1 and the other of radius R_2 ($r_1 < r_2$) placed co-axially with centres coinciding with each other : **[½]**



[½]

Consider two circular coils : S_1 and S_2 of same length l, such that coil S_2 surrounds coil S_1 completely.

Let,

n_1 – Number of turns per unit length of S_1

n_2 – Number of turns per unit length of S_2

I_1 – Current passed through solenoid S_1

Φ_{21} – Flux linked with S_2 due to current flowing through S_1

$\Phi_{21} \propto I_1$

$\Phi_{21} = M_{21}I_1$ **[½]**

Where, M_{21} is the co-efficient of mutual induction of the two coils.

When current is passed through S_1, an emf is induced in S_2.

Magnetic field produced inside S_1 on passing current through it,

$B_1 = \mu_0 n_1 I_1$

Magnetic flux linked with each turn of S_2 will be equal to B_1 times the area of cross-section of S_1.

Magnetic flux linked with each turn of the $S_2 = B_1A$
Therefore, total magnetic flux linked with the S_2,

$\Phi_{21} = \mu_0 n_1 n_2 A_1 I_1$

$\therefore M_{21} = \mu_0 n_1 n_2 A_1$ [½]

Similarly, the mutual inductance between the two coils, when current is passed through coil S_2 and induced emf is produced in coil S_1, is given by

$M_{12} = \mu_0 n_1 n_2 A_1$

$\therefore M_{12} = M_{12} = M$ (say)

Hence, co-efficient of mutual induction between the two coils will be $M = \mu_0 n_1 n_2 A_1$

(b) We know that flux is given by

$\phi = NBA \cos\theta$ [½]

We also know,

$$e = -\frac{d\phi}{dt} = \left[-NBA(-\sin\theta)\frac{d\theta}{dt}\right]$$

$$= NBA\sin\theta(2\pi f)$$

For maximum induced emf = sin θ = 1

$$e_{max} = \left(-NBA\ \sin\theta\frac{d\theta}{dt}\right) = NBA\ (2\pi f)$$

Hence proved. [½]

Some Commonly Made Errors

- **Generally, Students confuse in self-inductance or mutual inductance.**
- **Inverse error propagation and model identification for coupled dynamic problems with application to electromagnetic metal forming.**

EXPERT ADVICE

- *First clear your chapter concepts. Then learn and memorise all the formula and most important try to derive new formula from the older one. In every question, there is a key word try to sort out that. Then solve your question.*
- *You must try to do as many problem as problem as you can. When you have done enough problems you will understand how to solve problem easily. When you have a good understanding of the lesson, use Fundamentals of Physics.*
- *First, determine the units of the quantity you're trying to find and the quantities you have. Only use base units (meters, kilograms, seconds, charge), not compound units (Force is measured in Newton's, which are just kg. m/s^2).*

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CHAPTER

7 ALTERNATING CURRENT

Chapter Objectives

This chapter will help you understand :

- ***Alternating current :*** *Introduction; AC voltage applied to a resistor; Representation of AC current and voltage by rotating vectors- Phasors; AC voltage applied to an inductor; AC voltage applied to a capacitor; and AC voltage applied to a series LCR circuit.*
- ***Power in AC circuits :*** *The power factor; LC oscillations; and Transformers.*

TOPIC - 1
Alternating Current P. 149

TOPIC - 2
Power in AC Circuit P. 167

TOPIC-1
Alternating Current

Quick Review

- Electricity was introduced to Ethiopia in 1896 after Emperor Menelik II ordered two newly invented electric chairs as a form of humane capital punishment and realised they were useless in his country without electricity.
- Albertville, France's electricity is powered by Beaufort cheese. Since whey is unnecessary to make Beaufort cheese, bacteria is added to the whey. This transforms the whey into biogas. This gas is then fed through an engine which heats water to 90°C (194°F) to generate 2,800 MWh per year of electricity.
- In 1746, Jean-Antoine Nollet conducted an experiment in which 200 monks formed a circle (1.6 km in circumference) and were linked by iron wire. He then had electricity pass through them which shocked all the monks simultaneously. He concluded that the speed of electricity was very high.
- Huge amounts of renewable energy can be stored over a long period of time by using Pumped Storage Hydropower, where water is pumped up a hill with renewable electricity then sent back down the hill to generate on demand clean electricity at up to 80% efficiency.
- In 1963, Quebec government bought out all the private power companies and nationalised electricity. Today, 96% of Quebec's power is from hydro-electricity, and it has some of the cheapest electricity rates in North America while earning billions in revenue.
- A typical micro-wave oven consumes more electricity powering its digital clock than it does heating food.
- The word "electrocute" is a combination of the words electro and execute, meaning you were killed by electricity. So if you don't die, you were not electrocuted, you were shocked.
- Benjamin Harrison, the 23rd president of the United States (US), was the first president to have electricity in the White House but he never touched any light switches because he was too scared of getting electrocuted.

TIPS...

- *Calculate the power used by an AC circuit with different terms like active power, reactive power and apparent power.*
- *Learn all the theorems to analyse the circuits.*
- *Differentiate reactance, resistance, inductance, conductance, etc.*

TRICKS...

- *Relate the formulas from one to another by changing the parameters.*
- *Always make sure to put all the formulas on paper before solving numerical problems.*
- *One general trick is to visual simplify it into sub-circuits. If you see an inductor and capacitor in parallel, it might be a resonant circuit.*
- *If you see a capacitor going to a resistor to ground, it might be an AC coupling circuit. If there is a coupling capacitor, you can often split it into two sub-circuits.*

Know the Links

- https://en.wikipedia.org/wiki/Alternating_current
- https://www.britannica.com/science/alternating-current
- https://www.encyclopedia.com/science-and-technology/computers-and.../alternating
- https://physics.info/current-alternating/

Multiple Choice Questions (1 mark each)

Q. 1. If the rms current in a 50 Hz AC circuit is 5 *A*, the value of the current 1/300 s after its value becomes zero is

(a) $5\sqrt{2}\ A$ **(b)** $5\sqrt{\frac{3}{2}}A$

(c) $\frac{5}{6}$A **(d)** $\frac{5}{\sqrt{2}}A$

[NCERT Exemp. Q. 7.1, Page 40]

Ans. Correct option : (b)

Explanation :

Here, $I_{rms} = 5\ A$, $\nu = 50$ Hz and $t = \frac{1}{300}s$

$$I_0 = \text{Peak value} = \sqrt{2}I_{rms} = \sqrt{2} \times 5 = 5\sqrt{2}\ A$$

Now, $I = I_0 \sin \omega t = 5\sqrt{2} \sin 2\pi\nu t$

$$= 5\sqrt{2} \sin 2\pi \times 50 \times \frac{1}{300} = 5\sqrt{\frac{3}{2}}\ A$$

Q. 2. An alternating current generator has an internal resistance R_g and an internal reactance X_g. It is used to supply power to a passive load consisting of a resistance R_g and a reactance X_L. For maximum power to be delivered from the generator to the load, the value of X_L is equal to

(a) zero **(b)** X_g

(c) $-X_g$ **(d)** R_g

[NCERT Exemp. Q. 7.2, Page 40]

Ans. Correct option : (c)

Explanation : As internal resistance of generator is already equal to external resistance R_g. So to deliver maximum power, *i.e.*, to make reactance equal to zero, the reactance in external circuit will be –Xg. In order to deliver maximum power, the generator to the load, the total reactance must be equal to zero, *i.e.*, $X_L + X_g = 0$, $X_L = -X_g$.

Q. 3. When a voltage measuring device is connected to AC mains, the meter shows the steady input voltage of 220 V. this means

(a) Input voltage cannot be AC voltage, but a DC voltage.

(b) Maximum input voltage is 220 V.

(c) The meter reads not v but (v^2) and is calibrated to read $\sqrt{(v^2)}$.

(d) The pointer of the meter is stuck by some mechanical defect.

[NCERT Exemp. Q. 7.3, Page 41]

Ans. Correct option : (c)

Explanation : The voltmeter in AC circuit reads value $<v^2>$ and meter is calibrated to rms value $<v^2>$ which is multiplied by $\sqrt{2}$ to get V_{rms}. In other words, voltmeter connected to the AC main read root mean square value of AC voltage, *i.e.*, $\sqrt{<v^2>}$.

Q. 4. To reduce the resonant frequency in an *L-C-R* series circuit with a generator

(a) the generator frequency should be reduced

(b) another capacitor should be added in parallel to the first

(c) the iron core of the inductor should be removed

(d) dielectric in the capacitor should be removed

[NCERT Exemp. Q. 7.4, Page 41]

Ans. Correct option : (b)

Explanation : The resonant frequency of *L-C-R* series circuit is $v_o = \frac{1}{2\pi\sqrt{LC}}$

So to reduce resonant frequency, we have either to increase *L* or to increase C.

To increase capacitance another capacitor must be connected in parallel with the first.

Q. 5. Which of the following combinations should be selected for better tuning of an *L-C-R* circuit used for communication?

(a) $R = 20\ \Omega$, $L = 1.5\ H$, $C = 35\ \mu F$

(b) $R = 25\ \Omega$, $L = 2.5\ H$, $C = 45\ \mu F$

(c) $R = 15\ \Omega$, $L = 3.5\ H$, $C = 30\ \mu F$

(d) $R = 25\ \Omega$, $L = 1.5\ H$, $C = 45\ \mu F$

[NCERT Exemp. Q. 7.5, Page 41]

Ans. Correct option : (c)

Explanation : Quality factor (*Q*) of an *L-C-R* circuit is given by, $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$

Tuning of an *L-C-R* circuit depends on quality factor of the circuit. Tuning will be better when quality factor of the circuit is high.

As, quality factor (*Q*) of an *L-C-R* circuit is given by,

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

For *Q* to be high, *R* should be low, *L* should be high and *C* should be low. Therefore, option (c) is most suitable.

Tick Two or More Options

Q. 6. As the frequency of an ac circuit increases, the current first increases and then decreases. What combination of circuit elements is most likely to comprise the circuit?

(a) Inductor and capacitor (b) Resistor and inductor

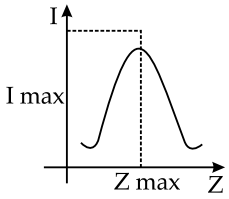
(c) Resistor and capacitor (d) Resistor, inductor and capacitor

[NCERT Exemp. Q. 7.8, Page 42]

Ans. Correct options : (a) and (d)

Explanation : Impedance is given by,

$$Z=\sqrt{R^2+\left(X_L-X_C\right)^2}=\sqrt{R^2+\left(2\pi\upsilon L-\frac{1}{2\pi\upsilon C}\right)^2}$$



At resonance frequency, Z is minimum and current is maximum. For any other value of frequency, the value of current will decrease (when compared to current at resonance).

Clearly circuit must have inductor and capacitor. However, it may be possible that the circuit also has resistor besides inductor and capacitor.

Q. 7. In an alternating current circuit consisting of elements in series, the current increases on increasing the frequency of supply. Which of the following elements are likely to constitute the circuit?

(a) Only resistor

(b) Resistor and an inductor

(c) Resistor and a capacitor

(d) Only a capacitor [NCERT Exemp. Q. 7.9, Page 42]

Ans. Correct options : (c) and (d)

Explanation : Here in question on increasing the frequency (υ) the current also increases. So capacitive reactance of circuit decreases and resistance does not depend on frequency. So on increasing frequency, X_C decreases to increase current in circuit. Therefore, circuit must have a capacitor; however, it may or may not have a resistor.

Q. 8. For an *LCR* circuit, the power transferred from the driving source to the driven oscillator is $P = I^2Z \cos\phi$.

(a) Here, the power factor $\cos\phi \geq 0$, $P \geq 0$.

(b) The driving force can give no energy to the oscillator ($P = 0$) in some cases.

(c) The driving force cannot syphon out ($P < 0$) the energy out of oscillator.

(d) The driving force can take away energy out of the oscillator. [NCERT Exemp. Q. 7.11, Page 42]

Ans. Correct options : (a), (b) and (c)

Explanation : As per question $P = I^2Z\cos\phi$

Power factor $\cos\phi = \frac{R}{Z}$

As $R > 0$ and $Z > 0$

So, $\cos\phi = \frac{R}{Z}$ is positive $\Rightarrow P > 0$

Q. 9. When an AC voltage of 220 V is applied to the capacitor *C*,

(a) the maximum voltage between plates is 220 V.

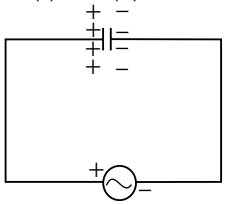
(b) the current is in phase with the applied voltage.

(c) the charge on the plates is in phase with the applied voltage.

(d) power delivered to the capacitor is zero.

[NCERT Exemp. Q. 7.12, Page 42]

Ans. Correct options : (c) and (d)



Explanation : On application of AC voltage into the capacitor, the charge on plates is in phase with the applied voltage because the charge on plates is in phase with applied voltage.

The current developed leads the applied voltage by a phase angle of 90°.

Therefore, average power delivered to the capacitor per cycle is, $P_{av} = V_{rms}\, I_{rms} \cos 90° = 0$.

Q. 10. The line that draws power supply to your house from street has

(a) zero average current.

(b) 220 V average voltage.

(c) voltage and current out of phase by 90°.

(d) voltage and current possibly differing in phase ϕ such that $|\phi| < \frac{\pi}{2}$.

[NCERT Exemp. Q. 7.13, Page 43]

Ans. Correct options : (a) and (d)

Explanation : AC current is used in power supply to our house from street. AC currents are having zero average value over a cycle.

As the power line is having some resistance, so power factor, $\cos\phi = \frac{R}{Z} \neq 0$.

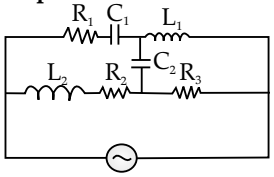
It also means, $\phi \neq \frac{\pi}{2}$ and $|\phi| < \frac{\pi}{2}$. So that, phase angle between voltage and current lies between 0 and $\frac{\pi}{2}$.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. If a *L-C* circuit is considered analogous to a harmonically oscillating spring block system, which energy of the *L-C* circuit would be analogous to potential energy and which one analogous to kinetic energy?
[NCERT Exemp. Q. 7.14, Page 43]

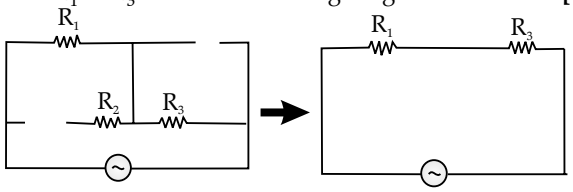
Ans. An *L-C* circuit is considered analogous to a harmonically oscillating spring block system in which Magnetic energy $\left(\frac{1}{2}LI^2\right)$ is analogous to kinetic energy and electrical energy due to charging of capacitor $\left(\frac{1}{2}CV^2\right)$ is analogous to potential energy. **[1]**

Q. 2. Draw the effective equivalent circuit of the circuit shown in figure, at very high frequencies and find the effective impedance.



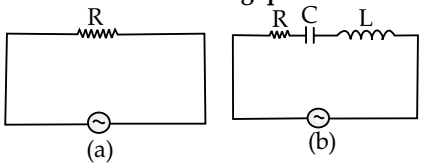
[NCERT Exemp. Q. 7.15, Page 43]

Ans. At high frequencies, capacitor ≈ short circuit (low reactance) and inductor ≈ open circuit (high reactance). Therefore, the equivalent circuit $Z \approx R_1 + R_3$ as shown in the figure given below. **[1]**



[1]

Q. 3. Study the circuits shown in figure (a) and figure (b) and answer the following questions.



(a) Under which conditions would the rms currents in the two circuits be the same?

(b) Can the rms current in circuit (b) be larger than that in (a)? **[NCERT Exemp. Q. 7.16, Page 43]**

Ans. (I_{rms}) in (a) $= \frac{V_{rms}}{R}$

(I_{rms}) in (b) $= \frac{V_{rms}}{Z} = \frac{V_{rms}}{\sqrt{R^2 + (X_L - X_C)}}$ **[½]**

(a) $(I_{rms})_a = (I_{rms})_b$

$$\frac{V_{rms}}{R} = \frac{V_{rms}}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\therefore R = \sqrt{R^2 + (X_L - X_C)^2}$$

Squaring both sides, we have

$$R^2 = R^2 + (X_L - X_C)^2$$

or $(X_L - X_C)^2 = 0$

$$X_L = X_C$$

So I_{rms} in circuits (a) and (b) will be equal if $X_L = X_C$. **[½]**

(b) $(I_{rms})_b > (I_{rms})_a$

$$\frac{V_{rms}}{\sqrt{R^2 + (X_L - X_c)^2}} > \frac{V_{rms}}{R}$$

As $V_{rms} = V$ so, $\sqrt{R^2 + (X_L - X_c)^2} < R$

Squaring both sides, we have

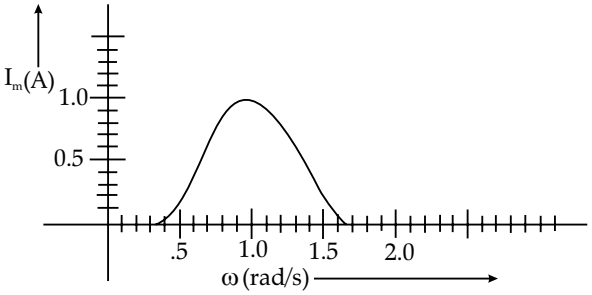
$R^2 + (X_L - X_C)^2 < R^2$

$(X_L - X_C)^2 < 0$ **[½]**

Square of any number can never be negative. Reactance of X_L and X_C cannot be negative.

So the rms current in circuit (b) cannot be larger than that in (a). **[½]**

Q. 4. In series *LCR* circuit, the plot of I_{max} vs ω is shown in figure. Find the bandwidth and mark in the figure.



[NCERT Exemp. Q. 7.18, Page 43]

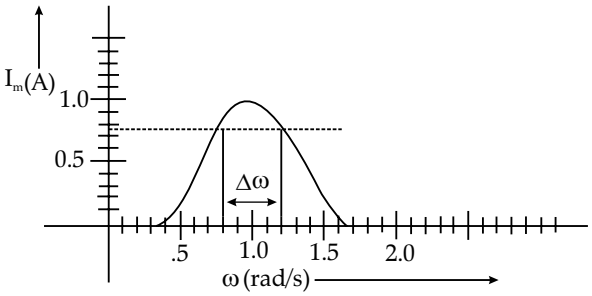
Ans. Bandwidth corresponds to frequencies at which

$I_m = \frac{I_{max}}{\sqrt{2}} \approx 0.707\, I_{max}$ **[½]**

It is shown in the Figure given below

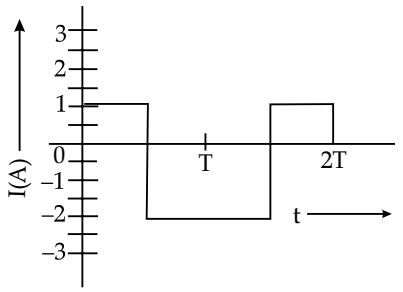
It is shown in the Fig.

$\Delta\omega = 1.2 - 0.8 = 0.4$ rad/s **[½]**



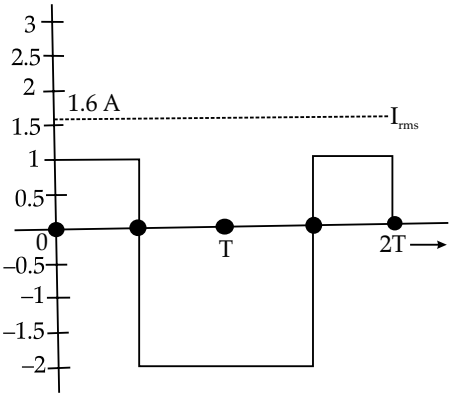
[1]

Q. 5. The alternating current in a circuit is described by the graph shown in figure. Show rms current in this graph.



[NCERT Exemp. Q. 7.19, Page 44]

Ans. $I_{rms} = \sqrt{\frac{1^2 + 2^2}{2}} = \sqrt{\frac{5}{2}} \cong 1.6\ A$ **[½]**



I_{rms} = 1.6 A (shown in Figure by dotted line). **[1½]**

Q. 6. How does the sign of the phase angle ϕ, by which the supply voltage leads the current in an *LCR* series circuit, change as the supply frequency is gradually increased from very low to very high values. [NCERT Exemp. Q. 7.20, Page 44]

Ans. We know that,

$$\tan\phi = \frac{X_L - X_C}{R} = \frac{2\pi\nu L - \frac{1}{2\pi\nu C}}{R}$$

If ν is small $X_C > X_L$ so $[2\pi\nu L - \frac{1}{2\pi\nu C}]$ is negative, so $\tan\phi < 0$. **[1]**

The phase angle 'ϕ ' by which voltage leads the current in *LCR* series circuit where $X_L > X_C$.

For ν is large, $X_L > X_C$

So X_L–X_C is positive or tan ϕ > 0.

For $X_L = X_C$, *i.e.*, at resonant frequency

X_L–X_C = 0 so tan ϕ=0.

So phase angle in series *LCR* ac circuit will change from a negative to zero and then zero to positive value. **[1]**

Q. 7. A 100 ω resistor is connected to a 220 V, 50 Hz ac supply.

(a) What is the rms value of current in the circuit?

(b) What is the net power consumed over a full cycle? [NCERT Ex. Q. 7.1, Page 266]

Ans. Given that :

R = 100 Ohms

V = 220 V

ν = 50 Hz

(a) We know that

$$I_{rms} = \frac{V_{rms}}{R}$$

Substituting the values,

$$I_{\text{rms}} = \frac{220}{100} = 2.2\ \text{A}$$ **[1]**

(b) Power = VI

Or power = 220 × 2.2

Or power = 484 W **[1]**

Q. 8. (a) The peak voltage of an ac supply is 300 V. What is the rms voltage?

(b) The rms value of current in an ac circuit is 10 A. What is the peak current? [NCERT Ex. Q. 7.2, Page 266]

Ans. **(a)** We know that

$$V_{rms} = \frac{V_{peak}}{1.414}$$

$$V_{rms} = \frac{300}{1.414}$$

or V_{rms} = 212.13 V **[1]**

(b) Using above identity for current,

$I_{peak} = 1.414 \times I_{rms}$

or

I_{peak} = 1.414×10 = 14.14 A. **[1]**

Q. 9. A 44 mH inductor is connected to 220 V, 50 Hz ac supply. Determine the rms value of the current in the circuit. [NCERT Ex. Q. 7.3, Page 266]

Ans. Given that

L = 44 mH

V = 220 V

ν = 50 Hz

Inductive reactance, $X_L = \omega L = 2\pi\nu L$

I_{rms} is given by, $\frac{V}{X_L}$

Determining inductive reactance

$X_L = 2 \times 3.14 \times 50 \times 44 \times 10^{-3}$

X_L = 13.82 ohms **[½]**

Therefore,

I_{rms} = 220/13.82

or I_{rms} = 15.92 A. **[½]**

Q. 10. A 60 μF capacitor is connected to a 110 V, 60 Hz ac supply. Determine the rms value of the current in the circuit. [NCERT Ex. Q. 7.4, Page 266]

Ans. Given that

C = 60 microfarads

V = 110 volts

ν = 60 Hz

$$I_{rms} = \frac{V}{X_C}$$

Now,

$$X_c = \frac{1}{2 \times 3.14 \times 60 \times 60 \times 10^{-6}}$$

X_c = 44.248 Ohms

Hence

$$I_{rms} = \frac{110}{44.248} = 2.488\ \text{A}.$$ **[2]**

Q. 11. In NCERT Exercises 7.3 and 7.4, what is the net power absorbed by each circuit over a complete cycle. Explain your answer.
[NCERT Ex. Q. 7.5, Page 266]

Ans. Zero. As the phase difference between voltage and current in pure capacitive or inductive circuit is 90°. So, the average power is zero and power is absorbed only by resistance in the circuit. **[1]**

Q. 12. Obtain the resonant frequency ω_r of a series *LCR* circuit with $L = 2.0$H, $C = 32\ \mu F$ and $R = 10\ \omega$. What is the *Q*-value of this circuit?
[NCERT Ex. Q. 7.6, Page 266]

Ans. Given that

$L = 2$ H
$C = 32\ \mu$F
$R = 10$ Ohms

Resonant frequency, $\omega_r = \dfrac{1}{\sqrt{LC}}$

$\omega_r = 125\ s^{-1}$ **[1]**

Now, Q-value $= \dfrac{\omega_r L}{R}$ or $\dfrac{1}{R}\sqrt{\dfrac{L}{C}}$

Putting the desired values gives us Q-value = 25. **[1]**

Q. 13. *A* series *LCR* circuit with $R = 20\ \omega$, $L = 1.5$ H and $C = 35\ \mu F$ is connected to a variable-frequency 200 *V* ac supply. When the frequency of the supply equals the natural frequency of the circuit, what is the average power transferred to the circuit in one complete cycle? [NCERT Ex. Q. 7.9, Page 266]

Ans. Given that

$R = 20$ ohms
$L = 1.5$ H
$C = 35\ \mu$F
$V = 200$ volts

Natural frequency $= \dfrac{1}{(LC)^{1/2}} = 138\ s^{-1}$.

At natural frequency,

$Z = R$

So, $I = \dfrac{V}{R} = \dfrac{200}{20} = 10$ A **[1]**

Thus

$P = I^2R$

Or $P = 10 \times 10 \times 20 = 2{,}000$ W. **[1]**

Q. 14. A radio can tune over the frequency range of a portion of *MW* broadcast band : (800 kHz to 1,200 kHz). If its *LC* circuit has an effective inductance of 200 μH, what must be the range of its variable capacitor? [Hint : For tuning, the natural frequency, *i.e.*, the frequency of free oscillations of the *LC* circuit should be equal to the frequency of the radio wave.] [NCERT Ex. Q. 7.10, Page 266]

Ans. By using the relation, $\dfrac{1}{(LC)^{1/2}} = 2\pi\nu$

Solving for C,

$C = \dfrac{1}{4\pi^2\nu^2 L}$

For $\nu = 800$ kHz,
$C' = 197.8$ pF
For $f = 1{,}200$ kHz,
$C'' = 87.9$ pF
Range : 88 pF to 198 pF. **[2]**

Q. 15. Define 'quality factor' of resonance in series *LCR* circuit. What is its SI unit?
[CBSE Board, Delhi Region, 2016]

Ans. Quality Factor : Quality factor relates the maximum or peak energy stored in the circuit (the reactance) to the energy dissipated (the resistance) during the cycle of oscillation. **[½]**

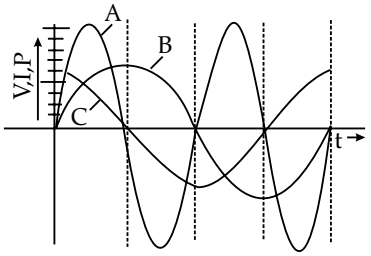
Quality factor is expressed as,

$Q = \dfrac{\omega_o L}{R}$

Q-factor is a dimensionless quantity. **[½]**

Short Answer Type Questions **(3 or 4 marks each)**

Q. 1. A device '*X*' is connected to an a.c. source. The variation of voltage, current and power in one complete cycle is shown in Figure :



(a) Which curve shows power consumption over a full cycle?
(b) What is the average power consumption over a cycle?
(c) Identify the device '*X*'.
[NCERT Exemp. Q. 7.21, Page 44]

Ans. **(a)** We know that $P=VI$, the amplitude of power will be equal to multiplication of V and I. So it must be maximum amplitude as compared to V and I. So curve A shows the power consumption over a cycle. The power of a circuit will become zero, when either $V = 0$ or $I = 0$ or both become zero which is clear from graph 'A is zero (on dotted lines) at these positions. **[1]**

(b) The graph of power is represented by A. the graph 'A' is symmetric with X-axis, *i.e.*, area of graph on positive and negative side are equal. So net area of power graph is zero. So power consumption in circuit is zero, which tallies with that average power of A.C. circuit over a cycle is zero. **[1]**

(c) As the average power of device is zero so power factor $\cos\phi = 0$, *i.e.*, $\phi = 90°$. Change in phase between current and voltage is in either inductor or capacitor. So device may be either capacitor or

inductor or the combination of both inductor and capacitor. **[1]**

Q. 2. Both alternating current and direct current are measured in amperes. But how is the ampere defined for an alternating current?

[NCERT Exemp. Q. 7.22, Page 44]

Ans. For direct current 'One Ampere' is equal to one coulomb charge flowing per sec.

An AC current changes direction with the source frequency and the attractive force would average to zero. Thus, the AC ampere must be defined in terms of some property that is independent of the direction of current. **[1½]**

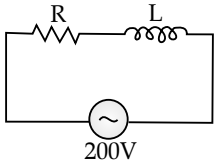
Joule's heating effect is such a property and hence Joule's heating law is used to define one ampere current in AC. According to Joule's heating effect, one ampere current in AC is the current which can produce same quantity of heat per second as the direct current can produce in 1 Ω resistance. **[1½]**

Q. 3. A coil of 0.01 Henry inductance and 1 Ω resistance is connected to 200 volt, 50 Hz ac supply. Find the impedance of the circuit and time lag between maximum alternating voltage and current.

[NCERT Exemp. Q. 7.23, Page 44]

Ans. Given that

$R = 1\ \Omega$, $L = 0.01$ H, $V = 200$ V, $\nu = 50$ Hz.



[½]

Impedance of the circuit is given as :

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + (2\pi\upsilon L)^2}$$

$$= \sqrt{(1)^2 + (2 \times 3.14 \times 50 \times 0.01)^2}$$

$$= \sqrt{1 + 9.86} = \sqrt{10.86}$$

$$= 3.3\ \Omega.$$ **[½]**

$\therefore$ For phase angle, $\tan\phi = Z/R$

$$\tan\phi = \frac{X_L}{R} = \frac{2\pi\nu L}{R} = \frac{2 \times 3.14 \times 50 \times 0.01}{1}$$

$$\tan\phi = 3.14$$

$$\phi = \tan^{-1} 3.14 = 72°$$ **[½]**

Phase difference,

$$\phi = \frac{72 \times \pi}{180^o} \text{rad}$$ **[½]**

$\phi = 1.20$ radian.

Time lag between alternating voltage and current,

$\phi = \omega t$

$$t = \frac{\phi}{\omega} = \frac{72 \times \pi}{180 \times 2 \times \pi \times 50} = \frac{1}{250} \text{ sec}$$ **[1]**

Q. 4. Explain why the reactance provided by a capacitor to an alternating current decreases with increasing frequency. [NCERT Exemp. Q. 7.25, Page 45]

Ans. When AC current is applied across a capacitor plate, the plates of the capacitor get charge and discharge alternately. Current through the capacitor is a result of this charging and discharging by AC voltage or current. **[1]**

A capacitor does not allow a direct current (having zero frequency) to pass through it. But as frequency of current increases, capacitor will pass more current through it as on increasing frequency, the charging and discharging happens at fast rate. It implies that the reactance offered by capacitor decreases on increasing frequency.

So the reactance of capacitor can be written as

$$X_C = \frac{1}{\omega C}$$ **[2]**

Q. 5. Explain why the reactance offered by an inductor increases with increasing frequency of an alternating voltage.

[NCERT Exemp. Q. 7.26, Page 45]

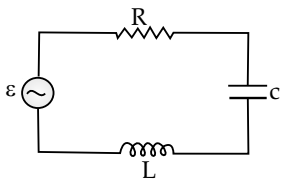
Ans. According to Lenz's law, when current in an inductor change, the direction of induced emf will oppose the change in current in inductor. The magnetic flux will be in opposite direction with the magnetic flux produced by changing emf or current in coil and vice-versa, $X_C = \frac{1}{2\pi\upsilon C}$ **[1]**

Since the induced emf is directly proportional to the rate of change of current, so an inductor produces greater reactance to flow of current through it. If direct current passes through an inductor the reactance produced by inductor is zero. So on applying an AC current, its reactance increases with increasing the frequency of AC. **[1½]**

So the reactance is directly proportional to frequency or reactance of inductor is

$X_L = 2\pi\nu L = \nu L$. **[½]**

Q. 6. Figure shows a series *LCR* circuit connected to a variable frequency 230 V source. $L = 5.0$ H, $C = 80\ \mu$F, $R = 40\ \Omega$.



(a) Determine the source frequency which drives the circuit in resonance.

(b) Obtain the impedance of the circuit and the amplitude of current at the resonating frequency.

(c) Determine the rms potential drops across the three elements of the circuit. Show that the potential drop across the *LC* combination is zero at the resonating frequency.

[NCERT Ex. Q. 7.11, Page 266]

Ans. Given that

$V = 230$ V

$L = 5$ H

$C = 80\ \mu$F

$R = 40$ Ohms

(a) Source frequency at resonance $= \frac{1}{(LC)^{1/2}}$

Putting respective values, the frequency is 50 rad/s. **[1]**

(b) At resonance,

Impedance (Z) = Resistance (R)

$\Rightarrow Z = R = 40$ Ohms

Now rms value of current, $I = \frac{V}{R}$

Or $I = \frac{230}{40}$

Hence, $I = 5.75$ A

Amplitude of this value of current

$= 1.414 \times I = 1.414 \times 5.75 = 8.13$ A. **[1]**

(c) Now taking into consideration the rms potential drops :

Across Resistance, $V_R = IR = 5.75 \times 40 = 230$ V.

Across Capacitance, $V_C = IX_C = 1437.5$ V.

Across Inductance, $V_L = IX_L = 5.75 \times 50 \times 5$

$= 1437.5$ V.

Across LC combination, $V_{LC} = I(X_L - X_C) = 0$

(at resonating frequency) **[1]**

Q. 7. Obtain the resonant frequency and *Q*-factor of a series *LCR* circuit with $L = 3.0$ H, $C = 27\ \mu$F, and $R = 7.4\ \Omega$. It is desired to improve the sharpness of the resonance of the circuit by reducing its 'full width at half maximum' by a factor of 2. Suggest a suitable way. [NCERT Add. Ex. Q. 7.21, Page 268]

Ans. Given that

Inductance, $L = 3.0$ H

Capacitance, $C = 27\ \mu\text{F} = 27 \times 10^{-6}$ F

Resistance, $R = 7.4\ \Omega$

At resonance, angular frequency of the source for the given *LCR* series circuit is given as :

$$\omega_r = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{3 \times 27 \times 10^{-6}}} = \frac{10^3}{9} = 111.11 \text{ rad s}^{-1}$$ **[1]**

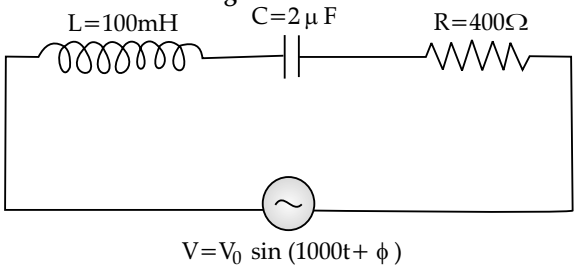
Q-factor of the series :

$$Q = \frac{\omega_r L}{R}$$

$$= \frac{111.11 \times 3}{7.4} = 45.0446$$ **[1]**

To improve the sharpness of the resonance by reducing its 'full width at half maximum' by a factor of 2 without changing ω_r, we need to reduce R to half, *i.e* Resistance $= \frac{R}{2} = \frac{7.4}{2} = 3.7\Omega$. **[1]**

Q. 8. (i) Find the value of the phase difference between the current and the voltage in the series *L*-*C*-R circuit shown below. Which one leads in phase : current or voltage?



(ii) Without making any other change, find the value of the additional capacitor C_1 to be connected in parallel with the capacitor *C*, in order to make the power factor of the circuit unity.

[CBSE Board, Delhi Region, 2017]

Ans. Given that

$L = 100$ mH $= 100 \times 10^{-3}$ H,

$C = 2\ \mu\text{F} = 2 \times 10^{-6}$ F, $\omega = 1000$

(i) For phase difference,

$$\tan\phi = \frac{\left(\omega L - \frac{1}{\omega C}\right)}{R}$$ **[½]**

[ϕ is phase difference between current and voltage.]

$$\omega = 1000 \Rightarrow \omega L = 1000 \times 100 \times 10^{-3} = 100\ \Omega$$

$$\frac{1}{\omega} = \frac{1}{1000 \times 2 \times 10^{-6}} = \frac{1}{2 \times 10^{-3}} = 500\ \Omega$$

$$\Rightarrow \tan\phi = \frac{100 - 500}{400} = \frac{-400}{400} = -1$$

$$\Rightarrow \phi = \tan^{-1}(-1)$$

$$\Rightarrow \phi = 135°$$

Since, $\omega L < \frac{1}{\omega C}$ or $X_L < X_C$

Therefore, current is leading in phase by a phase angle 135°. **[1]**

(ii) For unit power factor,

$\cos\phi = 1$

$$\Rightarrow \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C'}\right)^2}} = 1$$ **[½]**

Where, C' is the total capacitance.

$$\Rightarrow R^2 + \left(\omega L - \frac{1}{\omega C'}\right)^2 = R^2$$

$$\Rightarrow \omega L = \frac{1}{\omega C'}$$

$$\Rightarrow \omega L = 100 = \frac{1}{\omega C} = \frac{1}{1000} C'$$

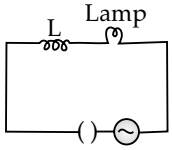
$$C' = \frac{1}{10^s} = 10^{-5} = 10\mu\text{F}$$

Additional capacitance required in parallel series :

$= 10\ \mu\text{F} - 2\ \mu\text{F} = 8\ \mu\text{F}$ **[1]**

Q. 9. (i) When an AC source is connected to an ideal inductor show that the average power supplied by the source over a complete cycle is zero.

(ii) A lamp is connected in series with an inductor and an AC source. What happens to the brightness of the lamp when the key is plugged in and an iron rod is inserted inside the inductor? Explain.



[CBSE Board, All India Region, 2016]

Ans. **(i)** The average power supplied by the source over a complete cycle is given as :

$P = VI \cos\phi$

where, $\cos\phi$ is called the power factor.

For pure inductive circuit, the phase difference between current and voltage is $\frac{\pi}{2}$.

$$\Rightarrow \cos\phi = \cos\frac{\pi}{2} = 0$$

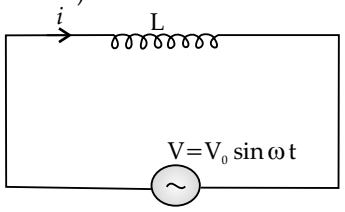
Therefore, the average power dissipated is zero. **[2]**

(ii) If an iron rod is inserted in the inductor, then the value of inductance L increases. As such, the current through the bulb will decrease, thus, decreasing the brightness of the bulb. **[1]**

Q. 10. A source of ac voltage $V = V_0 \sin \omega t$, is connected across a pure inductor of inductance L. Derive the expressions for the instantaneous current in the circuit. Show that average power dissipated in the circuit is zero.

[CBSE Board, Foreign Scheme, 2017]

Ans. Figure below shows the circuit in which voltage source $V = V_0 \sin\omega t$ eq. (1) is applied to pure inductor (zero resistance) coil of inductance L.



[½]

Figure : A.C. Circuit containing inductor of inductance L

As the current through the inductor varies and opposing induced emf is generated in it and is given by, $e = -L\frac{di}{dt}$

From Kirchhoff's loop rule :

$$V_0 \sin\omega t - L\frac{di}{dt} = 0 \quad \textbf{(1)}$$

Or,

$$di = \frac{V_0}{L}\sin\omega t\, dt$$

Integrating on both the sides, we get

$$i = \frac{-V_0}{\omega L}\cos\omega t + C$$

[½]

Where C is the constant of integration. This integration constant has dimensions of current and is independent of time. Since source has an emf which oscillates symmetrically about zero, the current it sustains also oscillates symmetrically about zero, so there is no time independent component of current that exists. Thus, constant $C = 0$.

$$i = \frac{-V_0}{\omega L}\cos\omega t$$

$$= \frac{V_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

$$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right) \quad ...(2)$$

where

$$i_0 = \frac{V_0}{\omega L} \quad ...(3)$$

[½]

is the peak value of current.

From instantaneous values of current and voltage (equations 1 and 2) we see that in pure inductive circuit the current lags behind emf by a phase angle of $\frac{\pi}{2}$.

This phase relationship is graphically shown below in the figure :

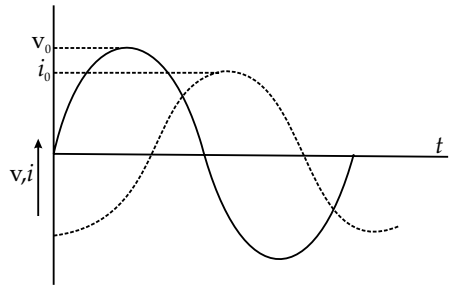


Figure (a)

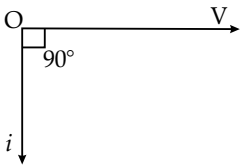


Figure (b) **[½]**

Figure (a) Sinusoidal representation

Figure (b) Phasor diagram representation

From equation (3) peak value of current in the coil is

$$i_0 = \frac{V_0}{\omega L} \text{ or } V_0 = (\omega L)\, i_0$$

Comparing it with the Ohm's law we find product ωL has dimension of resistance and it can be represented by $X_L = \omega L$, where X_L is known as reactance of the coil which represents the effective opposition of the coil to the flow of alternating current. $X_L = \omega L$ is zero for DC for which $\omega = 0$ and increases as the frequency of current increases.

Power Dissipated : Pure inductor refers that it has no internal resistance. If internal resistance in inductor is zero, so voltage and current will always be 90° out of phase with each other. You can see it in the formula :

$S = P + jQ$

In that case S is the complex power, P is the active power which is the average power, and Q is the reactive power, that means, the power consumed by the reactive elements. In each case :

$P = VI\cos\phi$

$Q = VI\sin\phi$

where ϕ is equal to the phase difference between current and voltage, so, as you can see when $\phi = 90°$, the active power or average power equals zero. **[1]**

Long Answer Type Questions (5 marks each)

Q. 1. An electrical device draws 2 kW power from AC mains [voltage 223 V(rms) = $\sqrt{50,000}$ V]. The current differs (lags) in phase by $\phi\left(\tan\phi = \frac{-3}{4}\right)$ as compared to voltage. Find :

(i) R

(ii) $X_C - X_L$, and

(iii) I_M.

Another device has twice the values for R, X_C and X_L. How are the answers affected?

[NCERT Exemp. Q. 7.27, Page 45]

Ans. $P = 2{,}000$ W

Current lags the voltage so

$V^2 = 50{,}000$ V $\quad X_C > X_L$

$$\tan\phi = \frac{-3}{4}$$

$$I_m = I_0$$

$$P = \frac{V^2}{Z}$$

$$2{,}000 = \frac{50{,}000}{Z}$$

$$Z = \frac{50{,}000}{2{,}000} = 25\ \Omega$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$ **[1]**

(i) $R^2 + (X_C - X_L)^2 = 25^2$ …(I)

$$R^2 + (X_C - X_L)^2 = 625$$

$$\tan\phi = \frac{-3}{4}$$

$$\frac{X_C - X_L}{R} = \frac{-3}{4}$$

$$X_C - X_L = \frac{-3}{4}R$$

$$(X_C - X_L)^2 = \frac{9}{16}R^2$$ …(II) **[1]**

Put the value of $(X_C - X_L)^2$ in (I)

$$R^2 + \frac{9}{16}R^2 = 625$$

$$\frac{25}{16}R^2 = 625$$

$$R^2 = 16 \times 25$$

$$R = 4 \times 5 = 20\ \Omega$$

(ii) $$\frac{X_C - X_L}{R} = \frac{-3}{4}$$

$$X_C - X_L = \frac{-3}{4} \times 20$$

$$X_C - X_L = -15\ \Omega$$ **[1]**

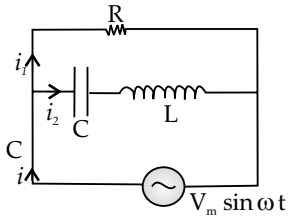
(iii) $$I_{rms} = \frac{V}{Z} = \frac{223}{25} \cong 9\ A$$

$$I_0 = \sqrt{2} I_{rms} = \sqrt{2} \times 9$$

$I_0 = 12.6$ A **[1]**

As (i) R, X_C, X_L all are double tan $\tan\phi = \frac{X_L - X_C}{R}$ will remain same. (ii) Z will become double then $I = \frac{V}{Z}$ become half as value of V does not change. As I become half $P = VI$ will become again half as voltage remains same. **[1]**

Q. 2. Consider the *LCR* circuit shown in figure. Find the net current i and the phase of i. Show that $I = \frac{V}{Z}$. Find the impedance Z for this circuit.



[NCERT Exemp. Q. 7.29, Page 45]

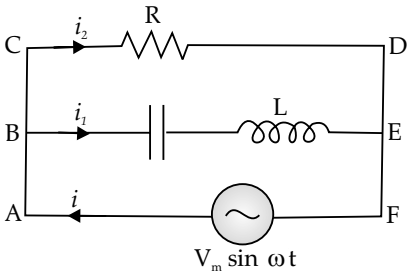
Ans. Total current i from the source $V_m \sin \omega t$ is divided at B in two part, i_1 through capacitor and inductor and part i_2 through resistance.

Potential across R = Potential of source

P.D. across $R = V_m \sin \omega t$

$i_2 R = V_m \sin \omega t$

$$i_2 = \frac{V_m \sin \omega t}{R}$$ ...(I) **[½]**



[½]

q_1 is charge on the capacitor at any time t, then for series combination of C, L, applying Kirchhoff's voltage law in loop $ABEFA$,

$V_C + V_L = V_M \sin \omega t$

$$\frac{q_1}{C} + L\frac{di}{dt} = V_m \sin\omega t$$

$$\frac{q_1}{C} + L\frac{d^2 q_1}{dt^2} = V_m \sin \omega t$$ …II

Let $q_1 = q_m \sin(\omega t + \phi)$ …III

$$i_1 = \frac{dq_1}{dt} = q_m \omega \cos(\omega t + \phi)$$ …IV

$$\frac{d^2 q_1}{dt^2} = -q_m \omega^2 \sin(\omega t + \phi)$$ …V **[1]**

Substitute the values of equations (*III*) and (*V*) in equation (*II*)

$$\frac{q_m \sin(\omega t+\phi)}{C} - Lq_m\omega^2 \sin(\omega t+\phi) = V_m \sin\omega t$$

$$q_m \sin(\omega t+\phi)\left[\frac{1}{C} - L\omega^2\right] = V_m \sin\omega t$$

at $\phi = 0$,

$$q_m\left[\frac{1}{C} - L\omega^2\right]\sin\omega t = V_n \sin\omega t$$

$$q_m\left[\frac{1}{C} - L\omega^2\right] = V_m$$

$$q_m = \frac{V_m}{\omega\left[\frac{1}{C\omega} - L\omega\right]}$$

…(VI)

[½]

Appling Kirchhoff's junction rule at junction *B*, $i = i_2 + i_1$ using relation *I* and *IV*,

$$i = \frac{V_m \sin\omega t}{R} + q_m \omega\cos(\omega t+\phi) \qquad \textbf{[½]}$$

Now using relation VI for q_m and at $\phi = 0$

$$i = \left[\frac{V_m \sin\omega t}{R} + \frac{V_m\omega\cos\omega t}{\omega\left[\frac{1}{\omega C} - \omega L\right]}\right]$$

$$i = \frac{V_m}{R}\sin\omega t + \frac{V_m}{\left(\frac{1}{\omega C} - \omega L\right)}\cos\omega t$$

Let

$$A = \frac{V_m}{R} = C.\cos\phi \qquad \text{...(VII)}$$

$$B = \frac{V_m}{\frac{1}{\omega C} - \omega L} = C.\sin\phi \qquad \text{…(VIII)}$$

$i = C\cos\phi \sin\omega t + C\sin\phi\cos\omega t$

$= C[\cos\phi \sin\omega t + \sin\phi\cos\omega t]$

$i = C\sin(\omega t + \phi)$

Squaring and adding (*VII*), (*VIII*)

$A^2 + B^2 = C^2\cos^2\phi + C^2\sin^2\phi$

$= C^2[\cos^2\phi + \sin^2\phi]$

$A^2 + B^2 = C^2$

or $C = \sqrt{A^2 + B^2}$ **[1]**

$$\phi = \tan^{-1}\frac{B}{A} = \tan^{-1}\frac{\frac{V_m}{\frac{1}{\omega C} - \omega L}}{\frac{V_m}{R}}$$

$$\therefore \tan\phi = \frac{R}{\left(\frac{1}{\omega C} - \omega L\right)}$$

$$\because C^2 = A^2 + B^2 = \frac{V_m^2}{R^2} + \frac{V_m^2}{\left(\frac{1}{\omega C} - \omega L\right)^2}$$

$$C = \left[\frac{V_m^2}{R^2} + \frac{V_m^2}{\left(\frac{1}{\omega C} - \omega L\right)^2}\right]^{\frac{1}{2}} \qquad \textbf{[1]}$$

$$\therefore i = \left[\frac{V_m^2}{R^2} + \frac{V_m^2}{\left(\frac{1}{\omega C} - \omega L\right)^2}\right]^{\frac{1}{2}} \sin(\omega t+\phi)$$

$$i = V_m\left[\frac{1}{R^2} + \frac{1}{\left(\frac{1}{\omega C} - \omega L\right)^2}\right]^{\frac{1}{2}} \sin(\omega t+\phi) \qquad \text{…(IX)}$$

And

$$\phi = \tan^{-1}\frac{R}{\left(\frac{1}{\omega C} - \omega L\right)}$$

$$\because I = \frac{V}{R} \text{ or } i = \frac{V}{Z}$$

For ac

$$i = \frac{V}{Z}\sin(\omega t+\phi) \qquad \text{…(X)}$$

Comparing (IX) and (X)

So

$$\frac{1}{Z} = \left[\frac{1}{R^2} + \frac{1}{\left(\frac{1}{\omega C} - \omega L\right)^2}\right]^{\frac{1}{2}} \qquad \textbf{[1]}$$

This is the impedance Z for the circuit.

Q. 3. For an *LCR* circuit driven at frequency ω, the equation reads

$$L\frac{di}{dt} + Ri + \frac{q}{c} = V_i = V_m \sin\omega t.$$

(a) Multiply the equation by *i* and simplify where possible.

(b) Interpret each term physical.

(c) Cast the equation in the form of a conservation of energy statement.

(d) Integrate the equation over one cycle to find that the phase difference between v and *i* must be acute. [NCERT Exemp. Q. 7.30, Page 45]

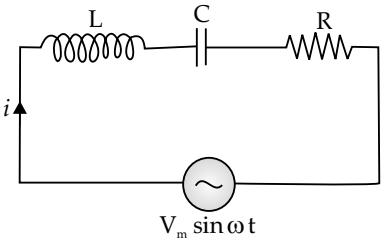
Ans. Consider *L*-*C*-*R* series circuit with AC supply,

$V = V_m \sin\omega t$

Applying voltage Kirchhoff's law over the circuit,

$\therefore V_L + V_C + V_R = V_m \sin\omega t$

$$L\frac{di}{dt} + Ri + \frac{q}{c} = V_i = V_m \sin\omega t. \qquad \textbf{[½]}$$



[½]

(a) Multiply the above equation by i in both the sides.

$$iL\frac{di}{dt}+Ri^2+\frac{q}{c}i=V_i=iV_m\sin\omega t$$

Multiply above equation $\frac{1}{2}$ on both sides,

$$\frac{1}{2}Li\frac{di}{dt}+i\frac{q}{2C}+\frac{i^2R}{2}=\frac{1}{2}V_m i\ \sin\omega t \qquad \left(\because i=\frac{dq}{dt}\right)$$

$$\frac{d\left(\frac{1}{2}Li^2\right)}{dt}+i\frac{1}{2C}\frac{dq^2}{dt}+\frac{i^2R}{2}=\frac{1}{2}V_m i\ \sin\omega t$$

…(I)

[½]

(b) **(i)** $\frac{d\left(\frac{1}{2}Li^2\right)}{dt}$ represents the rate of change of potential energy in inductance L. [½]

(ii) $\frac{d}{dt}\frac{q^2}{2C}$ represents energy stored in dt time in the capacitor. [½]

(iii) i^2R represents Joules heating loss. [½]

(iv) $\frac{1}{2}V_m i\ \sin\omega t$ is the rate at which driving force pours in energy. It goes into Ohmic loss and increase of stored energy in capacitor and inductor. [½]

(c) Here equation (I) is in the form of conservation of energy statement. [½]

(d) Integrating equation (I) both sides with respect to dt over a cycle we get

$$\int_0^T\frac{d\left(\frac{1}{2}Li^2\right)}{dt}dt+\int_0^T\frac{dq^2}{dt2C}dt+\int_0^T\frac{i^2R}{2}dt=\frac{1}{2}\int_0^T V_m i\ \sin\omega t dt$$

$$\int_0^T\frac{d}{dt}\left[\frac{1}{2}Li^2+\frac{q^2}{2C}\right]dt+\frac{1}{2}\int_0^T i^2Rdt=\frac{1}{2}\int_0^T Vi\ dt$$

$$[\therefore V=V_m\sin\omega t]$$

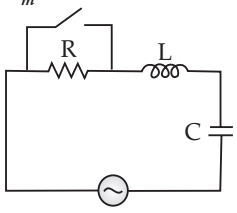
$$0+\frac{1}{2}i^2RT=\frac{1}{2}\int_0^T Vi\ dt$$

$$i^2RT=\int_0^T Vi\ dt$$

As i^2RT is $+ve$. [$\therefore i^2$, R and T can never be negative]

So, $\int_0^T Vi\,dt$ is positive which is only possible if phase difference is constant and the angle is acute. **[1]**

Q. 4. In the *LCR* circuit shown in figure, the ac driving voltage is $V = V_m \sin \omega t$.



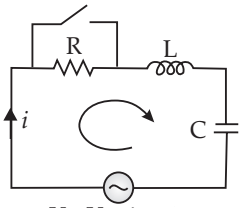
(i) Write down the equation of motion for q (t).

(ii) At $t = t_0$, the voltage source stops and R is short circuited. Now write down how much energy is stored in each of L and C.

(iii) Describe subsequent motion of charges.

[NCERT Exemp. Q. 7.31, Page 46]

Ans. **(i)** Consider series LCR circuit and tapping key K to short circuit R. Let i be the current in circuit.



$V=V_m\sin\omega t$ [½]

Then by Kirchhoff's voltage law, when key K is open,

$$V_R+V_L+V_C=V_m\sin\omega t$$

$$iR+L\frac{di(t)}{dt}+\frac{q(t)}{C}-V_m\ \sin\omega t=0$$

$$[\because i(t)=i=I_m\ \sin[\omega t+\phi]]\ [½]$$

$\Rightarrow$ As charge $q(t)$ changes in circuit with time in AC, Then

$$i=\frac{dq(t)}{dt}$$

$$\frac{di}{dt}=\frac{d^2q(t)}{dt^2}$$

(Differentiating again)

$$R\frac{dq(t)}{dt}+L\frac{d^2q(t)}{dt^2}+\frac{q(t)}{C}=V_m\sin\omega t$$

$$L\frac{d^2q(t)}{dt}+R\frac{dq(t)}{dt}+\frac{q(t)}{C}=V_m\sin\omega t$$

[½]

This is the equation for variation of motion of charge with respect to time.

(ii) Let time dependent charge in circuit is at phase angle with voltage then $q=q_m\cos(\omega t+\phi)$.

$$i=\frac{dq}{dt}=\omega q_m\ \sin(\omega t+\phi) \qquad \text{…(I)}$$

$$i_m=\frac{V_m}{Z}=\frac{V_m}{\sqrt{R^2+(X_C-X_L)^2}} \qquad \text{…(II)}$$

$$\tan\phi=\frac{(X_C-X_L)}{R}$$

[½]

At $t = t_0$, R is short circuited, then energy stored in L and C, when K is closed will be,

$$U_L=\frac{1}{2}Li^2 \qquad \text{…(III)}$$

At $t = t_0$,

$$i=i_m\sin(\omega t_0+\phi) \qquad \text{…(IV)}$$

From (II),

$$i=\frac{V_m}{\sqrt{R^2+(X_C-X_L)^2}}\ \sin(\omega t_0+\phi) \qquad \text{…(V)}$$

$$\therefore\ U_L=\frac{1}{2}L\left[\frac{V_m}{\sqrt{R^2+(X_C-X_L)}}\right]^2\ \sin^2(\omega t_0+\phi)$$

$$U_C=\frac{q^2}{2C}=\frac{1}{2C}[q_m^2\ \cos^2(\omega t_0+\phi)] \qquad [1]$$

Comparing (IV) and (I), $I_m=q_m\omega$

$$\therefore q_m = \frac{i_m}{\omega}$$

$$\therefore U_C = \frac{1}{2C}\cdot\frac{i_m^2}{\omega^2}\cos^2(\omega t_0+\phi) = \frac{i_m^2}{2C\omega^2}\cos^2(\omega t_0+\phi)$$ [½]

Using equation (*II*),

$$\therefore U_C = \frac{1}{2C\omega^2}\left[\frac{V_m^2}{R^2+(X_C-X_L)^2}\right]^2\cos^2(\omega t_0+\phi)$$ [½]

(iii) When R is short circuited, the circuit becomes *L*-*C* oscillator. The capacitor will go on discharging and all energy will transfer to *L*, and back and forth. Hence there is oscillation of energy from electrostatic to magnetic and vice versa. **[1]**

Q. 5. A coil of inductance 0.50 H and resistance 100 Ω is connected to a 240 V, 50 Hz ac supply.

(a) What is the maximum current in the coil?

(b) What is the time lag between the maximum voltage and the maximum current?

[NCERT Add. Ex. Q. 7.13, Page 267]

Ans. Inductance of the inductor, $L = 0.50$ H

Resistance of the resistor, $R = 100\ \Omega$

Potential of the supply voltage, $V = 240$ V

Frequency of the supply, $\nu = 50$ Hz

(a) Peak voltage is given as :

$$V_0 = \sqrt{2}\,V$$

$$= \sqrt{2}\times 240 = 339.41\ V$$ [½]

Angular frequency of the supply, $\omega = 2\pi\nu$

$= 2\pi \times 50 = 100\,\pi$ rad/s [½]

Maximum current in the circuit is given as :

$$I_0 = \frac{V_0}{\sqrt{R^2+\omega^2L^2}}$$

$$= \frac{339.41}{\sqrt{(100)^2+(100\pi)^2(0.50)^2}} = 1.82\ \text{A}$$ [½]

(b) Equation for voltage is given as :

$V = V_0 \cos \omega t$ [½]

Equation for current is given as :

$I = I_0 \cos(\omega t - \phi)$ [½]

Where,

ϕ = Phase difference between voltage and current at time, $t = 0$.

$V = V_0$ (voltage is maximum)

For $\omega t - \phi = 0$, *i.e.*, at time, $t = \frac{\phi}{\omega}$ [½]

$I = I_0$ (current is maximum)

Hence, the time lag between maximum voltage and maximum current is $\frac{\phi}{\omega}$. [½]

Now, phase angle ϕ is given by the relation,

$$\tan\phi = \frac{\omega L}{R} = \frac{2\pi\times 50\times 0.5}{100} = 1.57$$

$$\phi = 57.5° = \frac{57.5\pi}{180}\text{rad}$$

$$\omega t = \frac{57.5\pi}{180}$$

$$t = \frac{57.5}{180\times 2\pi\times 50}$$

$$= 3.19\times 10^{-3}\ \text{s}$$

$$= 3.2\ \text{ms}$$ **[1]**

Hence, the time lag between maximum voltage and maximum current is 3.2 ms. [½]

Q. 6. Obtain the answers (a) to (b) in Exercise 7.13 if the circuit is connected to a high frequency supply (240 V, 10 kHz). Hence, explain the statement that at very high frequency, an inductor in a circuit nearly amounts to an open circuit. How does an inductor behave in a dc circuit after the steady state? [NCERT Add. Ex. Q. 7.14, Page 267]

Ans. Inductance of the inductor, $L = 0.5$ Hz

Resistance of the resistor, $R = 100\ \Omega$

Potential of the supply voltages, $V = 240$ V

Frequency of the supply, $\nu = 10$ kHz $= 10^4$ Hz

Angular frequency, $\omega = 2\pi\nu = 2\pi \times 10^4$ rad/s

(a) Peak voltage, $V_0 = \sqrt{2}\times V = 240\sqrt{2}\ V$ **[1]**

Maximum current, $I_0 = \frac{V_0}{\sqrt{R^2+\omega^2L^2}}$ [½]

$$= \frac{240\sqrt{2}}{\sqrt{(100)^2+(2\pi\times 10^4)^2\times(0.50)^2}} = 1.1\times 10^{-2}\,\text{A}$$ [½]

(b) For phase difference Φ, we have the relation :

$$\tan\phi = \frac{\omega L}{R}$$

$$= \frac{2\pi\times 10^4\times 0.5}{100} = 100\pi$$

$$\phi = 89.82° = \frac{89.82\pi}{180}\text{rad}$$

$$\omega t = \frac{89.82\pi}{180}$$

$$t = \frac{89.82\pi}{180\times 2\pi\times 10^4} = 25\mu\text{s}$$ **[2]**

It can be observed that I_0 is very small in this case. Hence, at high frequencies, the inductor amounts to an open circuit. [½]

In a dc circuit, after a steady state is achieved, $\omega = 0$. Hence, inductor *L* behaves like a pure conducting object. [½]

Q. 7. A 100 μF capacitor in series with a 40 Ω resistance is connected to a 110 V, 60 Hz supply.

(a) What is the maximum current in the circuit?

(b) What is the time lag between the maximum current and the maximum voltage?

[NCERT Add. Ex. Q. 7.15, Page 267]

Ans. Capacitance of the capacitor, $C = 100\ \mu\text{F} = 100\times 10^{-6}$ F

Resistance of the resistor, $R = 40\ \Omega$

Supply voltage, $V = 110$ V

Frequency of oscillations, $\nu = 60$ Hz

(a) Angular frequency, $\omega = 2\pi\nu = 2\pi\times 60$ rad/s [½]

For a *RC* circuit, we have the relation for impedance as :

$$Z = \sqrt{R^2+\frac{1}{\omega^2C^2}}$$ [½]

Peak voltage, $V_0 = V\sqrt{2} = 110\sqrt{2}$ V [½]

Maximum current is given as :

$$I_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}}$$

$$= \frac{100\sqrt{2}}{\sqrt{(40)^2 + \frac{1}{(120\pi)^2 \times (10^{-4})^2}}}$$

$$= \frac{110\sqrt{2}}{\sqrt{1600 + \frac{10^8}{(120\pi)^2}}} = 3.24 \text{ A}$$

[1]

(b) In a capacitor circuit, the voltage lags behind the current by a phase angle of ϕ. This angle is given by the relation : **[½]**

$$\therefore \quad \tan\phi = \frac{\frac{1}{\omega C}}{R} = \frac{1}{\omega CR} = \frac{1}{120\pi \times 10^{-4} \times 40} = 0.6635$$

$$\phi = \tan^{-1}(0.6635) = 33.56°$$

$$= \frac{33.56\pi}{180} \text{ rad}$$

$$\therefore \text{ Time lag} = \frac{\phi}{\omega} = \frac{33.56\pi}{180 \times 120\pi} = 1.55 \times 10^{-3} \text{ s} = 1.55 \text{ ms}$$

[1+1]

Hence, the time lag between maximum current and maximum voltage is 1.55 ms.

Q. 8. Obtain the answers to (a) and (b) in Exercise 7.15 if the circuit is connected to a 110 V, 12 kHz supply? Hence, explain the statement that a capacitor is a conductor at very high frequencies. Compare this behaviour with that of a capacitor in a dc circuit after the steady state.

[NCERT Add. Ex. Q. 7.16, Page 267]

Ans. Capacitance of the capacitor, $C = 100\ \mu\text{F} = 100 \times 10^{-6}$ F

Resistance of the resistor, $R = 40\ \Omega$

Supply voltage, $V = 110$ V

Frequency of the supply, $\nu = 12$ kHz $= 12 \times 10^3$ Hz

Angular Frequency,

$\omega = 2\pi\nu = 2 \times \pi \times 12 \times 10^3 = 24\pi \times 10^3$ rad/s **[½]**

Peak voltage, $V_0 = V\sqrt{2} = 110\sqrt{2}\ V$ **[½]**

Maximum current, $I_0 = \frac{V_0}{\sqrt{\left(R^2 + \frac{1}{\omega^2 C^2}\right)}}$ **[½]**

$$= \frac{110\sqrt{2}}{\sqrt{(40)^2 + \frac{1}{(24\pi \times 10^3 \times 100 \times 10^{-6})^2}}}$$

$$= \frac{110\sqrt{2}}{\sqrt{1600 + \left(\frac{10}{24\pi}\right)^2}} = 3.9 \text{ A}$$

[½]

For an *RC* circuit, the voltage lags behind the current by a phase angle of ϕ given as :

$$\tan\phi = \frac{\frac{1}{\omega C}}{R} = \frac{1}{\omega CR} = \frac{1}{24\pi \times 10^3 \times 100 \times 10^{-4} \times 40}$$

$$\tan\phi = \frac{1}{96\pi}$$

$$\therefore \quad \phi = 0.2°$$

$$\therefore \quad \phi = 0.2°$$

$$= \frac{0.2\pi}{180} \text{rad}$$

$$\therefore \text{ Time lag} = \frac{\phi}{\omega} = \frac{0.2\pi}{180 \times 24\pi \times 10^3} = 0.04\ \mu\text{s}$$ **[2]**

Hence, ϕ tends to become zero at high frequencies. At a high frequency, capacitor *C* acts as a conductor. In a dc circuit, after the steady state is achieved, $\omega = 0$. Hence, capacitor *C* amounts to an open circuit. **[1]**

Q. 9. Keeping the source frequency equal to the resonating frequency of the series *LCR* circuit, if the three elements, *L*, *C* and R are arranged in parallel, show that the total current in the parallel *LCR* circuit is minimum, at this frequency. Obtain the current rms value in each branch of the circuit for the elements and source specified in Exercise 7.11 for this frequency.

[NCERT Add. Ex. Q. 7.17, Page 267]

Ans. An inductor (*L*), a capacitor (*C*), and a resistor (*R*) is connected in parallel with each other in a circuit where,

$L = 5.0$ H

$C = 80\ \mu\text{F} = 80 \times 10^{-6}$ F

$R = 40\ \Omega$

Potential of the voltage source, $V = 230$ V

Impedance (*Z*) of the given parallel *LCR* circuit is given as :

$$\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2}$$ **[1]**

Where, ω = Angular frequency

At resonance, $\frac{1}{\omega L} - \omega C = 0$

$$\therefore \omega = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = 50 \text{ rad/s}$$

[1]

Hence, the magnitude of *Z* is the maximum at 50 rad/s. As a result, the total current is minimum. rms current flowing through inductor *L* is given as : **[½]**

$$I_L = \frac{V}{\omega L}$$

$$= \frac{230}{50 \times 5} = 0.92 \text{ A}$$ **[½]**

rms current flowing through capacitor *C* is given as :

$$I_C = \frac{V}{\frac{1}{\omega C}} = \omega CV = 50 \times 80 \times 10^{-6} \times 230 = 0.92 \text{ A}$$ **[1]**

rms current flowing through resistor *R* is given as :

$$I_R = \frac{V}{R} = \frac{230}{40} = 5.75 \text{ A}$$ **[1]**

Q. 10. A series *LCR* circuit with $L = 0.12$ H, $C = 480$ nF, $R = 23\ \omega$ is connected to a 230 V variable frequency supply.

(a) What is the source frequency for which current amplitude is maximum? Obtain this maximum value.

(b) What is the source frequency for which average power absorbed by the circuit is maximum? Obtain the value of this maximum power.

(c) For which frequencies of the source is the power transferred to the circuit half the power at resonant frequency? What is the current amplitude at these frequencies?

(d) What is the *Q*-factor of the given circuit?

[NCERT Add. Ex. Q. 7.20, Page 268]

Ans. Inductance, $L = 0.12$ H;
Capacitance, $C = 480$ nF $= 480 \times 10^{-9}$ F;
Resistance, $R = 23\ \Omega$;
Supply voltage, $V = 230$ V;
Peak voltage is given as :

$V_0 = \sqrt{2} \times 230 = 325.22$ V

(a) Current flowing in the circuit is given by the relation,

$$I_0 = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)}}$$

Where, I_0 = maximum at resonance at resonance, we have ω_R, Resonance angular frequency

$$\therefore \omega_R = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.12 \times 480 \times 10^{-9}}} = 4166.67 \text{ rad/s}$$

$\therefore$ Resonant frequency,

$$\omega_0 = \frac{\omega_R}{2\pi} = \frac{4166.67}{2 \times 3.14} = 663.48\ Hz$$

Hence the resonant frequency is 663.48 Hz. And, maximum current

$$(I_0)_{max} = \frac{V_0}{R} = \frac{325.22}{23} = 14.14 \text{ A}$$

[1]

(b) Maximum average power absorbed by the circuit is given as :

$$(P_N)_{max} = \frac{1}{2}(I_0)^2_{max} R = \frac{1}{2} \times (14.14)^2 \times 23 = 2299.3 \text{ W}$$

[1]

(c) The power transferred to the circuit is half the power at resonant frequency.
Frequencies at which power transferred is half,

$= \omega_R \pm \Delta\omega$

$= 2\pi(\nu_R \pm \Delta\nu)$

Where,

$$\Delta\omega = \frac{R}{2L} = \frac{23}{2 \times 0.12} = 95.83 \text{ rad/s}$$

Hence, change in frequency,

$$\Delta\nu = \frac{1}{2\pi}\Delta\omega = \frac{95.83}{2\pi} = 15.26 \text{ Hz}$$

$$\therefore \nu_R + \Delta\nu = 663.48 + 15.26 = 678.74 \text{ Hz}$$

And, $V_R - \Delta V = 663.48 - 15.26 = 648.22$ Hz
Hence, at 648.22 Hz and 678.74 Hz frequencies, the power transferred is half. **[1]**
At these frequencies, current amplitude can be given as :

$$I' = \frac{1}{\sqrt{2}} \times (I_0)_{max} = \frac{14.14}{\sqrt{2}} = 10\ A$$

[1]

(d) *Q*-factor of the given circuit can be obtained using the relation,

$$Q = \frac{\omega_R L}{R} = \frac{4166.67 \times 0.12}{23} = 21.74$$

Hence, the *Q*-factor of the given circuit is 21.74. **[1]**

Q. 11. Answer the following questions :

(a) In any ac circuit, is the applied instantaneous voltage equal to the algebraic sum of the instantaneous voltages across the series elements of the circuit? Is the same true for rms voltage?

(b) A capacitor is used in the primary circuit of an induction coil.

(c) An applied voltage signal consists of a superposition of a dc voltage and an ac voltage of high frequency. The circuit consists of an inductor and a capacitor in series. Show that the dc signal will appear across *C* and the ac signal across *L*.

(d) A choke coil in series with a lamp is connected to a dc line. The lamp is seen to shine brightly. Insertion of an iron core in the choke causes no change in the lamp's brightness. Predict the corresponding observations if the connection is to an ac line.

(e) Why is choke coil needed in the use of fluorescent tubes with ac mains? Why can we not use an ordinary resistor instead of the choke coil?

[NCERT Add. Ex. Q. 7.22, Page 268]

Ans. **(a)** This statement is not right because it is applicable only in that condition in which any ac circuit, the applied voltage is equal to the average sum of the instantaneous voltages across the series elements of the circuit. However, this is not true for rms voltage because voltages across different elements may not be in phase. **[1]**

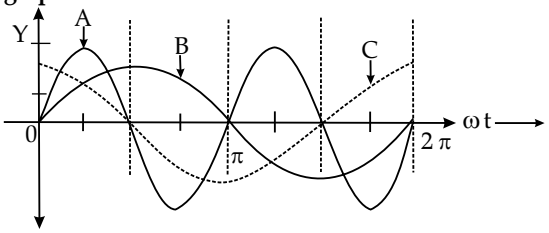
(b) High induced voltage is used to charge the capacitor. A capacitor is used in the primary circuit of an induction coil. This is because when the circuit is broken, a high induced voltage is used to charge the capacitor to avoid sparks. **[1]**

(c) The dc signal will appear across capacitor *C* because for dc signals, the impedance of an inductor (*L*) is negligible while the impedance of a capacitor (*C*) is very high (almost infinite). Hence, a dc signal appears across *C*. For an ac signal of high frequency, the impedance of *L* is high and that of *C* is very low. Hence, an ac signal of high frequency appears across *L*. **[1]**

(d) If an iron core is inserted in the choke coil, which is in series with a lamp connected to the ac line, then the lamp will glow dimly. This is because the choke coil and the iron core increase the impedance of the circuit. **[1]**

(e) A choke coil is needed in the use of fluorescent tubes with ac mains because it reduces the voltage across the tube without wasting much power. An ordinary resistor cannot be used instead of a choke coil for this purpose because it wastes power in the form of heat. **[1]**

Q. 12. A device '*X*' is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph :



(a) Identify the device '*X*'.

(b) Which of the curves *A*, *B* and *C* represent the voltage, current and the power consumed in the circuit? Justify your answer.

(c) How does its impedance vary with frequency of the ac source? Show graphically.

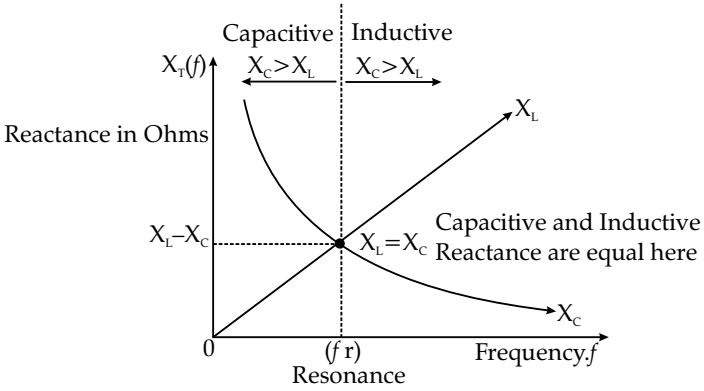
(d) Obtain an expression for the current in the circuit and its phase relation with ac voltage.

[CBSE Board, All India Region, 2017]

Ans. (a) The device '*X*' is a capacitor. **[½]**

(b) Curve *A*: Power consumed in the circuit
Curve *B*: Voltage
Curve *C*: Current
This is because current leads voltage in a capacitive circuit. **[1]**

(c) Impedance: $X_C = \dfrac{1}{2\pi fC}$



Therefore, impedance is directly proportional to frequency. **[2]**

(d) Voltage applied to the circuit is $V = V_0 \sin \omega t$

Due to this voltage, a charge will be produced which will charge the plates of the capacitor with positive and negative charges. If the potential difference across the plates of the capacitor is *V*, then

$$V = \frac{q}{C}$$

$$\therefore \quad q = CV$$

Therefore, the instantaneous value of the current in the circuit is

$$i = \frac{dq}{dt} = \frac{d(\mathrm{CV})}{dt} = \frac{d}{dt}(\mathrm{CV_0 \sin \omega t})$$

$$\therefore \quad i = \mathrm{CV_0}\, \omega \cos \omega t = \frac{V_0}{\dfrac{1}{\omega C}} \sin\left(\omega t + \frac{\pi}{2}\right)$$

Thus the peak value of current is

$$i_0 = \frac{V_0}{\dfrac{1}{\omega C}}$$

$$\therefore \quad i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$$

Hence current leads voltage by $\dfrac{\pi}{2}$. **[1½]**

Q. 13. A device *X* is connected across an ac source of voltage $V=V_0 \sin \omega t$. The current through *X* is given as $I=I_0 \sin (\omega t + \frac{\pi}{2})$.

(a) Identify the device *X* and write the expression for its reactance.

(b) Draw graphs showing variation of voltage and current with time over one cycle of ac, for *X*.

(c) How does the reactance of the device *X* vary with the frequency of the ac? Show this variation graphically.

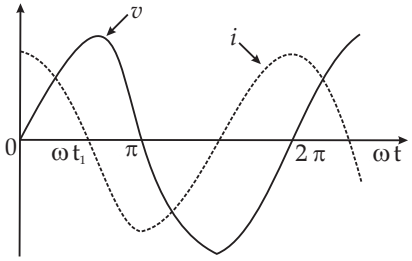
(d) Draw the phasor diagram for the device *X*.

[CBSE Board, Delhi Region, 2018]

Ans. **(a)** According to the question, the current is leading the voltage by 90° degree, so the device *X* is capacitor. **[1]**

The expression for reactance is $X_C = \frac{1}{\omega C} = \frac{1}{2\pi \nu C}$ **[1]**

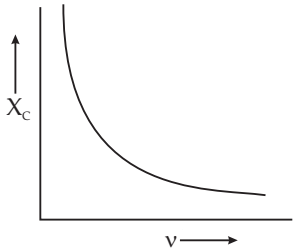
(b) Graph showing variation of voltage and current with time over one cycle of ac :



[1]

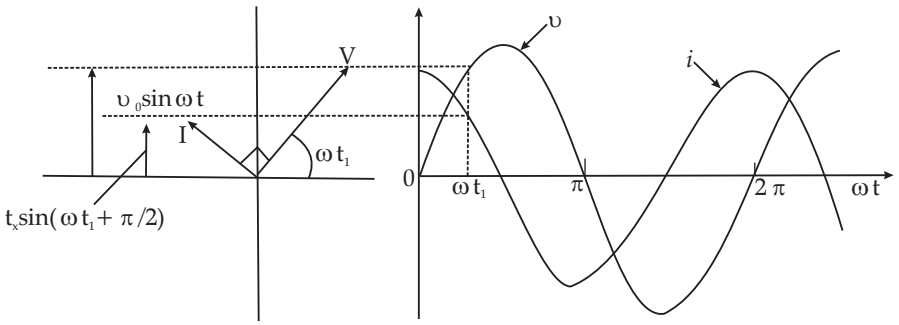
(c) $X_C \propto \frac{1}{\nu}$ **[½]**

The reactance varies inversely with frequency.



[½]

(d) Phasor diagram for the device :



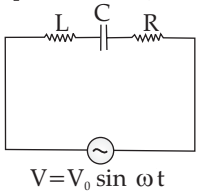
[1½]

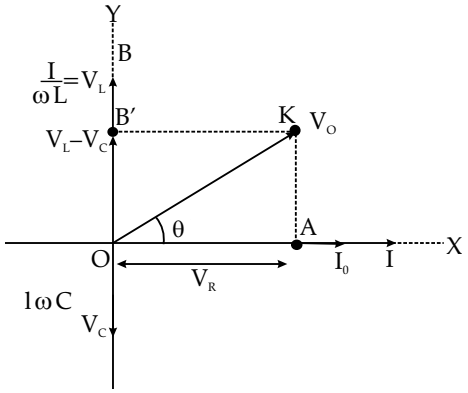
Q. 14. Answer the following :

(i) An a.c. source of voltage is connected to a series combination of *L*, *C* and *R*. Use the phasor diagram to obtain expression for impedance of a circuit and the phase angle between voltage and current. Find the condition when current will be in phase with the voltage. What is the circuit in this condition called?

(ii) In a series *LR* circuit, $X_L = R$ and power factor of the circuit is P_1. When capacitor with capacitance *C* such that $X_L = X_C$ is put in series, the power factor becomes P_2. Calculate $\frac{P_1}{P_2}$.

Ans. **[CBSE Board, Delhi Region, 2016]**





[½]

Voltage of the source is given as $V = V_0 \sin \omega t$

Let current of the source be $I = I_0 \sin \omega t$.

The maximum voltage across R is $\vec{V}_R = \vec{V}_0 R$, represented along OX.

The maximum voltage across L is $\vec{V}_L = \vec{I}_0 X_L$, represented along OY and is 90° ahead of I_0.

The maximum voltage across C is $\vec{V}_C = \vec{I}_0 X_C$, represented along OC and is lagging behind I_0 by 90°.

The voltage across L and C has a phase difference of 180°.

Hence, reactive voltage is $\vec{V}_L = \vec{V}_L$ represented by OB'. **[1]**

The vector sum of $\vec{V}_R$, $\vec{V}_L$ and $\vec{V}_C$, is resultant of OA and OB', represented along OK.

$$OK = V_0 = \sqrt{OA^2 + OB^2}$$

$$\Rightarrow \quad V_0 = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(I_0 R)^2 + (I_0 X_L - V_C)^2}$$

$$\Rightarrow \quad V_0 = I_0 \sqrt{R^2 + (X_L - X_C)^2}$$

[1]

The impedance can be calculated as follows :

$$Z = \frac{V_0}{I_0} = \sqrt{R^2 + (X_L - X_C)^2}$$

When $X_L = X_C$, the voltage and current are in the same phase. In such a situation, the circuit is known as non-inductive circuit. **[½]**

(ii) Given : Power factor P_1 (When $X_L = R$)

New power factor P_2 (when $X_L = X_C$) :

$$P_1 = \frac{R}{Z}$$

$$\Rightarrow \quad P_1 = \frac{R}{\sqrt{R^2 + X^2}} = \frac{R}{\sqrt{2R^2}} = \frac{1}{\sqrt{2}}$$

$$P_2 = \frac{R}{Z}$$

$$\Rightarrow \quad P_2 = \frac{R}{\sqrt{R^2 (X_L - X_C)}} = 1 \qquad \textbf{[1½]}$$

Thus,

$$\frac{P_1}{P_2} = \frac{1}{\sqrt{2}} \qquad \textbf{[½]}$$

Q. 15. A 2 μF capacitor, 100 ω resistor and 8 H inductor are connected in series with an AC source.

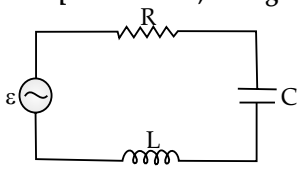
(i) What should be the frequency of the source such that current drawn in the circuit is maximum? What is this frequency called?

(ii) If the peak value of e.m.f. of the source is 200 V, find the maximum current.

(iii) Draw a graph showing variation of amplitude of circuit current with changing frequency of applied voltage in a series *LRC* circuit for two different values of resistance R_1 and R_2 ($R_1 > R_2$).

(iv) Define the term 'Sharpness of Resonance'. Under what condition, does a circuit become more selective? [CBSE Board, Foreign Scheme, 2016]

Ans.



[½]

Figure : Series *L-C-R-* circuit with ac source

(i) The series *RLC* circuit is the phenomenon of resonance. The phenomenon of resonance is common among systems that have a tendency to oscillate at a particular frequency. This frequency is called the system's 'natural frequency'. **[1]**

Given that,

$$R = 100\ \Omega; \quad C = 2\ \mu F; \quad L = 8\ H$$

For L-C-R Circuit,

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

At resonance condition, $X_C = X_L$

$$So, Z = \sqrt{R^2} = \sqrt{100} = 10\ \Omega$$

We know that,

$$i_m = \frac{v_m}{Z} = \frac{200}{10} = 20\ \text{Amp.}$$

Now, for resonance frequency **[1]**

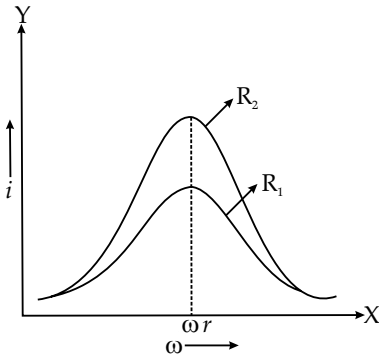
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 2 \times 10^{-6}}}$$

$$= \frac{1}{.004} = 250\ Hz$$

(ii) At resonant frequency, the current amplitude is maximum :

$$i_m = \frac{v_m}{R} = \frac{200}{100} = 2\ \text{Amp.} \qquad \textbf{[½]}$$

(iii) The variation of current with angular frequency for the two resistances R_1 and R_2 is showing in the graph.

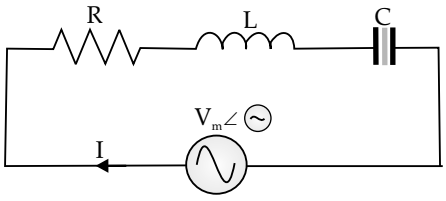


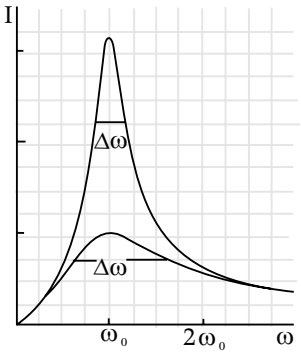
[½]

We can see that the resonance for the resistance R_2 is sharper than resistance R_1 because resistance R_2 is less than resistance R_1. Therefore, at resonance, the value of peak current will rise more abruptly for a lower value of resistance.

(iv) **Sharpness of Resonance :** It is defined as the ratio of the voltage developed across the inductance (L) or capacitance (C) at resonance to the voltage developed across the resistance (R). **[½]**

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$





[½]

If the response of the circuit is more narrowly peaked around the chosen frequency, we say that the circuit has higher "selectivity". *A* "quality factor" *Q*, as described below, is a measure of that selectivity, and we speak of a circuit having a "high *Q*" if it is more narrowly selective. **[½]**

TOPIC-2
Power in AC Circuit

Quick Review

- The idea of a transformer was introduced back in 1880. Then in the year 1950, about 70 years later, there was an introduction of 400 kV electrical power transformer in high-voltage electrical power system. Then in early 1970s, a large unit rating of 1,100 MVA was produced, before 800+ kV class transformers started to be manufactured 10 years later in 1980.
- Modern ac transmission technology requires the voltage to be higher such as 11 kV, 33 kV, 133 kV and 232 kV. This is required to minimise the transmission loss and making the transmission line lighter, hence less copper is needed. On the other hand, consumer end voltage level is 400, 200, 120, etc.
- Rectifier transformer or HVDC (High voltage DC transformer) – This types of power transformer is used in high voltage DC network where there is AC network. HVDC transformer is the combination of (a) regular transformer + rectifier circuit (to convert AC to DC) in primary or secondary; or (b) regular transformer + inverter circuit (to convert DC to AC) in primary or secondary.
- Traction type transformer- Transformer used in the electric rail services to supply electric power to the rail or tram from the overhead electrical power cable.
- An LC circuit, also called a resonant circuit, tank circuit, or tuned circuit. This circuit can act as an electrical resonator, an electrical analogue of a tuning fork, storing energy oscillating at the circuit's resonant frequency.
- Circuits with two energy storage elements (capacitors or inductors) are called second-order systems. In second-order systems, the voltages and currents rock back-and-forth, or oscillate. This article is an intuitive description of how this happens.

TIPS...

- *Parallel operation of transformers with different ratings. The transformers' reactance and resistance ratio are the same.*
- *The transformers must have the same transformation ratios; E_{pri}/E_{sec}, i.e. for both primary and secondary voltages.*
- *The transformers must have the same % regulation, i.e. the reduction in the voltage level of both transformers in the primary and secondary are the same when they are loaded.*
- *Utilising a 3phase transformer for 1phase service.*

TRICKS...

- *Transformers are rated in volt-ampere (VA) or kilovolt-ampere (kVA).*
- *Parallel the two windings of the transformer - say Phase-A and Phase-B - taking care that polarity markings are observed; parallel their corresponding secondary's in same manner.*
- *The maximum load that can be utilized for 1ph service is $\frac{2}{3} \times 100\% = 66.67\%$*

Know the Links

- https://www.electronics-tutorials.ws/transformer/transformer-basics.html
- https://www.electrical4u.com/what-is-transformer-definition-working-principle-of-tra...
- https://www.globalspec.com/learnmore/electrical.../transformers/transformers
- https://www.electronicshub.org/lc-oscillator-basics/

Multiple Choice Questions (1 mark each)

Q. 1. An inductor of reactance 1 Ω and a resistor of 2 Ω are connected in series to the terminals of a 6 *V* (rms) AC source. The power dissipated in the circuit is

(a) 8 W **(b) 12 W**

(c) 14.4 W **(d) 18 W**

[NCERT Exemp. Q. 7.6, Page 41]

Ans. Correct option : (c)

Explanation : Here

$E_{\text{rms}} = 6\ V, X_L = 1\ \Omega, R = 2\ \Omega, P_{av} = ?$

We know that, average power dissipated in the circuit is given by,

$P_{av} = E_{\text{rms}} I_{\text{rms}} \cos\phi$

$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{E_{rms}}{Z}$

$Z = \sqrt{R^2 + X_L^2} = \sqrt{4+1} = \sqrt{5}\ \Omega$

$\therefore I_{rms} = \frac{6}{\sqrt{5}}\ A$

$\cos\phi = \frac{R}{Z} = \frac{2}{\sqrt{5}}$

$P_{av} = 6 \times \frac{6}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = 14.4\ \text{W}$

Q. 2. The output of a step-down transformer is measured to be 24 V when connected to a 12 W Light bulb. The value of the peak current is

(a) $1/\sqrt{2}\ A$ **(b)** $\sqrt{2}\ A$

(c) $2\ A$ **(d)** $2\sqrt{2}\ A$

[NCERT Exemp. Q. 7.7, Page 41]

Ans. Correct option : (a)

Explanation : Given,

Power associated with secondary, $P_s = 12$ W

Secondary voltage, V_s=24 V

Current in the secondary, $I_s = \frac{P_s}{V_s} = \frac{12}{24} = 0.5\ A$

Peak value of the current in the secondary, $I_0 = I_s\sqrt{2} = (0.5)(1.414) = 0.707$ or $\frac{1}{\sqrt{2}}\ A$.

Tick Two or More Options

Q. 3. Electrical energy is transmitted over large distances at high alternating voltages. Which of the following statements is (are) correct?

(a) For a given power level, there is a lower current.

(b) Lower current implies less power loss.

(c) Transmission lines can be made thinner.

(d) It is easy to reduce the voltage at the receiving end using step-down transformers.

[NCERT Exemp. Q. 7.10, Page 42]

Ans. Correct options : (a), (b) and (d)

Explanation : As we know that Power is equal to $I^2_{\text{rms}}R$

So to decrease power loss, I_{rms} and R must be lower for a constant power supply. To decrease I_{rms}, V_{rms} must be increased by step up transformer to get same power in step up transformer

Output power=Input power

$V_s I_s = V_p I_p$

$V_S > V_P$, so $I_P >> I_S$ so loss of power during transmission become lower. We can again reduce voltage by using step down transformer.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Can the instantaneous power output of an ac source ever be negative? Can the average power output be negative?

[NCERT Exemp. Q. 7.17, Page 43]

Ans. Let the applied emf = $E = E_0 \sin(\omega t)$

$I = I_0 \sin(\omega t - \phi)$

Instantaneous power output of ac source,

$P = EI$

$P = E_0 \sin\omega t\ I_0\ \sin(\omega t - \phi)$

$P = E_0 I_0 \sin\omega t\ \sin(\omega t - \phi)$

We know that

$$\sin A.\sin B = \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]$$

$$P = E_0 I_0 \frac{1}{2}[\cos(\omega t - \omega t + \phi) - \cos(\omega t + \omega t + \phi)]$$

$$P = \frac{E_0 I_0}{2}[\cos\phi - \cos(2\omega t - \phi)] \quad \textbf{[1]}$$

Taken phase angle $\phi, \pm ve$.

Instantaneous Power P,

$P = \frac{E_0 I_0}{2}[\cos\phi \pm \cos(2\omega t - \phi)]$ as $\cos\phi = \frac{R}{Z}$,

R and Z can never be negative and value of $\cos\theta(\theta = 2\omega t \pm \phi)$ can vary from (1 to 0 to -1) in any case P can never be negative.

We know that average power of *LCR* series ac circuit is

$$P_{av} = \frac{E_0 I_0}{2}\cos\phi$$

Again as $\cos\phi = \frac{R}{Z}$ is always positive, because R and Z, the reactance are always positive. So P_{av} can never be negative. **[1]**

Q. 2. A charged 30 μF capacitor is connected to a 27 mH inductor. What is the angular frequency of free oscillations of the circuit?

[NCERT Ex. Q. 7.7, Page 266]

Ans. Given that

$C = 30\ \mu$F

$L = 27$ mH

Angular frequency of free oscillations = $\frac{1}{(LC)^{1/2}}$ **[½]**

Substitution results,

Angular frequency = 1111.11 s^{-1}. **[½]**

Q. 3. Suppose the initial charge on the capacitor in Exercise 7.7 is 6 mC. What is the total energy stored in the circuit initially? What is the total energy at later time?

[NCERT Ex. Q. 7.8, Page 266]

Ans. As we know that

Initial energy, $U_i = \frac{(q_m)^2}{2C}$ **[1]**

Solving,

$$U_i = \frac{1}{2}\frac{(6\times10^{-3})^2}{30\times10^{-6}}$$

$U_i = 0.6$ J **[½]**

The energy will remain constant at all times. **[½]**

Short Answer Type Questions **(3 or 4 marks each)**

Q. 1. A 60 W load is connected to the secondary of a transformer whose primary draws line voltage. If a current of 0.54 A flows in the load, what is the current in the primary coil? Comment on the type of transformer being used.

[NCERT Exemp. Q. 7.24, Page 44]

Ans. Given that :

$P_S = 60$ W, $I_S = 0.54$ A, $I_P = ?$

We know that, Power in secondary coil is given as : **[½]**

$P_S = V_S I_S$

$60 = V_s \times 0.54$

$$\frac{60}{0.54} = V_s$$

$V_s = 111.10$ Volt **[½]**

In multiple of 11

$V_s \cong 110$ Volt

Ratio factor of transformer 'r'=output voltage/ input voltage

$$r = \frac{V_s}{V_p} \Rightarrow r = \frac{110\,\text{Volt}}{220\,\text{Volt}} = \frac{1}{2}$$ **[1]**

Or $r < 1$, so transformer is step down transformer. **[½]**

In transformer, output power = Input power **[½]**

$I_s V_s = I_p V_p$

$$I_p = \frac{I_s V_s}{V_p} = \frac{0.54\times110}{220}$$

$I_p = 0.27$ Amp. **[1]**

Q. 2. A power transmission line feeds input power at 2,300 V to a step-down transformer with its primary windings having 4,000 turns. What should be the number of turns in the secondary in order to get output power at 230 V?

[NCERT Add. Ex. Q. 7.23, Page 268]

Ans. Given that,

Input voltage, $V_i = 2300$ V

Number of turns in primary coil, $n_1 = 4000$

Output voltage, $V_0 = 230$ V

Number of turns in secondary coil = n_2

Voltage is related to the number of turns as :

$$\frac{V_i}{V_0} = \frac{n_1}{n_2}$$

$$\frac{2,300}{230} = \frac{4,000}{n_2}$$

$$\therefore \quad n_2 = \frac{4,000\times230}{2300} = 400$$ **[2]**

Hence, there are 400 turns in the second winding. **[1]**

Q. 3. At a hydroelectric power plant, the water pressure head is at a height of 300 m and the water flow available is 100 $m^3 s^{-1}$. If the turbine generator efficiency is 60%, estimate the electric power available from the plant (g = 9.8 ms^{-2}).

[NCERT Add. Ex. Q. 7.24, Page 268]

Ans. Height of water pressure head, $h = 300$ m

Volume of water flow per second, $V = 100\ m^3/s$

Efficiency of turbine generator, $n = 60\% = 0.6$ **[½]**

Acceleration due to gravity, $g = 9.8\ m/s^2$

Density of water, $\rho = 10^3\ kg/m^3$ **[½]**

Electric power available from the plant

$= \eta \times h\rho g V$ **[1]**

$= 0.6 \times 300 \times 10^3 \times 9.8 \times 100$

$= 176.4 \times 10^6$ W

$= 176.4$ MW **[1]**

Q. 4. A capacitor of unknown capacitance, a resistor of 100 Ω and an inductor of self-inductance $L = (4/\pi^2)$ henry are connected in series to an ac source of 200 V and 50 Hz. Calculate the value of the capacitance and impedance of the circuit when the circuit is in phase with the voltage. Calculate the power dissipated in the circuit.

[CBSE Board, Delhi Region, 2016]

Ans. Current in phase with the voltage means angle between emf and current is 0. This is resonance condition in resonance, **[½]**

Inductive reactance = Capacitive reactance **[½]**

$\Rightarrow X_L = X_C$

$\omega L = \dfrac{1}{\omega C}$

$\therefore C = \dfrac{1}{\omega^2 L}$

Substituting the value of L

$\omega = 2\pi f$

$= 2\pi \times 50$

$= 100\ \pi$ rad

$C = \dfrac{1 \times \pi^2}{(100\pi)^2 \times 4} = \dfrac{1}{40,000}$ Farad

Impedance of resonating circuit = Resistance

$= 100\ \Omega$

Power = VI $= \dfrac{V^2}{R} = \dfrac{(200)^2}{100} = 400$ watt **[2]**

Long Answer Type Questions **(5 marks each)**

Q. 1. 1 MW power is to be delivered from a power station to a town 10 km away. One uses a pair of Cu wires of radius 0.5 cm for this purpose. Calculate the fraction of Ohmic losses to power transmitted if.

(i) power is transmitted at 220 V. Comment on the feasibility of doing this.

(ii) a step-up transformer is used to boost the voltage to 11,000 V, power transmitted, then a step-down transformer is used to bring voltage to 220 V. ($P_{Cu} = 1.7 \times 10^{-8}$ Ωm)

[NCERT Exemp. Q. 7.28, Page 45]

Ans. **(i)** When power is transmitted at 220 V.

Power lost in transmission $P = I^2R$

$P = VI$

$P = 1$ MW $= 10^6$ W

$V = 220$ Volt

$I = \dfrac{P}{V} = \dfrac{10,00,000}{220}$

$I = \dfrac{50,000}{11}$ A

$R = \rho \dfrac{1}{A}$

$\Rightarrow R = \dfrac{\rho l}{\pi r^2}$

$l = 10\ \text{km} \times 2 = 20,000$ m

$\therefore A = \pi r^2$

$r = 0.5$ cm $= 0.5\ \text{cm} \times 10^{-2} = 5 \times 10^{-3}$ m **[1]**

$\rho_{Cu} = 1.7 \times 10^{-8}\ \Omega$ m

$R = \dfrac{\rho l}{A}$

$\therefore \quad R = \dfrac{1.7 \times 10^{-8} \times 20,000}{3.14 \times 5 \times 10^{-3} \times 5 \times 10^{-3}}$

$= \dfrac{170 \times 20,000 \times 10^{-8+6}}{314 \times 5 \times 5}$

$= \dfrac{170 \times 20,000}{314 \times 25 \times 100}\ \Omega$

$\rho_{Cu} = 1.7 \times 10^{-8}\ \Omega$ m

$R = \dfrac{\rho l}{A}$

$\therefore \quad R = \dfrac{1.7 \times 10^{-8} \times 20,000}{3.14 \times 5 \times 10^{-3} \times 5 \times 10^{-3}}$

$= \dfrac{170 \times 20,000 \times 10^{-8+6}}{314 \times 5 \times 5}$

$= \dfrac{170 \times 20,000}{314 \times 25 \times 100}\ \Omega$

$R = \dfrac{170 \times 800}{314 \times 100} = \dfrac{170 \times 4}{157} = \dfrac{680}{157} \cong 4\ \Omega$

$\therefore$ Power loss $= I^2R$

$= \dfrac{50,000}{11} \times \dfrac{50,000}{11} \times 4 = \dfrac{100 \times 10^8}{121}$

$= 8.26 \times 10^7$ **[1]**

Power loss in heating = 82.6 MW

As 82.6 MW > 1 MW

So this method cannot be used to transmit the power. **[½]**

(ii) When power is transmitted at 11,000 V

$P = 10^6$ W

$VI = 10,00,000$

11,000 I = 10,00,000

$I = \dfrac{10,00,000}{11,000} = \dfrac{1,000}{11}$

$\therefore$ Power loss $= P = I^2R$ **[1]**

$P = \dfrac{1,000}{11} \times \dfrac{1,000}{11} \times 4$

$= \dfrac{4,000}{121} \times 4^4$

$P = 3.3 \times 10^4$ Watt **[½]**

Fraction power loss $= \dfrac{3.3 \times 10^4}{10^6} = \dfrac{3.3}{1,00} = 0.033$ **[½]**

Power loss in % = 3.3% **[½]**

Q. 2. An *LC* circuit contains a 20 mH inductor and a 50 μF capacitor with an initial charge of 10 mC.

The resistance of the circuit is negligible. Let the instant the circuit is closed be $t = 0$.

(a) What is the total energy stored initially? Is it conserved during *LC* oscillations?

(b) What is the natural frequency of the circuit?

(c) At what time is the energy stored (i) completely electrical (*i.e.*, stored in the capacitor)? (ii) completely magnetic (*i.e.*, stored in the inductor)?

(d) At what times is the total energy shared equally between the inductor and the capacitor?

(e) If a resistor is inserted in the circuit, how much energy is eventually dissipated as heat?

[NCERT Add. Ex. Q. 7.12, Page 267]

Ans. Inductance of the inductor, $L = 20\ \text{mH} = 20 \times 10^{-3}\ \text{H}$

Capacitance of the capacitor, $C = 50\ \mu\text{F} = 50 \times 10^{-6}\ \text{F}$

Initial charge on the capacitor, $Q = 10\ \text{mC}$

$= 10 \times 10^{-3}\ \text{C}$

(a) Initially total energy stored in the circuit is given as :

$$E = \frac{1}{2}\frac{Q^2}{C}$$

$$= \frac{(10\times10^{-3})^2}{2\times50\times10^{-6}} = \text{I J}$$

[½]

Hence, the total energy stored in the *LC* circuit will be conserved because there is no resistor connected in the circuit. **[½]**

(b) Natural frequency of the circuit is given by the relation,

$$\nu = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi\sqrt{20\times10^{-3}\times50\times10^{-6}}}$$

$$= \frac{10^3}{2\pi} = 159.24\ \text{Hz}$$ **[½]**

Natural angular frequency,

$$\omega, = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{20\times10^{-3}\times50\times10^{-6}}} = \frac{1}{\sqrt{10^{-6}}} = 10^3\ \text{rad/s}$$

Hence, the natural frequency of the circuit is 10^3 rad/s. **[½]**

(c) **(i)** For time period $\left(T = \frac{1}{\upsilon} = \frac{1}{159.24} = 6.28\ \text{ms}\right)$, total charge on the capacitor at time t,

$$Q' = Q\cos\frac{2\pi}{T}t$$

[½]

For energy stored is electrical, we can write $Q' = Q$. Hence, it can be inferred that the energy stored in the capacitor is completely electrical at time,

$$t = 0, \frac{T}{2}, T, \frac{3T}{2}, ...$$

[½]

(ii) Magnetic energy is the maximum when electrical energy, Q' is equal to 0.

Hence, it can be inferred that the energy stored in the capacitor is completely magnetic at time,

$$t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}.....$$

[½]

(d) Q_1 = Charge on the capacitor when total energy is equally shared between the capacitor and the inductor at time t.

When total energy is equally shared between the inductor and capacitor, the energy stored in the capacitor = $\frac{1}{2}$ (maximum energy),

$$\Rightarrow \frac{1}{2}\frac{(Q_1)^2}{C} = \frac{1}{2}\left(\frac{1}{2}\frac{Q^2}{C}\right) = \frac{1}{4}\frac{Q^2}{C}$$

$$Q_1 = \frac{Q}{\sqrt{2}}$$

But $Q_1 = Q\cos\frac{2\pi}{T}t$.

$$\frac{Q}{\sqrt{2}} = Q\cos\frac{2\pi}{T}t$$

$$\cos\frac{2\pi}{T}t = \frac{1}{\sqrt{2}} = \cos(2n+1)\frac{\pi}{4}: \quad \text{where } n=0,1,2,\ldots$$

$$t = (2n+1)\frac{T}{8}$$ **[1]**

Hence total energy is equally shared between the inductor and the capacity at time,

$$t = \frac{T}{8}, \frac{3T}{8}, \frac{5T}{8}.....$$

(e) If a resistor is inserted in the circuit, then total initial energy is dissipated as heat energy in the circuit. The resistance damps out the *LC* oscillation. **[½]**

Q. 3. A circuit containing a 80 mH inductor and a 60 μF capacitor in series is connected to a 230 V, 50 Hz supply. The resistance of the circuit is negligible.

(a) Obtain the current amplitude and rms values.

(b) Obtain the rms values of potential drops across each element.

(c) What is the average power transferred to the inductor?

(d) What is the average power transferred to the capacitor?

(e) What is the total average power absorbed by the circuit?

[Hint : 'Average' implies 'averaged over one cycle'.]

[NCERT Add. Ex. Q. 7.18, Page 267]

Ans. Given that,

Inductance, $L = 80\ \text{mH} = 80 \times 10^{-3}\ \text{H}$

Capacitance, $C = 60\ \mu\text{F} = 60 \times 10^{-6}\ \text{F}$

Supply voltage, $V = 230\ \text{V}$

Frequency, $\nu = 50\ \text{Hz}$

Angular frequency, $\omega = 2\pi\nu = 100\ \pi\ \text{rad/s}$

Peak voltage, $V_0 = V\sqrt{2} = 230\sqrt{2}\ V$

(a) Hence, *rms* value of current,

$$I_{rms} = \frac{V_0}{\left(L - \frac{1}{C}\right)}$$

$$= \frac{230\sqrt{3}}{\left(100\pi\times80\times10^{-3} - \frac{1}{100\pi\times60\times10^{-6}}\right)}$$

$$= \frac{230\sqrt{2}}{\left(8\pi - \frac{1000}{6\pi}\right)} = -11.63\ \text{A}$$ **[1]**

The negative sign appears because $\omega L < \frac{1}{\omega C}$

Amplitude of maximum current,

$|I_0| = 11.63$ A

$I = \frac{I_{rms}}{\sqrt{2}} = \frac{-11.63}{\sqrt{2}} = -8.22$ A **[1]**

(b) Potential difference across the inductor,

$V_L = I \times \omega L$ **[½]**

$= 8.22 \times 100\,\pi \times 80 \times 10^{-3}$

$= 206.61$ V **[½]**

Potential difference across the capacitor,

$V_e = I \times \frac{1}{\omega C} = 8.22 \times \frac{1}{100\pi \times 60 \times 10^{-6}} = 436.3$ V **[½]**

(c) Average power consumed by the inductor is zero as actual voltage leads the current by $\frac{v}{2}$. **[½]**

(d) Average power consumed by the capacitor is zero as voltage lags current by $\frac{\pi}{2}$. **[½]**

(e) The total power absorbed (averaged over one cycle) is zero. **[½]**

Q. 4. Suppose the circuit in Exercise 7.18 has a resistance of 15 Ω. Obtain the average power transferred to each element of the circuit, and the total power absorbed. [NCERT Add. Ex. Q. 7.19, Page 267]

Ans. Average power transferred to the resistor = 788.44 W

Average power transferred to the capacitor = 0 W

Total power absorbed by the circuit = 788.44 W

Inductance of inductor, $L = 80$ mH $= 80 \times 10^{-3}$ H

Capacitance of capacitor, $C = 60\,\mu$F $= 60 \times 10^{-6}$ F

Resistance of resistor, $R = 15\ \Omega$

Potential of voltage supply, $V = 230$ V

Frequency of signal, $\nu = 50$ Hz

Angular frequency of signal, $\omega = 2\pi\nu = 2\pi \times (50)$

$= 100\,\pi$ rad/s

The elements are connected in series to each other. Hence, impedance of the circuit is given as : **[½]**

Current flowing in the circuit,

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$= \sqrt{(15)^2 + \left(100\pi(80 \times 10^{-3}) - \frac{1}{(100\pi \times 60 \times 10^{-6})}\right)^2}$$

$$= \sqrt{(15)^2 + (25.12 - 53.08)^2} = 31.728\ \Omega$$

$I = \frac{V}{Z} = \frac{230}{31.728} = 7.28$ A **[1½]**

Average power transferred to resistance is given as :

$P_R = I^2R = (7.25)^2 \times 15 = 788.44$ W **[½]**

Average power transferred to capacitor,

$P_C = 0$ **[½]**

Average power transferred to inductor,

$P_L = 0$ **[½]**

Total power absorbed by the circuit :

$= P_R + P_C + P_L = 788.44 + 0 + 0 = 788.44$ W **[1]**

Hence, the total power absorbed by the circuit is 788.44 W. **[½]**

Q. 5. A small town with a demand of 800 kW of electric power at 220 V is situated 15 km away from an electric plant generating power at 440 V. The resistance of the two wire line carrying power is 0.5 Ω per km. The town gets power from the line through a 4,000–220 V step-down transformer at a sub-station in the town.

(a) Estimate the line power loss in the form of heat.

(b) How much power must the plant supply, assuming there is negligible power loss due to leakage?

(c) Characterize the step up transformer at the plant.

[NCERT Add. Ex. Q. 7.25, Page 268]

Ans. Total electric power required, $P = 800$ kW

$= 800 \times 10^3$ W

Supply voltage, $V = 220$ V

Voltage at which electric plant is generating power, $V' = 440$ V

Distance between the town and power generating station, $d = 15$ km

Resistance of the two wire lines carrying power

$= 0.5\ \Omega$/km

Total resistance of the wires, $R = (15 + 15)0.5$

$= 15\ \Omega$ **[½]**

A step-down transformer of rating 4,000 − 220 V is used in the sub-station.

Input voltage, $V_1 = 4{,}000$ V

Output voltage, $V_2 = 220$ V

rms current in the wire lines is given as :

$I = \frac{P}{V_1} = \frac{800 \times 10^3}{4{,}000} = 200A$ **[½]**

(a) Line power loss $= I^2R$

$= (200)^2 \times 15$

$= 600 \times 10^3$ W **[1]**

$= 600$ kW

(b) Assuming that the power loss is negligible due to the leakage of the current :

Total power supplied by the plant = 800 kW + 600 kW = 1,400 kW **[1]**

(c) Voltage drop in the power line $= IR = 200 \times 15 =$ 3,000 V **[½]**

Hence, total voltage transmitted from the plant = 3,000 + 4,000 = 7,000 V **[½]**

Also, the power generated is 440 V. Hence, the rating of the step-up transformer situated at the power plant is 440 V – 7,000 V. **[1]**

Q. 6. Do the same exercise as above with the replacement of the earlier transformer by a 40,000–220 V step-down transformer (Neglect, as before, leakage losses though this may not be a good assumption any longer because of the very high voltage transmission involved). Hence, explain why high voltage transmission is preferred?

[NCERT Add. Ex. Q. 7.26, Page 268]

Ans. The rating of a step-down transformer is

40,000 V − 220 V.

Input voltage, $V_1 = 40{,}000$ V

Output voltage, $V_2 = 220$ V

Total electric power required, $P = 800$ kW

$= 800 \times 10^3$ W

Source potential, $V = 220$ V
Voltage at which the electric plant generates power, $V' = 440$ V
Distance between the town and power generating station, $d = 15$ km
Resistance of the two wire lines carrying power $= 0.5\ \Omega$/km
Total resistance of the wire lines, $R = (15 + 15)0.5$
$= 15\ \Omega$ [½]

$P = V_1 I$
rms current in the wire line is given as :

$$I = \frac{P}{V_1} = \frac{800 \times 10^3}{40,000} = 20\ A$$

[½]

(b) Assuming that the power loss is negligible due to the leakage of current.
Hence, power supplied by the plant = 800 kW + 6kW
= 806 kW [½]

(c) Voltage drop in the power line = $IR = 20 \times 15$
= 300 V [½]
Hence, voltage that is transmitted by the power plant
= 300 + 40,000 = 40,300 V [½]
The power is being generated in the plant at 440 V.
Hence, the rating of the step-up transformer needed at the plant is 440 V – 40,300 V.
Hence, power loss during transmission =

$$\frac{6}{806} \times 100 = 0.744\%$$

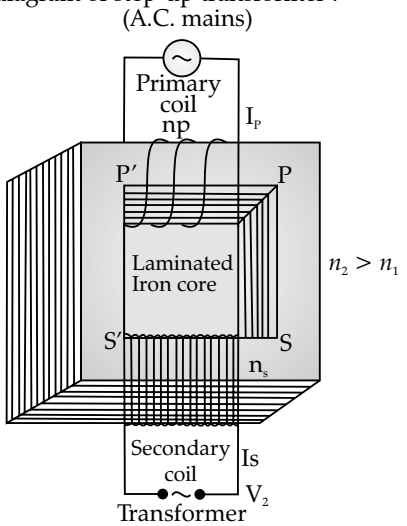
[1]
In the previous exercise, the power loss due to the same reason is

$$\frac{600}{1,400} \times 100 = 42.8\%$$

[1]
Since the power loss is less for a high voltage transmission, high voltage transmissions are preferred for this purpose. [½]

Q. 7. Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary to primary voltage in terms of number of turns and currents in the two coils. [CBSE Board, Delhi Region, 2017]

Ans. The diagram of step-up transformer :



[½]

Principle : A transformer is based on the principle of electromagnetic mutual induction, *i.e.*, whenever the amount of magnetic flux linked with a coil changes, an emf is induced in the neighbouring coil. [½]

Working : Alternating emf is supplied to the primary coil PP'. The resulting current induces a current in secondary coil. [½]
Let the alternating current, alternating voltage and the number of turns in primary and secondary coils are (I_1, I_2), (V_1, V_2) and (n_s, n_p) respectively. Let ϕ be the flux in each turn in the core at time 't' due to current in the primary when a voltage V_2 is applied to it. [½]
Magnetic flux linked with primary is also linked with the secondary. The induced emf in each turn of the secondary is equal to that induced in each turn of the primary. [½]
Let
E_p – Alternating emf applied to primary
n_p – Number of turns in the primary
$\frac{d\phi}{dt}$ – Rate of change of flux through each turn primary coil

$$E_P = -n_P \frac{d\phi}{dt} \quad ...(1)$$

[½]

E_s – Alternating emf of secondary
n_s – Number of turns in secondary

$$E_s = -n_s \frac{d\phi}{dt} \quad ...(2)$$

[½]

Dividing equation (2) by (1)

$$\frac{E_s}{E_p} = \frac{n_s}{n_p} = k$$

[½]
For step-up transformer, $K > 1$

$$\therefore E_s > E_p$$

[½]
If transformer is 100% efficient, then there is no loss of energy in the process. [½]

Q. 8. A power transmission line feeds input power at 2,200 V to a step-down transformer with its primary windings having 3,000 turns. Find the number of turns in the secondary winding to get the power output at 220 V.
[CBSE Board, Delhi Region, 2017]

Ans. Given that,
Given, input power (V_p) = 2,200V
Number of turns (n_1) = 3,000
Output power (V_s) = 220 V **[1]**

$$\text{As,} \frac{V_s}{V_p} = \frac{n_2}{n_1}$$

$$\Rightarrow \frac{220}{2,200} = \frac{n_2}{3,000}$$

$$\Rightarrow \quad n_2 = \frac{220}{2,200} \times 3,000$$ **[1]**

$\Rightarrow 300$ turns
$\therefore$ Number of turns in the secondary winding, $n_2 = 300$ turns. **[1]**

Q. 9. Answer the following :

(i) Write the function of a transformer. State its principle of working with the help of a diagram. Mention various energy losses in this device.

(ii) The primary coil of an ideal step-up transformer has 100 turns and the transformation ratio is also 100. The input voltage and power are 220 V and 1,100 W, respectively. Calculate the following :

(a) number of turns in secondary

(b) current in primary

(c) voltage across secondary

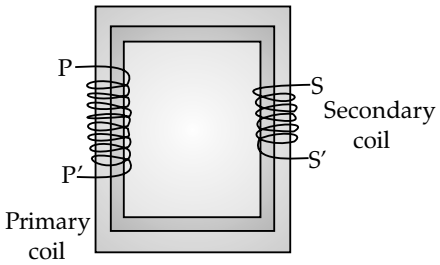
(d) current in secondary

(e) power in secondary

[CBSE Board, Delhi Region, 2016]

Ans. **(i)** A transformer is a device that changes or transforms a low alternating voltage into a high alternating voltage or vice-versa. **[½]**

It works on the principle of mutual induction. Here, a changing alternating current in primary coil produces a changing magnetic field, which in turn, induces a changing alternating current in secondary coil. **[½]**



[½]

Energy losses :

(a) Flux leakage, due to poor design of the core and air gaps in the core. **[¼]**

(b) Resistance of the windings, which leads to loss of energy due to heat produced. **[¼]**

(c) Eddy currents, due to alternating magnetic flux in the iron core which leads to loss due to heating. **[¼]**

(d) Hysteresis, frequent and periodic magnetisation and demagnetisation of the core, leading to loss of energy due to heat. **[¼]**

(ii) **(a)** Number of turns in secondary coil is given by

$$\frac{N_s}{N_p} = n$$

$$\Rightarrow \frac{N_s}{100} = 100$$

$$\Rightarrow \quad N_s = 10,000$$ **[½]**

(b) Current in primary is given by

$$I_P V_P = P$$

$$\Rightarrow I_P = \frac{1,100}{220} = 5\ A$$ **[½]**

(c) Voltage across secondary is given by

$$\frac{V_s}{V_P} = \frac{N_s}{N_p} = n$$

$$\Rightarrow V_s = 100 \times 220 = 22,000\ V$$ **[½]**

(d) Current in secondary is given by

$$V_S I_S = P$$

$$\Rightarrow I_S = \frac{P}{V_S} = \frac{1,100}{22,000} = 0.05\ A$$ **[½]**

(e) Power in secondary is equal to power in primary in an ideal transformer.

$= 1,100$ W **[½]**

Q. 10. (i) Draw a labelled diagram of a step-down transformer. State the principle of its working.

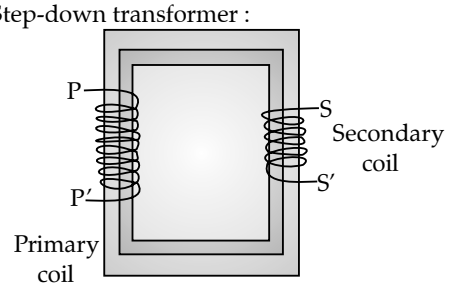
(ii) Express the turn ratio in terms of voltages.

(iii) Find the ratio of primary and secondary currents in terms of turn ratio in an ideal transformer.

(iv) How much current is drawn by the primary of a transformer connected to 220 V supply when it delivers power to a 110 V – 550 W refrigerator?

[CBSE Board, All India Region, 2016]

Ans. Step-down transformer :



[½]

Principle : A transformer is based on the principle of electromagnetic mutual induction, *i.e.*, whenever the amount of magnetic flux linked with a coil changes, an emf is induced in the neighbouring coil. **[½]**

Working : Alternating emf is supplied to the primary coil PP'. The resulting current induces a current in secondary coil. **[½]**

Magnetic flux linked with primary is also linked with the secondary. The induced emf in each turn of the secondary is equal to that induced in each turn of the primary. **[½]**

Let

E_p – Alternating emf applied to primary

n_p – Number of turns in the primary

$\frac{d\phi}{dt}$ – Rate of change of flux through each turn primary coil

$$E_P = -n_P \frac{d\phi}{dt} \quad ...(1)$$

[½]

E_s – Alternating emf of secondary

n_s – Number of turns in secondary

$$E_s = -n_s \frac{d\phi}{dt} \quad ...(2)$$

[½]

Dividing equation (2) by (1)

$$\frac{E_s}{E_p} = \frac{n_s}{n_p} = k$$ **[½]**

For step-up transformer, $K > 1$

$\therefore E_s > E_p$

For step-down transformer, $K < 1$

$\therefore E_s < E_p$

(ii) Turn ratio, $n = \frac{N_s}{N_p} = \frac{E_s}{E_p}$ [½]

(iii) According to law of conservation of energy,

Input electrical power = Output electrical power

$$E_p I_p = E_s I_s$$

$$\frac{E_s}{E_p} = \frac{I_p}{I_s}$$

$$\frac{I_p}{I_s} = \frac{E_s}{E_p} = \frac{N_s}{N_p}$$

$$\Rightarrow \quad \frac{I_p}{I_s} = n$$ [½]

(iv) We have

$$I_p V_p = I_s V_s = 500 \text{ W}$$

$$V_p = 220 \text{ V}$$

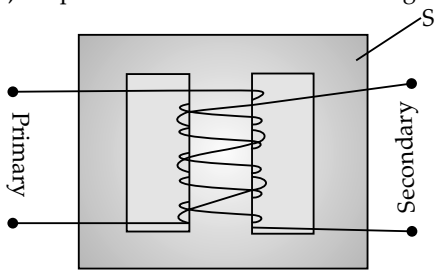
$$I_p = \frac{550}{220} = \frac{5}{2} = 2.5 \ A$$ [½]

Q. 11. (a) Describe, with the help of a suitable diagram, the working principle of a step-up transformer. Obtain the relation between input and output voltages in terms of the number of turns of primary and secondary windings and the currents in the input and output circuits.

(b) Given the input current 15 A and the input voltage of 100 V for a step-up transformer having 90% efficiency, find the output power and the voltage in the secondary if the output current is 3 A.

[CBSE Board, Foreign Scheme, 2017]

Ans. (a) Step-down Transformer Circuit Diagram :



[½]

Principle : The transformer is based on the principle of electro-magnetic mutual induction. [½]

1. When the current flows through the primary coil changes, an emf is induced in the secondary coil due to change in magnetic flux linked with the primary coil. [½]
2. Turn ratio in terms of voltage is,

$$n = \frac{N_s}{N_p} = \frac{V_s}{V_p}$$ [½]

3. For an ideal transformer, according to the law of conservation of energy,

Input electrical power=Output electrical power

$I_P V_P = I_S V_S$

$$\frac{V_s}{V_p} = \frac{I_p}{I_S}$$

$$\therefore \quad \frac{I_p}{I_S} = \frac{V_s}{V_p} = \frac{N_S}{N_p}$$

i.e., $\Rightarrow \frac{I_p}{I_S} = n$

[½]

(b) Given that,

$I_S = I_{out} = 3$ Amp.; $I_P = I_{inp} = 15$ Amp.; $V_P = V_{inp} = 100$ V; Efficiency = 90%; $V_{out} = ?$

We know that,

$$\text{Efficiency} = \frac{p_{out}}{P_{inp}}$$ [½]

But we know that in case of 100% efficiency, input power is always equal to output power. That means $P_{out} = P_{inp}$. [½]

We know that $P = VI$.

$$\frac{90}{100} = \frac{V_P I_P}{V_S I_S}$$

$$\frac{90}{100} = \frac{15 \times 100}{3 \times V_S}$$

$$V_S = \frac{15 \times 100 \times 100}{3 \times 90}$$

So, $V_S = 555.55 V$ **[1½]**

So the output voltage will be 555.55 Volt.

Some Commonly Made Errors

- Careless errors occur simply because students do not pay attention, or they are working too fast. Some examples might be : Dropping a negative sign; Not following the directions of the current.
- Making one computational mistake in a multistep problems means the rest of their work will be wrong and the final solution wrong.
- Conceptual errors occur because students have misunderstood the underlying concepts or have used incorrect logic.
- Students mainly confused when they are solving transformer-related numerical problems either it is Step-up Transformer or Step-down Transformer.
- Draw wrong phasor diagram in case of leading or lagging voltage or current in the circuit analysis.

EXPERT ADVICE

- ☞ *Always required students to show all their steps because if they have used the correct procedure that they understood the concept.*
- ☞ *Students should have clear all the conceptual and computational topics and formulae, according to the given cases in terms of Current and Voltage equations.*
- ☞ *Study all the phasor diagram of current and voltage in case of Inductor, Capacitor and Resistor.*
- ☞ *Study all the case in ac circuits reared with inductor, resistor and capacitor.*
- ☞ *Learn all the resonance conditions of the Circuit Element.*

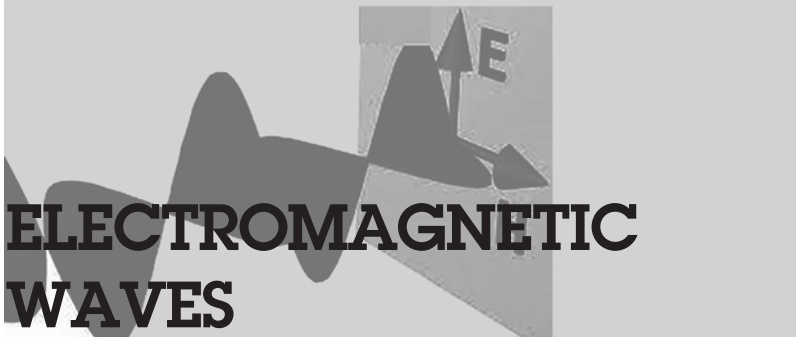
OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/q-DKc4KHPqU **Or Scan the Code**	*https://youtu.be/cS83YUpcEsY* **Or Scan the Code**
https://youtu.be/vN9aR2wKv0U **Or Scan the Code**	*https://goo.gl/PYVxw1* **Or Scan the Code**

CHAPTER

8 ELECTROMAGNETIC WAVES



Chapter Objectives

This chapter will help you understand :

➢ ***Electromagnetic waves :*** *Introduction; Displacement current; Electromagnetic waves; Electromagnetic spectrum.*

Quick Review

- ❖ One of the most amazing aspects of physics is the electromagnetic spectrum—radio waves, microwaves, infrared light, visible light, ultraviolet light, X-rays, and gamma rays—as well as the relationship between the spectrum and electromagnetic force.
- ❖ In 1888, Heinrich Hertz, in Karlsruhe, published his experimental validations of Maxwell's equations and forced a conceptual revolution in European theoretical physicists by showing that electromagnetic effects propagate at a finite speed. He also discovered the existence of radio waves.
- ❖ Heinrich Hertz, born to a wealthy Hamburg family, earned a doctorate in physics at the unusually young age of 23. Only a few years later, in 1886, he began to research electromagnetic waves, testing Maxwell's theories, which had gained little traction in continental Europe.
- ❖ Hertz's oscillator was composed of two rods to serve as a receiver and a spark gap as a receiving antenna. When the antennae picked up a radio wave, a spark would jump. Hertz found that these signals had the properties of electromagnetic waves. These radio waves had the same velocity as light.

TIPS...

- *The problem is solved under the physical assumptions $a \ll d \ll \lambda$, where a is the characteristic size of the bodies, d is the minimal distance between neighboring bodies, $\lambda = 2\pi/k$ is the wave length and k is the wave number.*
- *An improved matrix method of calculating the solutions of wave equations in a horizontally stratified anisotropic medium is described.*
- *An inhomogeneous medium is divided into a number of thin horizontal and homogeneous slabs.*

TRICKS...

- *In calculating the wave fields, a matrix is divided into two matrices for two independent solutions and Gram-Schmidt orthogonal zing process is applied to prevent numerical sampling.*
- *Comparisons are made between this and other full wave methods regarding step size and computer time, and this method is found to be more efficient than the full wave method especially at high frequency wave fields.*
- *The magneto ionic modes separated from the resultant wave are shown for a model ionosphere.*

Know the Links

- ☞ *https://imagine.gsfc.nasa.gov/science/toolbox/emspectrum1.html*
- ☞ *https://www.greenfacts.org/glossary/def/electromagnetic-spectrum.htm*
- ☞ *https://www.sciencedirect.com/topics/neuroscience/electromagnetic-radiation*
- ☞ *https://www.britannica.com/science/electromagnetic-spectrum*

Multiple Choice Questions

(1 mark each)

Q. 1. One requires 11 eV of energy to dissociate a carbon monoxide molecule into carbon and oxygen atoms. The minimum frequency of the appropriate electromagnetic radiation to achieve the dissociation lies in

(a) visible region. **(b) infrared region.**
(c) ultraviolet region. **(d) microwave region.**

[NCERT Exemp. Q. 8.1, Page 47]

Ans. Correct option : (c)

Explanation : $E = 11 \text{ eV} = 11 \times 1.6 \times 10^{-19} \text{ J}$

$h = 6.62 \times 10^{-34}$ Js

$E = h\nu$

$$\nu = \frac{E}{h} = \frac{11 \times 1.6 \times 10^{-19}}{6.62 \times 10^{-34}} = \frac{8.8 \times 10^{-19+34}}{3.31} = \frac{880}{331} \times 10^{15} \text{ Hz}$$

$= 2.65 \times 10^{15}$ Hz

This frequency radiation belongs to the ultraviolet region.

Q. 2. A linearly polarized electromagnetic wave given as $E = E_0 \hat{i}$ cos(kz–ωt) is incident normally on a perfectly reflecting infinite wall at $z = a$. Assuming that the material of the wall is optically inactive, the reflected wave will be given as

(a) $E_r = -E_0 \hat{i}$ **cos (kz-ωt).** **(b)** $E_r = E_0 \hat{i}$ **cos (kz +ωt).**

(c) $E_r = -E_0 \hat{i}$ **cos (kz+ωt).** **(d)** $E_r = E_0 \hat{i}$ **sin (kz-ωt).**

[NCERT Exemp. Q. 8.2, Page 47]

Ans. Correct option : (b)

The phase of a wave changes by 180° or π radian after got reflected from a denser medium. But the type of waves remains identical.

Therefore, for the reflected wave, we have

$\hat{z} = -\hat{z}$, $\hat{i} = -\hat{i}$ and additional phase of π in the incident wave.

It is given that the incident electromagnetic wave. Then,

$E = E_0(-\hat{i})\cos(kz - \omega t)$

Therefore, the reflelected electromagnetic wave is given by

$$\mathrm{E}_r = E_0(-\hat{i})\cos[k(-z) - \omega t + \pi] \quad [\because \hat{z} = -\hat{z} \text{ and } \hat{i} = -\hat{i}]$$

$$= -E_0\hat{i}\cos[\pi - (kz + \omega t)]$$

$$= -E_0\hat{i}[-\cos\{(kz + \omega t)\}]$$

$$= E_0\hat{i}\cos(kz + \omega t)$$

Q. 3. Light with an energy flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence. If the surface has an area of 30 cm^2, the total momentum delivered (for complete absorption) during 30 minutes is

(a) 36 × 10^{-5} kg m/s. **(b) 36 × 10^{-4} kg m/s.**

(c) 108 × 10^4 kg m/s. **(d) 1.08 × 10^7 kg m/s.**

[NCERT Exemp. Q. 8.3, Page 48]

Ans. Correct option : (b)

Explanation : Energy flux$= \phi = 20$ W/cm^2

Area $A = 30$ cm^2, time $t = 30 \times 60$ sec

U= Total energy falling in t sec= Energy flux × Area × time = ϕAt

$U = 20 \times 30 \times 30 \times 60$ J

$$\text{Momentum of the incident light} = \frac{u}{c} = \frac{20 \times 30 \times 30 \times 60}{3 \times 10^8}$$

$= 36 \times 10^{-4}$ kg-ms^{-1}

As no reflection from the surface and for complete absorption so momentum of reflected radiation is zero.

Momentum delivered to surface = Change in momentum

$= p_f - p_i = 0 - 36 \times 10^{-4}$ kgm/s

$= -36 \times 10^{-4}$ kg m/s

(-) sign shows the direction of momentum.

Q. 4. The electric field intensity produced by the radiations coming from 100 W bulb at a 3 m distance is *E*. The electric field intensity produced by the radiations coming from 50 W bulb at the same distance is

(a) $\frac{E}{2}$ **(b)** $2E$

(c) $\frac{E}{\sqrt{2}}$ **(d)** $\sqrt{2}E$

[NCERT Exemp. Q. 8.4, Page 48]

Ans. Correct option : (c)

Explanation : We know that,

$E_0 \propto \sqrt{P_{\text{av}}}$

$$\frac{(E_0)_1}{(E_0)_2} = \sqrt{\frac{(P_{\text{av}})_1}{(P_{\text{av}})_2}} = \sqrt{\frac{100 \text{ W}}{50 \text{ W}}} = \frac{\sqrt{2}}{1}$$

$$\therefore (E_0)_2 = \frac{(E_0)_1}{\sqrt{2}}$$

Q. 5. If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of the electromagnetic wave, the direction of propagation of electromagnetic wave is along

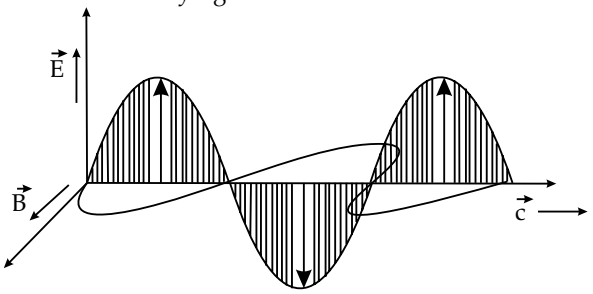
(a) $\vec{E}$. **(b)** $\vec{B}$.

(c) $\vec{B} \times \vec{E}$. **(d)** $\vec{E} \times \vec{B}$.

[NCERT Exemp. Q. 8.5, Page 48]

Ans. Correct option : (d)

Explanation : The direction of propagation of electromagnetic wave is perpendicular to $\vec{E}$ and $\vec{B}$ both and by right thumb rule.



The direction of propagation of electromagnetic wave is perpendicular to both the electric field vector $\vec{E}$ and $\vec{B}$ magnetic field vector B, *i.e.*, in the direction of $\vec{E} \times \vec{B}$.

Here, electromagnetic wave is along the z-direction which is given by the cross product of $\vec{E}$ and $\vec{B}$.

Q. 6. The ratio of contributions made by the electric field and magnetic field components to the intensity of an EM wave is

(a) c **: 1** **(b)** c^2 **: 1**

(c) 1 : 1. **(d)** $\sqrt{c}$ **:1**

[NCERT Exemp. Q. 8.6, Page 48]

Ans. Correct option : (c)

Explanation : Average energy by electric field E_0 is U_{av}

$$U_{av} = \frac{1}{2}\varepsilon_0 E_0^2$$

But $E_0 = cB_0$

$$(U_{av})_{\text{electric field}} = \frac{1}{2}\varepsilon_0(cB_0)^2 = \frac{1}{2}\varepsilon_0 c^2 B_0^2$$

$$= \frac{1}{2}\varepsilon_0 \cdot \frac{1}{\mu_0\varepsilon_0}(B_0)^2 \qquad \because c^2 = \frac{1}{\mu_0\varepsilon_0}$$

$$(U_{av})_{electric\ field} = \frac{1}{2\mu_0} B_0^2 (U_{av})_{(magnetic\ field)}$$

$$\text{Ratio} = \frac{(U_{av})\text{electric field}}{(U_{av})\text{magnetic field}} = \frac{1}{1}, i.e., 1:1$$

Q. 7. An EM wave radiates outwards from a dipole antenna, with E_0 as the amplitude of its electric field vector. The electric field E_0 which transports significant energy from the source falls off as

(a) $\frac{1}{r^3}$ **(b)** $\frac{1}{r^2}$ **(c)** $\frac{1}{r}$ **(d) remains constant**

[NCERT Exemp. Q. 8.7, Page 48]

Ans. Correct option : (c)

Explanation : A diode antenna radiates the electromagnetic waves outwards. The amplitude of electric field vector (E_0) which transports significant energy from the source falls intensity inversely as the distance (r) from the antenna. As we know that electromagnetic waves are radiated from dipole antenna and radiated energy $E \propto \frac{1}{r}$.

Tick Two or More Options

Q. 8. An electromagnetic wave travels in vacuum along + z direction

$E = (E_1\hat{i} + E_2\hat{j})\cos(kz - \omega t)$.

Choose the correct options from the following :

(a) The associated magnetic field is given as

$B = \frac{1}{c}(E_1\hat{i} + E_2\hat{j})\cos(kz - \omega t)$.

(b) The associated magnetic field is given as

$B = \frac{1}{c}(E_1\hat{i} - E_2\hat{j})\cos(kz - \omega t)$.

(c) The given electromagnetic field is circularly polarised.

(d) The given electromagnetic wave is plane polarised. **[NCERT Exemp. Q. 8.8, Page 49]**

Ans. Correct options : (a) and (d)

Explanation : We are given that the electric field vector of an electromagnetic wave travels in a vacuum along z-direction as,

$$\vec{E} = (E_1\hat{i} + E_2\hat{j})\cos(kz - \omega t)$$

The magnitude of the electric and the magnetic fields in an electromagnetic wave are related as

$$B_0 = \frac{E_0}{c}$$

$$\vec{B} = \frac{\vec{E}}{c} = \frac{E_1\hat{i} + E_2\hat{j}}{c}\cos(kz - \omega t)$$

Also, $\vec{E}$ and $\vec{B}$ are perpendicular to each other and the propagation of electromagnetic wave is perpendicular to $\vec{E}$ as well as $\vec{B}$, so the given electromagnetic wave is plane polarised.

Q. 9. An electromagnetic wave travelling along z-axis is given as $E = E_0$ cos (kz-ωt). Choose the correct options from the following :

(a) The associated field is given as

$\vec{B} = \frac{1}{c} = \hat{k} \times \vec{E} = \frac{1}{\omega}(\hat{k} \times \vec{E})$

(b) The electromagnetic field can be written in terms of the associated magnetic field as $\vec{E} = c(\vec{B} \times \hat{k})$

(c) $\hat{k} \cdot \vec{E} = 0, \hat{k} \cdot \vec{B} = 0$

(d) $\hat{k} \times \vec{E} = 0, \hat{k} \times \vec{B} = 0$

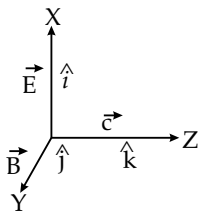
[NCERT Exemp. Q. 8.9, Page 49]

Ans. Correct options : (a), (b), and (c)

Explanation : **(a)** The direction of propagation of an electromagnetic wave is always along the direction of vector product $\vec{E} \times \vec{B}$. Refer to Figure.

$$\vec{B} = B\hat{j} = B(\hat{k} \times \hat{i}) = \frac{E}{c}(\hat{k} \times \hat{i})$$

$$= \frac{1}{c}(\hat{k} \times E\hat{i}) = \frac{1}{c}(\hat{k} \times \vec{E}) \quad \left(\text{as } \frac{E}{B} = c\right)$$



(b) $\vec{E} = E\hat{i} = cB(\hat{j} \times \hat{k}) = c(B\hat{j} \times \hat{k}) = c(\vec{B} \times \hat{k})$

(c) $\hat{k} \cdot \vec{E} = \hat{k} \cdot (E\hat{i}) = 0, \vec{k} \cdot \vec{B} = \hat{k} \cdot (B\hat{j}) = 0$

(d) $\hat{k} \times \vec{E} = \hat{k} \times (E\hat{i}) = E(\hat{k} \times \hat{i}) = E\hat{j}$

and $\hat{k} \times \vec{B} = \hat{k} \times (B\,\hat{j}) = B(\hat{k} \times \hat{j}) = -B\hat{i}$

Q. 10. A plane electromagnetic wave propagating along x direction can have the following pairs of E and B

(a) E_x, B_y. **(b)** E_y, B_z.

(c) B_x, E_y. **(d)** E_z, B_y.

[NCERT Exemp. Q. 8.10, Page 49]

Ans. Correct options : (b) and (d)

Explanation : As the EM wave is plane polarised and its propagation is in + x direction. So, direction of $\vec{E}$ and $\vec{B}$ will be in either y and z direction or z and y direction. So, verifies answers (b) and (d).

Q. 11. A charged particle oscillates about its mean equilibrium position with a frequency of 10^9 Hz. The electromagnetic waves produced :

(a) will have frequency of 10^9 Hz.

(b) will have frequency of 2×10^9 Hz.

(c) will have a wavelength of 0.3 m.

(d) fall in the region of radio waves.

[NCERT Exemp. Q. 8.11, Page 50]

Ans. Correct options : (a), (c), and (d)

Explanation : Here it is given that the frequency by which the charged particles oscillates about its mean equilibrium position, it is equal to 10^9 Hz. The frequency of electromagnetic waves produced by a charged particle is equal to the frequency by which it oscillates about its mean equilibrium position. So, frequency of electromagnetic waves produced by the charged particle is $\upsilon = 10^9$ Hz.

Wavelength $\lambda = \frac{c}{v} = \frac{3\times10^8}{10^9} = 0.3$ m

The frequency of 10^9 Hz falls in the region of radio waves.

Q. 12. The source of electromagnetic waves can be a charge

(a) moving with a constant velocity.

(b) in a circular orbit.

(c) at rest.

(d) falling in an electric field.

[NCERT Exemp. Q. 8.12, Page 50]

Ans. Correct options : (b) and (d)

Explanation : An electromagnetic wave can be produced by accelerated or oscillating charge. An oscillating charge is accelerating continuously; it will radiate electromagnetic waves continuously. Electromagnetic waves are also produced when fast moving electrons are suddenly stopped by a metal target of high atomic number.

Motion of a particle in circular orbit is accelerated motion verifies answer (b).

When a charge particle falls in electric field, the velocity of charge particle changes so its motion becomes accelerated and can produce E.M. wave. It verifies answer (d).

Q. 13. An E.M. wave of intensity I falls on a surface kept in vacuum and exerts radiation pressure *p* on it. Which of the following are true?

(a) Radiation pressure is *I/c* if the wave is totally absorbed.

(b) Radiation pressure is *I/c* if the wave is totally reflected.

(c) Radiation pressure is $\frac{2I}{c}$ if the wave is totally reflected.

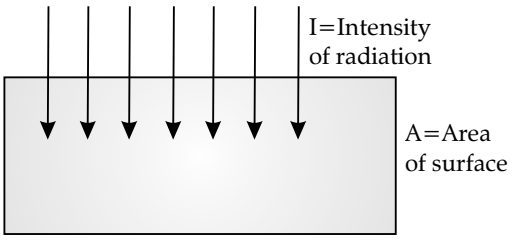
(d) Radiation pressure is in the range of $\frac{I}{c} < p < \frac{2I}{c}$ for real surfaces. **[NCERT Exemp. Q. 8.13, Page 50]**

Ans. Correct options : (a), (c) and (d)

Explanation : Radiation pressure is the force exerted by particles (dual nature of particle) on unit area, due to the change in momentum of radiated particles per unit area per sec $= \frac{1}{c}$.

I= intensity of radiation

c= velocity of radiation



Radiations are absorbed, so momentum per unit area per second $= \frac{I}{c}$, verify the answer (a).

When radiation is reflected back, the momentum becomes double as in earlier case, so discards answer (b) and verifies answer (c).

Therefore, variation of radiation pressure p comes between the range $\frac{I}{c} < p < \frac{2I}{c}$,verifies answer (d).

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Why is the orientation of the portable radio with respect to broadcasting station important?

[NCERT Exemp. Q. 8.14, Page 50]

Ans. Transmitted carrier wave signals are plane polarised and if the intensity of signal is poor then receiving antenna of radio must be parallel to the component of either electric or magnetic field. **[1]**

Because energy is only due to amplitudes of electric and magnetic components in EM wave, magnitude of amplitude is in particular direction perpendicular to each other and perpendicular to wave propagation. **[1]**

Q. 2. Why does microwave oven heats up a food item containing water molecules most efficiently?

[NCERT Exemp. Q. 8.15, Page 50]

Ans. In microwave oven, molecules of food item start to vibrate by driven force due to microwaves with the frequency of microwave. But the natural frequency of water molecules matches with microwave frequency which causes resonance (more amplitude) which further causes increase in temperature. **[1]**

Q. 3. The charge on a parallel plate capacitor varies as $q=q_0$ cos 2πνt. The plates are very large and close together (area = *A*, separation = *d*). Neglecting the edge effects, find the displacement current through the capacitor?

[NCERT Exemp. Q. 8.16, Page 50]

Ans. The displacement current I_d through the capacitor is given by,

$$I_d = I_c = \frac{dq}{dt} \quad ...(i)$$

Here we are given, $q = q_0 \cos 2\pi\nu t$

Putting this value in *Eq* (i), we get

$$I_d = I_c = -q_0 \sin 2\pi\nu t \times 2\pi\nu$$

$$I_d = I_c = -2\pi\nu q_0 \sin 2\pi\nu t \quad \textbf{[2]}$$

Q. 4. A variable frequency a.c source is connected to a capacitor. How will the displacement current change with decrease in frequency?

[NCERT Exemp. Q. 8.17, Page 50]

Ans. Capacitive reactance, $X_C = \frac{1}{2\pi fC}$ **[½]**

So that, $X_C \propto \frac{1}{f}$

As frequency decreases, X_C increases and the conduction current is inversely proportional to $X_C \left(\because I \propto \frac{1}{X_C}\right)$ **[½]**

So, the displacement current decreases and frequency decreases when the conduction current is equal to displacement current. **[1]**

Q. 5. The magnetic field of a beam emerging from a filter facing a floodlight is given by $B = 12\times10^{-8}$ sin$(1.20\times10^{7}z - 3.6\times10^{15}t)$ T. What is the average intensity of the beam?

[NCERT Exemp. Q. 8.18, Page 50]

Ans. We know that the equation of magnetic field can be expressed as $B = B_0 \sin \omega t$.

So, $B = 12\times10^{-8}\sin(120\times10^{7}\,z - 3.60\times10^{15}t)T$

On comparing this equation with standard equation, we get

$B_0 = 12\times10^{-8}$ T

The average intensity of the beam

$$I_{av} = \frac{1}{2}\frac{B_0^2}{\mu_0}\cdot c = \frac{1}{2}\times\frac{(12\times10^{-8})^2\times3\times10^{8}}{4\pi\times10^{-7}}$$

$$= 1.71 \text{ W/m}^2$$ **[2]**

Q. 6. Pointing vectors S is defined as a vector whose magnitude is equal to the wave intensity and whose direction is along the direction of wave propagation. Mathematically, it is given by $S = \frac{1}{\mu_0}\left(\vec{E}\times\vec{B}\right)$. Show the nature of S versus t graph.

[NCERT Exemp. Q. 8.19, Page 50]

Ans. Consider an electromagnetic wave. Let electric field (E) of EM wave varies along y-axis the propagation of wave is along x-axis, then $\left(\vec{E}\times\vec{B}\right)$ will give the direction of flow of energy in electromagnetic wave along x-axis. **[½]**

Let $\vec{E} = E_0\sin(\omega t - kx)\hat{j}$

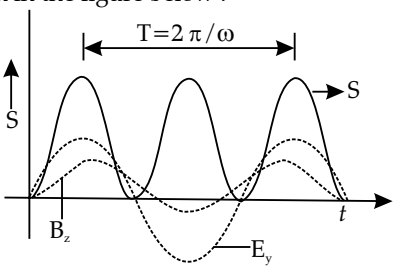
$\vec{B} = B_0\sin(\omega t - kx)\hat{k}$

$$S = \frac{1}{\mu_0}(\vec{E}\times\vec{B}) = \frac{1}{\mu_0}E_0B_0\sin^2(\omega t - kx)(\hat{j}\times\hat{k})$$

$$\Rightarrow \quad S = \frac{E_0B_0}{\mu_0}\sin^2(\omega t - kx)\hat{i} \quad (\text{As } \hat{j}\times\hat{k} = \hat{i})$$ **[½]**

Since, $\sin^2(\omega t - kx)$ is never negative, $\vec{S}(x,t)$ always points in the positive x-direction, *i.e.*, in the direction of wave propagation. **[½]**

The variation of $|S|$ with the time T will be given as given in the figure below :



[½]

Q. 7. Professor C.V Raman surprised his students by suspending freely a tiny light ball in a transparent vacuum chamber by shining a laser beam on it. Which property of EM waves was exhibiting? Give one more example of this property.

[NCERT Exemp. Q. 8.20, Page 51]

Ans. We know that EM wave carries both energy and momentum. Due to this change in momentum (by direction or velocity of wave), EM wave exert pressure on the surface, by reflection and refraction. This property of EM wave helped professor CV Raman to surprise his students by suspending freely a tiny light ball in transparent vacuum chamber by shining a laser beam on it. The tails of the comets are also due to radiation pressure. **[1]**

Electromagnetic radiations can pass even through vacuum and has particle nature. Mobile phone placed in evacuated transparent chamber can ring up which can be seen through transparent chamber, but sound of ringtone cannot be heard, proves the propagation of electromagnetic wave in vacuum. **[1]**

Q. 8. For X-rays of wavelength 10^{-10} m, red light of wavelength 6800 Å and radio waves of wavelength 500 m, what physical quantity could be the same?

[NCERT Ex. Q. 8.3, Page 286]

Ans. The speed of light (3×10^8 m/s) in a vacuum is the same for all wavelengths. It is independent of the wavelength in the vacuum. **[1]**

Q. 9. What can be understood about the directions of magnetic and electric field vectors of a plane electromagnetic wave travelling in vacuum along z-direction. What is the wavelength of the electromagnetic wave when its frequency is 30 MHz? **[NCERT Ex. Q. 8.4, Page 286]**

Ans. The electromagnetic wave travels in a vacuum along the z-direction. The electric field (E) and the magnetic field (H) are in the x-y plane. They are mutually perpendicular. Frequency of the wave, $\upsilon = 30$ MHz $= 30\times10^6$ s^{-1}

Speed of light in vacuum, c = 3×10^8 m/s

Wavelength of a wave is given as :

$$\lambda = \frac{c}{\nu}$$

$$= \frac{3\times10^8}{30\times10^6} = 10 \text{ m}$$ **[2]**

Q. 10. What is the wavelength band of a radio that can tune in to any station in the 7.5 MHz to 12 MHz band? **[NCERT Ex. Q. 8.5, Page 286]**

Ans. A radio can tune to minimum frequency, ν_1=7.5 MHz = 7.5×10^6 Hz

Maximum frequency, ν_2=12 MHz=12×10^6 Hz

Speed of light, $c = 3\times10^8$ m/s

Corresponding wavelength for ν_1 can be calculated as :

$$\lambda_1 = \frac{c}{\nu_1} = \frac{3\times10^3}{7.5\times10^6} = 40 \text{ m}$$

Corresponding wavelength for υ_2 can be calculated as :

$$\lambda_2 = \frac{c}{\nu_2} = \frac{3\times10^3}{12\times10^6} = 25 \text{ m}$$

Thus, the wavelength band of the radio is 40 m to 25 m. **[2]**

Q. 11. What is the frequency of the electromagnetic waves produced by the oscillator which oscillates a charged particle about its mean equilibrium position with a frequency of 10^9 Hz?

[NCERT Ex. Q. 8.6, Page 286]

Ans. The frequency of an electromagnetic wave produced by the oscillator is same as that of a charged particle oscillating about its mean position, *i.e.*, 10^9 Hz. **[1]**

Q. 12. What is the amplitude of the electric field part of the harmonic electromagnetic wave whose amplitude of the magnetic field part in vacuum is B_0=510 nT? [NCERT Ex. Q. 8.7, Page 286]

Ans. Amplitude of magnetic field of an electromagnetic wave in a vacuum,
B_0=510 nT=510×10^{-9} T
Speed of light in vacuum, $c = 3\times10^8$ m/s
Amplitude of electric field of an electromagnetic wave is given by the relation,
$E=cB_0=3\times10^8 \times 510 \times 10^{-9} = 153$ N/C
Therefore, the electric field part of the wave is 153 N/C. **[2]**

Q. 13. Why are microwaves considered suitable for radar systems used in aircraft navigation? [CBSE Board, Delhi Region, 2016]

Ans. Microwaves are considered suitable for radar systems used in aircraft navigation because they have a short wavelength range which makes them suitable for long-distance communication. **[1]**

Q. 14. Do electromagnetic waves carry energy and momentum? [CBSE Board, All India Region, 2017]

Ans. Yes, EM waves carry both energy and momentum. Any type of particle or wave that is in motion carries momentum. Electromagnetic waves like any other wave carry energy as they travel through empty space. If light didn't carry energy, it wouldn't be able to heat stuff up or generate photo current in photo cells. **[1]**

Q. 15. Identify the electromagnetic waves whose wavelengths vary as
(a) 10^{-12} m < λ < 10^{-8} m (b) 10^{-3} m < λ < 10^{-1} m
Write one use for each. [CBSE Board, All India Region, 2017]

Ans. **(a)** X-rays with wavelengths 10^{-12} m < λ < 10^{-8} m are used for medical purposes. **[1]**
(b) Microwaves with wavelengths 10^{-3} m < λ < 10^{-1} m are used in radar systems. **[1]**

Q. 16. Why are infrared waves often called heat waves? Explain. [CBSE Board, Delhi Region, 2018]

Ans. Infrared waves are often called heat waves because water molecules present in most of the materials readily absorb infrared waves (many other molecules, for example, CO_2, NH_3, also absorb infrared waves). After absorption, their thermal motion increases, *i.e.*, they heat up and heat their surroundings. **[1]**

Q. 17. What do you understand by the statement, "Electromagnetic waves transport momentum"? [CBSE Board, Delhi Region, 2018]

Ans. Electromagnetic waves transport momentum means when these waves strike a surface, a pressure is exerted on the surface. Electromagnetic wave means light. Light contains photons that have momentum p=hf/c. In the laws of physics, momentum and energy should be conserved for any interaction. In photoelectric effect and Compton scattering, a photon transports it's momentum and energy to the electron. **[1]**

Q. 18. (i) Which segment of electromagnetic waves has highest frequency? How are these waves produced? Give one use of these waves.
(ii) Which EM waves lie near the high frequency end of visible part of EM spectrum? Give its one use. In what way this component of light has harmful effects on humans? [CBSE Board, Foreign Scheme, 2016]

Ans. **(i)** The gamma rays are just like other electromagnetic rays which are emitted by different sources under different circumstances. These are emitted by the hottest part of our universe and also by the radioactive decay of certain material, like Cobalt-60, Zinc-65, Cesium-137, and Radium-226. Owing to their shorter wavelength they travel larger distances and can also penetrate more than other types of electromagnetic rays. Example : It is used for treatment of cancerous cell in our body without the use of the surgery. **[1]**
(ii) Ultraviolet rays
Example : Used in LASIK eye surgery, UV lamps to kill germs in water purifier.
Side Effects : Causes sunburn/skin cancer/harms eyes when exposed to direct UV rays **[1]**

Q. 19. How is the speed of electromagnetic waves in vacuum determined by the electric and magnetic fields? [CBSE Board, Delhi Region, 2017]

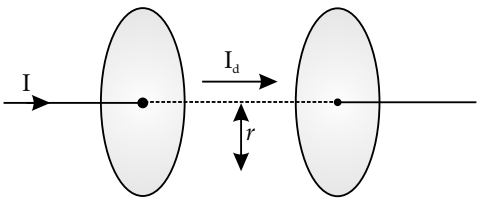
Ans. To determine the speed of electromagnetic waves in vacuum we use following formula :
$c = \frac{E_0}{B_0}$. **[1]**

Short Answer Type Questions

(3 or 4 marks each)

Q. 1. Show that the magnetic field B at a point in between the plates of a parallel-plate capacitor during charging is $\frac{\mu_r \varepsilon_r}{2}\frac{dE}{dt}$ (symbols having usual meaning). [NCERT Exemp. Q. 8.21, Page 51]

Ans. Let I_d be the displacement current in the region of magnetic field between two plates of parallel plate capacitor. **[½]**



[½]

The magnetic field induction at a point in a region between two plates of a capacitor at a perpendicular distance from the axis of plate is : **[1]**

$$B = \frac{\mu_0 2I_d}{4\pi r} = \frac{\mu_0}{2\pi r} I_d = \frac{\mu_0}{2\pi r} \times \varepsilon_r \frac{d\phi_E}{dt} \quad \left[\because I_d = \frac{\varepsilon_r d\phi_E}{dt}\right]$$

$$\Rightarrow B = \frac{\mu_0 \varepsilon_r}{2\pi r} \frac{d}{dt}(E\pi r^2) = \frac{\mu_0 \varepsilon_r}{2\pi r} \pi r^2 \frac{dE}{dt} \quad \left[\because \phi_E = E\pi r^2\right]$$

$$\Rightarrow B = \frac{\mu_0 \varepsilon_r}{2} \frac{dE}{dt} \quad \textbf{[1]}$$

Q. 2. Electromagnetic waves with wavelength

(i) λ_1 is used in satellite communication.

(ii) λ_2 is used to kill germs in water purifies.

(iii) λ_3 is used to detect leakage of oil in underground pipelines.

(iv) λ_4 is used to improve visibility in runways during fog and mist conditions.

(a) Identify and name the part of electromagnetic spectrum to which these radiations belong.

(b) Arrange these wavelengths in ascending order of their magnitude.

(c) Write one more application of each.

[NCERT Exemp. Q. 8.22, Page 51]

Ans. (a)(i) Microwave is used in satellite communications so, λ_1 is the wavelength of microwave. It is used in microwave oven. **[½]**

(ii) Ultraviolet rays are used to kill germs in water purifier so, λ_2 is the wavelength of ultraviolet rays. **[½]**

(iii) X-rays are used to detect leakage of oil in underground pipelines, so λ_3 is in X-ray region. It is also used to detect cracks in machinery and to detect fracture in bones of body. **[½]**

(iv) Infrared rays λ_4 are used to improve visibility due to larger wavelength of low scattering. Infrared rays are used in optical communication. **[½]**

(b) The arrangement of the wavelengths in ascending order are $\lambda_3 < \lambda_2 < \lambda_4 < \lambda_1$. **[½]**

(c) Uses are along with part (a).

Radiation	Uses
γ-rays	Provide information on nuclear structure, medical treatment, etc.
X-rays	Medical diagnosis and treatment study of crystal structure, industrial radiograph.
UV-rays	Preserve food, sterilising the surgical instruments, detecting the invisible writings, finger prints, etc.
Visible light	To see objects.
Infrared rays	To treat, muscular strain for taking photography during the fog, haze, etc.
Microwave and Radio Wave	In radar and communication

[1½]

Q. 3. Show that average value of radiant flux density 'S' over a single period 'T' is given by $S = \frac{1}{2c\mu_0} E_0^2$.

[NCERT Exemp. Q. 8.23, Page 51]

Ans. We know that radiant flux density is given as :

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) = c^2 \varepsilon_0 (\vec{E} \times \vec{B}) \quad \left[\because c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}\right] \quad \textbf{[½]}$$

Let electromagnetic waves be propagating along x-axis, if electric field vector of electromagnetic wave be along y-axis, then magnetic field vector be along z-axis. Therefore,

$$\vec{E} = E_0 \cos(kx - \omega t)$$

and $\vec{B} = B_0 \cos(kx - \omega t)$

$$\vec{E} \times \vec{B} = (E_0 \times B_0) \cos^2 (kx - \omega t)$$

$$S = c^2 \varepsilon_0 (\vec{E} \times \vec{B})$$

$$= c^2 \varepsilon_0 (E_0 \times B_0) \cos^2 (kx - \omega t) \quad \textbf{[1]}$$

Average value of the magnitude of radiant flux density over complete cycle is

$$S_{av} = c^2 \varepsilon_0 \,|\, E_0 \times B_0 \,|\, \frac{1}{T} \int_0^T \cos^2(kx - \omega t) dt$$

$$\Rightarrow \quad S_{av} = c^2 \varepsilon_0 E_0 B_0 \times \frac{1}{T} \times \frac{T}{2} \quad \textbf{[½]}$$

$$\text{As} \left[\int_0^T \cos^2(kx - \omega t) dt = \frac{T}{2}\right]$$

$$\Rightarrow \quad S_{av} = \frac{c^2}{2} \varepsilon_0 E_0 \left(\frac{E_0}{c}\right) \quad \left[\text{As } c = \frac{E_0}{B_0}\right]$$

$$\Rightarrow \quad S_{av} = \frac{c}{2} \varepsilon_0 E_0^2$$

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \text{ or } \varepsilon_0 = \frac{1}{c^2 \mu_0}$$

$$\Rightarrow \quad S_{av} = \frac{E_0^2}{2\mu_0 c}.$$

Hence proved. **[1]**

Q. 4. You are given a 2 μF parallel plate capacitor. How would you establish an instantaneous displacement current of 1mA in the space between its plates? [NCERT Exemp. Q. 8.24, Page 51]

Ans. The capacitance of capacitor $C = 2$ μF,

Displacement current $I_d = 1$ mA

Charge on capacitor, $q = CV$

$I_d dt = CdV \qquad [\because q = it]$

or $I_d = C\frac{dV}{dt}$

$$1 \times 10^{-3} = 2 \times 10^{-6} \frac{dV}{dt}$$

or $\frac{dV}{dt} = \frac{1}{2} \times 10^3 = 500$ V/s

Hence by applying a varying potential difference of 500 V/s, we would produce a displacement current of desired value. **[3]**

Q. 5. Show that the radiation pressure exerted by an EM wave of intensity I on a surface kept in vacuum is $\frac{I}{C}$. [NCERT Exemp. Q. 8.25, Page 51]

Ans. Let us consider a surface exposed to electromagnetic radiation. The radiation is falling normally on the surface. Further, intensity of radiation is I and area of surface exposed to radiation is A. **[½]**

Pressure on surface due to particle nature of wave $P = \frac{F}{A}$ **[½]**

Force= Rate of change of momentum

If mass (m) of radiant particle for wave of velocity c then,

$E = mc^2$

Let $E = U$

$U = mc^2$

$U = pc$ **[½]**

Where, momentum $p=mc$

Now differentiating w.r.t time (t) we get,

$$\frac{dU}{dt} = c.\frac{dp}{dt}$$

$$\frac{1}{c}.\frac{dU}{dt} = \frac{dp}{dt}\left[\because \frac{dp}{dt} = F(\text{by Newton's second law})\right]$$

$$\therefore \quad F = \frac{1}{c}\frac{dU}{dt} \text{ or } \frac{F}{A} = \frac{1}{A}.\frac{1}{c}.\frac{dU}{dt}$$ **[1]**

Pressure (P) on surface due to EM wave radiation and $P = \frac{F}{A}$

$$P = \frac{1}{c}.\left[\frac{1}{A}.\frac{dU}{dt}\right]$$

We know that intensity of radiation is equal to the radiant energy (U) falling on unit surface per second.

$$\therefore I = \frac{1}{A}.\frac{dU}{dt}$$

or

$$P = \frac{1}{c}$$

Hence proved. **[½]**

Q. 6. What happens to the intensity of light from a bulb if the distance from the bulb is doubled? As a laser beam travels across the length of a room, its intensity essentially remains constant.
What geometrical characteristic of LASER beam is responsible for the constant intensity which is missing in the case of light from the bulb?
[NCERT Exemp. Q. 8.26, Page 51]

Ans. Bulb spreads its light in all ground spherically and symmetrically. So, if the distance from the bulb is doubled, the surface area covered by radiations change from $4\pi r^2$ to $4\pi(2r)^2$, *i.e.*, 2^2 or four times decreased in straight laser. But in a bulb it decreased by $4(4\pi)$, *i.e.*, 16π decreased. So the intensity becomes one-fourth the initial value in straight line. Since, $I \propto \frac{1}{r^2}$ and for spherical source $I \propto \frac{1}{4\pi r^2}$. **[2]**

As the distance is doubled, so the intensity becomes one-fourth the initial value. But in case of laser it does not spread, so its intensity remains same. **[½]**

Some geometrical characteristics of LASER beam which are responsible for the constant intensity is

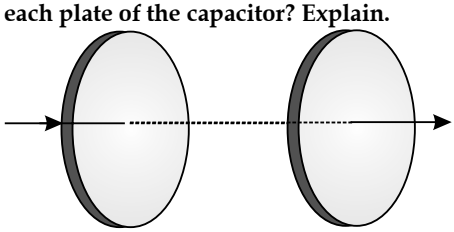
(i) Unidirectional
(ii) Monochromatic
(iii) Coherent light
(iv) Highly collimated **[1]**

These characteristics are missing in the case of normal light from the bulb. **[½]**

Q. 7. Even though an electric field E exerts a force qE on a charged particle yet the electric field of an EM wave does not contribute to the radiation pressure (but transfers energy). Explain.
[NCERT Exemp. Q. 8.27, Page 52]

Ans. The electric field of an electromagnetic wave is an oscillation field. It exerts electric force on a charged particle, but this electric force averaged over an integral number of cycles is zero, since its direction changes every half cycle. Hence, electric field is not responsible for radiation pressure though it transfers the energy. In fact, radiation pressure appears as a result of the action of the magnetic field of the wave on the electric currents induced by the electric field of the same wave. As the electric field vibrates, the radiation of energy will take place only due to vibrating electric and magnetic field. **[3]**

Q. 8. Figure shows a capacitor made of two circular plates each of radius 12 cm, and separated by 5.0 cm. The capacitor is being charged by an external source (not shown in the figure). The charging current is constant and equal to 0.15 A.
(a) Calculate the capacitance and the rate of change of potential difference between the plates.
(b) Obtain the displacement current across the plates.
(c) Is Kirchhoff's first rule (junction rule) valid at each plate of the capacitor? Explain.



[NCERT Ex. Q. 8.1, Page 285]

Ans. According to question we know that,

Radius of each circular plate (r) = 12 cm = 0.12 m

Distance between the plates (d) = 5 cm = 0.05 m

Charging current (I) = 0.15 A

Permittivity of free space, $\varepsilon_0 = 8.85\times10^{-12}\,C^2N^{-1}m^{-2}$

(a) Capacitance between the two plates is given by the relation,

$$C = \frac{\varepsilon_0 A}{d}$$

Where,

A = Area of each plate = πr^2

$$C = \frac{\varepsilon_0 \pi r^2}{d}$$

$$= \frac{8.85\times10^{-12}\times\pi(0.12)^2}{0.05}$$

$$= 8.0032 \times 10^{-12}\ \text{F}$$

$$= 80.032\ \text{pF}$$

Charge on each plate, $q = CV$
Where,
V = Potential difference across the plates
Differentiation on both sides with respect to time (t) gives :

$$\frac{dq}{dt} = C\frac{dV}{dt}$$

But, $\frac{dq}{dt}$ = Current (I)

$$\therefore \quad \frac{dV}{dt} = \frac{I}{C}$$

$$\Rightarrow \frac{0.15}{80.032 \times 10^{-12}} = 1.87 \times 10^{9} \text{ V/s}$$

Therefore, the change in potential difference between the plates is 1.87×10^9 V/s. **[2]**

(b) The displacement current across the plates is the same as the conduction current. Hence, the displacement current, i_d is 0.15 A. **[1]**

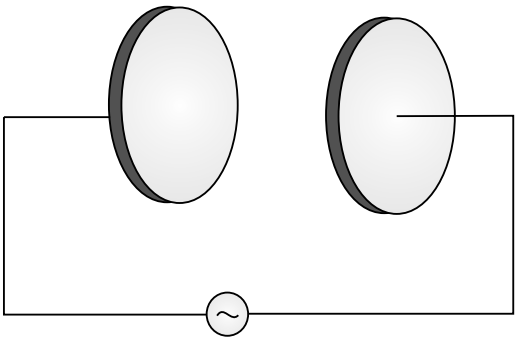
(c) Yes, Kirchhoff's first rule is valid at each plate of the capacitor provided that we take the sum of conduction and displacement for current. **[1]**

Q. 9. A parallel plate capacitor made of circular plates each of radius R = 6.0 cm has a capacitance C = 100 pF. The capacitor is connected to a 230 V ac supply with a (angular) frequency of 300 rad s^{-1}.

(a) What is the rms value of the conduction current?

(b) Is the conduction current equal to the displacement current?

(c) Determine the amplitude of B at a point 3.0 cm from the axis between the plates.



[NCERT Ex. Q. 8.2, Page 286]

Ans. Given that,
Radius of each circular plate, R = 6.0 cm = 0.06 m
Capacitance of a parallel plate capacitor, C = 100 pF = 100×10^{-12} F
Supply voltage, V = 230 V
Angular frequency, $\omega = 300$ rad s^{-1}

(a) RMS value of conduction current, $I = \frac{V}{X_c}$
Where,
X_c = Capacitive reactance

$$X_C = \frac{1}{\omega C}$$

$$\therefore I = V \times \omega C$$

$$= 230 \times 300 \times 100 \times 10^{-12}$$

$$= 6.9 \times 10^{-6} \text{ A}$$

$$= 6.9\ \mu\text{A}$$

Hence, the RMS value of conduction current is 6.9 μA. **[1½]**

(b) Yes, conduction current is equivalent to displacement current., *i.e.*, $i_c = i_d$ **[½]**

(c) Magnetic field is given as :

$$B = \frac{\mu_0 r}{2\pi R^2} I_0$$

Where,

μ_0 = Permeability of free space = $4\pi \times 10^{-7}\ TmA^{-1}$
I_0 = Maximum value of current = $\sqrt{2}\ I$
r = Distance between the plates from the axis
= 3.0 cm = 0.03 m

$$\therefore B = \frac{4\pi \times 10^{-7} \times 0.03 \times \sqrt{2} \times 6.9 \times 10^{-6}}{2\pi \times (0.06)^2}$$

$$= 1.63 \times 10^{-11} \text{ T}$$

Hence, the magnetic field at that point is 1.63 $\times 10^{-11}$ T. **[1]**

Q. 10. Determine,

(a) B_0, ω, k and λ supposing that the electric field amplitude of an electromagnetic wave is E_0=120 N/C and that its frequency is ν = 50 MHz.

(b) Also find expressions for E and B.

[NCERT Ex. Q. 8.8, Page 286]

Ans. Given that,
Electric field amplitude, E_0=120 N/C
Frequency of source, ν = 50 MHz = 50×10^6 Hz
Speed of light, $c = 3 \times 10^8$ m/s

(a) Magnitude of magnetic field strength is given as :

$$B_0 = \frac{E_0}{c}$$

$$= \frac{120}{3 \times 10^8}$$

$$= 4 \times 10^{-7}\ T = 400 \text{ nT}$$

Angular frequency of source is given by :

$$\omega = 2\pi v = 2\pi \times 50 \times 10^6$$

$$= 3.14 \times 10^8 \text{ rad/s}$$

Propagation constant is given as :

$$k = \frac{\omega}{c}$$

$$= \frac{3.14 \times 10^8}{3 \times 10^8} = 1.05 \text{ rad/m}$$

Wavelength of wave is given by :

$$\lambda = \frac{c}{\nu}$$

$$= \frac{3 \times 10^8}{50 \times 10^6} = 6.0 \text{ m}$$ **[2]**

(b) Suppose the wave is propagating in the positive x direction. Then, the electric field vector will be in the positive y-direction and the magnetic field vector will be in the positive z-direction. This is because all three vectors are mutually perpendicular.
Equation of electric field vector is given as :

$$\vec{E} = E_0 \sin(kx - \omega t)\hat{j}$$

$$= 120 \sin[1.05x - 3.14 \times 10^8 \text{t}]\hat{j}$$

$\overline{B} = B_0 \sin(kx - \omega t)\hat{k}$

$\overline{B} = (4\times10^{-7}) \sin(1.05x - 3.14\times10^{8}t)\hat{k}$ **[2]**

Q. 11. Obtain the photon energy in units of eV for different parts of the electromagnetic spectrum using the formula $E = h\nu$ (for energy of a quantum of radiation : photon). How are they obtained different scales of photon energies related to the sources of electromagnetic radiation?

[NCERT Ex. Q. 8.9, Page 286]

Ans. Energy of a photon is given as :

$$E = h\upsilon = \frac{hc}{\lambda}$$

Where,

h = Planck's constant = 6.6×10^{-34} Js

c = Speed of light = 3×10^8 m/s

λ = Wavelength of radiation

$$\therefore E = \frac{6.6\times10^{-34}\times3\times10^{8}}{\lambda} = \frac{19.8\times10^{-26}}{\lambda} \text{J}$$

$$= \frac{19.8\times10^{-26}}{\lambda\times1.6\times10^{-19}} = \frac{12.375\times10^{-7}}{\lambda} \text{ eV}$$ **[1]**

The given tables lists the photon energies for different parts of an electromagnet spectrum for different λ.

λ (m)	10^3	1	10^{-3}	10^{-6}	10^{-8}	10^{-10}	10^{-12}
E (eV)	12.375×10^{-10}	12.375×10^{-7}	12.375×10^{-4}	12.375×10^{-1}	12.375×10^{1}	12.375×10^{3}	12.375×10^{5}

[1½]

The photon energies for the different parts of the spectrum of a source indicate the spacing of the relevant energy levels of the source. **[½]**

Q. 12. About 5% of the power of a 100 W light bulb is converted to visible radiation. What is the average intensity of visible radiation?

(a) at a distance of 1 m from the bulb?

(b) at a distance of 10 m?

Assume that the radiation is emitted isotropically and neglect reflection.

[NCERT Add. Ex. Q. 8.12, Page 287]

Ans. Power rating of bulb, P = 100 W

It is given that about 5% of its power is converted into visible radiation.

$\therefore$ Power of visible radiation,

$$P' = \frac{5}{100}\times100 = 5 \text{ W}$$

Hence, the power of visible radiation is 5 W. **[1]**

(a) Distance of a point from the bulb, d_1 = 1 m

Hence, intensity of radiation at that point is given as :

$$I = \frac{P'}{4\pi d_1^{\,2}}$$

$$= \frac{5}{4\pi(1)^2} = 0.398 \text{ W/m}^2$$ **[1]**

(b) Distance of a point from the bulb, d_2 = 10 m

Hence, intensity of radiation at that point is given as :

$$I = \frac{P'}{4\pi(d_2)^2}$$

$$= \frac{5}{4\pi(10)^2} = 0.00398 \text{ W/m}^2$$ **[1]**

Q. 13. Use the formula $\lambda_m T$ = 0.29 cm K to obtain the characteristic temperature ranges for different parts of the electromagnetic spectrum. What do the numbers that you obtain tell you?

[NCERT Add. Ex. Q. 8.13, Page 287]

Ans. A body at a particular temperature produces a continuous spectrum of wavelengths. In case of a black body, the wavelength corresponding to maximum intensity of radiation is given according to Planck's law. **[1]**

It can be given by the relation,

$$\lambda_m = \frac{0.29}{T} \text{ cmK}$$

Where,

λ_m = Maximum wavelength

T = Temperature

Thus, the temperature for different wavelengths can be obtained as :

For $\lambda_m = 10^{-4}\text{cm}; T = \frac{0.29}{10^{-4}} = 2900°\text{K}$

For $\lambda_m = 5\times10^{-5}\text{cm}; T = \frac{0.29}{5\times10^{-5}} = 5800°\text{K}$

For $\lambda_m = 10^{-6}$ cm; $T = \frac{0.29}{10^{-6}} = 290000°\text{K}$ and so on. **[2]**

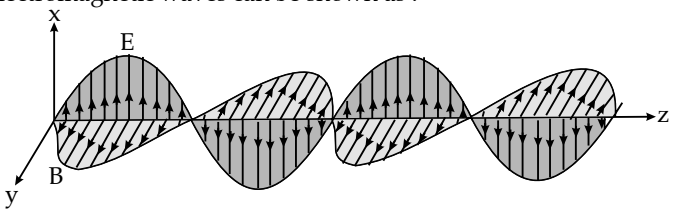
The numbers obtained tell us that temperature ranges are required for obtaining radiations in different parts of an electromagnetic spectrum. As the wavelength decreases, the corresponding temperature increases. **[1]**

Q. 14. How are EM waves produced by oscillating charges? Draw a sketch of linearly polarized EM waves propagating in the Z-direction. Indicate the directions of the oscillating electric and magnetic fields.

[CBSE Board, Delhi Region, 2016]

Ans. A moving charge produces both electric and magnetic fields, and an oscillating charge produces oscillating magnetic and electric fields. These oscillating electric and magnetic fields with respect to space and time produce electromagnetic waves. **[1½]**

The propagation of electromagnetic waves can be shown as :



[1½]

Q. 15. (i) Identify the part of the electromagnetic spectrum which is :

(a) suitable for radar system used in aircraft navigation.

(b) produced by bombarding a metal target by high speed electrons.

(ii) Why does a galvanometer show a momentary deflection at the time of charging or discharging a capacitor? Write the necessary expression to explain this observation.

[CBSE Board, All India Region, 2016]

Ans. (i) (a) Microwaves are suitable for radar system in aircraft navigation.

(b) X-rays are produced by bombarding a metal target by high speed electrons. **[1]**

(ii) As the capacitor is connected to the battery, the electrons start moving towards the plate connected to the negative terminal of the battery and electrons leave from the plate connected to the positive terminal of the battery. This happens until the potential of the capacitor becomes equal to that of the battery. As this happens very quickly, the charging current produces a deflection. The reverse process is repeated for discharging, *i.e.*, the charge is lost. As the galvanometer can be considered as a resistance, the circuit behaves like a R-C circuit having time constant equal to RC. Hence, the expression governing this phenomenon is $q = q_0\left(e^{\frac{-t}{RC}}\right)$. **[2]**

Q. 16. How are electromagnetic waves produce? What is the source of energy of these waves? Write mathematical expressions for electric and magnetic fields of an electromagnetic wave propagating along the z-axis. Write any two important properties of electromagnetic waves.

[CBSE Board, Delhi Region, 2016]

Ans. Electromagnetic waves are produced by accelerated or oscillating charges and do not need any medium for propagation. The source of energy of these waves are energy of oscillating charge particle. **[½]**

Mathematical expressions for electric and magnetic fields of an electromagnetic wave propagating along the z-axis.

$E_x = E_0 \sin [kz - \omega t]$

$B_y = B_0 \sin [kz - \omega t]$ **[1]**

Properties of electromagnetic waves

(i) The direction of E, B, and direction of propagation of waves are mutually perpendicular to one another.

(ii) The amplitude ratio of E and B gives velocity of light.

(iii) Electromagnetic waves are not deflected by electric and magnetic fields. **[1½]**

Q. 17. (a) How are electromagnetic waves produced? Explain.

(b) A plane electromagnetic wave is travelling through a medium along the +ve z-direction. Depict the electromagnetic wave showing the directions of the oscillating electric and magnetic fields. [CBSE Board, Foreign Scheme, 2017]

Ans. (a) A charged particle produces an electric field. The electric field exerts a force on charged particles. Positive charges accelerate in the direction of the field and negative charges accelerate in the direction of the field and negative charges accelerate in a direction opposite to the direction of the field. Also a moving charged particle produces a magnetic field. This field exerts a force on other moving charges. The force on those charges is always perpendicular to the direction of their velocity and therefore only changes the direction of velocity. Since, electromagnetic waves are electric and magnetic field traveling through empty space at the speed of light, then an accelerating charged particle produces an electromagnetic wave. **[1½]**

(b) In an electromagnetic wave, the electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$ perpendicular to the direction of the propagation of wave and also perpendicular to each other .

Since, the wave is travelling along z-direction, $\vec{E}$ and $\vec{B}$ will show their variation in the *x-y* plane

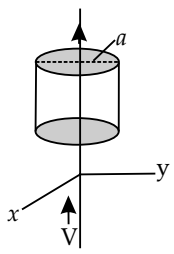
$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8 m/s}{30 \times 10^6 s^{-1}} = 10 \text{ m}$$

along x–y direction

$\lambda = 10\ m$ **[1½]**

Long Answer Type Questions **(5 marks each)**

Q. 1. An infinitely long thin wire carrying a uniform linear static charge density λ is placed along the z-axis in figure. The wire is set into motion along its length with a uniform velocity $\vec{v} = v\hat{K}_z$. Calculate the Pointing vector $S = \frac{1}{\mu_0}\left(\vec{E} \times \vec{B}\right)$.



[NCERT Exemp. Q. 8.28, Page 52]

Ans. Consider a cylindrical Gaussian surface in such a way that the axis of cylinder lies on wire. Electric field intensity due to long straight wire at a distance a and charge density λ C/m. **[1]**

The electric field due to infinitely long thin wire

$$\vec{E} = \frac{\lambda}{2\pi\varepsilon_0 a}\hat{j}$$ **[1]**

Magnetic field due to the wire, $\vec{B} = \frac{\mu_0 i}{2\pi a}\hat{i}$ **[1]**

Equivalent current flowing through the wire,

$$i = \frac{q}{t} = \frac{\lambda l}{t} = \lambda \upsilon$$

Hence $\vec{B} = \frac{\mu_0 \lambda v}{2\pi a}\hat{i}$ **[1]**

$$\therefore \quad \vec{s} = \frac{1}{\mu_0}[\vec{E}\times\vec{B}] = \frac{1}{\mu_0}\left[\frac{\lambda}{2\pi\varepsilon_0 a}\hat{j}\times\frac{\mu_0\lambda v}{2\pi a}\hat{i}\right]$$

$$\vec{s} = \frac{\lambda^2 v}{4\pi^2 a^2\varepsilon_0}\left(\hat{j}\times\hat{i}\right) = \frac{-\lambda^2 v}{4\pi^2 a^2\varepsilon_0}\hat{k}$$ **[1]**

Q. 2. Sea water at frequency ν = 4 × 10^8 Hz has permittivity ε = 80 ε_o, permeability $\mu \approx \mu_o$ and resistivity ρ = 0.25 Ω–m. Imagine a parallel plate capacitor immersed in sea water and driven by an alternating voltage source $V(t) = V_o$ sin (2πνt). What fraction of the conduction current density is the displacement current density?

[NCERT Exemp. Q. 8.29, Page 52]

Ans. Suppose distance between the parallel plates of capacitor is 'd' and the applied voltage :

$V(t) = V_0 \sin(2\pi\nu t)$ **[½]**

Thus, electric field, $E = \frac{V(t)}{d}$

$$\Rightarrow \quad E = \frac{V_0}{d}\sin(2\pi\nu t)$$ **[1]**

Now using Ohm's law, the conduction current density, $J_c = \frac{E}{\rho} = \frac{1}{\rho}\frac{V_0}{d}\sin(2\pi\nu t)$

$$\Rightarrow \quad J_c = \frac{E}{\rho d}\sin(2\pi\nu t) = J_{c0}\sin 2\pi\nu t$$ **[1]**

Here, $J_0^c = \frac{V_0}{\rho d}$

The displacement current density is given as :

$$J_d = \varepsilon\frac{dE}{dt} = \varepsilon\frac{d}{dt}\left[\frac{V_0}{d}\sin(2\pi\nu t)\right]$$

$$= \frac{\varepsilon 2\pi\nu V_0}{d}\cos(2\pi\nu t)$$

$$\Rightarrow \quad J_d = J_{d0}\cos(2\pi\nu t)$$ **[1]**

Let, $J_0^d = \frac{2\pi\nu\varepsilon V_0}{d}$

$$\Rightarrow \quad \frac{J_0^d}{J_0^c} = \frac{\frac{2\pi\nu\varepsilon V_0}{d}}{\frac{V_0}{\rho d}} = 2\pi\nu\varepsilon\rho$$

$$= 2\pi\times 80\varepsilon_0\nu\times 0.25 = 4\pi\varepsilon_0\nu\times 10$$

$$\Rightarrow \quad \frac{J_0^d}{J_0^c} = \frac{10\times 4\times 10^8}{9\times 10^9} = \frac{4}{9}$$ **[1½]**

Q. 3. A long straight cable of length l is placed symmetrically along z-axis and has radius a ($a << l$). The cable consists of a thin wire and a co-axial conducting tube. An alternating current $I(t) = I_o$ sin (2πνt) flows down the central thin wire and returns along the co-axial conducting tube. The induced electric field at a distance s from the wire inside the cable is

$$\vec{E}(s,t) = \mu_0 I_0 \nu\cos(2\pi\nu t)\log_e\left(\frac{s}{a}\right)\hat{k}.$$

(i) Calculate the displacement current density inside the cable.

(ii) Integrate the displacement current density across the cross-section of the cable to find the total displacement current I_d.

(iii) Compare the conduction current I_0 with the displacement current $I_o{}^d$.

[NCERT Exemp. Q. 8.30, Page 52]

Ans. **(i)** We know that, the induced electric field at a distance r from the wire inside the cable.

$$\vec{E}(s,t) = \mu_0 I_0 \nu\cos(2\pi\nu t)\ ln\left(\frac{s}{a}\right)\hat{k}$$

The displacement current density is given by

$$\vec{J}_d = \varepsilon_0\frac{d\vec{E}}{dt}$$

$$= \varepsilon_0\frac{d}{dt}\left[\mu_0 I_0\nu\cos(2\pi\nu t)\ ln\left(\frac{s}{a}\right)\hat{k}\right]$$

$$= \varepsilon_0\mu_0 I_0\nu\frac{d}{dt}[\cos 2\nu t]\ ln\left(\frac{s}{a}\right)\hat{k}$$

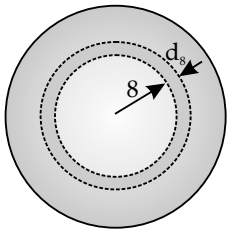
$$= \frac{1}{c^2}I_0\nu^2 2\pi[-\sin 2\pi\nu t]\ ln\left(\frac{s}{a}\right)\hat{k}$$

$$= \frac{\nu^2}{c^2}2\pi I_0\sin 2\pi\nu t\ ln\left(\frac{a}{s}\right)\hat{k}$$

$$= \frac{1}{\lambda^2}2\pi I_0\ ln\left(\frac{a}{s}\right)\sin 2\pi\nu t\hat{k}$$

$$\Rightarrow \vec{J}_d = \frac{2\pi I_0}{\lambda^2}\ ln\left(\frac{a}{s}\right)\sin 2\pi\nu t\,\hat{k}$$ **[1½]**

(ii) Total displacement current, $I_d = \int J_d 2\pi s ds$



[½]

$$I_d = \int J_d.\text{sds}\ d\theta = \int_0^a J_d.s\ ds\int_0^{2\pi} d\theta = \int_0^a J_d.\ sds[2\pi]$$

$$= 2\pi\int_0^a \frac{2\pi I_0}{\lambda^2}\left[\log_e\left(\frac{a}{s}\right)\sin(2\pi\nu t)\hat{k}\right].sds$$

$$I_d = \left(\frac{2\pi}{\lambda}\right)^2 I_0\sin(2\pi\nu t)\hat{k}\int_0^a \log\left(\frac{a}{s}\right).sds$$

Integration of $\int_0^a \log\left(\frac{a}{s}\right).sds$

$$= \left[\log\left(\frac{a}{s}\right)\cdot\int_0^a s\,ds\right]_0^a - \int_0^a\left[\frac{d}{ds}\left[\log\left(\frac{a}{s}\right)\right]\cdot\int s\,ds\right]ds$$

$$= \left[\log\left(\frac{a}{s}\right)\frac{s^2}{2}\right]_0^a - \int_0^a\left(\frac{s}{a}\right)\cdot\frac{s^2}{2}ds$$

$$= \left[\log\left(\frac{a}{a}\right)\cdot\frac{a^2}{2} - 0\right] - \frac{1}{2a}\int_0^a s^3 ds \qquad \textbf{[2]}$$

$$= 0 - \frac{1}{2a}\cdot\left[\frac{s^4}{4}\right]_0^a = -\frac{a^3}{8} \qquad [\because \log_e 1 = 0]$$

$$\therefore I_d = -\frac{a^3}{8}\cdot\left(\frac{2\pi}{\lambda}\right).I_0\sin 2\pi\nu t\,\hat{k}$$

$$= -\frac{a}{2}\cdot\frac{a^2}{4}\left(\frac{2\pi}{\lambda}\right)^2 I_0\sin(2\pi\nu t)\hat{k}$$

The negative sign shows that the displacement current I_d is opposite to the conduction current I_C.

$$\therefore I_d = \frac{a}{2}\left(\frac{2\pi a}{2\lambda}\right)^2 I_0\sin(2\pi\nu t)(-\hat{k})$$

$$I_d = \frac{a}{2}\left(\frac{\pi a}{\lambda}\right)^2 I_0\sin(2\pi\nu t)(-\hat{k})$$

I_d is in-Z direction and I_c is in $+Z$ direction.

(iii) $$I_d = \frac{a}{2}\left(\frac{\pi a}{\lambda}\right)^2 I_0\sin(2\pi\nu t)(-\hat{k})$$

$$I_d = I_0^d\sin(2\pi\nu t)$$

Where $I_0^d = \frac{a}{2}\left(\frac{\pi a}{\lambda}\right)^2 I_0$

Required ratio, $\frac{I_0^d}{I_0} = \frac{\frac{a}{2}\left(\frac{\pi a}{\lambda}\right)^2.I_0}{I_0} = \frac{a}{2}\left(\frac{\pi a}{\lambda}\right)^2$

$$\frac{I_0^d}{I_0} = \frac{\pi^2 a^3}{2\lambda^2} \qquad \textbf{[1]}$$

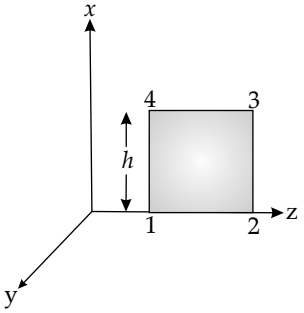
Q. 4. A plane EM wave travelling in vacuum along z-direction is given by $\vec{E} = E_0\sin(kz - \omega t)\,\hat{i}$ and $\vec{B} = B_0\sin(kz - \omega t)\,\hat{j}$.

(i) Evaluate $\int E.dl$ over the rectangular loop 1234 shown in figure.

(ii) Evaluate $\int B.ds$ over the surface bounded by loop 1234.

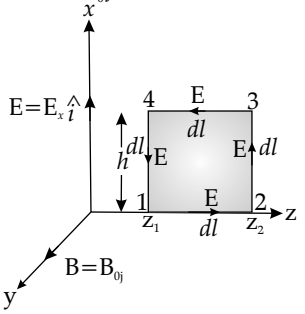
(iii) Use equation $\oint E.dl = \frac{-d\phi_B}{dt}$ to prove $\frac{E_0}{B_0} = c$.

(iv) By using similar process and the equation $\oint B.dl = \mu_0 I + \varepsilon_0\frac{d\phi_E}{dt}$, prove that $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$.



[NCERT Exemp. Q. 8.31, Page 52]

Ans. **(i)** Let electromagnetic wave is propagating along z-axis, in this case electric field vector $(\vec{E})$ be along x-axis and magnetic field $(\vec{B})$ along y-axis, *i.e.*, $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{j}$.



The line integral of $\vec{E}$ over the closed rectangular path 1234 in x-z plane of the figure is :

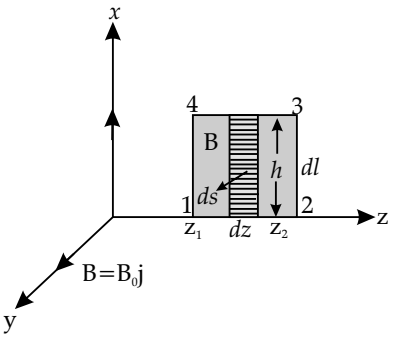
$$\oint\vec{E}\cdot\vec{dl} = \int_1^2\vec{E}\cdot\vec{dl} + \int_2^3\vec{E}\cdot\vec{dl} + \int_3^4\vec{E}\cdot\vec{dl} + \int_4^1\vec{E}\cdot\vec{dl}$$

$$= \int_1^2 E\cdot dl\,\cos 90° + \int_2^3 E\cdot dl\cos 0° + \int_3^4 E\cdot dl\,\cos 90° + \int_4^1 E\cdot dl\,\cos 180°$$

$$\oint\vec{E}\cdot\vec{dl} = E_0 h[\sin(kz_2 - \omega t) - \sin(kz_1 - \omega t)] \qquad \ldots(i)$$

[1]

(ii) Now let us evaluate $\int\vec{B}\cdot\vec{ds}$, let us consider the rectangle 1234 to be made of strips of are $ds = hdz$ each.



$$\int\vec{B}\cdot\vec{ds} = \int B\cdot ds\,\cos 0° = \int B\cdot ds$$

$$= \int_{Z_1}^{Z_2} B_0\sin(kz - \omega t)hdz$$

$$\int\vec{B}\cdot\vec{ds} = \frac{-B_0 h}{k}[\cos(kz_2 - \omega t) - \cos(kz_1 - \omega t)] \qquad \ldots(ii)$$

[1]

(iii) We know that $\oint E\cdot dl = \frac{-d\phi_B}{dt} = -\frac{d}{dt}\oint B\cdot ds$

Substituting the values from Eqs. (i) and (ii), we get

$$E_0 h[\sin(kz_2 - \omega t) - \sin(kz_1 - \omega t)]$$

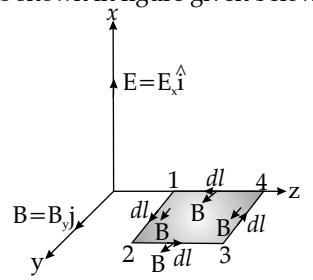
$$= \frac{-d}{dt}\left[\frac{B_0 h}{k}\{\cos(kz_2 - \omega t)\} - \cos(kz_1 - \omega t)\right]$$

$$= \frac{B_0 h}{k}\omega[\sin(kz_2 - \omega t) - \sin(kz_1 - \omega t)]$$

$$\Rightarrow E_0 = \frac{B_0\omega}{k} = B_0 c \qquad \left(\because \frac{\omega}{k} = c\right)$$

$$\Rightarrow \frac{E_0}{B_0} = c \qquad \textbf{[1]}$$

(iv) For evaluating $\oint \vec{B}\cdot\vec{dl}$, let us consider a loop 1234 in *y*-*z* plane as shown in figure given below.



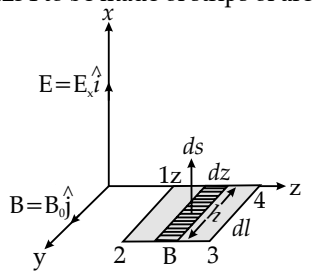
$$\oint \vec{B}\cdot\vec{dl} = \int_1^2 \vec{B}\cdot\vec{dl} + \int_2^3 \vec{B}\cdot\vec{dl} + \int_3^4 \vec{B}\cdot\vec{dl} + \int_4^1 \vec{B}\cdot\vec{dl}$$

$$= \int_1^2 B\cdot dl\cos 0° + \int_2^3 B\cdot dl\cos 90°$$

$$+ \int_3^4 B\cdot dl\cos 180° + \int_4^1 B\cdot dl\cos 90°$$

$$\oint \vec{B}\cdot\vec{dl} = B_0 h[\sin(kz_2 - \omega t) - \sin(kz_1 - \omega t)] \quad \ldots(iii)$$

Now to evaluate $\phi_E = \int \vec{E}\cdot\vec{ds}$, let us consider the rectangle 1234 to be made of strips of area *hds* each.



$$\phi_E = \int \vec{E}\cdot\vec{ds} = \int E ds\cos 0 = \int E ds$$

$$= \int_{z_1}^{z} E_0 \sin(kz_1 - \omega t) h dz$$

$$\oint \vec{E}\cdot\vec{ds} = -\frac{E_0 h}{k}[\cos(kz_2 - \omega t) - \cos(kz_1 - \omega t)]$$

$$\therefore \frac{d\phi_E}{dt} = \frac{E_0 h\omega}{k}[\sin(kz_1 - \omega t) - \sin(kz_2 - \omega t)] \quad \ldots(iv)$$

Let $\oint B\cdot dl = \mu_0\left(I + \frac{\varepsilon_0\phi_E}{dt}\right)$ where I = Conduction current

= 0 in vacuum

$$\therefore \oint B\cdot dl = \mu_0\varepsilon\frac{d\phi_E}{dt}$$

Using relations obtained in Eqs, (iii) and (iv) and simplifying, we get

$$B_0 = E_0\frac{\omega\mu_0\varepsilon_0}{k}$$

$$\Rightarrow \frac{E_0}{B_0}\frac{\omega}{k} = \frac{1}{\mu_0\varepsilon_0}$$

But $\frac{E_0}{B_0} = c$ and $\omega = ck \Rightarrow c\cdot c = \frac{1}{\mu_0\varepsilon_0}$

Therefore, $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$ **[2]**

Q. 5. A plane EM wave travelling along *z*-direction is described by
$E = E_0 \sin(kz - \omega t)\hat{i}$ and $B = B_0 \sin(kz - \omega t)\hat{j}$.
Show that

(i) the average energy density of the wave is given by

$$u_{av} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\frac{B_0^2}{\mu_0}$$

(ii) the time averaged intensity of the wave is given by

$$I_{av} = \frac{1}{2}c\varepsilon_0 E_0^2. \quad \textbf{[NCERT Exemp. Q. 8.32, Page 53]}$$

Ans. **(i)** The electromagnetic wave carries energy which is due to electric field vector and magnetic field vector. In electromagnetic wave, *E* and *B* vary with time.

The energy density due to electric field $\vec{E}$ is

$$U_E = \frac{1}{2}\varepsilon_0 E^2.$$

The energy density due to magnetic field *B* is

$$U_B = \frac{1}{2\mu_0}B^2$$

Total average energy density of electromagnetic wave,

$$U = U_E + U_B = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0}B^2$$

$$E = E_0 \sin(kz - \omega t)\hat{i}$$

$$B = B_0 \sin(kz - \omega t)\hat{j}$$

Average value of E^2 over a cycle $\frac{E_0^2}{2}$

The average value of B^2 over a cycle $= \frac{B_0^2}{2}$

$$\therefore U_{av} = \frac{1}{2}\varepsilon_0\frac{1}{2}E_0^2 + \frac{1}{2\mu_0}\frac{(B_0^2)}{2}$$

$$U_{av} = \frac{1}{4}\left[\varepsilon_0 E_0^2 + \frac{B_0^2}{\mu_0}\right] \quad \textbf{[2½]}$$

(ii) We know that $E_0 = cB_0$ and $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$.

$$\frac{1}{4}\frac{B_0^2}{\mu_0} = \frac{1}{4}\frac{E_0^2/c^2}{\mu_0} = \frac{E_0^2}{4\mu}\mu_0\varepsilon_0 = \frac{1}{4}\varepsilon_0 E_0^2$$

$$U_B = U_E$$

$$U_{av} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\frac{B_0^2}{\mu_0}$$

$$\because \frac{E_0}{B_0} = c \Rightarrow \frac{E_0^2}{B_0^2} = c^2$$

$$\Rightarrow \frac{E_0^2}{B_0^2} = \frac{1}{\mu_0\varepsilon_0} \Rightarrow B_0^2 = \mu_0\varepsilon_0 E_0^2$$

$$\therefore U_{av} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{\mu_0\varepsilon_0^2 E_0^2}{4\mu_0} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\varepsilon_0 E_0^2$$

$$U_{av} = \frac{1}{2}\varepsilon_0 E_0^2$$

$$U_{av} = \frac{1}{4}\varepsilon_0\cdot\frac{B_0^2}{\mu_0\varepsilon_0} + \frac{B_0^2}{4\mu_0}\left[\because E_0^2 = \frac{B_0}{\mu_0\varepsilon_0}\right]$$

$$= \frac{B_0^2}{4\mu_0} + \frac{B_0^2}{4\mu_0}$$

$$U_{av} = \frac{B_0^2}{2\mu_0}$$

$$\therefore U_E = U_B$$

The average intensity of wave,

$$I_{av} = U_{av} \cdot c = \frac{1}{2}\frac{B_0^2 c}{\mu_0} = \frac{1}{2}\varepsilon_0 E_0^2 c$$

[2½]

Q. 6. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude 48 V m^{-1}.

(a) What is the wavelength of the wave?

(b) What is the amplitude of the oscillating magnetic field?

(c) Show that the average energy density of the *E* field equals the average energy density of the *B* field. [$c = 3\times10^8$ m s^{-1}.]

[NCERT Ex. Q. 8.10, Page 286]

Ans. Given that,

Frequency of the electromagnetic wave,

$\nu = 2 \times 10^{10}$ Hz

Electric field amplitude, E_0=48 Vm^{-1}

Speed of light, $c = 3\times10^8$ m/s

(a) Wavelength of a wave is given as :

$$\lambda = \frac{c}{\nu} = \frac{3\times10^8}{2\times10^{10}} = 0.015 \text{ m}$$ **[1]**

(b) Magnetic field strength is given as :

$$B_0 = \frac{E_0}{c} = \frac{48}{3\times10^8} = 1.6\times10^{-7}\ T$$ **[1]**

(c) Energy density of the electric field is given as :

$$U_E = \frac{1}{2}\varepsilon_0 E^2$$

And, energy density of the magnetic field is given as :

$$U_B = \frac{1}{2\mu_0}B^2$$

Where,

ε_0 = Permittivity of free space

μ_0 = Permeability of free space

$E = cB$ …(i)

Where,

$$c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}B$$

Squaring on both sides, we get

$$E^2 = \frac{1}{\varepsilon_0\mu_0}B^2$$

$$\Rightarrow \varepsilon_0 E^2 = \frac{B^2}{\mu_0}$$

Multiplying both sides by $\frac{1}{2}$ **[3]**

$$\frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\frac{B^2}{\mu_0}$$

$$\Rightarrow U_E = U_B$$

Q. 7. Suppose that the electric field part of an electromagnetic wave in vacuum is $E = \{(3.1 \text{ N/C}) \cos[(1.8 \text{ rad/m})\, y + (5.4\times10^6 \text{ rad/s})t]\}\hat{i}$.

(a) What is the direction of propagation?

(b) What is the wavelength λ?

(c) What is the frequency ν?

(d) What is the amplitude of the magnetic field part of the wave?

(e) Write an expression for the magnetic field part of the wave. [NCERT Add. Ex. Q. 8.11, Page 287]

Ans. **(a)** From the given electric field vector, it can be inferred that the electric field is directed along the negative x-direction. Hence, the direction of motion is along the negative y-direction, *i.e.*, $\hat{j}$. **[1]**

(b) It is given that,

$$= 3.1 \text{ N/C} \cos\left[(1.8 \text{ rad/m})y + (5.4\times10^8 \text{ rad/s})t\right] \quad \ldots(1)$$

The general equation for the electric field vector in the positive x-direction can be written as :

$$\vec{E} = E_0 \sin(kx - \omega t)\hat{i} \quad \ldots(2)$$

Comparing equations (1) and (2), we get

Electric field amplitude, E_0 = 3.1 N/C

Angular frequency, ω = 5.4×10^8 rad/s

Wave number, k = 1.8 rad/m

Wavelength, $\lambda = \frac{2\pi}{1.8} = 3.490$ m **[1]**

(c) Frequency of wave is given as :

$$\nu = \frac{\omega}{2\pi} = \frac{5.4\times10^8}{2\pi} = 8.6\times10^7 \text{ Hz}$$ **[1]**

(d) Magnetic field strength is given as :

$$B_0 = \frac{E_0}{c}$$

Where,

c = Speed of light = 3×10^8 m/s

$$\therefore B_0 = \frac{3.1}{3\times10^8} = 1.03\times10^{-8} \text{ T}$$ **[1]**

(e) On observing the given vector field, it can be observed that the magnetic field vector is directed along the negative z-direction. Hence, the general equation for the magnetic field vector is written as :

$$\vec{B} = B_0 \cos(ky + \omega t)\hat{k}$$

$$= \{(1.03\times10^{-8} \text{ T})\cos[(1.8 \text{ rad/m})y + (5.4\times10^6 \text{ rad/s})t]\}\hat{k}$$ **[1]**

Q. 8. Given below are some famous numbers associated with electromagnetic radiations in different contexts in physics. State the part of the electromagnetic spectrum to which each belongs.

(a) 21 cm (wavelength emitted by atomic hydrogen in interstellar space).

(b) 1,057 MHz (frequency of radiation arising from two close energy levels in hydrogen; known as Lamb shift).

(c) 2.7 K [temperature associated with the isotropic radiation filling all space-thought to be a relic of the 'big-bang' origin of the universe].

(d) 5890–5896 Å [double lines of sodium]

(e) 14.4 Kev [energy of a particular transition in ^{57}Fe nucleus associated with a famous high resolution spectroscopic method (Mossbauer spectroscopy)].

[NCERT Add. Ex. Q. 8.14, Page 287]

Ans. **(a)** Radio waves : It belongs to the short wavelength in end of the electromagnetic spectrum. **[1]**

(b) Radio waves : It belongs to the short wavelength end. **[1]**

(c) Temperature, T = 2.7°K

λ_m is given by Planck's law as :

$$\lambda_m = \frac{0.29}{2.7} = 0.11 \text{ cm}$$

This wavelength corresponds to microwaves. **[1]**

(d) This is the yellow light of the visible spectrum. **[½]**

(e) Transition energy is given by the relation, $E = h\nu$ **[½]**

Where,

h = Planck's constant = 6.6×10^{-34} Js

ν = Frequency of radiation

Energy, E = 14.4 keV

$$\therefore \nu = \frac{E}{h} = \frac{14.4 \times 10^3 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 3.4 \times 10^{18} \text{ Hz}$$

This corresponds to X-rays. **[1]**

Q. 9. Answer the following questions :

(a) Long distance radio broadcasts use short-wave bands. Why?

(b) It is necessary to use satellites for long distance TV transmission. Why?

(c) Optical and radio telescopes are built on the ground but X-ray astronomy is possible only from satellites orbiting the Earth. Why?

(d) The small ozone layer on top of the stratosphere is crucial for human survival. Why?

(e) If the Earth did not have an atmosphere, would its average surface temperature be higher or lower than what it is now?

(f) Some scientists have predicted that a global nuclear war on the Earth would be followed by a severe 'nuclear winter' with a devastating effect on life on Earth. What might be the basis of this prediction? [NCERT Add. Ex. Q. 8.15, Page 287]

Ans. **(a)** Long distance radio broadcasts use short-wave bands because only these bands can be refracted by the ionosphere. **[½]**

(b) It is necessary to use satellites for long distance TV transmissions because television signals are of high frequencies and high energies. Thus, these signals are not reflected by the ionosphere. Hence, satellites are helpful in reflecting TV signals. Also, they help in long distance TV transmissions. **[1]**

(c) With reference to X-ray astronomy, X-rays are absorbed by the atmosphere. However, visible and radio waves can penetrate it. Hence, optical and radio telescopes are built on the ground, while X-ray astronomy is possible only with the help of satellites orbiting the Earth. **[1]**

(d) The small ozone layer on the top of the atmosphere is crucial for human survival because it absorbs harmful ultra-violet radiations present in sunlight and prevents it from reaching the Earth's surface. **[1]**

(e) In the absence of an atmosphere, there would be no greenhouse effect on the surface of the Earth. As a result, the temperature of the Earth would decrease rapidly, making it chilly and difficult for human survival. **[½]**

(f) A global nuclear war on the surface of the Earth would have disastrous consequences. Post-nuclear war, the Earth will experience severe winter as the war will produce clouds of smoke that would cover maximum parts of the sky, thereby preventing solar light from reaching the atmosphere. Also, it would lead to the depletion of the ozone layer too. **[1]**

Some Commonly Made Errors

- Generally, students confuse in Maxwell Equations.
- Students cannot define the Right Wave Direction.
- They do not try to take correct assumptions in Wave Propagation.
- They do not remember the limits of the given closed surface if any like circular, plane sheet or cylindrical.

EXPERT ADVICE

- *The problem should be solved under the physical assumptions ka << 1, where a is the characteristic size of the bodies and 'k' is the wave number.*
- *This problem should be solved asymptotically and numerical experiments are provided to illustrate the idea of the method.*
- *Theory for solving electromagnetic wave scattering problem by many small perfectly conducting and impedance bodies was developed.*
- *The basic difference between various types of electromagnetic waves lies in their wavelengths or frequencies since all of them travel through vacuum with the same speed. Consequently, the waves differ considerably in their mode of interaction with matter.*

OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/8zcibqTc4IM **Or Scan the Code**	*https://goo.gl/bGUDzZ* **Or Scan the Code**
https://goo.gl/URQeUB **Or Scan the Code**	*https://goo.gl/Xtzy3N* **Or Scan the Code**

CHAPTER

9 RAY OPTICS AND OPTICAL INSTRUMENTS

Chapter Objectives

This chapter will help you understand :

- ***Reflection and refraction of light :*** *Introduction; Reflection of light; Reflection of light by spherical mirrors; Refraction of light; and Refraction of light through spherical lenses.*
- ***Prism and optical instruments :*** *Total internal reflection; Refraction through prism; Dispersion by a prism; Some natural phenomena due to sunlight; and Optical instrument.*

TOPIC - 1
Reflection and Refraction of Light **P. 194**

TOPIC - 2
Prism and Optical Instruments **P. 213**

TOPIC-1
Reflection and Refraction of Light

Quick Review

- Ray optics is also known as geometrical optics.
- It describes the propagation of light in terms of "rays". Light rays are defined to propagate in a straight path as they travel in a homogeneous medium.
- The origins of the word 'lens' can be traced back to the seventeenth Century.
- The word 'lens' comes from lēns, the Latin name of the lentil, because a double-convex lens is lentil-shaped. The lentil plant also gives its name to a geometric figure.
- The working of a lens is based on the refraction of light when they pass through it.
- Measure of fraction of reflection by a surface is known as the Reflectance.
- The "spherical" eye has the same curve in all directions, and forms one image on the back of the eye (the retina).

Know the Links

- *https://physics.info/lenses/*
- *http://www.physicsinsights.org/simple_optics_spherical_lenses-1.html*
- *https://www.britannica.com/science/physics-science*

TIPS...

- *Understand the important terms related to Spherical mirrors.*
- *Understand relation between all the terms related to spherical mirrors.*
- *Draw and understand the ray diagrams, in order to learn image formation using spherical mirrors.*
- *Understand the sign convention.*
- *Understand the difference between actual reflection and appearing reflection that generates real and virtual images respectively.*

TRICKS...

- *Follow the laws of reflections to solve the numerical.*
- *Understand and remember the mirror formula and magnification to solve numerical problems.*
- *Practice ray diagrams and learn the difference between real and virtual images.*

Multiple Choice Questions (1 mark each)

Q. 1. The phenomena involved in the reflection of radio waves by ionosphere are similar to

(a) reflection of light by a plane mirror.

(b) total internal reflection of light in air during a mirage.

(c) dispersion of light by water molecules during the formation of a rainbow.

(d) scattering of light by the particles of air.

[NCERT Exemp. Q. 9.7, Page 55]

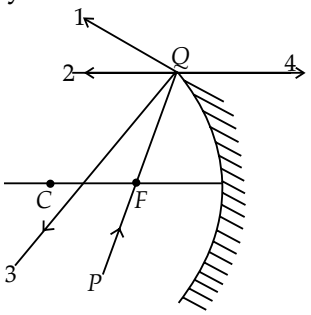
Ans. Correct option : (b)

Explanation : Radiowaves are reflected by a layer of atmosphere called the Ionosphere, so they can reach distant parts of the Earth. The reflection of radiowaves by ionosphere is due to total internal reflection. It is the same as total internal reflection of light in air during a mirage, that is, angle of incidence is greater than critical angle.

Ionosphere is transparent optical medium and radiowave is reflected back.

Reflection through transparent surface is total internal reflection so that internal reflection of radiowave takes place.

Q. 2. The direction of ray of light incident on a concave mirror is shown by *PQ* while directions in which the ray would travel after reflection is shown by four rays marked 1, 2, 3 and 4 (Figure). Which of the four rays correctly shows the direction of reflected ray?



(a) 1 **(b) 2**

(c) 3 **(d) 4**

[NCERT Exemp. Q. 9.8, Page 56]

Ans. Correct option : (b)

Explanation : Incidence ray *PQ* is coming through principal focus *F* so it must be parallel to principal axis, that is, either 2 or 4.

As it is a concave mirror so, ray cannot go behind the mirror so ray (4) is discarded.

So, ray 2 is the reflected ray that verifies answer (b)

Q. 3. A car is moving with at a constant speed of 60 km h^{-1} on a straight road. Looking at the rear-view mirror, the driver finds that the car following him is at a distance of 100 m and is approaching with a speed of 5 kmh^{-1}. In order to keep track of the car in the rear, the driver begins to glance alternatively at the rear and side mirror of his car after every 2 s till the other car overtakes. If the two cars were maintaining their speeds, which of the following statement(s) is/are correct?

(a) The speed of the car in the rear is 65 km/h.

(b) In the side mirror the car in the rear would appear to approach with a speed of 5 km h^{-1} to the driver of the leading car.

(c) In the rear view mirror the speed of the approaching car would appear to decrease as the distance between the cars decreases.

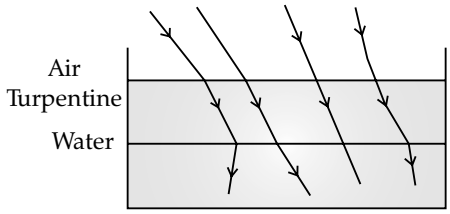
(d) In the side mirror, the speed of the approaching car would appear to increase as the distance between the cars decreases.

[NCERT Exemp. Q. 9.10, Page 56]

Ans. Correct option : (d)

Explanation : So, when rear car approaches, initially it appears at rest as image is formed at focus. When car approaches nearer this speed will appear to increase so answer is (d).

Q. 4. The optical density of turpentine is higher than that of water while its mass density is lower. Figure shows a layer of turpentine floating over water in a container. For which one of the four rays incident on turpentine in Figure, the path shown is correct?



(a) 1 **(b) 2**

(c) 3 **(d) 4**

[NCERT Exemp. Q. 9.9, Page 56]

Ans. Correct option : (b)

Explanation : $\mu_A < \mu_T > \mu_W$

As incidence ray passes from air to turpentine to water it means, from rare to denser then denser to rarer so first it bends towards normal then away from normal so the path shown is correct for ray (2).

Q. 5. A short pulse of white light is incident from air to a glass slab at normal incidence.

After travelling through the slab, the first colour to emerge is

(a) blue. **(b) green.**

(c) violet. **(d) red.**

[NCERT Exemp. Q. 9.2, Page 54]

Ans. Correct option : (d)

Explanation : As we know that the velocity of wave is :

$v = \nu\lambda$

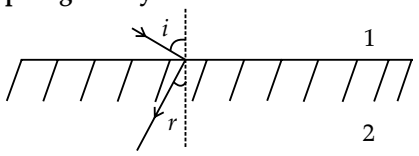
When light ray goes from one medium to other medium, the frequency of light remains unchanged.

So, $v \propto \lambda$ or greater the wavelength, greater the speed.

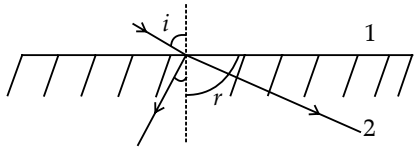
And, the light of red colour is of highest wavelength and therefore of highest speed. So, after travelling through the slab, the red colour emerges first.

Q. 6. There are certain materials developed in laboratories which have a negative refractive index (Figure). A ray incident from air (medium 1) into such a medium (medium 2) shall follow a path given by :

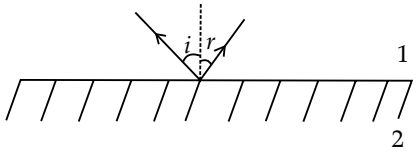
(a)



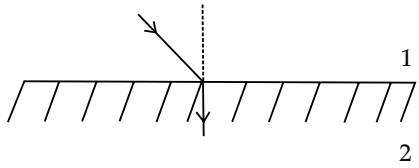
(b)



(c)



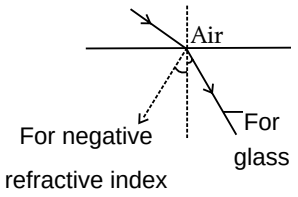
(d)



[NCERT Exemp. Q. 9.11, Page 57]

Ans. Correct option : (a)

Explanation : The negative refractive index materials are those in which incident materials are those in which incident ray from air (medium 1) to them refract or bends differently or opposite and symmetric to normal to that of positive refractive index medium.



Q. 7. An object approaches a convergent lens from the left of the lens with a uniform speed 5 m/s and stops at the focus. The image

(a) moves away from the lens with an uniform speed 5 m/s.

(b) moves away from the lens with an uniform acceleration.

(c) moves away from the lens with a non-uniform acceleration.

(d) moves towards the lens with a non-uniform acceleration. [NCERT Exemp. Q. 9.3, Page 55]

Ans. Correct option : (c)

Explanation : As the object approaches a convergent lens from the left of the lens with a uniform speed of 5 m/s, so the image will move away from the lens with a non-uniform acceleration, and the image moves slower in the beginning and faster later on from F to 2*F* and when the object moves from 2*F* to *F*, the image will move from 2*F* to infinity. At 2*F*, the speed of the object and image will be equal.

Q. 8. The radius of curvature of the curved surface of a planoconvex lens is 20 cm. If the refractive index of the material of the lens be 1.5, it will

(a) act as a convex lens only for the objects that lie on its curved side.

(b) act as a concave lens for the objects that lie on its curved side.

(c) act as a convex lens irrespective of the side on which the object lies.

(d) act as a concave lens irrespective of side on which the object lies. [NCERT Exemp. Q. 9.6, Page 55]

Ans. Correct option : (c)

Explanation : As we know the relations between *f*, μ, R_1 and R_2 is known as lens maker's formula :

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$R_1 = \infty,\ R_2 = -R$

$$f = \frac{R}{(\mu - 1)}$$

Given that,

$R = 20$ cm,

$\mu = 1.5$

Put the values;

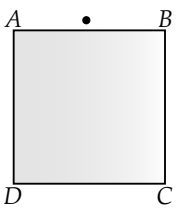
$$f = \frac{R}{\mu - 1} = \frac{20}{15 - 1} = 40 \text{ cm}$$

As $f > 0$, it means converging nature of the lens.

So, lens act as a convex lens irrespective of the side on which the object lies.

Tick Two or More Options

Q. 9. A rectangular block of glass *ABCD* has a refractive index 1.6. A pin is placed midway on the face *AB* (Figure). When observed from the face *AD*, the pin shall



(a) appear to be near *A*.

(b) appear to be near *D*.

(c) appear to be at the centre of *AD*.

(d) not be seen at all.

[NCERT Exemp. Q. 9.13, Page 57]

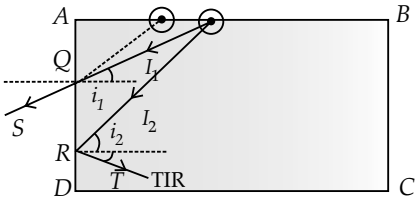
Ans. Correct options : (a) and (b)

Explanation : As we know that,

$$\text{Sin}\, C = \frac{1}{\mu}$$

So that, $\text{Sin}\, C = \frac{1}{1.6}$

$$C = 38.7^\circ$$



A point P is on the mid-point of face AB. When seen through face AD, near to point A, the angle of incidence (i_1) will be smaller than critical angle, $i_c = 38.7°$

So image of P will be at P' and image can be seen at P'. P' is nearer to both A and D as compared to P verifies options (a) and (b).

When seen near point D through face AD, angle of incidence, $i_2 > i_c$.

So total internal reflection takes place and object cannot be observed. But object can be seen when viewed near to A so option (d) not verified.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Explain the following, giving reasons :

(i) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.

(ii) When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a reduction in the energy carried by the wave?

[CBSE Board, All India Region, 2016]

Ans. (i) Frequency and wavelength of light are related as $v = \nu\lambda$

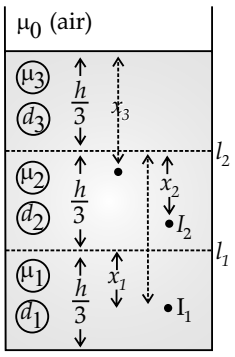
where v is the velocity of light. As the wave travels from one medium to another, the wavelength of light changes with change in the speed of light with it. The frequency remains constant. **[1]**

(ii) The energy of wave depends only on its frequency. Since the frequency does not change as wave travels from one medium to another, therefore, the energy of the wave remains same and does not decrease. **[1]**

Q. 2. Three immiscible liquids of densities $d_1 > d_2 > d_3$ and refractive indices $\mu_1 > \mu_2 > \mu_3$ are put in a beaker. The height of each liquid column is $h/3$. A dot is made at the bottom of the beaker. For near normal vision, find the apparent depth of the dot.

[NCERT Exemp. Q. 9.20, Page 58]

Ans. The liquids are immiscible, so liquids arranged from bottom to top d_1, d_2 and d_3 with their refractive indices μ_1, μ_2, and μ_3 respectively separated by layers l_1 and l_2 is given in figure.



Consider the layer l_1. Let object is at its bottom at P then distance (x_1) of image of P by liquid of refractive index μ_1 from layer l_1.

$$x_1 = \frac{\frac{h}{3}}{{}_2\mu_1}$$

As we know,

$${}_2\mu_1 = \frac{\mu_1}{\mu_2}$$

So,

$$x_1 = \frac{h}{3\frac{\mu_1}{\mu_2}} = \frac{h}{3}\frac{\mu_2}{\mu_1}$$

This image I_1 acts as object for liquid of μ_2 and form the image at I_2.

The distance, x_2 below level I_2.

$$\text{Real depth} = \left(\frac{h}{2} + x_1\right)$$

$$x_2 = \frac{\text{Real depth}}{{}_3\mu_2} = \frac{\text{Real depth}}{\mu_2/\mu_3}$$

$$x_2 = \frac{\mu_3}{\mu_2}\left[\frac{h}{3} + x_1\right]$$

$$= \frac{\mu_3}{\mu_2}\left[\frac{h}{3} + \frac{h}{3}\frac{\mu_2}{\mu_1}\right]$$

$$= \frac{h}{3}\frac{\mu_3}{\mu_2}\left[1 + \frac{\mu_2}{\mu_1}\right]$$

$$= \frac{h}{3}\left[\frac{\mu_3}{\mu_2} + \frac{\mu_3}{\mu_2}\frac{\mu_2}{\mu_1}\right] = \frac{h}{3}\left[\frac{\mu_3}{\mu_2} + \frac{\mu_3}{\mu_1}\right]$$ **[1]**

As the point P is seen from outside

$$x_3 = \frac{\mu_0}{\mu_3}\left[\frac{h}{3} + x_2\right]$$

$$= \frac{1}{\mu_3}\left[\frac{h}{3} + \frac{h}{3}\left(\frac{\mu_3}{\mu_2} + \frac{\mu_3}{\mu_1}\right)\right] \quad \left[\begin{matrix}\mu_0 = 1 \text{ refractive} \\ \text{index of air}\end{matrix}\right]$$

$$= \frac{h}{3}\left[\frac{1}{\mu_3} + \frac{1}{\mu_3}\left(\frac{\mu_3}{\mu_2} + \frac{\mu_3}{\mu_1}\right)\right]$$

$$= \frac{h}{3}\left[\frac{1}{\mu_3} + \frac{\mu_3}{\mu_3}\left(\frac{1}{\mu_2} + \frac{1}{\mu_1}\right)\right] = \frac{h}{3}\left[\frac{1}{\mu_3} + \frac{1}{\mu_2} + \frac{1}{\mu_1}\right]$$

x_3 is apparent depth. **[1]**

Q. 3. A small pin fixed on a table top is viewed from above from a distance of 50 cm. By what distance would the pin appear to be raised if it is viewed

from the same point through a 15 cm thick glass slab held parallel to the table? Refractive index of glass = 1.5. Does the answer depend on the location of the slab?

[NCERT Ex. Q. 9.16, Page 347]

Ans. Given that,

Actual depth of the pin, $d = 15$ cm

Apparent depth of the pin = d'

Refractive index of glass, $\mu = 1.5$

Ratio of actual depth to the apparent depth is equal to the refractive index of glass,

$$\mu = \frac{d}{d'}$$

$$d' = \frac{d}{\mu}$$

$$= \frac{15}{1.5}$$

$$= 10 \text{ cm}$$

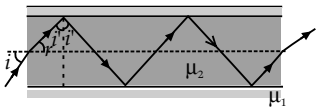
The distance at which the pin appears to be raised $= d' - d$

$= 15 - 10 = 5$ cm

For a small angle of incidence, this distance does not depend on the location of the slab. **[1]**

Q. 4. (a) Figure shows a cross-section of a 'light pipe' made of a glass fibre of refractive index 1.68. The outer covering of the pipe is made of a material of refractive index 1.44. What is the range of the angles of the incident rays with the axis of the pipe for which total reflections inside the pipe take place, as shown the figure.

(b) What is the answer if there is no outer covering of the pipe?



[NCERT Ex. Q. 9.17, Page 347]

Ans. **(a)** Given that,

Refractive index of the glass fibre, $\mu_1 = 1.68$

Refractive index of the outer covering of the pipe, $\mu_2 = 1.44$

Let the,

Angle of incidence = i

Angle of refraction = r

Angle of incidence at the interface = i'

So, the refractive index (μ) of the inner core-outer core interface is :

$$\mu = \frac{\mu_2}{\mu_1}$$

$$= \frac{1}{\sin i'}$$

$$\sin i' = \frac{\mu_1}{\mu_2}$$

$$= \frac{1.44}{1.68}$$

$$= 0.8571$$

So,

$$i' = 59^\circ$$

For the critical angle, total internal reflection (TIR) takes place only when $i > i'$,

So that, $i > 59^\circ$

Maximum angle of reflection, $r_{max} = 90^\circ - i'$

$$= 90^\circ - 59^\circ = 31^\circ$$

Let, i_{max} be the maximum angle of incidence.

The refractive index at the air-glass interface, $\mu_1 = 1.68$

We have the relation for the maximum angles of incidence and reflection as :

$$\mu_1 = \frac{\sin i_{max}}{\sin r_{max}}$$

$$\sin i_{max} = \mu_1 \sin r_{max}$$

$$= 1.68 \sin 31^\circ$$

$$= 1.68 \times 0.5150$$

$$= 0.8652$$

Therefore, $i_{max} = \sin^{-1} 0.8652 \approx 60^\circ$

So that, all the rays incident at angles lying in the range $0 < i < 60^\circ$ will suffer total internal reflection. **[1]**

(b) If the outer covering of the pipe is not present, then :

Refractive index of the outer pipe, μ_1 = refractive index of air = 1

For the angle of incidence $i = 90^\circ$,

As we now the Snell's law at the air-pipe interface as :

$$\frac{\sin i}{\sin r} = \mu_2 = 1.68$$

$$\sin r = \frac{\sin 90^\circ}{1.68} = \frac{1}{1.68}$$

$$r = \sin^{-1}(0.5952) = 36.5^\circ$$

$$\therefore \quad i' = 90^\circ - 36.5^\circ = 53.5^\circ$$

Since $i' > r$, So that all incident rays will suffer total internal reflection. **[1]**

Q. 5. Will the focal length of a lens for red light be more, same or less than that for blue light?

[NCERT Exemp. Q. 9.17, Page 58]

Ans. As we know that $\mu_v > \mu_R$,

So, $\frac{1}{f}$ will be large for blue light and smaller for red coloured light. That's why, f will be larger for red coloured light. **[1]**

Q. 6. The image of a small electric bulb fixed on the wall of a room is to be obtained on the opposite wall 3 m away by means of a large convex lens. What is the maximum possible focal length of the lens required for the purpose?

[NCERT Ex. Q. 9.19, Page 347]

Ans. Given that,

Distance between the object and the image, $d = 3$ m

Maximum focal length of the convex lens = f_{max}

For real images, the maximum focal length is given as :

$$f_{max} = \frac{d}{4} = \frac{3}{4} = 0.75 \text{ m}$$

So, for the required purpose, the maximum possible focal length of the convex lens is 0.75 m. **[1]**

Q. 7. A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different locations separated by 20 cm. Determine the focal length of the lens.
[NCERT Ex. Q. 9.20, Page 347]

Ans. Given that,
Distance between the image (screen) and the object, $D = 90$ cm
Distance between two locations of the convex lens, $d = 20$ cm
Let the focal length of the lens $= f$
Focal length is related to d and D as :

$$f = \frac{D^2 - d^2}{4D}$$
$$= \frac{(90)^2 - (20)^2}{4 \times 90}$$
$$= \frac{770}{36}$$
$$= 21.39 \text{ cm}$$

So, the focal length of the convex lens is 21.39 cm. **[2]**

Short Answer Type Questions (3 marks each)

Q. 1. (a) Calculate the distance of an object of height h from a concave mirror of radius of curvature 20 cm, so as to obtain a real image of magnification 2. Find the location of image also.
(b) Using mirror formula, explain why a convex mirror always produces a virtual image.
[CBSE Board, Delhi Region, 2016]

Ans. (a) Given that,
Radius of curvature of mirror, $R = 20$ cm
So, focal length of mirror, $f = \frac{R}{2} = -10$ cm
It is given that the image is real, so that means magnification of image, $m = -2$

$$m = \frac{-v}{u}$$
$$-2 = \frac{-v}{u}$$
$$v = 2u$$

Using mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$
$$= \frac{1}{2u} + \frac{1}{u} = \frac{3}{2u}$$
$$u = \frac{3}{2} f = \frac{3}{2} \times (-10) = -15 \text{ cm}$$

Therefore, $v = 2u = -30$ cm

So, the distance of the object is 15 cm in front of the mirror and the position of the image is 30 cm in front of the mirror. **[1½]**

(b) As we know
For a convex mirror,
Focal length, $f > 0$
Position of object, $u < 0$
From mirror formula, we have :

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$
$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$
$$v > 0$$

Therefore, $v > 0$

This means that the image formed by a convex lens is always behind the mirror and hence, virtual. **[1½]**

Q. 2. A small candle, 2.5 cm in size is placed at 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Describe the nature and size of the image. If the candle is moved closer to the mirror, how would the screen have to be moved?
[NCERT Ex. Q. 9.1, Page 345]

Ans. Given that,
Size of the candle, $h = 2.5$ cm
Image size $= h'$
Object distance, $u = -27$ cm
Radius of curvature of the concave mirror, $R = -36$ cm
As we know that the focal length of the concave mirror,

$$f = \frac{R}{2} \quad \text{[½]}$$
$$= -18 \text{ cm}$$

Let the image distance $= v$
The image distance can be obtained using the mirror formula :

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$
$$= \frac{1}{-18} - \frac{1}{-27}$$
$$= \frac{-3 + 2}{54}$$
$$= -\frac{1}{54}$$

Hence, $v = -54$ cm **[½]**

So, the screen should be placed 54 cm away from the mirror to obtain a sharp image.
The magnification of the image is :

$$m = \frac{h'}{h} = -\frac{v}{u}$$

Therefore, $h' = -\frac{v}{u} \times h$

$$= -\left(\frac{-54}{-27}\right) \times 2.5 = -5 \text{ cm} \quad \textbf{[1]}$$

The height of the candle's image is 5 cm. The negative sign indicates that the image is inverted and real.

If the candle is moved closer to the mirror, then the screen will have to be moved away from the mirror in order to obtain the image. **[1]**

Q. 3. A 4.5 cm needle is placed 12 cm away from a convex mirror of focal length 15 cm. Give the location of the image and the magnification. Describe what happens as the needle is moved farther from the mirror.

[NCERT Ex. Q. 9.2, Page 345]

Ans. Given that, Height of the needle, $h_1 = 4.5$ cm

Object distance, $u = -12$ cm

Focal length of the convex mirror, $f = 15$ cm

Let the image distance $= v$

As we know the mirror formula,

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

$$=\frac{1}{15}+\frac{1}{12}=\frac{4+5}{60}=\frac{9}{60}$$

Therefore, $v=\frac{60}{9}=6.7$ cm **[1]**

So, the image of the needle is 6.7 cm away from the mirror. Also, it is on the other side of the mirror.

The image size is given by the magnification formula :

$$m=\frac{h_2}{h_1}=-\frac{v}{u}$$

$$\therefore\ h_2=-\frac{v}{u}\times h_1=\frac{-6.7}{-12}\times 4.5=+2.5 \text{ cm}$$

Hence, magnification of the image,

$$m=\frac{h_2}{h_1}=\frac{2.5}{4.5}=0.56$$ **[1]**

The height of the image is 2.5 cm. The positive sign indicates that the image is erect, virtual and diminished.

If the needle is moved farther from the mirror, the image will also move away from the mirror, and the size of the image will reduce gradually. **[1]**

Q. 4. Use the mirror equation to deduce that

(a) an object placed between *f* and 2*f* of a concave mirror produces a real image beyond 2*f*.

(b) a convex mirror always produces a virtual image independent of the location of the object.

(c) the virtual image produced by a convex mirror is always diminished in size and is located between the focus and the pole.

(d) an object placed between the pole and focus of a concave mirror produces a virtual and enlarged image. [NCERT Ex. Q. 9.15, Page 346]

Ans. **(a)** For a concave mirror, the focal length (f) is negative.

So, $f < 0$

When the object is placed on the left side of the mirror, the object distance (u) is negative.

So, $u < 0$

For image distance v, according to the lens formula :

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u} \quad ...(1)$$

So the object lies between f and $2f$.

$\therefore\ 2f < u < f$ ($\because u$ and f are negative)

$$\frac{1}{2f}>\frac{1}{u}>\frac{1}{f}$$

$$-\frac{1}{2f}<-\frac{1}{u}<-\frac{1}{f}$$

$$\frac{1}{f}-\frac{1}{2f}<\frac{1}{f}-\frac{1}{u}<0 \quad ...(2)$$

Using equation (1), we get :

$$\frac{1}{2f}<\frac{1}{v}<0$$

Since, $\frac{1}{v}$ is negative, *i.e.*, v is negative.

$$\frac{1}{2f}<\frac{1}{v}$$

$$2f > v$$

$$-v > -2f$$

Therefore, the image lies beyond $2f$. **[1]**

(b) For a convex mirror, the focal length (f) is positive.

$\therefore f > 0$

When the object is placed on the left side of the mirror, the object distance (u) is negative.

$\therefore u < 0$

For image distance v, according to the mirror formula :

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

Using equation (2), we can conclude that :

$$\frac{1}{v}<0$$

$$v > 0$$

Thus, the image is formed on the back side of the mirror.

So, a convex mirror always produces a virtual image, regardless of the object distance. **[½]**

(c) For a convex mirror, the focal length (f) is positive.

So, $f > 0$

When the object is placed on the left side of the mirror, the object distance (u) is negative,

Therefore, $u < 0$

For image distance v, for this the mirror formula :

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

But we have $u < 0$

Therefore, $\frac{1}{v}>\frac{1}{f}$

$$v < f$$

So, the image formed is diminished and is located between the focus (f) and the pole. **[½]**

(d) For a concave mirror, the focal length (f) is negative.

So, $f < 0$

When the object is placed on the left side of the mirror, the object distance (u) is negative.

So, $u < 0$

It is placed between the focus (f) and the pole.

Therefore,

$$f > u > 0$$

$$\frac{1}{f} < \frac{1}{u} < 0$$

$$\frac{1}{f} - \frac{1}{u} < 0$$

For image distance v, we have the mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

Therefore, $\frac{1}{v} < 0$

$$v > 0$$

The image is formed on the right side of the mirror. So that, it is a virtual image.

For $u<0$ and $v>0$,

$\frac{1}{u} > \frac{1}{v}$

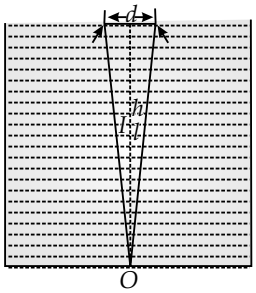
So,

$v > u$

Magnification, $m = \frac{v}{u} > 1$

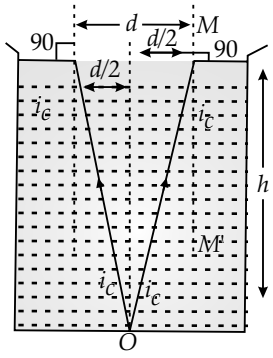
Therefore, the formed image is enlarged. **[1]**

Q. 5. A jar of height h is filled with a transparent liquid of refractive index μ (figure). At the centre of the jar on the bottom surface is a dot. Find the minimum diameter of a disc, such that when placed on the top surface symmetrically about the centre, the dot is invisible.



[NCERT Exemp. Q. 9.26, Page 59]

Ans. The point O will be invisible if the light ray coming from O does not come out or it get graze the surface of liquid or gets total internal reflection as shown in the figure,

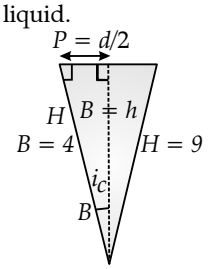


[1]

Ray OA is incident at A with critical angle i_C and its angle of refraction will be 90°.

For other ray if incident angle is more than i_c it will get total internal reflection from the surface so will not come out from liquid.

So disc of diameter d is required to stop the rays from 'O' out of liquid. **[1]**



$$\tan i_c = \frac{d/2}{h}$$

or

$$\tan i_c = \frac{d}{2h}$$

$$\Rightarrow \quad d = 2h \tan i_c$$

$$\Rightarrow \quad \sin i_c = \frac{1}{\mu} = \frac{P}{H} \qquad \text{[From figure]}$$

$$B^2 = H^2 - P^2$$

$$B = \sqrt{\mu^2 - 1}$$

$$\text{Therefore, } \tan i = \frac{P}{B} = \frac{1}{\sqrt{\mu^2 - 1}}$$

$$\text{Hence, } \quad d = 2h \times \frac{1}{\sqrt{\mu^2 - 1}}$$ **[1]**

Q. 6. A tank is filled with water to a height of 12.5 cm. The apparent depth of a needle lying at the bottom of the tank is measured by a microscope to be 9.4 cm. What is the refractive index of water? If water is replaced by a liquid of refractive index 1.63 up to the same height, by what distance would the microscope have to be moved to focus on the needle again? [NCERT Ex. Q. 9.3, Page 345]

Ans. Given that,

Actual depth of the needle in water, h = 12.5 cm

Apparent depth of the needle in water, h = 9.4 cm

Refractive index of water = μ

The value of μ can be obtained as follows :

$$\mu = \frac{h_1}{h_2}$$

$$= \frac{12.5}{9.4} \approx 1.33 \qquad \textbf{[1]}$$

So, the refractive index of water is about 1.33.

Water is replaced by a liquid of refractive index, $\mu' = 1.63$

As we know that the actual depth of the needle remains the same, but its apparent depth changes. So, let y be the new apparent depth of the needle.

So, we can write the relation :

$$\mu' = \frac{h_1}{y}$$

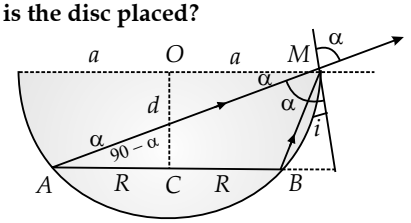
Therefore, $y = \frac{h_1}{\mu'}$

$$= \frac{12.5}{1.63} = 7.67 \text{ cm} \qquad \textbf{[1]}$$

Thus, the new apparent depth of the needle is 7.67 cm. It is less than h_2. Therefore, to focus the needle again, the microscope should be moved up.

So, distance by which the microscope should be moved up = 9.4 − 7.67 = 1.73 cm **[1]**

Q. 7. A circular disc of radius 'R' is placed co-axially and horizontally inside an opaque hemispherical bowl of radius 'a' (Figure). The far edge of the disc is just visible when viewed from the edge of the bowl. The bowl is filled with transparent liquid of refractive index μ and the near edge of the disc becomes just visible. How far below the top of the bowl is the disc placed?



[NCERT Exemp. Q. 9.23, Page 59]

Ans. In figure, AM and BM are the rays from the ends of disc AB reaching at one end of bowl at M. MN is tangent at M, so MN perpendicular to AB, that is, $\angle N = 90°$

Taking incidence ray BM and refracted ray MD

$BN = CN - CB = OM - CB = a - R$

$$MB = \sqrt{d^2 + (a-R)^2}$$

$$\therefore \sin i = \frac{BN}{BM} = \frac{(a-R)}{d^2 + (a-R)^2}$$

$$\angle r = \angle \alpha = \angle AMN$$

$$\sin r = \cos(90^\circ - \alpha) = \frac{AN}{AM} = \frac{a+R}{\sqrt{d^2 + (a+R)^2}}$$

For incidence ray BM to the horizontal level of liquid MP, MN will be normal at M. $\angle i$ and $\angle r$ will be incidence and refracted angles when ray BM passes from liquid (μ) to air. By Snell's law, as ray passes from liquid to air,

$$_1\mu_0 = \frac{\sin i}{\sin r}$$

$$\Rightarrow \frac{\mu_0}{\mu_1} = \frac{\sin i}{\sin r} \qquad \begin{bmatrix} \mu_0 \text{ for air} = 1 \text{ and} \\ \mu_1 = \mu \text{ for liquid} \end{bmatrix} \qquad \textbf{[1½]}$$

$$\Rightarrow \frac{1}{\mu_1} = \frac{\sin i}{\sin r}$$

$$\frac{1}{\mu_1} = \frac{\frac{a-R}{\sqrt{d^2 + (a-R)^2}}}{\frac{(a+R)}{\sqrt{d^2 + (a+R)^2}}}$$

$$= \frac{(a-R)\sqrt{d^2 + (a+R)^2}}{(a+R)\sqrt{d^2 + (a-R)^2}}$$

$$d = \frac{\mu(a^2 - d^2)}{\sqrt{(a+r)^2 - \mu(a-r)^2}} \quad (\because \mu_1 = \mu) \qquad \textbf{[1½]}$$

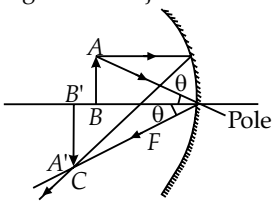
Q. 8. (a) Draw a ray diagram to show image formation when the concave mirror produces a real, inverted and magnified image of the object.

(b) Obtain the mirror formula and write the expression for the linear magnification.

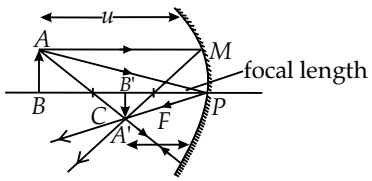
[CBSE Board, Delhi Region, 2018]

Ans. (a) A ray diagram to show image formation when the concave mirror produces a real, inverted and magnified image of the object is shown below :

(a)



(b)



[1]

$\Delta A'B'F$ and MPF are similar

So

$$\frac{A'B'}{MP} = \frac{B'P}{FP} \qquad \cdots\text{(i)}$$

ΔABP and A'B'P are similary

$$\frac{A'B'}{BP} = \frac{B'P}{BP} \qquad \cdots\text{(ii)}$$

$$MP = AB \qquad \text{(By geometry)}$$

$$\frac{A'B'}{AB} = \frac{B'F}{FB} \qquad \cdots\text{(iii)}$$

Equate (ii) and (iii)

$$\frac{B'P}{BP} = \frac{B'F}{FP}$$

$$\frac{(-v)}{(-u)} = \frac{B'P - FP}{FP}$$

$$\frac{v}{u} = \frac{(-v) - (-f)}{-f}$$

$$\frac{-vf}{uvf} = \frac{-uv + uf}{uvf}$$

$$-\frac{1}{u} = -\frac{1}{f} + \frac{1}{v}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{A'B'}{AB} = \frac{B'P}{BP}$$

$$\frac{-h'}{h} = \frac{-v}{-u}$$

$$m = \frac{h'}{h} = \frac{-v}{u}$$

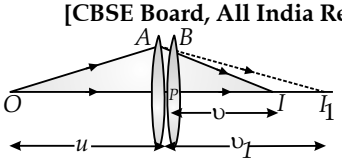
$$\boxed{m = -\frac{v}{u}} \qquad \begin{pmatrix} h' \to \text{Height of image,} \\ h \to \text{Height of object} \end{pmatrix}$$

$$\boxed{m = \frac{h'}{h}}$$ **[2]**

Q. 9. (a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.

[CBSE Board, All India Region, 2018]

Ans.



[1]

Two thin lenses, of focal length f_1 and f_2 are kept in contact. Let O be the position of object and let u be the object distance. The distance of the image (which is at I_1), for first lens is v_1.

This image serves as object for the second lens.

Let the final image be at I. We then have

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u}$$

Adding, we get

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Therefore, $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

Hence, $P = P_1 + P_2$ **[2]**

Q. 10. A converging lens has a focal length of 20 cm in air. It is made of a material of refractive index 1·6. If it is immersed in a liquid of refractive index 1.3, find its new focal length.

[CBSE Board, Foreign Scheme, 2017]

Ans. Given that,

Refractive index of the material, ${}^{a}\mu_g = 1.5$

Focal length of the lens in air, $f_{air} = 20$ cm

Using Lens Maker's formula,

$$\frac{1}{f_{air}} = \left({}^{a}\mu_g - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{10}$$

Refractive index of the liquid, ${}^{a}\mu_l = 1.3$ **[1½]**

Refractive index of the glass with respect to liquid, ${}^{a}\mu_g / {}^{a}\mu_l = 1.15$

Let the focal length of the lens in liquid be f_l

$$\frac{1}{f_l} = \left({}^{l}\mu_g - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_l} = 0.015$$

$$\frac{1}{f_l} = 66.67 \text{ cm}$$ **[1½]**

Q. 11. A thin convex lens of focal length 25 cm is cut into two pieces 0.5 cm above the principal axis. The top part is placed at (0, 0) and an object placed at (–50 cm, 0). Find the coordinates of the image.

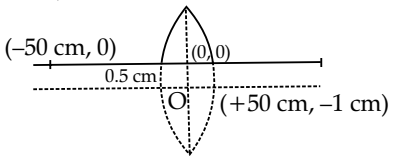
[NCERT Exemp. Q. 9.24, Page 59]

Ans. Given that,

Object is placed 0.5 cm above principal axis.

$u = -50$ cm

$f = +25$ cm,



$$\frac{1}{v} - \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-50} = \frac{1}{25}$$

$$\frac{1}{v} = \frac{1}{25} - \frac{1}{50}$$

$$= \frac{2-1}{50} = \frac{1}{50}$$

$v = 50$ cm **[1½]**

$$m = \frac{+v}{u} = \frac{+(50)}{-50} = -1$$

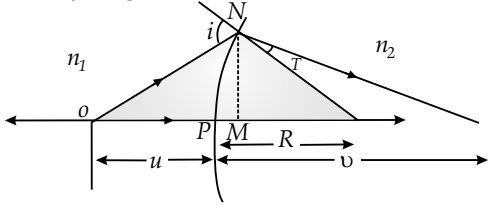
So the size of image is equal to that of object, m is negative so image is inverted.

So image is at (50 cm, −1 cm) and 0.5 cm below the X-X' axis. **[1½]**

Q. 12. (a) A point object is placed on the principal axis of a convex spherical surface of radius of curvature R, which separates the two media of refractive indices n_1 and n_2 ($n_2 > n_1$). Draw the ray diagram and deduce the relation between the object distance (u), image distance (v) and the radius of curvature (R) for refraction to take place at the convex spherical surface from rarer to denser medium.

[CBSE Board, Foreign Scheme, 2017]

Ans. The ray diagram is shown below :



The rays are incident from medium of refractive index n_1 to another medium of refractive index n_2. NM is taken to be nearly equal to the length of the perpendicular from the point N on the principal axis. We can have approximation for small angles. We get.

$$\tan NOM = \frac{MN}{OM} \simeq \angle NOM$$

$$\tan NCM = \frac{MN}{MC} \simeq \angle NCM$$

$$\tan NIM = \frac{MN}{MI} \simeq \angle NIM$$

Consider, ΔNOC

Where, i is the exterior angle.

So, $i = \angle NOM + \angle NCM$

$$\Rightarrow \quad i = \frac{MN}{OM} + \frac{MN}{MC}$$

Similarly, in ΔNCI

$$r = \angle NMC - \angle NIM$$

$$i.e., \; r = \frac{MN}{MC} - \frac{MN}{MI}$$

Now, using Snell's law, we have **[1½]**

$$n_1 \sin i = n_2 \sin r$$

For small angles, $n_1 i = n_2 r$

Now, putting the value of i and r from equations (i) and (ii),

$$n_1\left(\frac{MN}{OM} + \frac{MN}{MC}\right) = n_2\left(\frac{MN}{MC} - \frac{MN}{MI}\right)$$

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$$

From the Cartesian sign,

$OM = -u, MI = +v, MC = +R$

Putting this in the above equation,

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$ **[1½]**

Q. 13. Double-convex lenses are to be manufactured from a glass of refractive index 1.55, with both faces of the same radius of curvature. What is the radius of curvature required if the focal length is to be 20 cm? [NCERT Ex. Q. 9.7, Page 346]

Ans. Given that,

Refractive index of glass, $\mu = 1.55$

Focal length of the double-convex lens, $f = 20$ cm

Radius of curvature of one face of the lens $= R_1$

Radius of curvature of the other face of the lens $= R_2$

Radius of curvature of the double-convex lens $= R$

Therefore, $R_1 = R$ and $R_2 = -R$

The value of R can be calculated as :

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$\frac{1}{20} = (1.55 - 1)\left[\frac{1}{R} + \frac{1}{R}\right]$$

$$\frac{1}{20} = 0.55 \times \frac{2}{R}$$

Therefore, $R = 0.55 \times 2 \times 20 = 22$ cm

So that, the radius of curvature of the double-convex lens is 22 cm. **[3]**

Q. 14. An object of size 3.0 cm is placed 14 cm in front of a concave lens of focal length 21 cm. Describe the image produced by the lens. What happens if the object is moved further away from the lens? [NCERT Ex. Q. 9.9, Page 346]

Ans. Given that,

Size of the object, $h_1 = 3$ cm

Object distance, $u = -14$ cm

Focal length of the concave lens, $f = -21$ cm

Let the Image distance $= v$

According to the lens formula :

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = -\frac{1}{21} - \frac{1}{14} = \frac{-2-3}{42} = \frac{-5}{42}$$

Therefore, $v = -\frac{42}{5} = -8.4$ cm **[1½]**

So, the image is formed on the other side of the lens, 8.4 cm away from it. The negative sign shows that the image is erect and virtual.

The magnification of the image is :

$$m = \frac{\text{Image height } (h_2)}{\text{Object height } (h_1)} = \frac{v}{u}$$

Therefore, $h_2 = \frac{-8.4}{-14} \times 3 = 0.6 \times 3 = 1.8$ cm

Thus, the height of the image is 1.8 cm.

If the object is moved further away from the lens, then the virtual image will move towards the focus of the lens, but not beyond it. So, the size of the image will decrease with the increase in the object distance. **[1½]**

Q. 15. What is the focal length of a convex lens of focal length 30 cm in contact with a concave lens of focal length 20 cm? Is the system a converging or a diverging lens? Ignore thickness of the lenses. [NCERT Ex. Q. 9.10, Page 346]

Ans. Given that,

Focal length of the convex lens, $f_1 = 30$ cm

Focal length of the concave lens, $f_2 = -20$ cm

Focal length of the system of lenses $= f$

The equivalent focal length of a system of two lenses in contact is :

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f} = \frac{1}{30} - \frac{1}{20} = \frac{2-3}{60} = -\frac{1}{60}$$

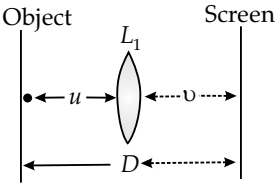
Therefore, $f = -60$ cm

Thus, the focal length of the combination of lenses is 60 cm. The negative sign indicates that the system of lenses acts as a diverging lens. **[3]**

Q. 16. In many experimental set-ups the source and screen are fixed at a distance say *D* and the lens is movable. Show that there are two positions for the lens for which an image is formed on the screen. Find the distance between these points and the ratio of the image sizes for these two points. [NCERT Exemp. Q. 9.25, Page 59]

Ans. $u = -(D - v)$

$v = v$



$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{v} + \frac{1}{(D-v)}$$

$$= \frac{D - v + v}{v(D-v)} = -\left[\frac{D}{vD - v^2}\right]$$

$$fD = vD - v^2$$

$$v^2 - vD + fD = 0$$

$$v = \frac{+D \pm \sqrt{D^2 + 4Df}}{2} = \frac{D}{2} \pm \frac{\sqrt{D^2 - 4Df}}{2} \quad \ldots(I)$$

$$u = -(D - v) = -\left[D - \left\{\frac{D}{2} \pm \frac{\sqrt{D^2 - 4Df}}{2}\right\}\right]$$

$$\mu = -\left[\frac{D}{2} \mp \frac{\sqrt{D^2 - 4Df}}{2}\right] \quad \ldots(II)$$

From II, when the position of object

$$u_2 = \frac{D}{2} + \frac{\sqrt{D^2 - 4Df}}{2} \quad \text{[In front of } L_2 \text{ (lower sign)]}$$

Then from I, the position of image

$$v_2 = \frac{D}{2} - \frac{\sqrt{D^2 - 4Df}}{2} \quad \text{[Lower sign]}$$

Similarly, when position of object

$$u_1 = \frac{D}{2} - \frac{\sqrt{D^2 - 4Df}}{2} \quad \text{[From II upper sign]}$$

Then the position of image

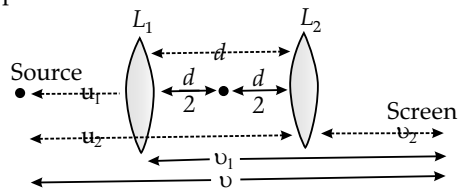
$$v_1 = \frac{D}{2} - \frac{\sqrt{D^2 - 4Df}}{2} \quad \text{[From I upper sign]}$$

The distance between two positions of lend

$d = v_1 - v_2$

$$d = \frac{D}{2} + \frac{\sqrt{D^2 - 4Df}}{2} - \left[\frac{D}{2} - \frac{\sqrt{D^2 - 4Df}}{2}\right] \quad \textbf{[1½]}$$

$d = \sqrt{D_2 - 4Df}$ is the distance between two positions of lenses.



In first case of L_1

$$u_1 = \frac{D}{2} - \frac{d}{2}$$

$$v_1 = \frac{D}{2} + \frac{d}{2}$$

So that,

$$m_1 = \frac{v_1}{u_1} = \frac{\frac{D}{2} + \frac{d}{2}}{\frac{D}{2} - \frac{d}{2}} = \frac{D + d}{D - d}$$

In second case:

$$u_2 = \frac{D}{2} + \frac{d}{2}$$

$$v_2 = \frac{D}{2} - \frac{d}{2}$$

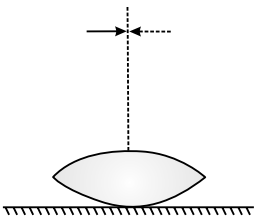
So,

$$m_2 = \frac{v_2}{u_2} = \frac{\frac{D}{2} - \frac{d}{2}}{\frac{D}{2} + \frac{d}{2}} = \frac{D - d}{D + d}$$

$$\frac{m_2}{m_1} = \frac{\frac{D-d}{D+d}}{\frac{D+d}{D-d}} = \left(\frac{D-d}{D+d}\right)^2$$

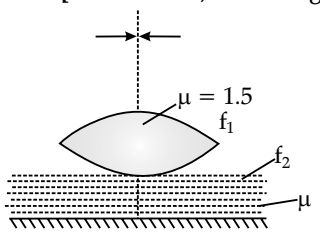
$\boxed{\frac{m_2}{m_1} = \left(\frac{D-d}{D+d}\right)^2}$ is the required ratio of size of images in both cases. **[1½]**

Q. 17. A symmetric biconvex lens of radius of curvature *R* and made of glass of refractive index 1.5, is placed on a layer of liquid placed on top of a plane mirror as shown in the figure. An optical needle with its tip on the principal axis of the lens is moved along the axis until its real, inverted image coincides with the needle itself. The distance of the needle from the lens is measured to be *x*. On removing the liquid layer and repeating the experiment, the distance is found to be *y*. Obtain the expression for the refractive index of the liquid in terms of *x* and *y*.



[CBSE Board, Delhi Region, 2018]

Ans.



In first case, there are two lens in combination double convex ($\mu = 1.5$) and plane concave lens.

The image overlapped at object when it placed on focus :

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

The distance x given so $f_1 = x$

$$\frac{1}{x} = \frac{1}{f_1} + \frac{1}{f_2} \qquad \text{...(i)}$$

When liquid removed distance given, y

So

$$\frac{1}{f_1} = \frac{1}{y}$$

$$f_1 = y$$

$$\frac{1}{f_2} = \frac{1}{x} - \frac{1}{f_1}$$

$$\frac{1}{f_2} = \frac{1}{x} - \frac{1}{f_1}$$

$$\frac{1}{f_1} = \frac{1}{x} - \frac{1}{y}$$

where

$$\frac{1}{y} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{y} = (1.5 - 1)\left(\frac{1}{R} + \frac{1}{R}\right)$$

$$\frac{1}{y} = \frac{1}{2} \times \frac{2}{R}$$

$$y = R$$

For f_2

$$\frac{1}{f_2} = (\mu - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right)$$

$$\frac{1}{f_2} = (\mu - 1)\left(\frac{-1}{R}\right)$$

$$\frac{1}{f_2} = \frac{(\mu - 1)}{-R}$$

$$\frac{1}{f_2} = \frac{1 - \mu}{R}$$

$$\frac{1}{f_2} = \frac{1 - \mu}{y} \qquad (y = R) \qquad \textbf{[1½]}$$

From equation (i)

$$\frac{1}{x} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{x} = \frac{1}{y} + \frac{1 - \mu}{y}$$

$$\frac{1}{x} = \frac{1 + 1 - \mu}{y}$$

$$\frac{1}{x} = \frac{2 - \mu}{u}$$

$$\frac{y}{x} = 2 - \mu$$

$$\boxed{\mu = 2 - \frac{y}{x}} \qquad \textbf{[1½]}$$

Long Answer Type Questions (5 marks each)

Q. 1. The mixture a pure liquid and a solution in a long vertical column (*i.e.*, horizontal dimensions << vertical dimensions) produces diffusion of solute particles and hence a refractive index gradient along the vertical dimension. A ray of light entering the column at right angles to the vertical is deviated from its original path. Find the deviation in travelling a horizontal distance $d << h$, the height of the column.

[NCERT Exemp. Q. 9.29, Page 60]

Ans. Consider a long vertical column of transparent liquid of infinite height (h) and thickness (dx). Let a ray AB that enters at an angle θ into liquid of height y in column of liquid and emerges at an angle ($\theta + d\theta$) at height ($y + dy$). **[1]**

From Snell's law,

$$\mu(y)\sin\theta = \mu(y + dy)\sin(\theta + d\theta)$$

$$\mu(y)\sin\theta \cong \left[\mu(y) + \frac{d\mu}{dy}.dy\right]\begin{pmatrix}\sin\theta\cos d\theta + \\ \cos\theta\sin d\theta\end{pmatrix}$$

As $d\mu$, $d\theta$ are very small tends to zero.

$\therefore \sin d\theta = d\theta$, $\cos d\theta \cong 1$

$\mu(y) = \mu$(μ at height Y constant)

$$\therefore \mu \sin\theta \cong [\mu + d\mu](\sin\theta + \cos\theta.d\theta)$$

$$\mu\sin\theta \cong \mu\sin\theta + \mu\cos\theta d\theta + d\mu\sin\theta + \cos\theta.d\theta d\mu$$

Again, $d\theta d\mu \cong 0$

$$\mu\cos\theta.d\theta = -d\mu\sin\theta$$

$$d\theta = -\frac{1}{\mu}d\mu\frac{\sin\theta}{\cos\theta} = -\frac{1}{\mu}d\mu\tan\mu$$

$\therefore \tan(180 - \theta) = \dfrac{dx}{dy}$ [From figure]

$$\tan\theta = \frac{dx}{dy} \qquad \textbf{[3]}$$

$$\therefore \quad d\theta = \frac{1}{\mu}\frac{d\mu}{dy}.dx$$

By integration on both sides

$$\int_0^\theta d\theta \cong \frac{-1}{\mu}\int_0^d \frac{d\mu}{dy}dx = -\frac{1}{\mu}\frac{d\mu}{dy}.\int_0^d dx \quad \left[\begin{array}{l}\because \mu \text{ and } y \text{ does not} \\ \text{change horizontally} \\ \text{and } dx, dy \text{ are constant} \\ \text{horizontally.}\end{array}\right]$$

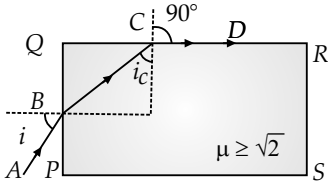
$$\theta \cong \frac{-1}{\mu}\frac{d\mu}{dy}d \qquad \textbf{[1]}$$

Q. 2. Show that for a material with refractive index $\mu \geq \sqrt{2}$, light incident at any angle shall be guided along a length perpendicular to the incident face.

[NCERT Exemp. Q. 9.28, Page 60]

Ans. Let assume a rectangular slab of refractive index $\mu \geq \sqrt{2}$. An incidence ray incidence at an angle i on face PQ at incidence point. Refracted ray BC strike at face QR which is perpendicular to PQ with incidence angle i_c so that refracted ray CD passes

normal to the face PQ as per required in question. So i_c must be critical angle. **[1]**



$$\mu = \frac{1}{\sin i_c} \qquad \text{(Snell's law at } C\text{)}$$

$$\sin i_c \geq \frac{1}{\mu} \qquad \left[\sin i_c = \frac{1}{\mu}\right]$$

$$\sin(90^\circ - r) \geq \frac{1}{\mu} \qquad \left[\because r + 90^\circ + i_c = 180^\circ\right]$$

$$\cos r \geq \frac{1}{\mu}$$

$$\cos^2 r \geq \frac{1}{\mu^2} \qquad \text{(Squaring both sides)}$$

$$-\cos^2 r \leq -\frac{1}{\mu^2}$$

$$1 - \cos^2 r \leq 1 - \frac{1}{\mu^2}$$

$$\sin^2 r \leq 1 - \frac{1}{\mu^2} \qquad \text{...(I)}$$

$$\frac{\sin i}{\sin r} = \mu \qquad \text{(By Snell's law)}$$

$$\sin i = \mu \sin r$$

or

$$\sin^2 i = \mu^2 \ \sin^2 r \qquad \text{(Squaring both sides)}$$

$$\frac{1}{\mu^2}\sin^2 i = \sin^2 r \qquad \text{...(II)}$$

[1½]

Put (II) in (I)

$$\frac{1}{\mu^2}\sin^2 i \leq 1 - \frac{1}{\mu^2}$$

$$\sin^2 i \leq \mu^2 - 1 \quad \left[\text{On multiplying by } \mu^2 \text{ both sides}\right]$$

For smallest angle, *i.e.*, $i = 90^\circ$ **[1½]**

$$\therefore \sin^2 90 \leq \mu^2 - 1 \qquad \left[\mu \geq \sqrt{2}\right]$$

$$1 + 1 \leq \mu^2$$

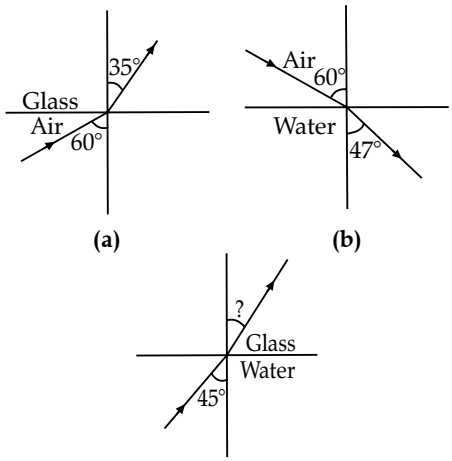
$$2 \leq \mu^2$$

Taking square root,

$$\sqrt{2} \leq \mu$$

Thus proved. **[1]**

Q. 3. Figures (*a*) and (*b*) show refraction of a ray in air incident at 60° with the normal to a glass-air and water-air interface, respectively. Predict the angle of refraction in glass when the angle of incidence in water is 45° with the normal to a water-glass interface [Figure (*c*)].



[NCERT Ex. Q. 9.4, Page 345]

Ans. Given that,
For the glass-air interface :
Angle of incidence, $i = 60°$
Angle of refraction, $r = 35°$
The relative refractive index of glass with respect to air is given by Snell's law as :

$$^a\mu_g = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 35^\circ} = \frac{0.8660}{0.5736} = 1.51 \qquad \text{...(i)}$$

[1]

Given that,
For the air-water interface :
Angle of incidence, $i = 60°$
Angle of refraction, $r = 47°$
The relative refractive index of water with respect to air is given by Snell's law as :

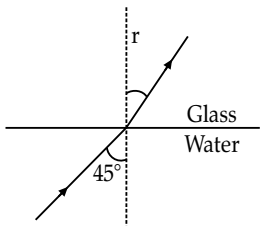
$$\mu_w^a = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 47^\circ} = \frac{0.8660}{0.7314} = 1.184 \qquad \text{...(2)}$$

[1]

Using (1) and (2), the relative refractive index of glass with respect to water can be obtained as :

$$\therefore \mu_g^w = \frac{\mu_g^a}{\mu_w^a} = \frac{1.51}{1.184} = 1.275$$

The following figure shows the situation involving the glass-water interface.



[1½]

Angle of incidence, $i = 45^\circ$
Angle of refraction = r
From Snell's law, r can be calculated as :

$$\frac{\sin i}{\sin r} = \mu_g^w$$

$$\frac{\sin 45^\circ}{\sin r} = 1.275$$

$$\sin r = \frac{\frac{1}{\sqrt{2}}}{1.275} = 0.5546$$

$$\therefore \quad r = \sin^{-1}(0.5546) = 38.68^\circ$$

Hence, the angle of refraction at the water-glass interface is 38.8°. **[1½]**

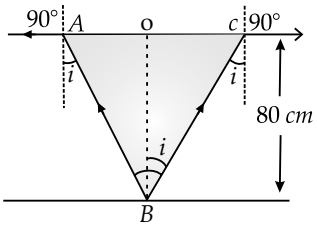
Q. 4. A small bulb is placed at the bottom of a tank containing water to a depth of 80 cm. What is the area of the surface of water through which light from the bulb can emerge out? Refractive index of water is 1.33. (Consider the bulb to be a point source.) [NCERT Ex. Q. 9.5, Page 346]

Ans. Given that,

Actual depth of the bulb in water, $d = 80$ cm $= 0.8$ m

Refractive index of water, $\mu = 1.33$

According to the given situation is shown in the below figure :



Where,

i = Angle of incidence

r = Angle of refraction = 90°

Since the bulb is a point source, the emergent light can be considered as a circle of radius,

$$R = \frac{AC}{2} = AO = OB \qquad \textbf{[2]}$$

$$\mu = \frac{\sin r}{\sin i}$$

$$1.33 = \frac{\sin 90^\circ}{\sin i}$$

$$\therefore \quad i = \sin^{-1}\left(\frac{1}{1.33}\right) = 48.75^\circ$$

Using the given figure, we have the relation :

$$\tan i = \frac{OC}{OB} = \frac{R}{d_1} \qquad \textbf{[1½]}$$

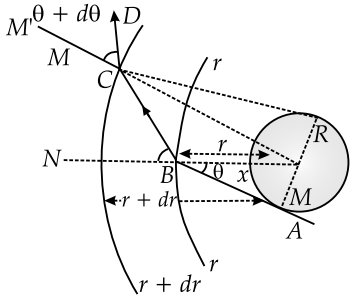
$\therefore R = \tan 48.75° \times 0.8 = 0.91$ m

$\therefore$ Area of the surface of water $= \pi R^2 = \pi\ (0.91)^2 = 2.61\ m^2$

Thus, the area of the surface of water through which the light from the bulb can emerge is approximately 2.61 m^2. **[1½]**

Q. 5. If light passes near a massive object, the gravitational interaction causes a bending of the ray. This can be thought of as happening due to a change in the effective refractive index of the medium given by $n(r) = 1 + \frac{2GM}{rc^2}$ where r is the distance of the point of consideration from the centre of the mass of the massive body, G is the universal gravitational constant, M the mass of the body and c the speed of light in vacuum. Considering a spherical object find the deviation of the ray from the original path as it grazes the object. [NCERT Exemp. Q. 9.30, Page 60]

Ans. Let assume two spherical surfaces at r and $(r + dr)$ distance from the centre of massive object of mass m and radius R.



A ray $ABCD$ incident at B and C on two surfaces at r and $(r+dr)$ then, by Snell's law

$$\mu(r)\sin\theta \cong \mu(r+dr)\sin(\theta+d\theta)$$

$$\mu(r)\sin\theta \cong \left[\mu(r) + \frac{d\mu}{dr}dr\right]\left[\because \sin\theta.\cos d\theta + \cos\theta\sin d\theta\right]$$

[$\therefore d\theta \to 0$ then $\cos d\theta = 1$ and $d\theta = 0$.]

$$\therefore \mu(r)\sin\theta \cong \left[\mu(r) + d\mu\right]\left[\sin\theta + \cos\theta d\theta\right]$$

$$\mu(r)\sin\theta \cong \mu(r)\sin\theta + d\mu\sin\theta + \mu(r)\cos\theta.d\theta + d\mu d\theta.\cos\theta$$

$d\mu.d\theta\cos\theta$ is very small as $d\mu$ and $d\theta$ are very small

$$\therefore d\ \mu d\theta\cos\theta \cong 0$$

$$\therefore \quad \mu(r)\sin\theta \cong \mu(r)\sin\theta + d\mu\sin\theta + \mu\cos\theta.d\theta$$

$$0 \cong d\mu\sin\theta + \mu(r)\cos\theta.d\theta$$

$$-d\mu\sin\theta = \mu(r)\cos\theta.d\theta$$

Dividing both sides by dr

$$\frac{-d\mu}{dr}\sin\theta = \mu(r)\cos\theta\frac{d\theta}{dr} \qquad \text{...(I)}$$

[1½]

$$\mu(r) = 1 + \frac{2GM}{c^2}\frac{1}{r} \quad \text{(Given)}$$

$$\frac{d\mu}{dr} = 0 + \frac{2GM}{c^2}(-1)r^{-2}$$

$$-\frac{d(\mu)}{dr} = \left[\frac{+2GM}{c^2r^2}\right]$$

$$\frac{2GM}{r^2c^2}\sin\theta = \left[1 + \frac{2GM}{rc^2}\right]\cos\theta.\frac{d\theta}{dr} \qquad \text{...(II)}$$

As the G is very small and c^2 is very large so,

$$\frac{2GM}{rc^2} \to 0$$

$$\frac{2GM}{r^2c^2}\frac{\sin\theta}{\cos\theta} = \frac{d\theta}{dr}$$

$$\frac{2GM}{r^2c^2}\tan\theta \cong \frac{d\theta}{dr}$$

$$\frac{2GM}{c^2}\int\frac{\tan\theta}{r^2}dr = \int_0^{\theta_0} d\theta$$

$$r^2 = x^2 + R^2 \qquad \text{[From figure]} \quad \textbf{[1½]}$$

Differentiate with respect to both sides

$2rdr = 2xdx$

$$dr = \frac{x}{r}dx$$

$$So,\ \int_0^{\theta_0} d\theta = \frac{2GM}{c^2}\int_{-\alpha}^{\alpha}\frac{1}{r^2}\frac{R}{r}dx \quad \left(\tan\theta = \frac{R}{x}\right)$$

$$= \frac{2GM}{c^2}\int_{-\alpha}^{+\alpha}\frac{R}{xr^3}xdx = \frac{2GM}{c^2}\cdot\int_{-\alpha}^{\alpha}\frac{R}{\left(x^2+R^2\right)^{3/2}}dx$$

$$\int_0^{\theta_0} d\theta = \frac{2GM}{c^2}\int_{-\alpha}^{\alpha} R\frac{dx}{\left(x^2+R^2\right)^{3/2}}$$

$\because$ x=R tan θ

dx=R sec^2 θ.dθ

$$\theta_0 = \frac{2GM}{c^2}\int_{-\pi/2}^{\pi/2}\frac{RR\sec^2\theta d\theta}{\left(R^2\tan^2\theta+R^2\right)^{3/2}}$$

$$= \frac{2GM}{c^2}\int_{-\pi/2}^{\pi/2}\frac{R^2\sec^2\theta d\theta}{\left[R^2\left(\tan^2\theta+1\right)\right]^{3/2}}$$

$$\int_0^{\theta_0} d\theta = \frac{2GM}{c^2}\int_{-\pi/2}^{\pi/2}\frac{R^2\sec^2\theta}{R^3\sec^3\theta}d\theta$$

$$= \frac{2GM}{Rc^2}\int_{-\pi/2}^{\pi/2}\frac{1}{\sec\theta}$$

$$= \frac{2GM}{Rc^2}\int_{-\pi/2}^{\pi/2}\cos\theta d\theta$$

$$= \frac{2GM}{Rc^2}\left[\sin\theta\right]_{-\pi/2}^{\pi/2}$$

$$= \frac{2GM}{Rc^2}\left[\sin\frac{\pi}{2} - \mathrm{Sin}\left(\frac{-\pi}{2}\right)\right]$$

$$\theta_0 = \frac{2GM}{Rc^2}\left[1+\sin\frac{\pi}{2}\right]$$

$$= \frac{2GM}{Rc^2}\left[1+1\right]$$

$$\theta_0 = \frac{4GM}{Rc^2}$$

[2]

Q. 6. Answer the following questions :

(a) You have learnt that plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances? Explain.

(b) A virtual image, we always say, cannot be caught on a screen. Yet when we 'see' a virtual image, we are obviously bringing it on to the 'screen' (*i.e.*, the retina) of our eye. Is there a contradiction?

(c) A diver under water, looks obliquely at a fisherman standing on the bank of a lake. Would the fisherman look taller or shorter to the diver than what he actually is?

(d) Does the apparent depth of a tank of water change if viewed obliquely? If so, does the apparent depth increase or decrease?

(e) The refractive index of diamond is much greater than that of ordinary glass. Is this fact of some use to a diamond cutter?

[NCERT Ex. Q. 9.18, Page 347]

Ans. (a) Yes,

Plane and convex mirrors can produce real images as well. If the object is virtual, that is, if the light rays are converging at a point behind a plane mirror (or a convex mirror) are reflected to a point on a screen placed in front of the mirror, then a real image will be formed. **[1]**

(b) No,

A virtual image is formed when light rays diverge. The convex lens of the eye causes these divergent rays to converge at the retina. In this case, the virtual image serves as an object for the lens to produce a real image. **[1]**

(c) The diver is in the water and the fisherman is on land (*i.e.*, in air). Water is a denser medium than air. It is given that the diver is viewing the fisherman. This indicates that the light rays are travelling from a denser medium to a rarer medium. So, the refracted rays will move away from the normal. As a result, the fisherman will appear to be taller. **[1]**

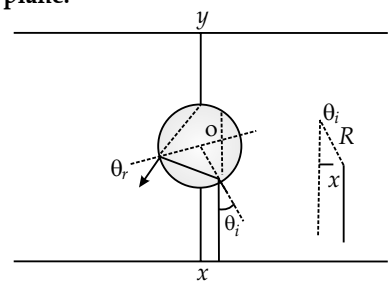
(d) Yes; Decrease

The apparent depth of a tank of water changes when viewed obliquely. This is because light bends on travelling from one medium to another medium. The apparent depth of the tank when viewed obliquely is less than the near-normal viewing. **[1]**

(e) Yes,

The refractive index of diamond (2.42) is more than that of ordinary glass (1.5). The critical angle for diamond is less than that for glass. A diamond cutter uses a large angle of incidence to ensure that the light entering the diamond is totally reflected from its faces. This is the reason for the sparkling effect of a diamond. **[1]**

Q. 7. An infinitely long cylinder of radius *R* is made of an unusual exotic material with refractive index –1 (Figure). The cylinder is placed between two planes whose normal are along the *y* direction. The centre of the cylinder *O* lies along the *y*-axis. A narrow laser beam is directed along the *y* direction from the lower plate. The laser source is at a horizontal distance *x* from the diameter in the *y* direction. Find the range of *x* such that light emitted from the lower plane does not reach the upper plane.



[NCERT Exemp. Q. 9.31, Page 60]

Ans. As the cylinder is made of refractive index (−1) and is placed in air of refractive index = 1 so, when ray *AB* is incident at *B* to cylinder, θ_r will be negative, that is, refractive ray will get reflected from normal. Similar thing happens at incidence point *C* and angle of refraction and incident at *B* and *C* will be equal (θ_r) as *OB* = *OC* = *R* and of refraction at *C* is θ_r. **[2½]**

$\theta_1 = |\theta_i| = |\theta_r| = |\theta'_r|$ as reflection takes place

The total deviation of out-coming ray from the incoming ray $4\theta_1$. Rays shall not reach the receiving plane if $\frac{\pi}{2} \le 4\theta \le \frac{3\pi}{2}$ angles measured clockwise from the y axis.

Or $\frac{\pi}{8} \le \theta \le \frac{3\pi}{8}$ (On dividing by 4 to all sides)

$\sin\theta_1 = \frac{x}{R}$

$\frac{\pi}{8} \le \sin^{-1}\frac{x}{R} \le \frac{3\pi}{8}$ or $\frac{\pi}{8} \le \frac{x}{R} \le \frac{3\pi}{8}$

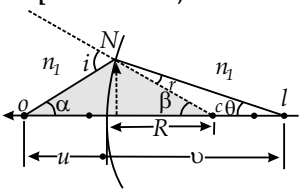
So, for light emitted from the source shall not reach the receiving plane if $\frac{R\pi}{8} \le x \le \frac{3R\pi}{8}$. **[2½]**

Q. 8. (i) Derive the mathematical relation between refractive indices n_1 and n_2 of two radii and radius of curvature R for refraction at a convex spherical surface. Consider the object to be a point since lying on the principle axis in rarer medium of refractive index n_1 and a real image formed in the denser medium of refractive index n_2. Hence, derive lens maker's formula.

(ii) Light from a point source in air falls on a convex spherical glass surface of refractive index 1.5 and radius of curvature 20 cm. The distance of light source from the glass surface is 100 cm. At what position is the image formed?

[CBSE Board, All India Region, 2016]

Ans. (i)



The above figure shows the geometry of formation of real image *I* of an object *O* and the principal axis of a spherical surface with centre of curvature *C* and radius of curvature *R*.

Assumptions :

(i) The aperture of the surface is small compared to other distance involved.

(ii) *NM* will be taken to be nearly equal to the length of the perpendicular from the point *N* on the principal axis.

$\tan NOM = \frac{MN}{OM}$

$\tan NCM = \frac{MN}{MC}$

$\tan NIM = \frac{MN}{MI}$

For ΔNOC, angle '*i*' is the exterior angle.

Assuming that the incident ray is very close to the principal axis, all the angles are very small.

So, for very small angles,

$\tan x = x = \sin x$,

So, $i = \angle NOM + \angle NCM$

$$i = \frac{MN}{MC} + \frac{MN}{MC} \quad ...(i)$$

Similarly, $r = \angle NCM - \angle NIM$

So that, $$r = \frac{MN}{MC} - \frac{MN}{MI} \quad ...(ii)$$

According to Snell's law,

$n_1 \sin i = n_2 \sin r$

For small angles, $\sin i = i$ and $\sin r = r$

$n_1 i = n_2 r$

Substituting the values of *i* and *r* from equations (i) and (ii), we get

$$n_1\left(\frac{MN}{OM} + \frac{MN}{MC}\right) = n_2\left(\frac{MN}{MC} - \frac{MN}{MI}\right)$$

Or, $$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC} \quad ...(iii)$$

Applying new Cartesian sign conventions,

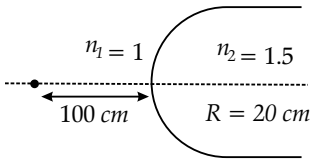
$OM = -u$, $MI = +v$, $MC = +R$

Substituting these in equation (iii), we obtain,

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad ...(vi)$$

[3]

(ii)



$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

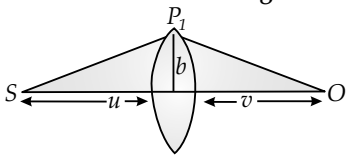
$$\Rightarrow \quad \frac{1.5}{v} - \frac{1}{(-100)} = \frac{1.5-1}{20} = \frac{1}{40}$$

$$\Rightarrow \frac{1.5}{v} = \frac{1}{40} - \frac{1}{100} = \frac{5-2}{200} = \frac{3}{200}$$

$$\Rightarrow \quad v = \frac{1.5 \times 200}{3} = 100 \text{ cm}$$

So, the image is formed 100 cm in the denser medium. **[2]**

Q. 9. (i) Consider a thin lens placed between a source (*S*) and an observer (*O*) (Figure). Let the thickness of the lens vary as $W(b) = W_0 - \frac{b^2}{\alpha}$ where *b* is the vertical distance from the pole. w_0 is a constant. Using Fermat's principle, *i.e.*, the time of transit for a ray between the source and observer is an extremum, find the condition that all paraxial rays starting from the source will converge at a point '*O*' on the axis. Find the focal length.



(ii) A gravitational lens may be assumed to have a varying width of the form

$$W(b) = K_1 In\left(\frac{K_2}{b}\right) \quad (b_{\min} < b < b_{\max})$$

$$= K_1 In\left(\frac{K_2}{b_{\min}}\right) \quad b < b_{\min}$$

Show that an observer will see an image of a point object as a ring about the centre of the lens with an angular radius

$$\beta = \sqrt{\frac{(n-1)k_1 \dfrac{u}{v}}{u+v}}$$ **[NCERT Exemp. Q. 9.32, Page 60]**

Ans. **(i)** The time taken by ray from S to $P_1 = \dfrac{SP_1}{c}$

$$t_1 = \frac{\sqrt{u^2+b^2}}{c} = \frac{u}{c}\left(1+\frac{b^2}{u^2}\right)^{1/2}$$

$$t_1 = \frac{u}{c}\left[1+\frac{b^2}{2u^2}\right] \quad \text{(Assuming } b <<< u\text{)}$$

Similarly, time required by ray from P_1 to $O = \dfrac{P_1O}{c}$

$$t_2 = \frac{v}{c}\left[1+\frac{b^2}{2v^2}\right]$$

Time required by ray to travel through the lens,

$$t_3 = \frac{(\mu-1)W(b)}{c}$$

So the total time by ray from S to O, $t = t_1 + t_2 + t_3$

$$t = \frac{u}{c}\left(1+\frac{b^2}{2u^2}\right)+\frac{v}{c}\left(1+\frac{b^2}{2v^2}\right)+\frac{(\mu-1)W(b)}{c}$$

$$= \frac{1}{c}\left[u+\frac{b^2}{2u}+v+\frac{b^2}{2v}+(\mu-1)W(b)\right]$$

$$= \frac{1}{c}\left[u+v+\frac{b^2}{2}\left(\frac{1}{u}+\frac{1}{v}\right)+(\mu-1)W(b)\right]$$

If $\dfrac{1}{D} = \dfrac{1}{u}+\dfrac{1}{v}$

$$\therefore t = \frac{1}{c}\left[u+v+\frac{b^2}{2D}+(\mu-1)\left(W_0-\frac{b^2}{\alpha}\right)\right]$$

By Fermat's Principle,

$$\frac{dt}{db} = \frac{1}{c}\left[0+0\frac{2b}{2D}+(\mu-1)\left[0-\frac{2b}{\alpha}\right]\right]$$

$$= \frac{b}{cD}-(\mu-1)\frac{-2b}{\alpha c}$$

But time from S to O remains constant, so, $\dfrac{dt}{db} = 0$

or $\dfrac{d}{cD}-\dfrac{2(\mu-1)b}{\alpha c} = 0$

$$\frac{2(\mu-1)b}{\alpha c} = \frac{b}{cD}$$

$$\alpha = 2(\mu-1)D$$

So, the convergence lens is formed $\alpha = 2(\mu-1)D$.
It is independent of B and so all paraxial rays from S will converge at O (*i.e.*, for rays $b <<< u$ and $b << v$).
So that,

$\dfrac{1}{D} = \dfrac{1}{u}+\dfrac{1}{v}$ is equivalent to focal length f

$$\frac{1}{f} = \frac{1}{v}-\frac{1}{u}$$

So that, $t = \dfrac{1}{c}\left[u+v+\dfrac{b^2}{2D}+(\mu-1)W(b)\right]$

where $KW(b) = K_1 \log\left(\dfrac{K_2}{b}\right)$ **[2½]**

(ii) $t = \dfrac{1}{c}\left[u+v+\dfrac{b^2}{2D}+(\mu-1)K_1 \log\dfrac{K_2}{b}\right]$

$$\frac{dt}{db} = \frac{1}{c}\left[0+0+\frac{2b}{2D}+(\mu-1)K_1\frac{b}{K_2}\right]\frac{d}{db}\left(\frac{K_2}{b}\right)$$

$$= \frac{1}{c}\left[\frac{b}{D}+(u-1)\frac{K_1}{K_2}b.K_2(-1)\frac{1}{b^2}\right]$$

As time from S to O is constant, so, $\dfrac{dt}{db} = 0$

So, $$0 = \frac{1}{c}\left[\frac{b}{D}-(\mu-1)\frac{K_1}{b}\right]$$

and, $(\mu-1)\dfrac{k_1}{b} = \dfrac{b}{D}$ $\left(\text{as}\dfrac{1}{c}\neq 0\right)$

$$b^2 = K_1D(\mu-1)$$

$$b = \sqrt{(\mu-1)K_1D}$$

Thus all the rays passing at a height b shall contribute to the image. The ray path makes an angle,

$$\beta = \frac{b}{v}$$

$$= \frac{\sqrt{(\mu-1)K_1D}}{v}$$

$$= \sqrt{\frac{(\mu-1)K_1D}{v^2}} \qquad \left(as,\ \frac{1}{D} = \frac{1}{v}+\frac{1}{u}\right)$$

$$= \sqrt{\frac{(\mu-1)K_1uv}{v^2(\mu+v)}} \qquad \left(D = \frac{uv}{u+v}\right)$$

$$\beta = \sqrt{\frac{(\mu-1)K_1u}{v(u+v)}}$$ **[2½]**

Q. 10. A beam of light converges at a point *P*. Now a lens is placed in the path of the convergent beam 12 cm from *P*. At what point does the beam converge if the lens is (*a*) a convex lens of focal length 20 cm, and (*b*) a concave lens of focal length 16 cm?
[NCERT Ex. Q. 9.8, Page 346]

Ans. Given that,
According to the given situation, the object is virtual and the image formed is real.
Object distance, $u = +12$ cm

(a) Focal length of the convex lens, $f = 20$ cm
Image distance $= v$
According to the lens formula, we have the relation,

$$\frac{1}{v}-\frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v}-\frac{1}{12} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20}+\frac{1}{12} = \frac{3+5}{60} = \frac{8}{60}$$

So,

$$v = \frac{60}{8} = 7.5 \text{ cm}$$

So that, the image is formed 7.5 cm away from the lens, towards its right. **[2½]**

(b) Focal length of the concave lens, $f = -16$ cm

Image distance = v

According to the lens formula,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=-\frac{1}{16}+\frac{1}{12}$$

$$\frac{1}{v}=\frac{-3+4}{48}=\frac{1}{48}$$

So,

$$v = 48 \text{ cm}$$

So, the image is formed 48 cm away from the lens, towards its right. **[2½]**

Q. 11. (a) Determine the 'effective focal length' of the combination of the two lenses in Exercise 9.10, if they are placed 8.0 cm apart with their principal axes coincident. Does the answer depend on which side of the combination a beam of parallel light is incident? Is the notion of effective focal length of this system useful at all?

(b) An object 1.5 cm in size is placed on the side of the convex lens in the arrangement (*a*) above. The distance between the object and the convex lens is 40 cm. Determine the magnification produced by the two-lens system, and the size of the image.

[NCERT Ex. Q. 9.21, Page 347]

Ans. Given that,

Focal length of the convex lens, $f_1 = 30$ cm

Focal length of the concave lens, $f_2 = -20$ cm

Distance between the two lenses, $d = 8.0$ cm

(a) **(i)** When the parallel beam of light is incident on the convex lens :

According to the lens formula,

$$\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f_1}$$

Where,

u_1 = Object distance = ∞

v_1 = Image distance

$$\frac{1}{v_1}=\frac{1}{30}-\frac{1}{\infty}=\frac{1}{30}$$

$$\therefore v_1 = 30 \text{ cm}$$

The image will act as a virtual object for the concave lens.

Applying lens formula to the concave lens,

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2}$$

Where,

u_2 = Object distance

$=(30 - d) = 30 - 8 = 22$ cm

v_2 = Image distance

$$\frac{1}{v_2}=\frac{1}{22}-\frac{1}{20}=\frac{10-11}{220}=\frac{-1}{220}$$

$$\therefore v_2 = -220 \text{ cm}$$

The parallel incident beam appears to diverge from a point that is

$$\left(220-\frac{d}{2}=220-4\right)=216 \text{ cm}$$

from the centre of the combination of the two lenses.

(ii) When the parallel beam of light is incident, from the left, on the concave lens :

According to the lens formula,

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2}$$

$$\frac{1}{v_2}=\frac{1}{f_2}+\frac{1}{u_2}$$

Where,

u_2 = Object distance = ∞

v_2 = Image distance

$$\frac{1}{v_2}=\frac{1}{-20}+\frac{1}{-\infty}=-\frac{1}{20}$$

$$\therefore v_2 = -20 \text{ cm}$$

The image will act as a real object for the convex lens.

Applying lens formula to the convex lens,

$$\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f_1}$$

Where,

u_1 = Object distance

$= -(20 + d) = -(20+8) = -28$ cm

And,

v_1 = Image distance

$$\frac{1}{v_1}=\frac{1}{30}+\frac{1}{-28}=\frac{14-15}{420}=\frac{-1}{420}$$

$$\therefore v_1 = -420 \text{ cm}$$

So, the parallel incident beam appears to diverge from a point that is (420 – 4 cm) = 416 cm from the left of the centre of the combination of the two lenses.

The answer does depend on the side of the combination at which the parallel beam of light is incident. The notion of effective focal length does not seem to be useful for this combination. **[2½]**

(b) Height of the image, $h = 1.5$ cm

Object distance from the side of the convex lens, $u_1 = -40$ cm

$|u_1| = 40$ cm

According to the lens formula :

$$\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f_1}$$

Where,

v_1 = Image distance

$$\frac{1}{v_1}=\frac{1}{30}+\frac{1}{-40}=\frac{4-3}{120}=\frac{1}{120}$$

$$\therefore v_1 = 120 \text{ cm}$$

Magnification, $m=\left|\frac{v_1}{u_1}\right|$

$$=\frac{120}{40}=3$$

So, the magnification due to the convex lens is 3.

The image formed by the convex lens acts as an object for the concave lens.
According to the lens formula :

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2}$$

Where,
u_2 = Object distance
$= +(120 - 8) = 112$ cm.
v_2 = Image distance

$$\frac{1}{v_2}=\frac{1}{-20}+\frac{1}{112}=\frac{-112+20}{2240}=\frac{-92}{2240}$$

$$\therefore v_2=\frac{-2240}{92}\text{ cm}$$

$$\text{Magnification, } m'=\left|\frac{v_2}{u_2}\right|=\frac{2240}{92}\times\frac{1}{112}=\frac{20}{92}$$

The magnification produced by the combination of the two lenses is calculated as :

$$= m\times m'=3\times\frac{20}{92}=\frac{60}{92}=0.652$$

The magnification of the combination is given as :

$$\frac{h_2}{h_1}=0.652$$

$$h_2=0.652\times h_1$$

Where,
h_1 = Object size = 1.5 cm
h_2 = Size of the image

$$\therefore h_2=0.652\times1.5=0.98\text{ cm}$$

So, the height of the image is 0.98 cm. **[2½]**

TOPIC-2
Prism and Optical Instruments

Quick Review

- Ray optics can be used to describe the geometrical aspects of imaging, including optical aberrations.
- Geometrical optics provides rules, which may depend on the colour (wavelength) of the ray (or light), for propagating these rays through an optical system.
- Glass prisms can be shaped to produce total internal reflection and as such are employed in binoculars, periscopes, telescopes and other optical instruments. Light rays may be conducted over long, twisting paths by multiple total internal reflections in glass or plastic rods or fibres.
- Periscope, optical instrument used in land and sea warfare, submarine navigation and elsewhere to enable an observer to see his surroundings while remaining under cover, behind armour, or submerged.
- A Keplerian telescope has a converging lens eyepiece and a Galilean telescope has a diverging lens eyepiece.
- The resolving power or resolution of an optical instrument is its ability to separate the images of two objects, which are close together.

TIPS...
- *Study the general terms related to dispersion of Light through prism.*
- *Understand the thin lens formula and Lens Maker's formula.*
- *Understand the working of Human eye.*

TRICKS...
- *Practice the diagram of Optical instruments.*
- *Make flow chart to remember all important terms related to Prism and Optical instruments.*

Know the Links

- *https://www.britannica.com/science/physics-science*
- *http://electron6.phys.utk.edu/light/4/instruments.htm*
- *http://physics.bu.edu/~duffy/PY106/Instruments.html*

Multiple Choice Questions **(1 mark each)**

Q. 1. You are given four sources of light each one providing a light of a single colour –red, blue, green and yellow. Suppose the angle of refraction for a beam of yellow light corresponding to a particular angle of incidence at the interface of two media is 90°. Which of the following statements

is correct if the source of yellow light is replaced with that of other lights without changing the angle of incidence?

(a) The beam of red light would undergo total internal reflection.

(b) The beam of red light would bend towards normal while it gets refracted through the second medium.

(c) The beam of blue light would undergo total internal reflection.

(d) The beam of green light would bend away from the normal as it gets refracted through the second medium.

[NCERT Exemp. Q. 9.5, Page 55]

Ans. Correct option : (c)

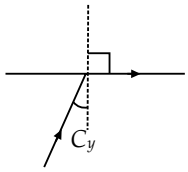
Explanation : According to the Cauchy relationship,

$$\lambda \propto \frac{1}{\mu}$$

Smaller the wavelength higher the refractive index and so that smaller the critical angle

So, light rays are passing from denser to rarer medium.

$$\text{As } \sin c = \frac{1}{\mu} \text{ so, } c \propto \frac{1}{\mu} > \mu_v > \mu_g > \mu_y > \mu_R$$



So, critical angle for $C_v < C_g < C_Y < C_R$, that is, critical angle of blue and green light is smaller than that of yellow and it is greater for red colour light. As the angle of refraction for yellow light is for a particular incident angle, this incidence angle is critical angle for yellow. Let it be C_Y.

As we know that, $C_R > C_v$.

So, it will not get total internal reflection and $C_V < C_Y$, $Cg < C_Y$.

So, light of blue and green colour get total internal reflection. So, correct answer is (c).

Q. 2. A ray of light incident at an angle θ on a refracting face of a prism emerges from the other face normally. If the angle of the prism is 5° and the prism is made of a material of refractive index 1.5, the angle of incidence is :

(a) 7.5°. **(b) 5°.**

(c) 15°. **(d) 2.5°.**

[NCERT Exemp. Q. 9.1, Page 54]

Ans. Correct option : (a)

Explanation : Given that,

$A = 5°$

$\mu = 1.5$

$i_2 = 0°$

$r_2 = 0°$

As we know that,

$r_1 + r_2 = A$

$r_1 = A - r_2$

$= 5 - 0 = 5°$

From Snell's law :

$$\mu = \frac{\sin i_1}{\sin r_1}$$

$\sin i_1 = \mu \sin r_1$

$= 1.5 \times \sin 5°$

$= 1.5 \times 0.087$

$= 0.1305$

$= 7.5°$

Q. 3. A passenger in an aeroplane shall

(a) never see a rainbow.

(b) may see a primary and a secondary rainbow as concentric circles.

(c) may see a primary and a secondary rainbow as concentric arcs.

(d) shall never see a secondary rainbow.

[NCERT Exemp. Q. 9.4, Page 55]

Ans. Correct option : (b)

Explanation : A passenger in an aeroplane may see primary and secondary rainbow as concentric circles due to the dispersion of sunlight.

Tick Two or More Options

Q. 4. Consider an extended object immersed in water contained in a plane trough. When seen from close to the edge of the trough the object looks distorted because

(a) the apparent depth of the points close to the edge are nearer the surface of the water compared to the points away from the edge.

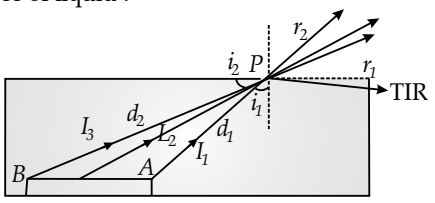
(b) the angle subtended by the image of the object at the eye is smaller than the actual angle subtended by the object in air.

(c) some of the points of the object far away from the edge may not be visible because of total internal reflection.

(d) water in a trough acts as a lens and magnifies the object. **[NCERT Exemp. Q. 9.12, Page 57]**

Ans. Correct options : (a), (b) and (c)

Explanation : We know that shifting (h) of image of an object immersed in liquid from object is directly proportional to the real distance of object from the surface of liquid :



$$h = t\left(1 - \frac{1}{\mu}\right)$$

Where t = Real depth or distance of the object from the surface of liquid of refractive index μ

If the object is seen from one edge of trough the relative differences of depth (distance) in water between two ends of objects is larger than if it seen from the top or away from edge.

By above formula or distortion of nearer end is smaller than farther, verifies the option (a).

The angle subtended by an object is larger than its image in water, as its image shifts upwards verifies option (b).

Rays coming out from object to observer passes from denser to rarer medium, and angle of incidence for rays from farther end *B* of object is larger than near end *A*. The incidence angles for rays coming from end 3 may have incidence angle more than critical angle and can cause total internal reflection, so will not reach to observer, cause non-visible of farther end verifies option (c).

Q. 5. Between the primary and secondary rainbows, there is a dark band known as Alexandar's dark band. This is because

(a) light scattered into this region interfere destructively.

(b) there is no light scattered into this region.

(c) light is absorbed in this region.

(d) angle made at the eye by the scattered rays with respect to the incident light of the sun lies between approximately 42° and 50°.

[NCERT Exemp. Q. 9.14, Page 57]

Ans. Correct options : (a) and (d)

Explanation : The Alexandar's dark band lies between the primary and secondary rainbows, formed due to light scattered into this region interfere destructively. The primary rainbows subtend an angle nearly 41°–42° at observer's eye, whereas secondary rainbows subtends an angle nearly 51°–54° at observer's eye with respect to incident light ray.

So, the scattered rays with respect to the incident light of the sun lies between approximately 42° and 50°.

Q. 6. A magnifying glass is used, as the object to be viewed can be brought closer to the eye than the normal near point. This results in

(a) a larger angle to be subtended by the object at the eye and hence viewed in greater detail.

(b) the formation of a virtual erect image.

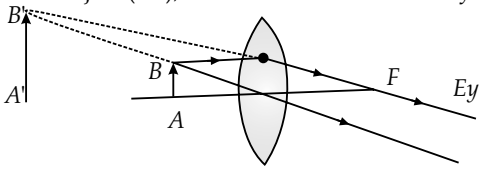
(c) increase in the field of view.

(d) infinite magnification at the near point.

[NCERT Exemp. Q. 9.15, Page 58]

Ans. Correct options : (a) and (b)

Explanation : In magnifying glass, the object is placed within the focal length and the image formed is magnified and erect. As (*A'B'*) image is magnified so it subtends larger angle at the eye than object (*AB*), so can be seen more clearly.



Q. 7. An astronomical refractive telescope has an objective of focal length 20 m and an eyepiece of focal length 2 cm.

(a) The length of the telescope tube is 20.02 m.

(b) The magnification is 1000.

(c) The image formed is inverted.

(d) An objective of a larger aperture will increase the brightness and reduce chromatic aberration of the image. [NCERT Exemp. Q. 9.16, Page 58]

Ans. Correct options : (a), (b) and (c)

Explanation : The length of the telescope

$L = f_o + f_e$

$= 20 + 00.02 = 20.02$ m

And

$$m = \frac{f_o}{f_e} = \frac{20\text{ m}}{0.02\text{ m}}$$

$$= \frac{2000}{2} = 1000$$

The final image formed in telescope (refracting) is inverted, virtual and smaller than object.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is three-fourth of the angle of prism. Calculate the speed of light in the prism.

[CBSE Board, All India Region, 2018]

Ans. Given that,

Angle of prism, $A = 60^\circ$

Refractive index of prism = 1.5

Angle of incidence,

$$i_1 = i_2 = \frac{3}{4}A$$

$$= \frac{3}{4} \times 60^\circ = 45^\circ$$

As, $A + \delta = i_1 + i_2$

$\Rightarrow 60^\circ + \delta = 45^\circ + 45^\circ$

So, the angle of minimum deviation,

$\delta = 90^\circ - 60^\circ = 30^\circ$

We know that, velocity of light in vacuum $= 3 \times 10^8$ m/s

Now, using the formula,

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}} = \frac{c}{v}$$

That is,

$$v = c\frac{\sin A/2}{\sin\left(A + \delta_m\right)/2} = 3 \times 10^8 \frac{\sin 30^\circ}{\sin(60^\circ + 30^\circ)/2}$$

$$= 3 \times 10^8 \frac{\sin 30^\circ}{\sin 45^\circ} = \frac{3 \times 10^8 \times 0.5000}{0.7071} = 2.12 \times 10^8 \text{ m/s}$$ **[2]**

Q. 2. How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced by red light? Give reason.
[CBSE Board, All India Region, 2017]

Ans. If the incident violet light is replaced with red light, then the angle of minimum deviation of a glass prism decreases.

$n_{violet} > n_{red}$ **[1]**

Q. 3. For a glass prism ($\mu = \sqrt{3}$) the angle of minimum deviation is equal to the angle of the prism. Find the angle of the prism. [NCERT Exemp. Q. 9.21]

Ans. The required relation for minimum angle of deviation :

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}} \qquad [\because \delta_m = \text{A (Given)}]$$

$$\therefore \mu = \frac{\sin A}{\sin\frac{A}{2}}$$

$$= \frac{2\sin\frac{A}{2}\cos\frac{A}{2}}{\sin\frac{A}{2}}$$

$$= 2\cos\frac{A}{2}$$

$$\sqrt{3} = 2\cos\frac{A}{2}$$

$$\Rightarrow \cos\frac{A}{2} = \frac{\sqrt{3}}{2}$$

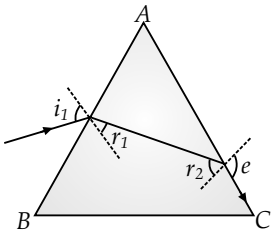
$$\cos\frac{A}{2} = \cos 30^\circ$$

$$\Rightarrow \frac{A}{2} = 30^\circ$$

$A = 60^\circ$ is angle of prism. **[2]**

Q. 4. At what angle should a ray of light be incident on the face of a prism of refracting angle 60° so that it just suffers total internal reflection at the other face? The refractive index of the material of the prism is 1.524. [NCERT Ex. Q. 9.22, Page 348]

Ans. The incident, refracted and emergent rays associated with a glass prism *ABC* are shown in the below figure.



Given that,
Angle of prism, $\angle A = 60°$
Refractive index of the prism, $\mu = 1.524$
i_1 = Incident angle
r_1 = Refracted angle
r_2 = Angle of incidence at the face *AC*
e = Emergent angle = 90°

According to Snell's law, for side *AC*,

$$\frac{\sin e}{\sin r_2} = \mu$$

$$\sin r_2 = \frac{1}{\mu} \times \sin 90^\circ = \frac{1}{1.524} = 0.6562$$

$$\therefore \quad r_2 = \sin^{-1} 0.6562 \approx 41^\circ$$

It is clear from the figure that angle, $A = r_1 + r_2$

$$\therefore \quad r_1 = A - r_2 = 60 - 41 = 19^\circ \quad .$$ **[1]**

According to Snell's law, we have the relation :

$$\mu = \frac{\sin i_1}{\sin r_1}$$

$$\sin i_1 = \mu \sin r_1 = 1.524 \times \sin 19^\circ = 0.496$$

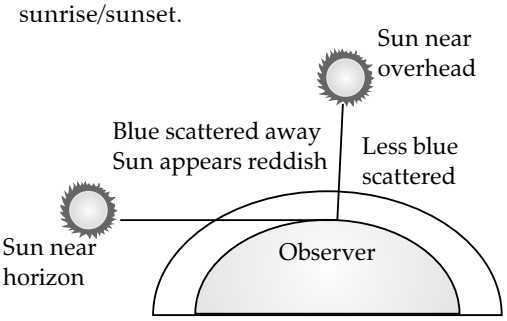
$$\therefore \quad i_1 = 29.75^\circ$$

So, the angle of incidence is 29.75°. **[1]**

Q. 5. Why does sun appear red at sunrise and sunset?
[CBSE Board, All India Region, 2016]

Ans. The sky appears reddish during sunrise/sunset but it appears white at noon. During noon, the sunlight has to travel less distance to reach us. Most of the colours reaching us get scattered. Due to this, sky appears white at noon. **[½]**

Colours near the red end of the spectrum scatter the least. During sunset, and sunrise, sunlight needs to travel more distance to reach us. Red colour is able to reach us because it is scattered the least. Hence, sky appears reddish during sunrise/sunset.



[½]

Q. 6. The near vision of an average person is 25 cm. To view an object with an angular magnification of 10, what should be the power of the microscope?
[NCERT Exemp. Q. 9.18, Page 58]

Ans. Given that,
The final image at distinct vision, D = 25 cm
For eyes lens $u = -f$ and $v = -25$

$$m = \frac{+v}{u}$$

$$10 = \frac{-25}{-f}$$

$$f = +\frac{25}{10} = +2.25 \text{ cm} = 0.025 \text{ m}$$

$$P = \frac{1}{f} = \frac{1}{0.025} = \frac{1000}{25} = 40\ D$$ **[1]**

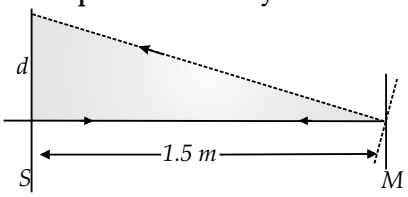
Q. 7. A person looking at a person wearing a shirt with a pattern comprising vertical and horizontal lines is able to see the vertical lines

more distinctly than the horizontal ones. What is this defect due to? How is such a defect of vision corrected? [NCERT Ex. Q. 9.27, Page 348]

Ans. In the given case, the person is able to see vertical lines more distinctly than horizontal lines. This means that the refracting system (cornea and eye lens) of the eye is not working in the same way in different planes. This defect is called astigmatism. **[1]**

The person's eye has enough curvature in the vertical plane. However, the curvature in the horizontal plane is insufficient. So that, the sharp images of the vertical lines are formed on the retina, but horizontal lines appear blurred. This defect can be corrected by using cylindrical lenses. **[1]**

Q. 8. Light incident normally on a plane mirror attached to a galvanometer coil retraces backwards as shown in Figure A current in the coil produces a deflection of 3.5° of the mirror. What is the displacement of the reflected spot of light on a screen placed 1.5 m away?



[NCERT Ex. Q. 9.37, Page 349]

Ans. Given that,

Angle of deflection, $\theta = 3.5°$

Distance of the screen from the mirror, $D = 1.5$ m

The reflected rays get deflected by an amount twice the angle of deflection, $2\theta = 7.0°$

The displacement (d) of the reflected spot of light on the screen is given as :

$$\tan 2\theta = \frac{d}{1.5}$$

$$\therefore \quad d = 1.5 \times \tan 7^\circ = 0.184 \text{ m} = 18.4 \text{ cm}$$

So, the displacement of the reflected spot of light is 18.4 cm. **[2]**

Q. 9. An unsymmetrical double convex thin lens forms the image of a point object on its axis. Will the position of the image change if the lens is reversed? [NCERT Exemp. Q. 9.19, Page 58]

Ans. As we know that the position of image in convex lens changes either by changing 'u' of 'f', where, 'u' is constant formula used,

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_1}\right) \quad \textbf{[1]}$$

Where, and curvature of the lens are same so 'f' is constant. So, on reversing the lens, position and nature of the image will not change. **[1]**

Q. 10. Does the magnifying power of a microscope depend on the colour of the light used? Justify your answer.

[CBSE Board, Foreign Scheme, 2017]

Ans. Yes, the magnifying power of a microscope depends on the colour of the light used.

Refractive index (n) is different for different colours of light.

Lens Maker formula :

$$f = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

So for different wavelength (colour) of light refractive index will be different and so focal length of the lens (f) will be different.

$$m = \frac{f}{(f + u)}$$

For microscope 'u' can also consider, $m = 1 + d/f$

By this formula different 'f' will have different magnifications and so colour of light used does have effect on the magnitude of magnification. **[1]**

Q. 11. (a) State the condition under which a large magnification can be achieved in an astronomical telescope.

(b) Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope.

[CBSE Board, Foreign Scheme, 2017]

Ans. In astronomical telescope as light is incident from infinity on the objective lens it forms the image at the focus of the objective lens and for large magnification the length between the lens is so adjusted that the final image is formed at least distance of distinct vision from the eyepiece.

Magnification in this case is,

$$M = \frac{f_0}{f_e}\left(1 + \frac{f_e}{D}\right) \quad \textbf{[1]}$$

(b) The reflecting telescopes are preferred over refracting type because of the following reasons :

(i) There is no chromatic aberration in case of reflecting telescopes as the objective is a mirror. **[½]**

(ii) Spherical aberration is reduced in case of reflecting telescopes by using mirror objective in the form of a paraboloid. **[½]**

Q. 12. Explain two advantages of a reflecting telescope over a refracting telescope.

[CBSE Board, Delhi Region, 2018]

Ans. Advantages of a reflecting telescope over a refracting telescope :

(i) Reflecting telescope has height resolving power due to a large aperture of mirror. **[1]**

(ii) Due to availability of paraboloidal mirror, the image is free from chromatic and spherical aberration. **[1]**

Short Answer Type Questions **(3 marks each)**

Q. 1. You are given prisms made of crown glass and flint glass with a wide variety of angles. Suggest a combination of prisms which will

(a) deviate a pencil of white light without much dispersion,

(b) disperse (and displace) a pencil of white light without much deviation.

[NCERT Ex. Q. 9.23, Page 348]

Ans. **(a)** Place the two prisms beside each other. Make sure that their bases are on the opposite sides of the incident white light, with their faces touching each other. When the white light is incident on the first prism, it will get dispersed. When this dispersed light is incident on the second prism, it will recombine and white light will emerge from the combination of the two prisms.

Mathematically, the total angular dispersion produced must be equal to zero. So,

$(\mu_v - \mu_R)A + (\mu'_v - \mu'_R)A' = 0$ **[1½]**

(b) Take the system of the two prisms as suggested in answer (a). Adjust (increase) the angle of the flint-glass prism so that the deviations due to the combination of the prisms become equal. This combination will disperse the pencil of white light without much deviation.

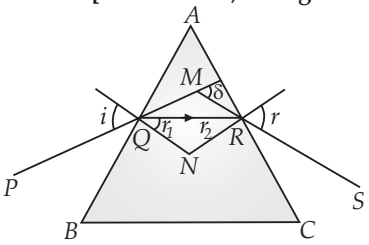
Mathematically, the total deviation produced by the combination must be equal to zero. So,

$(\mu - 1)A + (\mu' - 1)A' = 0$ **[1½]**

Q. 2. Draw the ray diagram showing refraction of light through a glass prism and hence obtain the relation between the refractive index μ of the prism, angle of prism and angle of minimum deviation.

[CBSE Board, Foreign Scheme, 2017]

Ans.



[1]

From ΔMQR, $(i - r_1) + (e - r_2) = \delta$

So $(i + e) - (r_1 + r_2) = \delta$

From ΔPQN,

$r_1 + r_2 + \angle QNR = 180^\circ$

Also

$A + \angle QNR = 180^\circ$

So that,

$A = r_1 + r_2$

And, $i + e - A = \delta$

At minimum deviation,

$i = e$,

$r_1 = r_2 = r$ and $\delta = \delta_m$

$\Rightarrow \quad i = \dfrac{A + \delta_m}{2}$

and $\quad r = \dfrac{A}{2}$

Also $\quad \mu = \dfrac{\sin i}{\sin r}$

Thus, $\quad \mu = \dfrac{\sin\left(\dfrac{A+\delta_m}{2}\right)}{\sin\left(\dfrac{A}{2}\right)}$ **[2]**

Q. 3. A myopic adult has a far point at 0.1 m. His power of accommodation is 4 dioptres.

(i) What power lenses are required to see distant objects?

(ii) What is his near point without glasses?

(iii) What is his near point with glasses? (Take the image distance from the lens of the eye to the retina to be 2 cm).

[NCERT Exemp. Q. 9.27, Page 59]

Ans. **(i)** Power of lens required to see clearly the object placed at infinity. $u = -\infty$

$\dfrac{1}{f} = \dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{-10} - \dfrac{1}{\infty}$	$P = \dfrac{1}{f}\text{m}$
$\dfrac{1}{f} = \dfrac{1}{-10}$	$P = \dfrac{1}{-0.1}\text{m}$
$f = -10\text{ cm} = -0.1\text{ m}$	$P = -10\ D$

[1]

(ii) When no corrective lens used : Let assume powers of eye when object is at far point, near point are P_f and P_n respectively and power

$\therefore P_n = P_f + P_a$

When object is at far point its clear image is formed at retina 2 cm from eye lens

$\therefore u = -10\text{ cm} = -0.1\text{ m},\ v = 2\text{ cm} = 0.02\text{ m}$

If f is focal length of eye lens focused at far point then

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{0.02} - \frac{1}{(-0.1)}$$

$$\frac{1}{f} = 50 + 10 = 60$$

$$P_f = 60D$$

$$\therefore P_n = P_f + P_a = 60 + 4 = 64D$$

Let the near point be x_n

$u = -x_n\ (v = 2\text{ cm} = 0.02\text{ m})$

$$\therefore \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{0.02} + \frac{1}{x_n} = \text{Power } (P_n)$$

$$50 + \frac{1}{x_n} = 64$$

$$\frac{1}{x_n} = 64 - 50 = 14\ D$$ **[1]**

Near point without glass, $x_n = \dfrac{1}{4}\text{ m} = \dfrac{100}{14}\text{ cm} \simeq 7\text{ cm}$ (Approx).

(iii) When used corrective lens : When corrective lens is used then eye can see the object at infinity. Power of eye lens in this situation is P_∞

$u = \infty$ and $v = 2\text{ cm} = 0.02\text{ m}$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$P_\infty = \frac{1}{0.02} - \frac{1}{\infty} = 50$$

$$P_\infty = 50$$

$$P_\infty = 50\ D$$

If P_n = Power of eye at near point when corrective lens is used.

$P_n = P_\infty + P_a = 50 + 4 = 54\ D$

Let near point in this situation is x_n

$u = -x_n$ m

$v = +2$ cm $= 0.02$ m

$$\frac{1}{f} = 54$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{0.02} + \frac{1}{x_n} = 54 \qquad \text{(All distance are in 'm'.)}$$

$$50 + \frac{1}{x_n} = 54$$

$$\frac{1}{x_n} = 4$$

$$x_n = \frac{1}{4}\ \text{m} = 0.25\ \text{m} \qquad \textbf{[1]}$$

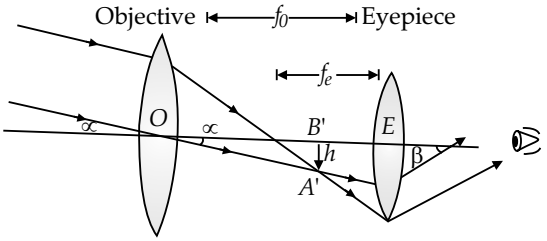
Q. 4. (a) Draw a ray diagram depicting the formation of the image by an astronomical telescope in normal adjustment.

(b) You are given the following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope? Give reason.

Lenses	Power (D)	Aperture (cm)
L_1	3	8
L_2	6	1
L_3	10	1

[CBSE Board, All India Region, 2017]

Ans. (a) Ray diagram of astronomical telescope :



[2]

(b) Objective lens : Lens L_1

Eyepiece lens : Lens L_2

Reason : The objective should have large aperture and large focal length while the eyepiece should have small aperture and small focal length. **[1]**

Q. 5. For a normal eye, the far point is at infinity and the near point of distinct vision is about 25 cm in front of the eye. The cornea of the eye provides a converging power of about 40 dioptres, and the least converging power of the eye lens behind the cornea is about 20 dioptres. From this rough data estimate the range of accommodation (*i.e.*, the range of converging power of the eye lens) of a normal eye. [NCERT Ex. Q. 9.24, Page 348]

Ans. Given that,

Least distance of distinct vision, $d = 25$ cm

Far point of a normal eye, $d' = \infty$

Converging power of the cornea, $P_c = 40\ D$

Least converging power of the eye-lens, $P_e = 20\ D$

To see the objects at infinity, the eye uses its least converging power.

Power of the eye-lens, $P = P_c + P_e = 40 + 20 = 60\ D$ **[1]**

Power of the eye-lens is given as :

$$P = \frac{1}{\text{Focal length of the eye lens } (f)}$$

$$f = \frac{1}{P} = \frac{1}{60D} = \frac{100}{60} = \frac{5}{3}\ \text{cm}$$

To focus an object at the near point, object distance $(u) = -d = -25$ cm

Focal length of the eye-lens = Distance between the cornea and the retina

= Image distance **[1]**

So, image distance, $v = 5/3$ cm

According to the lens formula,

$$\frac{1}{f'} = \frac{1}{v} - \frac{1}{u}$$

Where,

f' = Focal length

$$\frac{1}{f'} = \frac{3}{5} + \frac{1}{25} = \frac{15+1}{25} = \frac{16}{25}\ \text{cm}^{-1}$$

$$\text{Power, } P' = \frac{1}{f'} \times 100$$

$$= \frac{16}{25} \times 100 = 64\ D$$

Thus, power of the eye-lens = 64 − 40 = 24 D

So, the range of accommodation of the eye-lens is from 20 D to 24 D. **[1]**

Q. 6. Does short-sightedness (myopia) or long-sightedness (hyper-metropia) imply necessarily that the eye has partially lost its ability of accommodation? If not, what might cause these defects of vision? [NCERT Ex. Q. 9.25, Page 348]

Ans. A myopic or hyper-metropic person can also possess the normal ability of accommodation of the eye-lens. Myopia occurs when the eye-balls get elongated from front to back. **[1½]**

Whereas, hyper-metropia occurs when the eye-balls get shortened. When the eye-lens loses its ability of accommodation, the defect is called presbyopia. **[1½]**

Q. 7. A man with normal near point (25 cm) reads a book with small print using a magnifying glass : a thin convex lens of focal length 5 cm.

(a) What is the closest and the farthest distance at which he should keep the lens from the page so that he can read the book when viewing through the magnifying glass?

(b) What is the maximum and the minimum angular magnification (magnifying power) possible using the above simple microscope?

[NCERT Ex. Q. 9.28, Page 348]

Ans. (a) Given that,

Focal length of the magnifying glass, $f = 5$ cm

Least distance of distance vision, $d = 25$ cm
Closest object distance $= u$
Image distance, $v = -d = -25$ cm
According to the lens formula,

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{u}=\frac{1}{v}-\frac{1}{f}$$

$$=\frac{1}{-25}-\frac{1}{5}=\frac{-5-1}{25}=\frac{-6}{25}$$

$$\therefore u=-\frac{25}{6}=-4.167 \text{ cm}$$

So, the closest distance at which the person can read the book is 4.167 cm.

For the object at the farthest distant (u'), the image distance ($v' = \infty$)
According to the lens formula,

$$\frac{1}{f}=\frac{1}{v'}-\frac{1}{u'}$$

$$\frac{1}{u'}=\frac{1}{\infty}-\frac{1}{5}=-\frac{1}{5}$$

$$\therefore u' = -5 \text{ cm}$$

So, the farthest distance at which the person can read the book is 5 cm. **[1½]**

(b) Maximum angular magnification is given by the relation :

$$\alpha_{max}=\frac{d}{|u|}=\frac{25}{\frac{25}{6}}=6$$

Minimum angular magnification is :

$$\alpha_{min}=\frac{d}{|u'|}=\frac{25}{5}=5$$ **[1½]**

Q. 8. A myopic person has been using spectacles of power –1.0 dioptre for distant vision. During old age he also needs to use separate reading glass of power +2.0 dioptres. Explain what may have happened. [NCERT Ex. Q. 9.26, Page 348]

Ans. The power of the spectacles used by the myopic person, $P = -1.0\ D$
Focal length of the spectacles,

$$f=\frac{1}{P}=\frac{1}{-1\times10^{-2}}=-100 \text{ cm}$$

Thus, the far point of the person is 100 cm. He might have a normal near point of 25 cm. When he uses the spectacles, the objects placed at infinity produce virtual images at 100 cm. He uses the ability of accommodation of the eye-lens to see the objects placed between 100 cm and 25 cm. **[2]**
During old age, the person uses reading glasses of power; $P' = +2D$

so, $f = \frac{1}{P} = 50$ cm is the ability of accommodation lost in old age. This defect is called presbyopia. As a result, he is unable to see clearly the objects placed at 25 cm. **[1]**

Q. 9. A card sheet divided into squares each of size 1 mm^2 is being viewed at a distance of 9 cm through a magnifying glass (a converging lens of focal length 9 cm) held close to the eye.
(a) What is the magnification produced by the lens? How much is the area of each square in the virtual image?
(b) What is the angular magnification (magnifying power) of the lens?
(c) Is the magnification in (*a*) equal to the magnifying power in (*b*)? Explain.
[NCERT Ex. Q. 9.29, Page 348]

Ans. **(a)** Given that,
Area of each square, $A = 1$ mm^2
Object distance, $u = -9$ cm
Focal length of a converging lens, $f = 9$ cm
For image distance v, the lens formula can be written as :

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{10}=\frac{1}{v}+\frac{1}{9}$$

$$\frac{1}{v}=-\frac{1}{90}$$

$$\therefore v = -90 \text{ cm}$$

Magnification, $m=\frac{v}{u}$

$$=\frac{-90}{-9}=10$$

$\therefore$ Area of each square in the

virtual image $=(10)^2 A$

$$=10^2\times1=100 \text{ mm}^2=1 \text{ cm}^2$$ **[1]**

(b) Magnifying power of the lens

$$=\frac{d}{|u|}=\frac{25}{9}=2.8$$ **[1]**

(c) The magnification in (*a*) is not the same as the magnifying power in (*b*).

The magnification magnitude is, $\left(\left|\frac{v}{u}\right|\right)$ and the magnifying power is, $\left(\frac{d}{|u|}\right)$.

The two quantities will be equal when the image is formed at the near point (25 cm). **[1]**

Q. 10. (a) At what distance should the lens be held from the figure in Exercise 9.29 in order to view the squares distinctly with the maximum possible magnifying power?
(b) What is the magnification in this case?
(c) Is the magnification equal to the magnifying power in this case? Explain.
[NCERT Ex. Q. 9.30, Page 348]

Ans. **(a)** Given that,
The maximum possible magnification is obtained when the image is formed at the near point ($d = 25$ cm).
Image distance, $v = -d = -25$ cm
Focal length, $f = 10$ cm

Let the object distance = u

According to the lens formula,

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{u}=\frac{1}{v}-\frac{1}{f}$$

$$=\frac{1}{-25}-\frac{1}{10}=\frac{-2-5}{50}=-\frac{7}{50}$$

$$\therefore u=-\frac{50}{7}=-7.14 \text{ cm}$$

So, to view the squares distinctly, the lens should be kept at 7.14 cm away from them. **[1]**

(b) Magnification $=\left|\frac{v}{u}\right|=\frac{25}{\frac{50}{7}}=3.5$ **[1]**

(c) Magnifying power $=\frac{d}{u}=\frac{25}{\frac{50}{7}}=3.5$

Since the image is formed at the near point (25 cm), the magnifying power is equal to the magnitude of magnification. **[1]**

Q. 11. What should be the distance between the object in Exercise 9.30 and the magnifying glass if the virtual image of each square in the figure is to have an area of 6.25 mm^2. Would you be able to see the squares distinctly with your eyes very close to the magnifier? [NCERT Ex. Q. 9.31, Page 348]

Ans. Given that,

Area of the virtual image of each square, $A = 6.25 \text{ mm}^2$

Area of each square, $A = 1$ mm

So, the linear magnification of the object can be calculated as,

$$m=\sqrt{\frac{A}{A_0}}=\sqrt{\frac{6.25}{1}}=2.5$$

But $m=\frac{\text{Image distance } (v)}{\text{Object distance } (u)}$

$$\therefore v = mu = 2.5\, u \quad \ldots(1)$$

Focal length of the magnifying glass, $f = 10$ cm

According to the lens formula, we have the relation,

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{10}=\frac{1}{2.5u}-\frac{1}{u}=\frac{1}{u}\left(\frac{1}{2.5}-\frac{1}{1}\right)=\frac{1}{u}\left(\frac{1-2.5}{2.5}\right)$$

$$\therefore \quad u=\frac{1.5\times 10}{2.5}=-6 \text{ cm}$$

And $v = 2.5\, u$

$$=2.5\times 6=-15 \text{ cm}$$ **[2]**

The virtual image is formed at a distance of 15 cm, which is less than the near point (*i.e.*, 25 cm) of a normal eye. Thus, it cannot be seen by the eyes distinctly. **[1]**

Q. 12. A person with a normal near point (25 cm) using a compound microscope with objective of focal length 8.0 mm and an eyepiece of focal length 2.5 cm can bring an object placed at 9.0 mm from the objective in sharp focus. What is the separation between the two lenses? Calculate the magnifying power of the microscope. [NCERT Ex. Q. 9.12, Page 346]

Ans. Given that,

Focal length of the objective lens, $f_o = 0.8$ cm

Focal length of the eyepiece, $f_e = 2.5$ cm

Object distance for the objective lens, $u_o = -9.0$ mm $= -0.9$ cm

Least distance of distant vision, $d = 25$ cm

Image distance for the eyepiece, $v_e = -d = -25$ cm

Object distance for the eyepiece = u_e

Using the lens formula,

$$\frac{1}{v_e}-\frac{1}{u_e}=\frac{1}{f_e}$$

$$\frac{1}{u_e}=\frac{1}{v_e}-\frac{1}{f_e}$$

$$=\frac{1}{-25}-\frac{1}{2.5}$$

$$=\frac{-1-10}{25}$$

$$=\frac{-11}{25}$$

$$\therefore \quad u_e=-\frac{25}{11}=-2.27 \text{ cm}$$ **[1]**

We can also obtain the value of the image distance for the objective lens, v_o using the lens formula :

$$\frac{1}{v_0}-\frac{1}{u_0}=\frac{1}{f_0}$$

$$\frac{1}{v_0}=\frac{1}{f_0}-\frac{1}{u_0}$$

$$=\frac{1}{0.8}-\frac{1}{0.9}=\frac{0.9-0.8}{0.72}=\frac{0.1}{0.72}$$

$$\therefore \quad v_0 = 7.2 \text{ cm}$$

The distance between the objective lens and the eyepiece $= |u_e| + v_0$

$= 2.27 + 7.2$

$= 9.47$ cm

The magnifying power of microscope is calculated as :

$$m=\frac{v_0}{|u_0|}\left(1+\frac{d}{f_e}\right)$$

$$=\frac{7.2}{0.9}\left(1+\frac{25}{2.5}\right)=8(1+10)=88$$

So that, the magnifying power of the microscope is 88. **[2]**

Q. 13. A small telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6.0 cm. What is the magnifying power of the telescope? What is the separation between the objective and the eyepiece? [NCERT Ex. Q. 9.13, Page 346]

Ans. Given that,

Focal length of the objective lens, $f_o = 144$ cm

Focal length of the eyepiece, $f_e = 6.0$ cm

The magnifying power of the telescope is :

$$m = \frac{f_0}{f_e}$$
$$= \frac{144}{6} = 24$$ **[1½]**

The separation between the objective lens and the eyepiece is calculated as :

$$f_0 + f_e = 144 + 6 = 150 \text{ cm}$$

Thus, the magnifying power of the telescope is 24 *D* and the separation between the objective lens and the eyepiece is 150 cm. **[1½]**

Q. 14. (a) A giant refracting telescope at an observatory has an objective lens of focal length 15 m. If an eyepiece of focal length 1.0 cm is used, what is the angular magnification of the telescope?

(b) If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is 3.48 × 10^6 m, and the radius of lunar orbit is 3.8 × 10^8 m. [NCERT Ex. Q. 9.14, Page 346]

Ans. Given that,

Focal length of the objective lens,

f_o = 15 m = 15 × 100 cm = 1500 cm

Focal length of the eyepiece, f_e = 1.0 cm

(a) The angular magnification of a telescope is given as :

$$\alpha = \frac{f_0}{f_e}$$
$$= \frac{15 \times 10^2}{1.0} = 1500$$

Thus, the angular magnification of the given refracting telescope is 1500. **[1]**

(a) Given that,

Diameter of the moon, $d = 3.48 \times 10^6$ m

Radius of the lunar orbit, $r_o = 3.8 \times 10^8$ m

Let '*d*' is the diameter of the image of the moon formed by the objective lens.

The angle subtended by the diameter of the moon is equal to the angle subtended by the image.

$$\frac{d}{r_0} = \frac{d'}{f_0}$$
$$\frac{3.48 \times 10^6}{3.8 \times 10^8} = \frac{d'}{15}$$

So,

$$d = \frac{3.48}{3.8} \times 10^{-2} \times 15$$
$$= 13.74 \times 10^{-2} \text{ m}$$
$$= 13.74 \text{ cm}$$

Thus, the diameter of the moon's image formed by the objective lens is 13.74 cm **[2]**

Q. 15. A small telescope has an objective lens of focal length 140 cm and an eyepiece of focal length 5.0 cm. What is the magnifying power of the telescope for viewing distant objects when :

(a) the telescope is in normal adjustment (*i.e.*, when the final image is at infinity)?

(b) the final image is formed at the least distance of distinct vision (25 cm)?

[NCERT Ex. Q. 9.34, Page 349]

Ans. Given that,

Focal length of the objective lens, f_o = 140 cm

Focal length of the eyepiece, f_e = 5 cm

Least distance of distinct vision, *d* = 25 cm

(a) When the telescope is in normal adjustment, its magnifying power is given as :

$$m = \frac{f_0}{f_e}$$
$$= \frac{140}{5} = 28$$

(b) When the final image is formed at d, the magnifying power of the telescope is given as :

$$\frac{f_0}{f_e}\left[1 + \frac{f_e}{d}\right]$$
$$= \frac{140}{5}\left[1 + \frac{5}{25}\right]$$
$$= 28\left[1 + 0.2\right]$$
$$= 28 \times 1.2 = 33.6$$ **[3]**

Q. 16. (a) For the telescope described in Exercise 9.34 (*a*), what is the separation between the objective lens and the eyepiece?

(b) If this telescope is used to view a 100 m tall tower 3 km away, what is the height of the image of the tower formed by the objective lens?

(c) What is the height of the final image of the tower if it is formed at 25 cm?

[NCERT Ex. Q. 9.35, Page 349]

Ans. Given that,

Focal length of the objective lens, f_o = 140 cm

Focal length of the eyepiece, f_e = 5 cm

(a) In normal adjustment, the separation between the objective lens and the eyepiece :

$$= f_0 + f_e = 140 + 5 = 145 \text{ cm}$$ **[1]**

(b) Given that,

Height of the tower, *h* = 100 m

Distance of the tower (object) from the telescope, *u* = 3 km = 3000 m

The angle subtended by the tower at the telescope is given as :

$$\theta = \frac{h_1}{u}$$
$$= \frac{100}{3000} = \frac{1}{30} \text{rad}$$

The angle subtended by the image produced by the objective lens is given as :

$$\theta = \frac{h_2}{f_0} = \frac{h_2}{140} \text{ rad}$$

Where,

h_2 = Height of the image of the tower formed by the objective lens

$$\frac{1}{30} = \frac{h_2}{140}$$
$$\therefore h_2 = \frac{140}{30} = 4.7 \text{ cm}$$

Thus, the objective lens forms a 4.7 cm tall image of the tower. **[1]**

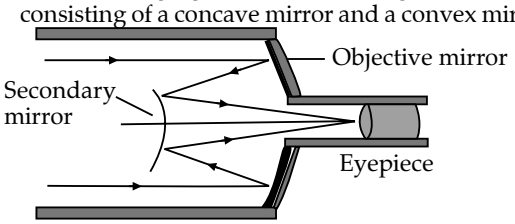
(c) Image is formed at a distance, $d = 25$ cm
The magnification of the eyepiece is :

$$m = 1 + \frac{d}{f_e} = 1 + \frac{25}{5} = 1 + 5 = 6$$

Height of the final image $= mh_2 = 6 \times 4.7 = 28.2$ cm
So that, the height of the final image of the tower is 28.2 cm. **[1]**

Q. 17. A Cassegrain telescope uses two mirrors as shown in Figure. Such a telescope is built with the mirrors 20 mm apart. If the radius of curvature of the large mirror is 220 mm and the small mirror is 140 mm, where will the final image of an object at infinity be? [NCERT Ex. Q. 9.36, Page 349]

Ans. The following figure shows a Cassegrain telescope consisting of a concave mirror and a convex mirror.



Given that,
Distance between the objective mirror and the secondary mirror, $d = 20$ mm
Radius of curvature of the objective mirror, $R_1 = 220$ mm
Radius of curvature of the secondary mirror, $R_2 = 140$ mm
So the focal length of the objective mirror,

$$f_1 = \frac{R_1}{2} = 110 \text{ mm}$$

Radius of curvature of the secondary mirror, $R_2 = 140$ mm
So the focal length of the secondary mirror,

$$f_2 = \frac{R_2}{2} = 70 \text{ mm}$$

The image of an object placed at infinity, formed by the objective mirror, will act as a virtual object for the secondary mirror.
Thus, the virtual object distance for the secondary mirror, $u = f_1 - d = 110 - 20 = 90$ mm **[1]**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f_2}$$

$$\frac{1}{v} = \frac{1}{f_2} - \frac{1}{u} = \frac{1}{70} - \frac{1}{90} = \frac{9-7}{630} = \frac{2}{630}$$

Therefore,

$$v = \frac{630}{2} = 315 \text{ mm}$$

So, the final image will be formed 315 mm away from the secondary mirror. **[2]**

Q. 18. A short object of length L is placed along the principal axis of a concave mirror away from focus. The object distance is u. If the mirror has a focal length f, what will be the length of the image? You may take $L << |v - f|$.
[NCERT Exemp. Q. 9.22, Page 59]

Ans. As the mean distance of object from mirror is u.

$$\therefore u_1 = u - \frac{L}{2} \text{ and } u_2 = \left(u + \frac{L}{2}\right)$$

Let the image of the two ends of object form at distance v_1 and v_2 ($v_1 > v_2$).
So, length of image on principal axis is $L' = (v_1 - v_2)$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \text{ or } \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{u - f}{uf}$$

$$\Rightarrow v = \frac{uf}{u - f}$$

$$\therefore L' = v_1 - v_2$$

$$= \frac{\left(u - \frac{L}{2}\right) f}{\left(u - \frac{L}{2}\right) - f} - \frac{\left(u + \frac{L}{2}\right) f}{\left(u + \frac{L}{2}\right) - f}$$

$$\Rightarrow L' = f\frac{\left(u - \frac{L}{2}\right)}{\left(u - f - \frac{L}{2}\right)} - \frac{\left(u + \frac{L}{2}\right)}{\left(u - f + \frac{L}{2}\right)} \quad \textbf{[1½]}$$

$$\Rightarrow L' = f\left[\frac{\left(u - \frac{L}{2}\right)\left(u - f + \frac{L}{2}\right) - \left(u + \frac{L}{2}\right)\left(u - f - \frac{L}{2}\right)}{\left(u - f - \frac{L}{2}\right)\left(u - f + \frac{L}{2}\right)}\right]$$

$$= \frac{f\left[u^2 - uf + \frac{uL}{2} - \frac{uL}{2} + \frac{fL}{2} - \frac{L^2}{4} - \left(u^2 - uf - \frac{uL}{2} + \frac{uL}{2} - \frac{fL}{2} - \frac{L^2}{4}\right)\right]}{(u - f)^2 - \frac{L^2}{4}}$$

Hence, $L << (u - f)$

So that, $\frac{L^2}{4} << (u - f)$

So, neglecting the terms $\frac{L^2}{4}$

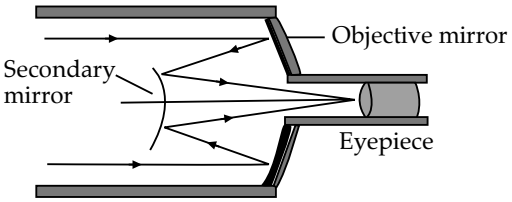
$$L^2 = f\left[\frac{\frac{fL}{2} + \frac{fL}{2}}{(u - f)^2}\right]$$

$$L' = \frac{f\,fL}{(u - f)^2} \Rightarrow L' = \frac{Lf^2}{(u - f)^2}$$

It is the length of image f. **[1½]**

Q. 19. Draw a schematic ray diagram of a reflecting telescope showing how rays coming from a distant object are received at the eyepiece. Write its two important advantages over a refracting telescope.
[CBSE Board, Delhi Region, 2016]

Ans. The schematic diagram of reflecting telescope is shown below :



[1½]

The two advantages of a reflecting telescope over a refracting telescope are as follows :

(i) There is no chromatic aberration in case of reflecting telescopes as the objective is a mirror.

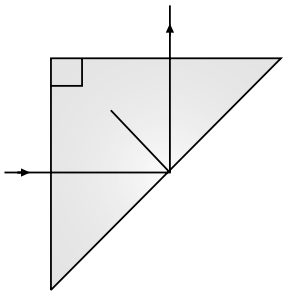
(ii) Spherical aberration is reduced in case of reflecting telescopes by using mirror objective in the form of a paraboloid. **[1½]**

Long Answer Type Questions **(5 marks each)**

Q. 1. (i) Plot a graph to show variation of the angle of deviation as a function of angle of incidence for light passing through a prism. Derive an expression for refractive index of the prism in terms of angle of minimum deviation and angle of prism.

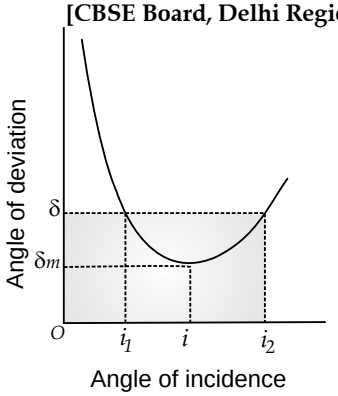
(ii) What is dispersion of light? What is its cause?

(iii) A ray of light incident normally on one face of a right isosceles prism is totally reflected, as shown in fig. What must be the minimum value of refractive index of glass? Give relevant calculations.



[CBSE Board, Delhi Region, 2016]

Ans. (i)



[1]

If the angle of incidence is increased gradually, then the angle of deviation first decreases, attains a minimum value (δ_m) and then again starts increasing. When angle of deviation is minimum, the prism is said to be placed in the minimum deviation position. There is only one angle of incidence for which the angle of deviation is minimum.

When $\delta = \delta_m$ [prism in minimum deviation position]

$e = i$ and $r_2 = r_1 = r$

$\because r_1 + r_2 = A$ ...(i)

From (i), we get $r + r = A$

$$r = \frac{A}{2}$$

Also, we have

$A + \delta = i + e$

Substituting $\delta = \delta_m$ and $e = i$,

$$A + \delta_m = i + i$$

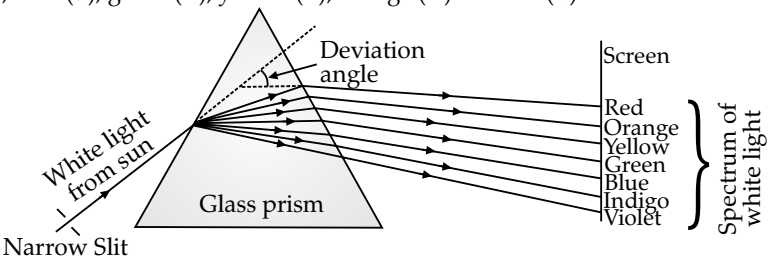
$$i = \frac{(A + \delta_m)}{2}$$

$$\because \quad \mu = \frac{\sin i}{\sin r}$$

$$\therefore \quad \mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$ **[1]**

(ii) Dispersion of light is the phenomenon of splitting of light into its component colours.

When a narrow beam of sunlight is incident on a glass prism, the emergent light splits into seven colours, namely, violet (*V*), indigo (*I*), blue (*B*), green (*G*), yellow (*Y*), orange (*O*) and red (*R*).



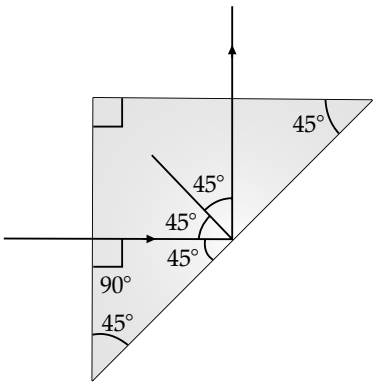
Causes of dispersion : Different colours of white light have different wavelengths. The wavelength of violet light is smaller than that of red light. The refractive index of a material in terms of the wavelength of the light is given by Cauchy's expression.

$$\mu = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} .....$$

where, *A*, *B* and Care constants for the material.

Thus, a material offers different refractive indices to lights of different wavelengths. This is the cause of dispersion of light. **[1½]**

(iii)



We know that light incident normally on one face of a right isosceles prism is totally reflected.

$$\mu = \frac{1}{\sin C} \qquad \text{(Total internal reflection)}$$

C is that critical angle, which is 45°.

$$\mu = \frac{1}{\sin 45°} = \sqrt{2} = 1.41$$

So, the minimum value of the refractive index of the glass is 1.41. **[1½]**

Q. 2. A prism is made of glass of unknown refractive index. A parallel beam of light is incident on a face of the prism. The angle of minimum deviation is measured to be 40°. What is the refractive index of the material of the prism? The refracting angle of the prism is 60°. If the prism is placed in water (refractive index 1.33), predict the new angle of minimum deviation of a parallel beam of light.

[NCERT Ex. Q. 9.6, Page 346]

Ans. Given that,

Angle of minimum deviation, $\delta_m = 40°$

Angle of the prism, $A = 60°$

Refractive index of water, $\mu = 1.33$

Refractive index of the material of the prism = μ'

The angle of deviation is related to refractive index as :

$$\mu' = \frac{\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{60°+40°}{2}\right)}{\sin\frac{60°}{2}} = \frac{\sin 50°}{\sin 30°} = 1.532$$

[2½]

So, the refractive index of the material of the prism is 1.532.

Since the prism is placed in water, let $\delta_{m'}$ be the new angle of minimum deviation for the same prism.

The refractive index of glass with respect to water is given by the relation :

$$\mu_g^W = \frac{\mu'}{\mu} = \frac{\sin\left(\frac{A+\delta'_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sin\left(\frac{A+\delta'_m}{2}\right) = \frac{\mu'}{\mu}\sin\left(\frac{A}{2}\right)$$

$$\sin\left(\frac{A+\delta'_m}{2}\right) = \frac{1.532}{1.33} \times \sin\left(\frac{60°}{2}\right) = 0.5759$$

$$\left(\frac{A+\delta'_m}{2}\right) = \sin^{-1}\ 0.5759 = 35.16°$$

$$60° + \delta'_m = 70.32°$$

$$\therefore \quad \delta'_m = 70.32° - 60° = 10.32°$$

So, the new minimum angle of deviation is 10.32°. **[2½]**

Q. 3. A compound microscope consists of an objective lens of focal length 2.0 cm and an eyepiece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at (*a*) the least distance of distinct vision (25 cm), and (*b*) at infinity? What is the magnifying power of the microscope in each case?

[NCERT Ex. Q. 9.11, Page 346]

Ans. Given that,

Focal length of the objective lens, $f_1 = 2.0$ cm

Focal length of the eyepiece, $f_2 = 6.25$ cm

Distance between the objective lens and the eyepiece, $d = 15$ cm

(a) Given that,

Least distance of distinct vision, $d' = 25$ cm

Image distance for the eyepiece, $v_2 = -25$ cm

Object distance for the eyepiece = u_2

According to the lens formula,

$$\frac{1}{v_2} - \frac{1}{u_2} = \frac{1}{f_2}$$

$$\frac{1}{u_2} = \frac{1}{v_2} - \frac{1}{f_2}$$

$$= \frac{1}{-25} - \frac{1}{6.25} = \frac{-1-4}{25} = \frac{-5}{25}$$

$$\therefore \quad u_2 = -5 \text{ cm}$$

Image distance for the objective lens, $v_1 = d + u_2 = 15 - 5 = 10$ cm

Object distance for the objective lens = u_1

According to the lens formula,

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{u_1} = \frac{1}{v_1} - \frac{1}{f_1} = \frac{1}{10} - \frac{1}{2} = \frac{1-5}{10} = \frac{-4}{10}$$

$$\therefore u_1 = -2.5 \text{ cm}$$ **[2½]**

Magnitude of the object distance, $|u_1| = 2.5$ cm

The magnifying power of a compound microscope is :

$$m = \frac{v_1}{|u_1|}\left(1 + \frac{d'}{f_2}\right) = \frac{10}{2.5}\left(1 + \frac{25}{6.25}\right) = 4(1+4) = 20$$

(b) The final image is formed at infinity.

Therefore, image distance for the eyepiece, $v_2 = \infty$

Object distance for the eyepiece = u_2

According to the lens formula, we have the relation,

$$\frac{1}{v_2} - \frac{1}{v_2} = \frac{1}{f_2}$$

$$\frac{1}{\infty} - \frac{1}{u_2} = \frac{1}{6.25}$$

$$\therefore \quad u_2 = -6.25 \text{ cm}$$

Image distance for the objective lens, $v_1 = d + u_2$
$= 15 - 6.25 = 8.75$ cm
Object distance for the objective lens = u_1
According to the lens formula,

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{u_1} = \frac{1}{v_1} - \frac{1}{f_1} = \frac{1}{8.75} - \frac{1}{2.0} = \frac{2 - 8.75}{17.5}$$

$$\therefore \quad u_1 = -\frac{17.5}{6.75} = -2.59 \text{ cm}$$

Magnitude of the object distance, $|u_1| = 2.59$ cm
The magnifying power of a compound microscope is,

$$m = \frac{v_1}{|u_1|}\left(\frac{d'}{|u_2|}\right) = \frac{8.75}{2.59} \times \frac{25}{6.25} = 13.51$$

Thus, the magnifying power of the microscope is 13.51. **[2½]**

Q. 4. Answer the following questions :

(a) The angle subtended at the eye by an object is equal to the angle subtended at the eye by the virtual image produced by a magnifying glass. In what sense then does a magnifying glass provide angular magnification?

(b) In viewing through a magnifying glass, one usually position's one's eyes very close to the lens. Does angular magnification change if the eye is moved back?

(c) Magnifying power of a simple microscope is inversely proportional to the focal length of the lens. What then stops us from using a convex lens of smaller and smaller focal length and achieving greater and greater magnifying power?

(d) Why must both the objective and the eyepiece of a compound microscope have short focal lengths?

(e) When viewing through a compound microscope, our eyes should be positioned not on the eyepiece but a short distance away from it for best viewing. Why? How much should be that short distance between the eye and eyepiece?

[NCERT Ex. Q. 9.32, Page 349]

Ans. **(a)** Though the image size is bigger than the object, the angular size of the image is equal to the angular size of the object. A magnifying glass helps one see the objects placed closer than the least distance of distinct vision (*i.e.*, 25 cm). A closer object causes a larger angular size. A magnifying glass provides angular magnification. Without magnification, the object cannot be placed closer to the eye. With magnification, the object can be placed much closer to the eye. **[1]**

(b) Yes, the angular magnification changes. When the distance between the eye and a magnifying glass is increased, the angular magnification decreases a little. This is because the angle subtended at the eye is slightly less than the angle subtended at the lens. Image distance does not have any effect on angular magnification. **[1]**

(c) The focal length of a convex lens cannot be decreased by a greater amount. This is because making lenses having very small focal lengths is not easy. Spherical and chromatic aberrations are produced by a convex lens having a very small focal length. **[1]**

(d) The angular magnification produced by the eyepiece of a compound microscope is $\left[\left(\frac{25}{f_e}\right) + 1\right]$.

where,
f_e = Focal length of the eyepiece
It can be inferred that if f_e is small, then angular magnification of the eyepiece will be large.
The angular magnification of the objective lens of a compound microscope is given as :

$$= \frac{1}{[|u_0| f_0]}$$

Where,
u_0 = Object distance for the objective lens
f_0 = Focal length of the objective
The magnification is large when $u_o > f_o$.
In the case of a microscope, the object is kept close to the objective lens. So, the object distance is very little. Since u_o is small, f_o will be even smaller. Therefore, f_e and f_o are both small in the given condition. **[1]**

(e) When we place our eyes too close to the eyepiece of a compound microscope, we are unable to collect much refracted light. As a result, the field of view decreases substantially.
So, the clarity of the image gets blurred.
The best position of the eye for viewing through a compound microscope is at the eye-ring attached to the eyepiece. The precise location of the eye depends on the separation between the objective lens and the eyepiece. **[1]**

Q. 5. An angular magnification (magnifying power) of 30× is desired using an objective of focal length 1.25 cm and an eyepiece of focal length 5 cm. How will you set up the compound microscope?

[NCERT Ex. Q. 9.33, Page 349]

Ans. Given that,
Focal length of the objective lens, $f_o = 1.25$ cm
Focal length of the eyepiece, $f_e = 5$ cm
Least distance of distinct vision, $d = 25$ cm
Angular magnification of the compound microscope = 30×
Total magnifying power of the compound microscope, $m = 30$
The angular magnification of the eyepiece is given by the relation,

$$m_e = \left(1 + \frac{d}{f_e}\right) = \left(1 + \frac{25}{5}\right) = 6$$ **[1½]**

The angular magnification of the objective lens (m_0) is related to m_e as :

$$m = m_o . m_e$$

$$m_e = \frac{m}{m_e} = \frac{30}{6} = 5$$

And we know that,

$$m_e = \frac{\text{Image distance for the objective lens } (v_0)}{\text{Object distance for the objective lens } (-u_0)}$$

$$5 = \frac{v_0}{-u_0}$$

$$\therefore v_0 = -5u_0 \quad ...(1)$$

Applying the lens formula for the objective lens,

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{1.25} = \frac{1}{-5u_0} - \frac{1}{u_0} = \frac{-6}{5u_0}$$

$$\therefore \quad u_0 = \frac{-6}{5} \times 1.25 = -1.5 \text{ cm}$$

And $v_0 = -5u_0$

$= -5 \times (-1.5) = 7.5$ cm **[1½]**

The object should be placed 1.5 cm away from the objective lens to obtain the desired magnification.

Applying the lens formula for the eyepiece :

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

where,

v_e = Image distance for the eyepiece

$= -d = -25$ cm

u_e = Object distance for the eyepiece

$$\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e}$$

$$= \frac{-1}{25} - \frac{1}{5} = -\frac{6}{25}$$

So,

$u_e = -4.17$ cm

Seperation between the objective lens and the eyepiece, $= |u_e| + |v_0|$

$= 4.17 + 7.5$

$= 11.67$ cm **[1]**

So, the separation between the objective lens and the eyepiece should be 11.67 cm. **[1]**

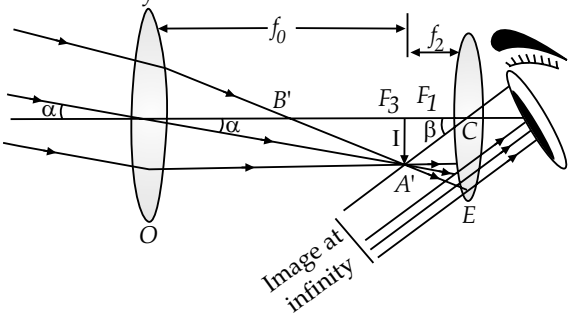
Q. 6. (a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.

(b) You are given three lenses of power 0.5 *D*, 4 *D* and 10 *D* to design a telescope.

(i) Which lenses should be used as objective and eyepiece? Justify your Answer.

(ii) Why is the aperture of the objective preferred to be large? [CBSE Board, All India Region, 2016]

Ans. (a) In normal adjustment, the final image is formed at infinity.



Magnifying power or angular magnification of astronomical telescope : It is defined as the ratio of the angle subtended at the eye by the final image to the angle subtended at the eye, by the object directly, when the final image and the object, both are at infinity.

Angular magnification, $M = \frac{\beta}{\alpha}$

β and α are very small.

$\therefore \beta \approx \tan\beta$

$\alpha \approx \tan\alpha$

$$M = \frac{\tan\beta}{\tan\alpha} \quad \textbf{[1]}$$

I is the image formed by the objective. f_0 and f_e are the focal lengths of the objective and eyepiece, respectively.

Where,

$$\tan\alpha = \frac{1}{f_0}$$

$$\tan\beta = \frac{1}{-f_e}$$

(Distance of the image from the eyepiece is taken as negative.)

$$M = \frac{\frac{-I}{f_e}}{\frac{I}{f_0}} = \frac{-f_o}{f_e} \quad \textbf{[1½]}$$

(b) (i) Lens with smallest power or largest focal length should be used as the objective, that is, lens with power 0.5 *D*. Lens with largest power or smallest focal length should be used as the eye-piece, that is, lens with power 10 *D*.

(ii) The aperture is preferred to be large so that the telescope can collect as much as light coming from the distant object as possible. **[2½]**

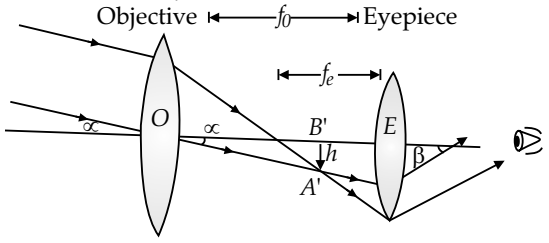
Q. 7. (i) Draw a labelled schematic ray diagram of astronomical telescope in normal adjustment.

(ii) Which two aberrations do objectives of refracting telescope suffer from? How are these overcome in reflecting telescope?

(iii) How does the resolving power of a telescope change on increasing the aperture of the objective lens? Justify your answer.

[CBSE Board, Foreign Scheme, 2016]

Ans. (i) The image formed by an astronomical telescope in normal adjustment is shown below :



In normal adjustment, the image is formed at infinity. Magnification power is defined as the angle subtended at the eye, by the final image, to the angle

subtended at the eye, by the object directly, when the final image and the object both are at infinity.

Mathematically, it is given by

Magnification, $m = \frac{\beta}{\alpha}$

Since α and β are very small,

$\beta = \tan\beta$

$\alpha = \tan\alpha$

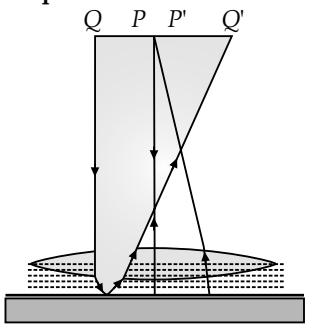
$\Rightarrow m = \frac{\tan\beta}{\tan\alpha}$ **[2]**

(ii) All refractors telescopes suffer from an effect called the chromatic aberration that produces a rainbow of colours around the image, one way to reduce this problem is to use the multiple compensating lenses to counteract the effects of this kind of aberration.

The refractor telescopes can suffer from a defect called the spherical aberration so that not all of the light is *focused* to the same point. **[2]**

(iii) When the aperture of the objective lens is increased, the resolving power of telescope increases. **[1]**

Q. 8. Figure shows an equiconvex lens (of refractive index 1.50) in contact with a liquid layer on top of a plane mirror. A small needle with its tip on the principal axis is moved along the axis until its inverted image is found at the position of the needle. The distance of the needle from the lens is measured to be 45.0 cm. The liquid is removed and the experiment is repeated. The new distance is measured to be 30.0 cm. What is the refractive index of the liquid?



[NCERT Ex. Q. 9.38, Page 350]

Ans. Given that,

Focal length of the convex lens, f_1 = 30 cm

The liquid acts as a mirror. So, focal length of the liquid = f_2

Focal length of the system (convex lens + liquid), f = 45 cm

For a pair of optical systems placed in contact, the equivalent focal length is given as :

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f_2} = \frac{1}{f} - \frac{1}{f_1}$$

$$= \frac{1}{45} - \frac{1}{30} = \frac{1}{90}$$

Therefore, f_2 = –90 cm **[1½]**

Let the refractive index of the lens be μ_1 and the radius of curvature of one surface be R. Therefore, the radius of curvature of the other surface is $-R$. R can be obtained using the relation,

$$\frac{1}{f_1} = (\mu_1 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right)$$

$$\frac{1}{30} = (1.5 - 1)\left(\frac{2}{R}\right)$$

$$\therefore R = \frac{30}{0.5 \times 2} = 30 \text{ cm}$$

[1½]

Let μ_2 be the refractive index of the liquid.

Radius of curvature of the liquid on the side of the plane mirror = ∞

Radius of curvature of the liquid on the side of the lens, $R = -30$ cm

The value of μ_2 can be calculated using the relation,

$$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right)$$

$$\frac{-1}{90} = (\mu_2 - 1)\left[\frac{1}{+30} - 0\right]$$

$$\mu_2 - 1 = \frac{1}{3}$$

$$\therefore \mu_2 = \frac{4}{3} = 1.33$$

Therefore, the refractive index of the liquid is 1.33. **[2]**

Some Commonly Made Errors

- Students confuse to differentiate the ray diagrams of spherical Mirrors and spherical Lenses.
- Generally, students get confused with the term 'Dispersion', 'Refraction' and 'Scattering'.
- Students confuse to differentiate between schematic diagrams of simple Microscope to Telescope.

EXPERT ADVICE

☞ *Don't ignore to study optical instruments.*

☞ *Apply method of elimination for MCQ questions. Look at the answers before even starting to solve the questions and see if you can eliminate any answer. When calculating answer for a big question you may not need to solve the problem fully in order to choose the right answer because you could eliminate the other three.*

- ☞ *Try to solve numerical by making suitable diagram, do not skip to draw ray diagrams.*
- ☞ *Understand the Cartesian sign convention.*
- ☞ *Don't skip to understand the working of Human Eye.*
- ☞ *Don't forget to understand the refraction through a spherical interface*
- ☞ *Understand the concept of 'Thin lens formula' and 'Lens Maker's Formula'.*
- ☞ *Draw suitable ray diagram to explain image formation due to different-different object location.*
- ☞ *Don't skip any important topics like- Total internal reflection (TIR), Magnifying power, magnification of a simple microscope and telescope, Natural phenomenon due to sunlight and optical instruments.*

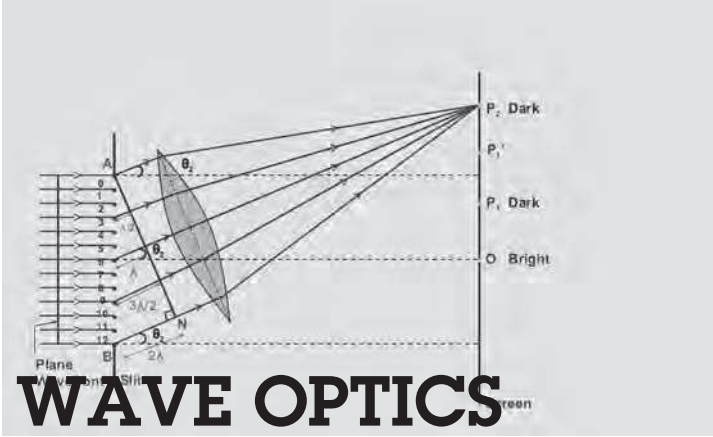
OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/95V-QJYZ2Dw **Or Scan the Code**	*https://goo.gl/JcmwZR* **Or Scan the Code**
https://goo.gl/WMNx1q **Or Scan the Code**	*https://goo.gl/vAyJ5V* **Or Scan the Code**

CHAPTER

10 WAVE OPTICS



Chapter Objectives

This chapter will help you understand :

- ***Wave optics :*** *Huygens' principle; Refraction and Reflection of plane waves; Coherent and Incoherent addition of waves; Young's experiment; Interference of light; Diffraction; and Polarisation.*

Quick Review

- The development of optics between 1450 and 1789 can be conveniently divided into two phases bridged by the optical work of Johannes Kepler (1576–1630) and distinguished by a radical change in analytic focus.
- During the first phase, that focus was primarily on sight, not light. During the second, it shifted completely from sight to light. Reflecting this shift, the following essay consists of three sections, the first dealing with pre-Keplerian optics, the second with the Keplerian transition, and the third with post-Keplerian developments.
- By 1450 two ostensibly contradictory models of sight were available to European thinkers. The first and simpler of the two harks back to the visual-ray theory of Euclid (fl. c. 300 b.c.e.). Brought to maturity by Ptolemy (c. 170 C.E.), this theory assumes that a constant stream of visual flux emanates from the center of the eye through the pupil to form a cone.
- The second model of vision harks back to Alhacen (965–1040) and his Perspectivist disciples, Roger Bacon (fl. c. 1265), Witelo (fl. c. 1275), and John Pecham (fl. c. 1280). Rejecting visual rays as functionally pointless, these theorists raised light to primacy in the visual process, supposing it to be an intrinsic quality of self-luminous or illuminated bodies.

TIPS...

- *Study about the Huygens' theorem properly for the working of wave.*
- *Study about all the cases of Refraction and Reflection of lights on plane surface.*
- *Understand the concept of Snell's Law of refraction.*
- *Study the Doppler's Effect.*

TRICKS...

- *Distinguish between Coherent and Incoherent sources for solving numerical problems.*
- *Try to make sure that all fringe pattern according to interference in single slit or double slit using Young's experiment.*
- *Study Snell's Law and do practicing on Huygens' theorem based problems.*

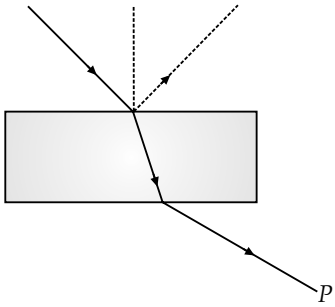
Know the Links

- *https://www.emedicalprep.com/study-material/.../wave-optics/introduction-wave-optic*
- *https://www.cliffsnotes.com/study-guides/physics/light/wave-optics*
- *https://en.wikipedia.org/wiki/Physical_optics*

Multiple Choice Questions

(1 mark each)

Q. 1. Consider a light beam incident from air to a glass slab at Brewster's angle as shown in Figure. A polaroid is placed in the path of the emergent ray at point *P* and rotated about an axis passing through the centre and perpendicular to the plane of the polaroid



(a) For a particular orientation there shall be darkness as observed through the polaroid.

(b) The intensity of light as seen through the polaroid shall be independent of the rotation.

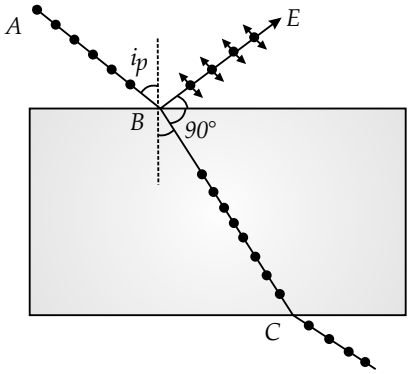
(c) The intensity of light as seen through the polaroid shall go through a minimum but not zero for two orientations of the polaroid.

(d) The intensity of light as seen through the polaroid shall go through a minimum for four orientations of the polaroid.

[NCERT Exemp. Q. 10.1, Page 62]

Ans. Correct option : (c)

Explanation : When ray *ABCD* of light passes through prism in such a way that angle between reflected ray *BE* and refracted ray *BC* is 90° then only reflected ray is plane polarised. So polaroid rotated in the way of *CD* the intensity will never be zero but varies in one complete rotation so, it verifies answer (c).



Q. 2. Consider sunlight incident on a slit of width 10^4 Å. The image seen through the slit shall

(a) be a fine sharp slit white in colour at the centre.

(b) a bright slit white at the centre diffusing to zero intensities at the edges.

(c) a bright slit white at the centre diffusing to regions of different colours.

(d) only be a diffused slit white in colour.

[NCERT Exemp. Q. 10.2, Page 62]

Ans. Correct option : (a)

Explanation : Given that, width of slit 10^4 Å=10,000 Å. Wavelength of visible light varies from 4,000 Å to 8,000 Å. As the width of slit 10,000 Å is comparable to that of wavelength of visible light, *i.e.*, 8,000 Å. Hence the diffraction occurs with maxima at the centre. So at the centre all colours appear, *i.e.*, white colour appears at the centre.

Q. 3. Consider a ray of light incident from air onto a slab of glass (refractive index n) of width *d*, at an angle θ. The phase difference between the ray reflected by the top surface of the glass and the bottom surface is

(a) $\frac{4\pi d}{\lambda}\left(1-\frac{1}{n^2}\sin^2\theta\right)^{\frac{1}{2}}+\pi$

(b) $\frac{4\pi d}{\lambda}\left(1-\frac{1}{n^2}\sin^2\theta\right)^{\frac{1}{2}}$

(c) $\frac{4\pi d}{\lambda}\left(1-\frac{1}{n^2}\sin^2\theta\right)^{\frac{1}{2}}+\frac{\pi}{2}$

(d) $\frac{4\pi d}{\lambda}\left(1-\frac{1}{n^2}\sin^2\theta\right)^{\frac{1}{2}}+2\pi$

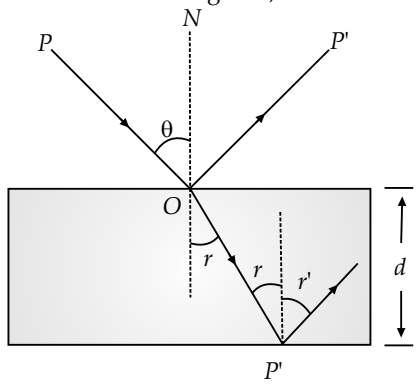
[NCERT Exemp. Q. 10.3, Page 63]

Ans. Correct option : (a)

Explanation : If slab of a glass is placed in air, the wave reflected from the upper surface (from a denser medium) suffers a sudden phase change of π, while the wave reflected from the lower surface (from a rarer medium) suffers no such phase change.

Now consider the diagram, the ray (*P*) is incident at an angle θ and gets reflected in the direction *P'* and refracted in the direction *P''*. Due to reflection from the glass medium, there is a phase change of π.

Time taken to travel along *OP''*,



$$\Delta t=\frac{OP''}{v}=\frac{\frac{d}{\cos r}}{\frac{c}{n}}=\frac{nd}{c\times c\cos r}$$

From Snell's law, $n=\frac{\sin\theta}{\sin r}$

$$\sin r=\frac{\sin\theta}{n}$$

$$\cos r=\sqrt{1-\sin^2 r}$$

$$=\sqrt{1-\frac{\sin^2\theta}{n^2}}$$

Phase difference,

$$\Delta\phi = \frac{2\pi}{T} \times \Delta t$$

$$\Rightarrow \Delta\phi = \frac{2\pi nd}{\lambda}\left(1 - \frac{\sin^2\theta}{n^2}\right)^{\frac{-1}{2}}$$

So, net phase difference = $\Delta\phi + \pi$

$$\Rightarrow \qquad \Delta\phi_{net} = \frac{4\pi d}{\lambda}\left(1 - \frac{1}{n^2}\sin^2\theta\right) + \pi$$

Q. 4. In a Young's double-slit experiment, the source is white light. One of the holes is covered by a red filter and another by a blue filter. In this case,

(a) there shall be alternate interference patterns of red and blue.

(b) there shall be an interference pattern for red distinct from that for blue.

(c) there shall be no interference fringes.

(d) there shall be an interference pattern for red mixing with one for blue.

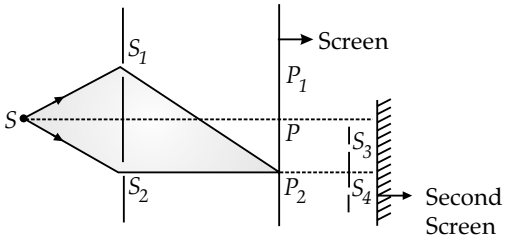
[NCERT Exemp. Q. 10.4, Page 63]

Ans. Correct option : (c)

Explanation : For sustained interference, the source must be coherent and should emit the light of same frequency.

In this problem, one hole is covered with red and other with blue, which has different frequency, so no interference takes place.

Q. 5. Figure shows a standard two-slit arrangement with slits S_1, S_2. P_1, P_2 are the two minima points on either side of P shows in Figure. At P_2 on the screen, there is a hole and behind P_2 is a second screen, 2-slit arrangement with slits S_3 and S_4 and a second screen behind them.



(a) There would be no interference pattern on the second screen but it would be lighted.

(b) The second screen would be totally dark.

(c) There would be a single bright point on the second screen.

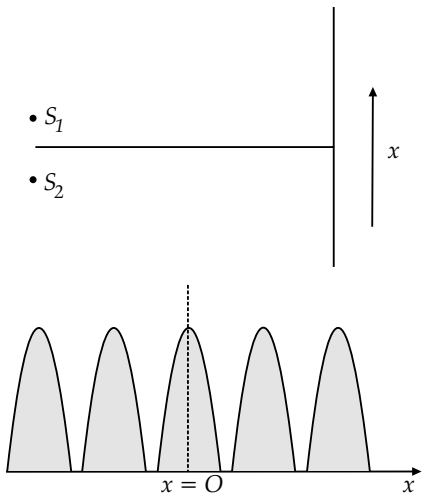
(d) There would be a regular two slit pattern on the second screen. [NCERT Exemp. Q. 10.5, Page 63]

Ans. Correct options : (d)

Explanation : At P_2 is minima due to two wave fronts in opposite phase coming from, two slits S_1 and S_2, but there is wave fronts from S_1, S_2. So P_2 will act as a source of secondary wavelets. Wave front starting from P_2 reaches at S_3 and S_4 slits which will again act as two monochromatic or coherent sources and will form pattern on second screen.

Tick Two or More Options

Q. 6. Two source S_1 and S_2 of intensity I_1 and I_2 are placed in front of a screen in Figure (a). The pattern of intensity distribution seen in the central portion is given by Figure (b). In this case, which of the following statements are true?



(a) S_1 and S_2 have the same intensities.

(b) S_1 and S_2 have a constant phase difference.

(c) S_1 and S_2 have the same phase.

(d) S_1 and S_2 have the same wavelength.

[NCERT Exemp. Q. 10.6, Page 64]

Ans. Correct options : (a), (b) and (c)

Explanation : **(i)** As the intensity at dark fringe is zero, so intensities of S_1 and S_2 are equal.

(ii) As the graph of maxima and minima is symmetric. So the waves from S_1 and S_2 are at same phase difference or zero phase difference.

For two coherent sources, the resultant intensity is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi$$

Resultant intensity at the point of observation will be maximum.

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

$$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$$

Resultant intensity at the point of observation will be minimum.

$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$$

Q. 7. Consider sunlight incident on a pinhole of width 10^3 Å. The image of the pinhole seen on a screen shall be

(a) a sharp white ring.

(b) different from a geometrical image.

(c) a diffused central spot, white in colour.

(d) diffused coloured region around a sharp central white spot. [NCERT Exemp. Q. 10.7, Page 64]

Ans. **Correct options :** (b) and (d)

Explanation : Diffraction of light can be observed only if the size of obstacle/aperture is less than the wavelength of light.

Given, width of pinhole = 10^3 Å = 1,000 Å

We know that wavelength of sunlight ranges from 4,000 Å to 8,000 Å. Clearly, wavelength X is less than width of the slit.

Hence, light is diffracted from the hole. Due to diffraction from the slit the image formed on the screen will be different from the geometrical image.

Q. 8. Consider the diffraction pattern for a small pinhole. As the size of the hole is increased

(a) the size decreases.

(b) the intensity increases.

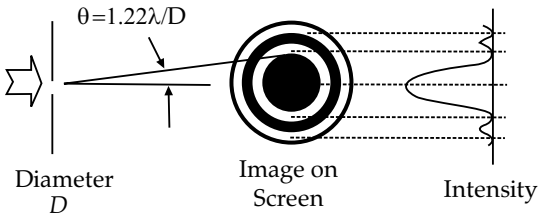
(c) the size increases.

(d) the intensity decreases.

[NCERT Exemp. Q. 10.8, Page 64]

Ans. **Correct options :** (a) and (b)

Explanation : The shadow of hole of diameter 'D' is spread out over an angle $\Delta\theta = 1.22\frac{\lambda}{D} \Rightarrow \Delta\theta \propto \frac{1}{D}$.



The central bright disc is known as Airy's disc. As the size of the hole is increased, AO will decrease and size of Airy's disc will decrease.

As the size of the hole is increased, the width of the central maximum of the diffraction pattern of hole decreases. Since the same amount of light is now distributed over a small area, as intensity tends to infinity or 1/area, the area is decreasing so intensity increases.

Q. 9. For light diverging from a point source

(a) the wavefront is spherical.

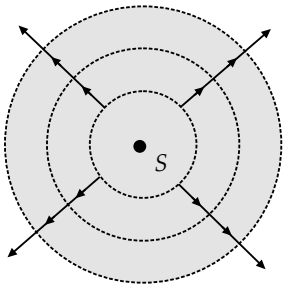
(b) the intensity decreases in proportion to the distance squared.

(c) the wavefront is parabolic.

(d) the intensity at the wavefront does not depend on the distance. **[NCERT Exemp. Q. 10.9, Page 64]**

Ans. **Correct options :** (a) and (b)

Explanation : Light from point source emits in all around the source with same speed, so forms a spherical surface of wavefront or spherical wavefront.



As the intensity (I) always decreases as the reciprocal of square of distance,

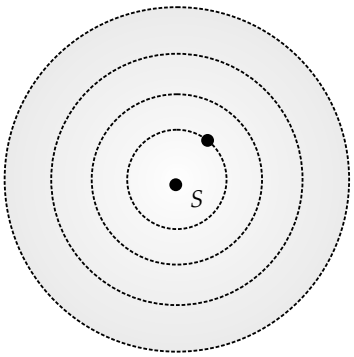
$$I = \frac{P}{4\pi r^2}$$

r= Radius of spherical wave front at any time ($r = vt$).

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Is Huygens' principle valid for longitudinal sound waves? **[NCERT Exemp. Q. 10.10, Page 64]**

Ans. Consider a source of sound formed with the compressions and rarefactions forward in all directions with same velocity. So longitudinal waves propagate with spherical symmetry in all directions as the wavefront in light waves. So Huygens' principle is valid for longitudinal sound waves also. **[1]**

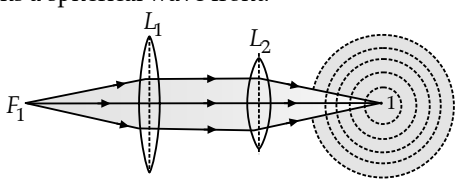


On a surface of sphere there will be either compression or rarefaction and that part can also behave like a source of sound but with low intensity. **[1]**

Q. 2. Consider a point at the focal point of a convergent lens. Another convergent lens of short focal length is placed on the other side. What is the nature of the wavefronts emerging from the final image?

[NCERT Exemp. Q. 10.11, Page 64]

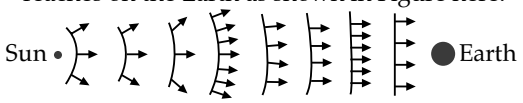
Ans. Consider a point 'F' on focus of converging lens L_1. The light rays from F, becomes parallel after refraction through L_1. When these parallel rays fall on converging lens L_2 placed co-axial on the other side of F of L_1. L_2 converges the rays at its focus at I. It now behaves like a point source of rays and forms a spherical wave front. **[1]**



[1]

Q. 3. What is the shape of the wavefront on Earth for sunlight? **[NCERT Exemp. Q. 10.12, Page 64]**

Ans. As the Sun is very-very far away from the Earth so it can be considered at infinity and the Sun can be considered as a point source which gives spherical wavefront. The size of the Earth is very small as compared to distance of the Sun from the Earth and the size of the Sun. So the plane wavefront reaches on the Earth as shown in Figure here. **[1]**



Spherical wavefront Plane wavefront **[1]**

Q. 4. Why is the diffraction of sound waves more evident in daily experience than that of light wave? [NCERT Exemp. Q. 10.13, Page 65]

Ans. We know that frequencies of sound wave vary from 20 Hz to 20,000 Hz, so its corresponding wavelength varies from 15 m to 15 mm, respectively. The size of slit (almost) becomes comparable to wavelength of sound so diffraction of sound wave takes place easily. But the wavelength of visible light varies from $0.4\,\mu$ to $0.7\,\mu$ which is very small. So the size of most of the slits is not comparable with wavelength of visible light, due to this diffraction of light cannot take place. **[2]**

Q. 5. The human eye has an approximate angular resolution of $\varphi = 5.8 \times 10^{-4}$ rad and a typical photo printer prints a minimum of 300 dpi (dots per inch, 1 inch = 2.54 cm). At what minimal distance z should a printed page be held so that one does not see the individual dots?

[NCERT Exemp. Q. 10.14, Page 65]

Ans. Angular separation $= 5.8 \times 10^{-4}$ radian

The average distance between any two dots,

$$\text{Arc} = \frac{2.54}{300} = 0.85 \times 10^{-2}\,\text{cm}$$

At the distance z cm, angle subtended

$$= \frac{\text{Arc}}{\text{Rad.}} = \frac{0.85 \times 10^{-2}}{z}$$

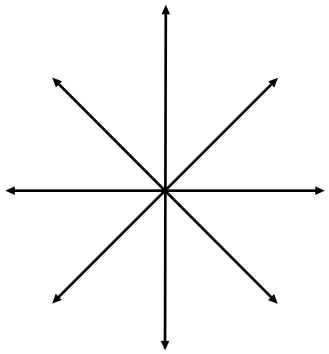
Resolution angle for human

$$= 5.8 \times 10^{-4}\ \text{rad} = \frac{0.85 \times 10^{-2}}{z}$$

Maximum distance up to which human eye cannot see two dots distinctly, $z = \dfrac{0.85 \times 10^{-2}}{5.8 \times 10^{-4}} = 14.5\,\text{cm}$, which is less than distance of distinct vision. So a normal person cannot see the dots. **[2]**

Q. 6. A polaroid (I) is placed in front of a monochromatic source. Another polaroid (II) is placed in front of this polaroid (I) and rotated till no light passes. A third polaroid (III) is now placed in between (I) and (II). In this case, will light emerge from (II). Explain. [NCERT Exemp. Q. 10.15, Page 65]

Ans. Polaroid III is placed between two crossed polaroids I and II and no light passes through II polaroid. Polaroid III is now placed between I, II and now retakes keeping I, II in no rotation. Let angle between Polaroid I and III be θ and intensity of plane polarize light after first polaroid is I_0. Then $I_1 = I_0$. **[½]**



Unpolarised light **[½]**

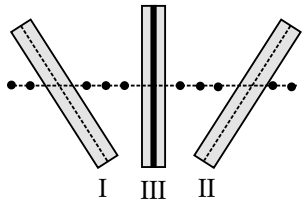
$I_3 = I_0 \cos^2\theta$ is intensity of light after polaroid III

$$I_2 = I_3 \cos^2\theta = I_0 \cos^2\theta . \cos^2(90° - \theta)$$

$$I_2 = I_0 \cos^2\theta . \sin^2\theta$$

$$I_2 = I_0 (2\cos\theta \sin\theta)^2 = I_0 (\sin 2\theta)^2$$

$$I_2 = \frac{I_0}{4} \sin^2 2\theta$$



θ is angle between I and III polaroids.

No light will pass through II polaroid from above equation. When

$$\sin^2 2\theta = 0$$

$$\sin 2\theta = 0$$

$$\sin 2\theta = \sin 0$$

$$\theta = 0°$$

Therefore, no light will pass when angle between I and III polaroids is zero, *i.e.*, plane of polaroid area is parallel. Only in the special cases when the pass axis of (III) is parallel to (I) or (II) there shall be no light emerging. In all other cases there shall be light emerging because the pass axis of (II) is no longer perpendicular to the pass axis of (III). **[1]**

Q. 7. Monochromatic light having a wavelength of 589 nm from the air is incident on a water surface. Find the frequency, wavelength and speed of

(i) reflected and

(ii) refracted light

[Given : 1.33 is the Refractive index of water.]

[CBSE Board, All India Region, 2017; NCERT Ex. Q. 10.1, Page 383]

Ans. Monochromatic light incident having wavelength, $\lambda = 589\ \text{nm} = 589 \times 10^{-9}\ \text{m}$

Speed of light in air, $c = 3 \times 10^8\ \text{ms}^{-1}$

Refractive index of water, $\mu = 1.33$

(i) In the same medium through which incident ray passed the ray will be reflected back.

Therefore, the wavelength, speed and frequency of the reflected ray will be the same as that of the incident ray.

Frequency of light can be found from the relation :

$$v = \frac{c}{\lambda} = \frac{3\times10^{8}}{539\times10^{-9}} = 5.09\times10^{14}\ \text{Hz}$$

Hence, the speed, frequency and wavelength of the reflected light are :

$c = 3 \times 10^{8}$ ms^{-1}, 5.09×10^{14} Hz and 589 nm, respectively. **[1]**

(ii) The frequency of light, which is travelling, never depends upon the property of the medium. Therefore, the frequency of the refracted ray in water will be equal to the frequency of the incident or reflected light in air.

Refracted frequency, $v = 5.09 \times 10^{14}$ Hz

Speed of light in water is related to the refractive index of water as :

$$v = \frac{c}{\mu} \Rightarrow v = \frac{3\times10^{8}}{1.33} = 2.26\times10^{8}\ \text{ms}^{-1}$$

Wavelength of light in water can be found by the relation :

$$\lambda = \frac{v}{\nu} = \frac{2.26\times10^{8}}{5.09\times10^{14}} = 444.007\times10^{-9}\ m = 444.01\ \text{nm}$$

Hence the speed, frequency and wavelength of refracted light are : 2.26×10^{8} m/s, 5.09×10^{14} Hz and 444.01 nm respectively. **[1]**

Q. 8. Find the shape of the wavefront in each of the following cases :

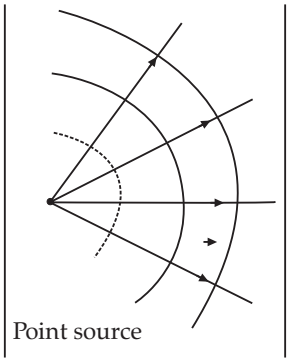
(i) Light diverging from a point source.

(ii) Light emerging out of a convex lens when a point source is placed at its focus.

(iii) The portion of the wavefront of light from a distant star intercepted by the Earth.

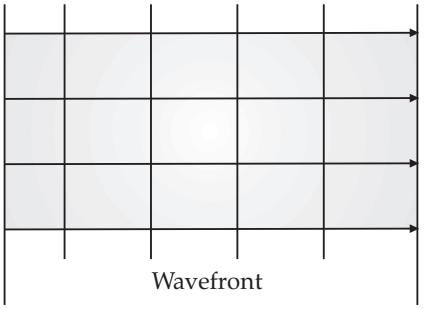
[NCERT Ex. Q. 10.2, Page 383]

Ans. **(i)** The shape of a wavefront is spherical in case of a light diverging from a point source is spherical. The wavefront is shown in the figure :



[½]

(ii) The shape of a wavefront is a parallel grid in the case of a light emerging out of a convex lens when a point source is placed at its focus.



[1]

(iii) The shape of the wavefront is a plane when a portion of the wavefront of light from a distant star intercepted by the Earth. **[½]**

Q. 9. (a) The refractive index of glass is 1.5. What is the speed of light in glass? Speed of light in vacuum is 3.0×10^{8} ms^{-1}.

(b) Is the speed of light in glass independent of the colour of light? If not, which of the two colours red and violet travels slower in a glass prism?

[NCERT Ex. Q. 10.3, Page 383]

Ans. **(a)** Refractive index of glass, $\mu = 1.5$

Speed of light, $c = 3 \times 10^{8}$ m/s

Speed of light in glass is given by the relation,

$$v = \frac{c}{\mu} = \frac{3\times10^{8}}{1.5} = 2\times10^{8}\ \text{m/s}$$

Hence, the speed of light in glass is 2×10^{8} m/s. **[1]**

(b) The speed of light in glass is not independent of the colour of light.

The refractive index of a violet component of violet component is greater than the refractive index of a red component. Hence, the speed of violet light is less than the speed of red light in glass. Hence, violet light travels slower than red light in a glass prism. **[1]**

Q. 10. In a Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment? [NCERT Ex. Q. 10.4, Page 383]

Ans. Distance between the slits, $d = 0.28$ mm

$= 0.28 \times 10^{-3}$ m

Distance between the slits and the screen, $D = 1.4$ m

Distance between the central fringe and the fourth ($n = 4$) fringe,

$u = 1.2$ cm $= 1.2 \times 10^{-2}$ m

In case of a constructive interference, we have the relation for the distance between the two fringes as :

$$u = n\lambda\frac{D}{d}$$

Where, n = Order of fringes =4

λ=Wavelength of light

$$\therefore \lambda = \frac{ud}{nD}$$

$$= \frac{1.2\times10^{-2}\times0.28\times10^{-3}}{4\times1.4}$$

$$= 6\times10^{-7}$$

$$= 600\ \text{nm}$$

Hence, the wavelength of the light is 600 nm. **[2]**

Q. 11. In Young's double-slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen where path difference is λ, is K units. What is the intensity of light at a point where path difference is $\frac{\lambda}{3}$? [NCERT Ex. Q. 10.5, Page 383]

Ans. Let I_1 and I_2 be the intensity of the two light waves. Their resultant intensities can be obtained as :

$$I' = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

[½]

Where,
ϕ = Phase difference between the two waves
For monochromatic light waves :
$I_1 = I_2$
Therefore $I' = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$ [½]
$= 2I_1 + 2I_1\cos\phi$

Phase difference $= \frac{2\pi}{\lambda} \times$ Path difference
Since path difference = λ, Phase difference, $\phi = 2\pi$ and $I' = K$ [Given]
Therefore, $I_1 = \frac{K}{4}$ ...(i)
When path difference $= \frac{\lambda}{3}$
Phase difference, $\phi = \frac{2\pi}{3}$
Hence, resultant intensity,
$I_g^{'} = I_1 + I_1 + 2\sqrt{I_1 I_1}\cos\frac{2\pi}{3}$ [½]
Using equation (i), we can write :
$I_g^{'} = I_1 = \frac{K}{4}$
Hence, the intensity of light at a point, which has path difference ($\frac{\lambda}{3}$), is $\frac{K}{4}$ units. [½]

Q. 12. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes in a Young's double-slit experiment.
(a) Find the distance of the third bright fringe on the screen from the central maximum for wavelength 650 nm.
(b) What is the least distance from the central maximum where the bright fringes due to both the wavelengths coincide?
[NCERT Ex. Q. 10.6, Page 383]

Ans. Wavelength of the light beam, λ_1 = 650 nm
Wavelength of another light beam, λ_2 = 520 nm
Distance of the slits from the screen = D
Distance between the two slits = d

(a) Distance of the nth bright fringe on the screen from the central maximum is given by the relation,
$x = n\lambda_1\frac{D}{d}$
If the third bright firnge : $n = 3$
So, $x = 3 \times 650\frac{D}{d} = 1950\frac{D}{d}$ nm **[1]**

(b) Let, the nth bright fringe due to wavelength λ_2 and $(n-1)^{th}$ bright fringe due to wavelength λ_1 coincide on the screen. We can equate the conditions for bright fringes as :
$n\lambda_2 = (n-1)\lambda_1$
$520n = 650n - 650$
$650 = 130n$
Therefore, $n = 5$
Hence, the least distance from the central maximum can be obtained by the relation :
$x = n\,\lambda_2\frac{D}{d} = 5 \times 520\frac{D}{d} = 2600\frac{D}{d}$nm
[Note : The values of d and D are not given in question.] **[1]**

Q. 13. In a double-slit experiment the angular width of a fringe is found to be 0.2° on a screen placed 1 m away. The wavelength of light used is 600 nm. What will be the angular width of the fringe if the entire experimental apparatus is immersed in water? Take refractive index of water to be 4/3.
[NCERT Ex. Q. 10.7, Page 383]

Ans. Distance of the screen from the slits, D = 1 m
Wavelength of light used, λ_1 = 600 nm
Angular width of the fringe in air, $\theta_1 = 0.2°$
Angular width of the fringe in water = θ_2
Refractive index of water, $\mu = \frac{4}{3}$
Refractive index is related to angular width as :
$\frac{\theta_2}{\theta_1} = \mu = \frac{4}{3}$
$\theta_2 = \frac{4}{3} \times \theta_1 = \frac{4}{3} \times 0.2 = 0.15^o$
Therefore, the angular width of the fringe in water will reduce to 0.15°. **[2]**

Q. 14. What is the Brewster angle for air to glass transition? (Refractive index of glass=1.5.)
[NCERT Ex. Q. 10.8, Page 383]

Ans. Refractive index of glass, μ=1.5
Let, Brewster angle=θ
Brewster angle is related to refractive index as :
$\tan\theta = \mu$
$\theta = \tan^{-1}(1.5) = 56.31°$
Hence, the Brewster angle for air to glass transition is 56.31°. **[2]**

Q. 15. Estimate the distance for which ray optics is good approximation for an aperture of 4 mm and wavelength 400 nm.
[NCERT Ex. Q. 10.10, Page 383]

Ans. Fresnel's distance (Z_F) is the distance for which the ray optics is a good approximation. It is given by the relation,
$Z_F = \frac{a^2}{\lambda}$
Where,
Aperture width, a = 4 mm=4×10^{-3} m
Wavelength of light, λ = 400 nm = 400×10^{-9} m
$Z_F = \frac{(4\times10^{-3})^2}{400\times10^{-9}} = 40$ m
Therefore, the distance for which the ray optics is a good approximation is 40 m. **[2]**

Q. 16. The 6,563 Å H_α line emitted by hydrogen in a star is found to be red shifted by 15 Å.
Estimate the speed with which the star is receding from the Earth.
[NCERT Add. Ex. Q. 10.11, Page 384]

Ans. Wavelength of H_α line emitted by hydrogen, λ = 6,563 Å = $6,563 \times 10^{-10}$ m.
Star's red-shift, (λ'–λ) =15 Å=15×10^{-10} m
Speed of light, $c = 3\times10^8$ m/s
Let the velocity of the star receding away from the Earth be v.

The red shift is related with velocity as :

$$\lambda' - \lambda = \frac{v}{c}\lambda$$

$$v = \frac{c}{\lambda} \times (\lambda' - \lambda)$$

$$= \frac{3 \times 10^8 \times 15 \times 10^{-10}}{6563 \times 10^{-10}} = 6.87 \times 10^5 \text{ m/s}$$

Therefore, the speed with which the star is receding away from the Earth is 6.87×10^5 m/s. **[2]**

Q. 17. Let us list some of the factors, which could possibly influence the speed of wave propagation :

(i) Nature of the source.

(ii) Direction of propagation.

(iii) Motion of the source and/or observer.

(iv) Wave length.

(v) Intensity of the wave.

On which of these factors, if any, does

(a) The speed of light in vacuum,

(b) The speed of light in a medium (say, glass or water), depend?

[NCERT Add. Ex. Q. 10.14, Page 384]

Ans. **(a)** The speed of light in a vacuum is 3×10^8 m/s (approximately) is a universal constant. It is not affected by the motion of the source, the observer, or both. Hence, the given factor does not affect the speed of light in a vacuum. **[1]**

(b) Out of the listed factors, the speed of light in a medium depends on the wavelength of light in that medium. **[1]**

Q. 18. For sound waves, the Doppler formula for frequency shift differs slightly between the two situations : (i) source at rest; observer moving, and (ii) source moving; observer at rest. The exact Doppler formulae for the case of light waves in vacuum are, however, strictly identical for these situations. Explain why this should be so. Would you expect the formulas to be strictly identical for the two situations in case of light travelling in a medium? [NCERT Add. Ex. Q. 10.15, Page 384]

Ans. No, sound waves can propagate only through a medium. The two given situations are not scientifically identical because the motion of an observer relative to a medium is different in the two situations. Hence, the Doppler formula for the two situations cannot be the same. **[1]**

In case of light waves, sound can travel in a vacuum. In a vacuum, the above two cases are identical because the speed of light is independent of the motion of the observer and the motion of the source. When light travels in a medium, the above two cases are not identical because the speed of light depends on the wavelength of the medium. **[1]**

Q. 19. In double-slit experiment using light of wavelength 600 nm, the angular width of a fringe formed on a distant screen is 0.1°. What is the spacing between the two slits?

[NCERT Add. Ex. Q. 10.16, Page 384]

Ans. Wavelength of light used, λ = 6,000 nm = 600×10^{-9} m

Angular width of fringe,

$$\theta = 0.1° = 0.1 \times \frac{\pi}{180} = \frac{3.14}{1,800} \text{rad}$$

Angular width of a fringe is related to slit spacing (d) will be given by :

$$\theta = \frac{\lambda}{d}$$

$$d = \frac{\lambda}{\theta} = \frac{600 \times 10^{-9}}{\frac{3.14}{1,800}} = 3.44 \times 10^{-4} \text{ m}$$

Therefore, the spacing between the slits is 3.44×10^{-4} m. **[2]**

Q. 20. Two towers on top of two hills are 40 km apart. The line joining them passes 50 m above a hill halfway between the towers. What is the longest wavelength of radio waves, which can be sent between the towers without appreciable diffraction effects?

[NCERT Add. Ex. Q. 10.18, Page 385]

Ans. Distance between the towers, d = 40 km

Height of the line joining the hills, d = 50 m.

Thus, the radial spread of the radio waves should not exceed 50 km.

Since the hill is located halfway between the towers, Fresnel's distance can be obtained as :

Z_P = 20 km = 2×10^4 m

Aperture can be taken as :

$a = d$ = 50 m

Fresnel's distance is given by the relation,

$$Z_P = \frac{a^2}{\lambda}$$

Where, λ = Wavelength of radio waves

$$\therefore \lambda = \frac{a^2}{Z_P} = \frac{(50)^2}{2 \times 10^{-4}} = 1,250 \times 10^{-4} = 0.1250 \text{ m} = 12.5 \text{ cm}$$

Therefore, the wavelength of the radio waves is 12.5 cm. **[2]**

Q. 21. A parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 m away. It is observed that the first minimum is at a distance of 2.5 mm from the centre of the screen. Find the width of the slit.

[NCERT Add. Ex. Q. 10.19, Page 385]

Ans. Wavelength of light beam, λ = 500 nm = 500×10^{-9} m

Distance of the screen from the slit, D = 1 m

For first minima, n = 1

Distance between the slits = d

Distance of the first minimum from the centre of the screen can be obtained as :

x = 2.5 mm = 2.5×10^{-3} m

It is related to the order of minima as :

$$n\lambda = x\frac{d}{D}$$

$$d = \frac{n\lambda D}{x} = \frac{1 \times 500 \times 10^{-9} \times 1}{2.5 \times 10^{-3}} = 2 \times 10^{-4} \text{ m} = 0.2 \text{ mm}$$

Therefore, the width of the slits is 0.2 mm. **[2]**

Q. 22. Answer the following questions :

(a) When a low flying aircraft passes overhead, we sometimes notice a slight shaking of the picture on our TV screen. Suggest a possible explanation.

(b) As you have learnt in the text, the principle of linear superposition of wave displacement is basic to understanding intensity distributions in diffraction and interference patterns. What is the justification of this principle?

[NCERT Add. Ex. Q. 10.20, Page 385]

Ans. (a) Weak radar signals sent by a low-flying aircraft can interfere with the TV signals received by the antenna. As a result, the TV signals may get distorted. Hence, when a low-flying aircraft passes overhead, we sometimes notice a slight shaking of the picture on our TV screen. **[1]**

(b) The principle of linear superposition of wave displacement is essential to our understanding of intensity distributions and interference patterns. This is because superposition follows from the linear character of a differential equation that governs wave motion. If y_1 and y_2 are the solutions of the second-order wave equation, then any linear combination of y_1 and y_2 will also be the solution of the wave equation. **[1]**

Q. 23. In deriving the single slit diffraction pattern, it was stated that the intensity is zero at angles of $\frac{n\lambda}{a}$. Justify this by suitably dividing the slit to bring out the cancellation.

[NCERT Add. Ex. Q. 10.21, Page 385]

Ans. Consider that a single slit of width d is divided into n smaller slits.

$\therefore$ Width of each slit, $d' = \frac{d}{n}$

Angle of diffraction is given by the relation,

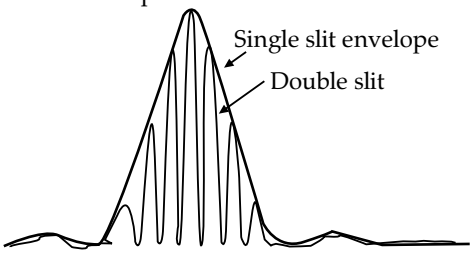
$$\theta = \frac{\frac{d}{d}\lambda}{d} = \frac{\lambda}{d}$$

Now, each of these infinitesimally small slit sends zero intensity in direction θ. Hence, the combination of these slits will give zero intensity. **[2]**

Q. 24. Draw the intensity pattern for single slit diffraction and double slit interference. Hence, state two differences between interference and diffraction patterns.

[CBSE Board, All India Region, 2017]

Ans. Under the Fraunhofer conditions, a single slit will exhibit a light curve following the single-slit diffraction intensity expression. The narrower the slit, the broader the peaks of light. The shape or "envelope" of this light curve will serve to set limiting intensities for multiple slit arrangements, assuming that all the slits are identical. Under the Fraunhofer conditions, the light curve (intensity vs position) is obtained by multiplying the multiple slit interference expression times the single-slit diffraction expression. **[½]**



[½]

The differences in between Interference and diffraction patterns :

Interference refers to the phenomenon of the interaction of the waves. In this phenomenon, two different wavefronts originate from two different coherent sources.

Diffraction refers to the phenomenon where the waves spread into a pattern after diffracting through an aperture. It constitutes the secondary wavelets that originate from the same wave.

In the interference, the region of minimum intensity is defined by perfect darkness and it is not perfectly dark in diffraction. The width of fringes is not equal in diffraction and it is equal in interference. **[1]**

Q. 25. Unpolarised light is passed through a polaroid P_1. When this polarised beam passes through another polaroid P_2 and if the pass axis of P_2 makes angle θ with the pass axis of P_1, then write the expression for the polarised beam passing through P_2. Draw a plot showing the variation of intensity when θ varies from 0 to 2θ.

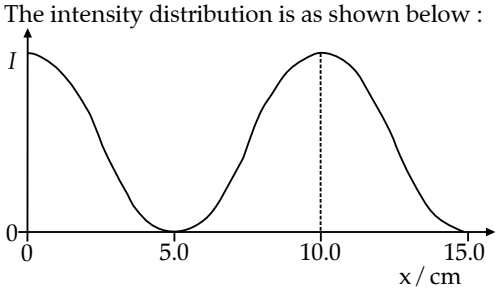
[CBSE Board, All India Region, 2017]

Ans. When unpolarised light passes through the polariser P_1 the intensity drops to half that is $\frac{I_0}{2}$ and when this polarised light passes through the polariser P_2 the intensity of the emitted light is

$$I = \frac{I_0}{2}\cos^2\theta$$

(θ is the angle between polariser P_1 and P_2) **[1½]**

The intensity distribution is as shown below :



[½]

Q. 26. Find the intensity at a point on a screen in Young's double slit experiment where the interfering waves of equal intensity have a path difference of (i) $\frac{\lambda}{3}$ (ii) $\frac{\lambda}{4}$

[CBSE Board, Foreign Scheme, 2017]

Ans. **(i)** When path difference is $\frac{\lambda}{3}$

$$\Rightarrow \frac{2\pi}{\lambda}\times\frac{\lambda}{3} = \frac{2\pi}{3}$$

So,

$$I' = I\cos^2\left(\frac{2\pi}{3}\right) = \frac{I}{4}$$

[1]

(ii) When path difference is $\frac{\lambda}{4}$

$$\Rightarrow \frac{2\pi}{\lambda}\times\frac{\lambda}{4} = \frac{\pi}{2}$$

So,

$$I' = I\cos^2\left(\frac{\pi}{2}\right) = 0$$

[1]

Q. 27. State Brewster's Law. The value of Brewster angle for a transparent medium is different for light of different colours. Give reason.

[CBSE Board, Foreign Scheme, 2016]

Ans. Brewster's Law : According to this law when unpolarised light is incident at polarising angle (i) on an interface separating a rarer medium from a denser medium, of refractive index μ such that $\mu = \tan i_p$, then light reflected in the rarer medium is completely polarised. Reflected and refractive rays are perpendicular to each other. **[1]**

The refractive index of a material depends on the colour or wavelength of light. As the polarising angle depends on refractive index ($\mu = \tan i_p$), so it also depends on wavelength of light. **[1]**

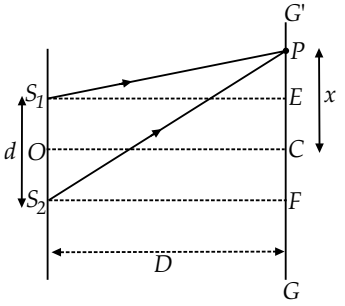
Short Answer Type Questions (3 or 4 marks each)

Q. 1. (i) In Young's double-slit experiment, deduce the condition for (a) constructive and (b) destructive interference at a point on the screen. Draw a graph showing variation of intensity in the interference pattern against position 'x' on the screen.

(ii) Compare the interference pattern observed in Young's double-slit experiment with single-slit diffraction pattern, pointing out three distinguishing features.

[CBSE Board, Foreign Scheme, 2016]

Ans. (i) Expression for fringe width in Young's double-slit experiment.



[½]

Let S_1 and S_2 be two slits separated by a distance d.

GG' is the screen at a distance D from the slits S_1 and S_2.

Point C is equidistant from both of the slits.

The intensity of light will be maximum at this point because the path difference of the waves reaching this point will be zero.

At point P, the path difference between the rays coming from the slits is given by,

$S = S_2P - S_1P$

Now, $S_1S_2 = d$, $EF = d$, and $S_2F = D$

In ΔS_2PF

$$S_2P = \sqrt{S_2F^2 + PF^2}$$

$$S_2P = \left[D^2 + \left(x + \frac{d}{2}\right)^2\right]^{\frac{1}{2}}$$

$$S_2P = D\left[1 + \frac{\left(x + \frac{d}{2}\right)^2}{D^2}\right]^{\frac{1}{2}}$$

Similarly, in ΔS_1PF,

$$S_1P = D\left[1 + \frac{1}{2}\frac{\left(x + \frac{d}{2}\right)^2}{D^2}\right] - D\left[1 + \frac{1}{2}\frac{\left(x - \frac{d}{2}\right)^2}{D^2}\right]$$

On expanding binomially, we get

$$S_2P - S_1P = \frac{1}{2D}\left[4 \times \frac{d}{2}\right] = \frac{xd}{D}$$

[1]

For constructive interference, the path difference is an integral multiple of wavelengths, that is, path difference is n.

Therefore,

$$n\lambda = \frac{xd}{D}$$

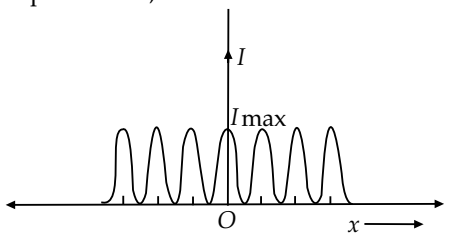
$x = \frac{n\lambda D}{d}$; where $n = 0, 1, 2, 3, ...$

Similarly for destructive interference,

$$x_n = (2n-1)\frac{\lambda}{2}\frac{D}{d}$$

[½]

Graph of intensity distribution in Young's double-slit experiment is,



[½]

(ii) Three distinguishing features observed in Young's double-slit experiment as compared to single-slit diffraction pattern is,

1. In the interference pattern, all the bright fringes have the same intensity. The bright fringes are not of the same intensity in a diffraction pattern.
2. In interference pattern, the dark fringes have zero or small intensity so that the bright and dark fringes can be easily distinguished. While in diffraction pattern, all the dark fringes are not of zero intensity.
3. In interference pattern, the widths of all fringes are almost the same, whereas in diffraction pattern, the fringes are of different widths. **[1½]**

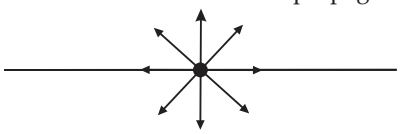
Q. 2. Distinguish between polarised and unpolarised light. Does the intensity of polarised light emitted

by a polaroid depend on its orientation? Explain briefly.

The vibrations in a beam of polarised light make an angle of 60° with the axis of the polaroid sheet. What percentage of light is transmitted through the sheet? [CBSE Board, Foreign Scheme, 2016]

Ans. **(i)** Unpolarised light are those light waves whose planes of vibrations are randomly oriented about the direction of propagation of the wave.

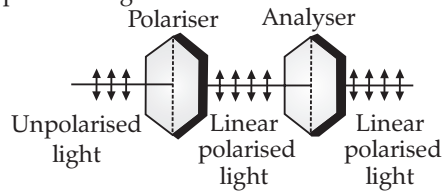
In a beam of unpolarised light, the vibrations of light vectors are in all directions in a plane perpendicular to the direction of propagation. **[½]**



[½]

Polarised light waves are those light waves in which planes of vibrations can occur in one particular plane only.

Polarised light can be distinguished, from unpolarised light, when it is allowed to pass through a polaroid. Polarised light does can show the change in its intensity, on passing through a polaroid; intensity remains same in case of unpolarised light. **[½]**



[½]

Yes, the intensity of polarised light emitted by a polaroid depends on its orientation. When polarised light is incident on a polaroid, the resultant intensity of light transmitted from varies directly as the square of the cosine of the angle between polarisation direction of light and axis of the polaroid. **[½]**

The intensity of light transmitted can be given using Malu's law as :

$$I = I_0 \cos^2\theta = I_0 \cos^2 60° = \frac{I_0}{4}$$ **[½]**

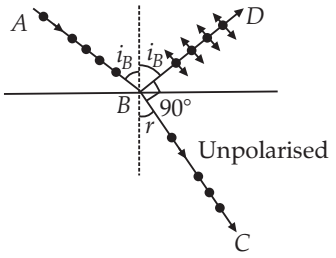
(ii) Percentage of light is transmitted through the polaroid sheet is $\frac{I_0 - \frac{I_0}{4}}{I_0} \times 100 = 75\%$

In the linearly polarised light, the vibrations of light take place in a particular direction, perpendicular to the direction of wave motion. **[1]**

Q. 3. Can reflection result in plane polarised light if the light is incident on the interface from the side with higher refractive index?

[NCERT Exemp. Q. 10.16, Page 65]

Ans. When a ray of light passes from a medium (air) of refractive index, μ_1 to another medium of refractive index, μ_2, more than μ_1 with incidence angle is equal to Brewster's angle (i_B), then transmitted ray BC will be unpolarised light and the reflected light BD will be plane polarised. Angle DBC between refracted and reflected ray is 90°. **[½]**



[½]

$\angle r + \angle i_B + 90° = 180°$

So $\angle r + \angle i_B + 90°$ or $\angle r = 90° - i_B$

By Snell's law, $\mu_2 = \frac{\sin i_B}{\sin r} = \frac{\sin i_B}{\cos i_B} = \tan i_B$

$\therefore \frac{\mu_2}{\mu_1} = \tan i_B$ ...(I)

$\left(\begin{array}{l} \because \mu_2 > \mu_1 \\ \therefore \tan i_B > 1 \\ i_B > 45° \end{array}\right)$ **[1]**

$\sin i_c = \frac{1}{\mu_2}$ when light passes from medium 2 to medium 1

$\sin i_c = \frac{\mu_1}{\mu_2}$ $(\because \mu_2 > \mu_1)$

so $\sin i_c < 1$ ...(II)

or $i_c < 90°$

From *Eq.* (I), $\tan i_B > 1$

From *Eq.* (II), $1 > \sin i_c$

Or $\tan i_B > 1 > \sin i_c$ or $|\tan i_B| > |\sin i_c|$

Because $45° < i_B$

$i_c < 90°$

So $45° < i_B < 90°$

$0 < i_c < 90°$

So $i_B > i_c$.

Thus, polarization by reflection takes place. **[2]**

Q. 4. For the same objective, find the ratio of the least separation between two points to be distinguished by a microscope for light of 5,000 Å and electrons accelerated through 100 V used as the illuminating substance. [NCERT Exemp. Q. 10.17, Page 65]

Ans. λ = 5,000 Å = 5,000 × 10^{-10} m

In microscope,

$R.P. = \frac{1}{d} = \frac{2\sin\beta}{1.22\lambda}$

Limit of resolution by light of 5,000 Å

$d_{\min} = \frac{1.22\lambda}{2\sin\beta}$

or

$d_{\min} = \frac{1.22 \times 5{,}000 \times 10^{-10}}{2\sin\beta}$ **[1]**

The de Broglie wavelength λd of illuminated light,

$\lambda_d = \frac{1.22}{\sqrt{V}}$ nm

$\lambda_d = \frac{1.22}{\sqrt{100}}$ nm $= \frac{1.22}{10} \times 10^{-10}$ m **[1]**

The limit of resolution by 100 V light will be :

$$d'_{min} = \frac{1.22\lambda_d}{2\sin\beta}$$

$$d'_{min} = \frac{1.22 \times 1.22 \times 10^{-10}}{2\sin\beta} \quad \textbf{[1]}$$

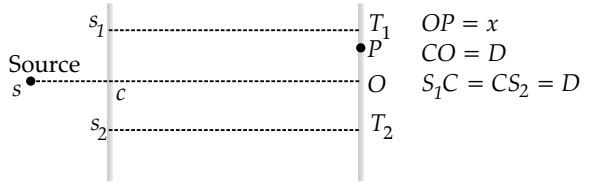
The required ratio will be given as :

$$= \frac{d'_{min}}{d_{min}} = \frac{\frac{1.22 \times 1.22 \times 10^{-10}}{2\sin\beta}}{\frac{1.22 \times 5,000 \times 10^{-10}}{2\sin\beta}} = \frac{1.22}{5,000}$$

The required ratio will give as :

$$= \frac{122}{500} \times 10^{-3} = 0.244 \times 10^{-3} \quad \textbf{[1]}$$

Q. 5. Consider a two slit interference arrangements in figure such that the distance of the screen from the slits is half the distance between the slits. Obtain the value of *D* in terms of λ such that the first minima on the screen falls at a distance *D* from the centre *O*.



[NCERT Exemp. Q. 10.18, Page 65]

Ans. According to figure,

$x = D$ (Given) …(I)

$D = \frac{1}{2}d$ (Given) …(II)

$d = 2D$

Path difference at $P = S_2P - S_1P$

$$= \sqrt{D^2 + \left(x + \frac{d}{2}\right)^2} - \sqrt{D^2 + \left(x - \frac{d}{2}\right)^2} \quad \textbf{[½]}$$

Substitute the value of d and x from equations (I) and (II),

$$= \sqrt{D^2 + (D+D)^2} - \sqrt{D^2 + (D-D)^2}$$

$$= \sqrt{5D^2} - \sqrt{D^2} \quad \text{…(III)}$$

$p = D(\sqrt{5} - 1)$ **[½]**

The path difference for nth dark fringe from central maxima 'O' is $(2n-1)\frac{\lambda}{2}$. **[½]**

$\therefore$ For first minima, $p = \frac{\lambda}{2}$ **[½]**

Put the value of p in *Eq.* (III)

$$\frac{\lambda}{2} = D(\sqrt{5} - 1) \quad \text{...(IV)}$$

$$D = \frac{\lambda}{2(\sqrt{5} - 1)} \quad \textbf{[1]}$$

Rationalizing the denominator, we get

$$D = \frac{\lambda}{2(\sqrt{5} - 1)} \times \frac{(\sqrt{5} + 1)}{(\sqrt{5} + 1)} = \frac{(2.236 + 1)}{2 \times (5 - 1)}\lambda = \frac{3.236}{2 \times 4}\lambda$$

$$= \frac{3.236}{8}\lambda = 0.404\lambda \quad \textbf{[1]}$$

Q. 6. Light of wavelength 5,000 Å falls on a plane reflecting surface. What are the wavelength and frequency of the reflected light? For what angle of incidence is the reflected ray normal to the incident ray? [NCERT Ex. Q. 10.9, Page 383]

Ans. Wavelength of incident light, λ = 5,000 Å = 5,000 × 10^{-10} m

Speed of light, $c = 3 \times 10^8$ m/s

Frequency of incident light is given by the relation,

$$v = \frac{c}{\lambda} = \frac{3 \times 10^8}{5,000 \times 10^{-10}} = 6 \times 10^{14} \text{ Hz}$$

The wavelength and frequency of incident light is the same as that of reflected ray. Hence, the wavelength of reflected light is 5,000 Å and its frequency is 6×10^{14} Hz. When reflected ray is normal to incident ray, the sum of the angle of incidence, $\angle i$ and angle of reflection, $\angle r$ is 90°. **[1]**

According to the law of reflection, the angle of incidence is always equal to the angle of reflection. Hence, we can write the sum as :

$\angle i + \angle r = 90°$

i.e., $\angle i + \angle i = 90°$

Hence, $\angle i = \frac{90}{2} = 45°$

Therefore, the angle of incidence for the given condition is 45°. **[2]**

Q. 7. Explain how Corpuscular theory predicts the speed of light in a medium, say, water, to be greater than the speed of light in vacuum. Is the prediction confirmed by experimental determination of the speed of light in water? If not, which alternative picture of light is consistent with experiment?

[NCERT Add. Ex. Q. 10.12, Page 384]

Ans. No; wave theory.

Newton's corpuscular theory of light states that when light corpuscles strike the interface of two media from a rarer (air) to a denser (water) medium, the particles experience forces of attraction normal to the surface. Hence, the normal component of velocity increases while the component along the surface remains unchanged. **[1]**

Hence, we can write the expression :

$c \sin i = v \sin r$ ...(i)

Where,

i = Angle of incidence

r = Angle of reflection

c = Velocity of light in air

v = Velocity of light in water

We have the relation for relative refractive index of water with respect to air as :

$$\mu = \frac{v}{c} \quad \textbf{[1]}$$

Hence, equation (i) reduces to

$$\frac{v}{c} = \frac{\sin i}{\sin r} = \mu \quad \text{…(ii)} \quad \textbf{[1]}$$

But, $\mu > 1$

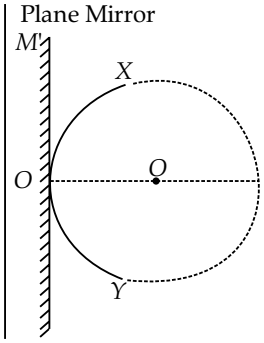
Hence, it can be inferred from equation (ii) that $v > c$. This is not possible since this prediction is opposite to the experimental results of $c > v$.

The wave picture of light is consistent with the experimental results. **[1]**

Q. 8. You have learnt in the text how Huygens' principle leads to the laws of reflection and refraction. Use the same principle to deduce directly that a point object placed in front of a plane mirror produces a virtual image whose distance from the mirror is equal to the object distance from the mirror.

[NCERT Add. Ex. Q. 10.13, Page 384]

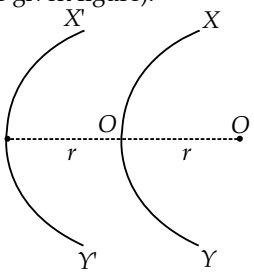
Ans. Let an object at O be placed in front of a plane mirror MO' at a distance r (as shown in the given figure). **[½]**



[½]

A circle is drawn from the centre (O) such that it just touches the plane mirror at point O'. According to Huygens' Principle, XY is the wave front of incident light. **[½]**

If the mirror is absent, then a similar wavefront $X'Y'$ (as XY) would form behind O' at distance r (as shown in the given figure).



[½]

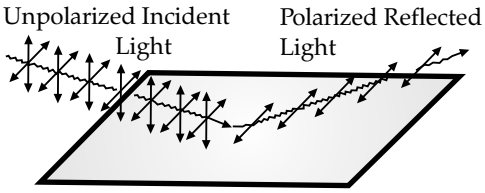
$X'Y'$ can be considered as a virtual reflected ray for the plane mirror. **[½]**

Hence, a point object placed in front of the plane mirror produces a virtual image whose distance from the mirror is equal to the object distance (r). **[½]**

Q. 9. Show using a proper diagram how unpolarised light can be linearly polarised by reflection from a transparent glass surface.

[CBSE Board, Delhi Region, 2018]

Ans. Unpolarised light can also undergo polarisation by reflection off of non-metallic surfaces. The extent to which polarisation occurs is dependent upon the angle at which the light approaches the surface and upon the material that the surface is made of. Metallic surfaces reflect light with a variety of vibrational directions; such reflected light is unpolarised. However, non-metallic surfaces such as asphalt roadways, snowfields and water reflect light such that there is a large concentration of vibrations in a plane parallel to the reflecting surface.



Reflection of light off of non-metallic surfaces results in some degree of polarization parallel to the surface.

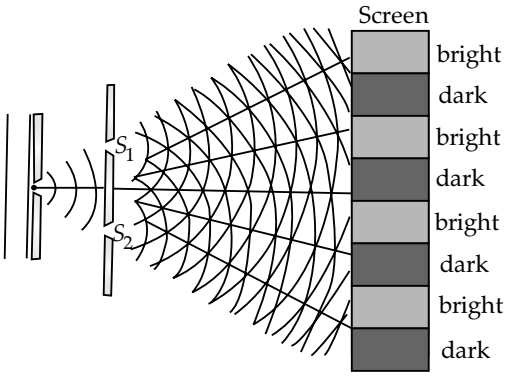
A person viewing objects by means of light reflected off of non-metallic surfaces will often perceive a glare if the extent of polarisation is large. Fishermen are familiar with this glare since it prevents them from seeing fish that lie below the water. Light reflected off a lake is partially polarised in a direction parallel to the water's surface. Fishermen know that the use of glare-reducing sunglasses with the proper polarisation axis allows for the blocking of this partially polarised light. By blocking the plane-polarised light, the glare is reduced and the fisherman can more easily see fish located under the water. **[3]**

Q. 10. If one of two identical slits producing interference in Young's experiment is covered with glass, so that the light intensity passing through it is reduced to 50%, find the ratio of the maximum and minimum intensity of the fringe in the interference pattern.

[CBSE Board, Delhi Region, 2018]

Ans. If only one slit is open the intensity (equal to rate of arrival of photons somewhere) becomes evenly distributed (no fringes) and at one quarter of the peak intensity from when both slits are open. The peak probability amplitude is halved, and the intensity goes as the square of the modulus of the probability amplitude; the new even intensity is a quarter of the old peak intensity value. **[1]**

Note that the number (or rate) of particles detected (anywhere in the screen), when only one slit is open, is half the number detected when both slits are open. This is because half the particles are now absorbed at the blocked slit. (Let assumed that the slits are wide enough so that we can ignore the single slit diffraction pattern, which is irrelevant here from a conceptual point of view.) **[1]**



[1]

It is given by

$$\frac{I_{max}}{I_{min}} = \frac{\left[\sqrt{\frac{I_1}{I_2}}+1\right]^2}{\left[\sqrt{\frac{I_1}{I_2}}-1\right]^2};\ \frac{I_1}{I_2}=\frac{1}{2}$$

$$\frac{I_{max}}{I_{min}} = \frac{\left[\frac{1}{2}+1\right]^2}{\left[\frac{1}{2}-1\right]^2} = \frac{\left(\frac{3}{2}\right)^2}{\left(\frac{1}{2}\right)^2} = 9:1$$

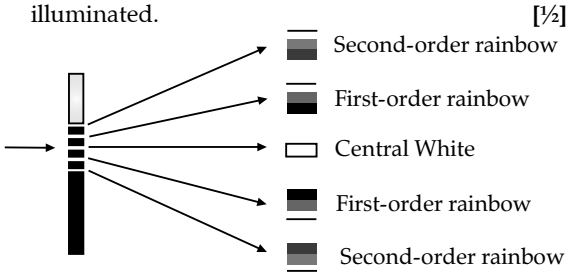
[1]

Q. 11. What kind of fringes do you expect to observe if white light is used instead of monochromatic light? [CBSE Board, Delhi Region, 2018]

Ans. White light consists of waves of innumerable wavelengths starting from violet to red colour. Therefore, if monochromatic light in Young's interference experiment is replaced by white light, then the waves of each wavelength form their separate interference patterns. The resultant effect of all these patterns is obtained on the screen. **[1]**

The path difference between waves starting from S_1 and S_2 at the location (M) of central fringe is zero, *i.e.*, for point M of screen $S_1(M) - S_2(M)=0$, *i.e.*, the waves of all colours reach at mid-point, M in the same phase. Therefore, the central fringe (at M) is white. As fringe width, *i.e.*, wavelength increases in order of colours denoted by **VIBGYOR** therefore on either side of it some coloured fringes are obtained in order of colour VIBGYOR. That is the violet (V) fringe appears first and the red (R) the last. After this the fringes of many colours overlap at each point of the screen and so the screen appears uniformly illuminated. **[1]**

Thus if we use white light in place of monochromatic light the central fringe is white, containing on either side a few coloured fringes (in order VIBGYOR) and the remaining screen appears uniformly illuminated. **[½]**



[1½]

Q. 12. In a single-slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band? Explain.

[CBSE Board, Delhi Region, 2018]

Ans. When width (a) of single slit is made double, the half angular width of central maximum which is λ/a, reduces to half. The intensity of central maximum will become 4 times. This is because area of central diffraction band would become 1/4th. **[1]**

The width of central fringe extends up to first secondary minima on both the side, *i.e.*

$$\beta_0 = 2y_1$$ **[½]**

We know,

$$y_1 = \frac{D\lambda}{a}$$

Or, $\beta_0 = \frac{2D\lambda}{a}$ **[1]**

So, if we double the slit size the width of the central maxima decreases by half. **[½]**

When the width of the central maxima is reduced to half, the area of the central diffraction bank will become one-fourth. Therefore, the intensity becomes four times. **[1]**

Q. 13. When a tiny circular obstacle is placed in the path of the light from a distance source, a bright spot is seen at the centre of the obstacle. Explain why.

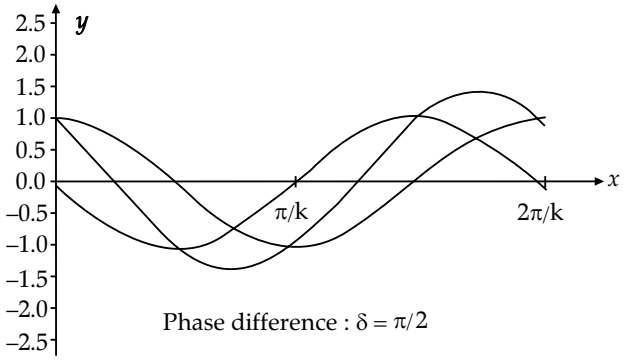
[CBSE Board, Delhi Region, 2018]

Ans. This phenomenon is based on constructive interference. A pair of light or sound waves will experience interference when they pass through each other. The individual waves will add together (superposition) so that a new wavefront is created. Constructive interference occurs when the maxima of two waves add together (the two waves are in phase), so that the amplitude of the resulting wave is equal to the sum of the individual amplitudes. Equivalently, the minima of the waves would be aligned. **[1]**

As shown in the waveforms of the image-the effects of constructive interference between two waves with the same amplitude and frequency (ω) described by the equations :

$y_1 = \cos(kx - \omega t)$ and $y_2 = \cos(kx - \omega t + \delta)$ **[1]**

where δ is the phase difference between the waves, k is the wave number, x is the wave position and t is time.



[1]

When a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. This is because light waves are diffracted from the edge of the circular obstacle, which interferes constructively at the centre of the shadow. This constructive interference produces a bright spot. **[1]**

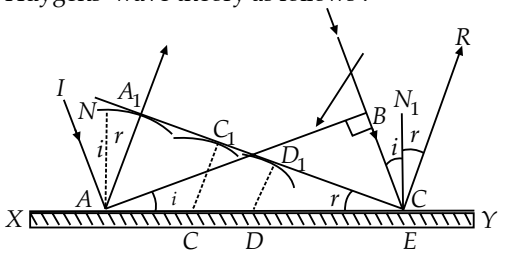
Q. 14. Define wavefront. Using Huygens' principle, verify the laws of reflection at a plane surface.

[CBSE Board, Delhi Region, 2018]

Ans. A wavefront is an imaginary surface over which an optical wave has a constant phase. The shape of a

wavefront is usually determined by the geometry of the source. **[½]**

We can prove the laws of reflection on the basis of Huygens' wave theory as follows :



[1]

In the given figure, *AB* is the wavefront incident on a reflecting surface *XY* with an angle of incidence, *i* as shown in figure. According to Huygens' Principle, every point on *AB* acts as a source of secondary wavelets. At first, wave incidents at point *A* and then to points *C*, *D* and *E*. They form a sphere of radii AA_1, CC_1 and DD_1 as shown in figure. **[½]**

A_1E represents the tangential envelop of secondary wavelet in forward direction.

In ΔABE and ΔAA_1E,

$\angle ABE = \angle AA_1E = 90°$

Side AE = Side AE

$AA_1 = BE$ = Distance travelled by wave in same time

So, these triangles are congruent.

So, $\angle BAE = i$ and $\angle BEA = r$

Thus, $i = r$

Hence, angle of incidence is equal to the angle of reflection. This is the **First Law of Reflection**.

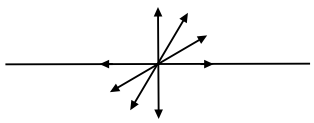
Since, the incident wavefront *AB*, normal and reflected wavefront A_1E lies in same plane, it verifies the **Second Law of Reflection**.

In this way, Laws of Reflection is verified by Huygens' Principle. **[1]**

Q. 15. Distinguish between unpolarised light and linearly polarised light. How does one get linearly polarised light with the help of a polaroid?

[CBSE Board, Delhi Region, 2017]

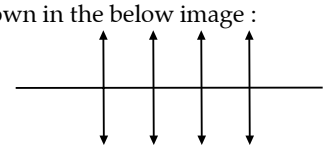
Ans. **Unpolarised light :** A light wave in which electric vector oscillates in all possible directions in a plane perpendicular to the direction of propagation is called unpolarised light as shown in the below image :



[½]

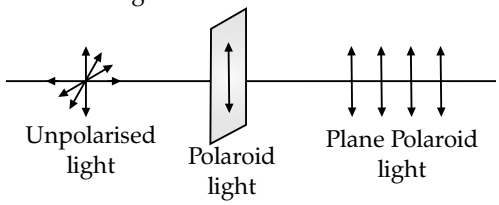
Linearly polarised : If the electric vector's vibration is restricted to one direction only in a plane perpendicular to the direction of propagation, the light is called linearly polarised. **[½]**

As shown in the below image :



[1]

Polaroid is a special crystalline solid which contains a special axis called optic axis. When we make an unpolarised light to fall on a polaroid, only vibration parallel to optic axis is allowed to pass through and all other vibration are cut. Thus, the output is a plane polarised light. As shown in the below image : **[1]**



[1]

Q. 16. (a) A monochromatic light of wavelength 500 nm is incident normally on a single slit of width 0.2 mm to produce a diffraction pattern. Find the angular width of the central maximum obtained on the screen.

(b) Estimate the number of fringes obtained in Young's double-slit experiment with fringe width 0.5 mm, which can be accommodated within the region of total angular spread of the central maximum due to single slit. [CBSE Board, Delhi Region, 2017]

Ans. Given that, $\lambda = 500$ nm; $w = 0.2$ mm

(a) The angular width of the central maximum obtained on the screen,

$$w = \frac{2\lambda}{a} = \frac{2 \times 500 \times 10^{-9}}{0.2 \times 10^{-3}} = 0.5 \times 10^{-3} \text{ rad}$$

Angular width of central maximum in diffraction pattern is 0.5×10^{-3} m for double slit. **[2]**

(b) The region of total angular spread of the central maximum due to Young's double-slit experiment will be,

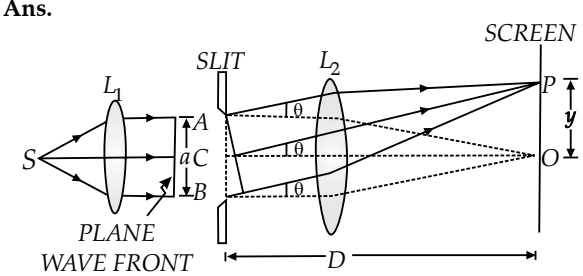
$$\frac{\text{Angular width}}{\text{Fringe width}} = \frac{5 \times 10^{-3}}{0.5 \times 10^{-3}} = 10 \text{ Fringes}$$

Number of fringes will be 10. **[2]**

Q. 17. A monochromatic light of wavelength λ is incident normally on a narrow slit of width '*a*' to produce a diffraction pattern on the screen placed at a distance *D* from the slit. With the help of a relevant diagram, deduce the conditions for obtaining maxima and minima on the screen. Use these conditions to show that angular width of central maximum is twice the angular width of secondary maximum.

[CBSE Board, Foreign Scheme, 2017]

Ans.



[1]

The wavelets from points *A* and *B* will have a path difference equal to *BN*.

From the right-angled ΔANB, we have,

$BN = AB \sin\theta$

Or

$BN = a \sin\theta$

Suppose that the point P on the screen is at such a distance from the centre of the screen that $BN = \lambda$ and the angle $\theta = \theta_1$.

$\lambda = a \sin \theta_1$

Or

$$\sin\theta_1 = \frac{\lambda}{a}$$

Such a point on the screen will be the position of first secondary minimum.

For n^{th} minimum at point P,

$$\sin\theta_n = \frac{n\lambda}{a} \quad \ldots(1)$$

[1]

If y_n is the distance of the nth minimum from the centre of the screen and D is the distance between the slit and the screen, then from right angled ΔCOP,

We have,

$$\tan\theta_n = \frac{OP}{CO}$$

Or

$$\tan\theta_n = \frac{y_n}{D} \quad \ldots(2)$$

In case θ_n is small,

$\sin\theta_n \approx \tan\theta_n$

Therefore, from the equations (1) and (2), we have,

$$\frac{y_n}{D} = \frac{n\lambda}{a}$$

or $$y_n = \frac{nD\lambda}{a} \quad \ldots(3)$$

The width of the secondary maximum,

$$\beta = y_n - y_{n-1} = \frac{nD\lambda}{d} - \frac{(n-1)D\lambda}{a}$$

$$\beta = \frac{D\lambda}{a}$$

Since β is in dependent of n, all the secondary maxima are of same width β. The central maximum extends up to the distance y_1 (the distance of first secondary minimum) on both sides of the centre of the screen. **[1]**

Therefore, the width of the central maximum, $\beta_0 = 2y_1$

From equation (3), setting $n = 1$, we have,

$$y_1 = \frac{D\lambda}{a}$$

Therefore, $$\beta_0 = \frac{2D\lambda}{a} \quad \ldots(4)$$

It follows that $\beta_0 = 2\beta$, *i.e.*, the central maximum is twice as wide as any other secondary maximum or minimum. **[1]**

Q. 18. (a) In Young's double-slit experiment, two slits are 1 mm apart and the screen is placed 1 m away from the slits. Calculate the fringe width when light of wavelength 500 nm is used.

(b) What should be the width of each slit in order to obtain 10 maxima of the double slits pattern within the central maximum of the single-slit pattern? [CBSE Board, All India Region, 2016]

Ans. Suppose,

The width of each slit = a

Now we have a formula for the linear separation between m bright fringes in the double-slit experiment :

$$y_m = \frac{m\lambda D}{d}$$ **[1]**

Since $y << D$

The angular separation between m bright fringes will be :

$$\theta_m = \frac{y_m}{D} = \frac{m\lambda}{d}$$ **[1]**

Here we have $m = 10$

So $$\theta_{10} = 10\frac{\lambda}{d} \quad \ldots(i)$$

Again the angular width of the principal maximum in the diffraction pattern due to a slit of width 'a'

$$= \frac{2\lambda}{a} \quad \ldots(ii)$$

Now equating (i) and (ii),

$$\frac{10\lambda}{d} = \frac{2\lambda}{a}$$

Or $$a = \frac{d}{5} = \frac{1}{5} = 0.2 \text{ mm}$$ **[2]**

Q. 19. (i) State law of Malus.

(ii) Draw a graph showing the variation of intensity (I) of polarised light transmitted by an analyser with angle (θ) between polariser and analyser.

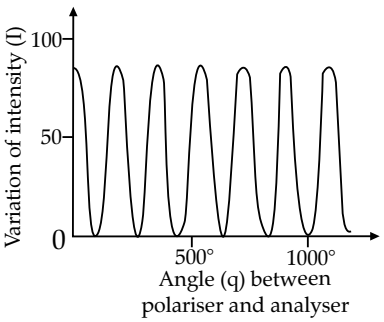
(iii) What is the value of refractive index of a medium of polarising angle 60°?

[CBSE Board, All India Region, 2016]

Ans. (i) Law of Malus : It states that when a completely plane polarised light beam is incident on an analyser, the intensity of the emergent light varies as the square of the cosine of the angle between the plane of transmission of the analyser and the polariser. **[1]**

$I = I_0\cos^2\theta$. **[1]**

(ii)



[1]

(iii) According to the Brewster law,

$$\mu = \tan i_p$$

$$\mu = \tan 60°$$

$$\mu = \sqrt{3}$$

The refractive index of the material is 1.73. **[1]**

Q. 20. Define the term wavefront. State Huygens' principle. Consider a plane wavefront incident on a thin convex lens. Draw a proper diagram

to show how the incident wavefront traverses through the lens and after refraction focuses on the focal point of the lens, giving the shape of the emergent wavefront.

[CBSE Board, All India Region, 2016]

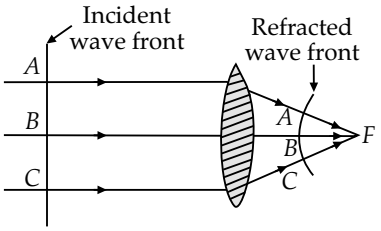
Ans. Wavefront is defined as the locus of all the points in space that reach a particular distance by a propagating wave at the same instant. **[½]**

Huygens' Principle is based on the following assumptions :

(1) Each point on the primary wavefront acts as a source of secondary wavelets, sending out disturbances in all directions in a similar manner as the original source of light does. **[1]**

(2) The new position of the wavefront at any instant (called secondary wave front) is the envelope of the secondary wavelets at that instant. **[1]**

(3) Incident wavefront traversing through the lens and after refraction focusing on the focal point of the lens is shown in the figure below. **[½]**



[1]

Q. 21. Explain the following, giving reasons :

(i) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency.

(ii) When light travels from a rarer to a denser medium, the speed decreases. Does this decrease in speed imply a reduction in the energy carried by the wave?

(iii) In the wave picture of light, intensity of light is determined by the square of the amplitude of the wave. What determines the intensity in the photon picture of light?

[CBSE Board, All India Region, 2016]

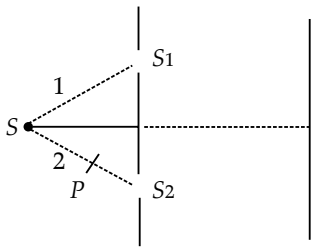
Ans. **(i)** Frequency, n and wavelength, λ of light are related as $\nu = \frac{v}{\lambda}$ where v is the velocity of light. As the wave travels from one medium to another, the wavelength of light changes changing the speed of light with it. The frequency remains constant. **[1]**

(ii) The energy of wave depends only on its frequency. Since the frequency does not change as wave travels from one medium to another, therefore, the energy of the wave remains same and does not decrease. **[1]**

(iii) In photon picture of light, the intensity of light is determined by the number of photons. **[1]**

Long Answer Type Questions **(5 marks each)**

Q. 1. Figure shows a two-slit arrangement with a source which emits unpolarised light. P is a polariser with axis whose direction is not given. If I_0 is the intensity of the principal maxima when no polariser is present, calculate in the present case, the intensity of the principal maxima as well as of the first minima.



[NCERT Exemp. Q. 10.19, Page 65]

Ans. The resultant amplitude of wave reaching on screen will be the sum of amplitude of either wave in perpendicular and parallel polarisation.

Amplitude of the wave in perpendicular polarisation,

$$A_\perp = A_\perp^1 + A_\perp^2 = A_\perp^0 \sin\,(kx-\omega t) + A_\perp^0 \sin\,(kx-\omega t+\phi)$$

$$= A_\perp = A_\perp(\sin\,(kx-\omega t) + \sin\,(kx-\omega t+\phi)) \quad \textbf{[1]}$$

Amplitude of the wave in parallel polarization,

$$A_\parallel = A_\parallel^{(1)} + A_\parallel^{(2)}$$

$$\Rightarrow \quad A_\parallel = A_\parallel^0\left[\sin\,(kx-\omega t) + \sin\,(kx-\omega t+\phi\right] \quad \textbf{[1]}$$

$\therefore$ Average intensity on the screen,

$$I = \left\{\left|A_\perp^0\right|^2 + \left|A_\parallel^0\right|^2\right\}[\sin^2\,(kx-\omega t)(1+\cos^2\phi + 2\sin\phi) +$$

$$\sin^2\,(kx-\omega t)\sin^2\phi]_{\text{average}}$$

$$= \left\{\left|A_\perp^0\right|^2 + \left|A_\parallel^0\right|^2\right\}\left(\frac{1}{2}\right)\cdot 2(1+\cos\phi)$$

$$\Rightarrow I = 2\left|A_\perp^0\right|^2(1+\cos\phi) \text{ since, } \left|A_\perp^0\right|_{av} = \left|A_\parallel^0\right|_{av} \quad \textbf{[1]}$$

With polariser P,

Assume $A_\perp^2$ is blocked.

$$\text{Intensity} = (A_\parallel^1 + A_\parallel^2)^2 + (A_\perp^1)^2$$

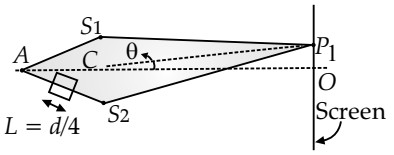
$$= \left|A_\perp^0\right|^2(1+\cos\phi) + \left|A_\perp^0\right|^2\cdot\frac{1}{2}$$

Given, $I_0 = 4\left|A_\perp^0\right|^2$ = Intensity without polariser at principal maxima.

Intensity at first minima with polariser,

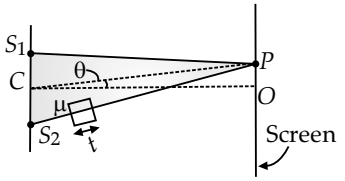
$$= \left|A_\perp^0\right|^2(1-1) + \frac{\left|A_\perp^0\right|^2}{2} = \frac{I_0}{8}. \quad \textbf{[2]}$$

Q. 2. A small transparent slab containing material of μ = 1.5 is placed along AS_2 in given Figure. What will be the distance from O of the principal maxima and of the first minima on either side of the principal maxima obtained in the absence of the glass slab.



[NCERT Exemp. Q. 10.20, Page 66]

Ans. When we introduce a thin transparent plate in front of one of the slits, the fringe pattern shifts towards the side where the plate is present.



$$(S_2O')_{new} = (S_2O' - t)_{air} + t_{plate}$$
$$= (S_2O' - t)_{air} + \mu t_{air}$$
$$(S_2O')_{new} = S_2O' + [\mu(t-1)]$$

Path difference,

$$\Delta x = (S_2O')_{new} - S_1O'$$
$$= (S_2O' - S_1O') + [\mu(t-1)]$$

$$\Rightarrow \quad \Delta x = d\sin\theta + [\mu(t-1)] \quad \ldots \text{(i)}$$

$$\Rightarrow \quad \text{Additional path difference} = (\mu - 1)t$$

Here the separation between the slits = $S_1S_2 = 2d$. Hence for calculating path difference, equation (i) becomes $\Delta x = 2d\sin\theta + [\mu(t-1)]$.

For the principal maxima,

$$\Delta x = 2d\sin\theta_0 + [\mu(t-1)] = 0 \quad \begin{bmatrix}\text{Path difference}\\ \text{is zero.}\end{bmatrix}$$

$$\sin\theta_0 = -\frac{L(\mu-1)}{2d} = \frac{-L(0.5)}{2d} \qquad [\because L = \frac{d}{4}]$$

or

$$\Rightarrow \sin\theta_0 = \frac{-1}{16}$$

θ_0 is the angular position corresponding to the principal maxima.

$$\Rightarrow \quad OP = D\tan\theta_0 \approx D\sin\theta_0 = \frac{-D}{16}$$

[2½]

For the first minima, the path difference is $\pm\frac{\lambda}{2}$.

$$\Delta x = 2d\sin\theta_1 + 0.5L = \pm\frac{\lambda}{2}$$

$$\sin\theta_1 = \frac{\pm\frac{\lambda}{2} - 0.5L}{2d} = \frac{\pm\frac{\lambda}{2} - \frac{d}{8}}{2d}$$

$$\Rightarrow \sin\theta_1 = \frac{\pm\frac{\lambda}{2} - \frac{\lambda}{8}}{2\lambda} = \pm\frac{1}{4} - \frac{1}{16}$$

[$\because$ The diffraction occurs if the wavelength of waves is nearly equal to the side width (d).]

On the positive side $\sin\theta_1^{+} = +\frac{1}{4} - \frac{1}{16} = \frac{3}{16}$

On the negative side $\sin\theta_1^{-} = \frac{1}{4} - \frac{1}{16} = -\frac{5}{16}$

The first principal maxima on the positive side is at distance,

$$D\tan\theta_1'^{+} = D\frac{\sin\theta_1'^{+}}{\sqrt{1-\sin^2\theta_1'}} = D\frac{3}{\sqrt{16^2-3^2}} = \frac{3D}{\sqrt{247}}$$

above point O

The first principal minima on the negative side is at distance,

$$D\tan\theta_1^{+} = \frac{5}{\sqrt{16^2-5^2}} = \frac{3D}{\sqrt{231}}$$ below point O. **[2½]**

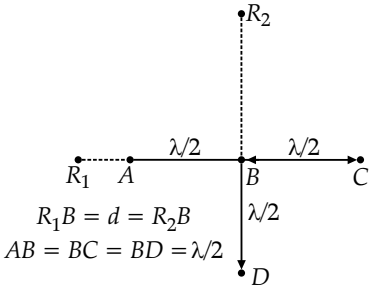
Q. 3. Four identical monochromatic sources *A*, *B*, *C*, and *D* as shown in the figure produce waves of the same wavelength λ and are coherent. Two receivers R_1 and R_2 are at great but equal distances from *B*.

(i) Which of the two receivers picks up the larger signal?

(ii) Which of the two receivers picks up the larger signal when *B* is turned off?

(iii) Which of the two receivers picks up the larger signal when is turned off?

(iv) Which of the two receivers can distinguish which of the sources *B* or *D* has been turned off?



[NCERT Exemp. Q. 10.21, Page 66]

Ans. Consider all the disturbances at R_1

Let the wave from source A has zero path difference at R_1.

$\therefore\ y_A = a\cos\omega t$ …(i)

Path difference between A and B is λ/2 or π so wave of B at R_1 is

$\therefore y_B = -a\cos\omega t$ …(ii)

Path difference between wave A and C is λ or 2π.

Thus, the wave equation at R_1, because of C is

$y_C = a\cos\omega t = a\cos(\omega t - 2\pi) = a\cos\omega t$ … (iii)

The path difference between the signals from D with that of A is :

$$\Delta x_{R_1} = \sqrt{d^2 + \left(\frac{\lambda}{2}\right)^2} - \left(d - \frac{\lambda}{2}\right)$$

$$= d\left(1 + \frac{\lambda^2}{4d^2}\right)^{\frac{1}{2}} - d + \frac{\lambda}{2}$$

$$= d\left(1 + \frac{\lambda^2}{8d^2}\right) - d + \frac{\lambda}{2} \approx \frac{\lambda}{2}\left(\text{neglecting } \frac{\lambda^2}{8d}(\because d >> \lambda)\right)$$

Therefore, phase difference is π.

$\therefore\ y_D = a\cos(\omega t - \pi) = -a\cos\omega t$ …(iv)

The resultant signal picked up at R_1, from all the four sources is the summation of all four waves,

$y_{R_1} = y_A + y_B + y_C + y_D$

$y_{R_1} = a\cos\omega t - a\cos\omega t + a\cos\omega t - a\cos\omega t = 0$

Thus, the signal picked up at R_1 is zero.

Now let us consider the resultant signal received at R_2. Let the equation of wave at R_2, because of B be,

$y_B = a_1 \cos \omega t$ …(v)

The path difference of the signal from D with that from B is $\frac{\lambda}{2}$ and hence, the phase difference,

$$\Delta\phi = \frac{2\pi}{\lambda} \times (\text{Path difference})$$

$$= \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

Thus, the wave equation at R_2, because of D is :

$y_B = a_1 \cos(\omega_1 - \pi) = -a_1 \cos \omega t$ …(vi)

The path difference between signal at A and that at B is :

$$\Delta x_{R_2} = \sqrt{(d)^2 + \left(\frac{\lambda}{2}\right)^2} - d$$

$$= d\left(1 - \frac{\lambda^2}{4d^2}\right)^{\frac{1}{2}} - d \simeq \frac{\lambda^2}{8d^2}$$

As $d >> \lambda$, therefore this path differences $\Delta x_{R_2} \to 0$ and phase difference,

$$\Delta\phi = \frac{2\pi}{\lambda} \times (\text{path difference})$$

$$= \frac{2\pi}{\lambda} \times 0 \to 0(\text{very small}) = \phi \text{ (say)}$$

Hence, $y_B = a_1 \cos(\omega_t - \phi)$

The resultant signal picked up at R_2, from all the four sources is the summation of all four waves,

$y_{R_2} = y_A + y_B + y_C + y_D$

$y_{R_2} = a_1 \cos \omega t - a_1 \cos \omega t + a_1 \cos\ (\omega t - \phi)$

$+ a_1 \cos\ (\omega t - \phi)$

$= 2a_1 \cos\ (\omega t - \phi)$

$\therefore$ Signal picked up by R_2 is $y_{R_2} = 2a_1 \cos\ (\omega t - \phi)$

$\therefore \left|v_{R_2}\right|^2 = 4a_1^2 \cos^2(\omega t - \phi) \to \left\langle I_{R_2} \right\rangle = 2a_1^2$

Thus, R_2 picks up the larger signal.

(ii) If B is switched off.

R_1 picks up $y = a \cos \omega t$

$$\therefore \left\langle I_{R_1} \right\rangle = a^2 \ <\cos^2 \omega t> = \frac{a^2}{2}$$

R_2 picks up y = $a \cos \omega t$

$$\therefore \left\langle I_{R_2} \right\rangle = a^2 \ <\cos^2 \omega t> = \frac{a^2}{2}$$

Thus, R_1 and R_2 pick up the same signal.

(iii) If D is switched off,

R_1 picks up $y = a \cos \omega t$

$$\therefore \left\langle I_{R_1} \right\rangle = \frac{1}{2}a^2$$

R_2 picks up $y = 3a \cos \omega t$

$$\therefore \left\langle I_{R_2} \right\rangle = 9a^2 \ <\cos^2 \omega t> = \frac{9a^2}{2}$$

Thus, R_2 picks up larger signal compared to R_1.

(iv) Thus, a signal at R_1 indicates B has been switched off and an enhanced signal at R_2 indicates D has been switched off. **[5]**

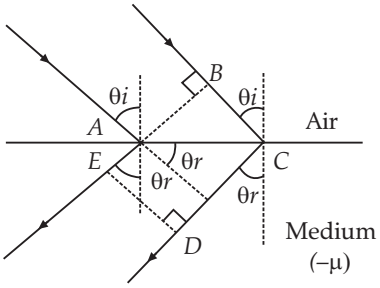
Q. 4. The optical properties of a medium are governed by the relative permittivity ε_r and relative permeability (μ_r). The refractive index is defined as $\sqrt{\varepsilon_r \mu_r} = \mu$. For ordinary material $\varepsilon_r > 0$ and $\mu_r > 0$ and the positive sign is taken for the square root. In 1964, a Russian Scientist V. Veselago postulated the existence of material with $\varepsilon_r < 0$ and $\mu_r < 0$. Since then such 'metamaterials' have been produced in the laboratories and their optical properties studied. For such materials $\mu = -\sqrt{\varepsilon_r \mu_r}$. As light enters a medium of such refractive index the phases travel away from the direction of propagation.

(i) According to the description above show that if rays of light enter such a medium from air (refractive index = 1) at an angle θ in second quadrant, then the refracted beam is in the third quadrant.

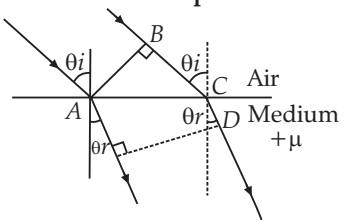
(ii) Prove that Snell's law holds for such a medium.

[NCERT Exemp. Q. 10.22, Page 67]

Ans. **(i)** Let the postulates $\mu = -\sqrt{\varepsilon_r \mu_r}$ and $\mu = +\sqrt{\varepsilon_r \mu_r}$ are true then two parallel rays would proceed as shown in figure (a) and figure (b), respectively.



(a) meta-materials with positive refractive index



(b) Ordinary-materials with positive refractive index

[1]

Let two parallel rays at incidence angle θ_i from air would proceed in medium as shown in figures above. ED shows a wavefront, then all the points on ED will remain in same phase. All the points with the same optical path length must have the same phase.

So $-\sqrt{\mu_r \varepsilon_r} AE = BC - \sqrt{\mu_r \varepsilon_r} CD$

$$BC = \sqrt{\mu_r \varepsilon_r}(CD - AE)$$

$BC > 0; CD > AE$

As showing that the postulate is reasonable if however, the light proceeded in the sense it does for ordinary material, *e.g.*, refracted rays are in IVth quadrant in figure (b) then

$-\sqrt{\varepsilon_r \mu_r} AE = BC - \sqrt{\mu_r \varepsilon_r} CD$

$BC = \sqrt{\mu_r \varepsilon_r}(CD - AE)$

$AE > CD$ hence $BC < 0$ **[2]**

Figure showing that $BC < 0$, this is not possible. Hence the given postulate is correct.

(ii) From fig. (a)

$BC = AC \sin\theta_i$

$CD\text{-}AE = AC \sin \theta_r$

$BC = \sqrt{\mu_r \varepsilon_r}$ (From Figure, $CD - AE = BC$)

$AC \sin \theta i = \sqrt{\mu_r \varepsilon_r}\ AC \sin \theta_r$

$$\frac{\sin\theta_i}{\sin\theta_r} = \sqrt{\mu_r \varepsilon_r}$$

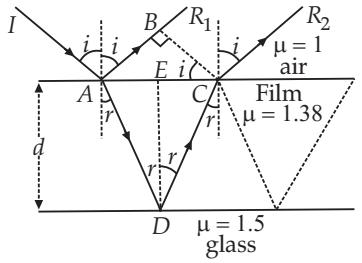
This proves the Snell's law. **[2]**

Q. 5. To ensure almost 100% transitivity, photographic lenses are often coated with a thin layer of dielectric material. The refractive index of this material is intermediated between that of air and glass (which makes the optical element of the lens). A typically used dielectric film is MgF_2 (n = 1.38). What should the thickness of the film be so that at the centre of the visible spectrum (5,500 Å) there is maximum transmission.

[NCERT Exemp. Q. 10.23, Page 67]

Ans. In the given figure, incidence ray I_A incident at point A from air to film surface with incident angle i. here at A, it gets partial reflection and refraction, passes through paths AR_1 and AD, respectively. At D it again gets partial reflection (and refractions) from the glass and film interface surface. At C the interface surface of film and air and finally after refraction from C pass through path CR_2 parallel to AR_1. **[1]**

The amplitude (intensity) of wave during refraction and reflection decreases. If the interference due to two reflected rays AR_1 and CR_2 is destructive interference, then the reflected rays AR, and CR_2 will not be dominant.



[1]

Both reflections are from lower to higher refractive index surfaces. So, optical path difference between AR_1 and CR_2 will be :

$\mu(AD + CD) - AB$ ...(I)

If d is the thickness of film then,

$$AD = AC = \frac{d}{\cos r} \quad \text{...(II)}$$

$AB = AC \sin i$

$$\text{or } \tan r = \frac{\frac{AC}{2}}{d}$$

$$\therefore\ d\tan r = \frac{AC}{2}$$

$AC = 2d \tan r$

$AB = 2d \tan r \sin i$ ...(III)

So the optical path difference from (I)

$= \mu AD + \mu AD - AB$ $(\because\ AD = CD)$

$= 2\mu AD - AB$

$= \dfrac{2\mu d}{\cos r} - 2d\tan r\ \sin i$ (From II and III)

$= 2 \cdot \dfrac{\sin i}{\sin r} \cdot \dfrac{d}{\cos r} - 2d \cdot \dfrac{\sin r}{\cos r} \cdot \sin i$

$= \dfrac{2d\sin i}{\cos r}\left[\dfrac{1}{\sin r} - \dfrac{\sin r}{1}\right]$

$= \dfrac{2d\sin i(-\sin^2 r)}{\cos r \sin r}$

$= \dfrac{2d\sin i\cos^2 r}{\sin r\cos r}$

Optical path difference between AR_1 and AR_2

$= 2d\mu \cdot \cos r$ $\left(\because \mu = \dfrac{\sin i}{\sin r}\right)$

For two rays AR_1 and CR_2 to interfere destructively, path difference should be $\dfrac{\lambda}{2}$.

$$\therefore 2d\mu\cos r = \frac{\lambda}{2}$$

$$\mu d\cos r = \frac{\lambda}{4}$$

For photographic lenses the sources are vertical planes, *i.e.*, rays incident at very small angle.

So $i = r \approx o$

$$\therefore \mu d \approx \frac{\lambda}{4} \qquad (\because \cos\ 0 = 1)$$

$$d = \frac{\lambda}{4\mu} = \frac{5{,}500\ \text{Å}}{1.38 \times 4}$$

$$d = 1{,}000\ \text{Å}$$

[3]

Q. 6. Answer the following questions :

(a) In a single-slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band?

(b) In what way is diffraction from each slit related to the interference pattern in a double-slit experiment?

(c) When a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. Explain why?

(d) Two students are separated by a 7 m partition wall in a room 10 m high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other even though they can converse easily.

(e) Ray optics is based on the assumption that light travels in a straight line. Diffraction effects (observed when light propagates through small apertures/slits or around small obstacles) disprove this assumption. Yet the ray optics assumption is so commonly used in understanding location and several other properties of images in optical instruments.
What is the justification?

[NCERT Add. Ex. Q. 10.17, Page 84]

Ans. **(a)** In a single-slit diffraction experiment, if the width of the slit is made double the original width, then the size of the central diffraction band reduces to half and the intensity of the central diffraction band increases up to four times. **[1]**

(b) The interference pattern in a double-slit experiment is modulated by diffraction from each slit. The pattern is the result of the interference of the diffracted wave from each slit. **[1]**

(c) When a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. This is because light waves are diffracted from the edge of the circular obstacle, which interferes constructively at the centre of the shadow. This constructive interference produces a bright spot. **[1]**

(d) Bending of waves by obstacles by a large angle is possible when the size of the obstacle is comparable to the wavelength of the waves. On the one hand, the wavelength of the light waves is too small in comparison to the size of the obstacle. Thus, the diffraction angle will be very small. Hence, the students are unable to see each other. On the other hand, the size of the wall is comparable to the wavelength of the sound waves. Thus, the bending of the waves takes place at a large angle. Hence, the students are able to hear each other. **[1]**

(e) The justification is that in ordinary optical instruments, the size of the aperture involved is much larger than the wavelength of the light used. **[1]**

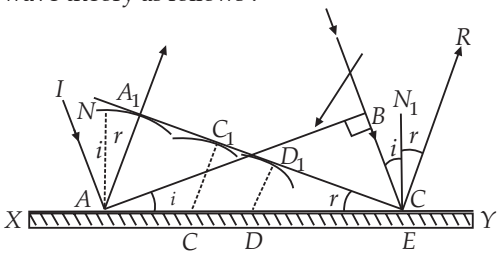
Q. 7. (a) Define wavefront. Use Huygens' principle to verify the laws of refraction.

(b) How is linearly polarised light obtained by the process of scattering of light? Find the Brewster angle for air-glass interface, when the refractive index of glass = 1·5.

[CBSE Board, All India Region, 2017]

Ans. **(a) Wavefront :** A wavefront is an imaginary surface over which an optical wave has a constant phase. The shape of a wavefront is usually determined by the geometry of the source.

The laws of reflection on the basis of Huygens' wave theory as follows :



In the given figure, AB is the wavefront incident on a reflecting surface XY with an angle of incidence i as shown in figure. According to Huygens' principle, every point on AB acts as a source of secondary wavelets. At first, wave incidents at point A, and then to points C, D and E. They form a sphere of radii AA_1, CC_1 and DD_1 as shown in figure.

A_1E represents the tangential envelop of secondary wavelet in forward direction.

In ΔABE and ΔAA_1E,

$\angle ABE = \angle AA_1E = 90°$

Side AE = Side AE

$AA_1 = BE$ = Distance travelled by wave in same time

So, these triangles are congruent.

So, $\angle BAE = i$ and $\angle BEA = r$

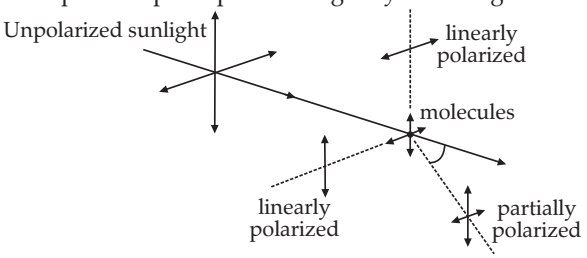
Thus, $i = r$

Hence, angle of incidence is equal to the angle of reflection. This is the **First Law of Reflection**.

Since, the incident wavefront AB, normal and reflected wavefront A_1E lies in same plane, it verifies the **Second Law of Reflection**.

In this way, Laws of Reflection are verified by Huygens' principle. **[2½]**

(b) When a beam of light passes through a medium, it gets scattered constituting the medium, provided the size of the particles is of the order of the wavelength of the light. The scattered light, viewed in a direction perpendicular to the direction of the beam of light, is found to be plane polarised. It gives us a method to produce plane polarised light by scattering.



When the light is incident at the polarising angle at the interface of a refracting medium, the refractive index of the medium is equal to the tangent of the polarising angle.

$\mu = \tan p$

Here,

$\mu \rightarrow$ Refractive index of the refracting medium

$p \rightarrow$ Polarising angle

Given, $\mu = 1.5$

Therefore,

$\tan p = 1.5$

$p = \tan^{-1}(1.5)$

$p = 56.31°$ **[2½]**

Q. 8. (i) State the essential conditions for diffraction of light.

(ii) Explain diffraction of light due to a narrow single slit and the formation of pattern of fringes on the screen.

(iii) Find the relation for width of central maximum in terms of wavelength 'λ', width of slit 'a', and separation between slit and screen 'D'.

(iv) If the width of the slit is made double the original width, how does it affect the size and intensity of the central band?

[CBSE Board, Foreign Scheme, 2016]

Ans. **(i)** The essential conditions for diffraction of light are given as :

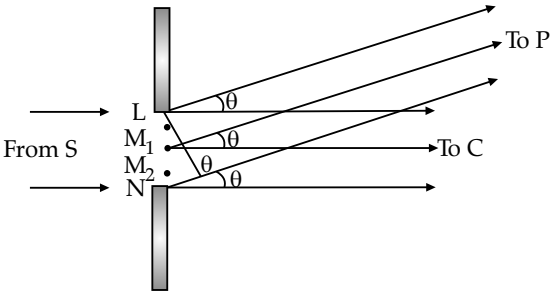
(a) Size of slit (diffraction effect depends upon the size of obstacle)

(b) Aperture must be smaller than of the order of wavelength of light (diffraction of light takes place if the size of obstacle is comparable to the wavelength of light). **[1]**

(ii) Single-slit diffraction is explained by treating different parts of the wavefront at the slit as sources of secondary wavelets. At the central point C on the screen, θ is zero. All path differences are zero and hence all the parts of the slit contribute in phase and give maximum intensity at C.

At any other point P, the path difference between two edges of the slit is given as :

$NP - LP = NQ = a\sin\theta \approx a\theta$



Any point P, in direction θ, is a location of minima if $a\theta = n\lambda$

This can be explained by dividing the slit into even number of parts. The path differences between waves from successive parts are 180° out of phase, and hence, cancel each other leading to minima.

Any point P, in direction Q, is a location of maxima if

$$a\theta = \left(n + \frac{1}{2}\right)\lambda$$

This can be explained by dividing the slit into odd number of parts. The contributions from successive parts cancel in pairs because of 180° phase difference. The unpaired part produces intensity at P, leading to a maxima. **[2]**

(iii) If θ is the direction of first minima, then

$$a\theta = \lambda \Rightarrow \theta = \frac{\lambda}{a}$$

Angular width of central maxima, $2\theta = \frac{2\lambda}{a}$

Linear width of central maxima, $\beta = 2\theta.\mathrm{D} = \frac{2\lambda D}{a}$ **[1]**

(iv) Let the width of the slit be 'a' when this is doubled then the half angular width of central maximum which is $\frac{\lambda}{a}$, reduces to half. Hence the intensity of central maximum will become four times, because the area of central diffraction band would be one-fourth. **[1]**

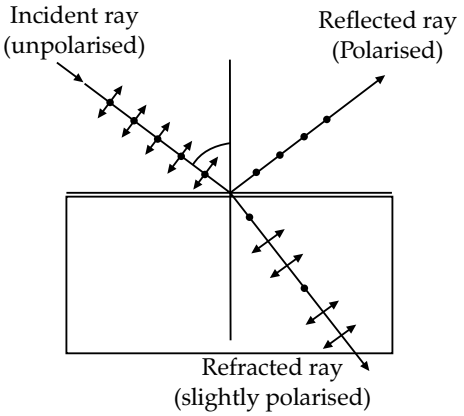
Q. 9. (a) Explain why the intensity of light coming out of a polaroid does not change irrespective of the orientation of the pass axis of the polaroid.

(b) State, using a proper diagram, the condition when unpolarised light incident on the boundary between two transparent media produces polarised light. Explain briefly.

Hence show that the angle of incidence 'i_β' is related to the refractive index μ by the relation, $\mu = \tan i_\beta$. [CBSE Board, All India Region, 2016]

Ans. (a) When unpolarised light passes through a polariser, vibrations perpendicular to the axis of the polaroid are blocked. Unpolarised light has vibrations in all directions. Hence, for all positions of polaroid, half of the incident light always get transmitted. Hence, the intensity of the light does not change. **[1]**

(b) In unpolarised light, the light vibrates in all possible planes but for polarised light the vibrations occur in a single plane. The reflected wave will be totally polarised if reflected wave and refracted wave are perpendicular to each other. **[1]**



[1]

If the incident angle is i_p and refracted angle 'r' then,

$i_\beta + 90° + r = 180°$

$i_\beta + 90° + r = 180°$

$\Rightarrow \quad r = 90° - i_\beta$

Then according to Snell's law

$$\mu = \frac{\sin i_p}{\sin r}$$

$$\Rightarrow \quad \mu = \frac{\sin i_\beta}{\sin(90° - i_\beta)}$$

$$\Rightarrow \quad \mu = \frac{\sin i_\beta}{\cos i_\beta}$$

$$\Rightarrow \quad \mu = \tan i_\beta$$

where μ is the refracted index of the transparent medium. This is called Brewster's law and i_p is called the Brewster angle. **[2]**

Some Commonly Made Errors

- Generally, students confuse between Snell's law and Snell's Law using Huygens' Principle.
- Don't take right incident angle or reflected angle/refracted angle.
- Students draw wrong diagram of Fringes and Intensity Pattern.
- Generally, students confuse in Phase Difference and Path Difference.

EXPERT ADVICE

- ☞ *Strategize and select the appropriate equations from relative topics.*
- ☞ *Be careful for subscript and superscript of numerical value.*
- ☞ *Learn all the universal constant values for making solution easy.*
- ☞ *Learn all the topics on conceptual basis, e.g., Huygens' Principle; Young's Experiment.*
- ☞ *Before doing any derivation, please make sure to draw the diagram for your convenience.*

OSWAAL LEARNING TOOLS

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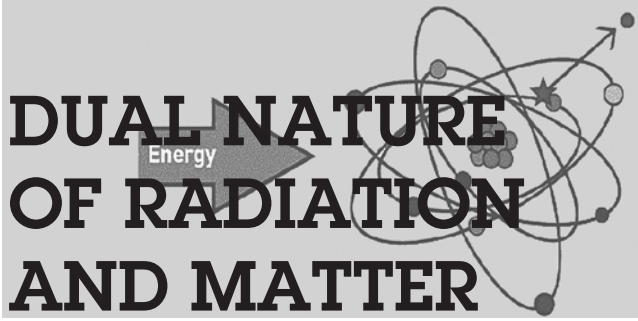
CHAPTER

11

DUAL NATURE OF RADIATION AND MATTER



Chapter Objectives

This chapter will help you understand :

- ***Dual Nature of Radiation and Matter :*** *Electron emission; Photoelectric effect; Photoelectric effect and wave theory of light; Einstein's photoelectric equation; Particle nature of light; Wave nature of matter; Davisson and Germer experiment.*

Quick Review

- The photoelectric effect was discovered in 1887 by Heinrich Hertz and explained by Albert Einstein in work for which he received the Nobel Prize.
- The energies of photons range from high-energy gamma rays and X-rays to low-energy infrared and radio waves, though all travel at the speed of light.
- In 1900, Max Planck, a German theoretical physicist, found that heat radiation is emitted and absorbed in distinct units, which he called quanta. He received the 1918 Nobel Prize in physics.
- Electromagnetic radiation can be described in terms of a stream of photons.
- There are three types of photons : shadow photons, global photons, and caustic photons.
- Gilbert Lewis coined the term photon in 1926.
- In the 1800s, the wave properties of light became glaringly obvious and scientists essentially threw the particle theory of light out the window. It wasn't until Albert Einstein explained the photoelectric effect and realised that light energy had to be quantised that the particle theory returned.
- The photon torpedo of Star Trek is a self-propelled missile consisting of a small quantity of matter and antimatter bound together in a magnetic container.

TIPS...

- *Try to solve objective problems to make yourself efficient in solving 1-mark question.*
- *Practice the emission and other important diagrams carefully.*
- *Revise the radiation of photoelectrons concepts regularly.*
- *Do not forget to mention the units if any of all physical entities in radiation.*

TRICKS...

- *You must first clear the concepts related to radiation topics.*
- *You should prepare your all formulas and values to solve the problems.*
- *You should learn techniques to minimise the calculation steps by relating one to another formula.*

Know the Links

- *https://profmattstrassler.com/articles...matter.../matter-and-energy-a-false-dichotomy/*
- *www.fizika.unios.hr/of4/wp.../sites/47/.../quantum-physics-2nd-Eisberg_Resnick.pdf*
- *www.opjsrgh.in/Content/2017/DUAL_NATURE_OF_MATTER_RADIATION.pdf*

Multiple Choice Questions

(1 mark each)

Q. 1. A particle is dropped from a height H. The de-Broglie wavelength of the particle as a function of height is proportional to

(a) H **(b)** $H^{1/2}$

(c) H^0 **(d)** $H^{-1/2}$

[NCERT Exemp. Q. 11.1, Page 68]

Ans. Correct option : (d)

Explanation : Velocity v, of freely falling body after falling from a height H, will be :

$H = v = \sqrt{2gH}$

We know that de-Broglie wavelength $\lambda = \frac{h}{p}$

$$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{2gH}}$$

h, m, and g are constant

$$\therefore \frac{h}{m\sqrt{2g}} \text{ is constant} \Rightarrow \lambda \propto \frac{1}{\sqrt{H}} \Rightarrow \lambda \propto H^{-1/2}$$

Q. 2. The wavelength of a photon needed to remove a proton from a nucleus which is bound to the nucleus with 1 MeV energy is nearly

(a) 1.2 nm **(b) 1.2×10^{-3} nm**

(c) 1.2×10^{-6} nm **(d) 1.2×10 nm**

[NCERT Exemp. Q. 11.2, Page 68]

Ans. Correct option : (b)

Explanation : Energy of the photon must be equal to the binding energy of proton

So, energy of photon= 1 MeV= $10^6 \times 1.6 \times 10^{-19}$ J

$$\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-13}} = \frac{6.63 \times 3}{1.60} \times 10^{-26+13}$$

$$= \frac{19.89}{1.60} \times 10^{-13} = 12.4 \times 10^{-13} = 1.24 \times 10^1 \times 10^{-13}$$

$$= 1.24 \times 10^{-9} \times 10^{-3} = 1.24 \times 10^{-3} \text{ nm}$$

Q. 3. Consider a beam of electrons (each electron with energy E_0) incident on a metal surface kept in an evacuated chamber. Then,

(a) no electrons will be emitted as only photons can emit electrons

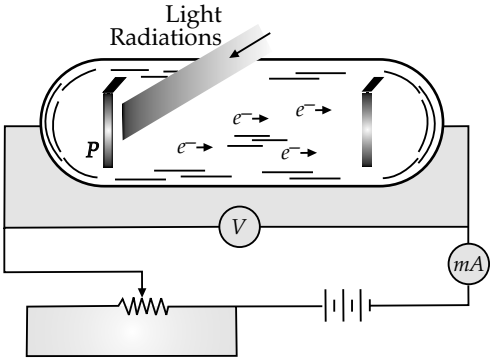
(b) electrons can be emitted but all with an energy, E_0

(c) electrons can be emitted with any energy, with a maximum of $E_0 - \phi$ (ϕ is the work function)

(d) electron can be omitted with energy, with a maximum of E_0 **[NCERT Exemp. Q. 11.3, Page 68]**

Ans. Correct option : (d)

Explanation : If a beam of electrons of having energy E_0 is incident on a metal surface kept in an evacuated chamber.



The electrons can be emitted with maximum energy E_0 (due to elastic collision) and with any energy less than E_0, when part of incident energy of electron is used in liberating the electrons from the surface of metal.

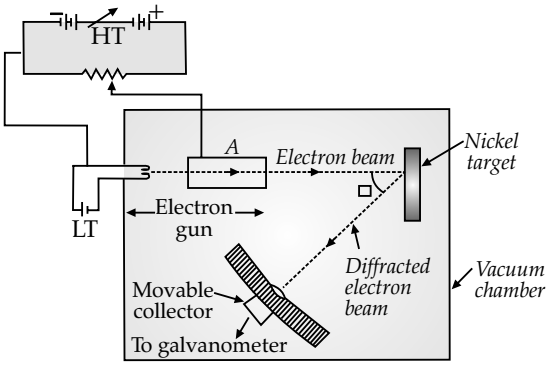
Q. 4. Consider the figure given below. Suppose the voltage applied to A is increased. The diffracted beam will have the maximum at a value of θ that

(a) will be larger than the earlier value

(b) will be the same as the earlier value

(c) will be less than the earlier value

(d) will depend on the target.



[NCERT Exemp. Q. 11.4, Page 69]

Ans. Correct option : (c)

Explanation : Davisson and Germer experiment

In Davisson-Germer experiment, the de-Broglie wavelength of diffracted beam of electrons

$$\lambda_d = \frac{12.27}{\sqrt{V}} \text{ Å} \quad \text{...(i)}$$

V is the applied voltage. If there is a maxima of the diffracted electrons at an angle θ, then

$$2d \sin\theta = \lambda \quad \text{...(ii)}$$

From equation (i), as the applied voltage in this experiment increases, the wave length λ_d decreases in turn sinθ or θ decreases by relation (ii). Hence, verifies the option (c).

Q. 5. A proton, a neutron, an electron and an α-particle have same energy. Then, their de-Broglie wavelengths compare as

(a) $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$ **(b) $\lambda_\alpha < \lambda_p = \lambda_n > \lambda_e$**

(c) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$ **(d) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$**

[NCERT Exemp. Q. 11.5, Page 69]

Ans. Correct option : (b)

Explanation : Matter waves (de-Broglie waves)

According to de-Broglie, a moving material particle sometimes acts as a wave and sometimes as a particle.

De- Broglie wavelength $\lambda_d = \frac{h}{p}$

$$E_p = E_n = E_e = E_x$$

$$K.E. = K = \frac{1}{2}mv^2$$

$$2K = mv^2$$

$$2Km = m^2v^2$$

$$2mK = p^2$$

$$\sqrt{2mK} = p$$

$$\therefore \quad \lambda_d = \frac{h}{p}$$

$$\lambda_d = \frac{h}{\sqrt{2mK}}$$

or $\lambda_d = \frac{h}{\sqrt{2mK}}$ [as h and E (K.E.) is constt.]

$$\therefore \lambda \propto \frac{1}{\sqrt{m}}$$

$$m_a > m_p = m_n > m_e$$

$$\therefore \lambda_a < \lambda_p = \lambda_n < \lambda_e$$

Q. 6. An electron is moving with an initial velocity $v = v_o \hat{i}$ and is in a magnetic field $B = B_o \hat{j}$. Then, its de-Broglie wavelength

(a) remains constant.

(b) increases with time.

(c) decreases with time.

(d) increases and decreases periodically.

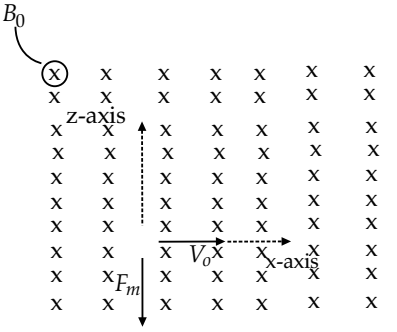
[NCERT Exemp. Q. 11.6, Page 69]

Ans. Correct option : (a)

Explanation : Given, $v = v_o \hat{i}$ and $B = B_o \hat{j}$

Magnetic force on moving electron

$= -e[v_o i \times B_o j] = -ev_o B_o k$



So, the force is perpendicular to v and B both as the force is perpendicular to the velocity. So, the magnitude will not change v or mv ($p = mv$ momentum) so, the de-Broglie wavelength remains same.

Q. 7. An electron (mass m) with an initial velocity $v = v_o \hat{i}$ ($v_0 > 0$) is in an electric field $E = -E_0 \hat{i}$ (E_0 = constant > 0). Its de-Broglie wavelength at time t is given by

(a) $\dfrac{\lambda_0}{\left[1 + \dfrac{eE_0}{m}\dfrac{t}{v_0}\right]}$

(b) $\lambda_0\left[1 + \dfrac{eE_0 t}{mv_0}\right]$

(c) λ_0

(d) $\lambda_0 t$ **[NCERT Exemp. Q. 11.7, Page 69]**

Ans. Correct option : (a)

Explanation : The wave associated with moving particle is called matter wave or de-Broglie wave and it propagates in the form of wave packets with group velocity. According to de-Broglie theory, the wavelength of de- Broglie wave is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

As initial velocity of the electron is $v_0 \hat{i}$, the initial de-Broglie wavelength of electron,

$$\lambda = \frac{h}{mv_0} \quad ...(i)$$

Electrostatic force on electron in electric field is,

$$\vec{F}_e = -e\vec{E} = -e\left[-E_0 \hat{i}\right] = eEz_0 \hat{i}$$

Acceleration of electron, $\vec{a} = \dfrac{\vec{F}}{m} = \dfrac{eE_0 \hat{j}}{m}$

Velocity of the electron after time t,

$$\vec{v} = v_0 \hat{i} + \left(\frac{eE_0 \hat{i}}{m}\right)t = \left(v_0 + \frac{eE_0}{m}t\right)\hat{i}$$

$$\Rightarrow \quad \vec{v} = v_0\left(1 + \frac{eE_0}{mv_0}\right)\hat{i}$$

de-Broglie wavelength associated with electron at time t is $\lambda = \dfrac{h}{mv}$

$$\Rightarrow \quad \lambda = \frac{h}{m\left[v_0\left(1 + \dfrac{eE_0}{mv_0}t\right)\right]} = \frac{\dfrac{h}{mv_0}}{\left(1 + \dfrac{eE_0}{mv_0}t\right)}$$

$$\Rightarrow \quad \lambda = \frac{\lambda_0}{\left[1 + \dfrac{eE_0}{mv_0}t\right]} \text{ As } \lambda_0 = \frac{h}{mv_0}$$

Q. 8. An electron (mass m) with an initial velocity $v = v_0 \hat{i}$ is in an electric field $E = E_0 \hat{j}$. If $\lambda_0 = \dfrac{h}{mv_0}$, its de-Broglie wavelength at time t is given by

(a) λ_0 **(b)** $\lambda_0\sqrt{1 + \dfrac{e^2E_0^2t^2}{m^2v_0^2}}$

(c) $\dfrac{\lambda_0}{\sqrt{1 + \dfrac{e^2E_0^2t^2}{m^2v_0^2}}}$ **(d)** $\dfrac{\lambda_0}{\left(1 + \dfrac{e^2E_0^2t^2}{m^2v_0^2}\right)}$

[NCERT Exemp. Q. 11.8, Page 70]

Ans. Correct option : (c)

Explanation : According to the problem de-Broglie wavelength of electron at time t=0 is $\lambda_0 = \dfrac{h}{mv_0}$

Electrostatic force on electron in electric field is

$$\vec{F}_e = -e\vec{E} = -eE_0 \hat{j}$$

The acceleration of electron, $\vec{a} = \dfrac{\vec{F}}{m} = -\dfrac{eE_0}{m}\hat{j}$

It is acting along negative y-axis.

The initial velocity of electron along x-axis, $v_{x_0} = v_o \hat{i}$

This component of velocity will remain constant as there is no force on electron in this direction. Now considering y-direction. Initial velocity of electron along y-axis, $v_{y0} = 0$

Velocity of electron after time t along y-axis

$$v_y = 0 + \left(-\frac{eE_0}{m}J\right)t = -\frac{eE_0}{m}t\hat{j}$$

Magnitude of velocity of electron after time t is

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{v_0^2 + \left(\frac{-eE_0}{m}t\right)^2}$$

$$\Rightarrow \qquad = v_0\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}$$

de-Broglie wavelength, $\lambda_0 = \frac{h}{mv_0}$

$$= \frac{h}{mv_0\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}} = \frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}}$$

$$\Rightarrow \qquad \lambda' = \frac{\lambda_0}{\left(1+\frac{e^2E_0^2t^2}{m^2v_0^2}\right)}$$

Tick Two or More Options

Q. 9. Relativistic corrections become necessary when the expression for the kinetic energy 1/2 mv^2, becomes comparable with mc^2. where m is the mass of the particle. At what de-Broglie wavelength, will relativistic corrections become important for an electron?

(a) λ=10 nm **(b)** $\lambda = 10^{-1}$ nm

(c) $\lambda = 10^{-4}$ nm **(d)** $\lambda = 10^{-6}$ nm

[NCERT Exemp. Q. 11.9, Page 70]

Ans. Correct options : (c) and (d)

Explanation : De-Broglie or matter wave is independent of the charge on the material particle. It means, matter wave of de-Broglie wave is associated with every moving particle (whether charged or uncharged). The de-Broglie wavelength at which relativistic corrections become important that the phase velocity of the matter waves can be greater than the speed of the light (3×10^8 m/s).

The wavelength of de-Broglie wave is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Here, $h = 6.6 \times 10^{-34}$ Js and for electron, $m = 9 \times 10^{-31}$ kg

To approach these types of problem we use hit and trial method by picking up each option one by one.

In option (a), λ_1=10 nm=10×10^{-9} m=10^{-8} m

$$\Rightarrow \qquad v_1 = \frac{6.6\times10^{-34}}{\left(9\times10^{-31}\right)\times10^{-8}}$$

$$= \frac{2.2}{3}\times10^5 \approx 10^5 \text{ m/s}$$

In option (b), $\lambda_2 = 10^{-1}$ nm = $10^{-1}\times10^{-9}$ m = 10^{-10} m

$$\Rightarrow \qquad v_2 = \frac{6.6\times10^{-34}}{\left(9\times10^{-31}\right)\times10^{-10}} \approx 10^7 \text{ m/s}$$

In option (c), $\lambda_3 = 10^{-4}$ nm = $10^{-4}\times10^{-9}$ m = 10^{-13} m

$$\Rightarrow \qquad v_3 = \frac{6.6\times10^{-34}}{\left(9\times10^{-31}\right)\times10^{-13}} \approx 10^{10} \text{ m/s}$$

In option (d), $\lambda_4 = 10^{-6}$ nm = $10^{-6}\times10^{-9}$ m = 10^{-15} m

$$\Rightarrow \qquad v_4 = \frac{6.6\times10^{-34}}{\left(9\times10^{-31}\right)\times10^{-15}} \approx 10^{12} \text{ m/s}$$

Thus, option (c) and (d) are correct as v_3 and v_4 is greater than 3 x 10^8 m/s.

Q. 10. Two particles A_1 and A_2 of masses m_1, m_2 ($m_1 > m_2$) have the same de-Broglie wavelength. Then,

(a) their momenta are the same.

(b) their energies are the same.

(c) energy of A_1 is less than the energy of A_2.

(d) energy of A_1 is more than the energy of A_2.

[NCERT Exemp. Q. 11.10, Page 70]

Ans. Correct options : (a) and (c)

Explanation :

We know the de-Broglie wavelength $\lambda = \frac{h}{mv}$ where, $mv = p$ (momentum) of the particle

$\Rightarrow$ But we can express wavelength $\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda}$

Hence, $p \propto \frac{1}{\lambda} \Rightarrow \frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1}$

But particles have the same de-Broglie wavelength. ($\lambda_1 = \lambda_2 = \lambda$).

Then. $\frac{P_1}{P_2} = \frac{\lambda}{\lambda} = 1 \Rightarrow p_1 = p_2$

Thus, their moment is same.

Also. $E = \frac{1}{2}mv^2 = \frac{1}{2}mv^2\frac{m}{m}$

$$= \frac{1}{2}\frac{m^2v^2}{m} = \frac{1}{2}\frac{p^2}{m}$$

As p is constant, $E \propto \frac{1}{m}$

$\therefore \frac{E_1}{E_2} = \frac{m_2}{m_1} < 1 \Rightarrow E_1 < E_2$

Q. 11. De-Broglie wavelength associated with uncharged particles : For Neutron De-Broglie wavelength is given as $v_e = \frac{c}{100}$. Then

(a) $\frac{E_e}{E_p} = 10^{-4}$

(b) $\frac{E_e}{E_p} = 10^{-2}$

(c) $\frac{P_e}{m_e c} = 10^{-2}$

(d) $\frac{P_e}{m_e c} = 10^{-4}$

[NCERT Exemp. Q. 11.11, Page 70]

Ans. Correct options : (b) and (c)

Explanation :

$$\text{De-Broglie wavelength}, \lambda = \frac{h}{p}$$

$$\therefore \lambda_e = \frac{h}{m_e v_e} = \frac{h}{m_e \frac{c}{100}}$$

$$= \frac{100h}{m_e c} \quad \text{...(i)}$$

$$\text{Now } K.E. = \frac{1}{2}mv^2 \Rightarrow E = \frac{m^2v^2}{2m} \quad \text{...(ii)}$$

$$\Rightarrow \ \mathrm{m}_e v_e = \sqrt{2m_e E_e}$$

$$\text{Now } \lambda_e = \frac{h}{m_e v_e} = \frac{h}{\sqrt{2m_e E_e}} \quad [\text{from (ii)}]$$

$$\Rightarrow \quad E_e = \frac{h^2}{2m_e \lambda_e^{\,2}} \quad \text{...(iii)}$$

$$\text{For proton } \lambda_p = 2\lambda_e [\text{given}]$$

$$\therefore \quad \mathrm{E}_p = \frac{hc}{\lambda_p} = \frac{hc}{2\lambda_e}$$

$$\text{Now}, \frac{E_p}{E_e} = \frac{hc}{2\lambda_e} \times \frac{2m_e\lambda_e^{\,2}}{h^2} = \frac{m_e \lambda_e c}{h} \quad [\text{from (iii)}]$$

$$= \frac{100h}{m_e c} \times \frac{m_e c}{h} = 100$$

$$\Rightarrow \quad \frac{E_e}{E_p} = 10^{-2} [\text{verifies Ans.(b)}]$$

$$\text{Now}, p_e = m_e v_e = m_e \times \frac{c}{100} \left(\because \mathrm{v}_e = \frac{c}{100} \right)$$

$$\Rightarrow \quad \frac{p_e}{m_e c} = \frac{1}{100} = 10^{-2} [\text{verifies Ans.(c)}].$$

Q. 12. Photons absorbed in a matter are converted to heat. A source emitting *n* photon/sec of frequency ν is used to convert 1 kg of ice at 0°C to water at 0°C. Then, the time *T* taken for the conversion

(a) decreases with increasing *n*, with ν fixed.

(b) decreases with n fixed, ν increasing.

(c) remains constant with *n* and ν changing such that *n*ν = constant.

(d) increases when the product *n*ν increases.

[NCERT Exemp. Q. 11.12, Page 71]

Ans. Correct options : (a), (b) and (c)

Explanation : We know that heat spent to convert ice into water

$$E_{absorb} = \text{mass} \times \text{latent heat}$$

$$\therefore \quad E = mL = (1000' \text{ g}) \times (80 \text{ cal/g})$$

$$E = 80000 \text{ cal}$$

Energy of photons used $= nT \times E = nT \times h\nu \ [\because E = h\nu]$

$$\text{So, } nTh\nu = mL \Rightarrow T = \frac{mL}{nh\nu}$$

$$\therefore \quad T \propto \frac{1}{n} \text{ when } \nu \text{ is constant.}$$

Thus, time taken for conversion decreases with increasing *n*, with *v* kept constant.

$T \propto \frac{1}{\nu}$ when *n* remains constant.

Thus, time taken for conversion decreases with increasing ν, with *n* kept constant.

$$\Rightarrow T \propto \frac{1}{n\nu}$$

Thus, time taken for conversion decreases with increase in product of *nv* and *T* is constant, if *nv* is constant.

Q. 13. A particle moves in a closed orbit around the origin, due to a force which is directed towards the origin. The de-Broglie wavelength of the particle varies cyclically between two values λ_1, λ_2 with $\lambda_1 > \lambda_2$ Which of the following statements are true?

(a) The particle could be moving in a circular orbit with origin as centre.

(b) The particle could be moving in an elliptic orbit with origin as its focus.

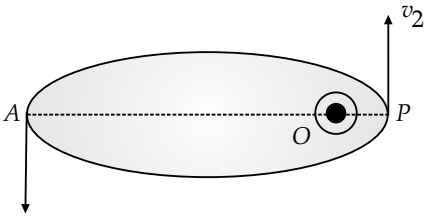
(c) When the de-Broglie wavelength is λ_1 the particle is nearer the origin than when its value is λ_2.

(d) When the de-Broglie wavelength is λ_2 the particle is nearer the origin than when its value is λ_1.

[NCERT Exemp. Q. 11.13, Page 71]

Ans. Correct options : (b) and (d)

Explanation : As the de-Broglie wavelength of particle varies cyclically between two values λ_1 and λ_2. It is possible when the particle is moving in elliptical orbit with origin as one of its focus. Because if λ_1 and λ_2 are equal, their speed must be equal and particle must move in circular orbit. Hence, it verifies answer (b).



Let v_1 and v_2 be the speeds of particle at *A* and *B*, respectively and origin is at *O*. if λ_1 and λ_2 are the de- Broglie wavelengths associated with particle at *A* and *B*, respectively, then

$$\lambda_1 = \frac{h}{mv_1} \text{ and } \lambda_2 = \frac{h}{mv_2}$$

$$\text{So } \frac{\lambda_1}{\lambda_2} = \frac{v_2}{v_1} \qquad [\because \lambda_1 > \lambda_2] \text{ (Given)}$$

$$\text{So } v_1 < v_2 \qquad \text{...(I)}$$

By the law of conservation of angular momentum, the speed of the particle will be more when it is closer to focus. It is verified by I.

So, the object is close to B and A or the particle is nearer to the origin when wavelength is λ_2 than when wavelength is λ_1. Hence, verifies answer (d).

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. A proton and an a-particle are accelerated, using the same potential difference. How are the de-Broglie wavelengths λ_p and λ_α related to each other? [NCERT Exemp. Q. 11.14, Page 71]

Ans. As we know that both the particles are accelerated at the same potential difference, so their K.Es. will be equal, that is, [½]

$$K_1 = K_2 = K = qV \qquad (\because V = \frac{W}{q}) \quad [½]$$

$$\text{So, } \lambda = \frac{h}{\sqrt{2mK}} \text{ or } \lambda = \frac{h}{\sqrt{2mqV}}$$

$$\therefore \frac{\lambda_p}{\lambda_\alpha} = \frac{h}{\sqrt{2m_p q_p V_p}} \times \frac{\sqrt{2m_\alpha q_\alpha V_\alpha}}{h} \quad [½]$$

$$m_\alpha = 4m_p; q_\alpha = 2e;\ q_p = e;\ V_p = V_\alpha = V$$

(applied potential difference)

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{2 \times 4m_p 2eV}{2mpeV}}$$

$$\lambda_p = \sqrt{8}\lambda_\alpha \quad [½]$$

So, the de-Broglie wavelength of proton $\sqrt{8}$ times of alpha (α) particle.

Q. 2. (i) In the explanation of photoelectric effect, we assume one photon of frequency ν collides with an electron and transfers its energy. This leads to the equation for the maximum energy E_{max} of the emitted electron as $E_{max} = h\nu - \phi_0$ where ϕ_0 is the work function of the metal. If an electron absorbs 2 photons (each of frequency ν), what will be the maximum energy for the emitted electron?

(ii) Why is this fact (two photon absorption) not taken into consideration in our discussion of the stopping potential?

[NCERT Exemp. Q. 11.15, Page 71]

Ans. **(i)** Here, 2 photons transfer its energy to one electron as $E = h\nu$

According to the question, an electron absorbs the energy of two photons each with frequency ν then $\nu' = 2\nu$ where ν' is the frequency of emitted electron.

$$\therefore \quad E_e = E_p$$

$$h\nu_e = 2h\nu$$

$$\therefore \quad \nu_e = 2\nu \quad [½]$$

Maximum energy of emitted electron is

$$E_{max} = h\nu_e - \phi_0 = h(2\nu) - \phi_0 = 2h\nu - \phi_0 \quad [½]$$

(ii) The probability of absorbing two photons by the same electron is very low. Hence, such emission will be negligible. **[1]**

Q. 3. There are materials which absorb photons of shorter wavelength and emit photons of longer wavelength. Can there be stable substances which absorb photons of larger wavelength and emit light of shorter wavelength?

[NCERT Exemp. Q. 11.16, Page 72]

Ans. We know that as the wavelength of photon increases, its frequency decreases or energy increases.

(i) When the materials which absorb photons of shorter wavelength has the energy of the incident photon on the material is high and the energy of emitted photon is low when it has a longer wavelength or in short we can say that energy given out is less than the energy supplied. **[1]**

(ii) The energy of the incident photon is low for the substances which has to absorb photons of larger wavelength and energy of emitted photon is high to emit light of shorter wavelength. This means in this statement material has to supply the energy for the emission of photons. But this is not possible for a stable substance. **[1]**

Q. 4. Do all the electrons that absorb a photon come out as photo electrons?

[NCERT Exemp. Q. 11.17, Page 72]

Ans. **Photoelectric effect :** The photoelectric effect is the emission of electrons called photoelectron when light strikes a surface. To escape from the surface, the electron must absorb enough energy from the incident radiation to overcome the attraction of positive ions in the material of the surface. **[1]**

The photoelectric effect is based on the **principle of conservation of energy**.

(i) Two conducting electrodes, the anode (Q) and cathode (P) are enclosed in an evacuated glass tube.

(ii) The battery or other source of potential difference creates an electric field in the direction from anode to cathode.

(iii) Light of certain wavelength or frequency falling on the surface of cathode causes a current in the external circuit called photoelectric current.

(iv) As potential difference increases, photoelectric current also increases till saturation is reached. **[1]**

Q. 5. There are two sources of light, each emitting with a power of 100 W. One emits X-rays of wavelength 1 nm and the other visible light at 500 nm. Find the ratio of number of photons of X-rays to the photons of visible light of the given wavelength.

[NCERT Exemp. Q. 11.18, Page 72]

Ans. There are three essential requirements for the production of X-rays.

(i) A source of electron

(ii) An arrangement to accelerate the electrons

(iii) A target of suitable material of high atomic weight and high melting point on which these high speed electrons strike. **[1]**

Let E_x and E_v are the energies given by one photon in X-rays and visible rays then,

$E_x = h\nu_x$ and $E_v = h\nu_v$

Let n_x and n_v are number of photons X-rays and visible light to give equal energies as both sources (X-rays and visible) emitting same power 100 W each so,

$$n_x h\nu_x = n_v h\nu_v$$

$$\frac{n_x}{n_v} = \frac{\nu_v}{\nu_x} = \frac{\frac{1}{\lambda_v}}{\frac{1}{\lambda_x}} = \frac{\lambda_x}{\lambda_v}\left(\because \nu = \frac{c}{\lambda} or\ \nu \propto \frac{1}{\lambda}\right)$$

$$\frac{n_x}{n_v} = \frac{1\text{nm}}{500\text{ nm}}\left(\because \lambda_x = 1\text{ nm} \text{ and } \lambda_v = 500\text{ nm}\right) \text{ [Given]}$$

$$n_x : n_v = 1 : 500$$

[1]

Q. 6. When is H_α line in the emission spectrum of hydrogen atom obtained?
Calculate the frequency of the photon emitted during this transition.
[CBSE Board, Delhi Region, 2016]

Ans. The H_α line in the emission spectrum of the hydrogen atom is obtained when the electron transition takes place between $n = 2$ to $n = 1$. **[½]**

We know that the energy level in nth level is

$$E_n = \frac{-13.6}{n^2}$$

$$\therefore E_2 - E_1 = \frac{-13.6}{2^2} - \frac{-13.6}{1^2}$$

$$\therefore E_2 - E_1 = \frac{-13.6}{2^2} + 13.6$$

$$\therefore E_2 - E_1 = 13.6\left(-\frac{1}{4} + 1\right) = 13.6 \times \frac{3}{4} = 10.2 \text{ eV}$$

$$\therefore E_2 - E_1 = 10.2 \times 16 \times 10^{-19} \text{ J}$$

$$\therefore E_2 - E_1 = 16.32 \times 10^{-19} \text{ J}$$

Now, this difference in energy gives the energy of the photon emitted

$$\therefore E_2 - E_1 = h\nu$$

$$\therefore \upsilon = \frac{E_2 - E_1}{h} = \frac{16.32 \times 10^{-19}}{6.36 \times 10^{-34}}$$

$$\therefore \upsilon = 2.46 \times 10^{15} \text{ Hz}$$ **[1½]**

Q. 7. Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from $n = \infty$ to n = 1. [CBSE Board, Delhi Region, 2016]

Ans. The electron in the hydrogen atom jumps from $n_2 = \infty$ to $n_1 = 1$.

Now, we know that

$$\frac{1}{\lambda} = \text{R}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

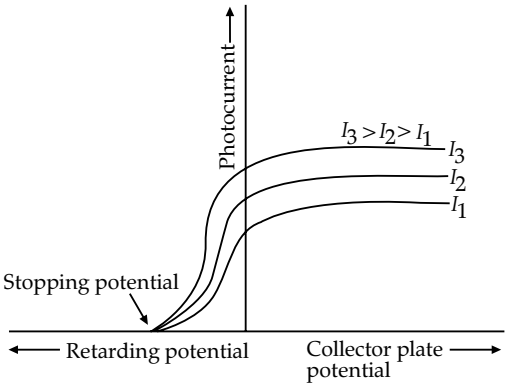
$$\therefore \frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{1}{1} - \frac{1}{\infty}\right) = 1.097 \times 10^7$$

$$\therefore \lambda = \frac{1}{1.097 \times 10^7} = 9.116 \times 10^{-8} \text{ m} = 911.6 \times 10^{-10} \text{ m}$$

$$\therefore \lambda = 911.6 \text{ Å}$$ **[2]**

Q. 8. Draw graphs showing variation of photoelectric current with applied voltage for two incident radiations of equal frequency and different intensities. Mark the graph for the radiation of higher intensity.[CBSE Board, Delhi Region, 2018]

Ans. Graph showing variations of photoelectric current with applied voltage for two incident radiations of equal frequency and different intensities : **[½]**



[½]

The stopping potential is defined as the potential necessary to stop any electron (or, in other words, to stop even the electron with the most kinetic energy) from reaching the other side.

Q. 9. If light of wavelength 412.5 nm is incident on each of the metals given below, which ones will show photoelectric emission and why?

Metal	Work Function
Na	1.92
K	2.15
Ca	3.20
Mo	4.17

[CBSE Board, Delhi Region, 2018]

Ans. A metal will show photoelectric emission in below situation :

E ≥ work function

Now according to the relation below :

$$\text{E} = h\left(\frac{c}{\lambda}\right)$$

$$\text{E} = \frac{1240}{\lambda}$$

$$\lambda = \frac{1240}{412.5} \text{ nm}$$

λ = 3.006 eV

So here, work function of *k* and Na are less than incident energy of photon. Hence, only potassium (K) and sodium (Na) will show photoelectric emission. **[2]**

Q. 10. Name the electromagnetic radiations use for
(a) water purification, and
(b) eye surgery. [CBSE Board, Delhi Region, 2018]

Ans. **(a)** Water purification : Ultraviolet radiation **[1]**

(b) Eye surgery : Infrared radiation **[1]**

Q. 11. An electron jumps from fourth to first orbit in an atom. How many maximum number of spectral lines can be emitted by the atom? To which series these lines correspond?
[CBSE Board, Foreign Scheme, 2016]

Ans. You can directly use the formula :

$$\text{Number of lines} = \frac{n(n-1)}{2}$$

Instead of looking at how each electron can make the jump(s) through levels, let's look at how an

electron in a given state can make a single jump. This gives us, from your example,

(i) For an electron in the 4th level, we have 3 single jump possibility, that is,

$4 \rightarrow 3$

$4 \rightarrow 2$

$4 \rightarrow 1$

(ii) Similarly,

$3 \rightarrow 2$

$3 \rightarrow 1$

(iii) And finally,

$2 \rightarrow 1$

That gives us a total of $3 + 2 + 1 = 6$.

If you look at, let's say 6 levels, you'll see that the number of possible lines will be of the $5 + \ldots + 1 = 15$. So, for a jump from kth state, the trend can be simplified as $\sum_{n=1}^{k-1} n$ **[1]**

Q. 12. The wavelength λ of a photon and the de-Broglie wavelength of an electron have the same value. Show that energy of a photon is $\left(2\lambda\frac{mc}{h}\right)$ times the kinetic energy of electron; where *m*, *c* and *h* have their usual meaning.

[CBSE Board, Foreign Scheme, 2016]

Ans. Kinetic energy of an electron is $E_e = \frac{p^2}{2m} = \frac{h^2}{2\lambda^2 m}$

Energy of a photon is $E_p = \frac{hc}{\lambda}$

So, $\frac{E_p}{E_e} = \frac{\frac{hc}{\lambda}}{\frac{h^2}{2m\lambda^2}} = \frac{2m\lambda c}{h}$ (proved) **[2]**

Q. 13. Calculate the de-Broglie wave length of the electron orbiting in the *n* = 2 state of hydrogen atom. [CBSE Board, Delhi Region, 2016]

Ans. Given,

$n=2$

K.E for the second state, E_k.

$\frac{13.6}{n^2} = \frac{13.6}{4}$

$= 3.4 \times 1.6 \times 10^{-19}$ J

De-Broglie wavelength, $\lambda = \frac{h}{\sqrt{2mE_k}}$

$= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 3.4 \times 1.6 \times 10^{-19}}}$

$= 0.067$ m **[2]**

Q. 14. Define ionization energy. How would the ionisation energy change when electron in hydrogen atom is replaced by a particle of mass 200 times that of electron but having the same charge?

[CBSE Board, All India Regions, 2016]

Ans. Ionisation energy : The minimum amount of energy required to remove an electron from the outermost orbit of a neutral atom in its ground state is known as the ionisation energy of that atom. **[1]**

The ionisation energy of hydrogen atom is

$E_{H'} \propto m_e$ ...(1)

When mass of electron is replaced by a particle having mass = $200m_e$, then

$E_{H'} \propto 200\ m_E$ ...(2)

From (1) and (2), we have

$E_{H'} = 200\ E_H = 200 \times (-13.6)$

$= -2720$ eV ($\because E_H = -13.6$ eV) **[1]**

Q. 15. Calculate the shortest wavelength of the spectral lines emitted in Balmer series.

[Given Rydberg constant, R = 10^7 m $^{-1}$]

[CBSE Board, All India Regions, 2016]

Ans. For Balmer series, wavelength is given as

$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$

For shortest wavelength, $n = \infty$

$\therefore \frac{1}{\lambda} = R\left(\frac{1}{2^2} - 0\right) = \frac{R}{4}$

$\Rightarrow \lambda = \frac{4}{R} = 4 \times 10^{-7}$ m **[2]**

Q. 16. A proton and an α particle are accelerated through the same potential difference. Which one of the two has,

(i) greater de-Broglie wavelength, and (ii) less kinetic energy? Justify your answer.

[CBSE Board, Delhi Region, 2016]

Ans. (i) The de-Broglie wavelength is related to the accelerating potential as

$\lambda = \frac{h}{\sqrt{2m_e V}}$

$\therefore \quad \lambda_p = \frac{h}{\sqrt{2m_p e_p V}}$

And, $\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha e_\alpha V}}$

$\therefore \quad \frac{\lambda_p}{\lambda_\alpha} = \frac{h}{\sqrt{2m_p e_p V}} \times \frac{\sqrt{2m_\alpha e_\alpha V}}{h} = \frac{\sqrt{m_\alpha e_\alpha}}{\sqrt{m_p e_p}}$ **[1]**

Now, mass and charge of α- particular is greater than that of proton.

$\therefore \lambda_p > \lambda_\alpha$ for the same potential difference.

(ii) The kinetic energy is related to accelerating potential as K.E. = eV

$\therefore$ K.E.$_p = e_p$V

And, K.E.$_\alpha = e_\alpha$V

$\therefore \frac{\text{K.E.}_p}{\text{K.E.}_\alpha} = \frac{e_p}{e_\alpha}$

Since, the charge on proton is less than that on alpha particle, K.E.$_p$ < K.E.$_\alpha$ for the same potential difference. **[1]**

Q. 17. Name the phenomenon which shows the quantum nature of electromagnetic radiation.

[CBSE Board, All India Region, 2017]

Ans. Photoelectric effect shows the quantum nature of electromagnetic radiation. **[1]**

Q. 18. A photosensitive surface emits photoelectrons when red light falls on it. Will the surface emit photoelectrons when blue light is incident on it? Give reason.

[CBSE Board, Foreign Scheme, 2017]

Ans. Photoemission takes place when red light is incident means

$$\frac{hc}{\lambda_{red}} > \text{Work function} \quad \text{...(i)}$$ **[½]**

When blue colour light is incident then energy imparted to the electron is $E = \frac{hc}{\lambda_{blue}}$

as $\lambda_{red} > \lambda_{blue}$

so $\frac{hc}{\lambda_{blue}} > \frac{hc}{\lambda_{red}}$ ...(ii) **[½]**

from relation (1) and (2)

$$\frac{hc}{\lambda_{blue}} > W$$

So, photoelectron will be emitted when blue light is incident.

Q. 19. The photoelectric cut-off voltage in a certain experiment is 1.5 V. What is the maximum kinetic energy of photoelectrons emitted?

[NCERT Ex. Q. 11.3, Page 407]

Ans. Photoelectric cut-off voltage, $V_0 = 1.5$ V

The maximum kinetic energy of the emitted photoelectrons is given as :

$K_e = eV_o$

Where,

e = Charge on an electron = 1.6×10^{-19} C

$\therefore K_e = 1.6 \times 10^{-19} \times 1.5 = 2.4 \times 10^{-19}$ J

Therefore, the maximum kinetic energy of the photoelectrons emitted in the given experiment is 2.4×10^{-19} J. **[2]**

Q. 20. The energy flux of sunlight reaching the surface of the earth is 1.388×10^3 W/m^2. How many photons (nearly) per square metre are incident on the Earth per second? Assume that the photons in the sunlight have an average wavelength of 550 nm. [NCERT Ex. Q. 11.5, Page 407]

Ans. Energy flux of sunlight reaching the surface of earth, $\phi = 1.388 \times 10^3$ W/m^2

Hence, power of sunlight per square metre, $P = 1.388 \times 10^3$ W

Speed of light, $c = 3 \times 10^8$ m/s

Planck's constant, $h = 6.626 \times 10^{-34}$ Js

Average wavelength of photons present in sunlight, $\lambda = 550$ nm $= 550 \times 10^{-9}$ m

Number of photons per square metre incident on earth per second = n

Hence, the equation for power can be written as : $P = nE$

$$\therefore \quad n = \frac{P}{E} = \frac{P\lambda}{hc} = \frac{1.388 \times 10^3 \times 550 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8}$$
$$= 3.84 \times 10^{21} \text{ photons/m}^2\text{s}^{-1}$$

Therefore, every second, 3.84×10^{21} photons are incident per square metre on earth. **[2]**

Q. 21. In an experiment on photoelectric effect, the slope of the cut-off voltage versus frequency of incident light is found to be 4.12×10^{-15} V s. Calculate the value of Planck's constant.

[NCERT Ex. Q. 11.6, Page 407]

Ans. The slope of the cut-off voltage (V) versus frequency (ν) of an incident light is given as :

$$\frac{V}{v} = 4.12 \times 10^{-15} \text{ Vs}$$

V is related to frequency by the equation : $h\nu = eV$

Where,

e = Charge on an electron = 1.6×10^{-19} C

h = Planck's constant

$$\therefore h = \frac{eV}{} = 1.6 \times 10^{-19} \times 4.12 \times 10^{-15} = 6.592 \times 10^{-34} \text{ Js}$$

Therefore, the value of Planck's constant is 6.592×10^{-34} Js. **[2]**

Q. 22. Answer the following questions :

(a) Quarks inside protons and neutrons are thought to carry fractional charges $\left[\left(+\frac{2}{3}\right)e; \left(-\frac{1}{3}\right)e\right]$. Why do they not show up in Millikan's oil-drop experiment?

(b) What is so special about the combination e/m? Why do we not simply talk of e and m separately?

(c) Why should gases be insulators at ordinary pressures and start conducting at very low pressures?

(d) Every metal has a definite work function. Why do all photoelectrons not come out with the same energy if incident radiation is monochromatic? Why is there an energy distribution of photoelectrons?

(e) The energy and momentum of an electron are related to the frequency and wavelength of the associated matter wave by the relations :

$E = h\nu, p = \frac{h}{\lambda}$

But while the value of λ is physically significant, the value of ν (and therefore, the value of the phase speed νλ) has no physical significance. Why? [NCERT Add Ex. Q. 11.37, Page 411]

Ans. (a) Quarks inside protons and neutrons carry fractional charges. This is because nuclear force increases extremely if they are pulled apart. Therefore, fractional charges may exist in nature; observable charges are still the integral multiple of an electrical charge. **[2]**

(b) The basic relations for electric field and magnetic field are

$\left(eV = \frac{1}{2}mv^2\right)$ and $\left(eBv = \frac{mv^2}{r}\right)$ respectively.

These relations include e (electric charge), v (velocity), m (mass), V (potential), r(radius), and B (magnetic field). These relations give the value of velocity of an electron as $\left[v = \sqrt{2V\left(\frac{e}{m}\right)}\right]$ and $\left[v = Br\left(\frac{e}{m}\right)\right]$ respectcively.

It can be observed from these relations that the dynamics of an electron is determined

not by e and m separately, but by the ratio $\frac{e}{m}$. **[2]**

(c) At atmospheric pressure, the ions of gases have no chance of reaching them
respective electrons because of collision and recombination with other gas molecules.
Hence, gases are insulators at atmospheric pressure. At low pressures, ions have a
chance of reaching their respective electrodes and constitute a current. Hence, they
conduct electricity at these pressures. **[2]**

(d) The work function of a metal is the minimum energy required for a conduction
electron to get out of the metal surface. All the electrons in an atom do not have the
same energy level. When a ray having some photon energy is incident on a metal surface, the electrons come out from different levels with different energies. Hence,
these emitted electrons show different energy distributions. **[2]**

(e) The absolute value of energy of a particle is arbitrary within the additive constant.
Hence, wavelength (λ) is significant, but the frequency (ν) associated with an
electron has no direct physical significance.
Therefore, the product νλ (phase speed) has no physical significance.
Group speed is given as :

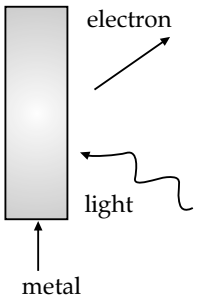
$$v_g = \frac{dv}{dk}$$

$$= \frac{dv}{d\left(\frac{1}{\lambda}\right)} = \frac{dE}{dp} = \frac{d\left(\frac{p^2}{2m}\right)}{dp} = \frac{p}{m}$$

This quantity has a physical meaning. **[2]**

Short Answer Type Questions **(3 or 4 marks each)**

Q. 1. Consider Figure for photo-emission. How would you reconcile with momentum conservation? Note light (photons) have momentum in a different direction than the emitted electrons.



[NCERT Exemp. Q. 11.19, Page 72]

Ans. When photons strike the metal surface, photons transfer their momentum to atoms of metal by decreasing its own speed up to zero. This momentum of photons transfers to nucleus and electrons of the metal. At microscopic level, atoms of a metal absorb the photon and its momentum is transferred mainly to the nucleus and electrons.
The excited electrons emit approximately opposite to the direction of photons. But total momentum transferred by the photons will be equal to the momentum of all electrons and nucleus. **[3]**

Q. 2. Consider a metal exposed to light of wavelength 600 nm. The maximum energy of electron doubles when light of wavelength 400 nm is used. Find the work function in eV?

[NCERT Exemp. Q. 11.20, Page 72]

Ans. Work function (or threshold energy) (W_o) : The minimum energy of incident radiation required to eject the electrons from metallic surface is defined as work function of that surface. **[1]**

$W_0 = h\nu_0 = \frac{hc}{\lambda_0}$ Joules; ν_0 = Threshold frequency;

λ_0 = Threshold wavelength

Work function in electron volt, $W_0(\text{eV}) = \frac{hc}{e\lambda_0} = \frac{12375}{\lambda_0^0(A)}$

Einstein's photoelectric equation is $E = W_0 + K_{max}$
Maximum energy $= h\nu - \phi$ **[1]**
According to the problem for first condition wavelength of light λ = 600 nm and for the second condition, wavelength of light λ′ = 400 nm.
Also, maximum kinetic energy for the second condition is equal to the twice of the kinetic energy in the first condition. **[1]**

that is, $K'_{max} = 2K_{max}$

then $K'_{max} = \frac{hc}{\lambda} - \phi$

$\Rightarrow \quad 2K_{max} = \frac{hc}{\lambda'} - \phi_0$

$\Rightarrow \quad 2\left(\frac{1230}{600} - \phi\right) = \left(\frac{1230}{400} - \phi\right)$ [$\because hc \simeq 1240$ eV nm]

$\Rightarrow \quad \phi = \frac{1230}{1230} = 1.02$ eV **[1]**

Q. 3. Assuming an electron is confined to a 1 nm wide region, find the uncertainty in momentum using Heisenberg uncertainty principle ($\Delta x \times \Delta p = h$). You can assume the uncertainty in position Δx as 1 nm. Assuming $p = \Delta p$, find the energy of the electron in electron volts.

[NCERT Exemp. Q. 11.21, Page 72]

Ans. Given that,
$\Delta x = 1$ nm $= 10^{-9}$ m, we have to find Δp. As we know
$\Delta x \,.\, \Delta p = h$

Therefore, $\Delta p = \frac{h}{\Delta x} = \frac{h}{2\pi \Delta x}$

$\Rightarrow \quad \frac{6.62 \times 10^{-34} \text{ Js}}{2 \times (22/7)(10^{-9}) \text{ m}}$

$= 1.05 \times 10^{-25} \text{ kg m/s}$

Energy, $E = \frac{p^2}{2m} = \frac{(\Delta p)^2}{2m}$ $[\because\ p \approx \Delta p]$

$= \frac{\left(1.05 \times 10^{-25}\right)^2}{2 \times 9.1 \times 10^{-31}}$ J

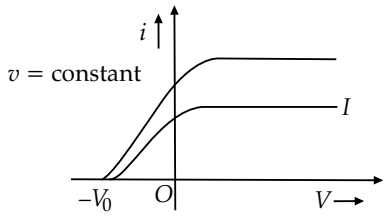
$= \frac{(1.05 \times 10^{-25})^2}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}}$ eV

$= 3.8 \times 10^{-2}$ eV **[3]**

Q. 4. Two monochromatic beams *A* and *B* of equal intensity *I*, hit a screen. The number of photons hitting the screen by beam A is twice that by beam *B*. Then, what inference can you make about their frequencies? [NCERT Exemp. Q. 11.22, Page 72]

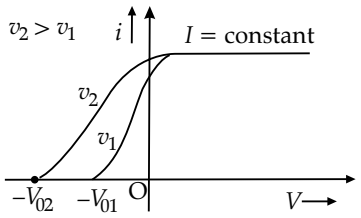
Ans. **Effect of intensity :** If the intensity of light is incensed (while its frequency is kept the same) the entrance levels off at a higher value, showing that more electrons are being emitted per unit time but the stopping potential V_0. doesn't change, that is, **[1]**

Intensity ∝ at number of incident photon ∝ number of emitted photoelectron per time ∝ photo current.



[½]

Effect of frequency : If frequency of incident light increases, (keeping intensity constant) stopping potential increases but there is no change in photoelectric current.



[½]

Let us assume n_A is the number of photons falling per second of beam *A* and n_B is the number of photons falling per second of beam *B*.

And it is given that the number of photons hitting the screen by beam A is twice that by beam *B*. $n_A = 2n_B$

Energy of falling photon of beam $A = h\nu_A$

Energy of falling photon of beam $B = h\nu_B$ **[1]**

Now, according to question, intensity of A is equal to intensity of B

Therefore, $I = n_A \nu_A = n_B \nu_B$

$\Rightarrow \quad \frac{\nu_A}{\nu_B} = \frac{n_B}{n_A} = \frac{n_B}{2n_B} = \frac{1}{2}$

$\Rightarrow \quad \nu_B = 2\nu_A$ **[1]**

Q. 5. Two particles *A* and *B* of de-Broglie wavelengths *A*, and combine to form a particle *C*. The process conserves momentum. Find the de-Broglie wavelength of the particle *C*. (The motion is one-dimensional). [NCERT Exemp. Q. 11.23, Page 73]

Ans. De-Broglie wavelengths

$\lambda = \frac{h}{p}$ or $p = \frac{h}{\lambda} \Rightarrow p_1 = \frac{h}{\lambda_1}$, $p_2 = \frac{h}{\lambda_2}$ and $p_3 = \frac{h}{\lambda_3}$

$p_1 + p_2 = p_3$

$\frac{h}{\lambda_1} + \frac{h}{\lambda_2} = \frac{h}{\lambda_3}$ (λ_3 = the wavelength of particle C)

$\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$

$\frac{\lambda_2 + \lambda_1}{\lambda_1 \lambda_2} = \frac{1}{\lambda_3}$ **[½]**

Case I : When p_1 and p_2 are positive then

$\lambda_3 = \frac{\lambda_1 \lambda_1}{\lambda_1 + \lambda_2}$ **[½]**

Case II : When p_1 and p_2 both are negative

$\lambda_3 = \left|\frac{-\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}\right| = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ **[1]**

Case III : $p_A > 0$, $p_B < 0$ or

$\frac{h}{\lambda_3} = \frac{h}{\lambda_1} - \frac{h}{\lambda_2}$ or $\frac{1}{\lambda_3} = \frac{\lambda_2 - \lambda_1}{\lambda_1 \lambda_2}$

$\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1}$ **[1]**

Case IV : $p_A < 0$, $p_B > 0$

$\frac{h}{\lambda_3} = \frac{h}{\lambda_2} = \frac{h}{\lambda_1} = \frac{(\lambda_1 - \lambda_2)h}{\lambda_1 \lambda_2}$

$\therefore\ \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$ **[1]**

Q. 6. A neutron beam of energy *E* scatters from atoms on a surface with a spacing $d = 0.1$ nm. The first maximum intensity in the reflected beam occurs at $\theta = 30°$. What is the kinetic energy *E* of the beam in eV? [NCERT Exemp. Q. 11.24, Page 73]

Ans. By Bragg's law of diffraction, condition for nth maxima is

$2d \sin\theta = n\lambda$

$n = 1$ so $\lambda = 2\text{d} \sin\theta$ [$\theta = 30°$(Given)]

$= 2 \times 0.1 \times 10^{-9} \sin 30°$ ($\because\ d = 0.1$ nm)

$p = \frac{h}{\lambda} = \frac{6.6 \times 10^{-34}}{10^{-10}} = 6.6 \times 10^{-24}$ kg m/s

$E = \frac{1}{2} mv^2 = \frac{1}{2} \frac{m^2 v^2}{m} = \frac{p^2}{2m}$

$\therefore\ E = \frac{6.6 \times 6.6 \times 10^{-24} \times 10^{-24}}{2 \times 1.6 \times 10^{-27}}$ J

$= \frac{6.6 \times 6.6 \times 10^{-48}}{2 \times 1.6 \times 10^{-27} \times 1.6 \times 10^{-19}} eV = \frac{66 \times 66 \times 10^{-48+46}}{2 \times 16 \times 16}$

$E = \frac{33 \times 33}{126} \times 10^{-2} = 8.5 \times 10^{-2} = 0.085$ eV **[3]**

Q. 7. A hydrogen atom initially in the ground level absorbs a photon, which excites it to the $n = 4$ level. Determine the wavelength and frequency of the photon. [CBSE Board, Delhi Region, 2018]

Ans. For ground level, $n_1 = 1$

Let E_1 be the energy of this level. It is known that E_1 is related with $n_1$as :

$$E_1 = -\frac{13.6}{n_1^2}\ \text{eV}$$

$$= -\frac{13.6}{1^2} = -13.6\ \text{eV} \quad \textbf{[½]}$$

The atom is excited to a higher level, $n_2 = 4$.

Let E_2 be the energy of this level.

$$\therefore E_2 = -\frac{13.6}{n_2^2}\ \text{eV}$$

$$= -\frac{13.6}{4^2} = -\frac{13.6}{16}\ \text{eV} \quad \textbf{[½]}$$

The amount of energy absorbed by the photon is given as :

$$E = E_2 - E_1$$

$$= \left(\frac{-13.6}{16}\right) - \left(\frac{-13.6}{1}\right)$$

$$= 13.6 \times \frac{15}{16}\ \text{eV}$$

$$= \left(13.6 \times \frac{15}{16}\right) \times 1.6 \times 10^{-19} = 2.04 \times 10^{-18}\ \text{J} \quad \textbf{[1]}$$

For a photon of wavelength λ, the expression of energy is written as :

$$E = \frac{hc}{\lambda}$$

Where,

h = Planck's constant = 6.6×10^{-34} Js

c = Speed of light = 3×10^8 m/s

$$\therefore \lambda = \frac{h_c}{E} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{2.04 \times 10^{-18}}$$

$$= 9.7 \times 10^{-8}\ \text{m} = 97\ \text{nm} \quad \textbf{[1]}$$

And, frequency of a photon is given by the relation,

$$v = \frac{c}{\lambda}$$

$$= \frac{3 \times 10^8}{9.7 \times 10^{-8}} \approx 3.1 \times 10^{15}\ \text{Hz}$$

Hence, the wavelength of the photon is 97 nm while the frequency is 3.1×10^{15} Hz. **[1]**

Q. 8. How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation? [CBSE Board, Delhi Region, 2017]

Ans. Einstein photoelectric equation :

(i) A part of energy is used to overcome the surface barrier and come out of the metal surface. This part of energy is called "**work function**". It is expressed as $\phi_o = h\nu_o$ **[1]**

(ii) The remaining part of energy is used in giving a velocity v to the emitted photoelectron. This is equal to the maximum kinetic energy of photo electrons $\left(\frac{1}{2}mv_{\max}^2\right)$ **[1]**

(iii) According to the law of conservation of energy

$$h\nu = \phi_0 + \frac{1}{2}mv^2{}_{max} = h\nu_0 + \frac{1}{2}mv^2{}_{max}$$

$$\therefore \frac{1}{2}mv^2{}_{max} = k_{max} = h(\nu - \nu_0) = h\nu - \phi_0$$

$$\therefore \boxed{k_{max} = h\nu - \phi_0}$$

This equation is called Einstein photoelectric equation. **[1]**

Q. 9. The work function of the following metals is given as Na = 2.75 eV, K = 2.3 eV, Mo = 4.17 eV and Ni = 5.15 eV. Which of these metals will not cause photoelectric emission for radiation of wavelength 3300 Å from a laser source placed 1 m away from these metals? What happens if the laser source is brought nearer and placed 50 cm away? [CBSE Board, Delhi Region, 2017]

Ans. Energy of the incident radiation of wavelength λ,

$$\text{E} = \frac{hc}{\lambda} \quad \textbf{[½]}$$

$$\frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{3300 \times 10^{-10}}$$

$$= \frac{19.8 \times 10^{-16}}{3.3 \times 10^3} = 6 \times 10^{-19}\ \text{J}$$

$$= 3.75\ \text{eV} \quad \textbf{[1]}$$

This energy of the incident radiation is greater than the work function of Na and K but less than those of Mo and Ni, so photoelectric emission will occur only in Na and K metals and not in Mo and Ni. If the laser is brought closer, the intensity of incident radiation increases. This does not affect the result regarding Mo and Ni metals, while photoelectric current from Na and K will increase in proportion to intensity. **[1½]**

Q. 10. Write three characteristic features in photoelectric effect that cannot be explained on the basis of wave theory of light, but can be explained only using Einstein's equation. [CBSE Board, Delhi Region, 2016]

Ans. Following are the three characteristic features in photoelectric effect that cannot be explained on the basis of the wave theory of light but can be explained only using Einstein's equation :

(i) For a given metal and frequency of incident radiation, the number of photoelectrons ejected per second is directly proportional to the intensity of the incident light. **[1]**

(ii) For a given metal, there exists a certain minimum frequency of the incident radiation below which no emission of photoelectrons takes place. This frequency is called threshold frequency. **[1]**

(iii) Above the threshold frequency, the maximum kinetic energy of the emitted photoelectron is independent of the intensity of the incident light and is dependent only upon the frequency (or wavelength) of the incident light. **[1]**

Q. 11. Define the term "cut off frequency" in photoelectric emission. The threshold frequency of a metal is *f*. When the light of frequency 2*f* is incident on the metal plate, the maximum velocity of photo-electrons is v_1. When the frequency of the incident radiation is increased to 5*f*, the maximum velocity of photo-electrons is v_2. Find the ratio $v_1 : v_2$.

[CBSE Board, Foreign Scheme, 2016]

Ans. That particular frequency of incident radiation for which the minimum negative potential V_o given to a plate for photoelectric current becomes zero is called the cut-off frequency. **[1]**

Given that threshold frequency of metal is *f*. For light of frequency 2*f*, using Einstein's equation for photoelectric effect, we can write

$$h(2f - f) = \frac{1}{2}mv_1^2 \quad ...(i)$$

Similarly, for light having frequency 5*f*, we have

$$h(5f - f) = \frac{1}{2}mv_2^2 \quad ...(ii)$$

Using (1) and (2), we find

$$\frac{f}{4f} = \frac{v_1^2}{v_2^2} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{1}{4}}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{1}{2}$$

[2]

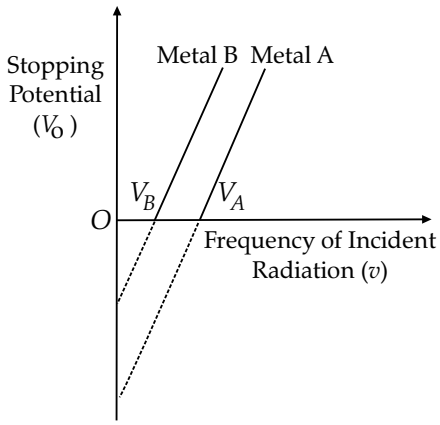
Q. 12. Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials *A* and *B* having threshold frequencies $\nu_A > \nu_B$.

(i) In which case is the stopping potential more and why?

(ii) Does the slope of the graph depend on the nature of the material used? Explain.

[CBSE Board, All India Region, 2016]

Ans. **(i)** Graph between stopping potential and the frequency of the incident radiation :



[½]

From the graph, we note :

(i) The stopping potential is inversely proportional to the threshold frequency. Hence, the stopping potential is higher for metal *B*. **[1]**

(ii) The slope of the graph does not depend on the nature of the material used.

As we know,

$K_{max} = h\nu - \Phi_o = eV_o$

Dividing the whole equation by '*e*', we get

$$\frac{h\nu}{e} - \frac{\phi_o}{e} = V_o$$

[½]

From the above equation, the slope of the graph is $\frac{h}{e}$ (on comparing with the straight line equation).

Thus, we see that the slope is independent of the nature of the photoelectric material. **[1]**

Q. 13. State two important properties of photon which are used to write Einstein's photoelectric equation. Define

(i) stopping potential and

(ii) threshold frequency, using Einstein's equation and drawing necessary plot between relevant quantities. [CBSE Board, Delhi Region, 2016]

Ans. Characteristic properties of photons :

(i) Energy of photon is directly proportional to the frequency (or inversely proportional to the wavelength). **[½]**

(ii) In photon-electron collision, total energy and momentum of the system of two constituent's remains constant. **[½]**

(iii) In the interaction of photons with the free electrons, the entire energy of photon is absorbed. Cut-off or stopping potential is that minimum value of negative potential at anode which just stops the photo electric current. For a given material, there is a minimum frequency of light frequency is called as threshold frequency. **[1]**

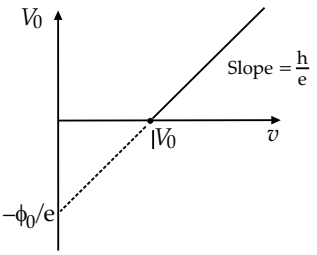
By Einstein's photoelectric equation

$$KE_{max} = \frac{hC}{\lambda} - \phi_0 = h\nu - h\nu_0 \quad \text{where } \phi_0 = h\nu_0$$

$$eV_0 = h\nu - h\nu_0$$

$$V_0 = \frac{h}{e}\nu - \frac{h}{e}\nu_0$$

[½]



[½]

Q. 14. Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory.

[CBSE Board, All India Region, 2017]

Ans. When a photon of energy '$h\nu$' falls on a metal surface, the energy of the photon is absorbed by the electrons and is used in the following two ways :

(i) A part of energy is used to overcome the surface barrier and come out of the metal surface. This part of energy is known as a work function and is expressed as $\phi_o = h\nu_o$ **[1]**

(ii) The remaining part of energy is used in giving a velocity '*v*' to the emitted photoelectron which

is equal to the maximum kinetic energy of photo electrons $\left(\frac{1}{2}mv^{2}{}_{\max}\right)$ **[½]**

(iii) According to the law of conservation of energy,

$$h\nu = \phi_o + \frac{1}{2}mv^2{}_{max}$$

$$h\nu = h\nu_o + \frac{1}{2}mv^2{}_{max}$$

$$\therefore \frac{1}{2}mv^2{}_{max} = KE_{max} = h\nu - h\nu_o$$

$$KE_{max} = h(\nu - \nu_o)$$

or $KE_{max} = h\nu - \phi_o$

This equation is called Einstein photoelectric equation. **[½]**

Features of photoelectric effect which cannot be explained by wave theory :

(i) Maximum kinetic energy of emitted electrons is independent of the intensity of incident light.

(ii) There exists a 'threshold frequency' for each photosensitive material.

(iii) 'Photoelectric effect' is instantaneous in nature. **[1]**

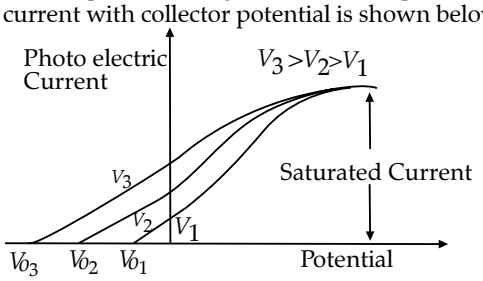
Q. 15. (a) Draw a plot showing the variation of photoelectric current with collector potential for different frequencies but same intensity of incident radiation.

(b) Use Einstein's photoelectric equation to explain the observations from this graph.

(c) What change will you observe if intensity of incident radiation is changed but the frequency remains the same?

[CBSE Board, Foreign Scheme, 2017]

Ans. (a) The plots showing the variation of photoelectric current with collector potential is shown below.



Stopping potential is directly proportional to the frequency of incident radiation. **[½]**

(b) Photoelectric emission does not take place by continuous absorption of energy from radiation. Radiation energy is built up of discrete units called quanta of energy of radiation. Each quantum of radiant energy has energy $h\nu$, where h is Planck's constant and ν is the frequency of light. In photoelectric effect, an electron absorbs a quantum of energy ($h\nu$) of radiation. If this quantum of energy absorbed exceeds the minimum energy needed for the electron to escape from the metal surface (work function ϕ_0), the electron is emitted with maximum kinetic energy

$K_{max} = h\nu - \phi_0$ **(i)**

Equation (i) is known as Einstein's photoelectric equation. According to Eq. (i), K_{max} depends linearly on ν, and is independent of intensity of radiation, in agreement with observation. This has happened because in Einstein's picture, photoelectric effect arises from the absorption of a single quantum of radiation by a single electron. The intensity of radiation (that is proportional to the number of energy quanta per unit area per unit time) is irrelevant to this basic process. Since, K_{max} must be non-negative, Eq. (i) implies that photoelectric emission is possible only if

$h\nu > \phi_0$

or $\nu > \phi_0$, where $\nu_0 = \frac{\phi_0}{h}$ ...(ii)

Equation (ii) shows that the greater the work function ϕ_0, the higher the minimum or threshold frequency ν_0 needed to emit photoelectrons. **[1½]**

(c) If the frequency of the incident wave is increased, then the kinetic energy of the photoelectrons increases. If the kinetic energy of electrons increases then their velocity also increases, if their velocity increases then the number of electrons passing an area in 1 second should also increase, that is, the current should increase. **[1]**

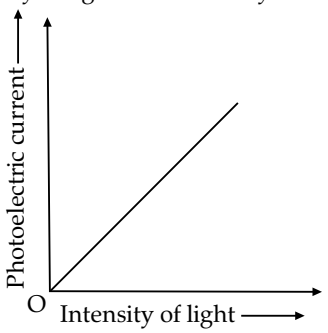
Q. 16. Plot a graph showing the variation of photoelectric current with intensity of light. The work function for the following metals is given :

Na : 2.75 eV and Mo : 4.175 eV.

Which of these will not give photoelectron emission from a radiation of wavelength 3300A^{o} from a laser beam? What happens if the source of laser beam is brought closer?

[CBSE Board, Foreign Scheme, 2016]

Ans. The plot between photoelectric current and intensity of light varies linearly as shown below :



[1]

Given that $\lambda = 3300 \times 10^{-10}$ m, $\phi_{Na} = 2.75\ eV$, $\phi_{Mo} = 4.175\ eV$

Then energy of the laser beam is

$$E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 10^{8}}{3300 \times 10^{-10} \times 1.6 \times 10^{-19}} = 3.75\ eV$$

[1]

Since, $E < \phi_{Mo}$, therefore there will be no emission of photoelectrons for Molybdenum (Mo). Bringing the source nearer will bring no change as photoelectric effect is independent of the distance of the source **[1]**

Q. 17. Find the

(a) maximum frequency, and

(b) minimum wavelength of X-rays produced by 30 kV electrons. [NCERT Ex. Q. 11.1, Page 407]

Ans. Potential of the electrons, $V = 30$ kV $= 3 \times 10^4$ V

Hence, energy of the electrons, $E = 3 \times 10^4$ eV

Where,

e = Charge on an electron $= 1.6 \times 10^{-19}$ C

(a) Maximum frequency produced by the X-rays = ν
The energy of the electrons is given by the relation :
$E = h\nu$
Where,
h = Planck's constant = 6.626×10^{-34} Js
So,

$$\nu = \frac{E}{h}$$
$$= \frac{1.6\times10^{-19}\times3\times10^{4}}{6.626\times10^{-34}}$$
$$= 7.24\times10^{18} \text{ Hz}$$

Hence, the maximum frequency of X-rays produced is 7.24×10^{18} Hz

(b) The minimum wavelength produced by the X-rays is given as :

$$\lambda = \frac{c}{v}$$
$$= \frac{3\times10^{8}}{7.24\times10^{18}}$$
$$= 4.14\times10^{-11} \text{ m}$$
$$= 0.0414 \text{ nm}$$

Hence, the minimum wavelength of X-rays produced is 0.0414 nm. **[3]**

Q. 18. *A* 100 W sodium lamp radiates energy uniformly in all directions. The lamp is located at the centre of a large sphere that absorbs all the sodium light which is incident on it. The wavelength of the sodium light is 589 nm.

(a) What is the energy per photon associated with the sodium light?

(b) At what rate are the photons delivered to the sphere? [NCERT Ex. Q. 11.7, Page 407]

Ans. Power of the sodium lamp, P = 100 W
Wavelength of the emitted sodium light, λ = 589 nm = 589×10^{-9} m
Planck's constant, $h = 6.626 \times 10^{-34}$ Js
Speed of light, $c = 3 \times 10^{8}$ m/s

(a) The energy per photon associated with the sodium light is given as :

$$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{589\times10^{-9}} = 3.37\times10^{-19} \text{ J}$$
$$E = \frac{3.37\times10^{-19}}{1.6\times10^{-19}} = 2.11 \text{ eV}$$

(b) Number of photons delivered to the sphere = n.
The equation for power can be written as :
$P = nE$

$$\therefore n = \frac{P}{E} = \frac{100}{3.37\times10^{-19}} = 2.96\times10^{20} \text{ photons/s}$$

Therefore, every second, 2.96×10^{20} photons are delivered to the sphere. **[3]**

Q. 19. The threshold frequency for a certain metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz is incident on the metal, predict the cut-off voltage for the photoelectric emission.
[NCERT Ex. Q. 11.8, Page 408]

Ans. Threshold frequency of the metal, $V_0 = 3.3 \times 10^{14}$ Hz
Frequency of light incident on the metal, V = 8.2×10^{14} Hz
Charge on an electron, $e = 1.6 \times 10^{-19}$ C
Planck's constant, $h = 6.626 \times 10^{-34}$ Js
Cut-off voltage for the photoelectric emission from the metal = V_0
The equation for the cut-off energy is given as :
$eV_0 = h(\nu - \nu_0)$

$$\therefore V_0 = \frac{h(\nu-\nu_0)}{e} = \frac{6.626\times10^{-34}\times(8.2\times10^{14}-3.3\times10^{14})}{1.6\times10^{-19}}$$
$$= 2.0292 \text{ V}$$

Therefore, the cut-off voltage for the photoelectric emission is 2.0292 V **[3]**

Q. 20. The work function for a certain metal is 4.2 eV. Will this metal give photoelectric emission for incident radiation of wavelength 330 nm?
[NCERT Ex. Q. 11.9, Page 408]

Ans. No Work function of the metal, $\phi_0 = 4.2$ eV
Charge on an electron, $e = 1.6 \times 10^{-19}$ C
Planck's constant, $h = 6.626 \times 10^{-34}$ Js
Wavelength of the incident radiation, λ = 330 nm = 330×10^{-9} m
Speed of light, $c = 3 \times 10^{8}$ m/s
The energy of the incident photon is given as :

$$\therefore = \frac{hc}{} = \frac{6\ 626\times10^{\ 34}\times3\times10^{8}}{330\times10} = 6.0\times10^{\ 19} \text{ J}$$
$$= \frac{6\ 0\times10^{\ 19}}{1\ 6\times10^{\ 19}} \quad 3.76 \text{ eV}$$

It can be observed that the energy of the incident radiation is less than the work function of the metal. Hence, no photoelectric emission will take place. **[3]**

Q. 21. Light of frequency 7.21×10^{14} Hz is incident on a metal surface. Electrons with a maximum speed of 6.0×10^{5} m/s are ejected from the surface. What is the threshold frequency for photoemission of electrons? [NCERT Ex. Q. 11.10, Page 408]

Ans. Frequency of the incident photon, v = 488 nm = 488×10^{-9} m
Maximum speed of the electrons, $v = 6.0 \times 10^{5}$ m/s
Planck's constant, $h = 6.626 \times 10^{-34}$ Js
Mass of an electron, $m = 9.1 \times 10^{-31}$ kg
For threshold frequency ν_0, the relation for kinetic energy is written as :

$$\frac{1}{2}mv^2 = h(\nu - \nu_0)$$
$$\therefore \quad \nu_0 = \nu - \frac{mv^2}{2h}$$
$$\nu_0 = 7.21\times10^{14} - \frac{9.1\times10^{-31}\times(6\times10^{5})^2}{2\times6.626\times10^{-34}}$$
$$= 7.21\times10^{14} - 2.472\times10^{14}$$
$$= 4.738\times10^{14} \text{ Hz}$$

Therefore, the threshold frequency for the photoemission of electrons is 4.738×10^{14} Hz. **[3]**

Q. 22. Light of wavelength 488 nm is produced by an argon laser which is used in the photoelectric effect. When light from this spectral line is

incident on the emitter, the stopping (cut-off) potential of photoelectrons is 0.38 V. Find the work function of the material from which the emitter is made. [NCERT Ex. Q. 11.11, Page 408]

Ans. Wavelength of light produced by the argon laser, $\lambda = 488 \text{ nm} = 488 \times 10^{-9} \text{ m}$

Stopping potential of the photoelectrons, $V_0 = 0.38\ V$

$1eV = 1.6 \times 10^{-19} \text{ J}$

$$V_0 = \frac{0.38}{1.6 \times 10^{-19}} \text{ eV}$$

Planck's constant, $h = 6.6 \times 10^{-34}$ Js

Charge on an electron, $e = 1.6 \times 10^{-19}$ C

Speed of light, $c = 3 \times 10^8$ m/s

From Einstein's photoelectric effect, we have the relation involving the work function ϕ_0 of the material of the emitter as :

$$eV_0 = \frac{hc}{\lambda - \varphi_0}$$

$$\therefore \varphi_0 = \frac{hc}{\lambda - \text{eV}_0}$$

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 488 \times 10^{-9} - \left(1.6 \times 10^{-19} \times \frac{0.38}{1.6 \times 10^{-19}}\right)}$$

$$= 2.54 - 0.38$$

$$= 2.16 \text{ eV}$$

Therefore, the material with which the emitter is made has the work function of 2.16 eV. **[3]**

Q. 23. Show that the wavelength of electromagnetic radiation is equal to the de Broglie wavelength of its quantum (photon).

[NCERT Ex. Q. 11.18, Page 408]

Ans. The momentum of a photon having energy (hν) is given as :

$$p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{p} \quad \text{...(i)}$$

Where,

λ = Wavelength of the electromagnetic radiation

c = Speed of light

h = Planck's constant

De Broglie wavelength of the photon is given as : $\lambda = \frac{h}{mv}$

But $p = mv$

Therefore, $\lambda = \frac{h}{p}$ ...(ii)

Where, m = Mass of the photon

v = Velocity of the photon

Hence, it can be inferred from equations (i) and (ii) that the wavelength of the electromagnetic radiation is equals to the de Broglie wavelength of the photon. **[3]**

Q. 24. (a) An X-ray tube produces a continuous spectrum of radiation with its short wavelength end at 0.45 Å. What is the maximum energy of a photon in the radiation? (b) From your answer to (a), guess what order of accelerating voltage (for electrons) is required in such a tube?

[NCERT Add Ex. Q. 11.23, Page 409]

Ans. (a) Wavelength produced by an X-ray tube, $\lambda = 0.45\,\mathring{A} = 0.45 \times 10^{-19} \text{ m}$

Planck's constant, $h = 6.626 \times 10^{-34}$ Js

Speed of light, $c = 3 \times 10^8$ m/s

The maximum energy of a photon is given as :

$$E = \frac{hc}{\lambda}$$

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{0.45 \times 10^{-10} \times 1.6 \times 10^{19}}$$

$$= 27.6 \times 10^3 \text{ eV}$$

$$= 27.6 \text{ keV}$$

Therefore, the maximum energy of an X-ray photon is 27.6 keV.

(b) Accelerating voltage provides energy to the electrons for producing X-rays. To get an X-ray of 27.6 keV, the incident electrons must possess at least 27.6 keV of kinetic electric energy. Hence, an accelerating voltage of the order of 30 keV is required for producing X-rays. **[3]**

Q. 25. In an accelerator experiment on high-energy collisions of electrons with positrons, a certain event is interpreted as annihilation of an electron-positron pair of total energy 10.2 BeV into two γ-rays of equal energy. What is the wavelength associated with each γ-ray? (1BeV = 10^9 eV) .

[NCERT Add Ex. Q. 11.24, Page 409]

Ans. Total energy of two γ-rays : $E = 10.2 \text{ BeV} = 10.2 \times 10^9 \text{ eV} = 10.2 \times 10^9 \times 1.6 \times 10^{-10}$ J

Hence, the energy of each γ-ray :

$$E' = \frac{E}{2}$$

$$= \frac{10.2 \times 1.6 \times 10^{-10}}{2}$$

$$= 8.16 \times 10^{-10} \text{ J}$$

Planck's constant, $h = 6.626 \times 10^{-34}$ Js

Speed of light, $c = 3 \times 10^8$ m/s

Energy is related to wavelength as:

$$E' = \frac{hc}{\lambda}$$

$$\therefore \lambda = \frac{hc}{E'} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{8.16 \times 10^{-10}}$$

$$\lambda = 2.436 \times 10^{-16} \text{ m}$$

Therefore, the wavelength associated with each gamma-ray is 2.436×10^{-16} m **[3]**

Q. 26. Compute the typical de Broglie wavelength of an electron in a metal at 27 °C and compare it with the mean separation between two electrons in a metal which is given to be about 2×10^{-10} m.

[Note : Exercises 11.35 and 11.36 reveal that while the wave-packets associated with gaseous molecules under ordinary conditions are non-

overlapping, the electron wave packets in a metal strongly overlap with one another. This suggests that whereas molecules in an ordinary gas can be distinguished apart, electrons in a metal cannot be distinguished apart from one another. This indistinguishability has many fundamental implications which you will explore in more advanced Physics courses.]

[NCERT Add Ex. Q. 11.36, Page 411]

Ans. Temperature, T = 27°C = 27 + 273 = 300 K

Mean separation between two electrons,

$$r = 2 \times 10^{-10}\ \text{m}$$

De Broglie wavelength of an electron is given as :

$$\lambda = \frac{h}{\sqrt{3mk\text{T}}}$$

where,

h=Planck's constant=6.6×10^{-34} Js

m=mass of an electron=9.11×10^{-31} kg

k=Boltzmann constant=1.38×10^{-23} $\text{Jmol}^{-1}\text{K}^{-1}$

$$\therefore \lambda = \frac{6.6\times10^{-34}}{\sqrt{3\times9.11\times10^{-31}\times1.38\times10^{-23}\times300}}$$

$$\lambda \approx 6.2\times10^{-9}\ \text{m}$$

Hence, the de Broglie wavelength is much greater than the given inter-electron separation. **[3]**

Long Answer Type Questions (5 marks each)

Q. 1. Consider a thin target (10^{-2} cm square, 10^{-3} m thickness) of sodium, which produces a photocurrent of 100 μA when a light of intensity 100 W/m^2 (λ = 660 nm) falls on it. Find the probability that a photoelectron is produced when a photon strikes a sodium atom. [Take density of Na = 0.97 kg/m^3]

[NCERT Exemp. Q. 11.25, Page 73]

Ans. According to the problem,

Area of the target $A = 10^{-2}\ \text{cm}^2 = 10^{-4}\ \text{m}^2$ and thickness, $d = 10^{-3}$ m

Photo current, $i = 100\times10^{-6}$ A $= 10^{-4}$ A

Intensity, $I = 100\ W/\text{m}^2$

$\Rightarrow$ $\lambda = 660$ nm $= 660\times10^{-9}$ m

$\rho_{\text{Na}} = 0.97\ \text{kg/m}^3$

Avogadro's number $= 6\times10^{23}$ kg atom

Volume of sodium target

$= A\times d$

$= 10^{-4}\times10^{-3}$

$\Rightarrow$ $= 10^{-7}\ \text{m}^3$ **[½]**

We know that 6×10^{23} atoms of sodium weights = 23 kg

Density of sodium = 0.97 kg/m^3

Hence the, volume of 6×10^{6} sodium atoms $= \frac{23}{0.97}\ \text{m}^3$

Volume occupied by one sodium atoms $\frac{23}{0.97\times(6\times10^{23})} = 3.95\times10^{-23}\ \text{m}^3$ **[½]**

Number of sodium atoms in target (N_{sodium})

$$= \frac{10^{-7}}{3.95\times10^{-23}} = 2.53\times10^{16}$$ **[½]**

Let n be the number of photons falling per second on the target.

Energy of each photon $= \frac{hc}{\lambda}$

Total energy falling per second on target $= \frac{nhc}{\lambda} = IA$

$$\therefore \quad n = \frac{IA\lambda}{hc}$$

$$= \frac{100\times10^{-4}\times(660\times10^{-9})}{(6.62\times10^{-34})\times(3\times10^{8})} = 3.3\times10^{16}$$ **[½]**

Let P be the probability of emission per atom per photon.

The number of photoelectrons emitted per second.

$$N = P\times n\times(\text{N}_{\text{sodium}})$$

$$= P\times(33\times10^{16})\times(2.53\times10^{16})$$ **[1]**

Now, according to question,

$$i = 100\mu A = 100\times10^{-6} = 10^{-4}\ \text{A}$$

Current, $i = Ne$

$$\therefore 10^{-4} = P\times(3.3\times10^{16})\times(2.53\times10^{16})\times(1.6\times10^{-19})$$ **[1]**

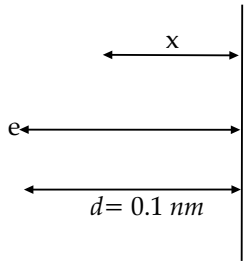
$$\Rightarrow\ P = \frac{10^{-4}}{(3.3\times10^{16})\times(2.53\times10^{16})\times(1.6\times10^{-19})}$$

$$= 7.48\times10^{-19}$$ **[1]**

Then, the probability of photoemission by a single photon on a single atom is very much less than 1. Because, the absorption of two photons by an atom is negligible.

Q. 2. Consider an electron in front of metallic surface of a distance d. Assume the force of attraction by the plate is given as $\frac{1}{4}\frac{q^2}{4\pi\varepsilon_0 d^2}$. Calculate work in taking the to an infinite distance from the plate. Taking d=0.1 nm. find the work done in electron volts? [NCERT Exemp. Q. 11.26, Page 73]

Ans. Consider the figure in which an electron is displaced slowly by a distance x by the means of an external force which is,



[1]

$$F = \frac{q^2}{4\times4\pi\varepsilon_0 d^2}$$

where, d=0.1 nm = 10^{-10} m

Let the electron be at distance x from metallic surface. Then, force of attraction on it is

$$F_x = \frac{q^2}{4\times 4\pi\varepsilon_0 d^2}$$ **[1]**

Work done by external agency in taking the electron from distance d to infinity is

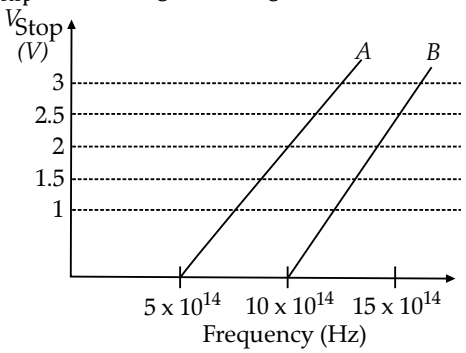
$$W = \int_d^\infty F_x dx = \int_d^\infty \frac{q^2 dx}{4\times 4\pi\varepsilon_0}\frac{1}{x^2}$$

$$= \frac{q^2}{4\times 4\pi\varepsilon_0}\left[\frac{1}{d}\right]$$

$$= \frac{(1.6\times 10^{-19})^2 \times 9\times 10^9}{4\times 10^{-10}}\ \text{J}$$

$$= \frac{(1.6\times 10^{-19})^2 \times (9\times 10^9)}{(4\times 10^{-10})\times(1.6\times 10^{-19})}\text{eV} = 3.6\ \text{eV}$$ **[3]**

Q. 3. A student performs an experiment on photoelectric effect, using two materials *A* and *B*. *A* plot of V_{stop} versus v is given in figure.



(i) Which material *A* or *B* has a higher work function?

(ii) Given the electric charge of an electron = 1.6×10^{-19} *C*, Find the value of *h* obtained from the experiment for both *A* and *B*.

Comment on whether it is consistent with Einstein's theory

[NCERT Exemp. Q. 11.27, Page 73]

Ans. Threshold Frequency (n_o): The minimum frequency of incident radiations, required to eject the electron from metal surface is defined as threshold frequency. **[½]**

If incident frequency $\nu < \nu_o$ = No photoelectron emission

For most metals the threshold frequency is in the ultraviolet (corresponding to wavelengths between 200 nm and 300 nm), but for potassium and caesium oxides, it is in the visible spectrum (λ between 400 nm and 700 nm)

Here we are given threshold frequency of *A*

$$\nu_{OA} = 5\times 10^{14}\,\text{Hz}$$ **[½]**

For *B*,

$$\nu_{OB} = 10\times 10^{14}\,\text{Hz}$$ **[½]**

We know that

Work function, $\phi = h\nu_0$ or $\phi \propto \nu_0$ **[1]**

So, $\frac{\phi_{OA}}{\phi_{OB}} = \frac{5\times 10^{14}}{10\times 10^{14}} < 1$ $\Rightarrow \phi \propto \nu_0$

$\Rightarrow \quad \phi_{OA} < \phi_{OB}$ **[½]**

Thus, work function of *B* is higher that *A*.

(ii) For metal *A*, slope $= \frac{h}{e} = \frac{2}{(10-15)10^{14}}$

or $h = \frac{2e}{5\times 10^{14}} = \frac{2\times 1.6\times 10^{-19}}{5\times 10^{14}}$

$= 6.4\times 10^{-34}$ Js **[1]**

For metal *B*, slope $= \frac{h}{e} = \frac{2.5}{(15-10)10^{14}}$

or $h = \frac{2.5\times e}{5\times 10^{14}} = \frac{2.5\times 1.6\times 10^{-19}}{5\times 10^{14}}$

$= 8\times 10^{-34}$ Js **[1]**

Since, the value of h from experiment for metals *A* and *B* is different. Hence, experiment is not consistent with Einstein's theory.

Q. 4. A particle *A* with a mass m_A is moving with a velocity v and hits a particle *B* (mass m_B) at rest (one dimensional motion). Find the change in the de-Broglie wavelength of the particle *A*. Treat the collision as elastic.

[NCERT Exemp. Q. 11.28, Page 73]

Ans. The laws of conservation of momentum and kinetic energy are obeyed because the collision is elastic. **[½]**

By the law of conservation of momentum,

$$m_A v + m_B 0 = m_A v_1 + m_B v_2$$

$$\Rightarrow \quad m_A(v - v_1) = m_B v_2$$ **[½]**

And the law of conservation of energy,

$$\frac{1}{2}m_A v^2 = \frac{1}{2}m_A v_1^2 + \frac{1}{2}m_B v_2^2 \quad \cdots(\text{i})$$

$$\Rightarrow \quad m_A\left(v^2 - v_1^2\right) = m_B v_2^2$$

$$\Rightarrow \quad m_A(v - v_1)(v + v_1) = m_B v_2^2 \quad \cdots(\text{ii})$$ **[1]**

Dividing *Eq.* (ii) by *Eq.* (i),

we get $v + v_1 = v_2$ or $v = v_2 - v_1$ $\cdots$(iii) **[½]**

Solving Equations. (i) and (iii), we get

$$v_1 = \left(\frac{m_A - m_B}{m_A + m_B}\right)v \text{ and } v_2 = \left(\frac{2m_A}{m_A + m_B}\right)v$$

$$\lambda_{initial} = \frac{h}{m_A v}$$

$$\lambda_{final} = \frac{h}{m_A v_1} = \frac{h(m_A + m_B)}{m_A(m_A - m_B)v}$$

$$\Delta\lambda = \lambda_{final} - \lambda_{initial} = \frac{h}{m_A v}\left[\frac{m_A + m_B}{m_A - m_B} - 1\right]$$ **[2½]**

Q. 5. Consider a 20 W bulb emitting light of wavelength 5000 Å and shining on a metal surface kept at a distance 2 m. Assume that the metal surface has work function of 2 eV and that each atom on the metal surface can be treated as a circular disk of radius 1.5 Å.

(i) Estimate number of photons emitted by the bulb per second. [Assume no other losses]

(ii) Will there be photoelectric emission?

(iii) How much time would be required by the atomic disk to receive energy equal to work function (2 eV)?

(iv) How many photons would atomic disk receive within time duration calculated in (iii) above?

(v) Can you explain how photoelectric effect was observed instantaneously?

[NCERT Exemp. Q. 11.29, Page 74]

Ans. Given that in the problem,

$P = 20W$

$\lambda = 5000\ \overset{0}{\text{A}} = 5000 \times 10^{-10}$ m

$d = 2$ m

work function $\phi_0 = 2$ eV

radius $r = 1.5\ \overset{0}{\text{A}} = 1.5 \times 10^{-10}$ m

Now, Number of photon emitted by bulb per second, $n' = \frac{dN}{dt}$ [½]

(i) Number of photon emitted by bulb per second is

$$n' = \frac{P}{hc/\lambda} = \frac{P\lambda}{hc}$$ [½]

$$= \frac{20 \times (5000 \times 10^{-10})}{(6.62 \times 10^{-34}) \times (3 \times 10^8)}$$

$$\Rightarrow \quad n' = 5 \times 10^{19} / \text{sec}$$

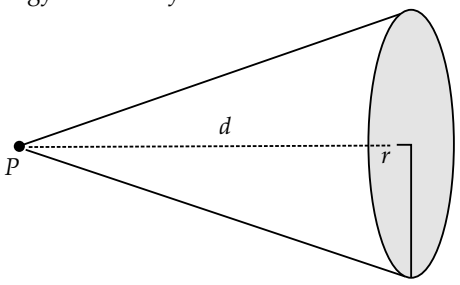
(ii) Energy of the incident photon

$$= \frac{hc}{\lambda} = \frac{(6.62 \times 10^{-34})(3 \times 10^8)}{5000 \times 10^{-10} \times 1.6 \times 10^{-19}} = 2.48 \text{ eV}$$ [½]

As this energy is greater than 2 eV (*i.e.*, work function of metal surface), hence, photoelectric emission takes place.

(iii) Let Δt be the time spent in getting the energy $\phi =$ (work function of metal).

Consider the figure, if P is the power of source then energy received by atomic disc.



[½]

$$\frac{P}{4\pi d^2} \times \pi r^2 \Delta t = \phi_0$$

$$\Rightarrow \quad \Delta t = \frac{4\phi_0 d^2}{P r^2}$$

$$= \frac{4 \times (2 \times 1.6 \times 10^{-19}) \times 2^2}{20 \times (1.5 \times 10^{-10})^2} = 28.4 \text{ s}$$

[1]

(iv) Number of photons received by atomic disc in time Δt is

$$N = \frac{n' \times \pi r^2}{4\pi d^2} \times \Delta t$$

$$= \frac{n' r^2 \Delta t}{4d^2}$$

$$= \frac{(5 \times 10^{19}) \times (1.5 \times 10^{-10})^2 \times 28.4}{4 \times (2)^2} = 2$$

[1]

Now let us discuss the last part in detail. As time of emission of electrons is 11.04 sec. Hence, in this problem, the photoelectric emission is not instantaneous.

(v) In photoelectric emission, there is an oscillation between incident photon and free electron of the metal surface, which last for very short interval of time (= 10^{-9} sec), hence, we can say that photoelectric emission is instantaneous. **[1]**

Q. 6. The work function of cesium metal is 2.14 eV. When light of frequency 6×10^{14} Hz is incident on the metal surface, photoemission of electrons occurs. What is the

(a) maximum kinetic energy of the emitted electrons,

(b) Stopping potential, and

(c) maximum speed of the emitted photoelectrons?

[NCERT Ex. Q. 11.2, Page 407]

Ans. Work function of cesium metal , $\phi_0 = 2.14$ eV

Frequency of light, $\nu = 6.0 \times 10^{14}$ Hz

(a) The maximum kinetic energy is given by the photoelectric effect as :

$K = h\nu - \phi_0$

Where,

h = Planck's constant = 6.626×10^{-34} Js

$$\therefore K = \frac{6.626 \times 10^{34} \times 6 \times 10^{14}}{1.6 \times 10^{-19} - 2.14} = 0.345 \text{ eV}$$

Hence, the maximum kinetic energy of the emitted electrons is 0.345 eV.

(b) For stopping potential, V_0 we can write the equation for kinetic energy as :

$K = eV_0$

$$\therefore V_0 = \frac{K}{e} = \frac{0.345 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-19}} = 0.345 \text{ V}$$

Hence, the stopping potential of the material is 0.345 V.

(c) Maximum speed of the emitted photoelectrons = v

Hence, the relation for kinetic energy can be written as :

$$K = \frac{1}{2} m\text{v}^2$$

Where,

m = Mass of an electron = 9.1×10^{-31} kg

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 0.345 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}}$$

$$\therefore v = 3.323 \times 10^5 \text{ m/s}$$

Hence, the maximum speed of the emitted photoelectrons is 3.323 m/s. **[5]**

Q. 7. Monochromatic light of wavelength 632.8 nm is produced by a helium-neon laser. The power emitted is 9.42 mW.

(a) Find the energy and momentum of each photon in the light beam,

(b) How many photons per second, on the average, arrive at a target irradiated by this beam? (Assume the beam to have uniform cross-section which is less than the target area), and

(c) How fast does a hydrogen atom have to travel in order to have the same momentum as that of the photon? [NCERT Ex. Q. 11.4, Page 407]

Ans. Wavelength of the monochromatic light, $\lambda = 632.8$ nm $= 632.8 \times 10^{-9}$ m
Power emitted by the laser, $P = 9.42$ mW $= 9.42 \times 10^{-3}$ W
Planck's constant, $h = 6.626 \times 10^{-34}$ Js
Speed of light, $c = 3 \times 10^{8}$ m/s
Mass of a hydrogen atom, $m = 1.66 \times 10^{-27}$ kg

(a) The energy of each photon is given as :

$$E=\frac{hc}{\lambda}=\frac{6.626\times10^{-34}\times3\times10^{8}}{632.8\times10^{-9}}=3.141\times10^{-19}\ \text{J}$$

The momentum of each photon is given as :

$$p=\frac{h}{\lambda}=\frac{6.626\times10^{-34}}{632.8\times10^{-9}}=1.047\times10^{-27}\ \text{kgms}^{-1}$$ **[1]**

(b) Number of photons arriving per second, at a target irradiated by the beam = n Assume that the beam has a uniform cross-section that is less than the target area. Hence, the equation for power can be written as :

$P = nE$

$$\therefore n=\frac{p}{E}=\frac{9.42\times10^{-3}}{3.141\times10^{-19}}=3\times10^{16}\ \text{photons/sec(approx).}$$ **[2]**

(c) Momentum of the hydrogen atom is the same as the momentum of the photon,
$p = 1.047 \times 10^{-27}$ kg m s^{-1}
Momentum is given as :
$p = mv$
Where,
v = Speed of the hydrogen atom

$$\therefore v=\frac{p}{m}=\frac{1.047\times10^{-27}}{1.66\times10^{-27}}=0.621\ \text{m/s.}$$ **[2]**

Q. 8. Calculate the
(a) momentum, and
(b) de Broglie wavelength of the electrons accelerated through a potential difference of 56 V.
[NCERT Ex. Q. 11.12, Page 408]

Ans. Potential difference, $V = 56$ V
Planck's constant, $h = 6.6 \times 10^{-34}$ Js
Mass of an electron, $m = 9.1 \times 10^{-31}$ kg
Charge on an electron, $e = 1.6 \times 10^{-19}$ C

(a) At equilibrium, the kinetic energy of each electron is equal to the accelerating potential, *i.e.*, we can write the relation for velocity (v) of each electron as :

$$\frac{1}{2}mv^2 = e\text{V}$$

$$v=\sqrt{\frac{2e\text{V}}{\text{m}}}$$

$$v=\sqrt{\frac{2\times1.6\times10^{-19}\times56}{9.1\times10^{-31}}}$$

$$=\sqrt{19.69\times10^{12}}$$

$$=4.44\times10^{6}\ \text{m/s}$$

The momentum of each accelerated electron is given as :
$p = mv$
$= 9.1 \times 10^{-31} \times 4.44 \times 10^{6}$
$= 4.04 \times 10^{-24}$ kg m s^{-1}
Therefore, the momentum of each electron is 4.04×10^{-24} kg m s^{-1}. **[3]**

(b) De Broglie wavelength of an electron accelerating through a potential V is given by the relation :

$$\lambda=\frac{12.27}{\sqrt{V}}\ \overset{o}{A}$$

$$=\frac{12.27}{\sqrt{56}}\times10^{-10}\ \text{m}$$

$$=0.1639\ \text{nm}$$

Therefore, the de Broglie wavelength of each electron is 0.1639 nm. **[2]**

Q. 9. What is the
(a) momentum,
(b) speed, and
(c) de Broglie wavelength of an electron with kinetic energy of 120 eV. [NCERT Ex. Q. 11.13, Page 408]

Ans. Kinetic energy of the electron, $E_k = 120$ eV
Planck's constant, $h = 6.6 \times 10^{-34}$ Js
Mass of an electron, $m = 9.1 \times 10^{-31}$ kg
Charge on an electron, $e = 1.6 \times 10^{-19}$ C

(a) For the electron, we can write the relation for kinetic energy as :

$$E_k=\frac{1}{2}mv^2$$

Where, v = Speed of the electron

$$\therefore v=\sqrt{\frac{2e\text{E}_\text{k}}{m}}$$

$$=\sqrt{\frac{2\times1.6\times10^{-19}\times120}{9.1\times10^{-31}}}$$

$$=\sqrt{42.198\times10^{12}}$$

$$=6.496\times10^{6}\ \text{m/s}$$

Momentum of the electron,
$p = mv$
$= 9.1 \times 10^{-31} \times 6.496 \times 10^{6}$
$= 5.91 \times 10^{-24}$ kg ms^{-1}
Therefore, the momentum of the electron is 5.91×10^{-24} kg m s^{-1}. **[2]**

(b) Speed of the electron, v $= 6.496 \times 10^{6}$ m/s. **[1]**

(c) De Broglie wavelength of an electron having a momentum p is given as :

$$\lambda=\frac{h}{p}=\frac{6.6\times10^{-34}}{5.91\times10^{-24}}=1.116\times10^{-10}\ \text{m}$$

$$\lambda = 0.112\ \text{nm}$$

Therefore, the de Broglie wavelength of the electron is 0.112 nm. **[2]**

Q. 10. The wavelength of light from the spectral emission line of sodium is 589 nm. Find the kinetic energy at which
(a) an electron, and
(b) a neutron, would have the same de Broglie wavelength. [NCERT Ex. Q. 11.14, Page 408]

Ans. Wavelength of light of a sodium line, $\lambda = 589$ nm $= 589 \times 10^{-9}$ m
Mass of an electron, $m_e = 9.1 \times 10^{-31}$ kg
Mass of a neutron, $m_n = 1.66 \times 10^{-27}$ kg
Planck's constant, $h = 6.6 \times 10^{-34}$ Js

(a) For the kinetic energy K, of an electron accelerating with a velocity v, we have the relation :

$$K_e = \frac{1}{2} m_e v^2 \quad ...(i)$$

We have the relation for de Broglie wavelength as :

$$\lambda = \frac{h}{m_e v}$$

Squaring both side,we get

$$\therefore v^2 = \frac{h^2}{\lambda^2 m_e^2} \quad ...(ii)$$

Substituting equation (ii) in equation (i), we get the relation :

$$K_e = \frac{1}{2}\frac{m_e h^2}{\lambda^2 m_e^2} = \frac{h^2}{2\lambda^2 m_e} \quad ...(iii)$$

$$= \frac{1}{2}\frac{(6.626\times10^{-34})^2}{(589\times10^{-9})^2 \times 9.1\times10^{-31}}$$

On solving we get,

$$K_e \approx 6.9\times10^{-25} \text{ J}$$

$$K_e = \frac{6.9\times10^{-25}}{1.6\times10^{-19}} = 4.31\times10^{-6} \text{ eV}$$

$$K_e = 4.31 \ \mu\text{eV}$$

Hence, the kinetic energy of the electron is 6.9×10^{-25} J or 4.31 μeV. **[2½]**

(b) Using equation (3), we can write the relation for the kinetic energy of the neutron as :

$$K_n = \frac{h^2}{2\lambda^2 m_n} = \frac{(6.626\times10^{-34})^2}{2\times(589\times10^{-9})^2 \times 1.66\times10^{-27}}$$

On solving we get,

$$K_n = 3.78\times10^{-28} \text{ J}$$

$$K_n = \frac{3.78\times10^{-28}}{1.6\times10^{-19}} = 42.36\times10^{-9} \text{ eV}$$

$$K_n = 2.36 \text{ neV}$$

Hence, the kinetic energy of the neutron is 3.78×10^{-28} J or 2.36 neV. **[2½]**

Q. 11. What is the de Broglie wavelength of

(a) a bullet of mass 0.040 kg travelling at the speed of 1.0 km/s,

(b) a ball of mass 0.060 kg moving at a speed of 1.0 m/s, and

(c) a dust particle of mass 1.0×10^{-9} kg drifting with a speed of 2.2 m/s? [NCERT Ex. Q. 11.15, Page 408]

Ans. **(a)** Mass of the bullet, $m = 0.040$ kg

Speed of the bullet, $v = 1.0$ km/s $= 1000$ m/s

Planck's constant, $h = 6.6 \times 10^{-34}$ Js

De Broglie wavelength of the bullet is given by the relation :

$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}}{0.040\times1000} = 1.65\times10^{-35} \text{ m}$$

(b) Mass of the ball, m = 0.060 kg

Speed of the ball, v = 1.0 m/s

De Broglie wavelength of the ball is given by the relation :

$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}}{0.060\times1} = 1.1\times10^{-32} \text{ m}$$

(c) Mass of the dust particle, $m = 1 \times 10^{-9}$ kg

Speed of the dust particle, $v = 2.2$ m/s

De Broglie wavelength of the dust particle is given by the relation :

$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}}{2.2\times1\times10^{-9}} = 3.0\times10^{-25} \text{ m}$$

[5]

Q. 12. An electron and a photon each have a wavelength of 1.00 nm. Find

(a) their momenta,

(b) the energy of the photon, and

(c) the kinetic energy of electron.

[NCERT Ex. Q. 11.16, Page 408]

Ans. Wavelength of an electron (λ_e) and a photon (λ_p),

So, $\lambda_e = \lambda_p = \lambda = 1$ nm $= 1 \times 10^{-9}$ m

Planck's constant, $h = 6.63 \times 10^{-34}$ Js

(a) The momentum of an elementary particle is given by de Broglie relation :

$$\lambda = \frac{h}{p}$$

$$\text{So, } p = \frac{h}{\lambda}$$

It is clear that momentum depends only on the wavelength of the particle. Since the wavelengths of an electron and a photon are equal, both have an equal momentum.

$$p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{1\times10^{-9}} = 6.63\times10^{-25} \text{ kg m s}^{-1}$$

(b) The energy of a photon is given by the relation :

$$E = \frac{hc}{\lambda}$$

where,

Speed of light, $c = 3 \times 10^8$ m/s

$$\text{E} = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3\times10^{8}}{1\times10^{-9}\times1.6\times10^{-19}} = 1243.1 \text{ eV}$$

Therefore, the energy of the photon is 1.243 keV.

(c) The kinetic energy (K) of an electron having momentum p, is given by the relation :

$$K = \frac{1}{2}\frac{p^2}{m}$$

Where,

m = Mass of the electron $= 9.1 \times 10^{-31}$ kg

$p = 6.63 \times 10^{-25}$ kg m s^{-1}

$$K = \frac{1}{2}\frac{(6.63\times10^{-25})^2}{9.1\times10^{-31}} = 2.415\times10^{-19} \text{ J}$$

$$K = \frac{2.415\times10^{-19}}{1.6\times10^{-19}} = 1.51 \text{ eV}$$

Hence, the kinetic energy of the electron is 1.51 eV. **[5]**

Q. 13. (a) For what kinetic energy of a neutron will the associated de Broglie wavelength be 1.40×10^{-10} m?

(b) Also find the de Broglie wavelength of a neutron, in thermal equilibrium with matter, having an average kinetic energy of (3/2) kT at 300 K.

[NCERT Ex. Q. 11.17, Page 408]

Ans. **(a)** De Broglie wavelength of the neutron, $\lambda = 1.40 \times 10^{-10}$ m
Mass of a neutron, $m_n = 1.66 \times 10^{-27}$ kg
Planck's constant, $h = 6.6 \times 10^{-34}$ Js
Kinetic energy (K) and velocity (v) are related as :

$$K = \frac{1}{2} m_n v^2 \quad ...(i)$$

De Broglie wavelength (λ) and velocity (v) are related as :

$$\lambda = \frac{h}{m_n v} \quad ...(ii)$$

Using equation (ii) in equation (i), we get :

$$K = \frac{1}{2}\frac{m_n h^2}{\lambda^2 m_n^2} = \frac{h^2}{2\lambda^2 m_n}$$
$$= \frac{\left(6.63\times10^{-34}\right)^2}{2\times(1.40\times10^{-10})^2\times1.66\times10^{-27}}$$
$$= 6.75\times10^{-21} \text{ J}$$
$$= \frac{6.75\times10^{-21}}{1.6\times10^{-19}}$$
$$= 4.219\times10^{-2} \text{ eV}$$

Hence, the kinetic energy of the neutron is 6.75×10^{-21} J or 4.219×10^{-2} eV. **[2½]**

(b) Temperature of the neutron, $T = 300$ K
Boltzmann constant, $k = 1.38 \times 10^{-23}$ kg m^2 s^{-2} K^{-1}
Average kinetic energy of the neutron :

$$K' = \frac{3}{2}kT$$
$$= \frac{3}{2}\times1.38\times10^{-23}\times300$$
$$= 6.21\times10^{-21} \text{ J}$$

The relation for the de Broglie wavelength is given as :

$$\lambda' = \frac{h}{\sqrt{2K'm_n}}$$

where,

$$m_n = 1.66\times10^{-27} \text{ kg}$$
$$h = 6.6\times10^{-34} \text{ Js}$$
$$K' = 6.75\times10^{-21} \text{ J}$$
$$\therefore \lambda' = \frac{6.63\times10^{-34}}{\sqrt{2\times6.21\times10^{-21}\times1.66\times10^{-27}}}$$
$$= 1.46\times10^{-10}$$
$$= 0.146 \text{ nm}$$

Therefore, the de-Broglie wavelength of the neutron is 0.146 nm. **[2½]**

Q. 14. What is the de-Broglie wavelength of a nitrogen molecule in air at 300 K? Assume that the molecule is moving with the root-mean square speed of molecules at this temperature. (Atomic mass of nitrogen = 14.0076 u)

[NCERT Ex. Q. 11.19, Page 408]

Ans. Temperature of the nitrogen molecule, T = 300 K
Atomic mass of nitrogen = 14.0076 u
Hence, mass of the nitrogen molecule, $m = 2 \times 14.0076 = 28.0152$ u
But 1 u = 1.66×10^{-27} kg
Therefore, $m = 28.0152 \times 1.66 \times 10^{-27}$ kg
Planck's constant, $h = 6.63 \times 10^{-34}$ Js
Boltzmann constant, $k = 1.38 \times 10^{-23}$ J K^{-1}
We have the expression that relates mean kinetic energy (3/2 kT) of the nitrogen molecule with the root mean square speed (v_{rms}) as :

$$\frac{1}{2}mv^2_{rms} = \frac{3}{2}kT$$
$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

Hence, the de Broglie wavelength of the nitrogen molecule is given as:

$$\lambda = \frac{h}{mv_{rms}} = \frac{h}{\sqrt{3mkT}}$$
$$= \frac{6.63\times10^{-34}}{\sqrt{3\times28.0152\times1.66\times10^{-27}\times1.38\times10^{-23}}}$$
$$= 0.028\times10^{-9} \text{ m}$$
$$= 0.028 \text{ nm}$$

Therefore, the de Broglie wavelength of the nitrogen molecule is 0.028 nm. **[5]**

Q. 15. (a) Estimate the speed with which electrons emitted from a heated emitter of an evacuated tube impinge on the collector maintained at a potential difference of 500 V with respect to the emitter. Ignore the small initial speeds of the electrons. The specific charge of the electron, i.e., its e/m is given to be 1.76×10^{11} C kg^{-1}.

(b) Use the same formula you employ in (a) to obtain electron speed for a collector potential of 10 MV. Do you see what is wrong? In what way is the formula to be modified?

[NCERT Add Ex. Q. 11.20, Page 409]

Ans. **(a)** Potential difference across the evacuated tube, V= 500 V Specific charge of an electron,

$\frac{e}{m} = 1.76 \times 10^{11}$ C kg^{-1}.

The speed of each emitted electron is given by the relation for kinetic energy as

$$KE = \frac{1}{2}mv^2 = eV$$
$$v = \sqrt{\frac{2eV}{m}} = \sqrt{2V\times\frac{e}{m}}$$
$$v = \sqrt{2\times500\times1.76\times10^{11}}$$
$$v = 1.327\times10^{7} \text{ m/s}$$

Therefore, the speed of each emitted electron is 1.327×10^7 m/s. **[2]**

(b) Potential of the anode, V= 10 MV = 10×10^6 V
The speed of each electron is given as :

$$v = \left(2V\times\frac{e}{m}\right)^{\frac{1}{2}}$$
$$= (2\times10^7\times1.76\times10^{11})^{\frac{1}{2}}$$
$$= 1.88\times10^9 \text{ m/s}$$

This result is wrong because nothing can move faster than light. In the above formula, the expression $\left(\frac{mv^2}{2}\right)$ for energy can only be used in the non-relativistic limit, *i.e.*, for $v << c$. For very high speed problems, relativistic equations must be considered for solving them. In the relativistic limit, the total energy is given as :

$E = mc^2$

Where, m = Relativistic mass

m_0 = Rest mass

$$m = m_0\left[1-\frac{v^2}{c^2}\right]^{\frac{1}{2}}$$

$$\text{Relative Momentum } p = mv = \frac{m_0 v}{\sqrt{1-\frac{v^2}{c^2}}} \quad ...(i)$$

$$\text{Total Energy: } E = mc^2 = \frac{m_0 c^2}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$\Rightarrow E = m_0c^2\left|1-\frac{v^2}{c^2}\right|^{-\frac{1}{2}} \quad ...(ii)$$

$$\Rightarrow E = m_0c^2\left|1+\frac{v^2}{2c^2}+\frac{1}{2}\cdot\frac{3}{4}\frac{v^4}{c^4}+.....\right|$$

$$\Rightarrow E = m_0c^2+\frac{1}{2}m_0v^2+.....$$

Thus $E - m_0c^2$, *i.e.*, $mc^2 - m_0c^2$ is called Kinetic Energy.

When $v << c$ then higher order can be neglected.

$$\therefore K.E. = \frac{1}{2}m_0v^2$$ **[3]**

Q. 16. (a) A mono energetic electron beam with electron speed of 5.20×10^6 m s^{-1} is subject to a magnetic field of 1.30×10^{-4} T normal to the beam velocity. What is the radius of the circle traced by the beam, given e/m for electron equals 1.76×10^{11} C kg^{-1}?

(b) Is the formula you employ in (a) valid for calculating radius of the path of a 20 MeV electron beam? If not, in what way is it modified?

[Note : Exercises 11.20(b) and 11.21(b) take you to relativistic mechanics which is beyond the scope of this book. They have been inserted here simply to emphasize the point that the formulas you use in part (a) of the exercises are not valid at very high speeds or energies. See answers at the end to know what 'very high speed or energy' means.]

[NCERT Add Ex. Q. 11.21, Page 409]

Ans. **(a)** Speed of an electron, $v = 5.20 \times 10^6$ m/s

Magnetic field experienced by the electron, $B = 1.30 \times 10^{-4}$ T

Specific charge of an electron, $\frac{e}{m} = 1.76 \times 10^{11}$ C kg^{-1}

Where,

e = Charge on the electron = 1.6×10^{-19} C

m = Mass of the electron = 9.1×10^{-31} kg^{-1}

The force exerted on the electron is given as :

$$F = e\left|\vec{v}\times\vec{B}\right|$$
$$= evB\sin\theta$$

θ = Angle between the magnetic field and the beam velocity.

The magnetic field is normal to the direction of beam.

The beam traces a circular path of radius, r. It is the magnetic field, due to its bending

nature, that provides the centripetal force for the bea :m :

$$F = \frac{mv^2}{r} \quad ...(1)$$

Hence, equation (1) reduces to :

$$evB = \frac{mv^2}{r}$$

$$\therefore\ r = \frac{mv}{eB} = \frac{v}{\left(\frac{e}{m}\right)B}$$
$$= \frac{5.20\times10^6}{(1.76\times10^{11})\times1.30\times10^{-4}}$$
$$= 0.227 \text{ m}$$
$$= 22.7 \text{ cm}$$

Therefore, the radius of the circular path is 22.7 cm. **[2½]**

(b) Energy of the electron beam, $E = 20\,MeV = 20\times10^6 \times 1.6\times10^{-19}\,J$

The energy of the electron is given as :

$$E = \frac{1}{2}mv^2$$

$$\therefore v = \left(\frac{2E}{m}\right)^{\frac{1}{2}}$$
$$= \sqrt{\frac{2\times20\times10^6\times1.6\times10^{-19}}{9.1\times10^{-31}}}$$
$$= 2.652\times10^9 \text{ m/s}$$

This result is incorrect because nothing can move faster than light. In the above formula,

the expression $\left(\frac{mv^2}{2}\right)$ for energy can only be used

in the non-relativistic limit, *i.e.*, for

$v << c$. When very high speeds are concerned, the relativistic domain comes into consideration.

In the relativistic domain, mass is given as :

$$m = m_o\left[1-\frac{v^2}{c^2}\right]^{1/2}$$

m_o = Mass of the particle at rest

Hence, the radius of the circular path is given as :

$$r = \frac{mv}{eB}$$

$$= \frac{m_0 v}{eB\sqrt{\frac{c^2 - v^2}{c^2}}}$$ **[2½]**

Q. 17. An electron gun with its collector at a potential of 100 V fires out electrons in a spherical bulb containing hydrogen gas at low pressure (10^{-2} mm of Hg). A magnetic field of 2.83 × 10^{-4} T curves the path of the electrons in a circular orbit of radius 12.0 cm. (The path can be viewed because the gas ions in the path focus the beam by attracting electrons, and emitting light by electron capture; this method is known as the 'fine beam tube' method. Determine e/m from the data.

[NCERT Add Ex. Q. 11.22, Page 409]

Ans. Potential of an anode, V = 100 V

Magnetic field experienced by the electrons, B = 2.83 × 10^{-4} T

Radius of the circular orbit r = 12.0 cm = 12.0 × 10^{-2} m

Mass of each electron = m

Charge on each electron = e

Velocity of each electron = v

The energy of each electron is equal to its kinetic energy, *i.e.*,

$$eV = \frac{1}{2}mv^2$$

$$\therefore v^2 = \left(\frac{2eV}{m}\right) \quad ...(i)$$

It is the magnetic field, due to its bending nature, that provides the centrifugal force $\left(F = \frac{mv^2}{r}\right)$ for the beam. Hence, we can write :

Centrifugal force = Magnetic force

$$\frac{mv^2}{r} = evB$$

$$eB = \frac{mv}{r}$$

$$v = \frac{eBr}{m} \quad ...(ii)$$

Putting the value of v in eq. (i), we get :

$$\frac{2eV}{m} = \frac{e^2B^2r^2}{m^2}$$

$$\frac{e}{m} = \frac{2V}{B^2r^2}$$

$$= \frac{2 \times 100}{(2.83 \times 10^{-4})^2 \times (12 \times 10^{-2})^2}$$

$$= 1.73 \times 10^{11} \text{ C kg}^{-1}$$

Therefore,

The specific charge ratio $\left(\frac{e}{m}\right)$ is 1.73×10^{11} C kg^{-1} **[5]**

Q. 18. Estimating the following two numbers should be interesting. The first number will tell you why radio engineers do not need to worry much about photons! The second number tells you why our eye can never 'count photons', even in barely detectable light. (a) The number of photons emitted per second by a Medium wave transmitter of 10 kW power, emitting radio waves of wavelength 500 m. (b) The number of photons entering the pupil of our eye per second corresponding to the minimum intensity of white light that we humans can perceive (~10^{-10} W m^{-2}). Take the area of the pupil to be about 0.4 cm^2, and the average frequency of white light to be about 6 × 10^{14} Hz. [NCERT Add Ex. Q. 11.25, Page 409]

Ans. **(a)** Power of the medium wave transmitter, P = 10 kW = 10^4 W = 10^4 J/s.

Hence, energy emitted by the transmitter per second, E = 10^4

Wavelength of the radio wave, λ = 500 m

The energy of the wave is given as :

$$E_1 = \frac{hc}{\lambda}$$

Where, h = Planck's constant = 6.6×10^{-34} Js

c = Speed of light = 3×10^8 m/s

$$\therefore E_1 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{500} = 3.96 \times 10^{-28} \text{ J}$$

Let n be the number of photons emitted by the transmitter.

$$\therefore nE_1 = E$$

$$n = \frac{E}{E_1}$$

$$= \frac{10^4}{3.96 \times 10^{-28}}$$

$$= 2.525 \times 10^{-31}$$

$$n \approx 3 \times 10^{31}$$

The energy (E_1) of a radio photon is very less, but the number of photons (n) emitted per second in a radio wave is very large. The existence of a minimum quantum of energy can be ignored and the total energy of a radio wave can be treated as being continuous. **[2½]**

(b) Intensity of light perceived by the human eye,

I = 10^{-10} W m^{-2} Area of a pupil,

A = 0.4 cm^2 = 0.4 × 10^{-4} m^2

Frequency of white light, ν = 6 × 10^{14} Hz

The energy emitted by a photon is given as :

$$\therefore E = h\nu$$

Where, h = Planck's constant = 6.6×10^{-34} Js

$$\therefore E = 6.6 \times 10^{-34} \times 6 \times 10^{14} = 3.96 \times 10^{-19} \text{ J}$$

Let n be the total number of photons falling per second, per unit area of the pupil.

The total energy per unit for n falling photons is given as :

$$\therefore E = n \times 3.96 \times 10^{-19} \text{ Js}^{-1}\text{m}^{-2}$$

The energy per unit area per second is the intensity of light.

$$\therefore E = I$$

$$n \times 3.96 \times 10^{-19} = 10^{-10}$$

$$n = \frac{10^{-10}}{3.96 \times 10^{-19}}$$

$= 2.52 \times 10^8 \text{ m}^2 s^{-1}$

The total number of photons entering the pupil per second is given as :

$n_A = n \times A$

$= 2.52 \times 10^8 \times 0.4 \times 10^{-4}$

$= 1.008 \times 10^4 \text{ s}^{-1}$

This number is not as large as the one found in problem (a), but it is large enough for the human eye to never see the individual photons. **[2½]**

Q. 19. Ultraviolet light of wavelength 2271 Å from a 100 W mercury source irradiates a photocell made of molybdenum metal. If the stopping potential is −1.3 V, estimate the work function of the metal. How would the photo-cell respond to a high intensity (10^5 W m^{-2}) red light of wavelength 6328 Å produced by a He-Ne laser?

[NCERT Add Ex. Q. 11.26, Page 410]

Ans. Wavelength of ultraviolet light, λ = 2271 Å = 2271 × 10^{-10} m

Stopping potential of the metal, V_0 = 1.3 V

Planck's constant, $h = 6.6 \times 10^{-34}$ Js

Charge on an electron, $e = 1.6 \times 10^{-19}$ C

Work function of the metal = ϕ_o

Frequency of light = ν

We have the photo-energy relation from the photoelectric effect as :

$$\phi_o = h\nu - eV_0$$

$$\phi_0 = \frac{hc}{\lambda} - eV_0$$

Where, h = Planck's constant = 6.6×10^{-34} Js

c = Speed of light = 3×10^8 m/s

$$\phi_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{2271 \times 10^{-10}} - 1.6 \times 10^{-19} \times 1.3$$

$$= 8.72 \times 10^{-19} - 2.08 \times 10^{-19}$$

$$= 6.64 \times 10^{-19} \text{ J}$$

$$= \frac{6.64 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$= 4.15$ eV

let ν_0 be the threshold frequency of the metal.

$$\therefore \phi_0 = h\nu_0$$

$$\nu_0 = \frac{\phi_0}{h} = \frac{6.64 \times 10^{-19}}{6.6 \times 10^{-34}} = 1.006 \times 10^{-15} \text{ Hz}$$

Wavelength of red light, $\lambda_r = 6328$ Å

$$\therefore \text{Frequency of red light, } v_r = \frac{c}{\lambda_r} = 6328 \times 10^{-10} \text{ m}$$

$$= \frac{3 \times 10^8}{6328 \times 10^{-10}} = 4.74 \times 10^{14} Hz$$

Since $\nu_0 > \nu_r$, the photocell will not respond to the red light produced by the laser. **[5]**

Q. 20. Monochromatic radiation of wavelength 640.2 nm (1nm = 10^{-9} m) from a neon lamp irradiates photosensitive material made of cesium on tungsten. The stopping voltage is measured to be 0.54 V. The source is replaced by an iron source and its 427.2 nm line irradiates the same photo-cell. Predict the new stopping voltage.

[NCERT Add Ex. Q. 11.27, Page 410]

Ans. Wavelength of the monochromatic radiation,

λ = 640.2 nm = 640.2 × 10^{-9} m

Stopping potential of the neon lamp, V_0 = 0.54 V

Charge on an electron, $e = 1.6 \times 10^{-19}$ C

Planck's constant, $h = 6.6 \times 10^{-34}$ Js

Let be the work function ϕ_o and ν be the frequency of emitted light.

We have the photo-energy relation from the photoelectric effect as :

$$eV_0 = h\nu - \phi_0$$

$$\phi_0 = \frac{hc}{\lambda} - eV_0$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{640.2 \times 10^{-9}} - 1.6 \times 10^{-19} \times 0.54$$

$$= 3.093 \times 10^{-19} - 0.864 \times 10^{-19}$$

$$= 2.229 \times 10^{-19} \text{ J}$$

$$= \frac{2.229 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$= 1.39$ eV

Wavelength of the radiation emitted from an iron source, λ' = 427.2 nm = 427.2 × 10^{-9} m

Let V'_o be the new stopping potential. Hence, photo-energy is given as :

$$eV_0' = \frac{hc}{\lambda'} - \phi_0$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{427.2 \times 10^{-9}} - 2.229 \times 10^{-19}$$

$$= 4.63 \times 10^{-19} - 2.229 \times 10^{-19}$$

$$= 2.401 \times 10^{-19} \text{ J}$$

$$= \frac{2.401 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$= 1.5$ eV

Hence, the new stopping potential is 1.50 eV. **[5]**

Q. 21. A mercury lamp is a convenient source for studying frequency dependence of photoelectric emission, since it gives a number of spectral lines ranging from the UV to the red end of the visible spectrum. In our experiment with rubidium photo-cell, the following lines from a mercury source were used : λ_1 = 3650 Å, λ_2 = 4047 Å, λ_3 = 4358 Å, λ_4 = 5461 Å, λ_5 = 6907 Å, the stopping voltages, respectively, were measured to be : V_{01} = 1.28 V, V_{02} = 0.95 V, V_{03} = 0.74 V, V_{04} = 0.16 V, V_{05} = 0 V Determine the value of Planck's constant *h*, the threshold frequency and work function for the material.

[Note : You will notice that to get h from the data, you will need to know e (which you can take to be 1.6 × 10^{-19} C). Experiments of this kind on Na, Li, K, etc. were performed by Millikan, who, using his own value of e (from the oil-drop experiment) confirmed

Einstein's photoelectric equation and at the same time gave an independent estimate of the value of *h*.]

[NCERT Add Ex. Q. 11.28, Page 410]

Ans. Einstein's photoelectric equation is given as :

$$eV_o = h\nu - \phi_o$$

$$V_o = \frac{h}{e}\nu - \frac{\phi_o}{e} \quad ...(i)$$

Where,

V_0 = Stopping potential

h = Planck's constant

e = Charge on an electron

ν = Frequency of radiation

ϕ_o = Work function of a material

It can be concluded from equation (1) that potential V_0 is directly proportional to frequency ν.

Frequency is also given by the relation :

$$\nu = \frac{\text{Speed of light(c)}}{\text{Wavelength}(\lambda)}$$

This relation can be used to obtain the frequencies of the various lines of the given wavelengths.

$$\nu_1 = \frac{c}{\lambda_1} = \frac{3\times10^8}{3650\times10^{-19}} = 8.219\times10^{14} \text{ Hz}$$

$$\nu_2 = \frac{c}{\lambda_2} = \frac{3\times10^8}{4047\times10^{-19}} = 7.412\times10^{14} \text{ Hz}$$

$$\nu_3 = \frac{c}{\lambda_3} = \frac{3\times10^8}{4358\times10^{-19}} = 6.884\times10^{14} \text{ Hz}$$

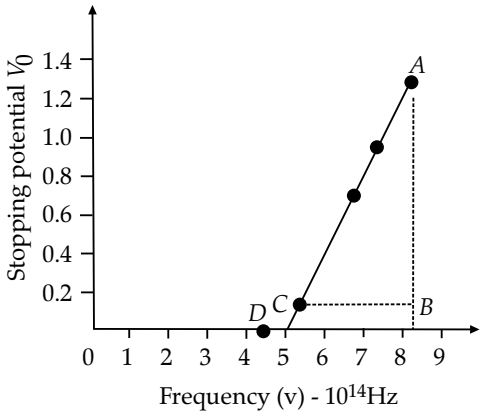
$$\nu_4 = \frac{c}{\lambda_4} = \frac{3\times10^8}{5461\times10^{-19}} = 5.493\times10^{14} \text{ Hz}$$

$$\nu_5 = \frac{c}{\lambda_5} = \frac{3\times10^8}{6907\times10^{-19}} = 4.343\times10^{14} \text{ Hz}$$

The given quantities can be listed in tabular form as :

Frequency×10^{14} Hz	8.219	7.412	6.884	5.493	4.343
Stopping Potential V_0	1.28	0.95	0.74	0.16	0

The following figure shows a graph between ν and V_0.



It can be observed that the obtained curve is a straight line. It intersects the ν-axis at 5×10^{14} Hz, which is the threshold frequency (ν_0) of the material. Point D corresponds to a frequency less than the threshold frequency. Hence, there is no photoelectric emission for the λ_5 line, and therefore, no stopping voltage is required to stop the current.

$$\text{Slope of the straight line} = \frac{AB}{CD}$$

$$= \frac{1.28 - 0.16}{(8.214 - 5.493)\times10^{14}} \quad ...(1)$$

From eq. (1), the slope $\frac{h}{e}$ can be written as:

$$\frac{h}{e} = \frac{1.28 - 0.16}{(8.214 - 5.493)\times10^{14}}$$

$$\therefore h = \frac{1.12\times1.6\times10^{-19}}{2.726\times10^{14}}$$

$$= 6.573\times10^{-34} \text{ Js}$$

The work function of the metal is given as:

$$\phi_0 = h\nu_0$$

$$= 6.573\times10^{-34}\times5\times10^{14}$$

$$= 3.286\times10^{-19} \text{ J}$$

$$= \frac{3.286\times10^{-19}}{1.6\times10^{-19}}$$

$$= 2.054 \text{ eV}$$ **[5]**

Q. 22. The work function for the following metals is given :

***Na* : 2.75 eV; *K* : 2.30 eV; *Mo* : 4.17 eV; *Ni* : 5.15 eV. Which of these metals will not give photoelectric emission for a radiation of wavelength 3300 Å from a He-Cd laser placed 1 m away from the photocell? What happens if the laser is brought nearer and placed 50 cm away?**

[NCERT Add Ex. Q. 11.29, Page 410]

Ans. Mo and Ni will not show photoelectric emission in both cases

Wavelength for a radiation, λ = 3300 Å = 3300 × 10^{-10} m

Speed of light, $c = 3\times10^8$ m/s

Planck's constant, $h = 6.6\times10^{-34}$ Js

The energy of incident radiation is given as :

$$E = \frac{hc}{\lambda}$$

$$= \frac{6.6\times10^{-34}\times3\times10^8}{3300\times10^{-10}}$$

$$= 6\times10^{-19} \text{ J}$$

$$= \frac{6\times10^{-19}}{1.6\times10^{-19}}$$

$$= 3.158 \text{ eV}$$

It can be observed that the energy of the incident radiation is greater than the work function of Na and K only. It is less for Mo and Ni. Hence, Mo and Ni will not show photoelectric emission.

If the source of light is brought near the photocells and placed 50 cm away from them,

then the intensity of radiation will increase. This does not affect the energy of the

radiation. Hence, the result will be the same as before. However, the photoelectrons

emitted from Na and K will increase in proportion to intensity. **[5]**

Q. 23. Light of intensity 10^{-5} W m^{-2} falls on a sodium photo-cell of surface area 2 cm^2. Assuming that the top 5 layers of sodium absorb the incident energy, estimate time required for photoelectric emission in the wave-picture of radiation. The work function for the metal is given to be about 2 eV. What is the implication of your answer?

[NCERT Add Ex. Q. 11.30, Page 410]

Ans. Intensity of incident light, $I = 10^{-5}$ W m^{-2}

Surface area of a sodium photocell, $A = 2\ cm^2 = 2 \times 10^{-4}\ m^2$

Incident power of the light, $P = I \times A$

$= 10^{-5} \times 2 \times 10^{-4}$

$= 2 \times 10^{-9}$ W

Work function of the metal, = 2 eV

$= 2 \times 1.6 \times 10^{-19}$

$= 3.2 \times 10^{-19}$ J

Number of layers of sodium that absorbs the incident energy, $n = 5$

We know that the effective atomic area of a sodium atom, Ae is $10^{-20}\ m^2$.

Hence, the number of conduction electrons in n layers is given as :

$$n' = n \times \frac{A}{A_e} = 5 \times \frac{2 \times 10^{-4}}{10^{-20}} = 10^{17}$$

The incident power is uniformly absorbed by all the electrons continuously. Hence, the

amount of energy absorbed per second per electron is :

$$E = \frac{P}{n'} = \frac{2 \times 10^{-9}}{10^{17}} = 2 \times 10^{-26}\ \text{J/s}$$

Time required for photoelectric emission:

$$t = \frac{\phi_0}{E} = \frac{3.2 \times 10^{-19}}{2 \times 10^{-26}} = 1.6 \times 10^{7}\ \text{s}$$

$t \approx 0.507$ years

The time required for the photoelectric emission is nearly half a year, which is not

practical. Hence, the wave picture is in disagreement with the given experiment. **[5]**

Q. 24. Crystal diffraction experiments can be performed using X-rays, or electrons accelerated through appropriate voltage. Which probe has greater energy? (For quantitative comparison, take the wavelength of the probe equal to 1 Å, which is of the order of interatomic spacing in the lattice) (me = 9.11 × 10^{-31} kg).

[NCERT Add Ex. Q. 11.31, Page 410]

Ans. An X-ray probe has a greater energy than an electron probe for the same wavelength.

Wavelength of light emitted from the probe, λ = 1 Å = 10^{-10} m

Mass of an electron, $m_e = 9.11 \times 10^{-31}$ kg

Planck's constant, $h = 6.6 \times 10^{-34}$ Js

Charge on an electron, e = 1.6×10^{-19} C

The kinetic energy of the electron is given as :

$$E = \frac{1}{2} m_e v^2$$

$$m_e v = \sqrt{2Em_e}$$

where,

v = Velocity of the electron

$m_e v$ = Momentum (p) of the electron

According to the de Broglie principle, the de Broglie wavelength is given as :

$$\lambda = \frac{h}{p} = \frac{h}{m_e v} = \frac{h}{\sqrt{2Em_e}}$$

$$\therefore E = \frac{h^2}{2\lambda^2 m_e}$$

$$= \frac{(6.6 \times 10^{-34})^2}{2 \times (10^{-10})^2 \times 9.11 \times 10^{-31}} = 2.39 \times 10^{-17}\ \text{J}$$

$$= \frac{2.39 \times 10^{-17}}{1.6 \times 10^{-19}} = 149.375\ \text{eV}$$

Energy of a photon, $E' = \frac{hc}{\lambda e}$ eV

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{10^{-10} \times 1.6 \times 10^{-19}}$$

$= 12.375 \times 10^3$ eV

= 12.375 keV

Hence, a photon has a greater energy than an electron for the same wavelength. **[5]**

Q. 25. (a) Obtain the de Broglie wavelength of a neutron of kinetic energy 150 eV. As you have seen in Exercise 11.31, an electron beam of this energy is suitable for crystal diffraction experiments. Would a neutron beam of the same energy be equally suitable? Explain. (m_n = 1.675 × 10^{-27} kg)

(b) Obtain the de Broglie wavelength associated with thermal neutrons at room temperature (27 °C). Hence explain why a fast neutron beam needs to be thermalized with the environment before it can be used for neutron diffraction experiments.

[NCERT Add Ex. Q. 11.32, Page 410]

Ans. **(a)** De Broglie wavelength = 2.327×10^{-12} m; neutron is not suitable for the diffraction experiment

Kinetic energy of the neutron, K = 150 eV

$= 150 \times 1.6 \times 10^{-19}$

$= 2.4 \times 10^{-17}$ J

Mass of a neutron, $m_n = 1.675 \times 10^{-27}$ kg

The kinetic energy of the neutron is given by the relation :

$$K=\frac{1}{2}m_n v^2$$

$$m_n v=\sqrt{2\,\mathrm{K}m_n}$$

Where,

v = Velocity of the neutron

$m_n v$ = Momentum of the neutron

De-Broglie wavelength of the neutron is given as :

$$\lambda=\frac{h}{m_n v}=\frac{h}{\sqrt{2\mathrm{K}m_n}}$$

It is clear that wavelength is inversely proportional to the square root of mass. Hence, wavelength decreases with increase in mass and vice versa.

$$\lambda=\frac{6.6\times10^{-34}}{\sqrt{2\times2.4\times10^{-17}\times1.675\times10^{-27}}}$$

$$=2.327\times10^{-12}\ \mathrm{m}$$

It is given in the previous problem that the inter-atomic spacing of a crystal is about 1 Å,

i.e., 10^{-10} m. Hence, the inter-atomic spacing is about a hundred times greater. Hence,

a neutron beam of energy 150 eV is not suitable for diffraction experiments. **[3]**

(b) De Broglie wavelength $=1.477\times10^{-10}$ m

Room temperature, $T = 27°\mathrm{C} = 27 + 273 = 300$ K

The average kinetic energy of the neutron is given as :

$$E=\frac{3}{2}k\mathrm{T}$$

where,

k = Boltzmann Constant$=1.38\times10^{-23}\ \mathrm{J\,mol^{-1}\,K^{-1}}$

The wavelength of the neutron is given as:

$$\lambda=\frac{h}{\sqrt{2m_n\mathrm{E}}}=\frac{h}{\sqrt{3m_n\mathrm{kT}}}$$

$$=\frac{6.6\times10^{-34}}{\sqrt{3\times1.675\times10^{-27}\times1.38\times10^{-23}\times300}}$$

$$=1.447\times10^{-10}\ \mathrm{m}$$

This wavelength is comparable to the inter-atomic spacing of a crystal. Hence, the high energy neutron beam should first be thermalized, before using it for diffraction. **[2]**

Q. 26. An electron microscope uses electrons accelerated by a voltage of 50 kV. Determine the de Broglie wavelength associated with the electrons. If other factors (such as numerical aperture, etc.) are taken to be roughly the same, how does the resolving power of an electron microscope compare with that of an optical microscope which uses yellow light? [NCERT Add Ex. Q. 11.33, Page 411]

Ans. Electrons are accelerated by a voltage, $V = 50\,\mathrm{kV} = 50 \times 10^3$ V

Charge on an electron, $e = 1.6 \times 10^{-19}$ C

Mass of an electron, $m_e = 9.11 \times 10^{-31}$ kg

Wavelength of yellow light $= 5.9 \times 10^{-7}$ m

The kinetic energy of the electron is given as :

$E = eV$

$= 1.6 \times 10^{-19} \times 50 \times 10^3$

$= 8 \times 10^{-15}$ J

De-Broglie wavelength is given by the relation :

$$\lambda=\frac{h}{\sqrt{2m_n E}}$$

$$=\frac{6.6\times10^{-34}}{\sqrt{2\times9.11\times10^{-31}\times8\times10^{-15}}}$$

$$=5.467\times10^{-12}\ \mathrm{m}$$

This wavelength is nearly 10^5 times less than the wavelength of yellow light.

The resolving power of a microscope is inversely proportional to the wavelength of light used. Thus, the resolving power of an electron microscope is nearly 10^5 times that of an optical microscope. **[5]**

Q. 27. The wavelength of a probe is roughly a measure of the size of a structure that it can probe in some detail. The quark structure of protons and neutrons appears at the minute length scale of 10^{-15} m or less. This structure was first probed in early 1970's using high energy electron beams produced by a linear accelerator at Stanford, USA. Guess what might have been the order of energy of these electron beams. (Rest mass energy of electron = 0.511 MeV.)

[NCERT Add Ex. Q. 11.34, Page 411]

Ans. Wavelength of a proton or a neutron, $\lambda \approx 10^{-15}$

m Rest mass energy of an electron :

$m_0c^2 = 0.511$ MeV

$= 0.511 \times 10^6 \times 1.6 \times 10^{-19}$

$= 0.8176 \times 10^{-13}$ J

Planck's constant, $h = 6.6 \times 10^{-34}$ Js

Speed of light, $c = 3 \times 10^8$ m/s

The momentum of a proton or a neutron is given as :

$$p=\frac{h}{\lambda}$$

$$=\frac{6.6\times10^{-34}}{10^{-15}}$$

$$=6.6\times10^{-19}\ \mathrm{kg\,m/s}$$

The relativistic relation for energy (E) is given as :

$$E^2=p^2c^2+m_0^2c^4$$

$$=(6.6\times10^{-19}\times3\times10^8)^2+(0.8176\times10^{-13})^2$$

$$=392.04\times10^{-22}+0.6685\times10^{-26}$$

$$\approx392.04\times10^{-22}$$

$$\therefore E=1.98\times10^{-10}\ \mathrm{J}$$

$$=\frac{1.98\times10^{-10}}{1.6\times10^{-19}}$$

$$=1.24\times10^{9}\ \mathrm{eV}$$

$$=1.24\ \mathrm{BeV}$$

Thus, the electron energy emitted from the accelerator at Stanford, USA might be of the order of 1.24 BeV. **[5]**

Q. 28. Find the typical de Broglie wavelength associated with a He atom in helium gas at room temperature (27°C) and 1 atm pressure; and compare it with the mean separation between two atoms under these conditions.

[NCERT Add Ex. Q. 11.35, Page 411]

Ans. De Broglie wavelength associated with He atom = 0.7268×10^{-10} m

Room temperature, T = 27°C = 27 + 273 = 300 K

Atmospheric pressure, P = 1 atm = 1.01×10^5 Pa

Atomic weight of a He atom = 4

Avogadro's number, $N_A = 6.023 \times 10^{23}$

Boltzmann constant, $k = 1.38 \times 10^{-23}$ J mol^{-1} K^{-1}

Average energy of a gas at temperature T, is given as :

$$E = \frac{3}{2}kT$$

De Broglie wavelength is given by the relation :

$$\lambda = \frac{h}{\sqrt{2mE}}$$

where,

$$m = \text{ mass of a He atom} = \frac{\text{Atomic weight}}{N_A}$$

$$= \frac{4}{6.023 \times 10^{23}}$$

$$= 6.64 \times 10^{-24} \text{ g}$$

$$= 6.64 \times 10^{-27} \text{ kg}$$

$$\therefore \lambda = \frac{h}{\sqrt{3mkT}}$$

$$= \frac{6.6 \times 10^{-34}}{\sqrt{3 \times 6.64 \times 10^{-27}} \times 1.38 \times 10^{-23} \times 300}$$

$$= 0.7268 \times 10^{-10} \text{ m}$$

We have the ideal gas formula :

$PV = RT$

$PV = kNT$

$$\frac{V}{N} = \frac{kT}{P}$$

Where,

V = Volume of the gas

N = Number of moles of the gas

Mean separation between two atoms of the gas is given by the relation :

$$r = \left(\frac{V}{N}\right)^{\frac{1}{3}} = \left(\frac{kT}{P}\right)^{\frac{1}{3}}$$

$$= \left[\frac{1.38 \times 10^{-23} \times 300}{1.01 \times 10^5}\right]^{\frac{1}{3}}$$

$$= 3.35 \times 10^{-9} \text{ m}$$

Hence, the mean separation between the atoms is much greater than the de Broglie wavelength. **[5]**

Some Commonly Made Errors

- Students get confuse between electron emission and photon emission.
- Students don't know that free electrons in a metal do not all have the same energy. They have different energy levels according to material.
- Students get confuse in between wave-picture and photon-picture of photoelectric effect.
- Student don't remember that phase velocity v_p has no physical significance. Whereas, the group velocity of the matter wave is physically meaningful and equals the velocity of the particle.

EXPERT ADVICE

- *Make notes on important topics like work function, photoelectric effect, Einstein's photoelectric equation, photon, wave nature of matter, etc.*
- *Make a complete list of derivations, formulae, and experiments in your syllabus and keep that list in your hand.*
- *While solving a derivation, try and comprehend the logic behind the derivations.*
- *During the preparation of syllabus, try to write your descriptive answers in points and give pictorial or graphical illustrations wherever possible. It enhances visibility.*

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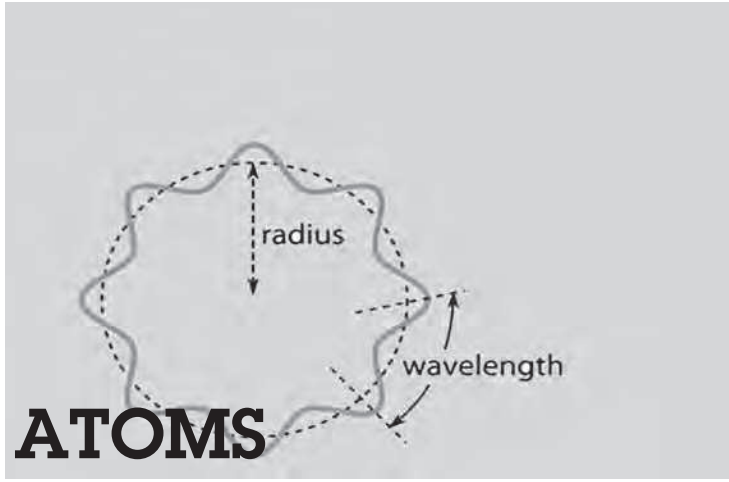
CHAPTER

12 ATOMS



Chapter Objectives

This chapter will help you understand :

- ***Atoms :*** *Introduction; Alpha-particle scattering; Rutherford's Nuclear Model; Atomic spectra; Bohr's model of hydrogen atom; The line spectra of the hydrogen atom; de-Broglie's explanation of Bohr's second postulate of quantisation.*

Quick Review

- Matter has mass and takes up space. Atoms are the basic building blocks of matter and cannot be chemically sub-divided by ordinary means.
- The most abundant types of atom in the universe is the hydrogen atom. Almost 74% of the atoms in the Milky Way galaxy are hydrogen atoms.
- **The Wave Model and Quantum Theory :** The Bohr's model was only able to explain the very simplest atoms, like hydrogen. But in the modern day theory is based on mathematics and the properties of waves.
- The wave model forms the basis for the Quantum Theory. This theory gives the probability of locating electrons in a particular location, unlike assuming electron's orbit the nucleus as in the Bohr's model.
- The Bohr's model of the atom (1913), which looks like a solar system, has been replaced by the more accurate Quantum Model of the Atom since 1927.
- In other words, **Schrodinger's model of the atom** (the quantum model that shows "electron clouds ") is the most accurate representation of an atom.
- Balmer was a mathematics teacher at a secondary school for girls in Switzerland; he was 60 years old when he wrote the paper on the spectral lines of hydrogen that made him famous. Balmer published only one other paper on the topic, which appeared when he was 72 years old.

TIPS...

- *Each electron shell has a maximum number of electrons that can reside in it. The first shell can only have two electrons, while the next two shells have a maximum of eight electrons. The third shell is capable of holding up to 18 electrons if necessary.*
- *The energy level occupied by an electron usually is called the ground state.*
- *These states of allowed electronic motion are those in which the angular momentum of an electron is an integral multiple of $h/2\pi$ or one can say that the angular momentum of an electron is quantised.*

TRICKS...

- *All lines in the visible region are of Balmer series but reverse is not true, i.e., all the Balmer's lines will not fall in the visible region.*
- *In the Rydberg's formula, when $n_2 = \infty$ the line produced is called the limiting line of that series.*

- The French Physicist Louis de Broglie, in 1924 suggested that besides the particle properties, particles may also exhibit wave properties. It took around 3 years to spot the wave nature of electrons.
- **Louis de Broglie** achieved a worldwide reputation for his discovery of the wave theory of matter, for which he received the **Nobel Prize** for physics in 1929. His work was extended into a full-fledged wave mechanics by Erwin Schrödinger and thus contributed to the creation of quantum mechanics.

Know the Links

- *http://www.3rd1000.com/chem101/chem104h.htm*
- *https://www.encyclopedia.com/science-and-technology/physics/physics/spectrum*

Multiple Choice Questions (1 mark each)

Q. 1. O_2 molecule consists of two oxygen atoms. In the molecule, nuclear force between the nuclei of the two atoms

(a) is not important because nuclear forces are short-ranged.

(b) is as important as electrostatic force for binding the two atoms.

(c) cancels the repulsive electrostatic force between the nuclei.

(d) is not important because oxygen nucleus have equal number of neutrons and protons.

[NCERT Exemp. Q. 12.5, Page 76]

Ans. Correct option : (a)

Explanation : As we know that the nuclear forces is too much stronger. Only attractive force as compared to electrostatic repulsive force and nuclear force decreases to zero on increasing distance.

So in case of oxygen molecule, the distance between atoms of oxygen is larger as compared to the distances between nucleons in a nucleus. So that, the force between the nuclei of two oxygen atoms is not important as nuclear forces are short-ranged forces.

Q. 2. A set of atoms in an excited state decays.

(a) in general, to any of the states with lower energy.

(b) into a lower state only when excited by an external electric field.

(c) all together simultaneously into a lower state.

(d) to emit photons only when they collide.

[NCERT Exemp. Q. 12.7, Page 76]

Ans. Correct option : (a)

Explanation : A set of atoms in an excited state decays in general to any of the states with lower energy.

Q. 3. Two H atoms in the ground state collide inelastically. The maximum amount by which their combined kinetic energy is reduced is

(a) 10.20 eV. **(b) 20.40 eV.**

(c) 13.6 eV. **(d) 27.2 eV.**

[NCERT Exemp. Q. 12.6, Page 76]

Ans. Correct option : (a)

Explanation : Total energy of two H-atom in ground state = $2(-13.6) = -27.2$ eV.

The maximum amount by which their combined kinetic energy is reduced when any one H-atom goes into first excited state after the inelastic collision, that is, the total energy of two H-atom after inelastic collision :

$$E = \frac{13.6}{n^2} + 13.6$$

$$= \frac{13.6}{2^2} + 13.61 \text{ [For excited state } (n = 2)]$$

$$= 3.4 + 13.6 = 17.0 \text{ eV}$$

So that the loss in kinetic energy due to inelastic collision will be,

$= 27.2 - 17.0 = 10.2$ eV

Q. 4. Taking the Bohr radius as $a_0 = 53$ pm, the radius of Li^{++} ion in its ground state, on the basis of Bohr's model, will be about

(a) 53 pm. **(b) 27 pm.**

(c) 18 pm. **(d) 13 pm.**

[NCERT Exemp. Q. 12.1, Page 75]

Ans. Correct option : (c)

Explanation : According to **Bohr's model of an atom**, radius of an atom in its ground state is

$$r = \frac{r_0}{Z}$$

where, r_0 is Bohr's radius and Z is atomic number.

As given that,

$r_0 = 53$ pm and atomic number of Lithium atom is 3

so, $r = \frac{53}{3} = 17.67 \text{ pm} \approx 18 \text{ pm}$

Q. 5. The binding energy of a H-atom, considering an electron moving around a fixed nuclei (proton), is $B = -\frac{me^4}{8n^2\varepsilon_0^2h^2}$ (m = electron mass). If one decides to work in a frame of reference where the electron is at rest, the proton would be moving around it. By similar arguments, the binding energy would be

$$B = -\frac{Me^4}{8n^2\varepsilon_0^2h^2} \ (M = \text{proton mass})$$

This last expression is not correct because

(a) n would not be integral.

(b) Bohr-quantisation applies only to electron

(c) the frame in which the electron is at rest is not inertial.

(d) the motion of the proton would not be in circular orbits, even approximately.

[NCERT Exemp. Q. 12.2, Page 75]

Ans. Correct option : (c)

Explanation : In a hydrogen atom, electrons revolving around a fixed proton nucleus have some centripetal acceleration. So that, its frame of reference is non-inertial. If the frame of reference, where the electron is at rest, the given expression is not true as it forms the non-inertial frame of reference.

As the mass of an electron is negligible as compared to proton, so the centripetal force cannot provide the electrostatic force,

$$F_p = \frac{m_p v^2}{r}$$

So the given expression is not true, as it forms non-inertial frame of reference due to $m_e <<< m_p$ or centripetal force on $F_e <<< F_p$.

Q. 6. The simple Bohr model cannot be directly applied to calculate the energy levels of an atom with many electrons. This is because

(a) of the electrons not being subject to a central force.

(b) of the electrons colliding with each other

(c) of screening effects

(d) the force between the nucleus and an electron will no longer be given by Coulomb's law.
[NCERT Exemp. Q. 12.3, Page 76]

Ans. Correct option : (a)
Explanation : The simple Bohr model cannot be directly applied to calculate the energy levels of an atom with many electrons because when we derive the formula for radius/energy levels, etc., we make the assumption that centripetal force is provided only by electrostatic force of attraction by the nucleus.
So that, this will only work for single electron atoms. In multi-electron atoms, there will also be repulsion due to other electrons. The simple Bohr model cannot be directly applied to calculate the energy levels of an atom with many electrons.

Q. 7. For the ground state, the electron in the H-atom has an angular momentum = h, according to the simple Bohr model. Angular momentum is a vector and hence there will be infinitely many orbits with the vector pointing in all possible directions. In actuality, this is not true,
(a) because Bohr model gives incorrect values of angular momentum.
(b) because only one of these would have a minimum energy.
(c) angular momentum must be in the direction of spin of electron.
(d) because electrons go around only in horizontal orbits. [NCERT Exemp. Q. 12.4, Page 76]

Ans. Correct option : (a)
Explanation : According to Bohr's second postulate of atomic model, angular momentum of revolving electron must be some integral multiple of $\frac{h}{2\pi}$. So the Bohr's model does not give correct value of angular momentum.

Q. 8. Choose the correct alternative from the clues given at the end of each statement :
(a) The size of the atom in Thomson's model is ..... the atomic size in Rutherford's model. (much greater than/no different from/much less than)
(b) In the ground state of ....., electrons are in stable equilibrium, while in ..... electrons always experience a net force. (Thomson's model/ Rutherford's model)
(c) A classical atom based on ..... is doomed to collapse. (Thomson's model/Rutherford's model)
(d) An atom has a nearly continuous mass distribution in ..... but has a highly non-uniform mass distribution in ..... . (Thomson's model/ Rutherford's model)
(e) The positively charged part of the atom possesses most of the mass in ..... . (Rutherford's model/ both the models) [NCERT Ex. Q. 12.1, Page 435]

Ans. (a) The sizes of the atoms taken in Thomson's model is **not different from** the atomic size in Rutherford's model.
(b) In the ground state of **Thomson's model**, the electrons are in stable equilibrium, while in Rutherford's model electrons always experience a net force.
(c) A classical atom based on **Rutherford's model** is doomed to collapse.
(d) An atom has a nearly continuous mass distribution in **Thomson's model**, but has a highly non-uniform mass distribution in Rutherford's model.
(e) The positively charged part of the atom possesses most of the mass in **both the models**.

Tick Two or More Options

Q. 9. Consider aiming a beam of free electrons towards free protons. When they scatter, an electron and a proton cannot combine to produce a H-atom,
(a) because of energy conservation.
(b) without simultaneously releasing energy in the form of radiation.
(c) because of momentum conservation.
(d) because of angular momentum conservation.
[NCERT Exemp. Q. 12.9, Page 77]

Ans. Correct option : (a) and (b)
Explanation : The binding energy of H-atom is larger as compared to the energy of free electron. Large amount of energy is required so that electron can reach near the proton which is possible when nuclear force (attractive) acts between 'e' and 'p' become of order of nuclear force, it is not possible without energy conservation (a) and (b).

Q. 10. An ionized H-molecule consists of an electron and two protons. The protons are separated by a small distance of the order of angstrom. In the ground state,
(a) the electron would not move in circular orbits.
(b) the energy would be $(2)^4$ times that of a H-atom.
(c) the electrons, orbit would go around the protons.
(d) the molecule will soon decay in a proton and a H-atom. [NCERT Exemp. Q. 12.8, Page 77]

Ans. Correct options (a) and (c)
Explanation : In H-molecule two nucleus of 2 H-atoms are separated of the order of angstrom, that is, of order of nuclear forces. One electron will revolve around the protons. Hence verifies answer (a) and (c).

Q. 11. The Bohr model for the spectra of a H-atom
(a) will not be applicable to hydrogen in the molecular from.
(b) will not be applicable as it is for a He-atom.
(c) is valid only at room temperature.
(d) predicts continuous as well as discrete spectral lines.
[NCERT Exemp. Q. 12.10, Page 77]

Ans. Correct option : (a) and (b)
Explanation : Not applicable for H molecule or He atom. It does not depend on minor change in temperature.

Q. 12. The Balmer series for the H-atom can be observed
(a) if we measure the frequencies of light emitted when an excited atom falls to the ground state.
(b) if we measure the frequencies of light emitted due to transitions between excited states and the first excited state.

(c) in any transition in a H-atom.

(d) as a sequence of frequencies with the higher frequencies getting closely packed.

[NCERT Exemp. Q. 12.11, Page 77]

Ans. Correct option : (b) and (d)

Explanation : When electron jumps from higher energy level to first energy level, the spectrum of light is called Balmer series.

Spectrum lines become closer if electron jumps from higher energy level to ground level. So, it verifies both answers (b) and (d).

Q. 13. Let

$$E_n = \frac{-1}{8\varepsilon_0^2}\frac{me^4}{n^2h^2}$$

be the energy of the nth level of H-atom. If all the H-atoms are in the ground state and radiation of frequency $\frac{(E_2 - E_1)}{h}$ falls on it,

(a) it will not be absorbed at all

(b) some of atoms will move to the first excited state.

(c) all atoms will be excited to the $n = 2$ state.

(d) no atoms will make a transition to the $n = 3$ state.

[NCERT Exemp. Q. 12.12, Page 77]

Ans. Correct option : (b) and (d)

Explanation : Let assume E_2 and E_1 be the energy corresponding to $n = 2$ and $n = 1$, respectively. If radiation of energy,

$\Delta E = (E_2 - E_1) = h\nu$

incident on a sample where all the H-atoms are in the ground state, according to Bohr model some of the atoms will move to the first excited state. As this energy is not sufficient for transition from $n = 1$ to $n = 3$, hence no atoms will make a transition to the $n = 3$ state.

Q. 14. The simple Bohr model is not applicable to He^4 atom because

(a) He^4 is an inert gas.

(b) He^4 has neutrons in the nucleus.

(c) He^4 has one more electron.

(d) electrons are not subject to central forces.

[NCERT Exemp. Q. 12.13, Page 78]

Ans. Correct option : (c) and (d)

Explanation : Bohr's atomic model is applicable only for one electron and in He^4 there are two electrons. Electrons are not subject to central forces due to longer distance than nuclear size, so it verifies answer (c) and (d).

Very Short Answer Type Questions **(1 or 2 marks each)**

Q. 1. A proton and an α-particle are accelerated through the same potential difference. Which one of the two has

(i) greater de-Broglie wavelength, and

(ii) less kinetic energy. Justify your answer.

[CBSE Board, Delhi Region, 2016]

Ans. Let assume that when a charge particle is accelerated through 'V' potential difference then its kinetic energy 'KE'

KE = qV (q = charge; V = potential difference)

Proton and α-particle accelerated through same potential difference so

As, $q_\alpha > q_{\text{proton}}$ $\quad (q_\alpha = +2e \text{ and } q_{\text{proton}} = +e)$

$\text{KE}_\alpha > \text{KE}_{\text{proton}}$

their de-Broglie wavelength, $\lambda = \frac{h}{\sqrt{2mqV}}$ **[½]**

V = same and

$m_\alpha q_\alpha > m_p q_p$ $\quad [m_\alpha = 4\, m_p]$

So,

$\lambda_\alpha < \lambda_p$

where, λ_α=De-Broglie wavelength of α-particle and

λ_p = De-Broglie wavelength of proton

So, that

(i) Proton have greater de-Broglie wavelength, but

(ii) Proton has lesser kinetic energy. **[1½]**

Q. 2. Suppose you are given a chance to repeat the alpha-particle scattering experiment using a thin sheet of solid hydrogen in place of the gold foil. (Hydrogen is a solid at temperatures below 14 K.) What results do you expect?

[NCERT Ex. Q. 12.2, Page 435]

Ans. In the alpha-particle scattering experiment, if a thin sheet of solid hydrogen is used in place of a gold foil, then the scattering angle would not be large enough. This is because the mass of hydrogen (1.67×10^{-27} kg) is less than the mass of incident α-particles (6.64×10^{-27} kg). **[1]**

So that, the mass of the scattering particle is more than the mass of target nucleus (hydrogen). As a result, the α-particles would not bounce back if solid hydrogen is used in the α-particle scattering experiment. **[1]**

Q. 3. Write two important limitations of Rutherford nuclear model of the atom.

[CBSE Board, Delhi Region, 2017]

Ans. Following are the important limitations of Rutherford nuclear model of the atom :

(i) Inconsistent : As the revolving electron- loses energy continuously, so that it must spiral inwards and eventually fall into the nucleus. **[1]**

(ii) Spectrum : Atoms should emit continuous spectrum, but what we observe is only a line spectrum. **[1]**

Q. 4. Answer the following questions, which help you understand the difference between Thomson's model and Rutherford's model better.

(a) Is the average angle of deflection of α-particles by a thin gold foil predicted by Thomson's model much less, about the same, or much greater than that predicted by Rutherford's model?

(b) Is the probability of backward scattering (*i.e.*, scattering of α-particles at angles greater than 90°) predicted by Thomson's model much less, about the same, or much greater than that predicted by Rutherford's model?

(c) Keeping other factors fixed, it is found experimentally that for small thickness *t*, the number of α-particles scattered at moderate angles is proportional to *t*. What clue does this linear dependence on t provide?

(d) In which model is it completely wrong to ignore multiple scattering for the calculation of average angle of scattering of α-particles by a thin foil?

[NCERT Add. Ex. Q. 12.11, Page 436]

Ans. (a) about the same

The average angle of deflection of α-particles by a thin gold foil predicted by Thomson's model is about the same size as predicted by Rutherford's model. This is because the average angle was taken in both models. **[½]**

(b) much less

The probability sof scattering of α-particles at angles greater than predicted by Thomson's model is much less than that predicted by Rutherford's model. **[½]**

(c) Scattering is mainly due to single collisions. The chances of a single collision increase linearly with the number of target atoms. Since the number of target atoms increase with an increase in thickness, the collision probability depends linearly on the thickness of the target. **[½]**

(d) Thomson's model

It is wrong to ignore multiple scattering in Thomson's model for the calculation of average angle of scattering of α-particles by a thin foil. This is because a single collision causes very little deflection in this model. So, the observed average scattering angle can be explained only by considering multiple scattering. **[½]**

Q. 5. The mass of a H-atom is less than the sum of the masses of a proton and electron. Why is this?

[NCERT Exemp. Q. 12.14, Page 78]

Ans. According to the mass energy equivalence established by Einstein,

$E = mc^2$ **[1]**

If B.E represents binding energy of hydrogen atom (=13 eV), the equivalent mass of this energy

$= \frac{\text{B.E}}{c^2}$

Mass of a H-atom $= m_p + m_e - \frac{\text{B.E}}{c^2}$

which is clearly less than sum of the masses of a proton and an electron. **[1]**

Q. 6. When is Hα line in the emission spectrum of hydrogen atom obtained? Calculate the frequency of the photon emitted during this transition.

[CBSE Board, Delhi Region, 2016]

Ans. It is clear from the question that-

H_α line is obtained when electron make transition from $n = 3$ to $n = 2$. **[1]**

According to Bohr :

$E_3 - E_2 = h\nu$

$$\Rightarrow -\frac{13.6}{3^2} - \left(-\frac{13.6}{2^2}\right) = \frac{h\nu}{1.6\times10^{-19}}$$

$$\Rightarrow \quad 13.6\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = \frac{h\nu}{1.6\times10^{-19}}$$

$$\Rightarrow \qquad 13.6\left[\frac{5}{36}\right] = \frac{6.63\times10^{-34}\,\nu}{1.6\times10^{-19}}$$

$$\nu = \frac{0.4558\times10^{-19}}{10^{-34}}$$

$\Rightarrow 0.4558\times10^{15}$ Hz

or 4.558×10^{14} Hz **[1]**

Q. 7. Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from $n = \infty$ to $n = 1$. [CBSE Board, Delhi Region, 2016]

Ans. Given that,

In hydrogen atom, wavelength of radiation for transition $n = \infty$ to $n = 1$.

It is given as :

$$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_L^2} - \frac{1}{n_H^2}\right] \qquad \left\{\begin{array}{l|l}\text{Here} & \text{For}\\ n_L = 1 & \text{H-atom}\\ n_H = \infty & Z = 1\end{array}\right\}$$

$$\frac{1}{\lambda} = R(1)^2\left[\frac{1}{1^2} - \frac{1}{\infty}\right]$$

$$\frac{1}{\lambda} = R \Rightarrow \lambda = \frac{1}{R}$$

$\lambda \approx 912$ Å $\qquad \left\{R = 1.09\times10^7\,\text{m}^{-1}\right\}$ **[1]**

Q. 8. Imagine removing one electron from He^4 and He^3. Their energy levels, as worked out on the basis of Bohr model will be very close. Explain why.

[NCERT Exemp. Q. 12.15, Page 78]

Ans. If we remove one electron from the isotopes of He^4 and He^3, both atoms will have one electron as in H-atom and the nucleus is much (four times) heavier than H-atom. **[1]**

So, stability can remain and these atoms are very close to the H-atom so the energy levels are as of hydrogen and these atoms will be very close. **[1]**

Q. 9. When an electron falls from a higher energy to a lower energy level, the difference in the energies appears in the form of electromagnetic radiation. Why cannot it be emitted as other forms of energy?

[NCERT Exemp. Q. 12.16, Page 78]

Ans. When charge, *i.e.*, electron jumps from higher to lower energy level; there is acceleration in charged particle. The accelerated charge particle can produce electromagnetic wave only. **[2]**

Q. 10. Would the Bohr formula for the H-atom remain unchanged if proton had a charge $\left(+\frac{4}{3}\right)e$ and electron a charge $\left(-\frac{3}{4}\right)e$,where $e = 1.6 \times 10^{-19}$

C. Give reasons for your answer.

[NCERT Exemp. Q. 12.17, Page 78]

Ans. Yes. According to Bohr, for an electron around a stationary nucleus the electrostatics force of attraction provides the necessary centripetal force.

$\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2} = \frac{mv^2}{r}$ **[1]**

So, the magnitude of electrostatic force,

$F \propto q_1 \times q_2$

If proton had a charge $\left(+\frac{4}{3}\right)e$ and electron a charge $\left(-\frac{3}{4}\right)e$, then the Bohr formula for the H-atom remains same, since the Bohr formula involves only the product of the charges which remain constant for given values of charges. **[1]**

Q. 11. Consider two different hydrogen atoms. The electron in each atom is in an excited state. Is it possible for the electrons to have different energies but the same orbital angular momentum according to the Bohr model?

[NCERT Exemp. Q. 12.18, Page 78]

Ans. In excited state of electrons of two H-atoms, electrons may be in orbit or energy level either n = 2, 3 …. and can have same energy but angular momentum by Bohr's model is $L = \frac{nh}{2\pi}$ **[1]**

As n for both may be different so both H-atom will have different angular momentum. **[1]**

Q. 12. In accordance with the Bohr's model, find the quantum number that characterises the earth's revolution around the sun in an orbit of radius 1.5×10^{11} m with orbital speed 3×10^4 m/s. (Mass of the earth = 6.0×10^{24} kg).

[NCERT Ex. Q. 12.10, Page 436]

Ans. Given that,

Radius of the orbit of the Earth around the Sun, $r = 1.5 \times 10^{11}$ m

Orbital speed of the Earth, $v = 3 \times 10^4$ m/s

Mass of the Earth, $m = 6.0 \times 10^{24}$ kg

According to Bohr's model, angular momentum is quantised and given as :

$$mvr = \frac{nh}{2\pi}$$ **[1]**

Where,

h = Planck's constant = 6.62×10^{-34} Js

n = Quantum number

As we know that,

$$n = \frac{mvr2\pi}{h}$$

$$= \frac{2\pi \times 6 \times 10^{24} \times 3 \times 10^4 \times 1.5 \times 10^{11}}{6.62 \times 10^{-34}}$$

$$= 25.61 \times 10^{73}$$

$$= 2.6 \times 10^{74}$$

So, the quantum number that characterises the Earth' revolution is 2.6×10^{74}. **[1]**

Q. 13. The ground state energy of hydrogen atom is –13.6 eV. What are the kinetic and potential energies of the electron in this state?

[NCERT Ex. Q. 12.5, Page 436]

Ans. Given that,

Ground state energy of hydrogen atom,

$E = -13.6$ eV

This is the total energy of a hydrogen atom.

Kinetic energy (KE) is equal to the negative of the total energy.

Kinetic energy $= -(-13.6) = 13.6$ eV **[1]**

Potential energy (PE) is equal to the negative of two times of kinetic energy.

Potential energy $= -2 \times (13.6) = -27.2$ eV **[1]**

Q. 14. What is the shortest wavelength present in the Paschen series of spectral lines?

[NCERT Ex. Q. 12.3, Page 436]

Ans. According to the Rydberg's formula :

$$\frac{hc}{\lambda} = 21.76 \times 10^{-19}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Where,

h = Planck's constant = 6.6×10^{-34} Js

c = Speed of light = 3×10^3 m/s

(n_1 and n_2 are integers) **[1]**

The shortest wavelength present in the Paschen series of the spectral lines is given for values $n_1 = 3$ and $n_2 = \infty$.

$$\frac{hc}{\lambda} = 21.76 \times 10^{-19}\left[\frac{1}{(3)^2} - \frac{1}{(\infty)^2}\right]$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8 \times 9}{21.76 \times 10^{-19}}$$

$$= 8.189 \times 10^{-7} \text{ m}$$

or

$$\lambda = 818.9 \text{ nm}$$ **[1]**

Q. 15. A difference of 2.3 eV separates two energy levels in an atom. What is the frequency of radiation emitted when the atom makes a transition from the upper level to the lower level?

[NCERT Ex. Q. 12.4, Page 436]

Ans. Separation of two energy levels in an atom,

$E = 2.3$ eV

$= 2.3 \times 1.6 \times 10^{-19}$

$= 3.68 \times 10^{-19}$ J

Let ν be the frequency of radiation emitted when the atom transits from the upper level to the lower level. **[1]**

As we the relation for energy as :

$E = h\nu$

Where,

h = Planck's constant= 6.62×10^{-34} Js

$$\therefore \nu = \frac{E}{h}$$

$$= \frac{3.68 \times 10^{-19}}{6.62 \times 10^{-34}} = 5.55 \times 10^{14} \approx 5.6 \times 10^4 \text{ Hz}$$

So that, the frequency of the radiation is 5.6×10^{14} Hz. **[1]**

Q. 16. The radius of the innermost electron orbit of a hydrogen atom is 5.3×10^{-11} m. What are the radii of the $n = 2$ and $n = 3$ orbits?

[NCERT Ex. Q. 12.8, Page 436]

Ans. The radius of the innermost orbit of a hydrogen atom, $r_1 = 5.3 \times 10^{-11}$ m

Let r_2 be the radius of the orbit at $n = 2$. It is related to the radius of the innermost orbit as :

$r_2 = (n)^2 r_1$

$= 4 \times 5.3 \times 10^{-11} = 2.12 \times 10^{-10}$ m **[1]**

For n = 3, we can write the corresponding electron radius as :

$r_3 = (n)^2 r_1$

$= 9 \times 5.3 \times 10^{-11} = 4.77 \times 10^{-10}$ m

So that, the radii of an electron for $n = 2$ and $n = 3$ orbits are 2.12×10^{-10} m and 4.77×10^{-10} m, respectively. **[1]**

Q. 17. If Bohr's quantisation postulate (angular momentum = $\frac{nh}{2\pi}$ is a basic law of nature, it should be equally valid for the case of planetary motion also. Why then do we never speak of quantisation of orbits of planets around the sun?

[NCERT Add. Ex. Q. 12.16, Page 437]

Ans. We never speak of quantisation of orbits of planets around the Sun because the angular momentum associated with planetary motion is largely relative to the value of Planck's constant (h). The angular momentum of the Earth in its orbit is of the order of 10^{70} h. **[1]**

This leads to a very high value of quantum levels n of the order of 10^{70}. For large values of n, successive energies and angular momenta are relatively very small. So, the quantum levels for planetary motion are considered continuous. **[1]**

Q. 18. A 12.5 eV electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted.

[CBSE Board, All India Region, 2017]

Ans. Given that,

The energy of the electron beam used to bombard gaseous hydrogen at room temperature is 12.5 eV.

And, the energy of the gaseous hydrogen in its ground state at room temperature is −13.6 eV.

When gaseous hydrogen is bombarded with an electron beam, the energy of the gaseous hydrogen becomes −13.6 + 12.5 eV = −1.1 eV.

Orbital energy related to orbit level (n) is,

$$E = \frac{-13.6}{(n)^2}\text{eV}$$

For $n = 3$,

$$\Rightarrow E = \frac{-13.6}{(3)^2}\text{eV}$$

$$\Rightarrow \frac{-13.6}{9}\text{eV}$$

$$\Rightarrow -1.5 \text{ eV}$$

This energy is approximately equal to the energy of gaseous hydrogen. That means the electron has jumped from $n = 1$ to $n = 3$ level.

During its de-excitation, electrons can jump from $n = 3$ to $n = 1$ directly, which forms a line of the Lyman series of the hydrogen spectrum.

Relation for wave number for the Lyman series is :

$$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$$

For first member $n = 3$

$$\frac{1}{\lambda_1} = R\left[\frac{1}{1^2} - \frac{1}{(3)^2}\right]$$

$$= R\left[\frac{1}{1} - \frac{1}{9}\right]$$

$$\frac{1}{\lambda_1} = 1.097 \times 10^7\left[\frac{9-1}{9}\right]\left(\begin{array}{l}\text{where Rydberg cons-}\\ \text{tant } R = 1.097 \times 10^7 \text{m}^{-1}\end{array}\right)$$

$$\frac{1}{\lambda_1} = 1.097 \times 10^7 \times \frac{8}{9}$$

$$\Rightarrow \lambda_1 = 1.025 \times 10^{-7}\text{m}$$ **[1]**

For $n = 2$

$$\frac{1}{\lambda_2} = \text{R}\left[\frac{1}{1^2} - \frac{1}{(2)^2}\right]$$

$$= \text{R}\left[\frac{1}{1} - \frac{1}{4}\right]$$

$$\therefore \frac{1}{\lambda_2} = 1.097 \times 10^7\left[\frac{4-1}{4}\right]\left(\begin{array}{l}\text{where Rydberg constant}\\ \text{R} = 1.097 \times 10^7 \text{m}^{-1}\end{array}\right)$$

$$\therefore \frac{1}{\lambda_2} = 1.097 \times 10^7 \times \frac{3}{4}$$

$$\Rightarrow \lambda_2 = 1.215 \times 10^{-7}\text{m}$$

Relation for wave number for the Balmer series is

$$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$$

For first member $n = 3$

$$\therefore \frac{1}{\lambda_3} = R\left[\frac{1}{2^2} - \frac{1}{(3)^2}\right] = 1.097 \times 10^7 \times \left[\frac{1}{4} - \frac{1}{9}\right]$$

$$\lambda_3 = 6.56 \times 10^{-7}\text{m}$$ **[1]**

Q. 19. Find the wavelength of the electron orbiting in the first excited state in hydrogen atom.

[CBSE Board, Delhi Region, 2017]

Ans. The energy of nth state of the hydrogen atom is given as;

$$E_n = -\frac{13.6}{n^2}$$ **[1]**

So, in first excited state, $n = 2$

Therefore :

$$E_2 = -\frac{13.6}{4}\text{ eV} = -3.4\text{ eV}$$

So,

$$\lambda = \frac{hc}{E}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3.4 \times 1.6 \times 10^{-19}}$$

$$= 3.656 \times 10^{-7}\text{ m}$$ **[1]**

Q. 20. Define ionization energy. How would the ionization energy change when electron in hydrogen atom is replaced by a particle of mass 200 times that of electron but having the same charge? [CBSE Board, All India Region, 2016]

Ans. Ionisation energy :

The minimum amount of energy required to remove an electron from the outermost orbit of a neutral atom in its ground state is known as the ionisation energy of that atom.

The ionisation energy of hydrogen atom is

$E_{H'} \propto m_e$ ...(i)

When mass of electron is replaced by a particle having mass = 200 m_e, then

$E_{H'} \propto 200\ m_e$ ...(ii)

[1]

From equation (i) and (ii), we get,

$E_{H'} = 200\ E_H$

$= 200 \times (-13.6)$

$= -2720$ eV (Because, $E_H = -13.6$ eV) **[1]**

Q. 21. Calculate the shortest wavelength of the spectral lines emitted in Balmer series.
[Given Rydberg constant, $R = 10^7\ m^{-1}$]
[CBSE Board, All India Region, 2016]

Ans. As we know the formula :

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$$

For short wavelength,

$$\frac{1}{\lambda} = \frac{R}{4}$$

$$\Rightarrow \lambda = \frac{4}{R} = 4 \times 10^{-7}\,\text{m}$$

$$\Rightarrow \lambda = 4 \times 10^{-7}\,\text{m}$$ **[2]**

Q. 22. (i) State Bohr postulate of hydrogen atom that gives the relationship for the frequency of emitted photon in a transition.

(ii) An electron jumps from fourth to first orbit in an atom. How many maximum numbers of spectral lines can be emitted by the atom? To which series these lines correspond?
[CBSE Board, Foreign Scheme, 2016]

Ans. **(i)** According to **Bohr,**

"Energy is radiated in the form of a photon when the electron of an excited hydrogen atom returns from higher energy state to the lower energy state."

In other words, energy is radiated in the form of a photon when electron in hydrogen atom jumps from higher energy orbit ($n = n_i$) where $n_i > n_f$.

The energy of the emitted radiation or photon is given by :

$$h\nu = E_{n_i} - E_{n_f}$$

As we know that,

$$E_n = \frac{-me^4}{8h^2\varepsilon_0^2 n^2}$$

$$\therefore\ h\nu = \frac{-me^4}{8\varepsilon_0^2 n_i^2 h^2} + \frac{me^4}{8\varepsilon_0^2 n_f^2 h^2}$$

$$h\nu = \frac{me^4}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\nu = \frac{me^4}{8\varepsilon_0^2 h^3}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$n = 4$ to $n = 3$.....Paschan series

$n = 4$ to $n = 2$.....Balmer series

$n = 4$ to $n = 1$.....Lyman series **[1]**

(ii) Number of lines produced when electron from nth shell drops to ground state

$$= \frac{n(n-1)}{2}$$

Now according to question lines produced when electron drops from 4th shell to ground state :

$$= \frac{4(4-1)}{2}$$

$$= 4\left(\frac{3}{2}\right)$$

$$= \frac{12}{2} = 6$$

Then, number of spectral lines when electron jumps from 4th orbit to first orbit is, six spectral lines and these lines correspond to Lyman series. **[1]**

Q. 23. Use de-Broglie's hypothesis to write the relation for the nth radius of Bohr orbit in terms of Bohr's quantization condition of orbital angular momentum. [CBSE Board, Foreign Scheme, 2016]

Ans. (i) According to Bohr's second postulate, we have

$$mvr_n = n\frac{h}{2\pi}$$

$$\Rightarrow 2\pi r_n = \frac{nh}{mv}$$

But according to the de-Broglie hypothesis

$$\frac{h}{mv} = \frac{h}{p} = \lambda$$

So that, $2\pi r_n = n\lambda$; where is the de-Broglie wavelength. **[2]**

Short Answer Type Questions **(3 marks each)**

Q. 1. Define the distance of closest approach. An α-particle of kinetic energy 'K' is bombarded on a thin gold foil. The distance of the closest approach is 'r'. What will be the distance of closest approach for an α-particle of double the kinetic energy?
[CBSE Board, Delhi Region, 2017]

Ans. When an α-particle is bombarded over a gold nucleus, it is repelled by electrostatic repulsion.

So that the kinetic energy of α-particle is converted into electrostatic potential energy. At a certain distance between the α-particle and nucleus at which the moving charge loses all its kinetic energy and becomes stationary momentarily.

This distance (r) is known as distance of closest approach. **[1½]**

In this process, all the kinetic energy (K) of moving particle is converted into potential energy.

From the given data,

Initially, $\frac{1}{4\pi\varepsilon_0} \times \frac{2e \times Ze}{r} = K$

Let r_0 be the new distance of closest approach for a twice energetic α – particle. Then,

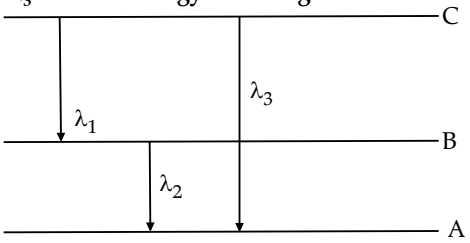
$$\frac{1}{4\pi\varepsilon_0}\times\frac{2e\times Ze}{r_0}=2K$$

Dividing equations (i) and (ii), we get

$$\frac{r_0}{r}=\frac{1}{2}\Rightarrow r_0=\frac{r}{2}$$ **[1½]**

Q. 2. (i) State Bohr's quantization condition for defining stationary orbits. How does the de-Broglie hypothesis explain the stationary orbits?

(ii) Find the relation between three wavelengths λ_1, λ_2 and λ_3 from the energy-level diagram shown below.



[CBSE Board, Delhi Region, 2016]

Ans. (i) Bohr's quantisation condition : According to Bohr, an electron can revolve only in certain discrete, non-radiating orbits for which total angular momentum of the revolving electron are an integral multiple of $\frac{h}{2\pi}$, where h is the Planck's constant.

$$mvr=\frac{nh}{2\pi}$$ **[1½]**

(ii) Using the Rydberg formula of the spectra of hydrogen atom,

$$\frac{1}{\lambda_1}=R\left(\frac{1}{n_2^2}-\frac{1}{n_3^2}\right) \quad ...(1)$$

$$\frac{1}{\lambda_2}=R\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right) \quad ...(2)$$

$$\frac{1}{\lambda_3}=R\left(\frac{1}{n_1^2}-\frac{1}{n_3^2}\right) \quad ...(3)$$

Adding (1) and (2), we find that

$$\frac{1}{\lambda_1}+\frac{1}{\lambda_2}=R\left(\frac{1}{n_1^2}-\frac{1}{n_3^2}\right)=\frac{1}{\lambda_3}$$

which is the required relation between the three wavelengths. **[1½]**

Q. 3. Positronium is just like a H-atom with the proton replaced by the positively charged anti-particle of the electron (called the positron which is as massive as the electron). What would be the ground state energy of positronium?

[NCERT Exemp. Q. 12.19, Page 78]

Ans. As in the new H-atom (positronium) proton is replaced by positron of mass, $m=\frac{m_e}{2}$ as under

Mass of positronium $= m = m_e^- + m_e^+$

$$m = m_e + m_e = 2m_e$$

$$m_e^+ = m_e^{-1} = \frac{m_e}{2}$$

As we know that,

$E_n = -13.6$ eV

So energy of positron will be,

$$E_n' = \frac{-m_e^+ e^4}{8\varepsilon_0 n^2 h^2}$$

$$= \frac{-\left(\frac{m_e}{2}\right)e^4}{8\varepsilon_0 n^2 h^2}$$

$$= \frac{-13.6}{2}$$

which is,

$$E_n' = \frac{13.6}{2}\text{eV} \qquad \left(m_e=\frac{m}{2}\right)$$

$E_n' = 6.8$ eV **[3]**

Q. 4. A hydrogen atom initially in the ground level absorbs a photon, which excites it to the $n = 4$ level. Determine the wavelength and frequency of photon. [NCERT Ex. Q. 12.6, Page 436]

Ans. For ground level, $n_1 = 1$

Let E_1 be the energy of this level. It is known that E_1 is related with n_1 as :

$$E_1 = \frac{-13.6}{n_1^2}\text{eV}$$

where $n_1 = 1$

$$= \frac{-13.6}{1^2}$$

$$= -13.6 \text{ eV}$$

The atom is excited to a higher level, $n_2 = 4$

Let E_2 be the energy of this level.

$$E_2 = \frac{-13.6}{n_2^2}\text{ eV}$$

$$= \frac{-13.6}{4^2} = \frac{13.6}{16}\text{ eV}$$ **[1]**

The amount of energy absorbed by the photon is given as :

$$E = E_2 - E_1$$

$$= \frac{-13.6}{16} - \left(-\frac{13.6}{1}\right)$$

$$= \frac{13.6\times 15}{16}\text{ eV}$$

$$= \frac{13.6\times 15}{16}\times 1.6\times 10^{-19}$$

$$= 2.04\times 10^{-18}\text{ J}$$ **[1]**

For a photon of wavelength λ, the expression of energy is written as :

$$E = \frac{hc}{\lambda}$$

Where,

h = Planck's constant = 6.6×10^{-34} Js

c = Speed of light = 3×10^8 m/s

$$\lambda = \frac{hc}{E}$$

$$= \frac{6.6\times 10^{-34}\times 3\times 10^8}{2.04\times 10^{-18}}$$

$$= 9.7\times 10^{-8}\text{ m}$$

$$= 97\text{ nm}$$

And, frequency of a photon is given by the relation,

$$\nu = \frac{c}{\lambda}$$

$$= \frac{3\times10^{8}}{9.7\times10^{-8}} \approx 3.1\times10^{15}\,\text{Hz}$$

So, the wavelength of the photon is 97 nm while the frequency is 3.1×10^{15} Hz. **[1]**

Q. 5. Assume that there is no repulsive force between the electrons in an atom but the force between positive and negative charges is given by Coulomb's law as usual. Under such circumstances, calculate the ground state energy of a He-atom.

[NCERT Exemp. Q. 12.20, Page 78]

Ans. For H-atom $Z = 1$ and $n = 1$

$$\therefore E_n = \frac{-m_e e^4}{8\varepsilon_0^2 1^2 h^2} = 13.6 \text{ eV} \qquad \textbf{[1½]}$$

For He-atom, $m = 4\,m_e$ and $n = 1$

$$E_n = \frac{-4m_e e^4}{8\varepsilon_0^2 1^2 h^2}$$

$$= -4\times13.6$$

$$= -54.4 \text{ eV} \qquad \textbf{[1½]}$$

Q. 6. The gravitational attraction between electron and proton in a hydrogen atom is weaker than the coulomb attraction by a factor of about 10^{-40}. An alternative way of looking at this fact is to estimate the radius of the first Bohr orbit of a hydrogen atom if the electron and proton were bound by gravitational attraction. You will find the answer interesting. [NCERT Add. Ex. Q. 12.12, Page 436]

Ans. Radius of the first Bohr orbit is given by the relation,

$$r_1 = \frac{4\pi\varepsilon_0\left(\frac{h}{2\pi}\right)^2}{m_e e^2} \qquad ...(i)$$

Where,

ε_0 = Permittivity of free space

h = Planck's constant = 6.63×10^{-34} Js

m_e = mass of electron = 9.1×10^{-31} kg

e = charge of an electron = 1.9×10^{-19} C

m_p = mass of proton = 1.67×10^{-27} kg

r = Distance between the electron and the proton

Coulomb attraction between an electron and a proton is given as :

$$F_c = \frac{e^2}{4\pi\varepsilon_0 r^2} \qquad ...(2)$$

Gravitational force of attraction between an electron and a proton is given as :

$$F_G = \frac{Gm_p m_e}{r^2} \qquad ...(3)$$

Where,

G = Gravitational constant = 6.67×10^{-11} N m^2/kg^2

If the electrostatic (Coulomb) force and the gravitational force between an electron and a proton are equal, then we can write :

$$F_G = F_C$$

$$\frac{Gm_p m_e}{r_2} = \frac{e^2}{4\pi\varepsilon_0 r^2}$$

$$\therefore \frac{e^2}{4\pi\varepsilon_0} = Gm_p m_e \qquad ...(4)$$

[1½]

Putting the value of equation (4) in equation (1), we get :

$$r_1 = \frac{\left(\frac{h}{2\pi}\right)^2}{Gm_p m_e^2}$$

$$= \frac{\left(\frac{6.63\times10^{-34}}{2\times3.14}\right)^2}{6.67\times10^{-11}\times1.67\times10^{-27}\times(9.1\times10^{-31})^2} \approx 1.21\times10^{29}\,\text{m}$$

It is known that the universe is 156 billion light years wide or 1.5×10^{27} m wide. Hence, we can conclude that the radius of the first Bohr orbit is much greater than the estimated size of the whole universe. **[1½]**

Q. 7. Using Bohr model, calculate the electric current created by the electron when the H-atom is in the ground state. [NCERT Exemp. Q. 12.21, Page 79]

Ans. Let for an electron of H-atom velocity in orbit = v m/s

Radius of orbit = a_0 = Bohr's radius

So the number of revolutions per second $\nu = \frac{2\pi a_0}{v}$

$$I = -e\nu \qquad \left(\nu = \frac{2\pi a_0}{v}\right)$$

$$= \frac{-e2\pi a_0}{v} \qquad \textbf{[3]}$$

Negative (−) sign shows that the direction of current is opposite to the direction of motion of electron.

Q. 8. Show that the first few frequencies of light that is emitted when electrons fall to the nth level from levels higher than n, are approximate harmonics (*i.e.*, in the ratio 1 :2 :3...) when $n >> 1$.

[NCERT Exemp. Q. 12.22, Page 79]

Ans. When an electron falls from $(n + n')$th to nth energy level, the frequency of radiations in spectrum H-atom like atoms is given as :

$$v = cRZ^2\left[\frac{1}{(n+n')} - \frac{1}{n^2}\right]$$

where $n >> n'$

n' = 1,2,3, ……..

R = Rydberg's constant

$$v = cRZ^2\left[\frac{1}{n^2\left[1+\frac{n'}{n^2}\right]^2} - \frac{1}{n^2}\right] = cRZ^2\left[\frac{1}{n^2}\left[1+\frac{n'}{n}\right]^{-2} - \frac{1}{n^2}\right]$$

Neglecting the higher terms as $n >> n'$

$$= cRZ^2\left[\frac{1}{n^2}\left(1-\frac{2n'}{n}\right) - \frac{1}{n^2}\right] = cRZ^2\left[\frac{1}{n^2} - \frac{2n'}{n^3} - \frac{1}{n^2}\right]$$

$$= \frac{-cRZ^2 2n'}{n^3} = \left(\frac{2cRZ^2}{n^3}\right)n'$$

[2]

So the first few frequencies of light that is emitted when electrons fall from $(n+n')$th to nth energy level are in the ratio of n' = 1 :2 :3……. when $n >>> 1$ **[1]**

Q. 9. What is the minimum energy that must be given to a H-atom in-ground state so that it can emit an

H_γ line in Balmer series. If the angular momentum of the system is conserved, what would be the angular momentum of such H_γ photon?

[NCERT Exemp. Q. 12.23, Page 79]

Ans. We know that H_γ spectral line in Balmer series formed when electron falls from $n = 5$ to $n = 2$.

Where the electron is in ground state, *i.e.*, $n = 1$ and must be taken to $n = 5$ for H line.

So the energy of

$$H_\gamma = E_5 - E_1 = \left(\frac{13.6}{5^2}\right) - (13.6)$$

$$= -0.54 + 13.6$$

$$= 13.06 \text{ eV}$$ **[1½]**

Since angular momentum is conserved, so the angular momentum of H-atom = change in angular momentum of electron

$$= L_5 - L_2 = \left[\frac{5h}{2\pi} - \frac{2h}{2\pi}\right]$$

$$= \frac{3h}{2\pi}$$

$$= 3 \times 1.06 \times 10^{-34}$$

$$= 3.16 \times 10^{-34} \text{ kgm}^2\text{s}^{-1}$$ **[1½]**

Q. 10. (a) Using the Bohr's model calculate the speed of the electron in a hydrogen atom in the n = 1, 2, and 3 levels.

(b) Calculate the orbital period in each of these levels.

[NCERT Ex. Q. 12.7, Page 436]

Ans. **(a)** Let v_1 be the orbital speed of the electron in a hydrogen atom in the ground state level, $n_1 = 1$. For charge (e) of an electron, v_1 is given by the relation,

$$v_1 = \frac{e^2}{n_1 4\pi\varepsilon_0\left(\frac{h}{2\pi}\right)} = \frac{e^2}{2\varepsilon_0 h}$$

Where,

$$e = 1.6 \times 10^{-19} \text{C}$$

$$\varepsilon_0 = \text{Permittivity of free space} = 8.85 \times 10^{-12} N^{-1}C^2m^{-2}$$

$$h = \text{Planck's constant} = 6.62 \times 10^{-34} \text{Js}$$

$$\text{So, } v_1 = \frac{(1.6 \times 10^{-19})^2}{2 \times 8.85 \times 10^{-12} \times 6.62 \times 10^{-34}}$$

$$= 0.0218 \times 10^8$$

$$= 2.18 \times 10^6 \text{ m/s}$$

For level $n_2 = 2$, we can write the relation for the corresponding orbital speed as :

$$v_2 = \frac{e^2}{n_2 2\varepsilon_0 h}$$

$$= \frac{(1.6 \times 10^{-19})^2}{2 \times 2 \times 8.85 \times 10^{-12} \times 6.62 \times 10^{-34}}$$

$$= 1.09 \times 10^6 \text{ m/s}$$

And, for the n =3, we can write the relation for the corresponding orbital speed as,

$$v_3 = \frac{e^2}{n_3 2\varepsilon_0 h}$$

$$= \frac{(1.6 \times 10^{-19})^2}{3 \times 2 \times 8.85 \times 10^{-12} \times 6.62 \times 10^{-34}} = 7.27 \times 10^5 \text{ m/s}$$

So, the speed of the electron in a hydrogen atom in $n = 1$, $n = 2$ and $n = 3$ is 2.18×10^6 m/s, 1.09×10^6 m/s, 7.27×10^5 m/s, respectively. **[1½]**

(b) Let T_1 be the orbital period of the electron when it is in level $n_1 = 1$.

Orbital period is related to orbital speed as :

$$T_1 = \frac{2\pi r_1}{v_1}$$

Where,

r_1 = Radius of the orbit

$$= \frac{n_1^2 h^2 \varepsilon_0}{\pi m_e e^2}$$

h = Planck's constant = 6.62×10^{-34} Js

e = Charge on an electron = 1.6×10^{-19} C

ε_0 = Permittivity of free space = $8.85 \times 10^{-12} \text{N}^{-1} \text{C}^2 \text{m}^{-2}$

m_e = Mass of an electron = 9.1×10^{-31} kg

$$T_1 = \frac{2\pi r_1}{v_1}$$

$$= \frac{2\pi \times (1)^2 \times (6.62 \times 10^{-34})^2 \times 8.85 \times 10^{-12}}{2.18 \times 10^6 \times \pi \times 9.1 \times 10^{31} \times (1.6 \times 10^{-19})^2}$$

$$= 15.27 \times 10^{-17}$$

$$= 1.527 \times 10^{-16} \text{s}$$

For level $n_2 = 2$, we can write the period as :

$$\therefore T_2 = \frac{2\pi r_2}{v_2}$$

Where,

r_2 = Radius of the electron in $n_2 = 2$

$$= \frac{(n_2)^2 h^2 \varepsilon_0}{\pi m_e e^2}$$

$$T_2 = \frac{2\pi \times (2)^2}{v_2}$$

$$= \frac{2\pi \times (2)^2 \times (6.62 \times 10^{-34})^2 \times 8.85 \times 10^{-12}}{1.09 \times 10^5 \times \pi \times 9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2}$$

$$= 1.22 \times 10^{-15} \text{s}$$

And, for level $n_3 = 3$, we can write the period as :

$$T_3 = \frac{2\pi r_3}{v_3}$$

Where,

r_3 = Radius of the electron in n_3

$$= 3\frac{(n_3)^2 h^2 \varepsilon_0}{\pi m_e e^2}$$

$$\therefore T_3 = \frac{2\pi r_3}{v_3}$$

$$= \frac{(n_3)^2 h^2 \varepsilon_0}{\pi m_e e^2}$$

$$= \frac{2\pi \times (3)^2 \times (6.62 \times 10^{-34})^2 \times 8.85 \times 10^{-12}}{7.27 \times 10^5 \times \pi \times 9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2}$$

$$= 4.12 \times 10^{-15} \text{s}$$

$$\therefore T_3 = \frac{2\pi r_3}{v_3}$$

So that, the orbital period in each of these level is 1.52×10^{-16} s, 1.22×10^{-15} s, and 4.12×10^{-15} s respectively. **[1½]**

Q. 11. Obtain an expression for the frequency of radiation emitted when a hydrogen atom de-excites from level *n* to level (*n* – 1). For large *n*, show that this frequency equals the classical frequency of revolution of the electron in the orbit.

[NCERT Add. Ex. Q. 13, Page 436]

Ans. It is given that a hydrogen atom de-excites from an upper level (n) to a lower level ($n-1$).

We have the relation for energy (E_1) of radiation at level n as :

$$E_1 = h\nu_1 = \frac{hm_e e^4}{(4\pi)^3 \varepsilon_0^2 \left(\frac{h}{2\pi}\right)^3} \times \left(\frac{1}{n^2}\right) \quad ...(i)$$

Where,

ν_1 = Frequency of radiation at level n

h = Planck's constant

m_e = Mass of hydrogen atom

e = Charge on an electron

ε_0 = Permittivity of free space

Now, the relation for energy (E_2) of radiation at level ($n-1$) is given as :

$$E_2 = h\nu_2 = \frac{hm_e e^4}{(4\pi)^3 \varepsilon_0^2 \left(\frac{h}{2\pi}\right)^3} \times \frac{1}{(n-1)^2} \quad ...(ii)$$

ν_2 = Frequency of radiation at level $(n-1)$

Energy (E) released as a result of de-excitation :

$E = E_2 - E_1$

$h\nu = E_2 - E_1$ …(iii)

Where,

ν = Frequency of radiation emitted

Putting values from equations (i) and (ii) in equation (iii), we get :

$$\nu = \frac{m_e e^4}{(4\pi)^3 \varepsilon_0^2 \left(\frac{h}{2\pi}\right)^3} \left[\frac{1}{(n-1)} - \frac{1}{n^2}\right]$$

$$= \frac{m_e e^4 (2n-1)}{(4\pi)^3 \varepsilon_0^2 \left(\frac{h}{2\pi}\right)^3 n^2 (n-1)^2}$$

For large n, we can write $(2n-1) - 2n$ and $(n-1) \simeq n$.

So,

$$\nu = \frac{m_e e^4}{32\pi^3 \varepsilon_0^2 \left(\frac{h}{2\pi}\right)^3 n^3} \quad ...(iv)$$

[1½]

Classical relation of frequency of revolution of an electron is given as :

$$v_c = \frac{v}{2\pi r}$$

Where,

Velocity of the electron in the *nth* orbit is given as :

$$v = \frac{e^2}{4\pi\varepsilon_0 \left(\frac{h}{2\pi}\right) n} \quad ...(vi)$$

And, radius of the *nth* orbit is given as :

$$r = \frac{4\pi\varepsilon_0 \left(\frac{h}{2\pi}\right)^2}{me^2} n^2 \quad ...(vii)$$

Putting the values of equations (vi) and (vii) in equation (v), we get :

$$v_c = \frac{m_e e^4}{32\pi^3 \varepsilon_0^2 \left(\frac{h}{2\pi}\right)^3 n^3} \quad ...(viii)$$

So, the frequency of radiation emitted by the hydrogen atom is equal to its classical orbital frequency. **[1½]**

Q. 12. The total energy of an electron in the first excited state of the hydrogen atom is about –3.4 eV.

(a) What is the kinetic energy of the electron in this state?

(b) What is the potential energy of the electron in this state?

(c) Which of the answers above would change if the choice of the zero of potential energy is changed?

[NCERT Add. Ex. Q. 12.15, Page 436]

Ans. **(a)** Total energy of the electron, $E = -3.4$ eV

Kinetic energy of the electron is equal to the negative of the total energy.

$K = -E = -(-3.4) = +3.4$ eV

So, the kinetic energy of the electron in the given state is +3.4 eV. **[1]**

(b) Potential energy (U) of the electron is equal to the negative of twice of its kinetic energy.

$U = -2\,\text{K} = -2 \times 3.4 = -6.8$ eV

So, the potential energy of the electron in the given state is −6.8 eV. **[1]**

(c) The potential energy of a system depends on the reference point taken. And here, the potential energy of the reference point is taken as zero. If the reference point is changed, then the value of the potential energy of the system also changes. Since total energy is the sum of kinetic and potential energies, total energy of the system will also change. **[1]**

Q. 13. Obtain the first Bohr's radius and the ground state energy of a muonic hydrogen atom [*i.e.*, an atom in which a negatively charged muon (μ) of mass about 207 m_e orbits around a proton].

[NCERT Add. Ex. Q. 12.17, Page 437]

Ans. Mass of a negatively charged muon, $m_\mu = 207\, m_e$

According to Bohr's model,

Bohr radius $r_e \propto \left(\frac{1}{m_e}\right)$

And, energy of a ground state electronic hydrogen atom, $E_e \propto m_e$,

Also, energy of a ground state muonic hydrogen atom, $E_\mu \propto m_e$,

We have the value of the first Bohr orbit, $r_e = 053\,\mathring{A} = 0.53 \times 10^{-10}$ m

Let r_μ be the radius of muonic hydrogen atom.

At equilibrium, we can write the relation as :

$$m_\mu r_\mu = m_e r_e$$

$$207 m_e \times r_\mu = m_e r_e$$

$$\therefore \quad r_\mu = \frac{0.53 \times 10^{-10}}{207} = 2.56 \times 10^{-13}\,\text{m}$$

So that, the value of the first Bohr radius of muonic hydrogen atom is 2.56×10^{-13} m.

As we that,

$E_e = -13.6$ eV

Take the ratio of these energies as :

$$\frac{E_e}{E_\mu} = \frac{m_e}{m_\mu} = \frac{m_e}{207 m_e}$$

$$E = 207 E_e$$

$$= 207 \times (-13.6) = -2.81 \text{ keV}$$

So that, the ground state energy of a muonic hydrogen atom is –2.81 keV. **[3]**

Q. 14. (a) State Bohr's postulate to define stable orbits in hydrogen atom. How does de Broglie's hypothesis explain the stability of these orbits?

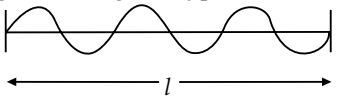
(b) A hydrogen atom initially in the ground state absorbs a photon which excites it to the $n = 4$ level. Estimate the frequency of the photon.

[CBSE Board, Delhi Region, 2018]

Ans. (a) For stable orbit the angular momentum of revolving electron is : $\frac{nh}{2\pi}$

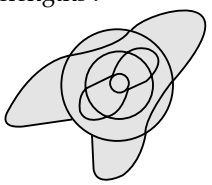
$$mvr = \frac{nh}{2\pi}$$

According to de-Broglie's hypothesis :



$n\lambda = \alpha l$

Revolving electron is stable orbit also associated with wave nature. So, a circular orbit can be taken to be stable orbit if it contains an integral number of de-Broglie wavelengths :



$$2\pi r = n\lambda$$

$$2\pi r = n\left(\frac{h}{p}\right)$$

$$pr = \frac{nh}{2\pi}$$

$$mvr = \frac{nh}{2\pi}$$

Only a certain number of wavelengths would fit in the electron orbits. **[1½]**

(b) $$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{4^2}\right)$$

$$\frac{1}{\lambda} = R\left(1 - \frac{1}{16}\right)$$

$$\frac{1}{\lambda} = R\left(\frac{15}{16}\right)$$

$$v = \frac{c}{\lambda}$$

$$v = c \times R\left(\frac{15}{16}\right)$$

$$v = 3 \times 10^8 \times 1.09 \times 10^7 \times \frac{15}{16}$$

$$v = 3 \times 10^{15} \text{ Hz}$$ **[1½]**

Q. 15. Calculate the de-Broglie wave length of the electron orbiting in the $n = 2$ state of hydrogen atom. [CBSE Board, All India Region, 2016]

Ans. As we know that,

$$mvr_2 = \frac{nh}{2\pi}$$

The de-Broglie wavelength is,

$$\lambda = \frac{h}{mv} = \frac{h}{nh} \times (2\pi r_2) = \pi r_2$$

As $r_2 = 0.529 \times n^2 \times 10^{-10}$ m

So that,

$$\lambda = \pi \times 0.529 \times 10^{-10} \times \text{n}^2$$

$$= 3.14 \times 0.529 \times 10^{-10} \times 2^2$$

$$= 6.64 \times 10^{-10} \text{m}$$ **[3]**

Long Answer Type Questions (5 marks each)

Q. 1. Deuterium was discovered in 1932 by Harold Urey by measuring the small change in wavelength for a particular transition in ^1H and ^2H. This is because, the wavelength of transition depends to a certain extent on the nuclear mass. If nuclear motion is taken into account then the electrons and nucleus revolve around their common centre of mass. Such a system is equivalent to a single particle with a reduced mass μ, revolving around the nucleus at a distance equal to the electron-nucleus separation. Here $\mu = \frac{m_e M}{(m_e + M)}$ where M is the nuclear mass and m_e is the electronic mass. Estimate the percentage difference in wave length for the 1st line of the Lyman series in ^1H and ^2H. (Mass of ^{1}Hnucleus is 1.6725×10^{-27} kg, Mass of ^2H nucleus is 3.3374×10^{-27} kg, Mass of electron = 9.109×10^{-31} kg.)

[NCERT Exemp. Q. 12.25, Page 79]

Ans. Total energy of electron in n^{th} stable orbit in H or like atom

$$E_n = \frac{\mu Z^2 e^4}{8\varepsilon_0^2 h^2 n^2}$$

μ = Reduced mass of electron, proton and neutron (mass defect)

$$E_H = \frac{\mu_H (1)^2 e^4}{8\varepsilon_0^2 h^2}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = \frac{\mu_H e^4}{8\varepsilon_0^2 h^2}\left[\frac{1}{1} - \frac{1}{2^2}\right] = \frac{\mu_H e^4}{8\varepsilon_0^2 h^2}\left[\frac{3}{4}\right]$$

$$E = h\nu = \frac{h}{\lambda} \text{ or } \lambda_H = \frac{h}{E_H}$$

$$\therefore h\nu_H = \frac{\mu_H e^4}{8\varepsilon_0^2 h^2}\frac{3}{4}$$

$$\nu_H = \frac{\mu_H e^4}{8\varepsilon_0^2 h^3}\cdot\frac{3}{4}$$

The percentage difference in the wavelength

$$= \frac{(\lambda_D - \lambda_H)}{\lambda_H}\times 100$$

Percent change in wavelength,

$$\%\text{ change } \Delta E = \left[\frac{\lambda_D}{\lambda_H} - 1\right]\times 100 \ (\because \Delta E = E_1 - E_2) \qquad ...(i)$$

$$h\nu = \frac{\mu e^4}{8\varepsilon_0^2 h^2}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\nu = \frac{\mu e^4}{8\varepsilon_0^2 h^3}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{c}{\lambda} = \frac{\mu e^4}{8\varepsilon_0^2 h^3}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

So,

$$\frac{1}{\lambda} = \frac{\mu e^4}{8\varepsilon_0^2 ch^3}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

As μ = mass defect, e, ε_0 c, and h are constants for an atom. **[2½]**

$$\lambda \propto \frac{1}{h}$$

So eqn. (i) can be written as percentage change in the wavelength $= \left[\frac{\mu_H}{\mu_D} - 1\right]\times 100$

$\because \mu = \frac{m_e M}{(M + m_\varepsilon)}$ (Given)

$\therefore$ Percentage change in wavelength

$$= \left[\frac{\frac{m_e M_H}{(M_H + m_e)}}{\frac{m_e + M_D}{(M_D + m_e)}} - 1\right]100$$

$$\frac{\Delta\lambda}{\lambda_H}\times 100 = \left[\frac{M_H}{M_D}\frac{(M_D + m_e)}{(M_H + m_e)} - 1\right]\times 100$$

$$= \left[\frac{M_H}{M_D}\frac{M_D\left(1 + \frac{m_e}{M_D}\right)}{M_H\left(1 + \frac{m_e}{M_H}\right)} - 1\right]\times 100$$

$$= \left[\left(1 + \frac{m_e}{M_D}\right)\left(1 + \frac{m_e}{M_H}\right)^{-1} - 1\right]\times 100$$

$$= \left[\left(1 + \frac{m_e}{M_D}\right)\left(1 - \frac{m_e}{M_H}\right) - 1\right]\times 100$$

$m_e << M_D$ so neglecting the higher degree term

$$\frac{\Delta\lambda}{\lambda_H}\times 100 = \left[1 - \frac{m_e}{M_H} + \frac{m_e}{M_D} - \frac{m_e m_e}{M_D \cdot M_H} - 1\right]\times 100$$

$$= \left[\frac{1}{M_D} - \frac{1}{M_H}\right]\times 100$$

$$= 9.1\times 10^{-31}\left[\frac{1}{3.3374\times 10^{-27}} - \frac{1}{1.6725\times 10^{-27}}\right]\times 100$$

$$= \frac{9.1\times 10^{-31+2}}{10^{-27}}\left[\frac{1.6725 - 3.3374}{3.3374\times 1.6725}\right]$$

$$\frac{\Delta\lambda\times 100}{\lambda_H} = \frac{-9.1\times 10^{-29+27}\times 0.6649}{3.3374\times 1.6725} = \frac{-6.05057\times 10^{-2}}{5.5180}$$

$$\frac{\Delta\lambda\times 100}{\lambda_H} = -1.084\times 10^{-2}\ \%\text{ decrease in wavelength}$$

Negative (–) sign shows that $\lambda_D < \lambda_H$. **[2½]**

Q. 2. The first four spectral lines in the Lyman series of a H-atom are λ = 1218 Å, 1028 Å, 974.3 Å and 951.4 Å. If instead of Hydrogen, we consider Deuterium, calculate the shift in the wavelength of these lines. [NCERT Exemp. Q. 12.24, Page 79]

Ans. Reduced mass of H atom (mass defect) μ_H then

$\frac{1}{\mu_H} = \frac{1}{m_e} + \frac{1}{M}$ (M is mass of H atom)

$$\frac{1}{\mu_H} = \frac{M + m_e}{M\cdot m_e} = \frac{M\left[1 + \frac{m_e}{M}\right]}{M\cdot m_e} \qquad (\text{As, } M >>> m_e)$$

So, $\mu_H = m_e\left[1 + \frac{m_e}{M}\right]^{-1} = m_e\left[1 - \frac{m_e}{M}\right]$

For deuterium M = 2M

Reduced mass of deuterium = μ_D

$$\mu_D = m_e\left[1 - \frac{m_e}{M}\right]\left[1 + \frac{m_e}{M}\right]$$

Total energy E_n of the electron revolving in an nth stationary orbit,

$$E_n = \frac{-me^4}{8n^2\varepsilon e_0^2 h^2}$$ **[2½]**

m is the reduced mass of the electron and proton in H atom.

So $h\nu = E_{n_i} - E_{n_f}$

$$v = \frac{m_e^4}{8\varepsilon_0^2 h^2}\left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right] = \frac{c}{\lambda}$$

$$\frac{1}{\lambda} \propto \mu \Rightarrow \lambda \propto \frac{1}{\mu}$$

For hydrogen and deuterium $= \frac{\lambda_H}{\lambda_D}$

So $\frac{\lambda_H}{\lambda_D} = \frac{\mu_H}{\mu_D} = \frac{m_e\left[1 - \frac{m_e}{M}\right]}{m_e\left[1 - \frac{m_e}{M}\right]\left[1 + \frac{m_e}{2M}\right]}$

$$\lambda_D = \left[1 + \frac{m_e}{2M}\right]^{-1}\lambda_H = \left(1 - \frac{m_e}{M}\right)\lambda_H$$

$$\lambda_D = \lambda_H(0.99973)$$

λ_H = 1218 Å, 1028 Å, 974 Å and 954 Å (Given)

$\lambda_{D1} = 0.9973\times 1218$ Å = 1214 Å

$\lambda_{D2} = 0.9973\times 1028 = 1025$ Å

$\lambda_{D3} = 0.9973\times 974 = 971$ Å

$\lambda_{D4} = 0.9973\times 954 = 951$ Å **[2½]**

Q. 3. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. What series of wavelengths will be emitted?
[NCERT Ex. Q. 12.9, Page 436]

Ans. Given that,
The energy of the electron beam used to bombard gaseous hydrogen at room temperature is 12.5 eV.
Also, the energy of the gaseous hydrogen in its ground state at room temperature is −13.6 eV.
And, When gaseous hydrogen is bombarded with an electron beam, the energy of the gaseous hydrogen becomes −13.6 + 12.5 eV (*i.e.*, −1.1 eV). **[1]**
Orbital energy is related to orbit level (n) as :

$$E = \frac{-13.6}{(n)^2} \text{ eV}$$

$$\text{For } n = 3,\ E = \frac{-13.6}{9} = -1.5 \text{ eV}$$

This energy is approximately equal to the energy of gaseous hydrogen. It can be concluded that the electron has jumped from $n = 1$ to $n = 3$ level.
During its de-excitation, the electrons can jump from $n = 3$ to $n = 1$ directly, which forms a line of the Lyman series of the hydrogen spectrum. **[1]**
We know the relation for wave number for Lyman series as :

$$\frac{1}{\lambda} = R_y\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$$

Where,
R_y = Rydberg constant = $1.097 \times 10^7\ m^{-1}$
λ = Wavelength of radiation emitted by the transition of the electron
For $n = 3$, we can obtain λ as :

$$\frac{1}{\lambda} = 1.097\times10^7\left(\frac{1}{1^2} - \frac{1}{3^2}\right)$$
$$= 1.097\times10^7\left(1 - \frac{1}{9}\right)$$
$$= 1.097\times10^7 \times \frac{8}{9}$$
$$\lambda = \frac{9}{8\times1.097\times10^7}$$
$$= 102.55 \text{ nm}$$ **[1]**

If the electron jumps from $n = 2$ to $n = 1$, then the wavelength of the radiation is given as :

$$\frac{1}{\lambda} = 1.097\times10^7\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$$
$$= 1.097\times10^7\left(1 - \frac{1}{4}\right)$$
$$= 1.097\times10^7 \times \frac{3}{4}$$
$$\lambda = \frac{4}{1.097\times10^7\times3}$$
$$= 121.51 \text{ nm}$$

If the transition takes place from $n = 3$ to $n = 2$, then the wavelength of the radiation is :

$$\frac{1}{\lambda} = 1.097\times10^7\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$
$$= 1.097\times10^7\left(\frac{1}{4} - \frac{1}{9}\right)$$
$$= 1.097\times10^7 \times \frac{5}{36}$$
$$\lambda = \frac{36}{5\times1.097\times10^7}$$
$$= 656.33 \text{ nm}$$ **[1]**

This radiation corresponds to the Balmer series of the hydrogen spectrum. So, in Lyman series, two wavelengths (*i.e.*, 102.5 nm and 121.5 nm) are emitted. And in the Balmer series, one wavelength (*i.e.*, 656.33 nm) is emitted. **[1]**

Q. 4. If a proton had a radius R and the charge was uniformly distributed, calculate using Bohr theory, the ground state energy of a H-atom when
(i) R = 0.1 Å, and (ii) R = 10 Å.
[NCERT Exemp. Q. 12.26, Page 79]

Ans. **(i)** Consider in H atom nucleus as point charge electron is revolving around nucleus with speed v and radius r_A. The Coulombian force provides centripetal force to revolve around nucleus.

$$\therefore \frac{m_e v^2}{r_A} = \frac{-ke^2}{r_A^2} \quad \text{...(I)}$$

$$\text{where } k = \frac{1}{4\pi\varepsilon_0}$$

Negative (–) sign shows the force of attraction.
By Bohr's postulate, angular momentum

$$= \frac{nh}{2\pi}$$
$$mvr_A = \frac{nh}{2\pi}$$
$$v = \frac{nh}{2\pi m r_A}$$
$$\frac{mn^2h^2}{4\pi^2m^2r_A^2r_A} = \frac{ke^2}{r_A^2} \text{ [From I]}$$
$$r_A = \frac{n^2h^2}{4\pi^2mke^2} \quad \text{...(II)}$$

For ground state $n = 1$

$$r_A = \frac{h^2}{4\pi^2mke^2}$$

Put the values,

$$= \frac{6.63\times10^{-34}\times6.63\times10^{-34}}{(2\times3.14)^2\times9.1\times10^{-31}\times9\times10^9\times1.6\times10^{-19}\times1.6\times10^{-19}}$$
$$= \frac{6.63\times6.63\times10^{-68+38+31-9}}{9.19\times1.6\times1.6\times4\times3.14\times4\times3.14}$$
$$\Rightarrow r_A = 0.53 \text{ Å}$$

Potential energy,

$$\text{P.E} = \frac{-ke^2}{r_A}$$
$$= \frac{-9\times10^9\times1.6\times10^{-19}\times1.6\times10^{-19}}{0.53\times10^{-10}} \text{J}$$
$$= \frac{-9\times1.6\times1.6\times10^{-19}\times10^{-19}\times10^9}{0.53\times10^{-10}\times1.6\times10^{-19}} \text{J}$$

$$\text{P.E.} = \frac{14.4}{0.53} = 27.17 = 27.2 \text{ eV}$$

Kinetic energy, $\text{K.E} = \frac{1}{2}mv^2$

$= \frac{1}{2}\frac{m \cdot n^2h^2}{4\pi^2m^2r^2} = \frac{1}{2}\frac{h^2}{4\pi^2mr^2}$ ($n = 1$ for ground state)

$= \frac{1}{2}\frac{6.62\times10^{-34}\times6.62\times10^{-34}}{4\times3.14\times3.14\times9\times10^{-31}\times0.53\times10^{-10}\times0.53\times10^{-10}}$ J

$= \frac{6.62\times6.62\times10^{-68+31+10+10}}{4\times3.14\times3.14\times18\times0.53\times0.53\times1.6\times10^{-19}}$ eV

$= \frac{6.62\times6.62\times10^{-68+51+19}}{4\times3.14\times3.14\times18\times0.53\times0.53}$

K.E. = 0.1373 × 10^2 eV = 13.7 eV

P.E. = 27.2 eV **[2½]**

(ii) Now for spherical nucleus of radius, R, electron moves charge inside the nucleus $R >> r_b$ then electron moves inside the nucleus. Then (r_b is radius of new Bohr's orbit of revolving electron).

$$\text{Charge} = \frac{e\cdot\left(\frac{4}{3}\pi r_b^3\right)}{\frac{4}{3}\pi R^3}$$

$$e' = q_2 = \frac{er_b^3}{R^3}$$

$$q_1 = e$$

$$\frac{mv^2}{r_b} = \frac{kee'}{r_b^2} \text{ (By Coulomb's law)}$$

$$mvr_b = \frac{nh}{2\pi}$$

$$\Rightarrow v_b = \frac{nh}{2\pi mr_b} \text{ (By Bohr's postulate)}$$

$$\therefore \frac{m}{r_b}\frac{n^2h^2}{4\pi^2m^2r_b^2} = \frac{kee'}{r_3^2}$$

$$r_b = \frac{n^2h^2}{4\pi^2mkee'}$$

Now for ground state of H, $n = 1$ and $e' = \frac{er_A^3}{R^3}$, then

$$\therefore\ r_b = \frac{h^2}{4\pi^2mk\cdot e\cdot e\cdot\frac{r_b^3}{R^3}} = \left(\frac{h^2}{4\pi^2mke^2}\right)\times\frac{R^3}{r_b^3} = r_A\times\frac{R^3}{r_b^3}$$

$\left[\because r_A = \frac{h^2}{4\pi^2mke^2} = 0.53 \text{ Å calculated in part (i)}\right]$

$$r_b = r_A\left(\frac{R}{r_b}\right)^3$$

$$r_b^4 = r_AR^3 = 0.53 \text{ Å}\times(10 \text{ Å})^3$$

$$r_b^4 = 0.53\times1000 \text{ (Å)}^4 \qquad (\because r_A = 0.53 \text{ Å})$$

$$r_b = [530 \text{ Å}^4]^{1/4} = 4.8 \text{ Å} < R = 10 \text{ Å}$$

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$= \frac{m}{2}\cdot\frac{h^2}{4\pi^2m^2r_b^2}$$

$$= \frac{h^2}{8\pi mr_b^2} \qquad \left[\because v^2 = \frac{n^2h^2}{4\pi^2m^2r_b^2}\right]$$

$$= \frac{6.62\times6.62\times10^{-34}\times10^{-34}}{8\times3.14\times3.14\times9.1\times10^{-31}\times4.8\times4.8\times10^{-20}\times1.6\times10^{-19}} \text{ eV}$$

$$= \frac{6.62\times6.62\times10^{-68+31+20+19}}{8\times3.14\times3.14\times9.1\times4.8\times4.8\times1.6} \text{ eV}$$

$$= \frac{43.8244}{26460.2}10^{-58+70} = 0.001656\times10^2 \text{ eV}$$

K.E. = 0.167 eV

$$\text{P.E.} = \frac{e^2}{4\pi\varepsilon_0}\frac{(r_b^2 - 3R^2)}{R^3} \qquad \left(\text{P.E} = \frac{kq_1q_2}{r}\right)$$

$$\text{P.E.} = \left[\frac{e^2}{4\pi\varepsilon_0r_A}\right]\frac{r_A(r_b^2 - 3R^2)}{R^3} \qquad \left[\text{multiplied by}\frac{r_A}{r_A}\right]$$

From part (i)

$$\text{P.E.} = \frac{e^2}{4\pi\varepsilon_0r_A} = 27.2 \text{ eV}$$

$$\therefore \text{P.E.} = 27.2\left[\frac{0.53(\sqrt{530} - 300)}{1000}\right] \quad \left(\because r_b = (530)^{1/4} \text{ Å and } R = 10 \text{ Å}\right)$$

$$= \frac{27.2 \text{ eV}\times0.53(23.02 - 300) \text{ Å}^3}{1000 \text{ Å}^3}$$

$$= 27.2\times\frac{0.53(-276.9)}{1000} \text{ eV}$$

$$\text{P.E.} = \frac{3992.9}{1000}$$

$$= -3.99 \text{ eV}$$

K.E. = 0.167 eV **[2½]**

Q. 5. In the Auger process an atom makes a transition to a lower state without emitting a photon. The excess energy is transferred to an outer electron which may be ejected by the atom. (This is called an Auger electron). Assuming the nucleus to be massive, calculate the kinetic energy of an $n = 4$ Auger electron emitted by Chromium by absorbing the energy from a $n = 2$ to $n = 1$ transition.

[NCERT Exemp. Q. 12.27, Page 79]

Ans. The energy E_n of the nth state

$$E_n = +Z^2R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = Z^2R\left(\frac{1}{1} - \frac{1}{4}\right) \text{(for } n_1 = 1, n_2 = 2)$$

$Z = 24$

R_y = Rydberg constant

$$\therefore E_n = \frac{3}{4}Z^2R$$

The energy required to eject an electron from $n = 4$ state is

$$E_4 = Z^2R\frac{1}{4^2} = \frac{1}{16}Z^2R$$

Energy given to electron is converted into K.E., of ejected electron. **[2½]**

So, the K.E. of Auger (ejected) electron = $E_n \cdot E_4$

$$\text{K.E} = Z^2R\frac{3}{4} - \frac{1}{16}Z^2R = \frac{11}{16}Z^2R = \frac{11}{16}\times24\times24\times13.6 \text{ eV}$$

K.E. = 11 × 36 × 13.6 = 5385.6 eV **[2½]**

Q. 6. The inverse square law in electrostatics is

$$|\mathbf{F}| = \frac{e^2}{(4\pi\varepsilon_0)\cdot r^2}$$

for the force between an electron and a proton.

The $\left(\frac{1}{r^2}\right)$ dependence of |F| can be understood in quantum theory as being due to the fact that the 'particle' of light (photon) is massless. If photons had a mass m_p, force would be modified to

$$|F| = \frac{e^2}{(4\pi\varepsilon_0)r^2}\left[\frac{1}{r^2}+\frac{\lambda}{r}\right].\exp(-\lambda_r)$$

where $\lambda = \frac{m_p c}{h}$ **and** $\hbar = \frac{h}{2\pi}$.

Estimate the change in the ground state energy of a H-atom if m_p were 10^{-6} times the mass of an electron.

[NCERT Exemp. Q. 12.28, Page 80]

Ans. Mass of photon $= 9.1 \times 10^{-31} \times 10^{-6}$ kg
$= 9.1 \times 10^{-37}$ kg

Wavelength associated with a photon $= \frac{h}{m_p c}$

$$\lambda = \frac{6.62\times10^{-34}}{9.1\times10^{-37}\times3\times10^{8}}$$

$$= \frac{6.62}{9.1\times3}\times10^{-34+37-8}$$

$$= 2.4\times10^{-7} >> r_A$$

$$\lambda << \frac{1}{r_A} < e\lambda_{r_A} << 1$$

$$U(r) = \frac{-e^2 e^{-\lambda_r}}{\pi\varepsilon_0 r}$$

$$mvr = \frac{h}{2\pi} = \hbar \text{ or } v = \frac{\hbar}{mr} \quad ...(I)$$

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0}\left[\frac{1}{r^2}+\frac{\lambda}{r}\right]$$

$$\left[\because F = \frac{e^2}{(4\pi\varepsilon_0)r^2}\left[\frac{1}{r^2}+\frac{\lambda}{r}\right]e^{-\lambda_r} \text{ given}\right]$$

$$\frac{m}{r}\cdot\frac{\hbar^2}{m^2r^2} = \frac{e^2}{4\pi\varepsilon_0}\left[\frac{1+\lambda r}{r^2}\right]$$

$$\frac{\hbar^2}{mr} = \frac{e^2}{4\pi\varepsilon_0}(1+\lambda r)$$

$$\frac{\hbar^2 4\pi\varepsilon_0}{me^2} = (r+\lambda r^2)$$

If $\lambda = 0, r = r_A = \frac{\hbar^2 4\pi\varepsilon_0}{me^2}$ [Neglecting r^2]

$$\frac{\hbar}{m} = \frac{e^2}{4\pi\varepsilon_0}r_A \quad (r_A \approx r+\lambda r^2)$$

$\because \lambda_A >> r_B$ and $r = r_A + \delta$

Taking $r_A + r + \lambda r^2$ $[\because r = (r_a + \delta)]$

$$r_A = (r_A+\delta)+\lambda(r_A+\delta)(\text{put } r = r_A+\delta^2)$$

$$0 = \delta + \lambda r_A^2 + 2r_A\delta\lambda$$

(Neglecting small term δ^2)

$$0 = \delta + 2r_A\delta\lambda + \lambda r_A^2$$

$$\Rightarrow \delta[1+2r_A\lambda] = -\lambda r_A^2$$

$$\delta = \frac{-\lambda r_A^2}{(1+2r_A\lambda)} = \lambda r_A^2(1+2r_A\lambda)^{-1}$$

$$\delta = -\lambda r_A^2[1-2r_A\lambda] = -\lambda r_A^2 + 2r_A^3\lambda^2$$

$\therefore \lambda$ and $r_A << 1$ so $r_A^3\lambda^2$ is very small so by neglecting it we get,

$$\delta = -\lambda r_A^2$$

$$V(r) = \frac{-e^2}{4\pi\varepsilon_0} = \frac{e^{(-\lambda\delta-\lambda y_\gamma)}}{(r_A+\delta)} = \frac{-e^2}{4\pi\varepsilon_0}\cdot\frac{e^{-\lambda(\delta+\gamma)}}{r_A\left(1+\frac{\delta}{r_A}\right)}$$

$$= \frac{-e^2}{4\pi\varepsilon_0 r_A}e^{-\lambda\gamma}\left(1+\frac{\delta}{r_A}\right)^{-1} \quad (\because r = r_A+\delta^2)$$

$$= \frac{-e^2 e^{-\lambda\gamma}}{4\pi r\varepsilon_0 r_A}\left(1-\frac{\delta}{r_A}\right) \quad \textbf{[2½]}$$

$V(r) = -27.2$ eV remains unchanged.

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m\left(\frac{\hbar}{mr}\right)^2$$

$$= \frac{1}{2}\frac{\hbar^2}{mr^2} \quad \left(\text{From I, } v = \frac{\hbar}{mr}\right)$$

$$= \frac{-\hbar^2}{2m(r_A+\delta)^2}$$

$$= \frac{-\hbar^2}{2mr_A^2\left(1+\frac{\delta}{r_A}\right)^2}$$

$$= \frac{\hbar^2}{2mr_A^2}\left(1-\frac{2\delta}{r_A}\right)$$

$$= \frac{\hbar^2}{2mr_A}\left(1-\frac{-\lambda r_A^2}{r_A}\right) \quad (\because \delta = -\lambda r_A^2)$$

$$= \frac{\hbar^2}{2mr_A}(1+2\lambda r_A)$$

$$= 13.6 \text{ eV } (1 + 2\lambda r_A)$$

$$\text{Total energy} = \frac{-e^2}{4\pi\varepsilon_0 r_A} + \frac{\hbar^2}{2mr_A^2}(1+2\lambda r_A)$$

$$= [-27.2 + 13.6(1+2\lambda r_A)]\text{eV}$$

$$= -27.2 + 13.6 + 27.2\lambda r_A \text{ eV}$$

$$\text{Total E} = -13.6 + 27.2\lambda r_A$$

$$\text{Change in energy} = -13.6 + 27.2\lambda r_A - (-13.6)$$

$$= 27.2\lambda r_A \text{ eV}$$

[2½]

Q. 7. Classically, an electron can be in any orbit around the nucleus of an atom. Then what determines the typical atomic size? Why is an atom not, say, thousand times bigger than its typical size? The question had greatly puzzled Bohr before he arrived at his famous model of the atom that you have learnt in the text. To simulate what he might well have done before his discovery, let us play as follows with the basic constants of nature and see if we can get a quantity with the dimensions of length that is roughly equal to the known size of an atom ($\sim 10^{-10}$m).

(a) Construct a quantity with the dimensions of length from the fundamental constants e, m_e, and c. Determine its numerical value.

(b) You will find that the length obtained in (a) is many orders of magnitude smaller than the atomic dimensions. Further, it involves c. But energies of atoms are mostly in non-relativistic domain where *c* is not expected to play any role. This is what may have suggested Bohr to discard *c* and look for 'something else' to get the right atomic size. Now, the Planck's constant *h* had already made its appearance elsewhere. Bohr's great insight lay in recognising that *h*, m_e, and *e* will yield the right atomic size. Construct a quantity with the dimension of length from *h*, me, and *e* and confirm that its numerical value has indeed the correct order of magnitude.

[NCERT Add. Ex. Q. 12.14, Page 437]

Ans. **(a)** m_e = Mass of electron = 9.1×10^{-31} kg

e = Charge of an electron = 1.9×10^{-19} C

c = Speed of light = 3×10^8 m/s

Let us take a quantity involving the given quantities as :

$$= \frac{e^2}{4\pi\varepsilon_0 m_e c^2}$$

Where,

ε_0 = Permittivity of free space

And, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{Nm}^2\text{C}^{-2}$

The numerical value of the taken quantity will be :

$$\frac{1}{4\pi\varepsilon_0} \times \frac{e^2}{m_e c^2}$$

$$= 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{9.1 \times 10^{-31} \times (3 \times 10^8)^2}$$

$$= 2.81 \times 10^{-15} \text{ m}$$

So, the numerical value of the taken quantity is much smaller than the typical size of an atom. **[2½]**

(b) Given that,

h = Planck's constant = 6.63×10^{-34} Js

m_e = Mass of electron = 9.1×10^{-31} kg

e = Charge of an electron = 1.9×10^{-19} C

Let us take a quantity involving the given quantities as

$$\frac{4\pi\varepsilon_0 \left(\frac{h}{2\pi}\right)^2}{m_e e^2}$$

Where, ε_0 = Permittivity of free space

And, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 Nm^2C^{-2}$

The numerical value of the taken quantity will be :

$$= 4\pi\varepsilon_0 \times \frac{\left(\frac{h}{2\pi}\right)^2}{m_0 e^2}$$

$$= \frac{1}{9 \times 10^9} \times \frac{\left(\frac{6.63 \times 10^{-34}}{2 \times 3.14}\right)^2}{9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2}$$

$$= 0.53 \times 10^{-10} \text{m}$$

So, the value of the quantity taken is of the order of the atomic size. **[2½]**

Q. 8. The Bohr model for the H-atom relies on the Coulomb's law of electrostatics. Coulomb's law has not directly been verified for very short distances of the order of angstroms. Supposing Coulomb's law between two opposite charge $+q_1 - q_2$ is modified to

$$|\mathbf{F}| = \frac{q_1 q_2}{(4\pi\varepsilon_0)} \frac{1}{r^2}, r \geq R_0$$

$$= \frac{q_1 q_2}{4\pi\varepsilon_0} \frac{1}{R_0^2} \left(\frac{R_0}{r}\right)^{\varepsilon}, r \leq R_0$$

Calculate in such a case, the ground state energy of a *H* atom, if $\varepsilon = 0.1, R_0 = 1$ Å.

[NCERT Exemp. Q. 12.29, Page 80]

Ans. Case I : When $r \leq R_0 = 1$ Å

Let $\varepsilon = 2 + \delta$

$$F = \frac{q_1 q_2}{4\pi_0 R_0^2} \frac{R_0^{r+\delta}}{r^{2+\delta}} = \left(\frac{e \cdot (-e)}{4\pi\varepsilon_0}\right) \frac{R^\delta}{r^{2+\delta}} [q_1 = e, q_2 = -e]$$

$$|F| = -(1.6 \times 10^{-19})^2 \times 9 \times 10^9 \frac{R^\delta}{r^{2+\delta}} \left[\begin{array}{l}\text{Negative (–) ve sign}\\ \text{shows force of}\\ \text{attraction.}\end{array}\right]$$

$$|F| = 23.04 \times 10^{-29} \frac{R^\delta}{r^{2+\delta}}$$

The electrostatic force of attraction between positively charged nucleus and negatively charged electron provides necessary centripetal force.

$$\frac{mv^2}{r} = (23.04 \times 10^{-29}) \frac{R_0^\delta}{r^{2+\delta}}$$

Let $23.04 \times 10^{-29} = A$

$$\therefore v^2 = A \frac{R_0^\delta}{r^{2+\delta}} \cdot \frac{r}{m} = \frac{AR_0^\delta}{mr^{1+\delta}}$$

By Bohr's second postulate, angular momentum

$$L = \frac{nh}{2\pi} \text{ and } \hbar = \frac{h}{2\pi}$$

$$\therefore L = n\hbar$$

$$mvr = n\hbar$$

$$r = \frac{m\hbar}{mv} = \frac{n\hbar}{m} \sqrt{\frac{mr^{1+\delta}}{\Lambda R_0^{\ \delta}}} = \frac{\hbar}{m} \left[\frac{m}{\Lambda R_0^{\ \delta}}\right]^{1/2} r^{1/2+\delta/2}$$

$$r^{1-\frac{1}{2}-\frac{\delta}{2}} = \frac{n\hbar}{m} \left[\frac{m}{\Lambda R_0^\delta}\right]^{1/2}$$

$$r^{\frac{1}{2}(1-\delta)} = \frac{n\hbar}{m} \left[\frac{m}{\Lambda R_0^\delta}\right]^{1/2}$$

$$r_n = \left[\frac{n^2\hbar^2}{m\Lambda R_0^\delta}\right]^{\frac{1}{1-\delta}}$$

$$= \left[\frac{\hbar^2}{m\Lambda R_0^\delta}\right]^{\frac{1}{1-\delta}} \qquad (n = 1 \text{ for ground state})$$

$$r_1 = \left[\frac{(1.05 \times 10^{-34})^2}{9.1 \times 10^{-31} \times 23.04 \times 10^{-29} \times 10^{19}}\right]^{\frac{1}{29}}$$

$$\left[\because \hbar = \frac{h}{2\pi} = \frac{6.6 \times 10^{-34}}{2 \times 3.14} = (1.05 \times 10^{-34})\right]$$

Where $h = 1.05 \times 10^{-34}$ Js^{-1}

$R_0 = 10^{19}$ and $1 - \delta = 2.9$

$R_1 = 8 \times 10^{-11}$ m = 0.08 nm

This is a radius of orbit of electron in ground state of hydrogen atom.

Velocity of electron in ground state

By second postulate of Bohr's Atomic model

$$mv_n r_n = \frac{nh}{2\pi} \qquad \left(\hbar = \frac{h}{2\pi}\right)$$

$$v_n = \frac{n\hbar}{mr_n}$$

$$= \frac{n\hbar}{m}\left[\frac{m\Lambda R_0^{\delta}}{n^2\hbar^2}\right]^{\frac{1}{1-\delta}}$$

[2½]

Case II :

$$n = 1, v_1 = \frac{\hbar}{mr_1}$$

$$= \frac{1.05 \times 10^{-34}}{9.1 \times 10^{-31} \times 0.08 \times 10^{-9}}$$

$$= \frac{1.05}{0.728} \times 10^{-34+31+9}$$

$$= 1.44 \times 10^{-34+40}$$

$$= 1.44 \times 10^{6}\ \text{m/s}$$

$$v_1 = 1.44 \times 10^{6}\ \text{m/s}$$

$$\text{K.E.} = \frac{1}{2}mv_1^2 - 9.43 \times 10^{-19}$$

$$= \frac{1}{2} \times 9.1 \times 10^{-31} \times (1.44 \times 10^{6})^2 - 9.43 \times 10^{-19}\ \text{J}$$

$$= 5.9\ \text{eV}$$

P.E. from R_0 to $\frac{-A}{R_0}$

P.E. from R_0 to $r = +AR_0^{\delta}\int_{R_0}^{r}\frac{dr}{r^{2+\delta}}$

This is the P.E. of electron in ground state R_0 to $r = \frac{-A}{R_0}$

$$\text{P.E.} = -\frac{AR_0^{\delta}}{(1+\delta)}\left[\frac{1}{r^{1+\delta}} - \frac{1}{R_0^{1+\delta}}\right]$$

$$= \frac{-A}{(1+\delta)}\left[\frac{R_0^{\delta}}{r^{1+\delta}} - \frac{1}{R_0}\right] = \frac{-A}{(1+\delta)}\left[\frac{R_0^{\delta}}{r^{1+\delta}} - \frac{1}{R_0} + \frac{1+\delta}{R_0}\right]$$

Put $\delta = -1.9$

$$\text{P.E.} = \frac{-A}{(1+(-1.9))}\left[\frac{R_0^{-1.9}}{r^{-0.9}} - \frac{1}{R_0} - \frac{1.9}{R_0}\right]$$

$$\text{P.E.} = \frac{-A}{(1+(-1.9))}\left[\frac{R_0^{-1.9}}{r^{-0.9}} - \frac{1}{R_0} - \frac{1.9}{R_0}\right]$$

$$= \frac{-A}{-0.9}\left[\frac{R_0^{-1.9}}{r^{-0.9}} - \frac{1.9}{R_0}\right]$$

$$= \frac{23.04 \times 10^{-2.9}}{0.9}[(0.8)^{0.9} - 1.9]\ J$$

$$= -17.3\ \text{eV}$$

Total E = P.E. + K.E.

$$= (-17.3 + 5.9)\ \text{eV}$$

Total energy = −11.4eV

This is the required total energy of electron in ground state of H-atom. **[2½]**

Some Commonly Made Errors

- Students get confuse to understand the Excitation Energy, Ionisation Energy, Excitation Potential and Ionisation Potential.
- Generally, students get confused in the numerical, based on the different Spectral Series of Hydrogen Atom.

EXPERT ADVICE

- *Do not skip to study the limitations of Bohr's atomic model theory and Schrodinger's equation.*
- *Apply method of elimination for MCQ questions. Look at the answers before even starting to solve the questions and see if you can eliminate any answer. When calculating answer for a big question you may not need to solve the problem fully in order to choose the right answer because you could eliminate the other three.*
- *Learn and understand the Bohr's postulates and de-Broglie's hypothesis.*
- *Understand the basic concepts behind the topic before you start any topic.*
- *Convert all units to SI, with charges in Coulombs and distances in meters, so as to be consistent with the SI value of the Coulomb's constant.*
- *Try to solve numerical by making suitable diagram, do not skip any required diagrams.*
- *Learn the values of Universal constants.*
- *Draw suitable diagram to explain Spectral series.*
- *Don't skip any important topics like- Bohr's model, spectral lines and de-Broglie's hypothesis, etc.*
- *Review your notes frequently and you will have the material mastered by test time.*

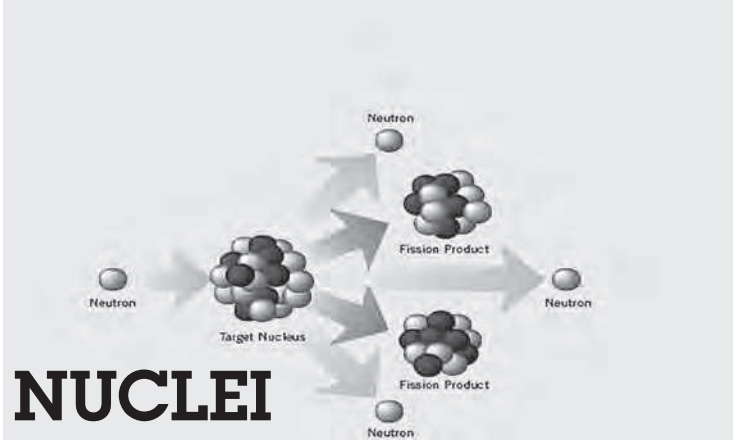
OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://goo.gl/rj2Q4i	*https://goo.gl/crB9fA*
Or Scan the Code	**Or Scan the Code**

CHAPTER

13 NUCLEI



Chapter Objectives

This chapter will help you understand :

- ***Nuclei :*** *Introduction; Terms related to nucleus; Mass defect; Nuclear binding energy; Nuclear force; Radio-activity; Life of a radio-active element; Activity of a radioactive element; Nuclear fission; Nuclear chain reaction; Nuclear reactor; and Nuclear fusion.*

Quick Review

- Nuclear physics is the study of atomic nuclei and their interactions.
- The largest atom (cesium) is approximately nine times bigger than the smallest atom (helium).
- Nuclear power plants use 'Nuclear fission' (the process of splitting an atom in two). 'Nuclear fusion' (the process of combining atoms into one) has the potential to be safer but has not yet been developed to operate within a large power plant.
- After Radium was discovered by Marie Curie, she died from radiation.
- All elements with more than 83 protons (i.e., an atomic number >83) are radio-active. Some radio-active isotopes also occur in elements with lower atomic numbers.
- Natural radio-active decay provides a source of heating in the Earth's interior that drives mantle dynamics and plate tectonics.

TIPS...

- *Study all terms related to nucleus.*
- *The atomic number is the number of protons and the atomic mass is equal to the number of protons added to the number of neutrons.*
- *Learn basic formulae in radio-activity decay.*
- *Understand the difference between Nuclear fission and Nuclear fusion.*

TRICKS...

- *Use natural log to solve questions related to radio-activity.*
- *During alpha decay, an alpha particle is emitted from the nucleus, it means the nucleus loses two protons and two neutrons.*
- *During beta decay, a neutron turns into a proton and emits an electron from the nucleus.*

Know the Links

- *http://hyperphysics.phy-astr.gsu.edu/hbase/NucEne/nucbin.html*
- *https://www.encyclopedia.com/science-and-technology/physics/physics/radioactivity*

Multiple Choice Questions

(1 mark each)

Q. 1. The gravitational force between a H-atom and another particle of mass m will be given by Newton's law :

$F = G\frac{M.m}{r^2}$, **where r is in km and**

(a) $M = m_{proton} + m_{electron}$

(b) $M = m_{proton} + m_{electron} - \frac{B}{c^2}$ **(B = 13.6 eV).**

(c) M **is not related to the mass of the hydrogen atom.**

(d) $M = m_{proton} + m_{electron} - \frac{|V|}{c^2}$ **($|V|$= magnitude of the potential energy of electron in the H-atom).**

[NCERT Exemp. Q. 13.2, Page 81]

Ans. Correct option : (b)

Explanation : During formation of H-atom some mass of nucleons convert into energy by the equation

$E=mc^2$

This energy is used to bind the nucleons along with nucleus. So mass of atom becomes slightly less than sum of actual masses of nucleons and electrons.

Actual mass of H atom

$= M_p + M_e - \frac{B.E.}{c^2}$ $\quad \frac{B}{c^2}$ = Binding energy

So, the binding energy of H atoms is 13.6 eV per atom.

Q. 2. Heavy stable nucleus have more neutrons than protons. This is because of the fact that

(a) neutrons are heavier than protons.

(b) electrostatic force between protons are repulsive.

(c) neutrons decay into protons through beta decay.

(d) nuclear forces between neutrons are weaker than that between protons.

[NCERT Exemp. Q. 13.6, Page 82]

Ans. Correct option : (b)

Explanation : Electrostatic force between proton-proton is repulsive which causes the instability of nucleus. So neutrons are larger than proton.

Q. 3. Suppose we consider a large number of containers each containing initially 10,000 atoms of a radioactive material with a half-life of 1 year. After 1 year,

(a) all the containers will have 5,000 atoms of the material.

(b) all the containers will contain the same number of atoms of the material but that number will only be approximately 5,000.

(c) the containers will in general have different numbers of the atoms of the material but their average will be close to 5,000.

(d) none of the containers can have more than 5000 atoms. **[NCERT Exemp. Q. 13.1, Page 81]**

Ans. Correct option : (c)

Explanation : Half-life time for a radio-active substance is defined as the time in which a radio-active atomic substance remains half of its original value of radio-active atom.

Given that,

Half-life = 1 year.

So, after 1 year means one half-life, that is, average atoms of radioactive substance remain after 1 year in each container is equal to half of 10,000 = 5,000 atoms (average).

Q. 4. When a nucleus in an atom undergoes a radioactive decay, the electronic energy levels of the atom

(a) do not change for any type of radioactivity.

(b) change for α and β radioactivity but not for *r* radioactivity.

(c) change for α radioactivity but not for others.

(d) change for β radioactivity but not for others.

[NCERT Exemp. Q. 13.3, Page 82]

Ans. Correct option : (b)

Explanation : As we know that,

α-particle : 2 unit of positive charge

β-particles : 1 unit positive charge

γ-particles : No charge

Due to this reason, energy levels of the atom change for α and β but not for others.

So the electronic energy level of the atom changes in emission of α and β particle, but not in Υ^* decay.

Q. 5. M_x and M_y denote the atomic masses of the parent and the daughter nuclei respectively in a radioactive decay. The *Q*-value for a β-decay is Q_1 and that for a β^+ decay is Q_2. If m_e denotes the mass of an electron, then which of the following statements is correct?

(a) $Q_1 = (M_x - M_y)c^2$ **and** $Q_2 = (M_x - M_y - 2m_e)c^2$

(b) $Q_1 = (M_x - M_y)c^2$ **and** $Q_2 = (M_x - M_y)c^2$

(c) $Q_1 = (M_x - M_y - 2m_e)c^2$ **and** $Q_2 = (M_x - M_y + 2m_e)c^2$

(d) $Q_1 = (M_x - M_y + 2m_e)c^2$ **and** $Q_2 = (M_x - M_y + 2m_e)c^2$

[NCERT Exemp. Q. 13.4, Page 82]

Ans. Correct option : (a)

Explanation : Let the parent nuclei ${}^A_zX$ is radioactive atom and decay β^- as :

$${}_ZX^A \rightarrow {}_{z+1}Y^A + {}_{-1}e^0 + \overline{\nu} + Q_1$$

$$Q_1 = \left[m_n({}_zX^A) - m_n({}_{z+1}Y^A) - m_e\right]c^2$$

$$= \left[m_n({}_zX^A) + m_eZ - m_n\left({}_{z+1}Y^A\right) - (Z+1)M_e\right]c^2$$

$$= \left[m({}_zX^A) - m({}_{z+1}Y^A)\right]C^2$$

Let the nucleus = ${}_zX^A$ radiate β decay

$${}_zx^A \rightarrow {}_{z-1}Y^A + {}_1e^0 + \nu + Q_2$$

$$Q_2 = [m_n({}_zX^A) - m({}_{z-1}Y^A) - 2m_e]c^2$$

$$= [m_n({}_zX^A) + m_eZ - m_n({}_{z-1}Y^A) - (Z-1)m_e - 2m_e]c^2$$

$$= [m({}_zX^A) - m_e({}_{z-1}Y^A) - 2m_e]c^2$$

$$= (M_x - M_y - 2m_e)c^2$$

Q. 6. Tritium is an isotope of hydrogen whose nucleus Triton contains 2 neutrons and 1 proton. Free neutrons decay into $p + \bar{e} + \nu$. If one of the neutrons in Triton decays, it would transform into ${}_2He^3$ nucleus. This does not happen. This is because

(a) triton energy is less than that of a He^3 nucleus.

(b) the electron created in the beta decay process cannot remain in the nucleus.

(c) both the neutrons in triton have to decay simultaneously resulting in a nucleus with 3 protons, which is not a He^3 nucleus.

(d) because free neutrons decay due to external perturbations which is absent in a triton nucleus.

[NCERT Exemp. Q. 13.5, Page 82]

Ans. Correct option : (a)

Explanation : Triton (${}^3_1H$) has 1 proton and 2 neutrons.

If a neutron decays as,

$n \rightarrow p + \bar{e} + \nu$

Then nucleus will have 2 protons and 1 neutron, *i.e.* triton atom converts in ${}_2He^3$ (2 proton and 1 neutron). Binding energy of ${}_1H^3$ is much smaller than ${}_2He^3$ so transformation is not possible energetically.

Q. 7. In a nuclear reactor, moderators slow down the neutrons which come out in a fission process. The moderator used has light nuclei. Heavy nuclei will not serve the purpose because

(a) they will break up.

(b) elastic collision of neutrons with heavy nuclei will not slow them down.

(c) the net weight of the reactor would be unbearably high.

(d) substances with heavy nuclei do not occur in liquid or gaseous state at room temperature.

[NCERT Exemp. Q. 13.7, Page 82]

Ans. Correct option : (b)

Explanation : For elastic collision masses of both must be equal so that they can exchange the velocities. To slow down the speed of neutron substance should be made up of 1 proton for perfectly elastic, that is, we need light nuclei not heavy.
For heavy nuclei only direction will change not the speed.

Tick Two or More Options

Q. 8. Samples of two radioactive nuclides A and B are taken. λ_A and λ_B are the disintegration constants of A and B respectively. In which of the following cases, the two samples can simultaneously have the same decay rate at any time?

(a) Initial rate of decay of A is twice the initial rate of decay of B and $\lambda_A = \lambda_B$.

(b) Initial rate of decay of A is twice the initial rate of decay of B and $\lambda_A > \lambda_B$.

(c) Initial rate of decay of B is twice the initial rate of decay of A and $\lambda_A > \lambda_B$.

(d) Initial rate of decay of B is same as the rate of decay of A at $t = 2h$ and $\lambda_B < \lambda_A$.

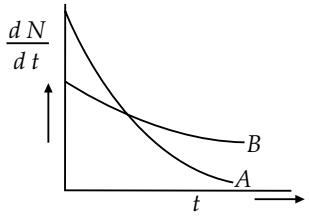
[NCERT Exemp. Q. 13.9, Page 83]

Ans. Correct option : (b) and (d)

Explanation : The two samples of the two radioactive nuclides A and B can simultaneously have the same decay rate at any time if initial rate of decay of A is twice the initial rate of decay of B and $\lambda_A > \lambda_B$.

Also, when initial rate of decay of B is same as rate of decay of A at $t = 2h$ and $\lambda_B < \lambda_A$.

Q. 9. The variation of decay rate of two radioactive samples A and B with time is shown in Figure. Which of the following statements are true?



(a) Decay constant of A is greater than that of B, hence A always decays faster than B.

(b) Decay constant of B is greater than that of A but its decay rate is always smaller than that of A.

(c) Decay constant of A is greater than that of B but it does not always decay faster than B.

(d) Decay constant of B is smaller than that of A but still its decay rate becomes equal to that of A at a later instant. [NCERT Exemp. Q. 13.10, Page 83]

Ans. Correct options : (c) and (d)

Explanation : From the figure, we can observe that the slope of curve A is greater than that of curve B. So, the rate of decay is faster for A than that of B. At point P shown in the diagram the two curves intersect. Hence at point P, the rate of decay for both A and B is the same.

Q. 10. Fusion processes, like combining two deuterons to form a He nucleus are impossible at ordinary temperatures and pressure. The reasons for this can be traced to the fact :

(a) nuclear forces have short range.

(b) nuclei are positively charged.

(c) the original nuclei must be completely ionized before fusion can take place.

(d) the original nuclei must first break up before combining with each other.

[NCERT Exemp. Q. 13.8, Page 83]

Ans. Correct options : (a) and (b)

Explanation : The process of fusion is impossible at ordinary pressure and temperature. The reason is nuclei are positively charged and nuclear forces are short range strongest forces.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Which one of the following cannot emit radiation and why? Excited nucleus, excited electron.

[NCERT Exemp. Q. 13.14, Page 84]

Ans. Excited electron because energy of electronic energy levels is in the range of eV, only not in MeV. as γ- radiation has energy in MeV. **[1]**

Q. 2. $_1He^3$ and $_2He^3$ nuclei have the same mass number. Do they have the same binding energy?

[NCERT Exemp. Q. 13.11, Page 84]

Ans. In $_2He^3$ there are two protons which give electrostatic force of repulsion but in $_1He^3$ between nucleons there is only nuclear attractive force. So binding energy of $_1He^3$ is larger than $_2He^3$. **[2]**

Q. 3. Four nuclei of an element undergo fusion to form a heavier nucleus, with release of energy. Which of the two - the parent or the daughter nucleus - would have higher binding energy per nucleon?

[CBSE Board, Delhi Region, 2018]

Ans. Nuclear fusion :

The process in which two light nuclei combined to form a single heavier nucleus is called nuclear fusion. **[½]**

The mass of the heavier nucleus formed is less than the sum of masses of the combining nuclei. So, mass defect is more in daughter nuclei so binding energy is more per nucleon in daughter nuclei. **[½]**

Q. 4. Obtain the binding energy (in MeV) of a nitrogen nucleus $_7N^{14}$, given m ($_7N^{14}$) = 14.00307 u.

[NCERT Ex. Q. 13.3, Page 462]

Ans. Atomic mass of nitrogen $_7N^{14}$, $m = 14.00307\ u$

A nucleus of $_7N^{14}$ nitrogen contains 7 neutrons and 7 protons.

So that, the mass defect of this nucleus, $\Delta m = 7m_H + 7m_n - m$

Where,

Mass of proton, $m_H = 1.007825\ u$

Mass of neutron, $m_n = 1.008665\ u$

$\Delta m = 7 \times 1.007825 + 7 \times 1.008665 - 14.00307$

$= 7.054775 + 7.06055 - 14.00307$

$= 0.11236\ u$

But $1\ u = 931.5\ \text{MeV/c}^2$

$\Delta m = 0.11236 \times 931.5\ \text{MeV/c}^2$ **[1]**

So that, the binding energy of the nucleus is :

$E_b = \Delta m c^2$

Where, c = Speed of light

$$E_b = 0.11236 \times 931.5 \left(\frac{MeV}{c^2}\right) \times c^2$$

$= 104.6634\ \text{MeV}$

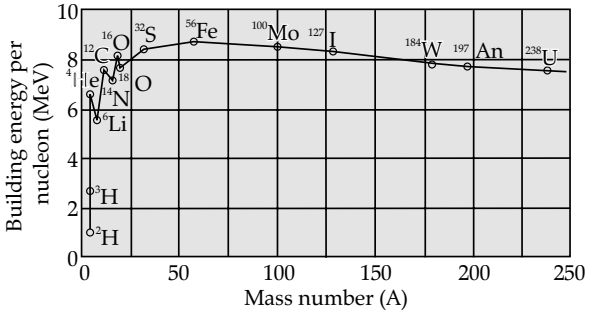
So that, the binding energy of a nitrogen nucleus is 104.66334 MeV. **[1]**

Q. 5. Explain the processes of nuclear fission and nuclear fusion by using the plot of binding energy per nucleon (*BE*/*A*) versus the mass number *A*.

[CBSE Board, Delhi Region, 2018]

Ans. **Nuclear fission :** In a nuclear fission, the sum of the masses of final products is less than the sum of the masses of the reactant components. The difference of masses called mass defect, is converted into energy. If we divide this energy with nucleons then it is called **binding energy per nucleons.** **[½]**

Nuclear fusion : The process in which two light nuclei combine to form a single heavier nucleus is called nuclear fusion. The mass of heavier nucleus formed is less than the sum of the masses of the combining nuclei. The mass defect is released as energy. If we divide this energy by nucleons than it is called binding energy per nucleons. **[½]**



[1]

Q. 6. A radioactive isotope has a half-life of 10 years. How long will it take for the activity to reduce to 3.12%? [CBSE Board, Delhi Region, 2018]

Ans. As we know that,

$N = N_0 e^{-\lambda t}$

Given that,

$N = 3.125\%$ of N_o

$N = \frac{3.125}{100} \times N_0$, or

$N = \frac{N_0}{32}$

$\frac{N_0}{32} = N_0 e^{-\lambda t}$

$N_0 \times 2^{-5} = N_0 e^{-\lambda t}$

$2^5 = e^{\lambda t}$

$5 \log 2 = \lambda t$

$t = \frac{5 \log 2}{\lambda}$ **[1]**

$\lambda = \frac{\log 2}{T_{1/2}}$

$\lambda = \frac{\log 2}{10}$

$t = \frac{5 \log 2}{\log 2} \times 10$

$t = 50$ years

Thus, the isotope will take about 50 years to reduce to 3.125% of its original value. **[1]**

Q. 7. Draw a graph showing the variation of decay rate with number of active nuclei.

[NCERT Exemp. Q. 13.12, Page 84]

Ans. By law of radio-active decay :

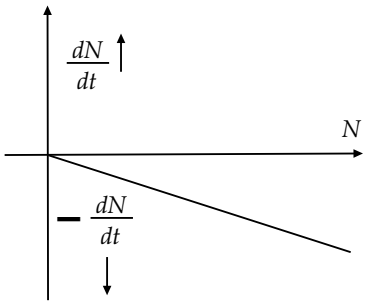
$-\frac{dN}{dt} \propto \lambda N$

Where,

N is the number of radioactive nuclei in the sample.

So, $\frac{dN}{dt}$ can be negative. **[1]**

The variation of decay rate with number of active nuclei is :



[1]

Q. 8. In pair annihilation, an electron and a positron destroy each other to produce gamma radiation. How is the momentum conserved?

[NCERT Exemp. Q. 13.15, Page 84]

Ans. When an electron and positron combine together coming from opposite directions they destroy each other by the emission of two γ-rays in opposite direction to conserve linear momentum as below : **[1]**

${}_0e^{-1} + {}_0e^{+1} \rightarrow 2\gamma - \text{ray photons.}$ **[1]**

Q. 9. Which sample, *A* or *B* shown in Figure has shorter mean-life? [NCERT Exemp. Q. 13.13, Page 84]

Ans. *B* has shorter mean life as λ is greater for *B*. This can be explained mathematically as :

From the given graph, at

$$t = 0, \left(\frac{dN}{dt}\right)_A = \left(\frac{dN}{dt}\right)_B \Rightarrow (N_0)_A = (N_0)_B$$

Considering any instant t by drawing a line perpendicular to time axis, we find that

$$\left(\frac{dN}{dt}\right)_A = \left(\frac{dN}{dt}\right)_B$$

$$\Rightarrow \lambda_A N_A > \lambda_B N_B$$

$\because \quad N_A > N_B$ (Rate of decay of *B* is slower)

$\because \quad \lambda_B > \lambda_A$

As, average life, $\tau = \frac{1}{\lambda}$

$\Rightarrow \tau_A > \tau_B$

So mean life time of sample *A* is greater than of *B*. **[2]**

Q. 10. The normal activity of living carbon-containing matter is found to be about 15 decays per minute for every gram of carbon. This activity arises from the small proportion of radioactive $_6C^{14}$ present with the stable carbon isotope $_6C^{12}$. When the organism is dead, its interaction with the atmosphere (which maintains the above equilibrium activity) ceases and its activity ceases and its activity begins to drop. From the known half-life (5,730 years) of $_6C^{14}$, and the measured activity, the age of the specimen can be approximately estimated. This is the principle of $_6C^{14}$ dating used in archaeology. Suppose a specimen from Mohenjo-Daro gives an activity of 9 decays per minute per gram of carbon. Estimate the approximate age of the Indus-Valley civilisation. [NCERT Ex. Q. 13.8, Page 462]

Ans. Given that,

Decay rate of living carbon-containing matter, $R = 15$ decay/min

Half-life of $_6C^{14}$, $T_{1/2} = 5{,}730$ years

Let N be the number of radio-active atoms present in a normal carbon-containing matter.

The decay rate of the specimen obtained from the Mohenjo-Daro site :

$R' = 9$ decays/min

Let N' be the number of radio-active atoms present in the specimen during the Mohenjo-Daro period.

So, the decay constant, λ and time, t as :

$$\frac{N'}{N} = \frac{R'}{R} / R = e^{-\lambda t}$$

$$e^{-\lambda t} = \frac{9}{15} = \frac{3}{5}$$

$$-\lambda t = \log_e \frac{3}{5} = -0.5108$$

$$\text{where, } t = \frac{0.5108}{\lambda}$$

$$\text{But } \lambda = \frac{0.693}{T_{\frac{1}{2}}} = \frac{0.693}{5730}$$

$$t = \frac{0.5108}{\frac{0.693}{5730}} = 4223.5 \text{ years}$$

So that, the approximate age of the Indus-valley civilisation is 4223.5 years. **[2]**

Q. 11. Write nuclear reaction equations for

(i) α-decay of $_{88}Ra^{226}$

(ii) α-decay of $_{94}Pu^{242}$

(iii) β^--decay of $_{15}P^{32}$

(iv) β^--decay of $_{83}Bi^{210}$

(v) β^+-decay of $_6C^{11}$

(vi) β^+-decay of $_{43}Tc^{97}$

(vii) Electron capture of $_{54}Xe^{120}$

[NCERT Ex. Q. 13.6, Page 462]

Ans. Alpha (α) is a nucleus of helium $_{42}He$ and β is an electron (e^- for β^- and e^+ for β^+). In every α decay, there is a loss of 2 protons and 4 neutrons. In every β^+ decay, there is a loss of 1 proton and a neutrino is emitted from the nucleus. In every β^- decay, there is a gain of 1 proton and an anti-neutrino is emitted from the nucleus.

For the given cases, the various nuclear reactions can be written as :

For the given cases, the various nuclear reactions can be written as :

(i) $^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + ^{4}_{2}\text{He}$

(ii) $^{242}_{94}\text{Pu} \rightarrow ^{238}_{92}\text{U} + ^{4}_{2}\text{He}$

(iii) $^{32}_{15}\text{P} \rightarrow ^{32}_{16}\text{S} + e^- + \bar{\nu}$

(iv) $^{210}_{83}\text{B} \rightarrow ^{210}_{84}\text{PO} + e^- + \bar{\nu}$

(v) $^{11}_{6}\text{C} \rightarrow ^{11}_{5}\text{B} + e^+ + \nu$

(vi) $^{97}_{43}\text{Tc} \rightarrow ^{97}_{42}\text{MO} + e^+ + \nu$

(vii) $^{120}_{54}\text{Xe} + e \rightarrow ^{120}_{53}\text{I} + \nu$ **[2]**

Q. 12. The half-life of $_{38}Sr^{90}$ is 28 years. What is the disintegration rate of 15 mg of this isotope?

[NCERT Ex. Q. 13.10, Page 463]

Ans. Half-life of $_{38}Sr^{90}$, $t_{\frac{1}{2}} = 28$ years

$$= 28 \times 365 \times 24 \times 60 \times 60$$

$$= 8.83 \times 10^8 \text{ s}$$

Mass of the isotope, $m = 15$ mg

90 g of $_{38}Sr^{90}$ atom contains 6.023×10^{23} (Avogadro's number) atoms.

Therefore, 15 mg of $_{38}Sr^{90}$ contains :

$$= \frac{6.023 \times 10^{23} \times 15 \times 10 \times 10^{-3}}{90}$$

$= 1.038 \times 10^{20}$ number of atoms

Rate of disintegration, $\frac{dN}{dt} = \lambda N$

Where,

$$\lambda = \text{decay contant} = \frac{0.693}{8.83 \times 10^8} \text{ s}^{-1}$$

$$\frac{dN}{dt} = \frac{0.693 \times 1.0038 \times 10^{20}}{8.83 \times 10^8}$$

$$= 7.878 \times 10^{10} \text{ atoms/s}$$

So that, the disintegration rate of 15 mg of $_{38}Sr^{90}$ is 7.878×10^{10} atoms/s. **[2]**

Q. 13. Obtain approximately the ratio of the nuclear radii of the gold isotope $_{79}Au^{197}$ and the silver isotope $_{47}Ag^{107}$. [NCERT Ex. Q. 13.11, Page 463]

Ans. Given that,

Nuclear radius of the gold isotope $_{79}Au^{197} = R_{Au}$

Nuclear radius of the silver isotope $_{79}Ag^{107} = R_{Ag}$

Mass number of gold, $A_{Au} = 197$

Mass number of silver, $A_{Ag} = 107$

The ratio of the radii of the two nuclei is related with their mass numbers :

$$\frac{R_{Au}}{R_{Ag}} = \left(\frac{R_{Au}}{R_{Ag}}\right)^{\frac{1}{3}}$$

$$= \left(\frac{197}{107}\right)^{\frac{1}{3}}$$

$$= 1.2256$$

So, the ratio of the nuclear radii of the gold and silver isotopes is about 1.23. **[2]**

Short Answer Type Questions (3 marks each)

Q. 1. Are the nucleons fundamental particles, or do they consist of still smaller parts? One way to find out is to probe a nucleon just as Rutherford probed an atom. What should be the kinetic energy of an electron for it to be able to probe a nucleon? Assume the diameter of a nucleon to be approximately 10^{-15} m. [NCERT Exemp. Q. 13.19, Page 84]

Ans. To detect the properties of nucleons inside the nucleus the wavelength of particle, which may detect nucleons, that must be of size of nucleons (10^{-15} m).

So the wavelength of particle which can detect the nucleons must be equal to or less than 10^{-15}. **[1½]**

$$\lambda = 10^{-15}\text{m}$$

$$\lambda = \frac{h}{p}$$

$$E = h\nu = \frac{hc}{\lambda} \qquad [\because c = \nu\lambda]$$

$$\text{K.E.} = \text{P.E. of electron} = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{10^{-15}}\text{J}$$

$$\text{K.E.} = \frac{6.6 \times 3 \times 10^{-34+8+15}}{1.6 \times 10^{-19}} eV = \frac{99 \times 10^{-34+23+19}}{8}$$

$$= 12.4 \times 10^{-34+42} = 1.24 \times 10^{1} \times 10^{+8}$$

$$= 1.24 \times 10^{9}\ \text{eV}$$

So the K.E. of particle which may detect nucleon inside the nucleus must be of 1.24×10^9 eV per particle. **[1½]**

Q. 2. Why do stable nuclei never have more protons than neutrons? [NCERT Exemp. Q. 13.16, Page 84]

Ans. If in a stable nucleus number of protons is larger than neutrons the repulsive electrostatic force between proton-proton becomes larger in place of nuclear force of attraction between nucleons. **[1½]**

So for stability, repulsive force between proton-proton must be smaller than nuclear attractive force between nucleons. **[1½]**

Q. 3. A nuclide 1 is said to be the mirror isobar of nuclide 2 if $Z_1 = N_2$ and $Z_2 = N_1$.

(a) What nuclide is a mirror isobar of $_{11}Na^{23}$?

(b) Which nuclide out of the two mirror isobars have greater binding energy and why? [NCERT Exemp. Q. 13.20, Page 84]

Ans. **(a)** Here Z is atomic number and N is no. of neutron in $_{11}Na^{23}$

$$Z_1 = 11$$

$$N_1 = 23 - 11 = 12$$

Mirror isobar of $_{11}Na^{23}$ is

$$Z_2 = N_1 = 12$$

So Mg is isobar of $_{11}Na^{23}$.

So $_{12}Mg^{23}$ is the mirror isobar of $_{11}Na^{23}$. **[1½]**

(b) As the neutrons in $_{12}Mg^{23}$ are '11' and in $_{11}Na^{23}$ are '12' so, the number of neutrons in Na is larger than Mg and hence nuclear short range attractive forces in Na will be larger than repulsive electrostatic forces between proton-proton.

So that, $_{11}Na^{23}$ has more binding energy than $_{12}Mg^{23}$. **[1½]**

Q. 4. In a periodic table the average atomic mass of magnesium is given as 24.312 u. The average value is based on their relative natural abundance on earth. The three isotopes and their masses are $_{12}Mg^{24}$ (23.98504 u), $_{12}Mg^{25}$ (24.98584 u) and $_{12}Mg^{26}$ (25.98259 u). The natural abundance of $_{12}Mg^{24}$ is 78.99% by mass. Calculate the abundances of other two isotopes. [NCERT Ex. Add. Q. 13.23, Page 464]

Ans. Average atomic mass of magnesium, $m = 24.312$ u

Mass of magnesium isotope $_{12}Mg^{24}$, $m_1 = 23.98504$ u

Mass of magnesium isotope $_{12}Mg^{25}$, $m_2 = 24.98584$ u

Mass of magnesium isotope $_{12}Mg^{26}$, $m_3 = 25.98259$ u

Abundance of $_{12}Mg^{24}$, $\eta_1 = 78.99\%$

Abundance of $_{12}Mg^{25}$, $\eta_2 = x\%$

So that, abundance of

$_{12}Mg^{26}$, $\eta_3 = 100 - x - 78.99\% = (21.01 - x)\%$

We have the relation for the average atomic mass as :

$$m = \frac{m_1\eta_1 + m_2\eta_2 + m_3\eta_3}{\eta_1 + \eta_2 + \eta_3}$$

$$24.312 = \left(\frac{23.98504 \times 78.99 + 24.98584 \times x + 25.98259 \times (21.01 - x)}{100}\right)$$

$$2431.2 = 1894.5783096 + 24.98584x + 545.8942159 - 25.98259x$$

$$0.99675x = 9.2725255$$

$$\therefore x \approx 9.3\%$$

And $21.01 - x = 11.71\%$

Therefore, the abundance of $^{25}_{12}$Mg is 9.3% and that of $^{26}_{12}$Mg is 11.71%. **[3]**

Q. 5. (a) Lithium has two stable isotopes $_3Li^6$ and $_3Li^7$ have respective abundances of 7.5% and 92.5%. These isotopes have masses 6.01512 u and 7.01600 u, respectively. Find the atomic mass of lithium.

(b) Two stable isotopes of Boron, $_5B^{10}$ and $_5B^{11}$. Their respective masses are 10.01294 u and 11.00931 u, and the atomic mass of boron is 10.811 u. Find the abundances of $_5B^{10}$ and $_5B^{11}$.

[NCERT Ex. Q. 13.1, Page 462]

Ans. **(a)** Given that,

Abundance of $_3Li^7$ isotope = 7.5%

Abundance of $_3Li^7$ isotope = 92.5%

Mass of $_3Li^6$ = 6.01512 u

Mass of $_3Li^7$ = 7.016 u

The atomic mass of lithium atom will depend on the atomic masses of these two isotopes with their abundances as given in the question.

So,

Atomic mass of Li = (6.01512 × 7.5 + 7.016 × 92.5)/100 = 6.940934 u = 6.941 u **[1½]**

(b) Given that,

Mass of $_5B^{10}$ = 10.01294 u

Mass of $_5B^{11}$ = 11.00931 u

Atomic mass of B = 10.811 u

The abundances of $_5B^{10}$ and $_5B^{11}$ can be calculated by using the technique employed in solution of part (a) of this question.

Let the abundance of $^{10}{}_5B$ be y%.

Then, the abundance of $^{11}{}_5B$ = (100 – y)%

Now calculating weighted average

Atomic mass of boron =

$$\frac{[10.01294v + 11.00931(100 - y)]}{100}$$

Put the values,

1081.1 = 10.01294y + 1100.931 – 11.00931y

Simplifying this linear equation which gives

y = 19.9%

100 – y = 80.1%.

So the, abundance of $_5B^{10}$ = 19.9%

And, abundance of $_5B^{11}$ = 80.1% **[1½]**

Q. 6. The three stable isotopes of neon : $_{10}Ne^{10}$, $_{10}Ne^{21}$ and $_{10}Ne^{22}$ have respective abundances of 90.51%, 0.27% and 9.22%. The atomic masses of the three isotopes are 19.99 u, 20.99 u and 21.99 u, respectively. Obtain the average atomic mass of neon.

[NCERT Ex. Q. 13.2, Page 462]

Ans. Atomic mass of $_{10}Ne^{20}$, $m_1 = 19.99$ u

Abundance of $_{10}Ne^{20}$, $\eta_1 = 90.51\%$

Atomic mass of $_{10}Ne^{21}$, $m_2 = 20.99$ u

Abundance of $_{10}Ne^{21}$, $\eta_2 = 0.27\%$

Atomic mass of $_{10}Ne^{22}$, $m_3 = 21.99$ u

Abundance of $_{10}Ne^{22}$, $\eta_3 = 9.22\%$

The average atomic mass of neon is given as :

$$m = \frac{m_1\eta_1 + m_2\eta_2 + m_3\eta_3}{\eta_1 + \eta_2 + \eta_3}$$

$$= \frac{19.99 \times 90.51 + 20.99 \times 0.27 + 21.99 \times 9.22}{90.51 + 0.27 + 9.22}$$

$$= 20.1771 \text{ u}$$ **[3]**

Q. 7. Obtain the binding energy of the nuclei $_{26}Fe^{56}$ and $_{83}Bi^{209}$ in units of MeV from the following data : m ($_{26}Fe^{56}$) = 55.934939 u, m ($_{83}Bi^{209}$) = 208.980388 u.

[NCERT Ex. Q. 13.4, Page 462]

Ans. Atomic mass of $^{56}_{23}$Fe, m_1 = 55.934939 u

$^{56}_{23}$Fe nucleus has 26 protons and (56 − 26) = 30 neutrons

So, the mass defect of the nucleus,

$\Delta m = 26 \times m_H + 30 \times m_n - m_1$

Where,

Mass of a proton, m_H = 1.007825 u

Mass of a neutron, m_n = 1.008665 u

Put the values,

Δm = 26 × 1.007825 + 30 × 1.008665 − 55.93493

= 26.20345 + 30.25995 − 55.934939

= 0.528461 u

But 1 u = 931.5 MeV/c^2

Δm = 0.528461 × 931.5 MeV/c^2

The binding energy of this nucleus is :

$$E_{b1} = \Delta mc^2$$

Where, c = Speed of light

$$E_{b1} = 0.528461 \times 931.5\left(\frac{MeV}{c^2}\right) \times c^2$$

$$= 492.26 \text{ MeV}$$

Average binding energy per nucleon

$$= \frac{492.26}{56} = 8.79 \text{ MeV}$$

Atomic mass of $^{209}_{83}$Bi, m_2 = 208.980388 u

$^{209}_{83}$Bi nucleus has 83 protons and (209 – 83) + 126 neutrons.

Hence, the mass defect of this nucleus is given as :

$$\Delta m' = 83 \times m_H + 126 \times m_n - m_2$$

Where,

Mass of proton, m_H = 1.007825 u

Mass of neutron, m_n = 1.008665 u

$$\Delta m' = 83 \times 1.007825 + 126 \times 1.008665 - 208.980388$$

$$= 83.649475 + 127.091790 - 208.980388$$

$$= 1.760877 \text{ u}$$

But 1 u = 931.5 MeV/c^2

$$\Delta m' = 1.760877 \times 931.5 \text{ MeV/c}^2$$ **[2]**

So that, the binding energy of this nucleus is given as :

$$E_{b2} = \Delta m'c^2$$

$$= 1.760877 \times 931.5\left(\frac{MeV}{c^2}\right) \times c^2$$

$$= 1640.26 \text{ MeV}$$

Average binding energy per nucleon = $\frac{1640.26}{209}$ = 7.848 MeV **[1]**

Q. 8. A given coin has a mass of 3.0 g. Calculate the nuclear energy that would be required to separate all the neutrons and protons from each other. For

simplicity assume that the coin is entirely made of $^{63}_{29}Cu$ atoms (of mass 62.92960 u).

[NCERT Ex. Q. 13.5, Page 462]

Ans. Mass of a copper coin, $m' = 3$ g

Atomic mass of $_{29}Cu^{63}$ atoms, $m = 62.92960$ u

The total number of $_{29}Cu^{63}$ atoms in the coin,

$$N = \frac{N_A \times m'}{\text{Mass number}}$$

where,

N_A = Avogadro's number = 6.023×10^{23} atoms/g

Mass number = 63 g

$$\therefore N = \frac{6.023 \times 10^{23} \times 3}{63} = 2.868 \times 10^{22} \text{ atoms}$$

$_{29}Cu^{63}$ nucleus has 29 protons and (63 – 29) = 34 neutrons

$\therefore$ Mass defect of this nucleus, $\Delta m' = 29 \times m_H + 34 \times m_n - m$

Where,

Mass of a proton, $m_H = 1.007825$ u

Mass of a neutron, $m_n = 1.008665$ u

So,

$$\Delta m' = 29 \times 1.007825 + 34 \times 1.008665 - 62.9296$$
$$= 0.591935 \text{ u}$$

Mass defect of all atoms present in the coin,

$$\Delta m = 0.591935 \times 2.868 \times 10^{22}$$
$$= 1.69766958 \times 10^{22} \text{ u}$$

But 1 u = 931.5 MeV/c^2

So, $\Delta m = 1.69766958 \times 10^{22} \times 931.5$ MeV/c^2 **[2]**

So that, the binding energy of the nuclei of the coin is given as :

$$E_b = \Delta m c^2$$
$$= 1.69766958 \times 10^{22} \times 931.5 \left(\frac{\text{MeV}}{c^2}\right) \times c^2$$
$$= 1.581 \times 10^{25} \text{ MeV}$$

But 1 MeV $= 1.6 \times 10^{-13}$ J

$$E_b = 1.581 \times 10^{25} \times 1.6 \times 10^{-13}$$
$$= 2.5296 \times 10^{12} \text{ J}$$

This much energy is required to separate all the neutrons and protons from the given coin. **[1]**

Q. 9. (a) State the law of radioactive decay. Write the SI unit of 'activity'.

(b) There are $4\sqrt{2} \times 10^6$ radioactive nuclei in a given radioactive sample. If the half-life of the sample is 20 s, how many nuclei will decay in 10 s?

[CBSE Board, Foreign Scheme, 2017]

Ans. Law of Radio-active Decay :

Radio-activity is a nuclear phenomenon and when a nucleus disintegrates by emitting a particle (α and β) or by capturing an electron from the atomic shell, the process is called radio-active decay. This decay process is spontaneous. And its unit is Becquerel(*Bq*). **[½]**

Mathematical expression of law of Radio-active Decay :

The rate of decay (activity, *A*) is proportional to the number of parent nuclei (*N*) present.

$$A \propto N$$
$$A = \frac{dN}{dt}$$
$$\frac{dN}{dt} \propto N$$
$$\frac{dN}{dt} = -\lambda N$$

λ (lambda) is a positive constant called the decay constant. It has the unit s^{-1}.

The minus sign is included because *N* decreases as the time *t* in seconds (s) increases.

$$\frac{dN}{dt} = -\lambda N$$

So that,

$$\int \frac{dN}{N} = -\lambda$$
$$\log_e N = -\lambda t + C \quad \text{...(i)}$$

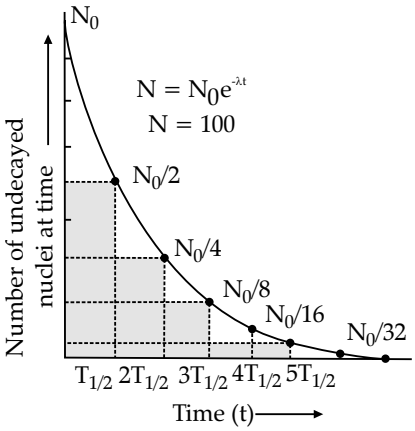
If the initial number of nuclei is N_o and $N = N_0$ when $t = 0$

Then (i) becomes,

$$\log_e N_o = C$$

Substituting for *C* into (i)

$$\log_e N = -\lambda t + \log_e N_0$$
$$\log_e N - \log_e N_0 = -\lambda t$$
$$\log_e \left(\frac{N}{N_0}\right) = -\lambda t$$
$$\Rightarrow \quad \frac{N}{N_0} = e^{-\lambda t}$$
$$\Rightarrow \quad N = N_0 e^{-\lambda t}$$



[1]

Decay curve for a radio-active element

(b) Given that,

Initial value of radio-active element, $N_o = 4\sqrt{2} \times 10^6$

Half-life, $t_{1/2} = 20$ sec

Value of radio-active element after time, $t = 10$ sec

As we know the relation :

$t_{1/2} = 0.693/\lambda$

So,

$\lambda = 0.693/t_{1/2}$

Put the value, $\lambda = 0.693/20$

$= 0.03465$

And we know the relation :

$N = N_o e^{-\lambda t}$

$N = 4\sqrt{2} \times 10^6 e^{-0.03465 \times 10}$

$N = 4\sqrt{2} \times 10^6 (0.70716)$

$N = 3.9996 \times 10^6$

Or

$N = 2.822 \times 10^6$

So, nuclei will decay in 10 s will be :

$= 4\sqrt{2} \times 10^6 - 2.82\sqrt{2} \times 10^6$

$= 1.18\sqrt{2} \times 10^6$ **[1½]**

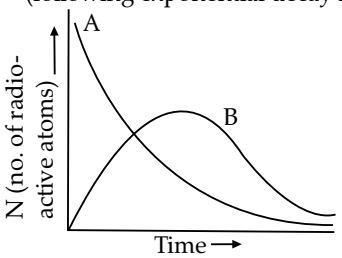
Q. 10. Consider a radioactive nucleus A which decays to a stable nucleus *C* through the following sequence :

$A \rightarrow B \rightarrow C$

Here *B* is an intermediate nuclei which is also radioactive. Considering that there are N_0 atoms of *A* initially, plot the graph showing the variation of number of atoms of *A* and *B* versus time.

[NCERT Exemp. Q. 13.17, Page 84]

Ans. At $t = 0$, $N_A = N_O$ while $N_B = 0$. As time increases, N_A falls off exponentially, the number of atoms of B increases, becomes maximum and finally decays to zero at ∞ (following exponential decay law). **[1½]**



[1½]

Q. 11. A radioactive isotope has a half-life of *T* years. How long will it take the activity to reduce to (a) 3.125%, (b) 1% of its original value?

[NCERT Ex. Q. 13.7, Page 462]

Ans. Given that,

Half-life of the radioactive isotope = T years

Original amount of the radioactive isotope = N_0

(a) After decay, the amount of the radioactive isotope = N

It is given that only 3.125% of N_0 remains after decay. So, we can write :

$$\frac{N}{N_0} = 3.125\% = \frac{3.125}{100} = \frac{1}{32}$$

But $\frac{N}{N_0} = e^{-\lambda t}$

Where,

λ = Decay constant

t = Time

$$\therefore -\lambda t = \frac{1}{32}$$

$$-\lambda t = \text{In } 1 - \text{In } 32$$

$$-\lambda t = 0 - 3.4657$$

$$t = \frac{3.4657}{\lambda}$$

Since $\lambda = \frac{0.693}{T}$

$$\therefore t = \frac{3.466}{\frac{0.693}{T}} = 5T \text{ years}$$

So that, the isotope will take about $5T$ years to reduce to 3.125% of its original value. **[1½]**

(b) After decay, the amount of the radioactive isotope = N

It is given that only 1% of N_0 remains after decay. So, we can write :

$$\frac{N}{N_0} = 1\% = \frac{1}{100}$$

But $\frac{N}{N_0} = e^{-\lambda t}$

$$\therefore e^{-\lambda t} = \frac{1}{100}$$

$$-\lambda t = \text{In } 1 - \text{In } 100$$

$$-\lambda t = 0 - 4.6052$$

$$t = \frac{4.6052}{\lambda}$$

Since, $\lambda = \frac{0.693}{T}$

$$\therefore t = \frac{4.6052}{\frac{0.693}{T}} = 6.645T \text{ years}$$

So that, the isotope will take about $6.645T$ years to reduce to 1% of its original value. **[1½]**

Q. 12. (i) A radioactive nucleus '*A*' undergoes a series of decays as given below :

$$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$$

The mass number and atomic number of A_2, are 176 and 71 respectively. Determine the mass and atomic numbers of A_4 and *A*.

(ii) Write the basic nuclear processes underlying β^+ and β^- decays.

[CBSE Board, All India Region, 2017]

Ans. (i) In α- decay, the mass number decreases by 4 and atomic number decreases by 2.

In β-decay, mass number remains the same but atomic number increases by 1.

In γ- decay, mass number and atomic number remains the same.

Therefore, the equation is,

$${}_{72}A^{180} \rightarrow {}_{70}A_1^{176} \xrightarrow{\beta} {}_{71}A_2^{176} \xrightarrow{\alpha} {}_{69}A_3^{172} \xrightarrow{\lambda} {}_{60}A_4^{172}$$

Mass number of A_4 is 172

Atomic number of A_4 is 69 **[1½]**

(ii) β decays :

Beta decay occurs when, in a nucleus with too many protons or too many neutrons, one of the protons or neutrons is transformed into the other.

β^- decay : In beta minus decay, a neutron decays into a proton, an electron, and an anti-neutrino :

$n \rightarrow p + \beta^+ + \bar{\nu}$(anti-neutrino)

(neutron) (proton)

decay

So that, ${}^{32}_{15}P \rightarrow {}^{32}_{16}S + e^- + \bar{\nu}$

β^+ decay : In beta plus decay, a proton decays into a neutron, a positron, and a neutrino :

$p \rightarrow n + \beta^- + \nu$(neutrino)

(proton decay) (neutron)

So that, ${}^{22}_{11}N \rightarrow {}^{22}_{10}Ne + e^+ + \bar{\nu}$

[1½]

Q. 13. Find the *Q*-value and the kinetic energy of the emitted α-particle in the α-decay of

(a) $_{88}R_a^{226}$ **(b)** $_{86}R_n^{220}$

Given

$m\left(_{88}R_a^{226}\right) = 226.02540$ **u,** $m\left(_{86}R_n^{222}\right) = 222.01750$ **u,**

$m\left(_{88}R_n^{222}\right) = 220.01137$ **u,** $m(_{84}P_o^{216}) = 216.00189$ **u.**

[NCERT Ex. Q. 13.12, Page 463]

Ans. **(a)** Alpha particle decay of $_{88}R_a^{226}$ emits a helium nucleus. As a result, its mass number reduces to 222 (226 – 4) and its atomic number reduces to 86 (88 – 2). This is shown in the following nuclear reaction.

$_{88}Ra^{226} \rightarrow {}_{86}Ra^{222} + {}_{2}^{4}He$

Q-value of emitted α-particle = (Sum of initial mass – Sum of final mass) c^2

Where, c = Speed of light

Given that,

$_{88}R_a^{226} = 226.02540$ u

$_{86}R_n^{222} = 222.01750$ u

$m(_2He^4) = 4.002603$ u

Q- value = $[226.02540 - (222.01750 + 4.002603)]\ uc^2$

$= 0.005297\ uc^2$

But 1 u = 931.5 MeV/c^2

$Q = 0.005297 \times 931.5 \approx 4.94$ MeV

Kinetic energy of the α-particle

$$= \left(\frac{\text{Mass number after decay}}{\text{Mass number before decay}}\right) \times Q$$

$$= \frac{222}{226} \times 4.94 = 4.85 \text{ MeV}$$ **[1½]**

(b) Alpha particle decay of $_{86}R_n^{220}$

Given that,

Mass of $_{86}R_n^{220} = 220.011737$ u

Mass of $_{84}P_o^{216} = 216.00189$ u

Q-value $= [220.01137 - (216.00189 + 4.00260)] \times 931.5$

≈ 641 MeV

Kinetic energy of the

$$\alpha - \text{particle} = \left(\frac{220-4}{220}\right) \times 6.41$$

$= 6.29$ MeV [1½]

Q. 14. The nucleus $_{10}N_e^{23}$ decays β^- by emission. Write down the β decay equation and determine the maximum kinetic energy of the electrons emitted. Given that :

$m(_{10}N_e^{23}) = 22.994466$ **u**

$m(_{11}N_a^{23}) = 22.989770$ **u.**

[NCERT Ex. Q. 13.14, Page 463]

Ans. In β-emission, the number of protons increases by 1, and one electron and an antineutrino is emitted from the parent nucleus.

β-emission from the nucleus $_{10}N_e^{23}$.

$_{10}Ne^{23} \rightarrow {}_{11}Na^{23} + e^- + \bar{\nu} + Q$

It is given that :

Atomic mass of $m(_{10}N_e^{23}) = 22.994466$ u

Atomic mass of $m(_{11}N_e^{23}) = 22.989770$ u

Mass of an electron, $m_e = 0.000548$ u

Q-value of the given reaction is given as :

$Q = \left[m(_{10}N_e^{23}) - \left[m(_{11}N_a^{23}) + m_e\right]\right]c^2$

There are 10 electrons and 11 electrons in $_{10}^{23}Ne$ and $_{11}^{23}Na$ respectively. So, the mass of the electron is cancelled in the *Q*-value equation. **[2]**

$Q = [22.994466 - 22.989770]\ c^2$

$= (0.004696\ c^2)$ u

But 1 u = 931.5 Mev/c^2

$Q = 0.004696 \times 931.5 = 4.374$ MeV

The daughter nucleus is too heavy as compared to e^- and $\bar{\nu}$. So, it carries negligible energy. And the kinetic energy of the antineutrino is nearly zero.

That's why, the maximum kinetic energy of the emitted electrons is almost equal to the *Q*-value, that is, 4.374 MeV. **[1]**

Q. 15. From the relation $R = R_0A^{1/3}$, where R_0 is a constant and A is the mass number of a nucleus, show that the nuclear matter density is nearly constant (*i.e.*, independent of A).

[NCERT Ex. Q. 13.21, Page 464]

Ans. As we know the expression for nuclear radius as :

$R = R_0A^{1/3}$ **[1]**

where,

R_0 = Constant.

A = Mass number of the nucleus

Nuclear matter density,

ρ = Mass of the nucleus/Volume of the nucleus

Let m be the average mass of the nucleus.

So, the mass of the nucleus = mA

Therefore, $\rho = \dfrac{mA}{\frac{4}{3}\pi R^3}$

$$= \frac{3mA}{4\pi(R_0A^{\frac{1}{3}})^3}$$

$$= \frac{3mA}{4\pi R_0^3A}$$

$$= \frac{3m}{4\pi R_0^3}$$

So that, the nuclear matter density is independent of A. It is nearly constant. **[2]**

Q. 16. For the β^+ (positron) emission from a nucleus, there is another competing process known as electron capture (electron from an inner orbit, say, the K−shell, is captured by the nucleus and a neutrino is emitted).

$e^+ + {}_Z^AX \rightarrow {}_{Z-1}^AY + \nu$

Show that if β^+ emission is energetically allowed; electron capture is necessarily allowed but not vice-versa. **[NCERT Ex. Q. 13.22, Page 464]**

Ans. Let assume the amount of energy released during the electron capture process be Q_1. The nuclear reaction can be written as :

$$e^+ + {}_Z^AX \rightarrow {}_{Z-1}^AY + \nu + Q_1 \quad ...(1)$$

Let the amount of energy released during the positron capture process be Q_2. The nuclear reaction can be written as :

$$e^+ + {}^A_Z X \rightarrow {}^A_{Z-1} Y + \nu + Q_2 \quad ...(2)$$

$m_N\left({}^A_Z X\right)$ = Nuclear mass of ${}^A_Z X$

$m_N\left({}^A_{Z-1} X\right)$ = Nuclear mass of ${}^A_{Z-1} Y$

$m\left({}^A_Z X\right)$ = Atomic mass of ${}^A_Z X$

$m_N\left({}^A_{Z-1} X\right)$ = Atomic mass of ${}^A_{Z-1} Y$

m_e = Mass of an electron

c = Speed of light

Q-value of the electron capture reaction is given as:

$$Q_1 = \left[m_N({}^A_Z X) - m_e - m_N({}^A_{Z-1} Y)\right]c^2$$
$$= [m_N({}^A_Z X) - Zm_e + m_N({}^A_{Z-1} Y) - (Z-1)m_e]c^2$$
$$= [m_N({}^A_Z X) - m_N({}^A_{Z-1} Y)]c^2 \quad ...(3)$$

Q-value of the positron capture reaction is given as:

$$Q_2 = [m_N({}^A_Z X) - m_N({}^A_{Z-1} Y) - m_e]c^2$$
$$= [m_N({}^A_Z X) - Zm_e - m_N({}^A_{Z-1} Y) + (Z-1)m_e - m_e]c^2$$
$$= [m_N({}^A_Z X) - m_N({}^A_{Z-1} Y) - 2m_e]c^2 \quad ...(4)$$

[2]

It can be inferred that if $Q_2 > 0$, then $Q_1 > 0$; Also, if $Q_1 > 0$, it does not necessarily mean that $Q_2 > 0$. It means, if β^+ emission is energetically allowed, then the electron capture process is necessarily allowed, but not vice-versa. This is because the Q-value must be positive for an energetically allowed nuclear reaction. **[1]**

Q. 17. A piece of wood from the ruins of an ancient building was found to have a ^{14}C activity of 12 disintegrations per minute per gram of its carbon content. The ^{14}C activity of the living wood is 16 disintegrations per minute per gram. How long ago did the tree, from which the wooden sample came, die? Given half-life of ^{14}C is 5,760 years.

[NCERT Exemp. Q. 13.18, Page 84]

Ans. As it is clear that,

Rate of disintegration in old wood sample of $C-14$ radio-active atoms is 12 atoms per min gm. Initially rate of disintegration of $C-14$ when the tree was live=16 atoms per min per gm.

Given that,

$T_{1/2}$ of C–14 nuclei = 5,760 years

According to radio-active decay law,

$$N = N_0 e^{-\lambda t} \text{ or } R = R_0 e^{-\lambda t} \quad \textbf{[1]}$$

$$12 = 16e^{-\lambda t}$$

$$e^{\lambda t} = \frac{16}{12}$$

$$\log_e e^{\lambda t} = \log_e \frac{4}{3}$$

$$\lambda t = \log_e \frac{4}{3}$$

$$t = \frac{2.303 \times \log_{10}\left(\frac{4}{3}\right)}{\lambda} \quad \left(\because \lambda = \frac{0.6931}{T_{1/2}}\right) \quad \textbf{[1]}$$

$$= \frac{2.303 \times (\log 4 - \log 3) \times 5760}{0.6931} \text{ year}$$

$$\therefore t = \frac{2.303(0.6020 - 0.4771)5760}{0.6931}$$

$$= 2391.20 \text{ years} \quad \textbf{[1]}$$

Q. 18. Obtain the amount of $^{60}_{27}Co$ necessary to provide a radioactive source of 8.0 mCi strength. The half-life of $^{60}_{27}Co$ is 5.3 years.

[NCERT Ex. Q. 13.9, Page 463]

Ans. As we know the strength of the radio-active source :

$$\frac{dN}{dt} = 8.0 \text{ mCi}$$

$$= 8 \times 10^{-3} \times 3.7 \times 10^{10}$$

$$= 29.6 \times 10^7 \text{ decay/s} \quad \textbf{[1]}$$

where,

N = Required number of atoms

Half-life of $^{60}_{27}Co$ $T_{\frac{1}{2}} = 5.3$ years

$$= 5.3 \times 365 \times 24 \times 60 \times 60$$

$$= 1.67 \times 10^8 \text{ s} \quad \textbf{[1]}$$

For decay constant λ, we have the rate of decay as :

$$\frac{dN}{dt} = \lambda N$$

$$\text{where, } \lambda = \frac{0.693}{T_{\frac{1}{2}}} = \frac{0.693}{1.67 \times 10^8} \text{ s}^{-1}$$

$$N = \frac{1}{\lambda}\frac{dN}{dt} = \frac{29.6 \times 10^7}{\frac{0.693}{1.67 \times 10^8}}$$

$$= 7.133 \times 10^{16} \text{ atoms}$$

For $^{60}_{27}$ Co :

Mass of 6.023×10^{23} (Avogadro's number) atoms = 60 g

Mass of atoms 7.133×10^{16} atoms $= \frac{60 \times 7.133 \times 10^{16}}{6.023 \times 10^{23}}$

$$= 7.106 \times 10^{-6} \text{ g}$$

So that, the amount of $^{60}_{27}Co$ necessary for the purpose is 7.106×10^{-6} g. **[1]**

Q. 19. The radionuclide ^{11}C decays according to

$^{11}_6C \rightarrow {}^{11}_5B + e^+ + \nu$: $T_{1/2} = 20.3$ min.

The maximum energy of the emitted positron is 0.960 MeV. Given the mass values :

$m(^{11}_6C) = 11.011434$ u and $m(^{11}_5B) = 11.009305$ u.

Calculate Q and compare it with the maximum energy of the positron emitted.

[NCERT Ex. Q. 13.13, Page 463]

Ans. The given nuclear reaction is :

$$^{11}_6C \rightarrow \frac{11}{5}B + e^+ + \nu$$

Half-life of $^{11}_6C$ nuclei, $T_{1/2} = 20.3$ min

Atomic mass of $m(^{11}_6C) = 11.011434$ u

Atomic mass of $m(^{11}_5B) = 11.009305$ u

Maximum energy possessed by the emitted positron = 0.960 MeV

The change in the Q-value (ΔQ) of the nuclear masses of the $^{11}_6C$

$$\Delta Q = [m(^{11}_6C) - [m'(^{11}_5B + m_e)]]c^2 \quad ...(1)$$

[1]

Where,

m_e = Mass of an electron or positron = 0.000548 u

c = Speed of light

m' = Respective nuclear masses

If atomic masses are used instead of nuclear masses, then we have to add 6 m_e in the case of ^{11}C and 5 m_e in the case of ^{11}B.

$= 11.011434$ u **[1]**

Now, equation (1) reduces to :

$\Delta Q = [m(^{11}_{6}C) - m(^{11}_{5}B) - 2m_e]c^2$

Where, $m(^{11}_{6}C)$ and $m(^{11}_{5}B)$ are the atomic masses.

$\Delta Q = [11.011434 - 11.009305 - 2 \times 0.000548]\, c^2$

$= (0.001033\, c^2)$ u

But 1 u = 931.5 Mev/c^2

$\Delta Q = 0.001033 \times 931.5 \approx 0.962$ MeV

The value of Q is almost comparable to the maximum energy of the emitted positron. **[1]**

Q. 20. The Q value of a nuclear reaction $A + b \rightarrow C + d$ is defined by

$Q = [m_A + m_b - m_C - m_d]c^2$

where the masses refer to the respective nuclei. Determine from the given data the Q-value of the following reactions and state whether the reactions are exothermic or endothermic.

(i) $^{1}_{1}H + ^{3}_{1}H \rightarrow ^{2}_{1}H + ^{2}_{1}H$

(ii) $^{12}_{6}C + ^{12}_{6}C \rightarrow ^{20}_{10}Ne + ^{4}_{2}H$

Atomic masses are given to be

$m(^{2}_{1}H) = 2.014102$ u

$m(^{3}_{1}H) = 3.016049$ u

$m(^{12}_{6}C) = 12.00000$ u

$m(^{20}_{10}Ne) = 19.992439$ u

[NCERT Ex. Q. 13.15, Page 463]

Ans. **(i)** The given nuclear reaction is :

$^{1}_{1}H + ^{3}_{1}H \rightarrow ^{2}_{1}H + ^{2}_{1}H$

It is given that :

Atomic mass $m(^{1}_{1}H) = 1.007825\ u$

Atomic mass $m(^{3}_{1}H) = 3.014049\ u$

Atomic mass $m(^{2}_{1}H) = 2.014102\ u$

According to the question, the Q-value of the reaction can be written as :

$Q = [m(^{1}_{1}H) + m(^{3}_{1}H) - 2m(^{2}_{1}H)]\, c^2$

$= [1.007825 + 3.016049 - 2 \times 2.014102]\, c^2$

$Q = (-0.00433\, c^2)$ u

But ,

1 u = 931.5 MeV/c^2

$Q = -0.00433 \times 931.5$

$= -4.0334$ MeV

The negative Q-value of the reaction shows that the reaction is endothermic. **[1½]**

(ii) The given nuclear reaction is :

$^{12}_{6}C + ^{12}_{6}C \rightarrow ^{20}_{10}Ne + ^{4}_{2}H$

It is given that :

Atomic mass of $m(^{12}_{6}C) = 12.00000\ u$

Atomic mass of $m(^{20}_{10}Ne) = 19.992439\ u$

Atomic mass of $m(^{4}_{2}He) = 4.002603\ u$

The Q-value of this reaction is given as :

$Q = [m(^{12}_{6}C) - m(^{20}_{10}Ne) - m(^{4}_{2}He) = 4.002603\ u]\, c^2$

$= [2 \times 12.0 - 19.992439 - 4.002603]\, c^2$

$= [0.004958\, c^2]$ u

$= 0.004958 \times 931.5 = 4.618377$ MeV

The positive Q-value of the reaction shows that the reaction is exothermic. **[1½]**

Q. 21. Suppose, we think of fission of a $^{56}_{26}Fe$ nucleus into two equal fragments, $^{28}_{13}Al$. Is the fission energetically possible? Argue by working out Q of the process.

Given

$m(^{56}_{26}Fe) = 55.93494$u

$m(^{28}_{13}Al) = 27.98191$u

[NCERT Ex. Q. 13.16, Page 463]

Ans. The fission of $^{56}_{26}Fe$ is given as :

$^{56}_{26}Fe \rightarrow 2\ ^{28}_{13}Al$

Given that :

Atomic mass of $m(^{56}_{26}Fe) = 55.93494$ u

Atomic mass of $m(^{28}_{13}Al) = 27.98191$ u

The Q-value of this nuclear reaction is :

$Q = [m(^{56}_{26}Fe) - 2m(^{28}_{13}Al)]c^2$

$= [55.93494 - 2 \times 27.98191]c^2$

$= (-0.02888\, c^2)$ u **[1]**

But 1 u = 931.5 MeV/c^2

$Q = -0.02888 \times 931.5 = -26.902$ MeV **[1]**

The Q-value of the fission is negative. Therefore, the fission is not possible energetically.

For an energetically-possible fission reaction, the Q-value must be positive. **[1]**

Q. 22. The fission properties $^{239}_{94}Pu$ of are very similar to those of $^{235}_{92}U$. The average energy released per fission is 180 MeV. How much energy is released if all the atoms in 1 kg of pure $^{239}_{94}Pu$ undergo fission?

[NCERT Ex. Q. 13.17, Page 463]

Ans. Average energy released per fission of $^{239}_{94}Pu$, $E_{av} = 180$ Mev

Amount of pure $^{239}_{94}Pu$, m = 1 kg = 1,000 g

N_A = Avogadro number = 6.023×10^{23}

Mass number of $^{239}_{94}Pu$ = 239 g

1 mole of $^{239}_{94}Pu$ contains N_A atoms.

So that, 1000 gm of

$^{239}_{94}Pu$ contains $\left(\frac{N_A}{\text{Mass Number}} \times m\right)$ atoms

$\frac{6.023 \times 10^{23}}{239} \times 1000 = 2.52 \times 10^{24}$ atoms

$\therefore$ Total energy released during the fission of 1 kg of $^{239}_{94}Pu$ is,

$E = E_{av} \times 2.52 \times 10^{24}$

$= 180 \times 2.52 \times 10^{24}$

$= 4.536 \times 10^{26}$ MeV

Hence, 4.536×10^{26} MeV is released if all the atoms in 1 kg of pure $^{239}_{94}Pu$ undergo fission. **[3]**

Q. 23. A 1000 MW fission reactor consumes half of its fuel in 5.00 y. How much $^{235}_{92}U$ did it contain initially? Assume that the reactor operates 80% of the time, that all the energy generated arises

from the fission of $^{235}_{92}U$ and that this nuclide is consumed only by the fission process.

[NCERT Ex. Q. 13.18, Page 464]

Ans. Half-life of the fuel of the fission reactor, $t_{1/2}=5$ years

$= 5 \times 365 \times 24 \times 60 \times 60$ s

In the fission of 1 g of $^{235}_{92}U$ nucleus, the energy released is equal to 200 MeV.

As we know that,

1 mole, *i.e.*, 235 g of $^{235}_{92}U$ contains 6.023×10^{23} atoms.

So, 1 g contains $^{235}_{92}U \rightarrow 6.023 \times 10^{23}/235$ atoms contain **[1]**

The total energy generated per gram of $^{235}_{92}U$ is calculated as :

$$E = \frac{6.023\times10^{23}}{235}\times 200\text{MeV}$$

$$= \frac{200\times6.023\times10^{23}\times1.6\times10^{-19}\times10^{6}}{235}$$

$$= 8.20\times10^{16}\ \text{J/g}$$

The reactor operates only 80% of the time.

So that, the amount of $^{235}_{92}U$ consumed in 5 years by the 1,000 MW fission reactor is calculated as :

$$= \frac{5\times80\times60\times60\times365\times24\times1000\times10^{6}}{100\times8.20\times10^{10}} \approx 1538\ \text{kg}$$

Initial amount of $^{235}_{92}U = 2\times1538=3076$ kg **[2]**

Q. 24. How long can an electric lamp of 100 W be kept glowing by fusion of 2.0 kg of deuterium? Take the fusion reaction as $^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{1}He + n + 3.27$ MeV. **[NCERT Ex. Q. 13.19, Page 464]**

Ans. Fusion reaction is :

$^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{1}He + n + 3.27$ MeV

Given that,

Amount of deuterium, $m = 2$ kg

So, 1 mole, *i.e.*, 2 g of deuterium contains 6.023×10^{23} atoms.

2.0 kg of deuterium contains

$$= \frac{6.023\times10^{23}}{2}\times2000 = 6.023\times10^{26}\,\text{atoms}$$

It can be inferred from the given reaction that when two atoms of deuterium fuse, 3.27 MeV energy is released. **[1]**

$\therefore$ Total energy per nucleus released in the fusion reaction :

$$E = \frac{[3.27\times6.023\times10^{26}]}{2}\text{MeV}$$

$$E = \frac{3.27}{2}\times6.023\times10^{26}\times1.6\times10^{-19}\times10^{6}$$

$$= 1.576 \text{ x } 10^{14}\ \text{J}$$

Power of the electric lamp, $P = 100\ W = 100$ J/s

Hence, the energy consumed by the lamp per second = 100 J **[1]**

The total time for which the electric lamp will glow is calculated as :

$$= \frac{1.576\times10^{14}}{100}\text{s}$$

$$= \frac{1.576\times10^{14}}{100\times60\times60\times24\times365} \approx 4.9\times10^{4}\ \text{years}$$

[1]

Q. 25. Calculate the height of the potential barrier for a head on collision of two deuterons. (Hint : The height of the potential barrier is given by the Coulomb repulsion between the two deuterons when they just touch each other. Assume that they can be taken as hard spheres of radius 2.0 fm.)

[NCERT Ex. Q. 13.20, Page 464]

Ans. As we know that,

When two deuterons collide head-on, the distance between their centres, d is given as :

Radius of first deuteron + Radius of second deuteron **[1]**

Radius of a deuteron nucleus = 2 fm = 2×10^{-15} m

$d = 2 \times 10^{-15} + 2 \times 10^{-15} = 4 \times 10^{-15}$ m

Charge on a deuteron nucleus = Charge on an electron = $e = 1.6 \times 10^{-19}$ C

Potential energy of the two-deuteron system : **[1]**

$$U = \frac{e^2}{4\pi\varepsilon_0 d}$$

Where,

ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}\ \text{Nm}^2\text{C}^{-2}$$

So that,

$$U = \frac{9\times10^{9}\times(1.6\times10^{-19})^2}{4\times10^{-15}}\ \text{J}$$

$$= \frac{9\times10^{9}\times(1.6\times10^{-19})^2}{4\times10^{-15}\times(1.6\times10^{-19})}\ \text{eV}$$

$$= 360\ \text{keV}$$

So, the height of the potential barrier of the two-deuteron system is 360 keV. **[1]**

Q. 26. Suppose India had a target of producing by 2020 AD, 200,000 MW of electric power, ten percent of which was to be obtained from nuclear power plants. Suppose we are given that, on an average, the efficiency of utilization (*i.e.* conversion to electric energy) of thermal energy produced in a reactor was 25%. How much amount of fissionable uranium would our country need per year by 2020? Take the heat energy per fission of ^{235}U to be about 200 MeV.

[NCERT Ex. Add. Q. 13.31, Page 466]

Ans. Given that,

Amount of electric power to be generated, $P=2 \times 10^5$ MW

10% of this amount has to be obtained from nuclear power plants.

Amount of nuclear power, $P_1 = \frac{10}{100}\times2\times10^{5}$

$= 2\times10^{4}$ MW

$= 2\times10^{4}\times10^{6}$ J/s

$= 2\times10^{10}\times60\times60\times24\times365$ J/y

Heat energy released per fission of a ^{235}U nucleus, $E = 200$ MeV

Efficiency of a reactor = 25% **[1½]**

So that, the amount of energy converted into the electrical energy per fission is :

$$\frac{25}{100}\times 200 = 50 \text{ MeV}$$

$$= 50\times 1.6\times 10^{-19}\times 10^{6}$$

$$= 8\times 10^{-12} \text{ J}$$

Number of atoms required for fission per year :

$$\frac{2\times 10^{10}\times 60\times 60\times 24\times 365}{8\times 10^{-12}} = 78840\times 10^{24} \text{ atoms}$$

1 mole, *i.e.*, 235 g of U^{235} contains 6.023×10^{23} atoms.

Mass of 6.023×10^{23} atoms of U^{235} = 235 g

$= 235 \times 10^{-3}$ kg

Mass of 78840×10^{24} atoms of U^{235}

$$= \frac{235\times 10^{-3}}{6.023\times 10^{23}}\times 78840\times 10^{24}$$

$$= 3.076\times 10^{4} \text{ kg}$$

So that, the mass of uranium needed per year is 3.076×10^4 kg. **[1½]**

Long Answer Type Questions **(5 marks each)**

Q. 1. Nuclei with magic no. of proton Z = 2, 8, 20, 28, 50, 52 and magic no. of neutrons N = 2, 8, 20, 28, 50, 82 and 126 are found to be very stable.

(i) Verify this by calculating the proton separation energy S_p, for $^{120}_{50}$Sn (Z = 50) and $^{121}_{51}$Sb (Z = 51). The proton separation energy for a nuclide is the minimum energy required to separate the least tightly bound proton from a nucleus of that nuclide. It is given by :
$S_p = (M_{Z-1,N} + M_H - M_{Z,N})\,c^2$.
Given ^{119}In = 118.9058 u, ^{120}Sn = 119.902199 u, ^{121}Sb = 120.903824 u, ^{1}H = 1.0078252 u.

(ii) What does the existence of magic number indicate? [NCERT Exemp. Q. 13.26, Page 86]

Ans. (i) Given ^{119}In = 118.9058 u, ^{120}Sn = 119.902199 u, ^{121}Sb = 120.903824 u, ^{1}H = 10078252 u.

$$S_{pSn} = (M_{119,70} + M_H - M_{120,70})c^2 = (118.9058 + 1.0078252 - 119.902199)c^2 = 0.0114362c^2$$

Similarly, $S_{pSp} = (M_{120,70} + M_H - M_{121,70})\,c^2$

$$= (119.902199 + 1.0078252 - 120.903822)c^2 = 0.0059912c^2$$

Since, $S_{pSn} > S_{pSb}$, Sn nucleus is more stable than Sb nucleus. **[2½]**

(ii) The existence of magic numbers indicates that the shell structure of nucleus is similar to the shell structure of atom. This also explains the peaks in binding energy per nucleon curve. **[2½]**

Q. 2. The neutron separation energy is defined as the energy required to remove a neutron from the nucleus. Obtain the neutron separation energies of the nuclei $^{41}_{20}$Ca and $^{27}_{13}$Al from the following data :
$m(^{40}_{20}\text{Ca})$ = 39.962591 u $m(^{41}_{20}\text{Ca})$ = 40.962278 u
$m(^{26}_{13}\text{Al})$ = 25.986895 u $m(^{27}_{13}\text{Al})$ = 26.981541 u
[NCERT Ex. Add. Q. 13.24, Page 464]

Ans. For $^{11}_{20}$Ca: Separation energy = 8.363007 MeV

For $^{23}_{13}$Al: Separation energy = 13.059 MeV

A neutron ($_0n^1$) is removed from a $^{41}_{20}$Ca nucleus. The corresponding nuclear reaction can be written as :

$$^{41}_{20}\text{Ca} \rightarrow {}^{40}_{20}\text{Ca} + {}^{1}_{0}\text{n}$$

It is given that :

Mass $m(^{40}_{20}\text{Ca}) = 39.962591$ u

Mass $m(^{41}_{20}\text{Ca}) = 40.962278$ u

Mass $m(_0n^1) = 1.008665$ u

The mass defect of this reaction is given as :

$$\Delta m = m(^{40}_{20}\text{Ca}) + (^{1}_{0}\text{n}) - m(^{41}_{20}\text{Ca})$$

$$= 39.962591 + 1.008665 - 40.962278 = 0.008978 \text{ u}$$

But 1 u = 931.5 MeV/c^2

$$\therefore \Delta m = 0.008978 \times 931.5 \text{ MeV}/c^2$$

So that, the energy require for neutron removal is,

$$E = \Delta mc^2$$

$$= 0.008978 \times 931.5$$

$$= 8.363007 \text{ MeV}$$ **[2½]**

For $^{22}_{13}$Al, the neutron removal reaction can be written as :

$$^{27}_{13}\text{Al} \rightarrow {}^{26}_{13}\text{Al} + {}^{1}_{0}\text{n}$$

It is given that :

Mass $m\left(^{27}_{13}\text{Al}\right) = 26.981541$ u

Mass $m\left(^{26}_{13}\text{Al}\right) = 25.986895$ u

The mass defect of this reaction is given as :

$$\Delta m = m\left(^{26}_{13}\text{Al}\right) + m\left(^{1}_{0}\text{n}\right) - m\left(^{27}_{13}\text{Al}\right)$$

$$= 25.986895 + 1.008665 - 26.9815141$$

$$= 0.014019 \text{ u}$$

$$= 0.014019 \times 931.5 \text{ MeV}/c^2$$

So that, the energy required for neutron removal is,

$$E = \Delta mc^2$$

$$= 0.014019 \times 931.5$$

$$= 13.059 \text{ MeV}$$ **[2½]**

Q. 3. Deuteron is a bound state of a neutron and a proton with a binding energy B = 2.2 MeV. A γ-ray of energy E is aimed at a deuteron nucleus to try to break it into a (neutron + proton) such that the n and p move in the direction of the incident γ-ray. If $E = B$, show that this cannot happen. Hence calculate how much bigger than B must E be for such a process to happen.
[NCERT Exemp. Q. 13.22, Page 85]

Ans. Given that,

Binding energy (B) of deuteron = 2.2 MeV

From conservation law of energy :

Some part of energy of γ-ray is used up against binding energy B = 2.2 MeV and the rest part will impart K.E. to neutron and proton.

$$E - B = K_n + K_p = \frac{p_n^2}{2m} + \frac{p_p^2}{2m} \quad ...(i)$$

From conservation of momentum,

$$p_n + p_p = \frac{E}{c} \quad ...(ii)$$

If, $E = B$, then the first equation gives $p_n = p_p = 0$ and hence the second equation cannot be satisfied, and the process cannot take place.

For the process to take place, Let $E = B + \lambda$, where λ would be $<< B$.

Then, substituting for p_n from Equation (2) into Equation (1). **[2½]**

Solve the above equation :

$$\lambda = \frac{1}{2m}(p_p^2 + p_n^2)$$

$$= \frac{1}{2m}\left(p_p^2 + \left(p_p - \frac{E}{c}\right)^2\right)$$

$$\therefore 2p_p^2 - \frac{2E}{c}p_p + \left(\frac{E^2}{c^2} - 2mX\right) = 0$$

$$\therefore p_p = \frac{\frac{2E}{c} \pm \sqrt{\frac{4E^2}{c^2} - 8\left(\frac{E^2}{c^2} - 2mX\right)}}{4}$$

Since the denominator must be positive for p_p to be real :

$$\frac{4E^2}{c^2} - 8\left(\frac{E^2}{c^2} - 2m\lambda\right) = 0$$

$$\text{or } 16m\lambda = \frac{4E^2}{c^2}$$

$$\lambda = \frac{E^2}{4mc^2} \approx \frac{B^2}{4mc^2}$$

[2½]

Q. 4. The deuteron is bound by nuclear forces just as H-atom is made up of p and e bound by electrostatic forces. If we consider the force between neutron and proton in deuteron as given in the form of a Coulomb potential but with an effective charge e'

$$F = \frac{1}{4\pi\varepsilon_0}\frac{e'^2}{r}$$

estimate the value of (e'/e) given that the binding energy of a deuteron is 2.2 MeV.

[NCERT Exemp. Q. 13.23, Page 85]

Ans. The binding energy of H atom in ground state :

$$E = \frac{me^4}{8\varepsilon_0^2 h^2} = 13.6 \text{ eV} \quad ...(I) \quad \textbf{[1]}$$

If proton and neutron had charge e' each and governed by the same electrostatic force, then in the above equation we would need to replace electronic mass m by the reduced mass m' of proton-neutron (as some mass of proton and neutron is used by binding energy) and electronic charge e is replaced by e'. **[1]**

$$m' = \frac{M}{2}$$

$$= \frac{1836 \text{ m}_e}{2}$$

$$= 918 \text{ m}_e$$

$$\text{Binding energy} = \frac{918\text{m}_e(e')^4}{8\pi\varepsilon_0^2 h^2} = 2.5\text{MeV} \quad ...(2)$$

Dividing equations ... (2) by ... (1), we get

$$918\left(\frac{e'}{e}\right)^4 = \frac{2.2\text{MeV}}{13.6\text{eV}} = \frac{22\times10^6}{13.6}$$

$$\Rightarrow \left(\frac{e'}{e}\right)^4 = \frac{2.2\text{MeV}}{13.6\times918} = 176.21$$

$$\Rightarrow \frac{e'}{e} = (176.21)^{1/4} = 3.64$$

[3]

Q. 5. Before the neutrino hypothesis, the beta decay process was thought to be the transition,

$$n \to p + \bar{e}$$

If this was true, show that if the neutron was at rest, the proton and electron would emerge with fixed energies and calculate them. Experimentally, the electron energy was found to have a large range. [NCERT Exemp. Q. 13.24, Page 85]

Ans. Before beta decay, neutron is at rest. So that,

$$p_n = p_p + p_e$$

$$\text{or } p_p + p_e = 0 \Rightarrow |p_p| = |p_e| = p$$

$$\text{Also, } E_p = (m_p^2c^4 + p_p^2\,c^2)^{\frac{1}{2}},$$

$$\Rightarrow E_e = (m_e^2c^4 + p_e^2\,c^2)^{\frac{1}{2}} = \left(m_e^2c^4 + p_e^2\,c^2\right)^{\frac{1}{2}}$$

From conservation of energy,

$$(m_p^2c^4 + p^2c^2)^{\frac{1}{2}} + (m_e^2c^4 + p^2c^2)^{\frac{1}{2}} = m_nc^2$$

$$m_pc^2 \approx 936\ MeV,$$

$$m_nc^2 \approx 938\ MeV,$$

$$m_ec^2 = 0.51\ MeV$$

Since, the energy difference between n and p is small, pc will be small, $pc << m_pc^2$, while pc may be greater than m_ec^2.

$$\Rightarrow m_pc^2 + \frac{p^2c^2}{2m_p^2c^4} \simeq m_nc^2 - pc$$

To first order

$$pc \simeq m_nc^2 - m_pc^2$$

$$= 938MeV - 936MeV$$

$$= 2MeV \quad \textbf{[4]}$$

This gives the momentum,

Then,

$$E_p = \left(m_p^2c^4 + p^2c^2\right)^{\frac{1}{2}} = \sqrt{936^2 + 2^2} \simeq 936\ MeV$$

$$E_e = \left(m_e^2c^4 + p^2c^2\right)^{\frac{1}{2}} = \sqrt{(0.51)^2 + 2^2} \simeq 2.06\ MeV \quad \textbf{[1]}$$

Q. 6. Sometimes a radioactive nucleus decays into a nucleus which itself is radioactive.

An example is :

$$^{38}\textbf{sulphur} \xrightarrow[-2.48\text{h}]{half-life} {}^{38}Cl \xrightarrow[-0.62\text{h}]{half-life} {}^{38}\textbf{Ar(stable)}$$

Assume that we start with $^{38}_{1000}S$ nuclei at time $t = 0$. The number of ^{38}Cl is of count zero at $t = 0$ and will again be zero at $t = \infty$. At what value of t, would the number of counts be a maximum?

[NCERT Exemp. Q. 13.21, Page 85]

Ans. Consider the chain reaction of two decays :

$$^{38}S \xrightarrow[-2.48\,h]{half-life} {}^{38}Cl \xrightarrow[=0.62\,h]{half-life} {}^{38}Ar$$

At any time t, let ^{38}S have N_1 (t) active nuclei and ^{38}Cl have N_2 active nuclei.

$$\frac{dN_1}{dt} = -\lambda_1 N_1 = \text{rate of formation of } Cl^{38}.$$

Also, $\frac{dN_2}{dt} = -\lambda_2 N_2 + \lambda_1 N_1$

But $N_1 = N_0 e^{-\lambda_1 t}$

$$\frac{dN_2}{dt} = \lambda_1 N_0\, e^{-\lambda_1 t} - \lambda_2 N_2 \qquad ...(i)$$

Multiplying by $e^{\lambda_2 t} dt$ and rearranging

$$e^{\lambda_2 t} dN_2 + \lambda_2 N_2 e^{\lambda_2 t} = \lambda_1 N_0 e^{(\lambda_2 - \lambda_1)t} dt$$

Integrating both sides

$$N_2 e^{\lambda_2 t} = \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} + C \qquad [e^{\lambda_2 t} dN_2 dt = 0]$$

Since, at $t = 0, N_2 = 0, C = -\frac{N_0 \lambda_1}{\lambda_2 - \lambda_1}$ **[2½]**

$$N_2 e^{\lambda_2 t} = \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1}\left[e^{(\lambda_2 - \lambda_1)t} - \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1}\right]$$

$$\frac{N}{e^{-\lambda_2 t}} = \frac{N_0 \lambda_1}{(\lambda_2 - \lambda_1)}[e^{(\lambda_2 - \lambda_1)t} - 1]$$

Multiplying $e^{-\lambda_2 t}$ to both sides, we get

$$N_2 = \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1}\left[e^{(\lambda_2 - \lambda_1 - \lambda_2)t} - e^{-\lambda_2 t}\right][\because e^{\circ} = 1]$$

$$N_2 = \frac{N_0 \lambda_1}{(\lambda_2 - \lambda_1)}[e^{-\lambda_1 t} - e^{-\lambda_2 t}]$$

$$N_2\lambda_2 - N_2\lambda_1 = \lambda_1 N_0 e^{-\lambda_1 t} - \lambda_1 N_0 e^{-\lambda_2 t}$$

For N_2 to be maximum, $\frac{dN_2}{dt} = 0$

$$\therefore\ N_2\lambda_2 - N_2\lambda_1 = \lambda_2 N_0 e^{-\lambda_1 t} - \lambda_1 N_2$$

$$N_2\lambda_2 - \lambda_1 N_2 + \lambda_1 N_2 = \lambda_2 e^{-\lambda_1 t}$$

$$N_2\lambda_2 = \lambda_1 e^{-\lambda_1 t}$$

$$N_2\lambda_2 = \lambda_1 N_0 e^{-\lambda_1 t}$$

$$N_2 = \frac{\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t}$$

$$\therefore \frac{\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t} e^{-\lambda_1 t} = \frac{N_0 \lambda_1}{(\lambda_2 - \lambda_1)}\ [e^{(\lambda_1 - \lambda_2)t} - 1]$$

By cross multiplication and multiplying both sides by $e^{-(\lambda_2 - \lambda_1)t}$

$$\frac{\lambda_2 - \lambda_1}{\lambda_2} = 1 - e^{-(\lambda_2 - \lambda_1)t}$$

$$1 - \frac{\lambda_1}{\lambda_2} = 1 - 1 - e^{-(\lambda_2 - \lambda_2)t}$$

$$\frac{\lambda_1}{\lambda_2} = e^{(\lambda_1 - \lambda_2)t}$$

$$\log e\left(\frac{\lambda_1}{\lambda_2}\right) = \log_e e^{(\lambda_1 - \lambda_2)t}$$

$$\text{or } \log e\left(\frac{\lambda_1}{\lambda_2}\right) = (\lambda_1 - \lambda_2)t$$

$$t = \frac{\log_e\left(\frac{\lambda_1}{\lambda_2}\right)}{\lambda_1 - \lambda_1} \qquad \left(\because\ \lambda = \frac{0.6931}{T_{1/2}}\right)$$

$$\therefore\ \frac{\lambda_1}{\lambda_2} = \frac{\frac{0.6931}{2.48}}{\frac{0.6931}{0.62}} = \frac{0.62}{2.48} = \frac{1}{4}$$

$$(\lambda_1 - \lambda_2) = \frac{0.6931}{2.48} - \frac{0.6931}{0.62}$$

$$= \frac{0.6931(-1.86)}{2.48 \times 0.62} = \frac{-0.6931 \times 1.86}{2.48 \times 0.62}$$

$$\therefore\ t = \frac{\log_e\left(\frac{1}{4}\right) \times 2.48 \times 0.62}{-0.6931 \times 1.86}$$

$$= \frac{-\log_e 4 \times 2.48 \times 0.62}{-0.6931 \times 1.86}$$

$$= \frac{2.303 \times 0.3010 \times 2.48 \times 0.62}{0.6931 \times 1.86}$$

$$= \frac{1.06586}{1.2892} = 0.8267h$$

Number Cl^{38} radioactive atoms will be maximum at $N_2 = 0.8267$ h.

Q. 7. The activity R of an unknown radioactive nuclide is measured at hourly intervals.

The results found are tabulated as follows :

t (hours)	0	1	2	3	4
R (mega Bq)	100	35.36	12.51	4.42	1.56

(i) Plot the graph of R versus t and calculate half-life from the graph.

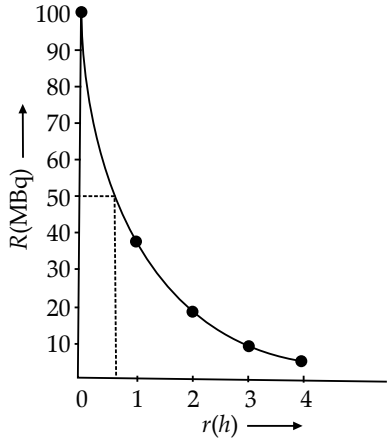
(ii) Plot the graph of log R/R_0 versus t and obtain the value of half-life from the graph.

[NCERT Exemp. Q. 13.25, Page 85]

Ans. In the table given below, we have listed values

$r(h)$	0	1	2	3	4
$R(MB_q)$	100	35.36	12.51	4.42	1.56
$\frac{R}{R_1}$	-	−1.04	−2.08	−3.11	−4.16

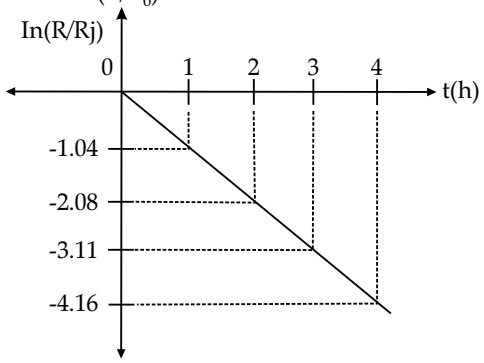
(i) When we plot the graph of R *versus* t, we obtain an exponential curve as shown.



From the graph we can say that activity R reduces to 50% in $t = OB \approx$. 40 min

So, $t_{1/2}$ is approx. 40 min. **[2½]**

(ii) The adjacent figure shows the graph of ln (R/R_0) versus t. In (R/R_0)



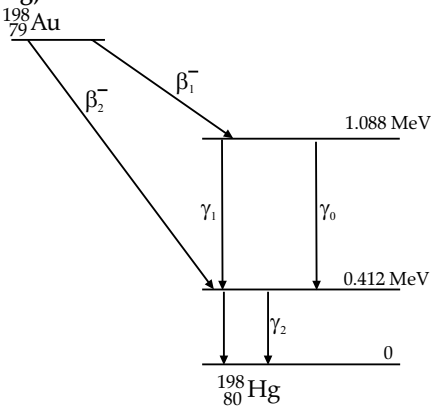
Slope of this graph=–λ

From the graph, $\lambda = -\left(\frac{-4.16-3.11}{1}\right) \Rightarrow -1.05\ h^{-1}$

$$\text{Half-life}, T_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{1.05} = 0.66h = 39.6\text{min} \approx 40\text{min}$$

[2½]

Q. 8. Obtain the maximum kinetic energy of β-particles, and the radiation frequencies of γ decays in the decay scheme shown in Figure. You are given that
$m(^{198}\text{Au}) = 197.968233$ u
$m(^{198}\text{Hg}) = 197.966760$ u



[NCERT Ex. Add. Q. 13.29, Page 465]

Ans. It can be observed from the given γ-decay diagram that γ_1 decays from the 1.088 MeV energy level to the 0 MeV energy level.

Hence, the energy corresponding to γ_1-decay is given as :

$E_1 = 1.088 - 0 = 1.088$ MeV

$h\nu_1 = 1.088 \times 1.6 \times 10^{-19} \times 10^6$ J

Where, h = Plank's constant$=6.6 \times 10^{-34}$ Js

ν_1 = Frequency of radiation radiated by γ_1-decay

$$\therefore \nu_1 = \frac{E_1}{h}$$

$$= \frac{1.088 \times 1.6 \times 10^{-19} \times 10^6}{6.6 \times 10^{-34}} = 2.637 \times 10^{20}\ \text{Hz}$$

It can be observed from the given γ-decay that γ_2 decays from the 0.412 MeV energy level to the 0 MeV energy level.

So that, the energy corresponding to γ_2-decay is given as :

$E_2 = 0.412 - 0 = 0.412$ MeV

$h\nu_2 = 0.412 \times 1.6 \times 10^{-19} \times 10^6$ J

Where,

ν_2 = Frequency of radiation radiated by γ_2-decay

$$\therefore \qquad v_2 = \frac{E_2}{h}$$

$$\frac{0.412 \times 1.6 \times 10^{-19} \times 10^6}{6.6 \times 10^{-34}} = 9.988 \times 10^{19}\ \text{Hz}$$ **[2]**

It can be observed from the given γ-decay that γ_3 decays from the 1.088 MeV energy level to the 0.412 MeV energy level.

So that, the energy corresponding to γ_3-decay is given as :

$E_3 = 1.088 - 0.412 = 0.676$ MeV

$h\nu_3 = 0.676 \times 10^{-19} \times 10^6$ J

Where,

ν_3=Frequency of radiation radiated by γ_3-decay

$$\therefore \nu_3 = \frac{E_3}{h}$$

$$= \frac{0.676 \times 1.6 \times 10^{-19} \times 10^6}{6.6 \times 10^{-34}}$$

$$= 1.639 \times 10^{20}\ \text{Hz}$$

Mass of $m\left(^{198}_{78}\text{Au}\right) = 197.968233$ u

Mass of $m\left(^{198}_{80}\text{Hg}\right) = 197.966760$ u **[1]**

1 u = 931.5 MeV/c^2

Energy of the highest level is given as :

$$E = \left[m\left(^{198}_{78}\text{Au}\right) - m\left(^{190}_{80}\text{Hg}\right)\right]$$

197.968233-197.966760=0.001473 u

$= 0.001473 \times 931.5 = 1.3720995$ MeV

β_1 decays from the 1.3720995 MeV level to the 1.088 MeV level

∴ Maximum kinetic energy of the β_1 particle

$= 1.3720995 - 1.088 = 0.2840995$ MeV

β_2 decays from the 1.3720995 MeV level to the 0.412 MeV level

∴ Maximum kinetic energy of the β_2 particle

$= .3720995 - 0.412 = 0.9600995$ MeV **[2]**

Q. 9. A source contains two phosphorous radio nuclides $^{32}_{15}P$ ($T_{1/2}$ = 14.3 d) and $^{33}_{15}P$ ($T_{1/2}$ = 25.3 d). Initially, 10% of the decays come from $^{33}_{15}P$. How long one must wait until 90% do so?

[NCERT Ex. Add. Q. 13.25, Page 464]

Ans. Given that,

Half-life of $^{32}_{15}P$, $T_{1/2} = 14.3$ days

Half-life of $^{33}_{15}P$, $T'_{1/2} = 25.3$ days

$^{33}_{15}P$ nucleus decay is 10% of the total amount of decay.

The source has initially 10% of $^{33}_{15}P$ nucleus and 90% of $^{32}_{15}P$ nucleus.

Suppose after t days, the source has 10% of $^{32}_{15}P$ nucleus and 90% of $^{33}_{15}P$ nucleus.

Initially :

Number of $^{33}_{15}P$ nucleus = N

Number of $^{32}_{15}P$ nucleus = 9 N

Finally :

Number of $^{33}_{15}P$nucleus = 9 N′

Number of $^{32}_{15}P$ nucleus = N′ **[2½]**

For $^{32}_{15}P$ nucleus, we can write the number ratio as :

$$\frac{N'}{9N} = \left(\frac{1}{2}\right)^{\frac{t}{T'_{1/2}}}$$

$$N' = 9N(2)^{\frac{-t}{14.3}} \quad ...(1)$$

For $^{33}_{15}P$, we can write the number ratio as :

$$\frac{9N'}{N} = \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}}$$

$$9N' = N(2)^{\frac{-t}{25.3}} \quad ...(2)$$

On dividing equation (1) by equation (2), we get :

$$\frac{1}{9} = 9 \times 2^{\left(\frac{t}{25.3} - \frac{t}{14.3}\right)}$$

$$\frac{1}{81} = 2^{-\left(\frac{11t}{25.3 \times 14.3}\right)}$$

$$\log 1 - \log 81 = \frac{-11t}{25.3 \times 14.3} \log 2$$

$$\Rightarrow \frac{-11t}{25.3 \times 14.3} = \frac{0 - 1.908}{0.301}$$

$$t = \frac{25.3 \times 14.3 \times 1.908}{11 \times 0.301}$$

$$\approx 208.5 \text{ days}$$

So that, it will take about 208.5 days for 90% decay of $_{15}P^{33}$. **[2½]**

Q. 10. Consider the fission of $^{238}_{92}U$ by fast neutrons. In one fission event, no neutrons are emitted and the final end products, after the beta decay of the primary fragments, are $^{140}_{58}Ce$ and $^{99}_{44}Ru$. Calculate Q for this fission process. The relevant atomic and particle masses are :

$$m\left(^{238}_{92}U\right) = 238.05079 \text{ u}$$

$$m\left(^{140}_{58}Ce\right) = 139.90543 \text{ u}$$

$$m\left(^{99}_{44}Ru\right) = 98.90594 \text{ u}$$

[NCERT Ex. Add. Q. 13.27, Page 465]

Ans. In the fission of $^{238}_{92}U$, 10 β^- particles decay from the parent nucleus. The nuclear reaction can be written as :

$$^{238}_{92}U + ^{1}_{0}n \rightarrow ^{140}_{58}Ce + ^{99}_{44}Ru + 10^{0}_{-1}e$$

It is given that :

Mass of nucleus $^{238}_{92}U$, $m_1 = 238.05079$ u

Mass of nucleus $^{140}_{58}Ce$, $m_2 = 139.90543$ u

Mass of nucleus $^{99}_{44}Ru$, $m_3 = 68.90594$ u

Mass of neutron $^{1}_{0}n$, $m_4 = 1.008665$ u

Q-value of the above equation,

$$Q = \left[m'(^{238}_{92}U) + m\left(^{1}_{0}n\right) - m'\left(^{140}_{58}Ce\right) - m'\left(^{99}_{44}Ru\right) - 10m_c\right]c^2$$

where,

m'= Represents the corresponding atomic masses of the nuclei

$$m'\left(^{238}_{92}U\right) = m_1 - 92m_e$$

$$m'\left(^{140}_{58}Ce\right) = m_2 - 58m_e$$

$$m'\left(^{99}_{44}Ru\right) = m_3 - 44m_e$$

$$m\left(^{1}_{0}n\right) = m_4$$

[3]

$$Q = [m_1 - 92m_e + m_4 - m_2 + 58m_e - m_3 + 44m_e - 10m_e]c^2$$

$$= [m_1 + m_4 - m_2 - m_3]c^2$$

$$= [238.0507 + 1.008665 - 139.90543 - 98.90594]c^2$$

But 1 u = 931.5 MeV/c^2

$$\therefore Q = 0.247995 \times 931.5 = 231.007 \text{ MeV}$$

So that, the Q-value of the fission process is 231.007 MeV. **[2]**

Q. 11. Calculate and compare the energy released by (a) fusion of 1.0 kg of hydrogen deep within Sun and (b) the fission of 1.0 kg of ^{235}U in a fission reactor.

[NCERT Ex. Add. Q. 13.30, Page 466]

Ans. **(a)** Amount of hydrogen, m=1000 g

1 mole, *i.e.*, 1 g of hydrogen ($^{1}_{1}H$) contains 6.023 × 10^{23} atoms.

$\therefore$ 1000 g of $^{1}_{1}H$ contains 6.023 × 10^{23} × 1000 atoms.

Within the Sun, four $^{1}_{1}H$ nuclei combine and form one $^{4}_{2}He$ nucleus. In this process 26 MeV of energy is released.

So that, the Energy released from the fusion of 1 kg $^{1}_{1}H$ is :

$$E_1 = \frac{6.023 \times 10^{23} \times 26 \times 10^3}{4}$$

$$= 39.1495 \times 10^{26} \text{ MeV}$$ **[2½]**

(b) Amount of $^{235}_{92}U$=1 kg = 1000 g

1 mole, *i.e.*, 235 g of $^{235}_{92}U$ contains 6.023 × 10^{22} atoms.

$\therefore$ 1000 g of $^{235}_{92}U$ contains

$$\Rightarrow \frac{6.023 \times 10^{23} \times 1000}{235} \text{ atoms}$$

It is known that the amount of energy released in the fission of one atom of $^{235}_{92}U$ is 200 MeV.

So that, energy released from the fission of 1 kg of $^{235}_{92}U$ is :

$$E_2 = \frac{6\times10^{23}\times1000\times200}{235}$$
$$= 5.106\times10^{26} \text{ MeV}$$
$$\frac{E_1}{\therefore E_2} = \frac{39.1495\times10^{26}}{5.106\times10^{26}} = 7.67 \approx 8$$

Therefore, the energy released in the fusion of 1 kg of hydrogen is nearly 8 times the energy released in the fission of 1 kg of uranium. **[2½]**

Q. 12. Consider the D–T reaction (deuterium–tritium fusion)

$_1^2\text{H} + _1^3\text{H} \rightarrow _2^4\text{He} + \text{n}$

(a) Calculate the energy released in MeV in this reaction from the data :

$m(_1^2\text{H}) = 2.014102$ u

$m(_1^3\text{H}) = 3.016049$ u

(b) Consider the radius of both deuterium and tritium to be approximately 2.0 fm. What is the kinetic energy needed to overcome the coulomb repulsion between the two nuclei? To what temperature must the gas be heated to initiate the reaction?

(Hint : Kinetic energy required for one fusion event = average thermal kinetic energy available with the interacting particles = 2(3 kT/2); k = Boltzman's constant, T = absolute temperature.)

[NCERT Ex. Add. Q. 13.28, Page 465]

Ans. **(a)** Take the D-T nuclear reaction :

$_1^2\text{H} + _1^3\text{H} \rightarrow _2^4\text{He} + \text{n}$

It is given that :

Mass of $_1^2\text{H}$, $m_1 = 2.014102$ u

Mass of $_1^3\text{H}$, $m_2 = 3.016049$ u

Mass of $_2^4\text{He}$, $m_3 = 4.002603$ u

Mass of $_0^1\text{n}$, $m_4 = 1.008665$ u

Q-value of the given D-T reaction is :

$$Q = [m_1 + m_2 - m_3 - m_4]\, c^2$$
$$= [2.014102 + 3.016049 - 4.002603 - 1.008665]\, c^2$$
$$= [0.018883\, c^2] \text{ u}$$

But 1 u = 931.5 MeV/c^2

$$\therefore Q = 0.01883\times931.5 = 17.59 \text{ MeV}$$ **[2½]**

(b) Radius of deuterium and tritium, $r \approx 2.0$ fm $= 2\times10^{-15}$ m

Distance between the two nuclei at the moment when they touch each other, $d = r + r = 4\times10^{-15}$ m

Charge on the deuterium nucleus = e

Charge on the tritium nucleus = e

So that, the repulsive potential energy between the two nuclei is given as :

$$u = \frac{e^2}{4\pi\varepsilon_0(d)}$$

Where,

ε_0 = Permittivity of free space

$$\frac{1}{4\pi\varepsilon_0} = 9\times10^9 \text{ N m}^2 \text{ C}^{-2}$$
$$\therefore \text{U} = \frac{9\times10^9\times(1.6\times10^{-19})^2}{4\times10^{-15}} = 5.76\times10^{-14} \text{ J}$$
$$= \frac{5.76\times10^{-14}}{1.6\times10^{-19}} = 3.6\times10^5 \text{ eV} = 360 \text{ keV}$$

So that, 5.76×10^{-14} J or 360 keV of kinetic energy (KE) is needed to overcome the Coulomb repulsion between the two nuclei.

However, it is given that :

$$\text{KE} = 2\times\frac{3}{2}kT$$

Where,

k = Boltzmann constant = 1.38×10^{-23} m^2 kg s^{-2} K^{-1}

T = Temperature required for triggering the reaction

$$\therefore T = \frac{KE}{3k}$$
$$= \frac{5.76\times10^{-14}}{3\times1.38\times10^{-23}} = 1.39\times10^9 \text{ K}$$

So that, the gas must be heated to a temperature of 1.39×10^9 k to initiate the reaction. **[2½]**

Q. 13. Under certain circumstances, a nucleus can decay by emitting a particle more massive than an (α-particle. Consider the following decay processes :

$_{88}^{223}\text{Ra} \rightarrow _{82}^{209}\text{Pb} + _6^{14}\text{C}$

$_{88}^{223}\text{Ra} \rightarrow _{86}^{219}\text{Rn} + _2^4\text{He}$

Calculate the Q-values for these decays and determine that both are energetically allowed.

[NCERT Ex. Add. Q. 13.26, Page 464]

Ans. Take a $_6^{14}$C emission nuclear reaction :

$_{88}^{223}\text{Ra} \rightarrow _{82}^{209}\text{Pb} + _6^{14}\text{C}$

We know that :

Mass of $_{88}^{223}$Ra, $m_1 = 223.01850$ u

Mass of $_{82}^{209}$Pb, $m_2 = 208.98107$ u

Mass of $_6^{14}$C, $m_3 = 14.00324$ u

So that, the Q-value of the reaction is given as :

$$Q = (m_1 - m_2 - m_3)\, c^2$$
$$= (223.01850 - 208.98107 - 14.00324)\, c^2$$
$$= (0.03419\, c^2) \text{ u}$$

But 1 u = 931.5 MeV/c^2

$$\therefore Q = 0.03419\times931.5$$
$$= 31.848 \text{ MeV}$$

So that, the Q-value of the nuclear reaction is 31.848 MeV. Since the value is positive, the reaction is energetically allowed.

Now take a $_2^4$He emission nuclear reaction :

$_{88}^{223}\text{Ra} \rightarrow _{86}^{229}\text{Rn} + _2^4\text{He}$

We know that :

Mass of $_{88}^{223}$Ra, $m_1 = 223.01850$

Mass of $_{82}^{219}$Rn, $m_2 = 219.00948$

Mass of $_2^4$He, $m_3 = 4.00260$ **[3]**

Q-value of this nuclear reaction is given as :

$$Q = (m_1 - m_2 - m_3)\, c^2$$
$$= (223.01850 - 219.00948 - 4.00260)\ c^2$$
$$= (0.00642\ c^2)\ u$$
$$= 0.00642 \times 931.5 = 5.98\ \text{MeV}$$

So that, the Q value of the second nuclear reaction is 5.98 MeV. Since the value is positive, the reaction is energetically allowed. **[2]**

Some Commonly Made Errors

- Students confuse to differentiate the Isotopes and Isobars.
- In all the nuclear equations, the following two things must occur :
  (i) The mass number (A) on the left and the right of the arrows must be equal,
  (ii) The atomic number (Z) on the left and the right of the arrows must be equal.
- Don't confuse with the term half-life and mean life of radioactive substance, both are different.
- Generally, students get confused with the term 'Radio-activity decay-alpha-decay, beta-decay and gamma decay'.

EXPERT ADVICE

- *Don't ignore to study terms related to nucleus because many fundamental questions come from here.*
- *Apply method of elimination for MCQ questions. Look at the answers before even starting to solve the question and see if you can eliminate any answer. When calculating answer for a big question you may not need to solve the problem fully in order to choose the right answer because you could eliminate the other three.*
- *Try to solve numerical by making suitable diagram, do not skip the diagrams if required.*
- *Explain the radio-active emission by proper required equations and diagram.*
- *Learn formulae and understand concept of the radio-active decay law.*
- *Understand the nature of radio-active emissions.*
- *Draw suitable diagram and equation to explain nuclear fission and nuclear fusion reactions.*
- *Convert all units to SI.*

OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/uqK2iYCw6WY **Or Scan the Code**	*https://youtu.be/qSQ5f-DlJow* **Or Scan the Code**
https://goo.gl/1XA8Om **Or Scan the Code**	

CHAPTER

14

SEMICONDUCTOR ELECTRONICS – MATERIALS, DEVICES AND SIMPLE CIRCUITS

Chapter Objectives

This chapter will help you understand :

- ***Semiconductors and diodes :*** *Classification of metals, Conductors and Semiconductors; Intrinsic semiconductors; Extrinsic semiconductor; p-n Junction; Semiconductor diodes; Application of junction diode as a rectifier; and Special purpose p-n junction diodes.*
- ***Transistors and electronic circuits :*** *Junction transistor; Digital electronics and Logic gates; and Integrated circuits.*

TOPIC - 1
Semiconductors and Diodes **P. 323**

TOPIC - 2
Transistors and Electronic Circuits **P. 339**

TOPIC-1
Semiconductors and Diodes

Quick Review

- In general, two types of diodes exist. Older diodes were vacuum tubes containing two metal components, while newer diodes are solid-state devices consisting of one *n*-type and one *p*-type semiconductor.
- Solid-state devices are electronic devices that take advantage of the special conducting properties of solids. Semiconductors are substances that conduct a feeble electric current.
- **Vacuum tube diode :** The working element in a vacuum tube diode is a metal wire or cylinder known as the cathode.
- Newer types of diodes are made from *n*-type semiconductors and *p*-type semiconductors. *n*-type semiconductors contain small impurities that provide an excess of electrons with the capability to move in the conductor.

TIPS...

- *Current Flow direction for Drift current, Charged particle motion in response to an Electric field.*
- *Current flow for Diffusion current, Particles tend to spread out or Redistribute from areas of high concentration to lower concentration*

TRICKS...

- *Sum of Drift current and Diffusion current is equal to Diode current.*
- *Students should try to understand the difference between Resistance and Resistivity or Conductance and Conductivity.*
- *Students should also know about the Difference between Electric field and Electric field intensity.*

Know the Links

- *https://www.electronics-tutorials.ws/diode/diode_1.html*
- *https://en.wikibooks.org/wiki/Semiconductor_Electronics*
- *https://www.elprocus.com/semiconductor-devices-types-and-applications/*

Multiple Choice Questions **(1 mark each)**

Q. 1. In an *n*-type silicon, which of the following statement is true :

(a) Electrons are majority carriers and trivalent atoms are the dopants.

(b) Electrons are minority carriers and pentavalent atoms are the dopants.

(c) Holes are minority carriers and pentavalent atoms are the dopants.

(d) Holes are majority carriers and trivalent atoms are the dopants. [NCERT Ex. Q. 14.1, Page 509]

Ans. Correct option : (c)
Explanation : In an *n*-type silicon the holes are the minority carriers. An *n*-type semiconductor is obtained when pentavalent atoms, such as phosphorus, are doped in silicon atoms.

Q. 2. Which of the statements given in Exercise 14.1 is true for *p*-type semiconductors?
[NCERT Ex. Q. 14.2, Page 509]

Ans. Correct option : (d)
Explanation : In a *p*-type semiconductor, the holes are the majority carriers, while the electrons are the minority carriers. A *p*-type semiconductor is obtained when trivalent atoms, such as aluminium, are doped in silicon atoms.

Q. 3. Carbon, silicon and germanium have four valence electrons each. These are characterised by valence and conduction bands separated by energy band gap respectively equal to $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$ Which of the following statements is true?

(a) $(E_g)_{Si} < (E_g)_{Ge} < (E_g)_C$ **(b)** $(E_g)_C < (E_g)_{Ge} > (E_g)_{Si}$

(c) $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$ **(d)** $(E_g)_C = (E_g)_{Si} = (E_g)_{Ge}$

[NCERT Ex. Q. 14.3, Page 509]

Ans. Correct option : (c)
Explanation : Above mentioned three given elements, the energy band gap of carbon is the maximum and that of germanium is the least. The energy band gaps of these elements are related as : $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$

Q. 4. In an unbiased *p-n* junction, holes diffuse from the *p*-region to *n*-region because
(a) free electrons in the *n*-region attract them.
(b) they move across the junction by the potential difference.
(c) hole concentration in *p*-region is more as compared to *n*-region.
(d) All the above. **[NCERT Ex. Q. 14.4, Page 509]**

Ans Correct option : (c)
Explanation : The diffusion of charge carriers across a junction takes place from the regions of higher concentration to lower concentration. In this case, the *p*-region has greater concentration of holes than the *n*-region. Hence, in an unbiased *p-n* junction, holes diffuse from the *p*-region to the *n*-region.

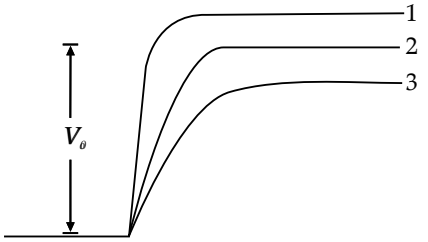
Q. 5. When a forward bias is applied to a *p-n* junction, it
(a) raises the potential barrier.
(b) reduces the majority carrier current to zero.
(c) lower the potential barrier.
(d) None of the above **[NCERT Ex. Q. 14.5, Page 510]**

Ans. Correct option : (c)
Explanation : When a forward bias is applied to a *p-n* junction, it lowers the value of potential barrier. In the case of a forward bias, the potential barrier opposes the applied voltage. Hence, the potential barrier across the junction gets reduced.

Q. 6. The conductivity of a semiconductor increases with increase in temperature because
(a) number density of free current carriers increases.
(b) relaxation time increases.
(c) both number density of carriers and relaxation time increase.
(d) number density of current carriers increases; relaxation time decreases but effect of decrease in relaxation time is much less than increase in number density.
[NCERT Exemp. Q. 14.1, Page 87]

Ans. Correct option : (d)
Explanation : In semiconductor, the density of charge carriers (electron, holes) are very small, so its resistance is high. When temperature increases the charge carriers (density) increases which increases the conductivity. As temperature of semiconductor increases, the speed of free electrons increases which decreases the relaxation time. As the density of charge carrier is small so there is small effect on decrease of relaxation time.

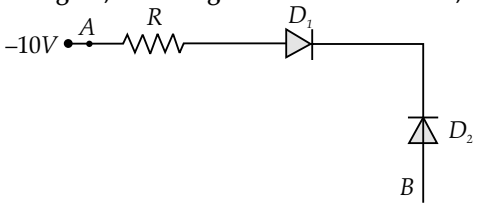
Q. 7. In Figure, V_o is the potential barrier across a *p-n* junction, when no battery is connected across the junction



(a) 1 and 3 both correspond to forward bias of junction
(b) 3 corresponds to forward bias of junction and 1 corresponds to reverse bias of junction
(c) 1 corresponds to forward bias and 3 corresponds to reverse bias of junction.
(d) 3 and 1 both correspond to reverse bias of junction.
[NCERT Exemp. Q. 14.2, Page 87]

Ans. Correct option : (b)
Explanation : when *p-n* junction is in forward bias, it compresses or decreases the depletion layer so potential barrier in forward bias decreases and in reverse layer so potential barrier in forward bias decrease and in reverse bias potential barrier increases.

Q. 8. In Figure, assuming the diodes to be ideal,

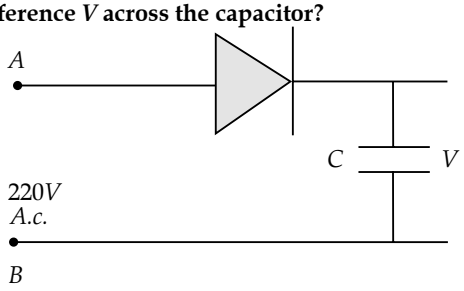


(a) D_1 is forward biased and D_2 is reverse biased and hence current flows from *A* to *B*
(b) D_2 is forward biased and D_1 is reverse biased and hence no current flows from *B* to *A* and vice versa.
(c) D_1 and D_2 are both forward biased and hence current flows from *A* to *B*.
(d) D_1 and D_2 are both reverse biased and hence no current flows from *A* to *B* and vice versa.
[NCERT Exemp. Q. 14.3, Page 88]

Ans. Correct option : (b)
Explanation : In circuit, *A* is at −10 V and *B* is at 0 V. So *B* is positive than *A*. So D_2 is in forward bias and

D_1 is in reverse bias so no current flows from A to B or B to A.

Q. 9. A 220 V A.C. supply is connected between points A and B in Figure. What will be the potential difference V across the capacitor?



(a) 220 V **(b) 110 V**

(c) 0 V **(d) $220\sqrt{2}$V**

[NCERT Exemp. Q. 14.4, Page 88]

Ans. Correct option : (d)

Explanation : Potential difference across capacitors will be peak voltage when diode is in forward bias. Diode will be in forward bias when end A is at positive potential of cycle.

So potential at C=peak value of

$V = V_{rms}\sqrt{2} = 220\sqrt{2}\text{V}$.

Q. 10. Hole is

(a) an anti-particle of electron.

(b) a vacancy created when an electron leaves a covalent bond.

(c) absence of free electrons.

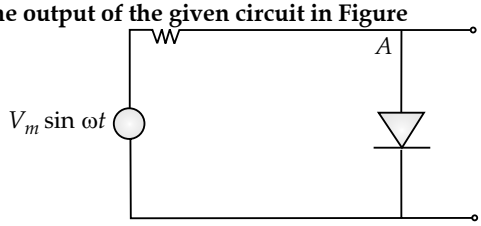
(d) an artificially created particle.

[NCERT Exemp. Q. 14.5, Page 88]

Ans. Correct option : (b)

Explanation : Atoms of semiconductor are binding by covalent bonds between the atoms of same or different type. Due to thermal agitation when an electron leaves its position and become free, then it leaves a vacancy of electron, this vacancy in the bond (covalent) is called hole.

Q. 11. The output of the given circuit in Figure



(a) would be zero at all times.

(b) would be like a half wave rectifier with positive cycles in output.

(c) would be like a half wave rectifier with negative cycles in output.

(d) would be like that of a full wave rectifier.

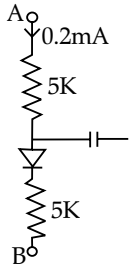
[NCERT Exemp. Q. 14.6, Page 88]

Ans. Correct option : (c)

Explanation : When positive cycle is at A, diode will be in forward bias and resistance due to diode is approximately zero so potential across diode will be about zero.

Similarly, when there is negative half cycle at A, diode will be in reverse bias and resistance will be maximum so potential difference across diode is $V_m \sin\omega t$ with negative at A. So we get only negative output at A so it behaves like a half wave rectifier with negative cycle at A in output, verifies the answer (c).

Q. 12. In the circuit shown in Figure, if the diode forward voltage drop is 0.3 V, the voltage difference between A and B is



(a) 1.3 V **(b) 2.3 V**

(c) 0 V **(d) 0.5 V**

[NCERT Exemp. Q. 14.7, Page 89]

Ans. Correct option : (b)

Explanation : In the middle right of the circuit the capacitor behaves like an open circuit for dc 0.2 mA current so current will flow from A to B only. Let potential across A and B is V, so by

Kirchhoff's loop law,

$V_{AB} = (5{,}000 \times 0.2 \times 10^{-3}) + 0.3 + 5{,}000 \times 0.2 \times 10^{-3}$

$V_{AB} = 1\text{ V} + 0.3\text{ V} + 1\text{ V}$

$V_{AB} = 2.3\text{ V}$

Q. 13. For a transistor amplifier, the voltage gain

(a) remains constant for all frequencies.

(b) is high at high and low frequencies and constant in the middle frequency range.

(c) is low at high and low frequencies and constant at mid frequencies.

(d) None of the above. **[NCERT Ex. Q. 7, Page 510]**

Ans. Correct option: (c).

Explanation: The voltage gain of a transistor amplifier is constant at mid frequency range only. It is low at high and low frequencies.

Tick Two or More Options

Q. 14. When an electric field is applied across a semiconductor

(a) electrons move from lower energy level to higher energy level in the conduction band.

(b) electrons move from higher energy level to lower energy level in the conduction band.

(c) holes in the valence band move from higher energy level to lower energy level.

(d) holes in the valence band move from lower energy level to higher energy level.

[NCERT Exemp. Q. 14.9, Page 89]

Ans. **Correct options :** (a) and (c)

Explanation : In the direction of electric field (E), the potential energy (PE) or energy of E decreases. So electrons always move opposite to direction of EF, that is, from lower energy level to higher energy level and holes move from higher energy level to lower energy level. In semiconductors, electrons are in conduction band and holes are in valence band.

Q. 15. Consider an *n-p-n* transistor with its base-emitter junction forward biased and collector base junction reverse biased. Which of the following statements are true?

(a) Electrons crossover from emitter to collector.

(b) Holes move from base to collector.

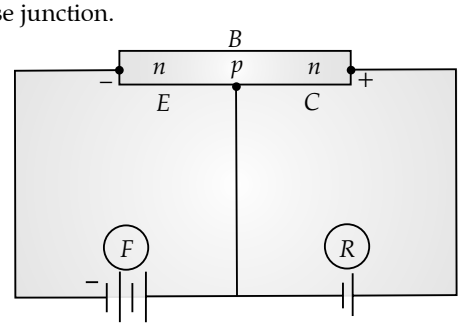
(c) Electrons move from emitter to base.

(d) Electrons from emitter move out of base without going to the collector.

[NCERT Exemp. Q. 14.10, Page 90]

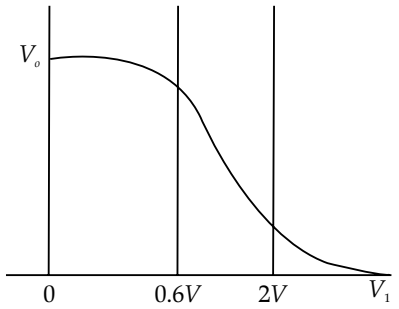
Ans. **Correct options :** (a) and (c)

Explanation : Electrons are repelled by forward bias from emitter to collector 5% of electrons combine with holes of base and rest 95% electrons of emitter are attracted by reverse bias of collector-base junction.



So electrons move from emitter to collector through base, *i.e.*, emitter to base verifies (a) and (c).

Q. 16. Figure shows the transfer characteristics of a base-biased CE transistor. Which of the following statements are true?



(a) At V_i = 0.4 V, transistor is in active state.

(b) At V_i = 1 V, it can be used as an amplifier.

(c) At V_i = 0.5 V, it can be used as a switch turned off.

(d) At V_i = 2.5 V, it can be used as a switch turned on.

[NCERT Exemp. Q. 14.11, Page 90]

Ans. **Correct options :** (b), (c) and (d)

Explanation : **(a)** At V_i = 0.4 V there is no collector current, only output voltage is V_o due to V_{ce} battery. So at 0.4 V transistor is not in active mode.

(b) when at V_i = 1 volt, transistor is between active region (0.6 V to 2 V). So it can be used as an amplifier.

(c) At V_i = 0.5 V, the transistor is in cut-off state and it can be used as a switch turned off.

(d) At V_i = 2.5 V, the transistor is beyond active region. The collector current is in saturated state and there is no effect on changing input voltage and the transistor can be used as a switch turned on. So, the statements (b), (c) and (d) are true.

Q. 17. In a *n-p-n* transistor circuit, the collector current is 10 mA. If 95% of the electrons emitted reach the collector, which of the following statements are true?

(a) The emitter current will be 8 mA.

(b) The emitter current will be 10.53 mA.

(c) The base current will be 0.53 mA.

(d) The base current will be 2 mA.

[NCERT Exemp. Q. 14.12, Page 90]

Ans. **Correct options :** (b) and (c)

Explanation : I_c = 10 mA

I_c = 95% of I_e or $I_c = \frac{95}{100} I_e$

$$I_e = \frac{10 \text{ mA} \times 100}{95} = 10.53 \text{ mA}$$

$$I_b + I_c = I_e$$

So $I_b = I_e - I_c = 10.53 - 10 = 0.53$ mA

Q. 18. In the depletion region of a diode

(a) there are no mobile charges

(b) equal number of holes and electrons exist, making the region neutral.

(c) recombination of holes and electrons has taken place.

(d) immobile charged ions exist.

[NCERT Exemp. Q. 14.13, Page 90]

Ans. **Correct options :** (a), (b), (c) and (d)

Explanation : During formation of *p-n* junction electrons from *n* side and holes from *p* side move towards each other and form a potential barrier across the junction.

Within the barrier there are holes and electrons or ions which cannot move. So in depletion layer no mobile charges but there are negative and positive ion; barrier layer was formed by recombination of electrons hand holes when *p-n* junction formed.

Q. 19. What happens during regulation action of a Zener diode?

(a) The current in and voltage across the Zener remains fixed.

(b) The current through the series Resistance (R_s) changes.

(c) The Zener resistance is constant.

(d) The resistance offered by the Zener diode changes.

[NCERT Exemp. Q. 14.14, Page 90]

Ans. **Correct options :** (b) and (d)

Explanation : During action of regulation, current R_s changes and resistance offered by Zener diode changes. The current through the Zener diode changes but voltage across the Zener remains constant.

Q. 20. To reduce the ripples in a rectifier circuit with capacitor filter

(a) R_L should be increased.

(b) input frequency should be decreased.

(c) input frequency should be increased.

(d) capacitors with high capacitance should be used.

[NCERT Exemp. Q. 14.15, Page 90]

Ans. Correct options : (a), (c) and (d)

Explanation : As the ripple factor

$$r = \frac{1}{4\sqrt{3}R_L C_V}$$

So to decrease r, R_L and C_V must be increased. Input frequency should be increased.

Q. 21. The breakdown in a reverse biased *p–n* junction diode is more likely to occur due to

(a) large velocity of the minority charge carriers if the doping concentration is small.

(b) large velocity of the minority charge carriers if the doping concentration is large.

(c) strong electric field in a depletion region if the doping concentration is small.

(d) strong electric field in the depletion region if the doping concentration is large.

[NCERT Exemp. Q. 14.16, Page 91]

Ans. Correct options : (a) and (d)

Explanation : In reverse biasing, minority charge carriers accelerate due to high electric field applied (due to reverse PD). So the minority charge carrier in both junctions accelerated toward depletion layer, which on striking with atoms causes ionization of atoms resulting the secondary electron and so more number of charge carriers constitute more current in reverse direction. More current in thin depletion layer is due to the heavy doping and large no. of ions in the depletion layer which give rise to strong electric field across the junction.

Q. 22. For transistor action, which of the following statements are correct:

(a) Base, emitter and collector regions should have similar size and doping concentrations.

(b) The base region must be very thin and lightly doped.

(c) The emitter junction is forward biased and collector junction is reverse biased.

(d) Both the emitter junction as well as the collector junction are forward biased.

[NCERT Ex. Q. 6, Page 510]

Ans. Correct option: (b) and (c).

Explanation: For a transistor action, the junction must be lightly doped so that the base region is very thin. Also, the emitter junction must be forward-biased and collector junction should be reverse biased.

Very Short Answer Type Questions (1 or 2 mark each)

Q. 1. Why are elemental dopants for Silicon or Germanium usually chosen from group XIII or group XV? [NCERT Exemp. Q. 14.17, Page 91]

Ans. The size of the dopant atom should be equivalent to the size of Si or Ge. So that the symmetry of pure Si or Ge, does not disturb and dopants can contribute the charge carrier on forming covalent bonds with Si or Germanium atoms. As the Si and Ge belong to XIVth group, so similar size of atom will be in XIII and XV groups of Modern Periodic Table. **[1]**

Q. 2. *Sn*, *C*, and *Si*, *Ge* are all group XIV elements. Yet, *Sn* is a conductor, *C* is an insulator while *Si* and *Ge* are semiconductors. Why?

[NCERT Exemp. Q. 14.18, Page 91]

Ans. A material will conduct current if there is no energy gap between conduction and valence band in energy band diagram of atom. This energy gap decreases from insulator to semiconductor and form semiconductor and from semiconductor to conductor. **[½]**

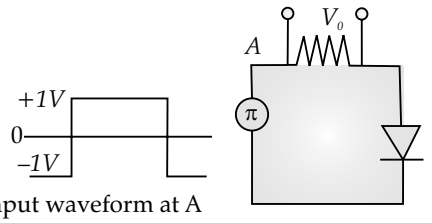
The energy gaps in *Sn*, *C*, *Si* and *Ge* are 0 eV, 0.54 eV, 1.1 eV and 0.7 eV, respectively related to their atomic size. So the Sn is a conductor, *C* is an insulator while *Si* and *Ge* are semiconductors. **[½]**

Q. 3. Can the potential barrier across a *p-n* junction be measured by simply connecting a voltmeter across the junction?

[NCERT Exemp. Q. 14.19, Page 91]

Ans. We cannot measure the potential barrier across *p-n* junction by voltmeter, because the resistance across depletion layer is smaller than resistance of voltmeter so current to measure the potential will not flow in voltmeter, and current passes through the junction. **[1]**

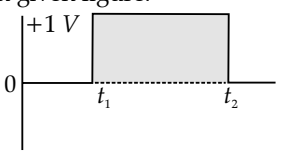
Q. 4. Draw the output waveform across the resistor Figure.



Input waveform at A

[NCERT Exemp. Q. 14.20, Page 91]

Ans. In given circuit, waveform is connected at *A* and waveform obtained across resistance diode conducts when diode is in forward biased so output will be only when input is +1 V is between t_1 to t_2. So, output waveform will be only t_1 to t_2 which is in given figure. **[½]**



[½]

Q. 5. The amplifiers *X*, *Y* and *Z* are connected in series. If the voltage gains of *X*, *Y* and *Z* are 10, 20 and 30, respectively and the input signal is 1 mV peak value, then what is the output signal voltage (peak value)

(a) if dc supply voltage is 10 V?

(b) if dc supply voltage is 5 V?

[NCERT Exemp. Q. 14.21, Page 91]

Ans. $A_{V_x} = 10$, $A_{V_y} = 20$ and $A_{V_z} = 30$

Signal $A_{V_i} = 1\text{ mV} = 10^{-3}$ V

$$A_V = \frac{V_0}{V_i}$$

$$\frac{V_0}{V_i} = A_{V_x} \times A_{V_y} \times A_{V_z}$$

$$V_0 = V_i \times A_{V_x} \times A_{V_y} \times A_{V_z}$$

$$= 10^{-3} \times 10 \times 20 \times 30$$

= 6 V (output voltage) **[1]**

(a) If DC supply voltage is 10 V and output voltage 6 V as the theoretical gain and practical gains are equal so output will be 6 V. **[½]**

(b) When DC supply voltage 5 V or *Vce* = 5 V then output peak cannot exceed by 5 V. So output will be 5 V. **[½]**

Q. 6. Write two points of difference between intrinsic and extrinsic semiconductors.

[CBSE Board, Foreign Scheme, 2017]

Ans. Two different points in intrinsic and extrinsic semiconductor are following :

Intrinsic Semiconductor :

(i) It is pure semiconducting material and no impurity atoms are added to it. Examples : Crystalline forms of pure silicon and germanium.

(ii) The number of free electrons in the conduction band and the number of holes in valence band is exactly equal and very small indeed. Its electrical conductivity is low. Its electrical conductivity is a function of temperature alone. **[1]**

Extrinsic Semiconductor :

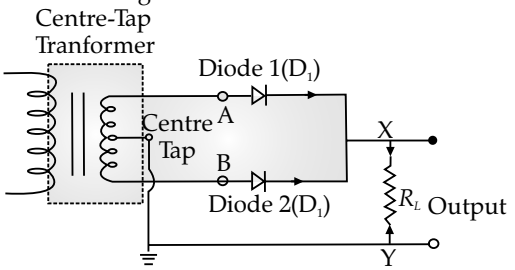
(iii) It is prepared by doping a small quantity of impurity atoms to the pure semiconducting material. Examples : silicon "Si" and germanium "Ge" crystals with impurity atoms of *As*, *Sb*, *P*, etc., or In *B*, *Aℓ*, etc.

(iv) The number of free electrons and holes is never equal. There is excess of electrons in *n*-type semiconductors and excess of holes in *p*-type semiconductors. Its electrical conductivity is high. Its electrical conductivity depends upon the temperature as well as on the quantity of impurity atoms doped the structure. **[1]**

Q. 7. A student wants to use two *p-n* junction diodes to convert alternating current into direct current. Draw the labelled circuit diagram she would use and explain how it works.

[CBSE Board, Delhi Region, 2018]

Ans. Circuit diagram of full-wave rectifier :



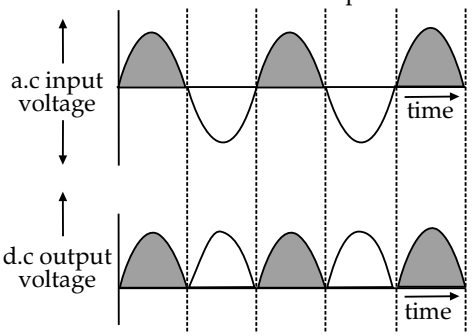
[½]

The circuit consists of two diodes as shown in the figure given above. It gives output-rectified voltage corresponding to both the positive as well as negative half of the AC cycle. Hence, it is known as full-wave rectifier.

Here the *p*-side of the two diodes is connected to the ends of the secondary of the transformer. The *n*-side of the diodes is connected together and the output is taken between this common point of diodes and the mid-point of the secondary of the transformer.

So for a full-wave rectifier, the secondary of the transformer is provided with a centre tapping and so it is called centre-tap transformer.

The voltage rectified by each diode is only half the total secondary voltage. Each diode rectifies only for half the cycle, but the two do so for alternate cycles. Thus, the output between their common terminals and the centre tap of the transformer becomes a full-wave rectifier output. **[1]**



[½]

Suppose the input voltage to A with respect to the centre tap at any instant is positive, at that instant, voltage at *B* being out of phase will be negative. So, diode D_1 gets forward biased and conducts (while D_2 being reversed biased is not conducting). Hence, during this positive half cycle we get an output current (and a output voltage across the load resistor R_L).

Q. 8. In half-wave rectification, what is the output frequency if the input frequency is 50 Hz. What is the output frequency of a full-wave rectifier for the same input frequency?

[NCERT Ex. Q. 8, Page 510]

Ans. Input frequency = 50 Hz

For a half-wave rectifier, the output frequency is equal to the input frequency.

$\therefore$ Output frequency = 50 Hz

For a full-wave rectifier, the output frequency is twice the input frequency.

$\therefore$ Output frequency $= 2 \times 50 = 100$Hz **[2]**

Q. 9. Two amplifiers are connected one after the other in series (cascaded). The first amplifier has a voltage gain of 10 and the second has a voltage gain of 20. If the input signal is 0.01 volt, calculate the output ac signal. [NCERT Ex. Q. 10, Page 512]

Ans. Voltage gain of the first amplifier, $V_1 = 10$

Voltage gain of the second amplifier, $V_2 = 20$

Input signal voltage, $V_i = 0.01$ V

Output AC signal voltage $= V_o$

The total voltage gain of a two-stage cascaded amplifier is given by the product of voltage gains of both the stages, *i.e.*,

$$V = V_1 \times V_2$$
$$= 10 \times 20$$
$$= 200$$

We have the relation:

$$V = \frac{V_0}{V_i}$$
$$V_0 = V \times V_i$$
$$= 200 \times 0.01$$
$$= 2\text{V}$$

Therefore, the output AC signal of the given amplifier is 2V. **[2]**

Short Answer Type Questions (3 or 4 marks each)

Q. 1. In a CE transistor amplifier there is a current and voltage gain associated with the circuit. In other words, there is a power gain. Considering power, a measure of energy, does the circuit violate conservation of energy? [NCERT Exemp. Q. 14.22, Page 91]

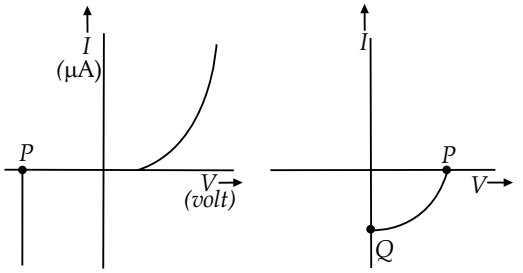
Ans. In CE transistor amplifier, the DC supply is connected to give energy to signal. Due to this fact, there is a large power gain in CE configuration amplifier. So, energy of output signal is equal to the sum of energy of signal (low) and energy supplied by DC source in output CE circuit (V_{CE}). **[1]**

Q. 2. Answer the following :

(i) Name the type of a diode whose characteristics are shown in Figure (a) and Figure (b).

(ii) What does the point *P* in Figure (a) represent?

(iii) What do the points *P* and *Q* in Figure (b) represent?



(a) **(b)**

[NCERT Exemp. Q. 14.23, Page 92]

Ans. **(i)** Diode in figure (a) is Zener diode and figure (b) shows solar cell. **[1]**

(ii) Zener break-down voltage is represented in figure (a). **[1]**

(iii) In figure (b), *Q* represents zero voltage and negative current. It means that light energy falling on solar cell with at least minimum threshold frequency which gives the current in opposite direction that due to a battery connected to the solar cell. But for point *P* the battery is short circuited and hence represents the short-circuit current.

In figure (b), point *P* represents some positive voltage on solar cell with zero current through solar cell. So there is battery connected to a solar cell which gives rise to the equal and opposite current to that in solar cell by virtue of light falling on it.

As current is zero for point *P*, hence we say *P* represents open circuit voltage. **[2]**

Q. 3. Three photo diodes D_1, D_2 and D_3 are made of semiconductors having band gaps of 2.5 eV, 2 eV and 3 eV, respectively. Which ones will be able to detect light of wavelength 6,000 Å? [NCERT Exemp. Q. 14.24, Page 92]

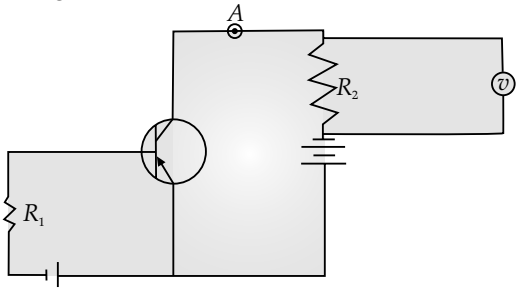
Ans. $\lambda = 6{,}000\ \text{Å} = 6{,}000 \times 10^{-10}$ m

$$E = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{6{,}000 \times 10^{-10} \times 1.6 \times 10^{-19}}\ eV$$
$$= \frac{6.62 \times 3 \times 10^{-34+8+10+19}}{6 \times 1.6 \times 10^{3}}\ eV = \frac{3.31 \times 10^{-34+37}}{1.60 \times 10^{3}}\ eV$$
$$= \frac{331}{160} \times \frac{10^{3}}{10^{3}}\ eV = 2.06\ eV$$

[2]

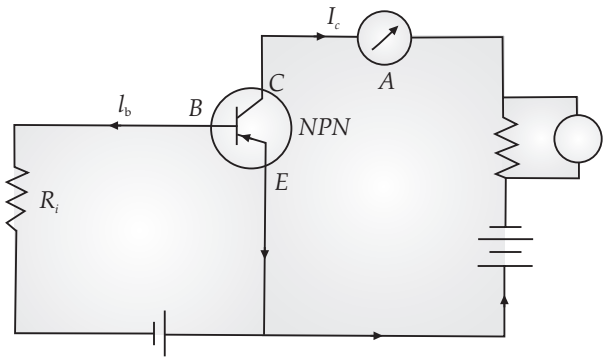
The incident radiation is detected by photodiode D_2 has band gap or knee voltage 2 eV only which is less than incident radiation of 2.06 eV. **[1]**

Q. 4. If the resistance R_1 is increased in Figure, how will the readings of the ammeter and voltmeter change?



[NCERT Exemp. Q. 14.25, Page 92]

Ans. Consider the figure given below to find out the change in reading of output *E*-*C* circuit



[1]

Current equation is given as of the given circuit,

$I_B = \frac{V_{BB} - V_{BE}}{R_i}$ **[1]**

As R_i increases the PD across R_i also increases so potential across BE decreases so the I_c will also decrease in turn reading of ammeter and voltmeter will decrease because $I_c = \beta_{Ib.}$ **[1]**

Q. 5. Explain why elemental semiconductor cannot be used to make visible LEDs

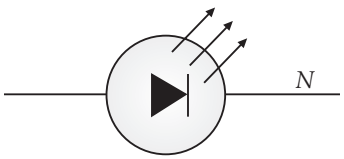
[NCERT Exemp. Q. 14.28, Page 92]

Ans. Specially designed diodes, which give out light radiations when forward biases, LED's are made of GaAsP, GaP, etc. These are forward-biased *p-n* junctions which emits spontaneous radiation. In elemental conductor, the band gap is such that the emissions are in infrared region and not in visible region **[1]**

$$\lambda = \frac{hc}{E_g} = \frac{1,242 \text{ eV}}{E_g} \text{nm}$$

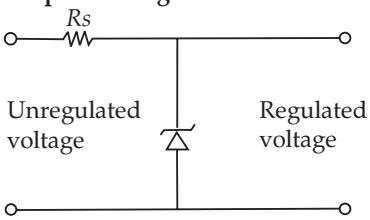
For Si; $E_g = 1.1$ eV, $\lambda = \frac{1,242}{1.1} = 1,129$ nm

For Ge; $E_g = 0.7$ eV, $\lambda = \frac{1,242}{0.7} = 1,725$ nm **[1]**



[1]

Q. 6. A Zener of power rating 1 W is to be used as a voltage regulator. If Zener has a breakdown of 5 V and it has to regulate voltage which fluctuated between 3 V and 7 V, what should be the value of R_s for safe operation figure?



[NCERT Exemp. Q. 14.30, Page 93]

Ans. Given that,

$P = 1$ Watt

Zener breakdown voltage=5 V

Minimum voltage V_{min}=3 V

Maximum voltage V_{max}=7 V

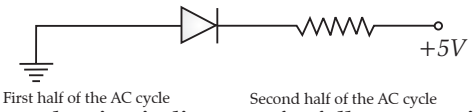
$\text{Current} = I_{max} = \frac{P}{V_z}$ **[1]**

$I_{Z\max} = \frac{1}{5} = 0.2$ A **[1]**

$R_s = \frac{V_{max} - V_{min}}{I_{Z\max}} = \frac{7-3}{0.2} = \frac{4}{0.2} = 20\ \Omega$ **[1]**

Q. 7. Answer these questions

(a) In the following diagram, is the junction diode forward biased or reverse biased?

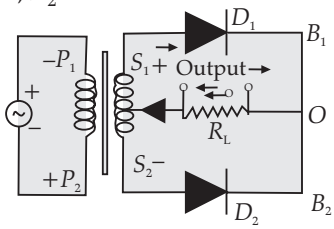


(b) Draw the circuit diagram of a full wave rectifier and state how it works.

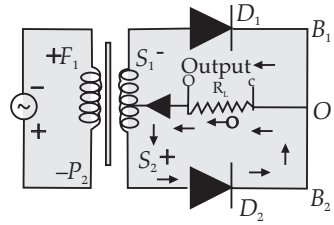
[CBSE Board, All India Region, 2017]

Ans. **(a)** The above diagram is the junction diode reverse biased. **[1]**

(b) This circuit uses two diodes and a centre-tap transformer whose secondary coil is divided into two equal parts. The voltages at any instant at S_1 and S_2 are out of phase with respect to each other. If the voltage at any instant at S_1 is positive, then diode D_1 conducts and because the voltage at S_2 is negative, D_2 does not conduct.

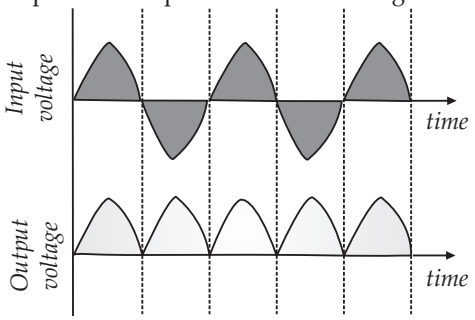


First half of the AC cycle



Second half of the AC cycle **[1]**

Input and Output Waveform of the given circuit



[1]

Working Principle of Full Wave Rectifier

1. During this positive half cycle, we get an output current, and hence, voltage across load resistance.
2. If the voltage at S_1 becomes negative, only D_2 conducts, and we get an output in the negative

half-cycle as well, which is however identical to the output in the positive half cycle. Thus, we find that during both halves, current flows in the same direction. **[1]**

Q. 8. Answer the following :

(a) Derive an expression for drift velocity of free electrons.

(b) How does a drift velocity of electrons in a metallic conductor vary with increase in temperature? Explain. [CBSE Board, All India Region, 2016]

Ans. **(a)** Expression for drift velocity : When conductor is subjected to an electric field E, each electron experiences a force.

$$F = \frac{-e}{E}$$

and acquires an accleration

$$a = \frac{F}{m} = \frac{-eE}{m} \quad \text{...(i)} \quad \textbf{[1]}$$

Here m = mass of election, e = charge, E = electric field.

The average time difference between two consecutive collisions is known as relaxation time of election.

$$F_e = -eE$$

$$\Rightarrow F_e = \frac{-eV}{I}$$

We know,

$$a = \frac{-F}{m} = \frac{-eV}{m}$$

Also,

$$v = u + at$$

Here,

$u = 0$;

Relaxation time, $t = T$

Therefore,

$$v_d = -aT$$

$$v_d = \frac{-eV}{m}T \quad \textbf{[2]}$$

(b) According to drift velocity expression, relaxation time is the time interval between successive collisions of an electron on increasing temperature, the electrons move faster and more collisions occur quickly. Hence, relaxation time decreases with increase in temperature which implies that drift velocity also decreases with temperature. **[1]**

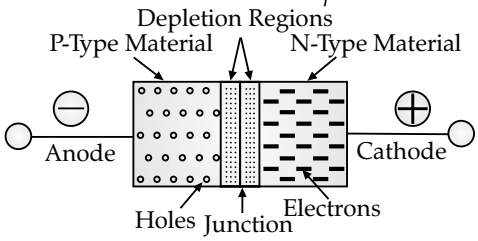
Q. 9. Answer the following

(a) Explain with the help of a diagram the formation of depletion region and barrier potential in a *p-n* junction.

(b) Draw the circuit diagram of a half wave rectifier and explain its working.

[CBSE Board, All India Region, 2016]

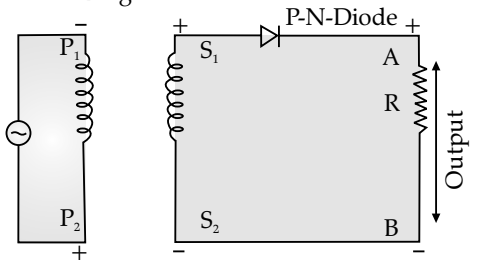
Ans. **(a)** In a *p-n* junction, a *p*-type and an *n*-type material are joined together. *p*-type has a higher concentration of holes and *n*-type has a higher concentration of electrons. Hence, there is a concentration gradient between the *p*-type and *n*-type materials. Therefore, holes drift from *p*-side to *n*-side by the process of diffusion due to concentration gradient. In a similar manner, electrons move from *n*-side to *p*-side. **[1]**



N-Type Material **[½]**

As a result of the movement of holes from *p*-side, they leave ionised spaces (negative charge) on *p*-side near the junction. Similarly, a positive space charge region is formed on the *n*-side due to the diffusion of electrons from inside. The two space charge regions on either side of the junction is called as depletion region. A potential difference is developed across the junction of two regions, which is known as the barrier potential. The reverse polarity of this potential opposes further low of carriers. [1]

(b) Circuit Diagram of Half-Wave Rectifier–



[½]

Working : When an alternating voltage is applied across a diode in series with a load, a pulsating voltage will appear across the load only during that half cycle on the AC input during which the diode is forward biased.

In the positive half cycle of AC input, a current flows through load resistance R_L and we get the output voltage. In the negative half cycle, there is no output. Therefore, the output voltage is restricted to only one direction and is said to be rectified. **[1]**

Q. 10. For a *C-E* Transistor Amplifier, the audio signal voltage across the collector resistance of 2 kΩ is 2 V. Suppose the current amplification factor of the transistor is 100, find the input signal voltage and base current, if the base resistance is 1 kΩ.

[NCERT Ex. Q. 14.9, Page 510; CBSE Board, All India Region, 2016]

Ans. Given that :

$R_C = 2\ \text{k}\Omega,\ V_C = 2\ \text{V}, \beta = 100$

$R_B = 1\ \text{k}\Omega$

Output current, $I_C = \frac{V_C}{R_C} = \frac{2}{200} = \frac{1}{100} A = 1\ \text{mA}$ **[1]**

Current gain, $\beta = \frac{I_C}{I_B}$ **[½]**

$\Rightarrow$ Thus, base current

$$I_B = \frac{I_C}{\beta} = \frac{10^{-3}}{100} = 10\ \mu\text{A} \quad \textbf{[½]}$$

Input signal voltage,

$V_B = I_B \times R_B = 10^{-5} \times 1{,}000$ V

$= 10$ mV **[1]**

Q. 11. Answer the following :

(a) Write the functions of three segments of a transistor.

(b) Draw a circuit diagram for studying the input and output characteristics of a *n-p-n* transistor in common emitter configuration. Using the circuit, explain how input, output characteristics are obtained. [CBSE Board, Delhi Region, 2016]

Ans. **(a)** The emitter emits the majority charge carriers. The collector collects these majority charge carriers. The interaction between the collector and the emitter is provided by the base. **[1]**

(b) Circuit Diagram of *n-p-n* transistor :

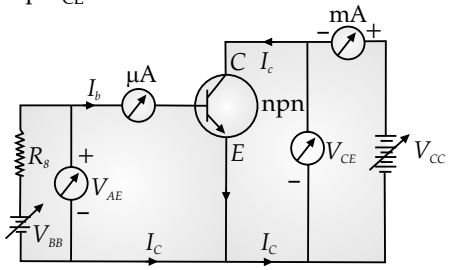
For input characteristics,

Study the dependence of I_B on V_{BE}

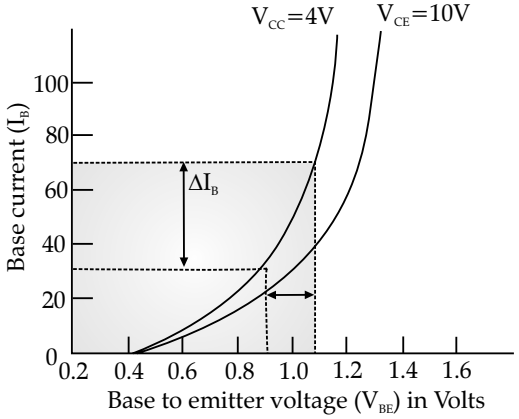
For output characteristics, keep I_B as constant

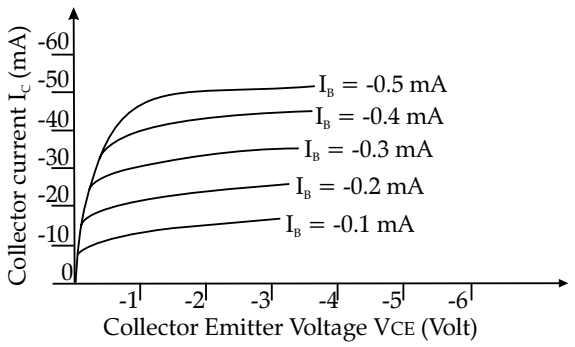
Study the dependence of I_C or V_{CE}

Keep V_{CE} as fixed value



[½]





[1]

Using the circuit, input characteristics are represented by a graph showing variation between the base current and the base-emitter voltage. This characteristic or graph is further used to evaluate input dynamic resistance of the transistor as the slope of the graph. **[½]**

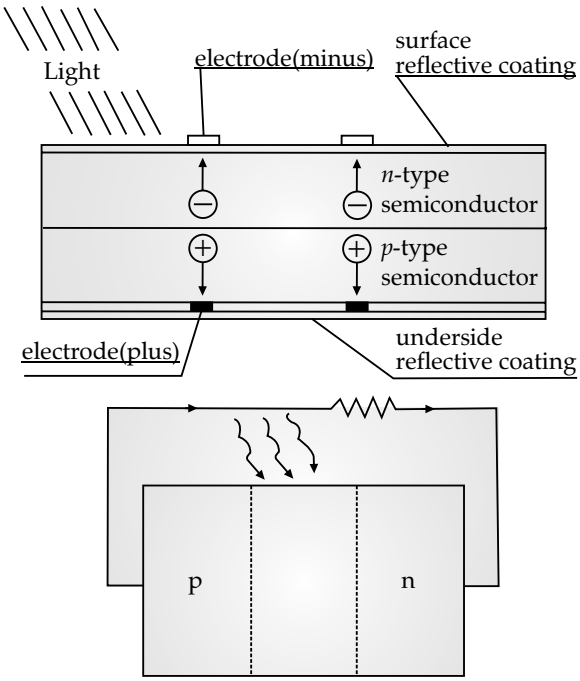
Similarly, the same circuit is used to find the output characteristics of the circuit using the graph representing the variation of the collector current and the emitter voltage, keeping base current fixed. Here, the output dynamic resistance is evaluated as the slope of the output characteristics graph. **[1]**

Q. 12. Answer the following :

(a) Describe the working principle of a solar cell. Mention three basic processes involved in the generation of emf.

(b) Why are *Si* and GaAs preferred materials for solar cells. [CBSE Board, Foreign Scheme, 2016]

Ans. **(a) Working Principle :** When sunlight hits the semiconductor, an electron springs up and is attracted toward the *n*-type semiconductor. This causes more negatives in the *n*-type semiconductors and more positives in the *p*-type, thus generating a higher flow of electricity. This is the photovoltaic effect.



[1]

Three basic processes involved in the generation of emf.

(i) Generation of *e-h* pairs due to light (with $h\nu > E_g$) close to the junction;

(ii) Separation of electrons and holes due to electric field of the depletion region. Electrons are swept to *n*-side and holes to *p*-side;

(iii) The electrons reaching the *n*-side are collected by the front contact and holes reaching *p*-side are collected by the back contact. Thus *p*-side becomes positive and *n*-side becomes negative giving rise to photo voltage. **[1]**

(b) The reasons may be :

(i) These materials are preferred because of their lower value of energy gap, so that they can be activated using solar radiation.

(ii) Their *V-I* characteristics are similar to a semiconductor. Solar cell means when exposed to light, can generate and support an electric current without being attached to any external voltage source and GaAs and *Si* are the semiconductors and preferred so. **[1]**

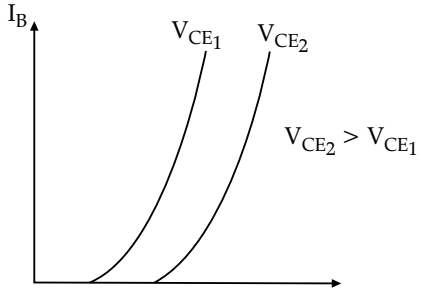
Q. 13. Draw the typical input and output characteristics of an *n-p-n* transistor in CE configuration. Show how these characteristics can be used to determine

(a) the input resistance (r_i), and

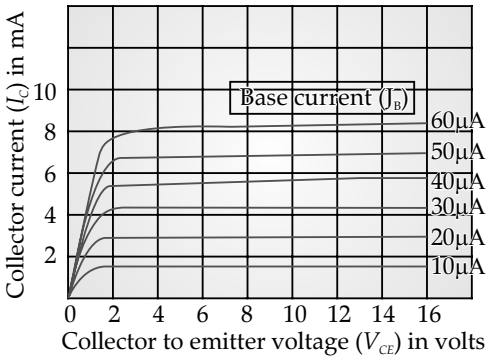
(b) current amplification factor (β)

[CBSE Board, Delhi Region, 2018]

Ans. Input Characteristics and output Characteristics of *n-p-n* CE configuration :



Input Characteristics for CE Configuration



[1]

(a) Input resistance (r_i) : This is defined as the ratio of change in base emitter voltage (ΔV_{BE}) to the resulting change in base current (ΔI_B) at constant collector-emitter voltage (V_{CE}). This is dynamic (ac resistance) and as can be seen from the input characteristic, its value varies with the operating current in the transistor

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}}$$

The value of R_i can be anything from a few hundreds to a few thousand ohms. **[½]**

(b) This is defined as the ratio of the change in collector current to the change in base current at a constant collector-emitter voltage (V_{CE}) when the transistor is in active state.

$$\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}}$$

This is also known as small signal current gain and its value is very large. **[½]**

If we simply find the ratio of I_C and I_B we get what is called DC β of the transistor. Hence,

$$\beta_{dc} = \frac{I_C}{I_B}$$

Since I_C increases with I_B almost linearly and $I_C = 0$ when $I_B = 0$, the values of both β_{dc} and β_{ac} are nearly equal. So, for most calculations β_{dc} can be used. Both β_{ac} and β_{dc} vary with V_{CE} and I_B (or I_C) slightly. **[1]**

Q. 14. The number of silicon atoms per m^3 is 5×10^{28}. This is doped simultaneously with 5×10^{22} atoms/m^3 of Arsenic and 5×10^{20} per m^3 atoms of Indium. Calculate the number of electrons and holes. Given that $n_i = 1.5 \times 10^{16}$ m^{-3}. Is the material *n*-type or *p*-type?

[NCERT Add Ex. Q. 12, Page 510]

Ans. Number of silicon atoms, $N = 5\times10^{28}$

Number of arsenic atoms, $n_A = 5\times10^{22}$ atoms / m^3

Number of indium atoms, $n_{In} = 5\times10^{20}$ atoms / m^3

Number of thermally-generated electrons,

$$n_i = 1.5\times10^{16} \text{ electrons / m}^{-3}$$

Number of electrons, $n_e = 5\times10^{22} - 1.5\times10^{16}$

$$= 4.99\times10^{22}$$

Number of holes = n_h

In thermal equilibrium, the concentrations of electrons and holes in a semiconductor are related as:

$$n_e n_h = n_i^2$$

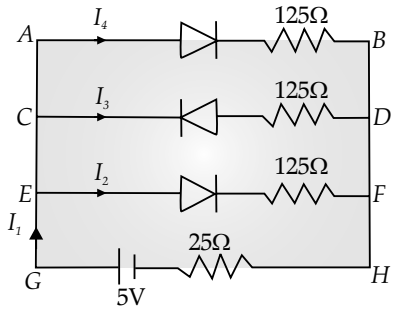
$$\therefore n_h = \frac{n_i^2}{n_e} = \frac{(1.5\times10^{16})^2}{4.99\times10^{22}}$$

$$n_h = 4.50\times10^9 \approx 4.51\times10^9$$

Therefore, the number of electrons is approximately 4.99×10^{22} m^{-3} and the number of holes are about 4.51×10^9 m^{-3}. Since the number of electrons is more than the number of holes, the material is an n-type semiconductor. **[3]**

Long Answer Type Questions **(5 marks each)**

Q. 1. If each diode in Figure has a forward bias resistance of 125 Ω and infinite resistance in reverse bias, what will be the values of the current I_1, I_2, I_3 and I_4?



[NCERT Exemp. Q. 14.31, Page 93]

Ans. Diode D_1 and D_3 are in forward bias and D_2 is in reverse bias so resistance in arm AB and EF = 125 + 25 = 150 Ω and Resistance in CD is infinite. **[½]**

$$\frac{1}{R_{24}} = \frac{1}{150} + \frac{1}{150} = \frac{2}{150} = \frac{1}{75}$$

$\therefore R_{24} = 75\ \Omega$ **[½]**

Total resistance in circuit = $R = 75 + 25 = 100\ \Omega$

So $V = IR$

$50 = I_1 100$ **[1]**

$$I_1 = \frac{5}{100} = 0.05 \text{ Amp}$$

$I_1 = I_2 + I_3 + I_4$ **[½]**

$\therefore$ Resistance R of CD $= \infty$

So $I_3 = 0$

So $I_1 = I_2 + I_4$

Here resistances are equal.

$\therefore I_2 = I_4$

Or $I_1 = 2I_2$

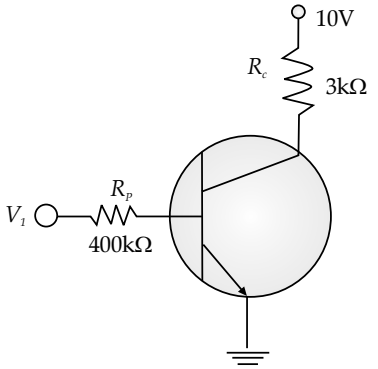
$\therefore\ I_2 = \frac{I_1}{2}$

$$\frac{0.05}{2} = I_2$$

$I_2 = 0.025$

$I_1 = 0.05$ Amp, $I_2 = 0.025$ Amp, $I_4 = 0.025$ Amp and $I_3 = 0$ Amp. **[2½]**

Q. 2. In the circuit shown in Figure, when the input voltage of the base resistance is 10 V, V_{be} is zero and V_{ce} is also zero. Find the values of I_b, I_c and β.



[NCERT Exemp. Q. 14.32, Page 93]

Ans. Voltage across $R_B = 10$ V

$$V_B = R_B I_B$$

$$I_B = \frac{V_B(\text{Voltage across } R_B)}{R_B}$$

$$= \frac{10}{400 \times 1{,}000} \text{A} = 2.5 \times 10^{-5} \text{A} = 25 \times 10^{-6} \text{A}$$

$$I_C = \frac{V_C(\text{Votage across } R_C)}{R_C} = \frac{10}{3 \times 1{,}000} = 3.3 \times 10^{-3} \text{A}$$

$$\text{Current gain } (\beta) = \frac{I_C}{I_B} = \frac{3.3 \times 10^{-3}}{25 \times 10^{-6}} = 133$$ **[5]**

Q. 3. Consider the circuit arrangement shown in Figure (a) for studying input and output characteristics of *n-p-n* transistor in *CE* configuration.

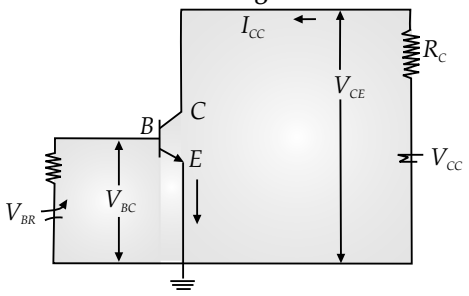


Figure (a)

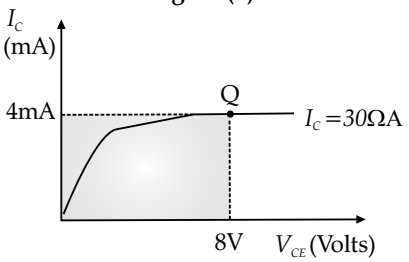


Figure (b)

Select the values of R_B and R_C for a transistor whose V_{BE} = 0.7 V, so that the transistor is operating at point Q as shown in the characteristics shown in Figure (b). Given that the input impedance of the transistor is very small and $V_{CC} = V_{BB}$ = 16 V, also find the voltage gain and power gain of circuit making appropriate assumptions.

[NCERT Exemp. Q. 14.34, Page 94]

Ans. For output characteristic at point Q

V_{CE} = 8 V, I_B = 30 μA, I_C = 4 mA, V_{BE} = 0.7 V

Applying Kirchhoff's law in collector-emitter loop

$$V_{CC} = V_{CE} + R_C I_C$$

$$R_C = \frac{V_{CC} - V_{CE}}{I_C} = \frac{16 - 8}{4 \times 10^{-3}} = \frac{8}{4} \times 10^3 = 2{,}000\ \Omega$$ **[½]**

$R_C = 2{,}000\ \Omega$ **[½]**

Now applying Kirchhoff's loop law in base-emitter circuit,

$$V_{BB} = I_B R_B + V_{BE}$$ **[½]**

$$R_B = \frac{V_{BB} - V_{BE}}{I_B} = \frac{16 - 0.7}{30 \times 10^{-6}} = \frac{15.6 \times 10^{-6}}{30} = 530\ k\Omega$$ **[½]**

$R_C = 2{,}000\ \Omega$

Voltage gain $= A_V = -\beta \frac{R_C}{R_B}$ and $\beta = \frac{I_C}{I_B}$ **[½]**

$$\beta = \frac{4 \times 10^{-3}}{30 \times 10^{-6}} = \frac{40}{30} \times 10^3 = \frac{4{,}000}{30} = \frac{400}{3}$$ **[½]**

Average Voltage A_V

$$A_V = -\beta \frac{R_C}{R_B}$$ [½]

Negative (–) sign shows change in phase angle of output is $180°$ by input voltage.

$$A_V = \frac{+400}{3} \times \frac{2000}{530.000} = \frac{80}{15.6} = 0.52$$ [½]

Power gain = VI

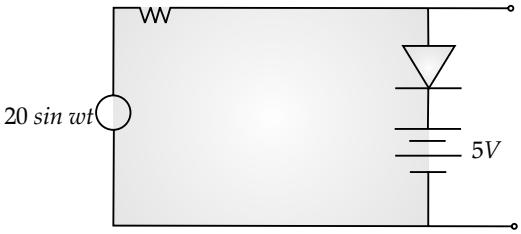
$$\text{Power gain} = \beta \times A_V = 0.52 \times \frac{400}{3}$$ [½]

$$\text{Power gain} = \frac{208}{3} = 69.33$$

or

$$P_{\text{gain}} = 69.33$$ [½]

Q. 4. Assuming the ideal diode, draw the output waveform for the circuit given in Figure. Explain the waveform.



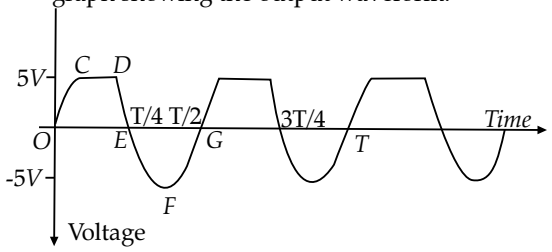
[NCERT Exemp. Q. 14.35, Page 94]

Ans. When signal 20 sin ωt gives input voltage less than 5 volt (because after 5 V diode will get positive voltage at its *p*-junction) then diode will be in reverse bias so resistance of diode remain infinity so input signal will not pass through diode and battery path so output across *A* and *B* will increase from 0 V to 5 V (Graph *OC*). **[1]**

Now when the input voltage 20 sin ωt increase beyond 5 V then path of diode 5 V battery will offer very low resistance, so the current passes through diode and battery and output (across *A* and *B*) remain 5 V (Graph *CD*). **[1]**

Now when the voltage decreases the diode will be in reverse bias and output will again fall from 5 V to 0 V as input changes (Graph *DE*). When input voltage becomes negative (there is opposite of 5 V battery in *p-n* junction input voltage becomes more than 5 V now) the diode is in reversed bias it will not conduct current through *CD*, and in output across *AB* will get same as input AC, *i.e.*, for negative cycle diode offer infinite resistance as compared to R in series graph *E*, *F*, *G*. **[1]**

Same repetition of input and output continues - graph showing the output waveform.



[2]

Q. 5. Suppose a 'n'-type wafer is created by doping Si crystal having 5×10^{28} atoms/m^3 with 1 ppm concentration of As. On the surface 200 ppm Boron is added to create '*p*' region in this wafer. Considering $n_i = 1.5 \times 10^{16}$ m^{-3},

(a) Calculate the densities of the charge carriers in the *n* and *p* regions.

(b) Comment which charge carriers would contribute largely for the reverse saturation Current when diode is reverse biased.

[NCERT Exemp. Q. 14.36, Page 94]

Ans. When As (pentavalent) is added to *Si* the *n*-type water is created. So the number of majority carriers in *n*-type wafer, **[½]**

$$N_e = (N_D \text{ and } D_{\text{Si}}) = \frac{1}{10^6} \times 5 \times 10^{28} = 5 \times 10^{22} \text{ per m}^3$$ **[½]**

For number of minority carriers n_h

$$n_e \cdot n_h = n_i^2$$

$$n_h = \frac{n_i^2}{n_e} = \frac{1.5 \times 10^{16} \times 1.5 \times 10^{16}}{5 \times 10^{22}}$$

We know that $n_i = 1.5 \times 10^{16}$ per m^3

$$n_h = 0.3 \times 1.5 \times 10^{32-22}$$ **[1]**

$$n_h = 0.45 \times 10^{10} \text{ per m}^3$$

When boron (trivalent) is implanted in *Si* crystal, *p*-type wafer is formed with number of holes,

$$n_h = (N_D \times n \text{ of Si})$$ **[½]**

$$= \frac{200}{10^6} \times 5 \times 10^{28} = 1000 \times 10^{28-6}$$

$$\therefore n_h = 1 \times 10^{25} \text{ per m}^3$$ **[½]**

Minority carrier in *p*-type water

$$n_e \cdot n_h = n_i^2$$ **[½]**

$$n_e = \frac{n_i^2}{n_h}$$

$$= \frac{1.5 \times 10^{16} \times 1.5 \times 10^{16}}{10^{25}}$$ **[½]**

$$= 2.25 \times 10^{32-25}$$

$$= 2.25 \times 10^7 \text{ per m}^3$$

When reversed bias is applied on *p-n* junction diode then the majority charge carrier moves towards depletion layer, *i.e.*, holes $n_h = 0.45 \times 10^{10}$ per m^3 from n side $n_e = 2.25 \times 10^7$ per m^3 from p side moves towards junction and makes the depletion layer thicker. **[1]**

Q. 6. Consider a box with three terminals on top of it as shown in Figure (a) :

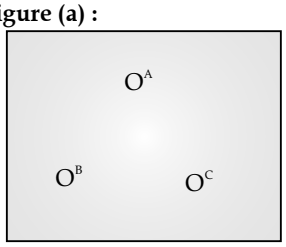


Figure (a)

Three components namely, two germanium diodes and one resistor are connected across these three terminals in some arrangement. A student performs an experiment in which any two of these three terminals are connected in the circuit shown in Figure (b).

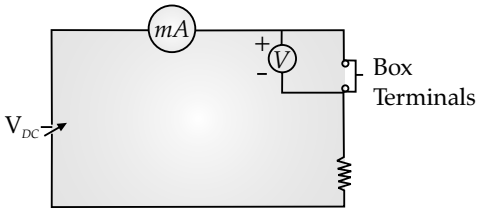


Figure (b)

The student obtains graphs of current-voltage characteristics for unknown combination of components between the two terminals connected in the circuit. The graphs are : From these graphs of current-voltage characteristic shown in Figures (c) to (*h*), determine the arrangement of components between *A*, *B* and *C*.

(c) When *A* is positive and *B* is negative.

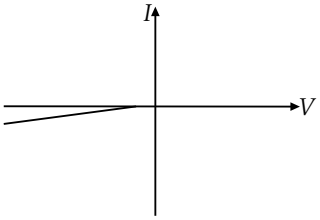


Figure (c)

(d) When *A* is negative and *B* is positive.

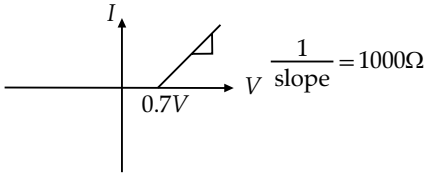


Figure (d)

(e) When *B* is negative and *C* is positive.

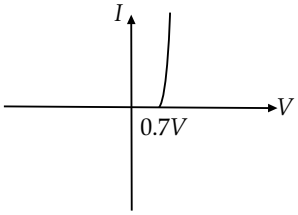


Figure (e)

(f) When *B* is positive and *C* is negative.

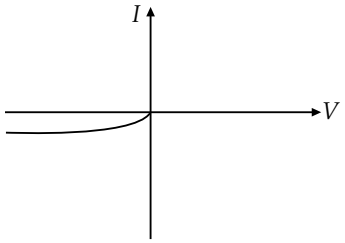


Figure (f)

(g) When *A* is positive and *C* is negative.

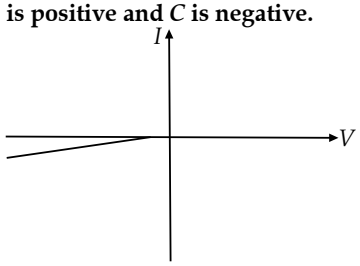


Figure (g)

(h) When *A* is negative and *C* is positive.

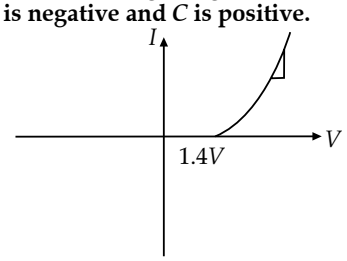
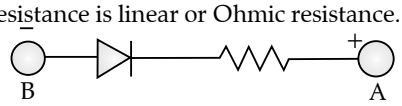


Figure (h)

[NCERT Exemp. Q. 14.38, Page 95]

Ans. **Case (a) :** Figure (c) shows the reverse bias characteristics of *p-n* junction. It is possible when *p-n* junction first is in a series combination with *p* side of diode towards *B* and *n* side towards a resistance. Its resistance is linear or Ohmic resistance. **[½]**



[½]

Case (b) : Figure (d) shows the forward bias characteristic of a transistor where 0.7 V is the knee voltage of a *p-n* junction first.

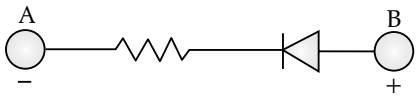
Given that, $\frac{1}{slope} = 1000\,\Omega$

$$\frac{1}{\text{Slope}} = \frac{1}{V/I}$$

$$1000 = \frac{1}{R}$$

$$\Rightarrow \quad R = \frac{1}{1000}\Omega$$

[½]

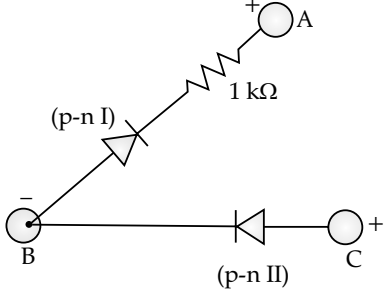


[½]

p side of *p-n* junction first is connected to *B* and *n* side to *A* terminal. Resistance is in series with diode.

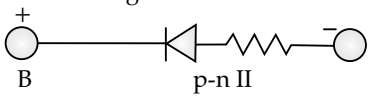
Case (c) : Figure (e) also shows the forward bias characteristics of *p-n* junction second having non ohmic as graph is not straight line. It also has knee voltage 0.7 V. **[½]**

Here *p* side *p-n* junction is connected to *C* and *n* side with *B* terminal as shown in figure.



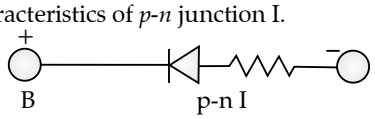
[½]

Case (d) : In figure (f) graph shows reverse bias characteristics of *p-n* junction II. *n* side with *B* and *p* side with *C* along with a resistance in series. **[½]**



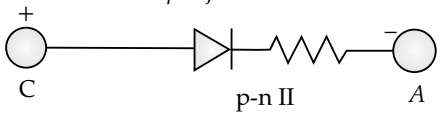
[½]

Case (e) : Figure (g) shows the reverse bias characteristics of *p-n* junction I.



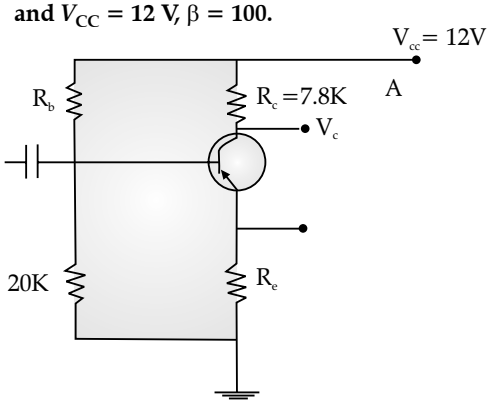
[½]

Case (f) : Figure (h) shows forward biased characteristics of *p-n* junction II.



[½]

Q. 7. For the transistor circuit shown in Figure, evaluate V_E, R_B, R_E given I_C = 1 mA, V_{CE} = 3 V, V_{BE} = 0.5 V and V_{CC} = 12 V, β = 100.



[NCERT Exemp. Q. 14.39, Page 97]

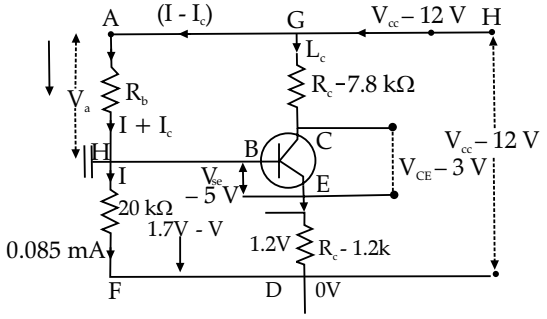
Ans. $I_C = I_B + I_E$

$\because I_B << I_C$

$\therefore I_C = I_E$

$I_C = 1$ mA (Given)

$\therefore I_E = I_C = 10^{-3}$ Amp. **[1]**



[1]

$\therefore I_C = I_E = 1$ mA

By Kirchhoff's loop law, in loop DGHJ

$I_C R_C + I_E R_E + V_{CE} = V_{CC}$

$I_C (R_C + R_E) + V_{CE} = 12$

$(R_C + R_E)1 \times 10^{-3} + 3 = 12$

$$R_C + R_E = \frac{9}{10^{-3}}$$

$7.8 \times 10^3 + R_E = 9000$

$R_E = 9000 - 7800 = 1200\ \Omega = 1.2\ k\Omega$

$V_E = I_E \times R_E = I_C \times R_E$

$= 10^{-3} \times 1200 = 1.2$ V

$V_{BD} = V_E + V_{BE} = 1.2 + 0.5$ V $= 1.7$ V

$V_{BE} = 0.5$ V and $V_{BE} = 1.2$ V

$$I = \frac{V_E}{20000} = \frac{1.7}{20000} A = 0.085 \text{ mA}$$

$$R_B = \frac{V_{BD}}{(I + I_B)} = \frac{V_{CC} - V_{BD}}{\left[1 + \left(\frac{I_C}{\beta}\right)\right]} \quad \left(\because I_B = \frac{I_C}{\beta}\right)$$

$$R_B = \frac{12 - 1.7}{\left[0.085 + \frac{1}{100}\right] \times 10^{-3}}$$

$$= \frac{10.3 \times 10^3}{(0.085 + 0.01)} = \frac{10.3 \times 10^3}{0.095}$$

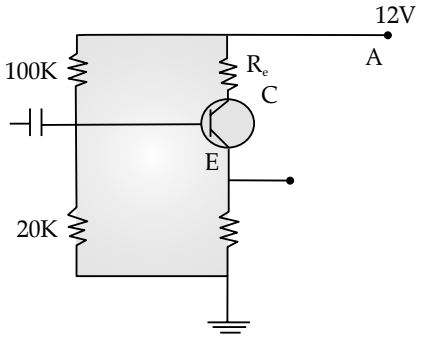
$= 108 \times 10^3\ \Omega = 108\ k\Omega$ **[3]**

Q. 8. In the circuit shown in Figure, find the value of R_C.

β = 100,

V_{BE} = 0.5 V

V_{CE} = 3 V

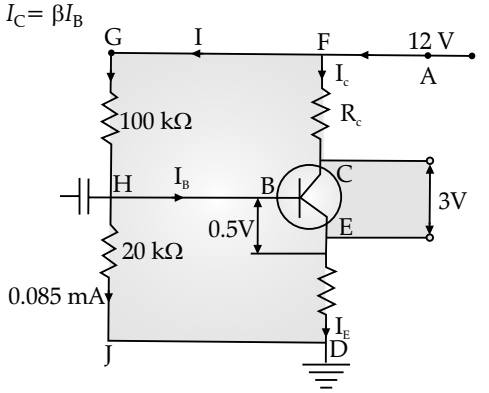


[NCERT Exemp. Q. 14.40, Page 97]

Ans. From figure applying Kirchhoff's junction rule at transistor

$I_E = I_C + I_B$

$I_C = \beta I_B$ **[1]**



[1]

Applying Kirchhoff's loop law to the loop AFCEDLA

$I_C R_C + V_{CE} + I_E R_E = V_{CC}$ …(i)

$RI_B + V_{BE} = I_E R_E = V_{CC}$ …(ii)

$\Rightarrow \quad I_E \approx I_C = \beta I_B$

$$(R+\beta R_E)I_B = V_{CC} - V_{BE}$$

$$\Rightarrow \quad I_B = \frac{V_{CC} - V_{BE}}{R + \beta R_E} = \frac{12 - 0.5}{80 + 1.2 \times 100} = \frac{11.5}{200} \text{ mA}$$

From equation (ii), we get

$$R_C + R_E = \frac{V_{CC} - V_{CE}}{I_C} = \frac{V_{CC} - V_{CE}}{\beta I_B} = \frac{200}{(11.5 \times 100)} \times (12 - 3)$$

$$= 1.56 \text{ k}\Omega$$

$$R_C = 1.56 - 1$$

$$= 0.56 \text{ k}\Omega$$ **[3]**

Q. 9. (a) Define the term drift velocity.

(b) On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend?

(c) Why alloys like constantan and manganin are used for making standard resistors?

[CBSE Board, Delhi Region, 2016]

Ans. **(a)** The average velocity of all the free electrons in the conductor with which they get drifted towards the positive end of the conductor under the influence of an external electric field is called the drift velocity. **[1]**

(b) $v_d = \frac{-eE\tau}{m}$

We have $E = -\frac{V}{l}$, where V is potential difference across the length 'ℓ' of the conductor

$$v_d = \frac{eV\tau}{m\ell}$$

Current flowing $I = neAv_d$

$$I = neAv_d \frac{eV\tau}{m\ell} = \frac{ne^2AV\tau}{m\ell}$$

$$\frac{I}{V} = \frac{ne^2A\tau}{m\ell} = \frac{1}{R} \quad ...(i)$$

Also, $R = \rho\frac{\ell}{A}$ ...(ii)

Comparing (i) and (ii)

$$\rho = \frac{m}{ne^2\tau}$$ **[2]**

Resistivity of the material of a conductor depends on the relaxation time, *i.e.*, temperature and the number of density of electrons. **[1]**

(c) Because constantan and manganin show very weak dependence of resistivity on temperature. **[1]**

Q. 10. A *p-n* photodiode is fabricated from a semiconductor with band gap of 2.8 eV. Can it detect a wavelength of 6000 nm?

[NCERT Ex. Q. 11, Page 510]

Ans. Energy band gap of the given photodiode, E_g = 2.8 eV

Wavelength, λ = 6000 nm = 6000×10^{-9} m

The energy of a signal is given by the relation:

$$E = \frac{hc}{\lambda}$$

where,

h = Planck's constant

$= 6.626 \times 10^{-34}$ Js

c = speed of light

$= 3 \times 10^8$ m/s

Now,

$$h = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{6000 \times 10^{-9}}$$

$$= 3.313 \times 10^{-20} \text{ J}$$

But 1.6×10^{-19} J = 1 eV

So, $E = 3.313 \times 10^{-20}$ J

$$E = \frac{3.313 \times 10^{-20}}{1.6 \times 10^{-19}} = 0.207 \text{ eV}$$

The energy of a signal of wavelength 6000 nm is 0.207 eV, which is less than 2.8 eV the energy band gap of a photodiode. Hence, the photodiode cannot detect the signal. **[5]**

Q. 11. In an intrinsic semiconductor the energy gap E_g is 1.2 eV. Its hole mobility is much smaller than electron mobility and independent of temperature. What is the ratio between conductivity at 600K and that at 300K? Assume that the temperature dependence of intrinsic carrier concentration ni is given by $n_i = n_o \exp\left(\frac{-E_g}{2k_BT}\right)$ where is a constant.

[NCERT Add. Ex. Q. 13, Page 510]

Ans. Energy gap of the given intrinsic semiconductor, E_g = 1.2 eV

The temperature dependence of the intrinsic carrier concentration is written as:

$$n_i = n_0 \exp\left(\frac{-E_g}{2k_BT}\right)$$

where k_B = Boltzman Constant = 8.62×10^{-5} eV/K

T = Temperature

n_0 = constant

Initial temperature, $T_1 = 300K$

The intrinsic carrier concentration at this temperature can be written as:

$$n_{i_1} = n_0 \exp\left[\frac{-E_g}{2k_B \times 300}\right] \quad ...(i)$$

Final temperature, T_2 = 600 K

The intrinsic carrier concentration at this temperature can be written as:

$$n_{i_2} = n_0 \exp\left[\frac{-E_g}{2k_B \times 600}\right] \quad ...(ii)$$

The ratio between the conductivities at 600 K and at 300 K is equal to the ratio between the respective intrinsic carrier concentration at these temperatures.

Therefore, the ratio between the conductivities.

$$\frac{n_{i_2}}{n_{i_1}} = \frac{n_0 \exp\left[\frac{-E_g}{2k_B \times 600}\right]}{n_0 \exp\left[\frac{-E_g}{2k_B \times 300}\right]}$$

$$= \exp\frac{-E_g}{2k_B}\left[\frac{1}{300} - \frac{1}{600}\right]$$

$$= \exp\left[\frac{1.2}{2\times 8.62\times 10^{-3}}\times\frac{2-1}{600}\right]$$

$$= \exp^{11.6}$$

$$\frac{n_{i_2}}{n_{i_1}} = 1.09\times 10^5 \quad \textbf{[5]}$$

Q. 12. In a *p-n* junction diode, the current *I* can be expressed as $I = I_0\exp\left(\frac{eV}{2k_BT}-1\right)$ where I_0 is called the reverse saturation current, *V* is the voltage across the diode and is positive for forward bias and negative for reverse bias, and *I* is the current through the diode, k_B is the Boltzmann constant (8.6×10^{-5} eV/K) and *T* is the absolute temperature. If for a given diode $I_0 = 5\times10^{-12}$ A and $T = 300$ K, then

(a) What will be the forward current at a forward voltage of 0.6 V?

(b) What will be the increase in the current if the voltage across the diode is increased to 0.7 V?

(c) What is the dynamic resistance?

(d) What will be the current if reverse bias voltage changes from 1 V to 2 V?

[NCERT Add. Ex. Q.14.13, Page 510]

Ans. Given that,

Reverse Saturation Current, $I_0 = 5\times10^{-12}$ A

Absolute temperature, $T = 300$ K

Boltzmann Constant, $k_B = 8.6\times10^{-5}$ eVK^{-1}

$= 8.6\times10^{-5}\times1.6\times10^{-19}$ JK^{-1}

(a) Forward Voltage, $V = 0.6$ V, then

$$\frac{eV}{k_BT} = \frac{1.6\times10^{-19}\times0.6}{8.6\times10^{-5}\times1.6\times10^{-19}\times300}$$

$$= 23.26$$

and

$$I = I_0\left[\exp\left(\frac{eV}{k_BT}-1\right)\right]$$

$$= 5\times10^{-12}\left[e^{23.26}-1\right]$$

$$= 5\times10^{-12}\left[1.259\times10^{10}-1\right]$$

$$= 5\times10^{-12}\times1.259\times10^{10}$$

$$= 0.063 \text{ A}$$

0.063 A is the required forward current.

(b) If the forward voltage is increased to $V = 0.7$ V, then

$$\frac{eV}{k_BT} = \frac{1.6\times10^{-19}\times0.7}{8.6\times10^{-5}\times1.6\times10^{-19}\times300} = 27.14$$

and

$$\text{Forward Current, } I = I_0\left[\exp\left(\frac{eV}{k_BT}-1\right)\right]$$

$$= 5\times10^{-12}\left[e^{27.14}-1\right]$$

$$= 5\times10^{-12}\left[6.07\times10^{11}-1\right]$$

$$= 5\times10^{-12}\times6.07\times10^{11}$$

$$= 3.035 \text{ A is the required forward current}$$

Thus, increase in current is given by,

$\Delta I = (3.035 - 0.063) = 2.972$ A

(c) As, $\Delta I = 2.972$ A, $\Delta V = 0.7 - 0.6 = 0.1$ V

$$\text{Therefore, Dynamic resistance} = \frac{\Delta V}{\Delta I} = \frac{0.1 \text{ V}}{2.972 \text{ A}}$$

$$= 0.0336\ \Omega$$

(d) For both the voltages, the current *I* will be almost equal to I_0. This implies that the circuit offers infinite dynamic resistance in the reverse bias. **[5]**

TOPIC-2
Transistors and Electronic Circuits

Quick Review

- A transistor is a solid-state electronic device used to control the flow of an electric current. The term 'solid state' refers to devices that take advantage of special properties of solids.
- Transistors were invented in the 1940s; they have come to revolutionise modern communications. They are found in an enormous variety of electrical devices, ranging from popular consumer items such as home computer games, pocket calculators and portable stereos to the complex electronic systems used by business and industry.
- Until World War II (1939–1945), most systems of communication used vacuum tubes for the amplification and control of electrical current. The demands of radar, in particular, during the war encouraged scientists to look for another method of amplifying and controlling electric current in communication devices.

TIPS...

- *Use Truth tables to explain – working of digital gates.*
- *Learn all the rules and identities of Digital logic of electronics to minimise the difficulty facing in Boolean expression minimisation.*
- *Learn the basic operations of Bi-polar Junction Transistor and Integrated circuits for solving Digital circuits working.*
- *To minimise the long Boolean expression, students have to divide the Boolean function into small functions and then merge them.*

- The discovery of the transistor was announced in 1948 by three scientists from the Bell Telephone Laboratories : William Shockley (1910–1989), John Bardeen (1908–1991) and Walter Brattain (1902–1987). The key to this discovery is a class of materials known as semiconductors.
- Integrated circuit (IC) : An electronic device that contains thousands or millions of microscopic-sized transistors etched on a single piece (chip) of material.
- The integrated circuits that operate only at a few defined levels instead of operating over all levels of signal amplitude are called as Digital ICs and these are designed by using multiple numbers of digital logic gates, multiplexers, flip flops and other electronic components of circuits.
- The integrated circuits that operate over a continuous range of signal are called as Analog ICs. These are subdivided into linear ICs and Radio-frequency Integrated Circuits (RF ICs).

TRICKS...

- *Always use 'Break the line and change the sign' in De-Morgan's Law.*
- *De Morgan's theorem allows large bars in a Boolean Expression to be broken up into smaller bars over individual variables.*
- *In order to reduce expressions with large bars, the bars must first be broken up*
- *Bubble pushing is a technique to apply De Morgan's theorem directly to the logic diagram.*
- *Change the logic gate (AND to OR and OR to AND).*
- *Add bubbles to the inputs and outputs where there were none, and remove the original bubbles.*

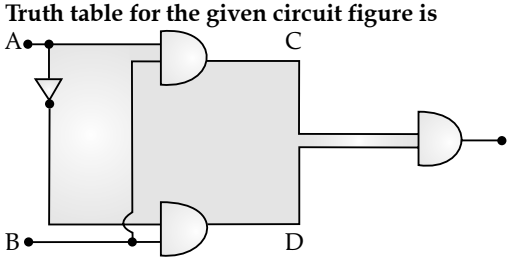
Know the Links

- *https://www.vidyakul.com/ncert-solutions-class-12-physics-chapter-14-semiconductor...*
- *www.dronstudy.com/book/chapter-notes-semiconductors-physics-class-12/*
- *www.examfear.com/notes/Class-12/Physics/Semiconductor.../1/introduction.htm*

Multiple Choice Question

(1 mark each)

Q. 1. Truth table for the given circuit figure is



(a)

A	B	E
0	0	1
0	1	0
1	0	1
1	1	0

(b)

A	B	E
0	0	1
0	1	0
1	0	0
1	1	1

(c)

A	B	E
0	0	0
0	1	1
1	0	0
1	1	1

(d)

A	B	E
0	0	0
0	1	1
1	0	1
1	1	0

[NCERT Exemp. Q. 14.8, Page 89]

Ans. Correct option : (c)

Explanation : From the given circuit diagram, we can see that this is AND Gate and their output will be multiple of its input. So the output of $C = AB$, and $D = \overline{A}B$. The Resultant output of E is equal to the AND multiplication of C and D Output as inputs of E.

We also know that NOT gate is also known as inverter that means if we apply '0' as a input then output will be '1' and vice versa.

Resultant Truth Table

A	B	$\overline{A}$	C	D	E
0	0	1	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	1	0	1	0	1

Tick Two or More Options

Q. 2. Which of these sets of logic gates are designated as universal gates?

(a) NAND **(b) NOR**

(c) EX-OR **(d) EX-NOR**

[OB Value Add.]

Ans. **Correct options :** (a) and (b)

Explanation : NAND gates are one of the two basic logic gates (the other being NOR logic) from which any other logic gates can be built. Due to this property, NAND and NOR gates are sometimes called "universal gates". However, modern integrated circuits are not constructed exclusively from a single type of gate.

Q. 3. How many entries will be in the truth table of a 3 input NAND gate and 2 input of NAND gate?

(a) 8 **(b) 6**

(c) 4 **(d) 10**

[OB Value Add.]

Ans. **Correct options :** (a) and (c)

Explanation : $Y = 2^n$

Where, Y is number of Entries in the truth table, while n is the number of inputs.

Q. 4. Which of the following is a digital device?

(a) Light switch **(b) Fan regulator**

(c) Traffic light **(d) Microphone**

[OB Value Add.]

Ans. **Correct options :** (a) and (c)

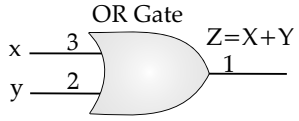
Explanation : A light switch and traffic signals have two conditions : ON and OFF, which is similar to that of digital conditions 1 and 0 which is the basic of Boolean algebra. Rest has Fuzzy logic.

Very Short Answer Type Question (1 or 2 marks each)

Q. 1. Draw logic symbol of an OR gate and write its truth table. [CBSE Board, Foreign Scheme, 2017]

Ans. OR Gate adds the input.

2 Input OR Gate



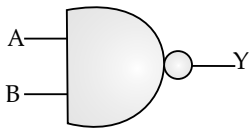
TRUTH TABLE

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

[1]

Q. 2. Give the truth table and circuit symbol for NAND gate. [CBSE Board, Delhi Region, 2018]

Ans. Symbol of logic gate : This circuit is known as NAND Gate because a NOT gate is added to the output of AND Gate. This logic inverts the result of AND Gate that means if AND Gate input gives the output is 1 then NAND will change the output is in form of 1.



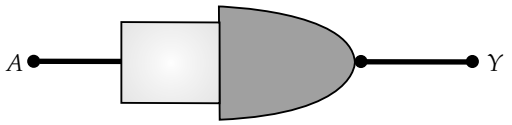
[½]

Truth Table :

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

[½]

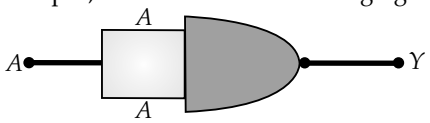
Q. 3. Write the truth table for a NAND gate connected as given in Figure.



Hence identify the exact logic operation carried out by this circuit.

[NCERT Add Ex. Q. 14.15, Page 511]

Ans. A acts as the two inputs of the NAND gate and Y is the output, as shown in the following figure.



NAND Gate **[½]**

Hence, the output can be written as :

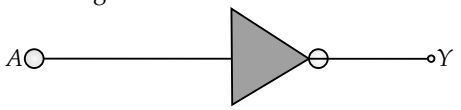
$Y = \overline{A \cdot A} = \overline{A} + \overline{A} = \overline{A}$ ...(i) **[½]**

The Truth Table for equation (i) can be drawn as :

A	Y (= $\overline{A}$)
0	1
1	0

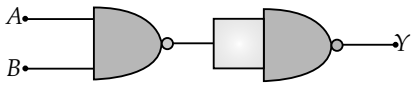
[½]

This circuit functions as a NOT gate. The symbol for this logic circuit is shown as :

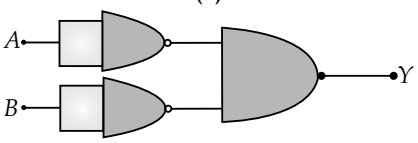


NOT Gate **[½]**

Q. 4. You are given two circuits as shown in Figure, which consist of NAND gates.



(a)

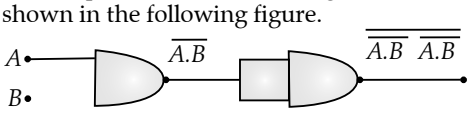


(a)

Identify the logic operation carried out by the two circuits. [NCERT Add Ex. Q. 16, Page 511]

Ans. In both the given circuits, A and B are the inputs and Y is the output.

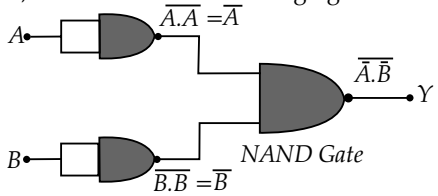
(a) The output of the left NAND gate will be $\overline{A.B}$, as shown in the following figure.



NAND Gate NAND Gate **[½]**

Hence, this circuit functions as an AND gate. **[½]**

(b) $\overline{A}$ is the output of the upper left of the NAND gate and $\overline{B}$ is the output of the lower half of the NAND gate, as shown in the following figure.



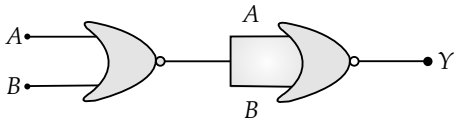
NAND Gate **[½]**

Hence, the output of the combination of the NAND gates will be given as :

$Y = \overline{A} \cdot \overline{B} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$

Hence, this circuit functions as an OR gate. **[½]**

Q. 5. Write the truth table for circuit given in Figure below consisting of NOR gates and identify the logic operation (OR, AND and NOT) which this circuit is performing.



(Hint : $A = 0, B = 1$ then A and B inputs of second NOR gate will be 0 and hence $Y = 1$. Similarly work out the values of Y for other combinations of A and B. Compare with the truth table of OR, AND, NOT gates and find the correct one.)

[NCERT Add Ex. Q. 17, Page 511]

Ans. A and B are the inputs of the given circuit. The output of the first NOR gate is $\overline{A+B}$. It can be observed from the following figure that the inputs of the second NOR gate become the output of the first one. **[½]**

Hence, the output of the combination is given as :

$\overline{\overline{A+B}+\overline{A+B}} = \overline{\overline{A}\cdot\overline{B}} + \overline{\overline{A}\cdot\overline{B}}$

$= \overline{\overline{A}\cdot\overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$ **[½]**

The Truth Table for this operation is given as :

A	B	$Y=(A+B)$
0	0	0
0	1	1
1	0	1
1	1	1

[½]

This is the truth table of an OR Gate. Hence, this circuit functions as an OR gate. **[½]**

Short Answer Type Questions

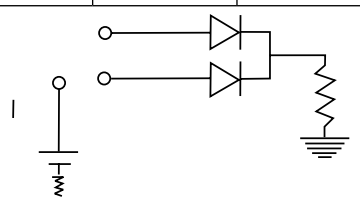
(3 or 4 marks each)

Q. 1. Two car garages have a common gate which needs to open automatically when a car enters either of the garages or cars enter both. Draw a circuit that resembles this situation, using diodes for this situation. [NCERT Exemp. Q. 14.26, Page 92]

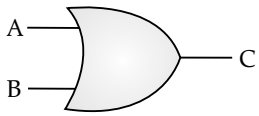
Ans. When a car enters the gate, any one or both are open. The device is shown in figure.

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

[1]



OR Gate



[2]

So, OR gate gives the desired output.

Q. 2. How would you set up a circuit to obtain NOT gate using a transistor?

[NCERT Exemp. Q. 14.27, Page 92]

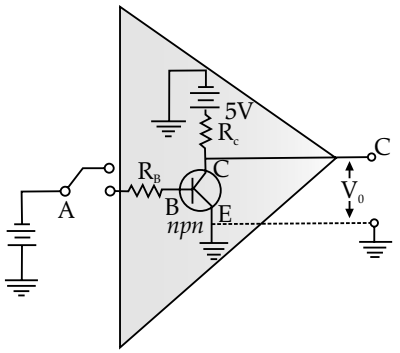
Ans. The NOT gate is a device which has only one input and one output, *i.e.*, $\overline{A} = Y$ means, Y equals to NOT A. Thus,

A	$Y = \bar{A}$
0	1
1	0

[1]

Here the base B of the transistor is connected to the input A through to the input A through the

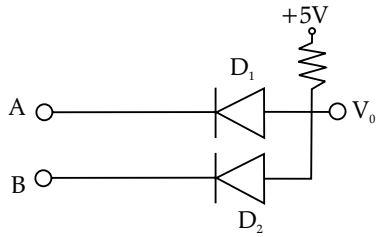
input resistance R_B and the emitter is earthed. The collector is connected to 5 V battery. The output Y is the voltage at C with respect to Earth. **[1]**



[½]

The resistance R_B and R_C are so chosen that if emitter-base junction is unbiased, the transistor is in cut-off region and if emitter-base junction is in forward bias by 5 V input at A, the transistor is in saturation state. **[½]**

Q. 3. Write the truth table for the circuit shown in Figure. Name the gate that the circuit resembles.



[NCERT Exemp. Q. 14.29, Page 93]

Ans. The circuit resembles AND gate. The Boolean expression of this circuit is $V_0 = A \cdot B$, *i.e.*, V_0 equals A and B. From the given circuit, we can see that when 5 V is applied, diode D_1 and diode D_2 will on and circuits behaves as short circuited and give output is '1' at A and B. and vice versa in case of 0 V. **[2]**

The Truth Table of this gate is as :

A	B	$V_0 = A.B$
0	0	0
0	1	0
1	0	0
1	1	1

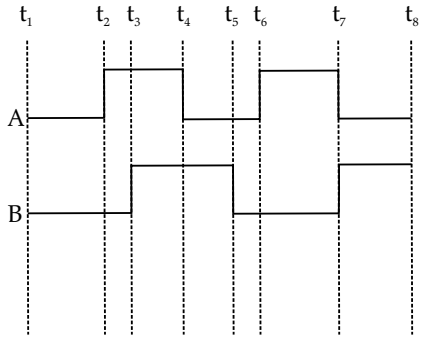
[1]

Q. 4. Answer the following

(a) Write the functions of the three segments of a transistor.

(b) The figure shows the input waveforms A and B for 'AND' gate.

Draw the output waveform and write the truth table for this logic gate.



[CBSE Board, All India Region, 2017]

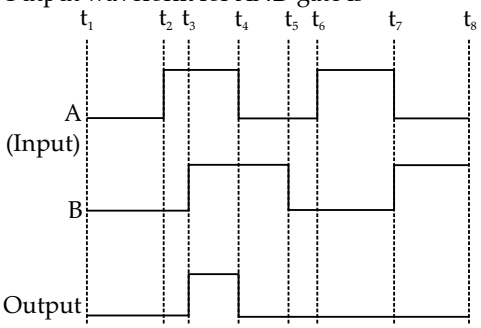
Ans. **(a)** Three segments of a transmitter are Emitter (E), Base (B) and Collector(C)

Emitter : It is of moderate size and heavily doped. It supplies a large number of majority carriers for the current flow through the transistor.

Base : It is the control segment and is very thin and lightly doped.

Collector : It is the segment, which collects a major portion of the majority carries, supplied by the emitter. It is moderately doped and large in size as compared to the emitter. **[1½]**

(b) Output waveform for AND gate is



[1½]

Truth Table for the same is

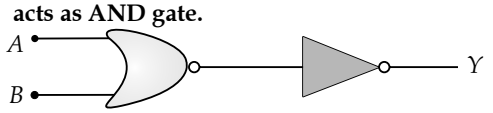
A	B	$Y = A.B$
0	0	0
0	1	0
1	0	0
1	1	1

[1]

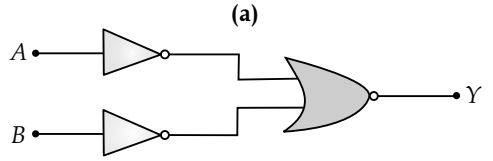
Q. 5. You are given the two circuits as shown in Figure. Show that circuit

(a) acts as OR gate while the circuit

(b) acts as AND gate.



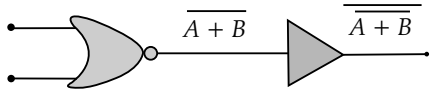
(a)



(b)

[NCERT Add Ex. Q. 14.14, Page 511]

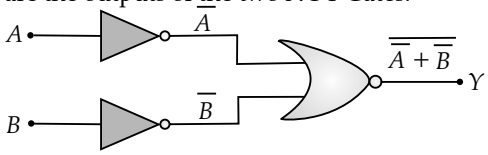
Ans. **(a)** A and B are the inputs and Y is the output of the given circuit. The left half of the given figure acts as the NOR Gate, while the right half acts as the NOT Gate. This is shown in the following figure. **[½]**



NOR Gates NOT Gate **[1]**

Hence, the output of the NOR Gate $= \overline{A+B}$
This will be the input for the NOT Gate. Its output will be $\overline{\overline{A+B}} = A+B$
$\therefore Y = A + B$
Hence, this circuit functions as an OR Gate. **[1]**

(b) A and B are the inputs and Y is the output of the given circuit. It can be observed from the following figure that the inputs of the right half NOR Gate are the outputs of the two NOT Gates.



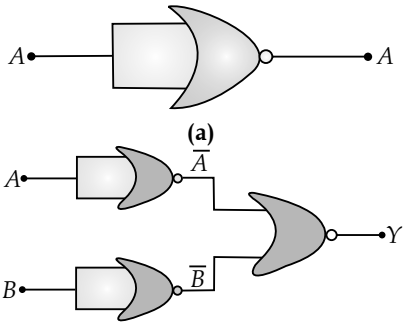
NOT Gates NOR Gate **[1]**

Hence, the output of the given circuit can be written as :

$Y = \overline{\overline{A}+\overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B$ **[1]**

Hence, this circuit functions as an AND Gate.

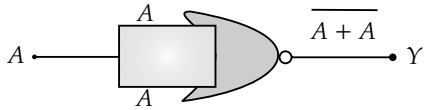
Q. 6. Write the truth table for the circuits given in Figure consisting of NOR gates only. Identify the logic operations (OR, AND, NOT) performed by the two circuits.



(a)

[NCERT Add Ex. Q. 14.18, Page 512]

Ans. **(a)** A acts as the two inputs of the NOR gate and Y is the output, as shown in the following figure. Hence, the output of the circuit is $\overline{A+A}$. **[½]**



NOR Gate **[1]**

Output, $Y = \overline{A+A} = \overline{A}$

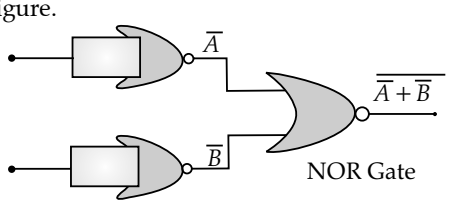
The Truth Table for the same is given as :

A	$Y = (\overline{A})$
0	1
1	0

[½]

This is the truth table of a NOT gate. Hence, this circuit functions as a NOT gate.

(b) A and B are the inputs and Y is the output of the given circuit. By using the result obtained in solution (a), we can infer that the outputs of the first two NOR gates are as shown in the following figure.



NOR Gate **[1]**

$\overline{A}$ and $\overline{B}$ are the inputs for the last NOR gate. Hence, the output for the circuit can be written as :

$Y = \overline{\overline{A}+\overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B$ **[½]**

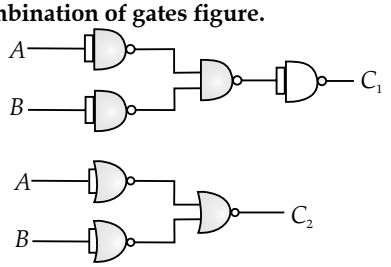
The Truth Table for the same can be written as :

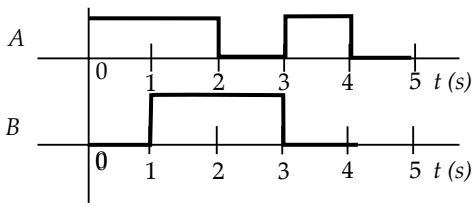
A	B	$Y = (A \cdot B)$
0	0	0
0	1	0
1	0	0
1	1	1

[½]

Long Answer Type Questions **(5 marks each)**

Q. 1. Draw the output signals C_1 and C_2 in the given combination of gates figure.





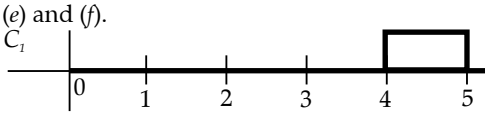
[NCERT Exemp. Q. 14.33, Page 93]

Ans. In circuit, Figure (b) $C_2 = \overline{\overline{A}+\overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B$ **[1]**

For the given signals shown in Figures (c) and (d), we have the following values for A, B, C_1 and C_2.

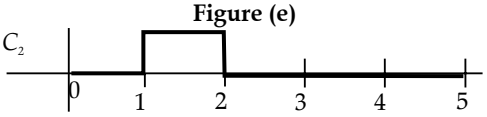
(i) For time 0–1 s, $A = 1, B = 0, C_1 = 0, C_2 = 0$.
(ii) For time 1–2 s, $A = 1, B = 1, C_1 = 0, C_2 = 1$
(iii) For time 2–3 s, $A = 0, B = 1, C_1 = 0, C_2 = 0$
(iv) For time 3–4 s, $A = 1, B = 0, C_1 = 0, C_2 = 0$
(v) For time 4–5 s, $A=0, B = 0, C_1 = 1, C_2 = 0$ **[2]**

The output signals C_1 and C_2 are shown in figures (*e*) and (*f*).



[1]

Figure (e)



[1]

Figure (f)

Q. 2. An X-OR gate has following Truth Table :

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

It is represented by following logic relation.

$Y = \overline{A}.B + A.\overline{B}$

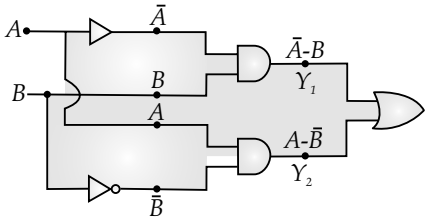
Build this gate using AND, OR and NOT gates.

[NCERT Exemp. Q. 14.37, Page 95]

Ans. In given logic relation, $Y = \overline{A}B + A\overline{B} = Y_1Y_2$ **[½]**

Y	$\overline{A}B$	$\overline{B}$	**A**	**B**	Y	$\overline{A}$	$A.\overline{B}$
0	0	1	0	0	0	1	0
1	0	0	0	1	1	1	1
1	1	1	1	0	1	0	0
0	0	0	1	1	0	0	0

[1]



[1]

Logic relation in given table is

$Y = \overline{A} \cdot B + A.\overline{B}$

$= Y_1 + Y_2$

$Y_1 = \overline{A} \cdot B$ and $Y_2 = A \cdot \overline{B}$

Y_1 can be obtained by $\overline{A}$ through NOT and B direct to AND gate.

So, $Y_1 = \overline{A} \cdot B$.

Y_2 can be obtained by $\overline{B}$ through NOT and A direct to AND gate.

So, $Y_2 = A \cdot \overline{B}$.

Now Y_1 and Y_2 are feed into the two terminals of OR gate to get

$Y = Y_1 + Y_2$ or $Y = \overline{A}B + A\overline{B}$ **[2½]**

Some Commonly Made Errors

- Students draw wrong circuit diagram of transistor configuration like common Emitter, Common Collector and Common Base. This makes solution wrong.
- Students always make wrong connection of elements in Circuit diagram analysis.
- Students get confused in making timing diagram of Digital Gates like AND Gate, OR Gate and NOT Gate.
- Students do not try to explain their answer by making suitable diagram.

EXPERT ADVICE

- ☞ *Learn how to use De Morgan's Theorem.*
- ☞ *To know the circuit analysis in solving transistor's problems, do not ignore basic operation of transistors like their current equations, voltage equation in different modes like cut-off region, saturation region and active region.*
- ☞ *Students have to study in deep levels about transistor operations.*
- ☞ *All the logic gates have to learn with Truth Table and mode of operation.*
- ☞ *Students have to deep study about the formation of NAND Gate, NOT Gate, Ex-NOR Gate and Ex-OR gate using Universal Gates.*
- ☞ *To study the timing diagram of logic gates, students have to study the logic of gates in the form of high level and low level.*
- ☞ *High logic is denoted by '1' and low logic is denoted by '0'. Digital circuits analysis totally depend on binary numbers.*

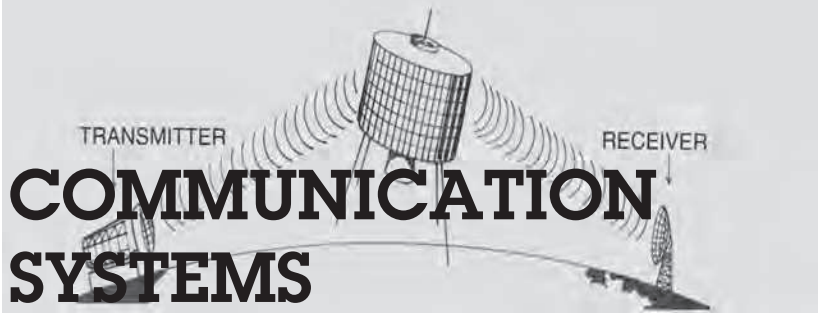
OSWAAL LEARNING TOOLS

***For* Suggested Online Videos**

https://youtu.be/jobn7AhrjCI **Or Scan the Code**	*https://youtu.be/w0-L-HIMa1Y* **Or Scan the Code**
https://youtu.be/ar7xDMR4P_U **Or Scan the Code**	*https://youtu.be/Xi18hI1LqAA* **Or Scan the Code**

CHAPTER

15 COMMUNICATION SYSTEMS



Chapter Objectives

This chapter will help you understand :

- ***Communication Systems :*** *Evolution in communication, communication system and modes of communication; Terminology, signal propagation terminology; Propagation of electromagnetic waves, Signals, noise and transmitter; Modulation and necessity, Band width and Amplitude modulation; Production of amplitude-modulated waves, Detection of amplitude-modulated waves, Demodulation and Repeater.*

Quick Review

- Electronic communication refers to the faithful transfer of information or message (available in the form of electrical voltage and current) from one point to another.
- An analog signal can be converted suitable into a digital signal and vice-versa.
- Every message signal occupies a range of frequencies. The bandwidth of a message signal refers to the band of frequencies, which are necessary for satisfactory transmission of the information contained in the signal.
- In modulation, some characteristic of the carrier signal like amplitude, frequency or phase varies in accordance with the modulating or message signal.
- In frequency modulation (m_f) frequency modulation index) is inversely proportional to modulating frequency f_m. while in *PM* it does not vary with modulating frequency. Moreover, *FM* is more noise immune.

TIPS...

- *Learn the concept of communication systems with the help of Flow chart.*
- *Practice Graphs and diagrams thoroughly as it help you to score good marks.*
- *Learn the importance of communication systems with the help of examples.*

TRICKS...

- *Make a flow chart for quick revision.*
- *Always make a flow chart to learn any classification.*
- *Also revise and practice a couple of step-by-step conversion questions from one compound to another.*

Know the Links

- *https://drive.google.com/file/d/0B8hXbvn1ab-BVWFNME9QQlU3bUk/view*
- *https://www.thinkiit.in/board-notes/english/class-12/physics/all/communication-systems/*
- *https://drive.google.com/file/d/0B8hXbvn1ab-BaEZrUkRCVTM4Tkk/view*
- *https://www.tiwariacademy.com/ncert-solutions-class-12-physics-pdf/*

Multiple Choice Questions

(1 mark each)

Q. 1. Three waves *A*, *B* and *C* of frequencies 1,600 kHz, 5 MHz and 60 MHz, respectively are to be transmitted from one place to another. Which of the following is the most appropriate mode of communication?

(a) *A* is transmitted via space wave while *B* and *C* are transmitted via sky wave.

(b) *A* is transmitted via ground wave, *B* via sky wave and *C* via space wave.

(c) *B* and *C* are transmitted via ground wave while *A* is transmitted via sky wave.

(d) *B* is transmitted via ground wave while *A* and *C* are transmitted via space wave.

[NCERT Exemp. Q. 15.1, Page 98]

Ans. Correct option : (b)

Explanation : The radio waves emitted from a transmitter antenna can reach the receiver antenna by the following mode of operation :

(i) Ground wave propagation
(ii) Sky wave propagation
(iii) Space wave propagation
(iv) Mode of communication frequency range :
(a) Ground wave propagation : 500–1,710 kHz
(b) Sky wave propagation : 2–40 MHz
(c) Space wave propagation : 54–42 MHz.
As, *A* is transmitted via ground wave, *B* via sky wave and *C* via sky wave.

Q. 2. A 100 m long antenna is mounted on a 500 m tall building. The complex can become a transmission tower for waves with λ

(a) ~400 m. **(b) ~25 m.**
(c) ~150 m. **(d) ~2,400 m.**

[NCERT Exemp. Q. 15.2, Page 98]

Ans. Correct option : (a)
Explanation : Length of the building (L) is = 500 m
And length of antenna = 100 m
And we know wavelength of the wave which can be transmitted by
$L = \frac{\lambda}{4}$, So, $\lambda \sim 4l = 4 \times 100 = 400$ m
Wavelength (λ) is nearly equal to 400 m.

Q. 3. A 1 kW signal is transmitted using a communication channel which provides attenuation at the rate of –2 dB per km. If the communication channel has a total length of 5 km, the power of the signal received is [Gain (in dB) = 10 log $\left(\frac{P_0}{P_1}\right)$]

(a) 900 W. **(b) 100 W**
(c) 990 W. **(d) 1010 W.**

[NCERT Exemp. Q. 15.3, Page 99]

Ans. Correct option : (b)
Explanation : Power of signal transmitted is P_i = 1 kW= 1,000 W

Rate of attenuation signal = $-2\ \frac{dB}{km}$

Length of total path = 5 km
Thus, loss suffered in the communication channel $= 5 \times (-2) = -10$ dB

Also, gain in $dB = 10 \log\left(\frac{P_i}{P_0}\right)$ (i)

Here P_0 is the power of the received signal.
Putting the given values in *Eq.* (i),

$$-10 = 10 \log\left(\frac{P_0}{P_i}\right) = -10 \log\left(\frac{P_i}{P_o}\right)$$

$$\Rightarrow \log\frac{P_i}{P_0} = r \Rightarrow \log\frac{P_i}{P_0} = \log 10$$

$$\Rightarrow \quad \frac{P_i}{P_0} = 10 \Rightarrow 1{,}000\ W = 10\ P_0$$

$$\Rightarrow \quad P_0 = 100\ W$$

Q. 4. A speech signal of 3 kHz is used to modulate a carrier signal of frequency 1 MHz, using amplitude modulation. The frequencies of the side bands will be

(a) 1.003 MHz and 0.997 MHz.
(b) 3,001 kHz and 2,997 kHz.
(c) 1,003 kHz and 1,000 kHz.
(d) 1 MHz and 0.997 MHz.

[NCERT Exemp. Q. 15.4, Page 99]

Ans. Correct option : (a)
Explanation : The process of changing the amplitude of a carrier wave in accordance with the amplitude of the audio frequency (AF) signal is known as amplitude modulation (AM).
In AM, frequency of the carrier wave remains unchanged.
Side band frequencies : The AM wave contains three frequencies f_C, $(f_C + f_m)$ and $(f_C - f_m)$, f_C is called carrier frequency, $(f_C + f_m)$ and $(f_C - f_m)$ are called side band frequencies.
$(f_C + f_m)$: Upper side band (USB) frequency
$(f_C - f_m)$: Lower side band (LSB) frequency
Side band frequencies are generally close to the carrier frequency.
According to the problem, frequency of the carrier signal is f_C = 1 MHz and frequency of speech signal = 3 kHz.
$= 3 \times 10^{-3}$ MHz = 0.003 MHz
We know that frequencies of side bands = $(f_C + f_m)$= (1 + 0.003) and (1 − 0.003)
So, side band frequencies are 1.003 MHz and 0.997 MHz.

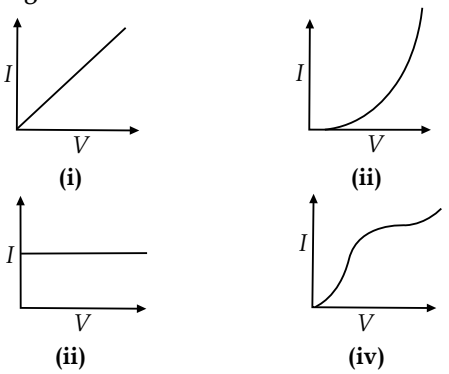
Q. 5. A message signal of frequency ω_m is superposed on a carrier wave of frequency ω_c to get an amplitude modulated wave (AM). The frequency of the AM wave will be

(a) ω_m **(b)** ω_c
(c) $\frac{\omega_c + \omega_m}{2}$ **(d)** $\frac{\omega_c - \omega_m}{2}$

[NCERT Exemp. Q. 15.5, Page 99]

Ans. Correct option : (b)
Explanation : The process of changing the amplitude of carrier wave in accordance with the amplitude of the audio frequency (AF) signal is known as amplitude modulation (AM).
In AM, frequency of the carrier wave remains unchanged or we can say that the frequency of modulated wave is equal to the frequency of carrier wave. Now, according to the problem, frequency of carrier wave is f_C.
Thus, the amplitude modulated wave also has frequency f_C.

Q. 6. I-V characteristics of four devices are shown in Figure.



Identify devices that can be used for modulation :

(a) 'i' and 'iii'.

(b) only 'iii'.

(c) 'ii' and some regions of 'iv'.

(d) All the devices can be used.

[NCERT Exemp. Q. 15.6, Page 99]

Ans. Correct option : (c)

Explanation : A square law modulator is the device which can produce modulated waves by the application of the message signal and the carrier wave.

Square law modulator is used for modulation purpose. Characteristics shown by (i) and (iii) corresponds to linear devices.

And by (ii) corresponds to square law device which shows non-linear relations. Some part of (iv) also follows square law.

Q. 7. A male voice after modulation-transmission sounds like that of a female to the receiver. The problem is due to

(a) poor selection of modulation index (selected $0 < m < 1$).

(b) poor bandwidth selection of amplifiers.

(c) poor selection of carrier frequency.

(d) loss of energy in transmission.

[NCERT Exemp. Q. 15.7, Page 100]

Ans. Correct option : (b)

Explanation : In this problem, the frequency of modulated signal received becomes more, due to improper selection of bandwidth.

This happens because bandwidth in amplitude modulation is equal to twice the frequency of modulating signal.

But, the frequency of male voice is less than that of female.

Q. 8. A basic communication system consists of (*A*) transmitter, (*B*) information source, (*C*) user of information, (*D*) channel and (*E*) receiver.

Choose the correct sequence in which these are arranged in a basic communication system :

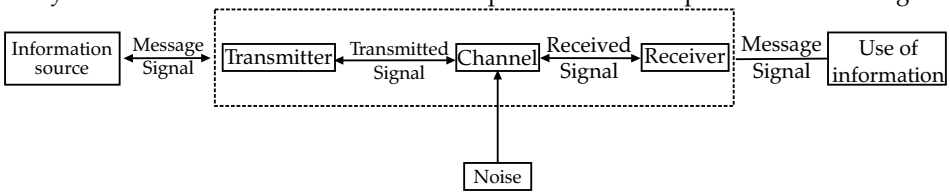
(a) *ABCDE*. **(b)** *BADEC*.

(c) *BDACE*. **(d)** *BEADC*. **[NCERT Exemp. Q. 15.8, Page 100]**

Ans. Correct option : (b)

Explanation : A basic communication system consists of an information source, a transmitter, a link (channel) and a receiver or a communication system is the set-up used in the transmission and reception of information from one place to another.

The whole system consists of several elements in a sequence. It can be represented as the diagram given below :



Q. 9. Identify the mathematical expression for amplitude modulated wave :

(a) $A_c \sin[\{\omega_c + k_1 v_m(t)\}t + \phi]$

(b) $A_c \sin\{\omega_c t + \phi + k_2 v_m(t)\}$

(c) $\{A_c + k_2 V_m(t)\}\sin(\omega_c t + \phi)$

(d) $A_c v_m(t)\sin(\omega_c t + \phi)$

[NCERT Exemp. Q. 15.9, Page 100]

Ans. Correct option : (c)

Explanation : Let the change in phase angle of the modulating signal ϕ. Now consider a sinusoidal modulating waveform signal $m(t)$ is represented by

$$m(t) = A_m \sin\omega_m t \quad \text{(i)}$$

Where, A_m = Amplitude of peak value of modulating signal.

ω_m = Angular frequency = $2\pi v_m = \phi(v_m)$

Where v_m is the amplitude of frequency.

Also consider a sinusoidal carrier wave $C(t)$ represented by

$$C(t) = A_c \sin\omega_c t \quad \text{(ii)}$$

Thus, modulated wave is given by

$$C_m(t) = (A_c + A_m \sin\omega_m t)\sin\omega_c t$$

$$= A_c\left(1 + \frac{A_m}{A_c}\sin\omega_m t\right)\sin\omega_c t$$

Here, $\frac{A_m}{A_c} = M$

$$C_m(t) = (A_c + A_c \times \mu \sin\omega_m t)\sin\omega_c t \quad \text{(iii)}$$

Now, we know that $A_c \times \mu = k$ [Wave constant]

And $\sin\omega_m t = v_m$

Thus Eq. (iii) becomes

$C_m(t) = (A_c + k \times v_m)\sin\ {}_c t$

Now consider change in phase angel by ϕ, then

$\sin\omega_c t \to \sin(\omega_c t + \phi)$

Thus, $C_m(t) = (A_c + Kv_m)(\sin\omega_c + \phi)$

Tick Two or More Options

Q. 10. An audio signal of 15 kHz frequency cannot be transmitted over long distances without modulation because

(a) the size of the required antenna would be at least 5 km which is not convenient.

(b) the audio signal cannot be transmitted through sky waves.

(c) the size of the required antenna would be at least 20 km, which is not convenient.

(d) effective power transmitted would be very low, if the size of the antenna is less than 5 km.

[NCERT Exemp. Q. 15.10, Page 100]

Ans. **Correct options :** (a), (b) and (d)

Explanation : Size of the antenna or aerial. For transmitting a signal, we need an antenna or an aerial. This antenna should have a size comparable to the wavelength of the signal (at least one-fourths in dimension) so that the antenna properly senses the time variation of the signal. For an electro-magnetic wave of frequency 20 kHz the wavelength, λ is 15 km. Obviously, such a long antenna is not possible to construct and operate. Hence direct transmission of such baseband signal is not practical. We can obtain transmission with reasonable antenna if transmission frequency is high (*e.g.*, if n is 1 MHz, then λ is 300 m).

Therefore, there is a need of translating the information contained in our original low-frequency baseband signal into high or radio-frequencies before transmission. Effective power radiated by an antenna; a theoretical study of radiation from a linear antenna (length, l) shows that power radiated is proportional to $(1/\lambda)^2$. This implies that for the same antenna length, the power radiated increases with decreasing λ, that is, increasing frequency. Hence, the effective power radiated by a long wavelength baseband signal would be small. For a good transmission, we need high powers and hence this also points out to the need of using high-frequency transmission.

According to the problem, frequency of the wave to be transmitted is

$v_m = 15 \text{ kHz} = 15 \times 10^3 \text{ Hz}$

Wavelength $\lambda_m = \frac{c}{\nu_m} = \frac{3\times10^8}{15\times10^3} = \frac{1}{5}\times10^5 \text{ m}$

Size of the antenna required, $L = \frac{\lambda}{4} = \frac{1}{4}\times\left(\frac{1}{5}\times10^5\right) = 5\times10^3 \text{ m} = 5 \text{ km}$

The audio signal cannot be transmitted through sky waves because in the sky wave propagation the frequency range is 2–40 MHz. Frequency of this wave is 15 kHz.

Effective power radiated by the antenna of length L. $P \propto \left(\frac{L}{\lambda}\right)^2$ if L decreases, P also decreases.

Q. 11. Audio sine waves of 3 kHz frequency are used to amplitude modulate a carrier signal of 1.5 MHz. Which of the following statements are true?

(a) The side band frequencies are 1,506 kHz and 1,494 kHz.

(b) The bandwidth required for amplitude modulation is 6 kHz.

(c) The bandwidth required for amplitude modulation is 3 MHz.

(d) The side band frequencies are 1,503 kHz and 1,497 kHz. **[NCERT Exemp. Q. 15.11, Page 101]**

Ans. **Correct options :** (b) and (d)

Explanation : Side band frequencies : The AM wave contains three frequencies f_C, $(f_C + f_m)$ and $(f_C - f_m)$, f_C is called carrier frequency, $(f_C + f_m)$ and $(f_C - f_m)$ are called side band frequencies.

$(f_C + f_m)$: Upper side band (USB) frequency

$(f_C - f_m)$: Lower side band (LSB) frequency

Side band frequencies are generally close to the carrier frequency.

According to the problem, frequency of audio sine wave $f_m = 3$ kHz and frequency of the carrier signal $f_C = 1.5$ MHz = 1,500 kHz.

Now, side band frequencies,

$f_C + f_m = (1{,}500 \pm 3) = 1{,}503$ kHz and 1,497 kHz

USB = 1,503 kHz and LSB = 1,497–1503 kHz

Also, band width = $2f_m = 2 \times 3 = 6$ kHz.

Q. 12. A TV transmission tower has a height of 240 m. Signals broadcast from this tower will be received by LOS communication at a distance of (assume the radius of earth to be 6.4×10^6 m)

(a) 100 km. **(b) 24 km.**

(c) 55 km. **(d) 50 km.**

[NCERT Exemp. Q. 15.12, Page 101]

Ans. **Correct options :** (b), (c) and (d)

Explanation : Distance, or range of transmission tower, $d_T = \sqrt{2Rh_T}$ where, R is the radius of the earth (~6,400 km), h_T is the height of transmission tower, d_T is also called the radio horizon of the transmitting antenna.

Height of tower $h = 240$ m

For LOS (line of sight) communication,

Range or the maximum distance on earth from the transmitter up to which a signal can be received is given by

$d = \sqrt{2Rh}$

Radius of earth, R – 6,400 km = 6.4×10^6 m (i)

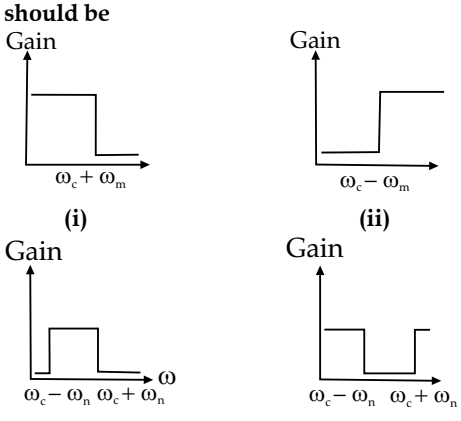
Putting all these value in equation (i), we get

$d = \sqrt{2Rh} = \sqrt{2 \times 6.4 \times 10^6 \times 240}$

$= 55.4 \times 10^3 \text{ m} = 55.4 \text{ km}$

Therefore, the range of 55.4 km covers the distance 24 km, 55 km and 50 km.

Q. 13. The frequency response curve (Figure) for the filter circuit used for production of AM wave should be



(a) (i) followed by (ii). **(b) (ii) followed by (i).**

(c) (iii). **(d) (iv).**

[NCERT Exemp. Q. 15.13, Page 101]

Ans. **Correct options :** (a), (b) and (c)

Explanation : (i) Side band frequencies : The AM wave contains three frequencies f_C, $(f_C + f_m)$ and $(f_C - f_m)$, f_C is called carrier frequency, $(f_C + f_m)$ and $(f_C - f_m)$ are called side band frequencies.

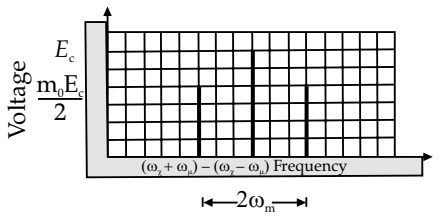
$(f_C + f_m)$: Upper side band (USB) frequency

$(f_C - f_m)$: Lower side band (LSB) frequency

Side band frequencies are generally close to the carrier frequency.

(ii) **Bandwidth :** The two side bands lie on either side of the carrier frequency at equal frequency interval ω_m.

So, bandwidth = $(\omega_C + \omega_m) - (\omega_C - \omega_m) = 2\omega_m$



To produce an amplitude modulated wave, bandwidth is given by the difference between USB frequency and LSB frequency. Bandwidth = $\omega_{USB} - \omega_{LSB} = (\omega_C + \omega_m) - (\omega_C - \omega_m)$.

Q. 14. In amplitude modulation, the modulation index m, is kept less than or equal to 1 because

(a) ***m* > 1, result in interference between carrier frequency and message frequency, resulting into distortion.**

(b) ***m* > 1 will result in overlapping of both side bands resulting into loss of information.**

(c) ***m* > 1 will result in change in phase between carrier signal and message signal.**

(d) ***m* > 1 indicates amplitude of message signal greater than amplitude of carrier signal resulting into distortion.**

[NCERT Exemp. Q. 15.14, Page 102]

Ans. **Correct options :** (b) and (d)

Explanation : Modulation index : The ratio of change of amplitude of carrier wave to the amplitude of original carrier wave is called the modulation factor or degree of modulation or modulation index (m_a).

$$m_a = \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} = \frac{kE_m}{E_e}$$

Where k = A factor which determines the maximum change in the amplitude for a given amplitude E_m of the modulating signal. If $k = 1$, then

$$m_a = \frac{E_m}{E_c} = \frac{E_{max} - E_{min}}{E_{max} + E_{min}}$$

If a carrier wave is modulated by several sine waves, the total modulated index m_t is given by,

$$m_t = \sqrt{m_1^2 + m_2^2 + m_3^2 + ...}$$

Frequency modulation index (m_f) :

The ratio of maximum frequency deviation to modulating frequency to the modulating frequency is called modulation index.

$$m_f = \frac{\delta}{f_m} = \frac{f_{max} - f_c}{f_m} = \frac{f_c - f_{min}}{f_m} = \frac{k_f E_m}{f_m}$$

So, $m = \frac{\text{Amplitude of message signal } (A_m)}{\text{Amplitude of carrier signal } (A_c)}$

If $m > 1$, then $A_m > A_c$.

There will be distortion of the resulting signal of amplitude modulated wave in this situation.

Maximum modulation frequency (m_f) of A_m wave is

$$m_f = \frac{\text{Carrier frequency deviation}}{\text{Modulating frequency}}$$

If $m_f > 1$, $\Delta f_{max} > f_m$. It means, there will be overlapping of both side bands of modulated wave resulting into loss of information.

Very Short Answer Type Questions (1 or 2 marks each)

Q. 1. Which of the following would produce analog signals and which would produce digital signals?

(i) A vibrating tuning fork

(ii) Musical sound due to a vibrating sitar string

(iii) Light pulse

(iv) Output of NAND gate

[NCERT Exemp. Q. 15.15, Page 102]

Ans. Analog and digital signals are the gateway of information or we can say that they are used to transmit information through electric signals. In both these signals, the information such as any audio or video is transformed into electric signals.

The difference between analog and digital technologies is that in analog technology information is translated into electric pulses of varying amplitude. In digital technology, translation of information is into binary format (0 or 1) where each bit is representative of two distinct amplitudes. So, output of a NAND gate and a light pulse produces a digital signal.

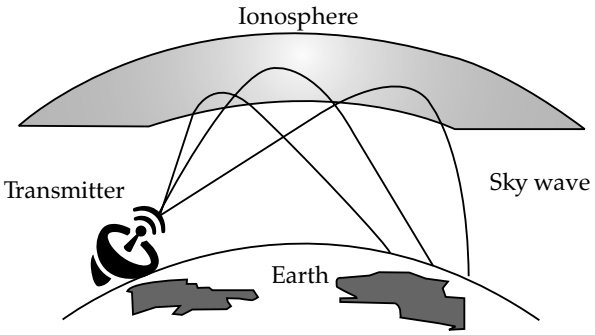
Thus, (*a*) and (*b*) would produce analog signal and (*c*) and (*d*) would produce digital signals. **[2]**

Q. 2. Would sky waves be suitable for transmission of TV signals of 60 MHz frequency?

[NCERT Exemp. Q. 15.16, Page 102]

Ans. A signal to be transmitted through sky waves must have a frequency range of 1,710 kHz to 40 MHz. But, here the frequency of TV signals is 60 MHz which is beyond the required range (frequency range : there is a maximum frequency of EM waves called critical frequency, above which wave cannot reflect back). So, sky waves will not be suitable for transmission of TV signals of 60 MHz frequency.

Important point : Sky wave propagation : These are the waves which are reflected back to the earth by ionosphere. Ionosphere is a layer of atmosphere having charged particles, ions and electrons and extended above 80–300 km from the earth's surface.



[2]

Q. 3. The maximum amplitude of an AM wave is found to be 15 V while its minimum amplitude is found to be 3 V. What is the modulation index?
[NCERT Exemp. Q. 15.18, Page 102]

Ans. The maximum amplitude of an AM wave is

$A_{max} = A_c + A_m = 15$ V (i)

And the minimum amplitude of an AM wave is

$A_{min} = A_c - A_m = 3$ V (ii)

Where, A_c and A_m be the amplitudes of carrier wave and modulating wave, respectively.

By adding Eqs. (i) and (ii), we get

$2A_c = 18$

or $A_c = 9$ V

And, $A_m = 15 - 9 = 6$ V

m or $\mu = \frac{\text{Amplitude of message signal } (A_m)}{\text{Amplitude of carrier signal } (A_c)}$

Modulating index of wave m or $\mu = \frac{A_m}{A_c} = \frac{6}{9} = \frac{2}{3}$ **[2]**

Q. 4. Compute the *LC* product of a tuned amplifier circuit required to generate a carrier wave of 1 MHz for amplitude modulation.
[NCERT Exemp. Q. 15.19, Page 102]

Ans. Frequency of tuned amplifier or resonance frequency is given by $\nu = \frac{1}{2\pi\sqrt{LC}}$, which generates a carrier wave of 1 MHz for amplitude modulation. So, we get

$$\frac{1}{2\pi\sqrt{LC}} = 1 \text{ MHz}$$

$$\sqrt{LC} = \frac{1}{2\pi \times 10^6}$$

Squaring both sides

$$LC = \frac{1}{(2\pi \times 10^6)^2} = 2.54 \times 10^{-14} \text{ s}$$

Thus, the product of LC is 2.54×10^{-14} s. **[2]**

Q. 5. Why is a AM signal likely to be more noisy than a FM signal upon transmission through a channel?
[NCERT Exemp. Q. 15.20, Page 102]

Ans. In case of AM, the instantaneous voltage of carrier waves is varied by the modulating wave voltage. So, during the transmission, noise signals can also be added and receiver assumes noise a part of the modulating signal.

In case of FM, the frequency of carrier waves is changed as the change in the instantaneous voltage of modulating waves. This can be done by mixing and not while the signal is transmitting in channel. So, noise does not affect FM signal.

Important point : In frequency modulation (m_f), (frequency modulation index) is inversely proportional to modulating frequency, f_m. while in PM it does not vary with modulating frequency. Moreover, FM is more noise immune. **[2]**

Q. 6. Which of the following frequencies will be suitable for beyond-the-horizon communication using sky waves?
(a) 10 kHz **(b) 10 MHz**
(c) 1 GHz **(d) 1,000 GHz**
[NCERT Ex. Q. 15.1, Page 530]

Ans. **Correct option :** (b)

Explanation : 10 MHz For beyond-the-horizon communication, it is necessary for the signal waves to travel a large distance. 10 kHz signals cannot be radiated efficiently because of the antenna size. The high energy signal waves (1−1,000 GHz) penetrate the ionosphere. 10 MHz frequencies get reflected easily from the ionosphere. Hence, signal waves of such frequencies are suitable for beyond-the-horizon communication.

Q. 7. Frequencies in the UHF range normally propagate by means of
(a) ground waves. **(b) sky waves.**
(c) surface waves. **(d) space waves.**
[NCERT Ex. Q. 15.2, Page 530]

Ans. **Correct option :** (d)

Explanation : Space waves owing to its high frequency, an ultra-high frequency (UHF) wave can neither travel along the trajectory of the ground, nor can it get reflected by the ionosphere. The signals having UHF are propagated through line-of-sight communication, which is nothing but space wave propagation.

Q. 8. Digital signals
(i) do not provide a continuous set of values,
(ii) represent values as discrete steps,
(iii) can utilise binary system, and
(iv) can utilise decimal as well as binary systems. Which of the above statements are true?
(a) (i) and (ii) only
(b) (ii) and (iii) only
(c) (i), (ii) and (iii) but not (iv)
(d) All of (i), (ii), (iii) and (iv).
[NCERT Ex. Q. 15.3, Page 530]

Ans. **Correct option :** (c)

Explanation : A digital signal uses the binary (0 and 1) system for transferring message signals. Such a system cannot utilise the decimal system (which corresponds to analogue signals). Digital signals represent discontinuous values.

Q. 9. A signal of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz. What are the frequencies of the side bands produced? [CBSE Board, All India Region, 2016]

Ans. Given frequency of carrier wave $f_c = 2$ MHz $= 2 \times 10^3$ kHz

Frequency of modulating signal $f_m = 5$ kHz

Frequency of lower side band (LSB) $= f_c - f_m$

LSB = $2 \times 10^3 - 5$ kHz = 1,995 kHz

Frequency of upper side band (USB) = $f_c + f_m$

USB = $2 \times 10^3 + 5$ kHz = 2,005 kHz **[1]**

Q. 10. Why is base band signal not transmitted directly? Give any two reasons.

[CBSE Board, All India Region, 2016]

Ans. The needs of modulation for transmission of a signal are given below :

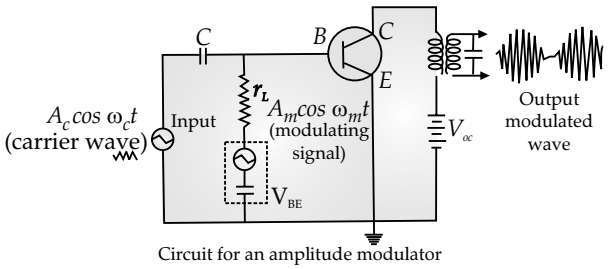
(i) The transmission of low-frequency signal needs antenna of height 4–5 km which is impossible to construct. So, there is need to modulate the wave in order to reduce the height of antenna to a reasonable height.

(ii) Effective power radiated by antenna for low wavelength or high-frequency wave as

$$P \propto \frac{1}{\lambda^2}$$

So, for effective radiation by antenna, there is need to modulate the wave.

Figure shows the amplitude modulation using a sinusoidal signal as the modulating signal.



Circuit for an amplitude modulator

[2]

Q. 11. Which mode of propagation is used by short wave broadcast services?

[CBSE Board, All India Region, 2018]

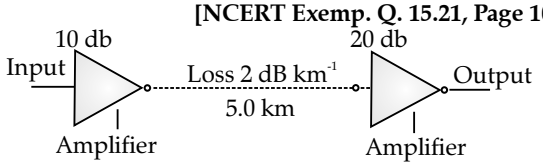
Ans. Sky wave propagation : When radio wave propagates from one place of earth to other after reflection by ionosphere, the range of frequencies from few MHz to 30 MHz gets reflected back by ionosphere. This range is also reflected as short-wave band. This mode of propagation is used by short-wave broadcast service. **[1]**

Short Answer Type Questions

(3 marks each)

Q. 1. Figure shows a communication system. What is the output power when input signal is of 1.01 mW? (Gain in $dB = 10 \log_{10}\left(\frac{P_o}{P_i}\right)$.

[NCERT Exemp. Q. 15.21, Page 103]



Ans. Given that, Loss suffered in path of transmission = 2 dB per km

And the distance travelled by the signal or path length is 5 km

So, total loss suffered in 5 km = $-2 \times 5 = -10$ dB

Total amplifier gain = 10 dB + 20 dB = 30 dB

Overall gain in signal = 30 dB – 10 dB = 20 dB

Here, it is given that gain in $dB = 10 \log_{10} \frac{P_0}{P_i}$

$$\therefore \quad 20 = 10 \log_{10} \frac{P_0}{P_i}$$

$$\text{or } \log_{10} \frac{P_0}{P_i} = 2$$

Input power is, P_i = 1.01 mW and P_0 is the out power.

$$\therefore \frac{P_0}{P_i} = 10^2 = 100$$

$$\Rightarrow P_0 = P_i \times 100 = 1.01 \times 100$$

P_0 = 101 mW

Thus, the output power is 101 mW. **[3]**

Q. 2. Two waves *A* and *B* of frequencies 2 MHz and 3 MHz, respectively are beamed in the same direction for communication via sky wave. Which one of these is likely to travel longer distance in the ionosphere before suffering total internal reflection? [NCERT Exemp. Q. 15.17, Page 102]

Ans. We know that refractive index μ of a layer is

$$\mu = \mu_0 \sqrt{1 - \frac{81.45\ \text{N}}{\nu^2}}$$

The refractive index of wave *B* is more than refractive index of wave A because frequency of wave *B* is more than wave A (as refractive index increases with frequency increases).

$\frac{\sin i}{\sin r} = \mu$ (Lesser the value of *r*, larger the value of μ)

For higher frequency wave (*i.e.*, higher refractive index) the angle of refraction is less, that is, bending is less. So, wave *B* travels longer distance in the ionosphere before suffering total internal reflection.

Importance point : Refractive index of a medium is that characteristic which decides speed of light in it. Dependence of refractive index :

(i) Nature of the media of incidence and refraction.

(ii) Colour of light or wavelength of light.

(iii) Temperature of the media : Refractive index decreases with the increase in temperature.

Total internal reflection : When a ray of light goes from denser to rarer medium it bends away from the normal and as the angle of incidence in denser medium increases, the angle of refraction in rarer medium also increases and at a certain angle, angle of refraction becomes 90°. This angle of incidence is called critical angle (*C*).

When angle of incidence exceeds the critical angle then light ray comes back into the same medium after reflection from interface. This phenomenon is called total internal reflection (TIR).

$$\mu = \frac{1}{\sin C} = \text{cosec } C$$

Conditions for TIR :

(i) The ray must travel from denser medium to rarer medium.

(ii) The angle of incidence i must be greater than critical angle C.

Dependence of critical angle :

(i) Colour of light (or wavelength of light) : Critical angle depends upon wavelength as $\lambda \propto \frac{1}{\mu} \propto \sin C$.

(a) $\lambda_R = \lambda_V \Rightarrow C_R > C_V$

(b) $\sin C = \frac{1}{{}_R\mu_D} = \frac{\mu_R}{\mu_D} = \frac{\lambda_D}{\lambda_R} = \frac{\nu_D}{\nu_R}$ (For two media)

(ii) Nature of the pair of the media : Greater than refractive index lesser will be the critical angle.

(a) For (glass-air) pair $\rightarrow C_{\text{glass}} = 42°$

(b) For (water-air) pair $\rightarrow C_{\text{water}} = 49°$

(c) For (diamond-air) pair $C_{\text{diamond}} = 49°$

(iii) **Temperature :** With temperature rise refractive index of the material decreases, therefore critical angle increases. **[3]**

Q. 3. (a) Give three reasons why modulation of a message signal is necessary for long distance transmission.

(b) Show graphically an audio signal, a carrier wave and an amplitude modulated wave.

[CBSE Board, All India Region, 2018]

Ans. (a) 1. Practical antenna length : To transmit a signal effectively, the height of antenna should be comparable to the wavelength of the signal at least $\frac{\lambda}{4}$ in length.

$$\text{Wavelength} = \frac{\text{Wave velocity}}{\text{Frequency}}$$

$$= \frac{3\times10^8}{20\times10^3 \text{ Hz}}$$

$$\lambda = 15\mu\text{m}$$

$$L = \frac{\lambda}{4} = 3.75 \text{ km}$$

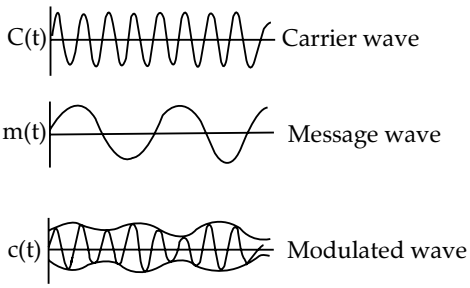
3.75 km antenna is practically not possible.

2. Effective power radiated by antenna :

$$\text{Power radiated} \propto \left(\frac{L}{\lambda}\right)^2$$

Thus, the power radiated by short wavelength would be large

3. **Multiplexing frequency :** In order to transmit different audio signals simultaneously through one antenna, each signal is translated to a low frequency range so that it is easily distinguishable from other signals at the receiving station. **[1½]**



[1½]

Q. 4. What is space wave propagation? State the factors which limit its range of propagation. Derive an expression for the maximum line of sight distance between two antennas for space wave propagation.

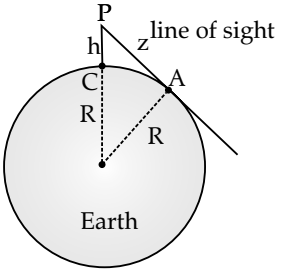
[CBSE Board, All India Region, 2016]

Ans. When a wave propagates in a straight line, from the transmitting antenna to the receiving antenna, its mode of propagation is called space wave communication.

Frequency range : Above 40 MHz.

Space waves are used for the line of sight (LOS) communication. Space wave communication involves the transmission from transmitter, traveling along a straight line in space, reaches to receiving antenna. The range of their frequencies is 40 MHz and above.

The range of space wave propagation is limited by line of sight distance between transmissions to receiver/repeater antenna.



$h <<< R$

h = Height of antenna

r = Radius of earth

Range = CA

As h is very small $CA \approx PA$

Using Pythagoras theorem

$PA^2 + R^2 = (R + h)^2$

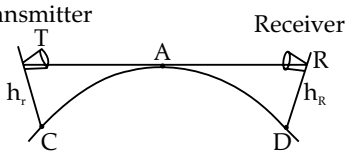
$PA^2 + R^2 = R^2 + 2Rh + h^2$

(Neglecting h^2)

$PA = \sqrt{2Rh}$

Range $\Rightarrow \sqrt{2Rh}$

On ground surface :

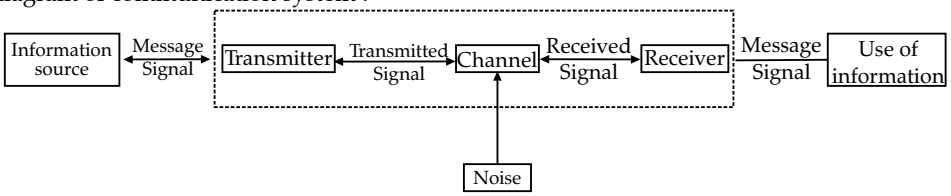


Total range on ground $= CA + AD$

$= \sqrt{2Rh_T} + \sqrt{2Rh_R}$ **[3]**

Q. 5. Draw a block diagram of a generalised communication system. Write the functions of each of the following :
(a) Transmitter (b) Channel
(c) Receiver [CBSE Board, All India Region, 2017]

Ans. Block diagram of communication system :



[1½]

(a) Transmitter : A transmitter is an arrangement that converts the message signal to a form suitable for transmission and then transmits it through some suitable communication channel.

(b) Channel : Channel is the medium through which the signal is transmitted from transmitter to receiver.

(c) Receiver : A receiver extracts the desired message signals from the received signals at the channel output. **[1½]**

Q. 6. (a) How is amplitude modulation achieved?

(b) The frequencies of two side bands in an AM wave are 640 kHz and 660 kHz respectively. Find the frequencies carrier and modulating signal. What is the bandwidth required for amplitude modulation?

[CBSE Board, All India Region, 2017]

Ans. (a) Amplitude modulation : In AM, the modulating wave is super-imposed on a carrier wave in such a manner that the frequency of the modulated wave is the same as that of the carrier wave but its amplitude varies in accordance with the instantaneous amplitude of the modulating wave.

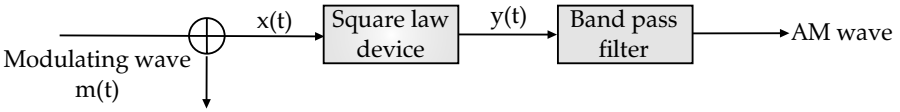


Figure. Block diagram for a simple modulator for obtaining an AM signal

Let the modulating signal be represent by

$m(t)=A_m\sin\omega_m t$

And carrier wave

$c(t)=A_c\sin\omega_c t$

When they added the resultant wave

$x(t)=A_m\sin\omega_m t+A_c\sin\omega_c t$

$$x(t)=A_c\left(1+\frac{A_m}{A_c}\sin\omega_m t\right)\sin\omega_c t$$

This can be further written as :

$x(t)=A_c\sin\omega_c t+\mu A_c\sin\omega_m t\times\sin\omega_c t$

$\mu=\frac{A_m}{A_c}$ is the modulation index.

In practice, $\mu \le 1$ to avoid distortion and it is represented in percentage.

Using trigonometric relation,

$$\sin A\ \sin B=\frac{1}{2}[\cos(A-B)-\cos(A+B)]$$

So,

$$x(t)=A_c\sin\omega_c t+\frac{\mu A_c}{2}\cos(\omega_c-\omega_m)t-\frac{\mu A_c}{2}\cos(\omega_c+\omega_m)t.$$

(b) Given : $\omega_c+\omega_m=660$ kHz …(1)

And $\omega_c-\omega_m=640$ kHz …(2)

By adding equation (1) and equation (2), we have

$2\omega_c=660+640$ kHz

$2\omega_c=1300$ kHz

$$\omega_c=\frac{1300}{2}=650\text{ kHz}$$

Then $\omega_m=650-640$

$\omega_m=10$ kHz

$\Rightarrow \quad \omega_c=2\pi f_c=650$ kHz $\quad (\because \omega=2\pi f)$

$\Rightarrow \quad f_c=\frac{650}{2\pi}$ kHz

$\Rightarrow \quad \omega_m=2\pi f_m=10$ kHz

$\Rightarrow \quad f_m=\frac{10}{2\pi}$ kHz

$\Rightarrow$ Band width required for amplitude modulation

= Upper side band – Lower side band

$=(f_c+f_m)-(f_c-f_m)=2f_m$ **[3]**

Q. 7. Is it necessary for a transmitting antenna to be at the same height as that of the receiving antenna for line-of-sight communication? A TV transmitting antenna is 81 m tall. How much service area can it cover if the receiving antenna is at the ground level? [NCERT Ex. Q. 15.4, Page 530]

Ans. Line-of-sight communication means that there is no physical obstruction between the transmitter and the receiver. In such communications, it is not necessary for the transmitting and receiving antennas to be at the same height.

Height of the given antenna, $h = 81$ m

Radius of earth, $R = 6.4 \times 10^6$ m

For range, $d = \sqrt{2Rh}$,

The service area of the antenna is given by the relation : $A = nd^2 = n(2Rh)$

$= 3.14 \times 2 \times 6.4 \times 10^6 \times 81$

$= 3255.55 \times 10^6\ m^2 = 3255.55 \sim 3256\ km^2$ **[3]**

Q. 8. A carrier wave of peak voltage 12 V is used to transmit a message signal. What should be the peak voltage of the modulating signal in order to have a modulation index of 75%?

[NCERT Ex. Q. 15.5, Page 530]

Ans. Amplitude of the carrier wave, A_c = 12 V
Modulation index, m = 75% = 0.75
Amplitude of the modulating wave = A_m
Using the relation for modulation index :

$$m=\frac{A_m}{A_c}$$

Therefore, $A_m = mA_c$

$$= 0.75 \times 12 = 9 \text{ V} \quad \textbf{[3]}$$

Q. 9. A TV transmission tower antenna is at a height of 20 m. How much service area can it cover if the receiving antenna is (i) at ground level, (ii) at a height of 25 m? Calculate the percentage increase in area covered in case (ii) relative to case (i).

[NCERT Exemp. Q. 15.22, Page 103]

Ans. Distance, or range of transmission tower, $d_T=\sqrt{2Rh_T}$ where, R is the radius of the earth (~6,400 km), h_T is the height of transmission tower, d_T is also called the radio horizon of the transmitting antenna.
In this problem, height of antenna h_T = 20 m
Radius of earth $d_T = 6.4 \times 10^6$ m

(i) Distance of range $d_T=\sqrt{2h_Td_T}=\sqrt{2\times20\times6.4\times10^6}$
= 1,6000 m = 16 km
Area covered A = $\pi(d)^2$
$=3.14 \times 16 \times 16 = 803.84 \text{ km}^2$

(ii) At a height of h_R = 25 m ground level
Distance of range $d_M=\sqrt{2h_T}+\sqrt{2h_R}$

$$=\sqrt{2\times20\times6.4\times10^6}+\sqrt{2\times25\times6.4\times10^6}$$
$$=16\times10^3+17.9\times10^3$$
$$=33.9\times10^3 \text{ m}$$
$$=33.9 \text{ km}$$

Area covered = $\pi(d)^2$
$= 3.14 \times 33.9 \times 33.9$
$= 3608.52 \text{ km}^2$

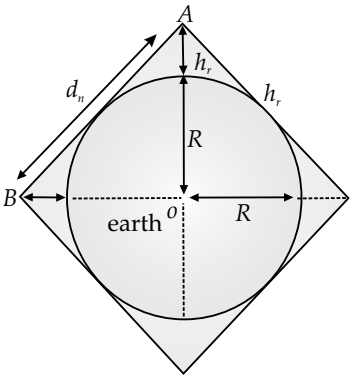
Hence, percentage increase in area

$$=\frac{\text{Difference in area}}{\text{Initial area}}\times100$$
$$=\frac{(3608.52-803.84)}{803.84}\times100$$
$$=348.9\% \quad \textbf{[3]}$$

Q. 10. If the whole earth is to be connected by *LOS* communication using space waves (no restriction of antenna size or tower height), what is the minimum number of antennas required? Calculate the tower height of these antennas in terms of earth's radius?

[NCERT Exemp. Q. 15.23, Page 103]

Ans. Distance or range of transmission tower, $d_T=\sqrt{2Rh_T}$ where, R is the radius of the earth (~6,400 km). h_T is the height of transmission tower,
Because d_T is also called the radio horizon of the transmitting antenna.
Let us consider the figure given below to solve this problem.



Assume the height of transmitting antenna or receiving antenna in order to cover the entire surface of earth through communication is h_T, that is, $h_T = h_R$ and radius of earth is. R. If dm is the line-of-sight distance between the transmission and receiving antennas, then maximum distance :

$$d_m=\sqrt{2h_TR}+\sqrt{2h_TR}=2\sqrt{2h_TR}$$
$$d_m^2=(R+h_T)^2+(R+h_T)^2$$
$$=2(R+h_T)^2$$
$$\Rightarrow \quad 8h_TR=2(R+h_T)^2$$
$$\Rightarrow \quad 4h_TR=R^2+2Rh_T+h_T^2$$
$$\Rightarrow R^2-2h_TR+h_T^2=0$$
$$\Rightarrow \quad (R-h_T)^2=0$$
$$\Rightarrow \quad R=h_T$$

As, space wave frequency is used, so $\lambda << h_T$ hence only tower height is to be taken into consideration. In three dimensions of earth, six antenna towers of each of height, $h_T = R$ would be used to cover the entire surface of earth with communication programme. **[3]**

Q. 11. The maximum frequency for reflection of sky waves from a certain layer of the ionosphere is found to be fmax = 9(N_{max})$\frac{1}{2}$, where N_{max} is the maximum electron density at that layer of the ionosphere. On a certain day it is observed that signals of frequencies higher than 5 MHz are not received by reflection from the F_1 layer of the ionosphere while signals of frequencies higher than 8 MHz are not received by reflection from the F_2 layer of the ionosphere. Estimate the maximum electron densities of the F_1 and F_2 layers on that day. **[NCERT Exemp. Q. 15.24, Page 103]**

Ans. Given that, the maximum frequency for reflection of sky waves is $f_{max}=9(N_{max})^{\frac{1}{2}}$
Where, N_{max} is a maximum electron density.
For layer F_1, f_{max} = 5 MHz
So, $5\times10^6=9(N_{max})^{\frac{1}{2}}$
This implies, $N_{max}=\left(\frac{5}{9}\times10^6\right)^2=3.086\times10^{11} \text{ per m}^3$
Then for layer F_2, f_{max} = 8 MHz
So, $8\times10^6=9(N_{max})^{\frac{1}{2}}$
This implies, $N_{max}=\left(\frac{8\times10^6}{9}\right)^2=7.9\times10^{11} \text{ per m}^3$ **[3]**

Q. 12. On radiating (sending out) an AM modulated signal, the total radiated power is due to energy carried by ω_c, $\omega_c - \omega_m$ and $\omega_c + \omega_m$. Suggest ways to minimise cost of radiation without compromising on information.

[NCERT Exemp. Q. 15.25, Page 103]

Ans. Side band frequencies : The AM wave contains three frequencies ω_C, $(\omega_C + \omega_m)$ and $(\omega_C - \omega_m)$, ω_C is called carrier frequency, $(\omega_C + \omega_m)$ and $(\omega_C - \omega_m)$ are called side band frequencies.

$(\omega_C + \omega_m)$: Upper side band (USB) frequency

$(\omega_C - \omega_m)$: Lower side band (LSB) frequency

Side band frequencies are generally close to the carrier frequency.

Only side band frequencies contain information in amplitude modulated signal [only $(\omega_C + \omega_m)$ and $(\omega_C + \omega_m)$]

Here, the total radiated power is due to energy carried by ω_C, $(\omega_C - \omega_m)$ and $(\omega_C + \omega_m)$.

For reduction of cost of radiation without compromising on information ω_C can be left and transmitting the frequencies $(\omega_C + \omega_m)$, $(\omega_C - \omega_m)$ or both $(\omega_C - \omega_m)$ and $(\omega_C + \omega_m)$. **[3]**

Q. 13. For an amplitude modulated wave, the maximum amplitude is found to be 10 V while the minimum amplitude is found to be 2 V. Determine the modulation index μ. What would be the value of μ if the minimum amplitude is zero volt?

[NCERT Ex. Q. 15.7, Page 531]

Ans. Maximum amplitude, $A_{max} = 10$ V

Minimum amplitude, $A_{min} = 2$ V

Modulation index μ, is given by the relation :

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

$$= \frac{10-2}{10+2} = \frac{8}{12} = 0.67$$

If $A_{min} = 0$

Then $\mu = \frac{A_{max}}{A_{max}} = \frac{10}{10} = 1$ **[3]**

Q. 14. Due to economic reasons, only the upper sideband of an AM wave is transmitted, but at the receiving station, there is a facility for generating the carrier. Show that if a device is available which can multiply two signals, then it is possible to recover the modulating signal at the receiver station.

[NCERT Ex. Q. 15.8, Page 531]

Ans. Let ω_c and ω_s be the respective frequencies of the carrier and signal waves.

Signal received at the receiving station, $V = V_1 \cos(\omega_c + \omega_s)t$

Instantaneous voltage of the carrier wave, $V_{in} = V_c \cos\omega_c t$

$$\therefore VV_{in} = V_1 \cos(\omega_c + \omega_s)t.(V_c \cos\omega_c t)$$

$$= V_1 V_c [\cos(\omega_c + \omega_s)t.\cos\omega_c t]$$

$$= \frac{V_1 V_c}{2} [2\cos(\omega_c + \omega_s)t.\cos\omega_c t]$$

$$= \frac{V_1 V_c}{2} [\cos\{(\omega_c + \omega_s)t + \omega_c t\} + \cos\{(\omega_c + \omega_s)t - \omega_c t\}]$$

$$= \frac{V_1 V_c}{2} [\cos\{(2\omega_c + \omega_s)t + \cos\omega_s t\}]$$

At the receiving station, the low-pass filter allows only high frequency signals to pass through it. It obstructs the low-frequency signal ω_s. Thus, at the receiving station, one can record the modulating signal $\frac{V_1 V_c}{2}\cos\omega_s t$ which is the signal frequency. **[3]**

Long Answer Type Questions (5 marks each)

Q. 1. (i) The intensity of a light pulse travelling along a communication channel decreases exponentially with distance x according to the relation $I = I_o e^{-\alpha x}$, where I_0 is the intensity at $x = 0$ and α is the attenuation constant. Show that the intensity reduced by 75% after a distance of $\left\{\frac{\ln(4)}{\alpha}\right\}$

(ii) Attenuation of a signal can be expressed in decibel (dB) according to the relation $dB = 10 \log_{10}\left\{\frac{I}{I_a}\right\}$.

What is the attenuation in dB per km for an optical fibre in which the intensity falls by 50% over a distance of 50 km?

[NCERT Exemp. Q. 15.26, Page 104]

Ans. (i) According to the problem, the intensity of a light pulse travelling along a communication channel is given by

$I = I_0 e^{-\alpha x}$

Where I_0 is the intensity at $x = 0$ and $\propto$ is attenuation constant.

If the intensity reduced by 75% that means,

I = 25% of I_0

So, we can write this as $I = \frac{25}{100} . I_0 = \frac{I_0}{4}$

By using the above relation mentioned in the problem,

$$I = I_0 e^{-\alpha x}$$

$$\frac{I_0}{4} = I_0 e^{-\alpha x}$$

Or, $\frac{1}{4} = e^{-\alpha x}$

Taking natural log on both sides, we get

$\ln(1) - \ln(4) = \alpha \times x \ln e$

$[\because \ln(e) = 1 \text{ and } \ln(1) = 0]$

$\Rightarrow -\ln(4) = -\alpha x$

$\Rightarrow \quad x = \frac{\ln(4)}{\alpha}$

Therefore, at distance $x = \frac{\ln 4}{\alpha}$, the intensity is reduced to 75% of initial intensity. **[2½]**

(ii) Here, α is the attenuation constant expressed in *dB* per km. If x is distance travelled by signal, then

$$10\log_{10}\left(\frac{I}{I_0}\right)=-\alpha x \quad ...(i)$$

Where, I_0 is the initial intensity.

Here 50% intensity reduced by distance 50 km. So, I=50% of I_0

$I-\frac{I_0}{2}$ and x=50 km

Substituting the value of x in Eq. (i)

$$10\log_{10}\frac{I_0}{2I_0}=-\alpha\times 50$$

$$10\,[\log(1)-\log(2)]=-50\alpha$$

$$\frac{10\times\log(2)}{50}=\alpha$$

$$\frac{0.3010}{5}=\alpha$$

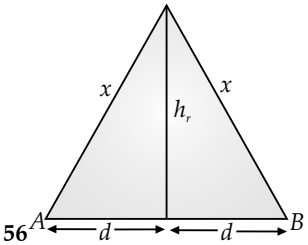
$\therefore$ The attenuation for given optical fibre

α=0.0602 dB per km **[2½]**

Q. 2. A 50 MHz sky wave takes 4.04 ms to reach a receiver via re-transmission from a satellite 600 km above earth's surface. Assuming re-transmission time by satellite negligible, find the distance between source and receiver. If communication between the two was to be done by Line of Sight (LOS) method, what should size and placement of receiving and transmitting antenna be?

[NCERT Exemp. Q. 15.27, Page 104]

Ans. Let the receiver is at point *A* and source is at *B*.



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Velocity of waves = 3×10^8 m/s

Time to reach a receiver = 4.04×10^{-3} s

Let the height of satellite is h_s =600 km

Radius of earth = 6,400 km

Size of transmitting antenna = h_r

$$\text{Velocity of waves}=\frac{\text{Distance travelled by wave}}{\text{Time}}$$

$$\frac{2x}{4.04\times10^{-3}}=3\times10^8$$

$$\text{Or, } x=\frac{3\times10^8\times4.04\times10^{-3}}{2}=6.06\times10^5=606\text{ km}$$

Using Pythagoras theorem,

$d^2=x^2-h_s^2=(606)^2-(600)^2=7{,}236$

or, d = 85.06 km

So, the distance between source and receiver $=2d$

$= 2\times85.06=170$ km

The maximum distance covered

$d=\sqrt{2Rh_T}$

Size of antenna or $\frac{d^2}{2R}=h_T$

$$\text{Or, } h_T=\frac{7{,}236}{2\times6{,}400}=0.565\text{ km}=565\text{ m}$$ **[5]**

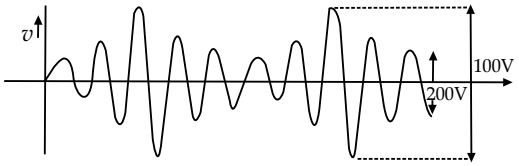
Q. 3. An amplitude modulated wave is as shown in Figure. Calculate :

(i) the percentage modulation,

(ii) peak carrier voltage and,

(iii) peak value of information voltage.

[NCERT Exemp. Q. 15.28, Page 104]



Ans. It is observed from the above diagram that

Maximum voltage $V_{max}=\frac{100}{2}=50$ V

Minimum voltage $V_{min}=\frac{20}{2}=10$ V

(i) Percentage modulation, $\mu=\frac{V_{max}-V_{min}}{V_{max}+V_{min}}\times100$

$$=\frac{50-10}{50+10}\times100$$

$$=\frac{40}{60}\times100=66.67\%$$ **[1½]**

(ii) Peak carrier voltage, $V_c=\frac{V_{max}+V_{min}}{2}$

$$=\frac{50+10}{2}=30\text{ V}$$ **[1½]**

(iii) Peak value of information voltage,

$$V_m=\mu V_c=\frac{66.67}{100}\times30=20\text{ V}$$ **[2]**

Q. 4. (i) Draw the plot of amplitude versus 'ω' for an amplitude modulated wave whose carrier wave ($\omega > \omega_c$) is carrying two modulating signals, ω_1 and ω_2 ($\omega_2 > \omega_1$).

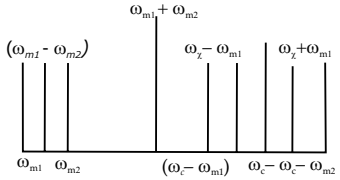
(ii) Is the plot symmetrical about ω_c? Comment especially about plot in region $\omega < \omega_c$.

(iii) Extrapolate and predict the problems one can expect if more waves are to be modulated.

(iv) Suggest solutions to the above problem. In the process can one understand another advantage of modulation in terms of bandwidth?

[NCERT Exemp. Q. 15.29, Page 105]

Ans. (i)



[2]

(ii) In the plotted graph shown, we note that frequency spectrum is not symmetrical about W_c. Crowding of spectrum is present for $\omega < \omega_c$. **[1]**

(iii) If more modulating signals are present then there will be more crowding in the modulation signal in the region $\omega < \omega_c$. That will result more chances of mixing of signal. **[1]**

(iv) To accommodate more signals, we should increase bandwidth and frequency carrier waves ω_c. This shows that large carrier frequency enables to carry more information (*i.e.*, more ω_m) and the same will in turn increase bandwidth. **[1]**

Q. 5. An audio signal is modulated by a carrier wave of 20 MHz such that the bandwidth required for modulation is 3 kHz. Could this wave be demodulated by a diode detector which has the values of *R* and *C* as

(i) $R = 1\ \text{k}\Omega, C = 0.01\ \mu\text{F}$.

(ii) $R = 10\ \text{k}\Omega, C = 0.01\ \mu\text{F}$.

(iii) $R = 10\ \text{k}\Omega, C = 0.1\ \mu\text{F}$

[NCERT Exemp. Q. 15.30, Page 105]

Ans. According to the problem, carrier wave frequency

$f_c = 20\ \text{MHz}$

$= 20 \times 10^6\ \text{Hz}$

Bandwidth required for modulation is

$$f_m = \frac{3\times10^3}{2} = 1.5\times10^3\ \text{Hz}$$

Demodulation of diode is possible if the condition

$$\frac{1}{f_c} << RC < \frac{1}{f_m} \text{ is satisfied}$$

Thus, $\frac{1}{f_c} = \frac{1}{20\times10^6}\text{Hz} = 0.5\times10^{-7}\ \text{s}$ ...(i)

And $\frac{1}{f_m} = \frac{1}{1.5\times10^3}\ \text{Hz} = 0.7\times10^{-3}\ \text{s}$ ...(ii)

Now, gain through all the options of R and C one by one, we get

$RC = 1\ k\Omega \times 0.01\ \mu\text{F}$

(i) $RC = 10^3\ \Omega \times (0.01\times10^{-6}\ \text{F})$

$= 10^{-5}\ \text{s}$

Here, condition $\frac{1}{f_c} << RC < \frac{1}{f_m}$ is satisfied.

Hence it can be demodulated. **[1½]**

$RC = 10\ k\Omega \times 0.01\ \mu\text{F}$

(ii) $RC = 10^4\ \Omega \times 10^{-8}\ \text{F}$

$= 10^{-4}\ \text{s}$

Here, condition $\frac{1}{f_c} << RC < \frac{1}{f_m}$ is satisfied.

Hence it can be demodulated. **[1½]**

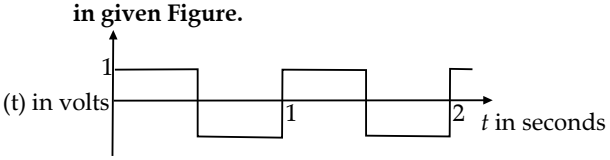
$RC = 10\ \text{K}\Omega \times 1\ \mu\text{F}$

(iii) $RC = 10^4\ \Omega \times 10^{-12}\ \text{F}$

$= 10^{-8}\ \text{s}$

Here, condition $\frac{1}{f_c} < RC$, so this cannot be demodulated. **[2]**

Q. 6. A modulating signal is a square wave, as shown in given Figure.



The carrier wave is given by $c(t) = 2\sin(8\pi t)$ volts

(i) Sketch the amplitude modulated waveform

(ii) What is the modulation index?

[NCERT Ex. Q. 15.6, Page 531]

Ans. It can be observed from the given modulating signal that the amplitude of the modulating signal, $A_m = 1\ \text{V}$

It is given that the carrier wave, $c(t) = 2\sin(8\pi t)$

$\therefore$ Amplitude of the carrier wave, $A_c = 2\ \text{V}$

Time period of the modulating signal, $T_m = 1\ \text{s}$

The angular frequency of the modulating signal is calculated as :

$$\omega_m = \frac{2\pi}{T_m}$$

$= 2\pi\ \text{rad s}^{-1}$ ...(i)

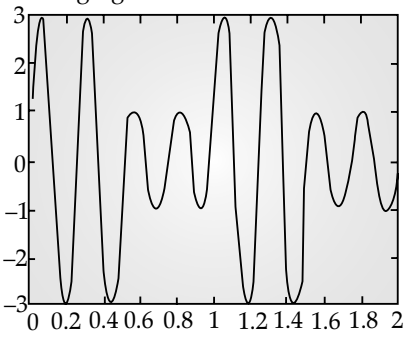
(i) The angular frequency of the carrier signal is calculated as :

$\omega_c = 8\pi\ \text{rad s}^{-1}$ ...(ii)

From equations (i) and (ii), we get :

$\omega_c = 4\ \omega_m$

The amplitude modulated waveform of the modulating signal is shown in the following figure.



[2½]

(ii) Modulation index,

$$m = \frac{A_m}{A_c} = \frac{1}{2} = 0.5$$

[2½]

Some Commonly Made Errors

- Students ignore to learn the initial concept of Communication systems.

EXPERT ADVICE

- ☞ *Try to make the flow chart or mind map to make the Concept/Derivations in easier way.*
- ☞ *Always study in Chronological rather than Priority order.*
- ☞ *Try to learn Qualitative than Quantitative.*
- ☞ *There are some concept-based questions also, so understand the concept thoroughly. It will help you in your exams while writing answers to questions.*
- ☞ *Don't leave any important topics from Communication systems.*
- ☞ *Solve a lot of question papers and sample papers to have a check on the knowledge you possess.*

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